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ABSTRACTS

CLINICAL VIGNETTES

SUCCESSFUL TREATMENT OF RECURRENT LEUKOCYTCLASTIC VASCULITIS WITH CELOCOXIB IN A CYSTIC FIBROSIS PATIENT. C. Vergara¹, R. Knauff²; ¹University of Connecticut Health Center, Farmington, CT; ²Hartford Hospital, Hartford, CT

LEARNING OBJECTIVES: Recognize the occurrence of leukocytoclastic vasculitis in cystic fibrosis patients. Recognize arthralgias and tendonitis associated with such conditions. Treat such conditions with Celocoxib - a COX 2 inhibitor nonsteroidal anti-inflammatory agent.

CASE INFORMATION: Case: We present a 24 year old white female with cystic fibrosis with recurrent pulmonary infections and associated rashes on her extremities. The rashes are on her legs as well as her arms. They are always associated with wax and wane with acute exacerbations of lung disease (infections). They are small circumscribed pruritic red lesions which became palpable within a week of appearance. This episode, in addition to the rash she had polyarthralgias and tendonitis of the distal left pronator tendons. Lab results: WBC 15,000/ μ L, Hemoglobin 13 gm/dL, Platelets 254,000/ μ L, serum chemistry within normal limits (WNL), coagulation parameters WNL, urinalysis WNL, ESR 100 mm/hr, serum ANA/Rf negative, C-ANCA negative, sputum cultures grew *Pseudomonas aeruginosa*, Complement C3 was low, Cryoglobulins negative. Biopsy Results: Skin biopsy results showed angiocentric segmental inflammation, endothelial cell swelling, fibrinoid necrosis of postcapillary venules with cellular infiltrate around and within dermal blood vessel walls composed mostly of neutrophils having fragmentation of nuclei (leukocytoclasia). Immunofluorescence and ultrastructural studies documented immunoglobulins (IgG), complement components (C1q, C3) and fibrin deposits within postcapillary venule walls. Treatment: She had previously been treated with high dose oral corticosteroids, IVIG G as an immune modulator, ibuprofen (800 three times a day) all with poor tolerance and many side effects. After treating her with Celocoxib 200mg a day, her arthralgias disappeared in 2 days and her purpura by the third day. Further recurrences treated similarly were also decreased in extent, duration and severity.

DISCUSSION: Leukocytoclastic vasculitis is a vasculitis of the post capillary venules resulting in palpable purpura and a hallmark histopathologic of angiocentric segmental inflammation, endothelial cell swelling, fibrinoid necrosis of post capillary venules. The cellular infiltrate around and within dermal blood vessel walls is composed largely of neutrophils showing fragmentation of nuclei (karyorrhexis or leukocytoclasia). Leukocytoclastic vasculitis is manifested clinically by a spectrum of cutaneous lesions, although "palpable purpura" is its clinical hallmark. Although removal of the cause is the definitive treatment, it can be symptomatically treated with a variety of immunomodulating agents and/or anti-inflammatory drugs. We found the use of celocoxib - a COX 2 inhibitory nonsteroidal anti-inflammatory agent without any significant gastrointestinal or antiplatelet actions to be very useful in the management of such a case.

UNILATERAL RETROPERITONEAL FIBROSIS AS ABDOMINAL MASS. M. Abouvousse¹; ¹University of Pittsburgh, Pittsburgh, PA

LEARNING OBJECTIVES: Recognizing Unilateral Retroperitoneal Fibrosis as a cause for hydronephrosis, and functional constipation.

CASE INFORMATION: A 47-year-old black male previously in good health presented with three weeks of left flank pain associated with constipation. His past medical history is significant for type 2 diabetes mellitus on metformin. Preliminary work up revealed increase in creatinine to 1.6. CAT scan left hydronephrosis and hydronephrosis, with soft tissue fullness, 4 cm in the largest diameter. A laparotomy was performed, with excision of fullness around the ureter. Final pathology of an abdominal lymph node and tissue from the mass revealed retroperitoneal. Twelve months later he remained asymptomatic and follow-up CT scan showed no evidence of recurrence.

SEVERE HYPERNATREMIA DUE TO A HIGH CALORIE MEAL SUPPLEMENT. D. Addamo¹; ¹Mount Sinai School of Medicine, New York, NY

LEARNING OBJECTIVES: 1) Raise index of suspicion for severe metabolic derangement in elderly patients. 2) Recognize clinical signs of hypernatremia. 3) Recognize limitations of using concentrated supplements as meal replacement.

CASE INFORMATION: A 79 year old woman with advanced Alzheimer's disease was brought to the E.D. due to a 4-week progressive decline in her overall level of function. She had been hospitalized 8 weeks prior for an aspiration pneumonia. Her daily fluid intake since that time consisted of six 8oz cans of a "balanced nutrition shake", along with some soup. 4 days prior she was given antibiotics by her primary physician. She showed no improvement and developed diarrhea. Physical exam revealed an obtunded woman. Her pulse was 120bpm and rectal temperature was 38.9 Celsius. She was normotensive. She had dry oral mucosa and skin tenting. Her neurological exam was nonfocal. Lab studies were significant for sodium of 185mEq/L, chloride of 148mEq/L, BUN of 124mg/dL and a creatinine of 3.0mg/dL. Urine osmolality was 819mOsmol per kg with urine sodium of 22mMol/L. Her WBC was 13K. Labs from her prior admission revealed sodium of 143mEq/L; chloride of 106mEq/L; BUN and creatinine were 7mg/dL and .7mg/dL respectively. Her free water deficit was calculated at 9 liters, and she was placed on D5W1/2NS at 150cc/hr. Her tachycardia and fever resolved. She began keeping her eyes open and speaking some words. On the sixth hospital day serum sodium was 146 with normal BUN and creatinine. The patient was discharged on a diet of purees and thickened liquids.

DISCUSSION: Patients with dementia are at a risk for hypernatremia due to their inability to verbalize a need for water or to obtain it for themselves. Hypernatremic dehydration can lead to mental status changes that are easily missed in a severely demented patient. While canned high calorie shakes are an attractive supplement for patients with poor oral intake, they are highly concentrated and contain little water. This renders them a poor substitute for total meal replacement.

SPINAL METASTASIS AS THE INITIAL PRESENTATION OF A NON-SECRETORY GLUCAGONOMA. A. Aggarwal¹, J. Brainard¹, D. Brotman¹; ¹Cleveland Clinic Foundation, Cleveland, OH

LEARNING OBJECTIVES: 1. Recognize that glucagonomas can present with spinal metastases. 2. Appreciate that the absence of cutaneous, endocrinologic and gastrointestinal manifestations does not rule out the presence of a glucagonoma. 3. Know that tissue staining for neuroendocrine peptides can confirm that a tumor is of pancreatic origin, even when there is no radiographically identifiable pancreatic mass.

CASE INFORMATION: A 38 year-old African-American woman presented to the emergency department with low back pain of three months duration. The pain involved both flanks and radiated to the buttocks and posterior thighs bilaterally, as well as to the right groin and medial right thigh. By exam, she was thin and appeared uncomfortable. Shotty nodes were palpable in multiple regions, but none were fixed or enlarged. There was percussion tenderness of the lumbar spine, but motor function and reflexes were normal in the lower extremities. Serum chemistries were notable for an elevated alkaline phosphatase but otherwise normal liver enzymes. Serum calcium and glucose were normal. CT of the abdomen and pelvis showed peri-pancreatic lymphadenopathy and numerous hepatic lesions suggesting metastases. MRI of the spine demonstrated a destructive lesion in the second lumbar vertebra with soft-tissue extension compressing the spinal cord (figure). Whole body bone scan showed diffuse osseous metastases to the upper and lower extremities as well as the axial skeleton. Ultrasound-guided biopsy of the peri-pancreatic adenopathy revealed a low-grade neuroendocrine tumor that stained positive for glucagon, but stained negative for serotonin, pancreatic polypeptide and gastrin. The serum glucagon level was normal. Due to widespread metastases, the patient was deemed a poor surgical candidate, however she did undergo radiation therapy to the spine and to painful lesions in the right humerus and left femur. Subsequently she was treated with systemic chemotherapy consisting of adriamycin and streptozocin with no initial reduction in tumor mass.

DISCUSSION: Glucagonomas have previously been reported to metastasize to bone, but to our knowledge there is only one other reported case of bony metastases as the initial manifestation

of the tumor. This may be because most glucagonomas come to early clinical attention via the effects of high serum levels of glucagon. It has previously been reported that systemic and endocrine manifestations of glucagonomas are more common in advanced disease and are directly related to tumor size. This is clearly not the case in our patient. Notably absent were the typical skin rash (necrolytic migratory erythema), diarrhea, hyperglycemia, stomatitis and chelosis associated with the classic glucagonoma syndrome. Another important clinical feature in our patient was the absence of a radiographically identifiable pancreatic mass despite widespread metastases. While the absence of a clear primary source is not an uncommon situation in cancer patients, the identification of the primary tumor carries particular importance for neuroendocrine malignancies. Metastatic small-cell carcinoma of the lung has a more rapid progression than both carcinoid and islet cell tumors, and the treatment is different for these three types of malignancies. In our patient, tissue staining for neuroendocrine markers allowed a specific diagnosis to be made and appropriate treatment to be instituted. We conclude that the absence of clinical evidence of hormonal activity should not dissuade the physician from considering the possibility of a pancreatic islet cell primary when a patient presents with metastatic neuroendocrine malignancy, even when there is no radiographic evidence of pancreatic mass. Stains for neuroendocrine peptides are of crucial importance to making the diagnosis in such patients.

ALL ASTHMA IS NOT ASTHMA: LUPUS MIMICKING AN ASTHMA EXACERBATION. G. Agrestia¹, J. Wiese¹, ¹Tulane University, New Orleans, LA

LEARNING OBJECTIVES: To recognize the limitations of pericardiocentesis in diagnostic evaluation.

CASE INFORMATION: An 48 y/o woman with a history of asthma presented with shortness of breath and a dry non-productive cough, fever, night sweats, and back pain. Her symptoms responded to bronchodilators and steroids. She had a blood pressure of 173/115 mmHg, a heart rate of 119 beats/min., a respiratory rate of 20 breaths/min. and a temperature of 37.5 C. There was a macular rash over sun-exposed areas, patchy alopecia and numbness of the lateral aspect of the right thigh. The lung exam was normal. The hemoglobin was 10mg/dl. The EKG showed sinus tachycardia with low voltage. Chest x-ray was significant for cardiomegaly. A CXR one year prior to admission showed no cardiomegaly. An echocardiogram demonstrated a large pericardial effusion with indentation of the right atrium. The ejection fraction was 50%. Pericardiocentesis returned one liter of exudative fluid. Cultures and cytology were negative. Serology was ANA positive (1:320 titer) in a nucleolar pattern. After the pericardial drain was removed, a repeat echo showed no signs of effusion. A diagnosis of systemic lupus erythematosus was made. The serositis and shortness of breath resolved with 20mg of prednisone. Systemic lupus is characterized by symptoms that wax and wane. A careful physical examination is essential in making the diagnosis, and distinguishing lupus from other common diagnoses. This patient's dyspnea had been diagnosed as asthma on two separate occasions. On neither occasion did the patient's physical examination suggest reactive airway disease, nor did she respond to bronchodilators as expected. The physical findings of a rash, alopecia, peripheral neuropathy and back pain were inconsistent with the diagnosis of asthma. These inconsistencies prompted evaluation of other diagnoses. Echo evaluation of the effusion allowed proper diagnosis. The patient had been treated with prednisone during her last visit, which likely accounted for her clinical improvement at that time. An echocardiogram performed when cardiomegaly was first identified on CXR may have led to an earlier diagnosis.

DISCUSSION: This patient underwent pericardial drainage that resulted in immediate improvement. This procedure, however, was probably unnecessary. Pericardial drainage in patients without tamponade or suspected purulent pericarditis has a low diagnostic yield and no clear therapeutic benefit.

INTRA-CEREBRAL HEMANGIOMAS DISGUISED. S. Agresta¹, J. Wiese¹, ¹Tulane University, New Orleans, LA

LEARNING OBJECTIVES: To recognize the role of clinical reasoning in diagnostic algorithms.

CASE INFORMATION: A 46 year-old woman presented with new onset seizures. Four months prior to admission, she had been started on estrogen after a total abdominal hysterectomy and bilateral salpingo-oophorectomy. Two months prior to admission, she began to experience fatigue and headaches. A screening mammogram at that time showed an 8 cm cystic-nodular area in the right breast and a 6 cm cystic lesion in the left breast. Her family history was significant for an older sister with breast cancer and another with an intra-cerebral hemorrhage. At presentation, a 2 cm mass was palpated in the right breast. There were no neurological deficits. The remaining physical examination was normal. A head CT showed a 2 cm mass in the left frontal lobe and a 1.5 cm mass in the right parietal lobe. A contrast enhanced abdominal CT revealed two, 1 cm liver lesions suggestive of hemangiomas. A repeat mammogram, however, was normal. A MRI performed to re-evaluate the intra-cerebral lesions revealed that they were consistent with cavernous hemangiomas. She was started on anti-convulsant therapy, and the estrogen was discontinued. She had no further seizure activity at one-month follow-up. A repeat mammogram and breast examination were normal.

DISCUSSION: The association of cavernous hemangiomas of the liver with estrogens has been described; this case suggests that a similar association between estrogen and intra-cerebral cavernous hemangiomas may exist. Diagnostic algorithms should complement clinical reasoning. When diagnostic data is inconsistent with the leading diagnosis, greater consideration should be given to alternative explanations. This patient's repeat mammogram demonstrated unexpected negative results. Despite the high pretest probability of metastatic breast cancer, the negative likelihood ratio for mammography, (-LR: 0.28), significantly reduced the post-test probability. The CT scan findings associated with metastatic cancer, mass effect and vasogenic edema, were not seen on this patient's study. This prompted consideration of other etiologies. The MRI demonstrated popcorn appearing reticular lesions with a peripheral rim of hemosiderin, and these are characteristics representative of cavernous hemangiomas.

ARTHRALGIAS, RASH, AND FEVER. S. Agresta¹, J. Wiese¹, ¹Tulane University, New Orleans, LA

LEARNING OBJECTIVES: To recognize the value of likelihood ratios in making difficult diagnoses.

CASE INFORMATION: A 31 year-old woman presented with four weeks of fever, arthralgias, and a rash. The arthralgias were localized to her knees, ankles, and wrists. The evanescent, salmon-colored rash started over her legs and became generalized, sparing the soles and palms. She reported a sore throat and 15-pound weight loss two weeks prior to the onset of illness. There was no history of blood transfusions, environmental exposures, or recent travel. Her examination was notable for a macular, salmon-colored rash on the legs, arms, and torso, sparing the palms and soles. There were no papules, nodules, vesicles, or pustules. The joint exam was normal, and she was afebrile. Initial laboratory evaluation revealed a WBC count of 20,000/uL with 90% neutrophils; AST of 56 U/L, ALT of 105 U/L, ALKP of 244 U/L, ferritin of 696 ng/ml, and a sedimentation rate of 97 mm/hr. She was given ceftriaxone and doxycycline. These were stopped after all cultures and serologic tests proved negative. Serologic, immunologic, and infectious evaluations were normal. These included viral hepatitis serologies, HIV, RPR, blood and urine cultures, ANA, and rheumatoid factor. Skin biopsy revealed a leucocytoclastic vasculitis. Methylprednisolone 1 g IV was initiated. Within 24 hours, symptoms resolved. The arthralgias and rash returned in one month. These symptoms regressed with aspirin 800 mg every eight hours.

DISCUSSION: Adult Still's has an annual incidence of one per 200,000 people. The diagnosis of Still's disease is one of exclusion, as infections, malignancies, and other immunologic disorders may present with similar symptoms. The major criteria are fever, rash, arthralgias, and leukocytosis. The minor criteria are sore throat, lymphadenopathy, splenomegaly, liver dysfunction, and the absence of positive rheumatoid factor and ANA. The diagnosis in this patient was entertained after diagnoses that are more common were excluded. The patient met all four major and four of six minor criteria. When a patient fulfills five of these criteria, the likelihood ratio for Still's disease is 12. Despite the low pre-test probability, the high likelihood ratio increased the post-test probability above treatment threshold. The dramatic response to steroids or aspirin seen in this patient is typical for Adult Still's. Interpreting the results of physical findings and laboratory studies by their likelihood ratios facilitates evaluation of difficult diagnoses.

WHEN THE PLUMBING DOES NOT WORK: A CASE OF MITRAL STENOSIS. A. Ayer¹, R. Granieri¹, ¹University of Pittsburgh, Pittsburgh, PA

LEARNING OBJECTIVES: 1. To recognize the clinical manifestations of mitral stenosis. 2. To recognize the need for echocardiography in assessing severity of mitral stenosis. 3. To distinguish cases of mitral stenosis that require aggressive therapy and intervention.

CASE INFORMATION: A 43 year-old female presented with a two year history of gradually increasing lower extremity edema, progressive dyspnea on exertion and generalized fatigue but no orthopnea, paroxysmal nocturnal dyspnea, chest pain, or syncope. She took no medications. Her physical exam revealed a BP 112/78, HR 99, RR 18, O₂ saturation of 96% on room air, JVP at 13cm and rales at both bases. Heart exam revealed a visible precordial heave, an opening snap, 4/6 diastolic rumble heard best at the apex, a loud S1, and a 4/6 holosystolic murmur heard at the apex, radiating to the axilla. There was abdominal distention, hepatomegaly, and 3+ lower extremity edema extending to the thighs. Serum Na was 125 and albumin was 2.9. All other labs were normal. EKG revealed right axis deviation, right ventricular hypertrophy, and biatrial enlargement. Echocardiogram showed severe mitral stenosis with moderate mitral regurgitation, an estimated mitral valve area of 1.4cm², and severe pulmonary hypertension. On cardiac catheterization, she had a 90% left main lesion, a 100% right coronary lesion, and a mitral valve area of 0.53cm². Due to the severity of her disease, she underwent coronary artery bypass surgery and mitral valve replacement but developed intraoperative complications, and expired.

DISCUSSION: Mitral stenosis (MS) is most commonly caused by rheumatic fever. The principal symptom is exertional dyspnea. Other symptoms of left-sided heart failure depend on left atrial pressure. Fatigue and weakness due to low cardiac output, hoarseness due to compression of the recurrent laryngeal nerve (Ortner's syndrome) and thromboembolism due to atrial fibrillation may also occur. Classic physical exam findings, as demonstrated by our patient, include opening snap followed by a diastolic murmur, a prominent S1, elevated JVP, sacral and peripheral edema, ascites, and a right ventricular heave. Echocardiography is the initial diagnostic tool for assessing the severity of MS through estimation of mitral valve area and determination of the applicability of mitral valvuloplasty. Patients with moderate to severe MS or symptoms should have surgical intervention. Medical management includes antibiotic prophylaxis for appropriate procedures, diuretics, and anticoagulation in the presence of atrial fibrillation as there is high risk of embolism in these patients.

SUBACUTE MYELITIS AS FIRST PRESENTATION OF LYME DISEASE. A. Alghadban¹, A. Hama², ¹SUNY at Stony Brook, Port Jefferson, NY; ²NUMC, Eastmeadow, NY

LEARNING OBJECTIVES: 1- Recognize myelitis as a rare complication of Lyme disease 2- Recognize the new MRI findings in the spinal cord in Lyme disease.

CASE INFORMATION: 23 years old white male with history of schizophrenia came with progressive weakness in his legs over two months period associated with severe instability that he was unable to stand or walk. He denied any sensory symptoms or urinary or fecal incontinence. Physical exam showed paraparesis (proximal and distal strength 4/5) with decrease vibration sensation, severe ataxic gait and no other sensory deficits. Laboratory tests were normal except for extremely elevated ELISA Lyme titer (1/12948). Western blot was positive for both IgM and IgG. Cerebrospinal fluid (CSF) Lyme titer on ELISA was also very high (1/12947) and Western blot was positive for IgM and IgG. The index of CSF/serum for Lyme titer was very high (1.63). MRI of the cervical spine showed increased signal in T2 weighted-image in patchy distribution from the 7th thoracic segment to the 10th. Treatment with intravenous ceftriaxone and physical therapy were started. The patient showed marked improvement and was able to walk with a walker in two weeks.

DISCUSSION: Lyme disease involves multisystems with dermatological, rheumatological, neurological and cardiac manifestations. It is caused by *Borrelia burgdorferi* and transmitted by Ixodes. Several neurological syndromes had been described including lymphatic meningitis, cranial neuropathy (commonly unilateral or bilateral Bell's palsy), painful radiculoneuritis, optic neuritis, mononeuritis multiplex, Guillain-Barre syndrome, encephalomyelitis, peripheral neuropathy and encephalopathy. Myelitis has recently been reported as an unusual complication of Lyme disease. Our case represents one of the few cases of this type. The unique feature in our case is that Lyme disease was first presented as myelitis.

CEREBRAL AUTOSOMAL DOMINANT ARTERIOPATHY WITH SUBCORTICAL INFARCT AND LEUCOENCEPHALOPATHY (CADASIL) SYNDROME. A. Alghadban¹, A. Hamad²; ¹SUNY at Stony Brook, Port Jefferson, NY; ²NUMC, Eastmeadow, NY

LEARNING OBJECTIVES: -recognize cerebral autosomal dominant arteriopathy with subcortical infarct and leucoencephalopathy (CADASIL) in the differential diagnosis for normal pressure hydrocephalus.

CASE INFORMATION: 53 years old white female presented with progressive cognitive dysfunction and dementia, gait disturbance, urinary incontinence over the last 3 years. She has long standing history of migraine. 3 years ago, she had a computed tomography (CT) of the brain that showed dilated ventricles and labeled as normal pressure hydrocephalus and the patient refused treatment. Family history was significant for early dementia and migraine in the mother.

Her physical exam showed severe dementia, severe ataxia on standing or walking despite normal muscular strength. Sensory exam was normal. Laboratory tests were unremarkable. Head CT and MRI showed severe atrophy with severe dilated ventricles and severe periventricular white matter disease. Skin biopsy showed characteristics PAS positive granular material in the media and narrowing of the lumen of the vessels which is typically seen in cerebral autosomal dominant arteriopathy with subcortical infarct and leucoencephalopathy (CADASIL). The patient was placed in a nursing home after explaining the nature of the disease to the family.

DISCUSSION: CADASIL can be considered as a new disease that had been described recently (1996) in few families. It predominantly affects the small vessels of the brain. The transmission is autosomal dominant linked to chromosome 19. The clinical features include migraine with or without aura (the gene for familial type of migraine is located on the same chromosome), strokes or stroke-like episodes, psychiatric symptoms and dementia. In our case the presentation was similar to the normal pressure hydrocephalus symptoms. We want physicians to be aware of this similarity as the management is completely different.

MULTIPLE CHRONIC NONHEALING ULCERS IN A 40 YEAR OLD PATIENT. A. Ali¹, T. Willingham¹, T. Takahashi²; ¹Mayo Clinic, Rochester, MN; ²Mayo Clinic, Rochester, MN

LEARNING OBJECTIVES: 1. Describe etiologies of nonhealing ulcers. 2. Describe common characteristics and manifestations of calciphylaxis.

CASE INFORMATION: 40 year-old woman presented to the hospital with nonhealing ulcers on her breast, thighs and toes. She developed systemic lupus at a young age that ultimately caused complete renal failure. She underwent renal transplant; however, she reinitiated hemodialysis after failure of the transplant. Her examination revealed necrotic ulcers in the left upper breast region. Radiologic evaluation of the breast, hands and chest revealed significant calcifications throughout the body. This pattern revealed systemic calciphylaxis. She underwent debridement and nerve block for pain control. She was dismissed; however, she eventually died three months after dismissal.

DISCUSSION: Nonhealing ulcers are a common source of morbidity in hospitalized patients. Common etiologies include ischemia from vascular causes, chronic edema, and small vessel disease. Uncommon etiologies like calciphylaxis differ from traditional chronic ulcers. The skin appears violaceous with a plaque-like or a nodular appearance. The ulcers cause extreme pain independent of size or appearance. The pathogenesis of calciphylaxis remains unknown. Biochemical tests often reveal abnormal calcium or phosphorus levels. Treatment options often include subtotal parathyroidectomy in some patients. Normalization of calcium and phosphorus remains the primary focus of medical treatment. The chronic wounds are often aggressively debrided and antibiotics are used in cases with infections. The use of immunosuppression for calciphylaxis has had mixed results. Despite aggressive wound care and calcium management, the prognosis of calciphylaxis remains poor at 60% mortality. Further research needs to be performed on this uncommon yet potentially lethal illness.

EMPHYSEMATOUS PYELITIS IN A WOMAN WITH DIABETES MELLITUS. A. Almahameed¹, S. Baghdasarian², S. Frost²; ¹Cleveland Clinic Foundation, Shaker Heights, OH; ²The Cleveland Clinic Foundation, Cleveland, OH

LEARNING OBJECTIVES: 1) Recognize the potential for severe complications in diabetic patients with urinary tract infection. 2) Diagnose renal emphysema. 3) Recognize that prompt medical and surgical treatment is essential to decrease the high rate of mortality associated with emphysematous pyelonephritis and pyelitis.

CASE INFORMATION: A 48 yo black woman with multiple sclerosis and diabetes mellitus presented with abdominal pain, fever, and dysuria 3 days after starting ciprofloxacin for presumed urinary tract infection (UTI). Physical exam demonstrated T=39C, HR=130, BP=150/80, and tenderness in the left lower quadrant without peritoneal signs. Laboratory investigation revealed WBC=9.91, Cr=1.2, and BUN=25. Urinalysis showed trace leukocyte esterase, WBC=6-10, 2+ blood, and RBC=0-3. Ampicillin and gentamicin were administered. CT scan of the abdomen was obtained demonstrating slight enlargement of the left kidney with stranding of the perinephric fat, and a dilated left renal pelvis and proximal ureter containing pockets of gas. Hydroureter, pyelonephritis, and emphysematous pyelitis were diagnosed and the patient promptly underwent intraoperative cystoscopy to evaluate for ureteral obstruction. Debris occluding the left ureter was visualized and a ureteral stent was deployed. Urine and blood cultures revealed *Proteus mirabilis* and *Aerococcus* *Urinae*. The patient

improved quickly and was discharged on the fourth day of hospitalization to complete antibiotic treatment with Augmentin.

DISCUSSION: UTI is a significant problem in diabetic patients, and complications such as pyelonephritis, renal or perirenal abscess, focal or multifocal bacterial nephritis, xanthogranulomatous pyelonephritis, and renal papillary necrosis can occur. Emphysematous pyelonephritis (air in the renal parenchyma), and emphysematous pyelitis (air in the collecting system) are complications that occur more frequently in diabetic than in non-diabetic patients. Early diagnosis of emphysematous complications is crucial due to the high rate of mortality in patients whose diagnosis is delayed. Radiographic confirmation is essential (preferably by abdominal CT scan) as the history, physical exam and laboratory findings are rarely diagnostic. Emphysematous pyelonephritis responds poorly to antibiotics alone, with mortality rates as high as 80%. Surgical intervention is virtually mandatory, with nephrectomy lowering mortality to 20% or less. Emphysematous pyelitis is less critical, with an overall mortality of 20%. It is often associated with urinary tract obstruction, and responds well to intravenous antibiotics and obstruction relief.

A 55-YEAR-OLD PHYSICIAN WITH LYMPHOCYTOSIS, FEVER, ABDOMINAL PAIN, AND PROFOUND THROMBOCYTOPENIA. A. Almahameed¹, A. Absi², D. Faray-berges², S. Rehm³; ¹Cleveland Clinic Foundation, Shaker Heights, Ohio; ²The Cleveland Clinic Foundation, shaker hts, Ohio; ³The Cleveland Clinic Foundation, Cleveland, Ohio

LEARNING OBJECTIVES: 1- Recognize infectious mononucleosis as a rare cause of severe thrombocytopenia. 2- Utilize EBV-specific antigens to diagnose infectious mononucleosis in monospot-negative patients. 3- Treat EBV-induced severe thrombocytopenia.

CASE INFORMATION: A previously healthy 55-year-old male physician reported 2 weeks of fatigue, malaise, and mild abdominal discomfort. On initial evaluation, a right submandibular lymph node was moderately enlarged and the CBC was normal. Several days later, he became febrile (39 C). He had a normal WBC count with lymphocytosis (50%) and atypical lymphocytes (25%), as well as transaminitis (ALT 239 U/L and AST 198 U/L). Monospot tests, CMV titers, HIV antibodies, and blood and urine cultures were all negative. Self-prescribed oral therapy with empiric clarithromycin and ciprofloxacin produced little improvement. One week later, his platelet count was 90,000/mm³. He was admitted three days later with petechial rash involving the upper extremities and the upper part of his trunk, and a platelet count of 8,000/mm³. He was afebrile, had diffuse non-palpable, non-blanching petechiae on his upper extremities, ecchymoses on his buccal mucosa, 3 mildly enlarged non-tender lymph nodes in the right submandibular area, and a palpable spleen tip. The WBC count was 10,000/mm³ with 26% polymorphs and 63% lymphocytes, of which 20% were atypical. His hemoglobin was 13.4 g/dl, and his platelet count was less than 1,000/mm³. Blood chemistries and coagulation parameters were normal. Occasional atypical, large lymphocytes and rare platelets were evident on a peripheral blood smear. Bone marrow biopsy showed normocellular composition with severe thrombocytopenia and adequate numbers of megakaryocytes. Flow cytometry of the blood found no malignant cells. Serologies were negative for toxoplasmosis, hepatitis C, hepatitis B, herpes simplex, and herpes virus 6, but the patient had elevated levels of IgM and IgG antibodies to Epstein-Barr virus viral capsid antigen (VCA) (IgG VCA: 152 IU/ml, normal <18, and IgM VCA 71 TV (normal <18). The patient was treated with high-dose prednisone and IVIG 1 g/kg/day for 2 days, and his platelet count rose gradually (11,000/mm³ on discharge and 180,000/mm³ 3 weeks later).

DISCUSSION: Infectious mononucleosis (IM) should be included in the differential diagnosis of severe thrombocytopenia in adults even when the initial monospot test is negative. Low-grade neutropenia and thrombocytopenia are common during the first month of illness. Heterophile antibodies are useful in diagnosing disease in many cases but are often not detectable in children under age 5, in the elderly, and in patients presenting with symptoms not typical of IM. An alternative is EBV-specific antibody testing. Titers of IgM and IgG antibodies to viral capsid antigen are elevated in the serum of more than 90% of patients at the onset of disease. IgM antibody to VCA is useful for the diagnosis of acute IM because levels are elevated only during the first 2 months of the disease. We can not entirely exclude the possibility that ciprofloxacin or clarithromycin therapy caused the thrombocytopenia. Given the clinical illness, the results of serologic testing, and the rarity of profound antibiotic-associated thrombocytopenia, we believe that acute Epstein Barr virus infection is the more likely cause of the abnormalities.

A CASE OF PANCYTOPENIA IN A 70 YEAR OLD VEGAN. S. Archuleta¹; ¹The Mount Sinai Medical Center, New York, NY

LEARNING OBJECTIVES: 1) Recognize the importance of a dietary history in the work-up of pancytopenia, 2) Diagnose and treat megaloblastic anemia due to vitamin B12 deficiency, 3) Distinguish among the different etiologies of B12 deficiency.

CASE INFORMATION: A 70 year old woman presented to the Emergency Room with a three month history of progressive fatigue, palpitations and dyspnea on exertion. She denied any melena, blood per rectum, change in her bowel habits, anorexia or weight loss. She had no past surgical history and was not taking any medications. She was vegan for approximately ten years and denied eating any meat, fish, eggs, dairy products or other foods of animal origin. She denied drinking alcohol. On presentation the patient's vital signs were stable and she was not orthostatic. She was markedly pale and had a smooth tongue. A soft flow murmur was present on cardiac exam. There was no lymphadenopathy, or hepatosplenomegaly. On rectal exam, she had heme negative brown stool. Neurologic exam revealed no difficulty with balance, no decrease in vibration or position sense. Lab values were significant for a pancytopenia with Hgb 4.8gm/dl, WBC 2,000, PLT 100,000, MCV 124, RDW 34%. The peripheral blood smear contained macro-ovalocytes and hypersegmented neutrophils. The reticulocyte count was 2.3%, LDH 3,700, total bilirubin 1.0. Iron studies were normal and red-cell folate mildly reduced at 135ng/ml. Serum B12 levels were <50pg/ml. The patient was started on intramuscular B12 injections 1,000mcg qd for two days while in the hospital, then was discharged home to receive further injections as an outpatient. She subsequently had a Schilling test that showed normal absorption of oral vitamin B12.

DISCUSSION: Vitamin B12 comes from the diet and is present in all food products of animal origin. Liver stores of B12 can last for several years after dietary ingestion ceases. Although dietary B12 deficiency is extremely rare, it can occur in vegans who abstain from all such foods for prolonged periods of time.

WHEN A PATIENT PRESENTS WITH WEAKNESS: IT MAYBE HELPFUL TO REVIEW YOUR PRESCRIPTION. A. Arshad¹, R. Keating¹; ¹Fairview Hospital, Cleveland, OH

LEARNING OBJECTIVES: 1) Diagnose drug-induced myositis from its presentation. 2) Recognize how adding multi drug antihyperlipidemic therapy increases the incidence of myositis.

CASE INFORMATION: A 72 year old white female with past medical history of diabetes mellitus, hypertension and hyperlipidemia who was in relatively good health presents with progressive weakness for the last one week. A week ago she had a yeast infection and took two tablets of diflucan. Shortly after she started feeling weak and the weakness progressed gradually. She is now unable to stand or walk has trouble feeding due to weakness in her arms and also occasionally gets short of breath. She denies any headache, double vision, blurred vision, fevers, chills, pain in her muscles or joints or any recent falls. She is currently on cerivastatin, insulin, lansoprazole, gemfibrozil, atenolol, aspirin, nortriptyline, sertraline, and lisinopril. On physical examination she is afebrile with stable vital signs. The muscle strength was severely decreased in all extremities, upper greater than lower and especially in the shoulder and pelvic girdles. No skin rash or mouth ulcers were seen. Laboratory work shows a creatine kinase of 4132 U/L, myoglobin titer of 3634. Liver enzymes were mildly elevated. Patient was started on steroids and she adequately recovered.

DISCUSSION: The onset of generalized progressive myopathy in a patient taking cholesterol lowering drugs should alert us to the diagnosis of drug induced myopathy. Several anti-hyperlipidemic drugs have caused myopathy including the statins, gemfibrozil, nicotinic acid and clofibrate. Lovastatin has been reported several times in literature to induce myositis. The concomitant use of gemfibrozil in patients taking statins increases the incidence of myositis to five percent. The creatine kinase and sedimentation rate is markedly elevated in these patients. Creatine kinase elevations are very specific and consist of only the MM fraction. Discontinuation of statin therapy in these patients is advisable because of the potential development of rhabdomyolysis and acute renal failure. Cautious use of statins is recommended in patients with prior history of liver disease, those with active liver disease and in heavy alcohol users. Hepatic and muscle enzyme levels should be tested periodically during statin treatment. Other medications known to increase the incidence of myopathy in patients taking statins are cyclosporine, nicotinic acid, prednisone and azathioprine. Treatment includes discontinuation of the medication and use of steroids. The use of diflucan was most likely incidental as there have not been cases of myopathy attributed to diflucan.

AN UNUSUAL CAUSE OF AN INTRACRANIAL ABSCESS. A. Arshad¹, K. Gopalakrishna¹; ¹Fairview Hospital, Cleveland, OH

LEARNING OBJECTIVES: 1) Diagnose epidural abscess when presenting as a space-occupying lesion.

2) Recognize *Propionibacterium acnes* as a cause of intra-cranial abscess. 3) Manage *Propionibacterium acnes* with surgical drainage and antibiotic monotherapy. **CASE INFORMATION:** A fifty-two year old white male presented with a lump on the head for over a year. Magnetic resonance imaging (MRI) showed a mass lesion in the parietal lobe. He subsequently underwent craniotomy for an epidural neoplasm and left parietal reconstruction with duragen. He then received two doses of radiotherapy. About three months later he presents with right sided weakness, focal seizures in the right hand and a fight facial palsy. After recurrent seizures on anticonvulsant therapy a repeat computed tomography scan and MRI scan was done which showed an epidural lesion with mass effect. The patient was afebrile and had a normal peripheral blood count. The patient was taken to surgery and an epidural abscess was drained. The abscess aspirate on gram stain revealed pleomorphic gram positive bacteria. Aerobic cultures were negative and so were the anaerobic cultures initially but after several days of incubation they grew out a pure culture of a pleomorphic gram positive organism which was identified as *Propionibacterium acnes*. He was started on intravenous ceftriaxone for 6 weeks and he recovered fully after completion of therapy.

DISCUSSION: This case illustrates a case of post-surgical epidural abscess caused by *Propionibacterium acnes*. *Propionibacterium acnes* is a corynebacterium usually isolated from the human skin where it plays a role in the pathogenesis of acne vulgaris. There have been reports in literature of clinically significant infections related to CNS shunts, intracranial abscesses related to dental sepsis, sepsis related to indwelling vascular catheters, and also a few cases of intracranial abscesses unrelated to foreign bodies. The above case is an addition to these rare cases of intracranial abscesses caused by this unusual pathogen. They are slow growing organisms and as their identification presents problems, they have been frequently overlooked as contaminants. It has been suggested that grain positive rods on gram stain and negative aerobic cultures of suspected infected fluids without organisms on gram stain should be cultured anaerobically. Most infections reported in literature were treated with a combination of antibiotics, though there have been reports of successful treatment with penicillin and surgical drainage alone. The above case also shows how ceftriaxone monotherapy is an effective treatment for such infections along with surgical evacuation of the abscess.

THROMBOSIS AND HIV: A CASE REPORT. D. Asuncion¹, E.F. Yee²; ¹University of California, Los Angeles, Los Angeles, CA; ²UCLA/San Fernando Valley Program, Sepulveda, CA

LEARNING OBJECTIVES: 1. Discuss the possible role of HIV as a cause of thrombosis. **CASE INFORMATION:** A 48 year-old male with HIV (CD4 of 343, undetectable viral load) was admitted to the hospital with shortness of breath. He was diagnosed with pneumocystis

pneumonia vs cocci pneumonia and treated with Bactrim and Fluconazole. One week later he returned with continuing dyspnea. A VQ scan showed high probability for pulmonary embolism. He was treated with heparin then coumadin and discharged. Three days later he returned with new left lower extremity swelling and pain. He had been taking Coumadin with an INR of 5.1. He denied shortness of breath, chest pain, IV drug use, smoking, ETOH, or recent travel. His medications included Abacavir, Efavirenz, Nelfinavir, Fluconazole, and Bactrim. Physical examination revealed a tender, swollen left lower extremity with a positive Homan's sign. An ultrasound showed thrombosis in the left popliteal vein. The swelling and pain resolved with heparin and he was discharged on Lovenox. A hypercoagulable evaluation revealed normal levels of protein S, activated protein C resistance, anti-thrombin III, and homocysteine. Protein C was slightly elevated. Lupus anticoagulant and anti-cardiolipin antibody were negative. An abdominal CT revealed no mass lesions.

DISCUSSION: Thrombosis is an uncommon complication of HIV infection that many primary care providers may not be aware of. Anecdotal evidence in the literature relates thrombosis and HIV, especially in patients with IV drug use, Kaposi's, cytomegalovirus (CMV) disease, megestrol use, protease inhibitor use (including nelfinavir), and hypercoagulable states such as anticardiolipin antibody and protein S deficiency. A recent epidemiologic study showed that HIV patients with thrombotic events had the following characteristics: age greater than 45 years, CMV disease, AIDS-defining opportunistic infection, hospitalization, megestrol use, and indinavir use. The only risk factors for thrombosis that this patient had were hospitalization and nelfinavir use. HIV itself may be a risk factor for thrombotic events and should be considered in cases where shortness of breath remains unresolved despite maximal medical therapy. The mechanism by which HIV causes thrombosis remains unknown but several theories have been proposed. This case illustrates that the link of HIV as a possible cause of thrombosis may have implications for prophylaxis in HIV patients. There are no consensus studies and prophylaxis remains at the discretion of the clinician.

REFRACTORY HYPERTENSION DUE TO FIBROMUSCULAR DYSPLASIA(FMD) AND POST-ANGIOPLASTY RENAL INFARCTS. M.J. Bair, DEPARTMENT OF MEDICINE, BROOKE ARMY MEDICAL CENTER, FORT SAM HOUSTON, TX. M.J. Bair¹; ¹Brooke Army Medical Center, Ft. Sam Houston, TX

LEARNING OBJECTIVES: 1. To diagnose FMD 2. To discuss the treatment of FMD 3. To discuss the management of a complicated hypertensive patient.

CASE INFORMATION: A 42-year-old African-American female presented with poorly controlled hypertension. She was taking nifedipine and lisinopril. There was no history of alcohol, tobacco, or drug use. Her BP was 199/116 in right arm and 190/116 in the left arm and HR 78. General physical examination was unremarkable. Initial laboratory evaluation included: normal electrolytes, BUN/creatinine, calcium, TSH, CBC, chest x-ray, serum renin and aldosterone, and 24 hour urine collection for metanephrines, and cortisol. Urinalysis(UA) demonstrated 10-14 RBC/hpf, 1-4 WBC/hpf, negative protein, and no casts. Renal ultrasound showed normal sized kidneys with increase in right main renal artery velocity. Arteriography was remarkable for intra-arterial "webbing" of her right renal artery consistent with FMD. Renal angioplasty was performed resulting in abrupt BP lowering. However, she developed severe right back and flank pain, BP 196/121, and maintained normal renal function. Abdominal CT findings were suggestive of a right renal infarct. Acetaminophen with codeine, clonidine patch and hydrochlorothiazide(HCTZ) were added. Subsequent clinic visits continued to show poorly controlled hypertension. Twenty-four hour ambulatory monitoring was confirmatory. She assured adherence to her medications. Further work-up included an echocardiogram notable for diastolic relaxation anomaly, but no left ventricular hypertrophy was detected. A repeat renal ultrasound was consistent with recurrent right renal artery stenosis that was confirmed by arteriogram. A second angioplasty was performed and complicated by another renal infarct. Her BP was 166/118 and labetalol was added. At the next clinic visit her BP read 218/122 and she was started on minoxidil, lasix and the HCTZ was stopped. Her regimen consisted of nifedipine 90mg twice a day, lisinopril 40mg/daily, lasix 20mg/daily, labetalol 600mg twice a day, clonidine patch 0.4mg every week, and minoxidil 10mg/daily. At the most recent appointment her BP measured 150/84.

DISCUSSION: FMD typically affects younger women and usually involves the distal main renal arteries. The gold standard for diagnosis of FMD is arteriography showing the "string of beads" appearance. Percutaneous angioplasty is the treatment of choice for FMD. Most reports find that 50-85 percent are cured, 30 to 35 percent improved (lower BP and fewer anti-hypertensive medications required), and less than 15 percent are failures. Furthermore, the restenosis rate following successful angioplasty is less than 10 percent. This case is unusual because of the recurrent episodes of restenosis after angioplasty and the poor treatment outcome in terms of medications required.

MISLEADING EKG IN AN 82 YEAR OLD FEMALE WITH AORTIC DISSECTION. P. Balingit¹, T. Spector¹, A. El-bialy¹; ¹UCLA San Fernando Valley Program, Sylmar, CA

LEARNING OBJECTIVES: 1. Consider non-cardiac etiologies of chest pain. 2. Recognize that abnormal EKGs are not necessarily specific for acute myocardial infarction (AMI). 3. Recognize that recurrent presentations for similar symptoms without a conclusive diagnosis should warrant further investigation.

CASE INFORMATION: An 82 year old female was admitted to the hospital with three days of increasing severity and frequency of anginal-type chest pain. Her cardiac risk factors included HTN, postmenopausal state, and multiple first-degree relatives with coronary artery disease. Three years prior, she presented with similar symptoms. Cardiac cath at that time revealed no flow-limiting lesions and a normal ejection fraction. She continued to have occasional chest pain, believed to be related to microvascular heart disease or Prinzmetal's angina, and was placed on anti-anginal medication. Her symptoms remained stable until 10 months prior to admission, after which she was admitted four times with chest pain, each time being ruled out for AMI and discharged to outpatient follow-up for risk factor modification.

Physical examination: Temp 36.9C, BP 118/74, pulse 63, respiratory rate 16. The patient was alert and oriented. No carotid bruits or elevated JVP were appreciated. Cardiac exam revealed a

regular heart rate, S4 gallop, and nondisplaced PMI. No murmurs or friction rubs were noted. Lungs fields were clear to auscultation and percussion. Abdominal exam was unremarkable. Distal pulses were equal, and no pretibial edema was present.

Laboratory studies: Electrolytes, BUN, creatinine, CBC, and coagulation studies were normal. Troponin I was 0.6 (normal <0.3). CXR revealed cardiomegaly, but no pulmonary infiltrates, effusions, or widened mediastinum. EKG revealed normal sinus rhythm, left axis deviation, Q waves in leads III and aVF, and poor R wave progression. There were no significant changes in the EKG findings noted from previous admissions. However, given her EKG findings and multiple cardiac risk factors, the presumptive diagnosis of unstable angina vs. AMI was again made.

The patient was given aspirin, morphine, and nitroglycerin with minimal relief of her chest pain. IV nitroglycerin and heparin were started, and the patient was admitted to the ICU. Subsequent echocardiogram showed no wall motion abnormalities and a normal ejection fraction. A CT scan of the thorax revealed a Type B aortic dissection originating past the subclavian bifurcation and extending distally to the left iliac artery.

DISCUSSION: This case illustrates that although the EKG findings were suggestive of myocardial ischemia or infarction, neither was supported by cardiac cath and echocardiogram data. The recurrent presentations without EKG changes over time should have lead to a broader differential diagnosis. This case is, therefore, a cautionary tale about the importance of excluding non-cardiac etiologies of chest pain.

ADRENAL CRISIS IN A PATIENT WITH POLYENDOCRINOPATHY. S.C. Baxi¹; ¹Division of General Internal Medicine, VA San Diego Health Care System, San Diego, CA

LEARNING OBJECTIVES: 1) Recognize danger of causing adrenal crisis if thyroid hormone is replaced prior to glucocorticoid hormone. 2) Recognize features of autoimmune polyendocrinopathy.

CASE INFORMATION: A previously healthy 39-yr old woman presented to another physician with typical symptoms of diabetes. He started her on insulin and levothyroxine. Although she was taking her medications she felt worse and came to us 3 days later. She had extreme weakness, dizziness, and severe postural hypertension. Patient was hospitalized. Laboratory tests—glucose 450 mg/dl, Na 123 mmol/L, potassium 5.9 mmol/L, normal anion gap, arterial blood gases—normal pH. After initial test results diabetic ketoacidosis was ruled out. In view of above presentation and additional diagnosis of adrenocortical insufficiency was suspected. Patient was given intravenous normal saline, stress dose hydrocortisone and intravenous insulin infusion. Within a few hours her symptoms and postural hypotension improved. Hydrocortisone dose was tapered to a maintenance dose of 20 mg in the morning and 10 mg in the evening. She was started on 3 injections of insulin-regular and NPH insulin before breakfast, regular insulin before dinner, and NPH insulin at bedtime. Outpatient follow-up a few days after discharge from hospital—T4 level was 7.9 µg/dl (normal 4 to 11 µg/dl), TSH was 14.4 (normal 0.5 to 6 µU/ml). ACTH stimulation test was done. Baseline cortisol was 4 µg/dl (normal am level 7 to 25 µg/dl). These tests confirmed the diagnoses of primary hypothyroidism and primary adrenocortical insufficiency or Addison's disease. She was started on 0.1 mg of levothyroxine. Replacement hydrocortisone dose was continued and insulin dose was adjusted. As control of her diabetes improved and her thyroid status returned to normal state, she continued to feel better.

DISCUSSION: Our patient presented with simultaneous onset of all 3 major features of type II autoimmune polyglandular syndrome (APS) type 1 diabetes, hypothyroidism and adrenocortical insufficiency. Thyroxine therapy prior to the correction of adrenocortical insufficiency increased the turnover of marginal levels of cortisol and precipitated adrenal crisis. Syndromes of hypofunction of multiple endocrine glands induced by autoimmunity occur in well-described patterns. When a patient has one or more features of APS, clinician should look for deficiencies of other endocrine glands by checking tropic hormones and autoantibodies.

SULFONYLUREA-INDUCED HYPOLYCEMIA FROM MEDICATION DISPENSING ERRORS. Y. Belu-john¹, B. Hoogwerf²; ¹Cleveland Clinic, Akron, Ohio; ²Cleveland Clinic Foundation, Cleveland, OH

LEARNING OBJECTIVES: 1. To illustrate clinical features of hypoglycemia from inadvertent administration of a sulfonylurea compared to an insulinoma. 2. To demonstrate the importance of carefully reviewing medications that patients with hypoglycemia are taking.

CASE INFORMATION: Two patients were referred for recurrent episodes of hypoglycemia thought to be due an insulinoma. The clinical presentations were characterized by precipitous drops in blood glucose (BG) associated with marked adrenergic symptoms. The first patient was a 69 yo man with a coronary artery disease, heart failure, hypertension, chronic renal insufficiency, and peripheral vascular disease. He had recurrent acute hypoglycemic reactions (including 3 hospital admissions) accompanied by adrenergic symptoms. He also had a recent myocardial infarction. Medications included warfarin, metoprolol, nifedipine, isosorbide dinitrate (IsoD), furosemide, famotidine, lisinopril, potassium, pentoxifylline, and colchicine. His chemistry panel was normal. Prior workup included normal cortrosyn stimulation test and abdominal CT scan, and negative serum sulfonylurea levels. A 36 hour fast (the patient refused to go beyond 36 hours) showed a BG of 59 mg/dl, a corresponding c-peptide of 3.5 ng/ml (0.7–3.0), and absent insulin antibody. Inspection of his medications revealed a bottle labeled for IsoD contained two types of light green pills: IsoD and glyburide 5 mg. The second patient was an 87 yo woman who was referred for surgical treatment for a presumed insulinoma. Over a 2 mo period she has had several episodes consisting of confusion, diaphoresis, and syncope with confirmed hypoglycemia (BG levels < 40mg/dl). Medications included cardizem, ranitidine, acetaminophen, and diazepam. Evaluation included a normal chemistry panels, somatomedin, and cortrosyn stimulation test. Serum insulin and c-peptide levels at the time of hypoglycemia (BG= 37 mg/dl) were 36.38 pmol/L and 1600 pmol/L respectively. Serum and urine SFU screens were negative. CT and octreotide scans of the abdomen showed no abnormalities. Identification of the patient's home medications was done. In a bottle labeled diazepam were light blue-green tablets identified as generic glyburide 5mg. To further confirm SFU induced hypoglycemia, a repeat 72 hour fasting was done. At a glucose level of 94mg/dl, the insulin and insulin antibody levels were 3.2uU/ml (4.0–24.0) and <5 %

respectively, and at a glucose level of 58mg/dl, the levels were <2 uU/ml and <5% respectively. An exploratory laparotomy was canceled.

In each case, no further hypoglycemic episodes occurred following discontinuation of the SFU. DISCUSSION: The biochemical features of hypoglycemia due to sulfonylureas and insulinomas are similar. Insulinomas are commonly associated neuroglycopenic symptoms that are insidious in onset, and recalcitrant hypoglycemia with adrenergic symptoms is uncommon. Presence of such symptoms mandates a search for other causes. SFU-induced hypoglycemia may mimic an insulinoma biochemically. Our two cases illustrate the importance of careful attention to history, and a thorough inspection of all medications in patients presenting with hypoglycemia before extensive, diagnostic tests, including invasive procedures or surgery are undertaken

REMEMBERING COCCIDIOIDOMYCOSIS AS A CAUSE OF PNEUMONIA. J. Bevan¹, E.F. Yee¹; ¹UCLA/San Fernando Valley Program, Sepulveda, CA

LEARNING OBJECTIVES: 1. Recognize Coccidioidomycosis as a cause of pneumonia. 2. Discuss the epidemiology and treatment of Coccidioidomycosis

CASE INFORMATION: A 63yo previously healthy Caucasian male was admitted to the hospital with a diagnosis of pneumonia. He had multiple symptoms including shortness of breath, nonproductive cough, fevers, night sweats, and a 10 pound weight loss. He was treated with Azithromycin, then Doxycycline upon discharge for a 12-day course. Initially, he felt better but his symptoms returned 4 days after finishing the antibiotics. He was evaluated in clinic, found to have an unresolved pneumonia, and prescribed Levofloxacin. Five days later, he returned with continuing symptoms. At this visit, Azithromycin was restarted and a CT scan ordered. He was also seen in pulmonary clinic where bronchoscopy was considered to rule out malignancy. Six days later, he returned for a third clinic visit with worsening shortness of breath and new headaches. He denied TB exposure, sick contacts, travel, or HIV risk factors. He had lived in the San Fernando Valley his entire life. Upon walking back to the exam room, he became dyspneic. Notable on physical exam were crackles at both lung bases. A CXR and chest CT showed lymphadenopathy and patchy lower lobe infiltrates. His PPD was negative and Coccidioidomycosis (Cocci) titers were 1:128. The patient was re-admitted to rule out disseminated disease because of his high titers and new headaches. A bone scan and lumbar puncture were negative. He was started on Fluconazole with resolution of symptoms. His titers decreased to 1:16.

DISCUSSION: Having a suspicion for Cocci, especially in endemic areas like the Southwest, is crucial to diagnosis, as many cases go unrecognized. The numbers of cases are increasing as the population and tourism expand in endemic areas. This case illustrates how Cocci pneumonia may be symptomatically difficult to distinguish from other pneumonias. The diagnosis can be made via serology, cultures, and infected tissue. While most cases are self-limited, the infection can be debilitating with substantial time lost from work. Antifungal treatment should be initiated in cases with prolonged symptoms, high titers, bilateral infiltrates, anergic skin testing, and in the immunocompromised. Prolonged illness and disseminated disease can be life threatening. Seemingly minor complaints like a new headache or joint pain may be a sign of dissemination, which occurs in 1:200 cases. Although dissemination can occur in healthy people, immunocompromised, pregnant, African American, and Filipino patients are unusually susceptible to serious complications of the disease.

OPIOID OVERDOSE LEADING TO ACUTE COMPARTMENT SYNDROME, RHABDOMYOLYSIS, AND ACUTE RENAL FAILURE. A. Biviano¹; ¹Columbia University, New York, NY

LEARNING OBJECTIVES: 1) Diagnose and manage acute compartment syndrome. 2) Recognize that acute compartment syndrome is a risk factor for the development of rhabdomyolysis and subsequent acute renal failure.

CASE INFORMATION: A 28 year-old previously healthy male presented to the Emergency Department with pain and the loss of sensation and motor strength in his left foot. The symptoms began after awakening from a heroin-induced unconscious state of approximately twelve hours. During this time, his left leg had been flexed at the knee and under a significant amount of body weight.

Physical examination of the left leg revealed increased capillary refill of 3–4 seconds, no DP/PT pulse by Doppler, and inability to dorsi-/plantarflex the foot/toes. The four left leg compartment pressures were all found to be elevated to > 70 mm Hg. The patient was diagnosed with acute compartment syndrome, and emergently taken to the OR for fasciotomy. The patient also had urine output < 10 cc/hour, a serum creatinine of 4.3, and a CK=146, 200. He was diagnosed with rhabdomyolysis and acute renal failure. His hospital course was marked by several leg debridements and one month of hemodialysis support. He was discharged approximately six weeks after presentation with nearly complete return of leg and renal function.

DISCUSSION: Acute compartment syndrome results from injury to muscle due to increased tissue pressure within a closed muscle compartment. This pressure compromises circulation and neuromuscular function and causes ischemic damage. Common etiologies include limb trauma and severe compression from coma or intoxication.

Symptoms of compartment syndrome include increased pain out of proportion to injury, paresthesias, pulselessness, and paralysis of the extremity. Diagnosis is confirmed by pressure recording of the muscle compartments. Fasciotomy is recommended when the tissue pressure rises to within 10–30 mm Hg of the diastolic pressure in patients with symptoms/signs of compartment syndrome. Complications of compartment syndrome include the following: partial/complete loss of neuromuscular function, contractures (first reported by Volkman in 1872), and rhabdomyolysis. The ischemia and necrosis of muscle tissue can cause syndromes ranging from asymptomatic elevations of muscle enzymes to life-threatening cases of rhabdomyolysis with electrolyte imbalances and renal failure. Treatment for rhabdomyolysis due to compartment syndrome includes: fasciotomy to relieve ongoing muscle damage, vigorous early IV fluid hydration (i.e., within six hours of the insult), and hemodialysis support. Most cases of renal failure resolve in several weeks.

A PATIENT WITH IGA NEPHROPATHY SUBSEQUENTLY DIAGNOSED WITH SERONEGATIVE SPONDYLOARTHRITIS. A. Bleistein¹, M. Wright²; ¹Wright State University, Dayton, OH; ²Wright State University, Beaver Creek, OH

LEARNING OBJECTIVES: 1) Recognize the presentation of seronegative spondyloarthropathy. 2) Recognize the association of spondyloarthropathy and IgA glomerulonephritis. 3) Realize the importance of periodic renal screening in patients with seronegative spondyloarthropathy.

CASE INFORMATION: MV, a 34 year-old white male with a history of IgA glomerulonephritis, presents with knee pain and swelling which resolved two days prior to his visit. The patient had been given a diagnosis of gout in the past, but had never had joint fluid analyzed for crystals. He reports having intermittent swelling and pain without erythema or warmth in his right knee and left ankle for many years. He has not had joint involvement of his upper extremities or of his metatarsal phalanges in the past. On further questioning, he reports intermittent back pain since his teenage years, which is worst in the morning and improves after an hour of activity. He complains of occasional heel pain. He denies history of urethritis although he reports having multiple urinary tract infections when he was a teen. He also denies eye infections, uveitis, oral lesions, and inflammatory bowel disease. On physical exam at the time of the visit, the patient has no joint swelling but does have decreased lumbar range of motion. The remainder of the exam is unremarkable. Laboratory studies reveal an IgA level of 368 mg/dL (normal: 81-463 mg/dL); AST, ALT are normal. ANA and Rheumatoid factor are both negative. Sedimentation rate is 6. Hepatitis B surface antigen and hepatitis C antibody are negative. HLA-B27 is positive. Twenty-four hour urine protein is elevated at 330 mg. X-rays of the pelvis reveal no sacroiliitis. X-rays of the ankles are normal. X-rays of the feet reveal bilateral calcaneal spurs. The patient's clinical presentation is consistent with undifferentiated seronegative spondyloarthropathy (SNSA) marked by asymmetric oligoarticular arthritis of the lower extremities, inflammatory spinal pain, and enthesopathy with correlating calcaneal spurring. The previous diagnosis of gout is less likely given the patient's age and the absence of the redness and warmth so notable in gout attacks.

DISCUSSION: Seronegative spondyloarthropathy presents clinically with inflammatory spinal pain and decreased lumbar spine range of motion, asymmetric oligoarticular arthritis predominantly of the lower extremities, and, in addition, may also present with one or more of the following: enthesopathy, inflammatory bowel disease, urethritis, cervicitis, or sacroiliitis. This patient has both SNSA and IgA nephropathy, an association that has been documented in the literature. The mechanism of the association is not clear; however, the diseases have some commonalities, including higher prevalence among men, high serum levels of IgA1 and IgA-1C, association with a previous infection, possible genetic link given specific HLA types, and impaired expression of receptor CD89, which may be related to an impaired immunoregulatory mechanism. Indeed, renal abnormalities are common enough in patients with seronegative spondyloarthropathy that some authors have recommended routine, periodic renal screening, including urine microscopy and analysis, as well as serum BUN and creatinine, on all such patients.

COLCHICINE INDUCED NEUROMYOPATHY: AN UNUSUAL CAUSE OF WEAKNESS IN THE ELDERLY. L. Bookman¹, S. Shadowitz¹; ¹University of Toronto, Toronto, Ontario

LEARNING OBJECTIVES: 1) Recognize the importance of a thorough clinical examination in the elderly patient presenting with weakness. 2) Recognize the clinical presentation of drug-induced neuromyopathy. 3) Demonstrate the potential toxic effects when a commonly-prescribed medication is used in the setting of chronic renal failure.

CASE INFORMATION: An 87 year old male presented to the hospital emergency room with a 2-day history of a viral-like diarrheal illness that had since resolved, but was immediately followed by a two-day history of rapidly progressive generalized weakness and fatigue. On presentation, his muscle weakness was significant such that he was unable to ambulate, but had no respiratory difficulties. His past medical history included gouty arthritis, and NSAID-induced chronic renal failure. His gout had been managed with colchicine over the preceding 9 months. On examination, the patient was not dehydrated. He had motor weakness, most pronounced in the lower limbs, and more proximally than distally. As well, the patient's reflexes were absent in the lower limbs, and decreased in the upper limbs. Laboratory investigations on admission: creatinine kinase (CK) 1998, creatinine 147, urea 13.3, uric acid 618. Serum electrolytes were normal. Lumbar puncture revealed normal CSF protein, glucose, and CBC. Stool culture was negative for campylobacter. Nerve conduction studies and EMG showed mixed changes of neuromyopathic nature. Biopsies of the deltoid and quadriceps muscle revealed vacuolar myopathy with neurogenic atrophy, findings consistent with colchine-induced neuromyopathy. The patient's colchine had been held on admission. Over the next ten days, he made gradual gains in his motor strength with the help of physiotherapy. He was discharged to a rehabilitation centre for three weeks, after which his muscle strength had returned to near-normal. His reflexes, however, were still diminished.

DISCUSSION: Colchine-induced neuromyopathy in the setting of chronic renal failure has been described in various rheumatology journals, but is not commonly seen. The history, physical examination, and laboratory investigations in this patient were consistent with both guillain-barré syndrome, as well as a reaction to colchine. A muscle biopsy was needed to confirm the diagnosis of colchine-induced neuromyopathy. This case illustrates an interesting presentation of a previously described adverse drug reaction. It highlights the role of a careful history and physical examination in the elderly, in particular consideration of patient medications. This point is even more crucial in the diagnosis of generalized weakness, especially in the setting of chronic renal failure.

AN UNUSUAL CASE OF GENERALISED SPASMS. N. Britwum¹, L. Mamilla¹, C.A. Smith²; ¹Cook County Hospital, Chicago, IL; ²Rush Presbyterian St. Luke's Medical Center, Chicago, IL

LEARNING OBJECTIVES: 1. Diagnose Cervical-spine myelopathy despite unusual presentation. 2. Recognize that severe disease can occur before any symptoms.

CASE INFORMATION: A 50 year-old man with a history of hypertension presented with three days of left-sided weakness and spasms in both lower extremities. These symptoms

worsened when he bent over. He had no history of trauma. EXAM: T-97.6oF P-86 BP-160/85 RR-20. Spasms wracked his body in myoclonic-like jerks. Mental status and cranial nerves were normal. He was hypertonic and hyperreflexic in all limbs with mild weakness (4/5) in the left limbs only. He had decreased proprioception in the lower extremities with no saddle anesthesia. His toes were up-going bilaterally. Rectal tone was normal. At this point in his hospitalization, the differential diagnosis included a cervical-spine compression, a basal ganglia lesion, toxins, tetanus, and acute dystonia. Because all 4 limbs were involved but his sensorium was spared, an emergent MRI was obtained to exclude cervical spinal cord compression. This showed massive cervical cord compression: disc bulging with central herniations and ligamentum flavum hypertrophy causing marked spinal stenosis in C3-C7; the stenosed segment resembled a corkscrew in appearance; there was associated cord edema in C3-C4. The diagnosis of severe cervical spondylosis with myelopathy was made, and the patient was started on dexamethasone prior to an urgent laminectomy of the C3-C7 vertebrae. The patient improved post-operatively with marked improvement in the spasms. He was subsequently discharged to a long-term rehabilitation facility.

DISCUSSION: Degenerative changes in the spinal canal can cause cervical spondylosis with associated myelopathy. The peak age of onset lies between the fourth and sixth decades; males predominate in a 3:2 ratio. Trivial injuries often precede the onset of symptoms, yet the course is typically an indolent one punctuated with exacerbations. Classic features include weakness (legs affected earlier than the arms) and up-going plantars. Motor pathway damage dominates early cord compression presumably due to edema and poor capillary circulation that develops in watershed areas of the cord; the corticospinal tracts dominate these areas. Accordingly, there are few sensory findings in early cord compression (though vibration loss may be found). This case was unusual because spasms dominated the clinical picture. This was likely due to damage of the vestibulospinal and rubrospinal tracts; normally, these tracts suppress muscle activity. Surgical decompression is required once neurological exacerbations develop.

SURVIVING THE ECO-CHALLENGE: PERILS OF ADVENTURE TRAVEL. R.R. Cader¹; ¹UCLA/West L.A. VA, West Los Angeles, CA

LEARNING OBJECTIVES: 1. Recognize the clinical manifestations of Leptospirosis 2. Increase awareness of adventure travel hazards.

CASE INFORMATION: A 32 y/o Latino male presented to clinic with a 2 day history of fevers to 104 F, diffuse body aches, and generalized weakness. He denied diarrhea, nausea or vomiting. His travel history was significant for recently going to Malaysian Borneo where he competed in the Eco-Challenge Sabah (ECS) 2000 competition. He took part in jungle trekking, caving in "bat caves", outrigger sailing, canoeing, scuba diving, mountain biking, and a 12 km swim in the local rivers. On examination, he was afebrile, normotensive, with pulse 77 (his normal resting pulse was 43). He appeared ill and mildly dehydrated. He had shotty cervical and axillary lymphadenopathy. He was anicteric with good skin turgor. Neck, lung, heart and abdominal examination were unremarkable. Skin exam was significant for multiple healing excoriations on the lower extremities (from leech bites). Labs showed a mild hepatitis with alt 53 and ast 49. Electrolytes, urinalysis, and CBC were all within normal limits. The patient was treated with doxycycline and quinine for empiric treatment of leptospirosis vs. malaria. In the following days, it became apparent that other ECS competitors were infected with an unknown organism. Sera sent to the CDC from my patient was positive for leptospirosis. His symptoms resolved after a 10 day course of doxycycline (quinine was stopped on day 4 secondary to tinnitus). At last visit, he was healthy with normalization of transaminases and resolution of lymphadenopathy. He is currently planning to go to Vietnam for another adventure expedition.

DISCUSSION: Leptospirosis is caused by a spirochete and is characterized by high fever, muscle aches, and jaundice. Transmission occurs through contact with water contaminated with urine of infected animals. Leptospirosis occurs worldwide in both rural and urban settings. The CDC has confirmed at least 7 out of 25 suspected cases of leptospirosis amongst American competitors of the ECS. Other adventure challenges have resulted in outbreaks of meningitis, hookworm, and Dengue fever. My patient's swimming in local rivers with skin lesions from leech bites may have put him at increased risk of contracting leptospirosis. His activities in the bat caves may have also placed him at higher risk. Prophylaxis with doxycycline 200 mg once a week would have prevented infection. With the globalization of extreme sporting events such as the Eco-Challenge, it should be realized that infectious diseases play an important role in the health of the athletes involved. It is imperative that primary care physicians be aware of the diseases that are endemic to different regions of the world. Not providing the appropriate medical counseling/prophylaxis can prove to be life threatening.

AMIODARONE AND THE THYROID: ITS NOT WHAT YOU USUALLY THINK. E. Caiola¹; ¹Department of Medicine, University of Rochester School of Medicine, Rochester, NY

LEARNING OBJECTIVES: 1. Recall that amiodarone therapy can cause both hypo and hyperthyroidism.

2. Recognize that amiodarone's beta-blocking effects can mask the clinical signs and symptoms of hyperthyroidism.

3. Recognize that symptoms of amiodarone induced-hyperthyroidism may be worsening ischemic heart disease or congestive heart failure.

CASE INFORMATION: A 67 year-old woman with a past medical history of hypertension, coronary artery disease (without infarct) and chronic atrial fibrillation presented with recurrent episodes of congestive heart failure. She was 2 years status post coronary artery bypass grafting and her atrial fibrillation was being treated with amiodarone and warfarin. The patient was readmitted for the third time in 6 weeks with worsening dyspnea on exertion. She denied any associated peripheral edema, chest pain, nausea, vomiting, diaphoresis, fever or cough. Myocardial ischemia was ruled out on each occasion and medical therapy was optimized with titration of an ACE inhibitor, Beta-blocker, long acting nitrate and furosemide. The patient reported compliance with all her medications. She denied any weight loss, tremor, or heat intolerance. Physical examination was notable for an elevated jugular venous pulse, smooth palpable non-tender thyroid, diffuse crackles on lung exam and an irregular pulse with a holosystolic murmur at the cardiac apex. Chest radiographs were consistent with pulmonary

edema. EKG showed atrial fibrillation with a ventricular rate of 69 and non-specific ST-T wave changes. Echocardiography demonstrated normal left ventricular function with an ejection fraction of 60%, moderate mitral valve insufficiency and left ventricular hypertrophy. Routine follow-up thyroid function studies obtained due to amiodarone therapy showed marked hyperthyroidism with an undetectable TSH (<0.05 mIU/ml) and an elevated free T4 4.1 (0.9–1.8 ng/dl). Previous thyroid studies had been normal. Thyroglobulin and anti-thyroglobulin antibodies were within normal limits. Serum Interleukin-6 was elevated at 12 pg/ml (<5). Propylthiouracil was started for presumed amiodarone-induced hyperthyroidism and the amiodarone was discontinued. The patient's congestive heart failure resolved over the following week and she was discharged home.

DISCUSSION: Amiodarone therapy may lead to sub-clinical hypothyroidism in up to 20% of treated patients. Amiodarone-induced hyperthyroidism, however, is uncommon, occurring in less than 3% of patients. The drug's beta-blocking effects usually mask the clinical symptoms of hyperthyroidism. Common presenting symptoms can be worsening atrial arrhythmias, exacerbation of ischemic heart disease, or congestive heart failure. Serum thyroglobulin and Interleukin-6 levels may also be elevated.

GLUTEN-SENSITIVE ENTEROPATHY IN AN ASYMPTOMATIC MALE PRESENTING WITH IRON DEFICIENCY ANEMIA AND OSTEOPOROSIS. M. Campbell¹, G.W. Rouan², N. Gilinsky³; ¹University of Cincinnati College of Medicine, Cincinnati, Ohio; ²University of Cincinnati, Cincinnati, OH; ³University of Cincinnati, Cincinnati, Ohio

LEARNING OBJECTIVES: 1) Define the pathophysiology of gluten-sensitive enteropathy (GSE), and

2) Recognize the clinical scenario for atypical presentation of GSE.

CASE INFORMATION: A 71 year old Caucasian male presented for an annual physical exam. The patient had no complaints and no new findings on physical exam. Past medical history was significant for hypertension, osteoporosis, osteoarthritis, and a hiatal hernia. Significant laboratory data included a Hgb of 12.1, Hct of 38.0, MCV of 98.5, iron of 33 (normal 50–160), TIBC of 437 (normal 245–400), % Fe Sat of 8 (normal 20–55), and folic acid level of 14.4 (normal 4–12).

Subsequent workup for iron deficiency anemia (IDA) included upper endoscopy with distal duodenal biopsy to rule out villous atrophy as a cause for the anemia. The biopsy demonstrated small intestinal mucosa with nearly complete villous atrophy consistent with GSE. No active inflammation or microorganisms were seen. The anti-endomysial antibody was positive at >40 (normal <5). The anti-gliadin antibodies were also positive with an IgA of 34 AU (normal <4 AU) and an IgG of 89 AU (normal <43 AU). The patient was educated about GSE and instructed to follow a gluten-free diet.

DISCUSSION: Most adult patients with GSE present with a typical malabsorption syndrome characterized by weight loss, abdominal distention and bloating, diarrhea, steatorrhea, and abnormal results on tests of absorptive function. In some patients with GSE, the disease is insidious in onset with mild or even absent symptoms. Furthermore, some patients present with isolated abnormalities that may not initially suggest the diagnosis of GSE. The diagnosis of GSE may be delayed in patients who do not complain of gastrointestinal symptoms but do have osteoporosis, easy bruising secondary to Vitamin K deficiency, or iron deficiency anemia of unknown etiology or resistance to treatment.

Studies suggest that 26–36% of patients with GSE have osteoporosis at the lumbar spine (compared to 5% of the general population), and that approximately 30% of cases isolated, refractory IDA are secondary to GSE. GSE is a treatable disorder. In patients with osteoporosis or IDA of unknown etiology and/or resistance to therapy, the clinician should have a high index of suspicion for GSE.

NOT ALL THAT SEIZES IS IN THE HEAD: CONGENITAL LONG QT SYNDROME PRESENTING AS SEIZURES. P. Canley¹, D.W. Brady²; ¹Emory University, Atlanta, GA; ²Emory University, Decatur, GA

LEARNING OBJECTIVES: 1) To recognize the need to consider an underlying cardiac etiology for seizure disorders when appropriate. 2) To recognize the importance of diagnosing and appropriately treating the congenital long QT syndrome. 3) To develop an understanding of the pathophysiology of the syndrome.

CASE INFORMATION: A 47 year old female presented to the emergency department with a chief complaint of palpitations. They occurred at rest, were rapid but regular, were associated with shortness of breath, and terminated spontaneously. She denied chest pain, heart failure symptoms, and cardiac risk factors. An ECG was performed and showed no ischemia. The patient was then put into the chest pain center to rule-out myocardial infarction with serial ECGs and cardiac enzymes. While in the chest pain center, the patient experienced an episode of pulseless ventricular tachycardia which terminated spontaneously after the patient seized. The patient was then admitted to the CCU. Further questioning revealed that the patient had been diagnosed several years prior with a seizure disorder. The work-up of the etiology of this disorder was unrevealing, with no seizure focus being discovered. As her prodrome consisted of palpitations and dyspnea, a holter monitor was performed and revealed only a sinus arrhythmia. She suffered from monthly episodes and had tried various anti-seizure medications to no avail. She denied a family history of seizures, arrhythmias, sudden cardiac death, or deafness. Physical exam was unremarkable. Her ECG, however, was significant for a corrected QT interval (QTc) of 529 milliseconds. The patient was started on a beta-blocker and was then transferred to another institution for placement of a defibrillator. She has not had a seizure since implantation of the defibrillator.

DISCUSSION: Congenital long QT syndrome is a disorder of cardiac ion channels resulting in prolongation of the QT interval and characteristic T wave abnormalities. It is believed that one in 7,000 to 10,000 people are carriers of this autosomal dominant disorder. An estimated 3,000 young adults die of the disorder annually. Approximately 2/3 of carriers will experience syncope episodes, with 30% of those experiencing sudden cardiac death, often with the first episode. Five percent of asymptomatic carriers will suffer sudden cardiac death. Treatment with beta-blockers and pacemakers or implantable cardiac defibrillators can reduce mortality to 6%. Unfortunately,

patients with the disorder often go undiagnosed. Additionally, diagnosis is not straightforward, as up to 12% of known carriers have a QTc less than 440 milliseconds and 30% have a QTc between 450 and 470 milliseconds. Current genetic research may reveal a more direct way to diagnose this disorder, but until such time arrives it is important that we as primary care providers be able to recognize and appropriately treat this disorder.

CHRONIC LUNG DISEASE IN SICKLE CELL PATIENTS: UNDERDIAGNOSED? M.P. Carson¹; ¹St. Peter's University Hospital, New Brunswick, NJ

LEARNING OBJECTIVES: 1) Diagnose chronic lung disease in patients with sickle cell disease.

2) Recognize the risk of pulmonary hypertension during pregnancy.

CASE INFORMATION: A 30-year-old G2P1 with a history of sickle cell disease presented to our institution at 26 weeks gestation with the complaint of preterm contractions. She was admitted and placed on magnesium as a tocolytic. She then began to complain of back and leg pain leading to a diagnosis of a sickle crisis. Her medical history was notable for 3 to 4 inpatient admissions for sickle crises a year. She had a port-a-cath placed 11 months prior to the admission. Her gall bladder was removed two years prior to admission.

Her evaluation included room air PaO₂ of 68 mmHg and pulse oximetry of 93%, and a chest x-ray revealed bilateral pulmonary infiltrates in an alveolar pattern. Given a normal urinalysis and urine culture, and no other identifiable trigger for the crisis or the preterm contractions, she was treated for pneumonia. Magnesium was discontinued. The improvement in her oxygenation seemed slower than expected prompting the high-risk team to order an echocardiogram. Her left ventricle and valves were normal, but the right-sided chambers were enlarged and her estimated pulmonary artery systolic pressure was 55 mmHg. V/Q scan was normal. This is the first time she was diagnosed as having pulmonary hypertension. Review of prior chest x-rays from another institution revealed that the pulmonary infiltrates had been present for at least 18 months.

DISCUSSION: A review of the literature revealed many series of sickle cell patients with chronic lung disease. These patients may be asymptomatic, but abnormalities on pulmonary function testing are an early and reliable marker of this complication. Patients with abnormal chest radiography, pulmonary function testing, or hypoxia during stable periods have an average life expectancy of 2.6 years. They are at risk for sudden death from subendocardial infarction without epicardial arterial disease.

We are surprised that more pregnant patients with long standing sickle cell disease do not have routine screening for lung disease by their physicians. This is particularly important during pregnancy because severe pulmonary hypertension (pressures >80 mmHg) from any cause has a post-partum mortality rate as high as 50%. The precise risk with less severe elevations is not known.

Given the prevalence of sickle cell disease in the general medical population we recommend that all patients with long standing sickle disease have pulmonary function testing and/or echocardiography to evaluate pulmonary arterial pressures. If chronic pulmonary disease is diagnosed in a woman considering pregnancy then consultation with experienced internists, pulmonary specialists, and perinatologists is warranted.

"HE'S ACTING STRANGE": A CASE OF THYROTOXICOSIS. A. Cassoobhoy¹, D.W. Brady²; ¹Emory University, Atlanta, GA; ²Emory University, Decatur, GA

LEARNING OBJECTIVES: 1. Recognize neuropsychiatric symptomatology as a presentation of thyrotoxicosis. 2. Consider thyrotoxicosis as a precipitant in the differential diagnosis of acute affective psychosis.

CASE INFORMATION: H. H. is a twenty one year old black male with a past medical history significant for fetal alcohol syndrome who was brought to the emergency room because his mother was concerned he was "acting and talking strange." His change in behavior was first noted two months prior when he was fired for sleeping on the job. Since then he has had labile mood swings with agitation, nervousness, anxiety, fatigue and poor sleep. He uncharacteristically disappeared for a week during this time without telling his family. His social history is significant for adoption, completion of high school in a special education program, and a history of occasional alcohol and marijuana abuse. On exam H.H. appeared to be younger than his stated age and was agitated, trying to undo his restraints. His conversation was tangential, exhibiting forced speech and disorganization of thought. He was alert and oriented to person and place with prompting but had poor insight into his situation. His affect was inappropriate, alternating between laughter and anxiety. His physical exam was significant for tachycardia, mild proptosis, and a diffuse enlarged goiter. H. H. was admitted with the preliminary diagnosis of altered mental status and thyrotoxicosis. Subsequently, thyroid function tests demonstrated a TSH of less than .005, free thyroxine of 7.77, and T3 of greater than 650. With treatment of his thyrotoxicosis, his mental status returned to baseline. Other evaluation was negative. At discharge he was to continue outpatient treatment of Graves disease.

DISCUSSION: Thyrotoxicosis may present as tense dysphoria, including symptoms of nervousness, restlessness, tension, anxiety, and irritability. Also, there may be subjective impaired cognition. Frank psychiatric symptoms are rare. Environmental stressors and psychiatric conditions may be associated with the onset and clinical course of thyrotoxicosis. Thus clinically distinguishing thyrotoxicosis from an anxiety disorder, panic attack, or manic phase may be difficult.

A RARE CAUSE OF DYSPNEA IN A YOUNG ADULT: UNDIAGNOSED CONGENITAL AORTIC COARCTATION. E. Chamoun¹; ¹Hennepin County Medical Center, Minneapolis, MN

LEARNING OBJECTIVES: 1) Congenital abnormalities such as coarctation should be thought of when hypertension is diagnosed in young adults.

2) Legs pulses should be checked in all young adults with hypertension.

CASE INFORMATION: A.C. is a 28 y.o. Hispanic man who was followed in our clinic for hypertension diagnosed in Mexico in his early 20's. He was a healthy young man with no

complaints or symptoms suggestive of cardiopulmonary diseases. On a follow up visit he reported dyspnea on minimal exertion of two months duration. He also reported chest pressure, dizziness and nausea with the dyspnea. He told his doctor that he has dyspnea even after speaking few sentences. His exam revealed BP= 146/90 Pulse 80 regular. Heart auscultation revealed a normal S1, split S2, and I-II/VI systolic murmur. Lungs were clear to auscultation. Abdomen was soft, no hepatosplenomegaly. No pedal edema. EKG was done in the clinic showed left ventricular hypertrophy. Chest X-ray was normal. Stress echocardiogram revealed a small apical wall motion abnormality worsening with stress. A cardiac catheterization revealed normal coronaries, no wall motion abnormality and left ventricular end diastolic pressure of 35 mmHg. The aortic root was bicuspid with severe coarctation of the aorta at the isthmus.

DISCUSSION: Coarctation of the aorta is an unusual cause of hypertension in young adults and is often unrecognized by general practitioners. Although the left ventricle is usually hypertrophied because of the hypertension, the heart is usually normal in size or only slightly enlarged until cardiac decompensation occurs late in the course of the disease. Our patient left ventricular end diastolic pressure was 35 mmHg, which might account to his dyspnea. In retrospect knowing the diagnosis of coarctation our patient has the body habitus of a chronic coarctation of the aorta with a large head and upper extremities and is otherwise short. He had diminished leg pulses. His blood pressure was comparable in upper extremities; however, there was a difference of 50 mmHg between upper and lower extremities. Coarctation of the aorta is a potentially lethal condition at all ages. Many patients first come to medical attention because of signs and symptoms related to hypertension. The diagnosis should be considered in patients with hypertension and diminished leg pulses, and we recommend the routine examination of leg pulses in young adults with hypertension.

HEMOLYSIS IN A RECENTLY HOSPITALIZED PATIENT. L. Chang¹; ¹UCLA/San Fernando Valley Program, Sepulveda, CA

LEARNING OBJECTIVES: 1) To recognize drug-induced hemolysis as a common cause of hemolytic anemia

2) To recognize that 2nd and 3rd generation cephalosporins are the most common cause of drug-induced hemolytic anemia

3) To recognize that cephalosporin-induced hemolytic anemia is often serious, and can be fatal.

CASE INFORMATION: A 62-year-old female discharged 7 days ago after being treated for Streptococcus pharyngitis presents to a follow-up clinic complaining of a 5-day history of nausea, nonbilious and nonbloody vomiting, and anorexia. She also complains of a generalized headache and subjective fevers. In her previous hospitalization, she presented to the emergency room complaining of fever, sore throat, and epigastric pain. At that time, she was found to have a WBC of 16 with 89% neutrophils, a Hct of 44, and a clean urinalysis. A CT abdomen was done revealing diverticulosis only. Her temperature was 38.4. She was given one dose of Cefotetan in the ER and admitted for further workup. On the medicine wards, further evaluation revealed an exudative tonsillitis with anterior cervical lymphadenopathy. She admitted to a history of recurrent pharyngitis. She was treated with penicillin VK, improved clinically, and sent home on penicillin. She had no known drug allergies.

Physical examination: Temp 38.5, BP 141/70, pulse 97. She was moderately ill appearing. Eyes were icteric. Oropharynx revealed tonsillar swelling, but was significantly improved from her previous hospitalization exam. Neck was supple. Heart and lung exams were normal. Abdomen revealed mild epigastric tenderness without peritoneal signs. Neurological exam was normal.

Laboratory studies: WBC 18.2, Hgb 11, Hct 30.6, Plt 300, MCV 89. Total bilirubin 13.2, direct bilirubin 0.5, AST 34, ALT 28, alkaline phosphatase 93, PTT 23, INR 1.04. LDH 1578, corrected reticulocyte count 1.8, haptoglobin <50. Direct Coombs was markedly positive.

DISCUSSION: The positive direct Coombs test was further investigated by the blood bank that, along with assistance from the Red Cross, determined that this patient had developed a Cefotetan-induced autoimmune hemolytic anemia (HA). It should be recognized that 1st-generation cephalosporins rarely cause HA, but 2nd and 3rd generation cephalosporins are the most common cause of drug-induced HA. Most cases of cephalosporin-induced HA revealed moderate-to-severe hemolysis with fatalities reported. Most patients required treatment with blood transfusion and steroids. Because of the common use of 2nd and 3rd generation cephalosporins by physicians, and the severity of the hemolysis that can be induced by them, recognition of this entity is of significant importance to the practicing clinician.

A CASE OF HEMOPTYSIS FOLLOWING PERCUTANEOUS TRANSLUMINAL CORONARY ANGIOPLASTY. S. Chebrolu¹, J. Addagatta¹, F. Ali¹; ¹St. Francis Hospital, Evanston, IL

LEARNING OBJECTIVES: 1) Recognize the occurrence of bleeding complications including pulmonary alveolar hemorrhage following the use of Group IIb-IIIa receptor inhibitors in acute coronary syndromes. 2) Diagnose and manage pulmonary alveolar hemorrhage.

CASE INFORMATION: A 68 year-old woman with a past medical history of diabetes mellitus and hypertension presented to the emergency room with chest pain. She was a non-smoker and had no known lung disease. Physical examination was normal. EKG showed an acute anteroseptal myocardial infarction. Emergent angiogram revealed a 100% proximal left anterior descending artery lesion. Percutaneous transluminal coronary angioplasty with placement of a multilink stent of the left anterior descending artery lesion was done. During the procedure, she was given 10,000U of heparin and 0.25 mg/kg of abciximab boluses followed by infusion of heparin at 700U/hr and abciximab at 0.125 mcg/kg/min after the procedure. An intra-aortic balloon pump was also placed. One hour later, she had hematemesis and hemoptysis and developed respiratory distress with hypoxia. Heparin and abciximab were discontinued and emergent intubation was done. A copious amount of blood was aspirated from the endotracheal tube. CXR revealed diffuse bilateral interstitial infiltrates. Hemoglobin dropped from 12.6g/dl to 10g/dl. A diagnosis of pulmonary alveolar hemorrhage was made. Bright red blood was suctioned from the endotracheal tube for a few hours. Hypoxemia improved rapidly with complete clearing of the CXR over the next two days. She was extubated on day three and was

discharged home on day seven on aspirin, clopidogrel, metoprolol and ramipril. She received two units of packed red blood corpuscles during her hospital course.

DISCUSSION: Pulmonary alveolar hemorrhage is an infrequent and rarely reported hemorrhagic complication of the use of Group IIb-IIIa receptor inhibitors in acute coronary syndromes. It should be considered in the differential diagnosis of a patient who receives these agents and presents with respiratory distress, pulmonary infiltrates and a drop in hemoglobin, with or without hemoptysis. It is usually seen in older patients and in those with increased pulmonary capillary wedge pressure and evidence of pulmonary edema. Management is mainly supportive with discontinuation of all anticoagulants and with transfusions of packed red blood corpuscles and platelets as needed.

ABDOMINAL PAIN IN THE FACE OF A NORMAL PHYSICAL-EXAMINATION. S. Chebrolu¹, D. Metha¹; ¹St. Francis Hospital, Evanston, IL

LEARNING OBJECTIVES: 1) Assess a patient with abdominal pain in the paucity of physical findings and suspect mesenteric ischemia. 2) Recognize the association of Mesenteric Vein Thrombosis (MVT) with gastrointestinal malignancy.

CASE INFORMATION: An 82 year-old man was admitted with complaints of dysphagia and constipation for one week. It was associated with anorexia and significant weight loss over one month duration. He denied any history of abdominal pain, hematemesis or melena. Past medical history included COPD and coronary artery disease. Past surgical history included CABG, appendectomy and bilateral inguinal hernia repair. Abdominal examination was benign but he was found to be fecal occult blood positive. Laboratory work up revealed evidence of iron deficiency anemia and a gastrointestinal malignancy was suspected. He underwent gastro-duodenoscopy that revealed hiatal hernia and a colonoscopy was planned. A few hours after the procedure, he complained of severe lower abdominal pain when he was noticed to have right lower quadrant tenderness and fullness but with no guarding or rigidity. CT scan of the abdomen was done which revealed a mass in the terminal ileum and a 5-cm mass in the left lobe of liver. It also revealed superior MVT with ischemic changes in the small bowel. He underwent emergent angiography that confirmed the presence of MVT and was given local t-PA for dissolution of the MVT. Following this, his abdominal pain worsened with development of peritoneal signs and he underwent exploratory laparotomy with the resection of terminal ileum, right hemicolectomy and biopsy of the liver mass. Histopathology of the mass and liver biopsy revealed malignant carcinoid tumor with mesenteric node and liver involvement.

DISCUSSION: MVT is characterized by pain out of proportion to physical findings, nausea, vomiting, constipation and occasional bloody diarrhea. Previous abdominal surgery, hypercoagulable states, cirrhosis and gastrointestinal malignancies are the most common conditions associated with MVT. Abdominal distention, though nonspecific, is the most frequent sign. Leukocytosis may be found in half of the patients but laboratory tests are not helpful in the diagnosis. CT is the diagnostic test of choice and demonstrates thrombus in the superior mesenteric vein with evidence of bowel ischemia. Anticoagulation with heparin followed by warfarin in the absence of contraindications is the mainstay of medical management. All patients with localized or diffuse peritonitis should undergo immediate exploratory laparotomy with resection of nonviable bowel. MVT has not been reported previously as a presenting feature of malignant carcinoid tumor as seen in our patient here.

RARE COEXISTENCE OF PRIMARY HYPERPARATHYROIDISM AND MULTIPLE MYELOMA IN A HYPERCALCEMIC PATIENT. S. Chebrolu¹, F. Zar¹; ¹St. Francis Hospital, Evanston, IL

LEARNING OBJECTIVES: 1) Recognize that Primary Hyperparathyroidism (PHP) and Multiple Myeloma (MM) are the most common causes of hypercalcemia in the elderly population and diagnose them. 2) Recognize that more than one independent cause of hypercalcemia is very rare and should be considered in a patient unresponsive to treatment.

CASE INFORMATION: A 76 year-old male was admitted with weakness, lethargy and occasional confusion of three months duration. His past medical history included sick sinus syndrome and valve replacement for aortic incompetence. His primary doctor discontinued chlorthalidone four weeks prior to this admission after he was found to be hypercalcemic (13.9mg/dl). Examination was unremarkable. Initial studies revealed calcium 11.8mg/dl (normal 8.8-10.4mg/dl), ionized calcium 6.77 mg/dl (normal 4.6-5.3mg/dl), and phosphorus 2.3 mg/dl (normal 2.4-4.5mg/dl). CT scan of head was normal. Patient was hydrated with normal saline and later diuresed with furosemide. An intact parathyroid hormone level by immunochemiluminometric assay was 12.1 pmol/L. C-terminal peptide was 323 pmol/L. Serum osteocalcin was 21U/L. A CT scan of the neck showed a mass suggestive of a parathyroid adenoma, confirmed by Technetium-99m sestamibi scan showing a persistent crescentic area of increased activity in the left lower neck in both immediate and delayed films. In addition, urine protein electrophoresis showed monoclonal kappa light chains. Serum protein electrophoresis revealed a monoclonal spike and IgG kappa monoclonal protein (2770mg/dl). Beta-2 microglobulin was 6.4mg/dl. A skeletal survey revealed multiple lytic defects throughout the skeletal system. Bone marrow aspiration showed focal increase in plasma cells consistent with MM. The patient was given pamidronate and started on chemotherapy with melphalan and he responded well.

DISCUSSION: Although a variety of conditions can give rise to hypercalcemia, nearly 90% of the cases can be attributed to either PHP or MM. PHP was diagnosed in our patient by finding elevated intact parathyroid hormone levels associated with elevated C-terminal peptide levels. Technetium-99m-sestamibi scan confirmed its presence. IgG kappa monoclonal paraprotein on serum and urinary protein electrophoresis associated with multiple lytic defects on the skeletal survey was highly suggestive of MM which was confirmed by bone marrow biopsy. However, more than one independent cause of hypercalcemia in a patient is rarely seen. Only eighteen cases of simultaneous occurrence of both PHP and MM were reported and our patient will be the nineteenth such case. Considering the rarity, it appears that this is a chance association. Persistent and unresponsive hypercalcemia may occur if only one of

causes is recognized in such rare instances and in a MM patient, this may falsely lead to a higher tumor grading.

METASTASIS-INDUCED ACUTE PANCREATITIS. S. Chebroly¹, A. Naidu¹, F. Zar¹; ¹St. Francis Hospital, Evanston, IL

LEARNING OBJECTIVES: 1) Recognize that metastatic tumors to pancreas may rarely induce acute pancreatitis referred to as Metastasis-Induced Acute Pancreatitis (MIAP). 2) Diagnose and manage MIAP.

CASE INFORMATION: A 37-year-old woman presented with a one-day history of epigastric pain radiating to the back associated with nausea and decreased appetite. One and half years ago, she had a missed abortion followed by a Placental Site Trophoblastic Tumor (PSST) which was treated with total abdominal hysterectomy and chemotherapy with methotrexate. Her beta-HCG was less than 2 IU/l six months prior to this presentation. She had no history of alcoholism. On examination, she was found to have epigastric tenderness. Laboratory investigations revealed amylase 309 IU/l, lipase 154 IU/l, lactate dehydrogenase 544 IU/l, calcium 8.5mg/dl, glucose 221mg/dl and hemoglobin 13.6g/dl. She was diagnosed as acute pancreatitis and treated medically. Her pain persisted and she had CT scan of the abdomen that showed a swollen and enlarged pancreas with multiple rounded lesions of variable sizes. There was also a lobulated 3x4cm mass behind the sigmoid colon with a surgical clip adjacent to it associated with retroperitoneal and inguinal lymphadenopathy. Beta-HCG was found to be 2320.6 IU. ERCP was unremarkable. Exploratory laparotomy with bilateral salpingoophorectomy and tumor dissection was done showing histologic evidence of recurrent PSST. A diagnosis of MIAP secondary to metastatic PSST was made and chemotherapy with etoposide, methotrexate and adriamycin was started with resolution of her symptoms.

DISCUSSION: MIAP refers to non-pancreatic tumors metastasizing to the pancreas and resulting in acute pancreatitis. It is a very rare cause of acute pancreatitis and only 42 such cases have been reported in the literature. Most of them were associated with bronchogenic carcinoma and lymphoma. Mechanisms implicated in the causation of MIAP include mechanical ductal obstruction and rupture with direct parenchymal tumor invasion or ischemia secondary to vascular occlusion and encasement of pancreatic vessels. The clinical and laboratory presentation is similar to the other causes of pancreatitis and MIAP remains a diagnosis of exclusion. An accurate history, supportive clinical evidence and radiological findings of pancreatic and peri-pancreatic metastatic lesions in the presence of a known primary tumor aid in the diagnosis. MIAP is managed conservatively with nasogastric tube suction, intravenous fluids and analgesics and this approach is helpful in most cases. If it persists, systemic chemotherapy for the metastatic tumor or palliative abdominal radiation may help. Overall, MIAP portends a bad prognosis and many patients succumb to the metastatic tumor sooner or later. PSST, a rare form of Gestational Trophoblastic Disease, has never previously been reported as a cause of MIAP and this is the first such instance.

EGG DONATION APOPLEXY. R. Chiu, **STANFORD UNIVERSITY SCHOOL OF MEDICINE, PALO ALTO, CA.** R. Chiu¹; ¹Stanford University, Palo Alto, CA

LEARNING OBJECTIVES: 1) Recognize risk of pituitary apoplexy from leuprolide use 2) Assess the need of imaging for meningeal symptoms.

CASE INFORMATION: A previously healthy 22 year old female Stanford graduate student presented with three days of fever and headache with associated nausea, neck stiffness, and photophobia. Past medical history significant for migraine headache and menstrual irregularity which responded well to oral contraceptives. Current medication includes three days of leuprolide injections for commercial egg donation. On exam, patient was febrile to 39.6 degrees Celsius with otherwise normal and stable vital signs. Patient was slightly lethargic and had slight photophobia and mild posterior neck tenderness on palpation, but otherwise completely normal exam. Ophthalmologic and neurologic exams were completely normal. Labs were normal except for Sodium 126 and Potassium 3.3. Lumbar puncture revealed opening pressure of 13cm. CSF studies revealed Glucose 44, Protein 51, WBC 27, RBC 510, and negative microscopic evaluation. Bacterial, viral and fungal cultures were eventually all negative.

Empiric antibiotics of ceftriaxone, vancomycin and acyclovir were started. Patient continued to have headache, nausea, and fever with no clinical improvement or worsening except for Sodium continue to drop to 115 over next three days. On hospital day three, patient developed seizure and head MRI revealed 2.3cm x 1.8cm sellar mass with hemorrhage and compression on optic chiasm. Surgical decompression of mass was complicated by right ACA/MCA territory infarct secondary to vasospasm and great sodium fluctuation due to SIADH and adrenal insufficiency. Patient eventually was discharged on thyroxine, cortisone, and DDAVP. She has made incredible recovery and will be resuming her PhD studies.

DISCUSSION: The risk of pituitary apoplexy with use of leuprolide has been described in the literature but not well screened. Detailed clinical history, exam, and laboratory investigation might identify patients at higher risk of pituitary apoplexy for selective imaging evaluation. In this patient, the history of menstrual irregularity controlled by oral contraceptives, the hyponatremia and loss of body temperature control could have been the starting point of investigation for pituitary adenoma and apoplexy. With increasing number of patients receiving elective leuprolide for fertility purpose, the medical community should be aware of potential risk in order to assess and advise patients accordingly.

SYMPTOMATIC EPIDURAL LIPOMATOSIS IN CUSHINGS DISEASE. S. Christopoulos¹, J. Minuk¹, P. Assimakopoulos¹, S.R. Kahn¹; ¹SMBD Jewish General Hospital, McGill University, Montreal, Quebec

LEARNING OBJECTIVES: 1) Recognize epidural lipomatosis as a complication of Cushing's syndrome, including Cushing's disease; 2) Consider epidural lipomatosis as a cause of back pain, weakness, radiculopathy, or myelopathy in Cushing's syndrome; 3) Diagnose and treat epidural lipomatosis.

CASE INFORMATION: A 63-year-old male with a history of diabetes mellitus, hypertension, heavy alcohol consumption, and a T6 vertebral fracture was admitted to hospital with a one year history of progressive lower extremity proximal muscle weakness. Concurrent with his worsening weakness, the patient noticed increasing central obesity and easy bruising. Physical examination revealed a classic cushingoid appearance. The neurological examination demonstrated symmetric polyneuropathy as well as lower extremity proximal muscle weakness and wasting. A T6 sensory level was present. The reflexes were 1+ and the plantar responses were flexor bilaterally. Laboratory investigations confirmed Cushing's syndrome and suggested a pituitary source of hypercortisolism. An MRI of the spine showed multiple vertebral fractures and epidural lipomatosis in the posterior epidural space, from T2 to T8, compressing the spinal cord maximally at T6. Given the fragility of this patient's spine and the ongoing investigations to determine the cause of hypercortisolism, the patient did not undergo emergent epidural fat debulking. A pituitary microadenoma was diagnosed and resected by a transphenoidal approach. Although repeat imaging was not performed to document regression of the spinal lipomatosis, there was clinical and laboratory evidence of resolution of the hypercortisolemia. The post-operative course was complicated by pulmonary emboli and recurrent sepsis; the patient suffered a fatal cardiac arrest 96 days after surgery.

DISCUSSION: Epidural lipomatosis has been described in steroid use and obesity, and more rarely in association with endogenous hypercortisolism such as Cushing's disease. This case emphasizes the difficulty in suspecting epidural lipomatosis in a patient with Cushing's syndrome in the setting of other neurologic processes. The diagnosis of epidural lipomatosis needs to be systematically considered in any patient with Cushing's syndrome who presents with back pain, lower extremity weakness, radiculopathy, or subtle signs of myelopathy. MRI of the spine is the diagnostic test of choice. Treatment modality (surgical debulking of epidural fat versus correction of the hypercortisolic state or weight loss) should be individualized based on the severity of neurological signs, the potential reversibility of the causative factor, and the estimated operative morbidity.



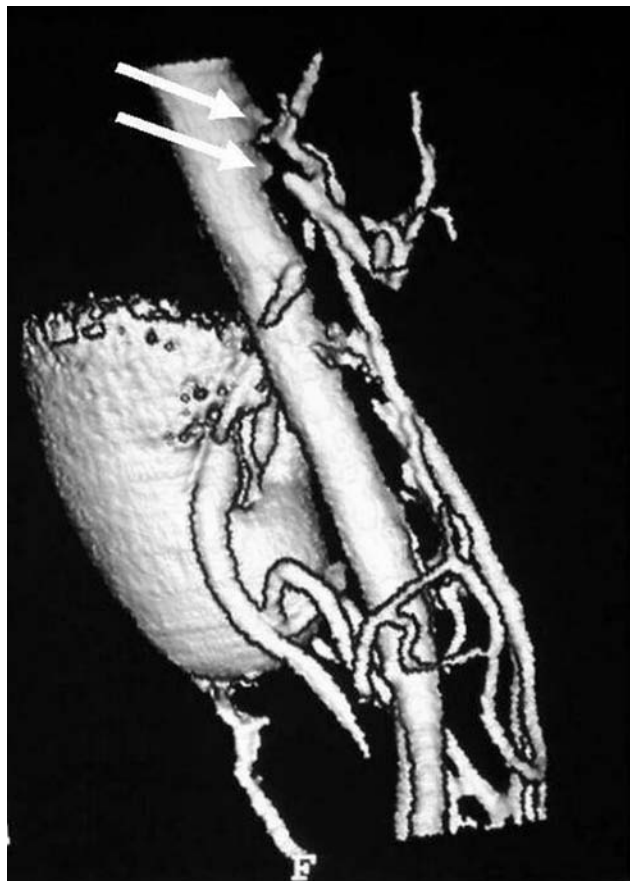
MRI demonstrating thoracic epidural lipomatosis with cord compression.

LIFE-THREATENING MESENTERIC ISCHEMIA IN A CHRONIC ERGOTAMINE USER. S. Christopoulos¹, A. Szilagyi¹, S.R. Kahn¹; ¹SMBD Jewish General Hospital, McGill University, Montreal, Quebec

LEARNING OBJECTIVES: 1) To recognize the importance of a careful medication history and a thorough review of side effects; 2) To recognize ergotism as a cause of mesenteric ischemia.

CASE INFORMATION: Over a two-year period, a 43-year-old woman presented to the emergency department on five separate occasions with complaints of severe abdominal pain. Extensive investigations including gastroscopy, colonoscopy, two CT scans and ultrasounds of the abdomen, small bowel follow-through contrast study, enteroclysis, as well as a complete gynecological work-up were all negative. Over this period of time, the patient's treating psychiatrist diagnosed her as having Somatic Pain Disorder. The patient was admitted to the medical service of our institution with severe malnutrition. Her past medical history included a longstanding history of migraine headaches treated with caffeine/ergotamine tartrate enemas, weekly or more often, since the age of 18. The patient described her abdominal pain as sharp, located primarily in the epigastric region, and radiating to the back. The episodes occurred one hour postprandially, and were associated with nausea and vomiting. She reported a 60 lb weight loss over a one-year period. Physical examination revealed cachexia. Serial abdominal examinations were unremarkable. Stool occult blood testing was negative. Tube feeding was initiated. A CT angiogram was requested, which revealed significant narrowing of both the celiac and superior mesenteric arteries with the presence of collaterals. A diagnosis of mesenteric ischemia was established. An angiogram was requested prior to intestinal bypass surgery, however the patient developed worsening abdominal pain, significant leucocytosis, a lactate level three times normal, and evidence of peritoneal irritation. She underwent urgent laparotomy, which revealed extensive ischemia of the small bowel and diffuse arterial disease of the abdominal aorta, the iliac, superior mesenteric, and celiac arteries. An aorto-superior mesenteric bypass was performed. The patient underwent a second look laparotomy within 24 hours and 30 cm of dusky small bowel were resected. The post-operative course was uneventful. Following the discontinuation of ergotamine use, the patient no longer experienced abdominal pain and gained a significant amount of weight.

DISCUSSION: Despite new modalities to treat migraine headaches, ergotamine is still used chronically by some migraine sufferers. The most recognized complication of ergotism is arterial vasospasm of the upper and lower extremities. More rarely, vasospasm is recognized in the carotid, retinal, renal, and mesenteric arteries. In our patient, failure to suspect this unusual cause of mesenteric ischemia led to delayed diagnosis, severe weight loss, and advanced intestinal infarction. The importance of a careful drug history cannot be over-emphasized.



3D Surface Shaded display showing high-grade stenosis of the celiac axis and superior mesenteric artery origins.

A RARE CAUSE OF LOWER GASTROINTESTINAL BLEEDING IN A YOUNG PATIENT.

L. Chung¹, L.J. Nicholson²; ¹Stanford University, Menlo Park, CA; ²Stanford University, Palo Alto, CA

LEARNING OBJECTIVES: Recognize the possibility of extra-pulmonary tuberculosis in foreign-born patients.

CASE INFORMATION: Various infections can cause gastrointestinal bleeding and must be considered in the differential diagnosis of patients with a significant travel history. A 32-year-old woman, originally from the Philippines, presented with a 9-month history of diarrhea and weight loss. Four months after her diarrhea began, she returned home and was diagnosed with an amebic infection, for which she was treated with a short course of antibiotics. At the time of presentation to our service, the patient's stools had become frankly bloody, associated with a significant drop in hematocrit. A 10-cm fungating, circumferential mass was found in the ascending colon on colonoscopy. Biopsies were consistent with a granulomatous colitis. In addition, the patient was found to have apical lung disease on chest x-ray with a positive tuberculin skin test. She was empirically treated with a 4-drug anti-tuberculosis regimen with complete resolution of the mass as well as her symptoms. Although lower gastrointestinal bleeding is an infrequent presentation of tuberculosis, the diagnosis of extra-pulmonary tuberculous infection should be considered in any patient born in or traveling to a high risk country.

EOSINOPHILIC MYOCARDITIS IN YOUNG HEALTHY FEMALE. AN UNUSUAL PRESENTATION OF CONGESTIVE HEART FAILURE. **M.A. Codina**¹, V. Martin¹; ¹University of Cincinnati, Cincinnati, OH

LEARNING OBJECTIVES:

- 1) Recognize and diagnose an unusual presentation of CHF, and
- 2) Manage a patient with eosinophilic myocarditis.

CASE INFORMATION: Although eosinophilic endocardial disease has been described, myocardial damage from eosinophilic infiltration is rarely diagnosed in life, usually due to its rapidly fatal course. Prompt diagnosis and treatment are essential to survival. This case report is of a 37 year old healthy white female who was referred to University Hospital from an outlying facility with a three week history of increasing shortness of breath, fatigue, musculoskeletal pains and intermittent chest pains. The patient was ultimately admitted with worsening symptoms.

PHYSICAL EXAM: BP 120/70; HR 135 reg; RR 18

CV: tachycardic with summation gallop; JVP of 8 cm

RESP: decreased breath sounds bilaterally at bases

PERTINENT LABORATORY DATA:

CBC: WBC 13,000 with Eosinophil count of 33%

CXR: Bilateral pleural effusions

ECHO: severe LV dysfunction with LVEF of 20%.

The patient underwent right and left heart catheterization which revealed normal coronary arteries, and the following pressures: RA = 11; RV = 35/12; PAWP = 17; PA 35/19. Further work up for etiology of CHF included viral serologies for CMV, EBV, Coxsackie, Varicella were all negative. A myocardial biopsy was finally obtained that showed a diffuse, severe eosinophilic infiltration in a relatively viable myocardium. The patient was started on Prednisone 50 mg BID, as well as ACE Inhibitor, Digoxin, and Carvedilol, with excellent clinical improvement. The patient was discharged to home on hospital day 10. Follow-up ECHO done three months post biopsy showed an LVEF of 60%.

DISCUSSION: Eosinophilic myocarditis is rarely diagnosed during life, likely due to its rapidly fatal course and non-specific clinical manifestation. Eosinophilic myocarditis has been seen associated with pronounced eosinophilia in the circulating peripheral blood like this patient, but the absence of profound eosinophilia does not rule out eosinophilic myocarditis. In the case of this young, previously healthy patient with an elevated peripheral eosinophil count, a myocardial biopsy proved to be the test which revealed the etiology of her CHF and subsequent appropriate treatment.

POLYCYSTIC OVARY SYNDROM AND SUBCLINICAL HYPOTHYROIDISM PRESENTING AS AMENORRHEA. **D.L. Conant**¹; ¹University of California, San Francisco, San Francisco, CA

LEARNING OBJECTIVES: 1. Diagnose polycystic ovary syndrome (PCOS). 2. Restore fertility in the setting of PCOS and subclinical hypothyroidism. 3. Recognize and manage the potential long-term sequelae of PCOS.

CASE INFORMATION: A 28 year-old G0P0 Asian woman presented for a new patient appointment concerned about 7 months of amenorrhea. She recalled that her menstrual cycles since menarche at age 13 had often been irregular-varying in length from 28 to 45 days-but she had never gone more than two months without a period. Her last menstrual period was shortly before an extended and stressful trip back to her family in Taiwan. After several months without a period, a basic workup in Taiwan revealed a negative pregnancy test, elevated blood pressure, and a slightly increased TSH. She returned to the U.S. and had no subsequent followup. She made her appointment when she and her husband had become increasingly concerned about her fertility. Her past medical history was remarkable only for elevated triglycerides several years ago. She took no medications and denied any use of tobacco, alcohol, or drugs. Born in Taiwan, she emigrated to the U.S. as a child. She lived with her husband and worked as a psychiatric nurse. Her family history was unremarkable. Review of systems was notable for weight gain of about 15 pounds over the last year, intermittent fatigue and mood swings, frequent frontal headaches and lower abdominal cramping, bilateral hand numbness at night, and increased hair loss. She denied heat or cold intolerance, fevers, night sweats, nausea, vomiting, diarrhea, dysuria, vocal or visual changes. On exam, she was mildly obese and well-appearing. Compared with her driver's license photo from two years before, she had apparently gained some weight but showed no obvious changes in her facial structure. Her blood pressure was 110/70 and her heart rate was 80. Her BMI was 30. Her thyroid gland was slightly enlarged. There were terminal hairs present on her upper back but no other evidence of hirsutism. Her abdomen was slightly obese without striae. Bimanual exam revealed slight enlargement of her ovaries and associated mild tenderness. Initial laboratory data included normal CBC and renal panel, undetectable serum beta-HCG, TSH 5.02, free T4 14, FSH 6.4, LH 11.5, prolactin 9.3, fasting glucose 103, total cholesterol 167, triglycerides 314, LDL 66, and HDL 38. Pelvic ultrasound revealed bilateral polycystic ovaries. Additional blood tests were obtained. Daily administration of 10mg of medroxyprogesterone acetate yielded a normal menstrual period on day nine. She was also started on thyroxine and a program of weight reduction.

DISCUSSION: PCOS is the most common endocrinopathy among women of reproductive age with an estimated prevalence of 4-7%. Though a clear consensus about diagnostic criteria for PCOS has not been achieved, the clinical hallmarks include evidence of hyperandrogenism, menstrual dysfunction, and polycystic ovaries. The diagnosis also requires the exclusion of other functional and neoplastic disorders with similar clinical features. Fertility may be restored in obese patients with PCOS through weight reduction alone. Metformin and clomiphene are other therapeutic options. In the unusual circumstance in which subclinical hypothyroidism may be contributing to menstrual disturbances, treatment with thyroxine has been demonstrated to restore normal function. Proper management of women with PCOS requires recognition of the long-term health risks associated with unopposed estrogen, insulin resistance, and lipid abnormalities including endometrial cancer, overt type 2 diabetes mellitus, and coronary artery disease.

EPHEDRINE NEPHROLITHIASIS ASSOCIATED WITH EPHEDRINE USE. **B.A. Costello**¹; ¹Mayo Clinic, Rochester, MN

LEARNING OBJECTIVES: 1) Awareness of nephrolithiasis as a potential complication of ephedrine overuse. 2) Recognition of importance of asking patients about over-the-counter (OTC) supplements.

CASE INFORMATION: A 34 yo female was referred to our institution following a diagnosis of nephrolithiasis. She reported 2-3 similar episodes/yr over the past 20 years. Prior to transfer, she was found to have a 5mm stone in her left ureter. Upon admission, analgesia and hydration were provided; urine was strained for debris. 24h urine for citrate, oxalate, calcium, uric acid, and cystine was normal. She passed a stone which was analyzed and found to be composed of ephedrine. Upon further questioning, she reported long-term use of ephedrine and pseudoephedrine for nasal congestion. In addition, she had recently been using herbal supplements for weight loss. Although she was unaware, we found an additional ephedrine stone on this patient in our referral lab files. We advised her to discontinue all these agents. She has not had further episodes after 6 months of follow-up.

DISCUSSION: Ephedrine nephrolithiasis has previously been reported in patients consuming ephedrine or ephedrine-containing herbal preparations (1,2). Our case suggests that chronic use of ephedrine can lead to recurrent nephrolithiasis. Herbal supplements used for dietary purposes commonly contain ephedrine and we suspect our patient was likely receiving additional ephedrine through the supplement. In 1997, the FDA did call into question the safety of supplements containing ephedrine (3), however, they remain readily available and are sometimes

hidden in dietary supplements. In 2000, the FDA described potential adverse effects of ephedrine alkaloids in a public meeting report. No mention of nephrolithiasis was made. This case report adds to previous findings of adverse effects related to ephedrine use.

PRIMARY KI-1-POSITIVE ANAPLASTIC LARGE CELL LYMPHOMA OF THE LUNG. E.J. Crane¹, E. Warm¹; ¹University of Cincinnati, Cincinnati, OH

LEARNING OBJECTIVES: 1) Document a case that, to our knowledge, has not been reported in the literature,

- 2) Recognize that lymphomas can present as primary lung lesions, and
- 3) Emphasize the importance of maintaining a broad differential diagnosis.

CASE INFORMATION: Anaplastic large cell lymphoma (ALCL) is a rare form of lymphoma, accounting for only 2–8% of all lymphomas. ALCL has been reported as a primary tumor in the gastrointestinal tract and the central nervous system. We present a case of primary lung Ki-1 positive ALCL in a young man. To our knowledge, no other cases have yet been presented in the literature. The patient is a 33-year-old male who presented with progressive fever, night sweats, and 60 pound weight loss. Initial chest x-ray demonstrated a round left upper lobe infiltrate that progressed to a 5% times 6.5 cm cavitary mass with satellite lesions and lingular involvement. Evaluation for an infectious etiology was negative. Tissue obtained via an open lung biopsy showed immuno-histochemical Ki-1 positive anaplastic large cell lymphoma. All lymph nodes obtained were negative for lymphoma, and CT scans of his body did not reveal any lymphadenopathy or splenomegaly.

DISCUSSION: Anaplastic large cell lymphoma (ALCL) was first described by Stein and colleagues in 1982. The Revised European-American lymphoma (REAL) classification system, the most recent system for classifying lymphomas, has recognized ALCL as being a distinct entity. Patients with this disease, compared to other large cell lymphomas, are younger, more likely to have B symptoms, and have better responses to therapy. During diagnosis, the Ki-1 antibody recognizes the membrane protein CD-30, which is also found on Reed Sternberg cells. Standard therapy for ALCL has been with combination chemotherapy of cyclophosphamide, doxorubicin, vincristine, and prednisone. Our patient was started on this regimen and had a positive initial response.

DIVERTICULITIS WITH A FRENCH TWIST: LEMIERRE'S SYNDROME OF THE PORTAL VEIN. W. Dale¹, R. Granieri¹; ¹University of Pittsburgh, Pittsburgh, PA

LEARNING OBJECTIVES: 1. To recognize clinical manifestations of fusobacterium infections 2. To recognize a proposed variant of Lemierre's syndrome: portal vein thrombosis as a sequelae of fusobacterium infection.

CASE INFORMATION: A 57 year old male with hypertension, alcohol dependence, hemorrhoids, and prostate cancer presented to out-patient clinic following the recent resection of multiple colonic adenomas. Symptoms began one month before as "the flu" (fatigue, malaise, gastrointestinal discomfort, decreased PO intake), and progressed to intermittent nausea and vomiting, non-melenic diarrhea, fevers to 102°F, and 4/10 LLQ, non-radiating, crampy pain unrelieved by over-the-counter medications. Patient denied lightheadedness, headache, visual changes, neck pain, hematemesis, or urinary symptoms. Medications included lansoprazole and sildenafil. On examination, temperature was 37.8°C, he appeared fatigued, and sclerae were anicteric; abdominal examination revealed LLQ tenderness without rebound or guarding, a liver span of 12cm, and hemoccult negative stool. WBC was 28.2 (79N, 15L, 5M, 1E, 0B), AST 51, ALT 83, alk phos 267, GGT 248, TBili 0.9, and Alb 2.5. Abdominal CT revealed peri-colonic stranding and a portal vein filling defect. Blood cultures were positive for Fusobacterium nucleatum. Treatment with intravenous piperacillin/tazobactam followed by oral amoxicillin/clavulanate produced symptomatic resolution. Patient was discharged with resolution of symptoms and normalization of leukocytosis.

DISCUSSION: The fusobacterium species are spindle-shaped, anaerobic, gram-negative bacilli indigenous to the GI tract. Two clinically important species exist: F. necrophorum and F. nucleatum. Areas with decreased perfusion and necrosis favor its growth as in vascular disease, shock, trauma, surgery, malignancy and edema. Common sites include bites, dental infections, aspiration pneumonia with empyema, postabortion infection, post-GI or gynecological surgery, appendicitis, diverticulitis, and septic thrombophlebitis. The well-described Lemierre's syndrome is associated with fusobacterium infections, with the usual agent being F. necrophorum. It is an oropharyngeal infection with a resulting septic thrombophlebitis causing necrotizing suppurative with thrombosis of the internal jugular vein. Two cases in the literature describe an associated septic portal vein thrombosis associated with fusobacterium infection, including F. nucleatum. We postulate that this patient developed Fusobacterium nucleatum diverticulitis after procedural (colonoscopic) trauma, leading to bacteremia, followed by the secondary development of portal vein thrombosis.

CHEST PAIN IN AN ANXIOUS WOMAN: THE IMPORTANCE OF A CAREFUL HISTORY AND PHYSICAL EXAMINATION. S. Daud¹, R. Granieri¹; ¹University of Pittsburgh, Pittsburgh, PA

LEARNING OBJECTIVES: 1) Recognize the clinical features and treatment of acute pericarditis.

- 2) Recognize the importance of physical examination in the evaluation of chest pain.

CASE INFORMATION: A 28 year-old female with a five year history of anxiety disorder was seen in the outpatient setting for a one week history of chest pain. She stated that whenever she felt anxious or stressed over the preceding week, she developed the chest pain, described as sharp, midsternal and without radiation. There were no associated palpitations, shortness of breath, diaphoresis, or nausea. She reported an upper respiratory infection two weeks prior to presentation, which she treated with over the counter medicines. She attributed the pain to home stressors, which included her four active children, and she requested a medicine to better control her anxiety. Her only chronic medication was sertraline. Her physical examination revealed a pleasant female who appeared in good health. A paradoxical pulse was not appreciated. The cardiac and lung examinations were normal in the upright position. When she was supine, however, her chest discomfort recurred, and she immediately sat upright with prompt relief of her

pain. A soft friction rub was heard while she was in this position which dissipated after a few seconds. An EKG showed ST segment elevation in I, II, V5, and V6. The diagnosis of acute pericarditis was made and the patient was treated with indomethacin. Her symptoms were relieved within two days. Coxsackie B titers that were drawn in the office returned positive one week later. **DISCUSSION:** Acute pericarditis is caused by inflammation of any of the layers of the pericardium. The syndrome has many etiologies, the most common being viral and idiopathic, with some cases due to uremia, myocardial infarction, and collagen vascular disease. Coxsackie B virus and echovirus are the main viruses implicated. Classic pericarditis is described as retrosternal or precordial chest discomfort. It may radiate to the arm, neck, or jaw, and is exacerbated by coughing, swallowing, or deep inspiration. In addition, it is highly postural, worse in the supine position, and relieved by sitting forward. A three component pericardial friction rub is pathognomonic for the disease but is not required for the diagnosis. The EKG usually shows diffuse ST segment and T wave abnormalities, differing from the regional changes of myocardial ischemia. Treatment is focused at the underlying cause. Aspirin and indomethacin are effective analgesics with corticosteroids used in refractory cases. The decision to admit patients to the hospital for cardiac monitoring is controversial.

ARRHYTHMOGENIC RIGHT VENTRICULAR DYSPLASIA AS A CAUSE OF SUDDEN CARDIAC DEATH. S. Daud¹, R. Granieri¹; ¹University of Pittsburgh, Pittsburgh, PA

LEARNING OBJECTIVES: 1) Recognize the clinical features of arrhythmogenic right ventricular dysplasia.

- 2) Recognize the need for implantable cardiac defibrillators in patients with arrhythmogenic right ventricular dysplasia.

CASE INFORMATION: A 45 year-old female was referred for evaluation of exertional dyspnea and palpitations. She reported episodic symptoms for over a year. She had a history of hypertension, but had been non-compliant with medications. She smoked cigarettes but did not use alcohol. Her family history was significant for arrhythmogenic right ventricular dysplasia (ARVD). Her father died at age 42 from sudden cardiac death. She had three affected sisters, two of whom had died from complications from ARVD. Another had undergone successful heart transplantation. Her physical examination was significant for jugular venous distention with a right atrial pressure of twelve cm of water, clear lung fields, and hepatomegaly. She was referred for exercise stress testing, during which time she had an eight-beat run of ventricular tachycardia. She was referred to an electrophysiologist, and a cardiac defibrillator was implanted. She was subsequently treated with diuretics and ACE inhibitors, and evaluated for heart transplantation. **DISCUSSION:** Arrhythmogenic right ventricular dysplasia (ARVD) is a rare, genetic disorder, characterized by progressive replacement of myocytes by collagen and adipose tissue. It most commonly occurs in young adulthood. Symptoms include peripheral edema, dyspnea on exertion, and abdominal fullness from right-sided failure. Syncope and palpitations are also common. Ventricular dysrhythmias are often refractory to amiodarone and sotalol and implantable cardiac defibrillators are necessary. Right sided failure is treated with sodium restriction, diuretics, and afterload reducing agents with moderate success. Cardiac transplantation is the definitive therapy for ARVD, though implantable defibrillators serve as a potential live-saving bridge until transplantation. Young adults presenting with right heart failure and with a family history of sudden cardiac death should be suspect for ARVD.

AN UNCOMMON ETIOLOGY OF GENITAL ULCERS: BEHCET'S DISEASE. S. Daud¹, R. Granieri¹; ¹University of Pittsburgh, Pittsburgh, PA

LEARNING OBJECTIVES: 1) Recognize the clinical features of Behcet's disease 2) Recognize the treatment of Behcet's disease.

CASE INFORMATION: A 43 year old female was referred for recurrent skin lesions and genital ulcers. She had noted painful, erythematous lesions on her labia lasting for three days on two separate occasions over the past year. She also noted painful oral ulcers, described as "canker sores", that had been present over this time. Additionally, she noted a single red lesion one day after she had her blood drawn two weeks previously. Her family doctor prescribed acyclovir for presumed herpes genitalis, without significant relief. She had no other medical problems and was not taking any medications. She had no risk factors for HIV and did not use alcohol or tobacco. There was no significant family history. She denied visual changes, hemoptysis, abdominal pain, or hematochezia. Her physical examination was significant for a pustular rash on the trunk. Her mouth was free of ulceration. The genital exam was notable for two shallow ulcers on the labia minora. The constellation of symptoms of aphthous ulcers, genital ulcers, skin rash and pathergy response was consistent with Behcet's syndrome. She was started on colchicine and topical triamcinolone with significant relief of her symptoms. Referrals to an ophthalmologist and gastroenterologist were arranged to rule out retinal and colonic involvement of the disease.

DISCUSSION: Behcet's disease is a clinical syndrome manifested by recurrent oral and genital ulcers. A papulopustular or acneiform rash involving the thighs or upper trunk is commonly seen. Ocular manifestations include uveitis, hypopyon, and occasionally retinal vasculitis. The pathergy response is the development of discrete pustules at sites of needle sticks. Less common features include cerebral vasculitis, deep vein phlebitis, and bowel involvement. The cause is unknown and there are no pathologic or serologic tests for Behcet's disease. Treatment includes topical corticosteroids for aphthous ulcers. Eye lesions are treated with colchicine. Thalidomide and chlorambucil are rarely used for refractory cases.

A CASE OF ACUTE EPIGLOTTITIS IN ADULTS. J.K. Dave¹, R.W. Kirby¹; ¹University of Illinois at Urbana-Champaign, Urbana, Illinois

LEARNING OBJECTIVES: 1. Recognize acute epiglottitis in adults as an emerging infectious disease. 2. Diagnose epiglottitis from its clinical and radiological manifestations. 3. Outline the differences in management of epiglottitis in adults compared to children.

CASE INFORMATION: A 63-year-old man presented with progressive dysphagia. He was well three days prior to admission, when he noticed difficulty swallowing solid foods associated

with hoarseness, dyspnea and sore throat. Since the morning of admission, the patient had drooling as he was unable to swallow saliva. His symptoms were aggravated when supine. He denied fever or cough. His vital signs were normal. Temperature was 36.8 C. Remainder of the examination was remarkable only for an erythematous posterior pharynx and stridor on auscultation of the neck. Laboratory evaluation revealed a white cell count of 21,700 with 87% neutrophils and a normal hematocrit, platelets and normal routine serum chemistries. Lateral view of the neck showed an edematous, thumb shaped epiglottis. Blood cultures revealed no bacterial growth, but throat culture grew *Haemophilus influenza* type b. The patient received intravenous methylprednisone and cefuroxime. His symptoms improved and he was discharged on the fifth hospital day.

DISCUSSION: Acute epiglottitis has increasingly become an adult disease, probably as result of immunization of children against *Haemophilus influenza* type b. A high index of suspicion is required in any patient with sudden onset of "4 Ds" dysphagia, dysphonia, dyspnea and drooling. The frequency of positive blood culture is lower in adults than in children. A lateral cervical soft tissue x-ray is a useful test with sensitivity of 79–90% and specificity of 86–90%. Indirect laryngoscopy is the confirmatory test needed only if diagnosis remains in doubt after initial evaluation. Conservative medical management consisting of antibiotics and corticosteroids with vigilant monitoring are the mainstay of therapy in adults. Prophylactic intubation is not recommended. Prognosis of adults with acute epiglottitis is good with early recognition and prompt treatment.

ORMOND'S DISEASE AS A CAUSE OF ANURIA. J.K. David¹, R.W. Kirby¹, ¹University of Illinois at Urbana-Champaign, Urbana, Illinois

LEARNING OBJECTIVES: 1. Recognize idiopathic retroperitoneal fibrosis (Ormond's disease) as a cause of anuria and acute renal failure. 2. Describe clinical, radiological and pathological features of idiopathic retroperitoneal fibrosis 3. Emphasize the importance of aggressive surgical treatment.

CASE INFORMATION: A 58-year-old man presented with anuria of 36 hours duration. He had a history of polyuria and polydipsia for two weeks but normal fasting blood sugars. He had lower back pain for 3 weeks. History was remarkable for sixty four-pack year history of smoking. On examination, pulse rate was 70 per minute and blood pressure was 184/102mm of Hg. Remainder of the examination was normal. Laboratory evaluation revealed creatinine of 10.2 mg/dl with blood urea nitrogen 56mg/dl. The white cell count was 14,300 with 60% neutrophil, normal hematocrit and platelet count. Computerized tomography scan showed bilateral hydronephrosis, retroperitoneal mass and no fluid within the urinary bladder suggestive of bilateral ureteral obstruction. The patient underwent bilateral ureteral stent placement. On exploratory laparotomy, there was medial deviation of both ureters, the left ureter was encased within the retroperitoneal mass beginning at the level of fourth lumbar vertebra and right ureter was encased in fibrous tissue. Bilateral ureteral lysis and intraperitonealization of both ureters was performed. The biopsy of retroperitoneal mass revealed fibrosis with mixed inflammatory cells consistent with retroperitoneal fibrosis. The tests for known etiologies for retroperitoneal fibrosis were negative. The patient was doing well eighteen months later with normal creatinine levels.

DISCUSSION: This case recognizes Ormond's disease as a reversible cause of acute renal failure. It should be suspected in a patient with history of back pain and bilateral ureteral obstruction. The medial deviation of the ureter is pathognomonic for retroperitoneal fibrosis. Early recognition and aggressive bilateral surgical treatment is essential.

ACHALASIA PRESENTING AS AIRWAY OBSTRUCTION. J.K. Dave¹, J.L. Hallett¹, ¹University of Illinois at Urbana-Champaign, Urbana, Illinois

LEARNING OBJECTIVES: 1. Recognize achalasia as a rare but potentially fatal cause of acute respiratory distress.

2. Discuss pathophysiology and management of airway obstruction as a complication of achalasia. **CASE INFORMATION:** An 84-year old woman was transferred from a nursing home for sudden onset of respiratory distress. She had no history of cardiac or pulmonary disorders. She was diaphoretic and cyanotic. She had esophageal dilatation for achalasia 18 months earlier. Pulse rate was 126 beats per minute, blood pressure was 152/90 mm Hg and respiratory rate was 36/min. There were bi-basilar rhonchi. Chest x-ray showed an enlarged mediastinum due to a dilated esophagus and clear lung fields. Computerized tomography scan confirmed a massively dilated esophagus compressing the trachea. The patient was intubated and esophagus was decompressed via esophgoscope, which showed tight lower esophageal sphincter and hiatus hernia. Botulinum toxin was injected in the lower esophageal sphincter. The next day she was treated with pneumatic esophageal dilation. Continued clinical improvement followed relief of the esophageal obstruction confirmed by endoscopy. She was discharged on the fourth hospital day in good condition.

DISCUSSION: Differential diagnosis of acute respiratory distress should include esophageal disorders in addition to various cardiac and respiratory disorders. Upper-airway obstruction is an infrequent but serious complication of achalasia. The pathophysiology of airway obstruction due to achalasia remains unknown. Suggested theories include forward displacement of the esophageal opening and cephalad protrusion of the esophagus posteriorly, cricopharyngeal dysfunction and loss of belch reflex. Radiograph of the chest is important in diagnosis. Prompt relief of obstruction is the key to the successful management.

CLOSTRIDIUM SEPTICUM INFECTION AND MALIGNANCY. J. Dave¹, T. Allison¹, S. Pulukurthy¹, L. Yekkiral¹, B. Valentine¹, ¹University of Illinois at Urbana-Champaign, Urbana, Illinois

LEARNING OBJECTIVES: 1. Recognize the important relationship between clostridium septicum and malignancy. 2. Describe the first reported case of clostridium septicum presenting as acute appendicitis.

CASE INFORMATION: A 43-year-old previously healthy man presented to emergency department with fever, pain in the left hip, and a bullous lesion on the left buttock. The patient had noted intermittent fever for two days prior to admission. On the day of admission, he had

noted pain in his left hip and a rash with swelling on his left buttock. Physical examination showed temperature of 98° F, pulse 96, blood pressure 102/74 and respiratory rate 20. Examination of the left gluteal region disclosed a 15-cm circular area of erythema and ecchymosis with small bullae. Crepitus was present on palpation. Computerized tomography scan revealed extensive gas dissecting from left knee to the left side of the lumbar spinous processes. He underwent and debridement. There was extensive necrosis of the left gluteal, vastus, and adjacent muscles. An exploratory laparotomy unveiled a ruptured appendix with pus extending through the left gluteal region, requiring right hemicolectomy and colostomy. Blood cultures grew clostridium septicum. The myonecrosis was resistant to multiple debridements and antibiotics, necessitating left hemipelvectomy on 11th day. He was discharged on the 26th hospital day. Eight months later patient presented with back pain and anuria. CT scan revealed large retroperitoneal masses; biopsy uncovered the masquerader, 'non-Hodgkin's lymphoma'. **DISCUSSION:** There is an association between Clostridium septicum infection and malignancy. Necrotic tumor tissue provides a favorable environment for growth of Clostridium septicum. Patients with Clostridium septicum must therefore undergo a vigorous search for malignancy following acute therapy. This is the first reported case of clostridium septicum infection presenting as acute appendicitis.

POSTPARTUM NECK PAIN AND DOUBLE VISION. N.B. Dave¹, R. Granieri¹, ¹University of Pittsburgh, Pittsburgh, PA

LEARNING OBJECTIVES: 1) To recognize the presentation of cavernous sinus thrombosis (CST) 2) To recognize the puerperium risk of thrombophilia.

CASE INFORMATION: Ms. P. is a 26-year-old 10 week postpartum female who presented with an 8 week history of unresolving neck stiffness and pain. Symptoms progressed to double vision, fevers, rigors, sharp left preauricular pain, drooping left eyelid, left eye tearing and dull retro-orbital pain. The patient denied upper respiratory infectious symptoms, paresthesias, weakness, or new rash. On physical, blood pressure was 110/palp, pulse 98, and temperature was 103.6 degrees (F). The patient was in mild distress. Other significant findings were an erythematous left eye without orbital swelling, left cranial nerve VI palsy and left ptosis. Laboratory data: WBC 20.2 (91% PMN, 3L), Hb 11.4, HCT 32.6. CSF revealed 21 WBC (81 PMN, 3L) and gram stain showed few WBCs but no organisms. An MRI was notable for an abnormality in the cavernous sinus. Consequently, a CT venogram was performed. A partial thrombosis of her left cavernous sinus and a 4 cm thrombosis of her left jugular vein were found. The patient was started on heparin. Subsequent laboratory testing identified Factor V Leiden mutation and anti-phospholipid antibody.

DISCUSSION: Of all the dural sinuses, the cavernous sinus most frequently becomes infected and thrombosed. The cavernous sinus is a direct continuation of the orbit through the superior orbital fissure. Blood from the superior orbital vein and cortical veins drains into the cavernous sinus and cranial nerves III, IV, V1, V2, and VI travel through it. Symptoms of CST include headaches, pain along the dermatomes of the ophthalmic and maxillary branches of cranial nerve V, diplopia, and orbital edema. Signs include fever, ptosis, proptosis, chemosis and ocular muscle paralysis. Treatment of a CST includes anticoagulation and antibiotics. Our patient developed a postpartum CST. Pregnancy and the postpartum period are known acquired risk factors for venous thromboembolism. Additional genetic defects that predispose to thromboembolism increase this relative risk significantly. A recent study reported the relative risk of thromboembolism was 9.3 for a woman with Factor V Leiden mutation who is pregnant or in the puerperal period, and women who present with a first episode of venous thromboembolism are 9 times as likely to have Factor V Leiden mutation. Therefore, women who present with their first venous thrombosis while pregnant or in the postpartum period warrant a hypercoagulable work-up, as was done in our patient.

TUMS—TOO MUCH OF A GOOD THING? A. Dechet¹, ¹University of California, San Francisco, San Francisco, CA

LEARNING OBJECTIVES: 1. Recognize the signs and symptoms of milk alkali syndrome. 2. Distinguish milk alkali syndrome from other causes of hypercalcemia 3. Manage milk alkali syndrome.

CASE INFORMATION: A 71 year old woman with history of diabetes and GERD presents to the emergency room with complaints of 2 weeks of general malaise, weakness, and fatigue of subacute onset. Denies any precipitating illnesses. Symptoms have made caring for self difficult. Also complains of nausea, constipation, heartburn, mild cough productive of white sputum, feelings of paranoia. Meds: glyburide, calcium carbonate (TUMS) prn. Social history: Patient lives alone. Recent altercation at outside hospital due to patient's aggressive and inflammatory behavior. Patient unwilling to delineate whether this is a personality change or not. Denies drug use. PE: Obese woman in no distress. Notable mood shifts, alternating between agreeable and irritable. Vital signs normal except for BP of 170/100. Diffuse white film on tongue, mildly dry mucus membranes. Chest, cardiac, abdominal, and neuro exams are normal. Labs: CBC, UA, CXR, and glucose are normal. Creatinine 2.9 (0.8 five months ago), BUN 30, Na 133, CO2 33. ECG shows normal sinus rhythm with shortened corrected QT interval of 390. Initial presumptive diagnosis and workup is for acute renal failure. Further studies: Ultrasound negative for hydronephrosis. Calcium 17.2, phosphorus 5.0. Further history: Denies bone pain, renal colic, weight loss, fevers, or night sweats. Consumes a minimum of 2 rolls of TUMS (calcium carbonate) per day "out of habit" and for relief of heartburn (equivalent to 9600 mg of calcium/day). New diagnosis: Hypercalcemia due to milk alkali syndrome.

DISCUSSION: Milk alkali syndrome: First described in 1949 when seen in conjunction with treatment for peptic ulcer disease. Due to high intake of calcium and alkali. Incidence now on the rise again with 12% of admissions for hypercalcemia due to milk alkali syndrome. Can occur with as little as 4 gms of calcium per day, especially in the setting of preexisting renal insufficiency. Classic triad is hypercalcemia, metabolic alkalosis, and renal failure. Symptoms of hypercalcemia include nausea, weakness, headache, irritability, personality changes, lethargy, and renal colic from renal calculi. Other causes of hypercalcemia such as parathyroid disease, cancer, granulomatous disease, thyrotoxicosis, and medication interactions can be delineated by

Careful history and exam and a few laboratory studies. Management of milk alkali syndrome includes cessation of calcium ingestion, several liters of hydration, loop diuretics ONLY after complete rehydration, and dialysis if necessary. Bisphosphonates are generally not needed. Outcome depends upon chronicity of syndrome. Hypercalcemia usually resolves within a few days, but permanent renal impairment not uncommon due to microcalcifications within renal parenchyma.

TAKAYASU'S ARTERITIS IN A 20-YEAR-OLD FEMALE. D.E. Delaet¹; ¹Columbia University, New York, NY

LEARNING OBJECTIVES: Review the clinical presentation, epidemiology, etiology, treatment, and disease course of Takayasu's Arteritis (TA).

CASE INFORMATION: A 20 year-old Latino female presented with a two-week history of rash, fever, and general malaise. In the four months prior to presentation, the patient was treated for several recurrent Staphylococcus aureus urinary tract infections. Two weeks prior to presentation, she noted the onset of a diffuse rash, accompanied by intermittent fever and general malaise. On initial physical examination, she was noted to be in no apparent distress with a low-grade fever and a resting heart rate of 130 bpm. The examination was otherwise remarkable only for numerous 2X2 cm erythematous, tender nodules distributed bilaterally on the arms and thighs. Laboratory analysis revealed an erythrocyte sedimentation rate of 125, normal serum complement levels, and a positive ANA titer of 1:4 with a speckled pattern. Subsequent magnetic resonance angiography (MRA) revealed diffuse wall thickening and enhancement of the aorta with mild to moderate stenosis of the proximal left common carotid artery, both main renal arteries, and both proximal common iliac arteries. Biopsy of a representative subcutaneous nodule revealed inflammatory changes consistent with TA. Therapy with corticosteroids and methotrexate was initiated. Repeat MRA at nine months follow-up revealed stability of the vascular lesions.

DISCUSSION: TA is a chronic inflammatory arteriopathy that characteristically involves the aorta and its primary branches. Although TA is a rare disease most commonly affecting women less than forty years of age, incidence may be as high as 2% among females in some Asian and Latin American countries. TA is marked in its early stages by evidence of systemic inflammation and later by stenotic vascular lesions of the aorta and its primary branches. While the etiology remains unknown, autoimmune processes triggered by infectious or other causative agents may play a pathophysiological role. Corticosteroid therapy remains the cornerstone of treatment, with cytotoxic agents being reserved for steroid-resistant cases. Patients should additionally receive antiplatelet agents as well as aggressive intervention for hypertension. Vascular surgery and/or angioplasty are often indicated to alleviate complications of vascular stenosis. Despite aggressive treatment, the disease is often progressive, with five-year mortality perhaps as high as 35 percent.

ACUTE ABDOMINAL PAIN: DIAGNOSIS - "MIMICITIS". J. Derby¹; ¹Creighton University, Omaha, NE

LEARNING OBJECTIVES: Recognize that advanced imaging techniques can help in differential diagnosis of abdominal pain.

Avoid unnecessary surgery for abdominal pain.

CASE INFORMATION: A 28 y.o. female presented with severe abdominal pain. The pain was in the right lower quadrant and epigastric area. It was refractory to several mg of morphine every two hours. She reported some radiation to her back. She has a history of GERD, and reports a recent uncomplicated upper respiratory infection. She denied fever or changes in her bowels. Her physical exam revealed marked tenderness with guarding in the right lower quadrant, no rebound and positive bowel sounds. Pelvic exam revealed an enlarged left ovary, extreme tenderness over the right adnexa, with some mild cervical motion tenderness. Lab revealed normal liver function tests, normal UA, normal CBC, negative B-HCG, negative cervical cultures, and negative blood cultures. A spiral CT was done to rule out appendicitis, it revealed extensive mesenteric and periaortic adenopathy. The pain resolved overnight after antibiotic treatment. Follow-up CT in six weeks revealed resolution of the adenopathy.

DISCUSSION: Mesenteric lymphadenitis is often labeled as "pseudo-appendicitis". It closely mimics acute appendicitis and PID. Before the development of enhanced imaging, it was only diagnosed at surgery with removal of a normal appendix. It is often caused by an infectious etiology with adenovirus (a possibility in this patient with her recent URI), streptococcus and staph infections, and classically with yersinia and tuberculosis. It is unclear why the antibiotics helped in this case, but the results were marked.

PERITONEAL TUBERCULOSIS: AN UNCOMMON EXTRAPULMONARY PRESENTATION OF TUBERCULOSIS REINFECTION. S. Dex¹, H. Liu¹; ¹Boston Medical Center, Boston, MA

LEARNING OBJECTIVES: (1) Recognize extrapulmonary tuberculosis (TB) as a cause of disease in the absence of pulmonary findings (2) Diagnose peritoneal TB and recognize the two types (3) Recognize reinfection as an increasingly significant cause of TB relapse.

CASE INFORMATION: A 49 year old homeless man with a history of TB, alcohol abuse, and chronic pancreatitis presented with fever of 104°F. He reported a 10-lb weight loss and chronic left upper quadrant discomfort for which he had been prescribed oxycodone/acetaminophen. There were no other abdominal complaints. He denied cough, night sweats, or hemoptysis. Four years prior he had been treated successfully for right upper lobe cavity TB. Physical examination was remarkable for mild guarding in the left upper quadrant of the abdomen without peritoneal signs. WBC was 4700. Chest x-ray showed left lower lobe atelectasis. The patient was admitted for a fever workup and continued to spike fevers daily. Blood cultures and HIV test were negative, and PPD revealed 10 mm of induration. An abdominal CT was ordered to evaluate for a source of fever. CT showed an infiltrative process of the greater omentum with a small amount of localized fluid. Paracentesis yielded ascitic fluid with WBC of 11,300 with lymphocytic predominance (60%). Ascitic acid-fast bacilli (AFB) smears were negative. A diagnostic laparoscopy and peritoneal biopsy were performed which revealed extensive peritoneal studding with trace ascites. Histological

analysis revealed caseating granulomas consistent with M. tuberculosis infection. The patient was later identified as one of a cluster of eight TB cases from the same homeless shelter, but he was not considered the index case. "DNA fingerprints" obtained on the isolates of M. tuberculosis from these eight cases were identical, thus indicating recent TB transmission at the shelter.

DISCUSSION: Peritoneal TB can present with fever, weight loss, abdominal swelling, vague abdominal pain, and bowel habit abnormalities. It is divided into two types: "wet" (ascites) or "dry plastic" (omental thickening). The diagnostic test of choice is laparoscopy with peritoneal biopsy. This case underscores the importance of extrapulmonary TB as a source of fever in a patient with TB risk factors but minimal pulmonary findings. However, peritoneal TB is uncommon and accounted for only 4% of extrapulmonary TB cases in the U.S. in 1991. Further, as the "DNA fingerprints" indicate TB transmission at the shelter, this case supports recent studies demonstrating that reinfection has a dominant role in the pathogenesis of TB relapse in homeless shelters and other high-incidence areas. This is in contrast to the conventional wisdom that TB relapse is invariably caused by reactivation. Preventing reinfection may require additional infection control measures, particularly to prevent reexposure of cured patients to active TB sources.

WRESTLING WITH A CASE OF RHABDOMYOLYSIS. R. Devasia¹, P. Williams¹, M.M. Burke Duke¹; University of Kentucky, Lexington, KY

LEARNING OBJECTIVES: 1) Review the clinical features of carnitine palmitoyltransferase II deficiency, the most common metabolic myopathy.

2) Consider a metabolic cause when rhabdomyolysis is recurrent.

CASE INFORMATION: A 25 year old male complained of back pain radiating to the abdomen and generalized muscle soreness after wrestling six matches at an indoor facility the previous day. He denied fasting or the use of performance enhancing drugs prior to the wrestling event. His physical exam was notable for multiple bruises on his arms and legs, muscle tenderness and a normal neurologic exam. His creatine kinase (CK) was 11,695 mg/dL, urinalysis revealed a large amount of blood, but only 2-4 red blood cells per high powered field. He was admitted and treated for rhabdomyolysis. His muscle soreness improved over the next four days and his BUN and creatinine, initially 25 and 3 mg/dL, respectively, improved to 6 and 1.2. The CK decreased to 5144 mg/dL by day 2, but rebounded to 10,330 mg/dL on day 3. Other studies of complement (C3, C4), thyroid function and electrolytes were normal. He reported two similar episodes after moderate exercise, the first at age 17 after a track meet and later at age 20 after a round of golf. On both occasions, he experienced muscle tenderness and elevated CK levels. With the first episode, a diagnosis of post-streptococcal glomerulonephritis was made, based on an elevated anti-streptolysin O titer, although C3, C4 levels were normal. We suspected an enzymatic deficiency of fatty acid oxidation or glycolysis. His clinical course was most consistent with carnitine palmitoyltransferase II (CPT II) deficiency, and a muscle biopsy was performed, to confirm this diagnosis.

DISCUSSION: An autosomal recessive disorder, CPT II deficiency is the most common of the metabolic myopathies. The enzyme deficiency affects fatty acid metabolism, leaving the fatty acids unusable as an energy source by the muscle cell. In states of glycogen depletion, this leads to anaerobic metabolism and eventual cell death. CPT II typically presents in males between 15-30 years of age who experience episodic rhabdomyolysis and myoglobinuria triggered by exercise. It may also occur during anesthesia or labor, if glycogen stores are depleted. Recurrent episodes of myoglobinuria lead to renal failure in 25% of patients. Serious complications may be avoided with accurate diagnosis and treatment. Thus, when patients experience recurrent rhabdomyolysis or rhabdomyolysis after only moderate exercise, enzymatic deficiencies should be considered. Muscle biopsy with enzyme assays provides a definitive diagnosis.

SPONTANEOUS INTRACRANIAL HYPOTENSION: A CLASSIC PRESENTATION OF AN UNUSUAL HEADACHE SYNDROME. J.F. Dewitt¹, M.A. Elliott¹; ¹Virginia Mason Medical Center, Seattle, WA

LEARNING OBJECTIVES: 1. Diagnosis and management of spontaneous intracranial hypotension (SIH). 2. Distinguish the headache of SIH from headache caused by intracranial bleed.

CASE INFORMATION: A previously healthy 43 year old woman developed a severe frontal and retro-orbital headache that was relieved when supine and exacerbated when upright. Associated symptoms included intractable nausea, vomiting, neck stiffness, and vertigo. She had no history of previous headaches, trauma, rhinorrhea, fevers, neurologic deficits, or mental status changes. Physical exam was unremarkable except for mild nuchal rigidity. Routine serum chemistry and complete blood count were normal. CT scan without contrast was normal. An MRI scan showed Chiari I malformation, diffuse pachymeningeal gadolinium enhancement and bilateral subdural collections without mass effect. The lumbar puncture revealed an opening pressure of 0 mmH2O, xanthochromia, RBC:112, WBC:29, protein:199, gram stain and culture negative. Nuclear medicine cysternogram showed rapid collection of radionuclide in the bladder suggestive of a cryptic cerebrospinal fluid (CSF) leak. The patient's headache was not responsive to narcotics or sumatriptan. She had slight improvement with epidural blood patch and IV caffeine. After five weeks of severe orthostatic headache, she noted sustained improvement of her symptoms.

DISCUSSION: Spontaneous intracranial hypotension (SIH) is a rare clinical syndrome caused by a cryptic CSF leak. Clinically SIH is characterized by orthostatic headache and must be distinguished from acute intracranial bleeding. A high clinical suspicion for SIH is confirmed by evidence of low CSF pressure with abnormal MRI scans showing diffuse pachymeningeal gadolinium enhancement. Reversible Chiari I malformation and subdural fluid collection and hemorrhage can also be found on MRI. Occasionally the CSF leak can be located by MRI, CT myelography or cysternography. The most effective treatment is epidural blood patch injected near the site of the leak. Bolus or continuous infusion of saline into the epidural space and intravenous caffeine have been reported to improve symptoms. Most cases will resolve over several weeks to months spontaneously.

TOO WARM FOR COMFORT—HEAT STROKE IN MINNESOTA. S. Dubey¹, J.W. Leatherman², ¹Hennepin County Medical Center, Minneapolis, MN; ²University of Minnesota

LEARNING OBJECTIVES: 1) To recognize the morbidity and mortality from heat stroke 2) To emphasize that heat stroke is preventable and screening should be incorporated into seasonal health care maintenance.

CASE INFORMATION: An 80 year old healthy male was found unresponsive by his wife in their sauna at home when he failed to come out in four hours time. Paramedics rushed him to the hospital where his core temperature was found to be 103 degrees F. He developed multiorgan damage including, respiratory failure requiring ventilatory support, myocardial injury, impaired hepatic and renal function, rhabdomyolysis and disseminated intravascular coagulation. With supportive treatment he made remarkable recovery and was discharged from the hospital on day six with a diagnosis of heat stroke.

DISCUSSION: Heat stroke is a medical emergency manifesting with core temperature greater than 106 degrees F and multiorgan injury. Heat stroke can be fatal with almost 200 deaths occurring during an average summer in the United States alone. Risk factors include advanced age, lower socioeconomic status, chronic illness, dehydration, strenuous exercise and hot weather. Prevention is of utmost importance. Treatment consists of cooling, which must begin in the field, and supportive care. The mortality rate is as high as 10% in spite of aggressive management. Since it is easily preventable, screening and counseling need to be integrated into health care maintenance during high risk seasons to reduce the burden inflicted by this illness.

IT QUACKS LIKE A DUCK, BUT IT ISNT A DUCK. S. Dubey¹, B. Marinelli¹, ¹Hennepin County Medical Center, Minneapolis, MN

LEARNING OBJECTIVES: 1) To recognize precordial changes on EKG as a manifestation of massive pulmonary embolism

2) To emphasize the importance of screening for major alternative diagnoses in patients who present with apparent acute coronary syndromes

CASE INFORMATION: A 44-year-old man, with history of coronary stent placement in the left anterior descending artery five months previously, presented with sudden chest pain and dyspnea. Relevant past medical history included a motor vehicle accident several months ago, which resulted in multiple lower extremity procedures and infections leading to reduced ambulation. EKG on admission was significant for precordial T wave inversion, and preliminary trans thoracic echocardiogram done in the emergency department showed normal left ventricular function. Considering stent closure, a coronary angiogram was done which showed patent coronaries. At that time, the formal reading from the already done echo was obtained and showed right ventricular enlargement and reduced right ventricular function. Immediate pulmonary angiogram was done and revealed extensive thrombus in the left pulmonary arterial system. Thrombolysis was initiated with t-PA.

DISCUSSION: This is an example of acute pulmonary embolus very convincingly masquerading as coronary stent closure. An important clue, namely, reduced ambulation due to recent leg injuries, lay hidden in the past medical history. Precordial T wave inversions on EKG in the setting of pulmonary embolism, has been well described in medical literature. Various theories, including true coronary insufficiency and catecholamine mediated global T wave inversions have been proposed. The presence of such T wave inversions correlates with increased severity of pulmonary embolism, and resolution over a few days correlates with improved prognosis. Treatment includes anticoagulation with or without thrombolysis. Such EKG changes warrant at least a detailed history and physical exam focusing on deep vein thrombosis and pulmonary embolism risk factors.

EVALUATION AND DIAGNOSIS OF APPENDICITIS. S. Elad¹, ¹UCLA San Fernando Valley/ Olive View Medical Center, Los Angeles, CA

LEARNING OBJECTIVES: 1. Recognize the clinical features of early appendicitis. 2. Identify atypical presentation of appendicitis. 3. Review common signs and symptoms of appendicitis.

CASE INFORMATION: A 30 year old African American male with no significant past medical history presents with a one day history of right lower quadrant pain and constipation. Pt denied nausea, vomiting and anorexia. Pt was able to eat and tolerate both breakfast and lunch without incident. He denied fevers, chills. He was not taking any medications. Family history was non-contributory. He denied allergies and did not use tobacco, alcohol or drugs. On exam the patient was afebrile, pulse rate was 80 and blood pressure was 120/70. The cardiovascular exam was normal and his lungs were clear bilaterally. Abdominal exam revealed normoactive bowel sounds and mild right lower quadrant tenderness to palpation. He had no psos and no murrphys sign. No rebound, no guarding. His stool was heme negative. Neurological exam was within normal limits. Abdominal x-ray revealed right colon full of stool with no air fluid levels and no fecalith. Lab studies: NA 138 K 3.8 Cre 1.2 ALT 29 WBC 8 Hgb 12 Hct 39 Alk P 73 AST 36 TBili 0.5 Pt was sent home with diagnosis of constipation and given milk of magnesia and fleets enema. Pt returned to ER the following morning with complaints of worsening abdominal pain and anorexia. CT of abdomen revealed para appendiceal stranding. Pt was taken to the operating room for emergent appendectomy. Pt was found to have a gangrenous appendix with micro-perforations. Pt required a five day hospitalization and was discharged on oral antibiotics.

DISCUSSION: The diagnosis of early appendicitis is often difficult to make. Classically appendicitis presents as a pentad of signs and symptoms consisting of RLQ pain, anorexia, vomiting, low grade fever and leukocytosis. Typically, there is a history of pain beginning in the peri-umbilical region that migrates to the RLQ. However, this presentation occurs in only 50% of patients. The location of the appendix may also alter the presentation of pain. A retrocecal appendix may cause flank pain whereas an appendix in the pelvis may elicit tenderness on rectal exam. Although the WBC is elevated in 80% of patients, patients with early appendicitis may have a normal WBC. The diagnosis of acute appendicitis remains a challenge to clinicians with up to a 20% missed diagnosis rate. Recently, studies have suggested that the introduction of focused appendiceal CT would reduce the cost of caring for patients with suspected appendicitis and aid in diagnosis. Prompt diagnosis of appendicitis can avert the development of

complications such as perforation and subsequent septicemia. Physicians should rely on both their clinical judgement and the aid of radiological evaluation in the diagnosis of acute appendicitis.

CLINICAL AND ETHICAL IMPLICATIONS OF GENETIC TESTING FOR HUNTINGTONS DISEASE. I.H. Farquhar¹, J. Grosser¹, E.F. Yee¹, A.G. Gomez², ¹UCLA-San Fernando Valley Program, Sepulveda, CA

LEARNING OBJECTIVES: 1. Recognize the clinical presentation and progression of Huntington's Disease 2. Discuss the clinical and ethical implications of genetic testing for Huntington's Disease in asymptomatic individuals

CASE INFORMATION: A 63 year-old male presented in clinic requesting genetic testing for Huntington's Disease (HD). Several of his family members had developed late-onset choreiform disorders. His nearest affected relative was an uncle who developed symptoms after age 60, progressively worsening until his death 5 years later. The patient's parents died of other causes and his siblings were well. The patient's physical exam was unremarkable. He demonstrated no involuntary movements; no abnormalities in sensation, motor strength, coordination, or gait; and no deficits in concentration, memory, or intellect. The patient was counseled as to the risks and benefits of testing. In particular, he was informed that no treatment exists to cure or forestall the course of the disease nor could the advent of symptoms be predicted. The patient continued to express a desire for testing, primarily to alleviate his anxiety with possibly harboring a genetic predisposition to HD. Laboratory testing revealed alleles with approximately 28 and 40 CAG repeats, consistent with high genetic burden for HD. 28 CAG repeats indicated a normal mutable HD allele which may expand upon transmission to offspring (27-35) while 40 CAG repeats was consistent with individuals who typically show features of HD (>39). The patient was informed of the results and their interpretation and referred to a neurologist for further discussion and counseling.

DISCUSSION: Huntington's disease is a degenerative brain disorder with autosomal dominant inheritance manifested by chorea and dementia. Initial involuntary movements may not be noticed by the patient, but symptoms progress to irregular, sudden movements of the limbs and trunk, which can be disabling. Deficits in attention, memory, and judgment can be present as well as depression, social withdrawal, and disinhibition. Genetic testing for HD carries with it several important clinical and ethical implications. A positive test result implies an inherited mutation that will lead to the disease at some unknown future date, with no available cure or even effective treatment. Protocols for pre- and post-test counseling have been proposed to address the potential psychological consequences, including anxiety, depression, family and marital stress, survivor's guilt, and even suicide. Anonymous testing has also been suggested to alleviate fears of discrimination in employment and insurance.

ALL THAT WHEEZES IS NOT ASTHMA. I.H. Farquhar¹, L.A. Mankin¹, A.G. Gomez¹, ¹UCLA-San Fernando Valley Program, Sepulveda, CA

LEARNING OBJECTIVES: 1. Recognize the presentation and diagnosis of tracheal malignancies. 2. Increase awareness of tracheal neoplasm as cause of wheezing

CASE INFORMATION: A 31 year-old female presented with several days of increasing cough, wheezing, and dyspnea. She was in good health until 18 months prior when she first developed a non-productive cough. Six months later, she noted intermittent wheezing that increased in frequency and intensity. Recently she developed dyspnea at night when lying down. Inhaled steroids and bronchodilators gave little improvement. She denied fevers, chills, abdominal pain, and symptoms of acid reflux. Her past medical history was significant only for hypothyroidism treated with thyroxine. She had never smoked and had no pets. On examination, the patient was afebrile and not tachypneic. She was able to speak in complete sentences and coughed frequently. A loud, monophasic, inspiratory wheeze was audible without a stethoscope. Auscultation of the lungs revealed stridor. Prior pulmonary functions tests were reported as normal. Inspiratory and expiratory PA and lateral radiographs of the chest revealed a 2 cm diameter soft tissue mass in the tracheal air column. This mass moved from the carina at inspiration to the level of the clavicular heads at expiration. A subsequent CT scan of the thorax confirmed a distal tracheal mass. The patient was admitted for bronchoscopy to remove the tumor. Pathological examination found that the tumor did not fit with any known entity and was conjectured to be a hamartoma-like lesion of the seromucinous glandular tissue normally present in the distal trachea.

DISCUSSION: Primary tracheal neoplasms are extremely rare, accounting for fewer than 0.1% of cancer deaths. They occur with equal frequency in men and women, most often between the ages of 30-50. The most common presenting symptoms relate to airway obstruction, notably cough, wheezing, dyspnea, and recurrent pneumonia. A "tracheal syndrome" of dyspnea, wheezing, stridor, hemoptysis, and voice change is present in up to 85 percent of patients with primary tracheal neoplasms. As this case illustrates, the diagnosis of primary tracheal neoplasms is typically marked by significant delay between the onset of symptoms and proper diagnosis. One study reported an average delay of eight months for malignant lesions and 25 months for benign tumors. Most patients are unsuccessfully treated for presumptive diagnoses of asthma or chronic bronchitis, with little improvement, before a diagnosis of a tracheal neoplasm is made. Other diagnoses, including tracheal neoplasms, should be considered when the usual treatment for asthma does not improve the symptoms.

DIFFICULT DYSURIA. S.B. Fazio¹, ¹Harvard University, Brookline, MA

LEARNING OBJECTIVES: 1. Characterize the differential diagnosis of dysuria 2. Distinguish prostatitis from other causes of rectal pain 3. Recognize atypical presentations of appendicitis

CASE INFORMATION: A 21 year old male presented to urgent care complaining of three weeks of dysuria and rectal pain. The pain was sharp and intermittent, brought on by urination or defecation, and radiated into the right inguinal area. He denied fever, nausea, vomiting, abdominal pain or tenesmus. Past medical history was unremarkable; he was on no

medications. He was sexually active in a monogamous heterosexual relationship. On exam, the patient was afebrile, abdominal exam was benign. GU exam revealed normal testes, no discharge or lesions, rectal exam with mildly tender prostate. Urinalysis revealed no wbc's, no bacteria. The patient was treated empirically for prostatitis. Urine culture as well as swab for GC and chlamydia were negative. Three weeks later he called with recurrent symptoms after finishing a two-week course of antibiotics and was given an additional four weeks of treatment. Symptoms largely resolved, but he still complained of mild intermittent discomfort. Exam was unchanged, with the exception of a non-tender prostate. One month later, he presented with worsening rectal pain and dysuria, now accompanied by a low-grade temperature to 100, as well as five days of abdominal discomfort. Evaluation revealed mild bilateral abdominal tenderness without rebound or guarding, and again a normal GU and rectal exam. Urine dipstick was negative. Wbc was elevated at 13.2, and the patient was referred to the Emergency Room for assessment. There he was felt to have either recurrent prostatitis or pyelonephritis, and was admitted for observation and antibiotics. The urology service was consulted, who questioned the diagnosis of prostatitis and recommended a renal/bladder ultrasound. Bladder and kidneys were normal, but in the right lower quadrant there was a non-compressible viscus structure with a focal area of echogenicity felt to be most consistent with acute appendicitis. Emergent CT scan was obtained, revealing a perforated appendix, contained, with an accompanying 7 cm periappendiceal abscess. He underwent CT-guided drainage of the abscess and was initiated on broad-spectrum antibiotics, followed by appendectomy. He did well post-operatively.

DISCUSSION: The differential diagnosis of dysuria is broad, but any inflammatory process close to the ureter can cause burning with urination. The absence of significant tenderness on rectal exam or pyuria should call the diagnosis of acute prostatitis into question. Chronic or recurrent prostatitis in the absence of an anatomical substrate is very unusual in this age group. Appendicitis is often a difficult diagnosis to make, mimicking a variety of other conditions. Current data indicate that the correct diagnosis is missed up to 20% of the time (NEJM, 1998 Jan 15). Signs may be subtle early in the disease, but early diagnosis and treatment are of paramount importance.

HIP FRACTURE IN A 51 YEAR OLD MALE WITH END STAGE RENAL DISEASE. R. Filippi¹, D.W. Brady²; ¹Emory University, Atlanta, GA; ²Emory University, Decatur, GA

LEARNING OBJECTIVES: 1. Recognize tertiary hyperparathyroidism as a cause of pathologic fracture in patients with renal disease. 2. Consider endocrinologic causes in the differential diagnosis of non-traumatic hip fractures in adults. 3. Review the diagnosis and management of hyperparathyroidism

CASE INFORMATION: Mr. K. is a 51-year-old man who presented with pain and decreased mobility of his left hip after transferring from his wheelchair to his car. The patient stated that he fractured this same hip 4 weeks ago while simply standing. Mr. K's past medical history is significant for end stage renal disease secondary to hypertension. In the last 10 years, the patient has suffered multiple fractures including right hip fracture at age 40. Physical exam demonstrated pain over the left hip, scalloping of the proximal phalanges in both hands, and deformity of the distal thumbs. Chest radiograph depicted absorption of the distal clavicles. Laboratory data showed an exponential 6-year rise in alkaline phosphatase, calcium, phosphorus, and parathyroid hormone levels. At the time of this admission, the patient's parathyroid hormone level was 2674. Mr. K's open parathyroid biopsy revealed a right adenomatous nodule and three hyperplastic glands. Parathyroidectomy with autologous transplantation to the right forearm was performed.

DISCUSSION: In a patient with renal failure, tertiary hyperparathyroidism is an important cause of pathologic fracture. Although combinations of bisphosphonates and phosphate binders are used to modulate parathyroid hormone release, the internist must recognize the potential for autonomous adenoma development. It is important to include this diagnosis among the causes of non-traumatic hip fracture in the young.

CRYOGLOBULINEMIA: A POTENTIALLY SEVERE EXTRA-HEPATIC COMPLICATION OF HEPATITIS C VIRUS INFECTION. C. Folds¹, M. Wiener¹, T.L. Simon¹; ¹Mount Sinai Hospital, New York, NY

LEARNING OBJECTIVES: 1. Diagnose hepatitis C virus associated cryoglobulinemia in the absence of clinical liver disease. 2. Recognize the many potential manifestations of cryoglobulinemia. 3. Manage hepatitis C associated cryoglobulinemia during a severe bacterial infection.

CASE INFORMATION: A 44 year old woman with a history of hypertension and anemia presented with a one month history of progressive dyspnea and fatigue and a 3-4 month history of relapsing rash. The rash consisted of non-blanching erythematous palpable and non-palpable macules and patches on the lower extremities. One day prior to admission, the patient developed severe dyspnea, left sided pleuritic chest pain, and a productive cough. The physical exam was significant for fever, tachycardia, tachypnea, and decreased breath sounds at the left lung base with egophony. Abnormal laboratory values were HGB 5.7, BUN/Creatinine 47/5.0, and bicarbonate 12. Urinalysis revealed >300 protein, RBCs, and fine granular casts. Chest x-ray revealed left upper lobe infiltrate with effusion. Ultrasound of the liver, kidneys, and bladder was normal. Further work-up revealed an empyema in the left chest, iron deficiency anemia, s. pneumoniae positive blood cultures, nephrotic range proteinuria, +HCVAb, +HCVRNA, -HIVAb, -pANCA, -cANCA, C3 76, C4 <2, -ASLO, -ANA, -dsDNA, -antiGMBAb, +rheumatoid factor (RF) 4000, and +cryoglobulins. Fractionation of the cryoglobulins revealed an IGM kappa monoclonal band. Biopsy of the purpura revealed a leukocytoclastic vasculitis. The hospital course was significant for worsening renal failure requiring hemodialysis and chest tube placement for loculated empyemas. Plasmapheresis decreased the cryoglobulin level to an unquantifiable level. Renal biopsy was not performed given the patient's bacteremia and persistent fevers. On discharge on hospital day 29 and on 5 month follow-up the creatinine was 5.1 and 3.5 respectively.

DISCUSSION: Hepatitis C virus (HCV) associated cryoglobulinemia is unusual in patients without evidence of clinical liver disease. This patient had a clinical diagnosis of

cryoglobulinemia given the high cryoglobulin titer, high RF titer, purpura, renal failure likely from membranoproliferative glomerulonephritis (MPGN), low C4, and absence of other disease. Due to the serious bacterial infection, she was neither treated with immunosuppressive nor cytotoxic agents. As general internists are more frequently diagnosing and co-managing HCV infected patients, it is important to be aware of extra-hepatic manifestations of HCV.

HEPATITIS C, GOITER AND SUBCLINICAL THYROIDITIS. R.K. Fox¹; ¹University of California, San Francisco, San Francisco, CA

LEARNING OBJECTIVES: (1) Recognize thyroid disease as an extrahepatic manifestation of chronic hepatitis C. (2) Manage thyroiditis associated with hepatitis C when the disease is either subclinical or mild. (3) Realize that thyroiditis can occur in patients with hepatitis C with or without the use of interferon therapy.

CASE INFORMATION: A 29 year old woman with a 15 year history of hepatitis C (HCV) presented to general internal medicine clinic to establish primary care. She had undergone initial evaluation of her HCV disease 3 years earlier but had declined liver biopsy or interferon therapy at that time and discontinued followup. She had recently returned to hepatology and had proceeded with liver biopsy. On questioning and complete review of systems, the patient had no complaints. Notable findings on exam included a symmetric, smooth, nontender goiter, 6 cm with no palpable nodules. Exam of the liver was normal. Studies from 3 years and 6 months earlier were reviewed. At both time points, TSH was below normal, ALT was moderately elevated and HCV viral quantitative measurements were low. Genotype was 1A. Liver biopsy revealed minimal disease with Grade I lobular inflammation and Stage 0 fibrosis. Updated studies were ordered and revealed a normal TSH, mildly elevated total T3, normal free T4 and elevated anti-thyroid antibodies. No treatment was prescribed. At followup 1 month later, she remained asymptomatic and her goiter had markedly resolved to a minimally palpable and visible gland. Repeat laboratories revealed an isolated mildly elevated total T3.

DISCUSSION: Chronic HCV infection has been associated with several extrahepatic conditions including thyroiditis. The mechanism may be related to HCV infection of mononuclear cells but the exact mechanism is still largely unknown. Recent case-control studies have found the prevalence of thyroid abnormalities and anti-thyroid antibodies significantly higher in patients than controls. Most commonly, there is no associated thyroid enlargement. Approximately 50% of those with anti-thyroid antibodies have subclinical thyroid disease and 50% have overt symptoms. Thyroid dysfunction has also been noted during interferon-alpha therapy in up to 12% of HCV patients but studies have found disease in both interferon-treated and untreated patients. Prospective studies show thyroid dysfunction may develop in patients on interferon therapy with or without antibodies. Those with antibodies do not show a significantly higher propensity to develop thyroid dysfunction during interferon therapy. In summary, as in the general population, management includes timely recognition of signs and symptoms of thyroid disease. In the HCV population, these symptoms can often be subtle or may be attributed to HCV itself. HCV patients with thyroiditis should be followed closely; however, they may not necessarily require intervention if thyroid disease is subclinical and the thyroiditis or goiter may remit spontaneously. Finally, pre-existing thyroiditis is not a contraindication to interferon therapy.

SPACE OCCUPYING LIVER LESIONS WITH FAMILIAR OCCUPANTS. E. Francois¹, L. Yatskar²; ¹New York University, Brooklyn, NY; ²New York University, New York, NY

LEARNING OBJECTIVES: 1. Develop a high index of suspicion for tuberculosis in refugees and immigrants 2. Recognize immunocompromised hosts as being at risk for extrapulmonary tuberculosis 3. Assess space occupying hepatic lesions in the setting of possible tuberculosis

CASE INFORMATION: A 35 year old Tibetan male with the Acquired Immune Deficiency Syndrome (AIDS), CD4 = 32/mm³ and viral load = 102,000 copies/ml, presented complaining of fever, chills, abdominal pain, diarrhea, and malaise of increasing severity over five days. Six years prior to this clinic visit the patient was treated with Isoniazid for 4 months after a positive purified protein derivative skin test. Two years prior to the visit he was diagnosed with AIDS and completed treatment for multiple episodes of infectious diarrhea. Two weeks prior to this current evaluation he began to experience abdominal discomfort associated with loose non bloody stools. A course of Metronidazole did not improve his symptoms. He had never been on antiretroviral therapy, and on examination was noted to be a thin febrile male in no acute distress with bitemporal wasting, oral thrush, nonicteric sclerae, moderate right upper quadrant tenderness to deep palpation, and hyperactive bowel sounds. The patient was admitted to the hospital for evaluation, at which time laboratory analysis was notable for a white blood count of 5.8, a hematocrit of 26.6, and an INR of 1.59. An abdominal CT revealed low attenuated lesions in the liver, the largest measuring 6cm in diameter, with a smaller lesion in the porta hepatis extending into the head of the pancreas. Percutaneous drainage of one of the lesions revealed acid fast organisms subsequently identified as Mycobacterium Tuberculosis (MTB) which was pan sensitive. A bronchoscopy did confirm concurrent pulmonary TB. There was radiographic evidence of resolution of the hepatic lesions over several months after initiating treatment with Isoniazid, Rifampin, Pyrazinamide, and Ethambutol.

DISCUSSION: One in three individuals worldwide is infected with MTB. In the U.S., 40% of cases occur among foreign born persons. Tuberculosis most commonly presents as pulmonary or lymph node disease, but can also affect other organs. Immunosuppressive therapy, malignancy, and AIDS are known risk factors for disseminated complications with the organism. Extrapulmonary disease is seen in 60% of AIDS patients with lung disease, and of those patients, 7.5% have tuberculosis of the liver. Hepatic tubercular involvement is most commonly described as granulomatous, but can also be seen as discrete space occupying lesions such as tuberculomas, and abscesses. Patients typically present with fever, night sweats, abdominal discomfort, weight loss, and hepatomegaly. Diagnosis is made by aspirating a lesion, and treatment is based on drug sensitivities.

SYMPTOMATIC BRADYCARDIA AND AV BLOCK ASSOCIATED WITH INITIATION OF CLONIDINE THERAPY FOR HYPERTENSION. A. Almameed¹, S. Ghaffari¹, B. Freda²; ¹Cleveland Clinic Foundation, Shaker Heights, Ohio; ²The Cleveland Clinic Foundation, Cleveland, Ohio

LEARNING OBJECTIVES: 1. Recognize that clonidine can result in symptomatic bradyarrhythmias presenting as near syncopal episodes. 2. Understand the pathophysiology of clonidine induced bradyarrhythmias. 3. Identify patients who might be at higher risk for such side effects.

CASE INFORMATION: A 57 year old black woman presented to the emergency department with new onset "dizzy spells" for three days. During her first episode, she "felt faint" while carrying groceries. There was no sensation of vertigo and the spell resolved after 1–2 minutes with no residual effects. She denied palpitations, chest discomfort, diaphoresis, visual changes, slurred speech or muscle weakness. Her daughter did not witness any seizure activity. The morning of admission she had another similar episode while walking in her house. She has a history of essential hypertension for which she began hydrochlorothiazide 25 mg qD three months ago. She denied any over the counter or herbal medicines. Clonidine 0.1mg BID was started four days prior to presentation. In the emergency department she was afebrile and breathing comfortably. Blood pressure was 162/52 and her pulse was 45 and regularly irregular. On exam there was no evidence of neurological deficits and her cardiopulmonary exam was negative for signs of valvular disease or left ventricular dysfunction. Twelve lead ECG showed a ventricular rate of 47, sinus rhythm with 2:1 AV block and LVH. CBC, blood glucose and electrolytes were normal. Troponin T was < 0.01. Her BUN was 22 and creatinine was elevated at 1.5. Her weight was 49 kg. CXR and CT scan of the head were unremarkable. She was admitted to the telemetry floor and her clonidine was held. She denied any more episodes while in the hospital and during ambulation. On the second day of admission, her rhythm was sinus bradycardia (rate = 50–60) with no further episodes of 2:1 block through discharge the following day.

DISCUSSION: Clonidine is a centrally acting alpha2-agonist used mainly in the treatment of hypertension. Stimulation of the alpha2 receptor decreases central sympathetic outflow from the medulla and decreases plasma norepinephrine levels. This decrease in sympathetic outflow results in a decrease in heart rate, blood pressure, and peripheral and renal vascular resistance. A small electrophysiology study found clonidine depressed both sinus and AV nodal function. Our patient developed symptomatic AV block with 2:1 conduction one day after starting clonidine therapy for hypertension. Her AV block resolved 24 hours after discontinuing the drug. The incidence of symptomatic bradycardia during treatment with clonidine has been reported at <0.3%. Besides sinus bradycardia and low-grade AV blocks, complete heart block, Mobitz type I and II and severe sinus bradycardia have been described in patients taking clonidine. Proposed risk factors for developing severe bradycardia during oral clonidine therapy include: renal insufficiency, clinical sinus node dysfunction, development of bradycardia on other sympatholytic drugs or concurrent usage of other sympatholytic agents. Our patient had significant renal insufficiency (estimated creatinine clearance of 32). Furthermore, the fact that she was unable to generate a heart rate greater than 50 bpm during brisk ambulation (rapidly climbing two flights of stairs) suggests that she may have underlying sinus node dysfunction. Careful review of the aforementioned risk factors should assist physicians in identifying and reducing the occurrence of clonidine-induced bradycardias.

PROGRESSIVE SWELLING IN A PATIENT WITH NEPHROTIC SYNDROME. C. Fung¹; ¹University of California, Los Angeles, LA, CA

LEARNING OBJECTIVES: 1) Recognize the risk of renal vein thrombosis in a patient with nephrotic syndrome. 2) Recognize the signs and symptoms of acute and chronic renal vein thrombosis. 3) Select the appropriate study to diagnose renal vein thrombosis.

CASE INFORMATION: A 57 year old male with a past medical history notable for a recent diagnosis of nephrotic syndrome secondary to membranous nephropathy presented with a chief complaint of right calf swelling. He had noted bilateral, asymmetric lower extremity swelling with tenderness of the right lower extremity, but no warmth or erythema of the extremities, fever, prolonged immobility, recent injury, dyspnea, or orthopnea. Medications included benazepril, furosemide, and prednisone. The temperature was 35.7; blood pressure 132/86; heart rate 66; and respiratory rate 20. Exam revealed a comfortable male with clear lungs, normal cardiac exam except a II/VI systolic murmur at the apex, jugular venous pressure of 7 cm water, and nontender back with trace sacral edema. His extremities had 4+ pitting edema, right greater than left, up to the level of the thighs without erythema, warmth, skin breakdown or palpable cord. The remainder of the exam was normal. Laboratory studies showed creatinine 0.9 mg/dl, WBC 15000 per cubic millimeter, and albumin 1.8 g/dl. Urinalysis was significant for protein greater than 300, and no white or red blood cells. In addition to his outpatient medications, the patient received intravenous steroids and furosemide (with albumin), with improvement to 2+ bilateral swelling. Doppler ultrasonography of the lower extremities and the renal veins was negative for deep vein thrombosis. A CT of the abdomen/pelvis with intravenous contrast, performed for reasons unrelated to his swelling, showed bilateral renal vein thrombosis and non-occlusive inferior vena cava thrombosis. He was started on intravenous heparin and eventually discharged on enoxaparin, warfarin, mycophenolate and his admission medications. Follow up revealed significant improvement in his lower extremity swelling.

DISCUSSION: Estimates of the incidence of renal vein thrombosis (RVT) in nephrotic patients range from 5% to 62%. Low antithrombin III levels among other hemostatic abnormalities contribute to this hypercoagulable state. While the acute presentation of RVT is associated with flank pain and tenderness, hematuria, and decline in renal function, patients with the chronic form are often asymptomatic or have subtle manifestations such as edema, nausea, apathy, and weakness. The gold standard for diagnosis is venography. CT and MRI with axial and sagittal views are also accepted as definitive diagnostic studies. Doppler ultrasonography is a useful noninvasive diagnostic study, although false negatives and positives may occur. Treatment includes anticoagulation. Thrombolytic therapy may be considered, although the exact role has not been established.

RECURRENT PULMONARY INFECTIONS: INCREASING CLINICAL SUSPICION FOR AN UNDERLYING IMMUNODEFICIENCY. D. Garcia¹, B. Chernof¹; ¹VA Greater LA Healthcare System, UCLA, Los Angeles, CA

LEARNING OBJECTIVES: 1. Recognize clinical features distinguishing a primary immune deficiency from common causes of recurrent pulmonary infections in the elderly. 2. Discuss the impact of diagnosis and treatment.

CASE INFORMATION: 71 year old male presented with fevers, chills, productive cough and progressive shortness of breath for 1 week. He has had recurrent pulmonary infections for 35 years. In the last five years, pulmonary infections have progressed where IV antibiotics are given at least three times per year and oral antibiotics are given 4–5 times per year. Hospitalizations have occurred at least once per year. In the last year, he was treated for chronic sinusitis, chronic otitis media complicated with mastoiditis. The past medical history is significant for rheumatoid arthritis, chronic obstructive lung disease (COPD), oxygen dependent, bronchiectasis, and esophageal stricture with reflux. His medications include prednisone 5 mg per day, inhalers, proton pump inhibitor, theophylline, and an oral quinolone. On examination, his skin appeared pale. The vital signs were normal except for a temperature of 39 C; O₂ saturation was 92% on 2 liters. Pulmonary exam revealed decrease breath sounds, rhonchi throughout all lung fields, some basilar crackles. There was no significant lymphadenopathy or an enlarged spleen. A CBC showed: Hgb, 11 g/dL; Hct, 33.6%; RBC, 4.09 million cells/uL; MCV, 82.3 fL; WBC, 8.19 × 1000 cells/uL and Plt, 254 × 1000/uL. CXR: revealed hyperinflation, left lower infiltrate. He was hospitalized for a presumed gram negative pneumonia. In the hospital, he improves on two-week therapy of intravenous antibiotics. Sputum cultures grew *Pseudomonas Aeruginosa*. On discharge, cultures continue to be positive, in spite of aggressive therapy. In clinic, serum immunoglobulins are found to be low: IgG 406 (694–1618), IgM 50 (57–238), IgA 22 (89–446). Subsequent studies including a bone marrow, serum protein electrophoresis, revealed no evidence of malignancy or HIV disease. He is started on IV IgG therapy every three weeks. Six months later, he reports feeling better. He has had fewer infections; Hgb has increased, and has gained weight. He still requires aggressive use of antibiotics for recurrent bronchial infections but has not been hospitalized.

DISCUSSION: Predisposing medical conditions increase susceptibility to infections in elderly adults with recurrent pulmonary infections. Diagnosing a primary immune disorder is not common and can be a challenge for the practicing internist. In this patient, COPD, rheumatoid arthritis, steroid use, esophageal dysfunction contribute to chronic, recurring pulmonary infections. There are other clinical features that increase suspicion for an immune deficiency disorder. *Pseudomonas Aeruginosa*, is an unusual agent in non-institutionalized adult patients. There is persistent, positive sputum cultures after prolonged aggressive antimicrobial therapy. In addition, there is a continued clinical deterioration with further weight loss, worsening obstructive lung disease, and increase shortness of breath. Serum blood tests that aid in the diagnosis of hypogammaglobulinemia include a CBC and immunoglobulin measurement. Numerous advances continue to be made in the identification and diagnosis of immunodeficiencies. This patient has an acquired hypogammaglobulinemia, which can be seen in HIV infection, congenital disorders (X-linked infantile, protein-losing enteropathy and in Common Variable Immunodeficiency. A diagnosis of hypogammaglobulinemia with immunodeficiency is made with the presence of recurrent pyogenic infections (at any age); an increased incidence of autoimmune disease; low total immunoglobulin levels; and commonly progresses with significant lung disease (oxygen dependence, bronchiectasis, COPD). The cause is unknown. Treatment consists of administration of appropriate antibiotics as clinically warranted and intravenous immunoglobulin every 2–4 weeks, based on symptoms. During acute illnesses, aggressive antibiotics and immunoglobulin should be given simultaneously.

SCURVY. O. Garcia¹, K. Ferumal¹; ¹Cook County Hospital, Dept of Medicine, Chicago, IL

LEARNING OBJECTIVE: Recognize the clinical manifestations of Vitamin C deficiency.

CASE INFORMATION: A 53 year old man with history of heavy alcohol abuse and smoker for 40 years presented with two months of progressive fatigue, weight loss, dysphagia, pain and swelling of the calves with petechial rash of the lower extremities. Diet consists of fast food (hot dogs), one meal a day and very rarely fruits and vegetables. Patient denied history of trauma. He had been treated for chronic arthralgia without clinical improvement.

On physical exam the patient appear cachectic. Vitals signs were unremarkable. He had poor dentition, gingival swelling and hemorrhage into the gums. There was a firm, non-tender and immobile right supraclavicular lymph node. The heart and lungs were normal. Findings in the extremities were dry skin; follicular hyperkeratosis; perifollicular hemorrhages and ecchymosis on the back of the calves. There was also non pitting edema of the legs with limited motion of the ankle joints. Neurologic examination was normal. Laboratory tests revealed normocytic, normochromic anemia. PT 11.0, INR 1.05, PTT 26.9, Platelets 340. Vitamin C level less than 0.12g/L (normal range 0.2–1.9).

A fine-needle aspiration of the lymph node revealed malignant cells consistent with squamous cell carcinoma. An EGD showed that the esophageal lumen was obliterated by a 3-cm mass. Biopsies were obtained corroborating the diagnosis of esophageal cancer.

The patient went for surgical resection and palliative radiotherapy. A G-tube was placed; feedings were restarted, including vitamin C supplementation.

DISCUSSION: Scurvy is an uncommon disease in our society today. However, it may still be encountered in selected situations. Cancer patients may have an increased risk for scurvy because decreased intake of nutrients arises from anorexia or gastrointestinal symptoms caused by their underlying malignancy, chemotherapy, radiation treatment, or depression. It is important to recognize the signs and symptoms of scurvy because it is easily treated with Vitamin C supplementation. The manifestations of scurvy tend to improve dramatically within 24 hours and disappear within few weeks. The rapid response seen in patients with scurvy helps improve the quality of life mostly in cancer patients.

PYRE IN THE BELLY: A CASE OF DISSEMINATED INTRA-ABDOMINAL INFECTION.

A.B. Garlin¹, A. Prasad²; ¹University of NM HSC, Albuquerque, NM, Albuquerque, NM; ²University of New Mexico, Albuquerque, NM

LEARNING OBJECTIVES: 1) To identify sub-clinical intra-abdominal infection as a cause of FUO 2) To explore the incidence of culture-negative disseminated infection 3) To recognize diverticular disease as a cause of pyogenic liver abscess

CASE INFORMATION: A 35 yo man presented to the ER complaining of "not feeling well." He described a 60 lb weight loss over 6 mos, which he attributed to diet limiting tooth decay. He had noticed subjective fevers but no chills, nausea, vomiting, diarrhea, or abdominal pain. He had no significant past medical/surgical history. He reported past heavy alcohol use and a heavy smoking history but denied IVU or blood transfusions. On physical exam he was cachectic and ill appearing with severe dental caries but no oral abscesses. His temperature was 39.1C and BP 105/68. His abdominal exam was totally normal. He had normal rectal tone and stool was heme negative. His WBC was 14.5 with 85 % neutrophils, hemoglobin was 8.3, and albumin 2.5. He had received one dose of PO doxycycline in the ER. Blood cultures were sent, then PO clindamycin started for infection presumed to be of dental origin. The next day, patient's temperature increased to 41C and he complained of episodic RUQ pain. On exam he had some abdominal guarding and mild tenderness to palpation. Antibiotics were changed to IV Amoxicillin-clavulanate. LFTs showed normal transaminases but an elevated alk phos of 548, and GGT of 259. A RUQ US done to investigate possible cholangitis showed multiple hyperechoic densities of 5-9 cm, consistent with liver abscesses vs. metastases. A CT scan confirmed this finding, and suggested free peritoneal fluid vs. ascites. Flagyl was added until amebic serology came back negative. Ultrasound-guided paracentesis was performed without success because of highly viscous fluid. One out of two admission blood cultures grew coagulase negative Staph; all other blood and fluid cultures remained negative. HIV test was negative. A concern for Pseudomyxoma peritonei was entertained. Subsequently, Interventional Radiology drained 500 cc of fluid from his abdomen, which showed PMNs on Gram's stain but did not yield cytologic abnormalities or culture growth. Sigmoidoscopy was attempted to investigate possible colonic source of metastases but passage was unsuccessful. A barium enema was ordered to resolve the possibility of colonic malignancy, and instead showed a diverticular abscess with a fistular tract. This was thought to be the origin of disseminated intraabdominal infection and he was transferred to the surgical service for diverting colostomy and abdominal washout. 2 liters of purulent fluid were drained from his abdominal cavity at exploratory laparotomy. Intraoperative cultures yielded no growth.

DISCUSSION: Intraabdominal infections are recognized as a source of FUO, and a clinically significant proportion of these may present with neither positive blood cultures nor abdominal signs or symptoms. A 16 year retrospective study of 142 patients with pyogenic liver abscess found 5 patients in whom diverticulitis was thought to be the cause through pylophlebotic spread. 60% of this cohort did not have symptoms referable to the RUQ. 15% had no bacterial pathogens isolated, all of which had received broad-spectrum antibiotics prior to treatment. In a 10- year retrospective study of pyogenic liver abscesses, positive blood cultures were found in only 40%. The persistently negative blood cultures and clinical improvement in our patient's condition on IV antibiotics dissuaded the surgical consultants from an aggressive approach, significantly delaying the treatment of choice for this overwhelming infection. A review from 1998 suggested that surgical clearance, not either source or microbiologic characteristics predicted outcome in intra-abdominal infections.

ATHEROEMBOLI AFTER A CARDIAC CATHETERIZATION. R. Gertner¹, R. Granieri¹;

¹University of Pittsburgh, Pittsburgh, PA

LEARNING OBJECTIVES: 1) To recognize atheroemboli as a common and devastating complication of cardiac catheterization. 2) To diagnose atheroemboli using common laboratory and clinical parameters. 3) To recognize the treatment of a patient with atheroemboli after a cardiac catheterization.

CASE INFORMATION: Mr. K is a 72 year old male with COPD, hypertension, chronic renal insufficiency, and diabetes who was hospitalized with a COPD exacerbation. During that hospitalization he suffered a subendocardial myocardial infarction. His cardiologist recommended cardiac catheterization, which he later underwent. Two days after the catheterization his creatinine was at his baseline of 1.6. Over the next several weeks he complained of cyanosis of his toes, petechial lesions on his feet, painful calves, and vomiting. Four weeks post catheterization the patient had a routine clinic appointment. Labs revealed a creatinine of 10.8, hgb 9.8, WBC 32000, eosinophil count 63%, and an ESR of 106. Pertinent exam findings included bilobar rales, an S3, livedo reticularis, cyanosis, peripheral edema, asterixis, and multiple small non-blanchable erythematous macules over the bilateral upper and lower extremities. Skin biopsy revealed vasculitis of a medium sized artery with eosinophils, thrombosis, and cholesterol clefts. Mr. K ultimately required hemodialysis.

DISCUSSION: The reported rate of atheroembolic disease after cardiac catheterization is 0.15% to 2%. The three clinical hallmarks of atheroemboli include (1) other signs of peripheral embolization such as cyanotic toes or livedo reticularis (2) peripheral eosinophilia or hypocomplementemia, and (3) renal failure beginning seven days after the catheterization.

Laboratory studies are helpful in making a diagnosis. In a retrospective study of 4920 patients undergoing cardiac catheterization, Izumi et. al. found eight patients with cholesterol embolization syndrome. Eosinophilia was present in all patients. Eosinophiluria may be an early clue for atheroemboli, and the ESR is characteristically elevated.

The therapy of cholesterol atheroemboli is largely supportive. Some investigators have used steroids, however results are inconsistent. Lovastatin may be beneficial according to case reports, and represents a reasonable approach given its minimal toxicity. Several studies suggest that withdrawing anticoagulant therapy improves outcome. Overall the prognosis is poor. In one retrospective study, the one-year mortality rate was 75%. Symptomatic therapy such as treating myocardial dysfunction, controlling blood pressure, and improving nutritional status can improve survival to as high as 60% in one year.

NEUROSARCOIDOSIS PRESENTING AS RECURRENT SEIZURES AND EPISODIC HEMIPLEGIA. S. Ghosh¹, P. Han¹, G. Rajupet¹;

¹University of Pittsburgh Medical Center Shadyside, Pittsburgh, PA

LEARNING OBJECTIVE: To recognize a potentially lethal neurological presentation of sarcoidosis.

CASE INFORMATION: A 36 year old African-American man, with a history of recurrent seizures and pulmonary sarcoidosis in remission for 9 years, presented with a one day history of increasing lethargy, confusion, drowsiness and progressive neurological deficits consisting of dysphagia, drooling, left sided facial droop and increasing weakness of the left arm and leg. Tonic-clonic seizures of the left arm and leg occurred 12 hours after onset of symptoms. Medications on admission included phenytoin 500 mg/day. The patient had a 5 pack-year cigarette smoking history and drank 3-4 cans of beer daily, but denied any history of illicit drug use. Physical examination revealed a moderately obese black male, confused and lethargic with a temperature of 38.5, pulse 90/min., respiratory rate 34/min., and BP 150/100mmHg. On neurological examination the patient was confused, disoriented, and drowsy but able to follow simple commands. He demonstrated a right gaze preference, left facial droop, absent gag reflex, and dysarthric speech. Hypotonia was noted in the left upper and lower extremities, with reduced motor strength (grade 1-3/5), reduced but symmetrical deep tendon reflexes, and negative Babinsky's sign. Laboratory and radiological investigations revealed a serum phenytoin level, adjusted for serum albumin, of 9 mcg/mL (sub-therapeutic). Unenhanced CT scan of the head showed patchy periventricular atrophy with no blood collection or mass. Gadolinium contrast MRI of the brain demonstrated bilateral periventricular white matter microinfarcts and patchy leptomeningeal enhancement near the optic chiasma and hypothalamus. Lumbar puncture revealed normal opening pressure, and clear CSF with glucose 56mg/dL, protein 99.9 mg/dL (elevated) and 6 WBC/mcL (all lymphocytes). High resolution CT scan of the thorax demonstrated old scars and bullae in the lungs, with no interstitial fibrosis or adenopathy. Serum calcium and ACE levels were normal. Finally, meningeal biopsy was performed, and revealed non-caseating granulomatosis with no vascular involvement, suggestive of CNS sarcoidosis. The patient improved slowly with high dose parenteral steroids, with complete resolution of his neurological deficits.

DISCUSSION: 5% of patients of Sarcoidosis present with neurological findings (Stem, Krumholz et al, Arch. Neurol., 1985). As in this case, the presentation may mimic seizure disorder, TIA or meningitis, though cranial neuropathy or neuro-endocrine disorders are more common. Seizure control alone is not sufficient for control of symptoms or prevention of further episodes. Neurosarcoidosis should be considered in the differential diagnosis in any patient with unexplained neurological deficits whenever a past history of sarcoidosis exists. Early recognition and institution of steroid/immunosuppressive therapy may be required to prevent significant disability or even death.

CALCIPHYLAXIS: MORE COMMON THAN YOU THINK? P. Dennen¹, C. Gilroy¹;

¹HealthONE Presbyterian St. Luke's Hospital, Denver, CO

LEARNING OBJECTIVES: 1) Diagnose calciphylaxis from its clinical manifestations, and 2) Recognize the changing parameters and prevalence of the disease.

CASE INFORMATION: A 79-year-old woman with end stage renal disease on dialysis for 3 years was admitted with cellulitis of bilateral lower extremities. Treatment with antibiotics for 2 weeks resulted in no clinical improvement, and her mental status began to deteriorate. On physical examination she had large necrotic eschars covering the medial, posterior aspect of both calves, with minimal surrounding erythema. Her calves were exquisitely tender to palpation. There was no surrounding edema and distal pulses were normal. Additionally, there were multiple areas of mottled, violaceous, reticulated lesions on both inner thighs. These areas were also extremely tender to light touch. Similar appearing skin lesions were noted on the lower aspect of her abdomen. Two of these were overlying areas very firm induration in the subcutaneous fat. Her exam was otherwise normal. Significant laboratory findings included an elevated WBC of 16.4, glucose 256, BUN 24, Creatinine 3.2, Calcium 11.9 (12.8,corrected), Phosphate 4.1, PTH 23. An abdominal CT revealed multiple pancreatic and renal calcifications and extensive vascular calcifications, and skin biopsy showed perivascular calcifications.

DISCUSSION: Selye originally described calciphylaxis as "a condition of induced systemic hypersensitivity in which tissues respond to appropriate challenging agents with calcium deposition". The condition includes soft-tissue calcification and cutaneous necrosis, as demonstrated in this case. Prevalence of calciphylaxis among hemodialysis patients has been reported as high as 4.1%. Previously thought to occur at a calcium phosphorus product (CAPP) of >70 mg²/dl², it has been documented in several case reports with a CAPP of only 40mg²/dl². Although the pathophysiology remains poorly understood, the most important risk factors are chronic hemodialysis and hyperparathyroidism (primary, secondary or tertiary). Type of dialysate and insulin-dependent diabetes have also been associated with an increased risk for calciphylaxis. Treatment options are limited and mortality rates are high, sometimes exacerbated by failure of early recognition. Although nephrologists and dermatologists most frequently encounter this condition, it is imperative that primary care physicians are aware of calciphylaxis and its presentation. Calciphylaxis should be considered in the differential diagnosis of any non-healing skin ulcer and painful superficial skin lesion. A high index of suspicion in chronic hemodialysis patients and early recognition may decrease mortality and morbidity.

CALCIFIC UREMIC ARTERIOLOPATHY: RISK FACTOR ANALYSIS TO PREVENT A DEADLY OUTCOME. P.D. Goggin¹, E. Warm¹;

¹University of Cincinnati, Cincinnati, OH

LEARNING OBJECTIVES:

- 1) Define Calcific Uremic Arteriopathy (CUA),
- 2) Discuss the risk factors associated with CUA, and
- 3) Discuss methods to prevent CUA.

CASE INFORMATION: A 67 year old obese white female with type II diabetes mellitus on hemodialysis, presented with two weeks of painful, erythematous, well-demarcated plaques on

her abdominal wall, breasts, thighs, and calves. The lesions were associated with central necrosis and peripheral mottling. She was alert, oriented, afebrile, and without leukocytosis. She had a calcium-phosphorus product of 75 and a parathyroid hormone level of 438. Despite aggressive wound care and antibiotics, her condition worsened precipitously over 72 hours. Her skin lesions became necrotic, ulcerative, and suppurative. She became febrile, hypotensive, and stuporous. Despite broadened antibiotics and aggressive fluid resuscitation, she became unresponsive, pulseless, apneic, and died.

DISCUSSION: Calcific Uremic Arteriolopathy (CUA) is characterized by calcification of subcutaneous arterioles and infarction of skin and subcutaneous adipose tissue. It occurs in a minority of chronic renal failure patients. In CUA, calcification of the media of arteriolar vessels leads to regional ischemia and then infarction of skin and subcutaneous tissue, especially in areas of high adipose content. Skin necrosis and ulceration develop and patients usually die as a result of ischemic or infectious complications. While the true pathogenesis of CUA is unknown, several factors have been associated with a higher incidence. A high calcium-phosphorus product and secondary hyperparathyroidism are often associated with an increased incidence of CUA. Other features such as female gender, white race, obesity, recent significant weight loss, insulin-requiring diabetes mellitus, and use of calcium carbonate as a phosphorus binder have been reported as predisposing factors. It is likely that modification of these factors in our patient, such as management of high calcium-phosphorus product with non-calcium containing phosphorus binders, reduction of calcium content in dialysate fluid, and gradual weight loss may have prevented this deadly outcome.

PNEUMOMEDIASTINUM IN A YOUNG MALE WITH AN ASTHMA EXACERBATION. N. Golechha¹, F. Ali¹, W.A. Harb¹, R.D. Hobbs¹; ¹Oakwood Healthcare System, Dearborn, MI

LEARNING OBJECTIVE: To recognize pneumomediastinum as a potential complication of bronchial asthma in adults.

CASE INFORMATION: A 20-year-old African-American male with a past history of asthma was admitted with acute shortness of breath and a three day history of dry cough. There was no history of fever, chest pain, nausea, vomiting, retching, chest trauma or any other symptoms. The examination revealed a temperature of 97.7 °F, blood pressure 150/ 70 mm Hg, pulse 94/min, respiratory rate 20/min. and an oxygen saturation of 94% on 2 liter/min O₂. The trachea was midline. There was bilateral air entry and diffuse ronchi present. No skin crepitus, cyanosis or clubbing was noted. Laboratory studies: WBC count 10,300, blood gas pH 7.45, pCO₂ 38 mm Hg, pO₂ 64 mm Hg, saturation 95%. Chest x-ray revealed hyperinflated lung fields with air outlining the carotids and no pneumothorax. The CT scan showed air in the mediastinum, surrounding the trachea, carotid vessels and extending into the prevertebral and retropharyngeal space. The patient was treated with a high FiO₂, intravenous steroids and inhaled bronchodilators. The thoracic surgeon, pulmonologist and gastroenterologist concurred that asthma was the cause of the pneumomediastinum. Serial x-rays showed complete resolution of the pneumomediastinum within three days.

DISCUSSION: Pneumomediastinum has been reported to result from such different causes as the Valsalva maneuver, strenuous exercise, vomiting, retching, esophageal rupture, cocaine use and barotrauma. It is an unusual complication of asthma in adults. In a study of the radiographic findings in 1,016 adults admitted with acute asthma over four years, only one case of pneumomediastinum was noted (Pickup CM et al, J Accid Emerg Med 1994;11:234). There is also a report of pneumomediastinum in a patient who was receiving liquid ventilation where the perfluorocarbon was seen surrounding the bronchi and vessels from the lung parenchyma to the hila, extending into the mediastinum. (Jamadar et al, J Comput Assist Tomogr 1996; 20: 309). It is not known why some patients develop pneumomediastinum versus a pneumothorax. Treatment with a high concentration of oxygen helps improve the patient's absorption of air from the soft tissue by decreasing the nitrogen concentration (79%) in the air within the tissues. Optimizing therapy for asthma should prevent build up of high airway pressures that could lead to pneumomediastinum.

CAVITARY PULMONARY LYMPHOMA IN A PATIENT WITH AIDS. M. Gonchar¹, C. Smith², L. Depalo¹; ¹Mount Sinai Hospital, New York, NY; ²North General Hospital, New York, NY

LEARNING OBJECTIVES: 1. Recognize the limited efficacy of common tests for making the diagnosis of primary pulmonary AIDS-related lymphoma (ARL). 2. Acknowledge the poor prognosis of patients with ARL.

CASE INFORMATION: The patient is a 33 year-old male with a history of HIV (CD4 cell count 110/mm³, viral load 4,982 copies/mL) and multiple opportunistic infections who presented to the clinic with a complaint of two weeks of low grade fever and productive cough. His medications included saquinavir, epivir, zidovudine, bacrim, and azithromycin. His exam was consistent with consolidation and chest x-ray showed a lobular mass in the superior segment of the left lower lobe with central lucency. Computerized tomography (CT) of the chest showed a lobular cavitated mass in the left lower lobe (8.2 cm X 4.4 cm). He underwent a bronchoscopy with broncho-alveolar lavage (BAL) and transbronchial biopsy. The biopsy showed necrotizing granulomas and the patient was started on treatment for mycobacterium tuberculosis (TB). The patient chose to discontinue anti-retroviral treatment at this time.

After being lost to follow-up for two months, he presented to the emergency room complaining of fever, chills, cough, and vomiting. All cultures from the previous bronchoscopy had been negative. A CT at this time showed no change in the mass. A repeat bronchoscopy with biopsy was non-diagnostic. The patient underwent a thoracotomy with left lower lobe resection. The resected lobe showed extensive necrosis. The pathology revealed a cavitating B cell lymphoma. There was no evidence of metastatic disease at this time. The patient was restarted on highly active anti-retroviral therapy (HAART) but refused chemotherapy at that time. He subsequently died 1.5 years after his diagnosis was established from complications of metastatic lymphoma.

DISCUSSION: This case clearly illustrates the difficulties in making the diagnosis of primary pulmonary ARL. Bronchoscopy, BAL, and transbronchial biopsy all have low yields in this setting. Percutaneous transthoracic needle biopsy has a higher yield, but is limited to cases with accessible lesions. The two procedures with highest diagnostic yield are open lung biopsy and pleural cytology (when an effusion is present).

Generally these patients have very poor prognoses. The average survival ranges from 4 months to 5.5 months. Institution of HAART, with or without chemotherapy, has not been shown to significantly improve this prognosis.

ABDOMINAL PAIN AND RASH. G.C. Goodear¹; ¹Tulane University, New Orleans, LA

LEARNING OBJECTIVES: 1. Recognizing diagnostic clues of hypertriglyceridemia pancreatitis.

2. Treating hypertriglyceridemia with heparin.

CASE INFORMATION: 35 year-old nonalcoholic man presented with two days of nausea and sharp epigastric pain. The pain radiated to the back and decreased in intensity with sitting forward. He also noted polyuria, polydipsia, and a rash for several months. On examination he had epigastric tenderness and involuntary guarding. There was a waxy papular white rash over his arms and legs. Repeated attempts at phlebotomy were unsuccessful, due to blood clotting in the syringe immediately after venopuncture. A large bore needle was required to draw samples. Shortly after being drawn and upon standing a lactescent layer formed in the serum tube. Initial laboratory results revealed a triglyceride of 9194 mg/dl, a cholesterol of 703 mg/dl, an amylase of 811 IU/L, and a lipase of 6457 IU/L. The serum glucose was 417mg/dl, the serum ketones were negative, and the urinalysis had protein > 300 mg/dl, glucose > 1000 mg/dl, and ketones > 80 mg/dl. A sonogram of the abdomen showed a normal gallbladder and bile ducts. CT scan of the abdomen demonstrated a swollen and edematous pancreas with significant inflammatory changes in the mesenteric fat.

He was admitted to the intensive care unit for aggressive hydration, pain management, and an insulin drip. Intravenous heparin was started in an effort to decrease the hypertriglyceridemia. The triglycerides decreased from 2349 mg/dl to 1887mg/dl. The heparin was stopped after one day because of gastrointestinal bleeding. After seven days of hospitalization, the patient's pain resolved and he was tolerating a full diet. At the time of discharge the triglyceride level was 293 mg/dl, with a total cholesterol of 259 mg/dl.

DISCUSSION: One to four percent of pancreatitis is caused by hypertriglyceridemia, usually when triglyceride levels are greater than 1000 mg/dl.¹ Although rare, it is an important etiology to consider in the differential diagnosis for abdominal pain. The diagnostic clues of xanthomas and highly viscous lactescent serum are important in making the diagnosis and selecting appropriate medical management. One example is the use of heparin to activate lipoprotein lipase, which was used in this case to decrease the hypertriglyceridemia.¹ Fortson MR, Freedman SN, Webster PD. Clinical assessment of hyperlipidemic pancreatitis. Am J Gastroenterol 1995;90(12):2134-2139

GENERALIZED AND SEVERE FOLLICULITIS ASSOCIATED WITH CONSTITUTIONAL SYMPTOMS AFTER ORAL MESALAMINE TREATMENT. M.I. Masroor¹, H.S. Gorr¹; ¹University of Louisville, Department of Medicine, Louisville, KY

LEARNING OBJECTIVES: The report of a case of unusual and severe folliculitis associated with constitutional symptoms that mimic systemic bacterial infection after oral mesalamine treatment.

CASE INFORMATION: A 26 -year-old female was hospitalized with a recent history of ulcerative colitis and toxic mega-colon that was treated with oral mesalamine, oral corticosteroids and Levaquin (Levofloxacin) respectively. Additional medication on the discharge date was Celexa (Citalopram) that she has been taking for several months for her depression. Six days after discharge from the hospital she presented to the Emergency Room with high-grade fever (maximum body temperature of 104.3 F), chills, and generalized pustule lesions on the entire body. The patient developed fever and chills 4 days after initiation of mesalamine treatment, followed by pustule lesions that started on the hands followed by lesions on the extremities that spread to the entire body. Her review of systems was unremarkable other than chills, febrile sensation, joint pain, weakness and bilateral swelling sensation in the parotid gland area. Skin pustule lesions (Figure 1) with erythematous base but without tenderness and warmth were noticed on the entire body but were more pronounced on the upper extremities and upper trunk. Her initial laboratory results showed elevation of white blood cell count up to 13.8 (thousand/ ml) with 83% granulocytes, elevated serum alanine aminotransferase (ALT) (170 U/L), mild hyponatremia (133 mmol/L), and mild elevation of amylase (327 U/L). Her remaining laboratory data was normal except for mild normocytic anemia. She was initially treated with Zosyn (piperacillin/tazobactam) for possible bacteremia caused by an immunocompromised condition. The mesalamine was stopped due to concern for drug-induced adverse events. The Levaquin and Celexa were also stopped to minimize any drug adverse events. However, oral prednisolone was continued to control her ulcerative colitis symptoms. She became afebrile within two days. The pustules slowly subsided after discontinuation of mesalamine and with continuation of corticosteroids. The antibiotic was discontinued when her blood culture did not show any growth in 48 hours. Her initial wound culture from the pustules also did not reveal any bacterial or virus growth. Her mono-spot test was negative. Computerized tomography of the head to evaluate the subjective tender parotid glands was negative for any abscess or lesion. White blood cell count also became normalized and ALT slowly decreased. A subsequent skin biopsy in the right arm showed pustular folliculitis with manifestation of drug-induced eruption.

DISCUSSION: Extraintestinal manifestations of ulcerative colitis include pyostomatitis vegetans and pyoderma vegetans, erythema nodosum, pyoderma gangrenosum, and bowel bypass syndrome (dermatitis-arthritis syndrome). Folliculitis has not been described as an extraintestinal manifestation of ulcerative colitis. Folliculitis is the inflammation of hair follicles commonly caused by infections. However, physical injury, chemical stimulation, and several medications including corticosteroids, halogenated containing sedative and expectorants, vitamins can also induce folliculitis. Mesalamine, known as 5-aminosalicylic acid, has been used for treatment of ulcerative colitis, proctitis, and proctosigmoiditis. Rash as adverse events due to mesalamine controlled-release capsules is only 1.3% (Control group showed 1.2%). Mesalamine was the most recent addition to the patient's treatment. Given a previous report that mesalamine induced folliculitis this is the most likely cause of the observed pustular folliculitis. The case presented here had similar pustular folliculitis but with more constitutional

symptoms such as fever, chills, joint pains, elevation of white blood cell counts and liver involvement. This complicated presentation could falsely lead the physician toward infectious rather than drug-induced problems.



3D Surface Shaded display showing high-grade stenosis of the celiac axis and superior mesenteric artery.

WHEN A DEPILETARY ALONE WONT DO: HIRSUTISM AND LATE ONSET CONGENITAL ADRENAL HYPERPLASIA SECONDARY TO 21-HYDROXYLASE (CYP21A2) DEFICIENCY. R. Granieri¹; ¹University of Pittsburgh Medical Center, Pittsburgh, PA

LEARNING OBJECTIVES: 1. To formulate an appropriate clinical evaluation of the patient presenting with hirsutism 2. To recognize the presentation of late onset congenital adrenal hyperplasia 3. To manage congenital adrenal hyperplasia secondary to 21-hydroxylase (CYP21A2) deficiency

CASE INFORMATION: RF is a 46 year old female who presented for routine care. She had no medical problems but was noted to have temporal balding and excessive facial hair. These changes began 9 years ago and had been gradual. She denied menstrual irregularity or change in voice. Menarche was at age 10 with normal sexual maturation. She was on no medications. FH was negative for hirsutism. On physical examination, she had temporal balding and excessive coarse lip, chin, arm and leg hair with a prominent escutcheon. Thyroid was nonpalpable and breast exam was normal. Pelvic was without clitoromegaly. Ovaries were normal size. Laboratory data revealed a normal sodium and potassium, testosterone 6.0 nmol/L (0.1–2.8), DHEA 2244 ng/dL (130–980), DHEA-sulfate 2710 ug/L (500–2500). As a next step, baseline 17-OH progesterone was drawn. An elevated level of 3612 ng/dl, rising to 4728 ng/dl after ACTH, was obtained. The diagnosis of late onset CAH secondary to 21 hydroxylase deficiency was made.

DISCUSSION: Hirsutism, the development of androgen-dependent terminal body hair in women, is due to excessive androgen production by the ovary or adrenal gland. CAH secondary to 21-hydroxylase deficiency has been reported in up to 2% patients with androgen excess and should be considered in the differential. 21-hydroxylase (CYP21A2) converts 17-hydroxyprogesterone to 11 deoxycorticosterone and progesterone to deoxycorticosterone. Several mutations in the CYP21A2 gene can occur. It is the extent of the mutations that determines the degree of enzyme dysfunction and clinical manifestations. Patients with minor mutations have impaired cortisol synthesis. Those with no enzyme activity develop salt wasting secondary to reduced aldosterone synthesis. ACTH is then increased and shifts the adrenal steroid pathway to excess androgen production. These women may be difficult to distinguish from those with PCOS but since treatment varies, precise lab diagnosis is essential. Hormonal abnormalities include elevated DHEA-S and testosterone but diagnosis is assured by the second step in the evaluation: obtaining a 17-OH progesterone. The most common treatment is a long acting glucocorticoid to reduce ACTH secretion and abort the cascade of excessive androgen production. In patients with salt losing form, mineralocorticoids may be necessary. Response to treatment is guided by hormonal levels and reduction of hirsutism.

CERVISTATIN STIFF SHOULDER. E. Green¹, R. Granieri¹; ¹University of Pittsburgh, Pittsburgh, PA

LEARNING OBJECTIVE: Identify and manage HMG-CoA reductase inhibitor induced myositis.

CASE INFORMATION: F.D. is a 73 year old male with coronary artery disease, hyperlipidemia and hypertension who presented with 1 week of progressive myalgias and stiffness in his shoulders and buttocks. He denied weakness, paresthesias, joint swelling, trauma, fevers, chills, malaise, or flu-like symptoms. Two weeks prior to onset of symptoms he was started on cerivastatin. Additional medications included lisinopril, clopidogrel, synthroid, and aspirin. He took no herbal or over the counter medications. Physical examination was notable for gluteal and rotator cuff muscle pain without weakness or limited range of motion. The remainder of the exam was normal. Laboratory evaluation was notable for a creatinine of 1.0, AST of 736, ALT 444, alkaline phosphatase of 53, GGT 22, CPK 26,940, CPK-MB 132, ESR 18, TSH 1.3, and a urinalysis with 3+ blood but 0 RBC. An EKG revealed normal sinus rhythm. Cerivastatin was discontinued and the patient was admitted for hydration and alkalinization of his urine. After 3 days his symptoms resolved and blood chemistries began to normalize.

DISCUSSION: HMG-CoA reductase inhibitors are popular lipid lowering agents. The most common adverse drug reaction is a transient chemical hepatitis of little clinical significance. Clinically significant rhabdomyolysis is extremely rare with an incidence of < 0.5%. HMG-CoA reductase inhibitors require metabolism by P-450 enzymes before they become active. Inhibition of the P-450 system leads to decreased metabolism of the statin drugs and higher plasma levels. The resultant increased delivery of the HMG-CoA reductase inhibitors to muscle cells decreases the production of cholesterol and cytochrome Q10 and makes muscles susceptible to oxidative damage and rhabdomyolysis. Cerivastatin is a third-generation HMG-CoA reductase inhibitor that is marketed as a "safer" drug because it is metabolized by two different P-450 isoenzymes, CYP3A4 and 2C8. Cases of rhabdomyolysis in patients treated with cerivastatin and gemfibrozil have been reported; however, there have been no published reports of cerivastatin alone causing rhabdomyolysis. The etiology of rhabdomyolysis in this patient is somewhat puzzling. He was on no potent inhibitors of CYP3A4 nor did he have pre-existing hepatic or renal dysfunction. It may be that he had an intrinsic deficiency of CYP2C8. The treatment of HMG-CoA reductase inhibitor induced rhabdomyolysis involves withdrawal of the drug and supportive care. The myositis usually resolves completely although renal failure has been reported.

AN UNUSUAL CASE OF FLANK PAIN. F. Green¹, R. Granieri¹; ¹University of Pittsburgh School of Medicine, Pittsburgh, PA

LEARNING OBJECTIVE: Identify and treat acute renal infarction.

CASE INFORMATION: J.C. is a 63 year old female with ischemic cardiomyopathy, mitral regurgitation, hypertension, and rheumatoid arthritis who developed spontaneous right flank pain that was constant, 8/10, sharp, and sufficient to wake her from sleep. The pain was associated with nausea, vomiting, and intermittent diaphoresis. There was no radiation to the groin, abdomen, or chest. There was no history of chest pain, palpitations, dysuria, abdominal trauma, vaginal discharge, or invasive vascular procedures. She has never had thromboembolic disease or atrial fibrillation. Medications included digoxin, losartan, prednisone, furosemide, hydroxychloroquine, aspirin, and methotrexate. Physical examination was notable only for compensated biventricular heart failure, mitral regurgitation, and right flank tenderness. Electrocardiogram revealed sinus tachycardia. Laboratory data was notable for BUN 35, Cr 1.2 WBC 14.1 (94% PMN), ALT 28, AST 37, GGT 180, Alk Phos 192, and LDH 386. Urinalysis revealed 4 RBC, 0 WBC, and trace protein. CT scan of the abdomen and pelvis with contrast showed an acute infarct of the inferior pole of the right kidney. The patient was admitted and anticoagulated. Echocardiogram demonstrated an ejection fraction of 25% and severe mitral regurgitation with no evidence of mural thrombus.

DISCUSSION: Renal infarction is an often overlooked clinical syndrome than can be mistaken for more common causes of flank pain. The etiology of most cases is embolic disease from underlying atrial fibrillation, valvular disease, or cardiomyopathy. No historical variable aside from risk factors for thromboembolic disease has been shown to be associated with renal infarction. Physical examination is non-specific although an incomplete infarction may cause a hyper-renin state with secondary hypertension. LDH is the best marker for renal infarction although initial measurements may be normal. Other lab abnormalities are inconsistently present. CT scans have replaced IVP and angiogram in making the diagnosis of renal infarct. While areas of infarction can be seen on non-contrast CTs, contrast is necessary to increase sensitivity, better define the infarct, and distinguish acute from completed infarcts. Surgery, angioplasty, fibrinolysis, and conservative medical management are therapeutic options. Revascularization appears to be most successful in early infarction (first 3 hours). Conservative therapy with anticoagulation (and additional antihypertensive therapy if needed) is thought to have lower morbidity than invasive therapy in most cases. Although the mortality from the renal infarct itself is low, these patients have a dismal one year prognosis because of further embolic disease and underlying cardiac disease.

STREPTOCOCCUS CONSTELLATUS INDUCED EMPYEMA AND PERICARDIAL EFFUSION. A. Haddad¹, A. Delara¹, N. Lekas¹, R.D. Hoobs¹; ¹Oakwood Healthcare System, Dearborn, MI

LEARNING OBJECTIVE: To recognize the unique clinical characteristics associated with Streptococcus constellatus infections.

CASE INFORMATION: A 42 year-old male roof repairman presented with shortness of breath for the past two weeks. He had no history of alcoholism, smoking or previous lung disease. He had been taking Azithromycin during the week prior to admission but had become worse. He had right-sided, pleuritic chest pain, fever, chills, yellow-greenish sputum, night sweats, and mild weight loss. Examination revealed moderate respiratory distress, a temperature of 101.1 °F and a pulse of 115. The mouth, teeth and oral pharynx were normal. The cardiac exam was normal except for tachycardia. Diminished breath sounds were noted on the right side of the chest. His WBC was 19.2 with 89% PMN's. A CXR and subsequent CT revealed a large loculated pleural effusion with right upper and middle lobe atelectasis. A sputum gram stain showed gram positive cocci in chains. A 2D echo showed a large peri-cardial effusion without evidence of tamponade. 1,200 cc of yellowish-greenish pus was drained following thoracentesis and placement of a chest tube. Cultures grew only Streptococcus constellatus which was sensitive to Clindamycin and Ampicillin. He recovered following these antibiotics with chest tube drainage.

DISCUSSION: Streptococcus constellatus is a gram positive microaerophilic coccus that grows in chains or pairs and is a member of the Streptococcus intermedius group. It is usually present as a harmless commensal organism in the mouth and GI tract but like other members of the S. intermedius group has the propensity to form abscesses. Intrinsic virulence factors (hydrolytic enzymes, different adhesins on cell surfaces and a polysaccharide capsule) facilitate adherence, decrease phagocytosis and increase pathogenicity. S. constellatus is frequently associated with thoracic infections and more prone to occur in males, alcoholics, people with previous pneumonia and in those at risk for aspiration. Mortality ranges between 15–30%. Although this patient had few risk factors for the development of this infection, the presence of a

dense empyema and pericardial effusion suggested *S. costellatus* or other organisms in the *S. intermedius* group as a possible cause.

FEVER OF UNKNOWN ORIGIN: AN UNUSUAL PRESENTATION OF INFERIOR VENA CAVA THROMBOSIS AND PROTEIN S DEFICIENCY IN A PATIENT WITH AIDS. A. Hamad¹, R. Clark¹, J. Singh¹; ¹Nassau University Medical Center, Eastmeadow, New York

LEARNING OBJECTIVES: 1- deep venous thrombosis (DVT) can cause of fever of unknown origin (FUO).

2- the importance of computed tomography of the abdomen during work up for FUO. 3- AIDS is associated with high incidence of thrombosis.

CASE INFORMATION: A 45 years old Haitian female presented with generalized weakness, weight loss, and low-grade fever. She had a history of AIDS with wasting syndrome and cerebral toxoplasmosis, and recurrent deep venous thrombosis in the lower extremities. She had a temperature of 100.0F, with no swelling in the lower extremities. Plain radiograph of the chest and Urinalysis were normal. Blood, stool, and urine cultures were negative. Stool for Clostridium difficile, ova and parasite, Cryptosporidium, acid-fast bacilli and Isospora belli was negative. Sputum for acid-fast bacilli smear and culture was negative. Hepatitis profile was negative. Sputum for acid-fast bacilli and Pneumocystis carinii were negative. Few days later, she complained of severe perineal pain, rectal exam was very painful and frank blood was found on a gloved finger. Sigmoidoscopy was not done because of the pain. She was treated for an anorectal fissure and the pain improved after few days but she continued to have fever. Computed tomographic scan of the head with contrast and cerebrospinal fluid analysis were unremarkable. Computed tomography of the abdomen was done as work up for FUO. It showed massive thrombosis extending from the right renal vein to the inferior vena cava up to the liver. Hematologic work up was significant for decreased total protein S (28%) with normal protein C (64%) and antithrombin III (102%). Anticoagulation therapy was started and fever improved after more than a week.

DISCUSSION: Our patient has massive thrombosis, which we think caused her prolonged fever, as an extensive work up ruled out other causes of fever. Total protein S is normal in AIDS patients with decrease in free protein S level. We believe that our patient has two separate disorders predisposing her to thrombosis (HIV infection and primary protein S deficiency). This belief depends on the fact that her total protein S was < 50% of the normal, consistent with the classic type of the hereditary protein S deficiency. Although acute thrombosis can decrease antithrombin III, it is less likely to cause a decrease in protein S or protein C. The decrease of protein S in our patient is not secondary to the thrombosis, as her antithrombin III and protein C levels were normal. We found only few reports of patients with HIV infection with protein S deficiency who presented with significant DVT.

COMPLICATED DISTAL RENAL TUBULAR ACIDOSIS. A. Hamad¹, L. Balsam¹, S. Kamath¹; ¹NYMC, Eastmeadow, New York

LEARNING OBJECTIVES: 1- diagnosis of renal tubular acidosis and differentiation of the distal from the proximal types.

2- simple treatment for distal renal tubular acidosis with bicarbonate can prevent serious complications.

CASE INFORMATION: An 18 year old Hispanic male with past medical history of transient paralysis of all extremities 5 years ago in El Salvador presented with weakness of all his limbs. Weakness started the night before presentation and deteriorated rapidly in the next day. He reported having polydipsia and polyuria in the last few days. The patient was afebrile with blood pressure of 110/70 and heart rate of 90 bpm. Physical exam was significant for short stature (155cm) and moderate weakness of all extremities. Laboratory tests showed severe hypokalemia (1.3 meq/dl) and hypophosphatemia (1meq/dl), low bicarbonate (9 meq/dl) and normal anion gap. Arterial blood gas showed acidosis (pH = 7.12) and pCO₂ of 28 mmHg. After receiving 280 meq of potassium supplements over 16 hours his potassium level improved to 2.7 meq/dl and he regained his power completely. The patient received replacement of potassium, phosphorus, and bicarbonate for 4 days period till his numbers became normal and was discharged home on potassium citrate supplements only. Work up during hospitalization revealed urine pH repeatedly >6.5, urine specific gravity of 1.010, urine anion gap of +40, urine pCO₂ of 50 mmHg after correction of the acidosis, daily requirement of 20 to 40 meq of bicarbonate only to maintain normal serum level, and fractional excretion of bicarbonate of 2.5% and phosphorus of 22%. Renal ultrasound revealed echogenic shadows distinct between the medulla and the cortex, consistent with calcification.

DISCUSSION: Distal and proximal renal tubular acidosis are uncommon diseases. The proximal type characterized by reduction in reabsorption of bicarbonate while the distal type results from inability to secrete hydrogen ion. This patient represents a typical case of type I RTA (distal) with all the manifestations and complications. Short stature, severe hypokalemia with secondary paralysis and polyuria (loss of urine concentrating ability), renal calcifications, positive urine anion gap, inability to acidify the urine despite severe acidosis (opposite to proximal type), decreased urine pCO₂ (which is usually higher than the plasma because of secretion of H⁺ in the distal tubule that binds the bicarbonate and become CO₂), fractional excretion of bicarbonate of 2.5% (in proximal RTA this number is > 15%) and minimal amount of bicarbonate replacement was enough to keep normal blood level. We want physicians to be aware of this rare disease that is very easy to treat but if left untreated can have detrimental consequences.

QT PROLONGATION SECONDARY TO HYPOCALCEMIA AFTER CORRECTION FOR ELEVATED SERUM ALBUMIN. A. Hamad¹, M. Zihlif¹, M. Salameh¹, A. Alghadban², D. Beinfeld¹; ¹Nassau University Medical Center, Eastmeadow, New York; ²SUNY at Stony Brook, Port Jefferson NY

LEARNING OBJECTIVE: 1- recognize that correction of calcium level for albumin should be done also if albumin is high.

CASE INFORMATION: 42 years old white male with no significant past medical history except alcoholism came to the emergency room with vomiting of small amount of blood associated with

mild epigastric pain. He was not taking any medications at home. He was afebrile, blood pressure was 123/83, heart rate was 126 bpm. Physical exam was unremarkable except for mild dehydration and tachycardia with no added sounds or murmur. Laboratory tests were significant for normal calcium level (9 mg/dl), magnesium of 2 meq/dl, serum albumin of 5.3 mg/dl, potassium of 3.3 meq/dl. Arterial blood gas showed normal pH of 7.44. Electrocardiogram showed regular sinus rhythm with prolonged QT interval (QTc = 44 msec). Because of this prolongation, the patient was given calcium gluconate intravenously as corrected calcium with elevated albumin was low (7.9 mg/dl). Ionized calcium was low (1.04 meq/dl). Repeat EKG showed return of QT interval to normal limit. The patient was treated with vitamins and anxiolytics for his mild alcohol withdrawal and was discharged home after 2 days.

DISCUSSION: QT prolongation can occur as a congenital or an acquired disorders. The acquired form is secondary to pharmacologic agents (mainly antiarrhythmics, tricyclic antidepressants, antihistamines and erythromycin), and electrolytes disturbances (hypokalemia, hypomagnesemia and hypocalcemia). We are presenting this case to stress on fact of correcting calcium level to the serum albumin, this correction can be by adding 0.8 mg to each 1 mg of albumin below the normal level (4mg/dl) or by subtracting 0.8 mg to each 1 mg of albumin above the normal limit.

NEUROMYELITIS OPTICA. I.P. Haque¹; ¹University of Virginia at Roanoke-Salem, Roanoke, VA

LEARNING OBJECTIVE: Diagnose the Different Variants of Multiple Sclerosis including Neuromyelitis Optica Treatment of Neuromyelitis Optica

CASE INFORMATION: Neuromyelitis optica or Devic's syndrome is an uncommon neurological illness characterized by optic neuropathy and myelopathy. This case presents a 39 year old African-American female with new onset of visual changes, periorbital pain, headache, malaise, lower extremity weakness, and back pain. The patient's symptoms began one week prior to admission. She previously presented with lower extremity pain and weakness six months prior and was suspected clinically to have multiple sclerosis and had resolution of her symptoms with corticosteroids. She had no changes on magnetic resonance imaging of brain and cerebral spinal fluid findings were nonspecific at the time. Her symptoms of visual changes and loss with periorbital pain were not present upon this prior admission.

On physical exam, she had papillitis, severe visual field defect, profoundly decreased lower extremity weakness and sensation, and Lhermitte's sign. On laboratory exam, cerebral spinal fluid showed increased protein and she had an elevated ESR. Magnetic resonance imaging of spine showed diffuse cervical cord swelling and cervical signal changes. This patient presented with bilateral optic neuritis and myelitis, known as Devic's syndrome or Neuromyelitis optica. She was treated with intravenous methylprednisolone and had gradual improvement of her symptoms.

DISCUSSION: Devic's syndrome generally has a rapid onset of optic neuritis and myelitis. Patients with acute or subacute Devic's syndrome often respond to corticosteroids as this patient did. In Devic's syndrome, acute spinal cord lesions can demonstrate diffuse swelling extending over several levels or the entire cord in either a continuous or patchy distribution. The optic nerve and chiasm can show either demyelinating lesions and/or necrotizing lesions. Approximately 35 percent of patients have a monophasic illness and 55 percent develop relapses, and rarely patients have a fulminantly progressive course without relapses or a course typical of multiple sclerosis.

AN UNUSUAL CAUSE FOR A NEW ONSET SEIZURE. D. Hassan¹, G. Tabas¹; ¹UPMC Shadyside, Pittsburgh, PA

LEARNING OBJECTIVES: (1) Recognize atypical presentations of infective endocarditis (IE). (2) Identify the indications for valve replacement in IE.

CASE INFORMATION: A 41-year-old white male with past medical history of migraine headaches presented to the emergency department for a seizure occurring in a restaurant. While sitting, the patient suddenly developed stiffness and rigidity in the right arm then the left arm. He then collapsed and had a generalized seizure. According to an eyewitness the patient lost consciousness for 3-4 minutes, after which he was confused but without incontinence or focal weakness. Prior to this event the patient experienced low-grade fever as high as 101°F and myalgias for one week but no other symptoms. Physical examination revealed a young ill looking male; temperature was 37°C. He was tachycardic and the blood pressure was 164/78 mmHg. Significant findings included abrasions over the left temple, evidence of an old infarct in the left parietal region thought to be embolic. Echocardiogram showed a dilated left ventricle with borderline left ventricular hypertrophy and good systolic function, severe aortic regurgitation and suspicion of an aortic valve vegetation. A transesophageal echocardiogram showed a bicuspid aortic valve with prolapse of one of the leaflets, 3+ aortic incompetence and evidence of a vegetation 4-5 mm in size on one of the leaflets of the aortic valve. Blood cultures initially revealed Gram + Cocci in clusters later identified as coagulase negative Staphylococcus. All the cultures drawn later were negative for any growth. Because of evidence of embolic disease on the CT scan of the brain and the recent generalized seizure (representing multiple embolic events), aortic valve replacement was indicated. The patient had a St. Jude HB mechanical valve replacement. He had an uneventful post-operative course. He was started post-operatively on heparin and then later on Coumadin. Blood cultures and cultures from the aortic and mitral valves were negative for any growth. The culture from the aortic vegetation was also negative. The patient was continued on ceftriaxone for 4 weeks to treat likely atypical organisms.

DISCUSSION: When considering the differential diagnosis of a new onset seizure, IE is a relative uncommon cause, but should be considered when fever and/or a heart murmur is present. When IE is associated with seizures, the seizures usually occur later in the course of the disease rather than as part of the initial presentation as in this case. The bicuspid aortic valve in our patient was a risk factor of IE but it is important to recognize that one third of patients with IE have no identifiable predisposing cardiac lesions. Negative cultures occur in 2-5% patients with IE. The indication for valve replacement in this case was evidence of recurrent embolic events. Mechanical valve was chosen because of the patient's young age.

DOCTOR, CAN YOU HELP ME DIE? J.M. Hauser¹; ¹University of Chicago, Chicago, IL

LEARNING OBJECTIVES: 1. Explore reasons for a request for physician-assisted-suicide 2. Learn how to respond to a requests for physician-assisted-suicide

CASE INFORMATION: Mr. Hayes [name changed] was a 57 yo man with advanced esophageal cancer who requested assistance in dying.

He initially presented with dysphagia and underwent endoscopy which revealed esophageal cancer. After an abdominal CT scan showed multiple liver masses consistent with metastases, he was referred to hospice. His predominant symptoms were abdominal pain treated with morphine, anxiety treated with lorazepam and dysphagia which required he stop eating solid foods. Mr. Hayes initially resisted medications because of possible drowsiness. It soon became clear, however, that his refusal was because medication could not address his most significant suffering: as a religious person, he told me, he was angry at God for not curing his illness and for letting him die without a wife. He did not know how to cope with this anger and did not think that either hospice or his own priest could help address it.

At the end of our second visit, after discussing his symptoms and his continuing spiritual distress, I asked Mr. Hayes "Is there any other way I can help you?" He answered: "You can get put a needle in my vein and help me end it." I asked what he meant. "You can help me die. It's in your oath to stop suffering." I explored with him how he was suffering, whether there were things he looked forward to in his future and whether there were ways we could help him short of ending his life. After discussions with many members of the hospice team, aggressive treatment of his symptoms, and help from two of his brothers helping care for him, his desire that we help end his life waned but never completely disappeared.

DISCUSSION: Requests for physician assisted suicide (PAS) are troubling, partly because they highlight intense and unrelieved suffering. This case illustrates how such a request can be a window into inadequately addressed suffering, can reveal misunderstandings about what physicians are legally permitted to do; and how palliation of many symptoms may not be adequate for some patients' suffering. Rather than a debate about the ethics of PAS, this case discussion will consider the reasons behind such requests and our possible responses to them.

"ICEBERG LETTUCE AND CARROTS": A MEAL IN SEARCH OF A DIAGNOSIS. R.D. Hobbs¹; ¹Oakwood Healthcare System, Dearborn, MI

LEARNING OBJECTIVE: To recognize the varied presentations of Pica as well as its clinical significance.

CASE INFORMATION: A 57 year-old woman presented with memory loss, a fear of becoming obese and a recent four-month history of cravings for "iceberg lettuce and carrots." She was eating two heads of lettuce and a pound of carrots daily. Her past history included hypothyroidism (1970), treated with oral replacement; a hysterectomy (1970); severe osteoarthritis with chronic left hip pain; fibromyalgia treated with amitriptyline; a gastric exclusion procedure for weight loss (1994); and severe personal stress due to the recent suicide of her 27 year old son six months prior. Her weight loss procedure was a partial gastrectomy with creation on a six-ounce pouch and anastomosis of the distal duodenum to the terminal ileum. This resulted in a "four foot common channel." She lost 117 pounds without incident. Physical exam was unremarkable except for mild memory loss, obvious osteoarthritis with a left sided limp and well healed surgical scars. She weighed 170 pounds.

DISCUSSION: Although there was obvious depression and a profound fear of weight gain, the cravings for "iceberg lettuce and carrots" were of recent onset. Because of the history of the gastric bypass procedure and its expected result of intestinal mal-absorption, a Pica variant was suspected. Although Pica has been associated with such bizarre food cravings as laundry starch, clay, chalk, ice, and paint chips, it has been associated with normal foodstuffs such as apricots, almonds, chocolate, carrots, etc. Because of the association between Pica and iron deficiency, laboratory tests for anemia were ordered. The results were compatible with iron deficiency anemia. Additional studies revealed zinc deficiency, early vitamin B12 deficiency, and vitamin D deficiency with hypocalcemia and secondary hyperparathyroidism. With these findings, the patient revealed that although she had taken nutritional supplements following her operation, she stopped them following the death of her son. With reinstitution of nutritional supplementation her behavior and laboratory studies have been improving.

AN INSIDIOUS DIABETES. J. Huang¹, L. Nicholson²; ¹Palo Alto, CA; ²Stanford, Stanford, CA

LEARNING OBJECTIVES:

1. Recognize (DI) diabetes insipidus clinically.
2. Confirm DI using standard diagnostic testing.
3. Through thorough review of its possible causes, establish the etiologic lesion for DI in a patient.

CASE INFORMATION: A 23 year old male came to the emergency room with a chief complaint of a nine day history of blood in his stool, lightheadedness, nausea, diarrhea, and profound weakness. He was admitted for dehydration, anemia, and bright red blood per rectum. On review of systems, it was noted that he had been drinking several gallons of water a day since 5 months prior to admission and had been urinating hourly.

Diabetes insipidus (DI) was suspected when it was observed that he had a clear dilute urine despite his tachycardia and orthostatic hypotension. His urinalysis revealed a low specific gravity of 1.002 and no glucose; serum chemistries were normal. A water deprivation test was consistent with diabetes insipidus. Subsequent workup revealed panhypopituitarism and ultimately the causative lesion, a midline germinoma. The rectal bleeding, which might have lead us toward a different evaluation, was hemorrhoidal. The diarrhea was most likely secondary to his hypoadrenal state.

DISCUSSION:

CERVICAL SPINE OSTEOMYELITIS CAUSED BY *STREPTOCOCCUS MILLERI*. M.R. Huber¹, P.S. Mueller¹; ¹Mayo Clinic, Rochester, Minnesota

LEARNING OBJECTIVES: #1 Recognize that poor dental hygiene can lead to serious complications, such as cervical spine osteomyelitis.

#2 Distinguish *Streptococcus milleri* (*S. milleri*) from other viridans streptococci by its tendency for suppurative infections.

#3 Recognize *S. milleri* as a rare but known cause of vertebral osteomyelitis.

CASE INFORMATION: A 61 year old male was found to have fractures of C5 and C6 with associated left sided radiculopathy after falling down stairs. The patient had no chronic medical conditions except for recurrent tonsillitis and tonsilloliths.

The patient was admitted to an outside hospital and found to have a WBC of $13,300 \times 10^9/L$ and a WESR of 67 mm/hr. Surgery was performed to stabilize the fracture and revealed cervical osteomyelitis with associated phlegmon and epidural abscess. The patient underwent debridement and internal fixation. Five of six cultures grew only pan-sensitive *S. milleri*. With ampicillin, the patient's WBC had normalized and WESR had improved. However, he experienced two internal fixation rod failures and developed a mediastinal abscess with associated cutaneous fistula. He was then transferred to our hospital.

Tissue cultures from the neck and mediastinal abscess grew several organisms, and as such, antibiotic coverage had to be broadened. The patient required three additional spine stabilizing operations before a satisfactory result was achieved. He had a number of complications including drug induced neutropenia, thrombocytopenia, rash, and atrial fibrillation.

At the time of this writing, the patient has made a nearly full recovery. He has a WBC of $4,300 \times 10^9/L$ and WESR of 0 mm/hr. He has been maintained on gatifloxacin and minocycline while he has metal hardware in the previously infected region.

DISCUSSION: *Streptococcus milleri* represents several biologically diverse, yet clinically homogenous, types of viridans streptococci. They are characterized by their tendency for suppurative infections. They can be found in abscesses of the mouth, brain, and liver as well as empyemas. Several cases of vertebral osteomyelitis have been previously reported. Of the two cases of cervical osteomyelitis, one involved an intradiscal steroid injection and the other had known severe periodontal disease. The patient in this case had known palatine tonsillar disease which may very well have led to his cervical spine osteomyelitis.

RENAL MASS IN A PATIENT WITH DEMENTIA: THE CASE FOR WEAK PATERNALISM. M.T. Hughes¹; ¹Johns Hopkins University, Baltimore, Maryland

LEARNING OBJECTIVES: 1. Assess the role of weak paternalism for patients lacking decision-making capacity, when no readily identifiable surrogate decision-maker is available.

2. Weigh the obligations of beneficence, respect for autonomy, and an ethic of care.

3. Determine the goals of care for cognitively impaired patients with a life-limiting illness.

CASE INFORMATION: 82 year old male with multiple medical problems including coronary artery disease, severe chronic obstructive pulmonary disease, and moderate, multi-factorial dementia presented with poor appetite after being lost to medical follow up for six months. Previous physicians had prescribed megestrol for anorexia. Left renal mass had been detected five years earlier and "watchful waiting" approach had been arranged with urologist two years prior to presentation. Six months prior to presentation, CT scan demonstrated enlargement of mass and findings consistent with renal cell carcinoma. Patient had been referred back to urology but appointment did not occur. Over course of current evaluation, when asked whether to pursue repeat CT scan, patient stated, "It don't bother me, so don't bother with it." Patient lived with his 84 year old wife, who likewise had moderate dementia and had recently undergone surgical resection of Stage III gastric carcinoma. Patient had lost contact with distant cousins in another state and denied any other family members who could help with decision-making. His day consisted of "chewing the fat" with friends at the local market. He enjoyed his life and wanted to stay at home with his wife as long as he could. CT scan and urology consultation were not pursued. One year later, the patient suffered a major stroke. Home hospice was arranged, but his wife was unable to care for him at home. On presentation to the emergency room, she had no recollection of his stroke and no recognition of his dense hemiplegia. The patient was transferred to an inpatient hospice, where he died two weeks later. His wife underwent psychiatric hospitalization for her dementia and was subsequently transferred to an assisted living facility.

DISCUSSION: Informed consent rests upon the principle of respect for autonomy and thus requires voluntariness and competence. This patient had impaired judgment, poor recall, and an inability to comprehend complex information. Informed consent was not possible. His wife lacked decision-making capacity. Together, both could live independently in the community, compensating for each other's deficits. Neither could weigh the benefits and burdens of a major surgery. In acting out of beneficence, the physician in this case needed to determine whether definitive treatment for the renal mass (nephrectomy with significant perioperative risk) was a benefit or a harm. Goal setting and understanding the patient as person resulted in a decision that did not strictly adhere to an autonomy model. An ethic of care, focusing on maintenance of the relationship between husband and wife, was judged to be more determinative than a principle-based ethic. This case demonstrates that in certain well-defined situations, the physician may be justified in invoking weak paternalism.

"PANCREATITIS, HEPATITIS, AND DERMATITIS" IN A 70 YEAR OLD FEMALE. L. Humphreys¹, E. Yee¹, A. Gomez¹; ¹UCLA San Fernando Valley Program, Sepulveda, Ca

LEARNING OBJECTIVES: 1) Recognize the clinical presentation of dermatomyositis, 2) Discuss the work up and treatment of dermatomyositis, 3) Discuss screening for occult malignancies associated with dermatomyositis.

CASE INFORMATION: A 70 yo Latina female with a history of cholecystectomy presented with 3 days of nausea, vomiting, and epigastric pain. She denied fever, chills, diarrhea and constipation. She took no medications and denied ETOH use. She was a thin female with T 36.1 C, BP 96/52, P 68, R 18. A violaceous rash was present on her forehead, eyelids, and nose. She had epigastric tenderness with guaiac negative stool. Extremities had subcutaneous calcifications with diffuse weakness proximally. Lab work was remarkable for AST 842, ALT 467, T bili 1.3, D bili 0.3, Alk Phos 352, lipase 9,759, and serial CK's of 5640, 4811, 3835. EKG was normal and troponin negative x 2. An U/S showed no biliary dilation, gallbladder, or stones, and a poorly visualized pancreas. She was hospitalized with the presumptive diagnosis of "pancreatitis, hepatitis, and dermatitis". Subsequent workup found normal triglyceride and calcium levels, negative viral hepatitis serologies, and an unremarkable

abdominal CT scan. An elevated aldolase of 24.9, EMG with myopathic changes, and muscle biopsy demonstrating myositis confirmed the diagnosis of dermatomyositis. She improved clinically with steroids. Work up for occult malignancy including CXR, PAP smear, mammography, CA-125 level, endoscopy, and colonoscopy were all normal. An etiology for her pancreatitis was never found.

DISCUSSION: Dermatomyositis (DM) is an idiopathic inflammatory myopathy that usually presents with proximal muscle weakness. Distinctive skin manifestations include a psoriatic rash over the knuckles (Gottron's papules), periorbital discoloration (heliotrope rash), and subcutaneous calcifications. DM occurs in approximately 5/1,000,000 people. Diagnosis is confirmed by elevated muscle enzymes, myopathic EMG findings, and muscle inflammation on biopsy. Presence of anti-Mi-2 antibody is specific for DM, but is found in only 25% of cases. First line therapy is systemic corticosteroids, with methotrexate, azathioprine, or cyclophosphamide reserved for refractory cases. Malignancy occurs between 7% and 34% in DM patients, with ovarian and colon cancers found most often. Cancer associated disease is more common in older patients and is associated with a worse prognosis. There may be a delay of up to five years before any associated malignancy develops. Although this patient had an extensive workup because of her symptoms, asymptomatic patients with DM should routinely undergo age specific screening tests for occult malignancies.

TITLE: CRANIAL NERVE III PALSY IN CEREBRAL ANEURYSM. G. JAFARI, OLIVE VIEW-UCLA MEDICAL CENTER, SYLMAR, CA. G. Jafari¹; ¹University of California, Los Angeles, Los Angeles, CA

LEARNING OBJECTIVE: To recognize the early signs of cerebral aneurysms and make an early diagnosis

CASE INFORMATION: 59-year-old male with history of well controlled hypertension and COPD presented with a severe headache 10/10 in intensity, throbbing, constant and gradual in onset which began 11 days ago in the occipital area and then extended to the right temporal/posterior orbital area. Two days prior to admission he started having diplopia, ataxia and nausea/vomiting. The patient denied fevers, chills, photophobia, phonophobia, neck stiffness, numbness, weakness, vertigo or seeing flashing lights. Past surgical history includes splenectomy after gun shot wound in 1965. Patient also with history of polysubstance abuse, last used 2 years ago. Family history is non-contributory. On initial examination blood pressure was 167/90, temperature 36.5, heart rate 72 and respiratory rate 18. The patient was alert and oriented and in no acute distress but did report a 10/10 headache. Eye examination was significant for a right pupil which was 5 mm and sluggish to react as opposed to a left pupil which was 2 mm and reactive. There was ptosis of the right eyelid, a deficit in medial and upward gaze on the right and a deficit in accommodation on the right side. There was no papilledema and pressures were 12 bilaterally. Visual acuity was normal and equal bilaterally. The neck was supple and there were no carotid bruits. Neurologically there were no other cranial nerve deficits and no other neurological deficits. CBC and chem.-7 were normal and patient had an ESR of 10. The patient had an MRA which confirmed the diagnosis of posterior communicating artery aneurysm compressing the third cranial nerve. The patient was then transferred to neurosurgery and underwent clipping of the aneurysm.

DISCUSSION: 90% of all intracranial aneurysms are congenital. Common locations include: anterior communicating artery, bifurcation of the middle cerebral artery, posterior communicating artery, internal carotid artery bifurcation and the basilar circulation. Unruptured aneurysms are usually < 6-7 mm and 20% of patients present with oculomotor nerve palsy. These patients usually have aneurysms of the internal carotid artery at the posterior communicating artery. Rupture most commonly causes subarachnoid hemorrhage. Symptoms of rupture include sudden severe headache, altered mental status, meningeal irritation, vomiting, diaphoresis, fever, focal neurological deficits, seizures and visual changes (diplopia, homonymous hemianopsia, miosis or mydriasis). The incidence of sudden death is greatest in the first week. Potential complications include rebleeding, cerebral swelling, hydrocephalus, and infarction secondary to vasospasm. The diagnosis can be confirmed with lumbar puncture where the CSF will show red blood cells, low glucose, elevated protein and a pressure usually > 150. A CT is also useful in detecting aneurysms > 5 mm, subarachnoid blood, infarction and edema. MRA is more sensitive and detects 80% or aneurysms. The treatment is usually surgical but medical therapy can be used to delay surgery and is surgery is contraindicated. Medical therapy includes lowering of blood pressure, bed rest, steroids, anticonvulsants and antifibrinolytic agents.

PRESENTATION OF FULMINANT MENINGOCOCCEMIA IN A YOUNG MALE. N. Jani¹, R. Granieri¹; ¹University of Pittsburgh, Pittsburgh, PA

LEARNING OBJECTIVES: 1) To recognize the clinical presentation of fulminant meningococemia and its associated complications. 2) To recognize the need for prompt treatment of fulminant meningococemia.

CASE INFORMATION: A 21 year-old male presented with a one week history of abdominal pain, nausea, and vomiting. The night prior to presentation, he was noted to be disoriented and lethargic. Several hours later, he was febrile and diaphoretic and was taken to the hospital. On physical exam, he was alert and responsive. His BP was 70/36, heart rate 132, respiratory rate 40, and temperature 35 degrees celsius. Significant findings also included edematous sclerae with petechiae, diffuse rales, a distended abdomen, and diffuse purpura over his extremities. ABG revealed a pH of 7.08, pCO₂ of 71, pO₂ of 77, and bicarbonate of 20. His WBC was 25,600 (28% bands), platelet count 18,000, BUN 36, Cr 4.2, and INR 3.2. Chest x-ray showed bilateral alveolar edema. Despite receiving several liters of saline, he continued to be hypotensive and became more hypoxicemic, requiring intubation. His cardiac rhythm deteriorated to pulseless electrical activity from which he was successfully resuscitated. The diagnoses of septic shock, acute renal failure, disseminated intravascular coagulation, and adult respiratory distress syndrome (ARDS) were made. He was started on Penicillin G for suspected meningococemia, dopamine, norepinephrine, continuous veno-venous hemodiafiltration (CVVHD), fresh frozen plasma and antithrombin III. His blood cultures grew *Neisseria meningitidis*. After 2 weeks, his renal function and coagulopathy resolved. His

left lower extremity became ischemic and nonviable and was amputated. One week later he was discharged to a rehabilitation facility.

DISCUSSION: Infections with *Neisseria meningitidis* range from mild upper respiratory infections to fulminant meningococemia in 10-20% of cases. Few other diseases present as catastrophically as fulminant meningococemia and prompt recognition and early treatment are crucial. Symptoms include sudden onset of fevers, chills, nausea, vomiting, and myalgias, progressing to shock and multisystem organ failure within hours. Purpura fulminans is present in 75% of patients, characterized as maculopapular or petechial lesions which can become hemorrhagic. In addition to antibiotics and coagulation factors, patients may also require ventilatory support. In order to remove toxic substances and cytokines, CVVHD is beneficial. The mortality rate with fulminant meningococemia is as high as 50%, but with prompt recognition and treatment, up to 80% of patients can have favorable outcomes.

HEPATIC ENCEPHALOPATHY MASKING A DEEPER ABNORMALITY. H. Jasti¹, R. Granieri¹; ¹University of Pittsburgh, Pittsburgh, PA

LEARNING OBJECTIVES: 1) Identify the clinical presentation of hemochromatosis (HC). 2) Recognize the diagnosis and management of this condition.

CASE INFORMATION: A 53-year-old female with hypertension, diabetes, and cirrhosis presented with worsening lethargy and confusion. Social history was remarkable for alcohol abuse, which ended two years before presentation. On physical, she had scleral icterus, hepatosplenomegaly, generalized jaundice with palmar erythema and spider angiomas. Neurologically, she was alert but intermittently confused. Asterixis was present. Laboratory studies were significant for elevated transaminases and ammonia levels. Subsequent lab work revealed a ferritin of 2000, Fe of 250, and transferrin saturation of 89%. A liver biopsy demonstrated increased iron deposition in the hepatocytes, confirming the diagnosis of HC. A consequent family discussion revealed that her younger brother (age 41) and nephew (age 36) both complained of arthralgias. Genetic testing confirmed HC in both of them.

DISCUSSION: Hemochromatosis is the most common inherited disorder in the U.S. It is an autosomal recessive condition in which there is increased absorption of iron that causes damage to the liver, heart, pancreas, joints, and other organs. The mean age of presentation is in the fifth decade. The classic triad of cirrhosis, diabetes, and skin hyperpigmentation is not usually the presenting sign; rather it represents late-stage disease. Arthropathy, especially in non-weight bearing joints, can be a common presenting complaint, as it is prevalent in 60% of symptomatic patients. Other early signs and symptoms include erectile dysfunction, amenorrhea, depression, and fatigue. Diabetes and cardiomyopathy may also be presenting signs in younger individuals. The liver is the most affected organ with iron deposition leading to functional abnormalities, fibrosis, and eventually cirrhosis. Liver failure is the most common cause of death. A combination of increased transferrin saturation (> 50%) and an elevated ferritin is about 95% accurate for the diagnosis. Confirmation is made with a liver biopsy. Genetic testing has also shown promise, especially to test the first-degree relatives of patients with HC. Treatment involves phlebotomy, which depletes iron stores. Given the high prevalence of the condition and effectiveness of early intervention, early recognition of the disease is critical for successful treatment and prevention of premature morbidity and mortality. Therefore, one should consider the diagnosis in those cases presenting with abnormal transaminases, idiopathic diabetes, or unusual presentations of arthritis.

AN UNWANTED FAMILY INHERITANCE: THE CASE OF THE RAPIDLY PROGRESSIVE DEMENTIA. H. Jasti¹, R. Granieri¹; ¹University of Pittsburgh, Pittsburgh, PA

LEARNING OBJECTIVES: 1) Recognize the clinical presentation of Creutzfeldt-Jakob disease (CJD). 2) Recognize the diagnosis of this condition.

CASE INFORMATION: A 44-year-old white female with no significant past medical history was in her usual state of health until two months prior to presentation when she began to experience an unsteady gait and problems with memory. One month prior to presentation, she experienced a seizure and then, two weeks later, she developed a rapid decline in her function, starting with short-term memory loss and progressing to unresponsiveness. Family history was significant for a paternal aunt who, at the age of 49 years, had a similar decline in cognitive function over 14 months. Physical exam revealed an awake woman who was unresponsive to commands. Vital signs were within normal limits. Neurological exam revealed dystonic posturing of the head with spontaneous conjugate eye movements, myoclonic activity of the upper extremities, and increased motor tone bilaterally. DTR's were normal bilaterally in the upper and lower extremities, with downgoing plantar reflexes. Bilateral grasp and glabellar signs were elicited. ANA, ESR, RPR, and Lyme titers were negative. An LP and a MRI were non-revealing. An EEG revealed periodic sharp-wave complexes. CSF was positive for a mutation of the prion protein gene, confirming familial CJD.

DISCUSSION: The prion diseases constitute a family of subacute, neurodegenerative diseases referred to as transmissible spongiform encephalopathies. Incubation periods last from months to years and symptoms gradually increase in severity and result in death over a period of months. CJD is the major transmissible spongiform encephalopathy in humans. While a sporadic form tends to occur after the sixth decade and has a rapidly progressive course of under a year, an inherited form usually manifests at a younger age and has a more protracted course. Cardinal clinical features include confusion and memory loss progressing to severe cortical dementia, ataxia, and myoclonus. The EEG typically demonstrates bilateral periodic discharges. Neuroimaging is helpful to exclude other etiologies, since there are usually no abnormalities. Elevated levels of the 14-3-3 protein in the CSF, which has a 96% sensitivity and specificity for CJD, are also helpful for the diagnosis. The defining pathology is the presence of "spongy" degeneration of cortical gray matter. At present, there is no effective treatment for CJD. The diagnosis of prion disease should be considered as part of the differential in cases presenting with dementia and an atypical movement disorder, especially if the rate of disease progression is rapid.

BEYOND THE OBVIOUS ABNORMAL LABORATORY VALUE. H. Jasti¹, R. Granieri¹; ¹University of Pittsburgh, Pittsburgh, Pennsylvania

LEARNING OBJECTIVES: 1) Recognize secondary causes of hypercholesterolemia. 2) Recognize the association between hypercholesterolemia and hypothyroidism.

CASE INFORMATION: A 47-year-old male with hypertension and diabetes presented for a new visit. He endorsed occasional myalgias and constipation. He denied fatigue, weakness, cold intolerance, or dry skin. On physical, blood pressure was 150/86 and pulse was 74. The remaining exam was unremarkable, including the absence of tendinous xanthomas and delayed DTRs. Old records revealed a diagnosis of hypercholesterolemia (295 mg/dL) and hypertriglyceridemia (1200 mg/dL), with subsequent initiation of gemfibrozil. His lipid levels continued to be elevated 6 months after initiation of the medication. Subsequent testing revealed an elevated TSH and abnormal TFTs, consistent with a picture of hypothyroidism. Levothyroxine was started. Upon follow-up in 2 months, the TSH levels had decreased; cholesterol levels had also declined modestly. Further follow-up 4 months later demonstrated normal TSH levels and significantly decreased cholesterol (220 mg/dL) and triglyceride (480 mg/dL) levels.

DISCUSSION: Significant hyperlipidemia may be present in the absence of clinical features; thus, secondary causes need to be considered before the initiation of lipid-lowering therapy. These include Type II diabetes, chronic renal insufficiency, nephrotic syndrome, cholestatic liver disease, obesity, smoking, alcohol, steroid, and oral contraceptive use. Hypothyroid patients also display a variety of patterns of lipid abnormalities, including elevated cholesterol, triglyceride, LDL levels. The main mechanism is a decrease in the number of LDL cell receptors, resulting in an increase in LDL levels. Other mechanisms include a diminished secretion of cholesterol into bile and reduced lipoprotein lipase activity, resulting in hypertriglyceridemia. The severity of the hypothyroidism may play a role in treatment, since many studies show a more significant improvement in the lipid levels in those with a more severe form of hypothyroidism. The evidence is less clear regarding the significance of the relationship between subclinical hypothyroidism and elevated cholesterol levels. In general, those with overt hypothyroidism seem to respond best to levothyroxine treatment and subsequently show a decline in their cholesterol levels, whereas those with subclinical hypothyroidism seem to need additional lipid-lowering agents. In conclusion, this case underscores the importance of identifying secondary causes of hypercholesterolemia before initiating treatment for an apparent lipid abnormality.

ITS NOT ALWAYS DKA. A. Tabas¹, G. Tabas¹; ¹University of Pittsburgh Medical Center Shadyside, Pittsburgh, PA

LEARNING OBJECTIVES: 1. Recognize lactic acidosis as a complication of metformin use. 2. Differentiate between diabetic ketoacidosis and metformin induced lactic acidosis.

CASE INFORMATION: A 22 year old black female was admitted with three days history of nausea, vomiting, headache and abdominal pain, preceded by two days of rhinorrhea, and diarrhea. She was diagnosed with insulin requiring diabetes three years ago. There was no history of pre-existing renal insufficiency, chronic pulmonary disease or heart disease. Her medication was changed from insulin to metformin 500 mg bid a week prior to hospitalization. Her physical exam revealed tachycardia, and tachypnea. Pulse oximetry revealed 100% saturation on room air. The rest of the exam was unremarkable. Laboratory data were significant for ABG on room air -PH-7.138, PCO₂-15, PO₂-115.4, HCO₃-5.2. Sodium-139 meq/L, Potassium-4.3 meq/L, Chloride-106 meq/L, CO₂-7 meq/L, BUN-5 mg/dl, Creatinine-0.8 mg/dl, Glucose-276mg/dl, serum acetone-positive. The anion gap was 26. CBC & differential, and chest x-ray were within normal limits. Lactate and metformin levels were not checked. The patient was treated with normal saline, insulin drip, sodium bicarbonate and metformin was held. The metabolic acidosis resolved and the patient was discharged on long acting insulin.

DISCUSSION: We do not know whether our patient had atypical diabetic ketoacidosis or metformin induced lactic acidosis. The presenting serum glucose in patients with diabetic ketoacidosis is usually between 450 mg/dl and 750 mg/dl. This patient's initial serum glucose of 276 mg/dl is somewhat uncommon for diabetic ketoacidosis, and suggests that other causes of high anion gap acidosis must be considered. The incidence of lactic acidosis is about 9 per 100,000 person-years in patients taking metformin. It usually occurs in patients who have pre-existing cardiac disease, congestive heart failure, chronic pulmonary disease with hypoxia, sepsis or old age. Metformin induced lactic acidosis in patients such as ours, with no pre-existing risk factors, is rare but has been reported in literature. The mechanism is unknown. In patients taking metformin, presenting with ketoacidosis, metformin induced lactic acidosis should be considered. Assessing risk factors, appropriate laboratory workup (arterial lactate level, metformin level), withdrawing metformin and supportive care are key components of management.

MORPHINE INDUCED RHABDOMYOLYSIS AFTER AN UNEVENTFUL SURGERY. A. Jha¹, K. Bernard¹, D. Wilson¹, R. Quinlin¹; ¹University of Pittsburgh Medical Center Shadyside, Pittsburgh, PA

LEARNING OBJECTIVES: 1) To recognize rhabdomyolysis due to morphine. 2) To consider rhabdomyolysis in the differential diagnosis of post-operative renal failure.

CASE INFORMATION: A 33-yr. old female underwent a laparoscopic 'Roux-en-Y' gastroplasty under general anesthesia for morbid obesity. The anesthesia was induced by Midazolam, Fentanyl, Rocuronium and Succinylcholine. There was no masseter rigidity, or hyperthermia noted. Maintenance of anesthesia was done by Nitrous oxide, Propofol and Fentanyl. The patient had an uneventful perioperative course. There was no hematoma, or muscle tenderness following surgery. Morphine PCA was started. She has a past medical history of sleep apnea, hypertension, tubal ligation and hysterectomy without any complications. Her medications include Sertraline, Hydrochlorothiazide, Enalapril, and Albuterol inhaler. Her family history was unremarkable. The next morning she was found unresponsive and hypoventilating. An ABG revealed acute respiratory acidosis. She received Naloxone and the

ABG normalized. Lab data revealed BUN -4 mg/dl, creatinine-2.0mg/dl (baseline - 0.8), calcium - 6.7 mg/dl, phosphorus-5.7mg/dl, magnesium-1.6mg/dl, sodium-138meq/L, potassium-4.1 meq/L, chloride-103 meq/L, CO₂-21meq/L, CPK-658 IU/L, CPK-MB-5.8 IU/L, Urine myoglobin- 374(<27mcg/L). EKG showed sinus tachycardia and an echocardiogram was normal. EEG, thyroid function, and serum Cortisol were normal. CPK levels increased to around 9400 IU/L and CPK-MB increased to 26 IU/L within two days. A diagnosis of morphine- induced rhabdomyolysis was made and patient was treated with D5W and NaHCO₃. Her Creatinine normalized the following day, and CPK, CPK-MB became normal on the 7th day.

DISCUSSION: Prolonged surgery, electrolyte disturbances, infection and drugs are common causes of rhabdomyolysis and myoglobin- induced renal failure in the post-operative period. Morphine infrequently causes rhabdomyolysis. Proposed mechanism for morphine-induced rhabdomyolysis is myohypoxic necrosis and myocompression due to sedation related immobility. Reversal of narcotic effect with naloxone, withdrawing morphine, and hydration are the key components of management.

PARKINSONS TREMOR MIMICKING MULTIPLE ARRHYTHMIAS. M.F. Jhandir¹, D. Brotman¹; ¹Cleveland Clinic Foundation, Cleveland, OH

LEARNING OBJECTIVES: 1. Rule out Parkinson's tremor first when pateints with Parkinson's disease present with symptoms or electrocardiogram changes suggestive of tachyarrhythmia. 2. Appreciate that holding anti-Parkinson's medications can be a simple provocative maneuver to determine whether the electrocardiogram abnormalities are of non-cardiac origin in such patients.

CASE INFORMATION: A 75 year old nursing home resident with a history of coronary artery disease and Parkinson's disease presented to the emergency department for evaluation of pre-syncope episodes and hypotension. He had a three day history of feeling lightheaded and unsteady while standing but had no true syncope. He denied chest pain, dyspnea and palpitations. There was no prior history of documented arrhythmia. His medications included aspirin, donepezil, levodopa/carbidopa, and lisinopril. On physical examination, he was not in acute distress. Heart rate was 68 with occasional premature beats. Blood pressure was 65/45 bilaterally with orthostatic changes. Cardiac exam was without murmurs or gallops. Neck veins were flat, lungs were clear and there was no peripheral edema. He had cogwheel rigidity but no tremor. The admission ECG showed sinus rhythm with occasional premature ventricular contractions. After intravenous crystalloids were given, the blood pressure normalized, and he was admitted to a monitored hospital bed for further observation. The following morning, the intern was called to the patient's bedside because of an apparent supraventricular tachycardia (SVT)(figure 2). Despite his abnormal ECG, the patient was asymptomatic, with a pulse of 64, and a blood pressure of 160/90. He had developed a pronounced bilateral upper extremity resting tremor at about 5 Hz, but otherwise his examination was unchanged. After the patient received his morning anti-Parkinson medications, the tremor resolved and his ECG returned to baseline. In order to prove that the tremor was responsible for the ECG changes, the next dose of levodopa/carbidopa was held; his tremor returned as did his pseudo-atrial flutter.

DISCUSSION: Because the classic Parkinson's tremor has a frequency of 4-5 Hz (260-300 cycles/min), it is not surprising that its electrical potentials on ECG can mimic atrial flutter, as in our case. Furthermore, Parkinson's disease is commonly associated with orthostatic hypotension and syncope, which may in turn prompt cardiac evaluations. Although there are previously documented cases of Parkinson's tremor leading to ECG artifacts resembling wide complex tachyarrhythmias such as ventricular tachycardia, we know of no other reports of Parkinson's tremor mimicking atrial flutter or SVT. Careful review of our patient's rhythm strip reveals that there are buried QRS complexes within the pseudo-SVT complexes (figure 2). The importance of correlating the ECG with the clinical exam cannot be overemphasized in this case, as the heart rate during the pseudo-SVT was normal, and the ECG artifacts were present only when the patient was tremulous. Finally, our case illustrates that if a patient with Parkinson's disease has transient ECG abnormalities suggestive of tremor-induced artifact, holding medicines can be a simple provocative maneuver to determine whether the ECG abnormalities are of non-cardiac origin. We conclude that Parkinson's tremor should be thoroughly ruled out when a patient with Parkinson's disease presents with symptoms and ECG changes suggestive of a tachyarrhythmia.



Figure 1

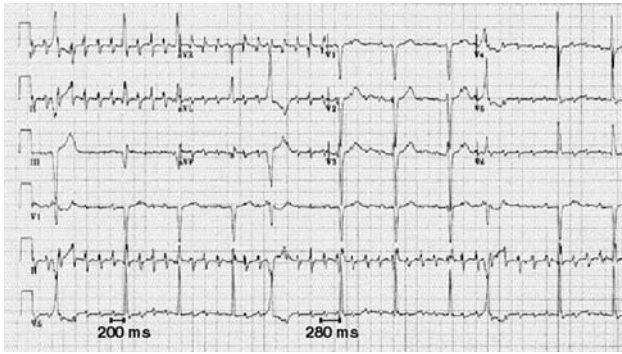


Figure 2

STRONGYLOIDIASIS AND THE HYPERINFECTION SYNDROME. H. Jneid¹, N. Correia¹, F. Michota¹; ¹Cleveland Clinic Foundation, Cleveland, Ohio

LEARNING OBJECTIVES: 1. List the differential diagnoses for immunocompromised patients with chronic obstructive pulmonary disease and progressive dyspnea. 2. Recognize the risk factors and clinical features of strongyloidiasis. 3. Recognize the hyperinfection syndrome of strongyloidiasis.

CASE INFORMATION: A 65-year-old retired coal miner presented with a 2-month history of progressive shortness of breath and nonproductive cough. He had a history of emphysema and coal worker's pneumoconiosis and had been using an inhaler and oxygen therapy. He had grown up in a rural Appalachian area of Ohio and stopped using tobacco 20 years ago. He had been started on cyclophosphamide and high-dose prednisone therapy for rapidly progressive glomerulonephritis 3 months previously. He was morbidly obese, with T 37.4°C, BP 134/65 mm Hg, Pulse 100, RR 22/min, and SpO₂ 91% on 4 L of oxygen. Lung auscultation showed decreased air entry and diffuse wheezing. Initial chest x-ray showed no parenchymal lung disease. Review of his kidney biopsy revealed chronic interstitial nephritis, cyclophosphamide was stopped, and prednisone was gradually tapered. His dyspnea and oxygen requirement increased despite aggressive therapy with bronchodilators. Venous duplex ultrasound of the upper and lower extremities showed no thrombosis. V-Q scan revealed no mismatches. Computed tomography of the chest subsequently showed diffuse alveolar ground-glass appearance. The patient was started on broad-spectrum antibiotics, including intravenous Bactrim and amphotericin. Bronchoscopy was done, and *Strongyloides stercoralis* was diagnosed by transbronchial biopsy. The patient was started on intravenous thiabendazole. Prednisone was tapered quickly and stopped. His respiratory status deteriorated, and the patient eventually died. Autopsy found disseminated infection, with panlobar pulmonary infestation and patchy small and large bowel strongyloidiasis.

DISCUSSION: Our immunocompromised patient had worsened respiratory symptoms despite inhaler therapy, which prompted us to seek other diagnoses. Initial work-up was negative for congestive heart failure and thromboembolism. His previous diagnosis of simple coal worker's pneumoconiosis was not supported by initial radiologic findings and would not have explained his rapid clinical deterioration and radiologic evolution. In this setting, opportunistic infections should be strongly suspected. *Strongyloides* is an intestinal parasite that is common in tropical and subtropical areas. Infection rates are highest among residents of the southern states and Appalachia, and among immigrants from endemic areas. Transmission occurs when infective larvae come in contact with the skin. The parasite has an autoinfection cycle that is usually contained by the host immune system; therefore, larval forms may persist in the body up to 50 years. Clinical manifestations of strongyloidiasis include abdominal upsets, lung manifestations such as wheezing and cough, and a pathognomonic skin eruption called larva currens. The autoinfection cycle accelerates inversely with the strength of the immune system and may yield to the highly lethal hyperinfection syndrome in immunosuppressed patients, characterized by massive dissemination to multiple organs and systems, such as the central nervous system, heart, liver, and endocrine glands. People at risk for *Strongyloides* infection should be screened before being given immunosuppressive therapy.

STROKE SECONDARY TO AIR EMBOLISM. H. Jneid¹, A. Abou-chebl¹, G. Hanes¹, S. Inamdar¹, A. Almahameed¹; ¹The Cleveland Clinic Foundation, Cleveland, Ohio

LEARNING OBJECTIVES: 1. List the steps in the diagnostic work-up of transient ischemic attack (TIA) and embolic stroke. 2. Recognize air embolism as an infrequent cause of stroke. 3. Recognize the predisposing conditions and the treatment of cerebral arterial air embolism.

CASE INFORMATION: A 50-year-old white woman presented with two transient 15-minute episodes of left upper extremity paresthesia, slurred speech, and diplopia. She had had inflammatory breast cancer 3 years previously, and had undergone right modified radical mastectomy, chemotherapy, and radiotherapy. Liver metastasis had recently been discovered. Neurologic exam on presentation was nonfocal. Computed tomography (CT) of the brain and lumbar puncture fluid analysis were normal. Magnetic resonance imaging found no leptomeningeal enhancement or focal lesions. Duplex ultrasound of her extracranial arteries showed no significant atherosclerotic stenoses, but 15 minutes of continuous Doppler monitoring showed active embolization (>1000 emboli/minute) to the brain. Venous duplex ultrasound of her upper and lower extremities showed no thrombosis, and CT scan of her abdomen showed no pelvic or inferior caval thrombosis. Holter monitoring showed no cardiac arrhythmias. Transesophageal echocardiography showed no valvular vegetations, intracardiac thrombi, or patent foramen ovale, but revealed spontaneous air bubbles in the left atrium. The patient's neurological status progressively worsened until she became comatose, entered respiratory arrest, and died. No autopsy was performed.

DISCUSSION: Cerebral arterial air embolism has been described with cardiac surgery, mechanical ventilation, underwater diving and bronchoscopic laser operations. Arterial embolization results from direct passage of air into the arterial circulation or from paradoxical embolization. The effect of an air embolus depends on its size as well as its rate of entry into the circulation. Arterial embolization is dangerous in that it can cause ischemic damage to organs. Because absorption of arterial emboli is inversely related to the arterial pressure of nitrogen, treatment with hyperbaric oxygen has been effective. This treatment also helps reduce the air bubble size and increases the oxygen content of blood. In our patient, no bubbles were detected in the right heart chambers, which makes paradoxical embolization unlikely. A plausible biological mechanism is cerebral air embolism through bronchovenous fistula, with possible pathogenetic implication of either micrometastases to the lung or the prior thoracic radiation therapy, but this remains to be proven.

PULMONARY ALVEOLAR PROTEINOSIS AND IMMUNOSUPPRESSION: A DEADLY MIX. H. Jneid¹, G.S. Hall¹, C. Farver¹, J. Tomford¹; ¹The Cleveland Clinic Foundation, Cleveland, Ohio;

LEARNING OBJECTIVES: 1. Recognize the clinical presentation and treatment of pulmonary alveolar proteinosis (PAP). 2. Recognize the described association of nocardiosis and PAP. 3. Recognize the possibility of multiple opportunistic infections in immunocompromised patients. **CASE INFORMATION:** A 40-year-old man presented to a hospital with a 5-month history of progressive dyspnea on exertion, productive cough, and gradual weight loss. He had a 52 pack-year history of smoking. He was treated with several courses of antimicrobials for bronchitis and was maintained on high-dose prednisone (60 mg daily) for several months. He presented to an outside hospital with respiratory distress, requiring 50% FiO₂. Chest x-ray showed bilateral alveolar infiltrates, and on computed tomography (CT), the chest had a diffuse ground-glass appearance. Bronchoscopy was nondiagnostic, so he underwent an open lung biopsy, which showed pulmonary alveolar proteinosis. He was transferred to our tertiary care facility where he was given normal saline lavage (17 L per lung) under general anesthesia. His respiratory status improved and he was discharged home on 6 L/min of oxygen and tapering doses of prednisone. He returned few days later with worsening dyspnea, fever, and increased cough, now productive of copious brown rusty sputum. His vitals: T 101.0°F, BP 95/56 mm Hg, Pulse 120, and RR 30/min. His lung auscultation revealed diffuse crackles. His PaO₂ on 6 L of oxygen was 57 mm Hg. He was intubated and started on broad-spectrum antibiotics. Sputum staining showed branching gram-positive filaments with weak acid-fast stain, and intravenous Bactrim (trimethoprim and sulfamethoxazole) was added for treatment of *Nocardia* pneumonia. Ophthalmologic exam and CT of the abdomen and brain showed no evidence of nocardial dissemination. His sputum stain subsequently showed septate hyphae, which was targeted with amphotericin B. The fungus was identified as *Trichosporon beigelii*. The patient eventually died of septic shock. Autopsy showed panlobar bilateral necrotizing pneumonia and the presence of silver-positive organisms consistent with *Nocardia*, *Trichosporon*, and *Aspergillus*.

DISCUSSION: Pulmonary alveolar proteinosis (PAP) is a diffuse lung disease characterized by dense accumulation of PAS-positive (periodic acid-Schiff) phospholipid material in the alveoli, with preservation of the lung architecture and absence of inflammation. Patients usually present with a subacute prodrome of cough and progressively decreasing exertional tolerance. Tissue biopsy, by open lung or thoracoscopy, is the gold standard for diagnosis, though transbronchial biopsy and fluid obtained by bronchoalveolar lavage may also be used. Whole lung lavage is the treatment of choice, and recently interest has grown about the potential therapeutic role of GM-CSF (granulocyte/macrophage colony stimulating factor) because defective alveolar macrophages are involved in the pathophysiology. There is no role for immunosuppressants in treatment, and, in fact, some suspect that corticosteroids may increase mortality. The association of pulmonary nocardiosis with PAP has been described frequently. *Nocardia* asteroides tends to disseminate from a pulmonary or cutaneous focus to any organ, notably the central nervous system, retina, heart, kidneys, and joints. The new onset of fever and change in the quality of sputum in our immunocompromised patient prompted us to think of superimposed opportunistic infections.

AN UNUSUAL CAUSE OF SYNCOPE. H. Jneid¹, S. Rohr², M. Rich², M. Garcia³; THE CLEVELAND CLINIC JOURNAL, Cleveland, Ohio, Summa Internal Medicine Center, ; THE CLEVELAND CLINIC FOUNDATION,

LEARNING OBJECTIVES: 1. List the differential diagnosis of spells. 2. Recognize systemic mastocytosis as an unusual cause of syncope. 3. Recognize the clinical presentation and treatment of systemic mastocytosis.

CASE INFORMATION: A thirty-three year old Caucasian man presented with history of recurrent spells. He described episodes of palpitations and hot flushes, followed by tunnel vision, pre-syncope feeling, and loss of consciousness. He had spontaneous recoveries, with no incontinence or postictal state. He reported loose stools, intermittent urticarial rash, and occasional wheezings associated with these episodes. He had an AICD (automatic implantable cardioverter-defibrillator) placed after the second spell for complete heart block that degenerated into ventricular fibrillation. Subsequent interrogation of the AICD showed no evidence of dysrhythmias despite recurrent spells. 24-hour EEG monitoring showed no epileptiform activity. He underwent a cardiac catheterization, which showed normal coronary arteries. CT scans of the head and the abdomen were non-revealing. Pulmonary angiography showed no evidence of pulmonary embolism. On presentation, his physical exam, including neurological exam, was completely normal, except for urticarial rash on the dorsum of his hands. Telemetry monitoring didn't reveal any arrhythmia. A tilt table test showed asymptomatic postural tachycardia. Urine metanephrine and 5-HIAA levels were within normal. Echocardiography showed no valvular lesions. Glucose levels and CBC were normal. A MIBG scan was also negative. The patient was discharged home on a beta-blocker. His spells recurred. Further work up included a twenty-four hour urine histamine collection, which was normal. Punch biopsy of his urticarial skin lesions, revealed urticaria pigmentosa, with perivascular and interstitial mast cells infiltration of the superficial dermis. Colonoscopy

showed mast cells infiltration of the lamina propria of the ileum and right colon. A bone marrow aspirate revealed 15% mast cell cellularity. The diagnosis was consistent with systemic mastocytosis. The patient was switched to a combination of H1 and H2 blockers, and later had a leukotriene receptor antagonist added to his regimen. He remained spell-free afterwards.

DISCUSSION: One needs to exclude other causes of spells, like cardiovascular, neurologic, endocrine, pharmacologic, and psychologic etiologies. Systemic mastocytosis is a disease characterized by mast cell infiltration of multiple organs. It presents with episodic flushing, diarrhea, abdominal pain, bronchospasm, alcohol intolerance, and syncope. Triggers include emotional stress, iodinated contrast agents, aspirin, and exercise. Urticaria pigmentosa is a characteristic skin lesion. Therapeutic armamentarium includes H1 and H2 blockers, cromolyn sodium and prostaglandin synthesis inhibitors. Epinephrine is used in severe cases associated with anaphylaxis, while interferon therapy is reserved for proliferative disease. Our patients had a dramatic presentation and a dramatic response after addition of a leukotriene antagonist.

AN ATYPICAL PRESENTATION OF MALIGNANT MESOTHELIOMA. H. Jneid¹, Z. Ghamra², S. Inamdar², A. Almahameed²; ¹THE CLEVELAND CLINIC FOUNDATION, Cleveland, Ohio; ²THE CLEVELAND CLINIC FOUNDATION; ³Cleveland Clinic Foundation, Shaker Heights, Ohio

LEARNING OBJECTIVES: 1. Recognize the risk factors and clinical features of malignant mesothelioma. 2. List the etiologic factors, diagnostic methods, and adverse outcomes of chylothorax. 3. Recognize an atypical presentation of malignant mesothelioma. **CASE INFORMATION:** A 35-year-old white woman presented with a 4-week history of progressive shortness of breath and persistent dull right-sided chest pain. She had never been a smoker. On admission, she was afebrile, RR 14, and had oxygen saturation of 95% on room air. Lung auscultation in the right lower lung fields found decreased breath sounds and decreased tactile fremitus. Chest x-ray confirmed the presence of right pleural effusion. Thoracentesis showed a milky fluid consistent with chylothorax (pH = 7.26, RBC = 79, WBC = 4.4, LDH = 398 u/L, protein = 1.6 g/dl, glucose = 51 mg/dl, triglyceride = 715 mg/dl, cholesterol = 26 mg/dl, and positive chylomicron screen). CT of the chest and abdomen revealed a large pleural mass emanating from the posterior hemithorax, with extension below the right diaphragm. Thoracoscopic biopsy of the mass revealed malignant mesothelioma. The patient had no occupational exposure to asbestos, but her father (who had no lung disease) had worked in the steel industry. The patient was switched to a medium-chain fatty acid diet, and her pleural drainage became serous, although it persisted. Eventually, a Pleurex catheter was inserted for self-administered therapeutic drainage as needed, and the patient was discharged home with hospice care.

DISCUSSION: Malignant mesothelioma is an insidious neoplasm that usually afflicts patients in their fifth to seventh decades of life. Patients usually present with dyspnea, nonpleuritic chest pain, and unilateral pleural effusion. Exposure to asbestos is elicited in only 70% to 90% of cases. Domestic exposure to asbestos through family members is well documented and could have been the cause in our patient. Multimodality therapeutic approaches (surgery with chemotherapy +/- radiotherapy) produce a slight improvement in quality of life and survival. Our patient's tumor extended across the diaphragm and was therefore nonresectable (stage IV). Chylothorax is usually due to either trauma such as surgery or nontraumatic causes such as malignancies, with lymphoma accounting for the majority of cases. Prolonged chyle loss can lead to nutritional depletion and immunologic deficiency. A triglyceride level above 110 mg/dL in a pleural effusion is very suggestive of chylothorax, and the presence of chylomicrons confirms the diagnosis. The presentation of a young patient with no asbestos exposure, with malignant mesothelioma and a chylothorax is extremely rare.

COMBINED ADRENAL CRISIS AND MYXEDEMA COMA DUE TO HYPOTHYROIDISM. K. Kamjo¹, N. Michail²; ¹University of California, Los Angeles, Sherman Oaks, Ca; ²UCLA

LEARNING OBJECTIVES: 1-To recognize the clinical presentation of severe hypopituitarism.

2-To understand the physiology of pituitary hormones.

3-Patient compliance as a cause of serious illness.

CASE INFORMATION: A 37 years old female with a history of pituitary adenoma status post trans-sphenoid resection in 1990, was brought to Emergency Department (ED). Pt was not responsive and paramedics' report indicated that patient had gotten worse over the past month with progressive lethargy, vomiting, and altered mental status. Patient had stopped her medications approximately one year prior to admission. In ED patient was comatose and her initial vitals were: Rectal temperature 33.3 C, pulse 64, respiration rate 14, blood pressure 80/44, and blood sugar of 26. On physical exam patient appeared pale and had marked facial puffiness. She had no heart murmur but was bradycardiac. Patient's neuro exam showed bilateral hyperreflexia with a definite delay in the relaxation phase of her deep tendon reflexes. Patient's skin was very dry and she had no axillary or pubic hair. CXR only showed mild cardiomegaly. EKG showed sinus bradycardia with diffuse low voltage and flat T-waves. Patient was given 2L of intravenous (IV) D5W and 100 mg of IV hydrocortisone. After administration of IV steroids, patient's core temperature increased to 37.2 C and her blood pressure was stable at 89/50. She had an O2 saturation of 100% on room air and became conscious and alert. Subsequently, she was admitted to step-down unit with a diagnosis of adrenal crisis due to untreated hypopituitarism. Patient was started on DDAVP, and hydrocortisone. Serum TSH and free T4 were pending at this time and patient was started on 0.1 mg of oral Synthroid. Patient was doing well after the first day of hospitalization. However, on the second day her mental status deteriorated again and became obtunded. She had increased puffiness in her face and marked edema of her lower and upper extremities. Patient's blood gas showed CO2 retention. Patient was diagnosed with myxedema coma and transferred to medical ICU. Patient was intubated and a single 300 mcg dose of IV Levothyroxine was administered, followed by 100 mcg of IV Levothyroxine daily. At this time serum TSH and free T4 drawn on admission showed TSH = 0.1 and free T4 < 0.40. The patient subsequently improved markedly and was discharged from hospital after 7 days.

DISCUSSION: Myxedema coma is a very rare disease occurring primarily in elderly patients with primary thyroid failure. This case illustrates the importance of early recognition that myxedema coma can occur in combination with adrenal crisis in patients with decompensated hypopituitarism. Non-compliance in this disease could lead to fatal consequences.

AN UNUSUAL CASE OF ERECTILE DYSFUNCTION. A. Karabanow¹; ¹University of Wisconsin-Madison, Madison, WI

LEARNING OBJECTIVE: To recognize Klinefelter Syndrome

CASE INFORMATION: A 56 year old male presented for routine primary care evaluation. His past medical history was significant only for mild COPD and tobacco use. He specifically denied any history of coronary artery disease, diabetes, hypertension or dyslipidemia. Review of systems identified long standing complaints of erectile dysfunction and fatigue. Specifically, the patient noted a frequent inability to reach orgasm and lower extremity fatigue when climbing stairs. Physical examination was notable for a blood pressure of 110/65, an arm span greater than height, intact lower extremity pulses, sparse male pattern hair distribution, gynecomastia, small penis size and small, firm testes.

Pulse volume recordings of the lower extremities and electrocardiogram were unremarkable. Laboratory evaluation found LDL=68, fasting glucose=104, Hct=48, TSH=1.08, free testosterone=2.6, LH=14, FSH=26 and prolactin=3.5. Subsequent chromosomal karyotyping indicated the presence of an 82% XXY pattern consistent with Klinefelter Syndrome.

The patient was treated with intramuscular testosterone. On follow-up, this patient reported improved energy levels and sexual performance. Indeed, the patient presented with new complaints consistent with bilateral epicondylitis. Apparently, following diagnosis and therapy, this patient was now hard at work buffing combines!

DISCUSSION: Klinefelter syndrome, a primary form of testicular failure, is a common disorder that affects 1 in 500 male patients. It results from nondisjunction during parental gametogenesis and yields an individual with an XXY karyotype. Individuals with Klinefelter syndrome do not progress through puberty and ultimately develop the clinical features of eunuchoidism. These features of androgen deficiency include: arm span greater than height, long legs, sparse male pattern hair distribution, gynecomastia, small testes and penis and infertility. Endocrine evaluation of patients with this syndrome indicates low free testosterone together with elevated gonadotropin levels (ie LH and FSH). Klinefelter syndrome is treated with testosterone supplementation. This produces positive changes in mood, behavior, muscle mass and bone density. Once testosterone supplementation is begun, patients must be routinely screened for erythrocytosis and lipid abnormalities.

RIGHT SIDED HEART FAILURE AS A PRESENTATION OF PULMONARY EMBOLISM. M. Zihlif¹, A. Hamad¹, M. Salameh¹, A. Karcic¹; ¹Departments of Medicine and Cardiology, Nassau University Medical Center, East Meadow, NY

LEARNING OBJECTIVE: 1. Consider chronic thromboembolic pulmonary hypertension (CTEPH) in patients presenting with right ventricular failure.

CASE INFORMATION: A 43-year-old man with history of recurrent spontaneous pneumothorax (he underwent pleurodesis 5 years ago), presented to our service complaining of 2 months of progressive abdominal swelling, leg edema, jaundice, and worsening shortness of breath. The patient is a chronic smoker who rarely consumes alcohol. On exam the patient was afebrile with a regular pulse of 110, blood pressure of 118/78, respiratory rate of 18/min, distended neck veins, S4 gallop, massive ascites and +3 leg edema. Other than high total/direct bilirubin 2.2/3.6, the labs were unremarkable. Echocardiogram revealed a markedly dilated right ventricle with moderate pulmonary hypertension, severe tricuspid regurgitation and left ventricle of normal size and function. Spiral chest CT revealed pulmonary embolism of both left lower and upper divisions of the pulmonary artery. Doppler ultrasound of both legs was negative. The patient was anticoagulated (heparin, then warfarin) and once stable discharged home.

DISCUSSION: CTEPH results from single or recurrent pulmonary emboli arising from sites of venous thrombosis. Most emboli completely resolve spontaneously, or they have minimal residual effects. In CTEPH, for unclear reasons, the emboli do not resolve completely. Rather, the emboli undergo abnormal organization and recanalization, leaving endothelium-covered residua that obstruct or significantly narrow major branches of the pulmonary artery. This is very different from primary pulmonary hypertension, where obstructive lesions appear in the small distal pulmonary arteries.

Most patients with CTEPH have history of venous thrombosis with embolism; however this history may not be elicited unless actively sought for. After the initial embolic event the patients' condition gradually improves and they usually resume their normal activities. This period of improvement is however a temporary "honeymoon period" frequently followed by more subclinical thromboembolic events. Recurrent pulmonary embolism destroys the pulmonary vascular bed and causes pulmonary hypertension. With time patients deteriorate and become dyspneic at lower levels of exertion. Clinically, they present with a triad of dyspnea, syncope and right heart failure. Initial treatment includes chronic anticoagulation; consider embolectomy for massive embolism.

RECURRENT CEREBROVASCULAR ISCHEMIC EVENTS AFTER SURGICAL CLOSURE OF A PATENT FORAMEN OVALE (PFO). M. Zihlif¹, A. Karcic¹, E. Karcic¹, F. Begum¹; ¹Departments of Internal Medicine and Cardiology, Naussau Medical Center, East Meadow, NY

LEARNING OBJECTIVE: Recurrent cerebrovascular ischemic events after surgical closure of PFO should prompt further evaluation to identify causes of cerebral vascular ischemia other than paradoxical embolism.

CASE INFORMATION: A 32-year-old woman with history of deep vein thrombosis while on oral contraceptives has had a left parietooctipital infarct three years ago with residual 4/5 right-sided weakness and paresthesia. At that time the patient had negative carotid duplex

and thrombotic and vasculitis work up. Transesophageal echocardiography (TEE) with agitated saline demonstrated a PFO, with significant right to left shunt, with transcranial Doppler demonstrating contrast reaching cerebral circulation. The patient underwent surgical closure of PFO by direct suturing. The patient now presented with the fourth episode of transient right-sided numbness and paresthesia in the last three years. TEE with agitated saline showed no residual shunt. Patient was anticoagulated (heparin, then warfarin) and once stable sent home.

DISCUSSION: Stroke causes more than 100,000 deaths in USA each year, and leaves many others with major disabilities. In 25 to 40% of strokes in young adults, an extensive evaluation fails to identify the cause; these are classified as cryptogenic strokes. Notably, patients with cryptogenic strokes have a higher incidence of PFO than patients with stroke of determined cause, even after correcting for presence of recognized stroke risk factors. Potential mechanism for PFO related ischemic event includes paradoxical embolism in which a thrombus dislodges from a venous thrombotic site, traverses the interatrial septum and reaches the brain. Other mechanisms may include a thrombus formation secondary to atrial arrhythmias or atrial septal aneurysm. The incidence of stroke will be higher as the size of a PFO increases and if it is associated with an atrial septal aneurysm. Treatment options for PFO may range from no therapy, to medical therapy, to surgical closure of PFO. Choosing the type of treatment would depend on the patient's age, surgical and anticoagulation risks. The average annual risk of recurrence of cardiovascular events is 3.8% for medical therapy (aspirin 250–500 mg/d, warfarin with target INR 2–3, or both together) and 2.5 % after percutaneous transcatheter closure of the PFO. Closure done through open heart surgery by direct suture or patch closure is the gold standard of therapy and offers permanent closure of the defect with minimal risk and avoidance of long term anticoagulation. Recurrence of cerebrovascular ischemic events is still possible after successful surgery. Having multiple neurological events before surgery is the only significant risk factor of recurrence after the procedure. One should try to identify causes other than paradoxical embolism in patients who experience recurrent cerebrovascular events after surgical closure of PFO.

WHY DOES MY PACEMAKER NOT WORK? (OR ABOUT BEATING A TIRED HORSE). A. Karcic¹, J. Dell'orlano¹, ¹Departments of Internal Medicine and Cardiology, State University of New York at Stony Brook and Nassau University Medical Center, East Meadow, NY

LEARNING OBJECTIVE: 1. To understand that ischemic myocardium has a decreased capability to conduct pacemaker beats.

CASE INFORMATION: A 73 years-old hypertensive man, had one year ago a pacemaker inserted for symptomatic sinus bradycardia (fainting spells). Coronary angiography performed at that time showed only mild nonobstructive coronary disease. The pacemaker was a dual chamber device, with leads in the right atrium and right ventricle, and it was functioning appropriately until a week ago, when the patient noticed chest pressure while shoveling snow. At that time his pulse rate, on self-exam was 40 bpm, so the patient went to see his internist, knowing that "there is something wrong with the pacemaker, since it was programmed to keep the heart rate above 60 bpm."

During hospitalization the patient had slightly positive cardiac enzymes. The ECG revealed sinus bradycardia and inferior wall ST depressions with regular vertical pacemaker spikes at 60 bpm, but failure of these paced stimuli to "capture" the myocardium and lead to ventricular beats.

Coronary angiography revealed a 100% obstructed right coronary artery (RCA) as the cause of ECG changes and ischemia. Angioplasty was performed on the RCA, and a stent inserted to keep the vessel open. Several hours after the procedure a repeat ECG revealed no ST depressions, with all the pacemaker spikes 'capturing' the myocardium, and pacing the heart at the set rate of 60 bpm.

DISCUSSION: Our patient had a complete RCA obstruction, leading to ischemia of the inferior areas of the heart, as seen on the ECG. It also caused ischemia of the myocardium where the right ventricular pacemaker lead was inserted. Ischemic myocardium (or dead myocardium — as a scar from an old myocardial infarction) has a decreased ability to conduct electrical stimuli, as those received from a pacemaker. It effectively acts as an obstacle to the conduction of these pacemaker beats to the surrounding healthy nonischemic myocardium. For at least some paced beats to be conducted through the ischemic myocardium, impulses at much higher energy levels need to be delivered by the pacemaker. This not only leads to faster pacemaker battery depletion, but also usually does not guarantee that all beats delivered by the pacemaker will be conducted, which is needed. The solution to this problem is relieving the ischemia, if possible (like in our patient), or by changing the position of the right ventricular lead, if it was located in myocardial scar tissue.

PRIMARY PULMONARY HYPERTENSION IN HIV PRESENTING AS ISOLATED RIGHT SIDED HEART FAILURE. A. Karcic¹, M. Zihlif¹, ¹Departments of Internal Medicine and Cardiology, Naussau Medical Center, East Meadow, NY

LEARNING OBJECTIVE: Diagnose primary pulmonary hypertension (PPH) in HIV positive patients.

CASE INFORMATION: A 30 year old woman, PPD positive (10 years), HIV positive (8 years), risk factor HIV positive hemophilic husband, with history of recurrent respiratory infections, was admitted for worsening dyspnea of several months duration, and two weeks of abdominal and leg swelling and oliguria. On exam she was tachypneic with a blood pressure 100/60 mm Hg and pulse rate of 100/ min. Lungs were clear to auscultation, while a right-sided gallop was audible. The jugular veins were distended, the liver painfully enlarged, and there was bipedal pitting edema. EKG revealed an incomplete RBBB. Chest radiograph showed mild cardiomegaly, and a prominent pulmonary artery, with clear lungs. Sonogram of the heart revealed large right-sided chambers, paradoxical septal motion and pulmonary hypertension. VQ scan was low probability for pulmonary embolism. Finally a right-sided cardiac catheterization excluded a left to right shunt and revealed severe pulmonary hypertension, with preserved left ventricular function.

DISCUSSION: PPH is a diagnosis of exclusion. In order for it to be diagnosed, no cause for the pulmonary hypertension should be found. Frequently, in HIV positive patients, pulmonary hypertension develops secondary to HIV related cardiomyopathy and left ventricular failure. Early reports of PPH in HIV involved hemophiliacs receiving large amounts of factor VIII concentrates, which over a period of time can cause pulmonary vascular insults leading to PPH. Only later came reports of non-hemophilic HIV positive patients with PPH. Intravenous drug users can develop pulmonary hypertension as tale emboli cause pulmonary fibrosis and vascular disease. Progressive pulmonary thromboembolism has also been reported to cause pulmonary hypertension in HIV positive patients. While mild and moderate pulmonary hypertension can develop in many cardiac and pulmonary diseases, it is proposed that a genetic predisposition (rare familial PPH), in concord with pulmonary vascular insults (shear stress, chronic hypoxia and chronic inflammation), leads to severe and fatal PPH. Patients with AIDS have a multitude of pulmonary vascular insults (infections, emboli, hypoxia etc.), but only 0.5% of them develop severe PPH. Our patient for example had frequent bronchitis. It appears that PPH in patients with HIV is multifactorial in origin, rather than being merely a coincidental finding. As a conclusion, one could mention that pulmonary hypertension in HIV which is labeled as primary may in fact be secondary to a multitude of factors that may remain unrecognized,

PACEMAKER INDUCED DIAPHRAGMATIC PACING. A. Karcic¹, J. Dell'orlano¹, S. Vlay¹, ¹Departments of Internal Medicine and Cardiology, SUNY at Stony Brook and Naussau Medical Center, East Meadow, NY

LEARNING OBJECTIVES: 1. To recognize diaphragmatic pacing. 2. To understand how a pacemaker causes diaphragmatic stimulation.

CASE INFORMATION: A 70 year old woman with a history of smoking and hypertension was admitted to the hospital with syncope. She was found to have episodes of significant sinus bradycardia for which she needed a pacemaker. A dual chamber pacemaker was inserted, with pacemaker leads placed in the right atrium and right ventricular apex. ECG, fluoroscopy and a chest radiograph confirmed the correct position of the leads, while postoperative pacemaker interrogation revealed that the right atrial and right ventricular pacing and sensing thresholds were within normal limits. Initially after the procedure the patient was feeling fine, but the same night she started complaining of persistent, rhythmic, frequent and annoying contractions on the right side of her upper abdomen, feeling somewhat like hiccups. There was nothing the patient could do in order to bring upon or abort these symptoms. After having interrogated the pacemaker, the cardiologist abandoned right atrial pacing and the symptoms resolved completely.

DISCUSSION: This patient had right hemidiaphragmatic contractions that were induced by pacemaker beats. The diaphragm is a muscle and hence can be stimulated by electrical stimuli from the pacemaker. In order to understand how a pacemaker induces diaphragmatic contractions, one should recall that the right phrenic nerve traverses the right atrium (atrial electrode location), and the left phrenic nerve the right ventricular apex (ventricular electrode location). If the pacemaker stimuli are of adequately high energy, not only the surrounding myocardium will be paced, but the paced beats will also be conducted through the overlying right or left phrenic nerve to cause contractions of the right or left hemidiaphragm respectively. In our patient the right hemidiaphragm was paced, telling us that the right phrenic nerve was paced via the right atrial electrode. Frequently there is a therapeutic window that enables us to pace the myocardium without pacing the overlying phrenic nerve. This is due to the fact that usually lower energy levels are needed to pace the myocardium. Gradual decrease of pacemaker energy output below that needed to pace the phrenic nerve, but still high enough to pace the myocardium is the perfect solution for this problem. If such a therapeutic window does not exist, the culprit electrode must be abandoned (atrial), or a new one can be placed in a different location (ventricular).

DELAYED DIAGNOSIS OF PERFORATED SIGMOID STERCORAL ULCER. M. Kasubhai¹, M. Singh¹, T. Saleem¹, A. Singh², V. Dimitrov¹, ¹Lincoln Medical and Mental Health Center., Bronx, NY; ²Cornell University

LEARNING OBJECTIVES: 1) Recognize that stercoral ulcers result from pressure necrosis of the mucosa by the direct effect of a mass of retained feces. 2) Fecal Impaction is commonly diagnosed and rapid and gentle care can prevent this morbid complication

CASE INFORMATION: 67 year old male presented with history of seizures, lower abdominal pain and constipation. Past history was relevant for seizure disorder, diabetes, stroke and bilateral above knee amputation. He was afebrile and hemodynamically stable. Left lower quadrant tenderness was noted. Rectal examination revealed fecal impaction and was manually disimpacted. The hemotocrit was 40 with leucocytosis. X-rays of chest and abdomen were normal. The patient subsequently developed fever and was treated for diverticulitis. Fever resolved with antibiotics. CT scan revealed thickening of the wall of the rectum and sigmoid, colonic dilatation and small bowel ileus. Colonoscopy performed two weeks later to evaluate the cause of constipation revealed an ulcer at the recto-sigmoid junction with biopsy revealing dysplastic changes. The other considerations of the ulcer being viral, bacterial, fungal or tuberculous were excluded by biopsy and culture. A stercoral ulcer was also considered and treated conservatively. After two weeks a repeat colonoscopy showed a persistent ulcer with acute and chronic inflammatory exudate. An exploratory laparotomy was performed because of persistent colonic dilatation and ileus. A sigmoid tear was noted over the ulcer, which was plugged by matted loops of small bowel. Sigmoid resection and colostomy was done. The patient recovered completely and was discharged home.

DISCUSSION: Stercoral ulcers are usually found in the rectum and sigmoid. Their size is variable and margin usually irregular. When symptomatic, they usually reveal themselves by bleeding or perforation. Relieving the fecal impaction manually can result in perforation of the ulcer. Fever and leucocytosis, as in our patient, should suggest perforation. Our patient also had unexplained localised ileus and colonic dilatation, which should have led to an early exploratory laparotomy. Fecal impaction is commonly diagnosed and the rapid and gentle care can prevent its morbid complications

HYPOKALEMIC PARALYSIS: PURSUING A DIAGNOSIS. J. Kent¹, W.P. Moran¹, L.B. Jones¹, J. Stopyra¹; ¹Wake Forest University School of Medicine, Winston-Salem, NC

LEARNING OBJECTIVES: 1) Recognize an uncommon presentation of Sjögren's Syndrome. 2) Use a literature search as a diagnostic aid. 3) Demonstrate the importance of pursuing a diagnosis to the end.

CASE INFORMATION: A 69 yo w female with past medical history of gastroesophageal reflux disease, hypertension, anxiety, arthritis and hypokalemic paralysis presented to the emergency department with a one-week history of progressive right-sided weakness. Her symptoms included difficulty using her right hand, impairment of gait, and a decrease in appetite. She had a history of right-sided paralysis due to hypokalemia nine months earlier requiring hospitalization, which had resolved with aggressive potassium replacement. She received neurologic and endocrine workups at that time without any etiology found, and she was discharged on oral potassium and spironolactone. With this second episode, her CT scan was negative for an acute CVA, and her potassium level was found to be 2.1. On exam, her oral mucous membranes were dry. Her lab values were suggestive of renal tubular acidosis. After doing a MEDLINE search with keywords including hypokalemia, paralysis, and acidosis, it was found that there have been over 20 case reports of patients with Sjögren's syndrome presenting with a hypokalemic paresis or paralysis. Upon further questioning, the patient did note that her eyes and mouth had felt unusually dry for the past year. Diagnostic testing included a positive Schirmer's test and a minor salivary gland biopsy that was strongly suggestive of Sjögren's syndrome. The hemiparesis completely resolved with replacement of her potassium, and she was discharged on potassium supplementation with spironolactone, as well as artificial tears and lozenges for treatment of her Sjögren's symptoms.

DISCUSSION: This patient's episode of hemiparesis/ paralysis due to hypokalemia was the second such episode within 9 months. During her first episode, it was also noted that she had symptomatic dryness of mouth and eyes and that her mucous membranes were dry. Sjögren's syndrome itself was not mentioned in a differential diagnosis. The main clinical problem appeared to be the hypokalemia and hemiparesis, which are not traditionally associated with Sjögren's. When the patient's hypokalemia and paralysis resolved, she was discharged with outpatient follow up. During this second episode, the patient's main symptoms were again relieved before a diagnostic etiology was found. However, the diligent search for a reason for her hypokalemia by the team resulted in the discovery of her underlying problem, Sjögren's Syndrome, which then allowed for more complete treatment. The use of a literature search and its finding of a series of case reports was a significant aid to the team in reaching the diagnosis.

THE IMPORTANCE OF A COMPLETE WORKUP FOR DELIRIUM IN. M. Khali¹, G.H. Tabas²; ¹UPMC Shadyside, Pittsburgh, PA; ²University of Pittsburgh, Pittsburgh, PA

LEARNING OBJECTIVES: 1. Recognize the importance of a complete workup for delirium. 2. Optimize treatment of delirium by recognizing reversible causes.

CASE INFORMATION: An 86 year old woman with no significant past medical history and taking no medications was admitted to the intensive care unit after a one day history of weakness, dizziness, nausea and vomiting. The patient was hypotensive with a blood pressure of 90/50 mmHg, without orthostatic changes. The pulse was 64; she was afebrile, and had dry oral mucosa. The remainder of the physical examination was normal. Laboratory workup was significant for pyuria and mild anemia (Hgb = 11.4 g/dL). Intravenous fluids were started and antibiotics were given for her urinary tract infection. During the hospital stay, the patient developed disorientation, hallucinations, and incoherent speech interpreted as delirium secondary to her infection and ICU psychosis. When her delirium did not resolve with hydration, antibiotics and transfer to a regular floor, a hepatic profile was ordered as part of the workup for metabolic causes for her delirium. This testing revealed an isolated elevated alkaline phosphatase (650 U/L). Abdominal ultrasound showed marked dilation of the common bile duct and subsequent abdominal CT scan showed no masses in the area of the gall bladder and liver. A 1.5 cm common bile duct stone was successfully extracted by endoscopic sphincterotomy and her mental status returned to baseline.

DISCUSSION: The causes of delirium include medication side effects or withdrawal, dehydration, infections, hypoxia, hypoglycemia, and other metabolic abnormalities. Many systemic illnesses may produce delirium in elderly patients, especially in combination with new medications, fever, or sleep deprivation. In this patient, a comprehensive evaluation for metabolic causes of delirium (despite lack of localizing symptoms or signs on physical examination led to a diagnosis of unsuspected common bile duct stone and institution of appropriate treatment. Since delirium is associated with adverse outcomes, making the correct diagnosis is important. It is essential to do a complete workup for delirium in order to determine all treatable causes, particularly in hospitalized elderly patients.

ANTICONSULSANT HYPERSENSITIVITY SYNDROME: A CALL FOR INCREASING AWARENESS. N. Khan¹, B. Cheong¹, A. Jaffer¹; ¹Cleveland Clinic Foundation, Cleveland, OH

LEARNING OBJECTIVES: 1. Diagnose the distinct entity of hypersensitivity reaction to anticonvulsants. 2. Recognize the prevalence of cross-reactivity between different anticonvulsant medications.

CASE INFORMATION: A 27 year-old lady with a history of multiple sclerosis was initiated on oral phenytoin for tonic spasms of extremities. Two weeks later she noted a mild maculopapular erythematous rash on her extremities. Concerns for drug induced rash and minimal response to treatment, led to a change in regimen from phenytoin to carbamazepine. One week later the patient was hospitalized with worsening pruritic maculopapular rash, more prominent on the trunk and extremities, and recurrent fevers. Except for a temperature of 101.5°F, her vital signs were within normal limits. Notable changes on physical exam included mild facial edema, with 1.5 to 2cm bilateral, tender, mobile jugulodigastric and submandibular lymphadenopathy (LAD). Laboratory abnormalities on admission included peripheral eosinophilia and elevated hepatic enzymes: AST=271 (7-40 U/L), ALT=324 (5-45 U/L). Other hepatic enzymes were normal. A CBC with differential revealed WBC=15.63 (4-11.0 K/uL) with neutrophils 59% (40-70%),

lymphocytes 27% (22-44%), eosinophils 11% (0-4%), monocytes 2% (0-7%), and atypical lymphocytes 1%. Carbamazepine level= 9.0 (8-12 ug/mL) Blood and urine cultures were normal. Titers for Epstein-Barr virus and cytomegalovirus were normal. Given the rash, pyrexia, eosinophilia, lymphadenopathy and recent treatment with anticonvulsants the patient was diagnosed with anticonvulsant hypersensitivity syndrome (AHS). Progressive resolution of the dermatological changes and hepatitis was noted with discontinuation of anticonvulsants and treatment with corticosteroids for 1week.

DISCUSSION: AHS is an adverse drug reaction associated with medications such as phenytoin, carbamazepine and phenobarbital. Originally cited prevalence of 15% is increasing with broader spectrum of use of anticonvulsants, such as for treatment of neuropathic pain. This patient exhibited a typical presentation including hepatitis and eosinophilia. The pathogenesis of AHS involves inadequate function of epoxide hydroxylase enzyme, leading to decreased inactivation of toxic metabolites of anticonvulsant drugs. Current data indicates allergic cross-reactivity between these medications to be as high as 80%. Treatment involves discontinuing the medication and initiating steroid therapy. With a mortality rate suggested at 40% (mostly due to hepatic necrosis), it is important for general practitioners to recognize AHS; its presentation, cross-reactivity of medications, and treatment options. A high index of suspicion based on the patient's history and exam will enable an astute primary care provider to make the diagnosis early.

HYPERPARATHYROIDISM IN PREGNANCY. R. Khurana¹; ¹University of Alberta, Edmonton, AB

LEARNING OBJECTIVES: 1) Recognize the changes in calcium metabolism that occur with pregnancy. 2) Manage hyperparathyroidism in pregnancy.

CASE INFORMATION: A 38-year old pregnant woman, G4P1 at 21 weeks gestation was seen for assessment of hypertension. Her blood pressure had been normal at the first prenatal visit, but had started to rise and had been high for 4 weeks. She was asymptomatic. Past medical history was significant for two miscarriages and one full-term pregnancy, all in Nigeria. There was no prior history of hypertension. Her only medications were prenatal vitamins. Physical exam was normal other than a blood pressure of 168/94 and a grade II/VI ejection systolic flow murmur. Laboratory screening for preeclampsia was normal. Laboratory screening for secondary causes of hypertension revealed hypercalcemia at 2.79 mmol/L (11.16 mg/dl) with an albumin of 36 g/L (3.6 mg/dl). An elevated ionized calcium confirmed hypercalcemia. A 24 hour urine calcium was elevated at 9.0 mmol/24h (360 mg/24h). 25-OH Vitamin D levels were normal, but her Parathyroid hormone (PTH) level was elevated at 15.0 pmol/L (normal 1.1-6.8 pmol/L). She was told to keep herself well hydrated with oral fluids and started on oral phosphate 500 mg bid. A 24 hour urine for metanephrines was ordered to rule out pheochromocytoma as part of a multiple endocrine neoplasia (MEN) syndrome in combination with the hyperparathyroidism. It was normal and she proceeded to exploratory surgery with removal of a right superior parathyroid adenoma. Post-operatively, her calcium levels normalized.

DISCUSSION: Primary hyperparathyroidism is rare during pregnancy. It may be masked because total calcium levels decrease secondary to a decrease in albumin from the volume expansion that occurs with pregnancy. However, ionized calcium levels remain unchanged despite decreases in PTH levels. This is likely due to the effects of PTH related peptide (PTH-rp) which is found to be increased during pregnancy. PTH-rp may also help regulate fetal calcium levels which are higher than maternal levels. Maternal hyperparathyroidism can cause significant morbidity for the fetus and neonate. There is an increased risk for stillbirth, intrauterine growth restriction, premature labor and neonatal hypocalcemic tetany. The diagnosis is often made on a retrospective basis postpartum after the development of neonatal tetany. Surgical management is usually preferred as it causes less morbidity than medical management. However, hyperparathyroidism can be associated with pheochromocytoma in Type 2 MEN and this should be ruled out prior to surgery.

DIABETIC NEUROPATHIC (CHARCOT) ARTHROPATHY: A FORGOTTEN DIAGNOSIS? A. Kim¹, Y. Braver¹, A. Jaffer¹; ¹The Cleveland Clinic Foundation, Cleveland, OH

LEARNING OBJECTIVES: 1) Recognize the clinical presentation of diabetic neuropathic arthropathy. 2) Review the management of acute-onset Charcot arthropathy.

CASE INFORMATION: A 51-year old African American female insulin-dependent diabetic of 12 years presented with a 2 month history of right lower extremity (RLE) swelling. This was associated with erythema, warmth, and minimal pain. She also reported a 3-day history of low-grade fever and generalized weakness. She had previously been treated with oral antibiotics for presumed cellulitis with no improvement. Diagnostic studies at that time revealed a normal WBC of 10 K/cmm with an elevated sedimentation rate (WSR) of 31 mm/hr. X-ray of the right foot/ankle showed soft tissue swelling without evidence of bony destruction. Lower extremity duplex revealed no deep vein thrombosis. CT scan of the extremity revealed only subcutaneous edema. A subsequent right ankle joint aspiration was performed which was negative for crystals and infection. The physical exam on admission was notable for an afebrile, obese female with 2+ pitting edema extending from the dorsal aspect of the right foot to the knee associated with warmth, erythema, and tenderness. There were no areas of skin breakdown or ulcers. Decreased sensation to pain, temperature, and proprioception was noted. The patient was admitted for cellulitis and started on IV antibiotics. Laboratory studies showed WBC of 8 K/cmm with an elevated WSR of 64 mm/hr and a CRP of 3.8 mg/dl. Blood cultures were negative. The patient remained afebrile with mild improvement of her symptoms. A MRI of the extremity revealed changes in the midfoot and forefoot consistent with an acute neuropathic arthropathy. Orthopedic consultation fitted the patient with a total-contact cast, instructed her on no weight bearing in the RLE with follow-up in 4 weeks.

DISCUSSION: Diabetes mellitus is now the most common cause of neuropathic (Charcot) arthropathy. The foot and ankle are the most commonly affected joints. Unilateral swelling, warmth, and erythema are typical clinical findings and it is usually painless. The pathogenesis of this disease is unclear but may be related to a combination of peripheral neuropathy and neurovascular insufficiency allowing repetitive microtrauma. This can result in chronic

should include HIV antibody as well as HIV RNA viral load or HIV DNA. Early treatment decreases viral burden as well as improves, and perhaps restores, host immune responses. HIV is highly contagious during the acute infectious stage. Early diagnosis is critical in decreasing HIV transmission and has significant impact on public health, treatment, and possibly prognosis. Because the majority of patients present to primary care physicians, they have an important role in recognition of early infection. Further education regarding symptoms and serologic testing, as well as screening strategies, may be needed to increase awareness of acute infection in primary care settings.

AN UNUSUAL CASE OF HYPERTENSION. S. Kripalani¹; Emory University, Atlanta, GA

LEARNING OBJECTIVES: 1) Explore causes of secondary hypertension in a young man. 2) Recognize the appearance and pathophysiology of Page kidney.

CASE INFORMATION: A 27-year-old narcotics agent presented for evaluation of previously untreated high blood pressure. Manual readings over the past year ranged from 140–151/96–110. He complained of daily bifrontal headaches, as well as occasional palpitations and diaphoresis, but only when working in high-stress situations. Review of systems was otherwise negative, as were past medical history and medication use, including OTC products. He denied smoking and illicit drug use, but he used chewing tobacco and had 2–3 alcoholic drinks per week. He used to play high school football and had been in a car accident years ago, with no apparent injury. His father received a heart transplant for idiopathic cardiomyopathy. Family history was otherwise negative. Blood pressure was 162/108 in the left arm, 152/104 in the right arm, and 154/106 in the right leg. Heart rate was 64, height 71 inches, and weight 196 pounds. He appeared comfortable and well-developed. Physical examination was unremarkable. EKG, chest x-ray, blood count, basic chemistry panel, TSH, urinalysis, and drug screen were normal.

The patient was started on amlodipine, to avoid interfering with anticipated testing, and his blood pressure responded appropriately. Initial 24-hour urine catecholamines were elevated, prompting a CT scan to help localize a potential pheochromocytoma. The CT showed normal adrenals, but unexpectedly, the left kidney had a 5x5x1 cm subcapsular fluid collection, known as Page kidney. An MRI/MRA confirmed the CT findings and demonstrated normal vasculature. Renin levels were slightly elevated — supine 1.7 ng/mL/hr (0.2–1.6) and upright 4.9 ng/mL/hr (0.5–3.3). Cortisol, aldosterone, and upright aldosterone/renin ratio were within normal limits. Head CT and repeat urine studies were normal. Given the imaging results, the initial catecholamine elevation was deemed a false positive.

DISCUSSION: Potential causes of secondary hypertension included renal parenchymal and renovascular diseases, coarctation of the aorta, mineralocorticoid excess (either through hyperaldosteronism or use of chewing tobacco or licorice), renin-producing tumors, pheochromocytoma, Cushing's syndrome, congenital adrenal hyperplasia, substance abuse, exogenous hormones or sympathomimetics, sleep apnea, hypo- and hyperthyroidism, hyperparathyroidism, acromegaly, carcinoid, and increased intracranial pressure. Page kidney is rare cause of hypertension resulting from compression of the renal parenchyma. High blood pressure is mediated by elevated renin levels, which are thought to be secondary to local ischemia or parenchymal scarring. In the model which he described in 1939, Page used cellophane wrapping to constrict the kidneys. However, most cases in clinical practice are due to a subcapsular hematoma resulting from blunt trauma, usually associated with contact sports or an accident. If blood pressure cannot be controlled medically, percutaneous drainage or nephrectomy can be curative.

NOT "JUST ANOTHER GI BLEED"! A. Krishnamurthy¹, R. Ruffner¹, D. Weber¹; ¹University of Pittsburgh Medical Center Shadyside, Pittsburgh, PA

LEARNING OBJECTIVES: 1) Consider small bowel tumors in the evaluation of GI bleeding with the upper endoscopy and colonoscopy are negative. 2) Small bowel series, enteroclysis and enteroscopy have a role in their diagnosis.

CASE INFORMATION: 84-year-old Asian male presented with his second episode of melena. He also complained of "fullness of the abdomen after meals". He had no nausea, vomiting, abdominal pain, constipation or diarrhea. The patient had been admitted two weeks before with melena and anemia requiring blood transfusions. He had both upper GI endoscopy (showed mild esophagitis) and colonoscopy (showed two polyps). PMH of hypertension. His medications included iron, famotidine and Metoprolol. He did not smoke or abuse alcohol. ROS: Negative. **Examination:** His temperature was 36.8, pulse 72 bpm, BP 120/60 and was not orthostatic. His conjunctiva was pale. Abdomen was soft with normal bowel sounds, no tenderness, no masses, liver/spleen not palpable. Rectal exam revealed heme positive stool. Examination of all other systems was normal. Labs: WBC 5.4, Hemoglobin 8.9, BUN 42 & Creatinine 1.2, Iron 28, TIBC 258 Ferritin 25. Small bowel enteroscopy was normal upto the duodenum but the scope could not be advanced beyond the fourth part of duodenum because of narrowing. Small bowel follow through showed an apple-core defect at the duodeno-jejunal junction. Surgical resection of lesion confirmed a 4-cm adenocarcinoma of jejunum without lymphnode involvement.

DISCUSSION: Small bowel neoplasms are relatively uncommon. **Small bowel accounts for less than 3% of GI malignancies despite contributing to more than 90% of the surface area of GI tract.** The possible reasons are:

1. Liquid nature of intestinal content may be less irritating to the mucosa
2. Rapid transit time reduces exposure to carcinogens.
3. Decreased bacterial population to produce carcinogens.
4. Alkaline pH.
5. Increased lymphoid tissue.
6. Enzyme benzyrene hydroxylase-detoxifies carcinogens. Types of Small Bowel Tumors: Adenocarcinoma (Duodenum & proximal jejunum), Lymphoma (jejunum & ileal), Sarcoma (jejunum) and Secondary adenocarcinomas (rare but reported from lung). Symptoms are nonspecific and only 19%–22% of these tumors are diagnosed preoperatively. The role of enteroscopy, enteroclysis and small bowel series will be discussed during the presentation.

SYNCOPE MASQUARADING AS SEIZURES — THE IMPORTANCE OF A GOOD HISTORY. A. Krishnamurthy¹, D. Weber¹, R. Ruffner¹; ¹University of Pittsburgh Medical Center Shadyside, Pittsburgh, PA

LEARNING OBJECTIVES: 1) Post-ictal confusion is the most specific symptom that is useful in differentiating seizures from syncope. 2) Brief myoclonus that might occur during syncope should not be mistaken for seizure activity.

CASE INFORMATION: 70 yoF with PMH hypertension, CAD presented with 6 months of recurrent loss of consciousness. She had 2 prior admissions for "unresponsiveness while sitting. Family observed twitching of her arms and staring into space for 10 minutes. The second episode was similar but with urinary incontinence. She regained consciousness after transfer to bed. She was worked up for syncope vs. seizures, EKG, holter, Echocardiogram, brain imaging & EEG all unrevealing. Presumptive treatment for seizures failed. A witnessed episode in hospital found her sitting, unresponsive, pale, pulse 55 bpm. She regained consciousness promptly. Holter monitor showed mild sinus bradycardia during the episode. Carotid sinus massage (5s) caused a 4.5 second pause without any symptoms. A Tilt table test revealed a drop in BP from 149/97 to 70/38. Heart rate remained 65 bpm. She became unresponsive with mild twitching of hands and awakened immediately on termination.

DISCUSSION: The history plays a vital role in differentiating syncope from seizure as the cause of transient loss of consciousness. A recent videometric analysis of 56 brief syncopes showed myoclonic activity in 90%, together with head turns, upward gaze or oral automatism in 60–80%. The occurrence of twitching should not be assumed as seizure activity. Patients with seizures are five times more likely to have post-episode disorientation than with syncope. Incontinence & trauma are not discriminative findings. The differentiation of syncope from seizures clinically is important as imaging/EEG have low yield in the work up of syncope. The tilt table test confirmed neurocardiogenic syncope in this patient.

BEWARE: VANCOMYCIN RESISTANT STAPHYLOCOCCUS AUREUS: A WORRISOME TREND. J. Kubrin¹, R. Granieri¹; ¹University of Pittsburgh, Pittsburgh, PA

LEARNING OBJECTIVES: 1) To recognize an emerging occurrence of vancomycin-resistant *Staphylococcus aureus* (VISA). 2) To recognize the potential role of quinupristin/dalfopristin in VRSA

CASE INFORMATION: A 53 year-old female with idiopathic thrombocytopenic purpura, atrial fibrillation, diabetes mellitus, and hypertension presented with abdominal pain. An exploratory laparotomy was performed, which revealed little. Her post-operative course was complicated by *Enterococcus faecalis* bacteremia and respiratory distress, leading to intubation. On day 35 of her hospitalization, she began to spike fevers to 40 degrees (C). Her central venous catheter line site was uninfected. Her WBC rose from 26,000 to 63,000 and blood cultures were positive for methicillin-resistant *Staphylococcus aureus* (MRSA). A new systolic murmur was discovered six days after her temperature spike, when a transesophageal echocardiogram showed new mitral regurgitation. Persistent MRSA bacteremia was attributed to endocarditis, based on Duke criteria, and vancomycin and gentamicin were initiated. Despite in vitro susceptibility of the organism to both antibiotics, blood cultures remained positive on day 9. Vancomycin and gentamicin were discontinued and quinupristin/dalfopristin was initiated. Blood cultures remained positive on day 5 of this therapy, but converted to negative on day 8. Rifampin was added, and she completed a 6-week course of antibiotics. Blood cultures remained negative 2 months later.

DISCUSSION: Although vancomycin is the antibiotic of choice for MRSA, numerous anecdotal reports have questioned its efficacy. Prolonged bacteremia, relapse and treatment failure of *S. aureus* bacteremia and endocarditis have occurred. There have been reports documenting the existence of strains of *S. aureus* that are intermediately resistant to vancomycin (VISA). Because of this concerning trend, new treatment options have been explored. Quinupristin/dalfopristin (30:70 ratio) is the first parenteral streptogramin antibacterial agent. It was approved for treatment of vancomycin-resistant enterococcal infections and skin infections due to methicillin-sensitive *Staphylococcus aureus* (MSSA). It has also been found to be effective against MRSA and VISA, achieving success rates up to 76%. However, in most cases of MRSA and VISA, patients were treated with multiple agents, which included quinupristin/dalfopristin. In our patient, blood cultures converted to negativity with treatment by quinupristin/dalfopristin alone. Although this is an anecdotal report, the experience implies that quinupristin/dalfopristin may be efficacious in patients with MRSA endocarditis who have failed conventional therapy.

SATURDAY NIGHT PARALYSIS. P.L. Kurtz¹; ¹University of Chicago, Oak Park, IL

LEARNING OBJECTIVES: 1) Diagnose thyrotoxic periodic paralysis from its clinical manifestations. 2) Recognize that 50% of US patients with thyrotoxic periodic paralysis show no signs or symptoms of hyperthyroidism. 3) Treat the syndrome of thyrotoxic periodic paralysis.

CASE INFORMATION: On a Saturday night in June a previously healthy 43 year old man experienced the sudden onset of sharp, cramping pain in his thighs followed by weakness and difficulty walking. He also developed cramping and weakness in his arms, but to a lesser degree. He went to bed, and had to crawl to the bathroom on his hands and knees early Sunday morning. His pain and weakness gradually subsided, and he was able to walk normally by Sunday afternoon. He presented for evaluation for Monday afternoon, 24 hours after the resolution of all symptoms. This was the third consecutive weekend that he had suffered with these identical symptoms. Physical exam was entirely unremarkable. Laboratory data revealed a markedly elevated total T4 of 19.3 (nl < 11.6), free T4 of 28.8 (nl < 10.5), and an undetectable TSH. The patient was started on propranolol and advised to avoid heavy carbohydrate meals. Graves Disease was diagnosed and treated with radioactive iodine. The episodes of paralysis resolved.

DISCUSSION: Thyrotoxic periodic paralysis occurs in 0.1% of U.S. patients with hyperthyroidism, 95% of whom are males. Only 50% of cases manifest overt signs or symptoms of hyperthyroidism. Attacks are seen nearly exclusively in spring and summer. Recurrent attacks on weekends are attributed to excesses in physical activity, diet, and alcohol ingestion. The precise pathophysiology of the paralysis, which results from rapid and reversible

shifts of potassium from the extracellular to the intracellular compartment, remains unexplained. Resolution of paralytic attacks is facilitated by oral potassium replacement. Treatment with propranolol, and avoidance of high carbohydrate meals prevent further paralytic episodes until the cause of the hyperthyroidism is diagnosed and treated.

IS COLLAGENOUS SPRUE A SEVERE VARIANT OF CELIAC SPRUE AFTER ALL? M. Lagowski¹, M. Taylor¹, G.C. Wickstrom¹; ¹Summa Health System, Akron, OH

LEARNING OBJECTIVES: 1. Identify collagenous sprue as a rare cause of malabsorption of the small bowel. 2. Present pathophysiology, clinical presentation, diagnosis and treatment of adult celiac sprue as compared with collagenous sprue. 3. Examine a possible relationship between collagenous sprue and adult celiac sprue based on serology and available literature.

CASE INFORMATION: A 71-year old woman presented to her primary care physician (in West Virginia) in November 1999 complaining of weight loss, bulky stools, fatigue, abdominal distention, and anorexia. Review of those hospital records revealed an extensive evaluation that yielded a diagnosis of "protein losing enteropathy". She improved with total parenteral nutrition (TPN). Small bowel biopsy and serologies were not reported, nor was there mention of treatment with a gluten-free diet. In May 2000, the patient presented to our institution with similar complaints and worsening lower extremity edema and weight loss. Physical examination revealed an extremely cachectic woman with flat affect, hypotension, tachycardia, inspiratory crackles, hepatomegaly, and 2+ pitting lower extremity edema. Laboratory exam showed severe hypoalbuminemia, hyponatremia, and hypocalcemia, mildly elevated transaminases and alkaline phosphatase, and iron deficiency anemia. Antigliadin and antiendomysial antibodies were elevated. Histologic examination of biopsies obtained from the proximal small bowel was consistent with collagenous sprue. Due to her severe presentation, short-term corticosteroids and TPN were initiated. She improved and was discharged with instructions to continue a gluten-free diet.

DISCUSSION: Adult patients with celiac sprue may present with a variety of gastrointestinal as well as extraintestinal symptoms, many of which were present in our patient. The diagnosis is usually made by histological evidence of villous atrophy in the small intestine along with symptomatic improvement on a gluten free diet. Serological tests (antiendomysial and antigliadin antibodies) further aid in diagnosis and screening of celiac sprue. Collagenous sprue was first described (in 1970) as continued malabsorption, failure of a gluten-free diet, and a subepithelial collagen layer on small bowel biopsies in a patient previously diagnosed with celiac sprue. Most published case reports of collagenous sprue have not detected these serological markers typical in celiac sprue. Recently, a relationship was postulated when serological markers were present in a patient with collagenous sprue. Our patient exhibited biopsy proven collagenous sprue, but also demonstrated positive serology for adult celiac sprue, supporting the notion that collagenous and adult sprue may be related. Whether this simply reflects an improvement of antibody assays remains to be established as more cases are evaluated.

ACUTE HEPATITIS B VIRUS INFECTION-RELATED POLYARTERITIS NODOSA PRESENTING WITH CARDIOGENIC SHOCK. M. Lagowski¹, J. Iturregui¹, G.C. Wickstrom¹; ¹Summa Health System, Akron, OH

LEARNING OBJECTIVES: Polyarteritis Nodosa (PAN) is a rare vasculitis that may be triggered by Hepatitis B Virus (HBV) infection. It affects small and medium vessels in many organ systems including the heart. It is important to consider secondary causes of PAN, such as HBV infection, as prognosis and treatment protocols differ from those of primary vasculitis.

Learning Objectives: 1. Identify PAN as a rare cause of cardiogenic shock, 2. Present the relationship between HBV and PAN, and 3. Review the current therapeutic strategies of PAN with acute HBV infection.

CASE INFORMATION: A 44-year-old African-American female presented with abdominal pain, nausea and emesis for two days, and 10 lb. weight loss over a month. Physical examination revealed a cachectic female in moderate distress, hypotension, tachycardia, and diffuse abdominal pain with rebound tenderness. Laboratory exam showed an anion gap metabolic acidosis with a mild lactic acidosis, acute renal failure due to acute tubular necrosis, hyponatremia, leukocytosis with bandemia, and elevated Troponin associated with anterior-lateral ischemic EKG changes. Swan-Ganz readings indicated cardiogenic shock. Emergent cardiac catheterization revealed diffuse "beading" of small and medium coronary vessels consistent with vasculitis. Renal angiography revealed a similar picture. A primary vasculitis workup was negative. Hepatitis studies were consistent with acute HBV: + Hb SAg, - Hb SAb, + Hb Core IgM. Quantitative HBV-DNA was 6290 pg/ml. Hepatitis A, C, and D studies were negative. Initially, high-dose parenteral steroids were started for suspected PAN, but when the HBV serologies became known steroids were discontinued and Lamivudine initiated. Hemodialysis was required. Clinically, the patient improved and was discharged home. Three months later the patient's condition deteriorated acutely, and an acute cardiorespiratory arrest lead to her death. Histological examination revealed a multisystem, segmental, necrotizing vasculitis of medium and small-size arteries (most notably the coronary and renal vasculature) in various phases of progression consistent with classic PAN.

DISCUSSION: HBV-related PAN is most commonly described with chronic HBV infections. In our case, laboratory evidence indicated acute HBV infection. Cardiopulmonary involvement of PAN is fairly common, but rarely associated with cardiogenic shock. Traditional treatment of PAN cannot be utilized when PAN is associated with an acute HBV infection. Good therapeutic outcomes have been reported with immunosuppressive agents along with plasma exchanges, or treatment with two weeks of high-dose steroids followed by immunosuppressives and plasma exchanges.

POST-CATHETERIZATION COMPLICATIONS MASQUERADING AS A URINARY TRACT INFECTION. M. Landry¹, K.B. Desalvo¹; ¹Tulane University, New Orleans, LA

LEARNING OBJECTIVES: Recognize possible complications of cardiac catheterization. Understand the risks and benefits of vascular sealing devices. Learn the management of pelvic hematoma.

CASE INFORMATION: A 45 year old woman with a history of hypertension, s/p total hysterectomy was admitted for unstable angina. She had a positive stress test and underwent a left heart catheterization (LHC) during her hospitalization that was normal. Her catheterization site was sealed with a vascular sealing device and was without external hematoma. Her pre-catheterization Hgb was 13.3 mg/dL. One day post-LHC, she complained of dysuria. A urinalysis was normal and she was discharged for outpatient evaluation of her chest pain. The patient developed worsening dysuria, frequency, and urgency and was prescribed antibiotics over the telephone for the treatment of a presumed UTI. Her symptoms improved but did not resolve and she was seen in the office 2 weeks later now also complaining of new constipation. Examination revealed normal vitals, a shuffling gait and rebound tenderness on abdominal exam. Pelvic exam revealed a tender, non-pulsating mass in the right lower quadrant. The catheterization site was identified as above the inguinal ligament and did not show any external signs of hematoma. An ultrasound and CT of the pelvis and abdomen revealed a 12x7x8cm mass overlying the iliac artery consistent with a hematoma and blood tracking up the right paracolic gutter. The bladder and colon were compressed by the mass. Doppler studies did not reveal active arterial flow from the iliac arteries. Her hemoglobin was 9.6 mg/dL. The patient was admitted for supportive care and her symptoms improved.

DISCUSSION: Cardiac catheterization is a common procedure with serious potential complications including a 1% risk of vascular injury including intraperitoneal bleeding. Risk factors for complications include female gender, congestive heart failure, percutaneous transluminal coronary angioplasty, valvuloplasty and peripheral vascular disease. The recent increased use of vascular sealing devices may decrease the risk of hemorrhage from anterior arterial perforation, but will not provide hemostasis from a posterior wall puncture. Thoughtful surveillance for post-LHC complications is essential. Management of peritoneal hematomas should include serial hemoglobins, diagnostic imaging and surgical consultation when necessary. In this patient, her post-LHC bleeding was misdiagnosed as a urinary tract infection. The Internist must recognize that post-LHC complications may masquerade as more common pathology leading to delayed diagnosis and potentially poor outcomes.

UNDIAGNOSED DEVELOPMENTAL DISABILITY AS A BARRIER TO CARE. K. Larsen¹; ¹Hennepin County Medical Center, Minneapolis, MN

LEARNING OBJECTIVES: 1. Recognize undiagnosed developmental disability in adult immigrants who were likely not screened for cognitive deficits as children. 2. Recognize the impact developmental disability may have as a barrier to healthcare delivery and compliance 3. Distinguish signs of developmental disability as distinct from mental illness

CASE INFORMATION: A 45 y.o. recently divorced Korean immigrant in the US for 20 years, presented to an outpatient clinic seeking primary care. Her complaints were depression and high blood pressure. Her primary language was Korean, but she spoke moderately fluent English. On exam she was oriented to person and place, but was hard to interview as she shifted from topic to topic. Depressive symptoms were not present, nor were symptoms of psychosis. Over the course of a few months she came often but erratically to clinic with vague complaints. She often discontinued her medications without any clear reasons. The nurses noted that although they understood her English, they could never tell what she wanted. They wondered if she was schizophrenic. After struggling to understand her clearly a Korean interpreter was obtained. This interpreter said that in Korean she was difficult to understand as well. After psychiatric evaluation which confirmed lack of mental illness, neuropsychometric testing was done with an interpreter. This revealed mental retardation with developmental disability especially in areas of language.

DISCUSSION: Mental retardation is often defined by a composite IQ of less than 70. Developmental disabilities are very prevalent and encompass overt mental retardation as well as less pervasive cognitive deficits. As in this case, these patients can have difficulty with reading and communicating clearly, which can be a barrier to effective healthcare. Most people in the US are screened for this in school or childhood clinic visits. Many adult immigrants likely were not screened. When this diagnosis is known accommodation can be made to improve compliance and effective communication. Also, a diagnosis of mental retardation may qualify a patient for government benefits and outpatient social services. This case highlights the need to diagnose developmental disability and differentiate it from mental illness especially in adult immigrants.

ELDER ABUSE: A GUIDE TO RECOGNITION AND MANAGEMENT IN THE OFFICE SETTING. A. Lent¹; ¹Hennepin County Medical Center, Minneapolis, MN

LEARNING OBJECTIVES: 1) Recognize the prevalence of elder abuse and the need for early intervention. 2) Recognize the warning signs of elder abuse in the office setting. 3) Provide resources and skills that enable a clinician to intervene in the office setting.

CASE INFORMATION: A 74 year-old African-American female with a history of mild dementia presents to a senior care clinic for routine health maintenance. She is accompanied by her adult son who provides the majority of the history. On exam, the patient is well-groomed yet cachectic and frail. Several large hematomas are noted on the left lower extremity. A small hematoma is noted on the right temple. Upon questioning, the son interjects stating the bruises are the result of a recent fall on the ice. Today her blood glucose is 390 which the son attributes to an unfilled prescription of insulin. The patient appears anxious, frequently requesting that the clinician expedite the interview. She asks few questions yet does inquire about the availability of a local nursing home. **DISCUSSION:** Elder abuse is a common problem that often goes unrecognized in a clinical setting. The warning signs may be subtle yet are important clues that enable a clinician to intervene and to prevent physical and emotional anguish in the elderly population.

"I KNOW HE LOVES ME AND HE DOESNT MEAN TO HURT ME": THE IMPORTANCE OF SCREENING FOR INTIMATE PARTNER ABUSE. Y. Levy¹, R. Granieri¹; ¹University of Pittsburgh, Pittsburgh, PA

LEARNING OBJECTIVES: 1. To recognize the prevalence of intimate partner abuse (IPA) in the community.

2. To assess patients for IPA
3. To recognize the preliminary management for IPA.

CASE INFORMATION: A 28 year old female presented requesting prenatal vitamins and offered no complaints. PMH was significant for exploratory laparoscopies for abdominal pain and endometriosis. The patient was in graduate school and had been married for 1.5 years. When asked "do you always feel safe at home?", she was noted to hesitate, whereas on previous questions, she answered rapidly. This question was immediately followed by "does your husband ever hit you, yell at you, or make you feel afraid?" She replied that while she knows her husband loves her, on occasions he does get "upset". They have frequent arguments, during which he breaks furniture and hits her. She stated "he never hits me hard", he "always apologizes immediately", and his hitting has "never gotten out of hand". She denied having a plan of action should that happen. There were no weapons in the house. Patient was immediately referred to a psychologist for counseling and who helped her to formulate a safety plan.

DISCUSSION: Intimate partner abuse (IPA) is common, with an estimated prevalence of 5–20% women per year. It is defined as a pattern of intentionally coercive and violent behavior toward an individual with whom there is or has been an intimate relationship. These behaviors can be used to establish control over an individual and include physical and sexual abuse, psychological abuse with verbal intimidation, progressive social isolation or deprivation, and economic control. IPV transcends social classes and so all women should be asked about safety in their current setting as part of routine primary care and when presenting for emergency care with traumatic injuries. A screening question can be "have you been hit, kicked, punched, or otherwise hurt by someone within the past year? If so by whom?" The physician must then make an assessment of the patient's safety, ascertaining the presence of weapons in the house, violence that is increasing in severity or intensity, threats to kill the woman or her children, and the woman having told her partner she is "planning to leave." It is important to give the patient validation and support and help her to develop a safety plan. Physicians should be aware of resources within their institution, which include social workers or domestic violence advocates. If these are unavailable the physician can provide counseling and give referral to a local domestic violence hotline.

BLOODY DIARRHEA IN A YOUNG WOMAN: A DIAGNOSIS BEYOND HER YEARS. Y. Lin¹, S. Shadowitz²; ¹University of Toronto, Toronto, Ontario

LEARNING OBJECTIVES: 1. Recognize ischemic colitis as a cause of lower gastrointestinal bleeding in younger patients. 2. Distinguish causes of ischemic colitis in a young person.

CASE INFORMATION: A 32 year old woman presented with bright red blood per rectum associated with tenesmus and urgency. She then developed left lower quadrant pain. There was no fever, rash, eye or joint pain. There was no foreign travel, unusual food intake, or history of antibiotic, NSAID or oral contraceptive use. Her past medical history was unremarkable. She had no known drug allergies, used alcohol on occasion, and did not use illicit drugs. Examination revealed postural hypotension, scant bowel sounds, and tenderness in the left lower quadrant. Rectal exam showed gross blood. Investigations revealed hemoglobin 141, platelet count 270, and wbc of 14.7. Electrolytes, INR, PTT, liver profile, amylase, and beta-HCG was negative. Abdominal films were normal. Micrologic examination of the stool was negative. Colonoscopy showed erythematous, friable mucosa from 75 cm (mid descending colon) to 35 cm distal to the anus. Pathology revealed patchy disease with inflammatory exudates on the surface, consistent with ischemic colitis. No crypt abscesses or granulomas were identified. A vasculitic workup was performed, with negative ANA, pANCA, cANCA, and normal complement levels. Hepatitis B and C serology were negative. Antithrombin III, Protein C, Protein S, homocysteine, Factor V Leiden genotype, and anticardiolipin antibody were all normal. Her diarrhea improved within 3 days of admission with conservative treatment, and she was discharged on ASA. She has had no recurrence.

DISCUSSION: Internists frequently consider ischemic colitis in the differential diagnosis of the elderly patient who presents with bloody diarrhea. However, ischemic colitis in young patients is more common than realized. Case series in younger patients have described an increased risk in women, and a lack of vascular risk factors, when compared to the typical older patient with ischemic colitis, who has multiple vascular risk factors. Etiologic agents that have been implicated include estrogen, collagen vascular disease, diabetes, hypercoagulable states, cocaine use and even long distance running. It is also common to find no identifiable etiologic cause. In an analysis of 68 consecutive cases diagnosed by early colonoscopy, 23 (34%) of the patients were younger than 50 and nineteen of those were women. Habu found that the associated factors for ischemic colitis were an increased incidence of chronic constipation and a prior history of abdominal surgery. Our case illustrates the importance of consideration of ischemic colitis in the differential diagnosis of bloody diarrhea in the younger patient.

BATS AND RABIES: IT DOESNT HAVE TO BE A BITE. D.S. Lindes¹; ¹University of California, San Francisco, San Francisco, CA

LEARNING OBJECTIVES: 1) Recognize the risk of rabies from non-bite exposure to bats. 2) Provide post-exposure prophylaxis against rabies infection.

CASE INFORMATION: A 40 year old woman living in the San Francisco Bay Area presented to her primary care physician with concern about the risk of rabies from contact with a bat. Three weeks prior to presentation, she had found a bat flying around in her living room. Her two cats had tried to catch it, without success. She used a window screen, held in front of herself and above her head, to gradually guide the bat towards an open window. The bat collided with this screen several times before it flew out the window, and the patient spent approximately 45 minutes in the room with the bat. She was not bitten, nor did she recognize any direct contact with the bat's saliva. At the time of presentation, she had no signs or symptoms suggestive of rabiesvirus infection.

DISCUSSION: Rabies is a rare but nearly always fatal viral infection. Its clinical course is characterized by a nonspecific prodrome, with associated paresthesias or fasciculations at the inoculation site, followed by an encephalitis with agitation, confusion, hyperesthesia, autonomic dysregulation, paralysis, and progressive brainstem dysfunction. Death generally results from respiratory center depression. Wild animals, including racoons, skunks, and bats, are the principal vectors for human rabies. Since 1980, 21 (58%) of the 36 human rabies cases

diagnosed in the United States have been associated with bats. Of these cases, there was definite history of a bite in only one or two. In another 10–12, contact but no bite was reported; the other 7–10 cases had no clear contact, but an undetected bat bite was the most plausible route of exposure. Rabid bats have been documented in all 49 Continental U.S. States. Current Centers for Disease Control and Prevention guidelines recommend consideration of post-exposure prophylaxis (PEP) against rabies whenever direct contact between human and bat has occurred, unless the exposed person can be certain that there was no bite, scratch, or mucous membrane exposure. Rabies PEP consists of the following measures: 1) thorough cleansing of all wounds with soap and water, as well as povidone-iodine irrigation if possible; 2) administration of rabies immune globulin, 20 IU/kg, with as much as possible infiltrated around the wound(s) and the remainder given intramuscularly at a site distant from vaccine administration (only needed for individuals not previously vaccinated); 3) rabies vaccine (human diploid cell vaccine [HDCV], rabies vaccine adsorbed [RVA], or purified chick embryo cell vaccine [PCEC]), 1.0 ml administered intramuscularly on days 0, 3, 7, 14, and 28. Because incubation periods of over one year have been reported, PEP is indicated regardless of the amount of time after exposure. If at all possible, the involved animal should be collected for rabiesvirus testing. In the above case, because of prolonged contact and possible mucous membrane exposure, the woman underwent PEP, as did her previously unvaccinated cats. Recognition of the significance of non-bite exposure to bats is important for primary care physicians because they are likely to encounter patients with such exposures, many of whom may be candidates for post-exposure prophylaxis.

PERIPARTUM CARDIOMYOPATHY: DIAGNOSIS AND IMPLICATIONS. C.L. Long¹, R. Deversa²; ¹University of Tennessee College of Medicine-Chatanooga unit, Chattanooga, TN

LEARNING OBJECTIVES: 1- Identify theories regarding etiology of peripartum cardiomyopathy. 2- Recognize physical findings of peripartum cardiomyopathy. 3-Recognize the morbidity and mortality associated with peripartum cardiomyopathy.

CASE INFORMATION: A 23 year old caucasian female, G 1 P1, who is status post caesarean section two weeks prior presented with increasing shortness of breath. She had elevated JVP, bibasilar crackles, and a S3 gallop. ECG showed no acute changes. CXR revealed cardiomegaly with pulmonary venous congestion. An echocardiogram revealed a dilated left ventricle and an ejection fraction of 15%. Despite appropriate treatment during admission, patient became hypotensive, and went into asystole. Aggressive resuscitation was unsuccessful.

DISCUSSION: Peripartum cardiomyopathy is defined as the development of heart failure in the last month of pregnancy or within five months after delivery without any determinable cause and without previous heart disease before the last month of pregnancy. The etiology of peripartum cardiomyopathy is unknown. Some actually question whether this is a distinct disorder or another form of idiopathic cardiomyopathy. Several ideas have been proposed including an immunologic response to the fetus, increased hemodynamic load of pregnancy, nutrition, and hormone response. Several studies have actually shown that these patients have a high incidence of myocarditis, possibly implicating a viral trigger like enterovirus. Signs and symptoms are typical for congestive heart failure. However, these can sometimes be confused with normal pregnancy. Therefore, peripartum cardiomyopathy may not be diagnosed until postpartum. Evaluation would include ECG, chest xray, and an echocardiogram. Cardiac catheterization should be considered in those at risk for atherosclerosis. Endomyocardial biopsy remains controversial. Serum can be tested for bacterial and viral cultures and Coxsackie B virus titers. Therapy consists of digoxin, diuretics, sodium restriction, and afterload-reducing agents. Thromboembolic phenomena are common and anticoagulation should be strongly considered. Cardiac transplantation may be considered as well. The mortality rate ranges from 25–50%. Nearly half of these deaths occur within the first 3 months postpartum. Death is often caused by chronic progressive congestive heart failure, an arrhythmia, or a thromboembolic event. If the patient's heart is to recover, it will usually do so within the first 6 months. If ventricular function does not improve, it is often recommended that the patient not become pregnant again due to the high risk of death.

WHEN ANTIBIOTICS DONT WORK, TIME TO THINK BEYOND BACTERIUM. S. Malhotra¹, K. Gopal¹; ¹Fairview Hospital, Fairview Park, OH

LEARNING OBJECTIVES: 1) Work up of non resolving pulmonary infiltrate. 2) Recognize the clinical manifestations of Disseminated Histoplasmosis. 3) Suspecting HIV disease in the presence of opportunistic infection

CASE INFORMATION: A 54 yrs. old white male presented to the ER with complaint of fever, shaking chills, myalgias, fatigue, and episodes of profound sweating. He had been sick for the past 4 weeks. His primary care physician ordered a chest radiograph that revealed an infiltrate & he was treated with Levofloxacin for 10 days. Patient continued to do poorly with fevers in the range of 100–104 F and was then treated with another 7–10 days of antibiotics. He lost 20 lb. of weight and hence presented to the ER. His other medical problems included hypertension and remote history of alcohol abuse. He denied smoking, current alcohol or drug use. He was divorced, had 3 children and worked as a social worker. Physical exam was remarkable for a fever of 102 F and mild tachycardia. A CXR on admission showed worsened left lung infiltrate. Initial Lab studies revealed WBC 3.0, HGB 11.0, HCT 31.8, MCV 81, PLT 192, Na 132, Alb 2.9. Other electrolytes, and LFT were normal. Patient was started on Piperacillin / Tazobactam. Serological studies for Mycoplasma, EBV, CMV, Legionella, Coccidioides, Blastomycetes, Histoplasma and Aspergillus were all negative. Blood & urine for cultures and histoplasma antigen were negative. A Chest CT showed severe left hilar and perihilar pneumonic consolidation, subcarinal and mediastinal lymphadenopathy, and splenomegaly. Patient continued to be leukopenic, anemic and febrile. A bronchial wash was negative for fungal cultures, AFB cultures and malignant cytology. A bone marrow biopsy revealed normocellular marrow and foci of non-necrotizing granulomatous inflammation. A video assisted thoracoscopic biopsy of left upper lobe revealed multifocal granulomatous inflammation and focal necrosis. Special stains revealed yeast forms suggestive of histoplasma capsulatum. Disseminated histoplasmosis was diagnosed and patient was commenced on IV Amphotericin B. An HIV test came back reactive and the CD4 count was 203. Patient was discharged home with

long term IV Amphotericin B & follow up at HIV clinic. Cultures eventually grew *Histoplasma capsulatum*.

DISCUSSION: A lung biopsy may be the last recourse to obtain a diagnosis and ascertain the correct treatment for a non resolving pulmonary infiltrate. Disseminated *Histoplasmosis* occurs in elderly and about 5% of AIDS patients in endemic areas and is characterized by fever, malaise, pulmonary infiltrates, hepatosplenomegaly, lymphadenopathy and weight loss. Where as asymptomatic infection is most common; CXR of patients with acute infection show focal mid lung infiltrates (30%), hilar or mediastinal lymphadenopathy (30%) or both (30%).

A COUGH THAT LINGERED: DIAGNOSING LEGIONELLA PNEUMONIA. R. Mann¹, D.W. Brady¹; ¹Emory University, Decatur, GA

LEARNING OBJECTIVES: 1. Diagnose legionella pneumonia from its clinical manifestations. 2. Recognize the limited utility of the legionella urinary antigen in excluding the diagnosis of legionella in a patient with a pneumonia that has been refractory to treatment. **CASE INFORMATION:** A 40 year old black male presented complaining of a 3-week history of cough (productive of dark yellow sputum), fevers, chills, nightsweats, and malaise. He initially was treated at an out of town facility with a 5 day course of azithromycin, with some improvement in symptoms; however, within 2–3 days after completing the medication, the symptoms returned. He presented to a second emergency room, where he received a chest xray and was discharged home with a prescription for guaifenesin. One week later, he presented again. A second chest xray revealed interval worsening of a right middle and right upper lobe pneumonia with a patchy airspace opacity in the left upper lobe consistent with pneumonia. On exam his blood pressure was 156/75, pulse 119, respiratory rate 30, temperature 36.9C, with bilateral rales throughout. Pulse ox-70% on room air (ABG on 100% FiO₂-pH 7.44; pCO₂ 30; pO₂ 100; WBC count 16.5 (77%N,6%Mono,16%L), Hct-34.4, Platelets-707, Na-137, K-5.3, Cl-99, Bicarb-18, BUN-13, Cr-0.9, Glucose-163, LFT's-nl, LDH-497, Alb-2.9; HIV test was negative. He was admitted and treated with IV ceftriaxone and erythromycin for the first 72 hours with worsening hypoxemia. Erythromycin was replaced with IV Azithromycin for the next 48 hours with no improvement. Levofloxacin 750mg IV qd was added to replace azithromycin to cover *Legionella*. A *Legionella* urine antigen was negative. After 48 hours of high dose Levofloxacin, the patient showed little clinical improvement. *Legionella*, mycoplasma, and influenza serologies were sent, as preparations were being made for bronchoscopy. On the 3rd day of high dose Levofloxacin the patient began to require lower FiO₂'s and subsequently the *Legionella* Pneumophila IgM serology was found to be positive (1:64). Within the next week, the patient was discharged home to complete a 21 day course of high dose Levofloxacin for treatment of *Legionella* pneumonia.

DISCUSSION: This patient's final diagnosis was made after a protracted outpatient and subsequent inpatient course of therapy. His pneumonia ultimately was found to have been caused by *Legionella pneumophila*; however, the legionella urinary antigen was negative. Clinicians should be aware that the legionella urinary antigen only tests for serogroup I of *L. pneumophila*, and a negative urinary antigen does not, in fact, rule out infection caused by *L. pneumophila* (the most common *Legionella* species causing community-acquired pneumonia in the U.S.). Often, serum antibodies are needed to confirm or exclude the diagnosis.

THROMBOTIC THROMBOCYTOPENIC PURPURA FOLLOWING THE FLU VACCINE. A. Markarian¹, R.D. Hobbs¹; ¹Oakwood Healthcare System, Dearborn, MI

LEARNING OBJECTIVE: To recognize thrombotic thrombocytopenic purpura (TTP) as a potential complication of the influenza vaccine.

CASE INFORMATION: A 58 year-old man with COPD, diabetes mellitus, and hypertension presented with mental status changes and oliguria. He had felt fatigued for several days and had noted a rash on his shins. His chronic medications were Metformin, Glipizide, Nitrobid, Metoprolol, Omeprazole, and Quinine. He received the flu vaccination two weeks prior to admission. He was afebrile but tachycardic. Physical exam showed flaccid paralysis, diminished deep tendon reflexes and bilateral Babinski signs. The heart, lungs and abdomen were unremarkable. A petechial skin rash was noted on both shins. Laboratory studies revealed mild anemia (Hgb 11.7), schistocytes on the peripheral smear, LDH (1927) and a low haptoglobin. The platelet count was 13,000. PT, PTT, INR, fibrinogen and fibrin degradation products were normal. The BUN was 84 and the creatinine 3.4. ANA was positive with a titer of 1:160; anti ds DNA and Lupus anticoagulant were negative. Total complement levels were normal. SSA/RO and SSB/LA were negative. Antiglo-merular basement membrane Ab, ANCA, ACLA-IgG and IgM, and C1 Esterase inhibitor were normal. HIV was non reactive. Cultures were negative for enteric pathogens. Von Willebrand factor activity was normal. Quinine dependant antibodies against platelet glycoprotein were negative but non drug dependant IgM antibody was positive. The patient was treated with plasmapheresis and hemodialysis and recovered.

DISCUSSION: The classic pentad of TTP includes microangiopathic hemolytic anemia, thrombocytopenia, neurologic symptoms, renal disease and fever. The etiology is poorly understood although underlying immune complex formation is frequently found. Platelet-associated antibodies and serum reactivity against endothelial cells are positive in some cases. Most cases are idiopathic. Associated conditions include infections (bacterial, viral, and fungal), autoimmune diseases, AIDS, cancer, vaccinations and drugs such as quinidine, quinine and ticlopidine. Cases of TTP following influenza vaccine have been reported (Brown et al, Br Med J 1973; 2(861):303; Symmers, Br Med J 1973; 2(866):614). In the present case, it is likely that the influenza vaccine triggered the antibody formation and immune complex deposition and led to the development of TTP.

CHARTLORE: WHEN NOT TO BELIEVE IT. C.T. Mccaleb¹, D.W. Brady¹; ¹Emory University, Atlanta, GA

LEARNING OBJECTIVES: 1. Recognize the potentially deceptive nature of "chartlore."
2. Recognize the importance of confirming the past medical history.

CASE INFORMATION: A 55-year old male post myocardial infarction was transferred for cardiac catheterization from a small community hospital. His past medical history per family included closed head trauma as a child, with residual cognitive impairment, hypertension, gout, and some hearing loss over the past 2 months. He has worked as a janitor for a local school for the past 13 years. He was described by his family as "slow" and had a 6th grade education. On hospital day 6, after stabilization, his disposition became a concern. That morning, he sat in bed watching cartoons, laughing, with the sound off. When examined, he became confused and cried. His answers to questions were intermittently unintelligible. He frequently answered questions not asked. An uncle confirmed that this was his baseline, and that he had exhibited odd behavior since a child; his family had attributed it to falling off a truck and hitting his head. His primary team asked psychiatry if evaluating the level of his cognitive functioning would help, and his supervisor at work was called. She said that he was an excellent and very dependable employee as long as he was shown exactly what to do. When asked if he would follow verbal orders, she again stated that he needed to be shown what to do. Suspecting severe hearing impairment, a stethoscope was placed in his ears. With the communicator shouting into the diaphragm, he could hear clearly and responded appropriately. Audiometry on day #10 revealed complete sensory hearing loss on the right, severe loss on the left, with 10% speech discrimination on the left, which increased to 20% when allowed to lip-read. Further questioning revealed that he remembered being able to hear to some degree as a very young child. He was not aware that his family did not know how profoundly deaf he was.

DISCUSSION: Chartlore, the process by which diagnoses are assigned via undocumented notes in the chart, is usually a product of loosely applied terms that, once used, tend to "stick." This particular example of chartlore is interesting in that the original misdiagnosis/assumption appears to have been made, not by the medical staff, but by the patient's own family, and tragically for him, perpetuated over decades, preventing its correction. Physicians must recognize the often deceptive nature of chartlore and the need to confirm history gleaned from medical records and family members. This fact is especially true when the "facts" obtained conflict.

MCARDLES DISEASE PRESENTING AS RHABDOMYOLYSIS. C. Mcdaniels¹, C. Mueller¹; ¹University of Cincinnati, Cincinnati, OH

LEARNING OBJECTIVES:

- 1) Recognize that McArdle disease is the number one glycogen storage disease diagnosed in adults,
- 2) Recognize the typical history of a patient with McArdle disease, and
- 3) Increase awareness of the disease for possible earlier detection.

CASE INFORMATION: CASE: CC is a 31-year-old male who presented with a one-day history of dark urine. CC had to swim strenuously when a barge he was on sank. He then developed diffuse body aches and dark urine. He had a long history of exercise intolerance, easy fatigability, and on several occasions had dark colored urine after exertion. CC's younger brother had a similar history but his symptoms were not as severe. Meds: none, otherwise negative PMH.

Pertinent on PE were BP 131/78, P 101, RR 24, WNWD male in moderate distress from diffuse whole body pain. CV: RRR with mild tachycardia, abd was normal with no HSM. The patient had diffuse muscular tenderness over the thighs, arms, forearms, anterior chest, and back. Labs BUN of 14, Cr of 1.2, AST of 461, ALT of 138, ALK PHOS of 97, LDH of 7,090, and a CK of 87,000. UA showed specific gravity of 1.015, pH 5.0, 4+ prot, 3+ blood, + LE, + NTT, 0–3 RBC's, and +3 myoglobin. The patient had a biceps muscle biopsy which showed the absence of myophosphorylase.

DISCUSSION: McArdle Disease or muscle phosphorylase deficiency results in limited ATP production from glycogenolysis and results in glycogen accumulation. It is the most common glycogen storage disease in adults. Inheritance is autosomal recessive, and the abnormal allele is on chromosome 11. Patients usually are diagnosed in the second or third decade but most have had symptoms since childhood. Exercise intolerance with muscle cramps, myoglobinuria, and a second wind phenomenon are common. Most patients have baseline elevated CK's. Diagnosis can be made by after ischemic exercise testing by measuring lactate levels and ammonia. Definitive diagnosis can be made by muscle biopsy and demonstrating myophosphorylase deficiency. DNA based diagnosis and carrier detection are available.

Prognosis is generally good since longevity is not affected. Patients should avoid strenuous exercise and know the symptoms for seeking medical care during an exacerbation. High protein diets and oral glucose or fructose may increase exercise endurance.

OH NO, NOT "NPO". UNCOVERING OCCULT DIABETES INSIPIDUS. S. Mcelhattan¹, R. Granieri¹; ¹University of Pittsburgh, Pittsburgh, PA

LEARNING OBJECTIVES: 1. To recognize that occult diabetes insipidus can be unmasked when a patient is no longer able to maintain adequate free water intake. 2. To recognize diagnosis and discuss pathophysiology of lithium-induced diabetes insipidus.

CASE INFORMATION: An 81 year-old male was admitted with aspiration pneumonia and dehydration two weeks after head and neck surgery. Postoperatively, he was maintained on nasogastric tube feedings but otherwise kept NPO. His past medical history was notable for bipolar disorder requiring lithium for 20 years. On physical, he was lethargic and orthostatic with dry mucous membranes. His sodium was 157 and creatinine was 2.6. The free-water deficit was calculated, and appropriate IVF therapy was initiated. However, despite increasing IVF administration and free-water boluses, hyponatremia and polyuria persisted. Urine osmolality was 171 and serum osmolality was 324. A water restriction test was not performed as the clinical occurrence of nephrogenic diabetes insipidus was suspected. The patient was given larger and more frequent free-water boluses with improvement.

DISCUSSION: Nephrogenic diabetes insipidus (DI) is resistance to endogenous ADH at the level of the kidney. Up to 20 percent of patients on chronic lithium therapy develop nephrogenic DI. Patients generally adjust their lifestyle and maintain normal fluid balance to keep up with the polyuria. Although the etiology is not totally understood, it is thought that lithium enters the renal collecting tubule cells via a sodium channel and inhibits adenylate cyclase. Lithium may also reduce the density of ADH receptors or downregulate the expression of aquaporin-2

molecules. The diagnosis is substantiated by a water restriction test. As water is restricted, urine and plasma osmolality are measured hourly. If a steady urine osmolality is measured for three successive hours or if the plasma osmolality reaches 300 and the urine osmolality remains below normal, exogenous ADH is given. If urine osmolality increases, central DI is diagnosed but if the urine osmolality does not change, a diagnosis of nephrogenic DI is made. In our patient, it may have remained clinically silent for many years if he had not been NPO. It is essential that all patients on lithium who are made NPO be monitored for the development of hypernatremia and fluid imbalance.

AN UNUSUAL CAUSE OF ABDOMINAL PAIN IN PRIMARY CARE: NOT JUST "HEARTBURN" ANYMORE. A.J. Mechaber¹, H.F. Mechaber¹; ¹University of Miami School of Medicine, Miami, FL

LEARNING OBJECTIVES: 1) Recognize an unusual cause of abdominal pain, and 2) Recognize the importance of an open-minded approach to clinical problem solving.

CASE INFORMATION: Abdominal pain is one of the most common ambulatory patient complaints. Its etiologies can be numerous, leaving the meticulous physician to pursue leads by the history, physical examination and laboratory or imaging tests. We present an unusual cause of abdominal pain that presented in the outpatient setting. **CASE:** A previously healthy 58 year-old male presented to the office with a two-day history of "heartburn." His abdominal pain worsened post-prandially and he noted early satiety, but denied nausea, vomiting, melena, or hematochezia. Alka-Seltzer and aspirin provided no relief. He denied usage of tobacco or illicit drugs, and reported occasional alcohol use, though none over the preceding days. His exam was significant only for mid-epigastric tenderness, without rebound or guarding. His rectal exam revealed brown colored stool, negative for occult blood. Laboratory data was significant only for a mildly elevated ALT. He was treated empirically with cimetidine for possible dyspepsia, gastritis, or peptic ulcer disease. Two days later, the patient was emergently seen for worsening abdominal pain. His exam was unchanged and laboratory data continued to be unremarkable. An urgent abdominal ultrasound surprisingly revealed acute portal, superior mesenteric, and splenic vein thromboses and an enlarged prostate. He was admitted for treatment and diagnostic work-up, which failed to reveal an underlying etiology for the thromboses.

DISCUSSION: Portal vein thrombosis (PVT) is a rare condition associated with inherited hypercoagulable states, underlying myeloproliferative disorders, neoplasms, infections, and other inflammatory processes. It remains idiopathic in 8% to 15% of cases. Abdominal pain is commonly reported when thrombosis also involves the superior mesenteric vessels and produces bowel ischemia, though not present in our patient. The treatment has traditionally focused on complications, most commonly acute gastrointestinal bleeding. The role of anticoagulation therapy remains controversial, but appears to be indicated in patients with acute thrombotic events without intestinal ischemia, as was the case with our patient. While rare, PVT remains an important cause of abdominal pain. A high index of clinical suspicion is necessary for successful diagnosis.

PULMONARY CAVITY IN AN IMMUNOSUPPRESSED PATIENT: WHATS INSIDE? J. Messler¹, V. Nunez¹, T. Jacobson¹; ¹Emory University, Atlanta, GA

LEARNING OBJECTIVES: 1. Learn the differential diagnosis of pulmonary cavities in HIV patients. 2. Recognize the importance of bronchoscopy for diagnosis. 3. Recognize the various pulmonary manifestations of cryptococcus.

CASE INFORMATION: A 41yo homeless man presents to the walk-in clinic with a chief complaint of cough. He complains of blood-tinged sputum for two months, worsening over the past three weeks. He denies fever or chills but reports night sweats, dizziness, decreased appetite, and weight loss. Except for an episode of "bronchitis" treated 2 months earlier, his past medical history is unremarkable. A former intravenous drug user, he has lived in and out of shelters for three years. A recent PPD was negative. He admits to multiple, unprotected bisexual contacts in the past year. He has never been tested for HIV. On admission, his temperature is 39.1C, pulse 98, respirations 16, and blood pressure 100/60, sitting and lying down. Significant findings on his exam include bitemporal wasting, pale conjunctiva and oral thrush. Chest x-ray on admission reveals a 3cm cavity in the right posterior upper lobe. His PPD is negative and sputum cultures for AFB are negative. He tests positive for HIV with a CD-4 count of 6mm3. Bronchoscopic washings grow budding yeast. Serum cryptococcal antigen returns positive at a titer of greater than 1:512. The opening pressure from the lumbar puncture is normal. Cerebrospinal fluid (CSF) analysis contains 4 WBC, glucose 40, and protein 50. CSF yields a positive india ink stain and CSF cryptococcal antigen titer is positive. Subsequently, he begins therapy for cryptococcal pneumonia and cryptococcal meningitis.

DISCUSSION: In HIV patients, the differential diagnosis for pulmonary cavities is extensive. The most common etiology is tuberculosis. However, as the CD-4 count falls below 200mm3, tuberculous cavities decrease in prevalence. Various other etiologies include atypical pneumocystis presentation, toxoplasmosis, invasive fungal infections, bacterial infections, and neoplasms including Kaposi's sarcoma. Sputum samples are a start to diagnosis, but bronchoalveolar lavage or biopsy would likely be required to obtain a definitive diagnosis for a cavity process. A cavity lesion involving cryptococcosis is a rare occurrence. Usual radiographic presentations for pulmonary cryptococcosis include interstitial infiltrates or pleural effusions. Additionally, there is no typical clinical picture. Interestingly, although the majority of cryptococcal patients are initially diagnosed with disseminated CNS disease, the pulmonary complaints of cryptococcosis often precede the diagnosis of meningitis by months. The variety of clinical and radiographic presentation underscore the need for definitive diagnosis, especially in the patient with clinically silent cryptococcal meningitis.

NOT JUST ANOTHER VIRAL INFECTION. L. Miller¹, D.W. Brady¹; ¹Emory University, Atlanta, GA

LEARNING OBJECTIVES: 1) Recognize the presentation of acute HIV infection in adolescents. 2) Diagnose acute HIV infection with serologic tests.

CASE INFORMATION: A 17 year old incarcerated male presented to our emergency department complaining of headache, nausea, vomiting, diarrhea, subjective fever and weight loss for one week. There were no sick contacts or recent travel. The patient had no prior medical history, and had tested negative for HIV six months before presentation. He was taking no medication and denied any family history. Sexual history revealed prior homosexual intercourse. On presentation, the patient's blood pressure was 99/44 mm/Hg with a temperature of 39.4 °C. Physical exam revealed seborrhea over the face and cervical, axillary and inguinal lymphadenopathy. Laboratory studies revealed a WBC count of 2,000, hematocrit of 48% and platelet count of 48,000. Urinalysis showed 2+ protein. Cerebrospinal fluid analysis, stool studies, and blood cultures were negative. Chest X-ray was normal. The patient underwent bone marrow aspiration, which revealed a hypoplastic marrow with decreased erythroid precursors, consistent with viral infiltration. The patient clinically improved and remained afebrile with resolution of all of his symptoms. His platelet count returned to normal. Parvovirus serologies were equivocal, and Epstein Barr serologies did not reveal acute infection. The patient underwent HIV testing which revealed a positive ELISA with an indeterminate Western blot. The CD4 count was 420. After consultation with the infectious disease team and the pathology department, we were permitted to perform viral load testing in the hopes of confirming the diagnosis of acute HIV infection. The viral load was greater than 750,000 copies per mL. **DISCUSSION:** Acute HIV infection presents with a variety of non-specific symptoms, including fever, lethargy, rash, myalgias and headache. Symptoms are present in 50-90% of patients with acute infection, and the complex is known as acute antiretroviral syndrome. Because of the non-specific symptoms, acute HIV infection in adolescents often resembles, and may be mistaken for, more common viral illnesses such as infectious mononucleosis. The diagnosis should be suspected in any patient at risk for HIV infection who presents with this constellation of symptoms or fever of unknown cause. Early diagnosis of HIV, particularly during acute infection, is crucial so that antiretroviral therapy may be initiated without delay. The diagnosis is based on a HIV-1 RNA level of greater than 50,000 copies per mL or positive p24 antigen in the absence of a positive ELISA and confirmatory Western blot test for HIV. If a patient is HIV enzyme immunoassay-negative and HIV RNA-positive, follow-up antibody testing should be performed 2-4 weeks after the resolution of symptoms to document seroconversion.

COMMON COUGH, UNCOMMON CAUSE. F. Millhouse¹, L.J. Schultz¹; ¹Emory University, Atlanta, GA

LEARNING OBJECTIVES: 1. Learn when to evaluate further young patients with "recurrent pneumonias." 2. Recognize the clinical presentation of pulmonary sequestration. 3. Learn the pathology and pathophysiology surrounding the formation of sequestration cysts.

CASE INFORMATION: A 26-year-old male presented with the complaint of a 2-3 week history of cough productive of approximately one cup of yellow green sputum per day. On the day prior to admission, he noted 1-2 teaspoons of blood with each cough. Five years earlier with a similar presentation, he was noted to have a cavity lesion on chest x-ray. He was treated with antibiotics for a presumed lung abscess and was asymptomatic until this admission. Vital signs on admission revealed a blood pressure of 137/76, heart rate of 137, respirations of 22, temperature of 38.4 degrees Celsius, and a pulse ox of 88%. On physical exam he was a young well-developed male in no apparent distress with dullness to percussion and tubular breath sounds at the right base. Chest x-ray showed an interval increase of the air-fluid level within a large right middle and lower lobe cystic lesion and a left mid lung infiltrate. The patient was admitted and started on antibiotics for pneumonia and presumed lung abscess. The patient was unable to tolerate a CT scan or bronchoscopy secondary to uncontrollable coughing and hemoptysis. With a high suspicion for pulmonary sequestration, an aorto-bronchial arteriogram was obtained that confirmed the diagnosis. It showed an aberrant vessel arising off the infradiaphragmatic aorta just above the level of the celiac artery supplying a right lower lobe sequestration.

DISCUSSION: Pulmonary sequestration is defined as an area of nonfunctioning lung tissue that lacks normal communication with the tracheobronchial tree and derives its blood supply from systemic vessels. Pulmonary sequestration is divided into two types: intralobar and extralobar sequestration. The former usually presents in early adulthood. The clinical presentation may range from asymptomatic to chest pain, chronic cough or hemoptysis. Radiographic findings can include either consolidation with irregular margins, multicystic lesions, or cavitation. The diagnosis should be considered in young patients with recurrent or persistent pneumonias. The treatment is surgical removal of the sequestration via segmentectomy or lobectomy.

SEVERE POSTURAL HYPOTENSION AS A DELAYED SEQUELA OF LIGHTNING INJURY. V. Mukerji¹, R. Nonneman¹; ¹Southern Illinois University, Springfield, Illinois

LEARNING OBJECTIVES: 1) Review the acute and delayed injuries that may result from lightning strike. 2) Recognize postural hypotension as a possible delayed sequela of lightning injury.

CASE INFORMATION: A 27-year-old woman was referred to us for increasing dizziness and palpitations. About 8 years ago while sitting indoors with her foot on the fireplace grill she was struck by lightning. She suffered a brief syncopal episode followed by severe lower extremity weakness and inability to walk. With physical therapy her weakness had almost cleared but she continued to experience increasing dizziness and palpitations. Her physical examination was unremarkable except for a resting heart rate of 110 beats per minute. The EKG and echocardiogram were normal. On tilt table testing the patient became dizzy and lightheaded as the blood pressure fell from 116 to 76/42 mm Hg and her heart rate rose from 94 to 150 beats per minute. Her symptoms cleared and the heart rate and blood pressure returned to normal as soon as the table was brought back to the horizontal position. The tilt table test was then repeated with both lower extremities wrapped with Ace bandage. This time there were only minor changes in her blood pressure and heart rate and the patient remained asymptomatic. Compression stockings were recommended for the patient with significant improvement in her symptoms.

DISCUSSION: Lightning is the second most common cause of environmental death in the United States. It is estimated that the number of human injuries may be as high as 5000 per year with 300 deaths. Permanent sequela may occur in 75% of survivors. The serious injuries from lightning strike are usually cardiovascular or neurologic. Most cases develop burns but they are rarely serious. Eye and ear injuries each occur in 50% of cases. Cardiovascular problems include asystole, dysrhythmias, cardiac contusion, pericardial effusion, EKG changes and cardiac enzyme elevations. Neurologic problems include syncope, confusion, paresthesias, and neuropsychiatric disorders. Respiratory center paralysis may occur. Less commonly seizures, cerebral edema, cranial nerve dysfunction and cerebellar ataxia have been reported. This patient developed autonomic dysfunction with severe disabling postural hypotension. The onset was gradual over several years after the actual lightning strike.

A PACEMAKER INFECTION WITH CORYNEBACTERIUM UREALYTICUM. K. Mullins¹;
¹UCLA San Fernando Valley Program, Sepulveda, CA

LEARNING OBJECTIVE: To recognize an unusual cause of a pacemaker pocket infection.
CASE INFORMATION: A 61 year old man was admitted with atrial fibrillation and congestive heart failure. Pharmacologic treatment to limit the ventricular rate was complicated by episodes of bradycardia. Heart rates as low as 30 beats per minute in the presence of continued tachycardia necessitated the implantation of a pacemaker. A pacemaker was implanted, and anticoagulants were restarted twenty-four hours after the surgery. A hematoma was noted at the site. Warmth and erythema developed, yet the patient was afebrile. Empiric antibiotics were started and an aspirate culture was drawn. The culture grew *Corynebacterium Urealyticum*. The patient was treated with vancomycin, and the pacemaker was explanted. Cultures of the pacemaker also grew *Corynebacterium Urealyticum*.

DISCUSSION: The incidence of pacemaker infections post-implantation ranges from one to six percent, with the majority of cases being caused by staphylococcus species. This case represents an unreported etiologic organism of pacemaker pocket infections. *Corynebacterium Urealyticum* is a slow growing, multi-drug resistant, gram positive aerobe that is often found in skin flora in hospitalized patients.

PERIPHERAL T-CELL LYMPHOMA-UNSPECIFIED. PRESENTING AS GENERALIZED ERYTHRODERMA IN A PATIENT WITH PSORIASIS. A RARE PRESENTATION OF A RARE DISEASE. M. Murdaza¹, V. Dimitrov¹, S. Wani¹, I. Mirza¹, J. Shum¹, A. Soni¹; ¹Lincoln Medical Center, Bronx, NY

LEARNING OBJECTIVES: 1. To recognize that Peripheral T-cell lymphoma- Unspecified may present as generalized erythroderma.
2. Generalized erythroderma is rare in Psoriasis and any such a change should prompt a biopsy and further evaluation.

CASE INFORMATION: Peripheral T-cell lymphoma is classified among aggressive group of non-Hodgkin lymphomas by R.E.A.L. classification and accounts for <15% of lymphomas in US. It presents as generalized disease with pruritis. Lymph nodes, skin, liver spleen and other viscera maybe involved. Presentation with generalized erythroderma is rare with peripheral T-cell lymphomas and is usually seen with primary cutaneous lymphomas. Case History: 78 year old Hispanic female with past medical history of hypertension, psoriasis and eczema for the last ten years presented to dermatology service with two month history of gradually worsening generalized erythematous macular rash with mild itching. Patient denied HIV infection or risk factors. Home medications included plendil, monoplir, aspirin and lidex oint. There was no change in her medications recently. Physical exam. revealed generalized erythematous non scaly macular rash with violaceous discoloration at some areas and cervical lymphadenopathy. CBC revealed leukocytosis with eosinophilia. LDH was 441. CXR was unremarkable. Skin biopsy was performed which revealed atypical lymphocytic infiltrate with abundant eosinophils. These cells were positive for CD3, CD4, CD5 (T cell markers) and negative for CD20 & CD30 (B cell markers). Lymph node and bone marrow biopsies were performed to further characterize the disease. Both specimens showed neoplastic infiltrates of lymphoid cells, which were positive for CD3, CD4, and CD5. Review at NIH confirmed the above findings and a diagnosis of peripheral T cell lymphoma- unspecified was made. Patient, who was visiting US from Santo Domingo, declined chemo-therapy and elected to return to her country.

DISCUSSION: Peripheral T cell lymphomas-unspecified is a rare kind of lymphoma. Whereas erythro-derma is not an uncommon presentation of primary cutaneous T cell lymphomas especially Sezary syndrome, it is very rare in peripheral T cell lymphoma-unspecified. In a patient with psoriasis, such as ours, generalized erythroderma is rare and such an evolution of the disease should prompt skin biopsy and further workup. Treatment of peripheral T cell lymphomas-unspecified is not clear due to rarity of this disease. These lymphomas are usually very aggressive and have only 20% cure rate.

MULTIPLE ENHANCING RING LESIONS IN THE BRAIN: ABSCESS OR METASTASIS? S. Narreddy¹, M. Snyder¹, P. Khan¹, N. Lekas¹, R.D. Hobbs¹; ¹Oakwood Healthcare System, Dearborn, MI

LEARNING OBJECTIVE: To recognize the dilemma posed by multiple enhancing ring lesions in the brain and emphasize the need for brain biopsy in equivocal cases.

CASE INFORMATION: A 67 year-old Arabic male presented with a one-month history of progressive weakness and sensory changes of the left arm and leg. He had lived in the US for thirty years before returning to Yemen two years ago to work on a farm. On examination, he had right homonymous hemianopia, left hemiparesis and left hemianesthesia but nothing that would suggest either an infectious process or a malignancy. Basic work-up including HIV test was negative. An MRI revealed multiple enhancing ring lesions in the left and right occipital lobes, the right parietal lobe, and left cerebellar hemisphere. The findings were interpreted as metastatic lesions from an unknown primary. When no primary focus was found a brain biopsy was performed. Gross pus was aspirated. Histology was consistent with

abscess. Smears revealed filamentous gram-positive organisms and cultures grew *Nocardia asteroides*.

DISCUSSION: Multiple ring enhancing lesions in the brain typically represent either metastatic lesions (3-15% of brain tumors); bronchogenic, breast, melanoma, renal, lymphoma are most common primaries) or abscesses. Differentiating between the two by clinical or radiographic means has been difficult, even with the development of CT and MRI scanning. Although advanced radiographic technologies (i.e., diffusion-weighted echo planar imaging, technetium-99m labeled leukocyte scintigraphy) accurately image brain appearance, to clarify treatment it is important to recognize that a brain biopsy is warranted when the diagnosis is in question and a primary focus cannot be identified.

MORE THAN JUST A COLD. M. Nguyen¹, J. Bevan¹, E.F. Yee²; UCLA San Fernando Valley Program, Sepulveda, CA; ¹VA GLAHC/Sepulveda ACC/UCLA, Los Angeles, CA

LEARNING OBJECTIVES: 1. Identify clinical presentation of Wegeners Granulomatosis.
2. Recognize the need for early diagnosis and treatment of Wegeners Granulomatosis.

CASE INFORMATION: A 44 year old Hispanic male presented to clinic with multiple complaints including: red itchy eyes, productive cough with occasional blood, rhinorrhea, nasal congestion, myalgias, abdominal pain, and subjective fevers and chills. Except for a gastric ulcer diagnosed 1 month earlier, he was otherwise healthy. He was diagnosed with a viral syndrome, reassured, and given symptomatic treatment. The patient returned for a second visit 2 weeks later without improvement in his symptoms. A CXR, PPD, and HIV test were all negative. An ophthalmology exam confirmed the diagnosis of viral conjunctivitis. The patient, concerned that his symptoms were not improving, returned a week later, appearing very ill, and was admitted for further work up. On physical exam, notable findings included a T = 38.8 C, scleral injection, nasal and oropharyngeal erythema, decreased breath sounds at both lung bases, and mild abdominal tenderness in the right upper quadrant. Labs were significant for proteinuria, hematuria, and elevated liver function tests. To work up his abdominal symptoms, a CT of the abdomen was obtained which showed gastric thickening and a right lung nodule. Subsequent testing included an EGD showing diffuse erythematous mucosa; a repeat CXR and CT of the chest revealing multiple opacities in both lungs with hilar lymphadenopathy; bronchoscopy showing erythematous friable mucosa; and a repeat slit lamp examination revealing bilateral episcleritis. C-ANCA titers were 1:160. A nasal biopsy was performed showing granulomatous disease with giant cells and chronic vasculitis confirming the diagnosis of Wegeners Granulomatosis. The patient was treated with Cytoxan and Prednisone with resolution of symptoms.

DISCUSSION: Wegeners Granulomatosis is a form of small vessel vasculitis with prevalence of 3:100,000. The mean age of onset is 40 years, with men affected slightly more than women. It classically involves upper and lower respiratory tracts and kidneys, but can involve virtually any organ system. This case illustrates how Wegeners can mimic colds and sinusitis, obscuring the diagnosis and delaying treatment. Early diagnosis is crucial because untreated Wegeners has a poor prognosis with median survival of 5 months and up to 90% mortality. Furthermore, renal failure can progress rapidly even in the absence of symptoms. Timely initiation of therapy with Cytoxan and Prednisone can be life saving and may markedly decrease morbidity and mortality.

REFRACTORY HYPOGLYCEMIA IN AN ALCOHOLIC. L. Orlando¹, S. Omoro¹, J. Wiese¹;
¹Tulane University, New Orleans, LA

LEARNING OBJECTIVES: To recognize the effect of alcohol on adrenal function. To illustrate the importance of clinical reasoning in solving complicated patient presentations.

CASE INFORMATION: A 52-year-old man presented with generalized weakness and syncope. On the morning of admission, he was unable to arise from bed. The weakness involved both upper and lower extremities and was associated with mild abdominal pain. After 5 hours, he alerted a family member who assisted him to his feet. Upon standing the patient lost consciousness. He denied vomiting or recent illness but noted decreased food intake for the 2 days prior to admission. His only past medical history was a penetrating knife wound to the forehead 20 years prior to admission. He was taking no medications and did not use intravenous drugs. The blood pressure was 90/54 mmHg, the heart rate 110 beats/min and the temperature 36C. He had a normal thyroid exam and there were no carotid bruits. Examination of the heart, lung and abdomen were normal. Reflexes were decreased throughout; proximal and distal muscle strength was diminished. There were no focal deficits; sensation and cerebellar functions were normal. His sodium, potassium, calcium, magnesium, and creatinine were normal. The blood glucose was 54 mg/dL, and despite multiple ampules of dextrose he remained hypoglycemic. Persistent hypotension and refractory weakness prompted an evaluation of his adrenal function. The random cortisol was 1.6 mg/dl; the ACTH was low.

DISCUSSION: The diagnosis of adrenal insufficiency should be considered when refractory hypoglycemia, orthostatic hypotension, and weakness simultaneously present. The causes of acute adrenal insufficiency include iatrogenic withdrawal of steroids, adrenal hemorrhage, tuberculosis and autoimmune disease. An important and under-recognized cause of adrenal insufficiency is inhibition of the hypothalamic-pituitary-adrenal axis by heavy alcohol ingestion. A repeat history revealed that the patient had a 6-day period of heavy alcohol and cocaine use. The normal electrolytes were clues that the adrenal insufficiency was not due to Addison's disease. The functional aldosterone axis and lack of hyperpigmentation implied a pituitary etiology which was confirmed by the low ACTH and robust rise in cortisol following cortrosyn stimulation. The weakness and hypoglycemia corrected with administration of hydrocortisone. Clinical reasoning identified the specific etiology of this patient's Addisonian syndrome and prevented unnecessary therapy.

PROGRESSIVE PNEUMONIA IN AN IMMUNOCOMPETENT HOST. L. Orlando¹; ¹Tulane Medical Center, New Orleans, LA

LEARNING OBJECTIVES: Suspect fungi in antibiotic refractory pneumonias and treat the presence of hyphae as invasive aspergillosis in these patients. Recognize that chronic diseases can increase the risk for invasive aspergillosis.

CASE INFORMATION: A 71 year-old man with severe COPD (not on steroids) presented with one week of a non-productive cough, fever and dyspnea. His exam revealed crackles and egophony over the right upper lobe (RUL). The chest X-ray confirmed a community acquired pneumonia (CAP), and he was started on Azithromycin. His symptoms did not improve. A chest CT showed extension of the RUL consolidation, but no cavities or endobronchial lesions. Blood and sputum gram stain, AFB, KOH, and cultures were negative, except one sputum with branching hyphae, which was dismissed as a contaminant. Fluconazole and Vancomycin were added, but his condition worsened. Acute renal failure occurred on day seven. Bronchoscopy was performed on day eight, and revealed black mucus and white nodules within the bronchioles. Lavage and biopsies were negative for organisms or cancer. The patient died from an acute lung hemorrhage on day ten. Autopsy revealed invasive *Aspergillus Niger* with an acute lung hemorrhage.

DISCUSSION: A progressive CAP despite antibiotic therapy implies one of the following: the wrong antibiotic selection, a post-obstructive pneumonia (POP), a fungal pneumonia, immunosuppression, alveolar proteinosis or BOOP. In our patient the acute presentation, imaging studies without fleeting infiltrates, and bronchoscopy without obstruction were clues against BOOP, alveolar proteinosis, and POP. The patient's brisk immune response, severe COPD and negative bacterial cultures were important hints towards a fungal infection. Although *Aspergillus pneumonia* usually occurs in the immunocompromised, it can occur in immunocompetent patients so the presence of branching hyphae on cultures should not be disregarded as colonization or contamination when a clinical pneumonia is present, especially in those with chronic diseases. Empiric Amphotericin B and lung biopsy should be considered when clinical evidence suggests *Aspergillus* since cultures and even bronchoscopy are frequently negative. This patient had *Aspergillus Niger* which produces oxalic acid creating a pathognomonic black acidic sputum and an immune mediated glomerulonephritis. Sputum and BAL cultures are positive in less than 10% and dissemination occurs in 25% of patients.

IMPENDING PARADOXICAL EMBOLUS. A. Pai¹, A.L. Riba¹, R.D. Hobbs¹; ¹Oakwood Healthcare System, Dearborn, MI

LEARNING OBJECTIVE: To recognize the unique radiologic and therapeutic features associated with impending paradoxical emboli.

CASE INFORMATION: A 61 year old female presented with an acute onset of intractable headaches, vertigo, dyspnea and melena. A CT scan confirmed an embolic CVA involving the right occipital lobe. The presence of an asymptomatic, swollen right leg, and an acute axis change on EKG lead to the suspicion of a DVT with PE. Large thrombi were seen in both pulmonary arteries on spiral CT. Additionally, a TEE demonstrated a large embolus straddling a patent foramen ovale. Therapy was problematic due to the patient's history of complications following a CABG in the past and active GI bleeding. A conservative treatment approach involved careful anticoagulation with heparin and Greenfield filter placement. The patient successfully recovered without further embolic episodes or complications.

DISCUSSION: Although 25% of patients at autopsy have a patent foramen ovale that can be detected by probing, the actual occurrence of paradoxical emboli (as with most pulmonary emboli) can safely be assumed to be under-diagnosed. Although newer technology (spiral CT, TEE) has increased our ability to diagnose such conditions, impending paradoxical embolus is still rarely reported in the literature. The optimal therapy is unknown although embolectomy with closure of the patent foramen ovale, thrombolysis and anticoagulation are the commonly available options. Treatment should be individualized depending on the patients age, existing co-morbidities and the availability of cardiothoracic surgical services.

POLYARTERITIS NODOSA PRESENTING AS AN ACUTE CEREBROVASCULAR ACCIDENT IN A YOUNG MALE. P.J. Bennett¹, J.J. Yum¹, M. Panda¹; ¹University of Tennessee - Chattanooga, Chattanooga, TN

LEARNING OBJECTIVES: To understand the differential diagnosis of CVA in young patients. To recognize clinical features and proper diagnosis of polyarteritis nodosa (PAN) **CASE INFORMATION:** A 24-year-old black male with a questionable history of multiple sclerosis presented to the ER with acute onset of right-sided weakness, slurred speech and dysphagia. The patient denied any visual symptoms, headache, fevers or recent illnesses. On exam the patient was afebrile, pulse 60 beats per minute, and BP 180/128mm/Hg, neck was supple without bruits, and a 2/6 systolic murmur was noted at the apex. No skin lesions were noted. On neurologic exam the patient had dysarthria, right facial droop and right hemiplegia. Routine lab work was unremarkable with a normal CBC, metabolic panel and sedimentation rate. Urinalysis revealed 2+ proteinuria with benign sediment. LP revealed no oligoclonal bands. An MRI of the brain revealed a recent deep left-sided infarct involving the internal capsule as well as a small aneurysm of a temporal branch of the right middle cerebral artery. Hypercoagulable work up was negative, p-ANCA was negative but myeloperoxidase antibody was positive. Echocardiogram revealed mild left ventricular hypertrophy with normal LV function. Cerebral and renal angiography revealed innumerable aneurysms involving almost all branches of medium-sized arteries. A biopsy of the sural nerve with accompanying vasculature confirmed the diagnosis of PAN. The patient was treated with steroids, monthly pulse dose Cytoan, and aggressive physical therapy. Nine months later he shows marked improvement and no further neurologic insults.

DISCUSSION: PAN is a systemic vasculitis involving small and medium sized arteries and is more common in middle-aged men. Any organ can be affected but skin, peripheral nerves, joints, intestinal tract and kidneys are most commonly involved. The onset of disease can be abrupt or gradual and usually involves symptoms such as fever, malaise, palpable purpura, infarctive ulcers, joint pain and livedo reticularis, although multiple mononeuropathies are the most typical neurologic manifestation. The sedimentation rate is usually elevated but, as in our case, can be completely normal. Antibody to hepatitis B surface antigen can be positive. Tissue biopsy confirms the diagnosis. Untreated the 5 year mortality rate is greater than 85%. High dose steroid therapy has improved the five-year mortality to 30-45% and the addition of immunosuppressive has reduced relapses. Plasmapheresis offers no additional advantages. Treatment of any associated viral illness (hepatitis, HIV) should be part of the initial therapy.

RAPIDLY PROGRESSIVE CARCINOID SYNDROME IN AN IMMUNOSUPPRESSED MALE. J.R. Pederson¹, J.J. Yum¹, M. Panda¹; ¹University of Tennessee-Chattanooga, Chattanooga, TN

LEARNING OBJECTIVES: 1. Recognize the increased risk for malignancy in an immunosuppressed patient 2. Recognize the signs and symptoms of carcinoid syndrome.

CASE INFORMATION: A 61 year old male, status post renal transplantation in 1995 for chronic renal insufficiency, was admitted in 9/1999 with complaints of weight loss, diarrhea, anorexia and early satiety. The patient noted a gradual onset of symptoms since 1996, with worsening over a six month period prior to presentation. He was on immunosuppressive therapy since 1995. Physical exam revealed a cachectic white male with redness and telangectasias of the face and neck. His physical exam was otherwise unremarkable. CBC, electrolytes and liver function tests were all within normal limits. Ultrasound of the abdomen, CT pelvis and colonoscopy in 1996 and panendoscopy in 1999 were normal. Work up was negative for malabsorption, *H. pylori*, Giardia and *C.difficile*. He was readmitted a month later for right hip fracture. He still had diarrhea and weight loss but also noted prominent bilateral facial, neck, and arm flushing. Punch biopsy from the skin of the neck was essentially unremarkable. Post-op CT of the abdomen revealed innumerable hepatic lesions in all lobes of the liver. Ultrasound guided liver biopsy revealed metastatic carcinoid tumors. Subsequent 24 hour urine collection revealed elevated 5-HIAA levels. The patient was treated with sandostatin and antihistamines. He rapidly deteriorated and died within several months of diagnosis.

DISCUSSION: Carcinoid accounts for 30 - 40% of all small intestinal tumors. Over 95% of all GI carcinoids occur in one of 3 sites: the rectum, appendix or small intestine (usually the ileum). Tumors less than 2 cm are less likely to metastasize. These tumors secrete vasoactive materials responsible for the clinical manifestations of carcinoid syndrome which is seen only in patients with hepatic metastasis. With advanced disease, treatment is symptomatic and true cure is seldom achieved. Rapid progression of indolent carcinoid has been reported in patients who are immunosuppressed. As our patients preliminary extensive GI work-up did not reveal an obvious primary, we postulate that our patient may have primary hepatic carcinoid which rapidly progressed due to his immunosuppressed state. There has been only one case reported in the literature of primary hepatic carcinoid in a renal transplant patient. With the world wide increase in the incidence of tumors among immunosuppressed patients, physicians should have a low threshold for early thorough workup of suspected tumors.

KIKUCHI'S DISEASE: AN UNCOMMON CAUSE OF FEVER OF UNKNOWN ORIGIN (FUO). I. Ahmad¹, S. Sheldon¹, M. Panda¹; ¹University of TN, Chattanooga, TN

LEARNING OBJECTIVE: 1. Recognize the diagnostic work-up of FUO

CASE INFORMATION: A 34 year old black male presented with a 2 week history of low grade fever, night sweats, generalized fatigue, weakness and myalgias. He had a waxing and waning rash on his face and arms. Past medical history was significant for hypertension since age 26 and a similar rash and fever about 12 years ago. On examination he was febrile (temperature of 103°F) and tachycardic. Physical examination was unremarkable except for a 2x2cm right submandibular lymph node, a few lymph nodes in the right posterior cervical triangle and a hyperpigmented maculopapular skin rash involving the upper extremities and the face. The laboratory data showed WBC: 2th/mm³, Hbg./Hct.-8.4gm/dl 25.8%. The blood cultures, urine cultures, chest x-ray, imaging studies of the head, neck, thorax, and abdomen were all negative. ESR: 98, CRP 8.4mg/dl. Rheumatologic workup, viral and rickettsial serology, RPR and HIV were negative. Bone marrow biopsy and culture was also negative. The right submandibular lymph node biopsy showed histiocytic necrotizing lymphadenitis. He was started on Vioxx for the fever and myalgia. The patient's fever subsided after a total of 5 weeks. He returned with vision problems and was found to have retinal hemorrhages bilaterally with normal disc margins. Steroids were initiated for ocular symptoms with some improvement.

DISCUSSION: Kikuchi's disease was first described in Japan in 1972 by Kikuchi and Fujimoto. It is a self limiting disease involving different ages, races and geographic regions. The etiology of this disease is uncertain, although its association with certain viruses like CMV, EBV, HSV, Parvo B-19, HIV, Yersinia and Toxoplasmosis has been described. These viruses or other antigens can induce immunological abnormalities in a susceptible individual with progression to autoimmune diseases like SLE and Still's disease. The most common clinical features of Kikuchi's disease are fever, weakness, fatigue, myalgia, arthralgia and lymphadenopathy (especially cervical). Skin rash is non-characteristic particularly involving the upper regions of the body. Hepatosplenomegaly and neurological manifestations like aseptic meningitis and ataxia have also been described. However this is the first case described with retinal changes. Kikuchi's syndrome needs to be differentiated from other diseases like malignant lymphoma, SLE, tuberculosis, sarcoidosis, and cat scratch disease. As the course of Kikuchi's disease is generally self limited, and histopathology is diagnostic it is important to include it in the differential diagnosis of FUO.

A CHALLENGING CASE OF DYSPNEA IN A PATIENT WITH BREAST CARCINOMA. P.J. Bennett¹, M. Panda¹; ¹University of Tennessee, Chattanooga, TN

LEARNING OBJECTIVES: Use of appropriate diagnostic testing to differentiate the pulmonary tumor emboli syndrome from a thromboembolic event. Understand the diagnosis and management of pulmonary tumor emboli

CASE INFORMATION: A 75 year old white female with a past history of poorly differentiated ductal adenocarcinoma of the right breast diagnosed in 7/95 presented with progressive shortness of breath for 4-5 days. On exam, she was afebrile, tachypneic and tachycardic. She had decreased breath sounds bilaterally with crackles posteriorly in the right lower hemithorax. ABG revealed: pH 7.46, PCO₂ 26mmHg, PaO₂ 43mmHg, HCO₃ 19meq, 82% saturation on room air. Chest films and CT were unchanged from previous of 8/98. There was no evidence of lymphangitic carcinomatosis. Echocardiogram revealed an EF of 72%, evidence of new moderate pulmonary hypertension and new tricuspid regurgitation, without tamponade. VQ scan revealed small unmatched peripheral deficits in the left upper lobe consistent with intermediate probability of pulmonary embolus. Subsequent pulmonary angiogram revealed no

evidence of pulmonary embolism. Due to worsening hypoxia, bronchoscopy with transbronchial biopsy was performed. Biopsy revealed metastatic poorly differentiated carcinoma with prominent intravascular growth consistent with a breast primary.

DISCUSSION: Pulmonary tumor embolism is more often a postmortem diagnosis in patients with known solid tumors such as breast, liver, prostate, and kidney. Potential mechanisms of tumor emboli to the lung include direct extension into the IVC, extension via the thoracic duct to pulmonary lymphatics or via the SVC & right side of the heart to the pulmonary arterioles. The tumor cells occlude the smaller arterioles without parenchymal involvement. Resulting intimal proliferation leads to increased pulmonary pressures and right ventricular strain. Progressive dyspnea is the most common symptom. Chest X-ray is often unremarkable. VQ scans may be normal, indeterminate, or reveal focal defects. Pulmonary angiography may only reveal slow blood flow. Occasionally, CT/MRI scans may reveal a tumor mass occluding the pulmonary artery. Diagnosis is proven by tissue biopsy. Treatment of patients with pulmonary tumor emboli is difficult. Tumor embolectomy may be an option in large proximal emboli. Chemotherapy and radiotherapy usually produce a poor response. Together with fat and amniotic fluid embolism, tumor embolism cause hypoxia with a normal pulmonary angiogram. As in our patient unrelenting hypoxia and new development of hypertension may be important diagnostic clues.

"IT IS NOT NICE TO FOOL WITH MOTHER NATURE":- CASE OF THE HERBAL INDUCED RASH. P.J. Bennett¹, M. Panda¹; ¹University of Tennessee, Chattanooga, TN

LEARNING OBJECTIVES: 1. Recognize the prevalence of herbal medication use and possible side effects. 2. Recognize the clinical presentation of eosinophilia-myalgia syndrome.

CASE INFORMATION: A 40 year old white female with no significant medical history presented with a 5 month history of a slowly expanding rash on her lower trunk and proximal extremities associated with generalized fatigue. The rash was painless, nonpruritic and not associated with arthralgias, fever or chills. The patient admitted to taking several herbal medications which she had recently stopped. Her physical exam was unremarkable except for a pink, blanchable, nontender, confluent, macular rash involving the lower abdomen and proximal portions of the arms and legs. There were no bullae or pustules. Laboratory evaluation revealed a negative ANA, rheumatoid factor, SS-A and SS-B antibodies. The patient had normal levels of aldolase, CPK and sedimentation rate. Her CBC was unremarkable except for the presence of 26% eosinophils. A deep fascial biopsy of one of the left thigh lesions revealed an inflammatory infiltrate of the fascia consistent with eosinophilic fasciitis. Upon review of the patient's herbal medication list and available literature, the probable causative agents were glucosamine sulfate and the fever few leaf. The patient was started on steroids with slow improvement.

DISCUSSION: Eosinophilia-myalgia syndrome was first described in 1989 when it was observed in 3 group of patients who developed scleroderma-like skin changes, myalgias, eosinophilia, and histologic findings similar to eosinophilic fasciitis after ingestion of L-tryptophan manufactured by a single company. However, as this case illustrates, other agents, including herbal medications can precipitate a similar syndrome. Other clinical manifestations include low grade fever, fatigue, dyspnea, cough, arthralgias/arthritis, erythematous rashes and myalgias. Myocarditis and pulmonary hypertension can occur. Usually significant peripheral eosinophilia is noted and striking tissue eosinophilia may be noted as well. Diagnosis is dependent on histologic findings from a deep biopsy including fascia and muscle. Response to treatment is variable but patients may respond to steroids, plaquenil, methotrexate, D-penicillamine and cimetidine. The course of the disease is poorly defined but many patients spontaneously regress or remain unchanged for years. Use of herbal medications is becoming very common as these products are widely advertised and readily available. However clinical trails and adequate information on efficacy and adverse reactions are limited. This case stresses the importance of incorporating questions on use of herbal medications routinely on medication history and the need to have a heightened awareness of their possible implication in disease.

ACUTE RENAL FAILURE: THE PRESENTING FEATURE IN PELVIC MALIGNANCY. S. Parikh¹, D.W. Brady¹; ¹Emory University, Decatur, GA

LEARNING OBJECTIVES: 1. Recognize urinary tract obstruction as a common and potentially reversible cause of acute renal failure. 2. Diagnose obstructive uropathy in the absence of hydronephrosis.

CASE INFORMATION: A 74 year old previously healthy male presents complaining of a 2 month history of progressively worsening scrotal swelling. The patient denied urinary hesitancy, frequency, hematuria, dysuria, or incontinence, but admits to mild decrease in urine output with occasional urgency. Past medical history was negative, and the patient denied using any medications including NSAIDs. Physical exam was significant for marked scrotal, suprapubic, penile shaft, and foreskin edema with normal testicular size. No masses or hernias were detected. Routine labwork showed BUN 67, Creatinine 9.6. When seen for the same complaint 5 weeks earlier, the patient had normal electrolytes and creatinine 1.2. Foley catheter was successfully placed with approximately 50cc of residual volume obtained. Renal ultrasound showed mild to moderate right hydronephrosis, mild prominence of the left collecting system without hydronephrosis. Percutaneous nephrostomy of the right kidney failed to improve renal function, and hemodialysis was initiated. Left renal biopsy revealed normal glomerular structure. CT scan of pelvis showed calcification of the prostate, with thickened mucosa of the sigmoid colon and rectum and bilateral hydroceles. Prostate biopsy was negative for malignancy. Rectal biopsy revealed poorly differentiated carcinoma of probable uroepithelial etiology. After 7 days, urine output began to improve, and dialysis was discontinued. Hepatitis serologies, urine and serum protein electrophoresis, ANCA, and anti-GBM antibodies were negative. Cystoscopy showed bladder thickening with locally invasive disease and bilateral ureteral obstruction. Results of bladder biopsy showed high grade papillary transitional cell carcinoma with vascular invasion and extension to the pelvic wall and rectum.

DISCUSSION: Approximately 4% of cases of renal failure due to obstructive uropathy present without dilation of the urinary tract. Of these cases, 60% are due to intrapelvic malignancy (prostate, bladder, colorectal). Proposed mechanisms for lack of dilation include ureteral encasement by tumor, ureteral edema, reabsorption, and interrupted peristalsis. Although renal

ultrasound is often used to rule out urinary tract obstruction, the absence of hydronephrosis does not exclude obstructive uropathy as the cause of acute renal failure. In the presence of high clinical suspicion, additional studies such as cystoscopy, intravenous pyelogram, or retrograde pyelography must be pursued to evaluate patency of the upper and lower urinary tracts.

ACQUIRED HEMOPHILIA IN SLE. J. Pedersen¹, J. Paty¹, M. Panda¹; ¹University of Tennessee College of Medicine - Chattanooga Unit, Chattanooga, TN

LEARNING OBJECTIVES: 1. Recognize the entity of acquired hemophilia secondary to anti-factor VIII antibodies and its association with autoimmune diseases and occult malignancy. 2. Understand the indications for treatment of acquired hemophilia.

CASE INFORMATION: A 76 year old white female with a past history of SLE and polymyalgia rheumatica, in remission on daily prednisone and Plaquenil, presented to the office with a one week history of spontaneous, extensive bruising of the right hip and thigh after arising from a chair. She denied trauma, a personal or family history of easy bruising or prolonged bleeding. Physical examination was unremarkable except for an extensive area of ecchymosis along the right inner and outer thigh. MRI revealed iliopsoas hemorrhage. Initial lab revealed a normochromic, normocytic anemia with a hemoglobin of 6 (one week prior to admission her hemoglobin was 10). She had a normal platelet count, prothrombin, thrombin and bleeding times, but an elevated partial thromboplastin time (57.1 seconds). The PTT partially corrected with the addition of normal plasma, so Factor VIII assay was done which revealed decreased activity at 8% with positive titers for factor VIII inhibitor (72 Bethesda, n1 0.1–0.8). Lupus anticoagulant and antiphospholipid antibodies were negative. Subsequently the patient had a spontaneous right retroperitoneal and left upper extremity hemorrhage and required factor VIII transfusions, high dose corticosteroids and oral cytoxin. Six months later she remains symptom free on 5mg of prednisone and Plaquenil 400mg daily. Her hemoglobin is 11.8; PTT-28.6 and there is no detectable Factor VIII inhibitor.

DISCUSSION: Hemophilia A, as a result of Factor VIII deficiency, can be inherited or acquired. Acquired inhibitors to Factor VIII are endogenous IgG/IgM immunoglobulins which interfere with normal blood coagulation. Acquired Factor VIII inhibitors can occur in patients with inherited hemophilia A, postpartum women, patients with collagen vascular diseases such as SLE, or idiopathically, especially in older patients. They are also associated with occult malignancy such as lymphoma and solid organ tumors. The syndrome usually presents with significant bleeding and soft tissue ecchymoses with minimal or no trauma. Treatment of active episodes of bleeding may involve administration of exogenous porcine or human Factor VIII or treatment with activated prothrombin complex concentrates as in our patient. Prophylactic treatment with steroids, with or without immunosuppressive agents, is indicated in patients with significant bleeding or with high titers of inhibitor.

RARE PRESENTATION OF PRIMARY EMPTY SELLA SYNDROME. K PERUMAL, S BELYAEV, AND C SMITH, DEPARTMENT OF INTERNAL MEDICINE, COOK COUNTY HOSPITAL, CHICAGO, IL. K. Perumal¹; ¹Cook County Hospital, Chicago, IL

LEARNING OBJECTIVES: 1. Recognize central causes of hypothyroidism.

2. Recognize primary empty sella syndrome as a rare cause of hypopituitarism.

CASE INFORMATION: 41 year-old African-American woman with no significant past medical history presented to clinic with complaints of bilateral lower extremity swelling, easy fatigability and generalized weakness of 2 weeks duration. She also reported increased sensitivity to cold, dryness of skin and menorrhagia. On exam, she was moderately obese, did not appear acutely ill. Vitals signs were normal. Physical exam remarkable for mild conjunctival pallor, soft heart sounds, slow relaxation phase of deep tendon reflexes and bilateral lower extremity non pitting edema.

EKG showed low voltage complexes in all leads. Echocardiogram demonstrated large pericardial effusion with no tamponade. Labs revealed normocytic, normochromic anemia.

Thyroid profile with TSH 2.07 (0.4–6) and free thyroxine 0.30 (0.6–1.7) pointed towards the diagnosis of central hypothyroidism. Other hormonal assays were done which also revealed coexisting adrenal insufficiency. MRI of brain showed flattened pituitary along the floor of sella and majority of the sella was filled with CSF. These findings were consistent with empty sella. No focal lesion in the pituitary gland were identified. It was diagnosed to have hypopituitarism due to primary empty sella syndrome. She was treated with thyroxine and steroids to which she responded appropriately. The pericardial effusion was presumed to be secondary to the hypothyroidism and pericardiocentesis was not performed as she was hemodynamically stable. **DISCUSSION:** Primary empty sella syndrome has been classically described in obese hypertensive women. It is characterized by the presence of an arachnoid herniation filled with fluid that compresses the pituitary against the sellar wall. It is often asymptomatic but may be associated with endocrine disorders. Less than 30 percent of these patients presents with symptoms suggestive of hormonal deficiency. Recent literature reports only about 10 percent of these patients present with hypopituitarism as in our patient. The most common causes of central hypothyroidism are pituitary adenomas, pituitary apoplexy, infiltrative lesions. Pericardial effusion is one of the common cardiovascular manifestation in hypothyroid patients but most of these cases are described in patients with primary hypothyroidism. There are only few case reports describing pericardial effusion as presenting manifestation in patients with central hypothyroidism. These pericardial effusions are usually large but rarely progress to tamponade. They usually respond well to thyroxine supplementation.

SMALL BOWEL INTUSSUSCEPTION DUE TO LYMPHOMA COMPLICATING COMMON VARIABLE IMMUNE DEFICIENCY. G. Prakash¹, O.R. Signori¹, J. Sunstrum¹; ¹Oakwood Healthcare System, Dearborn, MI

LEARNING OBJECTIVE: To improve recognition of lymphoreticular malignancies in patients with common variable immune deficiency (CVID).

CASE INFORMATION: A 51 year old white male presented with lower abdominal pain exacerbated by food. He had generalized weakness and had recently received a blood transfusion

for unexplained anemia (Hb 6 g/dl). The patient had a history of chronic diarrhea with weight loss secondary to a diagnosis of CVID in 1994. Multiple colonoscopies revealed only non-specific colitis. CT scan showed extensive mesenteric lymphadenopathy and thickening of the small bowel wall. He underwent EGD which showed healed duodenal erosions with copious bile in the esophagus and stomach. A small bowel follow-through revealed proximal bowel dilatation and a partially obstructing process in the mid-jejunum (see Figure). Exploratory laparotomy with partial small bowel resection revealed malignant lymphoma (large B-cell type). Bone marrow biopsy was normal. Tests for HIV antibody were repeatedly negative. The patient was discharged for outpatient oncological treatment.

DISCUSSION: GI lymphomas have been reported rarely in patients with CVID, but are not usually considered among the initial differential diagnoses. We present this case to call attention to non-specific symptomatic presentation of lymphomas of the GI tract (Lai Ping So and Mayer, *Semin Gastrointest Dis* 1997; 8:22; Gottesman et al., *Leuk Lymphoma* 1999; 32: 589). Figure Legend. Barium failed to progress rapidly to the distal jejunum (left panel), but was present after 15 minutes (right panel), indicating a partially obstructing process with considerable proximal bowel dilatation.

ADVANCE CARE PLANNING: MORE TALK, LESS MEDICINE. M.W. Rabow¹, K. Schanche¹; ¹University of California, San Francisco, San Francisco, CA

LEARNING OBJECTIVES: 1) Recognize that open communication is the foundation of appropriate advance care planning (ACP) among patients with end-stage illness; 2) Assess the patient's ability to participate in ACP decision-making; and 3) Recognize the importance of family caretakers in ACP.

CASE INFORMATION: An 84 year old woman with end-stage renal failure, congestive heart failure, insulin-dependent diabetes, depression, and osteoarthritis begins to express a desire to discontinue hemodialysis after many years. She feels sad, exhausted, and a burden to her family. The patient has been widowed for five years. Her medications include Celexa and Vicodin. She has four attentive children who all live in the same city. She is cared for primarily by her youngest daughter, who lives with the patient, administers medications and transports the patient to dialysis. Intent on caring well for her mother, and worried that pain was interfering with the patient's appetite, the patient's daughter has been administering increasing doses of Vicodin. While the patient has made a number of comments about discontinuing dialysis to each of her children, none of the children have talked with each other, hoping to avoid such discussions, as well as differences of opinion between them. The patient's thoughts have not been assessed during the last year by either her nephrologist or primary care physician. An end-of-life consultation team met with the patient and her daughter. Clarification of the appropriate use of pain medicine allowed the patient to control her pain with only occasional use of Vicodin, leading to increased energy, independence, and enthusiasm. Assessment of the patient's CESD depression score showed an increase from 14 (six months earlier) to 22. The patient's Celexa dose was adjusted. The patient's daughter was invited to attend support classes for family taking care of patients with end-stage illness and grew more comfortable talking about her fears and sense of responsibility. With renewed energy and improved mood, the patient withdrew her desire to discontinue dialysis for now. In a family meeting, all four children were able to recognize their own feelings and to better coordinate care.

DISCUSSION: Even among closely-knit families and well-supported patients, miscommunication is possible, especially around the highly-charged issues of advance care planning for patients with end-stage illness. Assessing for medication over- and under-prescribing, and ruling out depression as an etiology for withdrawal of support requests, are good first steps during discussions. Understanding the motivations and influences of relevant family members is necessary as well. The key to good advance care planning is to begin the discussions well in advance of the decisions as there is often much to learn and minor clinical interventions can have major effects on planning variables.

HYPERTROPHIC GASTROPATHY - ANEMIA. S. Ramalakshmi¹, J. Lloyd¹, R. Gregorio¹, H. Dubner¹; ¹UPMC Shadyside Hospital, Pittsburgh, PA

LEARNING OBJECTIVES: Hypertrophic gastropathy is a diverse disease characterized by giant mucosal hypertrophy of gastric rugae. The disease is usually confined to gastric body and fundus and can manifest with hypoalbuminemia. We present a case of an extensive hypertrophic gastropathy involving the antrum and body of the stomach associated with iron deficiency anemia. **CASE INFORMATION:** A 41-year-old male patient during a routine physical, with symptoms of fatigue was diagnosed with iron deficiency anemia. Physical exam was unremarkable with guaiac negative stools. Hemoglobin 6.0, Ferritin 1, Protein 4 and Albumin .9.

Work-up of anemia included endoscopy, which showed diffuse nodularity of the body and a nodular mass in the antrum, which obscured pylorus and extended into second portion of the duodenum. Biopsies of stomach and duodenum were inconclusive. Upper GI series confirmed endoscopic findings. Due to the extensive involvement of the lesion as well as uncertainty of the diagnosis, gastrotomy was done. Large polypoid lesions were seen in the gastric antrum prolapsing into the duodenum and were excised. The mucosal surface of the duodenum was normal. Pathology was consistent with hypertrophic gastropathy. Stains and serum antibodies were negative for H. pylori.

Patient was treated with acid suppressive therapy and iron supplementation. On follow-up patient was free of symptoms with Albumin 3.7 and Hemoglobin 12.2.

DISCUSSION: This case is a relatively new variant of the already known hypertrophic gastropathy, in regards to its site and clinical presentation.

HIV RELATED HEMOLYTIC UREMIC SYNDROME. M. Rashid¹, M. Murtaza², A. Singh³, M. Kasubhai⁴, V. Dimitrov²; ¹Lincoln Medical and Mental Health Center, New York, New York; ²Lincoln Medical Center, Bronx, NY; ³Cornell University; ⁴Lincoln Medical and Mental Health Center, Bronx, NY

LEARNING OBJECTIVES: 1.-To recognize HIV infection as a risk factor for HUS. 2.-To Suspect HIV infection in HUS of undetermined etiology

CASE INFORMATION: A 48-year-old HIV positive Hispanic lady with no history of AIDS defining illness came to the ER with 3 days history of shortness of breath and pedal edema. Physical exam in the ER showed a distressed lady with tachycardia and tachypnea. She had petechiae over the legs, 3+ bilateral pedal edema and a right pleural effusion. Labs revealed severe anemia (hematocrit 21); normal platelet count (227k/ml), increased BUN and creatinine (81/5.3); increased LDH (579) and near normal PT and PTT (14.8/28.6). Peripheral smear showed schistocytes. Factor V and VIII were normal. Hospital stay was initially represented by persistent anemia, worsening thrombocytopenia (down to 29k/ml), bleeding from multiple sites, worsening renal and pulmonary function. Neurological exam remained normal. LDH remained high whereas PT and PTT remained mostly unchanged. CD4 count and viral load sent at this time came back as 89 (x 106/L) and 42 (k copies/ml) respectively. A diagnosis of HUS was made on clinical and laboratory parameters. No known risk factors were identified. Patient underwent plasmapheresis and hemodialysis. Her condition however continued to deteriorate and she died seven weeks post admission. Autopsy confirmed HUS and ARDS.

DISCUSSION: HUS has been seen mostly in terminal AIDS patients and thrombotic microangiopathy in AIDS patients is increasingly being considered a consequence of direct pathogenic effect of the virus over glomerular capillaries and arterioles. We were unable to identify any other risk factor for HUS in our patient as well. Prior to this admission our patient did not have any AIDS defining illness. We concur with others that HUS may be considered as an AIDS defining illness. Plasmapheresis was not associated with survival in our patient. The role of this mode of therapy in HIV patients needs to be studied further.

SUSAC SYNDROME: A CASE REPORT OF THIS RARE DISEASE OF MICRO-ANGIOPATHY OF THE RETINA, COCHLEA AND BRAIN. J.R. Ready¹, L.P. Taylor², K.M. Gross²; ¹Virginia Mason Medical Center, Seattle, Washington; ²Virginia Mason Medical Center, Seattle, WA

LEARNING OBJECTIVE: Distinguish Susac syndrome from more common diagnoses such as Systemic Lupus Erythematosus and multiple sclerosis (MS).

CASE INFORMATION: A previously healthy 53-year-old woman lost hearing in her right ear in 1994 over a three-week period of time followed by a progressive hearing loss in the left ear. MRI showed mild periventricular white matter disease and audiometric testing showed severe bilateral vestibular defects. In late 1997 she presented with recurrent pulmonary infections and was diagnosed with common variable immunodeficiency (CVID). In October 1998 she began having 10–15 minute episodes of ascending right or left hand tingling and weakness followed by slurred speech and a reported mild facial droop that resolved with verapamil in December. In April 2000 she presented with declining memory, concentration problems and diarrhea. Work-up revealed secondary adrenal insufficiency, hyperprolactinemia and decreased insulin-like growth factor 1 (IGF-1).

Extensive testing over six years included three head MRIs, two lumbar punctures, audiology testing, psychosocial evaluation, bronchoscopy, muscle biopsy, abdominal CT and numerous blood tests. Positive studies included high CSF protein of 64, oligoclonal bands, cortisol base 3.2 µg/dL (6–25) after stimulation 14.6 µg/dL (≥18 at 60 minutes after injection) and ACTH 5, prolactin 109ng/mL (1–24), and IGF1 50 ng/mL (71–240).

The lack of linear plaques and atrophy of the corpus callosum on the MRI scans were inconsistent with MS and neurology and otolaryngology consultants concluded that her progression of symptoms, imaging studies, and lab testing were most consistent with a diagnosis of Susac Syndrome.

DISCUSSION: There have been less than 50 reported cases of Susac syndrome in the literature. This is the first reported case of immune deficiency and endocrine abnormalities accompanying Susac's syndrome which is described as a clinical course of hearing loss, encephalopathy, and visual changes consistent with a microangiopathy of the cochlea, brain and retina. High suspicion and extensive testing is required to rule out other etiologies of this triad of symptoms.

A REFRACTORY NEUTROPENIC PNEUMONIA. S. Regenbogen¹, A. Rosen²; ¹University of California, San Francisco, CA

LEARNING OBJECTIVE: 1) Diagnose and treat Bronchiolitis Obliterans Organizing Pneumonia (BOOP) in patients with hematologic malignancies who present with pneumonia.

CASE INFORMATION: A 59-year-old man with cryptogenic pancytopenia was admitted with 3 weeks of cough, fatigue and fever. Bone marrow was markedly hypocellular, but non-diagnostic. He had no hematuria, rheumatic symptoms, or toxin or radiation exposure, except for 6 months on a Naval nuclear submarine. He attributed his condition to myelosuppression from chronic ibuprofen use. Admission vitals were temp 39C, BP 133/71, pulse 78, 02 sat 97% on room air. He had no bleeding or bruising, and no lymphadenopathy. Lung exam showed crackles and egophony at the right mid-chest. Spleen and liver were not palpable. CBC showed WBC 0.4, Hct 34, Plt 102, and ANC 0.2. CXR showed dense right-sided infiltrates. Blood and induced sputum cultures were repeatedly negative and broncho-alveolar lavage was non-diagnostic. Despite treatment with broad-spectrum antibiotics and antifungals, his lung exam and CXR continued to worsen. His neutropenia persisted despite 3 weeks of stem cell growth factors. On day 15, immunoperoxidase staining confirmed the diagnosis of hairy cell leukemia. He underwent open splenectomy, which increased his WBC to 3.13, and ANC to 2.35. Open lung biopsy (OLB) on day 22 was consistent with BOOP. He was treated with high dose steroids and discharged on a prolonged prednisone taper.

DISCUSSION: BOOP is an inflammatory response to lung injury, with edematous plugs of granulation tissue and interstitial fibrosis. It is associated with infection, autoimmunity, hematologic cancers, and other conditions. Patients present with a subacute, febrile illness with dry cough and dyspnea that is unresponsive to antibiotics. CXR shows patchy alveolar infiltrates with airspace consolidation, and PFTs reveal a mixed restrictive/obstructive pattern. Trans-bronchial biopsy is often nondiagnostic, therefore requiring OLB for definitive diagnosis. BOOP is usually responsive to steroids, but relapse is common. In one series (*Am J Resp Crit Care Med* 161:723, 2000), among patients with hematologic cancers undergoing OLB for an unknown pulmonary process, BOOP was the most common finding—20% of cases where a diagnosis was made. BOOP accounted for 30% of the diagnoses missed by bronchoscopy but

found on OLB. Identification of a specific diagnosis on OLB significantly increased survival. In the setting of refractory pneumonia in a patient with a hematologic malignancy, the diagnosis of BOOP should be considered. OLB is advised if other procedures fail to identify a cause.

METASTATIC GASTROESOPHAGEAL ADENOCARCINOMA TO SKELETAL MUSCLE-A UNIQUE EVENT. S.U. Rehman¹, D.W. Cope², J.N. Basile²; ¹Medical University of South Carolina, MOUNT PLEASANT, SC; MEDICAL UNIVERSITY OF SOUTH CAROLINA, CHARLESTON, SC

LEARNING OBJECTIVES: 1. Participants will learn to recognize skeletal muscles as a rare metastatic site for the cancers.

2. Participants will learn diagnosis and treatment of esophageal cancer.

3. Participants will learn the incidence, pathophysiology, diagnostic strategy and treatment of skeletal muscle metastasis.

CASE INFORMATION: Introduction: Although comprising 50% of body mass with a very rich blood supply, skeletal muscle is rarely the site of metastatic disease with only 242 cases having previously been reported (1). Although primary cancers of the lung, colon, genitourinary tract and blood are most frequently involved (1,35), gastroesophageal adenocarcinoma has never been reported. We report a case of thigh muscle metastasis from primary adenocarcinoma of the gastroesophageal junction.

CASE REPORT A 71-year-old African American male presented with severe right thigh pain, causing him to be unable to walk. His medical diagnoses included Stage IV Adenocarcinoma of the Gastroesophageal junction, hypertension, COPD (chronic obstructive pulmonary disease), and GERD (gastroesophageal reflux disease). He was an active smoker and Ex-alcoholic. Physical examination was remarkable only for his cachectic state and an extremely tender right upper thigh. Despite NSAID (non-steroidal anti-inflammatory drug) and full dose narcotic analgesia, the patient continued to complain of severe pain in his thigh. Thigh X-rays showed no lytic or blastic lesions. Bone scan did not show any metastasis. A MRI was done and showed a deep 2 x 4-cm soft tissue muscle mass in the right thigh without bony involvement, just anterior and medial to the femur. Diagnostic considerations included a soft tissue metastasis or a sarcoma. A CT guided needle biopsy of the right thigh mass was positive for metastatic adenocarcinoma consistent with the primary esophageal cancer.

Radiation therapy was started to the patient's right thigh with a good response. At the time of discharge, the patient was walking without pain receiving narcotic analgesia. Upon returning to the clinic after 2 weeks for follow-up, he continued to walk without difficulty receiving outpatient radiation therapy.

DISCUSSION: The incidence of adenocarcinoma of the esophagus and esophagogastric junction has been increasing over the past 25 years (2,3,4,5), with cancer of the esophagus (used to be predominantly squamous cell carcinoma) ranking among the ten most frequent cancers in the world.

Although direct muscle invasion by carcinoma is well recognized, distant metastasis to skeletal muscle is uncommon (6). The present case is unique in that the localized thigh pain was produced by metastatic involvement to the thigh musculature without any osseous or perineural lymphatic extension.

Even though autopsy series report a 0.8% to 20% (8,9,10,11,12) incidence of microscopic intramuscular metastasis, only a few cases of visible metastasis to the muscle have been described in the literature (7,13,14,15,16). Accordingly, muscle metastasis often remains asymptomatic often undetected by both physical examination as well as diagnostic imaging procedures. The reported incidence, therefore, might be infrequent because of either a lack of recognition, underreporting, or infrequent autopsy evaluation for muscle metastasis. (7). Furthermore, it may be that only a fraction of patients with metastases to muscle survive long enough to allow clinical detection (17). Just why metastases to skeletal muscle is so rare is still unknown (17,18). Multiple factors, such as blood flow, intramuscular blood pressure, blood flow per unit of weight (milliliters/minute per gram); local changes in pH, as well as local temperature distribution may be involved. (6,9,11,19,20, 21,22). Organs with a high incidence of metastatic carcinoma such as lung, liver and bone have a constant blood flow. Although equally rich in its vasculature, it has been suggested that the blood flow in skeletal muscle is variable, is under the influence of Beta-adrenergic receptor control, and is subject to varying tissue pressure that may affect tumor implantation (20,23). Although some have suggested that protease inhibitors in the muscle's extracellular matrix may resist tumor cell invasion (24), it may be the production of lactic acid and other metabolites by skeletal muscle that inhibits tumor cell growth. (20,18)

Whereas two thirds of all cancers metastatic to muscle are carcinomas, about one third are from leukemias and lymphomas with rare cases originating from melanoma. (25). Accordingly, although factors in the recipient site may be responsible for the relatively low rates of metastasis to muscle, properties of the primary tumor may also be involved. Furthermore, the differentiation between a primary soft tissue sarcoma and metastatic carcinoma to muscle is important as the treatment and prognosis is markedly different.

Although the presence of a soft tissue mass caused by metastatic carcinoma may be misdiagnosed easily as a soft tissue sarcoma on physical examination and imaging studies (10), the current literature does not provide any clinical or radiographic characteristics that helps distinguish the two (17). Whereas 50% of carcinomas and sarcomas metastatic to muscle occur in the lower extremity, a greater percentage of upper extremity (26%) carcinomatous metastases occurs than has been reported for soft tissue sarcomas (10%). In addition, most of the cases reported are located in one muscle group or in one part of the body (26,27,28).

Although various imaging studies were used to identify metastases to muscle, none were specific for differentiating among carcinoma, sarcoma, or other muscle disorders. Plain radiographs, radionuclide scanning, and angiography have not been beneficial in differentiating carcinoma from sarcoma (17) although it has been thought that MRI imaging is superior to CT scanning (29,30). CT guided fine needle aspiration provide a rapid, minimally invasive means of diagnosis (17).

Treatment options include radiation therapy, surgery or a combination of the two. Reports suggest surgical resection followed by adjuvant radiation or chemotherapy provides excellent palliative results (31,32,33). Although solitary metastases less than 4 cm in diameter can be treated by excisional biopsy (34), other reports suggest a less invasive approach may be better

due to a low incidence of functional disability and because of the poor survival of these patients. Surgical resection is often reserved for those lesions that fail to be controlled locally with radiation or when the tumor growth results in neurologic deficit.

SUMMARY: Although metastasis to skeletal muscle is extremely uncommon, physicians must remain aware of its occurrence, as its detection often requires specific evaluation. Differentiation between a primary soft tissue sarcoma and metastatic carcinoma to muscle is important, as their treatment and prognosis are so markedly different. The current literature does not provide any clinical or radiographic characteristic (plane x-ray or bone scans) that distinguishes metastatic carcinoma to muscle from soft tissue sarcoma (17). MRI, however, appears promising, and it should be considered earlier in the diagnostic work up. Treatment continues to remain palliative as metastatic carcinoma to muscle continues to remain a late event, with an overall poor prognosis.

SHOULD WE SCREEN FOR HEPATOCELLULAR CARCINOMA? Q. Roehrig¹; ¹UCSF Primary Care Internal Medicine Program, San Francisco, CA

LEARNING OBJECTIVES: 1)Recognize screening methods for Hepatocellular carcinoma (HCC) and the evidence for or against them. 2)Identify who should be screened for HCC.

CASE INFORMATION: A 46 year old woman with a history of injection (IV) drug use and alcoholism presents for establishment of primary care, without specific medical complaints. Her last IV drug use was over 10 years ago, is recently HIV negative, and doesn't recall hepatitis testing. She reports 15 yrs of intermittent heavy drinking. Her past medical history is also significant for Diabetes mellitus Type 2 and obesity. Her medications include methadone and metformin. Review of systems: negative history of jaundice, ascities, bruising, or gastrointestinal bleeding. Exam: Obese woman with normal vital signs and no scleral icterus. She had normal cardiac and pulmonary exams and her abdomen was obese few spider angiomas. Initial lab values: Hepatitis C antibody positive, B negative, Hct: 35.8, Platelets: 109, INR 1.1, Albumin 3.0, AST: 94 (Normal <40), ALT: 87 (Normal <40), Total Bili 1.1, Alk Phos 174 (Normal <130). Given the lab results and spider angiomas, an abdominal ultrasound (U/S) was ordered which showed signs of cirrhosis as well as a 1.4 cm liver "hemangioma". An alpha-fetoprotein (AFP) subsequently obtained was 42.0 (normal <20). The patient was referred to Hepatology, but then lost to follow-up. Repeat AFP 6 months later was 316. An abdominal CT was obtained to better characterize the lesion showed growth to 2.7cm. A biopsy of the lesion was not done due to its location but a routine liver biopsy confirmed cirrhosis. The patient underwent transarterial chemoembolization for presumed HCC, her AFP at this time was 551.

DISCUSSION: The major risk factor for development of HCC is cirrhosis, at a rate of 1–7% per year. The most common etiologies of cirrhosis are viral Hepatitis (B or C) and alcohol. Data for time to development of HCC range from 15–25 years. Survival rates are 50% at 3 years for resectable lesions and 20% at one year for non-resectable HCC. Screening tools: AFP > 20 sensitivity of 39–70%, U/S = 58–85%, CT = 46–84%, AFP and U/S combined = 79–90%. Unofficial screening guidelines, followed by many surveyed by Chalasani et al, aren't necessarily supported through evidence-based medicine (EBM). The 6 month screening periods initially arose from data on tumor doubling time. Currently, most hepatologists do every 6 month AFP or U/S in patients with documented cirrhosis, with CT imaging alternatively being used. A recent cost-effective analysis by Sarasin et al concluded that screening offered minimal survival benefits for the majority. The most significant benefits were found in well-compensated Child's A cirrhosis, especially with resectable tumors. There is a logical argument for screening liver transplant candidates. In summary, there is currently no EBM to support screening beyond the select group of patients mentioned above. There is a need for more research to better establish the mortality benefit and cost-effectiveness of HCC screening.

DOC, MY BACK HURTS: A SERIOUS CAUSE OF A COMMON COMPLAINT. A. Rosen¹; A.B. Rosen¹; ¹University of California, San Francisco, CA

LEARNING OBJECTIVES: 1) Recognize clinical manifestations of spinal epidural abscess (SEA), 2) Review indications for emergent imaging to rule out SEA.

CASE INFORMATION: A 52 year old man with a history of low back pain and alcoholism presented to his PMD with severe low back pain. He was given Vicodin, but returned to clinic 5 days later with worsening pain, constipation, difficulty walking, and a fever. Pt was sent to a local hospital where he was found to be confused and febrile with bilateral chest infiltrates; last drink had been 2 days prior. He was treated for pneumonia and presumed alcohol withdrawal. 4/4 blood cultures grew out S. aureus, but TTE showed no valvular vegetations. 4 days later, pt was transferred to our hospital where he was febrile to 103, agitated, disoriented, unable to follow commands, but spontaneously moving all extremities. To the extent testable, neurologic exam was nonfocal except for marked low back pain on neck flexion and possible nuchal rigidity. He had no spinal tenderness. He was pan-cultured and a head CT showed no acute event. MRI and TEE were planned for the following morning to rule out SEA and endocarditis, respectively. Lumbar puncture was initially deferred because of concerns for possible SEA and because pt was already on meningitis-dose Nafcillin. The following morning (5 hrs after admission) the patient no longer spontaneously moved his lower extremities and his toes were upgoing bilaterally. Emergent MRI showed a large T9 epidural abscess with posterior and anterior extension. Patient underwent emergent neurosurgical evacuation and had slow but progressive improvement in lower extremity function over the next 2 weeks. One week after transfer to rehabilitation, he expired in the middle of the night. Autopsy showed reaccumulation of the epidural abscess with infarction of nearly half of his spinal cord.

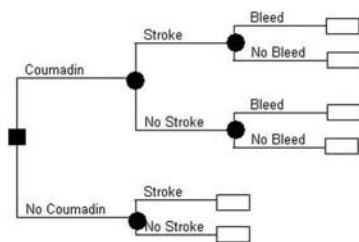
IMPLICATIONS/DISCUSSION: Spinal epidural abscess is a rare but serious infection with potentially disastrous consequences, as highlighted in this case. Entertaining the diagnosis early is the key to prompt discovery and treatment. Severe back pain in a febrile patient, particularly with other risk factors for SEA (IVDU, past back surgeries, bacteremia), warrants rapid workup. Spinal MRI should be done as soon as the diagnosis is entertained, ideally before any neurologic deficits develop. In most cases, spinal cord damage occurs suddenly secondary to cord infarction from thrombosis or interruption of the local venous or arterial supply rather than via cord compression. Management requires neurosurgical evacuation and antibiotics (etiology is S.

aureus in >60% of cases). Early surgical intervention (<24hrs) is associated with markedly better prognosis.

WHEN THE DATA ARE NOT ENOUGH: SHARED MEDICAL DECISION MAKING. A. Rosen¹; ¹University of California, San Francisco, CA

LEARNING OBJECTIVES: 1) Increase awareness of decision analysis as a tool for shared medical decision making (SMDM), 2) Recognize the importance of patient preferences in making medical decisions.

CASE INFORMATION: A 77yo male with paroxysmal atrial fibrillation and recent diagnosis of colon cancer with hepatic metastases presented to my clinic. His quality of life (QOL) and functional status were excellent and he desired no adjuvant chemotherapy unless his QOL significantly worsened. He presented questioning his need for coumadin. We constructed a "back-of-the-envelope" decision tree to address this question.



In conjunction with a review of the literature and discourse with the cardiology and oncology services, his risk of significant bleeding on coumadin with known liver metastases was thought equal to his risk of stroke off coumadin. In discussing his preferences for health states, his fear of mental incapacity and other potential sequelae of stroke far outweighed his distility for a bleeding complication of the coumadin. These preferences were so strong that coumadin remained the preferred treatment modality even when aspirin was presented as an option. While the patient initially desired cessation of coumadin, this decision changed following the presentation of a simple decision tree and open discussion of his preferences.

DISCUSSION: Decision analysis allows for the implicit decisions made everyday in clinical medicine to be made more explicit. The process of SMDM is often very illuminating, as it encourages better communication between patient and physician as they consider the consequences of a particular medical decision together. These consequences include the probabilities of health outcomes and the values, or utilities, that the patient places on these outcomes. Explicit knowledge of patient utilities may have a major impact on the medical decision particularly in a setting such as this one in which quality of life may be valued over life expectancy. SMDM is an excellent way to engage patients in the important dialogue needed to best inform medical decision making.

REVERSIBLE HYPOXEMIC CARDIOMYOPATHY. M. Saleem¹, N. Rahim², W. Lee³, J. McClung², S.J. Peterson⁴; ¹Yonkers, New York; ²Westchester Medical Center / New York Medical College, Valhalla, NY; ³None Given, Valhalla, NY; ⁴New York Medical College, New York, NY

LEARNING OBJECTIVE: Recognize that severe hypoxemia can cause profound myocardial depression which is reversible with aggressive therapeutic measures.

CASE INFORMATION: A 49-year-old woman with mild asthma developed acute severe dyspnea and wheezing while cleaning her home. She presented with status asthmaticus and was intubated. Admission chest x-ray did not show infiltrates or effusions. Antibiotics and intravenous steroids were started in addition to supportive care. On the second hospital day she developed bilateral alveolar infiltrates on chest film and increasing oxygen requirements. Pulmonary embolus was ruled out with a CT angiogram, and bronchoscopy revealed diffuse mucosal edema. Notable labs at this time were a white cell count of 23000, Hct 41, CPK 222, CK-MB 16 and Troponin-I 30 with normal electrolytes. Electrocardiogram showed T wave inversions in anterior and lateral leads. Transthoracic echocardiogram showed diffuse ventricular hypokinesis and a left ventricular function of less than 25%. Cardiac catheterization showed normal coronaries, right atrial pressure of 20, pulmonary capillary wedge pressure of 30, ejection fraction of 20% and severe hypokinesis of the left ventricle. She was started on digoxin, an ACE inhibitor and aggressively diuresed. Her oxygen requirements decreased significantly. A repeat transthoracic echocardiogram showed a normal LVEF and no hypokinesis. She was successfully extubated and transferred from the intensive care unit.

DISCUSSION: Acute reversible cardiomyopathy is uncommon. Most commonly it is associated with ischemia by causing stunning of myocardium. Reversible cardiomyopathy has been rarely described in association with pheochromocytoma, sub arachnoid hemorrhage and thyrotoxicosis. We describe a case wherein severe hypoxemia in the absence of coronary artery disease caused profound myocardial depression leading to heart failure. Complete echocardiographic recovery of heart function was seen on day 3 by reversing the hypoxemia. We believe this is the first reported case of reversible hypoxemic cardiomyopathy and this has far reaching implications. It could complicate any severe respiratory distress by causing

myocardial dysfunction, which worsens oxygenation, and create a vicious cycle leading to a fatal outcome. Aggressive therapeutic measures to reverse hypoxemia may prevent this downhill course.

PULMONARY THROMBOSIS IN BEHCET'S DISEASE: IN SITU PULMONARY VASCULITIS VERSUS HYPERCOAGULABILITY. M. Saleem¹, F. Rahim², S.J. Peterson³; ¹Yonkers, New York; ²Westchester Medical Center / New York Medical College, Valhalla, NY; ³New York Medical College, New York, NY

LEARNING OBJECTIVE: Pulmonary thrombosis in Behcet's disease can be secondary to in situ vasculitis or characteristic hypercoagulable state of Behcet's disease, so treatment should be directed at both pathogenic mechanisms (i.e. steroids and anticoagulation).

CASE INFORMATION: A 38-year-old male with a past history of Behcet's disease developed shortness of breath, pleuritic chest pain and dry cough one day prior to his presentation to our hospital. He denied hemoptysis, palpitations and urinary or bowel complaints. On examination vital signs were pulse 118, BP 130/80, temp. 101 and respiratory rate of 16. He had painful ulcers in the mouth, crackles at right lung base, a normal cardiovascular examination and erythema nodosum on lower extremities. An eye examination was normal and there were no genital ulcers. Laboratory work up showed white cell count of 13,900, hematocrit 34.8, ESR 95, CPK 43 with normal electrolytes and coagulation profile. His EKG showed normal sinus rhythm with a heart rate of 95 beats per minute and chest x-ray showed right lower lobe infiltrate and a small pleural effusion. An arterial blood gas showed pH of 7.45, pco2 of 36, po2 of 72 and oxygen saturations of 95% at room air with an A-a gradient of 33. Doppler ultrasound of the lower extremity was negative but CT scan showed pulmonary thrombosis in a segmental branch of the right pulmonary artery and right lower lobe atelectasis with small pleural effusion. He was treated with pulse steroids and anticoagulation.

DISCUSSION: BEHCET'S disease is an inflammatory disease of unknown cause, characterized by recurrent aphthous ulcers, uveitis, genital ulcers, and skin involvement. All these common manifestations are self-limiting except for uveitis that can lead to blindness. Involvement of the gastrointestinal tract, central nervous system, and large vessels is uncommon but can be life threatening.

The etiology of pulmonary thromboembolism in Behcet's disease may be due in part to vasculitis or the characteristic hypercoagulability associated with Behcet's disease. In studies of histopathology and immunohistopathology of Behcet's disease, patients have shown markedly impaired prostacyclin biosynthesis compared to controls, thus, tipping the balance in favor of hypercoagulable state. However, in other studies, histopathological findings were compatible with vasculitis. The immunohistology of vessel walls revealed perivascular infiltration of CD4+ T helper cells, CD8+ T suppressor cells and HLA DR+ cells thus indicating endothelial cell dysfunction due to possibly immune mediated vasculitis.

The question remains which particular mechanism is working at one point in time. Since it is not possible to biopsy every pulmonary thrombosis in patients who also have Behcet's disease. The treatment needs to be directed at both pathogenic mechanisms. This includes anticoagulation for thrombosis and anti-inflammatory medication for vasculitis (steroids or colchicine). We elected to treat with coumadin, steroids and colchicine.

PLEURAL CAVITY-A RARE SITE FOR CHLOROMA IN BLAST CRISIS OF CML. I. Saleem¹, M.H. Rashid¹, M. Singh², M. Kasubhai³, V. Dimitrov⁴; ¹Lincoln Medical and mental health center, New York, NY; ²Lincoln Medical Center, Bronx, NEW YORK; ³Lincoln Medical and Mental Health Center, Bronx, NY; ⁴Lincoln Medical Center, Bronx, NY

LEARNING OBJECTIVES: 1> To recognise a rare extramedullary site for blast crisis in CML 2> To recognise an unusual etiology of persistent pleural effusion in CML

CASE INFORMATION: A 51 years old Hispanic lady who had been diagnosed with CML (Philadelphia chromosome positive) a month ago was admitted to our hospital with complaint of left hypochondrial pain for 2 days. Upon physical examination she was hemodynamically stable, afebrile with no lymphadenopathy. There was mild splenomegaly and a left sided pleural effusion. Laboratory data showed Hb 13.6, platelets 579x109/liter and WBC count of 78.4x109/liter with 1% blast cells. Diagnostic pleurocentesis revealed pH 7.7, protein 3.7, LDH 743, Glucose 140 and a WBC count of 1155 with 56% monocytes, 19% polymorphs, 21% lymphocytes and 1% mesothelial cells. Pleural fluid to serum ratio of protein was 0.57 and of LDH was 0.55. Possible etiology of this pleural effusion was considered to be chloroma, mycobacterium tuberculosis or pneumonia. Pleural fluid cultures were negative for bacteria, fungi and mycobacterium tuberculosis. No malignant cells were seen on cytological examination and pleural biopsy did not reveal any specific lesion Patient was initially started on hydroxyurea and was subsequently discharged on interferon with a plan for bone marrow transplantation. Two months later during a follow up visit at the clinic a CT Scan of chest showed persistent pleural effusion with a density at the left pleural base. Patient did not consent for biopsy of the pleural mass. Within six months the CML progressed to the blast phase, despairing all hopes for cure.

DISCUSSION: CML usually changes it's clinical and hematological course, passing through an accelerated phase and terminating in acute leukemia — blast crises (either in the bone marrow or extramedullary). Very few cases of CML with infiltration of malignant cells in the pleural cavity have been reported in the literature. Preceding to the development of clinically overt blast crisis our patient had had a persistent massive pleural effusion with a pleural mass. This was considered to be, most likely the site of extramedullary blast crises — a very unusual presentation

SKIN NECROSIS: WARFARIN VERSUS HEPARIN. WHICH WAS THE CULPRIT? E. Sarpong¹; ¹COOK COUNTY HOSPITAL, CHICAGO, IL

LEARNING OBJECTIVES: 1. Recognize the possible risk factors for anticoagulant related skin necrosis. 2. Importance of early detection of skin necrosis in patients on anticoagulants.

CASE INFORMATION: 46 year old man presented with a 2 week history of right upper quadrant pain, weight loss and a cough productive of blood streaked sputum. Physical exams was unremarkable. A left upper quadrant ultrasound showed multiple solid appearing hypochoic

masses in the liver and a 4cm mass in the tail of the pancreas. VQ scan of the lungs was high probability for PE. A presumptive diagnosis of metastatic pancreatic carcinoma with pulmonary embolism was made.

Day 1, CT guided biopsy of the liver lesions done and IV unfractionated heparin started. Day 3, warfarin introduced. Day 5, platelet count dropped from 230 to 47. Heparin infusion was discontinued and warfarin was continued. On the same day he had a burning pain over his right shoulder and forearm. Day 6, well demarcated edema and erythema of his right shoulder and forearm and later purpura and skin necrosis occurred. Warfarin was discontinued and he was offered lepirudin. Biopsy of the skin confirmed anticoagulant related skin necrosis. Day 11, platelet count increased to 150. Platelet factor 4 antibody was positive confirming heparin induced thrombocytopenia (HIT). The liver biopsy showed metastatic adenocarcinoma.

DISCUSSION: Warfarin skin necrosis is a rare complication but could lead to extensive skin loss with its associated problems such as is seen in major burns. The patient usually complains of pain of the affected area and this should prompt the physician to consider it in any patient on anticoagulation who has unexplained cutaneous pain. It may be minimized by discontinuing the warfarin and substituting with heparin, heparinoids or lepirudin if anticoagulation is needed. If INR is high, reverse anticoagulation. Avoid initial "loading" dose of warfarin because this will lead to rapid depletion of protein C leading to a relatively hypercoagulable state. This risk appears to be particularly high in inherited hypercoagulable states and in patients with HIT. Heparin skin necrosis may occur with all types of heparin and could be at or away from the site of injection. Antibodies to heparin-platelet-factor 4 is suggestive. Give warfarin or lepirudin if needed. It is not possible to tell if this patient had warfarin or heparin skin necrosis. A method of distinguishing these two conditions is needed as treatment is different.

DELIRIOUS PATIENT, MADDENING DIFFERENTIAL. D. Satran¹, W. Stansbury-o'donnell¹; ¹Hennepin County Medical Center, Minneapolis, MN

LEARNING OBJECTIVES: 1) Assess delirium in an elderly patient. 2) Diagnose and treat theophylline toxicity.

CASE INFORMATION: A 73 year-old male with a history of type 2 diabetes, hypertension, asthma, and obstructive sleep apnea presented with a 3 week history of increasing confusion and 3 day history of right-sided frontal headache. His medications included thyroxine, oxybutynin, and theophylline. One week following recent knee surgery, he stated that his energy level was decreased and he was having persistent memory difficulties.

Physical examination was unremarkable except for delirium and a resting tremor. The patient was alert, but oriented only to self and location. He had difficulty cooperating with the examination and was easily distracted from tasks. He was admitted to the hospital, where outpatient medications were restarted. Chemistry panel, CBC, LFT's, thyroid function tests, blood cultures, and cardiac enzymes were all normal. Theophylline level was high normal (19.9 mg/L). An EKG was read as sinus arrhythmia. A head CT showed ventriculomegaly with no intracranial bleed or mass. Results of lumbar puncture, EEG, MRI of the brain, and overnight pulse oximetry were nondiagnostic.

On day 4 of hospitalization, confusion persisted. The patient experienced a 9 beat run of ventricular tachycardia and a cardiology consult was obtained. Review of EKG's and a rhythm strip revealed multifocal atrial tachycardia. A theophylline level was obtained and reported at 39 mg/L. Theophylline was discontinued. Over the next 7 days, the patient's mental status gradually cleared and he was discharged.

DISCUSSION: Theophylline has a narrow therapeutic index (10 to 20 mg/L). Hypotension, cardiac arrhythmias, and seizures have been reported in association with toxicity. Levels often do not correspond to symptoms in chronic toxicity. Treatment is mainly supportive, although dialysis is often used in management of severe toxicity.

The case illustrates: 1) the broad differential diagnosis and resource intensive evaluation of delirium in the elderly and 2) the insidious nature of chronic poisoning from a prescribed medication.

ROSIGLITAZONE AND EXACERBATION OF CONGESTIVE HEART FAILURE. M. Sattari¹, D.W. Brady²; ²Emory University, Decatur, GA

LEARNING OBJECTIVE: 1. Recognize the potential for congestive heart failure exacerbation by rosiglitazone.

CASE INFORMATION: A 70 year old male with past medical history of congestive heart failure (CHF), coronary artery disease, peripheral vascular disease, type 2 diabetes mellitus, chronic renal insufficiency, and anemia was started on rosiglitazone during a clinic visit. At that visit, chronic +1-2 pitting edema of his lower extremities was noted, but he denied chest pain, shortness of breath, or dyspnea upon exertion. Two weeks later, he presented to the emergency room, with a 2 week history of worsening lower extremity edema, dyspnea upon exertion, orthopnea, and increased abdominal girth. Physical exam revealed moderate distress with use of accessory muscles, jugular venous distention, sustained point of maximal impulse, murmur of mitral regurgitation, bilateral pulmonary crackles, abdominal distention, scrotal and presacral edema, and +3-4 pitting edema of lower extremities extending above the level of waist. These symptoms had not been relieved by increasing the dose of his outpatient oral lasix. Patient was admitted for treatment of CHF exacerbation with intravenous diuretics. Potential causes of CHF exacerbation such as myocardial infarction, arrhythmias, myocarditis, endocarditis, thyrotoxicosis, worsening renal failure, anemia, infection, medical or dietary non-compliance, pulmonary embolism, and uncontrolled hypertension were considered and systematically ruled out. An echocardiogram revealed an ejection fraction of 30-35%, a value comparable to 8 months earlier when the patient was compensated. No new wall motion or valvular abnormalities were detected. With discontinuation of the rosiglitazone, the patient exhibited a large diuresis for approximately 48 hours; however, the remainder of his symptoms resolved more slowly.

DISCUSSION: Rosiglitazone is a non-sulfonylurea insulin secretagogue that was recently approved by the FDA for use in diabetes. No cases have been reported yet regarding a possible relationship between this medicine and CHF exacerbation. The Medical Letter reports that its use is relatively contraindicated in CHF and that fluid retention is one of the potential side effects observed with this medicine. Troglitazone was another thiazolidinedione that was

recently taken off the market. A case of significant weight gain with Rezulin (troglitazone) therapy has been reported. It has been postulated that troglitazone caused plasma volume expansion by affecting renal fluid and electrolyte regulation. Potential questions relevant to use of the newest member in the thiazolidinedione family in patients with history of CHF are the mechanisms by which it causes fluid retention, whether it has any direct effect on cardiac function, and whether it exacerbates CHF by introducing a volume load to the compromised ventricles.

A DIAGNOSTIC DILEMMA — NONRESOLVING HYPOXIA. G. Sekhon¹, L. Rohr¹, A. Foulen²; ²Cook County Hospital / Rush University, CHICAGO, IL

LEARNING OBJECTIVES: 1) Recognize amiodarone toxicity as a cause of cough and dyspnea and differentiate from more common pulmonary disorders.

2) Recognize clinical findings of amiodarone toxicity

CASE INFORMATION: An 80-year-old woman with a history of hypertension, aortic and mitral valve replacement, and congestive heart failure was admitted with shortness of breath, cough with white sputum, orthopnea and PND for three weeks. Patient had been taking amlodipine, furosemide, amiodarone, captopril, hydrochlorothiazide, and warfarin for 5 years. On exam the temperature was 98.6, pulse 102, BP 161/78, RR 30. There was no JVD. CV exam-metallic click, normal S1, S2 no gallop or murmur. RS- mild bibasilar crackles. Her labs were significant for INR of 4.3. WBC - 5.9. Hemoglobin -12.2. ABG-7.48/pCO2-28/ pO2-46/ HCO3-20. Chest radiograph showed bilateral lower lobe infiltrates. EKG revealed sinus tachycardia. VQ scan was read as low probability. The presumptive diagnosis of congestive heart failure vs pneumonia was made. There was no improvement after 3 days of treatment with antibiotics and diuretics, prompting the team to search for an alternative diagnosis. Upon reexamination her eyes had yellowish brown spots around the cornea. Areas of scant bluish discoloration were noted on her shoulder. Her clinical presentation and lack of response to treatment raised the possibility of amiodarone toxicity. Pulmonary function testing revealed decreased DLCO and TLC. ECHO- Normal left ventricular systolic function. The offending drug was discontinued. Patient began to note symptomatic improvement within 5 days of discontinuing the drug.

DISCUSSION: The patient's clinical presentation and CXR findings were attributed to amiodarone toxicity after other pulmonary and cardiac causes were excluded. In several large series pulmonary toxicity was found to develop in 3.7-27% of patients taking amiodarone (Kennedy, Clinics of Chest Medicine, 3/90). The most common presentation is that of a relatively indolent subacute illness characterized by dyspnea and/or cough. The CXR usually reveals diffuse parenchymal infiltrates. Patients may present with a more abrupt onset of acute illness with fever cough CXR mimicking an infectious pneumonia or ARDS in extreme cases. BAL and biopsy are nonspecific, and diagnosis is made clinically. A DLCO reduction of 20% or greater and a decreased TLC support the diagnosis. The prognosis is generally good with the majority of patients seeing resolution of symptoms within 60-80 days of discontinuing the drug. In cases that fail to resolve, steroid therapy has been used.

AN UNUSUAL ETIOLOGY AND PRESENTATION OF CARDIAC TAMPONADE. G. Alghadban¹, A. Hamad¹; ¹Cook County Hospital / Rush University, CHICAGO, IL

LEARNING OBJECTIVES: 1) Recognize hypothyroidism as a cause of cardiac tamponade. 2) Diagnose cardiac tamponade in a case with an atypical clinical presentation.

CASE INFORMATION: A 53-year-old Filipino woman who had a total thyroidectomy eight years ago, on no hormone replacement, presented with symptoms of hypothyroidism, including cold intolerance, constipation, and alopecia. In addition, she noticed increasing dyspnea on minimal exertion. On exam, her temperature was 97 F with a blood pressure of 160/90 and heart rate of 56. There was no JVD or pulsus paradoxus. The heart sounds were clearly audible, and the lung fields were clear. No pedal edema was noted. The neuro exam demonstrated delayed relaxation of the deep tendon reflexes. Her labs were significant for a TSH of 165 IU/ml with an FT4 of 0.2 ng/dl. Her CXR showed cardiomegaly with normal pulmonary vascularity. Her EKG revealed low voltage in both the limb and precordial leads. Therefore, an echocardiogram was performed which showed a large pericardial effusion with right atrial and right ventricular collapse. Treatment with thyroxine was initiated. Right heart catheterization demonstrated equalization of diastolic pressures. Pericardiocentesis removed 750cc of pericardial fluid, and the intrapericardial pressure dropped from 16 to 2 mmHg. Tests for AFB, cytology, ANA, and gram stain were negative. The patient's dyspnea subsequently resolved, and her blood pressure normalized.

DISCUSSION: This patient's effusion was attributed to hypothyroidism, after other causes had been excluded. A recent prospective study of patients with large effusions revealed no cases related to hypothyroidism, but this is likely due to the typical early diagnosis and treatment of this condition. (Corey et. al., AJM, 1993). However, severely hypothyroid patients may be more likely to develop pericardial effusions. The literature suggests that hypothyroid effusions rarely cause tamponade, but this complication has been occasionally reported. In addition, this patient presented with subacute symptoms. She did not have JVD, hypotension, a pulsus paradoxus, or narrow pulse pressure, but was diagnosed with chronic tamponade and improved symptomatically with pericardiocentesis. Her hypertension resolved after the procedure, suggesting that the cause was increased peripheral vascular resistance due to low stroke volume. Atypical presentations of cardiac tamponade have been well documented, and it is prudent to consider this diagnosis in all patients with moderate to large effusions.

A CONTINUOUS CONUNDRUM: A RUPTURED ANEURYSM OF THE SINUS OF VALSALVA. G. Sekhon¹, B. Lucas²; ¹Cook County Hospital / Rush University, CHICAGO, IL; ²Cook County Hospital, Dept of Medicine, Chicago, IL

LEARNING OBJECTIVES: 1) recognize the typical presentation of a ruptured aneurysm of the sinus of Valsalva (ASV) is gradual in onset and 2) recognize the hallmarks of ASV on physical exam: a wide pulse pressure and a continuous murmur.

CASE INFORMATION: 27 year-old Hispanic woman without significant past medical history presented with four months of palpitations and non-exertional chest tightness. She denied any visual changes or weakness. She could walk briskly for ten blocks without dyspnea. She was taking no medicines. She denied a history of rheumatic fever, syphilis, or recent trauma. She did not use cigarettes or illicit drugs and had no family history of cardiac disease. On exam, BP 107/44, P 86, R 26, temp 98.6°C; no respiratory distress; no elevated JVP but bounding carotid upstrokes without bruit; systolic thrill across the upper sternum; a non-displaced, non-diffuse PMI; faint S2 and continuous (systolic-diastolic) murmur (crescendo mid-systolic 4/6; holodiastolic 3/4) loudest at left third intercostal space; normal breath sounds. ECG showed normal sinus rhythm without abnormalities. Chest X-ray revealed mild cardiomegaly without consolidation or effusion. Normal valves were seen on transthoracic echocardiogram (TTE) with moderate to severe aortic regurgitation and normal LV function. Subsequent transesophageal echocardiogram (TEE) confirmed the diagnosis of a ruptured ASV by demonstrating a regurgitant flow from an aneurysm of the right coronary sinus to the pulmonary outflow tract. Surgery was called for repair of the ruptured ASV.

DISCUSSION: Ruptured ASV are rare. They present at the median age of 30 years-old. They are thin-walled, tubular outpouchings that nearly always arise from the right sinus (> 90%) and rupture into the right ventricle. Though commonly associated with other cardiac anomalies (ventricular septal defects, bicuspid aortic valves) that may worsen the clinical presentation, if ASVs are isolated, they present gradually with symptoms of palpitations, fatigue, exertional dyspnea, or syncope. Exam is notable for a wide-pulse pressure and a continuous murmur loudest along the left sternal border (3rd intercostal space) that, unlike the murmur of a patent ductus arteriosus (PDA), does not extinguish as you move caudally down the anterior chest. Causes of diastolic-systolic murmurs include other continuous murmurs (PDA, coronary arteriovenous fistula) and the "to-and-fro" murmur of atrial septal defect with mitral stenosis. Our patient's presentation was typical for a ruptured ASV. The gradual onset of her symptoms together with a widened pulse pressure and a continuous murmur directed the initial work-up. However, the TEE results were confusing (a normal aortic valve with moderate to severe regurgitation in the presence of normal LV function); this led to a TEE that finally confirmed the diagnosis of a ruptured ASV.

GETTING INTO THE STREAM OF THINGS: A CASE OF IATROGENIC URINARY RETENTION. A.N. Shah¹, R. Granieri²; ¹9W-23 Montefiore University Hospital, Pittsburgh, PA; ²University of Pittsburgh, Pittsburgh, Pennsylvania

LEARNING OBJECTIVES: 1. To recognize overflow incontinence and its treatment. 2. To recognize the potential adverse effects of not taking a complete medication history.

CASE INFORMATION: AJ is a 72 year old male with benign prostatic hypertrophy who was evaluated in an emergent care setting for anuria. The patient had longstanding nocturia and dribbling but recently sought subspecialty urologic care for worsening of his original complaints. He was prescribed finasteride and prazosin. One week prior to this presentation, he was also seen by his primary care physician for upper respiratory symptoms. His current complaints included lower abdominal fullness and inability to void, despite the urge to, for 24 hours. Physical exam revealed an afebrile male in moderate distress with suprapubic distention. A bladder scan demonstrated a residual urine of greater than 800 cc; bladder catheterization produced one liter of clear, yellow urine. Urinalysis was within normal limits. A review of the patient's medication list revealed that he had most recently been prescribed phenylpropanolamine and guaifenesin and had also been taking oxybutin for "incontinence".

DISCUSSION: Benign prostatic hypertrophy (BPH) is a common problem, ranging from 40% - 50% in males 51-60 years old, to over 80% in males above 80 years of age. Males with BPH have symptoms suggestive of overflow incontinence. Pharmacologic therapy includes anti-adrenergic agents to assist with detrusor contraction and relaxation of the urinary sphincter. In addition, current literature provides some evidence for 5-alpha reductase inhibitors (finasteride) to decrease testosterone-dependent prostate gland enlargement. Phenylephrine, a popular over the counter and prescription medication, is commonly used as decongestant. It has been associated with acute urinary retention. It would follow that phenylpropanolamine (PPA), also a sympathomimetic, has similar capabilities. In this scenario, the patient's BPH in combination with PPA and the anticholinergic effect of the oxybutin precipitated acute bladder outlet obstruction and anuria. Although PPA has been taken off the market recently, other sympathomimetic drugs can be inferred as equally hazardous in these patients. One of the many ways to prevent adverse outcomes from polypharmacy is to take a medication history every time one sees a patient and to be cognizant of potential adverse medication interactions. By reviewing all current medications, we may reduce iatrogenesis.

GRAVE CONSEQUENCES OF HYPERTHYROIDISM AND A SWOLLEN LEG. N. Shah¹, R. Granieri¹; University of Pittsburgh, Pittsburgh, PA

LEARNING OBJECTIVES: 1. To recognize hyperthyroidism as a common predisposing factor to atrial fibrillation

2. To recognize an atypical manifestation of hyperthyroidism

3. To recognize the limitations of a doppler ultrasound for deep venous thrombosis

CASE INFORMATION: NP is a 47 year old female who suffered a right sided stroke one month prior to presentation. At that time she was found to be in atrial fibrillation and was anticoagulated to a therapeutic INR. One week prior to presentation, she had a CT scan of her head. She then presented with a several day history of right leg swelling, palpitations, and fatigue. She denied trauma, insect bites, or fever. An INR checked the previous week was therapeutic. A doppler ultrasound was consistent with deep venous thrombosis, but the suspicion of a compartment syndrome prompted a surgical consult. She was thought not to have a compartment syndrome and was admitted for systemic anticoagulation and venacaval filter placement. Her physical examination revealed pale conjunctiva, mild exophthalmos, lid lag and thyroid bruits. Her cardiac exam showed an irregular tachycardia at 125. Her right leg was swollen and tense, but nontender and nonerythematous. Her sensation was diminished. A distal pulse was dopplerable. Her labwork revealed H/H 8.5/25, platelets 158, PTT 47, PT > 100, an INR too high to be measured, TSH < .018 and T4 15. Doppler ultrasound was consistent with a

DVT but an MRI demonstrated extensive soft tissue bleeding. Anticoagulation was withheld and her coagulopathy resolved as the hyperthyroid state was treated.

DISCUSSION: Atrial fibrillation (AF) is a common arrhythmia, found in 1% of patients over age of 60. It is the most common cardiac condition associated with cerebral embolism. Underlying hyperthyroidism occurs in up to 12% of patients with this arrhythmia and therefore, a TSH is recommended in patients presenting with new onset AF. Our patient had an embolic right middle cerebral artery stroke due to AF and her AF was undoubtedly due to unrecognized hyperthyroidism. In addition, an atypical manifestation of hyperthyroidism was present. As has been reported in the literature, the hypermetabolic state of hyperthyroidism, exacerbated by a recent iodine load from her CT scan, led to consumption of clotting factors and a clinically relevant coagulopathy which resolved with treatment of the hyperthyroid state. Finally, in this patient, the extrinsic compression of the venous system by the bleed was initially incorrectly interpreted as a DVT. Only by careful physical exam and attention to pathophysiologic and laboratory details of the case was potentially harmful treatment with heparin avoided.

PHEOCHROMOCYTOMA RELATED CARDIOMYOPATHY. R. Shetty¹, E. Warm¹; ¹University of Cincinnati, Cincinnati, OH

LEARNING OBJECTIVES: 1) Recognize the cardiovascular manifestations of pheochromocytoma, and

2) Treat pheochromocytoma related cardiomyopathy.

CASE INFORMATION: A 36-year-old male without past medical history presented with two-day history of nausea, vomiting, lethargy, decreased appetite, and light-headedness. He also noted one week of constipation, bilateral pedal swelling, numbness and mottling. Echocardiogram on admission showed biatrial enlargement, mild right ventricular enlargement with diffuse hypokinesis, moderate concentric left ventricular hypertrophy, diffuse left ventricular hypokinesis, and estimated left ventricular ejection fraction of approximately 30%. During hospitalization the patient was noted to have jugular venous pressure of 6cm, irregularly irregular rhythm, soft S3 gallop audible at the apex, and grade III murmur at the left sternal border. After extensive work-up he was found to have an adrenal mass extending to the liver. Fine needle aspiration was consistent with pheochromocytoma. In addition he had elevated 24-hour urine metanephrine and normetanephrine. The patient was initially treated at an outside hospital with intravenous labetalol to control his heart rate. He was subsequently changed to phenoxybenzamine and titrated up to 80 mg twice a day. Repeat echocardiogram showed normal left ventricular systolic function.

DISCUSSION: Pheochromocytoma is a rare neuroendocrine tumor derived from enterochromaffin cells. Cardiovascular manifestations of this tumor include dilated and hypertrophic cardiomyopathy, acute pulmonary edema, coronary vasospastic syndrome, and supraventricular and ventricular arrhythmias.

Pheochromocytoma related cardiomyopathy has been linked to alpha-adrenergic receptor stimulation by high concentration of norepinephrine. In addition, reduction in coronary blood flow may have a contributing factor. The severity of injury has been linked to the extent of catecholamine surge. Complete reversal of this cardiomyopathy can be seen with both surgical removal of the tumor or aggressive medical management. This represents a case of reversal of left ventricular dysfunction after appropriate medical management with alpha-receptor blockade and intermittent use of beta-receptor blockers.

JOINT PAINS AND A RASH IN A YOUNG HOMOSEXUAL MALE. H. Short¹, R. Granieri²; ¹University of Pittsburgh, Pittsburgh, PA; ²University of Pittsburgh Medical Center, Pittsburgh, PA

LEARNING OBJECTIVES: 1) Recognize the presentation of disseminated gonococcal infection (DGI). 2) Recognize that gonococcal arthritis is the most common infectious arthritis in sexually active adults.

CASE INFORMATION: D.M. is a 45 year-old male who presented with arthralgias and rash. Three days prior, he developed an inflamed painful right second carpometacarpal joint. Over 2 days, symptoms progressed to include a painful left foot and right wrist, bilateral ankle swelling and a "purplish" rash on his feet. He denied trauma, fever, headache, photophobia, stiff neck or penile discharge, though he admitted to unprotected anal intercourse. Physical exam revealed erythema of the right second carpometacarpal joint, wrist and dorsum of his left foot, tenosynovitis of both ankles and hemorrhagic pustules on the extensor surface of his feet. WBC was 16.2 and ESR, 67. Gram stain of synovial fluid demonstrated gram negative diplococci. Blood cultures, urethral, pharyngeal and rectal swabs were negative. The patient was treated with ceftriaxone with significant improvement in 24 hours. He was discharged on the second hospital day.

DISCUSSION: Gonorrhea is the most frequently reported communicable disease in the United States. Disseminated gonococcal infection (DGI) is the most common cause of acute septic arthritis in sexually active adults and the most common reason for hospitalization due to infective arthritis. The classic triad is dermatitis, tenosynovitis, and migratory polyarthritides. Papules that progress to hemorrhagic pustules develop on the trunk and extensor surfaces of the distal extremities. Asymmetric joint involvement of the knees, elbows, wrists, MCPs and ankles occurs. The cutaneous and articular findings are due to an immune reaction to circulating gonococci and to immune-complex deposition. Synovial fluid cultures are consistently negative and blood cultures are positive in less than 45% of patients. If cultures are negative, alleviation of symptoms in 12 to 24 hours after antibiotic therapy supports a clinical diagnosis. Initial treatment is IV ceftriaxone. A similar arthritis-dermatitis syndrome may occur with *N. meningitidis*. If cultures are negative, this must be considered in the differential. However, gonococemia and meningococemia respond to similar antibiotic treatment.

A CASE OF PULMONARY HEMORRHAGE. H. Short¹, R. Granieri¹; ²University of Pittsburgh, Pittsburgh, PA

LEARNING OBJECTIVES: : 1) Recognize the association between propylthiouracil (PTU) and ANCA associated vasculitis/pulmonary hemorrhage. 2) Recognize that an antecedent flu-like syndrome may be followed by pulmonary hemorrhage in patients on PTU.

CASE INFORMATION: C.D. is a 44-year-old female with Grave's disease treated with propylthiouracil for two years. She presented with a 2 week history of productive cough, fevers, pleuritic chest pain and fatigue. She was diagnosed with right lower lobe pneumonia and admitted for IV antibiotics. Despite appropriate therapy, the patient's condition deteriorated and she developed hemoptysis and progressive hypoxemia that required intubation. CXR showed bilateral alveolar infiltrates. Bronchoscopy was consistent with diffuse alveolar hemorrhage. Admission CBC revealed pancytopenia. Her PTU was discontinued. Additional labs included serum perinuclear antineutrophil cytoplasmic antibody (p-ANCA) >160, proteinase 3 antibody 10 (>2 positive), myeloperoxidase antibody 8 (>6 positive), decreased C3 and C4, microscopic hematuria and negative ANA, anti-Smith, anti-RNP and HIV. The patient was diagnosed with p-ANCA vasculitis secondary to PTU and was treated with pulse dose steroids with stabilization of alveolar hemorrhage. She was extubated several days later. **DISCUSSION:** PTU is the most common antithyroid drug prescribed for Grave's disease. There have been 5 reported cases in the English literature of PTU-associated ANCA-positive syndrome with the development of diffuse alveolar hemorrhage. All patients had a preceding influenza-like prodrome, and all improved with discontinuation of PTU. Two were treated with corticosteroids and cyclophosphamide, and one with steroids alone. The mechanism of PTU-induced ANCA-positive vasculitis is unknown. It has been proposed that activated neutrophils in the presence of hydrogen peroxide release myeloperoxidase (MPO) which converts PTU into cytotoxic products. Alternatively, in the presence of MPO, PTU is converted to PTU-sulfonate which is immunogenic for T cells. The T cells activate B cells, which mediate vascular injury. A viral infection may trigger the cascade of events that leads to vascular injury. The discontinuation of PTU combined with corticosteroids and occasionally cyclophosphamide generally leads to full recovery.

DIARRHEA AND A LUNG MASS; OCCAMS RAZOR CUTS THROUGH THE MYSTERY. K. Shufelt¹, S. Shadowitz¹; ¹University of Toronto, Toronto, Ontario

LEARNING OBJECTIVES: 1. Recognize uterine angiosarcoma as a rare cause of pelvic malignancy. 2. Illustrate the importance of searching for a unifying diagnosis in a patient with multiple clinical findings.

CASE INFORMATION: A 47 year old female presented to the emergency department with a four week history of loose stool, malaise, low grade fever, and one day of crampy abdominal pain and explosive diarrhea. Past medical history was significant for hepatitis B and C contracted through past illicit drug use, alcohol abuse, and a thirty pack year smoking history. Examination revealed moderate volume contraction, diffuse abdominal tenderness, bilateral shoddy inguinal adenopathy, and clubbing of the fingernails. Labs revealed a WBC of 33.6, neutrophils 31.3, hemoglobin 121, platelets 430. Electrolytes, urea, creatinine, and liver enzymes were normal. A routine CXR unexpectedly revealed an opacification in the RUL. CT scan of the lung revealed a large mass in the RUL, extending to the chest wall. CT scan of the abdomen revealed a large pelvic mass which was closely opposed to the uterus. Transvaginal ultrasound showed the necrotic pelvic mass as separate from the ovaries. Bronchoscopy washings yielded cells consistent with non-small cell lung cancer. The patient's course was complicated by continuing diarrhea, worsening abdominal pain, an enlarging pelvic mass, and progressive anemia. At operation, significant hemorrhage into the tumor was seen. Pathology revealed high grade epithelioid angiosarcoma of the uterus. Reexamination of the bronchoscopy specimen revealed metastatic angiosarcoma. Three weeks later, the patient presented with headache, nausea and vomiting, and was found to have a right cerebellar lesion. Resection of the lesion revealed metastatic angiosarcoma. The patient's clinical status continued to deteriorate despite aggressive chemotherapy, and she died two months after her initial presentation.

DISCUSSION: Angiosarcoma of soft tissue is a malignant vascular tumor which most commonly arises in skin or superficial soft tissue. Primary uterine angiosarcoma is exceedingly rare, with only 20 reported cases since 1902. Among these cases, the common presenting findings are a pelvic mass, menorrhagia, and weight loss. This report adds a new case of this rare malignancy to the literature. The clinical presentation was atypical, and delayed until clinical symptoms of local tumor invasion and extrapelvic metastases were evident. This case confirms the highly aggressive nature of this malignancy and poor prognosis in patients with extrapelvic disease at diagnosis.

INCIDENCE OF PERIPARTUM CARDIOMYOPATHY IN A CITY HOSPITAL. A. Singh¹, M. Singh¹, T. Saleem¹, M. Rashid², V. Dimitrov¹; ¹Lincoln Medical Center, Bronx, NY; ²Lincoln Medical and Mental Health Center, New York, NY

LEARNING OBJECTIVES: 1. Recognize Peripartum Cardiomyopathy as a potential cause of Heart Failure and death in peripartum women. 2. Recognize that it is more common than generally appreciated. 3. Recognize that it is almost as common in hispanic as compared to African American Females.

CASE INFORMATION: A 22 year old African American Female 3 weeks post partum, S/P Twin delivery was brought in by EMS with complaints of difficulty breathing and was subsequently intubated for respiratory failure. CXR showed Cardiomegaly and congestion. EKG showed non specific ST-T changes. 2D echo showed Ejection Fraction of 35% with dilated right atrium right ventricle with right ventricular overload pattern. TEE showed no thrombi or vegetations. CT abdomen and pelvis failed to show any infectious etiology. Patient had a progressively downhill course continued to desaturate and had a cardiac arrest with unsuccessful resuscitation. Autopsy revealed cardiac dilatation with myocyte hypertrophy and organised diffuse alveolar damage of lungs. Coronary arteries, aorta, inferior vena cava were normal. Pulmonary arteries were free of thromboemboli.

A clinical review conducted of cases admitted to Lincoln Hospital in 6 months revealed 4 more such cases. One was African American and the other three were Hispanic females in the 26–28 age group. All were post partum 1–2 months. Their delivery was uneventful. They were intubated for respiratory failure. Investigations failed to reveal determinable causes of heart failure or demonstrable heart disease. Echo showed impairment of left ventricular systolic function. Patients were discharged home on Digoxin and Lasix with partial to complete recovery of physical activity.

DISCUSSION: Peripartum Cardiomyopathy is considered to be a rare disease occurring in 1:3,000–15,000 pregnancies with a higher incidence in africans. Given the approximately 4 million annual deliveries in the US, it is estimated that 250–1,350 women will have peripartum cardiomyopathy each year. Our series of cases show that these figures may be significantly higher and unless recognized early can lead to mortality. Also that Hispanics are affected almost as commonly as Africans. Prospective double blind trials are needed to help clarify the issues of management including drugs and cardiac transplantation. Primary care physicians have an important role in timely diagnosis and appropriate treatment of gravid patients with cardiac disorders

METASTATIC ENDOPHALMITIS IN A DIALYSIS PATIENT. M. Singh¹, A. Soni¹, T. Saleem¹, M. Rashid¹, M. Kasubhai²; ¹Lincoln Medical Center, Bronx, NY; ²Lincoln Medical and Mental Health Center, Bronx, NY

LEARNING OBJECTIVES: 1) Recognize metastatic endophthalmitis is a rare but devastating complication of septicemia in dialysis patients. Only 6 cases of endophthalmitis in dialysis patients have been reported in the medical literature so far. 2) ESRD patients with endophthalmitis may have a fulminant course. 3) Prompt treatment is required to prevent blindness.

CASE INFORMATION: A 53-year old male on maintenance dialysis for ESRD since one month presented to the ER with skin rash and fever for two days. He had a right subclavian catheter placed one month earlier for hemodialysis. On examination he had tenderness over the catheter site and a pericardial rub. There were purpuric macules on his trunk and extremities. Blood cultures were positive for gram-positive cocci, which were confirmed as staphylococcus aureus. He was started on IV Vancomycin and the catheter was removed. Patient started improving, but on the third day complained of pain in the left eye and erythema on examination. An urgent ophthalmologic consult revealed endophthalmitis. He was injected with intravitreal amikacin and vancomycin. Three days later pneumatic retinopathy was done for a small area of retinal detachment. Patient was discharged after two weeks of stay in the hospital. His vision in the left eye after three months of follow up was 20/25.

DISCUSSION: Metastatic endophthalmitis can occur despite therapeutic serum concentration of appropriate antibiotics. Dialysis patients are prone to staphylococcal endophthalmitis. ESRD patients may have a fulminating clinical course of metastatic bacterial endophthalmitis probably due to the immune incompetence associated with uremia (only 30% of the patients had full recovery of the vision). Early signs of endophthalmitis should be actively sought in dialysis patients with bacteremia as they can be easily overlooked. Prompt treatment, including consideration of intravitreal antibiotics is required to prevent vision loss.

DISSEMINATED INTRAVASCULAR COAGULATION DUE TO SHIGELLA FLEXNERI. M. Singh¹, M. Murtaza¹, M. Rashid¹, A. Singh¹, V. Dimitrov¹; ¹Lincoln Medical Center, Bronx, NY; ²Lincoln Medical and Mental Health Center, New York, New York; ³Cornell University

LEARNING OBJECTIVE: Recognise Disseminated intravascular coagulation due to shigella flexneri dysentery without bacteremia which has not been reported earlier in the literature.

CASE INFORMATION: A 21 year old hispanic female with no past medical history was admitted with history of diarrhea for two days and vomiting and fever for one day. Diarrhea was large volume without blood and mucus. On admission skin turgor was poor with BP of 90/60 and fever of 105° F. Abdomen was diffusely tender with no guarding or rigidity. Rectal and vaginal examination was normal. Stool guaiac was negative. She was admitted to ICU and resuscitated with IV fluids and was started on Ceftriaxone empirically. Admission laboratory investigations revealed normal PT/APTT and platelet count. Wright stain showed fecal leukocytes. Within six hours of admission platelets had dropped to 118K/cmm and PT/APTT increased to 20.1/81.3 respectively (INR of 3.2). Serum fibrinogen at this time was 179 and D-dimers were increased. Renal functions remained normal. Coagulopathy persisted for the next 36 hours. Subsequently stool cultures were positive for shigella flexneri. Blood cultures were negative.

A diagnosis of DIC due to non-bacteremic shigella flexneri was made. She started improving from the third day onwards and on the fifth day platelets were 221K/cmm with PT/APTT of 11.5/28.5. She was discharged from the hospital after seven days.

DISCUSSION: Shigella flexneri has persisted in the USA despite a steady decline and it is accompanied by a striking increase in the isolation rate among adult men. Hematological complications including thrombocytopenia, gastrointestinal complications, hemolytic anemia, HUS have been described earlier in association with shigella colitis mostly with shigella dysenteriae type-1. But DIC due to shigella flexneri in the absence of bacteremia has not been reported earlier as per our literature search. Hematological complications are associated with poorer prognosis, though in this case outcome was favorable and DIC resolved in three days with treatment of the underlying cause.

DIABETIC KETOACIDOSIS IN A GESTATIONAL DIABETIC DESPITE EUGLYCEMIA. J.A. Smith¹, E. Warm¹; ¹University of Cincinnati, Cincinnati, OH

LEARNING OBJECTIVES: 1) Recognize diabetic ketoacidosis (DKA) in the setting of pregnancy, 2) Discuss euglycemia and the pitfalls of diagnosis of DKA in pregnancy, and 3) Describe possible mechanisms of euglycemic DKA in pregnancy.

CASE INFORMATION: CASE: A 23-year-old G8P1 female with diet controlled gestational diabetes presented at 33 weeks gestation with dyspnea and productive cough. She was febrile, tachypneic and tachycardic. Her chest film revealed bilateral perihilar infiltrates and left lower lobe airspace disease. She was empirically treated for a community-acquired pneumonia. Initial laboratory results showed a HCO₃ of 9, an anion gap of 20, and a glucose of 94. Urinalysis showed greater than 80 ketones and no glucose. Arterial blood gas was 7.32/18/68. Subsequent investigation revealed a lactic acid level of 0.6 and large serum ketones. Shortly after admission, the patient exhibited worsening respiratory distress and polydipsia. Repeat labs showed a HCO₃ of 5, an anion gap of 24, a glucose of 113, and an ABG of 7.16/17/68. The patient was started on D5 1/2 NS and a continuous insulin infusion. Continued therapy led to resolution of the anion gap acidosis and clearance of serum ketones.

DISCUSSION: Pregnancy is a state of relative insulin resistance marked by enhanced lipolysis and ketogenesis. Under certain circumstances, this hormonal milieu may lead to DKA. DKA can present differently in the pregnant patient, making the diagnosis difficult. The mechanism responsible for insulin resistance in pregnancy involves human placental lactogen, progesterone, cortisol, and prolactin, all of which impair glucose uptake by insulin sensitive cells. The cause of DKA with euglycemia is not well understood, but may result from constant fetal removal of glucose, the expanded blood volume of pregnancy, or the rapid clearance of glucose due to increased glomerular filtration. As seen in our patient, DKA in pregnancy may not present classically. As such, high clinical suspicion of DKA is required in ill gestational diabetics to prevent adverse maternal and fetal outcomes.

ACUTE SPINAL COMPRESSION FRACTURES IN AN AIDS PATIENT. J. Sosman;
¹University of Wisconsin Medical School, Madison, WI

LEARNING OBJECTIVES: 1). Review the newly described association of osteoporosis and HIV disease 2). Identify the secondary causes of osteoporosis in AIDS patients.

CASE INFORMATION: A 39 year old man with AIDS presented to the emergency department with the acute onset of back pain over his thoracic spine. He denied trauma or radicular symptoms. His past medical history included Kaposi's Sarcoma of the lower legs and esophagus, CMV esophagitis, and adrenal insufficiency. He had received multiple antiretroviral regimens over the past 10 years. His current combination therapy included abacavir, lamivudine, and nelfinavir. His exam revealed marked tenderness over the thoracic spine from T4 to T10. His WBC was 3.1, hematocrit was 31%, electrolytes, renal and liver function tests were normal. A recent CD4 count was 10 cells/ul with a viral load of 85,000 RNA copies/ml. Spine radiographs revealed multiple new vertebral fractures at T5, T6 and T9 confirmed by MRI. Bone biopsy at T6, however, revealed normal cellularity without infiltrate, granuloma or obvious infection. A subsequent lumbar DEXA bone mineral density demonstrated a T score of -2.9, consistent with severe osteoporosis. Laboratory evaluation of secondary causes of osteoporosis demonstrated hypogonadism with a free testosterone of 0.8pg/ml (NL 47-244) and a total testosterone of 7ng/dl (NL 400-1080). The patient was managed with opioid analgesics, calcitonin, and calcium with vitamin D supplementation.

DISCUSSION: Metabolic bone disease may represent an important complication of HIV antiretroviral therapy. A few published reports have noted a prevalence of osteopenia or osteoporosis as high as 50% in some groups of HIV patients. This case, however, illustrates the multifactorial nature of osteoporosis and the importance of identifying all of the potential secondary causes. Since AIDS patients have a higher rate of GI malabsorptive syndromes, poor PO intake, immobility, and hypogonadism they are already at a substantial risk for osteoporosis regardless of the potential risk from antiretroviral therapies. As AIDS patients continue to live longer, the cumulative consequences of these risks may lead to a substantial rise in complications from osteoporosis. It is important for clinicians caring for HIV patients to be cognizant of this risk.

ACCESSORY SPLEEN MASQUERADING AS AN ISLET CELL TUMOR OF THE PANCREAS. J.L. Stem¹; ¹Western Pennsylvania Hospital, Pittsburgh, PA

LEARNING OBJECTIVES: 1. Identify the subtypes of pancreatic islet-cell tumors 2. Recognize accessory spleen as a rare cause of a pancreatic mass 3. Identify strategies to differentiate accessory spleen from a pancreatic hypervascular tumor

CASE INFORMATION: A 41 year old female with a past medical history of uric acid nephrolithiasis presented to the Emergency Department complaining of right flank pain and blood in her urine. Her only medication was allopurinol 300 mg daily. Physical exam revealed right flank tenderness to palpation. A spiral CT of the abdomen revealed two renal calculi in the right and one in the left kidney. In addition, a mass was seen in the tail of the pancreas. A CT scan of the abdomen with contrast confirmed a 4.1 x 4.8 cm hyperenhancing, hypervascular, well defined mass within the tail of the pancreas consistent with an islet cell tumor. Upon further questioning, the patient denied symptoms of peptic ulcer disease (gastrinoma), hypoglycemia (insulinoma), diarrhea and renal failure (VIPoma), hyperglycemia (glucagonoma), or diabetes, steatorrhea and cholelithiasis (somatostatinoma). A presumptive diagnosis of a nonfunctioning islet cell tumor was considered and the patient was referred to a surgeon. A distal pancreatectomy was performed and pathology revealed an intrapancreatic accessory spleen.

DISCUSSION: Accessory spleen has been recognized as a rare cause of an intrapancreatic mass. It is seldomly detected clinically due to its lack of symptoms. Due to the rise in abdominal imaging procedures performed, though, the detection of accessory splenic tissue in the pancreas can be expected to increase. Its importance lies in the fact that it can mimic a pancreatic hypervascular tumor. Strategies to differentiate an accessory spleen from a pancreatic hypervascular tumor include angiographic demonstration of splenic blood supply, similar enhancement of the accessory spleen and the spleen on CT or MRI, radionuclide testing or biopsy. In this case, the possibility of an accessory spleen should have been considered and potentially could have avoided unnecessary surgery.

PROCEED WITH CAUTION: THYROTOXICOSIS ENHANCES WARFARIN SENSITIVITY. S.L. Swenson¹; ¹Division of General Internal Medicine, University of California, San Francisco, CA

LEARNING OBJECTIVES: (1) to recognize that thyrotoxicosis can potentiate warfarin's anticoagulant effects; (2) to manage warfarin therapy in thyrotoxic patients.

CASE INFORMATION: A 59 yo woman with hypertension noted new-onset palpitations and "shakiness" at a routine clinic visit. She had no prior history of cardiac or thyroid disease and denied dizziness, loss of consciousness, chest pain, use of new medications, dietary supplements, drugs or alcohol. Current medications included benazepril and aspirin. Physical examination revealed a thin, anxious woman with a HR of 128 and BP of 100/60. She had a prominent, smooth, non-tender thyroid gland. Cardiopulmonary exam was notable for an irregularly irregular tachycardia, a normal S1 and S2 without gallops, and a II/VI short, systolic ejection murmur at the apex. An ECG showed rapid atrial fibrillation, prompting transfer to the

emergency department, where work-up revealed a normal PT/INR, creatinine, and hepatic function. After cardiopulmonary stabilization, she was discharged on metoprolol and warfarin 2.5 mg/d. Subsequent TSH < 0.03 mIU/L, free T4 > 71 pmol/L, total T3 of 667 ng/dL, elevated anti-TSH receptors, and abnormal thyroid uptake scan confirmed Graves' disease. The patient's palpitations resolved, but anti-coagulation proved challenging, with INR's above 4 on average warfarin doses of < 1.0 mg/d (Table I). She exhibited exquisite warfarin sensitivity until her free T4 levels normalized three weeks after starting methimazole.

DISCUSSION: Warfarin exhibits many drug-drug interactions, including an increased effect with l-thyroxine administration. Warfarin potentiation by endogenous thyroid hormone is a less appreciated phenomenon documented in case reports dating to 1972. As this patient's initial INR illustrates, thyrotoxicosis alone does not prolong bleeding times. Rather, thyroxine enhances the metabolic clearance of vitamin K-dependent clotting factors II, VII, IX, and X, resulting in relative hypoprothrombinemia and warfarin potentiation. As in this case, potentiation closely parallels free T4 levels and can occur after one dose. Thus, in thyrotoxic patients, appropriate initial warfarin doses may range from 1.0 mg QOD and require upward titration as hyperthyroidism improves. Awareness of such patients' potential enhanced warfarin sensitivity and judicious anti-coagulation may minimize hospital stays and prevent bleeding complications.

Table I: Relation of INR, Warfarin Dose, INR:Warfarin Ratio, and Free T4

	Day 1	5	9	12	19	22	25
INR	1.1	3.6	4.8	3.4	4.2	1.4	1.8
Dose	~2.5	2.5/1.0	1.0/0	0/1.0	1.0/0.5	0.5/1.0	1.0
Ratio		1.44	4.8	>5	4.2	2.8	1.8
Free T4		>71					26

TOXOPLASMA PNEUMONIA IN HIV INFECTION: B.O. Taiwo¹; ¹Lincoln Health Center, Durham, NC

LEARNING OBJECTIVES: 1. Consider Toxoplasma pneumonia in the differential diagnosis of bilateral pulmonary infiltrate in immunocompromised patients 2. Recognize that negative toxoplasma serology does not exclude a diagnosis of toxoplasmosis. 3. Educate susceptible immunocompromised persons on steps to prevent exposure to Toxoplasma gondii.

CASE INFORMATION: A 51 year-old male smoker with a history of crack cocaine use was treated with a five day course of oral Azithromycin for an acute illness characterized by non productive cough, shortness of breath, fever and night sweats. His symptoms worsened and in the third week of his illness, he presented to an emergency room. He denied weight loss and reported negative PPD test done three months earlier. In the ER, he appeared chronically ill with the following vital signs: BP 136/78, Pulse 103, Temp 98.7F, oxygen saturation 85% on room air, PO2-47. He had shotty cervical lymphadenopathy and bilateral coarse rhonchi. Additional data included WBC-3.1, HCT-34.8 and sputum-rare PMN and negative gram stain. CXR revealed scattered interstitial and air space densities especially the lower lobes (slide 1). He was placed in pulmonary isolation and started on high-dose Trimethoprim/Suphamethoxazole and Prednisone for presumptive diagnoses of PCP pneumonia and AIDS. HIV infection was confirmed by ELISA and Western blot, CD4 count was 26. Multiple sputum silver stains returned negative. AFB smears, cryptococcal antigen, fungal cultures were also negative. Expectorated sputum yielded cysts (slides 2 and 3) which were diagnostic for Toxoplasma pneumonia, although Toxoplasma IgM and IgG antibodies were negative. Treatment for Toxoplasmosis with sulfadiazine, folinic acid and pyrimethamine was initiated, which led to complete clinical and radiologic recovery (slide 4). He later recalled encountering many cats in a closed basement while working as a plumber a few weeks before he fell ill. Convalescent Toxoplasma IgM antibodies remain negative up to 13 months after presumed primary infection. He has weakly positive IgG antibodies at this time.

DISCUSSION: Toxoplasma gondii is an obligate intracellular parasite. About 25% of the United States have serologic evidence of prior exposure. Toxoplasma pneumonia is second to encephalitis, which is the most common clinical manifestation. Diagnosis depends on identification of Toxoplasma trophozoites or cysts in sputum, bronchoalveolar lavage, lung biopsy or autopsy. AIDS patients may not be able to mount antibody response, as was the case in this patient. There are guidelines (slide 5) on how to prevent the acquisition of Toxoplasma gondii.

PRIMARY HIV INFECTION. B.O. Taiwo¹; ¹Lincoln Health Center, Durham, NC

LEARNING OBJECTIVES: 1. Recognize the clinical features and differential diagnoses of primary HIV infection

2. Confirm primary HIV by ordering appropriate tests

CASE INFORMATION: A previously healthy 35 year-old hispanic male presented to an urgent care clinic 1 week after the onset of a febrile illness. Although he had sore throat, headache and muscle ache, he was most concerned about truncal rash that started 3 days prior to presentation. He denied taking any medication apart from Ibuprofen for fever. He did not have rhinitis or contact with anyone who had similar symptoms. His temperature was 102F, blood pressure-138/92, pulse-105/minute. He had some painful, shallow ulcers in his mouth and enlargement of his axillary and cervical lymph nodes. Cardiopulmonary examination was normal. His trunk and arms were covered with a macular eruption. His palms were normal. Examination of his abdomen, genitalia and neurologic system did not reveal any significant abnormality. STAT CBC was normal apart from a platelet count of 95,000. Rapid streptococcal test of throat swab, liver enzymes and Monospot tests were negative. HIV RNA by PCR was 55,000 copies/ml. HIV antibody tests were negative. A diagnosis of primary HIV infection was made. Repeat HIV antibody test, 1 month later was positive. His HIV RNA by PCR has dropped to a set point of 3,500 copies/ml.

DISCUSSION: Over 50% of patients with newly acquired HIV infection will develop an acute illness within a few weeks. Most of them are likely to contact primary care clinicians first. Unfortunately, the diagnosis is often missed because it resembles many other illnesses such as

infectious mononucleosis, streptococcal pharyngitis, rubella, drug reaction, acute herpes simplex and syphilis. Missing a diagnosis of PHI has profound implications in terms of lost opportunities to limit disease spread and perhaps alter the natural course of the illness in a newly infected patient.

WEAKNESS AND PAIN IN A 17 YEAR OLD WITH A HISTORY OF GROWTH HORMONE EXCESS. D. Takhtechian¹, P. Takahashi¹; ¹Mayo Clinic, Rochester, MN

LEARNING OBJECTIVES: 1) Diagnosis rhabdomyolysis from its clinical and biochemical presentation. 2) Recognize the clinical manifestations of growth hormone excess.

CASE INFORMATION: A 17-year-old male presented to the emergency room with a history of myalgias for two weeks. Two weeks prior to admission he underwent a dental extraction that required general anesthesia with succinylcholine. He developed symptoms of myalgias four hours after procedure that persisted until readmission. While in the emergency room, he described severe muscle pain in the upper and lower extremities. Enzyme analysis revealed a creatine kinase of 22,000, and he was admitted for rhabdomyolysis. The patient was treated with IV fluids and rest. His urinalysis was negative for myoglobinuria after hydration. He subsequently underwent a muscle biopsy that revealed a generalized myopathy. After hydration, pain control and initial rest, he gradually improved over the ensuing weeks.

DISCUSSION: Growth hormone excess in childhood results in an uncommon condition of gigantism. Gigantism causes inappropriate growth of both bones and muscles that results in a large individual. With the continued muscle growth, gigantism can cause an underlying myopathy. Succinylcholine provides muscle relaxation during anesthesia; however, it can cause rhabdomyolysis. This normally occurs in patients with an underlying muscle disorder. Succinylcholine may damage muscle in patients with muscular dystrophy or other primary muscle disorders. This is the first case of succinylcholine-induced rhabdomyolysis in a patient with primary myopathy from gigantism. While unusual, rhabdomyolysis from succinylcholine must be considered in all patients with potential myopathy. Treatment involves IV fluids and supportive care.

LEVOFLOXACIN AS A POSSIBLE CAUSE OF AUTOIMMUNE HEMOLYSIS. V.R. Tan¹, L. Pinter-brown¹, A.G. Gomez²; ¹UCLA-San Fernando Valley Program, Sepulveda, CA; ²University of California, Los Angeles, Sherman Oaks, CA

LEARNING OBJECTIVES: 1. To discuss the presentation and treatment of autoimmune hemolysis. 2. To address a possibility that levofloxacin can cause autoimmune hemolysis.

CASE INFORMATION: 40 yo male complained of one week of worsening epistaxis, petechiae and 2-3 days of hematuria. Two weeks prior to presentation, he had URI-like symptoms and was treated with Levofloxacin for a few days. He denied taking other drugs/medications/herbs and recalled no known drug allergies. He does not smoke, use ETOH, or IVDA. Examination showed a normotensive male with pale conjunctiva and confluent petechiae on the oral mucosa and extremity skin. No lymphadenopathy. No organomegaly. CBC showed platelets 4000, WBC 1900 with PMN16%, lymphocytes 41%, eosinophils 4%, basophil 5%, Hgb 8.4, Hct 22, MCV 86, RDW14.4. Reticulocytes corrected 1.9%. LDH 499. Coagulation panel, fibrinogen level, D-dimer, kidney panel, and liver function panel were normal. Urine hemosiderin, direct/indirect Coombs, HIV, hepatitis serology and IgG complement for infectious mono were negative. CMV-IgG was positive. ESR was 75. ANA was < 1:40. SPEP and UPEP showed no abnormal monoclonal spike. RUA showed small blood with no RBC. Sugar water test was negative. Occult mediastinal lymphadenopathy was not found with contrasted chest CT. Bone marrow biopsy showed normocellularity with blasts of less than 5%, myeloid with left shift and one lymphoid aggregate, with adequate numbers of megakaryocytes. The iron was absent. CD59 and CD55 cell markers were normal. Comprehensive studies for lymphoma/leukemia were normal. Autoimmune pancytopenia was confirmed. He was treated with platelet pheresis transfusion initially, which hemolyzed shortly. He responded to IVIG and solumedrol with normalized WBC and platelet count. He was discharged home on prednisone taper. One month follow up showed normal WBC and platelet, but with persistent iron deficient anemia. Pt remained asymptomatic.

DISCUSSION: There are three general causes of pancytopenia: splenic sequestration, insufficient production, and excessive destruction. In our case, drug-induced autoimmune processes were suspected after thorough work-up. Treatment for autoimmune hemolytic processes includes steroid, IVIG. If not effective, splenectomy may be an option. Immunosuppressive drugs such as azathioprine and cyclophosphamide, have been used with satisfactory results. Recently, we have had two patients present with autoimmune hematologic disorders within one to two weeks after using Levofloxacin. This association has not been reported previously in the literature. Submission of these cases may assist primary care providers recognize this adverse effect of Levofloxacin.

PROLONGED MOOD DISORDER AFTER 3,4-METHYLENEDIOXYMETH-AMPHETAMINE (MDMA) USE: A CASE REPORT. S.D. Thakur¹, L. Coberly¹; ¹University of Cincinnati, Cincinnati, OH

LEARNING OBJECTIVES: 1) Recognize the potential for prolonged neuropsychiatric illness after methylenedioxymeth-amphetamine use.

CASE INFORMATION: A 23-year-old male with no prior psychiatric history except polysubstance abuse, presented with sudden onset of anhedonia, mutism, avolition, feelings of guilt and worthlessness, and hypersomnia after an MDMA binge. The patient had severe paranoid delusions about eating and ultimately quit eating altogether. Hospitalization and a feeding tube were required for his malnutrition. The patient was diagnosed with major depressive disorder with psychotic features, and was started on venlafaxine, olanzapine, haloperidol, and benztropine. His symptoms have continued to persist for several months. He was discharged once but decompensated and attempted suicide within the week. Upon rehospitalization lithium augmentation therapy was implemented.

It is postulated that his prolonged psychiatric illness was precipitated by his MDMA use. MDMA ingestion has been linked to acute psychosis, mood disorder, memory and cognitive disorders. Recent reports have implicated its involvement in prolonged mental disturbances as well.

MDMA is thought to cause potent and possibly permanent neurotoxicity in the serotonin (5HT) pathways. These pathways are the same ones that underlie chronic neuropsychiatric illnesses such as major depressive disorder, psychosis, and panic disorder, supporting the view that MDMA can lead to prolonged neuropsychiatric problems.

As in our patient, recovery from the neuropsychiatric complications of MDMA tends to be slow and tedious. Improvement may be explained by compensatory metabolic changes within remaining 5HT and non 5HT neurons, or regeneration of damaged neurons. These mechanisms serve to reverse the 5HT deficit and lead to clinical improvement.

MDMA has gained popularity recently as a recreational drug in the high school and college populations. It is commonly referred to as "ecstasy", "X", "E", or "Adam". It has hallucinogenic and stimulant properties. Due to its potential for severe and chronic complications, it must be considered in the differential diagnosis of new onset psychiatric illness, especially in adolescents and young adults.

DISCUSSION:

TB OR NOT TB? A CASE OF PAINFUL TESTICULAR SWELLING. S. Thirumurthi¹, S. Lithgow¹; ¹Baylor College of Medicine, Houston, TX

LEARNING OBJECTIVES: 1) Diagnose the common and infrequent causes of epididymitis. 2) Recognize the presenting features of genitourinary tuberculosis.

CASE INFORMATION: A 55 year-old male presented to the ambulatory care clinic complaining of left testicular swelling. Two months prior, he noted pain and swelling of the left testicle that he attributed to heavy lifting at work. The pain lasted for two weeks and then subsided spontaneously, but the testicular swelling persisted. He denied dysuria, hematuria, urethral discharge, fever, sweats, weight loss, cough or sputum production. On physical exam, his temperature was 100.3 F. Lungs were clear to auscultation. His left testicle was nontender, with an enlarged, nodular epididymis. No inguinal hernias were detected. He refused examination of the prostate. Urinalysis and microscopy revealed a pH of 5.0, hematuria, bacteriuria and WBC clumps. He was treated empirically with ciprofloxacin for presumed bacterial epididymitis. Urine culture, however, was sterile for bacteria. A scrotal ultrasound revealed diffuse enlargement of the left epididymis with increased doppler flow, moderate left hydrocele and left testicular calcifications. IVP was unremarkable. We suspected tuberculous epididymitis based on the findings of sterile pyuria and epididymal enlargement. Further testing revealed a PPD with 50 mm induration and CXR with left upper lobe infiltrates. A bronchoscopy showed intense mucosal erythema of the left upper lobe. He was treated empirically with INH, rifampin, ethambutol and pyrazinamide for presumed tuberculosis. Urine cultures and sputum cultures from bronchial washings eventually grew *M. tuberculosis*.

DISCUSSION: Epididymitis is an inflammatory response to any of a number of infectious agents or local trauma. Sexually transmitted organisms are the likely culprits in sexually active men under the age of 35. Epididymitis in children and older men can usually be attributed to common urinary pathogens. Less commonly, infection with *Mycobacterium*, *Cryptococcus*, *Actinomyces*, or *Brucella* may cause epididymitis. Approximately 80% of male genital TB is associated with coexistent renal disease. Spread of *Mycobacterium* from renal foci usually involves the prostate, seminal vesicles, epididymis and testes. Genital TB without apparent renal involvement is thought to arise from lympho-hematogenous spread. Symptoms of the disease are twice as common in males and usually present between the ages of 20 and 50. They include dysuria, frequency, nocturia and urgency; few patients are completely asymptomatic. Sterile pyuria, microscopic hematuria and acid pH on urinalysis are suggestive of genitourinary TB. Acid-fast bacilli on a Ziehl-Neelsen prep or culture growth on Lowenstein-Jensen media make the definitive diagnosis.

AN ALCOHOLIC WITH HYPONATREMIA AND HYPOTENSION. E. Tiller¹, R. Granieri¹; ¹University of Pittsburgh, Pittsburgh, PA

LEARNING OBJECTIVES: 1. To recognize the presenting symptoms of hyponatremia. 2. To manage the initial clinical evaluation and laboratory testing in the patient with severe hyponatremia. 3. To outline the appropriate treatment of hyponatremia.

CASE INFORMATION: A 53 year old male with no known past medical history presented with a 4 day history of weakness, nausea, diarrhea and anorexia. He admitted to a 40 PYH of tobacco and up to 12 beers/day. On exam, his rectal temperature was 92 degrees, pulse 89 and BP 90/45 lying and 74/41 sitting. No icterus or jaundice was present. He had dry mucous membranes, normal JVP and no peripheral edema. Liver span was 10 cm, MMSE was 22/30, reflexes were symmetrically depressed and he exhibited dysmetria on finger-to-nose testing but no asterixis. He was globally weak and unable to walk. Initial laboratory data: sodium 111, potassium 5.1, chloride 81, bicarbonate 10.7, BUN/CR 58/3.1, WBC 13.7, H/H 15.6/46.9, MCV 101, platelets 143,000. Serum osmolality was 235, urine osmolality was 280 and urine sodium was <10. ABG on 2L NC revealed pH 7.46, pCO2 19, pO2 160 and bicarb 13.3. Given the patient's profound hyponatremia, hyponatremia, hyperkalemia and weakness, cortisol was drawn and he was empirically given steroids. Data otherwise supported a diagnosis of hypotonic hyponatremia secondary to hypovolemia and normal saline was begun. Acid-base status and renal failure corrected over 72 hours. Cortisol, TSH and ACTH stimulation test were normal. He was discharged home after alcohol counseling.

DISCUSSION: Hyponatremia is the most common electrolyte abnormality in hospitalized patients. Symptoms are related to CNS dysfunction and include headache, nausea, vomiting, anorexia, disorientation and depressed reflexes. Symptom severity is correlated with how rapidly the hyponatremia developed. Coma and seizures can occur with acute drops to less than 120 meq/L. The initial approach to diagnosis is measurement of serum and urine osmolality and urine sodium. The most common finding is hypotonic hyponatremia, which is further subdivided by volume status. Our patient was hypovolemic, and low urine sodium implied appropriate renal sodium retention. The diagnosis of beer potomania was also considered, although this is classically a euvolemic state with a low urine osmolality. Treatment should be initiated in all symptomatic patients and those with levels less than 120mEq/L. For hypovolemic hyponatremia, the treatment of choice is normal saline. Overly rapid correction is avoided to prevent central pontine myelinolysis.

FINDING WHAT YOU WEREN'T LOOKING FOR: A CASE OF INCIDENTAL THROMBOCYTOPENIA. E. Tilleros¹, R. Granieri¹; University of Pittsburgh, Pittsburgh, PA

LEARNING OBJECTIVES: 1. To recognize common causes of acquired thrombocytopenia. 2. To recognize the initial evaluation of thrombocytopenia. 3. To outline the treatment of idiopathic thrombocytopenic purpura.

CASE INFORMATION: A 64 y/o female was admitted with chest pain and worsening dyspnea. She noted easy bruising for 3 months and one episode of epistaxis. PMH included COPD, hypertension, and iron deficiency anemia. No other personal or family history of abnormal bleeding was known. Medications included albuterol and ipratropium inhalers, metoprolol, furosemide, potassium, lisinopril, fexofenadine, bupropion, and ginkgo biloba. Fexofenadine was recently changed from certrazine, due to her insurance formulary.

On admission labs, platelet count was 24,000; it was 156,000 six months earlier. On exam, there was no jaundice, icterus, ecchymoses or petechiae. Liver span was 8 cm and spleen was nonpalpable. Stool was heme negative. Peripheral smear showed decreased numbers of giant platelets. H&H was 9.5 and 27.7, WBC 4.0 (normal differential). Bone marrow biopsy showed mild megakaryocytosis. A diagnosis of idiopathic thrombocytopenic purpura (ITP) was made, and she was begun on prednisone. At 2 weeks, platelet count was 173,000 and at four weeks, 180,000.

DISCUSSION: Thrombocytopenia, defined as a platelet count under 150,000, may be an incidental finding on a routine CBC and generally requires further evaluation. The three major causes are increased destruction, increased sequestration and decreased production of platelets. A bone marrow biopsy may be indicated to rule out hematologic malignancies. If the bone marrow is normal and there is no clinical evidence of sepsis, DIC, vasculitis, viral infection, or TTP/HUS, a diagnosis of ITP can be made. Patients with ITP are often asymptomatic but may present with bleeding, bruising, fatigue or mild splenomegaly. Peripheral smear shows a decreased number of large platelets. Bone marrow biopsy may show mild megakaryocytosis. The etiology is unknown, but is thought to be the result of platelet specific autoantibodies. Treatment should be initiated if there is bleeding or if counts drop to less than 20–30,000. Initial therapy is prednisone, and response rates for adults range from 40–60%. If steroid therapy fails, treatment with IV IgG +/- splenectomy is indicated. For refractory cases, cytoxan and azathioprine have been used.

I'VE FALLEN AND I HAVE ASTHMA. P. Tinnis¹, R. Granieri¹; ¹University of Pittsburgh, Pittsburgh, PA

LEARNING OBJECTIVES: 1. To recognize the clinical presentation and natural history of Churg Strauss Syndrome (CSS) 2. To recognize the six clinical criteria of CSS proposed by the American Rheumatology Association 3. To recognize the differential diagnosis of CSS

CASE INFORMATION: A 56 year-old male with a longstanding history of steroid-dependent asthma and recurrent sinusitis presented with bilateral upper extremity weakness, beginning during a steroid taper. Symptoms then progressed to generalized weakness, malaise, anorexia, and a peripheral eosinophilia. He was given stress dose steroids with prompt resolution of both symptoms and eosinophilia and kept on prednisone 10 mg a day. He then began to develop wasting of his forearms, difficulty in performing tasks using his hands, and difficulty arising from a chair. Medications included salmeterol, fluticasone 220 mcg MDI, and prednisone. On physical examination, he was cushingoid. Lungs were clear. Pertinent neurologic findings included moderate proximal muscles weakness and motor weakness with atrophy in distribution of the left and right median and ulnar nerve, left radial nerve and bilateral peroneal nerves. Deep tendon reflexes were absent in the right biceps, brachioradialis and left achilles. White count was 22,000 with 30% eosinophils. ACTH stim test was consistent with adrenal insufficiency. Chest X-ray was clear. EMG was consistent with a mixed motor and sensory loss in the ulnar and median nerve distribution. Peroneal muscle biopsy showed eosinophilic infiltration of a blood vessel consistent with a vasculitis.

DISCUSSION: Our patient presented with adult-onset asthma, peripheral eosinophilia, and mononeuropathy multiplex. These features are consistent with Churg-Strauss Syndrome (CSS). The muscle biopsy showing eosinophilic infiltration of a blood vessel ensured the diagnosis. The American College of Rheumatology has developed classification criteria by comparing 20 patients who had CSS to 787 control patients with other forms of vasculitis. The six criteria more typical of CSS include: asthma, eosinophilia of greater than 10%, mononeuropathy or polyneuropathy, nonfixed pulmonary infiltrates on CXR, paranasal sinus abnormalities, and biopsy containing a blood vessel with extravascular eosinophils. The presence of any four or more yielded a sensitivity of 85% and a specificity of 99.7%. The natural history of the syndrome follows three clinical phases: (1) prodromal allergic phase (2) allergic rhinitis, nasal polyposis, and asthma and (3) peripheral and tissue eosinophilia. The differential diagnosis includes Wegener's granulomatosis, hypereosinophilic syndromes, polyarteritis nodosa, and microscopic polyangiitis.

RECOGNIZING A RARE DIAGNOSIS IN A SWOLLEN LEG. Y.M. Tjia¹, E.C. Chan¹; ¹ Department of Internal Medicine, The University of Texas-Houston Health Science Center, Houston, TX

LEARNING OBJECTIVES: 1) Recognize that an asymptomatic mass on an extremity can be a soft tissue sarcoma. 2) Consider an incisional biopsy in an extremity lesion, particularly if it is larger than 5 cm in diameter.

CASE INFORMATION: A 32-year-old African American woman with no significant past medical history presented with a 4 week history of right thigh pain, swelling and hemoptysis. She noted that her right thigh had been swollen without pain for several weeks prior to her presentation. On examination she had a swollen, tender, firm right thigh with right inguinal adenopathy. She had no skin lesions, clear lungs, and a normal pelvic exam. A pap smear was normal. Laboratory data were notable for a hematocrit of 27.7% and a serum iron of 24 ug/dl. Doppler ultrasound studies of the legs were negative for deep venous thrombosis. Chest x-ray showed multiple pulmonary nodules of variable sizes in both lungs, confirmed by a CT scan of the chest which also showed no mediastinal or hilar lymphadenopathy. MRI studies revealed a

6.5×4.5×4.5 cm soft tissue mass in the right thigh with lymphadenopathy in the right inguinal, external iliac and common iliac chains, and retrocaval region at the level of the renal veins. A biopsy of the right thigh mass revealed sheets of large monomorphic epithelioid cells which were positive by immunohistochemical studies for cytokeratin and vimentin. These findings were consistent with a diagnosis of epithelioid sarcoma metastatic to the lungs.

DISCUSSION: Soft tissue sarcomas account for fewer than one percent of all malignancies and often present as an asymptomatic mass on the extremities of young adults, most often in the third decade of life. Fewer than 20% present as metastatic disease. Because of their slow-growth (over months) and initial presentation, they may be mistaken for other entities (eg. lipomas). Compression of a nerve or blood vessel can cause symptoms. Sarcomas usually spread hematogenously to the lungs, although epithelioid sarcomas can also metastasize through the lymphatic system. Patients with soft tissue sarcomas <5 cm in diameter have a metastasis-free 5-year survival rate of 81%, regardless of tumor grade, depth, and location. In contrast, there is only a 20% 5-year survival rate for patients with lung metastasis. Diagnostic workup should include radiographs, MRI, referral for an incisional biopsy, and chest x-ray or chest CT to search for possible metastatic disease. Because sarcomas are rare and histological interpretation can be difficult, a primary care physician should refer patients to surgeons or centers experienced in the diagnosis of soft tissue sarcomas. The early consideration of this diagnosis can greatly improve patient outcome.

HYPOGLYCEMIA PRESENTING AS BIZARRE BEHAVIOR. T. Toffoli¹, D.W. Brady¹; ¹Emory University, Atlanta, GA

LEARNING OBJECTIVES: 1. To recognize and evaluate hypoglycemia in non-diabetic patients. 2. To recognize Whipple's triad and its role in hypoglycemia.

CASE INFORMATION: A 25 year-old male was seen in the Emergency Department two days prior to admission with bizarre, child-like behavior and confusion. During his evaluation and treatment for a urethritis, the patient was noted incidentally noted to have an accucheck of 54. On the day of admission, he was found in his home unresponsive. Emergency medical services were called, and they found his glucose to be 34. Glucose was administered en route to the hospital with an increase in his glucose to 154 and return of consciousness. He had no history of diabetes and denied use of exogenous insulin. Physical exam at the time of admission was normal. Routine labs were all normal except for a glucose of 52. The patient was admitted for a supervised 72-hour fast. Several hours into the fast, the patient's accucheck was <50, and he again began to exhibit bizarre, child-like behavior. Stat chemistries, serum insulin and C-peptide level were drawn revealing a glucose of 36, insulin 97.1 mU/ml and C-peptide 7.6ng/ml, values consistent with excess endogenous insulin production. A sulfonylurea level was ordered, and a CT scan was scheduled. The CT revealed a 5×2.5 cm mass in the body of the pancreas and numerous masses throughout the liver.

DISCUSSION: This case illustrates the importance of recognizing the signs and symptoms of hypoglycemia in the context of a low glucose value. Low glucose without neuroglycopenia or sympathoadrenal activation may not require further evaluation. However, the presence of these symptoms in an otherwise healthy-appearing, non-diabetic patient with low glucose necessitates a work-up for a hypoglycemic disorder. Careful measurement of insulin, C-peptide, glucose, and sulfonylurea levels can quickly narrow the differential. Whipples Triad (symptoms of hypoglycemia, a glucose of <50, and relief of symptoms with glucose) is present in the initial presentation of most hypoglycemic disorders.

CEREBRAL HERNIATION AND FATAL HYPERAMMONEMIA WITH NORMAL HEPATIC FUNCTION. R. Tripathi¹, E. Warm¹; ¹University of Cincinnati, Cincinnati, OH

LEARNING OBJECTIVES:

- 1) Determine differential diagnosis and treatment of severe hyperammonemia and microvesicular steatosis,
- 2) Describe the mechanism of action of a previously unreported enzyme deficiency (Hepatic Glutamine Synthetase Deficiency) that has fatal consequences, and
- 3) Discuss appropriate screening for this deficiency.

CASE INFORMATION: A 60 year-old white male with a past medical history significant only for chronic obstructive lung disease was admitted to the hospital with shortness of breath and pneumonia. He was treated with piperacillin/tazobactam, steroids and inhalers. On day 2 of hospitalization, the patient developed mental status changes and had an ammonia level of 41. By day 3 he developed seizures, and on day 4 he was comatose with an ammonia level of 2808. At that point, family withdrew all support. Throughout his course, the patient had essentially normal liver enzymes. Autopsy showed pulmonary embolism with infarction, ischemic colitis, cerebellar tonsil herniation, and multifocal areas of microvesicular steatosis in less than 50% of liver.

DISCUSSION: Microvesicular steatosis has a limited differential including Reye's syndrome and it is usually not associated with normal liver function enzymes and tremendous increases in ammonia. Similar histopathological and clinical findings as in our patient have been described in patients with heart-lung transplantation. A case series of these orthotopic lung transplant patients with fatal hyperammonemia have shown them to have a hepatic glutamine synthetase deficiency with normal urea cycle and liver function enzymes. The increased ammonia is shunted to the cerebrospinal fluid. Initial increase in ammonia is thought to be secondary to an increased protein load (i.e. gastrointestinal bleeds, TPN) and/or increased catabolic processes (i.e. major surgery). Patients with intact glutamine synthetase enzymes are able to handle this increased ammonia production and there is a pulmonary process in patients with the deficiency for clearing the excess ammonia. A combination of hepatic enzyme deficiency, increased protein load and acute pulmonary disease could be postulated to be the cause of rapid hyperammonemia, cerebral herniation and death in our patient.

This is a recently reported entity that may have been underdiagnosed in the critical care setting in the past. In the future, screening for this disorder may be important before patients are candidates for heart-lung transplantation or other pulmonary surgery. It may be important currently to screen patients with mental status changes and no known liver disease with ammonia levels because the only proven treatment modality for severe hyperammonemia is emergent hemodialysis.

FINDING YOURSELF DOWN IN THE BIG EASY. J.A. Troiano¹, M.D. Landry¹, J. Wiese¹; ¹Tulane University, New Orleans, Louisiana

LEARNING OBJECTIVES: Identify the causes of acute paralysis in the patient who uses drugs. Illustrate the importance of using the history and physical exam to quickly discern the correct diagnosis while definitive studies are pending

CASE INFORMATION: A 20 year-old man presented with acute onset of bilateral lower extremity weakness and anesthesia shortly after arriving in New Orleans. He reported passing out for 12 hours after injecting heroin and cocaine. When he regained consciousness, he could not ambulate and had to pull himself to the telephone to call for help. On physical exam, he was afebrile with normal vital signs. He had no heart murmurs and there were no skin ulcerations. He had bilateral loss of sensation in the L4-S2 distribution. He was unable to dorsiflex, plantarflex, invert or evert his ankles. Patellar reflexes were normal; ankle reflexes were absent bilaterally. Flexion of the hips and knees, and sphincter tone were normal. The remaining neurologic exam was normal. The urinalysis showed large blood, but few red blood cells. His serum creatinine was 4.1 mg/dL and the creatinine kinase was 43,900 U/dL. All other laboratory values were normal. The patient was diagnosed with rhabdomyolysis and a compression neuropathy of the peroneal nerves. He received aggressive intravenous hydration; his creatinine decreased to 1.3mg/dL at discharge. He regained sensation and progressive motor function in his lower extremities.

DISCUSSION: Acute paralysis is a medical emergency, which requires prompt diagnosis and treatment. The differential diagnosis in the intravenous drug user includes wound botulism, paradoxical embolism with stroke, HTLV-1 with spastic paraparesis, spinal cord compression from epidural abscess, nerve compression and rhabdomyolysis. The history and physical exam can quickly discern the correct diagnosis while definitive studies are pending. The absence of skin wounds and cranial nerve deficits excluded botulism; the distribution of nerve deficits were inconsistent with a cerebral stroke and there was no murmur on cardiac exam. The time course for the paralysis was too rapid for HTLV-1 infection, and the normal sphincter tone and hip function argued against spinal cord compression. Deductive reasoning led to the diagnosis of nerve compression and rhabdomyolysis, which was confirmed by laboratory testing. Upon further questioning, the patient reported finding himself with his buttocks on the floor, his knees draped over the metal railing of the shower doors and his torso slumped over his lower extremities. This resulted in compression of the peroneal and tibial nerves proximal to the popliteal fossa with subsequent paralysis.

SHOULD WE SCREEN YOUNG HYPERTENSIVE PATIENTS WITH ACUTE ISCHEMIC STROKES FOR HYPERHOMOCYSTEINEMIA? L. Vaidyanathan¹, K. Barnard¹; ¹University of Pittsburgh Medical Center Shadyside, Pittsburgh, PA

LEARNING OBJECTIVES: There is data suggesting that increased plasma homocysteine confers an independent risk for vascular disease (cardiac, cerebral and peripheral). It also (powerfully) increases the risk associated with smoking and hypertension. Therefore, consider obtaining homocysteine levels in young patients with strokes despite the presence of other risk factors.

CASE INFORMATION: A 40 year old hypertensive African American male was admitted to the hospital with sudden onset right sided weakness and slurred speech of 8 hours duration. Past medical history includes hypertension (diagnosed at 24 years). Patient is a nonsmoker. Initial evaluation revealed a blood pressure of 200/110 mmHg, right upper and lower extremity weakness with an upgoing right plantar reflex and dysarthria. MRI of the brain showed small vessel ischemia in the lenticulostriate distribution in the posterior limb of the left internal capsule extending into the anterior aspect of the thalamus and corona radiata. EKG showed normal sinus rhythm and there were no arrhythmias on the monitor. Transthoracic echocardiogram only showed left ventricular ejection fraction of 50%. Carotid and transcranial Dopplers were normal. Fasting homocysteine level was increased at 14.4 Umoles/liter (reference <9Umoles/liter).

DISCUSSION: More data is required to determine if hypertensives and smokers under 60 years with acute ischemic strokes, should be screened for hyperhomocysteinemia and if treating these patients with folate and vitamin B6 will decrease their risk of a recurrent cerebrovascular event. We treated our patient with aspirin, folate and Vitamin B6 for secondary stroke prevention.

TB, OR NOT TB; THAT IS THE QUESTION. E. Vanderveer¹, S. Fennessy¹, P. Davol²; ¹Providence St Vincent Medical Center, Portland, Oregon; ²OHSU

LEARNING OBJECTIVES: 1. Recognize the increasing prevalence of Tuberculosis (TB) in the U.S. due to the immigration of high-tech workers. 2. Highlight the diagnostic dilemma a physician can face when the traditional work-up for TB is negative. 3. Highlight the Adenosine Deaminase Test (ADA) as a quick and highly sensitive and specific test to aid in the diagnosis of TB.

CASE INFORMATION: A 26-year-old high-tech worker from Madras, India presented with new, severe headache. He denied cough or constitutional symptoms. The patient had a history of "Intestinal TB" two years prior, with 16 months of medical therapy, the details of which were not known at the time of admission. His physical exam was completely normal, his chest X-ray showed a diffuse miliary pattern, and a head CT showed multiple ring-enhancing lesions.

An extensive work-up for TB was undertaken, all of which was negative. Additionally, tests for fungi, HIV, bacteria, viruses and malignancy were negative. The patient's headache continued to worsen, so anti-tubercular therapy was started along with prednisone to cover for possible neurosarcoidosis. The patient did quite well and was discharged home soon thereafter.

One week later, the patient returned with altered mental status. A new CT scan showed massive hydrocephalus. A ventriculostomy was placed and the CSF was analyzed for TB; the AFB smears and PCR were once again negative. An Adenosine Deaminase Test of the CSF was positive. Treatment was initiated to cover for multi-drug resistant TB. An open lung biopsy was performed to harvest tissue for culture and a few AFB were identified by smear. The original sputum culture finally turned positive for TB eight weeks after the patient's initial presentation. The patient improved slowly over the next several weeks and was eventually discharged in stable but neurologically impaired condition.

DISCUSSION: CNS TB is uncommon, but is in the differential diagnosis of ring-enhancing lesions. It occurs in only 10% of immunocompetent patients and is highly lethal with mortality exceeding 25%. Even if a patient has widespread TB, the diagnosis of TB can be very difficult and require an extensive, invasive investigation. The ADA test is highly sensitive (100%) and specific (99%) for TB when compared to culture-proven disease in the CNS and lungs. It is an excellent test to aid in the diagnosis of TB and should be considered in the routine work-up whenever CSF or pleural fluid are analyzed.

WHEN NEW ONSET POLYURIA AND POLYDIPSIA DOES NOT EQUAL DIABETES. E.W. Vogel¹; ¹MCP Hahnemann University, Philadelphia, PA

LEARNING OBJECTIVES: 1. Recognize severe hypercalcemia as a cause of polyuria and polydipsia, and check for hypercalcemia in patients with polyuria when diabetes has been ruled out. 2. Recognize parathyroid carcinoma as a cause of severe hypercalcemia.

CASE INFORMATION: A 37 year old African American female with no significant past medical history presented with a two to three week history of increasing thirst, leading to intake of approximately three liters of water per day, associated with urination every 30 minutes. She also noted some polyphagia, dizziness, and mild blurred vision. She had a strong family history of type 2 diabetes, with her mother, brother, and uncle all having diabetes. On a brief initial physical examination, no significant abnormalities were detected. Somewhat surprisingly, her office finger stick glucose measurement was only 106 mg/dL. A basic seven-item chemistry panel, urinalysis, and hemoglobin A1C were sent to the lab from the office, and the patient was released to home. Although not specifically ordered initially, a calcium level included with the chemistry panel came back at 14.5 mg/dL; her glucose and hemoglobin A1C were normal. The patient was admitted to the hospital for treatment, and on re-examination was found to have a hard left thyroid mass. Her parathyroid hormone level was markedly elevated at 485 pg/mL, with normal thyroid function tests. After resolution of her hypercalcemia with intravenous normal saline and pamidronate, she underwent surgical neck exploration, and was found to have a tumor extending from her thyroid to local skeletal muscle and the esophagus. Pathological examination revealed parathyroid carcinoma. The patient later was found to also have metastatic disease in her lungs. Despite primary tumor resection, local neck irradiation, chemotherapy, and multiple outpatient doses of pamidronate, the patient has required several admissions for management of recurrent, severe hypercalcemia.

DISCUSSION: Severe hypercalcemia can have a number of clinical manifestations, including polyuria and polydipsia. Interestingly, this patient manifested none of the more common findings of hypercalcemia at the time of presentation, such as bone pain, renal colic from nephrolithiasis, or gastrointestinal or neurological symptoms. Clinicians should therefore consider hypercalcemia in the differential diagnosis of polyuria, especially when diabetes has been ruled out. In this case, the serum calcium was not ordered initially, but luckily this laboratory routinely includes a calcium level with the chemistry seven panel, which established the diagnosis. Parathyroid carcinoma is a rare cause of hypercalcemia; an underlying parathyroid cancer causes only 1 to 2 percent of cases of primary hyperparathyroidism. This patient had two of the warning signs for parathyroid carcinoma as the cause for primary hyperparathyroidism: severe hypercalcemia at the time of presentation (> 14 mg/dL), and a palpable neck mass. This patient also typifies the clinical course of parathyroid carcinoma that is not cured by primary surgery, in that she has developed recurrent episodes of severe hypercalcemia. This persistent hypercalcemia eventually results in renal and cardiac complications, accounting for most of the morbidity and mortality associated with this disease.

NOT ALL VOMITING IN PREGNANCY IS HYPEREMESIS. S. Vora¹, R.O. Powrie¹; ¹Brown University School of Medicine, Providence, RI

LEARNING OBJECTIVES: 1. Recognize the differential diagnosis of hyperemesis gravidarum and illustrate the role of a general internist in diagnosis and management of medical illness in pregnancy. 2. Demonstrate the clinical presentation of mesenteric thrombosis. 3. Emphasize the fact the pregnancy is a hypercoagulable state and briefly outline the management of thrombosis in pregnancy.

CASE INFORMATION: A 37 year-old primigravid Nigerian woman presented in her 19th week of an IVF pregnancy to the emergency room with complaints of two days of fever and constipation and three weeks of malaise, anorexia, and vomiting. She had been treated in the ER on three prior occasions with hydration and antiemetics for hyperemesis gravidarum. On this last presentation, she produced her photograph from one month prior to demonstrate the extent to which her appearance had deteriorated. Her past medical history was significant for only childhood malaria and sickle cell trait. She had not suffered from protracted nausea and vomiting of pregnancy during the first trimester. She was taking only prochlorperazine suppositories at the time of presentation. She had no personal history of smoking, alcohol or recreational drug use. Her examination was significant for a temperature of 101.6 F, a heart rate of 114 per minute, respiratory rate of 12 per minute, and BP 103/60. She was mildly jaundiced with sunken orbits. Her cardiopulmonary examination was normal. Her abdomen was markedly tender in the right upper quadrant with voluntary guarding but no rebound. No fluid wave was present and her rectal examination was normal. Her uterus was gravid and nontender to palpation. Her fetal heart tracing was reassuring. Neurologic examination was unremarkable. Her laboratory examination was remarkable for a leukocytosis of 36.6 with a left shift. Her chemistries revealed a hypochloremic metabolic alkalosis with serum bicarbonate of 34 mEq/L. She had a mild elevation of her serum AST to 40 U/L. The remainder of the laboratory data was unrevealing. A malarial blood smear was negative. An abdominal ultrasound revealed absent portal venous flow, but was otherwise normal. A subsequent CT scan of the abdomen demonstrated thrombosis of the portal vein and superior and inferior mesenteric venous branches. The patient was admitted, treated with intravenous unfractionated heparin, gentamycin and piperacillin and achieved significant improvement in pain. She defervesced by the second hospital day. Her liver enzymes normalized. Her hypercoagulable work-up revealed only a relative protein S deficiency. Once therapeutic on intravenous heparin, she was changed to low molecular weight heparin, which she continued until 36 weeks gestation. At that time her

therapy was changed to adjusted dose subcutaneous unfractionated heparin until her uncomplicated cesarean delivery of a healthy 2475 g girl.

DISCUSSION: Nausea and vomiting occurs frequently during pregnancy, especially during the first 20 weeks, afflicting 70-90% of all pregnant women. Causes can be physiologic, due to progesterone effects on the lower esophageal and pyloric sphincters, or due to mechanical compression of the diaphragm on the enlarging uterus. However pregnant women are also subject to all other causes of nausea and vomiting, such as severe gastroenteritis, cholecystitis, pyelonephritis, hyperthyroidism, primary hyperparathyroidism, liver dysfunction, and rarely, small bowel obstruction, mesenteric thrombosis or mesenteric ischemia. Internists can be instrumental in assisting in these less common medical diagnoses during pregnancy. Portal and mesenteric venous thromboses are rare conditions, even in nonpregnant patients. Most patients present with at least two weeks of abdominal pain, anorexia, vomiting, and change in bowel habits. Many patients are also febrile. In 20% and 40% of cases of mesenteric thrombosis no underlying cause can be found. Other investigated cases of mesenteric thrombosis during pregnancy have revealed inherited thrombophilias such as activated protein C resistance, antithrombin III deficiency, and protein C and S deficiency. The latter is a difficult diagnosis in the face of both active thrombosis and in pregnancy, since protein S normally decreases in both of these settings. The treatment of acute thrombosis involves heparin (either low molecular weight or unfractionated), as warfarin is contraindicated in pregnancy. Special care is required in the management of anticoagulation in the peripartum period.

SPONTANEOUS UNILATERAL NIPPLE DISCHARGE: WHEN SCREENING TESTS ARE NEGATIVE. D.L. Wahner-roedler¹; Mayo Clinic, Department of Internal Medicine, Rochester, MN

LEARNING OBJECTIVES: 1. Distinguish physiologic from pathologic nipple discharge. 2. Recognize the importance of ductography and fiberoptic ductoscopy, the latter being a new technology for nipple discharge.

CASE INFORMATION: A 45-year-old woman presented with a 1 day history of a spontaneous amber right nipple discharge. Breast examination revealed fibrocystic changes. A small amount of clear, non bloody fluid could be expressed from one of the right breast ducts. There was no palpable lymphadenopathy. A mammogram revealed fibroglandular densities. Ultrasound (US) of the right breast showed slightly dilated ducts. Ductography was performed as follows: by applying focal pressure a single draining orifice was identified, cannulated and 0.2 cc of Conray 60 were injected. Post- mammographic images showed the presence of a cystically dilated duct that contained a small 3–4 mm intraluminal filling defect, which now could also be visualized by repeat US examination. Multiple core biopsies were obtained under US guidance using a vacuum assisted biopsy (bx) needle. A small clip was deployed in the bx cavity following the procedure. Histology revealed high grade ductal carcinoma in situ (DCIS). The patient was seen in surgical consultation and underwent an attempted wide local excision, which revealed extensive DCIS, 5 cm in greatest extent. There were multiple microscopic foci of invasive cancer. The surgical resection margins were positive for DCIS after multiple re-excisions. After thorough discussion the patient elected to undergo bilateral mastectomies. Two sentinel lymph nodes (blue dye and lymphoscintigram) contained micrometastatic adenocarcinoma. The patient is presently undergoing adjuvant chemotherapy.

DISCUSSION: This case exemplifies the importance of an extensive evaluation of a pathologic nipple discharge. While a bilateral, inducible nipple discharge involving multiple ducts is a physiologic finding in many women, patients presenting with a spontaneous, unilateral discharge (bloody or watery) confined to one duct need careful evaluation for an underlying neoplasm (papilloma, carcinoma). A small malignancy, which in our patient had already involved axillary nodes, can be easily missed on physical examination and mammogram. In this setting an US examination in the "trigger zone" may identify an occult lesion. Cytologic evaluation of the discharge or ductal lavage fluid may be helpful. An important diagnostic tool is ductography, as performed in our patient. With the recent introduction of fiberoptic ductoscopy direct, real-time intraductal images, allowing detection of lesions as small as 0.2 mm can be obtained offering a safe alternative to ductography in guiding subsequent breast surgery in the treatment of nipple discharge. (Shen K-W. Breast Cancer Research and Treatment, Vol. 64, No 1, Nov. 2000, p. 30)

ACUTE METHEMOGLOBINEMIA SECONDARY TO TOPICAL BENZOCAINE SPRAY. Walker¹, H. Houston¹, S. Miller¹, G. Rouan¹; ¹University of Cincinnati, Cincinnati, OH

LEARNING OBJECTIVES: 1) Define the pathophysiology associated with methemoglobinemia, and 2) Recognize the clinical settings likely to be explained by methemoglobinemia.

CASE INFORMATION: JT is an 82-year-old white female with a history of HTN, paroxysmal atrial tachycardia, and hypothyroidism who was admitted for elective laparoscopic left adrenalectomy, which proved to be a benign adenoma. Post-operative course was uncomplicated until POD #3 when the patient developed nausea and vomiting with a distended abdomen and decreased bowel sounds. A diagnosis of paralytic ileus was made and a nasogastric tube was placed, preceded by a 3 second dose of Hurricane spray (20% Benzocaine, approximately 600–885 mg total dose) to the oropharynx. Within several minutes, the patient developed lethargy and confusion. The patient then became dyspneic, syncopal, and developed a cyanotic appearance. She did not respond to 100% O₂ by non-rebreather mask despite an arterial blood gas (ABG) with a PaO₂ of 310 and an O₂ saturation of 99.7%. A simultaneous pulse oximetry revealed an O₂sat of 86%. The arterial blood was extremely dark in appearance. An ECG revealed supraventricular tachycardia with a rate of 170–180 and ST depression, which converted after 6mg adenosine to sinus rhythm, with a rate of 120 and resolution of the ischemic changes. A diagnosis of methemoglobinemia was confirmed by co-oximetry, which revealed a methemoglobin level of 30.9% (normal = 0%). A 50 mg dose of methylene blue was administered, and over the course of 1 hour, the patient developed dramatic improvement in mental status and cyanosis. Methemoglobin levels decreased to 6.4%, 3.7% and 2.4% over the next 1, 3, and 6 hours, respectively.

DISCUSSION: Methemoglobinemia is a condition in which iron in hemoglobin becomes oxidized (Fe²⁺ to Fe³⁺) at an overwhelming rate, resulting in a drastic decline in the oxygen carrying capacity of the RBC. Pharmacological agents, such as benzocaine, are capable of inducing methemoglobinemia by direct or indirect oxidation of the hemoglobin molecule. The ABG is not an accurate indicator of oxygenation status in a patient suffering from acute methemoglobinemia. The treatment of choice is methylene blue (1mg/kg, IVP). This case of benzocaine-induced methemoglobinemia is similar to others that have been reported in the past and indicates the need for a high index of suspicion for this syndrome and a need for close supervision of patients receiving benzocaine spray for clinical signs and symptoms of cyanosis.

TRAUMATIC MEMORIES IN ADOLESCENT CHRONIC FATIGUE SYNDROME. R WANG-CHENG, A LIEBMAN, MEDICAL COLLEGE OF WISCONSIN, MILWAUKEE, WI. R.M. Wang-Cheng¹; ¹Medical College of Wisconsin, Milwaukee, WI

LEARNING OBJECTIVES:

- 1) Recognize the importance of the life narrative in the comprehensive understanding of the adolescent who presents with somatic distress and chronic fatigue.
- 2) Distinguish between diagnoses that denote a disease process (eg anemia) and those that are strongly influenced by cultural "memes" (eg chronic fatigue syndrome).

CASE INFORMATION: CASE 1: A 17 y/o white teenager developed increasing fatigue beginning at age 15. Other symptoms included daily fever spikes, visual changes, myalgias, and headaches. Her grades suffered greatly as a result of repeated absences from school, & she also had to quit her part-time job. At the age of 5, she witnessed her 3 y/o brother drown in their pool while her mother was in the house. Her parents subsequently divorced when she was 12, & she has a strained relationship with her alcoholic father. The patient received counseling after the drowning and at the time of the divorce.

CASE 2: A 18 y/o high school senior was diagnosed with chronic fatigue syndrome (CFS) at the age of 13. She was unable to attend school fulltime because of sleep disturbance, diffuse pain, & disabling fatigue. Both her parents were alcoholic, and her dad was verbally & physically abusive. She served as a caretaker for her wheelchair-bound mother. At the age of 12 she was raped by her half-sister's 29 y/o boyfriend. At the age of 17 while being treated with an antidepressant for depression, her psychiatrist would not allow concurrent oral contraceptives & she became pregnant and had an abortion.

DISCUSSION: Many hypotheses have been put forth to explain CFS, an illness in search of an etiology. Characterized by a variety of somatic symptoms, CFS often has an abrupt onset. We postulate that it may be a form of "post-traumatic stress disorder," particularly in young patients. Children with CFS have been reported to exhibit more psychological morbidity, such as anxiety and depression, than those with chronic disease, but these psychological categories are not necessarily helpful in patient management. We have found that the life narrative which will often include antecedent traumatic events serves to illuminate the unique personal issues of the adolescent with CFS. This approach can create a doctor-patient relationship that will engender trust and perhaps lead to better clinical outcomes.

SALMONELLA HEPATITIS, AN UNUSUAL MANIFESTATION OF TYPHOID FEVER. S. Wani¹, M. Singh¹, M. Murtaza¹, M. Kasubhai¹, A. Singh²; ¹Lincoln Medical Center, Bronx, NY; ²Cornell University

LEARNING OBJECTIVES: 1) Recognize Salmonella Hepatitis as a cause of acute hepatitis in a patient with typhoid fever. 2) Mortality rate is as high as 20% particularly with delayed treatment. The prognosis is good if treated early with specific antibiotic therapy.

CASE INFORMATION: A 22 year old female, immigrant from Bangladesh with no past medical history was admitted with complaints of fever, nausea, vomiting and headache of one week duration. The patient had visited Bangladesh three weeks back. Prior to this presentation she had two visits to the ER in the same week and was diagnosed as viral syndrome. On admission now the patient was delirious and agitated. Her BP was 112/62; Pulse-70; Temperature-105 deg. Fahrenheit. Physical Examination - Right hypochondriac tenderness, no hepatosplenomegaly, no neck rigidity, no icterus, no skin rash. Lumbar puncture and CT Scan of head were normal. USG Abdomen was normal and HIDA Scan showed no uptake. There was a moderate elevation in the Transaminase level with AST/ALT of 294/205 respectively, LDH 579, Total bilirubin of 1.3 and direct-0.5. Albumin PT/APTT and Reticulocyte counts were normal. WBC count was 8300 with 10% band cells. Blood cultures later came positive for Salmonella Typhi sensitive to ceftriaxone. Hepatitis serology was negative for A,B and C. The patient was started on ceftriaxone .LFT's improved and patient was discharged home after 7 days in stable condition.

DISCUSSION: Salmonella hepatitis is a rare condition and 20% of the patients with salmonella hepatitis may not be bacteremic. A diagnosis of salmonella hepatitis was made on the basis of 1)ALT/LDH < 4(0.35 in this case).This happens to be the best discriminator between Salmonella hepatitis and viral hepatitis as is appreciated in this case.2) Positive blood culture for Salmonella.3) Relative bradycardia. 4) High-grade fever.5) Left shift of WBC and 6) Positive travel history to an endemic area. The prognosis is good if treated early with specific antibiotic therapy, however the clinical course can be severe with a mortality rate as high as 20% particularly with delayed treatment.

SYSTEMIC LUPUS ERYTHEMATOSUS PRESENTING AS ACUTE FATAL PANCREATITIS. S. Wani¹, M. Murtaza¹, M. Kasubhai¹, M. Singh¹; ¹LINCOLN MEDICAL CENTER, Bronx, NY

LEARNING OBJECTIVES: To recognize acute pancreatitis as an uncommon manifestation of SLE. Corticosteroids can be used in such cases if required.

CASE INFORMATION: A 54 year old hispanic female presented with one week history of fatigue, malaise, rash over face, neck and trunk, arthralgia of shoulder and knee joint, vague generalised abdominal pain and recent onset of fever and confusion. Patient did not consume alcohol, drugs or any medications. She was febrile and disoriented. She had a macular rash on

the malar eminences, neck and trunk with bilateral axillary and submental lymphadenopathy. Neurological exam was normal and no joint deformity was noted. There was generalized abdominal tenderness with no guarding or organomegaly. Investigations revealed Hb-10.6 gm/dl and total WBC-3.4K/ul, BUN-23mg/dl, creatinine-0.9 mg/dl. Liver function tests showed albumin-2.3 g/dl, AST-54, ALT-26, GGT-53U/L, LDH-485 U/L. Serum amylase was 401 and serum lipase 358. Serum calcium was 8.9 with triglycerides of 187mg/dl. Urine analysis showed specific gravity of 1015 with 3+ proteinuria. CT scan of abdomen showed fullness of pancreatic head and ultrasound showed a normal sized CBD with no evidence of cholelithiasis. Biopsy of skin rash showed intravascular thrombi, vasculitis consistent with early lupus with atypical lymphocytic infiltrate. Diagnosis of SLE with acute pancreatitis was made on the basis of above and ANA of 1:1280 with homogenous pattern and Anti ds DNA positivity. Patient was started on intravenous steroids. Patient's clinical status continued to worsen and required mechanical ventilation. Repeat CTscan confirmed pancreatitis with bilateral pleural effusions and no evidence of necrosis or abscess formation. She improved over the next few days and was successfully extubated. However, two weeks later she developed diffuse and severe abdominal pain accompanied with fever. Repeat CTscan now showed Grade E pancreatitis(multiple fluid collections +/- gas in or adjacent to pancreas)with necrosis. CT guided aspirations of necrotic material was negative for gram stain but culture grew pseudomonas, MRSA and candida. The patient had an exploratory laprotomy with necroectomy. Despite aggressive antibiotic and antifungal treatment with multiple abdominal explorations the patient's condition deteriorated. She died of multiple organ failure.

DISCUSSION: Acute pancreatitis is an uncommon manifestation of SLE. It is even rarer for such patients to present with pancreatitis as the sole major organ affected. The role of corticosteroids in such setting is controversial. Review of literature suggests that corticosteroids do not cause pancreatitis in patients with SLE and they should be used during episodes of pancreatitis if required.

TESTICULAR ENLARGEMENT: AN UNUSUAL PRESENTATION OF SARCOIDOSIS. L. Ward¹; ¹University of Pennsylvania, Philadelphia, PA

LEARNING OBJECTIVES: 1.) Recognize testicular enlargement as an unusual presentation of sarcoidosis. 2.) Identify the indications for treatment of sarcoidosis. 3.) Appreciate the goals of treatment of sarcoidosis.

CASE INFORMATION: A 30 year old African-American male with no significant past medical history presented with the chief complaint of tender, enlarged testicles for 3 months. He also noted pruritic, non-tender skin lesions on his face and arms. He denied the use of any medications. Review of systems revealed a 10 lb. weight loss, blurry vision, nonproductive cough, hoarseness and mild dyspnea on exertion. He denied any HIV risk factors. Physical examination revealed widespread, small, hyperpigmented, nodular and papular lesions. A 3x2 cm subcutaneous nodule was noted on the left side of the neck. Breath sounds were decreased bilaterally, with dullness to percussion on the left. The testicles were each twice normal size and tender with enlarged epididymi. Vital signs and the remaining exam were unremarkable. Scrotal ultrasound confirmed the physical exam and showed no discrete masses. Chest x-ray and CT revealed a moderate left pleural effusion with mediastinal and hilar adenopathy but no infiltrates. PFTs were consistent with a mild restrictive pattern. Serum ACE level was elevated at 87. Skin biopsies revealed non-caseating granulomas, negative for AFB or fungi. Follow-up scrotal ultrasound after corticosteroid therapy showed regression of testicular abnormalities.

DISCUSSION: Sarcoidosis is a disease of unknown etiology, which can present with a wide variety of symptoms and involve virtually any organ system. In this unusual case, the patient presented for care due to testicular involvement but also had dermatologic and pulmonary manifestations, which he had ignored. While many cases of sarcoidosis spontaneously remit within 5 years and do not need treatment, therapy is indicated in patients such as this one, with symptomatic disease or systemic involvement. The goals of treatment include a reduction in symptoms and the avoidance of systemic complications. Despite attempts with steroid-sparing agents, the mainstay of treatment remains systemic corticosteroids. Monitoring with chest x-ray and PFT's is often warranted and patient education remains essential.

SARCOIDOSIS AND STONES. K. White¹, U. Mason¹; ¹Denver Health Medical Center, Denver, CO

LEARNING OBJECTIVES: 1) Diagnose renal manifestations of sarcoidosis, and 2) Recognize the need for monitoring calcium levels and renal function in sarcoidosis.

CASE INFORMATION: A 43-year-old man with a long history of cystic sarcoidosis presented with acute renal failure. He had chronic renal insufficiency with a baseline serum creatinine of 1.4 mg/dL. He had been treated with prednisone for several years and his dose had been tapered recently because his pulmonary status appeared stable. Review of systems ascertained a 10-15 pound weight loss and decreased energy for 2-3 months. Physical examination was unrevealing. Laboratory data was significant for serum creatinine of 2.5 mg/dL and calcium of 11.3 mg/dL. Renal ultrasound showed right hydronephrosis and bilateral nephrolithiasis. Computed tomography showed left staghorn calculus with caliectasis and right ureteropelvic junction stone with hydronephrosis, hydroureter and a more distal stone. A right percutaneous nephrostomy was placed and serum creatinine decreased to 1.8 mg/dL. Evaluation of the etiology of hypercalcemia included normal serum PTH, protein electrophoresis and PSA levels. Urinary calcium excretion was 252 mg/24 hours. The hypercalcemia was felt to be secondary to sarcoidosis and therefore the prednisone dose was increased. He underwent right nephrolithotomy, which was successful only for the proximal stone. Bilateral stone removal is planned for the future. Serum creatinine is back to baseline and serum calcium is normal.

DISCUSSION: Hypercalcemia - an uncommon complication of sarcoidosis--results from endogenous overproduction of 1,25 dihydroxyvitamin D. Nephrolithiasis is more common, occurring in about 10% of patients, and more frequently associated with hypercalciuria than with hypercalcemia. Nephrocalcinosis is much less common and is probably related to chronic

hypercalcemia. This patient had both nephrocalcinosis and nephrolithiasis, likely due to both long-standing hypercalcemia and hypercalciuria leading to chronic and then acute renal failure. Long-term management of patients with sarcoidosis includes frequent monitoring of calcium levels and renal function.

A PRESENTATION OF SHEEHAN'S SYNDROME 22-YEARS LATER. W. Whitwam¹, D. Stuart¹; ¹Hennepin County Medical Center, Minneapolis, MN

LEARNING OBJECTIVES: 1) Recognize complications of analgesics associated with unrecognized hypothyroidism. 2) Understand the complications of Sheehan's syndrome and panhypopituitarism.

CASE INFORMATION: A 62-year-old African-American female, with a past medical history notable only for a prior cesarean section, received an uneventful laparoscopic repair of a ventral hernia. Post-operatively, she received morphine analgesia by a patient controlled analgesia (PCA) pump. After developing an unanticipated stupor, her morphine was discontinued. She received several doses of naloxone without response. A computerized tomography (CT) of the head demonstrated no acute changes. With an arterial pCO₂ value of 62 mm Hg, she was placed on biphasic positive airway pressure (BiPAP) ventilation. She remained somnolent. A formalized evaluation of her delirium included a thyroid function panel and random cortisol level, both studies demonstrated abnormally low values. A magnetic resonance image (MRI) of her brain revealed a partially empty sella turcica. Her normal luteinizing hormone (LH) and follicle-stimulating hormone (FSH) levels were discordant with her post-menopausal status, and her prolactin level was below the normal reference range. After receiving thyroxine and hydrocortisone replacement therapy, her depressed consciousness resolved within two days. Upon a later interview, she revealed that 22-years prior she had a cesarean delivery of her seventh child, resulting in a large blood loss and subsequent hypotension. Unlike her six previous pregnancies, she failed lactation after delivery, and subsequently never resumed menstruation.

DISCUSSION: Sheehan's syndrome is an obstetric complication that occurs with excessive blood loss and ischemic shock following a complicated delivery. This results in an infarction of the pituitary's anterior lobe. As a result of her hypopituitarism, this patient developed secondary thyroid and adrenal failure as well as gonadal insufficiency. With an unrecognized secondary hypothyroidism, the exaggerated depressant effects of her analgesics resulted in a postoperative stupor. Careful attention to hypothyroidism should be noted with any patient who develops an excessive central nervous system response to anesthetics or analgesics.

EPISODIC HYPERTENSION ASSOCIATED WITH SITTING ON HARD SURFACES. R.M. Wittle¹; ¹Stanford University, Stanford, CA

LEARNING OBJECTIVES: 1. Distinguish between the potential causes of episodic hypertension. 2. Recognize the practical consequences of the complex relationship between pain and elevated blood pressure.

CASE INFORMATION: A 76-year-old Japanese-American male was referred to the hypertension clinic for a long-standing history of baseline normotension interrupted by hypertensive episodes associated only with sitting on hard surfaces. Home blood pressure measurements (consistent with measurements in clinic) ranged between 110-120/70-80 mm Hg, but rose to 180-190/100-110 mm Hg immediately after the patient sat on a hard surface such as an ordinary chair; no change in blood pressure from normal values was observed when the patient sat on a soft surface or was in any other position. Measured blood pressure was the same in both arms, and returned to a normal range immediately after the patient stopped sitting on a hard surface. The patient experienced no symptoms when sitting on a hard surface except for bilateral blurry vision; this too promptly returned to normal after getting up from the hard surface. At various times in his life, the patient had taken reserpine, lisinopril, verapamil, enalapril, atenolol, hydrochlorothiazide, and clonidine to control his hypertensive episodes, always without efficacy; he is currently taking no anti-hypertensive medications. Of possible relevance, the patient had chronic neck pain which was exacerbated by sitting on hard surfaces; however, the change in blood pressure reliably and reproducibly occurred before the pain. The patient had normal urinary catecholamines, vanillylmandelic acid, and metanephrine values, normal thyroid studies, no evidence of panic disorder, and no evidence of end-organ damage. A trial of opioid analgesics had no effect on the patient's blood pressure.

DISCUSSION: Episodic hypertension is a relatively uncommon clinical entity, most often associated with the "white-coat" phenomenon, pheochromocytoma, ingestion of sympathomimetic drugs, withdrawal from alcohol and other drugs, thyrotoxicosis, panic disorder, and intermittent pain. Prior studies have discovered a complex relationship between pain and blood pressure, with involvement at multiple areas of the central nervous system. This patient demonstrated an unusual temporal relationship between pain and blood pressure, with the rise in blood pressure occurring first; the hypothesis of a conditioned response should be considered.

BELLS PALS AND HYPERTENSION SECONDARY TO DISCONTINUATION OF ANTIHYPERTENSIVE TREATMENT. S. Yakoob¹, K. Barnard¹, J. Haretos¹; ¹UPMC Shadyside, Pittsburgh, PA

LEARNING OBJECTIVES: 1. Recognition of an association between facial nerve paralysis and accelerated hypertension. 2. High dose corticosteroid therapy often prescribed for facial nerve palsy has serious consequences in hypertensive patients. 3. Facial palsy resulting from severe hypertension may be distinguished from Bell's palsy by lack of pain and preservation of taste.

CASE INFORMATION: An 84-year-old male who presented with a two-day history of right-side facial drooping, difficulty in chewing and inability to close his right eye. His past medical history is significant for hypertension controlled on quinapril and bisoprolol/HCTZ. Two days prior to the onset of symptoms, the patient had discontinued his medications. Physical examination was remarkable for a blood pressure of 190/96 and signs of right lower motor

neuron facial nerve palsy. Diagnostic evaluation included CT scan of the brain that was negative for hemorrhage and a maxillofacial CT that showed thickening of paranasal sinuses. An EMG revealed decreased conduction over the right facial nerve. The patient's blood pressure was controlled by oral metoprolol and quinapril. High dose corticosteroid therapy was started and subsequently discontinued when the patient's blood pressure increased to 205/105 mm Hg. This case is similar to two other case reports in the literature of abrupt withdrawal of antihypertensive treatment resulting in an acute increase in the blood pressure and subsequent facial paralysis.

DISCUSSION:

BENZOCAINE INDUCED TOXIC METHEMOGLOBINEMIA DURING ENDOSCOPY. M.A. Zihlif¹, B. Khan¹, F. Begum¹, A. Karakurum¹, C. Pellcchia¹; ¹NASSAU UNIVERSITY MEDICAL CENTER, EAST MEADOW, NY

LEARNING OBJECTIVES: To recognize toxic methemoglobinemia as a rare but potentially lethal complication of local anesthetics so that prompt diagnosis and treatment can be initiated.

CASE INFORMATION: A 37 year old white female patient with negative past medical history, presented to our clinic with a complaint of epigastric pain and was scheduled for endoscopy. Benzocaine spray was administered topically in the oropharynx during endoscopy and the procedure was uneventful, but approximately 10 minutes after completion, the patient oxygen (O2) saturation began to fall and she became cyanotic despite administration of 100% O2 via a non-rebreathing mask. Intravenous Narcan and Flumazenil were administered without improvement. The O2 saturation continued to fall with the lowest level obtained was 81%. Arterial blood gas analysis showed PaO2 of 367mmHg and PaCO2 of 23mmHg so methemoglobinemia was suspected. Methemoglobin level was sent and later showed a level of 37.6%. Methylene blue was given intravenously, the patient started to improve gradually and within 20 minutes O2 saturation rose to 96% and cyanosis was resolved. The patient was transferred to intensive care unit, where her O2 saturation rose to 99% and methemoglobin level dropped to 1%. The remainder of hospital stay was unremarkable.

DISCUSSION: Methemoglobinemia is a rare complication of local anesthetics that has been reported with procedures including bronchoscopy, endoscopy, tracheal intubation and dental procedures. Methemoglobin is formed normally in the body because small amount of ferrous iron in hemoglobin is continuously oxidized to ferric iron which can't carry oxygen or carbon dioxide. Red cell defense against accumulated methemoglobin is methemoglobin NADH reductase which is responsible for 99% of in vivo reduction of methemoglobin to form normal hemoglobin. Methemoglobinemia has both acquired and congenital forms. Acquired methemoglobinemia can be associated with toxic shock or can result from ingestion or skin exposure to oxidizing agents which oxidizes hemoglobin to methemoglobin. Although cyanosis may appear with levels as low as 15%, acquired methemoglobinemia is rarely symptomatic when levels are below 20%. Lethargy, dizziness, lightheadedness and anxiety are present between levels of 30–40%. Coma, seizures, arrhythmias and acidosis could be caused with levels of 50–70%. Levels more than 70% are fatal. We were first alerted to the presence of methemoglobinemia by decrease in O2 saturation inspite of normal O2 tension (PaO2) which is almost a universal finding. Methylene blue given intravenously 1–2mg/Kg over 5 minutes acts a cofactor in the transfer of an electron from NADPH to ferric iron and methemoglobinemia should resolve within one hour, if not the dose may be repeated.

POST PARTUM DEPRESSION AS A PRESENTATION OF DURAL VENOUS SINUS THROMBOSIS. M. Zihlif¹, L. Wong¹, B. Flansbaum¹; ¹Nassau University Medical Center, East Meadow, NY

LEARNING OBJECTIVES: 1) To consider dural venous sinus thrombosis in the differential diagnosis of puerperal psychoneurotic symptoms. 2) Initiating management as soon as the diagnosis is recognized can cause major decrease in morbidity and mortality.

CASE INFORMATION: 21 year old female patient with no significant past medical history, had normal vaginal delivery 3 weeks prior to admission, presented with depressed mood, poor concentration and headache. Physical examination was negative. Brain computed tomography scan (CT) was done and found to be negative so the patient was diagnosed as a case of postpartum depression and was admitted to psychiatric floor. One day after admission patient became drowsy and her level of consciousness dropped. Head and neck CT scan with contrast showed thrombosis of the right jugular vein. Magnetic resonance image (MRI), magnetic resonance angiography (MRA) and magnetic resonance venography (MRV) were done and showed thrombosis of the superior sagittal sinus, right transverse and deep venous sinus. Patient developed respiratory distress and was intubated and was started immediately on Intravenous Heparin. After few days she was extubated and transferred to medical floor where she was switched to oral anticoagulants. Patient depressive symptoms improved and her mood returned back to normal.

DISCUSSION: Dural venous sinus thrombosis is an unusual disorder most often attributed to hematological disorders, oral contraceptives, association with pregnancy and puerperium, Behcet's disease, cardiac disease and post operative conditions. Patients usually present with nonspecific symptoms as headache, seizures, hemiplegia or other neurological symptoms. Brain swelling and bilateral involvement can produce lethargy or stupor early in the course. Diagnosis depend on recognition of impaired venous flow. On contrast CT scan a nonenhanced triangular area surrounded by contrast in the posterior sinus (the empty delta sign), should suggest the diagnosis. Cerebral angiography has been considered the gold standard for the diagnosis but MRI, MRA and MRV are becoming the diagnostic studies of choice. Management increasingly relies on the use of anticoagulation even in the presence of superimposed parenchymal hemorrhage. Venous occlusions are serious and often fatal but acute anticoagulation started as soon as the diagnosis recognized appears to lessen substantially the morbidity and mortality of the condition. Nonanticoagulated venous sinus thrombosis that is not complicated by infection carries a mortality rate of 25 to 40%. Uncontrolled series suggest that early heparin treatment can reduce mortality by more than a half.

HRSA SUBMISSIONS

SECTION 1: FACULTY DEVELOPMENT SUBMISSIONS

A WORKSHOP IN REFLECTIVE PRACTICE FOR FIRST YEAR MEDICAL STUDENTS. C. Baltimore¹; ¹University of Massachusetts Medical School (Worcester), Worcester, MA

OBJECTIVES: Reflective practice may enable physicians to become more competent and successful practitioners. This technique, however, is usually developed at the faculty level. The objective of this workshop is to begin to develop this skill at an earlier level. Students may then begin to utilize this skill for success at the student/resident level as well as honing this skill for use when at the faculty level, for both clinical practice and teaching.

METHODS: Students will be given a brief (approximately 1 1/2 hour) skill session in the use of reflective practice.

Intervention outcome will be assessed using the following: Pre/post test for knowledge, skills and attitudes regarding reflective practice. Randomization of the same class of students (Class of 2005) – 1/2 to intervention, 1/2 no intervention. Each group will then be given a reflective practice questionnaire following an OSCE exercise, to assess utilization of the skill.

RESULTS: None yet available.

CONCLUSION: None yet available.

"ITS THE TEACHING, STUPID!": LESSONS LEARNED FROM THE FIRST YEAR OF THE UCSF GENERAL INTERNAL MEDICINE FACULTY DEVELOPMENT PROGRAM. M.D. Feldman¹, A.B. Bindman¹; ¹University of California, San Francisco, San Francisco, CA

OBJECTIVES: The UCSF Division of General Internal Medicine (DGIM) faculty development program trains general internists to be effective teachers in primary care settings that serve a diverse patient population. During the first year of our program, we recruited general internists from community based health centers and faculty of the DGIM at UCSF. We developed and taught a 9-month long curriculum with six core content areas: teaching skills, evidence based medicine, cultural competency/caring for vulnerable patients, psychosocial medicine, population medicine, and leadership. Sessions were 3.5 hours long, two afternoons per month.

METHODS: Prior to implementing our program we conducted a baseline needs assessment among trainees. An external evaluator met with the trainees during and at the end of the year to elicit feedback and to assess if the program was meeting its' objectives and the trainee's learning goals.

RESULTS: Twelve trainees, 5 from under-represented minority groups, participated in the program last year. The evaluation concluded that the most valued aspect of the program for the trainees was the opportunity to learn and practice new teaching skills. The trainees also expressed interest in developing a tangible product, such as a new curriculum, at the completion of the program.

RESULTS: Primary care physicians in the SF Bay Area have adequate access to new medical knowledge, but have limited opportunities to learn and refine new teaching skills. Now in year 2, the program has been modified to put greater emphasis on learning and practicing teaching skills, and less time is devoted to teaching the trainees new medical knowledge. Trainees are now required to develop a project, with the guidance of a program mentor, that is relevant to their own teaching setting. The faculty development program has enabled community and academically based general internists to form a common bond around teaching.

EVOLUTION OF A PART-TIME, COMMUNITY-BASED FACULTY DEVELOPMENT PROGRAM. R.M. Butler¹, K. Busch², B.D. Friedman², E. Johnson¹, K. Dobbs², S. Pinheiro¹; ¹Michigan State University College of Osteopathic Medicine, Madison Heights, MI; ²Michigan State University College of Osteopathic Medicine Statewide Campus System, E. Lansing, MI

OBJECTIVES: Since 1993, Michigan State University College of Osteopathic Medicine Statewide Campus System (SCS) has received HRSA funding to develop a part-time, community-based faculty development program to improve the quality of medical education within the eighteen member hospitals.

METHODS: Instructional methods supporting individualized learning plans were enhanced by yearlong, on-site, one-on-one educational support from faculty development specialists. Curriculum has evolved to address the changing medical education environment. The program was examined within grant objectives, historical records, and archival data obtained through written surveys and focus groups.

RESULTS: Focus groups and survey results have tentatively identified that the one-on-one consultation enhanced the 59 participants' ability to provide curriculum instruction within SCS member hospitals. 88% of graduates continue to teach in SCS institutions, using skills obtained through the program. 33% of the participants have received teaching awards with two winning national poster awards. 95% of the participants have remained in primary care. Some graduates have moved to higher positions within medical education.

CONCLUSION: Developing part-time, community-based faculty is a multi-faceted endeavor. Faculty are challenged between providing service versus training others. The focus of providing a yearlong stipend supported program with one-on-one consultation support by specialists has proven to be a successful model in attracting and developing part-time, community-based faculty.

EFFECTS OF ADOPTING NEW CLINICAL TEACHING METHODS ON TEACHING EFFECTIVENESS RATINGS. R.L. Krippendorff¹, R.M. Wang-Cheng¹, C.A. Heidenreich¹, D.L. Schiendermayer¹, P.S. Lye¹, K.J. Marcandante¹, V.R. Rediske¹, D.E. Simpson¹; ¹Medical College of Wisconsin, Milwaukee, WI

OBJECTIVES: Primary care physicians are challenged to be clinically productive and effective teachers. A critical review of the literature revealed limited evidence regarding effectiveness of four often cited teaching methods: 1-2 focal teaching points; priming; teaching in the patient's presence [TIPP]; and feedback. To address this gap seven experienced clinician educator [CEs] faculty (4 general internists; 3 pediatricians), participated in a faculty development project to evaluate the effectiveness of the four selected clinical teaching methods.

METHODS: During monthly faculty development sessions, CEs were trained on the teaching methods using multiple strategies including simulations and written exercises. Each CE recorded use of these methods over a 10-month period on a Palm Pilot(tm) using a specially designed form. Baseline (July 1997–June 1998) and study period clinical teaching evaluation ratings for CEs (treatment) and non-participating faculty in their specialties (control) were compared using ANOVA.

RESULTS: CEs reported using all targeted teaching methods > 50% of time during recorded teaching sessions (N=1,176 sessions) with use of priming and feedback increasing over baseline while use of the other two methods remained constant. CEs reported that the teaching methods focused both the learner and the clinical teacher, making subsequent encounters more productive and efficient. CEs teaching evaluations were significantly higher ($p < .001$) on three items during the study period while control group's ratings showed no change (i.e., responded to student initiated learning issues; emphasized comprehension of concepts; instructor had sufficient data to assess learner performance). During the study period, two CEs were first time recipients of major department teaching awards with three CEs receiving highly competitive, college-wide teaching awards one-year post study period.

CONCLUSION: Experienced CEs who participate in a faculty development program can improve their clinical teaching ratings using four literature-based clinical teaching methods. Recognition by learners and peers of the CEs educational excellence through awards was an unanticipated outcome.

SHORT-TERM OUTCOMES OF OLDEST OLD PATIENTS (90 YEAR OLD AND MORE) TREATED IN AN EMERGENCY DEPARTMENT. N. Nakazato¹, Y. Tokuda¹; ¹Okinawa Chubu Hospital, Gushikawa, Okinawa

OBJECTIVES: The population of older persons is rapidly rising in developed countries like Japan. Oldest old patients aged 90-year-old and more had occasionally utilized acute care hospital emergency department in recent years. To our knowledge, there were no study evaluating clinical reasons and outcomes of these highly aged population.

METHODS: To determine the short-term outcomes and predictors of outcome, we reviewed medical records of elderly patient aged over 90-year-old visiting emergency department in Okinawa Chubu Hospital, acute care community hospital in rural Okinawa, Japan, in 1999.

RESULTS: Three hundred seventy one patients (95 men, 276 women: mean age 92.9 years old) were studied. Five common chief complaints were fever (65 cases), shortness of breath (46), consciousness disturbances (33), leg pain (29), and nausea or vomiting (21). Pulmonary diseases (94) were the commonest category of disorders, followed by orthopaedic diseases (43) and gastroenterologic diseases (42). Only 42(11%) patients were dead during hospitalization following admission from emergency department except for 14 cardiopulmonary arrests at arrival. In multiple logistic regression analysis, cognitive dysfunction was only significant predictor for short-term mortality.

CONCLUSION: Majority of oldest old patients had survived to discharge following emergency department admission. Cognitive function was important in terms of survival prediction.

AN INTEGRATED PARTNERSHIP BETWEEN THE UNIVERSITY OF CINCINNATI AND COMMUNITY HEALTH CENTERS. T. Redington¹; ¹College of Medicine, Cincinnati, OH

OBJECTIVES: 1. Identify the common interests of both academic health centers and community health centers that cause them to be natural partners. 2. Demonstrate the unprecedented level of support that the University of Cincinnati has shown in preserving the capacity of community health centers. 3. Describe the team approach adopted by the providers of the Cincinnati Health Network, Cincinnati Health Department, University of Cincinnati Hospital clinics, and Southern Ohio Health Network, which included a common eligibility system and data sharing throughout the delivery system. 4. Discuss this common approach to patient eligibility and clinical care includes disease management protocols for asthma, diabetes, hypertension and depression. This common clinical approach is expected to improve if not eliminate health disparities, consistent with the Bureau of Primary Health Care's 100% access, 0% disparities goal.

METHODS: The University of Cincinnati has budgeted 2 million dollars of in-kind support for all the center sites of the Cincinnati Health Network.

This support includes a pharmacy with pharmacist, radiology suite, midwives, and 12 full time providers, eleven are physicians. These physicians are extensively involved in medical student education at these center sites. Also the University led the successful effort to receive a million dollar grant (from HRSA the Community Access Program) to improve the infrastructure around the indigent and uninsured in Southern Ohio. The major partners include the University, a hospital of the Health Alliance of Greater Cincinnati, a five hospital system in southwestern Ohio, the Cincinnati Health Network, the Southern Ohio Health Network (both Federally Qualified Health Centers), and the Cincinnati Health Department. Together, these partners serve about 200,000 lives, 90% are at or below the minimum federal poverty level.

RESULTS: The quality of care standards around depression, hypertension, asthma and depression have been installed. The information technology infrastructure and eligibility systems are being developed and linked between different computing platforms. The measurement of our quality of care will begin later this calendar year. Medical education in these clinic sites is extensive, about three quarters of the medical student class are being educated in these sites.

CONCLUSION: There is a major partnership underway in southern Ohio, linking the University of Cincinnati's Hospital and College of Medicine with four large, previously independent health care systems, that have traditionally provided for the same patients, the indigent and uninsured. By the adoption of common eligibility and disease management standards, access will be improved and disparities reduced.

WHAT ARE THE NEEDS OF COMMUNITY BASED PRECEPTORS IN A RURAL STATE? A. Rubin¹, T. Bertsch¹; ¹University of Vermont College of Medicine, Burlington, VT

OBJECTIVES: To retain and recruit community-based preceptors in Vermont, we developed, distributed, and analyzed a survey to understand the needs of our community faculty and the barriers to precepting students in their offices.

METHODS: We mailed questionnaires to all 570 primary care physicians in Vermont. We received replies from 251, of whom 160 were preceptors. We asked questions about their needs to become better teachers, the infrastructure needed to teach, rewards for teaching, and the desired format for faculty development.

RESULTS: Areas of high importance for teaching include becoming more efficient, balancing student needs with patient time, increasing ability to teach clinical skills, giving feedback to students, dealing with difficult students, and finding ways of orienting students to practice. In terms of infrastructure, preceptors want more time and space to teach. They want rewards in the form of CME or CME credit and in hearing feedback about their students. They are happy to have faculty development sessions once a year, either at a central site or near their practice site.

CONCLUSION: As with our students, eliciting feedback from community based teachers can result in programmatic change. As a result of this survey, we have developed a teaching skills exercise using standardized students and presented it in three sites. We are also changing the ways we award CME credits.

A TEACHING EXERCISE FOR COMMUNITY PRECEPTORS USING STANDARDIZED STUDENTS. A. Rubin¹; ¹University of Vermont College of Medicine, Burlington, VT

OBJECTIVES: To assess the teaching skills needs of our community based teachers. To develop and present a workshop that would allow them to practice these skills.

METHODS: As part of our HRSA project we developed a needs assessment for our community based teachers. Of the areas in which they wanted to enhance their teaching skills, we chose four to learn and practice: Teaching Efficiently, Framing a Clinical Question, Giving Feedback, and The Difficult Student. We developed scripts and videos for a four station teaching exercise using a rolling role play and standardized students.

RESULTS: Preceptors rotated in groups of four through each half-hour station. In each, a brief didactic presentation was followed by a rolling role play with a standardized student, which gave each preceptor a chance to practice. A short debriefing followed. We have presented this workshop to 60 of our preceptors in three areas of the state.

CONCLUSION: Preceptors rate the workshop format as effective and "couldn't be better." They find the topics appropriate and practical. They especially like the student participation and the ability to practice what they learn in small groups. We don't yet know whether preceptors are using the skills they have learned. This is an area for further inquiry.

SECTION 2: RESIDENCY TRAINING SUBMISSIONS

STRENGTHENING GERIATRIC COMPETENCIES OF PRIMARY CARE RESIDENTS: ENHANCEMENTS IN CURRICULUM, CLINICAL EXPERIENCES AND EVALUATION. R.B. Baron¹, E.M. Aagaard¹, L. Burkholder¹; ¹University of California, San Francisco, CA

OBJECTIVES: Despite expansion of programs in geriatrics, it will not be feasible to train enough geriatricians to care for all elderly patients. Several organizations have suggested that improved training of primary care physicians in geriatrics is likely to have the greatest impact on improving medical care of geriatric patients. To address this need, we have enhanced several aspects of our curriculum in geriatrics to strengthen the clinical competencies of primary care residents. Our specific objectives are 1) to improve the didactic curriculum; 2) to increase the number and quality of ambulatory clinical experiences; and 3) to develop new methods for measuring the clinical competencies of residents in geriatrics.

METHODS: Primary care residents spend six months both years on structured ambulatory rotations. Two half days per week are devoted to structured didactics and eight half days to ambulatory rotations. Portions of both of these experiences are now devoted to enhanced geriatric training. Didactic curriculum: We have developed twenty specific learning objectives in geriatrics. These are taught in ten seminars accompanied by a syllabus of recent and sentinel articles. Ambulatory experiences: Ambulatory geriatric training begins in the first year on a home care rotation directed by geriatrics faculty. Second and third year residents may also work on home care and/or in a variety of geriatric health centers. Selected residents spend one half day per week during each ambulatory block in geriatric health centers as a second continuity experience. We have also begun to increase geriatric teaching in the residents' own primary care continuity clinics by recruiting more geriatrics-trained faculty to serve as core resident preceptors. Enhanced evaluation: We have enhanced our efforts at evaluating resident experiences in geriatrics with the use of a web-based evaluation system. We will also evaluate resident attitudes toward the care of older patients and the extent to which learning objective

have been met. Finally, we will develop new strategies in measuring resident competency in geriatrics with a geriatric clinical evaluation exercise (geri-CEX) and with structured medical record reviews.

RESULTS: We hypothesize that these methods will increase the knowledge, attitude and skills of residents in geriatrics and will result in improved competency in geriatrics for primary care physicians trained in this manner.

CONCLUSION: The curriculum of structured primary care residency programs can be modified to address specific deficiencies in health manpower. We have designed a strategy for specifically enhancing the competency of primary care physicians to care for the elderly.

AN ACADEMIC-COMMUNITY PARTNERSHIP AS A SETTING FOR MEDICAL STUDENTS AND RESIDENTS COLLABORATIVE RESEARCH: PRESENTATION OF A MODEL. D. Bor¹, R.J. Pels²; ¹Cambridge Health Alliance, Cambridge, MA; ²Harvard University, Newton, MA

OBJECTIVES: 1. Promote community-oriented primary care. 2. Teach medical students and residents about COPC through service learning in diverse underserved communities. 3. Develop role models in public health practice. 4. Engage the academic health center in its community. **METHODS:** 1. We established an "Institute for Community Health" in Cambridge and Somerville, dedicated to research and education in health promotion and disease prevention. Three health care networks (Cambridge Health Alliance, CareGroup and Partners) found common cause in a scholarly collaboration to improve the health of these communities. 2. The Community-oriented Primary Care (COPC) curriculum was chosen as one of three priority initiatives. The curriculum contains: a) experiential (research) elements: community defined projects; teams of a preceptor, medical resident and one or several medical students; year-long longitudinal projects. b) Didactic elements include a seminar series taught by faculty drawn from collaborating institutions. 3. We expect to enroll the first students in September.

RESULTS: 1. Trainees will be evaluated by faculty and their community partners - in accordance with defined competencies, the quality of their products (e.g., policies, guidelines, webpages, interventions, etc.) and the sustainability of their work. 2. The faculty and curriculum will be evaluated by the community on the basis of the contributions to the local community's health, spread to other communities (within the collaborators' catchment areas), and impact on students' careers; and by the trainees.

CONCLUSION: This educational model has the potential to positively influence the health of underserved communities, the careers of health professionals, and the missions of health care institutions - in Cambridge and Somerville, and beyond.

RESIDENT PARTICIPATION AT NATIONAL SGIM MEETINGS. D.W. Brady¹, L.J. Schultz¹, W.T. Branch²; ¹Emory University, Decatur, GA; ²Emory Healthcare, Atlanta, GA

OBJECTIVES: 1. Expose residents to general medicine career options by having them attend National SGIM meetings. 2. Encourage residents to participate in National SGIM meetings by submitting vignettes, abstracts, and workshops.

METHODS: Our residency training grant affords us the opportunity to send all 10 of our second-year primary care residents to the national SGIM meeting each year. We encourage the residents to actively participate in the meeting by collaborating with faculty on workshops and submitting their own research abstracts or clinical vignettes. By exposing them to generalists from various institutions and supporting their own scholarly activity, we hope they will pursue generalist careers and remain active in SGIM.

RESULTS: Beginning with the 1998 SGIM meeting in Chicago and including this year's meeting in San Diego, we have brought 40 second-year residents to national SGIM meetings. In 1998, all 10 residents collaborated on a workshop that won the David E. Rogers Award. In 1999, two of those residents returned as senior residents to present research posters. In 2000, three third-year residents presented research posters, and one second-year resident presented a clinical vignette poster. This year, one second-year resident is co-leading a workshop while the other 9 second-year residents are all submitting clinical vignettes. So far, 90% of our graduates have remained in general internal medicine, with approximately 50% obtaining academically-affiliated positions.

CONCLUSION: By exposing our residents to the myriad of people and opportunities afforded them at national SGIM meetings, we encourage them to remain excited about generalist careers and to pursue their own academic interests. We believe their attendance at the national SGIM meetings is an integral part of our internal medicine primary care residency program and our HRSA grant.

SELF-REFLECTION AS A TOOL TO ENCOURAGE CULTURAL COMPETENCY. D.W. Brady¹, I. Genao¹, J. Bussey-jones¹; ¹Emory University, Atlanta, GA

OBJECTIVES: 1. To promote cultural competency through self-reflection. 2. To encourage residents to examine their own cultural biases and prejudices. 3. To provide a safe environment where residents can ask difficult questions about other cultures in a spirit of inquiry.

METHODS: As part of our cross-cultural curriculum, we set aside one session to allow residents to self-reflect and ask questions they had about each other's cultures. The group consisted of 10 third-year primary care residents - six men and four women, including two African-Americans, two Asian-Americans, at least two evangelical Christians, one Jew, at least one atheist, and two homosexuals (these were some of the represented cultures known to the group). We asked them each to write down a question and place it in an envelope. They were asked to write questions that they had always wanted to ask a person of a different culture (race, ethnicity, religion, sexual orientation, etc.) but were either afraid to ask or thought was too stereotypical. The questions were not to be directed personally at any specific individual within the group. After collecting the questions, we passed the envelope back around the room and asked each person to take out one question and read it to the group. The intent was to create a safe environment where the residents could bring their "forbidden" questions into the open and to allow them to see the reaction of the group without being personally humiliated or attacked for asking their questions. Our hope was for each resident to become

more aware of his or her own cultural competency and to learn to hear "the other" in a spirit of inquiry.

RESULTS: The group rated this session higher than any other session in the curriculum. We originally intended for the group to just hear the questions and feel their impact. The group, however, at the beginning of the session decided by consensus to allow some discussion of each question by the group before proceeding. In evaluating the experience, the residents said that the discussion was integral to making the experience its most meaningful. They also said that late internship might be a good place to introduce this exercise as it would have helped them earlier in their training. They added, however, that introducing it too early, before the group has a chance to form, might have risked alienating members of the group, thus damaging their ability to develop cohesion.

CONCLUSION: The depth of learning and the residents' evaluation of the exercise have encouraged us to retain it as a part of our multicultural curriculum. Whenever we as physicians can afford trainees the opportunity to be self-reflective, especially in a cross-cultural context, we can enhance doctor-patient communication, learn more about ourselves and each other as human beings, and, ultimately, take better care of each individual patient and society as a whole.

FACILITATING CAREER DECISIONS FOR GENERALIST CAREERS. S. Brandenburg¹, L.J. Adams¹, B. Dwinell¹; ¹University of Colorado Health Sciences Center, Division of General Internal Medicine, Denver, CO. S.L. Brandenburg¹, L.J. Adams¹, B.G. Dwinell¹; ¹University of Colorado Health Sciences Center, Denver, CO

OBJECTIVES: To describe the efforts of one general internal medicine training program to promote the selection of generalist careers by residents. In the past couple of years, there has been a declining interest in generalist careers, yet primary care physicians play an important role in meeting the health care needs of the medically underserved.

METHODS: Multiple strategies for promoting selection of primary care careers have been piloted in the Primary Care track of our internal medicine residency program:

- Mentorship: Each primary care resident is paired with a general internal medicine faculty member based on common career or personal interests and goals. Mentorship pairs meet at least twice a year to review clinical performance, research, educational plan, and career goals.

- Continuity clinics: Primary care residents have two half days of clinic in settings which serve a large Medicaid and medically indigent population.

- Primary care rotations: Several primary care rotations including adolescent medicine, women's health, and outpatient HIV have been developed to improve the ability of residents to practice in a variety of outpatient settings.

- Preceptorships: All primary care residents are required to do a preceptorship during the R2 year. Rotations in medically underserved and rural areas are strongly encouraged.

- Career development seminars: Topics such as Finding a Job, Evaluating Contracts, and interaction with a panel of internists who practice in various settings are incorporated into the three-year primary care curriculum.

RESULTS: During HRSA funding (1997-2000), 23 of 25 graduates chose primary care careers; of the two choosing fellowships, one was in geriatrics. The career development sessions were consistently rated very positively; 4.85 to 5.00 on a 5-point scale. In focus groups, residents also reported high satisfaction with these sessions, primary care rotations, continuity clinics and the mentorship program.

CONCLUSION: While we have successfully promoted primary care career choices, we have been less successful in placing graduates in medically underserved areas. In attempt to increase graduates choosing careers in medically underserved areas, we are focusing recruitment efforts on students with documented interest in care of the medically underserved and supporting this interest throughout residency training. Due to the national trend of declining interest in primary care careers, these efforts will be extended to include categorical residents and medical students.

A REQUIRED ROTATION IN HOMELESS MEDICINE - PROMOTING ALTRUISM TO RESIDENTS. D.R. Buchanan¹, L. Rohr¹; ¹Cook County Hospital / Rush University, Chicago, IL

OBJECTIVES: Although the American Association of Medical Colleges (AAMC) considers altruism to be one of the basic qualities that physicians should possess, residency programs rarely incorporate required rotations aimed at meeting this goal. Our objective was to design and implement a required rotation to promote altruism and careers with underserved populations through teaching residents about homelessness.

METHODS: The course was designed to provide a balance between direct patient care and didactic sessions. Clinical experiences were supplemented by visits to providers of homeless services and by a lecture series. Residents worked in a variety of clinical settings, including performing medical outreach in locations around the city. The locations selected were meant to expose the residents to different facets of impoverished communities. In addition to the sites the residents visited as part of their clinical work, they also visited a number of medical and non-medical sites related to homelessness. Specific sites included Chicago's only respite center for medically ill homeless people, Chicago's largest homeless shelter and its adjoining health center, and a smaller homeless shelter. A lecture series was included to help provide a context for understanding the medical and social issues unique to this population. The lectures focused on causes of poverty and homelessness as well as their effects on health and access to healthcare. The series was also used as a time for discussions, allowing the residents to reflect on their experiences and on assigned readings.

RESULTS: The course was evaluated by pre and post course surveys. The surveys were designed to identify resident attitudes toward caring for homeless patients. We also used the surveys to identify barriers that residents feel toward pursuing careers in which they would care for underserved populations. Results of these surveys will be presented at the meeting.

CONCLUSION: This required rotation, "Caring for Homeless Patients", was implemented in the spring of 2001 for third year Primary Care Internal Medicine residents. Initial responses to the rotation have been positive. Future directions for this program include further assessment of whether the rotation is meeting its goal of promoting altruistic behavior among residents.

ESTABLISHING AN INNER-CITY COMMUNITY-BASED PRIMARY CARE TRAINING SITE TO ENHANCE CULTURAL COMPETENCY FOR INTERNAL MEDICINE RESIDENTS. T. Cavallieri¹, K. Welding¹, J. Ciesielski¹, J. Kaiser-smith¹, H. Dombrowski¹; ¹University of Medicine and Dentistry of New Jersey, Stratford, NJ

OBJECTIVES: To establish an inner-city community-based training site that will enhance primary care skills, build cultural competency, and improve internal medicine residents' skill in addressing the needs of an underserved population.

METHODS: The University of Medicine and Dentistry of NJ-School of Osteopathic Medicine (UMDNJ-SOM) established an affiliation with an existing primary care practice founded by the Diocese of Camden and the Society of Jesus. This practice, St. Luke's Catholic Medical Services, serves the poor, uninsured and underinsured community of Camden, NJ. It has a fluent bilingual staff and attracts a predominately Latino population. The physician at St. Luke's was a primary care internist/Jesuit priest who was volunteer faculty in UMDNJ-SOM's Department of Medicine. Initially, a full-time faculty member from UMDNJ-SOM was assigned part-time to assist him in providing patient care and to establish a training site for internal medicine residents. The Department then recruited a full-time primary care internist with a special interest and prior experience in serving the underserved for this facility. As part of this collaboration between UMDNJ-SOM, the Diocese and Jesuit Order, the Order funded the physician's participation in a Spanish immersion program in Mexico to improve language skills and cultural awareness. The physician has since initiated care at St. Luke's and been named Medical Director.

RESULTS: This new partnership with St. Luke's, attributable to the collaborative efforts of the University, Diocese and Jesuits, has led to the development of learning objectives for a primary care continuity experience for the residents in our HRSA funded primary care residency. The primary care residents have begun weekly half-day sessions at St. Luke's that sensitize them to the medical, cultural and psychosocial needs of this underserved community. As a result of this HRSA grant, the new affiliation with St. Luke's has created additional educational opportunities. It has provided a new training site for HRSA-funded geriatric fellows and UMDNJ-SOM's extern program as well as primary care residents.

CONCLUSION: This collaboration has provided the University with a unique training site and rewarding learning experience. Internists intending to serve underserved populations must be able to appreciate cultural diversity both within and between cultures; understand the impact that psychosocial, educational, cultural and environmental issues may have on client utilization and compliance; and respect the distinct cultural needs of various client populations. Collaboration between UMDNJ, the community and church, through HRSA support, has created a primary care training site to enhance cultural competency.

MAJOR DIFFICULTIES ENCOUNTERED BY RESIDENTS IN INTERNAL MEDICINE. L. Christophe¹, P. Arnaud, P. Eliane¹, A. Anne-françoise¹; ¹University Hospitals of Geneva. Dep Int Medicine, Geneva, Switzerland

OBJECTIVES: The first years of professional activity are critical for doctors in training. However, little is known about the difficulties encountered by residents.

METHODS: We put the following open question to 24 consecutive residents in a structured one year training in internal medicine: «Please, identify 2 to 3 major difficulties/concerns related to your practice of medicine within this hospital». Each resident gave confidential written answers. These were coded into 8 categories by content analysis by 3 researchers.

RESULTS: Characteristics of the residents were: female: 37%, mean age: 28 ± 2.2 years, mean duration of postgraduate training: 2.5 ± 1.3 years. The total number of answers was 122, with an average of 5.1 ± 1.3 per resident. 93% of the residents expressed difficulties in their relationships with their colleagues or felt not adequately supervised. 63% judged that they were not respected or recognized by senior staff. 63% admitted feelings of severe disarray and helplessness in dealing with patients. 58% complained about work overload. 50% felt burdened by the intensity of emotional investment and responsibility towards patients. Finally, 16% admitted fearing for their professional future and only 8% cited lack of theoretical knowledge about the diseases they were dealing with. Four answers did not fit into the 8 above categories.

CONCLUSION: Most of the interviewed residents in training expressed major difficulties within their professional relationships. Feelings of inadequate practical and relational training by medical school, of lack of recognition by senior staff, and of work pressure and overload were common. These elements challenge the organisation/coaching in our institution and should lead us to improve our teaching.

ENHANCING RESIDENT ABILITY TO ACCESS AND COMMUNICATE ABOUT PATIENT PROGNOSIS AT THE END-OF-LIFE. M. Conti¹, G. Iacono¹, Z.H. Piotrowski¹, B. Caraher¹; ¹West Suburban Health Care, River Forest, IL

OBJECTIVES: Increase resident: (1) knowledge about what factors most accurately estimate prognosis; (2) knowledge of how prognosis affects care plans; (3) skill in communicating information about prognosis and in enlisting active patient/family participation in making decisions about care.

METHODS: Internal medicine residents are exposed to curriculum in cancer prognosis using several instructional strategies including didactic/lecture, small group case discussions, role-playing and debriefing actual patient encounters. Additionally, some of the residents participate in an elective block rotation in end-of-life issues providing them with more intensive exposure by following patients in our hospice program. Along with relevant reading assignments for background, residents review sample case studies of terminally ill patients at latter stages of illness. After learning a system of estimating prognosis they discuss how this information may be used to design care plans so that the therapies that are most appropriate in light of known prognosis are chosen. Residents review basic principles of communicating with patients about prognosis and end-of-life issues that they then implement in role playing exercises.

RESULTS: Formative and summative evaluation methods are utilized to assess progress toward the objectives. These include individual faculty preceptor assessments of residents caring for terminally ill patients, knowledge and attitude questionnaires regarding end-of-life issues, scenarios highlighting communication issues, and resident self-assessment. In addition, role-plays are rated on several criteria and feedback is given to residents in a post session briefing.

CONCLUSION: Although end of life issues are receiving increased attention and concern, it remains an area physicians tend to feel insecure in their skills and poorly prepared by their medical training. Developing curriculum to increase physician's confidence and comfort with these issues improves the quality of care to patients and family members in the dying process.

COMBINING OSCES WITH THE MINI-CEX FOR RESIDENT ASSESSMENT. R.S. Crausman¹, J.P. Miskovsky¹; ¹Memorial Hospital of RI, Pawtucket, RI

OBJECTIVES: To assure a high standard of clinical competence throughout the continuum of medical education, it is imperative that appropriate, task-relevant evaluative tools and methods be continuously developed and applied. In response to the perceived inadequacies of conventional evaluative measures, the use of test patients for structured, standardized evaluation (Objective Structured Clinical Examination, OSCE) has gained favor for the assessment of medical students and residents. However, the role of physician faculty observer, when present, in these exercises is unclear. Physician supervisors bring a clinical perspective to the encounter different from, and complimentary to that of the standardized patient. The challenge is to provide a structured evaluative framework to improve the quality and reproducibility of assessment. In 1995 the American Board of Internal Medicine developed and validated the Mini-CEX. These observed resident-physician patient encounters consist of a single faculty member observing a resident conduct a focused history and physical examination. Residents and evaluators report satisfaction with the format, and reliability of the evaluation is thought to be superior to traditional clinical evaluation exercises. Here we describe our early experience combining the Mini-CEX with outpatient OSCE's for enhanced resident assessment.

METHODS: Second year medical residents (n = 10) in our Primary Care/General Internal Medicine Residency underwent a multi-station OSCE designed to assess their clinical skills.

Each 20 minute encounter was observed by a faculty physician who structured observation and assessment using the Mini-CEX instrument which uses a two page form with a 9 point rating for assessment. After each encounter both standardized patient (five minutes) and faculty physician (five minutes) provided structured feedback to the resident trainee. A time keeper assured that encounters were kept to 20 minutes.

RESULTS: A structured feedback session with all participants was conducted after the exercise and all reported high levels of satisfaction with the format. Standardized patients uniformly commented that the presence of a physician faculty evaluator represented an improvement and provided complimentary evaluative information. Faculty observers were pleased with the imposed structure that the Mini-CEX provided. Residents were satisfied that feedback was appropriate and valuable.

CONCLUSION: In response to the perceived inadequacies of conventional evaluative measures we have introduced a combination of outpatient OSCE's and the Mini-CEX towards developing a more flexible, reliable and task-relevant teaching and evaluation tool. Our early experience with this approach has been very positive.

COMMUNITY MEDICINE: A SEMINAR SERIES FOR PRIMARY CARE INTERNAL MEDICINE RESIDENTS. L. Francis¹, S. Glick¹; ¹Cook County Hospital, Chicago, IL

OBJECTIVES: One of the goals of our residency program is to improve residents' ability to care for the medically underserved. We expect that our graduates will not only provide excellent clinical care to individual patients, but will also integrate public health concerns into their medical care. In addition, we anticipate that our graduates will work with community residents, organizations and leaders to serve as advocates for the underserved. In order to expose our residents to this model of community-oriented primary care, we developed an eight-week course in community medicine.

METHODS: The Community Medicine seminar series was held 1/2-day each week for eight weeks and was attended by first, second and third year residents. The curriculum included a community inventory exercise in which residents obtained information about the community in which they saw continuity patients, a community assessment in which residents toured through a neighborhood on foot and described the people, businesses, housing and transportation they encountered, a community treasure hunt in which residents shopped at local businesses and map exercises in which residents plotted the location of their continuity patients' homes. A key component of the curriculum was an intranet website designed specifically for the Community Medicine course. Residents used the website to participate in on-line discussions of reading material and to access public health information and resources.

RESULTS: Residents were asked to complete an on-line assessment instrument at the conclusion of the course. 91% of the residents (10/11) found the seminar series to be very useful for their future practice of medicine. 91% (10/11) felt the course should be offered again in the future. 92% (11/12) felt they understood and could explain the concept of community-oriented primary care. 73% (8/11) of the residents felt the map exercises helped them understand where their patients lived and how they traveled to medical appointments. 100% (11/11) found the public health links on the intranet website useful for learning about COPC. 73% (8/11) residents found the community treasure hunt useful in terms of how to view a community from the standpoint of its people and institutions.

CONCLUSION: This Community Medicine course increased residents' knowledge of the communities in which they and their patients live. Future directions for investigation include determining whether this course impacts residents' decisions to practice primary care in underserved areas and to implement community-oriented primary care concepts after graduation from the program.

BARRIERS TO THE USE OF EMERGENCY CONTRACEPTION IN AN UNDERSERVED PATIENT POPULATION. E. Harleman¹, M. Schneiderman¹, C. Horton¹, D. Schillinger¹; ¹University of California, San Francisco, CA

OBJECTIVES: Emergency contraception (EC) is an effective but underused method of preventing unwanted pregnancy. In a 1997 national study, only 25% of college-educated women and 6% of women with high-school degrees knew key facts about EC. Only 5% of women had

heard about EC from their providers. While other studies have addressed barriers to EC use in the general population, few have focused on barriers in poor, urban populations. Three residents in a program designed to train physicians to care for the underserved looked at this issue in a community satellite clinic.

METHODS: As part of a Quality Improvement/Research curriculum in a primary care residency program, we conducted a focus group with eight providers at an urban, publicly-funded community health center in San Francisco. Provider beliefs/practices, perceived patient barriers, and possible interventions to increase emergency contraception use were discussed. The session transcript was then analyzed to identify a consensus surrounding specific barriers and interventions.

RESULTS: Providers identified a significant problem with unintended pregnancy at their clinic, estimated at 85–90% of all pregnancies. Despite this high rate, emergency contraception was prescribed only 2–3 times a year at the clinic. Providers identified the following obstacles to emergency contraception use: lack of patient knowledge regarding the availability of EC; the prevalence of social and economic stressors that made contraception a low priority for patients; patients' fatalistic beliefs surrounding pregnancy; and operational obstacles to obtaining emergency contraception prescriptions. Identified interventions to increase use/access were: patient and community education about emergency contraception; arranging to have a prepackaged product on-site to give to patients; and routine counseling by providers at all reproductive health visits.

CONCLUSION: In this urban, community-based primary care clinic, unintended pregnancy is common and emergency contraception rarely used. In the focus group session, providers clarified barriers and identified interventions that the clinic could employ to increase patient's access to and use of emergency contraception. Barriers to use by urban, underserved populations may need further study and targeted interventions before true access to emergency contraception becomes a reality for this vulnerable patient group.

SETTING UP A ONE-MONTH LECTURE BLOCK ROTATION IN INTEGRATIVE MEDICINE-IM. G. Iacono¹, C. Grijalva¹, M. Harris¹, E. Kroger¹, Z. Piotrowski¹; West Suburban Health Care, River Forest, IL

OBJECTIVES: Increase residents knowledge of common IM approaches (acupuncture, physical therapy/massage, mind-body medicine, herbal medicine, chiropractic) including what that approach entails, what conditions it is thought to be useful for treating, and any contraindications for its use; and ability to identify, access and critically evaluate evidenced-based research regarding treatment outcomes and efficacy of the particular approach.

METHODS: Residents are assigned a faculty mentor who oversees and provides guidance in the month; helps to monitor and direct the residents' learning and to address their objectives. Residents spend several 1/2 days each week in the office of IM providers. Each provider prepares a reading list the resident is responsible for working through in that area. Residents see patients along with the provider and have opportunity to discuss the patients from that IM perspective. Residents also spend time with a general internist who is well versed in IM at our Center for Integrative Medicine. They learn how primary care physicians incorporate IM into their practice. Residents attend several one-on-one classes. For example, residents meet with the librarian to learn how to search the web to find IM information and research and are introduced to IM databases in the area (i.e., Natural Medicine Database). An oncologist talks with the residents about the use of alternative techniques cancer patients often use; how the oncologist approaches discussions about IM with patients and the contraindications that may exist for various IM approaches within oncology. The residents also go on several site visits (walk through a pharmacy with the pharmacist looking at OTC herbal remedies as well as visiting a wellness center providing patient education, support groups and counseling to cancer patients).

RESULTS: Intensive interviews are conducted with each resident completing the rotation to evaluate quality of the experience and to elicit opinions for changes/additions to the experience. **CONCLUSION:** Setting up curriculum addressing nontraditional approaches in a western medical setting raises many issues regarding developing common standards for education, reimbursement for practitioner training time and credentialing, applying research standards to cross cultural scientific literature. The process is a challenging and yet a rewarding venture for both traditional and nontraditional medical learners and educators.

SETTING UP A ONE-MONTH LECTURE BLOCK ROTATION IN INTEGRATIVE MEDICINE (IM). G. Iacono¹, C. Grijalva¹, M. Harris¹, Z. Piotrowski¹; West Suburban Health Care, Oak Park, IL

OBJECTIVES: A one-month IM elective block rotation was developed for third year residents to more intensively, study IM approaches. It was hoped that this exposure would prepare residents to better integrate IM into their medical practice.

Increase residents knowledge of common IM approaches (acupuncture, physical therapy/massage, mind-body medicine, herbal medicine, chiropractic) including what that approach entails, what conditions it is thought to be useful for treating, and any contraindications for its use; and ability to identify, access and critically evaluate evidenced-based research regarding treatment outcomes and efficacy of the particular approach.

METHODS: Residents are assigned a faculty mentor who oversees and provides guidance in the month; helps to monitor and direct the residents' learning and to address their objectives. Residents spend several 1/2 days each week in the office of IM providers. Each provider prepares a reading list the resident is responsible for working through in that area. Residents see patients along with the provider and have opportunity to discuss the patients from that IM perspective. Residents also spend time with a general internist who is well versed in IM at our Center for Integrative Medicine. They learn how primary care physicians incorporate IM into their practice. Residents attend several one-on-one classes. For example, residents meet with the librarian to learn how to search the web to find IM information and research and are introduced to IM databases in the area (i.e., Natural Medicine Database). An oncologist talks with the residents about the use of alternative techniques cancer patients often use; how the oncologist approaches discussions about IM with patients and the contraindications that may exist for various IM approaches within oncology. The residents also go on several site visits

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A PUBLIC HEALTH CURRICULUM FOR PRIMARY CARE RESIDENTS. A.H. Jackson¹, L. Hu²; Boston Medical Center, Boston, MA; Boston University, Boston, MA

OBJECTIVES: In order to effectively care for patients and promote health, physicians must have a broader view of their role in society as "health promoters". Knowledge of public health resources as well as the process of policy development is especially important when caring for vulnerable patients. Because of the crowded curriculum of a residency program, we have developed a public health curriculum that is integrated into the existing framework of the residency program.

METHODS: A needs assessment was conducted via interviews with the Primary Care interns by an outside evaluator and a survey completed by Primary Care and traditional track house officers (80% response rate). Several areas within the existing curriculum were identified where public health concepts could be inserted — complementing, not replacing, "traditional medicine" topics. All sessions are evaluated. Residents will complete yearly surveys and an exit interview to assess the impact of this new curriculum.

RESULTS: The resident survey showed a significant level of interest in public health, greatest in the senior residents. Residents rated their own knowledge of many public health concepts as "average" to "low", and rated a broader knowledge of public health as "important" to "very important" to patient care. Faculty from the Boston University School of Public Health, Boston Public Health Commission and the Massachusetts Department of Public Health have agreed to participate, and run joint sessions with faculty from the Section of General Internal Medicine. Sessions include: 1) A monthly three-hour seminar during an ambulatory block, attended by all of the residents (both Primary Care and traditional track) that reviews the history of medicine and public health, highlighting areas of tension and opportunities for collaboration. 2) Ambulatory Morning Report: topics relevant to ambulatory medicine are presented by the Primary Care chief resident and internal medicine faculty, with monthly participation by public health faculty, adding the public health perspective with a 10–15 minute discussion. 3) A three-hour session in the Primary Care seminar, focused on the role of Departments of Public Health and an introduction to health policy and government. 4) An elective rotation at the Massachusetts Department of Public Health provides primary care residents with the opportunity to work directly with public health staff, participate in projects and learn how programs and policies are developed.

CONCLUSION: Our survey found that the residents were interested in public health and felt that it was relevant to clinical care. Public health faculty were very willing to participate in resident education and collaborate with curriculum development. Important public health concepts can be incorporated into a Primary Care residency curriculum without displacing existing "traditional topics".

VIRAL HEPATITIS AND HUMAN IMMUNODEFICIENCY VIRUS COINFECTION IN A PRIMARY CARE CLINIC: ARE WE LOOKING? A.B. Krain¹, J.P. Wisnivesky², T.G. McGinn¹, E. Garland²; ¹Mount Sinai School of Medicine, New York, New York; ²Mount Sinai Medical Center, New York, NY

OBJECTIVES: Since Hepatitis B (HBV), Hepatitis C (HCV) and HIV share the same high-risk behaviors, coinfection is frequent in clinical practice. In the US, 30–50% of HIV patients are coinfecting with chronic HCV. Almost 95% of HIV patients have serological evidence of HBV exposure. In the era of highly active antiretroviral therapy, death from rapidly progressive viral hepatitis is increasingly common in HIV patients. Since early detection and treatment of HIV and viral hepatitis has been shown to improve the outcome of patients, reactive serologies to HBV and/or HCV should provoke discussion of HIV testing in a general medicine clinic to reveal coinfection.

METHODS: Retrospective chart review of patients with serological exposure for viral hepatitis, identified or followed at a hospital based general medicine clinic in East Harlem, NY. A query was done on hepatitis serologies sent from 1/1/99 to 12/31/99 from the clinic to create 3 patient groups: 1) HBV exposure (Reactive HBc only) 2) HCV exposure (Reactive HCV only) 3) Both HBV and HCV exposure (Reactive HBc and HCV). Charts were reviewed to determine whether house staff physicians documented discussions of high risk HIV behavior after obtaining the hepatitis serologies. The number of HIV tests requested and results were noted. Patient return rate for HIV results was also noted.

RESULTS: 655 individual patient serologies for HBc Ab and 612 for HCV Ab were obtained in 1999. 29 were reactive to HCV, 72 to HBc and 81 to HBc and HCV. Charts were obtained for 24, 68, 73 patients, respectively. Discussions of HIV were documented in 6 of 24 (25%) HCV positive patients, 24 of 68 (35.3%) HBV positive patients and 29 of 73 (39.7%) HCV/HBV positive patients. HIV tests were performed in 4 of 24 (16.7%), 18 of 68 (26.4%) and 21 of 73 (28.7%), respectively. Predictors of HIV testing included male gender, sexual activity, multiple sex partners and history of sex with an HIV positive partner. (All with $p < .05$) History of injection drug use was not a significant predictor of HIV testing.

CONCLUSION: More awareness needs to be raised in Internal Medicine clinics about the high prevalence of coinfection and the consequences of delayed detection of HIV with viral hepatitis. Although HBV and HCV have clear epidemiological links with HIV, opportunities for diagnosis have been missed. Education of residents on the poor outcomes of coinfection and the need for HIV testing and counseling may increase rates of HIV discussion in primary care clinics.

DESIGNING A GUIDELINE TO PREVENT OSTEOPOROTIC FRACTURES IN A PUBLIC HEALTH SETTING. S. Lindner¹; ¹San Francisco, CA

OBJECTIVES: Most guidelines for prevention and treatment of osteoporotic fractures are neither evidence-based nor well-suited to the needs of the diverse patient population found in urban public health settings. This Quality Improvement project aimed to create a clinical guideline adapted to San Francisco's Community Health Network and its diverse population with limited resources. The Guideline aims to be based on current evidence wherever possible, and is based in part on existing evidence-based guidelines such as the National Osteoporosis Foundation (NOF) Guideline. Implementation of the guideline will follow published effective approaches to changing practice patterns.

METHODS: 1. Adaptation of currently existing evidence-based guidelines, such as the NOF guideline, and review of recent relevant literature to create a guideline appropriate for and tailored to San Francisco's Community Health Network (CHN.)

2. Distribution of these guidelines to CHN providers, ideally targeting those who see our at-risk patients, using a creative media approach with paper and electronic formats (including distribution of a file transferrable to handheld organizers such as the Palm OS) and additionally promoting the guideline in a didactic format.

3. Improving awareness of and adherence to these guidelines with such techniques as Clinical Alerts on the CHN computer record, and easy Web access on the CHN website.

RESULTS: We designed a guideline that was tailored to meet the needs of the CHN patient population. This guideline was evidence-based. Distribution was achieved via CHN website, computer-based reminders and triggers where appropriate, handheld organizer files, and didactic sessions in local clinics and conferences.

CONCLUSION: Few guidelines for prevention and treatment of osteoporotic fractures currently exist that are both evidence-based and tailored to meet the needs of an underserved urban public health community. Using good existing guidelines such as that published by the NOF, current literature can be adapted to design a guideline that is both based on good evidence and appropriate to the limited resources of an urban public health setting. Novel uses of computer-based and didactic tools can enhance distribution and awareness of clinical practice guidelines.

A CURRICULUM IN WOMENS HEALTH FOR INTERNAL MEDICINE RESIDENTS. M.A. Onwudike¹; ¹Wright State University, Dayton, OH

OBJECTIVES: To develop and implement a curriculum in women's health for internal medicine residents in a community hospital based university program.

METHODS: Participants are first year internal medicine residents in their ambulatory block month (N=16). The curriculum content was determined by a needs assessment conducted on previous and current residents. It consists of the following six units covered during three hourly didactic sessions: 1) Pretest and Introduction to Women's Health, 2) Selected Gynecological Conditions, 3) Obesity and Eating Disorders, 4) Menopause, Osteoporosis and Hormone Replacement, 5) Psychosocial Issues, 6) Resident Presentations and Posttest. Also included in the curriculum is a three-hour visit to the local domestic violence prevention center. All necessary reading materials are handed out during the first session after the pretest. The didactic sessions start with a brief case presentation by the participating resident(s); this is followed by an in-depth discussion of the topics of the day with expected resident participation. Each session except the introductory and resident presentation session ends with role-plays where the instructor plays the role of the patient and the resident the doctor. This enables the resident(s) to put into immediate practice what they have just learned. Clinical practice of women's health is expected to occur when residents see female patients in their primary care practices. To expand the scope of this curriculum beyond these limited topics both residents and faculty have been encouraged to incorporate into everyday teaching and learning how various conditions may differ or are unique in women. Participating residents are expected to make a formal thirty to forty-five minute presentation on a topic in women's health not covered in this curriculum. Learners are evaluated by the completion of a standard evaluation form used in the residency program. Evaluations are based on the residents' performance on the posttest, participation during the didactic sessions and their presentations. Learners evaluate the curriculum by completing a survey.

RESULTS: The curriculum was well received by all residents. All residents surveyed either agreed or strongly agreed that they would now be able to evaluate, manage or appropriately refer female patients as a result of knowledge gained during the month. The mean posttest score was 81%, an improvement over the mean pretest score of 66%. Residents stated particularly that role-plays helped improve their understanding of the topics. A survey of these residents in their third year regarding their perceptions on the women's health education they received in this program and how well they think it has prepared them for their future practices year is planned.

CONCLUSION: The curriculum improved the knowledge of residents in selected women's health topics. Using the same curricular model, topics can be adapted to meet the learning needs of residents in other residency programs.

IMPROVING THE HEALTH CARE RESPONSE TO DOMESTIC VIOLENCE: A SEMINAR SERIES FOR PRIMARY CARE RESIDENTS. K. Riordan¹, S. Love¹, C. Warshaw¹, S. Glick²; ¹Chicago Abused Women Coalition, Chicago, IL; ²Cook County Hospital, Chicago, Illinois

OBJECTIVES: Violence against women has been increasingly recognized as a major public health problem. Despite this, many physicians still find it difficult to ask patients about abuse. A major barrier that health care providers face in effectively identifying and treating victims of abuse is a lack of training on the prevalence and impact of domestic violence, ways to intervene appropriately and how to access available resources.

We integrated an innovative 8-week seminar series on domestic violence into our primary care residency training. The goals of our curriculum were to enable residents to: 1) understand the prevalence and dynamics of domestic violence; 2) identify the impact of domestic violence on women's physical and psychological health; 3) develop culturally respectful intervention skills to identify and assist abused patients; and 4) identify resources available to victims of abuse.

METHODS: The seminar series was led by the Hospital Crisis Intervention Project (HCIP), a collaborative program of the Cook County Bureau of Health Services and the Chicago Abused Women Coalition (CAWC). The seminars were held 1/2-day each week for eight weeks. Topics included An Overview of Domestic Violence; Identifying, Assessing and Intervening with Domestic Violence Victims; Working with Diverse Populations; The Batterer; Intervening with Sexual Assault Patients; Community Violence; and Child Abuse, Elder Abuse and Legal Issues. A highlight of the course was a site visit to CAWC's domestic violence shelter where the primary care residents were able to speak candidly with survivors of abuse. Seminars were led by domestic violence advocates, physicians, social workers and attorneys. Instructional format included lectures, interactive discussions, videotape reviews, problem-based case discussions and a community site visit.

RESULTS: Residents assessed each session and the course overall with written evaluations. 100% of the residents (9/9) found the course very effective at helping them learn more about the dynamics of domestic violence. 100% (12/12) found it very effective at helping them learn more about the impact of domestic violence. 100% (8/8) found it very effective at helping them work effectively with diverse populations. 92% (11/12) found the course very effective in helping them learn more about the resources available to victims of domestic violence.

CONCLUSION: This 8-week course on domestic violence improved residents' understanding of the dynamics and impact of domestic violence, culturally appropriate interventions and available resources. Future directions for evaluation include the use of simulated patients to determine whether this course improves residents' ability to identify and care for victims of violence in the clinical setting.

A BRIEF ROTATION IN ADDICTION MEDICINE FOR PRIMARY CARE RESIDENTS. J. Smith¹, S. Glick¹; ¹Cook County Hospital, Chicago, IL

OBJECTIVES: We designed a 2 week rotation in addiction medicine to promote the role of the primary care physician in prevention, early identification, and treatment of substance abuse. The objectives are to: 1) support positive attitudes towards patients with addiction disorders, 2) define the responsibilities of the primary care physician for substance abuse intervention, 3) practice new skills and apply new knowledge of substance abuse in a general medical setting, 4) provide access to physician role models.

METHODS: We adapted the structure and content of a successful 4 week elective rotation in addiction medicine into a 2 week core curriculum for all second year, primary care residents. Previous experience from the longer rotation informed us that residents unanimously chose as core learning objectives: screening and assessment, discussing substance abuse with a patient, and making a referral to community resources for substance abuse problems in general medical practice; and half also chose as a focus: developing a non-judgmental approach to addicted patients, providing preventive education, and management of withdrawal. We therefore made these the learning objectives of the brief course, adding: how to monitor and support patients in recovery. Modeled on the longer elective, the two weeks are structured around the residents' ambulatory primary care clinics, where they are responsible for practicing their new skills. They attend 4 half-day workshops, including case discussions of core topic areas and role play of office-based skills. Working with hospital-based substance abuse counselors, they perform substance abuse assessments and referrals for medicine and trauma patients; observe treatment groups of an out-patient treatment program for substance abusing persons with HIV; visit community-based residential treatment programs, a social-detox program, and a 12-step meeting.

RESULTS: We used one experienced faculty-member to organize the curriculum and make community contacts for the 2 week rotation. Workshops are delivered by a group of faculty including internists, psychiatrists, and chemical dependency professionals. We found that community-based, substance abuse treatment programs were willing to sponsor visits by small numbers of residents, and several would permit residents to observe treatment groups. Treatment programs were generally willing to arrange for residents to interview individuals clients.

CONCLUSION: A 2 week rotation in addiction medicine can be created that focuses on the role of the primary care physician in the prevention, early identification and treatment of substance abuse. Residency programs that do not have substance abuse services within their teaching hospitals or clinics can seek cooperation from community-based, substance abuse treatment programs to provide residents with clinical experiences.

IMPLEMENTATION OF MORNING REPORT IN A MULTIDISCIPLINARY AMBULATORY SETTING. S.H. Stewart¹, B. Clyburn¹, M.D. Silverstein¹; ¹Medical University of South Carolina, Charleston, SC

OBJECTIVES: Establish an ambulatory care morning report for residents, medical students, and other trainees in a multidisciplinary clinic, and encourage the integration of evidence based information into clinical practice.

METHODS: McClellan Banks Adult Primary Care Center at the Medical University of South Carolina serves as an ambulatory teaching site for internal medicine residents, medical students, physician assistant students, and pharmacy graduate students. Clinic faculty have appointments in general internal medicine, pharmacy, the physician assistant program, and psychiatry. We organized a forty-five minute, twice-weekly morning report for all residents and students on their monthly ambulatory rotations. The report takes place prior to clinic hours and is attended by four to eight residents and students and two to three faculty or fellows. Each student or resident is assigned a date to give a presentation describing one of their patients from the monthly rotation. The presenter is encouraged to focus on a common primary care topic, and to search for available evidence (guidelines, systematic reviews, clinical trials) on topics such as screening, diagnosis, prognosis, or treatment of the condition being discussed. The report begins with a brief history and physical, followed by a discussion of the diagnostic evaluation and management alternatives, and concludes with an evaluation of the evidence presented. Informal discussions, observation and exit interviews at the conclusion of the ambulatory rotation have assessed educational value and trainee and faculty satisfaction.

RESULTS: The program has been functioning for one year, and is a valued educational experience according to feedback from the trainees. The students have had excellent attendance, while residents have needed greater encouragement to attend the sessions. Faculty participation has been excellent, as the sessions provide greater opportunity to teach than is possible during busy clinic hours. An additional benefit has been the increased interaction of learners from the variety of disciplines represented. The timing of the report has not interfered with clinic operations.

CONCLUSION: Our ambulatory morning report enhances the quality of clinical education and is a valued addition to the ambulatory rotation. Plans are being discussed to include trainees from other clinical teaching sites. Future implementation should include prospective evaluation of impact on knowledge, attitudes, and the use of evidence in patient care.

EVALUATING INTERNAL MEDICINE RESIDENTS COMMUNITY-BASED LONGITUDINAL AMBULATORY CARE EXPERIENCE. G.H. Tabas¹, D.A. Lescisin¹, M. Elnicki¹; ¹University of Pittsburgh Medical Center Shadyside, Pittsburgh, PA

OBJECTIVES: As part of our HRSA-funded primary care grant, we increased the longitudinal ambulatory care requirement for our categorical internal medicine residents. With an increasing proportion of our residents having this ambulatory care experience in the community, we felt the need to determine whether the educational experience for these residents is comparable to that of residents in a more traditional hospital-based clinic setting. We developed a new evaluation tool to assess the experiences of residents in these two settings.

METHODS: In developing our Evaluation of Continuity Preceptor and Site instrument, we sought to assess those aspects of the resident's experience most important for optimal ambulatory education in both community and hospital-based sites. The evaluation instrument is divided into a group of questions concerning the preceptor and another group of questions regarding the ambulatory site. The 15 preceptor items include questions about role modeling, supervision, teaching and an overall rating. The 18 site items include questions about office systems (patient flow, medical records, billing), patient characteristics, continuity of care and an overall rating. Items were rated on a 5- point Likert scale, assessing frequency of preceptor and site attributes, using terms varying from never/almost never to always/almost always.

RESULTS: The instrument was piloted and yielded good internal consistency (Cronbach's alpha .88 for preceptor items and .83 for site items). The survey was then distributed to all categorical residents semiannually for two years. The response rate was 82%, and 132 surveys were analyzed. When tested for differences using the Mann-Whitney test, the group mean scores for site items were significantly higher for the community sites as compared to the hospital-based clinic site (4.08 vs 3.72; P=0.001). The group mean scores for the preceptor items were higher for the clinic-based faculty as compared with the community preceptors (4.58 vs 4.44), but the difference was not significant (p=0.08). Site items rated higher by residents in the community included questions about exposure to a broad range of patients (p=0.001) and to a full range of patient problems (p=0.07). Residents in the hospital-based clinic felt they had a higher level of responsibility for their patients (p < 0.001) and they rated their preceptors slightly higher in providing explanations and asking questions to enhance learning (p=0.13).

CONCLUSION: Internal medicine residents are being increasingly assigned to community sites for their longitudinal ambulatory care experience. In our survey, we found that residents in the community rated site items higher and residents in the hospital-based clinic rated preceptor items slightly higher. Awareness of the differences between the two types ambulatory care experiences is important for planning future ambulatory educational activities.

INCORPORATING INFORMATION LITERACY AND AN EVIDENCE BASED MEDICINE CURRICULUM INTO A COMMUNITY HOSPITAL RESIDENCY PROGRAM. S.S. Yen¹, M.Z. Cassey¹, J. Stielstra¹, H. Piotrowski¹, K. Weinstein¹, E. Kroger¹; ¹West Suburban Health Care, Oak Park, IL

OBJECTIVES: A web-based modular curriculum for information literacy was developed to support evidence-based clinical practice and refine residents' skills for preparing professional conference presentation materials. This curriculum was designed to fill the gaps in informatics knowledge necessary to subsequent assignments supporting an Evidence Based Medicine (EBM) residency curriculum.

METHODS: To assess readiness and assure curricular relevance, baseline data is gathered concerning essential EBM and information literacy skills. The initial one month curriculum consists of four two-hour class sessions supported by hands-on instruction and maintained on the open residency website. Content focuses on current and future informatics practice issues, orientation to significant information resources, emerging technology developments related to clinical practice, EBM searching strategies for the general Internet and quality medical sources, overview of quality patient education materials, government websites, and time saving shortcuts for common technology applications including browsers, word processing and presentation programs. The team of people who support and maintain this curriculum include a variety of Internal Medicine content experts, the medical librarian, a nursing informatics webmaster, and a research specialist.

RESULTS: The initial series of courses is completed within the first three months of the first post-graduate year. Continued exposure to EBM skill usage and the website is required during all three years of the resident's outpatient clinic via topic sessions. Final assignments of the second and third residency years include a digital presentation of a clinical topic in an Internal Medicine research forum conference.

CONCLUSION: Integration of information literacy skills into a residency curriculum must consider variations in learning curves and the need for changes in approaches to traditional problem solving techniques in clinical practice. Clinical faculty and residents need support and encouragement to adapt existing processes to include EBM approaches in daily practice. Traditional dogma states that a shift in institutional thinking may be needed to provide conveniently accessible Internet resources on the practice units and in the residency work areas. However, providing a readily accessible set of instructions and relevant EBM resources via a

residency website removes barriers to implementation of evidenced based clinical practice for residents and attending physicians.

INNOVATIONS IN MEDICAL EDUCATION

DEVELOPING A PORTFOLIO TO PROMOTE SELF-REFLECTION AND CHRONICLE MEDICAL PROFESSIONALISM. L.L. Blank¹, H.L. Lespoir¹; ¹American Board of Internal Medicine, Philadelphia, PA

STATEMENT OF PROBLEM OR QUESTION: Description: Reflecting upon what has been learned is the key element in constructing a portfolio. A portfolio is a collection of products prepared by the resident that provides evidence of learning and achievement related to a learning plan. [Reference: The ACGME Outcome Project Toolbox of Assessment Methods, Summer 2000.]

OBJECTIVES OF PROGRAM/INTERVENTION: Background: During 1999–2000, the Accreditation Council for Graduate Medical Education (ACGME) and the American Board of Medical Specialties (ABMS) identified six general competencies (patient care, medical knowledge, practice-based learning and improvement, interpersonal and communication skills, professionalism, and systems-based practice) for definition and implementation by all medical and surgical specialties. The ACGME/ABMS vision is the integration of life-long learning for physicians with the maintenance of Board certification. Their collective goal is to ensure that physicians are better prepared to provide compassionate, appropriate, and effective patient care in the changing health care environment.

DESCRIPTION OF PROGRAM/INTERVENTION: Purpose: Using this background as a framework, the ABIM Clinical Competence Program is exploring ways to foster self-reflection, self-assessment, and continuing professional development among internal medicine residents and subspecialty fellows throughout their training. Through a newly developed tool and template, the use of a portfolio designed and maintained by the physician, may facilitate achieving this goal and serve to chronicle medical professionalism.

FINDINGS TO DATE: What Comprises A Portfolio?: Multidimensional, qualitative and quantitative, portfolios encompass a broad spectrum of activity from definitive learning goals, performance assessment and reviews, to personal narratives and community service.

KEY LESSONS LEARNED: Conclusion: The professional portfolio is the tangible culmination of one's acquired and applied knowledge, learning experiences, and unique opportunities for self-assessment and self-reflection. The portfolio remains the practical and intellectual property of the creator. Through the creation of a portfolio, physicians are empowered to gain better insight into their own goals, personal and professional growth, and career development.

PHYSICIANS AS LEADERS: IDENTIFYING AND ADDRESSING UNIQUE MD/MBA COMPETENCIES. B. Chemol¹, V. Tabbush², J. Terwilliger¹, L. Wilkerson¹; ¹UCLA School of Medicine, Los Angeles, CA; ²Anderson School Of Management, UCLA, Los Angeles, CA

STATEMENT OF PROBLEM OR QUESTION: In 1997, UCLA's School of Medicine and Anderson School of Management created an integrated 5 year joint MD/MBA Program, one of seven in the country. We summarize our needs assessment of unique MD/MBA competencies and the integrated 5-year curriculum created in response to these needs.

OBJECTIVES OF PROGRAM/INTERVENTION: Our goals were to 1) identify unique MD/MBA competencies through the input of private sector and academic physician leaders; and 2) develop new curricular opportunities to meet these needs.

DESCRIPTION OF PROGRAM/INTERVENTION: Through a structured process, an advisory board comprised of 14 health care leaders identified six competency areas critical to MD/MBA graduates: clinician leadership skills in business environments, common business components of public and private health care delivery, a historical perspective on the US health care delivery infrastructure, the role of public policy in shaping health care delivery, an internal framework for ethical organizational decisionmaking and a population perspective in health care planning. These MD/MBA competencies represent additional skills required by graduates beyond the core competencies of each degree independently. During the first three years students complete all the standard MD curricular activities. MD students apply for the MD/MBA Program during their third year. Once accepted, students spend the fourth year completing the entire core and some elective requirements in the Business school. The fifth year curriculum was developed as a summation year, integrating the two Schools' curricula while explicitly addressing the unique dual degree competencies. Fifth year curricular elements include a management component integrated into standard clinical electives, a broadened range of electives in other Schools within the University such as Public Health, an evening Physician Leadership Colloquia Series, and a clinically oriented field study/research experience. Students completed an evaluation that covered both School's core competencies as well as the unique dual degree competencies using a Likert scale. Information was also elicited about students' short and long term plans.

FINDINGS TO DATE: All six members of the first cohort completed evaluations. Overall the curriculum addressed the majority of the unique MD/MBA competencies with a mean score of 3.0 or higher. Two competencies need further emphasis: developing an ethical framework for organizational decisionmaking (2.8) and an application of physician leadership skills in business environment (2.5).

KEY LESSONS LEARNED: An integrated five-year curriculum provides MD/MBA students with an effective opportunity to develop clinical and managerial skills. Both elements are critical for galvanizing a new model of physician leadership.

AN APPROACH TO EVALUATING THE HIDDEN CURRICULUM AND THE SOCIALIZATION OF MEDICAL STUDENTS. B. DWINNELL, G. BARLEY. UNIVERSITY OF COLORADO SCHOOL OF MEDICINE, DENVER, CO. B.G. Dwinnell¹; ¹University of Colorado Health Sciences Center, Denver, Colorado

STATEMENT OF PROBLEM OR QUESTION: Statement of the Problem: Professional behavior is often not a structured topic in the standard medical school curriculum. Much of the socialization for medical professionals occurs in what we refer to as the hidden curriculum. Frequently medical students may not have the understanding of what is considered appropriate professional behavior, yet they are exposed to positive and negative role models from which they may choose to emulate.

OBJECTIVES OF PROGRAM/INTERVENTION: Objectives: Our goal was to develop curriculum that begins to address some issues around the socialization that occurs for medical students during their clinical years of training. The purpose is to provide an informal, safe environment for students to discuss issues of professional behavior to which they are exposed during their third year of medical school. Information obtained from these gatherings is used to modify the experiences the students have during clinical training. In addition it serves the purpose of allowing students to share experiences with their colleagues.

DESCRIPTION OF PROGRAM/INTERVENTION: Description: In a course entitled "The Hidden Curriculum", junior medical students are divided into groups of 12 and have 6 scheduled meetings during the academic year with two facilitators (a faculty member and a senior medical student). For each session, the facilitators are provided with 2 to 3 questions concerning a variety of topics. Some of the topics include balance of professional and personal life, perceived medical student abuse and career goal exploration. A written summary report is generated from all of the data gathered from the small groups. The summary report is distributed to the clerkship directors, students and several of the Deans involved in medical student education.

FINDINGS TO DATE: Findings to Date: Medical student experiences are generally positive but they are continually exposed to some negative role models. The results from the curriculum thus far have lead to creating a more formal evaluation of professional behavior. In some instances specific individuals have received feedback concerning egregious behavior. Students view this as a very positive experience, finally allowing them a format that facilitates critical feedback about their experiences.

KEY LESSONS LEARNED: Key Lessons Learned: Medical students rarely receive formal training on what appropriate professional behavior entails. In many circumstances they do not feel comfortable providing honest, critical feedback to their superiors. Evaluation of professional behavior should be done in a more formal manner, addressing appropriate knowledge, skills and attitudes just as we do with clinical performance.

USING PATIENTS WITH CANCER TO EDUCATE RESIDENTS ABOUT GIVING BAD NEWS. N.J. Farber¹, A. Friedland¹, B.M. Aboff¹, D.B. Ehrenthal¹, T. Bianchetta¹; ¹Christiana Care Health System, Wilmington, DE

STATEMENT OF PROBLEM OR QUESTION: Communication skills, including the communication of bad news to patients, have been taught to residents using a variety of methods. The use of actual patients with life-threatening illnesses has not been used in resident education.

OBJECTIVES OF PROGRAM/INTERVENTION: Conduct a workshop with residents using patients with life-threatening illness in order to: 1) educate residents about how to communicate bad news to patients and 2) change residents' attitudes about communicating bad news and maintaining hope for patients.

DESCRIPTION OF PROGRAM/INTERVENTION: Four patients with cancer from the Wellness Community were recruited to participate in a 2 hour resident workshop on giving bad news to patients. The curriculum included patients using their own diagnoses and histories as the role play scenarios. Residents were instructed to give bad news as they usually do, then feedback from the patients, discussion about the role play and about the method as described by Buckman ensued. A second role play using the learned concepts was conducted and final debrief occurred. Prior to the workshop, residents completed an 11 item questionnaire about actions to be taken during the communication of bad news, with a similar post-test being used at the end of the workshop.

FINDINGS TO DATE: Twenty-five residents participated, with 15 (60%) completing pre and post tests. Attitudes toward ensuring hope is conveyed to patients ($p < 0.05$), starting the discussion by ascertaining the patient's understanding of the condition ($p < 0.01$), and encouraging the patient to express his/her feelings ($p < 0.01$) all significantly improved with the use of the workshop. Most of the improvement occurred in residents with previous training.

KEY LESSONS LEARNED: A role play workshop using untrained patients with cancer can significantly improve residents' attitudes about giving bad news. Volunteers are without cost to the institution, and have a positive influence in such workshops. Residents who are previously trained may hold counterproductive attitudes, and therefore all residents should participate in workshops using role play by actual patients about giving bad news.

TALKING MEDICINE: A COURSE IN MEDICAL HUMANISM. J.M. Hauser¹, M.L. Lypson¹; ¹University of Chicago, Chicago, IL

STATEMENT OF PROBLEM OR QUESTION: How do third year medical students view a self-reflection group exploring core concepts of humanism and professionalism?

OBJECTIVES OF PROGRAM/INTERVENTION: To create a series of small group discussions on humanism and professionalism where students can reflect on the process of becoming a physician and share personally or ethically difficult and rewarding cases with each other.

DESCRIPTION OF PROGRAM/INTERVENTION: Talking Medicine is predicated on the belief that humanism and professionalism come to students and others through understanding a number of core concepts and relationships complemented by self-reflection. The concepts, which have been described by ABIM's Project Professionalism, include ideas of integrity, respect

and others; the relationships include those with patients and families and with other health care professionals. Talking Medicine offers a consistent (every other week for 10 weeks) opportunity to share experiences in small groups (6-8), facilitated by two preceptors in a format driven by students' experiences. Although the focus is on students' experiences, readings are provided on basic topics and contexts in humanism and professionalism (e.g. end of life care, mistakes, spirituality in medicine, and boundaries between patients and doctors).

FINDINGS TO DATE: Talking Medicine began in summer, 2000 and half of the third year class has taken it. We surveyed students (total 54, response rate 63%) and found 94% felt "very" or "somewhat" comfortable in the course. 73% of students reported that the course increased their "connectedness" to classmates and 61% favored it occurring during all rotations. 59% reported that their interest in caring for patients improved and 53% reported their interest in internal medicine as a field improved. Answers to open-ended questions highlighted the importance of Talking Medicine as a forum to connect with others — both students and faculty.

KEY LESSONS LEARNED: These results suggest Talking Medicine may be most effective in helping classmates connect to and learn from each other, thereby setting a foundation for changes in how they interact with patients.

A CASE STUDY IN REMEDIATION: TEACHING A MEDICAL STUDENT COMMUNICATION SKILLS. C. Lin¹, F.W. Platt¹, G.E. Barley¹; ¹University of Colorado Health Sciences Center, Denver, CO

STATEMENT OF PROBLEM OR QUESTION: Medical students found to be deficient in communication and interview skills pose a difficult remediation challenge. There is no standard way to address such deficiencies. We describe the development and implementation of an intensive remedial curriculum.

OBJECTIVES OF PROGRAM/INTERVENTION: 1. Remediation of a second year medical student found to be deficient in communication and interviewing skills. 2. Development of a standardized intervention which is generally applicable to other students and at other institutions.

DESCRIPTION OF PROGRAM/INTERVENTION: A second year student found deficient in communication skills was held back for a year and paired with a clinical preceptor for intensive skills training. The interventions were modeled on Bayer Institute communication workshop materials and included a weekly precepted clinic, structured readings, standardized patient exercises, communications workshops, and end of year standardized clinical evaluations.

FINDINGS TO DATE: The student's self-assessment and the preceptor's assessments of communication skill gradually improved over the year. The student improved through a progression of standardized patient exercises focusing on specific communication skills. The student passed a final evaluation exercise with an excellent rating from the standardized patient and preceptor.

KEY LESSONS LEARNED: This multi-strategy approach to improve communication skills can be applied to other students and in other institutions. Based on the Bayer Institute communication workshops and integrated coaching techniques, the material forms a framework for the deficient student to become proficient in communication and interviewing skills.

THE WIT EDUCATIONAL INITIATIVE: EVALUATION OF A PROGRAM USING THE DRAMATIC ARTS IN MEDICAL EDUCATION. K.A. Lorenz¹, M.J. Steckart¹, M.J. Koren², K.A. Kaplan³, K.E. Rosenfeld²; ¹VA Greater Los Angeles Healthcare System, L.A., CA; ²Fan Fox and Leslie R. Samuels Foundation, New York, NY; ³Partnership for Caring, Washington, DC

STATEMENT OF PROBLEM OR QUESTION: End-of-life care is assuming increasing importance within medical education curricula. Traditional teaching methods may be inadequate to address the emotional, interpersonal, and spiritual aspects of palliative care. The performing arts offer an innovative approach to teach medical trainees in these essential but under-emphasized domains.

OBJECTIVES OF PROGRAM/INTERVENTION: We instituted a program using the dramatic arts to foster empathy for the terminally ill person's experience of illness, and self-reflection with regard to personal practices in caring for dying persons.

DESCRIPTION OF PROGRAM/INTERVENTION: The Wit Educational Initiative uses professional readings of the Pulitzer-Prize winning play "Wit" at medical training sites throughout North America. "Wit" narrates the personal and medical care experience of a patient dying from metastatic ovarian cancer. Medical students, housestaff, and ancillary providers attend readings of the play followed by structured discussions of the play's themes. Program participants are asked to complete a survey evaluating the program's acceptability and relevance.

FINDINGS TO DATE: To date, 9 of 14 program sites have returned surveys. An estimated 44% (614/1385) of program attendees completed the survey and 43% of respondents were medical students or residents. Eighty-four percent of trainees were emotionally moved "a great deal" or "pretty much" by the performance, (mean = 4.3; 1 = "not at all", 5 = "a great deal") and 64% reported that the play portrayed the emotions of dying patients in an "entirely real" or "very real" manner (mean = 4.3). Of trainees who provide direct patient care, 75% reported that the program was "extremely relevant" or "very relevant" to the care they provide (mean = 4.0). The program encouraged attendees to reflect on specific aspects of palliative care including helping patients live as long as possible (mean = 3.8, 1 = "not much at all", 5 = "very much"), talking about prognosis (mean = 4.4), addressing physical pain (mean = 4.4), addressing emotional and spiritual suffering (mean = 4.4), and talking with patients about treatment wishes (mean = 4.4). The majority of trainees rated the program a more useful training tool than didactic lectures on palliative care (83%), journal article readings (85%), and bedside rounds (57%).

KEY LESSONS LEARNED: The dramatic arts foster awareness of the patient experience of illness that is relevant to the care trainees provide terminally ill persons. Many trainees feel such experiences are more useful approaches to palliative care training than traditional educational methods.

TEACHING MANAGED CARE: VISITING A MANAGED CARE ORGANIZATION — A UME-21 PROJECT. J. Woolger, R. Hernandez, M. Rivo, M. O'Connell; University Of Miami School Of Medicine; Avmed Healthplan, Miami, FL

STATEMENT OF PROBLEM OR QUESTION: Like the general public, medical students have neutral to strongly negative attitudes towards managed care (MC).

OBJECTIVES OF PROGRAM/INTERVENTION: During 1 day, students observe, first hand, the operations of a MCO and learn how MCO's are organized and how MCO's coordinate and facilitate health care.

DESCRIPTION OF PROGRAM/INTERVENTION: The University of Miami School of Medicine has partnered with a not-for-profit managed care organization, AvMed Health Plan, Inc. Students visit the administrative offices of AvMed where presentations cover general concepts of MC, physician profiling, practice guidelines, outcome management, and disease management programs. Students also "round" in AvMed's four major departments; Pre-Authorization, Disease Management, Member Services, and Physician Services where they witness the daily operations of the health plan. A 14 item survey (5-point Likert scale) was completed by students at the beginning of the 2nd year, the beginning of the 3rd year and again immediately before (pre) and after (post) the day at the AvMed MCO.

FINDINGS TO DATE: The program has been evaluated favorably by more than 200 third-year medical students. There were no significant differences between mean responses at the beginning of the 2nd year of school and the beginning of the 3rd year (t-test). Of the 14 items, 8 responses were neutral and 6 were negative towards MC. The responses to the pre-seminar survey were also not different from the responses at the beginning of the 2nd or 3rd years. However, post-seminar responses showed significant ($p < .05$) changes favorable towards MC for 7 items and >20% of respondents abandoned their negative attitude towards MC for 5 survey items (e.g. MCO's have potential to improve quality, MCO's provide better care than the traditional system, I understand how MC functions).

KEY LESSONS LEARNED: Students' attitudes and self-assessed knowledge of MC can be favorably affected by curricular experiences at a MCO. Visiting the MCO is an important component of this experience.

"UNDERGRADUATE MEDICAL EDUCATION FOR THE 21ST CENTURY: A DEMONSTRATION OF CURRICULAR INNOVATIONS TO KEEP PACE WITH THE CHANGING HEALTH CARE ENVIRONMENT" (UME-21). AN OVERVIEW OF THE HEALTH RESOURCES AND SERVICES ADMINISTRATION SPONSORED FIVE YEAR PROJECT. D. Babbott¹, S.M. Bastacky², K.L. Pye³, J.M. Pascoe⁴, H.K. Rabinowitz⁵, J. Rising⁶, J. Rodak, Jr.², K.J. Veit⁷, D.L. Wood⁸; ¹University of Vermont, Burlington, VT; ²Health Resources & Services Administration, Rockville, MD; ³KLPye Consulting, Landenberg, PA; ⁴Wright State University, Dayton, OH; ⁵Jefferson Medical College of TJU, Philadelphia, PA; ⁶American Medical Student Association, Reston, VA; ⁷Philadelphia College of Osteopathic Medicine, Philadelphia, PA; ⁸American Association of Colleges of Osteopathic Medicine, Chevy Chase, MD

STATEMENT OF PROBLEM OR QUESTION: America's health care system is undergoing dramatic transformation in care management. Most medical schools, however, have not updated their curricula to prepare graduates for practice in this changing environment.

OBJECTIVES OF PROGRAM/INTERVENTION: Strategies to enhance student acquisition of the knowledge, skills, values and attitudes needed to practice in intensely managed and integrated health care systems; fostering partnerships between academic centers and appropriate health care organizations; enhancing interdisciplinary primary care education in ambulatory/community-based settings.

DESCRIPTION OF PROGRAM/INTERVENTION: Project requirements include: leadership among departments/divisions of general medicine, general pediatrics and family medicine; cooperation of participating partner organizations; and establishment of learning objectives in nine content areas: health systems financing, economics, organization and delivery; evidence-based medicine; ethics; communication skills; leadership; quality measurement; systems-based care; medical informatics, and wellness and prevention. Instruction is required for all students in their clinical years, and must begin no later than the third year.

FINDINGS TO DATE: The UME-21 projects within internal medicine include: Incorporating medical students in continuous quality improvement (CQI) measurements in community practices; students' use of palm pilots to monitor disease prevalence and adherence to practice guidelines in office settings; office-based instruction in physical diagnosis; web-based learning integrated into medical clerkships; utilization of simulated patients into medicine clerkships; interdisciplinary primary care clerkships; incorporation of managed care organization physicians and evidence-based medicine into medical clerkships; and epidemiology in community settings.

KEY LESSONS LEARNED: With strong leadership from general internists and their colleagues, U.S. medical schools are responding creatively to the challenges of curricular innovations in this national demonstration project.

TEACHING RESIDENTS THE IMPORTANCE OF COMMUNITY TO HEALTH. E. Jacobs¹, C. Kohrman², D. Vickers¹, M. Lemon¹; ¹Cook County Hospital, Chicago, IL; ²Westside Health Authority, Chicago, IL

STATEMENT OF PROBLEM OR QUESTION: Many residents care for patients from communities that are very different from their own, yet they rarely have any introduction to the lives, families, backgrounds, and cultural and social contexts of their patients. This is a crucial issue at Cook County Hospital (CCH) where many residents are international medical graduates who have had little exposure to the lives and socioeconomic struggles of our predominantly disadvantaged African American population.

OBJECTIVES OF PROGRAM/INTERVENTION: All first-year internal medicine and pediatrics residents participated in the course with the goals of (1) increasing understanding of the US system of health care for disadvantaged groups; (2) improving communication with patients from diverse cultures; (3) increasing appreciation of the complexity of patients' daily

lives and how it impacts their health and ability to access health care; (4) enhancing awareness of community and local services' impact on patients' well being.

DESCRIPTION OF PROGRAM/INTERVENTION: The curriculum consists of 5 teaching modules taught by faculty from CCH and the Westside Health Authority (WHA), a community health advocacy organization, in 4 half-day sessions over a month-long ambulatory rotation. The first module is a case-based overview of the financing of care in the US with a focus on Medicaid, Medicare, and the impact of insurance status on health. The second module addresses problems language or culture can create in patient encounters. The third module uses a case study of opening an ambulatory clinic in a diverse neighborhood to teach about community based primary care. The fourth and fifth modules focus on the community and its citizens and are taught by citizen leaders and a sociologist. In the fourth session, the citizen leaders lead a discussion about their community and their experiences with the health care system. In the fifth session, residents have lunch with citizen leaders in the community and are given an introduction to the neighborhood and to the mission and activities of WHA.

FINDINGS TO DATE: More than 50 internal residents participated in the curriculum. The evaluations of the program have been uniformly positive, with many residents indicating that it has changed the way they think about their patients and their health care. A majority has also asked that more teaching time be dedicated to these topics. An unexpected benefit has been the education and empowerment of the citizen leaders. They came to see physicians in a new light, as accessible human beings, and learned so much about health and health care that they became community health advocates.

KEY LESSONS LEARNED: Residents are eager to learn about the lives, cultural and social contexts of their patients and value this kind of teaching. Communities and residency programs can create mutually beneficial relationships in the common pursuit of improved patient care and community health.

DEVELOPMENT AND EVALUATION OF A COMMUNITY-BASED PRIMARY CARE EDUCATION PROGRAM. M.C. Reilly¹, G. Morrison¹, M. Cox¹; ¹University of Pennsylvania School of Medicine, Philadelphia, PA

STATEMENT OF PROBLEM OR QUESTION: Tertiary care has come to dominate the educational environment of most academic medical centers, and new training sites are needed to complement the traditional inpatient experience.

OBJECTIVES OF PROGRAM/INTERVENTION: A new curriculum (Curriculum 2000) — supported in part by the Undergraduate Medical Education for the 21st Century (UME-21) national demonstration project — and the acquisition of a large regional primary care network (Clinical Care Associates) provided the opportunity to introduce students to the community practice environment. Our primary goal is to develop a cohesive academic primary care network for both clerkship and residency education. The specific objective of the present component of this broad initiative was to assess medical students' satisfaction with the community training environment by comparing their evaluations of newly-recruited community preceptors with those of more experienced hospital-based preceptors.

DESCRIPTION OF PROGRAM/INTERVENTION: The core clinical clerkships in medicine (12 weeks) and pediatrics (6 weeks) are both equally divided between inpatient and outpatient experiences. Three weeks of the medicine clerkship are spent in a general internal medicine practice, three weeks in a family medicine practice, and six weeks in an inpatient setting. Pediatrics is comprised of three weeks in an outpatient setting and three weeks on an inpatient service. Outpatient preceptors work in 1:1 relationships with students.

FINDINGS TO DATE: In the 1995–96 academic year, primary care clerkship students spent a total 483 half-days in an outpatient environment, all as part of the internal medicine clerkship. Today, students spend 10,720 half-days annually in physicians' offices (internal medicine=3840, family practice=3840, and pediatrics=3040 half-days, respectively). Every student receives a minimum of 72 half-days of ambulatory (predominantly community-based) primary care training during their core clinical clerkship year. Using 9-point Likert scales in up to 15 different categories, students' ratings of newly recruited community internists and pediatricians compared very favorably with those of more experienced hospital-based preceptors. Community family physicians were more highly rated than a core group of residency-based family physicians.

KEY LESSONS LEARNED: Despite the increasing demand for clinical productivity, community faculty accepted medical students with enthusiasm, and students were delighted with the opportunity to work closely with community primary care preceptors. The positive responses from students indicate that the integration of community practice experiences into the undergraduate curriculum will provide medical schools with the opportunity to expose students to diverse role models and a wide spectrum of patients in very favorable settings.

THE SHAREWOOD PROJECT (SP) — A PRE-CLINICAL EXPERIENCE. J.A. Schneider¹, B.A. Lisse¹, C.C. Hung¹; ¹Tufts University School of Medicine, Boston, MA

STATEMENT OF PROBLEM OR QUESTION: Medical students can benefit from meaningful clinical experiences early in their education. A free clinic where students care for patients from an underserved community can provide a strong foundation of knowledge and experience for medical students still in their pre-clinical years.

OBJECTIVES OF PROGRAM/INTERVENTION: The goals of the SP are as follows: 1. To provide clinical experience to first and second year medical students. 2. To teach history taking and physical diagnosis skills early in medical education. 3. To improve the surrounding community's access to primary care services.

DESCRIPTION OF PROGRAM/INTERVENTION: The SP is a free health clinic that was started in 1996 by Tufts University medical students and volunteer physicians. The SP's mission is to improve the Chinatown community's access to health care by providing initial care and then facilitating referrals to ensure continuity of care. The clinic space is donated by a local church and laboratory services are donated by a local laboratory. A grant from the Massachusetts Medical Society, technical support from the New England Medical Center, and an annual, student-run auction, allows the SP to provide services free of charge. First and second year

medical students handle the clinic's management and administration; they act as physician's assistants, case managers and medical interpreters. Students are supervised by volunteer physicians. Since 1996, the clinic has continued to expand and now offers the following: General medical care every Tuesday, 6–10 PM, year round; Psychiatric and gynecological services; On-site EKG machine and phlebotomy; Free medication samples; Referrals to primary health care providers, specialists, dentists, and optometrists; Adult and child immunizations; Anonymous HIV counseling and testing.

FINDINGS TO DATE: Over the past 5 years, more than 400 medical students have volunteered at the SP. First-year medical students take over project management mid-year. The SP's success is a result of the dedication and innovation each class brings to the project. Medical students from Boston University and Harvard, as well as residents from the New England Medical Center have become involved and now contribute to the project. The following are demographics for the 1528 patient visits from 02/97–12/00: Ethnicity(%) Chinese=81.7, White=9.8, Black=2.7, Hispanic=1.3; Age of Patient(%) 0–19=4.7, 20–39=29.2, 40–59=32.2, 60+=33.9; Health Insurance(%) Yes=20, No=76.

KEY LESSONS LEARNED: The SP provides an educational model for medical schools around the country. It is an example of how medical students can establish and manage a free health care clinic and actively participate in patient care during their pre-clinical years. We have learned that it is important to stress the mission of the SP to its patients. The SP is not a substitute for primary care delivered to Chinatown residents. The SP is a place where medical students can begin to take responsibility for patient care and patients can begin to access the American health care system.

ATTENDING/RESIDENT COMMUNICATION PROCESS IMPROVEMENT. R. Sharan¹, E. Moukammal¹, M. Sivka¹, A. Tulsy¹; ¹Department of Internal Medicine, University of Pittsburgh Medical Center Shadyside, Pittsburgh, PA

STATEMENT OF PROBLEM OR QUESTION: In an ideal teaching environment medical interns and residents are expected to assume total care of teaching patients. When the residents are bypassed in the decision making process, it is likely due to a communication problem which leads to attending physicians writing orders directly. Despite repeated efforts to dissuade attendings from writing orders at our community based program we decided to look at what other factors may contribute to this.

OBJECTIVES OF PROGRAM/INTERVENTION: The goal was to improve communication between house staff and attendings and to ultimately decrease attending order writing on teaching patients. In addition we wanted to see an increase in satisfaction by the house staff, nursing staff, and attendings with the communication process.

DESCRIPTION OF PROGRAM/INTERVENTION: A team consisting of house staff, nursing staff, and attendings met on a regular basis to identify key issues impeding good communication. Using an interrelationship diagram the most important issue was identified as the nursing staff directly calling attendings on management issues, due to difficulty in identifying the house staff carrying their patient. The following interventions were developed to facilitate better identification and location of house staff. First, the intern's name and beeper number were entered on the nursing station board, as well as in the computer system which printed it daily on the nursing cardex. Second, the interns were issued personal non-cellular phones to provide direct communication with nurses and attendings. Third, the nursing station board was updated daily identifying the house officers on call. Fourth, the interns and residents were required to twice daily update their status on their beeper voice messages. Outcome measures used were orders written per patient by attendings as identified by a weekly log kept by the house staff. Second, there was a monthly survey of nursing staff, house staff, and attendings of their satisfaction with the communication process.

FINDINGS TO DATE: The average number of orders written per chart by attendings was 1.7 and showed no substantial variation on a run chart since the beginning of the intervention. Poor return on the satisfaction surveys has limited any useful interpretation at this time although efforts are being made to increase their return.

KEY LESSONS LEARNED: KEY LESSONS LEARNED: First, we learned that the major barrier to the communication process was the difficulty in identifying or locating the appropriate house staff. Second, we learned that the motivation for writing orders by attendings is multi factorial and often system related. We will continue to identify those other components that may contribute to order writing as we track the above outcomes.

COMMUNITY-BASED EDUCATION AND SERVICE: EXAMPLE OF A COMMUNITY-BASED PROGRAM IN KENYA THAT CAN BE USED AS A MODEL FOR TRADITIONAL WESTERN MEDICAL SCHOOLS. J. Sidle¹; ¹Indiana University Purdue University Indianapolis, Indianapolis, IN

STATEMENT OF PROBLEM OR QUESTION: Despite increasing demand for community-based or community-oriented programs in medical education in the US, traditional Western medical educational systems have had difficulties in developing and training community-oriented physicians.

OBJECTIVES OF PROGRAM/INTERVENTION: We outline the model of the Community-based Education and Service (COBES) program at Moi University Faculty of Health Sciences (MUFHS) in Eldoret, Kenya. This program serves as a model that provides a proven approach to community-based learning for medical students which is translatable to traditional Western medical educational systems.

DESCRIPTION OF PROGRAM/INTERVENTION: The COBES program implemented at MUFHS over the past ten years provides a valuable model for community-oriented education. The current program comprises approximately 27% of the curriculum for medical students at this nascent medical school in Eastern Africa. Under this program, students spend classroom time developing knowledge of research methodology, epidemiology and public health initiatives. During three years of the curriculum, they spend three to six weeks per year in rural communities learning community interaction skills, public health intervention, and health administration skills. In addition, they spend two years designing and implementing research projects within an urban community of their choice.

FINDINGS TO DATE: Focus groups with students and faculty have demonstrated overwhelming support for the value of this community-based educational program in Kenya. Students have gained valuable training and experience within communities, experienced cross-cultural interactions and have been able to conduct relevant descriptive and interventional research projects. COBES projects have won international awards for student research, but more importantly, the findings have been useful in creating public health initiatives for a variety of communities at a local level.

KEY LESSONS LEARNED: The COBES program at Moi University provides an important model for community-based education that can also be applied to the Western system of medical education. The community interaction skills, public health knowledge and research techniques gained by the students will benefit them in any future endeavors, whether in a subspecialty, research or in community practice. While the immediate value of these types of programs is obvious, further observation and research will be needed to demonstrate the longlasting effects of this model of community-based education on the makeup and practice of medical practitioners in the future.

DEVELOPMENT OF AN ELECTRONIC CONSULTATION NETWORK FOR COMMUNITY-BASED PROVIDERS OF HIV CARE. G. Sinclair¹, Y. Cantu², S. Duggan², P. Keiser¹;

¹Department of Internal Medicine, UTSW, Dallas, TX; ²Department of Pediatrics, UTHSC, San Antonio, TX

STATEMENT OF PROBLEM OR QUESTION: Rural and inner city physicians not specializing in HIV care often lack quick access to HIV specialists who provide diagnostic and treatment recommendations critical to the provision of quality care in this rapidly changing field.

OBJECTIVES OF PROGRAM/INTERVENTION: To test the effectiveness of different approaches designed to link community based physicians with a specialist at a tertiary care center.

DESCRIPTION OF PROGRAM/INTERVENTION: The Texas Oklahoma AIDS Education and Training Center (TOAETC), based at Parkland Memorial Hospital in Dallas, Texas, provides educational services for clinicians. The medical education director of TOAETC served as consulting physician (CP) for this project. In July 2000, a 1800 number was established to accept requests for patient specific consultations from providers in Texas. The number was advertised by: 1. Mailing brochures to potential HIV medication prescribers (n=17,000); 2. Distribution of brochures at TOAETC sponsored lectures (n=421); 3. Two day clinical trainings in which the CP engaged in hands on patient care with community based HIV providers, and personally distributed the TOAETC brochure (n=24). Electronic consultations involved Email, faxes, websites, digital photography, as well as phone conversations. Examples will be displayed. All consultations were completed within 24 hours of the request.

FINDINGS TO DATE: Forty-six patient specific consultations were performed between July 5, 2000 and January 5, 2001. The vast majority (87%) of requests for consultation came from community providers with whom the CP had engaged in a two day clinical training (n=40). Six requests came because of referrals from the CP's colleagues. No patient specific consultations stemmed from either the mass mailing or the direct appeals at TOAETC sponsored lectures. Total cost for the mass mailing was \$36,545. Total cost for the clinical trainings was \$6232. Interestingly, all requests came via the CP's private office number or his private email. The 1800 number was not utilized for its intended purpose of facilitating consultation.

KEY LESSONS LEARNED: Community HIV providers are willing and able to link with providers at tertiary care centers via electronic consultation. However, our experience suggests that they are not likely to utilize electronic consultation methods without first establishing a professional relationship between consultant and consultee, as can be accomplished in a cost effective two day clinical practicum.

BRINGING TOGETHER STUDENTS IN A COMMUNITY-BASED MEDICAL SCHOOL FOR AN EDUCATION DAY. D.J. Solomon¹, R. Martin¹, S. Smith¹; ¹Michigan State University, East Lansing, MI

STATEMENT OF PROBLEM OR QUESTION: Community-based educational programs pose challenges that include providing a high quality consistent curriculum across teaching sites and reducing student isolation.

OBJECTIVES OF PROGRAM/INTERVENTION: To provide an interactive all day educational experience for students in our community-based medical school. Education Day is designed to: 1) Provide consistent high quality instruction for all students in key areas of the clerkship curriculum; 2) Allow students to learn from our best teachers; 3) Provide a recruiting opportunity for our community-based residency programs; and 4) Demonstrate the fun and intellectual challenge of internal medicine.

DESCRIPTION OF PROGRAM/INTERVENTION: Students at our institution receive their clinical training in one of six community campuses located throughout the State. Students in the Medicine Clerkship travel to one campus for an all-day educational experience. The program includes three large group sessions and five small group workshops. Master teachers chosen from each community conduct the sessions. The three large-group sessions include: 1) Acid-Base Interpretation, an interactive lecture; 2) "Thieves Market" where students try to "steal" the diagnosis in a series of progressively disclosed cases; and 3) "Stump the Chump" where the department chairman "thinks aloud" working through a series of difficult clinical cases in a CPC-like format demonstrating the thought processes of an internist. Small groups of students rotate through each of five workshops, which are interspersed between the large group sessions. They include: 1) The Bedside Cardiac Exam using a Harvey (r) mannequin; 2) Starting IVs with Confidence using models and an interactive computer program; 3) Dynamic ECG Interpretation; 4) Chest X-Ray Interpretation; and 5) Utilizing the Peripheral Blood Smear.

FINDINGS TO DATE: We have conducted two iterations of Education Day that included 70 students. Written and oral feedback has been overwhelmingly positive with some suggestions for minor changes. Students expressed specific appreciation for the quality of the program, a chance to see their colleagues at other campuses, and a break from the tedium of their clinical responsibilities. Several students requested expanding the program to two days.

KEY LESSONS LEARNED: Education Day is one approach we have used to address the challenges of community-based education. Although it is not the entire answer, we are pleased with the initial results. While the feedback from students has been overwhelmingly positive, we look forward to assessing whether Education Day has achieved other desirable outcomes including improved shelf exam scores and an increased interest among our students in a career in internal medicine.

AN ELECTRONIC ORIENTATION FOR ROTATORS IN AN INTERNAL MEDICINE CLINIC. E.L. Brownfield¹, M. Mauldin¹, K.D. Rakes¹; ¹Medical University of South Carolina, Charleston, SC

STATEMENT OF PROBLEM OR QUESTION: Our Internal Medicine ambulatory rotation hosts a variety of learners with separate calendars and needs, and our many faculty members had their own sets of performance expectations. We had no consistent orientation process for these learners.

OBJECTIVES OF PROGRAM/INTERVENTION: 1) to have an orientation program for our ambulatory site listing expectations for medical students, physician assistant students, Internal Medicine residents and Gynecology residents 2) to develop an orientation program using a minimum of faculty time and available to learners at any point during the rotation

DESCRIPTION OF PROGRAM/INTERVENTION: Working with the Educational Technology Lab at MUSC, we developed an electronic, on-line orientation program for our Internal Medicine ambulatory block that can be navigated in a learner-specific fashion. This includes a virtual tour of the clinic, an overview of the patient population and expectations for medical students, physician assistant students, Internal medicine and Gynecology residents. We will present the orientation on computer notebook for perusal by attendees, and will discuss the development of the program.

FINDINGS TO DATE: The orientation has been well received by faculty, staff and students. The amount of faculty time spent in orientation has been minimal. We are currently gathering formal feedback that will be available for the presentation.

KEY LESSONS LEARNED: Learners must be given a set of expectations that should be communicated to them in a timely and consistent fashion. This on-line orientation is engaging, readily available and easy to update.

COMPUTER-BASED TESTING OF AUSCULTATORY AND OBSERVATIONAL SKILLS OF MEDICAL STUDENTS. A.S. Farajallah¹, A. Fitzgerald¹, R. Hopkins¹, J. Hart¹, L. Deloney¹, ¹University of Arkansas for Medical Sciences, Little Rock, AR

STATEMENT OF PROBLEM OR QUESTION: With the increased demands on teaching faculty time, the need for standardized methods to assess the acquisition of auscultatory and observational skills of students is required.

OBJECTIVES OF PROGRAM/INTERVENTION: This module was developed and administered to test clinical skills for medical students as part of their introduction to clinical medicine course, with focus on cardiac and chest auscultation as well as determination of radiological images findings.

DESCRIPTION OF PROGRAM/INTERVENTION: A computer software (Question mark Perception) was used to present five clinical scenarios to the students. Each followed by five to six multiple choice, one-best answer questions. Radiological images (.jpg format) and sound files (.wav format) were used to present common cardiac and period in the fall of 2000. A total of nine computers were available in the Learning Resource Center of the library. Library staff members lung findings. The exam was administered to 143 second year medical students over a one-week proctored exam. The exam was scored with the same software (Question Mark Perception)

FINDINGS TO DATE: The average time the students used to complete the test was 47:35 minutes. Minimum time was 4:13 minutes and maximum time was 2 hours and 21 minutes. Of the total class, 84 % of the students took the exam in the last two days. There was virtually no waiting time, although one student where he had to wait for 5 minutes for a computer to be available. The maximum score obtained was 97%; minimum score was 58%. The software program reported the mean score, the standard deviation and degree of difficulty of each question. One question was eliminated from scoring because the sound quality was poor.

KEY LESSONS LEARNED: Computer based testing of clinical skills needs further development, but is currently feasible. We plan in using the same software for an exam in spring 2001. This method of testing can also be expanded for use in clinical clerkships as well as assessing clinical skills of house staff.

NOVEL USE OF AN INTRANET WEBSITE TO TEACH COMMUNITY-ORIENTED PRIMARY CARE. L. Francis¹, S. Glick¹; ¹Cook County Hospital, Chicago, IL

STATEMENT OF PROBLEM OR QUESTION: Clinical practice in community sites is an integral part of our primary care internal medicine training program. Despite this, many of our residents remained unfamiliar with the practice of community-oriented primary care (COPC). In order to teach them about and encourage them to practice COPC, we developed an eight-week course in community medicine. In conjunction with this course, we designed an intranet website to facilitate rapid learning of COPC.

OBJECTIVES OF PROGRAM/INTERVENTION: The goals of our intranet site were to: 1) provide residents with readily accessible information about community medicine; 2) prompt residents to consider, discuss and learn about COPC on a near-daily basis; and 3) introduce distance learning concepts into the residency program.

DESCRIPTION OF PROGRAM/INTERVENTION: We located an internet website (<http://intranets.com/>) that supported the creation of free intranet sites. Once created, access to the intranet site was restricted to the primary care residents and selected faculty. The intranet site (<http://pcimcommunitymed.intranets.com>) featured discussion groups, documents and readings, links to important public health internet sites and a weekly course calendar.

Residents were oriented to the intranet site during the first session of the community medicine course. Each week, residents were asked to respond to an on-line discussion question. Homework exercises and additional reading assignments were also posted on-line. Residents were required to use several internet links to gather neighborhood public health data and to propose a COPC project.

FINDINGS TO DATE: 73% of the residents who participated in the community medicine course (11/15) responded to an anonymous post-course evaluation posted on the intranet website. 73% of the respondents (8/11) found the intranet site enhanced their learning of community-oriented primary care. 9% (1/11) found it neither enhanced nor detracted from their learning experience and 18% (2/11) felt it detracted from their learning. 40% of the respondents (4/10) used the intranet site more than once per week; 50% (5/10) used it once per week; 10% (1/10) used the site less than once every two weeks. The median number of visits to the intranet site during the eight-week Community Medicine course was 11 per resident (range 3 to 60). Seven residents did not participate in the on-line discussion groups. Key reasons for lack of participation included the amount of work assigned for the course (33%), lack of internet access at home (27%) and difficulty navigating the intranet site (20%).

KEY LESSONS LEARNED: Use of an intranet website shows promise as a tool to facilitate rapid learning of concepts in community-oriented primary care.

COMPUTERIZED TRACKING OF IM RESIDENTS INPATIENT EXPERIENCES: AN EDUCATIONAL AND QUALITY IMPROVEMENT TOOL. L. Halasyamani¹, J. Kubus¹, J. Sanfield¹; ¹St. Joseph Mercy Hospital, Ypsilanti, MI

STATEMENT OF PROBLEM OR QUESTION: A significant portion of graduate medical education is based on exposure to a variety of clinical experiences. For residents training in internal medicine the majority of these interactions occur in the inpatient setting. It is important not only to document the range of diagnoses that residents encounter, but also to track length of stay and readmission rates, two important markers of the quality of inpatient care. Quality markers may be linked to attendings supervising the residents and may indicate topics for faculty development.

OBJECTIVES OF PROGRAM/INTERVENTION: We developed a computerized database to 1) track inpatient diagnoses that IM residents encounter on a general medical service; 2) track length of stay, readmission rates, and other performance quality measures for particular diagnoses; and 3) link resident practice measures with supervising attendings to identify areas for faculty development.

DESCRIPTION OF PROGRAM/INTERVENTION: We created a computerized database in Microsoft Access. The database documents every patient admission to the internal medicine teaching service; the medical team (residents and attending) responsible for the patient's care; primary and secondary diagnoses listed for the patient (from the discharge summary); admission and discharge dates. The database user interface is organized with multiple data tables supporting "drop boxes" to click on the appropriate physician names and diagnoses. An administrative assistant enters the data, and any diagnoses not listed in the data tables are referred to a physician for clarification.

FINDINGS TO DATE: This computer-based information system allows tracking of all diagnoses seen by IM residents on a general medical service. The data can be used to compare each resident's exposure to specific diagnoses with his/ her performance on in-training exam sub-scale scores; educational interventions can be developed to enhance medical knowledge in those areas identified as having low exposure and low in-training score. The documentation of residents' experiences also fulfills the Residency Review Committee's requirement for documentation of IM residents' diagnostic experience. The database is used as part of hospital-wide disease-specific quality improvement initiatives to review length of stay and readmission rates on the teaching and non-teaching services, and to target physician education at all levels to improve patient outcomes. Future links between this database and pharmacy and laboratory utilization measures will permit even more detailed analyses of physician performance.

KEY LESSONS LEARNED: An accurate computerized tracking system is a useful educational and quality improvement tool for residents and attendings in internal medicine residency programs. This database recognizes the value of integrating measures of quality improvement into residency training, and may be used by training programs in other disciplines to track clinical performance.

AN INTERACTIVE MULTIMEDIA WEB-BASED PROGRAM TO ENHANCE THE LEARNING OF PHYSICAL EXAMINATION SKILLS BY MEDICAL STUDENTS. G. Kregar¹, D. Knutson¹, C. Fish¹, C. Stewart¹, T. Cain¹, L. Gabel¹; ¹Ohio State University, Columbus, OH

STATEMENT OF PROBLEM OR QUESTION: When faced with teaching physical examination skills to a large class of medical students (as well as coordinate the teaching efforts of a diverse group of physician teachers), it is a challenge to ensure that students are exposed to a common set of physical exam findings and find techniques that optimize their opportunity to develop physical exam skills.

OBJECTIVES OF PROGRAM/INTERVENTION: To expose students to certain common abnormal physical exam findings and enhance learning of physical examination skills through the integration of new technologies and traditional educational strategies.

DESCRIPTION OF PROGRAM/INTERVENTION: Computer-based technology is becoming an integral part of curriculum innovation at Ohio State University. As a part of this pedagogical effort, an innovative, highly interactive website was developed to promote active learning at a level appropriate for undergraduate medical students and to be compatible with the current Physical Examination Course curriculum. This web-site makes use of Macromedia Flash, video streaming, and SMIL technologies to bring to life portions of the physical exam and to make available common abnormal findings often difficult to demonstrate to medical students at any specific time. It emphasizes learning through reading, hearing, seeing, and doing in a sequence designed to capture the learner's attention and to make information

available through multiple sensory modalities. There are 6 core modules (HEENT, pulmonary, cardiac, abdominal, neurologic, musculoskeletal) under development. Each module includes the following: a Flash-based interactive component highlighting important anatomy and physiology; web text and video clips discussing and demonstrating essential physical examination skills; Flash-based interactive components where students hear findings or practice selected physical examination skills (e.g. taking a blood pressure and hearing Korotkoff sounds); a glossary of clinical terms; links to clinical pearls and additional explanations of techniques or abnormal findings; and clinical correlates where students practice integrating physical examination findings. This exhibit will showcase interactive portions of the modules via a CD-ROM presentation.

FINDINGS TO DATE: Pilot data were collected and of the nine respondents, all reported that this site was an excellent learning tool, desired further development of similar technology, and would recommend this site to other students. Eight of the nine students reported being able to navigate the site and use the interactive Flash components without difficulty, and stated they would be more likely to use this site than the textbook to prepare for practice sessions with faculty tutors. Overall, students found it helpful to hear findings, liked the links built into the site, and enjoyed the clinical correlates component. The only weakness identified was the time to download the video clips if access was obtained from modems as opposed to a high-speed internet connection.

KEY LESSONS LEARNED: Next Steps: Data are currently being collected with a larger sample concerning the use and effectiveness of this site to guide subsequent revision of current modules and to plan the design of future ones. Also being considered is the development of patient cases representing core disease presentations, as well as modules specifically designed for educators in this area.

A WEB EXERCISE IN EVIDENCE-BASED MEDICINE USING COGNITIVE THEORY. E.J. Lloyd¹, V.F. Reyna¹; ¹University of Arizona, Tucson, AZ

STATEMENT OF PROBLEM OR QUESTION: Advances in evidence-based research, together with unprecedented access to original sources of information, provide a unique challenge to medical decision-makers that may improve the quality of care provided to their patients—if they can find and correctly interpret this readily available information.

OBJECTIVES OF PROGRAM/INTERVENTION: Our aim was to instruct clinicians accessing and interpreting online sources of evidence-based research using a cognitive theory that has been applied to clinical decision-making.

DESCRIPTION OF PROGRAM/INTERVENTION: Decision-making tasks concerning chest pain evaluation in women were developed for medical students and internal medicine residents. The cognitive theory—fuzzy-trace theory—guided the selection of online sources (e.g. target articles) and decision-making tasks.

FINDINGS TO DATE: Thirty-four participants (12 students and 22 internal medicine residents) attended didactic conferences emphasizing search, evaluation, and clinical application of evidence. A 17-item Likert scale questionnaire assessed participants' evaluation of the instruction. Ratings for each of the 17 items differed significantly from chance in favor of this alternative approach to instruction.

KEY LESSONS LEARNED: Fuzzy-trace theory was found to be a useful guide for developing internal medicine learning exercises in medical informatics and decision making. However, studies with more learners and behavioral evidence of improved clinical decision-making skills and, ultimately, patient outcomes are needed.

E-REPORTS: MORNING REPORT SUMMARIES VIA E-MAIL. B. Lucas¹; ¹Cook County Hospital, Dept of Medicine, Chicago, IL

STATEMENT OF PROBLEM OR QUESTION: Scheduling conflicts prevent many residents, especially those on "non-ward" rotations, from attending morning report. Even when able to attend, however, residents may miss salient learning points presented during those conferences. In order to circumvent these problems, we created eReports, electronic summaries of morning report distributed to all of the internal medicine residents by E-mail.

OBJECTIVES OF PROGRAM/INTERVENTION: The goals of our intervention were: 1) to provide all of our residents with the learning opportunities of morning report even if unable to attend; 2) to model the practice of asking and answering patient-centered questions in an evidence-based way; and 3) to strengthen the educational impact of morning report by cogently synthesizing information presented, further emphasizing salient teaching points, and providing additional relevant information.

DESCRIPTION OF PROGRAM/INTERVENTION: Each day, we developed a summary of the material presented in morning report and distributed it to all of the residents by E-mail. These summaries, which we called eReports, mirrored the structure of our morning reports. Each eReport began with follow-up information that included clinical follow-up to previously presented cases and literature-based answers to questions generated by those cases. We then summarized the two new cases discussed in morning report. At the end of each eReport, a task (e.g., view the peripheral smear) or a central question (e.g., how helpful would a positive result of your planned diagnostic test be?) was generated for each new case.

FINDINGS TO DATE: Residents were asked to complete an anonymous survey prior to the dissemination of eReports. 81% of the residents responded. Of these, 71% "almost never" or only "occasionally" knew the content of morning report (cases or follow-up) when they were unable to attend. 88% "strongly agreed" that eReports would be educationally helpful. Further investigation is underway to determine if eReports were, in fact, helpful. For example, did they successfully clarify teaching points, stimulate self-directed learning, or contribute to improved memory of cases and information presented in morning report throughout the rotations studied?

KEY LESSONS LEARNED: eReports are a promising way to facilitate case-based learning among internal medicine residents. Not only do they ensure the dissemination of the material presented in morning report, they also afford a synthesis that strengthens the quality and content of that material.

WEB-BASED INTERACTIVE CURRICULUM FOR THE ACGME OUTCOMES PROJECT. N.B. Mehta¹, A.L. Hull¹, J.H. Isaacson¹, A. Jain¹; ¹Cleveland Clinic Foundation, Cleveland, OH

STATEMENT OF PROBLEM OR QUESTION: The ACGME has endorsed competencies in 6 core areas as a first step to emphasize educational outcome (rather than process) assessment for residency program accreditation. The guidelines require the residency programs to provide an appropriate educational experience in these areas, evaluate the residents' performance, provide feedback and achieve progressive improvements in residents' competence and performance. In addition the effectiveness of the educational program itself also needs to be evaluated.

To achieve the ACGME goals, residency programs must develop a new educational model that will engage residents in a time efficient manner. In addition to their areas of clinical training, the new curriculum should cover areas such as doctor patient communication, medical ethics and professionalism.

OBJECTIVES OF PROGRAM/INTERVENTION: We have designed and are now implementing a web-based interactive curriculum that would meet common and specific needs of various residency programs at our institution. The objectives of this web-based program are: 1. The curriculum is available at a convenient time and place convenient to the residents. 2. The residents' decide the order and pace of the curriculum. 3. The curriculum utilizes existing teaching materials as far as possible. 4. The new curriculum makes efficient utilization of educators' time and skills. 5. An efficient and effective evaluation and feedback system is an integral part of the new curriculum.

DESCRIPTION OF PROGRAM/INTERVENTION: We designed the web-based curriculum to be based on interactive, question and answer scenarios. The key features of this program are: 1. The curriculum is on a password protected web site. 2. The curriculum consists of brief interactive case scenarios that are grouped in the 6 core competency areas endorsed by the ACGME. 3. These case scenarios can be created by any educator by copying and pasting text from a text document into the text box of a separate development web site. Content is automatically converted into interactive HTML format and placed on the web site. Audio, video and graphic files can also be included. 4. Pretests, posttests, multiple-choice questions and free text entry areas can be included to capture the users responses, track performance and assess competency. In order to simulate a real-life scenario, the residents have an option to order consults and clinical tests to solve the cases. 5. The program provides extensive feedback including tracking amount of money spent while resolving a case scenario, amount of questions answered correctly at first attempt, number of attempts and the ability to compare themselves with other residents who have tried that particular scenario. 6. The program creates a customized recommended reading material based on the questions answered incorrectly. 7. If a user leaves a case scenario incomplete, the progress is saved and can be continued from that point when the user logs on the next time. 8. The program uses a database of house staff to create individualized reports for program directors on the participation and performance of their residents, and also is used to assign particular cases to a particular group of residents.

FINDINGS TO DATE: We plan to formally implement the program beginning in the July 2001 academic year. Pilot testing has shown this system to be acceptable to the house staff, attractive to the program directors and effective and efficient method of teaching for the educators.

KEY LESSONS LEARNED: As new requirements are placed on a medical education system strained for resources and funding, creative solutions are needed that will make efficient use of existing education material, to meet the needs of the students, educators and organizations using current technological advances in computers and the Internet. Using appropriate technology it is possible to create stimulating and educational material that also seamlessly combines tracking, feedback and evaluation and thus efficiently meet the ACGME goals.

WEB-BASED INTERACTIVE CURRICULUM FOR THE ACGME OUTCOMES PROJECT. N.B. Mehta¹, A. Hull¹, J.H. Isaacson¹, A. Jain¹; ¹Cleveland, Cleveland, OH

STATEMENT OF PROBLEM OR QUESTION: The ACGME has endorsed competencies in 6 core areas as a first step to emphasize educational outcome (rather than process) assessment for residency program accreditation. The guidelines require the residency programs to provide an appropriate educational experience in these areas, evaluate the residents' performance, provide feedback and achieve progressive improvements in residents' competence and performance. In addition the effectiveness of the educational program itself also needs to be evaluated. To achieve the ACGME goals, residency programs must develop a new educational model that will engage residents in a time efficient manner.

OBJECTIVES OF PROGRAM/INTERVENTION: 1. Implement a web-based interactive curriculum that would meet common and specific needs of residency programs. 2. The curriculum be available at a time and place convenient to the residents. 3. The residents' decide the order and pace of the curriculum. 4. Utilize existing teaching materials as far as possible. 4. Make efficient utilization of educators' time and skills. 5. Incorporate an evaluation and feedback system seamlessly in the new curriculum.

DESCRIPTION OF PROGRAM/INTERVENTION: 1. The curriculum is on a password protected web site and consists of brief interactive case scenarios (ICS) grouped in the 6 core competency areas. 3. The ICS can be created by an educator without programming knowledge by automatically converting text into interactive HTML format on the web site. Audio, video and graphic files can also be included. 4. Various question formats can be included to capture the users' responses and assess competency. The residents can order consults and clinical tests to solve the ICS. 5. Extensive feedback includes amount of money spent while resolving an ICS, number of attempts and questions answered correctly at first attempt and comparison with peers. 6. The program creates a customized recommended reading material based on performance. 7. If interrupted, the user's progress is saved for subsequent sessions. 8. The program creates individualized reports for program directors on the participation and performance of their residents.

FINDINGS TO DATE: We plan to formally implement the program beginning in the July 2001. Pilot testing has shown this system to be acceptable to house staff, attractive to program directors and effective and efficient method of teaching for the educators.

KEY LESSONS LEARNED: As new requirements are placed on a medical education system strained for resources and funding, creative solutions are needed that will make efficient use of existing education material, to meet the needs of the students, educators and organizations. Using current technological advances in computers and the Internet, one can create stimulating and educational material that combines tracking, feedback and evaluation and thus efficiently meet the ACGME goals.

DEVELOPMENT OF AN ONLINE TUTORIAL TO TEACH ELECTROCARDIOGRAM INTERPRETATION. R.G. Pinckney¹, M.B. Holmes¹, J.M. Tuthill¹; ¹University of Vermont, Burlington, VT

STATEMENT OF PROBLEM OR QUESTION: There is interest in using the Internet to assist in teaching medical clerks ECG reading skills, but multiple technical barriers have impeded efforts thus far. These include inability to place high quality ECG images on the Internet and failure to utilize software that allows interactivity yet is supported by most web browsers.

OBJECTIVES OF PROGRAM/INTERVENTION: To develop an internet-based, comprehensive ECG tutorial that is interactive and can be viewed by standard web browsers.

DESCRIPTION OF PROGRAM/INTERVENTION: We developed a new method for digitizing paper ECG files. We then digitized our complete ECG library used for our four session, clinical clerkship ECG tutorial. These high quality images were placed in web pages using multimedia development software that can be viewed with most web browsers.

FINDINGS TO DATE: An online ECG tutorial is now available to all internal medicine clerks. It covers 4 chapters of material and includes over 30 ECG cases. The interactivity allows users of various skill levels to choose the amount of assistance they receive in interpreting the cases.

KEY LESSONS LEARNED: A large amount of effort was required to overcome technical barriers. Once these issues were overcome, further work was very efficient. The solutions to these technical issues can be easily shared.

MORNING-REPORT.COM-NOVEL TEACHING MODALITIES, EDUCATIONAL RESOURCES, AND ELECTRONIC TOOLS FOR MORNING REPORT. G. Ramakrishna¹, F.S. McDonald¹, J.O. Ebbert¹, V. Montori¹, R.D. Ficalora¹, H.J. Schultz¹; ¹Mayo Clinic, Rochester, MN

STATEMENT OF PROBLEM OR QUESTION: Evidence-based medicine and Internet resources have become increasingly important in the organization and philosophy underlying morning report. Electronic media is easily available to house officers in real time and this often supercedes the limited, outdated content in standard textbooks. Managing that data and integrating it into an educational culture is a new challenge for residents and faculty alike.

OBJECTIVES OF PROGRAM/INTERVENTION: Prior to implementation of the new paradigm, the chief resident conducted morning report and the post-call senior resident presented the case. Faculty, other senior residents, interns, and medical students were present in the room. Qualitative surveys of residents at the time revealed concerns of knowledge stratification between interns and senior residents, need for more current handouts, and a desire for more involvement by the interns and medical students in discussion. In the new model, two chief residents conduct morning report and the two-day post call intern presents the case. No senior residents or faculty are present in the room.

DESCRIPTION OF PROGRAM/INTERVENTION: We describe a novel morning report format that:

1. Separates intern and senior morning reports to alleviate pressures of knowledge stratification and to enhance the learning experience for each group.
- 2.
3. Facilitates learner interaction by optimizing learning climate and geography (e.g., location, food, and open table settings).
- 4.
5. Implements two active facilitators to engage the group, maximize participation, and utilize complementary skill sets.
- 6.
7. Applies educational and computerized resources including electronic databases, networked computers, and data projectors to develop an efficient and educationally driven case selection process.
- 8.
9. Incorporates electronic tools ("E-tools") to adapt quickly to learner goals and needs, based on time of year, learner experience and learner interest.
- 10.

FINDINGS TO DATE: Anonymous intern evaluation data suggest that the new model has effectively addressed these issues. Based on current academic year data of almost 100 responses, all interns consistently felt either "challenged" or "interested". Almost 25% of interns in fact preferred a longer morning report. The use of electronic handouts has met with universal acceptance: 77% thought the handouts were "very good" and 23% considered them "good". Only one resident had a neutral opinion and there were no negative responses. Almost 96% of respondents agreed with exclusion of faculty and senior residents in morning report. Interns who were exposed to the morning report prototype (n=47) had a significant 9.7% increase in average ITE percentile scores ($p=0.015$, unpaired t-test) compared with interns not exposed to the model (n=96).

KEY LESSONS LEARNED: We conclude that our morning report model has increased the level of participation among interns, alleviated pressures of knowledge stratification between interns and residents, and successfully utilized electronic tools to enhance and adapt to a learner-centered environment. Areas for further study include successful adaptation of the model to the dynamic needs of report participants, correlation with the ITE, development of more specific outcome metrics, and impact on patient care.

"I'D LIKE TO USE A LIFELINE": DEVELOPMENT AND IMPLEMENTATION OF A CLERKSHIP COUNSELING HOTLINE. T.H. Gallagher¹, J. Munro¹; ¹Washington University in St. Louis, St. Louis, MO

STATEMENT OF PROBLEM OR QUESTION: The third year of medical school can be particularly stressful for students. Some students are reluctant to access their school's mental health resources for fear of appearing "weak" or unable to cope in a highly competitive environment.

OBJECTIVES OF PROGRAM/INTERVENTION: To help students cope with the stresses of their clinical clerkships, the Program for the Humanities in Medicine at Washington University School of Medicine instituted the Clerkship Counseling Hotline.

DESCRIPTION OF PROGRAM/INTERVENTION: The Clerkship Counseling Hotline was a confidential, anonymous service intended to provide third year students with a short-term mental health resource aimed at easing stress. The hotline was staffed by a masters level counselor around the clock.

FINDINGS TO DATE: 17 calls were received by the Hotline from August 1999 through November 2000. Issues prompting calls, in order of frequency, included a) disillusionment with medical environment; b) anxiety over performance/evaluation; c) personal/relationship problems; d) requests for information/referral; and e) anxiety over decision to go into medicine. 15 of the 17 callers were female. A year-end survey assessing the Hotline was returned by 83% of the students. 75% of all students said that continuing the Hotline was "somewhat important" or "very important," and 75% of students described the availability of the Hotline as "somewhat reassuring" or "very reassuring." 88% of women found the hotline reassuring, compared to 58% of men ($P=.002$).

KEY LESSONS LEARNED: This Clerkship Counseling Hotline was regularly used by our 3rd year students, who strongly endorsed its continuation. Female students used the Hotline substantially more frequently than did male students. Such a Hotline may provide a service that is complementary to the traditional mental health resources offered by most medical schools.

ESTABLISHING AND MAINTAINING AN INTERNAL MEDICINE RESIDENT SCHOLARLY ACTIVITY PROGRAM. B.F. Mandell¹, K.D. Saneto¹; ¹Cleveland Clinic Foundation, Cleveland, OH

STATEMENT OF PROBLEM OR QUESTION: What are the key components of an internal medicine residency's scholarly activity program (SAP)?

OBJECTIVES OF PROGRAM/INTERVENTION: 1. Establish a SAP to provide I.M. residents with tools for lifelong learning. 2. Comply with the ACGME-RRC scholarly activity requirement (SAR). 3. Evaluate the program.

DESCRIPTION OF PROGRAM/INTERVENTION: Key components of our SAP include: precise delineation of requirements; dedicated time of a research director and coordinator; core curriculum with didactics (trial design, test interpretation, meta-analysis, bioethics, medical writing, informatics, computer graphics), journal club focused on trial design, optional clinical teaching seminars, and an elective 10-hour biomedical writing course; optional research modules; SAP manual; staff mentors and institutional support. The SARs include: participation in didactic program and journal club; completion of a pre-approved written project; completion of a (staff evaluated) 30–60 minute oral presentation using computer-generated slides; and a 1:1 literature search session with a librarian. An annual research day includes: residents' poster presentations (in 2000: 33 posters) with a reception and a visiting professor lecture. Two selected research projects are presented by the residents at a medical grand rounds. Our 2001 budget of \$26,340 includes: resident travel to present at meetings, printing of SAP manual, biomedical writing course, Research Day expenses, and general operating costs. The budget does not include salaries for Research Director (10%) or Coordinator (50%).

FINDINGS TO DATE: Since inception of our SAP, resident publications and presentations have increased (in 2000: 10 published articles, 12 abstracts, 42 presentations at national or regional meetings). The SAP and the research mentor's department co-support residents' travel to present at meetings. The most common written project is an original research abstract; case reports and simple literature reviews are not acceptable. A survey of our graduating residents indicates acceptance of the SAP (20 of 29 PGY-3 residents felt the SAP to be "very valuable").

KEY LESSONS LEARNED: We believe the success/acceptance of our SAP are due to an established core curriculum, supportive personnel, dedicated funding (in particular for resident trips to meetings), availability of staff mentors, strong institutional resources, and meticulous tracking of residents' progress. ~50% of our residents take 1–2 research modules, which must be approved in advance. No relation exists between the type of project and whether residents took a research module.

RESIDENTS' OPINION REGARDING THE CLEVELAND CLINIC FOUNDATION GUIDELINES FOR ANTIMICROBIAL USAGE BOOK. S. Mossad¹, J. Long¹, M. Goldman¹, T. Keys¹; ¹Cleveland Clinic Foundation, Cleveland, Ohio

STATEMENT OF PROBLEM OR QUESTION: Guidelines for antimicrobial usage have been published and distributed to all house staff (HS) at the Cleveland Clinic Foundation (CCF) since 1991. These guidelines are intended to help in decisions regarding antimicrobial therapy in hospitalized patients.

OBJECTIVES OF PROGRAM/INTERVENTION: Formal assessment of the effects of these guidelines on the practice of HS has not been previously done.

DESCRIPTION OF PROGRAM/INTERVENTION: A questionnaire was sent to the mailboxes of all house staff in the Division of Medicine and an audience response system was also used to administer the questionnaire during a noon conference.

FINDINGS TO DATE: Thirty five HS participated in this survey; (48% GL-I, 24% GL-II, 14% GL-III, 7% GL-IV and 7% GL-V). Most of them thought the book was trustworthy (87%), accurate (86%), user friendly (57%), and overall useful for the management of patients (87%). Although most HS always carried the CCF book (73%), most also carried other

commercially available antibiotics recommendations book(s) (68%). There was no consensus which of these books was more useful for them. The CCF Book sections on guidelines for specific diseases and dosing information were referred to most often (37% & 40%; respectively). Requests were made to add more microbiology information and other clinically important antimicrobial drug interactions.

KEY LESSONS LEARNED: The CCF Guidelines for Antimicrobial Usage book is well respected among the HS. The book serves its purpose well in helping HS with their daily care of hospitalized patients. This survey helped us in implementing some changes that we believe will improve these guidelines and consequently, improve the quality of care for our patients.

DIVING FOR PERLS: USING PORTFOLIOS FOR EVALUATION AND REFLECTION ON LEARNING. K. Edwards¹, L.E. Pinsky¹; ¹University of Washington School of Medicine, Seattle, WA

STATEMENT OF PROBLEM OR QUESTION: Lifelong learning is an important goal of medical training. However, objective tools for teaching skills of self-assessment and for giving feedback on professional development are lacking. Medical training is an experiential process and reflection is necessary to develop expertise and professionalism from the practical experiences.

OBJECTIVES OF PROGRAM/INTERVENTION: Residents participating in the portfolio program will: 1) learn how to set achievable learning goals; 2) identify learning experiences that will facilitate achieving learning goals; 3) develop skills of self-assessment, identifying strengths and weaknesses 4) receive structured peer and supervisor feedback in a range of skill areas 5) develop important professional skills that support life-long learning, physician-patient communication, and excellence in patient care.

DESCRIPTION OF PROGRAM/INTERVENTION: Portfolios constitute an integrated system of learner-directed evaluation. The primary outcome of the evaluation project is for each resident to create a portfolio that includes a comprehensive collection of different types of feedback and self-assessment. The components of the portfolio may vary, but are based on specific activities to assure broad coverage of skill areas. They include: worksheets on goal setting, tracking and self summary of learning, teaching self-evaluation, colleague feedback; critical incident type narratives; mini-CEX observations; CD-ROM compilation of the resident's videotaped encounters with patients, directed self-evaluation, a summary of resident/attending meeting, and overall department evaluation. There is a Working Portfolio for the resident's use and work-in-progress, and a Performance Portfolio for highlighting core competencies and strengths at the end of the year.

FINDINGS TO DATE: After a pilot program with the internal medicine residents, a surprising number of the portfolio skills and objectives needed significant attention by the mentor faculty. Simple goal setting exercises presented challenges for some residents. This finding highlighted the need for the portfolio exercise within the residency program.

KEY LESSONS LEARNED: We identify 5 elements critical to the success of a meaningful and effective portfolio intervention: 1) distinguishing the goals of the Working and Performance portfolios; 2) establishing a collegial educational climate; 3) teaching and refining self-assessment acumen; 4) promoting reflection on goals and progress over time; and 5) supporting structured autonomy via iterative discussions with mentors. Without these elements, trainees may view a portfolio simply as another tedious exercise and we risk trivializing an innovative tool.

US-JAPAN COLLABORATION IN GRADUATE MEDICAL EDUCATION. Y. Tokuda¹, N. Nakazato¹, M. Maeshiro¹, S. Miyagi¹; ¹Okinawa Chubu Hospital, Gushikawa, Okinawa

STATEMENT OF PROBLEM OR QUESTION: Graduate medical education in Japan needs more effective training program.

OBJECTIVES OF PROGRAM/INTERVENTION: University of Hawaii John A. Burns School of Medicine had provided support for the development of graduate medical education in Okinawa Chubu hospital, Okinawa, Japan. Historically, it was started in 1966 under US military presence in postwar Okinawa to train graduates from mainland Japan medical schools. It is organized according to the residency training programs in the USA. Now it aims to achieve not only the best educational program in Japan but also one of the best hospitals in Japan.

DESCRIPTION OF PROGRAM/INTERVENTION: The University of Hawaii provides one-five full time American board certified faculty members and 10-15 short time consultants during a year. The current full time hospital professional staff includes about seventy physicians. Both Japanese hospital staff and American university faculty are actively involved in the clinical teaching, the basis of which is formed by admission rounds, ward rounds, daily conferences and didactic lectures.

The predominant feature of the training program is the responsibility given to the interns and residents in total care of the patients under the guidance and supervision of the staff. The trainees develop the skills of clinical examination, clinical procedural techniques, clinical problem solving, and eventual logical planning of patient management.

The program now has the exchange clinical clerkship program between Okinawa Chubu hospital and Kuakini Medical Center in Honolulu, Hawaii. Several residents from Okinawa Chubu hospital each year participate in clinical training in this University of Hawaii affiliated hospital. Post-trainees, who finish the training program and obtain the certificate of Educational Commission for Foreign Medical Graduates (Philadelphia, PA), are selected to join the two-year clinical fellowship training in mainland USA under the supervision of the program director.

FINDINGS TO DATE: In more than thirty years of its history this program had supplied 598 physicians to practice modern clinical medicine. The Ministry of Health and Welfare of Japanese government certify this program as the model of the training program for primary care in Japan. The Ministry of Education and Culture also recognize the program as a prototype model for the future graduate medical training in Japan.

KEY LESSONS LEARNED: To further promote medical cooperation between Okinawa and Hawaii is able to improve medical standard of a whole region of mainland Japan. It will consist of long-term supply of doctors capable of providing high quality medical care and model development of graduate medical education for other medical schools in Japan.

SING, SING A SONG: MAKE IT SIMPLE FOR MEDICAL EDUCATION. R.C. Anderson¹, R.M. Wang-Cheng¹, D. Simpson¹; ¹Medical College of Wisconsin, Milwaukee, WI

STATEMENT OF PROBLEM OR QUESTION: Adult learners retain concepts more readily when multiple modalities are employed. Singing words that have been written as capsule summaries of key educational concepts is a unique way to facilitating learning.

OBJECTIVES OF PROGRAM/INTERVENTION: To facilitate learner retention of educational concepts through the use of song.

DESCRIPTION OF PROGRAM/INTERVENTION: Songs were written by faculty to familiar tunes such as "These are a few of my favorite things." Key learning points were in areas such as Faculty Development, Evaluation, Mentoring and Research. These songs were then shared with learners and sung in group settings.

FINDINGS TO DATE: A song book, "Lyrical Teaching Tools for Academic Medicine," has been compiled. The book is available for use by faculty as a teaching and learning tool.

KEY LESSONS LEARNED: Composing and singing songs are a fun and innovative method for teaching, learning, and retaining major educational concepts.

COMMUNITY BASED FACULTY "TEACH THE TEACHERS ABOUT MEDICAL STUDENTS IN THE COMMUNITY BASED OFFICE. S. Baillie¹, B. Landres², H. Raskind¹; ¹University of California, Los Angeles, Los Angeles, CA

STATEMENT OF PROBLEM OR QUESTION: Community based ambulatory educational experiences have been increasingly emphasized in medical education due to the need to prepare medical students and residents to become more knowledgeable about primary care medicine in the era of managed care. At our institution, students start working in community based practices in the first year of medical school. Effectiveness of the students' educational experience in the community has frequently varied depending on the skills and interests in teaching of the physicians participating in the program. All physicians in our network are required to teach medical students at least 1/2 a day a week but little faculty development in teaching has been provided. How to effectively address the needs of these faculty is addressed in this innovative peer teaching program.

OBJECTIVES OF PROGRAM/INTERVENTION: In order to improve the overall experience for both students and faculty, our institution has developed a "Teach the Teachers" program. The primary objective of the program is having community based faculty associated with our institution's primary care network teach their peers skills in effective teaching and mentoring of medical students.

It is also an objective to develop a core of community based faculty who will be able to provide leadership in the faculty development arena for our community based teachers of medical students.

DESCRIPTION OF PROGRAM/INTERVENTION: Over the course of a year, the university staff collaborated with four affiliated community based primary care faculty to guide them in selecting the curriculum modules they felt would be most useful to office based teachers. The team also participated in the nationally organized General Internal Medicine Faculty Development Project (GIMFEL) where additional skills and training techniques were acquired. In order to develop a quality program, the community faculty selected four curriculum areas as well as a kick-off session for their program activities. They also worked with the Sr. Associate Dean for Medical education to make sure that the chairman of the Department of Medicine supported the program and would strongly encourage all of the university's primary care network of physicians (total of 91) to participate in at least 20 hours of faculty development over a two year time period. CME was also offered for all faculty development.

The program selected includes the following modules:

1. The kick off program (one time session for all participants), Developing Teaching Skills for the Primary Care Physician

A nationally known physician educator, Gary Ferenchick, MD, MS, was selected to present the basics of community based teaching to the entire group of primary care network physicians at one of their required dinner meetings. The speaker was chosen for his enthusiastic hands-on approach to office-based teaching. The speaker at the end of his talk then introduced the community based faculty development program and the four peer faculty.

Each of the four peer faculty developed an interactive module to be presented to each of four groups of teaching physicians at dispersed sites around our large urban service area. They selected these topics based on a needs survey and their own personal experiences in teaching. These topics include the following:

2. Orienting and Organizing the Student Experience
3. The One Minute Preceptor-Teaching on the Run
4. Arrows in the Quiver: Using Multiple Strategies to Reach Your Student
5. Developing Effective Techniques for Student Evaluation and Feedback

FINDINGS TO DATE: In an earlier program of training of community based faculty by our institution, training of community based faculty was very successful but the lack of departmental support for the training led to smaller numbers of faculty participating. This program reviewed the prior one and developed a different strategy.

The four community based faculty have spent two hours each month for the past year in preparation for the training program in addition to attending the three day GIMFEL conference on faculty development. They have reviewed major articles and materials on curriculum development in order to select those they felt most appropriate based on the initial needs assessment of their peers. They have obtained the Chair of Medicine's support for the program as well as for the publication of the four curriculum articles. The program times are scheduled and the kickoff session is set. Success of the program will be reported on at the conference.

KEY LESSONS LEARNED: Community based faculty need to be involved in leadership roles in faculty development for their peers. The topics may not be new to the field but having peers teaching peers is innovative and an effective strategy for encouraging the acceptance and adoption of new teaching skills.

Materials Produced and Available

A series of four curriculum articles written by the four community based faculty and workshop handouts will be available for those interested in implementing a similar type of peer training program. A video will also be available to demonstrate the interactive modules.

FACULTY DEVELOPMENT ON-LINE: PREPARING FACULTY TO MAKE OBSERVATIONS AND GIVE FEEDBACK ON COMMUNICATION SKILLS. R.W. Janicki¹, A. Kalet¹, M. Nachbar¹, S. Zabar¹; ¹NYU School of Medicine, New York, New York

STATEMENT OF PROBLEM OR QUESTION: Receiving feedback is an essential component of learning clinical skills. For this reason medical schools are devoting time and resources to prepare faculty for providing students with effective feedback. In an effort to reach more faculty and allow members to learn at their own pace, we designed a web-based faculty development module to address the relevant skills.

OBJECTIVES OF PROGRAM/INTERVENTION: -Improve the observation and feedback skills of faculty.

-Design an interactive web-based teaching module utilizing multimedia.

DESCRIPTION OF PROGRAM/INTERVENTION: The web-based module is the first of a series of teaching modules planned to support the Macy Initiative in Health Communication at NYU (<http://endeavor.med.nyu.edu/macy/nyumacy/>). The Macy Initiative is a three school (New York University, University of Massachusetts and Case Western University Schools of Medicine) project funded by the Macy Foundation to improve the communication skills of physicians. Part of the faculty development efforts at NYU include plans for several on-line teaching modules. The observation/feedback module contains three sections. In the first section participants are asked to watch a 1st year student interview and then to complete a checklist. During the second section, the participant's checklist data is compared with that of several expert faculty. In the last section, participants are taken stepwise through the feedback process. They begin by orienting the student, then by asking for the student's own self-assessment, followed by giving feedback and finishing with the closing. At each step participants are asked to enter their own comments and then watch a video clip of the same 1st year student receiving feedback from an expert faculty member. Seminar leaders for the current Physician, Patient and Society Course as well as Primary Care Residents will be asked to take the module. Participants fill out a pre-post survey to assess changes in attitudes and confidence in giving feedback and to comment on the module.

FINDINGS TO DATE: It is feasible to design an interactive web-based faculty development module. We are able to track which faculty have completed the module using a log-in page and retrieve faculty responses anonymously by having programmed the module to send them to a separate web-site without identifiers. Data on the changes in attitudes and confidence as well as feedback on the module will be available.

KEY LESSONS LEARNED: -Helpful collaborations can take place among Clinical and Academic Computing Departments. -These collaborations often require learning each others "languages" and involving project managers.

-Interactive digital media is a very exciting, innovative way to teach the medical interview.

A FACULTY DEVELOPMENT PROGRAM IN TEACHING QUALITY IMPROVEMENT. H.E. Mehaber¹, A.J. Mehaber¹, M. O'Connell¹; ¹University of Miami, FL

STATEMENT OF PROBLEM OR QUESTION: Quality improvement (QI) is an essential component of patient care. While issues of QI are encountered daily in clinical practice, the concept is often overlooked as one of the essential competencies for medical students and residents.

OBJECTIVES OF PROGRAM/INTERVENTION: The goals of our faculty development program were to enable participants to: 1) define key concepts of QI, 2) give examples of standards by which quality is measured, 3) rank QI as an important component of clinical care, and 4) incorporate QI issues in everyday clinical teaching at the bedside.

DESCRIPTION OF PROGRAM/INTERVENTION: Our program was comprised of an interactive discussion entitled "What Every Clinician Needs to Know About Quality" led by an international expert in QI, followed by a series of observed structured teaching experiences (OSTEs). These OSTEs were case-based teaching modules utilizing role-playing to address QI issues encountered in typical clinical cases. The program was evaluated through a formal discussion with participants at the conclusion of the seminar along with pre and post-program surveys.

FINDINGS TO DATE: Twenty faculty from the Departments of Medicine and Family Medicine participated in the program. The faculty was surveyed pre and post-program on their knowledge, competency to lecture about, and competency to teach clinical applications of QI. A 10-point scale was utilized, with 0 indicating no knowledge or competency, 5 being neutral, and 10 indicating superior knowledge or competency. In addition, attitudes towards teaching about QI were surveyed pre and post-program, using a 5-point Likert scale.

With an 85% response rate, perceived knowledge of QI increased from below neutral to above neutral in 29% of respondents. Competency to lecture on QI increased from below neutral to above neutral in 41% of respondents. Most importantly, competency to teach clinical applications of QI increased from below neutral to above neutral in 59% of respondents. The perceived value of teaching about QI also increased, ($p = 0.04$) and faculty uniformly commented on how this program reinforced the importance of (or value of) teaching this essential clinical competency.

KEY LESSONS LEARNED: QI issues are seen in daily clinical practice and need to be addressed with medical students and residents. Faculty development through interactive role-playing and discussions can be influential in changing attitudes and self-perceptions of teaching abilities.

TRAINING PRIMARY CARE FACULTY TO TRAIN OTHER PRIMARY CARE FACULTY IN HUMAN GENETICS: A WORK-IN-PROGRESS. E. Rich¹, W. Burke², K. Edwards²;

¹Creighton University, Omaha, NE; ²University of Washington, Seattle, WA

STATEMENT OF PROBLEM OR QUESTION: Equipping primary care faculty with the knowledge and skills to "bridge" primary care and genetics is a challenge.

OBJECTIVES OF PROGRAM/INTERVENTION: The federally funded Genetics in Primary Care (GPC): A Faculty Development Initiative has conducted Part I of a two-part GPC Training Program. Objectives of Part I were to enable faculty participants to understand the purpose of incorporating genetics into primary care teaching and practice, identify specific

teaming needs for home faculty and available resources, develop a 6-month implementation plan and acquire teaching techniques to incorporate genetics into primary care teaching and practice.

DESCRIPTION OF PROGRAM/INTERVENTION: Part I of the two-day GPC Training Program held in October 2000 included didactic and interactive plenary sessions, content-based breakout sessions, a poster session, and computer demonstrations. Twenty competitively selected faculty teams were paired to enhance interaction and to facilitate discussions with advisors (members of the GPC Advisory Committee or Genetics Education Consultant Committee) regarding specific implementation plans for faculty development. A draft curriculum manual, largely case-based in its organization with information concerning electronic and written resources, was prepared for use by participants and faculty in the course.

FINDINGS TO DATE: Findings from the external evaluation of Part I of the GPC Training Program show that the greatest self-reported increase in knowledge and skills are in the areas of resources, development of a specific implementation plan, and in applying teaching techniques for addressing ethical, legal, and social implications of genetics in medicine. Participants say they need more help with genetics content and with around specific methodologies for delivery of genetics-based faculty development to primary care faculty.

KEY LESSONS LEARNED: Meaningful incorporation of additional genetics content into Part 2 of the GPC Training Program is indicated. Additionally, more interaction time and opportunities for team members to observe and practice specific methods of delivering training may enhance participant teaming.

IMPROVING STUDENTS INTEREST IN INTERNAL MEDICINE AS A CAREER CHOICE — A BRIEF 2-WEEK AMBULATORY EXPERIENCE FOR JUNIOR MEDICAL STUDENTS. S.L. Davids¹, R.M. Wang-cheng¹; ¹Medical College of Wisconsin, Milwaukee, WI

STATEMENT OF PROBLEM OR QUESTION: To determine the effectiveness of a brief ambulatory experience in teaching junior medical students Internal Medicine.

OBJECTIVES OF PROGRAM/INTERVENTION: Our goals in providing an ambulatory experience to junior medical students were: 1) increase exposure to outpatient medicine thereby stimulating interest in Internal Medicine as a career choice. 2) improve shelf board exam scores 3) improve performance on OSCE examination in both physical exam and communication.

DESCRIPTION OF PROGRAM/INTERVENTION: During October 2000, 17 of 34 junior medical students were randomly assigned to participate in a 2-week ambulatory block during their 2 month required IM clerkship; the remaining 6 weeks were spent on inpatient wards. The remaining 17 students completed 2 months of inpatient wards. The settings included: community private clinics, VA Clinics, and faculty practices. The 2-week period was comprised of clinic 5 half days per week, and 4 half days of small group interactive sessions covering general medicine topics. Students were required to present an interesting case they were exposed to during the ambulatory portion of the clerkship.

FINDINGS TO DATE: Compared with inpatient students there was no difference on their performance.

Exam	Ambulatory	Inpatient	P
OSCE:Communication	96.53	96.69	NS
OSCE:Physical Exam	24.65	24.75	NS
Shelf Board Exam	66.06	68.06	NS

However, 60% were more likely to go into Internal Medicine, while only 33% stated it had no effect on their career choice. They enjoyed working in the outpatient setting, and saw on average 3-4 patients per 1/2 day. Approximately 50% watched procedures, while 20% were able to participate in procedures such as pelvic exams, joint injections and mole removal. They saw a wide range of problems including chronic medical diseases, skin diseases, smoking related issues, as well as preventative medicine. Overall, the students enjoyed their outpatient experience and would have preferred more time in the clinic setting.

KEY LESSONS LEARNED: Even though there was no statistically significant difference in how well the students performed on their examinations, they did view this two-week ambulatory block as a positive experience. In addition, we feel it is significant that 60% of the students reported they were more likely to go into Internal Medicine despite this very brief exposure to outpatient medicine.

DEVELOPING A MANAGED CARE CURRICULUM. C.A. Dimou¹, M. Anderson¹; ¹Rush Presbyterian St. Luke's Medical Center, Chicago, IL

STATEMENT OF PROBLEM OR QUESTION: The most common deficiency among our graduating residents was a lack of preparation for dealing with the business side of medicine. What is the best way to introduce third party payment systems and the intricacies of running an office practice to residents?

OBJECTIVES OF PROGRAM/INTERVENTION: We hope to create an ambulatory experience that would combine patient care with an educational focus on medical business. The residents will have an introduction to billing and coding. They will become familiar with the different payer sources and how they relate to patient care. The curriculum will allow residents to apply the basics of managed care (including formularies and referral) while taking care of patients. The residents should also develop a basic knowledge ethical conflicts that exists within insurance entities.

DESCRIPTION OF PROGRAM/INTERVENTION: The educational curriculum consists of one month rotations in ambulatory clinics with significant managed care exposure. The clinic experience is supplemented with lectures on billing, coding, insurance basics, and ethics. Residents provide input on deficiencies in their knowledge and which topics should be expanded. Residents also receive hands on experience by observing ancillary office staff in

their different duties (billers, referral staff, etc.) in order to help solidify their learning experience.

FINDINGS TO DATE: Residents show improvement in their ability to bill and code appropriately. They also are satisfied with their learning experience and think it should be expanded.

KEY LESSONS LEARNED: Knowledge of the business aspect of medicine has become an essential part of medical practice. Residency programs need to address this need in educating their residents. Residents are very receptive to lectures on medical business and see these topics to be an important supplement to medical education.

AN AMBULATORY PRIMARY CARE MUSCULOSKELETAL CLINIC FOR RESIDENTS. T. Houston¹, R. Connors¹, N. Cutler¹, M. Nidiry¹; ¹Johns Hopkins University, Baltimore, Maryland

STATEMENT OF PROBLEM OR QUESTION: Although a highly rated Rheumatology rotation exists, our residents rated their training in "primary care" musculoskeletal medicine (knee pain, biceps tendonitis, epicondylitis, ankle sprain, injections, etc.) as less than optimal. **OBJECTIVES OF PROGRAM/INTERVENTION:** Success in "Primary Care" Musculoskeletal Medicine training was limited by lacking primary care preceptors trained in musculoskeletal problems and by not having a concentrated collection of patients with musculoskeletal problems. Our **OBJECTIVE** was to establish an innovative primary care musculoskeletal medicine clinic precepted by general internists with additional training in joint issues.

DESCRIPTION OF PROGRAM/INTERVENTION: In the planning year, faculty attended special musculoskeletal training sessions sponsored by SGIM and Rheumatology organizations, attended orthopedic clinics, and assembled a musculoskeletal patient population. Triage nurses were instructed to only schedule patients with acute and chronic joint problems to a special clinic session at a community-based practice site. One half-day per week, residents interviewed patients, discussed diagnosis and management with faculty, and then were supervised during procedures. The curriculum also included group discussions focusing on diagnosis and treatment (medications, physical therapy, etc) of musculoskeletal problems not covered in the Rheumatology rotation, skills practice with injection models, and a reference syllabus.

FINDINGS TO DATE: Our evaluation of the musculoskeletal clinic within the first six months focused on the success in achieving a concentrated experience for the first six residents to participate in the rotation. Residents averaged seeing 4.3 patients with musculoskeletal complaints per half-day clinic. The most common musculoskeletal complaints were knee pain(25%), back pain(19%), shoulder pain(16%), and hip pain(9%). In treating these patients, residents performed a mean of 2.0 procedures per half-day clinic. The most common procedures included injections of the : knee, subacromial region, and trochanteric bursa. Residents have uniformly rated the usefulness of the clinic as very good/superb. An additional pre-post knowledge and skills assessment of residents will be completed at the end of the two-year curriculum.

KEY LESSONS LEARNED: Our needs assessment indicated that residents wanted additional training in musculoskeletal medicine, but preferred to learn in a primary care setting. Despite the logistic challenges, our curriculum has demonstrated the feasibility of a concentrated primary care musculoskeletal medicine experience.

TRAINING PHYSICIANS IN SMOKING CESSATION: USE OF VIDEOTAPE, ROLE PLAYS AND STANDARDIZED PATIENTS TO LEARN COUNSELING SKILLS. J. Humair¹, J. Cornuz²; ¹Geneva University Hospital, Geneva, GE; ²Lausanne University Hospital, Lausanne

STATEMENT OF PROBLEM OR QUESTION: Medical advice and nicotine replacement are effective smoking cessation interventions. However physicians have a limited impact as they miss many opportunities to help smokers quit and often lack the necessary counseling skills. As smoking cessation is a stepwise process, physicians should tailor counseling to each smoker's motivation to quit.

OBJECTIVES OF PROGRAM/INTERVENTION: We designed a program based on active learning methods and the "stages of change model" to train physicians in smoking cessation. At the end of training, physicians should be able to: (1) assess each smoker's motivation to quit; (2) advise all smokers with strategies matching their motivation to quit; (3) prescribe pharmacotherapy to smokers ready to stop.

DESCRIPTION OF PROGRAM/INTERVENTION: In two 4-hours sessions, active educational methods enable participants to progressively learn these new skills. In the first session, learners use a checklist to observe 3 videotaped encounters with smokers and identify the 3 main stages of motivation to quit. An interactive workshop follows for presentation and discussion of basic concepts and strategies of smoking cessation in relation to videotaped cases. Then participants are involved alternatively as physician, patient and observer in role plays with smokers at various motivation stages. In the second session, learners practice counseling skills with trained standardized patients who portray smokers with different profiles and readiness to quit. Trainees also receive a reference document, pocket algorithms, a record sheet for smokers, 5 stage-matched brochures for patients and instructions to patients about use of nicotine replacement.

FINDINGS TO DATE: We tested this training program in a randomized trial among residents in primary care clinics. Compared to the control group, trained residents provided smoking cessation interventions of higher quality (mean score: 4.0 vs. 2.7, p=0.01, range: 0–14) and expressed higher confidence in their skills 3 months after training (mean score: 7.7 vs. 5.2, p=0.002, range: 0–10). Moreover, 1-year smoking abstinence almost doubled among smokers visiting trained residents (13% vs. 7%, p=0.04).

KEY LESSONS LEARNED: We could develop a physician training program in smoking cessation based on active and progressive learning of counseling skills. This program was effective to improve physicians' practices and smokers' cessation rate. If this training is implemented at a large scale, physicians could contribute more effectively to reduce smoking prevalence and its major health consequences.

RESIDENTS AS TEACHERS OF DOMESTIC VIOLENCE SCREENING AND MANAGEMENT: A NOVEL CURRICULUM. J. Jaeger¹, J. Rosenblum¹, S. Goyal¹; ¹University of Pennsylvania, Philadelphia, PA

STATEMENT OF PROBLEM OR QUESTION: Internal Medicine (IM) residents should receive instruction in the screening and management of domestic violence, but there are few committed faculty with the time or inclination to do this well. Traditional IM residencies provide little forum for residents to learn formal teaching and public-speaking skills or to put to use their enthusiasm for patient advocacy.

OBJECTIVES OF PROGRAM/INTERVENTION: 1) To develop and implement a novel, multi-media, Domestic Violence (DV) Curriculum for IM residents; 2) To allow interested PGY-2 and -3 residents an opportunity to work closely with faculty in developing public-speaking, advocacy, and teaching skills.

DESCRIPTION OF PROGRAM/INTERVENTION: Two PGY-2 residents were identified who had an interest in DV. These residents helped to design a 3-hour curriculum which was implemented on a monthly basis for all interns during their Ambulatory Medicine rotation. The curriculum consisted of: 1) A portion of a video about DV survivors; 2) A lecture on the dynamics, screening, and management of DV; 3) A handout including articles on DV and an annotated bibliography; 4) A talk by a social worker on principles of counseling and local resources for survivors of DV; 5) A role play demonstrating screening strategies; 6) An interactive discussion of three cases of "positive" results of screening; 7) Small group sessions in which the interns were given case scenarios and asked to try the screening and management techniques they had learned. The lecture was given by the lead author, who is a faculty member. Elements 5–7 of the curriculum involved the PGY-2 residents to an increasing degree over the course of the spring.

FINDINGS TO DATE: The curriculum was well-received by the interns. Both the lecture and the portions led by the PGY-2's were consistently rated highly in feedback obtained at the conclusion of the sessions. The interns commented frequently on the benefits of the role playing. We were also able to identify several interns interested in DV who have been recruited to participate in the curriculum this spring.

KEY LESSONS LEARNED: Residents respond well to learning skills of outpatient medicine from their peers, whom they may feel have a better sense of the limitations and realities of their own practice. Many housestaff have an interest in both the care of vulnerable patients and advocacy that is rarely put to use in the traditional IM residency. Curriculum design and implementation can be made easier and more appealing by involving residents with specific interests. Faculty can serve as mentors in order to facilitate resident academic growth and development.

TRAINING RESIDENTS TO RECOGNIZE DOMESTIC VIOLENCE IN PRIMARY CARE. D.R. Korenstein¹, D. Thomas¹; ¹Mount Sinai Hospital, New York, NY

STATEMENT OF PROBLEM OR QUESTION: An estimated 2.5 million women experience domestic violence (DV) annually, with a reported one-year prevalence rate of 5.5%. It has myriad clinical presentations, but often goes unsuspected. Teaching about DV is challenging, with few educational interventions in the literature with documented efficacy.

OBJECTIVES OF PROGRAM/INTERVENTION: 1. To develop an evidence-based curriculum in DV for internal medicine residents. 2. To improve the ability of residents to appropriately suspect domestic abuse in their primary care patients.

DESCRIPTION OF PROGRAM/INTERVENTION: The educational intervention consisted of a 3-hour interactive seminar on DV, including discussion of video material, case discussion, an evidence-based literature review, and role-playing. The seminar was designed to heighten awareness of abuse as a problem in the primary care setting, and to teach the importance of screening and useful techniques for screening. After the seminar, residents were asked to screen all of their female patients for DV for the next 2 weeks, and return to discuss their findings in a follow up seminar. Six to 12 months after completing the intervention, all second and third year residents received a questionnaire containing eight modified essay questions concerning clinical scenarios. All PGY3 residents surveyed (intervention group) had received the educational intervention while none of the PGY2 residents (control group) had received it. The residents were unaware of the purpose of the questionnaire, and all responses were anonymous. Three of the eight cases described scenarios in primary care which were designed to stimulate a suspicion of DV, for which residents were asked to list up to five leading diagnoses. A response was counted as correct if it contained the words "violence" or "abuse." Questionnaires with at least 2 correct responses (to the 3 suspicious questions) were graded as positive for demonstrating adequate suspicion of DV.

FINDINGS TO DATE: The program was well-received, with an average rating of 4.6 on a 5-point scale (n=33). As of 12/29/00, response rates were 70% (26 of 37) in the intervention group and 51% (19 of 37) in the control group. The rates of "positive questionnaires" were 21% (4/19) for the control group and 58% (15/26) for the intervention group (p=.014).

KEY LESSONS LEARNED: An educational intervention in domestic violence using video, case discussion, evidence-based review and role-playing was well-received, changed physician awareness of domestic violence in primary care, and had a lasting effect six to 12 months after the intervention.

TO TEACH EBM OR NOT TO TEACH EBM?: HOW TO ENTHUSE RESIDENTS AND STAFF TO PRACTICE EVIDENCE BASED MEDICINE (EBM). W. Land¹, L. Copeland¹, N. Mehta¹, A.K. Jaffer¹, E. Caiola², T. Gildea¹; ¹The Cleveland Clinic Foundation, Cleveland, OH; ²University of Rochester, Webster, NY

STATEMENT OF PROBLEM OR QUESTION: Many methods exist to teach EBM. Which one is most effective? We present two methods of teaching EBM to primary care residents. These methods require staff participation and reinforce staff utilization of EBM practices.

OBJECTIVES OF PROGRAM/INTERVENTION: 1. To stimulate residents and staff to learn and practice EBM. 2. To present curriculum materials and offer ideas of how to integrate EBM programs into current curricula. 3. To demonstrate how to evaluate EBM teaching methods and to assess resident progress using these methods.

DESCRIPTION OF PROGRAM/INTERVENTION: We will demonstrate the session plans of two EBM courses. Course 1 consists of an introduction session in which staff present the fundamentals of asking clinical questions, searching for the best evidence, and critiquing articles. Afterwards, one resident prepares a patient case and presents it to the group. The resident then formulates a clinical question and searches for the best evidence to answer it. The resident must individually meet with a staff preceptor to review the answer. Each week a different resident presents a case, formulates the question, meets with a staff preceptor, and presents the findings to the group. EBM Course 2 consists of five didactic sessions. The first session is the same as for Course 1. The subsequent sessions include staff presenting methods on how to evaluate articles of diagnosis, therapy and prognosis. The students and staff then discuss cases, formulate clinical questions and go to the library together to search for answers. The group reconvenes to critique the search strategies.

FINDINGS TO DATE: Residents (N=12) in both courses completed the same 32-item questionnaire on the first and last sessions. Residents were asked the amount of searching done, comfort in searching/critiquing findings, applying findings to patient care, and what data source they'd search first to answer a particular clinical question. Pre and post differences were tested with the Wilcoxon Signed Ranks test. Comments from Course 1 regarding the need for more time to learn how to search prompted formal didactic sessions for Course 2. Course 2 residents significantly increased their frequency of searching for answers to clinical questions/applying EBM ($Z=2.02$, $p=.028$). They reported more confidence in their ability to conduct quality searches ($Z=2.21$, $p=.027$) and to critique the articles ($Z=2.23$, $p=.026$). Residents from both courses mentioned that they learned about many new sources of medical information on the Internet and library databases.

KEY LESSONS LEARNED: These courses may be effective means to increase residents' confidence to conduct EBM searches and to evaluate results. These courses are important ways to recruit staff to actively further their own education and application of EBM.

FOUNDATIONS OF DOCTORING: A NEW METHOD OF TEACHING MEDICAL STUDENTS THE FUNDAMENTALS OF COMMUNICATING WITH PATIENTS. C. Lin¹, F.W. Platt¹, G.E. Barley¹; ¹University of Colorado Health Sciences Center, Denver, CO

STATEMENT OF PROBLEM OR QUESTION: Students learning how to interview patients frequently use a high control, closed ended questioning method. What is the best way to teach students a patient-centered interview technique?

OBJECTIVES OF PROGRAM/INTERVENTION: 1. Develop a longitudinal interviewing skills curriculum in the first two years of medical school. 2. Create a standardized method to evaluate students interview performance before and after this intervention.

DESCRIPTION OF PROGRAM/INTERVENTION: Students in their first and second years of medical school participated in small group workshops modeled on the Bayer Institute workshops on communication during which they learned specific communications skills. Subsequently, they spent an afternoon session in groups of four students, one faculty coach and four standardized patients (SP's). Each student interviewed a patient, received verbal and written feedback from his/her peers, from the coach, and from the SP. Then they would re-do parts of the interview based on suggestions. At the end of the year, students interviewed a different SP case for a final exam, and were rated by instructors, the SP and themselves. The videotapes of these evaluations are then reviewed one on one with a faculty member trained in this process.

FINDINGS TO DATE: In contrast to group lectures and small group sessions, students were most highly engaged in the group practice sessions with the SP's. Many students used a high control interview, but with feedback, rapidly learned to use more patient-centered interview techniques.

KEY LESSONS LEARNED: After trying several didactic methods (lecture, small group workshop of 10-15 students, optional videotape review sessions), we found that students responded best to a carefully facilitated SP session within a very small group (4 students) with immediate feedback from student colleagues, coach and SP.

OFFICE BASED MANAGEMENT OF COMMON SPORTS MEDICINE PROBLEMS. J.D. Meitzel¹; ¹Hospital for Special Surgery, New York, NY

STATEMENT OF PROBLEM OR QUESTION: Internists in general practice are increasingly being asked to treat sports-related injuries. However, there is little formal training in the current curriculum geared towards the education of residents in the realm of musculoskeletal medicine. This interactive workshop will use a case-based approach to teach hands-on musculoskeletal examination skills.

(This workshop can run anywhere from 1 to 2 hours - I am not familiar with your program format, so I submitted this work here. However, if you think that it would be better-suited for another venue in this conference please let me know. I currently run the Sports Medicine elective for the Cornell Internal Medicine residents. Many thanks - Jordan Metzl, M.D.)

OBJECTIVES OF PROGRAM/INTERVENTION: 1) Teach the basics of evaluation and treatment of common sports injuries within general medical practice

2) Provide suggestions on how this material can be taught to residents in training

DESCRIPTION OF PROGRAM/INTERVENTION: Case based approach to common knee and shoulder injuries. Learning how to use patient history to narrow diagnostic possibilities, appropriate use of radiographs, and demonstration of physical examination of knee and shoulder (depending on time allowed for this workshop).

FINDINGS TO DATE:

KEY LESSONS LEARNED: 1) Better approach to the treatment of musculoskeletal problems

IMPROVING MEDICAL STUDENTS HISTORY TAKING SKILLS. G.S. Ogrinc¹, W.B. Weeks¹; ¹White River Junction VA Hospital, White River Junction, VT

STATEMENT OF PROBLEM OR QUESTION: The first 2 years of medical school are filled with a seemingly endless stream of new information. While the basic science classes are similar to prior learning experiences, clinical learning, which may be more satisfying (i.e., "feeling like a

doctor"), requires skills that are relatively new and different. History taking skills are often evaluated by Observed Structured Clinical Exams (OSCE) or by direct observation and feedback. The traditional method of understanding the medical history is often a static list of questions and categories that fail to direct the flow of the student-patient interaction. In addition, students often desire more feedback than is possible to provide in a busy office setting.

OBJECTIVES OF PROGRAM/INTERVENTION: This project seeks to create and implement a system by which medical students can understand, self-evaluate, and improve their history taking skills.

DESCRIPTION OF PROGRAM/INTERVENTION: The subjects were 8 first-year medical students at Dartmouth Medical School enrolled in the On Doctoring course - a longitudinal clinical care experience for all students in years 1 and 2. The students alternate one half-day per week with a facilitator in a small group setting, then one half-day one-on-one with a preceptor in an ambulatory setting. In the small group, the facilitator (GSO) and the students together built a process flow diagram of the medical history placing a strong emphasis on the transitions of the process. This process diagram was then used as a template for students to evaluate and improve their history taking skills. "History scorecards" were given to the students for use in their subsequent preceptor visits. At the preceptor's office, immediately after taking a history, the student rates several aspects of the interaction (e.g., introduction of self, agenda setting, use of silence) on a 10-point scale for each patient encounter.

FINDINGS TO DATE: On the subsequent preceptor visits, 25% of students used the "history scorecards" to evaluate their performance. Those who used the system found it easier to track their skills and to ask for directed feedback from their preceptor. Those who did not use the system either forgot to bring the scorecard, found it cumbersome, or did not realize that it was an "assignment."

KEY LESSONS LEARNED: In addition to learning history taking skills, students design a process diagram, analyze the process to find aspects that can be changed/improved, and collect and analyze data so as to monitor improvement. This novel way of teaching and learning the medical history incorporates elements of evidence based decision making for directed feedback and quality improvement skills that are important for students to understand.

DEVELOPING AN EVIDENCED-BASED WOMENS HEALTH CURRICULUM. H.G. Pursley¹, D.S. Kwolek¹; ¹University of Kentucky, Lexington, KY

STATEMENT OF PROBLEM OR QUESTION: The explosion of new knowledge in women's health and the need to teach this to internal medicine residents in an evidenced-based fashion has presented a great challenge to medical educators. To address this need, we developed and implemented an evidenced-based women's health curriculum.

OBJECTIVES OF PROGRAM/INTERVENTION: Residents are taught to define and utilize basic evidence-based medicine concepts to critically analyze women's health literature; to understand recent innovations in women's health; and to apply evidence-based medicine to the clinical practice of women's health.

DESCRIPTION OF PROGRAM/INTERVENTION: We defined our curriculum based on consideration of recommendations from the National Academy of Women's Health Medical Education, the American Board of Internal Medicine, the Fifth Report of the Council on Graduate Medical Education, and the results of needs assessments of internal medicine residents at our institution. Using Medline, an extensive literature search was performed on the following topics: osteoporosis, breast cancer, hormone replacement therapy, domestic violence, coronary artery disease in women, menopause, headaches, substance abuse in women, urinary incontinence, dementia, sexual dysfunction, and evidence-based medicine to create a bibliography of readings for residents who rotate through our women's health center. Peer-reviewed journal articles were compiled. Priority was given to data published since 1990, and randomized, double-blinded, placebo controlled studies. Faculty and residents review and analyze two to four articles weekly on a given subject. Content experts provide context and clinical expertise to discussions. Clinical questions, such as "What is the risk of my patient developing breast cancer on hormone replacement therapy?"; and "Should I prescribe hormone replacement therapy to my post-menopausal patient to help prevent coronary artery disease?" are addressed in each session. Evidence-based medicine core concepts are reviewed and applied, including the number needed to treat, absolute risk reduction, and relative risk.

FINDINGS TO DATE: Previous work from a needs assessment of residents at our institution found a discrepancy in perceptions and actual knowledge in women's health. Ninety-one percent of the residents rated inadequacies in the women's health curriculum. Our evidenced-based curriculum serves to bridge this gap of knowledge. Residents participating in our curriculum have expressed increased knowledge in the subjects discussed in our weekly conferences. Also, residents have stated that much of what is taught in this curriculum has not been covered elsewhere in their residency curriculum.

KEY LESSONS LEARNED: A gap exists to be filled between perceptions of curriculum inadequacies in women's health and actual knowledge. Our curriculum serves as a forum for an update in women's health literature, an exchange of ideas for the improvement of women's health as it is taught in internal medicine, and for further elucidation of the evidence behind what we practice and teach.

THE USE OF AN URGENT CARE CENTER FOR AMBULATORY EDUCATION AND CURRICULUM DESIGN. D.B. Wayne¹, S.B. Greenberg¹, D.A. Helmer¹, D.J. Hyman¹; ¹Baylor College of Medicine, Houston, TX

STATEMENT OF PROBLEM OR QUESTION: To be effective, General Internal Medicine physicians should be able to manage acute ambulatory "non-medical" problems, but it is unclear if residents gain adequate exposure to such conditions in training.

OBJECTIVES OF PROGRAM/INTERVENTION: 1) To improve ambulatory training in Educationally Targeted Specialty (ETS) conditions, (ENT, urology, dermatology, ophthalmology and orthopedics) we initiated an urgent care center (UCC). 2) To determine the suitability of the UCC for this goal, we reviewed the frequency of ETS conditions seen in the UCC and compared this to the resident continuity clinics (RC) and emergency room (ER).

DESCRIPTION OF PROGRAM/INTERVENTION: A UCC was developed at an urban teaching hospital and staffed with General Internal Medicine faculty. All charts from the UCC, RC and ER on randomly selected days over a 2-month period were prospectively reviewed (n=500 charts per location). Demographic data, past medical history and complaint(s) were recorded. Acute complaints were classified into 3 categories: General medicine, ETS conditions, and miscellaneous (upper respiratory infection, gynecology and psychiatry).

FINDINGS TO DATE: 476(95.2%) patients presenting to the UCC and 491(98.2%) patients in the ER had acute problems as compared to 236(47.2%) patients seen in the RC ($p < .001$). 17(3.4%) patients from the UCC, 5(1%) from the RC and 120(24%) from the ER were admitted to the hospital ($p < .001$). The total number of ETS conditions recorded were: 302(UCC), 104(RC) and 89(ER) ($p < .001$). The number of complaints in each category were as follows: ENT: 88(UCC), 17(RC), 19 (ER). Urology: 43(UCC), 10(RC), 14(ER). Dermatology: 41(UCC), 11(RC), 11(ER). Ophthalmology: 25(UCC), 7(RC), 10(ER). Orthopedics: 105(UCC), 59(RC), 35(ER).

KEY LESSONS LEARNED: Patients in the UCC had acute conditions, but were unlikely to require admission. Persons with ENT, urologic, dermatologic, ophthalmologic and orthopedic problems were seen more often by the residents in the UCC than in the other locations. Based upon these findings, a case-based and interactive curriculum has been developed. A UCC provides a useful venue for the clinical and didactic education of Internal Medicine Residents in areas of acute care which may be underrepresented in other outpatient rotations.

A RANDOMIZED TRIAL OF AN EDUCATIONAL INTERVENTION TO IMPROVE RESIDENTS USE OF CLINICAL PRACTICE GUIDELINES. G.C. Wickstrom¹, L. Clough², D.G. Jarjoura³, ¹Summa Health System, Akron, OH; ²Summa Health System, Akron, Ohio; ³Northeastern Ohio Universities College of Medicine, Rootstown, Ohio

STATEMENT OF PROBLEM OR QUESTION: Innovative methods are needed to sustain physician practice patterns that comply with standards of care. This exhibit will describe a model for evaluating clinical competency and educating physicians in the clinical practice of evidence-based guidelines for patient care.

OBJECTIVES OF PROGRAM/INTERVENTION: Develop tools to educate, promote and measure physician compliance with primary care guidelines for diabetes management and for care of the elderly.

DESCRIPTION OF PROGRAM/INTERVENTION: The educational intervention includes three components: 1) a videotaped clinical competency assessment using standardized patients with immediate faculty feedback; 2) individualized educational tutorials using the assessment videotapes and published guidelines; and 3) measurement and feedback of practice patterns relative to standards of care. The efficacy of this model was assessed through a randomized trial of educational interventions for two independent guidelines. After initial assessments with standardized patients, 28 internal medicine residents were randomly assigned to receive either the guideline intervention for elders at risk for depression or the guideline intervention for patients with diabetes. All residents in the study are reassessed through direct observation with standardized patients and through chart review of a panel of clinic patients, thus providing independent comparative measures of competence. The two randomized groups are being compared on a measure of the difference in performance with diabetic patients and elderly patients. This study design, using two interventions, has the potential to double the effect size producing a more powerful study than one with a single intervention for research involving small numbers of residents.

FINDINGS TO DATE: To date, half of the residents in the intervention have been reassessed using standardized patients. These results show that residents with the diabetic intervention performed better with their diabetic patients than with their elder patients, and the reverse was found for the residents trained in elder care. Difference between the diabetes trained (n=7) and elder care trained (n=6) residents on their difference in performance on two diabetes cases versus two elder care cases was large. The effect size of the two interventions (combined) was .72 standard deviation units (+/- .27 standard errors) and significant ($p = .02$). These results were consistent for the component score differences for history taking, physical examination, and diagnosis, but inconsistent for management plan.

KEY LESSONS LEARNED: The overall advantage of the intervention is an educational experience that involves faculty and residents in an interactive learning and evaluation setting. Added benefits are the development of new tools to measure quality and physician compliance with guidelines for patient care. Videotape review of standardized cases, tutorial outlines and chart review templates used in the study will be available at the exhibit.

IMPLEMENTING A WEB-BASED INTERDISCIPLINARY EVIDENCE-BASED MEDICINE OUTPATIENT TOPICS CURRICULUM IN A COMMUNITY HOSPITAL. S.S. Yen¹, M.Z. Cassey², J. Stielstra³, Z. Piotrowski⁴, ¹West Suburban Health Care, Oak Park, IL; ²West Suburban College of Nursing, Oak Park, IL; ³West Suburban Hospital Medical Center, Oak Park, Illinois; ⁴West Suburban Hospital Medical Center, River Forest, Illinois

STATEMENT OF PROBLEM OR QUESTION: How can impediments to evidence-based clinical practice and barriers to distribution of quality patient education materials be reduced in a community hospital Internal Medicine outpatient residency curriculum?

OBJECTIVES OF PROGRAM/INTERVENTION: At the conclusion of the Internal Medicine residency outpatient curriculum, residents will: 1. Effectively search for evidence-based medicine approaches to outpatient topics. 2. Incorporate evidence-based medicine rationale into outpatient care decisions. 3. Retrieve and distribute appropriate patient education materials to outpatients.

DESCRIPTION OF PROGRAM/INTERVENTION: Utilizing a traditional case-study methodology, the internal medicine faculty prepare the cases and questions for posting to a form on the program website. (<http://www.westsub.com/wshcre residency/optotopes/optotopes.htm>) A medical librarian applies EBM searching techniques to the case study topics and summarizes the most effective approach to obtaining current information from the OVID databases, MD Consult, www.guideline.gov and other online reference sources. The librarian generates a "tips sheet" to model searching behavior that will assist the learner in executing effective searches for

clinical decision-making activities for the case studies. To conclude the topic, the nursing informatics educator/webmaster identifies quality patient education resources for the specific outpatient topics. An online submission form sends case study answers via e-mail to the clinical faculty or chief resident for evaluation prior to the outpatient conference. The case study submission is also filed on the website server for future reference.

FINDINGS TO DATE: This approach has a strengthened the preparatory behavior of Internal Medicine residents in relation to the outpatient curriculum. Year 1 of a 3-year follow-up pre-post resident self-report suggests most changes in attitudes and EBM behaviors ($p < .10$). The lead resident submits information prior to the group presentation. The monitor receives the residents' answers via e-mail and addresses significant issues confidentially.

KEY LESSONS LEARNED: The outpatient website has increased the use of EBM clinical decision-making tools in the clinical practice settings. It has provided opportunities for improving residents' information literacy skills and knowledge of quality patient education resources. Attending physicians and monitors are requesting additional education related to information literacy and evidence-based medicine.

CLINICAL CLERKSHIP IN COMMUNITY MEDICINE IN GENEVA: LEARNING ABOUT AMBULATORY CARE AND HEALTH OF VULNERABLE POPULATIONS. J. Humair¹, H. Stalder¹, ¹Geneva University Hospital, Geneva

STATEMENT OF PROBLEM OR QUESTION: Undergraduate medical curriculum generally includes little time for education in ambulatory care and about health of vulnerable populations. However most medical students will later provide ambulatory care requiring to solve common clinical problems and respond to the needs of vulnerable groups of patients.

OBJECTIVES OF PROGRAM/INTERVENTION: Since 1995, Geneva medical school implemented a new undergraduate curriculum including a clinical clerkship in community medicine during the fourth or fifth year. Its main learning objectives are: (1) to identify common clinical problems in ambulatory care; (2) to provide comprehensive care to patients in their psycho-social context; (3) to be sensitized to health needs of vulnerable populations.

DESCRIPTION OF PROGRAM/INTERVENTION: This clerkship is organized for rotating groups of 12 students over 4 weeks. Clinical activities take place in 3 ambulatory settings: a primary care clinic serving mainly people of lower socio-economic status and migrants, a geriatric clinic delivering home care to fragile elderly patients, a unit caring for patients with alcohol or illicit drug abuse. Students attend 8 tutorials for clinical reasoning and problem solving regarding common ambulatory conditions. A workshop based on role plays sensitizes students to counseling for smoking cessation or alcohol use. All students must also present and comment a clinical case to their colleagues. Time is allocated for self-directed learning to prepare some activities. Students receive formative assessment based on their behavior in clinical activities and their ability to solve a new clinical problem.

FINDINGS TO DATE: At the end of the clerkship, students evaluate its various aspects on a 5-point scale. After the first year, 108 students expressed a high global satisfaction with the clerkship (mean score: 4.2). Mean scores were also high for key elements like achievement of learning objectives (4.3), organization (4.5) and quality of teachers (4.6).

KEY LESSONS LEARNED: We successfully developed a structured clinical clerkship in community medicine enabling students to learn about ambulatory care and health of vulnerable populations. This program met students' expectations and was highly appreciated. Further research should assess the impact of this clerkship on clinical competences in ambulatory care and career choices at the end of undergraduate training.

A MINI-RETREAT MODEL FOR TEACHING AND ASSESSING INTERN COMMUNICATION SKILLS. R.C. Anderson¹, R.M. Wang-Cheng¹, L. Cantwell¹, D.E. Weissman¹, ¹Medical College of Wisconsin, Milwaukee, WI

STATEMENT OF PROBLEM OR QUESTION: Based on self-assessment, Internal Medicine residents lack self-confidence and competence in end-of-life patient care, including breaking bad news, patient goal-setting, DNR discussions and death pronouncement. In 1999, we formed a working group to teach and evaluate these core communication skills in our Internal Medicine residency program.

OBJECTIVES OF PROGRAM/INTERVENTION: Demonstrate basic communication skills · Demonstrate how to give unwanted news · Demonstrate how to conduct a discussion with the "angry patient" · Demonstrate how to lead a discussion to establish goals/patient preferences, including DNR orders and changes in treatment from curative to palliative approach

DESCRIPTION OF PROGRAM/INTERVENTION: All interns attend 2 required afternoon retreats (8 hours total contact time). The first retreat focuses on basic communication skills such as empathy and attentive listening and breaking bad news. The second retreat addresses goal-setting and DNR decisions with the terminally ill patient and death pronouncement. We used a variety of formats: large group role-plays and debriefing discussions, brief handouts, pre-tests and videotaped, standardized patient stations with immediate, individual feedback from faculty observers.

FINDINGS TO DATE: 20 interns completed the first retreat training and 22 completed the second. Intern pre/post self-assessment shows significant improvement in degree of competency and knowledge. On a scale of 1 to 4 (4 being most competent/most knowledgeable), interns report greater knowledge in discussing goal setting (2.5 to 3.5; $p < 0.05$) and discussing DNR orders (2.4 to 3.5; $p < 0.05$) and self-perceived increased competence in giving bad news (3.1 to 3.7; $p < 0.05$) and death pronouncement procedures (2.9 to 3.7; $p < 0.05$). Interns praised the retreat for the open interaction with faculty experts, the opportunity to role play patient care situations and the exposure to topics not previously discussed in medical training. A few mentioned distaste for being videotaped.

KEY LESSONS LEARNED: A retreat format, using large group role-plays and individual standardized patient encounters with feedback, is a useful method of teaching and evaluating important end-of-life communication skills. Interns who needed extra help in communication and interpersonal skills were identified and individually reviewed these skills with faculty mentors.

A RESIDENT-AS-TEACHER CURRICULUM DURING AMBULATORY BLOCK. T.L. Simon¹; ¹Mount Sinai School of Medicine, New York, NY

STATEMENT OF PROBLEM OR QUESTION: Residents are expected to teach interns and medical students, yet receive no training to help improve their skills as teachers. An effective resident-as-teacher (RAT) program needs to utilize protected time and provide sufficient opportunity for skills practice, reflection, and reinforcement.

OBJECTIVES OF PROGRAM/INTERVENTION: (1) To prepare residents for their role as teachers in the medical setting. (2) To enable residents to examine their own teaching skills and behaviors, and take active steps to improve their teaching effectiveness. (3) To improve clinical teaching in the Department of Medicine.

DESCRIPTION OF PROGRAM/INTERVENTION: The RAT curriculum consisted of four consecutive 2.5 hour weekly workshops, repeated monthly, during the PGY2 ambulatory block rotation in the first half of the academic year. A group of 5-6 residents met with one faculty member for sessions utilizing interactive techniques such as facilitated discussion, videotape review, and role-play. The workshops covered the following content areas: (1) Introduction to Teaching Principles/Diagnosing the Learner, (2) Microskills of Clinical Teaching/Giving a Microlecture, (3) Resident as Team Leader, and (4) Evaluation and Feedback. Teaching homework assignments were completed between sessions and reviewed as part of the next workshop. Reinforcement materials were sent to the residents 1, 2, and 3 months after completion of the curriculum.

FINDINGS TO DATE: The program was well-received, with a mean "overall rating" score of 4.5 on a 5 point scale (n=33). A pre-test rating scale of teaching behaviors was also administered at the beginning of the curriculum, and a post-test will be distributed to each group four months after completion. Qualitative data was collected during workshop discussions, on topics such as "Overcoming Barriers to Teaching on the Wards" and "Characteristics of the Ideal Resident". In addition, residents completed self-assessment checklists of their own performance as clinic preceptors. Finally, there is a long term plan to analyze resident teaching evaluations from interns and medical students before and after the institution of this curriculum.

KEY LESSONS LEARNED: RAT programs have traditionally been done in either a single 1-2 day block, or "longitudinally" with discrete sessions in each year of training. Utilizing the ambulatory rotation allows for protected time with small groups of residents in a more relaxed setting. Weekly workshops provide the opportunity for practice teaching exercises, review, and reflection. Residents are eager for this type of training, participate enthusiastically, and rate the program highly. Whether this will translate into measurable improvement in teaching months to years later will be assessed.

A STANDARDIZED PATIENT CLINICAL EVALUATION EXERCISE. G.M. Talente¹, S.A. Haist¹, S.M. Quinlivan¹, J.F. Wilson¹; ¹University of Kentucky, Lexington, KY

STATEMENT OF PROBLEM OR QUESTION: The Clinical Performance Exercise (CEX) is an American Board of Internal Medicine (ABIM) requirement for internal medicine residents. Traditionally, faculty observe residents interviewing and examining patients and discuss further care with the resident. Time pressures have led to difficulty recruiting adequate numbers of faculty to participate in the traditional CEX at one institution. An alternative format became a necessity.

OBJECTIVES OF PROGRAM/INTERVENTION: Our primary objective was to develop a more reliable and reproducible test using standardized patients that successfully assessed the same skill areas as the traditional CEX. We also wanted to obtain data on resident competency that would be better suited to use for program evaluation.

DESCRIPTION OF PROGRAM/INTERVENTION: Four SP cases (chest pain/unstable angina, headache/tension, chest pain/pulmonary embolus and abdominal pain/dyspepsia) were developed. The residents were instructed to obtain a history of present illness, past medical and psychosocial history, perform a comprehensive physical examination (PE) and write an assessment and plan (AP). SPs completed the history (Hx) and PE checklists (yes/no, yes = done and done correctly, no = omitted or done incorrectly). The SP rated PE technique (6 questions) on a 5-point Likert scale e.g. "the resident minimized patient movements", interpersonal skills (5 questions) on a 5-point Likert scale, e.g. "the resident demonstrates a professional demeanor", and non-verbal communication skills (16 questions) e.g. "the resident's tone of voice was 1=unfriendly, cold, 7=friendly, warm". AP score was based on key developed by 5 faculty.

FINDINGS TO DATE: 50/52 intern and senior residents participated. Mean (SD) percent sub-scale scores were Hx 58.4 (7.0), PE 72.2 (15.8), AP 42.7 (18.5), technique 94.0 (4.9), interpersonal skills 94.9 (5.1), and non-verbal communication skills 83.8 (16.7). The scores on the Hx, PE, and AP were lower than expected and ratings of technique, interpersonal and non-verbal communication skills were acceptable.

KEY LESSONS LEARNED: The SP CEX is a feasible alternative to the traditional CEX. However, a considerable amount of time and effort is required to establish an examination using this format. A pre-existing SP program is a helpful to this process. The SP CEX enables residency programs to standardize the CEX and avoid variability from different raters (faculty). The SP CEX provides more detailed information on resident performance than the traditional format, which will be useful in providing more specific feedback to examinees and directing curricular development within a program.

AN INTERDISCIPLINARY DIABETES EDUCATION CONFERENCE AT A TEACHING COMMUNITY HEALTH CENTER: WHO ATTENDS? WHAT ARE THE BENEFITS? A.H. Balder¹, E.M. Benjamin²; ¹Mason Square Health Center, Springfield, MA; ²Baystate Medical Center, Springfield, MA

STATEMENT OF PROBLEM OR QUESTION: In the care of patients with chronic diseases, no doctor is an island. This is particularly true in group practices, clinics and health centers where a number of professionals and support staff interact with each patient. When a large resident component is added to the provider base, there is increased demand on all staff to understand the content and process of caring for individuals with common chronic diseases. Educating residents in the care of multisystem chronic conditions in outpatients is also vital to successful care, particularly with evolving changes in management.

OBJECTIVES OF PROGRAM/INTERVENTION: The Mason Square Neighborhood Health Center (MSNHC) is a hospital operated health center with a staff of 9 attendings, 8 RN's, 3 NP's, 32 med/peds residents, and 8 MA's. It serves 600 diabetic patients and employs a "staged diabetes management" guideline for care. MSNHC is challenged with keeping staff knowledgeable in the many facets of diabetes in order to improve care and teamwork. It must do so without a devoted diabetes educator to run such an effort.

DESCRIPTION OF PROGRAM/INTERVENTION: The authors developed a monthly diabetes lunch conference for all health center staff. Drawing on the resources of the hospital, each conference centered on one diabetes related specialty and featured a guest presenter. Topic areas have included vascular disease, pharmacology, renal, nutrition, etc. The format is case-based with 15 minutes left at the end for prepared comments in the presenter's field. Attendees are asked to complete a feedback form highlighting what they have learned in the session and listing questions for the next month's presenter.

FINDINGS TO DATE: Over 1/2 years, the average attendance has been 17 (3 attendings, 2 NP's; 4 residents, 2 RN's, 2MA's, 1 student). The most often cited key elements of the programs have been: lively discussion, a practical handout, good visual aids, and lunch. Most attendees have been able to specify 1-2 practice process changes they would engage in as a result of attending the conference. At one conference the group developed quality improvement goals for lipid management in diabetics.

KEY LESSONS LEARNED: ISSUES AND LESSONS FOR CONSIDERATION:

- 1) We have brought specialized knowledge to an interdisciplinary health center staff with a minimum investment of any one person's time.
- 2) Time pressures make staff involvement a challenge to sustain.
- 3) Our next step is to measure the long term effects of this intervention.
- 4) Providing lunch and same day reminders pay off!

AN INTERNAL MEDICINE WOMENS HEALTH RESIDENCY: AN INTERDISCIPLINARY MODEL FOR CURRICULAR CHANGE. L.M. Yanni¹, W.S. Klein¹; ¹Virginia Commonwealth University, Richmond, VA

STATEMENT OF PROBLEM OR QUESTION: Although women are the primary consumers of health care, physician training in women's health, particularly in Internal Medicine, has been inadequate. Curricular recommendations have been published. Practical implementation is clearly the next step. We report our experience with implementing an interdisciplinary Internal Medicine Women's Health Residency.

OBJECTIVES OF PROGRAM/INTERVENTION: This program was designed to address gender disparities in physician training through an interdisciplinary curricular model. Women's Health residents are trained in conditions unique to women, more common in women, and those that have manifestations, risk factors, or interventions that are different in women.

DESCRIPTION OF PROGRAM/INTERVENTION: Clinical components include an introductory workshop on the history and physical of the female patient using videotaped interviewing and surrogate patients, continuity clinics in gynecology and mental health, and clinical experience in metabolic bone clinic. A yearly community month at an interdisciplinary women's health care center and a breast health month with clinical exposure to oncology, pathology, radiology, and plastic surgery provide additional women's health training. Didactic components include a journal club on landmark women's health studies, women's health seminars and grand rounds, a primary care/women's health didactic month, and a CME women's health conference. Clinical research and community outreach projects are encouraged. A 36-month reading curriculum provides knowledge of core women's health topics. This consists of review articles from reputable journals, now accessible as a web-based curriculum. Clinical and didactic components are integrated into the Internal Medicine curriculum and all Women's Health residents are board-eligible in Internal Medicine at the completion of the program.

FINDINGS TO DATE: In July 1997, the program was piloted. Subsequently, highly qualified applicants have filled all 4 annually allotted positions. The ratio of applicants to positions continues to increase with a current ratio of 15:1. Of the graduating residents, 3 are pursuing fellowships, 2 are joining community practices, 1 has become an academic faculty member, and 1 has become a chief resident.

KEY LESSONS LEARNED: With both institutional support and interdepartmental collaboration, this program continues its success. It attracts highly motivated and qualified applicants, it is a vehicle of awareness for the medical community through the recognition of Women's Health residents, and it has begun to foster future leaders for the continued advancement of women's health research, writing, and education. This program is a model for integrated and comprehensive women's health training.

TEACHING THE FOUNDATIONS OF POPULATION-BASED MEDICINE. M. Cox¹, J.S. Schwartz¹, J.P. Metlay¹, K. Armstrong¹, J.A. Shea¹; ¹University of Pennsylvania School of Medicine, Philadelphia, PA

STATEMENT OF PROBLEM OR QUESTION: Mastery of the fundamentals of population-based medicine is increasingly required for effective clinical practice, but relatively few medical schools have moved beyond traditional epidemiology courses to include more broad instruction in the clinical evaluative sciences, the organization of the US health care system, and health policy development and analysis.

OBJECTIVES OF PROGRAM/INTERVENTION: A new curriculum (Curriculum 2000) – supported in part by the Undergraduate Medical Education for the 21st Century (UME-21) national demonstration project – provided the opportunity to introduce medical students to the foundations of population-based medicine upon entry to medical school. Our primary goal is to give students this fundamental skill set prior to their entry into the core clinical clerkships. In sequential fashion, students are introduced to: validity, uncertainty, resource allocation, financing and access, and managing care to enhance quality.

DESCRIPTION OF PROGRAM/INTERVENTION: Two new pre-clinical courses have been specifically developed for this purpose. Clinical Evaluative Sciences (CES) utilizes a

combined lecture and small-group, case-based format, and emphasizes problem solving, medical informatics and evidence-based medicine. Health Care Systems (HCS) utilizes lectures and expert panel discussions, and emphasizes the organization and financing of the US health care system. Both courses are taught by an interdisciplinary faculty from the Schools of Medicine and Nursing, The Wharton School, Aetna US Healthcare, and Health Partners, Inc. CES (70 hr) includes: (1) Clinical Epidemiology and Biostatistics; (2) Decision Science; and (3) Clinical Decision Making. HCS (45 hr) includes: (1) Scarcity, Trade-Offs and Values; (2) Health Care Financing and Workforce; and (3) Managing Care.

FINDINGS TO DATE: Students rate the curriculum in terms of content, presentation and teaching materials. Some courses are as highly rated as more traditional biomedical science courses. Unfavorable ratings correlate best with poor course organization. However, perceived "inappropriate" course content also plays a role. Student assessment maps changes in knowledge, values and attitudes, and includes a variety of traditional and more innovative modalities. Overall, students perform as well in these courses as in biomedical science courses. Preliminary data suggests that students attitudes about managing care may also be changing.

KEY LESSONS LEARNED: Interdisciplinary curriculum design, implementation and oversight – and extensive and ongoing input from students – have been critical elements in the success of this broad-based initiative. Additional interventions are planned during the clinical curriculum, where the principles introduced in the preclinical years will be reinforced and expanded. Future studies will assess whether this approach influences how students perform as clinical clerks and residents.

THE COMMUNITY MEDICINE ROTATION: TRAINING IN THE RECONCEPTUALIZATION OF MEDICAL PRACTICE. M.F. Gaden¹, ¹Tufts University, Springfield, Massachusetts

STATEMENT OF PROBLEM OR QUESTION: Physicians are more comfortable with the biomedical model of health and often neglect psychosocial and community issues in clinical encounters.

OBJECTIVES OF PROGRAM/INTERVENTION: The program aims to instruct residents: a) about the interplay between the biomedical and public health models of medical care, b) community resources which can assist in patient care, and the logistics of most effectively and efficiently utilizing them, and c) the role of the physician as a member of an interdisciplinary team which addresses health issues of concern to the local community.

DESCRIPTION OF PROGRAM/INTERVENTION: Residents spend 12–15 half day sessions over one month in the community with community based organizations, public health agencies, alternative care givers, and with a political health care lobbyist or member of a Congressional or Legislative staff. Each resident selects one site to evaluate in depth and compares it with similar initiatives nationally.

FINDINGS TO DATE: Residents were surveyed at the beginning of a second year rotation, and 6–15 months after the completion, about types of community resources available to them, non-clinical ways of interacting with the community, and thoughts about linkages between clinical medicine and public health. In its first year of evaluation, residents each described 2–3 new community resources and non-clinical ways of engaging with the community. Six/eight (6/8) demonstrated an improved understanding of the benefits for patient care of linking medicine and public health.

KEY LESSONS LEARNED: 1) Through direct contact with the life of the community in which one practices, physicians can better understand the role of the physical and psychosocial environment in health. 2) This can lead to an interdisciplinary approach to patient care in which there is a melding of medical and public health philosophies and resources.

THE MACY INITIATIVE IN HEALTH COMMUNICATION. M. Lipkin, Jr.¹, A. Lazare², T.V. Parran³, S. Wentz³, A. Kallet¹, M. Pugnare², K. Cole-kelly³, B.W. Janick¹, E. Ferrara², ¹NYU School of Medicine, New York, NY; ²University of Massachusetts Medical School, Worcester, MA; ³Case Western Reserve University Medical School, Shaker Heights, OH

STATEMENT OF PROBLEM OR QUESTION: Physician communication skills are important and can be taught, but are often left out of the medical school curriculum during the clinical years.

OBJECTIVES OF PROGRAM/INTERVENTION: The Josiah Macy Jr. Foundation, in a desire to improve physician communication skills, funded a three school collaborative curriculum development project, "The Macy Initiative in Health Communication". The goals of the Initiative are to improve the communication skills of physicians through the enhancement of curriculum for medical students and residents and through faculty development.

DESCRIPTION OF PROGRAM/INTERVENTION: The three schools are collaborating on defining a comprehensive list of Communication Competencies and on the Overall Project Evaluation. The main evaluation is a 10 station Communication Skills OSCE (Objective Structured Clinical Exam) conducted both before and after the 3rd year of medical school on both control and intervention groups. The three schools are working separately to develop and implement new curriculum and conduct faculty development.

FINDINGS TO DATE: -A comprehensive list of communication skills has been compiled in our Competency Document.

-All three schools have developed new curriculum for the current 3rd year class based on the Competency Document and taking into account each individual schools culture and needs.

-The evaluation team has completed the pre- and post-test of the control group and is collecting data on the intervention group.

KEY LESSONS LEARNED: -It is possible to collaborate on a large curriculum development project across three medical schools. It is important to define which aspects of the project the schools will collaborate on and which will be done individually.

-The competency document has helped drive specific curriculum at each school and has emphasized the evidence behind our curriculum.

-The teaching of communication skills is best done if it is integrated into clinical content and it provides active "hands on" experience.

-Rigorous, comprehensive evaluation is possible and integral to the success of the project.

TEACHING CROSS-CULTURAL COMMUNICATION. S. Mutha¹, C. Allen¹, M. Welch¹;

¹University of California, San Francisco, San Francisco, CA

STATEMENT OF PROBLEM OR QUESTION: Delivering quality health care to ethnically diverse populations is a growing challenge for health professionals. This curriculum, directed toward multidisciplinary clinicians, demonstrates an effective way of teaching culturally-sensitive communication skills.

OBJECTIVES OF PROGRAM/INTERVENTION: The goal of this curriculum is to improve to clinicians' ability to effectively care for diverse patient populations. At the conclusion of the training participants: 1) demonstrate, through personal examples, an increased awareness that health beliefs and practices are an integral part of the clinical encounter; 2) utilize new communication skills including effective use of interpreters, and 3) cross-cultural team building skills based on new communication models in the workplace.

DESCRIPTION OF PROGRAM/INTERVENTION: In this 2 1/2-day intensive workshop a multidisciplinary group of health care providers, including primary care physicians, explore health beliefs and values through a series of didactic and experiential exercises. Participants explore personal views of health and illness, develop communication skills to elicit culturally accurate information from patients, practice appropriate use of interpreter services and learn new communication techniques for cross-cultural communication and team building in the workplace. A critical component of the training is creating a safe and confidential setting for discussing personal beliefs and values about an oftentimes sensitive subject.

FINDINGS TO DATE: We have implemented the curriculum as workshop or in modules for approximately 60 diverse groups of clinicians including physicians, nurses and residents. Evaluations of the trainings have revealed high levels of agreement that the content was highly relevant to clinical practice and high satisfaction with the training sessions. A six-month follow-up evaluation is underway to assess the ways in which attendees have incorporated the contents of the training into their practices.

KEY LESSONS LEARNED: The curriculum content and design allow successful interaction among multidisciplinary health professionals. Participant evaluations underscore that simulations and experiential exercises are an especially powerful way to increase sensitivity to and awareness of diversity in the clinical setting. The sensitive nature of issues surrounding diversity such as prejudice as well as discrepancies in power and authority emphasizes the need to develop a cadre of clinician/educators who are committed to providing this type of training in clinical settings.

A STRUCTURED FELLOWSHIP FOR MEDICAL STUDENTS IN MEDICAL ORGANIZATIONAL AND ADMINISTRATIVE MANAGEMENT. J. Terwilliger¹, F. Chapman²,

B. Chernof³, E. Little², R. Miller⁴, J. Rosenthal³, G. Levey³; ¹University of California, Los Angeles, School of Medicine, Los Angeles, CA; ²UCLA Medical Center, Los Angeles, CA; ³UCLA School of Medicine, Los Angeles, CA; ⁴CIGNA Healthcare, Glendale, CA

STATEMENT OF PROBLEM OR QUESTION: Medical students have little or no exposure or instruction in the organizational and administrative management of health care organizations during their undergraduate years. They enter residency and practice with little understanding of how the larger health care system operates, and how their roles are influenced by a significant number of non-medical factors.

OBJECTIVES OF PROGRAM/INTERVENTION: 1) Through mentorship by senior administrative leaders, expose medical students to basic management activities of major health care organizations; 2) familiarize them with the language and objectives of management and administration; 3) allow for a broader perspective on their clinical activities that includes organizational and financial requirements; 4) create the basis for successful matriculation through a second degree (MBA or MPH) simultaneous with their MD.

DESCRIPTION OF PROGRAM/INTERVENTION: An eight-week program for 4-5 medical students between the M1 and M2 years, with each student linked directly to a senior administrator of a major academic medical center or private managed care organization. Each fellow is assigned a project, which is summarized into a presentation to all program faculty. Weekly journal club (focusing on health policy, administration, and financing) is also required. **FINDINGS TO DATE:** Evaluations by students of experience in the program have yielded uniformly positive results when asked to rate the quality of the program, as well as its importance to their careers as physicians. 6 of the 16 students participating in the program have applied for admission to UCLA's combined MD/MBA program upon completion of the fellowship. Students have gained increased understanding of the administrative aspects of health care organizations.

KEY LESSONS LEARNED: 1) Dedicated and experienced mentors are critical. 2) Students must tailor work projects to normal operations of organization; 3) Feedback and discussion with students during program is a critical success factor; 4) Most medical students could benefit from some type of exposure to the administrative

INNOVATIONS IN PRACTICE MANAGEMENT

COMPENSATING ACADEMIC EXCELLENCE AND PRODUCTIVITY IN A MULTISPECIALTY FACULTY PRACTICE GROUP. A.H. Balder¹; Health Center, Springfield, MA

STATEMENT OF PROBLEM OR QUESTION: Current fiscal reality has resulted in increased demands for clinical/ economic productivity on the part of all physicians. This is perhaps most acutely felt in academic practice groups that devote part of their activity to non-directly reimbursed teaching activity. With pressures to see more patients while teaching students and residents, time and excellence in teaching and research may suffer.

OBJECTIVES OF PROGRAM/INTERVENTION: After developing a variable (incentive) compensation program based on clinical productivity, the leadership of the Baystate Medical Education and Research Foundation (BMERF), at the strong urging of its membership, endeavored to develop a financial incentive program to reward and encourage excellence and productivity in research and teaching. As a multispecialty group, it was important to the group to treat all specialties fairly and equitably.

DESCRIPTION OF PROGRAM/INTERVENTION: The chairs of each academic department (Emergency Medicine, Medicine, OB/Gyn, Pathology, Pediatrics, Psychiatry and Surgery) nominated one member from each department to a Committee on Academic Excellence. Committee members identified a set of achievement criteria and then weighted them to form a quantitative measurement scale.

In the first year, major weighted criteria included but were not limited to: article publication; editor or author of text or journal; receipt of large research grants; achievement awards from national organizations. Junior faculty received additional consideration for more modest grant receipt; advancement to associate professor; and chairmanship of major committees in medical societies or at the medical school. Criteria were further refined the second year based on BMERF member feedback.

FINDINGS TO DATE: In the first year of 197 eligible members, 47 (24%) submitted material for consideration for compensation. 31 awards were given (66%), ranging upwards from \$1,000, few going to junior faculty (defined by the committee as 7 or fewer years out of residency). The second year there were 77 submissions (39%) and 41 awards (53%), with a minimum award of \$625. Three General Medicine Division members received awards, from a division membership of 24.

KEY LESSONS LEARNED: LESSONS AND ISSUES FOR CONSIDERATION:

- 1) The concept and execution of the program were well received.
- 2) There will be pressure to allocate more money in future years.
- 3) We need to further refine criteria for junior faculty and generalists.
- 4) It can be difficult to identify and reward excellence in teaching.

A SUCCESSFUL FACULTY CLINICAL INCENTIVE PROGRAM IN A HOSPITAL-BASED, PRIMARY CARE TEACHING PRACTICE. J.J. Heffernan¹, S. O'Neill¹, K. Brown¹, R.A. Parker¹, M. Aronson¹, R. Phillips¹, T. Delbanco¹; ¹Beth Israel Deaconess Medical Center, Boston, MA

STATEMENT OF PROBLEM OR QUESTION: Financial incentives may increase productivity and improve utilization of resources by physicians. In mixed populations of fee-for-service and capitated patients, incentive formulas grow more complex. Encouraging desirable clinical behavior among faculty physicians in teaching and research environments is an even greater challenge, given the broader mission and greater range of interests of such physicians.

OBJECTIVES OF PROGRAM/INTERVENTION: Healthcare Associates (HCA) is the hospital-based, primary care teaching practice of the Beth Israel Deaconess Medical Center. HCA serves 32,000 patients, 40% capitated, with 91,000 annual visits and is the primary ambulatory teaching site for 130 medical housestaff. We reward our clinicians with added compensation for high patient visit volume, panel growth, severity of illness, commitment to teaching (comanagement) and, most recently, panel management.

DESCRIPTION OF PROGRAM/INTERVENTION: Funded annually by the hospital, the incentive pool derives from an estimated payout per faculty FTE for practice-wide projected increases in visit volume and panel size. The core clinical payout is based on a point system. Promoting both visit volume and panel growth, new patient encounters earn 3 points, while all others earn 1 point. Incentive is paid for all points above a threshold of 8 patients/session, averaged over the year. To promote teaching, we incentivize comanagement sessions more heavily than individual practice sessions. As a proxy for severity of illness, we pay a third component based on the inpatient activity of a faculty member's panel. We are now adopting a more diversified model, with 25–35% of total incentive dollars linked to panel size, age and gender adjusted, to reflect the realities of clinical practice more accurately. We are also moving to include provider specific quality measures in the incentive system, and to extend incentive payments to non-physician and non-clinical staff.

FINDINGS TO DATE: Faculty visit volume productivity increased 30% in 3 years and overall panel has grown at a rate of over 400 new patients/month. In academic year 99–00, the faculty incentive paid a total of \$310,000 for an average payout of \$8,000 per faculty member, with a range of \$0–\$19,927, and we have been able to fill our comanagement teaching slots easily.

KEY LESSONS LEARNED: Financial incentives for clinical activity can succeed in a primary care teaching environment. Realistic targets and simple metrics are vital, matching payout to desirable behaviors.

REDUCING PHARMACY COST IN AN INDIGENT PATIENTS CLINIC. R. Cheung¹, M.J. Ault¹; ¹Cedars-Sinai Medical Center, Los Angeles, CA

STATEMENT OF PROBLEM OR QUESTION: Limited funding and resources are major obstacles in clinics that take care of the indigent populations.

OBJECTIVES OF PROGRAM/INTERVENTION: The goal is to decrease the pharmacy cost in the care of the indigent patients.

DESCRIPTION OF PROGRAM/INTERVENTION: One of our satellite clinics in our Internal Medicine Residency Training Program serves the indigent population. The patient income is equal to or less than 133 1/3% of the Federal poverty level. Currently, the clinic has about 650 continuity care patients. The patients receive free care from the clinic. Our on-site clinic dispensary usually provides medications to patients via a restricted formulary, medications samples and donations. Patients are seen by Resident Physicians. Faculty Physicians act as preceptors for the Residents. Using the existing Indigent Patient Drug Programs set up by various Pharmaceutical companies, medications are ordered by application forms with prescriptions for patients who are proved to be indigent and in need of the medications. In general, each Pharmaceutical company provides a three month supply of medications with each approved application. Some medications ordered include expensive medications such as HMG-CoA Reductase Inhibitors and Proton Pump Inhibitors, which are otherwise difficult to afford

by the clinic. A database was set up with a computer spreadsheet program to record information of each order. The database provides important data for patient care and functions as a monitoring device in the medications re-ordering process.

FINDINGS TO DATE: By using the above medications ordering program, the clinic saves about \$95,000 in the last three quarters in calendar year 2000.

KEY LESSONS LEARNED: By accessing individual Pharmaceutical companies' Indigent Patient Drug Programs, together with the existing restricted on-site Pharmacy formulary, along with use of medications samples and donations, we can demonstrate further decrease in Pharmacy costs in treating our indigent patients, while maintaining highest quality of care and teaching our Resident Physicians cost containment strategies.

THE FORUM: A CME APPROACH TO DISEASE MANAGEMENT. L. Armas¹, D.S. Balis¹, A. Ramirez¹, J. Munger¹; ¹Parkland Health and Hospital System, Dallas, TX

STATEMENT OF PROBLEM OR QUESTION: Problem: Parkland Health & Hospital System is a multi-site county health care system, with significant variability in clinical background and practice styles in primary care providers. System wide disease management initiatives and clinical practice guidelines have been created in an effort to standardize care but have been difficult to disseminate and implement. Traditional CME is individually chosen to accommodate the provider's preferences regardless of the system's needs. Furthermore, there is evidence that this approach rarely modifies physician behavior resulting in a delay of standard of care practices implementation. A program that combines disease management, continuing medical education, and peer review was designed to foster the creation, dissemination and evaluation of clinical practice guideline implementation for primary care issues, and improve system-wide provider communication.

OBJECTIVES OF PROGRAM/INTERVENTION: Objectives: To provide a forum for Adult Providers to exchange clinical interests and expertise.

To provide a forum to develop, implement and monitor quality improvement initiatives.

To develop a curriculum of clinical material relevant to primary care.

To promote collaboration among adult providers across the healthcare system.

To promote development of one standard of care throughout the system, that is both evidence based and cost effective.

DESCRIPTION OF PROGRAM/INTERVENTION: Description: An initial curriculum was developed based upon a survey of provider's interests and system needs identified by the quality management department. Prior to meetings planning committee reviews or creates disease specific standard indicators for appropriate measurement of guideline implementation. Monthly meetings consist of a lecture given by a recognized expert and a discussion of the perceived barriers to achieving that care in our institution. The indicators are then presented followed by a review of charts of identified patients with the discussed medical illness. Results of previous peer reviews are presented with a discussion of what could be done to improve them. Individualized provider specific feedback is distributed confidentially.

FINDINGS TO DATE: Results: Re-measurement of disease specific indicators have shown significant increase in provider compliance with established standards of care: a 24% increase in the use of ACE-Inhibitors in CHF, a 45% of increase in screening for microalbuminuria in diabetics. Yet, some of the newer disease management knowledge, such as use of beta-blocker in patients with CHF showed no significant change.

KEY LESSONS LEARNED: Key Lessons: A program that combines disease management and a continuing medical education involving self-evaluation through peer-reviews more effective to disseminate and implement system based clinical practice guidelines, also improves providers communication and involvement in those initiatives.

Problems Found: Chart gathering from multiple sites, provider turnover, and development of appropriate indicators

A COMMUNITY HOSPITAL-BASED HEART FAILURE PROGRAM DECREASES READMISSION RATES. H. Lee¹, D. Gamiss², R. Oliver³, C. Mccullogh⁴, D. Dulisse³; ¹Group Health Centre and McMaster University, Sault Ste Marie, Ontario; ²University of Ottawa, Ottawa, ON; ³Sault Area Hospitals, Sault Ste Marie, ON; ⁴Group Health Centre, Sault Ste Marie, Ontario

STATEMENT OF PROBLEM OR QUESTION: Congestive Heart Failure (CHF) is the leading cause of admission and re-admission in the Sault Area Hospital (SAH), a community referral centre serving a population of 125,000 in the Sault Ste Marie region. There is some evidence from university, tertiary care centres that a multi-disciplinary discharge transition program targetted at inpatients with CHF might decrease readmission and mortality. The Group Health Centre (GHC) is a not-for-profit, community-based health service organization serving 44,000 registered primary care patients. Could a collaborative CHF discharge transition program be effective in a community setting?

OBJECTIVES OF PROGRAM/INTERVENTION: (1) To develop a multi-disciplinary CHF education and follow-up program for SAH inpatients; (2) To pilot this program first to GHC patients, and then to all patients if proven effective.

DESCRIPTION OF PROGRAM/INTERVENTION: On admission, CHF inpatients are identified and a multi-disciplinary team consisting of a hospital pharmacist, dietician and discharge planner visit and educate the patient and family. Standardized education materials are provided. The GHC home-visiting nurse service, called Family Health Workers(FHW) also visits the CHF Project patient while in hospital. At discharge, the hospital pharmacy provides an updated medication list to the patient, the FHW, the patient's pharmacy and the GHC electronic medical record. The same FHW visits the patient within 48 hours of discharge, and arranges for further follow-up as needed.

FINDINGS TO DATE: [1] Over the first six months, the CHF Project has been well received by all. A crucial and well-appreciated role of the FHW is to sort out discharge medications at the initial home visit. [2] There have been a total of 174 CHF patient admissions, of which 57 belong to the GHC and were exposed to the intervention. [3] There has been a 68%, [95% confidence interval (CI) 9%–96%] decrease in readmission rates compared to historical controls and a non-significant 57% [95%CI -5 to 89%] decrease compared to concurrent

non-GHC controls. [4] There has also been a non-significant trend towards decreased mortality.

KEY LESSONS LEARNED: A community-based collaborative to help CHF patients in the crucial discharge transition period can be effective in a non-tertiary care setting. Further efforts are needed to expand access to the program for all patients, to study and improve upon separate components of the program and to sustain the improvements already achieved.

A PHARMACIST-BASED DIABETES MANAGEMENT MODEL CAN IMPROVE CARE FOR HIGH RISK DIABETIC PATIENTS. R. Rothman¹, M.P. Pignone¹, R. Malone¹, B. Bryant¹, C. Horlen¹; ¹University of North Carolina at Chapel Hill, Chapel Hill, NC

STATEMENT OF PROBLEM OR QUESTION: Conventional management has often failed to improve clinical outcomes in high risk diabetic patients with barriers to care.

OBJECTIVES OF PROGRAM/INTERVENTION: To improve self management skills and blood glucose control for underserved Type 2 diabetic patients.

DESCRIPTION OF PROGRAM/INTERVENTION: We developed a pharmacist led diabetes care program in an academic general medicine practice serving a predominantly indigent population. Clinic-based Pharmacists offered support for diabetes management including diabetes education, close phone follow-up, and the use of algorithms for titrating hypoglycemic medications. No specific interventions for management of blood pressure or lipids were used. To assess the impact of the program, we measured diabetes knowledge, blood pressure, lipids, and HgbA1c at enrollment and at 3–4 months follow-up.

FINDINGS TO DATE: Complete data are available for 129 of 159 (81%) high risk diabetic patients (HgbA1c > 8.0%) who were enrolled from September 1999–May 2000. Mean age was 57 (range 23–87); 69% of patients were African American; 60% were Female; 58% had less than a high school education; over 50% relied on Medicaid or pharmacy assistance for medications. Average duration of disease was 10.4 years (range 0–42). Although 81% of patient had hypercholesterolemia and 88% had hypertension, only 30% were on a lipid lowering agent and only 60% were on or had ever taken an ACE-inhibitor. Baseline HgbA1c at enrollment was 10.8% and did not differ significantly from values obtained 6–12 months prior to enrollment (mean 10.2%). At three to four months follow-up, the mean reduction in HgbA1c was 1.7% points ($p < .0001$). Baseline diabetes knowledge score on our 11 question diabetes knowledge test (DKT) averaged 36%. Of 69 patients who have retaken the DKT, there was a 23% point improvement from baseline to follow-up ($p < .0001$). Blood pressure and total cholesterol did not change importantly from baseline to follow-up. Multiple regression analysis shows that higher educational status, when adjusted for baseline HgbA1c, is associated with a significant improvement in HgbA1c. Other socioeconomic factors, such as race and gender were not significant predictors.

KEY LESSONS LEARNED: A pharmacist-assisted care program can improve patient's knowledge of diabetes and significantly reduce HgbA1c. We did not see improvement in blood pressure or lipid control - two areas that we did not specifically intervene upon. To further improve care and reduce the risk of macrovascular complications, we plan to test pharmacist-assisted, algorithm-based, hypertension and cholesterol management in a randomized controlled trial.

NON-VISIT CARE OF TARGETED POPULATIONS WAS USED TO IMPROVE STAFF DOCUMENTATION OF METERED DOSE INHALER PATIENT INSTRUCTION. R.E. White¹, D. Gray¹, B.J. Skipper²; ¹Albuquerque VAMC, Albuquerque, NM; ²University of New Mexico, Albuquerque, NM

STATEMENT OF PROBLEM OR QUESTION: Nursing staff in VA primary care clinics are required to document numerous patient education and screening activities, and they are using progressively more advanced software in the electronic patient record to accomplish this documentation. However, time limitations make it difficult to accomplish all these educational and screening mandates while the patient is in clinic, and staff may miss documenting activities which were previously accomplished but not quickly found in the record.

OBJECTIVES OF PROGRAM/INTERVENTION: We experimented with the process by which nurses documented that patients were taught the proper use of metered dose inhalers (MDIs). We chose to update our electronic documentation of this activity (1) at a time other than the patient visit and (2) by using a population based approach.

DESCRIPTION OF PROGRAM/INTERVENTION: At our facility, two teams of similar provider and support staff composition provide primary care in non-teaching clinics. Of the 6129 patients in group practice A, 640 use MDIs, and 572 of 5596 group practice B patients use MDIs. Information about this population of MDI users was compiled, and their names and telephone numbers were listed in the chronological order of their next appointment. Starting in May 2000, clerks began calling each patient as their next appointment approached, in order to remind patients to obtain MDI instruction when they attended clinic. In group practice B, nurses also used the list during lulls in routine clinic screening activities to contact patients, ask them about prior MDI education, and then update the electronic record accordingly. Five months later, electronic documentation of MDI patient education was extracted from the computer patient database. Documentation data was also obtained about one education/screening activity not targeted in this study (depression screening) because this activity applies to all group practice patients and are electronically documented in the same manner as MDI teaching.

FINDINGS TO DATE: In group practice B, MDI inhaler education and depression screening were documented on 84% and 60% of the 572 MDI patients. In group practice A, the same percentages were 52% and 65% of their 640 MDI patients. Chi square comparisons were made of patients receiving only MDI instruction or MDI teaching and depression screening. In each comparison, group practice B performance was significantly ($p < .05$) higher. Logistic regression also showed that group practice B was associated with a higher likelihood of MDI teaching documentation (odds ratio 3.54, $p < .001$).

KEY LESSONS LEARNED: As the number of routine patient education guidelines expands, there is not sufficient time during each clinic visit to complete and document them all.

Documentation may be better accomplished when a population of patients is targeted for review at times when they are not actually in clinic.

IMPROVING QUALITY OF CARE AT A HOMELESS CLINIC. T. Bu¹, M. Herbert¹, J. Baker¹, S. Connor¹, G. Lamsam¹, R. Cook¹; ¹The Program For Health Care To Underserved Populations, Division Of General Internal Medicine, And The School Of Pharmacy, University Of Pittsburgh, PA

STATEMENT OF PROBLEM OR QUESTION: It is a challenge to provide quality health care and to measure outcomes relevant to the homeless population in an outpatient setting that is entirely dependent on students, volunteer physicians and pharmacists.

OBJECTIVES OF PROGRAM/INTERVENTION: To assess the quality of care for chronic conditions as well as episodic acute care to a vulnerable patient population; To provide appropriate medications at low cost to the clinic; To integrate patients into the mainstream health care system through effective case management.

DESCRIPTION OF PROGRAM/INTERVENTION: We focused our efforts at a single, university-supported homeless clinic with approximately 1000 patient visits each year. Health care is provided by 18 volunteer physicians, 6 volunteer pharmacists, and over 160 health professions students. A full-time clinic coordinator oversees the daily operation together with 2 AmeriCorps volunteers who serve as case managers, a part-time lay health worker, and four medical students who coordinate service-learning activities. We developed an Access database to monitor clinical services, pharmaceutical utilization, and case management with a focus on preventive care, referral data, Medicaid enrollment, and pharmaceutical costs and utilization. Students in pharmacy, medicine, and nursing conducted quality assurance projects, including a patient satisfaction survey and chart reviews to assess hypertension management and antibiotic usage.

FINDINGS TO DATE: On-site dispensing was promoted over vouchers at a cost saving of \$21.34 per prescription with a total saving of about \$20,000 a year. Blood pressure control was achieved in 46% of hypertensive patients, but almost half failed to return for follow-up. Narrow-spectrum antibiotics were selected as first-line therapy 66% of the time, but over-prescribing for URI was also noted. In a predominantly uninsured (73%) clinic population, 27% of all patient encounters resulted in referrals with 30% of all referrals for dental care, 17% to primary care providers and medical specialists, and only 1% to behavioral health. Sub-optimal documentation was noted in chart reviews.

KEY LESSONS LEARNED: Based on our findings, we implemented an appointment reminder system, progress note templates, and a mechanism for tracking referrals. Providing routine feedback to volunteer providers on findings such as mental health referrals may promote more aggressive screening and counseling. In conclusion, quality assurance projects incorporated into volunteer clinic activities can identify unmet health needs and areas requiring improvement.

ESTABLISHING A PATIENT MEDICATION ASSISTANCE PROGRAM. M. Anderson¹, J. Koo¹, A. Ekpenyong¹, C. Dimou¹; ¹Rush Presbyterian St. Luke's Medical Center, Chicago, IL

STATEMENT OF PROBLEM OR QUESTION: Many patients are unable to pay for medications prescribed by physicians due to a lack of income and a lack of prescription medication coverage. How can physicians help patients obtain the medications that patients need?

OBJECTIVES OF PROGRAM/INTERVENTION: This program was initiated to utilize medication assistance programs offered by pharmaceutical companies to obtain medications at little or no cost to patients. In utilizing the medication assistance programs we wanted to aid our uninsured, underinsured, and indigent patients to obtain needed medications. We hoped the impact of free or inexpensive medications would increase compliance and decrease hospitalization, as well as improve our patients' quality of life.

DESCRIPTION OF PROGRAM/INTERVENTION: The Patient Medication Assistance Program utilizes established pharmaceutical company assistance programs. Patients are identified by their physician and an interpractice enrollment form is generated by the patient's physician. The coordinators then determine if there is a pharmaceutical program available and if the patient meets the criteria for the program. All applications are tracked by patient, time of submission, and time of receipt of medication. Once the medication has been received and logged in, patients are notified by the office staff. Medications are dispensed by nurses with written instructions.

FINDINGS TO DATE: The program was initiated in 1997, however patient enrollment was small until late 1999 and early 2000. Since early 2000 data has been compiled which shows benefit of \$3000–\$9000 per month in medication distributed to patients. Anecdotal data to date suggests decreased hospitalizations.

KEY LESSONS LEARNED: Preliminary data clearly shows a savings of \$3000–\$9000 per month in medication distributed to our population of enrolled patients. Anecdotal data may support an increase in compliance and a decrease in hospitalizations.

SAN FRANCISCO OFFICE-BASED OPIATE ADDICTION TREATMENT (OBOAT) PROGRAM. H. Palacio¹, B. Garcia¹, E. Bein¹, M.H. Katz¹; ¹San Francisco Department of Public Health, San Francisco, CA

STATEMENT OF PROBLEM OR QUESTION: There are an estimated 15,000 active heroin users in San Francisco. Heroin-related death rates in San Francisco is 3 times higher than the California State average. Substance abuse is the leading cause of death among San Francisco's homeless population, and heroin contributed to 87% of those drug-related deaths. Access to methadone maintenance (a highly efficacious treatment) is severely limited by restrictive federal and state regulations.

OBJECTIVES OF PROGRAM/INTERVENTION: Our objectives are: 1) to expand access to methadone and other effective medications for opiate addiction, 2) to support, enhance, and

work collaboratively with existing narcotic treatment programs, 3) to ensure quality opiate addiction treatment services.

DESCRIPTION OF PROGRAM/INTERVENTION: Phase I: Community-based planning process out of which a broad conceptual framework for the OBOAT Program was developed. Phase II: The OBOAT Feasibility Study, funded by the Substance Abuse and Mental Health Services Administration is the second and current stage of program development. Focus groups and advisory work groups with pharmacists, physicians and nurse practitioners, and counselors are being held to identify potential barriers to participation. Actuarial consultants are developing cost estimates and financial models for the proposed program. Project leaders work with state and federal regulatory and legislative officials to facilitate program planning and implementation. Phase III: Implementation of a pilot project is targeted for the summer of 2001.

FINDINGS TO DATE: The phase I community planning created program framework with the San Francisco Department of Public Health as holder of a central Narcotics Treatment Program (NTP) licensure for the OBOAT Program, in which interested and eligible public and private-sector physicians, counselors, and pharmacists would be trained to offer methadone and other approved medicines to treat opiate addiction in the setting of their regular clinical practices. Pharmacists view the opportunity for professional development as an incentive for potential program participation. They view lack of lack of methadone education in pharmacy school and inexperience with methadone dispensing to addicts as barriers to potential program participation. The planning process has challenged physicians in the narcotic treatment arena and those in the primary care arena to learn about each other's "culture" and communicate effectively to develop an innovative program that will serve patients in new way.

KEY LESSONS LEARNED: The planning process has involved reframing opiate addiction treatment as a medical model, rather than a regulatory model. Close collaboration with existing methadone clinics has been critical in moving program planning forward. Close work with state legislature has resulted in recent passage of enabling legislation as well as allocation of additional resources.

EFFECT OF COMPUTER GENERATED APPOINTMENT REMINDERS ON COMPLIANCE WITH CLINIC APPOINTMENTS. J. Witter¹, P.S. Mehler², D. McEwen³, A. Steele⁴; ¹Denver Health and Hospital Authority, Denver, CO; ²University of Colorado Health Sciences Center, Denver, CO; ³Denver Health & Hospital Authority, Denver, CO; ⁴University of Colorado Health Science Center, Denver, CO

STATEMENT OF PROBLEM OR QUESTION: Missed appointments are an important obstacle to the provision of effective health care. Computer generated appointment reminder systems may be a cost effective means of improving the kept appointment rate.

OBJECTIVES OF PROGRAM/INTERVENTION: To evaluate the effect of a computer generated appointment reminder system on the kept appointment rate in primary care clinics.

DESCRIPTION OF PROGRAM/INTERVENTION: Computer Telephony Integration (CTI) is a computer-based application that automatically calls patients to remind them of their appointments. This program is interfaced with the scheduling system to automatically download patient and appointment information. It uses an interactive voice response technology to call patients 24–72 hours before their appointment. If there is no answer, the patient will be called back up to 5 times. All patients with appointments scheduled at least 72 hours ahead of time at several Community Health Clinics over a period of 6 weeks were eligible for our study. 5,696 patients had appointments during this time and thus could have been called by the CTI system. During randomized alternating one week periods, patients at each of the participating clinics were assigned to receive automated appointment reminders. Patients not receiving calls were used as the control group. Kept and missed appointment rates were recorded for each of these groups.

FINDINGS TO DATE: 5,696 patients (3,038 in the appointment reminder group, 2,617 of whom consented to being called by the system; 2,658 in the control group) were prospectively followed over a 6 week period. The kept appointment rate in the appointment reminder group was 69.2%. The kept appointment rate in the control group was 64.1%. This result translated into an 8% increase in kept appointments when the automated system was used ($p < 0.01$). The increase occurred despite the fact that only 1,192 of 2,617 eligible patients were actually contacted.

KEY LESSONS LEARNED: A computerized telephone reminder system increased the kept appointment rate in our Community Health Clinics. This increase may have a considerable economic impact. Assuming an increase in the kept appointment rate of 8% translates into 8% more patient visits, this could represent 9,000 more visits a year to our clinics. If the average charge for an outpatient visit is \$177.35, 9,000 more visits represents \$1.5 million in increased charges. Implementation of such a system could have a large positive economic and health impact in a public health system such as ours.

IMPROVED GLYCEMIC CONTROL IN DIABETIC PATIENTS UNDERGOING CORONARY ARTERY BYPASS SURGERY. N. Ashri¹, R. Lippe¹, N. Rao¹; ¹UPMC Shadyside, Pittsburgh, PA

STATEMENT OF PROBLEM OR QUESTION: Management of blood sugar in the metabolically stressful postoperative period in a Diabetic patient undergoing major surgery such as coronary artery bypass (CABG) is often a challenging task. However, adequate glycemic control is necessary to prevent postoperative infections such as mediastinitis and delayed wound healing. Conventionally, the management involves insulin sliding scale and despite multiple calls to physicians and inconvenience to nursing staff, the glycemic control is not adequate.

OBJECTIVES OF PROGRAM/INTERVENTION: 1) To develop a simple, effective insulin infusion protocol, which will maintain Diabetic CABG patients capillary blood sugars (CBS) below 200 mg/dl in the first 48 hours postoperatively. 2) To decrease the rate of mediastinitis in postoperative period.

DESCRIPTION OF PROGRAM/INTERVENTION: The insulin infusion protocol is followed for first 48 hours in the postoperative period on all the Diabetic CABG Patients. The patients concurrently receive D5.45 NS with 20 meq KCL at 75 cc/hr. The infusion rates are determined based on hourly CBS :

CBS	Insulin rate
100 – 125	1 U/hr
126 – 150	2 U/hr
151 – 200	3 U/hr
201 – 250	6 U/hr
251 – 300	9 U/hr
351 – 400	11 U/hr
401 – 450	15 U/hr

CBS is checked every 3 hours if changes in infusion rates are less than 1 unit/hr for more than 2 hours.

FINDINGS TO DATE: To date more than 100 patients have been placed on this protocol and data has been collected from 77 patients. This data has been compared to 22 patients who were managed by insulin sliding scale and subcutaneous insulin. Patients on insulin infusion protocol had a lower mean capillary blood sugar of 155 mg% from 1765 values returned compared to 200 mg% from 261 values returned in non protocol patients. Only 11.2% of capillary blood sugars values have exceeded 200 mg% on protocol patients compared to 44.1% values in case of non protocol patients. No serious hypoglycemic episode has been reported so far. In addition, the protocol has been well received by the nursing staff and preliminary data has shown decline in the rate of mediastinitis.

KEY LESSONS LEARNED: Simplification and team approach lead to adequate glycemic control by continuous insulin infusion protocol in postoperative period in CABG patients.

CARING FOR THE UNDERSERVED: RESTRUCTURING A RESIDENT CLINIC TO IMPROVE CARE, SAVE MONEY AND TRAIN RESIDENTS BETTER. M.M. Nadkarni¹, J.B. Schorling¹; ¹University of Virginia, Charlottesville, VA

STATEMENT OF PROBLEM OR QUESTION: Medically underserved, chronically ill patient populations frequently receive care at internal medicine resident clinics. These clinics often have poor continuity of care, poor access, and "unmanaged" care leading to high per patient per month (ppm) costs and poor patient satisfaction. Residents may also feel they receive suboptimal education in this environment.

OBJECTIVES OF PROGRAM/INTERVENTION: 1) Integrate resident and faculty practices to improve cost-effectiveness, patient satisfaction and quality of care for an underserved chronically ill population at the University of Virginia. 2) Apply managed care principles and interventions to provide population based care. 3) Improve resident education in a continuity clinic setting.

DESCRIPTION OF PROGRAM/INTERVENTION: University Medical Associates is a resident continuity clinic providing care to a patient population of 9000 primarily indigent patients. Beginning in July 1998, we restructured the practice. Eight GIM faculty were matched with teams of 5 residents in an integrated faculty/resident practice which took responsibility for total care and medical utilization of this patient population. Methods of practice change included increased phone access and physician availability for patients, case management of high utilizing patients with a focus on collaborative care, close monitoring of emergency room use and hospitalized patients by primary physicians, practice profiling, incorporation of clinical practice guidelines and specific resident training on costs, coding and appropriateness of care. A resident walk-in clinic was eliminated so physicians had the opportunity to see their own patients for acute illnesses.

FINDINGS TO DATE: Compared to baseline Fiscal Year 1999, Fiscal Year 2000 data showed striking improvements in multiple outcomes: 1) Cost effective care: total overall PPM charges decreased from \$207.28 to \$164.35 (21% improvement) for global outpatient charge categories: (ambulatory visits, home health, E.D, lab, pharmacy, procedures, radiology, other); 2) E.D. utilization: reduced 33% from 1.11 to .73 visits ppm; 3) Continuity index: improved from 58.9% to 76.8%; 4) Phone access: proportion of calls answered increased from 59% to 89%; 4) Patient overall satisfaction: improved 4.2 to 4.8 on 5 point scale; 5) Resident satisfaction with clinic: improved.

KEY LESSONS LEARNED: 1) Innovative integration of separate resident and faculty practices into a team-based and population-focused practice can provide big dividends in cost effectiveness of care, quality of care, and improved resident education. 2) Introducing population based / managed care interventions in an uncaptured, underserved population can be achieved in a residency training setting. 3) Improving access and continuity of care are the most effective interventions. 4) Success requires substantial commitment to change from faculty, residents, residency director, departmental leadership and all hospital/ institutional parties involved.

A PHYSICIANS GUIDE TO TALKING ABOUT END-OF-LIFE CARE: A FOUR-STEP APPROACH. R.B. Balaban¹, G. Coste¹; ¹Cambridge Hospital / Harvard University, Cambridge, MA

STATEMENT OF PROBLEM OR QUESTION: Physicians are trained to maintain health and fight illness, but typically receive little guidance on how to communicate with dying patients and their families. Many physicians find it difficult to engage in end-of-life discussions. Poor communication can lead to suboptimal care for patients and their families.

OBJECTIVES OF PROGRAM/INTERVENTION: This workshop provides participants with a practical approach for addressing end-of-life issues with patients and their families. We discuss initially what dying patients and their families want and need from their doctor. Then we present a sensible and useful approach that enables physicians to be more effective and at ease in conducting these difficult discussions.

DESCRIPTION OF PROGRAM/INTERVENTION: A practical 4-step approach to conducting end-of-life discussions is presented. The four steps are based on structured interviews at a major university hospital with five faculty clinicians experienced in the care of dying patients. Their experiences and actual words have been synthesized into this 4-step approach. The four steps are:

- 1) Initiating Discussions
- 2) Clarifying Prognosis
- 3) Identifying End-of-Life Goals
- 4) Developing a Treatment Plan

FINDINGS TO DATE: Using the 4-step approach with your patients will result in improved communication. Good communication results in patient fears being allayed, pain and suffering minimized, and facilitates developing a treatment plan that is medically sound and concordant with the patient's wishes and values.

KEY LESSONS LEARNED: Providing good end-of-life care requires both an understanding of how patients and families experience the dying process, and a sensitive communication style. With these skills, physicians are able to conduct thoughtful discussions in which most decisions evolve comfortably and without controversy.

Providing care for a dying patient is challenging, and when done well, is a meaningful and gratifying experience for the physician. To help someone die in comfort, in peace, and with dignity is to give one final gift of life.

RESIDENTS' OPINION REGARDING THE CLEVELAND CLINIC FOUNDATION GUIDELINES FOR ANTIMICROBIAL USAGE BOOK. S. Mossad¹, J. Long¹, M. Goldman¹, T. Keys¹;

¹Cleveland Clinic Foundation, Cleveland, Ohio

STATEMENT OF PROBLEM OR QUESTION: Guidelines for antimicrobial usage have been published and distributed to all house staff (HS) at the Cleveland Clinic Foundation (CCF) since 1991. These guidelines are intended to help in decisions regarding antimicrobial therapy in hospitalized patients.

OBJECTIVES OF PROGRAM/INTERVENTION: Formal assessment of the effects of these guidelines on the practice of HS has not been previously done.

DESCRIPTION OF PROGRAM/INTERVENTION: A questionnaire was sent to the mailboxes of all house staff in the Division of Medicine and an audience response system was also used to administer the questionnaire during a noon conference.

FINDINGS TO DATE: Thirty five HS participated in this survey; (48% GL-I, 24% GL-II, 14% GL-III, 7% GL-V). Most of them thought the book was trustworthy (87%), accurate (86%), user friendly (57%), and overall useful for the management of patients (87%). Although most HS always carried the CCF book (73%), most also carried other commercially available antibiotics recommendations book(s) (68%). There was no consensus which of these books was more useful for them. The CCF Book sections on guidelines for specific diseases and dosing information were referred to most often (37% & 40%; respectively). Requests were made to add more microbiology information and other clinically important antimicrobial drug interactions.

KEY LESSONS LEARNED: The CCF Guidelines for Antimicrobial Usage book is well respected among the HS. The book serves its purpose well in helping HS with their daily care of hospitalized patients. This survey helped us in implementing some changes that we believe will improve these guidelines and consequently, improve the quality of care for our patients.

SUBACUTE CARE UNITS, THE PAST, PRESENT AND FUTURE. P. Mistry¹; ¹Cleveland Clinic Foundation, Cleveland, OH

STATEMENT OF PROBLEM OR QUESTION: Problems and challenges faced by post acute care units and attempts to preserve the quality of care in the face of financial cut-backs. **OBJECTIVES OF PROGRAM/INTERVENTION:** 1. Discuss rationale of subacute units and patient selection criteria. 2. Review financial issues relevant to subacute units. 3 Outline educational and research opportunities in subacute care.

DESCRIPTION OF PROGRAM/INTERVENTION: The Cleveland Clinic Subacute Care unit is a 59 bedded unit. On an average there are close to 1,500 patients seen in this unit annually, with a wide variety of medical, surgical and rehabilitation problems requiring skills of a physician trained in the care of these patients. This presentation is based on our experience and observation. The Lecture format will be as follows: Introduction: Advances in technology and an increase in life expectancy for the general population have resulted in an increasing demand for medical services. In the early 1980s, for example, Medicare introduced the Diagnosis Related Group (DRG) based system of payment for inpatient care. The DRG system created financial disincentives for the hospitals by imposing a fixed reimbursement rate for each condition unrelated to patient acuity or the actual costs of care incurred by the hospital. In response, many hospital systems created alternative discharge sites, thus shifting care to subacute units, nursing homes, and home-based care. Principles of subacute care The philosophy of Subacute or Transitional care units is the management of patients after acute exacerbation of an illness, providing a more healing environment and a change of focus away from high technologic interventions. While these patients do not need intense diagnostic work-up or invasive procedures, they still require frequent physician monitoring, nursing care, and rehabilitation. Patient Selection: The decision of appropriate referral to subacute unit is a key to both continuity of care and financial viability. Appropriate subacute care candidates must have a definitive goal and identified skilled needs. It is important to identify these patients as early as possible during their acute hospitalization in order to reduce the length of stay. Examples of patients appropriate for subacute units will be discussed in detail. Also difference between these patients and nursing home patients requiring custodial care will be discussed. Financial challenges: As the number and type of alternate care sites have increased, federal efforts at cost containment have shifted to limit financial reimbursement to post-acute care facilities as well. The Balance Budget Act of 1997 introduced the Prospective Payment System for these units, shifting the reimbursement from per-diem rate to a fixed rate which would include all the ancillary services received by a patient except physician visits. This fixed rate is calculated based on the Resource Utilization Group per Medicare guidelines. In many cases, the cost of providing care for medically complex patients far exceeds the reimbursement. Mounting economic pressure to contain treatment costs may

threaten the financial health of many subacute units especially as hospitals seek to admit only the sickest of the sick and treat them in the shortest possible amount of time. This section will include discussion on the problems created by these changes in reimbursement and the possible solution to maintain quality of care. Educational opportunities: With the decrease in length of stay on the acute inpatient site and increasing medical complexity, subacute units continue to provide care for a variety of patients. Members involved in their care require both medical and surgical skills. Our subacute unit is a site for residency training program and continues to offer wide variety of experience ranging from routine postoperative care to caring for the elderly and medically sick patients. Residents also learn the importance of maintaining functional integrity in elderly patients, concepts of rehabilitation working in an interdisciplinary team model and dealing with end of life decisions. They also learn to provide care in face of financial cut-backs, without compromising on the quality of care. Outcomes research continues to remain an unexplored area in this arm of health care. The value of this cannot be underestimated, especially in view of financial cut-backs, with the goal of maintaining a high standard of care for an enlarging, vulnerable patient population in this new environment.

FINDINGS TO DATE:

KEY LESSONS LEARNED:

PHYSICIAN RECORD KEEPING WAS ALTERED BY FOCUSED PROFILING. R.E. White¹, D. Gray¹; ¹Albuquerque VAMC, Albuquerque, NM

STATEMENT OF PROBLEM OR QUESTION: Since 1998, attending physicians in our general medicine, continuity clinics have instructed resident practitioners to enter patient problem lists into the electronic medical record. After eighteen months, however, only 45% of patients had an electronic problem list (EPL).

OBJECTIVES OF PROGRAM/INTERVENTION: We implemented a profiling system based exclusively on the electronic medical record database. It allowed one person to review 100% of patient and resident records and then provide peer compared feedback.

DESCRIPTION OF PROGRAM/INTERVENTION: Data about resident entry of EPLs was extracted from the computerized, medical record database and recurrently fed back to them between February and June 2000. Profile reports were posted in clinic, and all practitioners could compare their patient panel sizes and EPL completion rates with their peers. Several times during the intervention, each resident also received a list of his or her assigned patients who needed EPL action. The profiling intervention was suspended July through September 2000. Since all practitioners in the teaching group practice were profiled, no control group existed. Therefore, the presence of an EPL for each patient and the entry dates of individual problems contained in those lists were analyzed and compared across three time periods: the seven months before profiling (July 1999 through January 2000), the five months during the profiling intervention (February through June 2000), and after profiling ceased (July through September 2000).

FINDINGS TO DATE: During the five months of profiling, residents added Electronic Problem Lists to 1087 patients, raising the percentage of patients with EPLs from 45% in January to 88% in June. An unintended, side product of profiling was an 11% increase in the number of patients which residents identified as belonging to their panels, from a total of 2048 in January to 2272 in June. Besides starting more EPLs during profiling months, residents also entered significantly more individual problems into those lists during profiling. During the seven pre-profiling months, they entered an average of 321 individual patient problems per month, and during profiling they entered 757 per month (a 136% increase). Not only did individual residents increase their number of electronic problem entries, but also more residents entered problems. Prior to profiling, 18 of 71 residents did not enter a single problem over seven months, and only 30 residents each month entered problems. During profiling all 71 residents entered problems and 50 did so each month. Forty-two residents were present during all data collection months (July 1999 through September 2000). These 42 residents entered individual problems into EPLs at monthly rates of 220 before, 526 during and 285 after profiling. "Before" and "after" periods differed significantly from the profiling period (ANOVA $p < .005$) but not from each other. The number of residents adding problems each month averaged 20 before, 31 during and 26 after profiling (Chi-square $p < .02$ for the "before"/"during" comparison).

KEY LESSONS LEARNED: Profiling was associated with increased compliance with record keeping policy, and affected all residents. Repeated feedback to practitioners about their specific performance and with peer comparisons is more engaging and motivating than reminders by supervising attendings. Using a computer patient database to review 100% of all patient and physician documents offers an efficient means to enhance physician performance.

THE USE OF PALM HANDHELD COMPUTERS TO CARE FOR NURSING HOME PATIENTS. J.M. Previll¹; ¹East Carolina University, Greenville, NC

STATEMENT OF PROBLEM OR QUESTION: Our medical school nursing home service has had difficulty monitoring the care of our nursing home patients because different attendings and residents were involved in the care of these patients. These difficulties included not having access to the patient's diagnoses, code status, current medications, pending labs, imaging, or procedures scheduled for the patient, or the subsequent results. There was also the question of when patients must be seen to comply with Medicare and institutional requirements.

OBJECTIVES OF PROGRAM/INTERVENTION: The objective was to develop a system that would enable attendings and residents to track all the data on patients and provide better care with this information.

DESCRIPTION OF PROGRAM/INTERVENTION: A Palm Pilot database called HandBase was used to develop a database that had information on the nursing home patients concerning their location, dates seen, diagnoses, code status, allergies, medications, reports on labs and x-rays, and CPT and ICD-9 codes for billing. HandJet, an interface program, linked the database in the Palm Pilot to a Microsoft Access database located in the Brody School of Medicine computer network. Patient data is protected by passwords in the Palm Pilot and the

firewall in the Brody School of Medicine computer network. Changes to patient records can be made during time of visit or by telephone and disseminated to others by infrared beaming between Palm handhelds or hotsyncing to the computer network.

FINDINGS TO DATE: This system allows database backup, printing of work lists, and patient data notebooks. It also enables all physicians and staff involved in patient care to have access to information needed in the ongoing care of the nursing patients. Having a current and complete profile on each patient has given the doctors the ability to respond and care for these patients in a more timely manner. Paperwork is simplified for admissions, discharges and scheduling. Documentation is immediate and more detailed, making billing more efficient and accurate, as it is now a part of the report generator.

KEY LESSONS LEARNED: Maintaining a database that is mobile improves effectiveness and provides more timely patient care in the nursing care setting.

PATIENT PROFILE: A PRACTICE MANAGEMENT SYSTEM DEVELOPED USING AN OFF-THE-SHELF RELATIONAL DATABASE PROGRAM. T. Yackel¹, B. Slater²; ¹Oregon Health Sciences University, Portland, OR; ²George Washington University Medical Center, Washington, DC

STATEMENT OF PROBLEM OR QUESTION: The George Washington University Department of Health Care Sciences is part of a multi-specialty faculty practice with departmentally centralized paper medical records. An internal study revealed that patient chart availability at the time of care was as low as 66%. The use of electronic resources to obtain patient specific information was limited to laboratory results retrieval and patient scheduling. Laboratory results were available online only for tests drawn in the past 60 days.

OBJECTIVES OF PROGRAM/INTERVENTION: 1) Improve availability of patient information at the time of care. 2) Implement an electronic disease management system for patients with HIV and diabetes mellitus. 3) Accomplish these objectives in a short period of time with little funding.

DESCRIPTION OF PROGRAM/INTERVENTION: Using an off-the-shelf relational database product, Access 97 (Microsoft, Redmond, WA), we implemented an advanced clinical information system over a six-month period. The program was written during spare evenings and weekends by the authors, neither of whom have had formal training in database programming. The system includes sophisticated features such as a problem list linked to ICD-9 codes; a prescription manager that provides drug-drug interaction checking and patient-specific formulary information; templates for HIV and diabetes disease management; automatic importing of laboratory information with longitudinal views, including results from the past 3 years for HbA1C, PSA, cholesterol, HIV viral loads, and CD4 counts; a reminder system; the ability to import dictations; user-level security and encryption of data; and an easy-to-use interface.

FINDINGS TO DATE: Patient Profile has been in continuous use since June 1998 by a group of 3-5 users and has required little maintenance. The availability of the system exceeds 99%. Due to the program's simplicity, low-cost, and popularity among care providers, it became a prototype for a practice-wide electronic medical record project.

KEY LESSONS LEARNED: Using commercially available tools, clinicians with minimal experience in computer programming can develop and implement their own practice management systems for a fraction of the cost of commercial products. Because these programs are infinitely customizable, they may serve the specific needs of providers better than a system purchased from a vendor.

housestaff morale (mean $\pm$ SD= 4.54 $\pm$ 0.72), the academic reputation of the program (4.46 $\pm$ 0.78), a positive interview experience (4.10 $\pm$ 0.99), variety of clinical experiences by residents (4.05 $\pm$ 0.72) and location near spouse/significant other (4.04 $\pm$ 1.43). URM applicants (N=92 or 9%) were significantly more likely to identify ethnic diversity of the patients (3.78 $\pm$ 1.23 vs. 3.39 $\pm$ 1.25, p=.004), serving the medically indigent (3.79 $\pm$ 1.17 vs. 3.71 $\pm$ 1.20, p=.006), feeling of being wanted (3.76 $\pm$ 1.10 vs. 3.46 $\pm$ 1.20, p=.02), academic environment supportive of ethnic minorities (3.46 $\pm$ 1.35 vs. 2.51 $\pm$ 1.30, p=.0001), ethnic diversity of housestaff (3.32 $\pm$ 1.30 vs. 2.54 $\pm$ 1.27, p=.0001), and ethnic diversity of faculty (3.12 $\pm$ 1.34 vs. 2.42 $\pm$ 1.25, p=.0001) as important factors in selecting a program.

CONCLUSION: Applicants rate location and program-related factors as most important in influencing their decision to chose a particular Internal Medicine Residency Program. However, URM applicants also place significant importance on the ethnic diversity of patients, housestaff and faculty, and the support provided to ethnic minorities within the academic environment. Residencies must place an emphasis on improving these factors if they wish to recruit highly qualified minority applicants.

THREATS TO INTERNAL MEDICINE RESIDENTS IN THE WORKPLACE BY PATIENTS, THEIR FAMILIES, AND NON-PHYSICIAN STAFF. R.J. Abrams¹, S.B. Glick², L.S. Sadowski²; ¹Rush Presbyterian St. Luke's Medical Center, Chicago, IL; ²Cook County Hospital, Chicago, IL

PURPOSE: Although there is some literature on violence in the medical workplace, little has been written about threats physicians experience on the job. Our study examines the epidemiology of threats to resident physicians in internal medicine by patients, their families, and non-physician staff.

METHODS: We implemented a mass-administered survey to internal medicine residents in two large urban training programs. One program is in a public county hospital which serves predominantly uninsured patients. The other is in a private, tertiary care, academic medical center whose patient base is predominately insured by Medicare or managed care entities. We defined a threat as an actual or implied action intended to harm or torment someone. The survey measures the frequency, type and setting of threats. It examines the support structures residents use to deal with threats. We also assess resident attitudes towards receiving threats, particularly as they relate to patient care.

RESULTS: Our preliminary data analysis shows 24 of 69 (34.8%) residents at the public hospital and 15 of 40 (37.5%) of residents at the private hospital have experienced one or more threats (total number of threats 66, mean 1.7). The threats were most commonly from patients assigned to that resident (47%), 16% from patients they covered on call, 19% from family members of patients, and 18% from other non-physician staff. Eighty five percent of the threats were verbal in nature and 15% were physical. The location of the threat occurrence was usually on the wards (70%). The remainder occurred in the emergency department (24%), and 6% occurred in outpatient clinics. Residents reported that the threats made them feel angry, nervous, afraid, and helpless. Approximately three-quarters of residents (74%) sought some support for these episodes. This support came from a variety of sources including peers, family, attending physicians, and security. Of residents reporting a threat, one in five (21%) said the experience impacted on the care of the patient involved, while only 2 of 39 (5%) said the experience impacted on how they care for other patients. Eleven of 39 (28%) said that following the threat they are more likely to avoid certain situations at work to feel more safe. Residents strongly disagreed with the statement that being threatened by a patient or their family members is a part of their job.

CONCLUSION: Residents commonly experience threats from patients, their family members and non-physician staff in the workplace. The prevalence of threats was similar at both a public and private hospital. The most common reactions residents reported to these experiences include negative emotions and an impact on their care of the involved patient. An in-house support system needs to be developed to prevent and handle threats residents experience in the workplace.

SCIENTIFIC ABSTRACTS

LEARNER/TEACHER-CENTERED RESEARCH

MEDICAL STUDENTS CHOICE OF INTERNAL MEDICINE RESIDENCY PROGRAM: DO MINORITY APPLICANTS DIFFER FROM MAJORITY APPLICANTS? E.M. Aagaard¹, K. Julian¹, J. Dedier², I.D. Solomon³, J.H. Tillich⁴, E.J. Perez-stable¹; ¹University of California, San Francisco, San Francisco, CA; ²Massachusetts General Hospital, Boston, MA; ³New York Presbyterian Hospital, New York, NY; ⁴University of California Los Angeles

PURPOSE: To determine factors that most influence medical student selection of particular Internal Medicine Residency Programs and whether minority applicants differ from majority applicants with regard to these factors.

METHODS: We performed a cross-sectional survey of 4th year medical students applying during the 1999 National Residency Matching Plan in Internal Medicine to one or more of four Internal Medicine Residency Programs (University of California San Francisco, University of California Los Angeles, Massachusetts General Hospital, and New York Presbyterian Hospital). All students were mailed a questionnaire requesting they rate the importance of various factors in choosing a residency program using a 5 point Likert scale (1 = not important, 3 = somewhat important, and 5 = very important). Underrepresented minority (URM) applicants were defined as African American, Latino, or Native American.

RESULTS: Response rate was 36% (1005/2820), 55% were men, 30% were married and 61% self-rated in the top 25% of their class. The factors most important to all applicants were good

BALANCE GROUPS: FACILITATING AND MAINTAINING DISCUSSIONS ON PROFESSIONAL AND PERSONAL DEVELOPMENT IN INTERNAL MEDICINE RESIDENCY TRAINING. S.F. Babbott¹, S. Fischel¹; ¹Baystate Medical Center, Springfield, MA

PURPOSE: Training in Internal Medicine is an intense and stressful time for both professional and personal development. Residency programs have used support groups and Balint groups to address some of the stresses faced by trainees. We developed a monthly, faculty facilitated, hour-long session for residents on a ward or coronary care unit rotation. The goal of this session was to promote an opportunity and a safe environment for residents to explore clinical issues that had an impact on the feelings, attitudes, and challenges of becoming a physician. In each session, ground rules for safety were reiterated, as was the voluntary nature of this session. The group was held during one half of a scheduled attending rounds session in the third week of the month. The teaching attending (different from the facilitator) was welcome to participate.

METHODS: We developed a confidential and anonymous survey given to all medicine and medicine pediatrics residents at the end of the 1999-2000 academic year. The survey was designed to access resident attitudes towards participation, their acceptance of this discussion format and any potential behavior change as a result of participation in those groups.

RESULTS: Overall response rate was 75% (65% medicine and 35% medicine pediatrics). 62% of respondents were under age 30; 27% were age 30-35. 62% of respondents were female. Residents had attended an average of 1.6 sessions on the wards, and 1.1 sessions in the CCU. 90% of respondents indicated the timing was appropriate. Residents rated their comfort in these discussions with fellow residents on a 1 (low comfort) to 5 (high comfort) scale, with the mean response of 3.83. Residents were asked the following: The issues discussed in the balance groups are ones you may have thought about or discussed in the past. As a result of the discussions this past year, please rate your current activity from 1 (less activity) to 3 (same activity) to 5 (more activity) in: self reflection (mean response = 4.41), discussions with peers (mean response = 4.33) and discussions with family and friends (mean response = 4.17).

CONCLUSION: Balance groups in this Internal Medicine residency training program were well received regarding their timing, and the comfort felt among participants. Importantly,

residents indicated that they continued to reflect privately and with peers, family, and friends. Timing and attendance are significant barriers to successful groups; we found that scheduling this during a regular educational session, reinforcing appropriate ground rules and the voluntary nature of this session were important to its success. We recognize that some groups are longitudinal and benefit from the increased trust gained by the group. We chose to focus on the small group of each ward or unit team, and planned this discussion later in the month when some group identity and cohesion was more likely to have taken place.

PREDICTORS OF EFFECTIVE PHYSICAL DIAGNOSIS TEACHING. K. Barnard¹, D. Lescisin¹, N. Armistead¹, D. Elnicki¹; ¹University of Pittsburgh, Pittsburgh, PA

PURPOSE: Interns play an important role in the teaching of medical students. We sought to identify demographic characteristics and teaching behaviors that predict effective teaching of physical diagnosis skills.

METHODS: Self-administered surveys were completed by third year students at the end of their internal medicine clerkship at two medical schools from 7/00-11/00. Questions included demographic characteristics of interns and students, and occurrence of teaching behaviors, such as observation and feedback, reviewing cases, and establishing a comfortable learning climate. We also asked students to assess to what degree the intern contributed to their learning physical diagnosis skills (diagnosing heart murmurs, distinguishing pulmonary consolidation from pleural effusion, detecting hepatomegaly, examining the thyroid gland, detecting a joint effusion and lymphadenopathy). These items were selected from the CDIM-SGIM curriculum, and were combined to form a grouped physical diagnosis variable. A forward, step-wise linear regression model was built to determine which of the independent variables predicted effective physical diagnosis teaching. Adjusted p-values are shown here.

RESULTS: A total of 70 students were surveyed (95% response rate). The teaching behavior, "making helpful suggestions to improve performance", was most important in predicting effective teaching of physical diagnosis, as measured by the grouped variable, ($p=0.006$). The addition of two other variables, "corrected mistakes without making you feel belittled", and intern gender created a model with $R^2=0.35$. When the physical diagnosis variables were analyzed separately, the teaching behaviors, "observed you elicit physical examination findings" and, "demonstrated a breadth of knowledge in internal medicine" were important in predicting the ability to distinguish pleural effusions ($p=0.087$ and 0.075 respectively). Observing students eliciting findings was important to examination of the thyroid gland ($p=0.07$). The models for the individual physical examination skills explained 23%-59% of variances.

CONCLUSION: Teaching behaviors involving giving feedback and female gender were most important to students learning physical diagnosis skills from interns. Other teaching behaviors were important for individual physical examination skills. The reasons for the gender differences remain unclear.

PATIENT DIAGNOSIS AS A DETERMINANT OF RESIDENT SATISFACTION WITH PATIENTS IN THEIR CONTINUITY CLINIC. D.R. Barnett¹, P.F. Bass¹, J.F. Wilson¹, C.H. Griffith¹; ¹University of Kentucky, Lexington, Kentucky

PURPOSE: In order to prepare residents for primary care practice an understanding of what factors influence their satisfaction with the outpatient experience is necessary. However, few studies have addressed determinants of resident satisfaction in their continuity clinics. This study was conducted to determine these factors.

METHODS: Over a 2-month period, resident satisfaction was assessed through a self-administered questionnaire completed by each internal medicine resident in their outpatient continuity clinic. All 68 residents working in the clinic completed the questionnaire. The number of questionnaires completed by each resident ranged from 1 to 60 with a median of 23. The questionnaire was comprised of three 5-point Likert scale items that assessed resident satisfaction with each patient encounter. Resident satisfaction was defined as the mean of the 3 items. The resident satisfaction items had a Cronbach's alpha of 0.88. Other items assessed by the questionnaire included contextual aspects of the clinic, diagnoses of the patients, continuity, and demographic data.

RESULTS: Overall, the majority of clinic visits were satisfying to residents. The mean satisfaction score of all resident clinic visits was 4.02 (5-point scale) with a standard deviation of 0.91. However, patient diagnosis had a significant impact on resident satisfaction. The mean resident satisfaction score for patients diagnosed with general medical problems was 4.30. The mean resident satisfaction score for patients diagnosed with pain complaints was 3.58 ($p < 0.0001$) and for psychiatric disorders it was 3.27 ($p < 0.0001$). Clinic visits with patients diagnosed with pain complaints or psychiatric disorders were more satisfying to residents if the patients were male rather than female ($p < 0.05$). 1st year residents were less satisfied with patients diagnosed with pain complaints or psychiatric disorders than 2nd and 3rd year residents ($p < 0.05$). On the other hand, 1st year residents found visits with patients diagnosed with general medical problems more satisfying than 2nd and 3rd year residents ($p < 0.05$). A number of factors were not associated with resident satisfaction including resident gender, rotation, and call status.

CONCLUSION: Although resident satisfaction with patient encounters in the outpatient clinic is high, resident satisfaction is influenced by the patient diagnosis. Seeing patients with pain or psychiatric diagnoses is associated with decreased resident satisfaction in their continuity clinic. However, this appears to be influenced by the experience level of the resident. Future research can explore whether early resident training in the treatment of pain and psychiatric conditions may increase resident satisfaction with such patients.

INTERNAL MEDICINE RESIDENTS ABILITY TO IDENTIFY LOW LITERATE PATIENTS IN THEIR CONTINUITY CLINIC. P.F. Bass¹, D.R. Barnett¹, J.F. Wilson¹, C.H. Griffith¹; ¹University of Kentucky, Lexington, KY

PURPOSE: Approximately 25% of Americans are functionally illiterate and low literacy has been associated with poorer physical health, psychological health, and higher health care cost. This study was conducted to see if residents could identify patients with potential literacy

problems based solely on interactions with patients during a clinic visit. We hypothesized residents will overestimate patients' literacy abilities and not identify many patients at risk for potential literacy problems.

METHODS: Residents completed a questionnaire measuring a number of constructs regarding continuity clinic in general, and their perception of their continuity patients' literacy. The residents were asked, "Do you feel this patient has a literacy problem?" Patients in continuity clinic were given the Rapid Estimate of Adult Literacy in Medicine-Revised (REALM-R), a new instrument to quickly screen for potential literacy problems. The REALM-R has been previously correlated with the Wide Range Achievement Test-Revised (WRAT-R) and the Rapid Estimate of Adult Literacy in Medicine (REALM), two well-validated instruments but impractical for large scale screening in a busy clinic. The REALM-R asks patients to read a series of eight medical words and a correct response is given for each correct pronunciation. A score of six or less was used to identify patients with a potential literacy problem. Data were available from 182 patients who completed the REALM-R and whose residents completed the literacy question.

RESULTS: Patients ranged in age from 17 to 93, and 85% were Caucasian. Scores on the REALM-R ranged from 0 to 8 with a mean of 6.8 and S.D. of 2.1. Twenty-three percent of patients read at the eighth grade level or less according to the WRAT-R. Despite these baseline characteristics, residents believed only 10% (18) of patients had a literacy problem based on their clinical interactions. Residents perceived 90% (164) of patients to have no literacy problem, but 36% (59) of these patients scored 6 or less on the REALM-R. Conversely, residents identified three patients as having a literacy problem whom scored higher than six on the REALM-R, while the other 15 all scored six or less.

CONCLUSION: Residents perceived a significant number of patients to have no literacy problem, although these patients scored poorly on the REALM-R. Residents incorrectly identified only a few patients. Residents' overestimation of patients' literacy is concerning given that many patients will hide literacy problems from physicians and the poor health outcomes associated with poor literacy. A brief screening instrument like the REALM-R may help clinicians identify patients in whom literacy may complicate their health care.

LOOKING FORWARD TO PROMOTION: A PROSPECTIVE STUDY OF PROMOTION IN ACADEMIA (PSPA). B.W. Beasley¹, S.M. Wright²; ¹Saint Lukes Shawnee Mission Health System, Kansas City, MO; ²Johns Hopkins University, Baltimore, MD

PURPOSE: Recent studies have documented how department chairs and promotion committee chairs view the importance of different facets of a clinician-educator's performance. However, there is little data about what clinician-educators themselves believe is important for their promotion, and what kind of support they find useful in their careers.

METHODS: This data is derived from the intake questionnaire for participants in the PSPA study. We obtained names and addresses from the AAMC roster for faculty that entered academic medicine as Assistant Professor June through December of 1995. In 2000, 183 faculty members from 35 different states volunteered to participate. Participants indicated they were clinician-educators (CE) or clinician-investigators (CI). If participants indicated "other" or "clinician", we assigned them a priori to one of the two former categories based upon their percent effort allotted to various work activities.

RESULTS: Participants had a mean age of 40 years, and had been working at their current position for 4.7 years. 63% of the faculty were clinician-educators (CE) and 37% were clinician-investigators (CI). More CIs than CE's were minorities (30% vs. 14%, $p < .05$), but there was no significant difference in gender (35% female). Whereas 98% of CI's reported a publication expectation to be promoted, 25% of the CE's reported no such expectation ($p < .001$). Significantly more CI's had career mentors available than CE's (68% vs. 32%, $p < .001$), and 92% of CE's for whom they are not available indicated they would like one. 79% of CI's indicated >10% protected scholarly work time, compared to only 35% of CE's ($p < .001$). 53% of CI's vs 32% of CE's ($p < .01$) meet more often than yearly with their chief/ chair for performance review, and more CI's have seen written promotion guidelines (72% vs. 51%, $p < .01$). CE's and CI's assigned similar importance to clinical research, written scholarship, and reputation (top 3 out of 11 items in importance for both groups) as activities leading to their promotion. The top 3 important activities for CE's according to promotion committee chairs in our previous study (teaching skills, clinical skills, and mentoring), were ranked 4th, 10th, and 8th by CE's in this study. Both CE's and CI's agreed that CI's more likely get promoted (82% vs. 79%).

CONCLUSION: CE's indicated fewer performance reviews with their chair/chief, fewer mentors, and less knowledge of written promotion guidelines than their CI colleagues. Although university promotion chairs have indicated that teaching and clinical skills are most important in the promotion of CE's, CE's believe research and written scholarship are most critical to their promotion.

SOCIOCULTURAL AND LINGUISTIC BARRIERS IN AN URBAN ACADEMIC OUTPATIENT PRACTICE: OBSERVATIONS FROM DOCTORS IN TRAINING. J. Betancourt¹, A. Green¹, J. Harris³, C. Charles¹, B. Meyer², J. Carrillo²; ¹Weill Medical College of Cornell, New York, New York, ²Sarah Lawrence College

PURPOSE: The purpose of this study was to identify residents' perceptions of the socioculturally and linguistically based barriers they face in caring for a diverse patient population. We also attempted to determine whether, and in what ways, a required cross-cultural curriculum helped them deal with these barriers.

METHODS: We conducted structured interviews with the entire intern class of an urban academic internal medicine residency program ($n=40$). These were carried out with the stated goal of identifying general barriers to quality care for their patient population. Interviews were performed within six months of the completion of an eight-hour, case-based, cross-cultural curriculum. Residents were not aware of any connection between the research assistant who conducted the interview and the primary researchers who taught the curriculum. Probes for the interview focused on: general barriers to care, sociocultural and linguistic barriers to care, if and how they had been prepared to deal with these barriers and the type of training they had

received, and whether the cross-cultural curriculum had helped them and affected their attitude towards these issues. Interviews were recorded, transcribed, and coded by two independent outside researchers experienced in qualitative analysis.

RESULTS: Several barriers to delivering care were identified:

1. Language differences between patient and resident
2. Limited time to address important cross-cultural issues in medical encounter
3. Understanding socioculturally-based patient expectations and perceptions medical care (including mistrust)
4. Differences between patient beliefs and physician beliefs regarding disease and illness

Effect of cross-cultural curriculum on attitudes and practice:

1. Greater acknowledgement of the role of sociocultural issues
2. More probing into patients' beliefs and practices
3. Overall very helpful in addressing barriers

CONCLUSION: Residents cite several barriers to caring for their socioculturally and linguistically diverse patient populations. While they identified several benefits of a cross-cultural curriculum, they also highlighted the need for effective interpreter services and increased time to care these patients.

WOMENS HEALTH AMBULATORY CARE SKILLS: DO INTERNISTS FEEL PREPARED?

D.E. Bonds¹, R. Watkins¹, J.C. Mychalek¹,¹Wake Forest University, Winston-Salem, NC

PURPOSE: Previous studies have found that Internal Medicine faculty and residents rate their ability in ambulatory care procedures as poor. Our objective was to assess faculty and residents ability in those skills related to Women's Health, and whether that ability varies by gender or status (resident vs faculty).

METHODS: A descriptive survey was administered to all Internal Medicine residents and teaching faculty at one academic medical center. Respondants were asked to rate their ability to perform 5 ambulatory skills (breast exam, pelvic exam, wet mount/KOH, pap smear, endometrial biopsy) and comfort in obtaining 4 areas of history/counseling (sexual activity history, domestic violence history, preconception counseling, contraceptive counseling). A 4-point Likert scale was used.

RESULTS: 88 Internists completed the survey: 91% of residents (72/79) and 80% of faculty (16/20). Women comprised 32% (32% residents, 31% faculty). Greater than 70% felt able to perform all physical exam and procedural skills except endometrial biopsy (0% able). Conversely, less than 40% rated their comfort level as high for domestic violence history, and areas of counseling. Women rated their ability higher than men, and faculty higher than residents. See table 1 for full results (* indicates $p < 0.05$).

CONCLUSION: Faculty and residents rated their ability high for exam and procedural skills, but most felt uncomfortable in obtaining history or performing counseling. Women rated their skill level higher, as did faculty. Both faculty and residents need additional training in ambulatory skills related to women's health.

Table 1

	% Able to Perform	By Gender	By Status
		Female versus Male	Faculty vs Resident
Breast Exam	85%	96% vs 80%	100% vs 82%
Pelvic Exam	74%	89% vs 67%*	100% vs 68%
Wet Mount	70%	82% vs 65%	81% vs 68%
Pap Smear	90%	96% vs 87%	100% vs 88%
Endometrial Biopsy	0%	0% vs 0%	0% vs 0%
Sexual Activity Hx	70%	75% vs 68%	88% vs 67%
Domestic Violence	39%	54% vs 32%*	44% vs 38%
Preconception	21%	39% vs 12%*	27% vs 20%
Contraception	36%	46% vs 32%*	75% vs 28%*

IMPROVEMENT OF RESIDENTS PHYSICAL EXAMINATION SKILLS - PERCEIVED MOTIVATORS AND BARRIERS. **J.B. Bundrick¹, J.T. Li¹; Mayo Clinic, Rochester, MN**

PURPOSE: To assess the relative degrees to which a variety of factors are perceived to contribute positively and negatively towards motivating internal medicine residents to improve their physical examination skills.

METHODS: In May 2000, forty G1 internal medicine residents at the Mayo Clinic completed an objective, structured clinical examination (OSCE). Immediately following this exercise, they were anonymously surveyed regarding the degree to which certain factors tended to either increase or diminish their motivation for improving their skills in physical examination. Twelve potential motivators and thirteen potential barriers were ranked on a Likert scale from one to seven.

RESULTS: Those factors ranked as the strongest motivators for improvement had to do with the role of the physical examination (PE) in patient care decisions (mean score = 5.9), the regret of missed findings in the past (5.9), and the perceived contribution of strong PE skills to one's overall excellence as a clinician (6.0) or teacher (5.3). The need to enhance one's performance on formal measures such as the OSCE (3.6) or to fulfill the expectations of faculty (4.6) or peers (4.0) on rounds were rated as weaker motivators. Perceived barriers to improvement included lack of time to study (5.2) or apply (5.1) the methods of PE and a lack of faculty emphasis on the importance of PE (4.8). Although the sense of exam findings being eclipsed by test results in making clinical decisions was rated as a moderately strong demotivator (5.1), other factors

designed to look for overall "PE nihilism" were ranked as only weak hindrances to improvement (2.7). Likewise, a sense of discouragement over one's past performance and future prospects for improvement seemed to be a minor barrier (3.4), as did reliance on a perceived supervisory "safety net" (2.0).

CONCLUSION: Internal medicine residents in their first year of training seem to be most strongly motivated to improve their PE skills by factors which relate to their desire to improve the quality of patient care, to avoid regret, and to build personal confidence as an excellent clinician and teacher. The most significant demotivators are lack of time to study and apply the methods, and lack of faculty emphasis on the importance of these skills. The perception of test results eclipsing the contribution of the physical exam to clinical decisions also seems to demotivate at some level, and educational strategies which emphasize those situations wherein the exam makes an indispensable contribution to patient care may help to overcome this potential barrier.

TRAINING EXPECTATIONS AND SATISFACTION OF GENERAL INTERNAL MEDICINE FELLOWS: A WEB-BASED SURVEY. **E. Caiola¹, D.G. Litaker², M.G. Hewson², W. Land², N. Mehta², J.H. Isaacson²; ¹University of Rochester School of Medicine, Rochester, NY ; ²The Cleveland Clinic Foundation, Cleveland, OH**

PURPOSE: Fellowship training in general internal medicine (GIM) is becoming increasingly important in preparing generalists for academic careers. Fellowship programs have the stated goal of training future faculty and leaders in general internal medicine. Few studies, however, explore fellows' training expectations or their satisfaction with the program they select. We sought to evaluate these factors and identify areas of training that lead to a productive and satisfying fellowship experience.

METHODS: We developed a 43-item questionnaire that addressed fellows' initial training expectations, satisfaction with the quantity and perceived quality of training (teaching and research), availability of mentors, scholarly productivity (publications, grants) and anticipated career track. Question format included best answer and Likert scales. Directors of the 37 active fellowship programs listed in the 2000 SGIM Fellowship directory were contacted by e-mail and asked to forward our web-based survey to current fellows. To calculate an accurate response rate while maintaining respondent anonymity, each fellow used a program-specific code to initiate the survey.

RESULTS: Forty-three percent (65/152) of fellows completed the survey, with at least 1 respondent in 65% (24/37) of the programs. Nearly three-quarters (74%) anticipated careers as clinician-investigators. Eighty-two percent had high or very high initial expectations for research training, 26% for teaching and 20% for learning leadership skills. The availability of mentors (95%), program flexibility (94%) and protected time for research (91%) were rated as the most important factors in fellowship training. Most fellows were satisfied or very satisfied with the actual availability of mentors (80%), program flexibility (85%) and protected research time (88%). Over 80% agreed that their expectations had been met for both the quantity and quality of research opportunities, while 63% reported this for the quantity and 40% for quality of teaching opportunities. The availability of mentors and the perceived quality of research opportunities were significantly correlated with fellowship scholarly productivity, 0.507 and 0.539 respectively ($p < .001$).

CONCLUSION: This study suggests that the majority of GIM fellows enter training with the expectation for advanced research skills and are satisfied with most aspects of their experience. These data highlight the important correlation between mentoring, high quality research opportunities and scholarly productivity during fellowship. That relatively few fellows are satisfied with the quality of available teaching opportunities is concerning and deserves further exploration.

FACULTY DEVELOPMENT FOR TEACHING SKILLS IN U.S. TEACHING HOSPITALS. **J.M. Clark¹, T.K. Houston¹, D.E. Kern¹; ¹Johns Hopkins University, Baltimore, MD**

PURPOSE: To determine the current level of faculty development (FD) in teaching skills for Department of Medicine faculty in the United States.

METHODS: We designed a 10-page survey to determine the extent of faculty development (FD) activities that focused on "improving the teaching/educational skills" of faculty. For those hospitals with ongoing activities we asked about the subjects covered, the methods of teaching, the formats offered, and the evaluation of the programs. In April 2000, we mailed the survey to Department of Medicine chairmen at the 390 teaching hospitals with internal medicine residency training programs in the U.S. We classified teaching hospitals with membership in the Association of Professors of Medicine as university hospitals (N=114), and those without membership as non-university. We performed descriptive analyses of hospitals with and without ongoing FD, and used chi2 to detect differences in categorical variables and t-tests for differences in continuous variables.

RESULTS: Three hospitals responded that their training programs had closed. Of the remaining 387 hospitals, 279 responded for a response rate of 72%. Only 39% of teaching hospitals have ongoing FD, including 48% of university and 34% of non-university hospitals ($p=.007$). An additional 34% have occasional FD; 27% have no FD activities. At the hospitals with ongoing FD activities (N=109), an average of 10 content areas are covered, with university hospitals including more than non-university hospitals (11 vs. 9, $p=.036$). The most common content areas are general teaching principles (90%), giving feedback (88%), outpatient precepting (72%) and evaluation of learners (72%). FD is taught most commonly in small group sessions, with 65% of hospitals using them frequently/always, and in lectures, with 50% using them frequently/always. The most common format for FD is a 1/2 day workshop (78%), but 21% offer courses lasting more than a month. Compared to university hospitals, non-university hospitals are more likely to offer 1/2 day workshops only ($p=.008$), or courses < 7 days ($p=.013$), and less likely to offer longer courses ($p=.006$). Evaluation of FD programs is limited to forms filled out by participants at the end of the program. Most programs had trained: <50% of hospital-based and <25% of community-based faculty; <50% of general internal medicine faculty and <25% of subspecialist faculty. 2/3 of hospitals with FD, but only 37% of hospitals with no FD, arrange for faculty to go off-site for FD.

CONCLUSION: A minority of teaching hospitals have ongoing faculty development programs for improving their faculty's teaching skills and evaluations are primitive. University hospitals are more likely to have ongoing FD and to have programs that are more developed. Overall, only a limited percentage of faculty have been trained, especially community-based and subspecialist faculty. These gaps should be considered in future funding decisions to provide learning opportunities for all faculty.

RELATIONSHIP BETWEEN MEDICAL SCHOOL LECTURING AND CLINICAL TEACHING SKILLS. H. Copeland¹, A.L. Hull¹, M.G. Hewson¹, M.E. Mayer¹; ¹Cleveland Clinic Foundation, Cleveland, OH

PURPOSE: This study was done to determine the relationship between students' ratings of faculty on a clinical teaching evaluation form and a medical lecture evaluation form.

METHODS: Twenty-three faculty received ratings between July 1999 - December 2000 from medical students immediately after a didactic lecture on an 11-item lecture evaluation form that used a scale from 1 (not satisfied) to 7 (very satisfied) to assess lecture skills (Chronbach's alpha=.96) and at the end of a clinical rotation on a 15-item clinical teaching form that used a scale from 1 (never/poor) to 5 (always/superb) to assess attending/teaching behaviors in the patient care setting (Chronbach's alpha=.97). An average of 6 and 16 students per faculty completed the lecture and the clinical teaching forms respectively. Average students ratings on all items for each faculty member were calculated. Pearson correlation coefficients were computed between the average scores for each item on one form and the overall average scores on the other evaluation form.

RESULTS: Overall there is a correlation of .69 ($p < .01$) between the average scores on the two forms. Only correlation coefficients above .65 (to explain >45% of the variance) were considered meaningful. The following five items from the lecture form significantly correlated ($p < .01$) with the average clinical teaching scores: how well the course was organized ($r=.69$), how effectively time was used in class activities ($r=.67$), the clarity of the instructor's explanations ($r=.70$), the instructor's use of examples or illustrations to help clarify the material ($r=.67$), and having learned something which I consider valuable ($r=.67$). The following three items from the clinical teaching form significantly correlated ($p < .01$) with the average medical lecture scores: teaches diagnostic skills ($r=.73$), teaches effective patient/family communication skills ($r=.72$), and teaches principals of cost-appropriate care ($r=.70$). Interestingly, these three items were the only items that included the word "teaches" suggesting direct didactic instruction as opposed to the teaching strategies suggested in the other items (i.e., adjusts teaching, coaches, asks). Items from the lecture form that did not correlate with the average clinical teaching scores (e.g., specifying objectives, summarizing material, encouraging class discussion) are often considered fundamental in classroom teaching and may not easily translate to clinical teaching. Yet, the larger number of related items from the lecture form suggests that lecture techniques do crossover to clinical teaching, especially with providing explanations and using examples.

CONCLUSION: There appear to be some differences in the skills involved in clinical teaching and lecturing; only 47% of the variance is explained between them. Although requiring more work by students, using two forms to evaluate teaching may better enable instructors to identify specific strategies where they are strong or need improvement for the various teaching activities they perform.

AVAILABILITY OF SAMPLE MEDICATIONS AND RESIDENTS PRESCRIBING PATTERNS. J. Diaz¹, M.J. Fagan¹, A. Etienne¹; ¹Brown University, Providence, RI

PURPOSE: The use of sample medications in clinical teaching settings is controversial. One concern is that the presence of sample medications may affect physician compliance with treatment guidelines. This project examined the availability of anti-hypertension sample medications and the selection of specific medications by internal medicine residents to treat hypertension.

METHODS: From 12/4/00 to 12/20/00, charts of patients presenting to the residents' clinic at Rhode Island Hospital were reviewed if hypertension was listed as one of their diagnoses and the patients were receiving medications. All anti-hypertension medications prescribed, the date of visit and insurance status of each patient were recorded. During the same time period, inventory of the clinic's medication sample closet was taken for each clinic session and names of all anti-hypertension medications available were noted.

RESULTS: During the 3 week period, 288 patient charts met criteria for review. 56% (161/288) of patients had no insurance coverage for medications. A total of 41 different blood pressure medications were documented as being prescribed and 18 (44%) of these were available as samples. The 5 most commonly prescribed medications were hydrochlorothiazide (86/288), Norvasc (amlodipine-73/288), Accupril (quinapril-66/288), Vasotec (enalapril-42/288) and Toprol XL (metoprolol-32/288). Three of these, Norvasc (amlodipine), Accupril (quinapril), and Toprol XL (metoprolol) were available in the sample closet during the study period and one, Vasotec (enalapril), had recently been available. Except for diuretics, for each class of anti-hypertension medication, the most commonly prescribed agent was also available as a sample. Of 146 notations of ACE-Is, 65 (44.5%) were Accupril (quinapril). Of 118 calcium-channel blockers (CCBs) prescribed, 70 (59%) were Norvasc (amlodipine). Of 65 notations of beta-blocker use, 31 (48%) were Toprol XL (metoprolol). No diuretics were available as samples, but the diuretic most commonly prescribed was a generic agent, hydrochlorothiazide (70%, 84/120). Stratifying by medication coverage, a sample closet CCB and a sample closet ACE-I continued to be the medications most commonly prescribed in their drug classes. Norvasc (amlodipine) accounted for 54% (38/70) of CCBs prescribed to patients without medication coverage and 67% (32/48) of patients with coverage. Accupril (quinapril) accounted for 42% (36/85) of ACE-I prescribed to patients without coverage and 47% (29/61) of patients with coverage.

CONCLUSION: This pilot study suggests that residents selecting medications to treat hypertension may be influenced by the availability of sample medications. With the exception of hydrochlorothiazide, for each class of anti-hypertension medication, a drug available in the sample closet was the one most commonly prescribed. Although this reflects patterns at only one institution, residency programs should be aware of how sample medications may influence the prescribing patterns of physicians in training.

DEPRESSION IN INTERNAL MEDICINE HOUSE STAFF. R. Drayer¹, F. Pelzman¹; ¹New York Presbyterian Hospital, Weill Cornell Medical Campus, New York, NY

PURPOSE: Several studies over the past two decades have examined the issue of depression in house staff, but none have attempted to relate it to house staff performance. The ideal study to examine this question would be prospective and longitudinal, using validated psychiatric rating instruments and quality assurance data. Since IRB concerns have thus far precluded such a study, a pilot retrospective survey was developed to obtain preliminary data.

METHODS: After IRB approval was obtained for the pilot, the survey instrument was distributed to interns, junior residents, and senior residents in the internal medicine residency program at our institution. Participants were asked whether they had worked with someone they thought was depressed, if they had worked with a colleague whose performance had been affected by depression, and if they had ever been depressed themselves during the residency.

RESULTS: 45 members of the house staff (33%) replied to the survey. 34 respondents (76%) believed that they had worked with a depressed colleague during the residency, and 23 respondents (51%) believed that a coworker's performance had been affected by depression. Fifty-one percent of the respondents also reported having experienced depression themselves during their residency.

CONCLUSION: Although the sample size is small and these data are only preliminary, depression appears to be prevalent in some internal medicine residency programs. Furthermore, house staff depression may affect patient care. The current pilot study indicates the need for further research in this area.

TEACHING THE TEACHERS: A WORKSHOP FOR EVIDENCE-BASED MEDICINE JOURNAL CLUBS. D. Dupras¹, R. Ashton¹, C. Daniels¹, J.O. Ebbert¹, V.M. Montori¹, H.J. Schultz¹; ¹Mayo Clinic, Rochester, MN

PURPOSE: In 1999 and 2000 we conducted workshops at the Annual Meeting of the Royal College of Physicians and Surgeons in Canada. The purpose of the 90 minute workshop was to introduce concepts and methods of developing a successful journal club to teach EBM principles in the context of case-based clinical decision making (CDM). This study reports the participants rating of this workshop.

METHODS: Each workshop was open to a maximum of 25 individuals and involved three faculty, who functioned as facilitators. Approximately 1/3 of the time was a lecture format, the other 2/3rds was interactive. Each workshop included a case with a clinical question, the search strategy, and an article selected to answer the question. While in the small groups participants assessed the validity of the study, interpreted the results and applied the results of the study to the clinical question. We assessed the usefulness of the workshop, materials and effectiveness of the facilitators by asking participants to fill out an evaluation form using 5-point Likert scales to rate the individual items. The participants were asked to rate the workbook and its content, the facilitators, the format of the workshop, the didactic sessions, the interactive sessions, and if they would recommend the workshop.

RESULTS: Sixty-one participants completed the evaluation sheets. The median scores for all items were 4 or 5 for all items in both years. The workshop "practice session" was slightly higher rated in 2000, compared to 1999 ($p = 0.047$). There were no other significant differences between years.

CONCLUSION: A 90 minute workshop can effectively teach the principles and methods for the development of a successful EBM-CDM journal club to faculty members.

IS EVIDENCE BASED MEDICINE VALUED BY BOTH GENERALISTS AND SUBSPECIALISTS? S. Garg¹, M.F. Sasse¹, S. Abrams¹, R.J. Saqueton¹, A.T. Evans¹; ¹Rush Presbyterian St. Luke's Medical Center, Chicago, IL

PURPOSE: Evidence based medicine (EBM) is part of the curriculum of most medical schools and residency training programs in the US. We hypothesized that the concepts and skills may not be integrated into clinical practice because there is little support outside the classroom setting. We surveyed academic internal medicine generalists and subspecialists to assess whether they have different attitudes about evidence based medicine and whether they rate the importance of specific EBM skills differently.

METHODS: We surveyed a stratified random sample of academic internists and subspecialists from 11 medical centers in the Chicago area ($n=330$) in the Spring of 2000. Respondents completed a 39-item self-administered questionnaire, which was piloted and revised for internal consistency prior to use. The three survey questions required respondents to: 1) rate the importance of 9 core skills and 5 EBM skills on a 5-point Likert scale, 2) choose the five most important skills in residency training, and 3) indicate the level to which they agreed or disagreed with 15 attitudinal statements about EBM, also on a 5-point Likert scale. We also collected information on year of graduation from medical school, teaching responsibility, and exposure to a formal course on clinical epidemiology or evidence based medicine.

RESULTS: The response rate was 63%. Core medicine skills (clinical reasoning skills, differential diagnosis skills, physical exam skills, interviewing skills, and doctor-patient interaction skills) were scored highest, with all EBM skills rating at least 1 point lower: average core skill score 4.4 vs. average EBM skill score 3.5 ($p < 0.001$). None of the EBM skills were listed in the five most important skills. EBM skills were rated higher among younger faculty ($p=0.01$) and those who had previously taken an EBM course ($p=0.03$), but generalists and subspecialists rated the importance of EBM skills similarly ($p=0.56$). However, generalists had slightly more positive attitudes about EBM ($p=0.014$). Neither year of graduation nor exposure to an EBM course predicted faculty attitudes toward EBM.

CONCLUSION: Contrary to our initial hypothesis, generalists and subspecialists differ little in their attitudes about EBM and in their ratings of the importance of EBM skills in residency training. Although both groups rated EBM skills positively, the skills were prioritized much lower than all core skills. Perhaps EBM skills are not routinely reinforced outside formal teaching settings because they are not a high priority among academic faculty-generalists and subspecialists alike.

THE EFFECTIVENESS OF AN INTERNET-BASED TUTORIAL IN IMPROVING PRIMARY CARE PHYSICIANS SKIN CANCER TRIAGE SKILLS. B. Gerber¹, A. Bronstone¹, T. Maurer¹, T. Berger¹, S. Mcphee¹, N. Caspers¹; ¹Univ. of California, San Francisco, CA

PURPOSE: Few studies have examined the effectiveness of Internet-based medical educational programs for physicians. We wanted to determine whether a skin cancer triage intervention, developed and proved successful in a face-to-face, on-site application, could be effectively delivered over the Internet with the same positive results.

METHODS: Physicians were randomly assigned to an Intervention or Control Group. Intervention Group physicians completed modules which included a Pretest, Pretest Scores with Individualized Feedback, Instruction (text and hyperlinks to digitized images of skin lesions), and two Posttests. Control Group physicians completed the Pretest and Posttest I. The program could be completed in 3-4 hours, at participants' convenience. There were 14 outcome measures including diagnosis and evaluation planning for malignant melanoma, basal cell carcinoma, and squamous cell carcinoma.

RESULTS: 71 physicians completed the program through Posttest 1, and 46 (27 of 39 in the Intervention Group) completed the entire program. The Intervention Group showed significantly greater improvement than the Control Group in 9 of 14 outcome measures. By Posttest II, Intervention Group improvement was maintained in 5 of the 9 outcomes.

CONCLUSION: Our brief, Internet-based, educational intervention with individualized feedback was successful in improving the skin cancer diagnosis and evaluation planning test performance of primary care physicians.

MEDICAL STUDENTS AS AGENTS OF QUALITY IMPROVEMENT IN PRIVATE PRACTICE - CLINICAL AND ATTITUDINAL OUTCOMES. B.E. Gould¹, M. Grey¹, L. Abrahamson², C. Gruman³, E. Storey¹, C.G. Huntington², T. Van Hoof³, H. Karen³, M. Szydlowski⁴, M. Ferreira⁴, A.M. Conaty¹; ¹University of Connecticut, Farmington, CT; ²Central CT Area Health Education (AHEC) Program, Hartford, CT; ³Braceland Center for Mental Health and Aging, Hartford, CT; ⁴Qualidigm, Middletown, CT

PURPOSE: The IOM's report on medical errors focused attention on the importance of measuring quality and outcomes. As part of the UME-21 project funded by HRSA, the UCSOM developed a Quality Improvement Curriculum. This abstract examines the impact of student CQI projects on the quality of care delivered at community practices.

METHODS: Eighty second year students working in groups of 3-4, initiated a CQI study on Diabetes Mellitus at 23 community-based primary care practices collecting baseline data, implementing a results-specific intervention and re-measuring quality indicators 8 months later. Students attend community-based primary care continuity clinics one-half day/week, year I-III. Students were evaluated using Likert-scale rated attitudinal items as well as open ended questions.

RESULTS: 512 charts were abstracted for the baseline sample with 383 charts abstracted post-intervention. The number of documented foot and eye exams increased significantly from baseline to remeasurement (51% to 67.8%; $p < .001$ and 26.4% to 35.5%; $p = 0.003$, respectively). The mean value for glycohemoglobin dropped from 7.71% at baseline to 7.2% at remeasurement ($p = .0001$). Analysis of student attitudes revealed acknowledgement of the benefit of outcomes measurement in clinical practice despite frustration with the tedium of the chart abstraction process. Feedback was used to modify curriculum design for future classes.

CONCLUSION: Medical student-driven CQI projects can have significant impact on improving the quality of care at the practices in which they participate while introducing them to the process of quality measurement and improvement. Formative input from students should be used to optimize CQI experiences. The use of medical students to lead CQI efforts in private practices may represent an underutilized resource in our efforts to improve the quality of care provided to the public.

THE EFFECTS OF ENHANCED CAPITATION ON PRECEPTORS' ATTITUDES TOWARDS STUDENT TRAINING AND TOWARDS MANAGED CARE'S SUPPORT OF TEACHING MEDICAL STUDENTS. M.S. Grayson¹, M. Klein², R.H. Bernstein³, P. Visintainer²; ¹Saint Vincent's Hospital and Medical Center, NY, NY; ²New York Medical College, Valhalla, New York; ³Mount Sinai School of Medicine, Gladyne, PA

PURPOSE: New York Medical College (NYMC) and Aetna U.S. Health Care (AUSHC) launched the first program to financially support community-based teaching by enhancing the capitation level of primary care preceptors. We studied the effects of these enhanced capitation payments on preceptors' attitudes towards student training and towards managed care's support of teaching medical students.

METHODS: Participants were community-based preceptors who completed a questionnaire consisting of sections on attitudes towards teaching and attitudes towards managed care, given prior to the distribution of enhanced capitation payments (Time 1) and after receiving payments (Time 2). The instrument was given to three groups: AUSHC preceptors receiving enhanced capitation (86/93, 92.5% response rate) and control groups consisting of AUSHC preceptors who chose not to participate in the enhanced capitation program (57/66, 86.4% response rate) and non-AUSHC preceptors (62/68, 91.2% response rate). Responses were scored using a six-point Likert scale ranging from Strongly Disagree (1) to Strongly Agree (6). Statistical tests employed included chi-square for differences between the respondent groups, and principal factor analysis and t-tests for differences between Time 1 and Time 2.

RESULTS: Factor analysis identified three underlying factors: 1.) Attitudes towards managed care support of physicians, which became less positive, decreased from 3.58 to 3.43 ($p = 0.28$); 2.) Attitudes regarding the impact of teaching on their practices, remained positive, although slightly decreasing from 4.70 to 4.57 ($p = .043$); 3.) Attitudes towards their roles as trainers, which was very positive, remained unchanged (5.06 to 5.06, $p = .956$). No significant difference was found in attitudes when comparing those who received enhanced capitation to the control groups.

CONCLUSION: While the enhanced capitation program has provided community-based preceptors with additional financial support, it did not improve preceptors' attitudes concerning managed care's support of their teaching activities. We believe this program is an important first

step but more is required. Just as attitudes towards managed care are the result of many factors, initiatives to change those attitudes should be multifactorial.

HOW WE LEARN: MEDICAL RESIDENTS' EVALUATION OF DIFFERENT EDUCATION MODALITIES. E.H. Green¹, R. Granieri¹; ¹University of Pittsburgh School of Medicine, Pittsburgh, PA

PURPOSE: Internal medicine residency training involves exposure to different learning modalities including rounds, lectures, small group discussion, and independent study. Little is known about learners' views regarding different educational interventions. We administered a questionnaire to assess current medical residents' attitudes towards different learning modalities.

METHODS: Second and third year medical residents from an academic residency program and closely affiliated community residency program were surveyed ($n = 91$). Respondents were asked to rate, using a Likert scale and list ranking, the contribution of various educational activities to their understanding of clinical medicine, preparation for standardized tests, and overall education. Subgroup analysis was done according to level of training, career goals (primary care or subspecialty medicine; academic or community), gender, college major, and additional education.

RESULTS: A total of 50 residents (55%) responded. "Noon conference" (core lecture series) and morning report were highly valued, with 82 and 84% of residents rating these endeavors as moderately or very important to their overall education. Attending rounds were less important, with 68% rating them as moderately or very important. Further, 44% of residents felt "too much" time was devoted to attending rounds. Medical grand rounds had the least contribution to residency education with 33% rating it as moderately or very important. In contrast, 92% of residents rated independent learning with as moderately or very important. Independent study was the only educational activity that residents thought prepared them equally well for clinical medicine and standardized exams. Residents actively pursued independent study with a mean of 12.69 topics researched in the 3 months prior to the study. However, 52% of respondents felt they had "too little" time for this endeavor. Independent study and morning report were independently identified as the two most important educational activities during residency. The only significant finding from subgroup analysis was that residents with advanced degrees valued independent study more than their colleagues ($p = .038$).

CONCLUSION: In this study, internal medicine residents valued lectures and morning report over attending rounds or medical grand rounds. However, independent study is the most highly valued educational method, and house officers are very involved in this despite a perceived lack of time. Medical residents seem to be mastering the principles of adult, independent learning; future curricular reform should consider incorporating more time and opportunities for independent study.

HOUSESTAFF NON-VERBAL COMMUNICATION SKILLS AND PATIENT SATISFACTION. C.H. Griffith¹, J.F. Wilson¹, S. Langer², S. Haist¹; ¹University of Kentucky, Lexington, KY; ²University of Washington, Seattle, Washington

PURPOSE: A surprising limitation of studies of physician communication is that they tend either to focus on what was said (standardized patient studies, where interview "quality" is judged by a list of items asked or not asked), or how it was said (studies of various physician communication skills, both verbal and nonverbal behaviors), but studies do not investigate both what was said and how it was said. Our project's specific aim was to examine the association of nonverbal communication with patient satisfaction in the context of the "quality" of information provided or collected.

METHODS: We recruited 59 residents to agree to be videotaped in 3 different clinical encounters, encounters which we felt might require different skills. The scenarios included: 1) a straightforward primarily "medical" problem [chest pain]; 2) a patient with more psychosocial overlay [a depressed patient with a history of sexual abuse]; and 3) an encounter of primarily counselling [HIV risk factor reduction counselling]. Videotapes were then coded by trained raters for physician nonverbal behaviors (eg. body lean, open vs. closed body position, eye contact, smiling, tone of voice, nod, facial expressions, hands) in the three encounters. Using multiple regression approaches, we investigated the association of physician nonverbal behaviors with patient satisfaction in the context of the "quality" of the interview (SP checklist performance, measures of verbal communication skills), controlling for physician characteristics (gender, postgraduate year, etc).

RESULTS: Nonverbal communication skills independently predicted patient satisfaction to a varying degree for all 3 patient stations. The effect sizes were substantial, with nonverbal communication predicting 34% of the variance in patient satisfaction for the depression-sexual abuse station, 32% of the variance for the HIV counselling station, and 30% of the variance for the chest pain station, all highly statistically significant. Verbal communication skills predicted 12-16% of the variance in patient satisfaction. Percentage of checklist items was a significant predictor of patient satisfaction in the chest pain station only (4% variance).

CONCLUSION: Nonverbal communication is the primary determinant of patient satisfaction in a variety of different types of clinical encounters.

IN THE PALM OF YOUR HAND: THE USE OF PERSONAL DIGITAL ASSISTANTS AMONG MEDICAL HOUSESTAFF. P.S. Hahn¹, C. Kwee¹, F. Pelzman¹; ¹New York Presbyterian Hospital, Weill Medical College of Cornell University, New York, NY

PURPOSE: Rapid advances in health care technology have resulted in the prolific adoption of personal digital assistants (PDA's) among health care providers. The purpose of this study is to obtain objective data on PDA ownership demographics and application usage among internal medicine residents. Furthermore, the study examines the attitudes regarding the potential of PDA's to improve patient care, prevent medical errors, promote resident efficiency, and enhance medical education.

METHODS: The study was conducted as a non-blinded survey of the 133 members of the internal medicine residency program at New York-Presbyterian Hospital. The two-page surveys requested the following information: name and demographics; PDA model details; applications

used; whether PDA's replaced medical texts; and subjective responses regarding the impact of PDA's on patient care and medical education. 114 of the 133 surveys were returned (86% response rate).

RESULTS: 62% of medical residents owned PDA's (100%-based on the Palm operating system). The mean age of the cohort was 28 years, and 55% of PDA owners were male. Ownership was inversely correlated with both age and level of training. The most commonly-used medical applications were drug formulary references, medical formula calculators, medical note-taking, and reference texts. Interestingly, only 15% of PDA owners replaced their paper-based drug references with the PDA version. Relatively few residents used their PDA's for patient tracking, appointment scheduling, or to log procedures. The most-commonly used non-medical applications included the built-in address book, memo pad, to-do list, calculator, local city guides, and games. 83% of owners felt that PDA's improve efficiency, and a majority felt that they were useful in medical education. 69% responded favorably when asked if PDA's improve patient care, but only 48% believed that PDA's decrease medical errors. 37% of non-owners were committed to purchasing a PDA within 12 months.

CONCLUSION: The majority of internal medicine house officers at New York-Presbyterian Hospital already own PDA's. Most residents use drug references and medical calculators, perhaps because they provide point-of-care answers to direct questions. The majority of residents believe that PDA's are useful for improving efficiency, medical education, and patient care. However, more research is required to correlate these positive perceptions with objective improvements in clinical outcomes, as well as broadening the applicability of the results to other specialties. Furthermore, as PDA's and the availability of medical applications continue to proliferate, issues of medical privacy and quality-assurance of medical content will become increasingly important.

EFFECTS OF INCORPORATING ACTIVE LEARNING STRATEGIES INTO A RESIDENTS' LECTURE: RESULTS OF A CONTROLLED TRIAL. P. Haidet¹, B.F. Richards², R. Morgan¹, K. Wristers¹, B. Moran²; ¹ Houston VAMC/Baylor College of Medicine, Houston, TX; ²Baylor College of Medicine, Houston, TX

PURPOSE: Despite calls to incorporate active learning strategies into medical education, many teachers believe that efficient delivery of content can only be achieved through didactic lectures. The goal of this study was to examine the effects of reducing didactic lecturing and adding small group work, a commonly employed active learning strategy, on learner participation, attitudes, and knowledge acquisition.

METHODS: We recruited residents in internal medicine, pediatrics, and family medicine to attend a session on effective use of diagnostic tests, and assigned them to one of two groups. The control group received a standard 'slide and lecture' session where the lecturer spent a full hour delivering content. The intervention group received a session where residents were assigned to groups of 4-5 and completed a series of 3 small-group tasks with large-group lecturer-facilitated discussion between tasks; in this session, the lecturer delivered content only during large group discussions (approx. 30 minutes). Both sessions used the same lecturer and covered the same content. We measured resident demographics before and self-perception of active learning after both sessions. We measured attitudes toward the session content and knowledge before and after both sessions. We also employed trained observers, blinded to the study purpose, who used a standardized instrument to observe resident patterns of behavior during each session.

RESULTS: 27 residents completed the control session and 36 completed the intervention session. The study groups did not differ significantly in demographic variables, baseline attitudes about, or baseline knowledge of the session content. As measured by the trained observers, during the intervention session more students were communicating with other students ($p < .001$) than during the control session. Students in the intervention group also self-rated their participation in the session higher ($p=.001$). While both groups improved in both knowledge and attitude scores from pre- to post-session ($p < .001$ for all comparisons), there were no significant differences in the amount of improvement between groups.

CONCLUSION: In these sessions conducted in a controlled environment, we reduced the time spent 'lecturing' by the teacher by 50 percent and covered the same amount of content with no detrimental effects on knowledge acquisition or attitude enhancement. In addition, our incorporation of active learning strategies produced residents who both self-rated and were observed to be more actively engaged with each other and with the learning process. Our future work will measure the effects of this active engagement on long-term retention of content and attitudes.

MEDICAL STUDENT ATTITUDES TOWARD PATIENT-CENTERED CARE AND PATIENTS' PERCEPTIONS OF HUMANISM. P. Haidet¹, J.E. Dains², D.A. Patemiti³, T. Chang², E. Tseng², J.C. Rogers²; ¹ Houston VAMC / Baylor College of Medicine, Houston, TX; ² Baylor College of Medicine, Houston, Texas; ³University of California, Davis School of Medicine

PURPOSE: Existing data suggests that physicians who demonstrate patient-centered behaviors enjoy high patient satisfaction. However, little data exists to link physician attitudes about medical care with patient-centered outcomes. In this study, we explored associations between student physicians' attitudes toward the physician-patient relationship and patients' perceptions of students' humanism during the medical interaction.

METHODS: At the beginning of a primary care (PC) clerkship, we surveyed 293 third-year students using the Patient-Practitioner Orientation Scale (PPOS), a validated instrument that measures attitudes toward the physician-patient relationship. PPOS scores range from patient-centered (egalitarian, whole person-oriented) to physician-centered (paternalistic, less attuned to psychosocial issues). During the PC clerkship, all students performed a medical interview with a series of 5 standardized patients (SPs) who were trained to portray varying biopsychosocial issues. All SPs were unaware of the students' PPOS scores. After each interaction, each SP completed a 7-item validated instrument that measured their perceptions of the student physician's humanism. We calculated a total humanism score for each student and compared these to students' attitudes as measured by PPOS.

RESULTS: Mean humanism scores were lowest for students who scored in the most physician-centered quartile (79.6 ± 9.4) and highest for students who scored in the most patient-centered quartile (83.4 ± 7.0) on PPOS. In linear regression analysis controlling for month of the third-year on the primary care clerkship, patient-centered PPOS scores ($p=.01$) and female gender ($p=.05$) were found to be associated with higher total humanism scores.

CONCLUSION: Student physicians with more patient-centered attitudes demonstrated higher degrees of humanism as perceived by standardized patients. Patients' perceptions of humanism were also influenced by student gender. Further research is needed to explore the nature of associations between physicians' attitudes and patients' perceptions of care, as well as the impact of medical education on students' attitudes toward the physician-patient relationship.

A CONTROLLED STUDY OF PELVIC EXAMINATION TRAINING FOR INTERNS. J.E. Hebers¹, L. Wessel², J. El-bayoumi³, J.E. St. Onge⁴; ¹ VA Medical Center, Washington, DC; ² Women's Health Consultants, Washington, VA; ³ George Washington University, Washington, DC; ⁴ Georgetown University, Washington, DC

PURPOSE: Although training in the skills required to perform pelvic examinations is ubiquitous in medical schools, few residency programs assess or reinforce these skills, and only limited evidence justifies the substantial cost of post-graduate training. We report results of a controlled study of training for Internal Medicine interns utilizing professional instructors who also serve as models for the exam.

METHODS: 48 interns from two university Internal Medicine residencies completed questionnaires about their experiences with pelvic examinations. 23 participated in a two-hour training session after which their skills were assessed. At follow-up four months later, the skills of 13 of the trained interns were compared with that of 24 interns who had not been trained. Assessments were based on the observation of verifiable behaviors. Chi-square tests were used to assess the statistical significance of comparisons. Inter-rater reliability was assessed with the kappa statistic.

RESULTS: Mean age of the 37 interns in the controlled study was 28.4 (range 24-40), with 57% women and 89% graduates of U.S. medical schools. 70% of the interns described having received specific pelvic exam skills training in medical school, 78% felt comfortable performing the exam, and 91% felt they give patients clear instructions.

Interns who participated in the current training displayed 10 of 12 key behaviors more frequently when compared with interns who had not been trained. Overall, trained interns demonstrated 87% of key behaviors, compared with 63% for untrained interns. Specific items for which the largest improvements occurred were in providing an outline of the exam (100% for trained interns vs. 38% for those not trained) and explicitly giving permission to ask questions (69% vs. 17%; $p < 0.01$ for all three comparisons). Independent video and in-person observers agreed well with ratings of instructors (kappas > 0.7).

CONCLUSION: Most interns described exposure to medical school pelvic examination training, but many overestimated their skills. A program employing professional trainers can reliably evaluate and improve key skills and improvements are demonstrable months after training. Further work is needed to demonstrate consistent benefits in larger groups of interns and residents, to ascertain whether training efficiency could be improved, and to measure the impact of training on subsequent encounters with actual patients.

THE MINI-CLINICAL EVALUATION EXERCISE AND INTERACTIVE FEEDBACK: PRELIMINARY RESULTS. E. Holmboe¹, F.K. Williams², M. Yepes², J.J. Norcini³, L.L. Blank³, S. Huot⁴; ¹ Yale University, Cheshire, CT; ² George Washington University, Washington, DC; ³ American Board of Internal Medicine, Philadelphia, PA; ⁴ Yale University, New Haven, Connecticut

BACKGROUND: The American Board of Internal Medicine recently introduced the mini-Clinical Evaluation Exercise (mini-CEX) as an evaluation tool using observation of housestaff clinical performance by faculty. The mini-CEX can facilitate interactive feedback (FB) between faculty and residents to help residents correct deficiencies and reinforce good clinical skills. However, little is known about the nature of the FB that results after a mini-CEX is completed, and whether the FB is interactive.

PURPOSE: To assess how frequently interactive FB is given after completion of a mini-CEX with PGY-1s in an outpatient setting.

METHODS: Prospective cohort study of mini-CEXs performed with PGY-1s in the outpatient setting at three internal medicine residency programs: Yale University Primary Care, Washington Hospital Center, and the National Naval Medical Center. Feedback sessions were audiotaped after completion of the mini-CEX. All tapes were transcribed with all identifying information deleted. Transcripts were independently coded by two investigators (ESH, SH). Disagreements were resolved by consensus. Interactive FB was defined as: a) any recommendation(s) given to the PGY-1s; b) PGY-1s were asked to self-assess their mini-CEX performance (self-assessment); c) PGY-1s were given an opportunity to react to the FB (learner reaction); d) and faculty stated a specific plan to change PGY-1's future performance (action plan).

RESULTS: For this preliminary analysis, 30 attendings provided FB sessions to 41 different PGY-1s at the 3 residency programs. Fifty sessions have been transcribed and analyzed. Forty-five (95%) FB sessions included at least one recommendation. The median number of recommendations per session was 2 (range 0-9). Recommendations were given most commonly for communication skills, notably patient interviewing (31 sessions, 62%) and counseling (21 sessions, 42%). Feedback on components of the physical exam was given in 13 sessions (26%). Attendings asked PGY-1s to self-assess performance in 19 sessions (38%) and to specifically respond to the FB (learner reaction) in 32 sessions (64%). Despite the high frequency of recommendations, only 3 sessions (6%) included an attending's discussion for an action plan to help the PGY-1 correct noted deficiencies.

CONCLUSION: For this cohort, the mini-CEX effectively generated recommendations to improve clinical performance, especially in communication skills, and the majority of attendings asked PGY-1s to react to the FB. However, programs planning to use the mini-

CEX should also consider strategies to encourage their faculty to provide more FB on other observed skills, promote more self-assessment, and establish definitive action plans to correct deficiencies.

PREDICTORS OF PERCEIVED CLINICAL TEACHING EFFECTIVENESS. A.L. Hull¹, H. Copeland¹, M.G. Hewson¹; ¹ Cleveland Clinic Foundation, Cleveland, OH

PURPOSE: Clinicians' self-perceptions of teaching effectiveness will affect their interest and commitment to improving teaching ability. This IRB-approved study investigates the relationships among self-perception and characteristics of clinical teachers including demographics, interest and satisfaction with teaching, and perceived institutional support for teaching.

METHODS: Perceptions, attitudes, and characteristics of clinical teachers was obtained in a survey instrument sent to 164 Division of Medicine faculty who had taught and had teaching evaluation instruments completed Internal Medicine residents in 1998 and 1999. 94 (57%) of the surveys were completed after 2 reminders. The responses were coded and matched to the respondents' average of a 15-item clinical teaching evaluation (which has been judged to be reliable and valid) completed by residents for any rotation during the 2-year period. The dependent measure was a 5-point Likert-scale survey item evaluating the self-perceived teaching effectiveness (TE). The independent measures included faculty characteristics (age, gender, years in practice), satisfaction with teaching (enjoy teaching, career satisfaction, perceived institutional support for teaching), percent time spent teaching, involvement in teaching improvement activities (workshops, reading), and average of self-ratings of the 15-item clinical teaching evaluation. SPSS/PC was used to compute descriptive statistics, correlations between the dependent and independent measures, and a step-wise regression. Self-ratings of the 15-item teaching evaluation were not included in the regression analysis in order to maximize the effects of the other independent measures.

RESULTS: The average TE score was 3.85 ($\pm .63$). Statistically significant positive correlations ($p < .05$) were found between TE and average resident evaluation score ($r=.213$), percent time teaching ($r=.251$), average self-rating of the 15-item teaching evaluation ($r=.457$), and satisfaction from teaching residents ($r=.336$). All other characteristics were not significantly correlated with TE. Step-wise regression only resulted in reported enjoyment of teaching residents predicting TE ($R=.414$ [difference in N causes difference between r and R]).

CONCLUSION: Internal Medicine faculty believe that they effectively teach residents, and self-perceptions of teaching effectiveness are correlated with actual resident evaluations of teaching (suggesting accuracy of the faculty member's perception), and time spent and enjoyment from teaching (faculty self-select activities that they enjoy and are good at). Faculty self-perceptions in general are not correlated with demographic characteristics, effort at improving educational competence, and perceived support for teaching, suggesting that faculty development efforts may need to be more individualized and focused on faculty who elect to participate rather than institution or department-wide programs.

RESIDENTS AS TEACHERS. R.W. Janick¹, M.D. Schwartz¹, A. Kalet¹, S. Zabar¹; ¹ New York University School of Medicine, New York, New York

PURPOSE: We sought to measure residents' attitudes, preparation, and confidence about teaching and how these relate to their plans to teach.

METHODS: We conducted an anonymous, cross-sectional survey of Medicine and Psychiatry residents at NYU. The questionnaire assessed attitudes and preparedness to teach, self-perceived teaching knowledge and skills, and plans to teach.

RESULTS: Sixty-one residents (42 Psychiatry, 19 Internal Medicine) completed the questionnaire (79% response). Factor analysis confirmed scales for measuring residents' attitudes (Teaching Enthusiasm, Important Role in Department), preparedness to teach (Teaching Exposure, Teaching Capability) and self-perceived teaching knowledge and skills (Clinical Reasoning, Patient Management, and Procedural Skills). Scales were dichotomized to present the % of residents with high scores: Teaching Enthusiasm - 67% agree; Important Role in Department - 34% agree; Teaching Exposure - only 7% reported high amount; Teaching Capability - 23% high confidence; Knowledge of Clinical Reasoning - 30% confident; Patient Management - 49% confident; and Procedural Skills - 21% confident. When asked the frequency with which they plan to teach, 75% planned to directly observe learners, 71% planned to give feedback, and 74% planned to teach with the patient present, all at least once per week. Plans to teach was not associated with resident age, gender, year of training, or department. Nearly significant trends suggested that residents with high teaching enthusiasm were more likely to plan to observe their learners and those with high prior teaching exposure were more likely to plan to give feedback. Teaching capability and all 3 knowledge variables were correlated with plans to teach with the patient present ($p < .05$).

CONCLUSION: Residents value and look forward to their teaching roles despite having little preparation for teaching, and only modest confidence in their ability and knowledge to teach. Resident attitudes and self-perceived capabilities influenced their plans to teach. Programs to prepare residents as teachers that address attitudes, skills, and knowledge in these areas may lead to more teaching by residents.

CONVERTING UNITED STATES MEDICAL LICENSING EXAM SCORES INTO PERCENTILES. R. Jones¹, N. Desbiens¹; ¹University of Tennessee College of Medicine, Chattanooga Unit, Chattanooga, TN

PURPOSE: As opposed to the Comprehensive Osteopathic Medical Licensing Exam (COMLEX), the United States Medical Licensing Exam (USMLE) board does not convert scores into percentiles for its examinations to residency program directors or candidates, citing errors in interpretation when percentiles are reported. Nonetheless, percentiles are very important for program directors to allow them to better rank applicants.

METHODS: Scores on the USMLE exam were assumed to be normally distributed and scores and standard deviations for recent examinations reported to residency program applicants were used to generate tables of percentiles using the following S-Plus (Mathsoft, Inc.) code:

```
score <- 175:260
percentile <- pnorm(score,215,20)*
rows <- cbind(score,percentile)
rows
```

*205 was used for Step II

RESULTS: Score Part I %ile Part II %ile

180	4.0	10.6
200	22.7	40.1
220	59.9	77.3
240	89.4	96.0

CONCLUSION: Approximate percentiles for USMLE Parts I and II can be generated based on data provided to candidates by USMLE. However, USMLE should publish exact percentiles and instruct program directors and candidates how to use them wisely.

SPIRITUALITY EDUCATION OF PHYSICIANS: EXPERIENCES AND ATTITUDES. A.M. Kiloze¹, M. Franco², D. Bynum³, M. Monroe⁴; ¹ University of Florida, Gainesville, Florida; ² East Carolina University, Greenville, North Carolina; ³ University of North Carolina Faculty Development Program, Chapel Hill, North Carolina; ⁴ Carolinas Medical Center

PURPOSE: To determine what types and in what setting physicians received spirituality education. To examine physician attitudes on further training and the characteristics of physicians who desire further training.

METHODS: A spirituality/religiosity survey including a Spirituality Well Being Scale was administered to physicians at six centers four in NC, one in FL and another in VT.

RESULTS: Four hundred-forty four physicians responded to the questionnaire of which 74% were general internists, 24% family practitioners and 5% "other medical specialties." Forty percent of respondents received some form of spirituality training. Twenty-nine percent received this training while medical students compared to 18% as residents or as fellows and 23% as attending physicians. The training received in medical school was largely in the form of small group lecture series, but during residency and fellowship, the training consisted of informal clinic discussions. Attendings primarily received training through self-education (books and journals). Forty-two percent of respondents agreed that they would desire further spirituality training while 20% disagreed, and 38% were neutral. The physicians scoring highest (greater than 4 out of a possible 6 on the Spiritual Well-Being Scale) were more likely to desire further training. In addition, those physicians who attended worship service (48%) and those who agree to the statement "faith alone could cure disease" (60%) were more likely to desire further training (all p-values < .05). Also family physicians had a greater tendency to desire further training compared to general internists.

CONCLUSION: Formal spirituality training related to medical care is sparse in medical school and almost non-existent thereafter. Although physicians with stronger religious predilections desire formal training, nearly half of all primary care physicians who returned the survey felt that further training in spirituality and medicine was desirable. A curriculum concerning patient spirituality and medicine at the medical school and residency level appears to be warranted.

ADHERENCE TO COMMON GUIDELINES: IS THERE A TRAINING EFFECT? J.R. Kogan¹, E.E. Reynolds², J.A. Shea¹; ¹ University of Pennsylvania, Philadelphia, PA; ² Beth Israel, Boston, Massachusetts

PURPOSE: Chart audits are used to assess physician performance and quality of care. Typically, records are reviewed for practicing physicians. Only recently have resident report cards been described. The purpose of this study was to compare performances of interns, senior residents, and faculty on adherence to national guidelines regarding prevention and management of common outpatient medical conditions.

METHODS: The study was conducted at the outpatient practices of a university based internal medicine training program. For interns (n = 50), rosters of all new general medicine patients with 2 or more visits between July 1999 and April 2000 were generated. For senior residents (n = 39) and faculty (n = 30), the time frame extended from July 1998 to April 2000. A random sample of 9 patients per physician was identified. All visits with the designated provider through June 2000 were abstracted by nurse abstractors using a 115 item form. 91% (n = 965) of targeted charts were abstracted: 366 for interns, 349 for senior residents and 246 for faculty. Performance scores (the percent of indicated actions taken) were determined for each chart in the areas of cancer screening, immunizations, counseling, management of diabetes (DM), hypertension (HTN), asthma, coronary disease (CAD), and depression. Differences in intern, resident and faculty performance were assessed with ANCOVA and correlations.

RESULTS: The mean patient age was 52.9 (sd=17.6); 68% were women. Patients had a mean of 0.77 comorbidities and 4.1 visits with the provider. These factors were covariates in the ANCOVA due to group differences. Faculty performed better than interns and senior residents overall (p = .03) and in the areas of counseling (p = .0007), immunizations (p = .003) and HTN (p = .002). Unadjusted mean performance scores for interns, senior residents and faculty were as follows: overall (46%,49%,52%), counseling (26%,27%,37%), immunizations (36%,37%,43%), and HTN (44%,50%,57%). There were no differences between intern, resident and faculty performance in the other areas. Mean performances were: cancer screening (66%), management of DM (48%), asthma (56%), CAD (67%), and depression (41%). Higher performance scores were associated with more visits (r = .19, p=.0001), fewer comorbidities (r = -.08, p = .02) and younger age (r = -.12, p = .0002).

CONCLUSION: On only a few domains did performance improve with experience. In those domains, there was steady improvement through training. Average performance rates fell short of national guidelines for all groups, though some actions were likely not documented. Future

efforts should explore how to improve adherence in training programs, including among the teaching faculty.

IMPACT OF A NIGHT FLOAT SYSTEM ON HOUSESTAFF SATISFACTION. S. Kripalani¹, M.V. Williams¹; ¹ Emory University, Atlanta, GA

PURPOSE: In an effort to reduce workload, a night float (NF) rotation was established in July 1999. Three of twelve ward teams were on call each day and stopped admitting patients at 2300. They remained overnight to complete work-ups and cover their own service. From 2300 until 0700, NF teams performed all medical admissions and covered the other 9 teams. Two or three NF teams (consisting of an intern and resident) worked each night. In the morning, they handed-off the new patients to ward teams and attended a teaching conference before going home. We conducted this study to evaluate the impact of the NF system on satisfaction and quality of care.

METHODS: Ward residents (WR) and night float staff (NFS) completed a 2-page questionnaire at the end of each month from July 1999 to April 2000. Survey questions focused on the following domains: workload, educational activities, morale, and quality and continuity of care.

RESULTS: The overall response rate was 81% (n=92 WR and 51 NFS). WR got an average of 4.5 hours of sleep on call, compared with 2.4 hours the previous year, $p < .0001$. Nearly all (89%) felt more rested post-call, and 71% thought they could take better care of patients during the day because of the NF system. According to 87% of WR, the workload was significantly better, and 63% had more time for educational activities. Approximately 90% agreed that NF had improved housestaff morale and the overall experience of the ward rotation. Regarding the NF admissions, most WR appeared satisfied with the quality of history and physicals (63%), choice of diagnostic testing (72%), and initial treatment (71%). However, many WR disagreed with the assessment and changed the treatment plan about 25% of the time. One-third expressed some concern that continuity may suffer when patients are handed-off in the morning, but only 2% felt that overall patient care suffered, and 94% were glad that the NF system was in place. Each NF team received a mean of 3.9 admissions per night (range 2-11) and a median of 4.8 cross-cover calls, which required about 45 minutes to answer. Nearly 95% of NFS agreed that the overall workload and number of admissions were reasonable. However, only 66% were satisfied with the amount of learning/teaching on the NF rotation. Working nights for a month was difficult for 67% of NFS, and 80% felt it affected their social life. Daytime sleep was often unrefreshing, afternoon continuity clinics disrupted sleeping schedules, and NFS had limited time with friends and family. Nevertheless, 90% agreed that the month was a positive experience overall, and 96% were glad that the system was in place.

CONCLUSION: Residents on the ward teams reported improved morale, increased sleep, more time to read and attend conferences, and better quality of care delivery. In spite of complaints about the nocturnal lifestyle and educational content of the rotation, housestaff on the NF teams favored the system as well.

ASSESSING HAND HELD COMPUTER USE AMONG INTERNAL MEDICINE HOUSESTAFF. W. Lee¹, G. Sylvan¹, S.J. Peterson²; ¹ Westchester Medical Center, Valhalla, NY; ² New York Medical College, New York, NY

PURPOSE: Hand held computers (HHC) have become increasingly popular in recent years. While there has been interest regarding the potential for HHC use in medical education, no studies have described HHC use among internal medicine housestaff. We describe here the prevalence and patterns of HHC use among internal medicine housestaff at three academic teaching hospitals.

METHODS: All internal medicine housestaff at three academic teaching hospitals in the New York metropolitan area were surveyed. Basic demographic data, type of HHC, the operating system (OS) used, medical purposes for which the HHC was used, as well as the type of medical software installed on the HHC was obtained.

RESULTS: 172 surveys were given, with 102 respondents (response rate 59.3%). 32 of the respondents (31.4%) owned a HHC, and of these, most were PGY I level (53.1%). PGY II's and PGY III's made up 15.6% and 31.2% of the HHC users respectively. The majority of HHC users used a Palm Pilot device (78.1%) as opposed to a Handspring Visor (9.4%), Casio E series (3.1%) or other HHC (9.4%). 93.5% of HHC users used the Palm OS as opposed to the Windows CE OS (3.2%) or Other OS (3.2%). Medical formulas and drug references (24.8% and 23.8% respectively) made up most of the medical uses among HHC users. Medical text references (5.7%), inpatient tracking (3.8%) and evidence based medicine (0.00%) were poorly utilized if at all.

CONCLUSION: Most internal medicine housestaff do not own a hand held computer. Among those who do own a HHC, the majority are PGY I level, and most use either a Palm Pilot HHC, or a HHC compatible with the Palm OS. HHC users appear to rely on their computers for references, and there is little to no use of HHC's for either medical knowledge or evidence based medicine. There exists great potential for increasing the utility of hand held computers among internal medicine housestaff for medical education, particularly in the realms of medical knowledge and evidence based medicine.

TEACHING AND TRACKING COMMUNICATION SKILLS WITH THE CALGARY-CAMBRIDGE REFERENCED OBSERVATION GUIDES IN AN INTERNAL MEDICINE RESIDENCY. B.R. Leslie¹, R. Gonzales¹, L.J. Adams¹; ¹University of Colorado Health Sciences Center, Denver, Colorado

PURPOSE: Physician-patient communication skills are an essential part of medical practice. We implemented a clinic-based coaching model using rating forms and observed interviews. The purpose of this study is to describe the use and modification of the rating scales for coaching purposes, and to discuss the variations among specific communication skills by resident year and gender.

METHODS: The Calgary-Cambridge Referenced Observation Guides were selected as a teaching tool because of their specificity and ease of use during observed interviews. To provide

skill discrimination for each rating item, a 5 point Likert scale was created for each item ranging from poor to excellent performance. After obtaining patient consent, resident physicians were shadowed by the residency behavioral scientist (BRL) for one clinic afternoon or morning, and their skills were rated using the instrument. 25 resident physicians were observed for a total of 48 interviews. Likert scores were dichotomized into excellent (5) and less than excellent (1-4) categories for each item; and mean scores generated for the following domains: Initiating the session (5 items); Exploration of problems (7 items); Understanding the patient's perspective (5 items); Structuring the consultation (4 items); Building the relationship (7 items); and Explaining and planning (5 items). General Linear Models procedures were used to compare mean differences in domain scores by gender and year of resident training (R1, R2, R3).

RESULTS: No significant gender differences were found for any of the domains ($p < .05$). Overall, the resident physicians demonstrated the least proficiency with the domain "Understanding the patient's perspective" ($p = 0.01$, mean domain score (MDS) = 1.07). When scores were analyzed by resident year, significant maturation effects were observed for: "Exploration of problems" ($p = 0.01$, MDS: R1 = 2.6, R2 = 4.2, R3 = 4.8); "Understanding the patient's perspective" ($p = 0.01$, MDS: R1 = 0.28, R2 = 1.23, R3 = 1.94); and "Explaining and planning" ($p = 0.02$, MDS: R1 = 1.5, R2 = 2.1, R3 = 3.4). Resident evaluations of the coaching were uniformly positive.

CONCLUSION: The development of physician-patient communication skills is an essential part of resident education. The modified Calgary-Cambridge guides provide data that identifies areas of general strengths and weaknesses in communication skills. Using this instrument, we saw maturation effects in domains that require facilitation of emotional expression and shared decision-making. These results can be used to focus educational efforts, and may be useful for evaluating the impact of interventions to improve physician-patient communication.

FACTORIAL VALIDATION OF AN EDUCATIONAL FRAMEWORK FOR EVALUATING CLINICAL TEACHERS IN THE OUTPATIENT SETTING. D.K. Litzelman¹, K.M. Skeff², G.A. Stratos²; ¹Indiana University, Indianapolis, IN; ²Stanford University, Stanford, CA

PURPOSE: To examine whether the construct validity of the seven-category Stanford Faculty Development Program (SFDP) educational framework, previously demonstrated for inpatient clinical teachers, could be replicated for outpatient clinical teachers.

METHODS: Using a 58-item clinical teaching evaluation instrument previously employed to determine the construct validity of the framework for inpatient clinical teachers, a split sample of 4702 evaluations completed by 550 students rating 450 teachers was evaluated using factor analytic and standard item-reduction techniques. Students completed evaluations for any university- or community-based outpatient teacher with whom a minimum of three half-day teaching encounters had occurred during a one-month internal medicine ambulatory block rotation. Results were replicated using the second half of the data. Generalizability analyses were used to quantitatively determine the number of evaluations needed to obtain a stable estimate of a teacher's performance using the reduced instrument.

RESULTS: As in the construct validity testing with inpatient clinical teachers, the "knowledge" items statistically combined with "self-directed learning" items and "learning climate" items held up as a discrete construct; however, items from "control of session", "communication of goals", "understanding and retention", "evaluation", and "feedback" clustered into one remaining "global" teaching construct. These three factors explained 67 % of the variance. Following item reduction, the three constructs were compressed to contain three items each. The items remaining in the "global" construct were all "communication of goals" items. Overall internal consistency of the reduced instrument was .90 with internal consistency of constructs ranging from .90-.97. The overall inter-item correlation of the reduced instrument was .55 for sample one and .54 for sample two (optimal inter-item correlations are approx =.6). A minimum of 5 evaluations are needed per teacher to obtain a generalizability coefficient > .90 using conservative estimates from both samples.

CONCLUSION: For students having limited contact with outpatient teachers, setting the climate, communicating goals, and stimulating students' future learning emerge as focal areas. Further study is needed to see if this three-category framework persists as viewed by trainees with more extensive contact with outpatient preceptors.

DOES RESIDENCY CURRICULAR CONTENT EFFECT CAREER SATISFACTION? A SURVEY OF RECENT INTERNAL MEDICINE AND FAMILY PRACTICE GRADUATES. R.M. Lubitz¹, V.L. Dacey¹, M.A. Smith¹, M. Maryam¹, J.A. Monroe¹; ¹St. Vincent Hospital, Indianapolis, IN

PURPOSE: Professional isolation, poor financial remuneration, and insufficient training to meet patient needs decreases career satisfaction, making rural and inner city areas less desirable for primary care practice. Changes to Internal Medicine (IM) and Family Practice (FP) residency curriculum have attempted to address health care disparities by better preparing physicians for generalist careers. The effect of these curricular changes on career satisfaction has not been examined.

METHODS: We surveyed all IM (n=128) and FP (n=74) physicians 1 to 10 years after graduating from a large Mid-west community teaching hospital. The survey assessed physician and practice demographics (i.e., practice type and location, community size, case mix), career satisfaction, resource availability, and adequacy of curricular content for practice.

RESULTS: Survey response was 84% (105 IM, 65 FP, p=ns). Respondents were primarily male (66%), Caucasian (88%), and in solo (13%), group (42%), or multi-specialty group practice (20%). Thirty-one percent practiced in a community of <50,000 residents (9% <10,000) and 28% noted some or all subspecialties were unavailable in their community. Most respondents (88.5%) were somewhat to highly satisfied with their present career, and 81% would choose medicine again as a career (78% IM, 85% FP, p=ns); those practicing in emergency room settings were least satisfied with their present career (p=0.04). Greater career satisfaction was related to meeting expected financial remuneration (p=0.046), satisfaction with training in palliative care (p < 0.01), faculty role modeling of compassion (p < 0.05), and support/assistance provided by residency faculty (p=0.001). Career satisfaction was not related to community size (p=0.6), subspecialty availability (p=0.7), computer/internet searching skills (p=0.11), or time

spent in ambulatory training ($p=0.3$). Training better focused to practice demands in differential diagnosis, neurology, orthopedics, and hematology were associated with greater career satisfaction (all $p<0.05$). Using logistic regression, controlling for physician demographics, greater career satisfaction was associated with meeting financial remuneration expectations ($OR=3.9$, $p=0.01$), and better training in differential diagnosis ($OR=5.8$, $p=0.03$), and literature appraisal ($OR=3.0$, $p=0.04$).

CONCLUSION: Results suggest that at this institution, nationally mandated curricular reform in several areas is associated with greater career satisfaction. Satisfaction was also related to residency faculty role modeling and support. Whether additional curricular reform further improves career satisfaction warrants investigation.

COMMUNICATION AT THE END-OF-LIFE: A NOVEL CURRICULUM IMPLEMENTED BY MEDICAL STUDENTS. J.W. Magnani¹, M.A. Minor¹, J.M. Aldrich¹; ¹ Stanford University School of Medicine, Stanford, CA

PURPOSE: The AAMC has mandated the inclusion of end-of-life (EOL) and palliative care (PC) in undergraduate medical curricula. A significant number of medical schools teach about death and dying, yet focus predominantly on ethical aspects rather than the development of clinical skills in EOL and PC. We report here on the design and implementation of an EOL curriculum directed towards specific communication skills. This curriculum was developed and initiated by students at Stanford Medical School.

METHODS: A needs assessment of advanced clinical students demonstrated little formal training in EOL communication skills, while a literature review surveyed current EOL curricula on communication. These efforts were used to design a case-based curriculum that underwent faculty review. Next, faculty with national or local experience with EOL and palliative care training contributed to case development. Twelve of the faculty attended an orientation/training session on teaching the curriculum and then taught it in pairs to six groups of 10-14 medical students per group.

RESULTS: The four-hour curriculum described here was taught to students at the start of their clinical training. Topics entailed breaking bad news and discussing treatment options in the setting of life-limiting illness. In two successive sections, students (1) discussed and critiqued a physician modeling these areas [Weissman DE], (2) role-played a prepared case in dyads, (3) evaluated successful elements of communication skills, and (4) assessed their performance in the role-playing dyads.

CONCLUSION: Medical students require skill-based EOL and PC curricula that emphasize clinical skills specific to their level of training. This module taught early clinical students communication skills for the setting of PC and EOL care. Further interventions will be necessary to advance these students' communication skills as they progress in their clinical training. We propose next to evaluate the effectiveness of this curriculum by comparing students' EOL communication skills with standardized patients before and after this curriculum.

EFFECT OF THE CLINICAL SETTING OF AN EARLY LONGITUDINAL PRIMARY CARE CLINIC EXPERIENCE ON MEDICAL STUDENTS CAREER CHOICES. A.C. Maio¹, H. Sakowski¹, P. Turner¹, E.C. Rich¹; ¹Creighton University, Omaha, NE

PURPOSE: To determine whether the clinic setting of an early ambulatory care longitudinal clinic experience effected career choice.

METHODS: The longitudinal clinic experience at Creighton University consists of participating in the same clinic one-half day every other week of their 2nd year of medical school in a primary care setting (general internal medicine, family practice, or pediatrics). The 4th year medical students were surveyed regarding their 2nd year longitudinal clinic experience and their 1st choice for their future career. Nonparametric correlations (phi and point-biserial coefficients) were used to assess the associations between the student's actual choice of specialty to their self-reported effect of the longitudinal clinic on their specialty choice and the specific specialty (internal medicine, family practice, or pediatrics) of the longitudinal clinic. Hometown and MCAT scores were used as control variables.

RESULTS: Of the 106 students participating in the longitudinal clinic experience, 42% reported it had a positive effect on the career choice in primary care (e.g. internal medicine, family practice, or pediatrics), while 53% reported it had no effect, and the remaining 5% stated it had a negative effect. There was no association between the student's self-reported effect of the clinical experience on their actual choice of specialty ($\phi=0.248$, $p=.05$). There was also no association between a student's choice of specialty and whether they were involved in an internal medicine, family practice, or pediatric longitudinal clinic ($p>.05$ for all three comparisons).

CONCLUSION: The clinic setting had no effect on the students' career choice. Other factors (e.g. market factors such as surplus/shortage of primary care physicians or clinical experiences during their 3rd year) may also be influential in such decisions.

THE EFFECT OF A CD-ROM MULTIMEDIA TOOL ON THE CARDIAC AUSCULTATION ABILITY OF INTERNAL MEDICINE RESIDENTS. R.S. Mangrulkar¹, J.M. Watt², C.M. Chapman³, R.D. Judge³, D.T. Stern¹; ¹University of Michigan Health System and Ann Arbor VA Medical Center, Ann Arbor, MI; ² St. Joseph's Mercy Hospital, Ann Arbor, MI; ³ University of Michigan Health System, Ann Arbor, MI

PURPOSE: To determine if self study with a CD-ROM auscultation tool would increase internal medicine residents' knowledge and skills in cardiac auscultation.

METHODS: The interactive computer-based auscultation program was developed by two of the authors. It includes 20 multi-media cases that center on common cardiovascular conditions, including valvular abnormalities, congenital heart disease, coronary artery disease and congestive heart failure.

At 2 institutions, a total of 127 residents participating in their 1-month primary care rotation were enrolled in this fixed-group controlled trial from April 1999 through June 2000. Residents were assigned to the intervention or control arm based on their clinic site. Those allocated to the intervention group (n=56) were provided two hours per week during their month to use the program. Control residents (n=71) were exposed only to the content of their rotation.

Auscultation knowledge and skills were assessed by a computerized validated 40-item test administered before and after the rotation. Ten of these items tested the recognition and interpretation of heart sounds (skills component), and 30 tested facts about auscultation (knowledge component). Baseline and post-intervention scores were compared within groups using paired t-tests, and between groups with independent t-tests. When possible, residents who dropped out (n=10) were analyzed with their last outcome carried forward.

RESULTS: Pretest and posttest scores are shown in Table 1. At baseline, the two groups did not differ in demographic characteristics, pretest knowledge or total scores. The intervention arm had a lower mean pretest skills score than the control group. While both groups had improved scores after the rotation, residents in the intervention group had significantly higher improvements in skills, knowledge, and total scores than those not exposed to the CD-ROM based tool (* all $p<0.001$).

CONCLUSION: Protected time for internal medicine residents to pursue self-directed study with a multimedia auscultation tool during a primary care rotation enhanced both cardiac auscultation knowledge and skills.

Table 1. Results

	Skills Pretest	Skills Posttest	Knwldg Pretest	Knwldg Posttest	Total Pretest	Total Posttest
CD-ROM	57.6%	79.0%*	64.2%	73.5%*	62.6%	74.9%*
Control	66.2%	68.5%	64.6%	66.9%	65.0%	67.3%

THE EFFECTIVENESS OF A WORKSHOP TO IMPROVE EVIDENCE-BASED PRACTICE AMONG CLINICIANS. R.S. Mangrulkar¹, C.A. Smith², T.T. Force³; ¹Univ. of Michigan Health System & Ann Arbor VAMC, Ann Arbor, MI; ²Cook County Hospital/Rush Medical College, Chicago, IL; ³SGIM, Washington, DC

PURPOSE: To assess the impact of a pilot workshop designed to improve the knowledge, attitudes, and skills required for evidence-based practice by clinicians.

METHODS: To promote the real-time application of EBM by clinicians, we created a web-based desktop providing access to secondary sources of pre-appraised evidence. This served as the centerpiece for an 8-hour workshop delivered at 2 sites to 43 clinicians. Instruction centered on 4 skills: formulating clinical questions, searching evidence-based databases, understanding the results of searches, and applying them to patient care. We used computer labs to give direct experience with the desktop and provided access to the desktop for 3 months after the workshop. Outcomes were assessed using (1) a written survey administered before and after the workshop, and (2) monitored desktop use after the workshop. Survey questions were validated in previous studies and included (1) self-reported use of information resources, (2) self-reported comfort with EBM skills, and (3) demonstration of resource preferences and question formulation in case scenarios. Scoring of the quality of clinical questions were performed by 2 blinded raters using predetermined criteria ($\kappa=0.90$). Analysis of the change in scores after the workshop was performed using the Wilcoxon signed ranks test.

RESULTS: At baseline, participants reported using textbooks, colleagues, and expert consultants more frequently than computer-based resources ($p<0.05$), with at least 25% completely unfamiliar with Cochrane, Best Evidence, MD Consult and Up-To-Date. Participants expressed higher comfort levels after the workshop with all 4 EBM skills ($p<0.001$). Given a clinical scenario, participants were more likely to use Cochrane, Best Evidence and Up-To-Date after the workshop to answer their questions and less likely to consult a subspecialist or partner (all $p<0.02$). Use of MD Consult, Medline and textbooks were not affected. Finally, the workshop improved participants' ability to formulate a clinical question, improving total score ($p<0.001$) and percentage achieving a perfect score (2.6% vs. 15.3%, $p=0.025$). After the workshop, participants accessed the desktop for 198 sessions over 2 months (average of 92.5 minutes and 4.6 sessions per participant, 20 minutes per session). The use of this desktop peaked during the first month (61% of total use), with a decline in month 2 (36%). Up-to-Date and Best Evidence were the most frequently used resources, comprising over 3/4 of total usage time. Medline comprised only 0.9% of total use.

CONCLUSION: Participation in the workshop improved familiarity and comfort with EBM resources and skills, improved the quality of participants' clinical questions, and increased the likelihood of using EBM resources in practice.

DOES PRIMARY CARE RESIDENCY PREPARE PHYSICIANS TO PRACTICE PRIMARY CARE? A SURVEY OF RECENT INTERNAL MEDICINE AND FAMILY PRACTICE GRADUATES. M. Massoumi¹, R.M. Lubitz¹, V.L. Dacey¹, M.A. Smith¹, J.A. Monroe¹; ¹St. Vincent Hospital, Indianapolis, IN

PURPOSE: Recent changes to Internal Medicine (IM) and Family Practice (FP) residency curriculum have attempted to address health care disparities by refocusing training to better mirror primary care practice and encourage generalist careers. This focus has included moving clinical education to ambulatory settings, a shift more successfully made by family practice than other fields.

METHODS: To assess the effect of curricular change on practice patterns and satisfaction, we surveyed all Internal Medicine (n=128) and Family Practice (n=74) physicians 1 to 10 years after graduation from a large Mid-west community teaching hospital known for integrated primary care training. The survey assessed physician and practice demographics (i.e., practice type and location, community size, case mix), resource availability, and adequacy of residency training in curricular areas and with practice skills.

RESULTS: Survey response was 84% (105 IM, 65 FP, $p=0.27$). Respondents were primarily male (66% vs. 66%, $p=0.91$), Caucasian (85% vs. 94%, $p=0.075$), and in solo (17% vs. 6%, $p=0.036$), single-discipline group (41% vs. 45%, $p=0.675$) or multi-specialty group practice (17% vs. 22%, $p=0.495$). Compared with FPs, IMs were less likely to practice in an underserved community or in a community of <50,000 residents (22% vs. 45%, $p=0.0025$), spend time

seeing outpatients (70% vs. 86%, $p=0.0002$), utilize a nurse practitioner ($p=0.001$) or a hospitalist ($p=0.001$). FPs saw fewer adolescent patients (8% vs 13%, $p=0.0004$); while IMs were more likely to subspecialize (40% v 3%, $p=0.0001$), see more geriatric patients per week (21% vs. 41%, $p=0.0001$), and spend more time in the hospital (25% vs. 10%, $p=0.0001$) and ICU (9% vs. 1%, $p=0.0002$). IMs saw fewer outpatients daily (19 vs. 26, $p=0.0001$) but more inpatients (6 vs. 4, $p=0.0861$), and spent more time per week reading professional journals (2.9 vs. 1.8 hrs, $p=0.0142$). Using a list of curricular elements judged important by both IMs and FPs, IMs felt residency training was least adequate in ambulatory medicine, genetics, occupational medicine, practice management, preventive medicine, orthopedics and urology; FPs felt residency training was least adequate in cardiology, gastroenterology, infectious disease, pulmonary, rheumatology and differential diagnosis (all $p<0.05$). Career satisfaction was similar (somewhat to very satisfied, 86% vs. 89%, $p=NS$), although FPs were more satisfied with the amount of time and experience received in ambulatory care settings (68% vs. 86%, $p=0.010$). **CONCLUSION:** Results suggest that at this institution, nationally mandated curricular reform has not shifted career focus nor adequately met the needs of primary care practice for either discipline.

LEARNERS PERCEPTION OF THE AUDIENCE RESPONSE SYSTEM AS A LEARNING TOOL. R.G. Miller¹, B.H. Ashar¹, K.J. Getz¹; ¹Johns Hopkins University School of Medicine, Baltimore, MD

PURPOSE: The audience response system (ARS) is an electronic tool that, through use of individual keypads, provides learners with immediate feedback on the accuracy of their responses to questions compared with the group as a whole. While extremely popular, the ARS is expensive to purchase or rent. Such cost can be prohibitive for continuing medical education (CME) course directors. Our study attempts to measure the perceived value of the ARS by its users as a justification of the expense.

METHODS: A survey was administered to all participants attending a national internal medicine board review course in July 2000. The survey consisted of 18 questions designed to capture: 1) demographics, 2) learners' perceptions of the ARS regarding improvement of learning and reduction of test anxiety, and 3) ease of use of the ARS. Data was analyzed using SPSS software.

RESULTS: The response rate was 83.6% (158 of 189 attendees). Mean weighted age was 42 years, and 57.3% were male. Of the respondents, 53.8% were preparing for initial or recertification board exams. On a 5-point Likert scale, the learners favorably rated the ARS as in the table. In addition, over half the participants were more likely to choose a future CME course based upon ARS use (58.2%) and were willing to pay an additional registration fee for it (57.0%).

CONCLUSION: While expense remains a consideration, the audience response system is a valuable educational tool. Compared to their experience with prior courses, the learners in this broad cross-sectional group felt that the ARS enhanced learning and was easy to use. Course directors and other teachers may wish to consider incorporating the use of the ARS when planning future curricula and courses.

Learners' Perceptions of the ARS

	% Agree or Strongly Agree	% Disagree or Strongly Disagree
Easy to Use	96.8%	1.3%
Enhanced Learning	79.5%	5.1%
Decreased Test-Taking Anxiety	49.0%	13.4%
Helped Identify Weak Areas	85.3%	3.8%

LEARNING THE THYROID EXAMINATION — A MULTIMODALITY INTERVENTION FOR INTERNAL MEDICINE RESIDENTS. C.V. Soares-Welch¹, W.A. Houck, III¹, J.T. Li¹, Y.M. Montori¹; ¹Mayo Clinic, Rochester, MN

PURPOSE: Although thyroid examination is the least invasive and inexpensive method to detect thyroid pathology, most physicians lack the ability to detect thyroid abnormalities by examination. We developed a single and brief multimodality session to improve the thyroid examination skills of physicians-in-training and evaluated its efficacy in a controlled trial.

METHODS: Thirty-nine first-year internal medicine residents were randomly allocated to participate in a multimodality learning session on thyroid physical examination or to receive no intervention (control group). The 50-minute learning session consisted of a slide presentation, a computer graphics animation, videotaped and hands-on demonstrations of the appropriate technique, participant examination of a patient with thyroid abnormalities and a normal volunteer under preceptor observation, and an evidence-based handout. All the participants were assessed in two stations of an objective structured clinical examination (OSCE) for appropriate thyroid examination technique (through a 2-way mirror) and identification of thyroid abnormalities (using a scored participant report of findings). The OSCE preceptors and results analysts were blinded to participant allocation.

RESULTS: Of the 19 residents in the intervention group and 20 residents in the control group: 6 and 3 observed the neck for thyroid abnormalities (32% vs 15%; $p=0.3$); 17 and 16 residents used proper hand position (90% vs 80%; $p=0.7$); 13 and 15 residents had the patient swallow while palpating the neck (68% vs 75%, $p=0.7$), respectively. There was a significant difference between the mean scores on thyroid physical findings between the intervention and control groups (100 vs 52.5, $p=0.047$).

CONCLUSION: A single and brief multimodality learning session did not statistically improve the thyroid examination skills of first year Internal Medicine residents, but did positively affect their ability to detect abnormalities. Future research (with larger sample sizes) should test the effectiveness of iterative sessions or of increased emphasis on hands-on thyroid abnormality detection rather than appropriate technique.

LEARNERS PERSPECTIVE OF BEDSIDE PRESENTATION. S. Mossad¹; ¹Cleveland Clinic Foundation, Cleveland, OH

PURPOSE: During the last few decades, teaching medical students and residents how to care for patients has gradually moved from the bedside to the classroom. Many medical educators believe that bedside presentation (BSP) is more valuable for learners and patients. This study was performed to assess the validity of this belief.

METHODS: Two 2nd year, two 1st year medical residents & two 4th year medical students rotating on the inpatient infectious diseases consult service were asked to answer a 6-point questionnaire about their perception of BSP immediately after their initial evaluation of patients. Patients were then asked 2 questions regarding their perception of these encounters. **RESULTS:** Forty seven consecutive encounters were evaluated; percentages are shown in the table.

CONCLUSION: BSP was acceptable, more educative & perceived as efficient by medical residents & students. Most patients were comfortable & understood their illness better with BSP.

Results

Questions (1 – 6 answered by residents & students, 7&8 by patients)	Yes	No	Not sure
1. Where you comfortable presenting this case at the bedside?	100	0	0
2. Did you learn better by presenting this case at the bedside?	83	15	2
3. In your opinion, was the patient comfortable being presented at the bedside?	96	0	4
4. Do you think the patient understood his/her illness better by being presented at the bedside?	90	6	4
5. Was BSP better for this patient's care?	83	13	4
6. Was BSP more efficient than presenting away from the bedside?	96	2	2
7. Where you comfortable being presented at the bedside?	87	0	13
8. Did you understand your illness better this way?	87	0	13

INITIAL AND FINAL DIAGNOSIS ALIGNMENT. V. Navarro¹; ¹National Northeast University (UNNE), Resistencia, Chaco

PURPOSE: Diagnosis is a fundamental step to start the process of solving a patient's health problem. However its elaboration does not invariably produce a perfect result. Objectives : 1- To establish the alignment between the Initial (ID) and Final Diagnosis (FD). 2- To describe the factors that modified ID

METHODS: We analyzed 200 clinical records of patients who came to the Internal Medicine Department at the University's Hospital between July 1998 and June 2000. All the data was recorded in a single issue by the same researcher. Definitions: Total Agreement : when the ID is the same as the FD. Partial A : when there's no coincidence between ID and FD in some secondary diagnosis which are those not related to the cause of hospitalization , but they coincide with the main diagnostic. Partial B : when there's no coincidence at least in one main diagnostic between ID and FD , which are those related to the cause of hospitalization. No coincidence : when ID and FD are totally different. The statistic analysis of data was made with EPI 6 program.

RESULTS: Results : Of 200 clinical records evaluated , 88 (44%) belonged to female patients and 112 (56%) to male patients. Average age 52.7 ± 18.6 years old. The Concordance between ID and FD was Total 31% , Partial A 27.5 % , Partial B 29% and There was no coincidence in 12.5%. Factors that changed ID : Inferential reasoning 41 records (20.29%), Inferential reasoning + test 85 (42.7%), test 74 (37.01%). The test requested were Imaging studies, Biochemistry, Serology Microbiology, Consults to other specialties, Biopsy, Electrocardiograms, Therapy Test (see Table I).

CONCLUSION: Conclusions : Clinical data and physical exams are essential for ID. However complementary tests are important to determine FD. Among the tests, it's important to highlight the importance of CT scan , although we must re-evaluate others like Chest X-ray and Electrocardiograms which were taken to all patients but they were of poor value in patients without specific symptoms. We need to rationalize others like biochemistry tests and consults to others specialist. The biopsies and therapy tests were very useful. These results reflect what has happened with Medicine in the last two decades, in which the technological progress allowed a better diagnosis, although it was initially established by physicians at the patient's bedside.

Test requested

Diagnostic Tests	Number requested	Changed the Initial Diagnosis	Percentage that changed Diagnosis
Imaging Studies	431	181	42%
Chest X-ray	200	38	19%
CT scan	74	58	78.4%
Others	157	85	54%
Biochemistry,	1725	32	23.6%
Serology Microbiology			
Consults to other specialist	146	73	51%
Biopsy	20	13	65%
Electrocardiograms	200	43	21.5%
Therapy Test	10	10	100%

IS PATIENT SATISFACTION RATING OF MEDICAL STUDENT ASSOCIATED WITH STUDENTS COMPETENCE IN ELUCIDATING PATIENTS PROBLEMS IN OUTPATIENT CLINIC? Y. Oda¹; ¹Saga Medical School, Saga, Japan

PURPOSE: To determine whether patient satisfaction rating for humanistic quality of a medical student is associated with student's ability to gather and integrate clinical information obtained through history taking and physical examination in a university hospital outpatient setting.

METHODS: This study was conducted from April to October 2000, on last (sixth) year medical students while they were rotating through general medical outpatient clinic for 2 weeks. Faculty internists evaluated student's clinical competence observing student's ability to assemble clinical information gathered through medical history taking and physical examination and to formulate working diagnosis. Students were rated as follows; A, able to integrate pertinent information for initial assessment, B, able to gather pertinent information, C, information gathering is passive and superficial, and D, detrimental to patient. A, B and C were further divided into three subgroups according to individual performances to develop a 10 point Competence Score. Patient satisfaction ratings for students were also obtained at the same time by using a modified Japanese version of Patient Satisfaction Questionnaire developed by the American Board of Internal Medicine (m-PSQ). Multiple regression analysis was used for statistical comparison.

RESULTS: A total of 77 medical students were evaluated by 9 staff physicians and by 229 patients. The m-PSQ consisted of single factor with Cronbach's a coefficient of 0.90. Multiple regression test revealed that competence scores of students were strongly associated with the m-PSQ ratings ($p < 0.001$), but not with student's age and gender, Objective Structured Clinical Examination (OSCE) scores, patients' demographic characteristics such as age, sex, occupation status.

CONCLUSION: Patient satisfaction ratings of medical students in outpatient clinics were strongly associated with student's ability to gather and integrate clinical information obtained through history taking and physical examination in a university hospital outpatient clinic.

PALM PILOTS: UTILITY AND OBSTACLES FROM A RESIDENTS PERSPECTIVE. C. Tu¹, J. Oh², D. Elkin¹, L. Pollack³, H. Kerpen¹, A. Jotkowitz²; ¹LIJMC, New Hyde Park, NY; ²NUMC, East Meadow, NY; ³Bassett HC, Cooperstown, NY

PURPOSE: The use of Palm Pilots is rapidly expanding among residents and some training programs are even providing them for their housestaff. However, little is known about how residents view their utility. The purpose of this study is to determine 1) residents' perception of the utility of Palm, 2) potential obstacles toward their use, and 3) the influence of providing Palms to medical residents.

METHODS: We conducted a survey of all medicine residents at two hospitals. One of the hospitals heavily subsidized and encouraged residents' purchase of Palms. The survey included questions regarding perceptions toward the benefits of using Palms to help them record patient data (record) and improve the quality of their signouts (signout). We also surveyed for potential obstacles to their use: cost, difficulty in learning how to use (learning), fragility and effort required to enter data (enter). A Likert scale (1=strongly disagree to 5=strongly agree) was used. The survey was piloted for reliability at a third hospital.

RESULTS: 65 of 81 residents (80 %) in the hospital that did not subsidize the purchase of Palms (unsubsidized group) and 25 of 29 (86%) in the hospital that subsidized their purchase (subsidized group) returned the survey. 21 residents in the unsubsidized group (32%) and 24 residents in the subsidized group (96%) owned a Palm. The unsubsidized group rated questions on the utility to record patient data and signout higher than the subsidized group (table 1). Regarding potential obstacles the unsubsidized group felt that cost was a greater obstacle while the subsidized group felt that the fragility of the device and the effort to enter data was a greater obstacle than the control group. Analysis by ownership of a Palm independent of the site also failed to reveal any significant results with exception to cost and learning (less of an obstacle to Palm users).

CONCLUSION: Our study demonstrates that in a training program that subsidized the purchase of Palms, residents perceived them to be less useful and felt that the fragility of the device and the effort to enter data were greater obstacles to their use. These results imply that simply providing housestaff with Palms may not be beneficial. Studies are needed to see if providing medical software and additional training will increase their utility and decrease perceived obstacles.

Table 1

	record	signout	cost	learning	fragility	enter
Control	4.2	3.9	3.6	1.9	1.9	2.8
Palm	3.6	3.0	2.4	2.2	2.8	3.7
p	< 0.01	< 0.01	< 0.01	ns	< 0.001	< 0.001

PALM PILOTS: ARE THEY USEFUL TO MEDICAL RESIDENTS? J. Oh¹, U. Gavarasana², L. Pollack³, H. Kerpen², A. Jotkowitz²; ¹NUMC, East Meadow, New York; ²LIJMC, New Hyde Park, NY; ³Bassett HC, Cooperstown, NY

PURPOSE: Recently, we observed a rapid proliferation of Palm Pilots (Palms) in medical settings, particularly among residents and students. Some programs are even providing them for their housestaff. However, little is known about real life benefits of Palms. The purpose of this study is to determine how Palms influence medical residents' practice.

METHODS: We conducted a survey of all medicine residents at two tertiary care hospitals. One of these hospitals was selected because it heavily subsidized and encouraged residents to purchase Palms. The survey included questions regarding use of the following prediction rules: pre-op cardiac risk assessment, clinical likelihood of a DVT or PE and Pneumonia Severity Index. A Likert scale (1=never to 5=always) was used. For those who owned a Palm we also

inquired about the most common uses in patient care as well as the use of any medical programs. Information on training level and whether they owned a Palm Pilot was also obtained. The survey was piloted for test-retest reliability at a third hospital. Mann-Whitney U tests were performed to compare groups and logistic regression was performed to adjust for resident genre, level of training, and program site.

RESULTS: The survey was sent to 81 residents in the hospital that did not subsidize the purchase of Palm (unsubsidized group) and 29 residents in the hospital that subsidized their purchase (subsidized group). 65 residents (80 %) in the unsubsidized group and 25 (86%) in the subsidized group responded to the survey. 21 residents in the unsubsidized group (32%) and 24 residents in the subsidized group (96%) owned a Palm device. Palm owners in both groups responded that they used these devices to organize patient records (23%) and phone numbers used in patient care (82%), keep patient "to do" lists (39%), schedule events and appointments (75%), and as an aid for clinical calculations (86%). The most frequently used programs were pharmacopoeias (93%), medical reference (44%), and clinical calculators (18%). Ownership of Palm and training sites were not independent predictors of using clinical prediction rules more frequently on logistic regression.

CONCLUSION: Our study demonstrates that medical residents use Palms for a variety of purposes relating to patient care. However, owning Palm Pilots did not increase the likelihood of using clinical prediction rules and very few residents used them to organize patient records. From an educational perspective, at this point the available evidence does not support providing Palms to medical residents. Further research is still needed to determine if the use of Palm Pilots will result in better patient outcomes or resident efficiency.

A WOMENS HEALTH CURRICULUM FOR INTERNAL MEDICINE RESIDENTS. M.A. Onwudiwe¹; ¹Wright State University, Dayton, OH

PURPOSE: To develop and implement a curriculum in women's health for internal medicine residents in a community hospital based university program.

METHODS: Participants are first year internal medicine residents in their ambulatory block month (N=16). The curriculum content was determined by a needs assessment conducted on previous and current residents. It consists of the following six units covered during three hourly didactic sessions: 1) Pretest and Introduction to Women's Health, 2) Selected Gynecological Conditions, 3) Obesity and Eating Disorders, 4) Menopause, Osteoporosis and Hormone Replacement, 5) Psychosocial Issues, 6) Resident Presentations and Posttest. Also included in the curriculum is a three-hour visit to the local domestic violence prevention center. All necessary reading materials are handed out during the first session after the pretest. The didactic sessions start with a brief case presentation by the participating resident(s); this is followed by an in-depth discussion of the topics of the day with expected resident participation. Each session except the introductory and resident presentation session ends with role-plays where the instructor plays the role of the patient and the resident the doctor. This enables the resident(s) to put into immediate practice what they have just learned. Clinical practice of women's health is expected to occur when residents see female patients in their primary care practices. To expand the scope of this curriculum beyond these limited topics both residents and faculty have been encouraged to incorporate into everyday teaching and learning how various conditions may differ or are unique in women. Participating residents are expected to make a formal thirty to forty-five minute presentation on a topic in women's health not covered in this curriculum. Learners are evaluated by the completion of a standard evaluation form used in the residency program. Evaluations are based on the residents' performance on the posttest, participation during the didactic sessions and their presentations. Learners evaluate the curriculum by completing a survey.

RESULTS: The curriculum was well received by all residents. All residents surveyed either agreed or strongly agreed that they would now be able to evaluate, manage or appropriately refer female patients as a result of knowledge gained during the month. The mean posttest score was 81%, an improvement over the mean pretest score of 66%. Residents stated particularly that role-plays helped improve their understanding of the topics. A survey of these residents in their third year regarding their perceptions on the women's health education they received in this program and how well they think it has prepared them for their future practices year is planned.

CONCLUSION: The curriculum improved the knowledge of residents in selected women's health topics. Using the same curricular model, topics can be adapted to meet the learning needs of residents in other residency programs.

LISTENING IN: ASSESSING THE VALIDITY OF STANDARDIZED PATIENT RATINGS OF RESIDENT AND ATTENDING PERFORMANCE. J.W. Peabody¹, J. Luck², P. Glassman², O. Yeretsian³, C. Conti⁴, M. Bates³; ¹Society of General Internal Medicine, San Francisco, CA; ²UCLA, Los Angeles, CA; ³Institute for Global Health, San Francisco, CA; ⁴VA Greater Los Angeles Healthcare System, Los Angeles, CA

PURPOSE: Standardized patient (SP) reports are purported to be the gold standard to measure the quality of outpatient teaching and clinical care, but data on the validity of SP reporting in actual clinical settings are limited. Most validation studies are typically explicit observations of SPs and medical students done by medical faculty. In this study, we directly measured the validity of SP reports by covertly audio recording visits between consenting physicians and unannounced SPs.

METHODS: We placed SPs into the residents' and attendings' general internal medicine primary care clinics at two sites: a VA Medical Center and an academically affiliated teaching hospital in Northern California. Twelve professional actors were trained to be SPs and present with defined clinical scenarios for 2 common outpatient conditions (depression and chronic obstructive pulmonary disease (COPD)). Each physician subject saw a simple and a complex case for each disease. Fifteen physicians were randomly selected at each site, yielding a total of 120 visits. From these, we randomly selected 10 percent of these visits to audio record the SP physician interaction yielding 1 recording for each of the 12 SPs. To make the visits as realistic as possible, physicians consented not to be informed if a patient was an SP or if the visit was being recorded. SPs rated clinical care provided by physicians using closed-ended checklists that contained explicit quality criteria developed from national guidelines. Transcribed recordings

were scored identically using the same checklists to determine if SP reports were valid. We calculated the percentage of criteria where SP checklists and recordings were in agreement and disagreement. McNemar's chi-square test for paired proportions was used to determine the statistical significance of differences between the two methods.

RESULTS: The overall rate of agreement between SPs and recordings was 90.3 percent ($p=0.00$). The rate of agreement ranged from 86.9 percent ($p<0.01$) for the complex depression case to 95.3 percent ($p=0.375$) for the simple COPD case. Eighty-five percent of the disagreements between SPs and recordings occurred when SPs recorded that physicians performed actions that were not verified by the recording.

CONCLUSION: The high rate of agreement between SPs and recordings indicates that SPs are valid measures of the quality of outpatient care. The higher rate of agreement in the COPD cases as compared to depression may be related to two factors: 1) the larger number of items in the depression checklists, and 2) the subtlety of checklist items in the history section of these cases. The preponderance of overrecording vs. underrecording suggests that SPs may tend to give physicians the benefit of the doubt in the presence of ambiguity. Unannounced SPs are attractive quality measurement tools in the outpatient setting because they are inherently adjusted for case mix. These prospectively obtained results indicate that SP reports are valid measures of the quality of care, and may be useful for comparative evaluation of physician training across health care systems.

HOW EFFECTIVE ARE WE IN TEACHING HOUSESTAFF INDICATIONS FOR DRUG PRESCRIBING? P.V. Pepper¹, J. Bestoso¹; ¹Naval Medical Center San Diego, San Diego, CA

PURPOSE: Resident prescribing habits are often based on pharmaceutical detailing and advertising and not on FDA guidelines or cost-effective practices. This study seeks to examine resident physicians for knowledge on commonly used classes of medications.

METHODS: We surveyed all internal medicine housestaff at the Naval Medical Center in San Diego ($n=43$). Our questionnaire covered drug treatment of gastroesophageal reflux disease (GERD), allergic rhinitis, asthma, rheumatoid arthritis, nausea, diabetes, osteoarthritis; and the use of filgrastim, dihydropyridines, and angiotensin receptor blockers (ARBs).

RESULTS: Amongst R1–R3 residents, physicians were very well educated about appropriate drug therapy for GERD (90% correct response), asthma (78% correct response), rheumatoid arthritis (75% correct response), diabetes (78% correct response), osteoarthritis (63% correct response) and ARBs (83% correct response). Their performance for allergic rhinitis (50% correct response), nausea (15% correct response), usage of filgrastim (35% correct response), and dihydropyridines (10% correct response) was much less impressive. None of our interns and only 3 residents knew the FDA indication for a very commonly prescribed antihypertensive.

CONCLUSION: Increased education, especially for very popular drugs, is needed in the internal medicine curriculum. Further study of the impact of pharmaceutical detailing on resident drug prescribing is warranted.

THE INFLUENCE OF END-OF-LIFE CARE EXPERIENCES ON PRE-PROFESSIONAL ATTITUDES TOWARD END-OF-LIFE CARE. M.W. Rabow¹, J.J. Petersen¹; ¹University of California, San Francisco, San Francisco, CA

PURPOSE: Health care professional students are receiving increased training in providing end-of-life care. Little is known about changes in end-of-life care attitudes among medical and pharmacy students following experiences working with patients at the end of life.

METHODS: 8 medical or pre-medical students and 19 pharmacy students served as volunteer "patient advocates" ($n=27$), providing social support in the form of weekly phone calls and monthly visits to patients with end-stage congestive heart failure, chronic obstructive pulmonary disease, and cancer. Students worked in pairs with an individual patient for a 4-month period. In addition, students participated in a 10-hour series of physician-led discussions about end-stage illness, the patient and family experiences of illness, and the emotional and spiritual dimensions of end-of-life care. Using a confidential, self-administered questionnaire with 10-point Likert responses (1 = very negative; 10 = very positive), student attitudes about the end of life were assessed before and following their work as advocates and participation in the class. The mean values among all respondents pre- and post-training were compared and analyzed using the two-tailed t-test.

RESULTS: The response rate was 70% ($n=19$). Students felt that the quality of their experiences with patients at the end of life was more positive after the course than before (mean = 7.0 post vs. 5.7 pre; $p=0.013$). Students also felt more comfortable working with patients at the end of life (mean = 7.3 post vs. 6.4 pre; $p=0.045$). Additionally, following their experiences, students estimated the average quality of life for patients at the end of life as higher (mean = 6.3 post vs. 4.7 pre; $p=0.028$). Following their experiences, student belief in the possibility for growth at the end of life did not change significantly (mean = 8.7 post vs. 8.9 pre; $p=0.48$). Following their experience, students' rank of the professional importance of working with patients at the end-of-life remained high (mean = 9.0 post vs. 9.6 pre; $p=0.077$).

CONCLUSION: Direct contact with patients at the end of life and formal physician-led discussions about end-of-life care led to improved quality in students' experiences with people at the end of life, as well as increased comfort working with patients at the end of life. While end-of-life care experiences led students to estimate that patients' quality of life was higher than initially expected, following their work, students were no more likely to believe that growth at the end of life was possible. The influence of early end-of-life care experiences on pre-professional students is generally positive, but should be explored further as students receive increased training in end-of-life care.

MENTORSHIP DURING RESIDENCY: UNCOVERING THE GENDER GAP. R.A. Ramanan¹, W.C. Taylor¹, R.B. Davis¹, R.S. Phillips¹; ¹Division of General Medicine, Beth Israel Deaconess Medical Center, Harvard Medical School, Boston, MA

PURPOSE: Mentorship has been associated with greater career satisfaction among academic physicians, but has not been studied during residency training. While female physicians may

benefit more from mentorship, little is known about mentorship among female residents. We compared the prevalence and characteristics of mentorship between male and female residents in medicine.

METHODS: We developed a questionnaire to collect data on attitudes regarding mentorship, satisfaction with mentorship, and satisfaction with residency education. After pilot testing the questionnaire, we mailed it to residents in the five internal medicine residency programs affiliated with Harvard Medical School. We sent two additional mailings to nonrespondents. We compared responses between men and women, and then used logistic regression to identify factors significantly associated with having a mentor and being satisfied with mentorship or career guidance.

RESULTS: Among 329 respondents (response rate 65%), 47% were women. Of the women, 57% were white, 29% Asian and 9% underrepresented minorities. Although 50% of both men and women had a current mentor, and both men and women were equally likely to report satisfaction with mentorship (46% vs 38%, $p=0.16$), women were more likely to report that having a mentor was important during residency training (96% vs 90%, $p=0.04$). While women were more likely than men to want a mentor of the same gender (35% vs 8%, $p<0.01$) and were more likely to have a female mentor (52% vs 17%, $p<0.01$), having a female mentor was not associated with satisfaction with mentorship. In addition, regardless of whether they had a preference for a female mentor, women were equally likely to have a mentor and to be satisfied with mentorship. In multivariable models adjusting for ethnicity, year of training, being assigned a mentor and history of mentorship, women and men were equally likely to have a mentor and to be satisfied with mentorship. However, after adjusting for ethnicity, year of training and having a mentor, women were more likely than men to report receiving inadequate guidance with career decisions [adjusted odds ratio 1.8 (95% confidence interval 1.1, 2.8)] and were less likely than men to think that their residency was doing the best possible job preparing them for a future career (0.6 (0.4, 0.9)).

CONCLUSION: Although women in residency training developed mentoring relationships as often as men and were equally satisfied with mentorship, they were significantly more likely to judge guidance with career decisions as inadequate, and were less likely than men to believe that their residency was doing the best possible job preparing them for a future career.

SHOULD WE ASSIGN MENTORS TO RESIDENTS IN MEDICINE? R.A. Ramanan¹, W.C. Taylor¹, R.B. Davis¹, R.S. Phillips¹; ¹Division of General Medicine, Beth Israel Deaconess Medical Center, Harvard Medical School, Boston, MA

PURPOSE: Since mentorship has been associated with greater career satisfaction among academic physicians, some residency training programs assign mentors to all house officers. We compared characteristics of mentorship between residents who were assigned a mentor by their training program, and those who were not assigned a mentor.

METHODS: We developed a questionnaire to collect data on attitudes regarding mentorship, satisfaction with mentorship, and satisfaction with residency education. After pilot testing the questionnaire, we mailed it to residents in the five internal medicine residency programs affiliated with Harvard Medical School. We sent two additional mailings to nonrespondents. Using logistic regression, we identified factors associated with three different outcomes: having a mentor, being satisfied with mentorship, and being satisfied with career preparation.

RESULTS: Among the 329 respondents (response rate 65%), 34% were in residency programs that assigned mentors (assigned), and 66% were in programs that did not assign mentors (nonassigned). Most (87%) residents in programs that assigned mentors knew that they were assigned a mentor. Residents in assigned programs were significantly more likely to identify a mentor than those in nonassigned programs (82% vs 43%, $p<0.01$). Residents with mentors in assigned programs were less likely than residents with mentors in nonassigned programs to report their mentor was helpful with professional development (64% vs 77%, $p=0.05$) or personal development (45% vs 69%, $p<0.01$). However, residents with mentors in assigned programs were more likely to be satisfied with mentorship than residents with mentors in nonassigned programs (53% vs 36%, $p<0.01$). In multivariable models adjusting for sex, ethnicity, year of training and history of mentorship, residents in assigned programs were more likely to identify a mentor [adjusted odds ratio 8.1 (95% confidence interval 4.3, 15.3)] and to be satisfied with mentorship [2.2 (1.4, 3.6)] than residents in nonassigned programs. Regardless of whether residents were in a program that assigned mentors, multivariable analyses showed that residents who had a mentor during residency were more likely to think that their residency was doing the best job preparing them for their future career [2.0 (1.2, 3.2)] compared with residents who did not have a mentor.

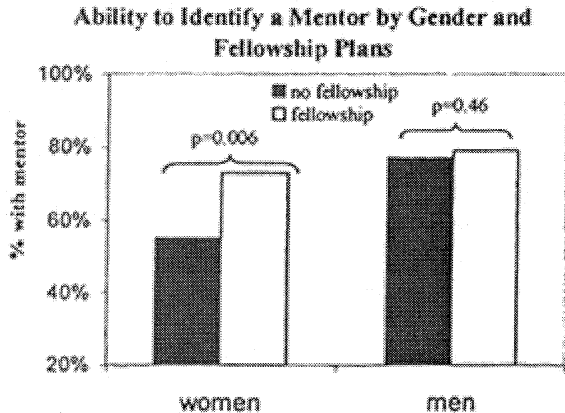
CONCLUSION: Internal medicine residents who were mentored were significantly more likely to report excellent preparation for future careers. Residents who were assigned a mentor were more likely to identify a mentor and to be satisfied with the mentorship they received. These data suggest that residency training programs should assign mentors to house officers.

GENDER DIFFERENCES IN THE ABILITY TO IDENTIFY A POTENTIAL MENTOR AT MORNING REPORT. A. Reisman¹, C. Gross¹; ¹Division of General Internal Medicine, Yale University School of Medicine, New Haven, CT

PURPOSE: Identifying potential mentors is an important part of residency training. Morning report (MR) provides exposure to a variety of attending physicians. We studied resident and program characteristics that were associated with the ability to identify a potential mentor at MR. **METHODS:** We performed a cross-sectional survey of 356 internal medicine residents from a convenience sample of thirteen residency programs. The instrument included questions about demographic characteristics, subspecialty fellowship plans and ability to identify a potential mentor in MR during the previous six months.

RESULTS: The response rate was 63% (38% female). Women were significantly more likely to have plans to go into general internal medicine (GIM) (43% vs. 58%, $p<0.001$). Overall, 73% of residents were able to identify a potential mentor at morning report within the previous six months. Women were significantly less likely than men to be able to identify a mentor (63% vs. 79%, $p<0.001$). Stratification by fellowship plans revealed that female residents not planning on subspecialty training were significantly less likely than male to be able to identify a potential

mentor (Figure). Residents from programs with >50% generalists at MR were less likely than those from programs with more specialists at MR to identify a potential mentor (57% vs 81%, $p < 0.001$).



CONCLUSION: Future generalists may face barriers to appropriate mentorship. Even though female residents were more likely to go into GIM, those who chose to do so were less likely to find a potential mentor at MR. Residents from programs with a majority of specialist attendings at MR were more likely to be able to identify a potential mentor, regardless of their career plans.

HOW MANY PATIENTS SHOULD RESIDENTS SEE IN CLINIC? THE RESIDENTS PERSPECTIVE. C. Rhoads¹, J. Watanabe¹; ¹University of Washington, Seattle, WA

PURPOSE: There is little information about what daily patient volume optimizes resident learning in the continuity clinic setting. By incorporating residents' perspectives on patient volume with the guidelines of the Internal Medicine Residency Review Committee and with information offered in the medical education literature, residency programs may be able to develop a better understanding of the optimal patient volume for continuity clinic sessions.

METHODS: Internal Medicine Residents attending continuity clinic at Harborview Medical Center during February 1997 and February 2000 completed a questionnaire at the end of each continuity clinic session which asked them to assess the number of patients they saw during that day's clinic using a 5 point Likert-type scale (1=too few, 3=just right, 5=too many). Chi-square analyses, Student's t-tests and linear regression modeling was used to evaluate the variables that contributed to optimal scheduling.

RESULTS: Residents completed questionnaires for 73% of their clinic sessions (N=105). Overall, they saw an average of 4.5 (SD=1.6) patients per session. 75% of the time, patient volumes below 3 patients were considered too low and volumes above 6 patients too high. During sessions where they rated patient volume as "just right" interns saw an average of 3.7 (SD=1.2) patients, second residents saw 4.6 (SD=0.7) patients and third year residents saw 5.3 (SD=1.2). Linear regression modeling adjusted for resident track, gender and study period, confirmed that second and third year residents saw more patients than interns, 0.7* and 1.5* patients, respectively; that contact with less than 1.2* patients per session below average constituted "too few" patients; and that contact with more than 1.8* patients above average constituted "too many". (*p-values < 0.0001)

CONCLUSION: Residents at all levels of training were comfortable seeing the minimum volume of patients recommended by the Internal Medicine Residency Review Committee (1 new and 3 follow-up patients). Patient volumes below 3 were too low and volumes above 6 were too high. In between these apparent limits, ideal volume depended on year of training, increasing by about 1 patient per session per year. While information from the medical education literature suggests that some aspects of the clinic educational program may suffer when patient volumes exceed 4 per session, we found that residents at more advanced levels of training preferred patient volumes higher than this. While one explanation for this observation is the higher skill level of the third year residents, it may also be that more advanced trainees derive greater educational benefit from direct patient care than from supplemental educational activities such as reviewing patient care with attendings. Important next steps will be to improve our understanding of the relationships between volume, type and complexity of patient visits, and the acquisition of expertise in specified aspects of ambulatory care, as well as to further delineate the relative educational value of the different clinic activities residents engage in.

PREPARATION FOR CLINICAL CLERKSHIPS: AN EVALUATION OF A CONTINUITY CLINICAL EXPERIENCE FOR MD/PHD STUDENTS. L.M. Rucker¹; ¹Albert Einstein College of Medicine/Jacobi Medical Center, Bronx, NY

PURPOSE: BACKGROUND: MD/Ph.D. students at the Albert Einstein College of Medicine have a 4.5 year hiatus (to do bench research) between the pre-clinical and clinical years of their medical education, during which time they lose touch with patient care. We developed a clinic where MD/Ph.D. students participated as lecturers and primary providers weekly for at least 1 year. The aims of our study are to evaluate whether this clinical experience: 1) serves as a bridge between the basic science years and the clinical years, maintaining clinical skills and confidence, and 2) improves students' knowledge of Ambulatory Medicine.

METHODS: At the end of the clinic year, the students took a standardized knowledge test and completed a validated questionnaire assessing their skills. MD/Ph.D. students who chose not to attend clinic and MD non-Ph.D. students also took the test and answered the questionnaire, serving as reference groups. To compare the groups, ANOVA was performed, followed by Tukey testing.

RESULTS: The MD group had proportionally more women and started medical school 2 years later than either MD/Ph.D. group ($p < .05$). The MD/Ph.D. clinic students (n=20) were significantly more confident than the MD students (n=49) about their skills in every category of the questionnaire (history and general competencies, readiness to function as a doctor, physical exam, diagnosis and treatment) ($p < .05$). They were more confident than the MD/Ph.D. non-clinic students (n=13) in all categories except physical diagnosis. There was a non-significant difference in test scores between the MD/Ph.D. clinic students and each of the other groups.

CONCLUSION: A weekly clinical experience, combining patient care and lecture can serve as a bridge between basic science and clinical years for MD/Ph.D. students. It does not improve knowledge of medicine.

DEVELOPING A STAGE-APPROPRIATE PROFESSIONALISM CURRICULUM: A BEGINNING. D. Rudy¹, C. Elam¹; ¹University of Kentucky, Lexington, Kentucky

PURPOSE: Throughout the medical curriculum, there are opportunities for the introduction of professionalism topics relevant to students' current level of experience. Although much of the focus on the teaching of professionalism is based in the patient care environment, we contend that the development of professionalism begins earlier in medical training, during the pre-clinical years. Patient-care related issues might not be relevant to pre-clinical students. To that end, we are asking our students to help us define stage-appropriate professional skills and attitudes. This information will be used to begin to develop a stage-appropriate professionalism curriculum.

METHODS: To introduce the American Board of Internal Medicine's professionalism framework (altruism, accountability, excellence, duty, honor and integrity, and respect for others) in a Introduction to the Medical Profession course, a video of faculty, residents, and students discussing professionalism in medicine was shown to first-year medical students of the entering class of 1999. Following review of the video, students were asked to list how a first-year student may exhibit professionalism. Responses were analyzed by themes by two raters. Differences were resolved via consensus.

RESULTS: Responses were received from 86 of 95 (91%) of students. Students listed a total of 376 responses; the average number of responses per student was 4.4. The top categories of responses were: dedication to learning 60% (diligence, lifelong learning, preparation for class); respect 58% (for classmates, faculty, patients, other health care professionals); patient confidentiality 43%, listening 31% (in class, to patients) honesty 35% (integrity, morality) dress and appearance 29%; courtesy 26%, team work 26% (helping other students learn) empathy 24%, favorable temperament 20% (maturity, positive attitude, patience), knowledge of limitations 20%, and punctuality 19%.

CONCLUSION: First-year students' perceptions of necessary professional skills focused upon behaviors relative to their current roles in their educational experience. The patient care related issues identified by students such as patient confidentiality and knowledge of limitations appeared to be appropriate for their level of training. Applying the ABIM format to the survey results shows that the majority of responses fall into the categories of excellence (dedication to learning), honor and integrity (honesty), and respect for others (respect). Presenting first-year students with a professionalism curriculum that is primarily focused upon patient-care related issues may not be as relevant as professionalism issues more closely related to their current level of training. Addressing issues such as dedication to learning, honesty, and respect may increase the relevance of professionalism to pre-clinical students. We are using this work to develop identify students' perceptions of stage-appropriate professional skills and behaviors to devise stage-appropriate educational interventions to teach professionalism.

EFFECT OF AN EARLY LONGITUDINAL PRIMARY CARE CLINIC EXPERIENCE ON STUDENT ATTITUDES AND INTEREST TOWARD PRIMARY CARE. H. Sakowski¹, A.C. Maio¹, P. Turner¹, E.C. Rich¹; ¹Creighton University, Omaha, NE

PURPOSE: This study evaluated the effect of a second year ambulatory primary care longitudinal clinic experience on medical student attitudes and interest in Primary Care.

METHODS: All second year medical students participated in a year-long ambulatory primary care experience, during which they spent a 1/2 day every other week in a primary care clinic (Family Medicine, General Pediatrics, or General Internal Medicine). An anonymous survey was administered to second year medical students at the beginning and again following the longitudinal clinic experience. First year students were also surveyed to serve as controls. The 21-item survey included demographic items, items dealing with attitudes toward primary care, items predicting interest in primary care as a career, and items to assess students' present career interests. Logistic and multi-nominal logistic regression analyses were used to assess changes in medical students' attitudes toward primary care over the academic year. We controlled for pretest responses, MCAT scores, and hometown locale (i.e. rural or urban).

RESULTS: Student responses showed generally favorable attitudes toward primary care careers. However, at the end of the academic year, second year medical students were 5.4 times less likely to be interested in pursuing a career in family medicine than first year students ($p=.018$) with no significant differences between the two groups regarding careers in either internal medicine ($p=.13$) or pediatrics ($p=.30$).

CONCLUSION: In spite of favorable attitudes toward primary care both before and after an early longitudinal primary care clinical experience, students' interest in some primary care careers may be adversely affected by this experience.

CAN FACULTY DEVELOPMENT SEMINARS IMPROVE FEEDBACK IN AMBULATORY TEACHING ENCOUNTERS? S. Salerno¹, J. Jackson¹, P. O'malley², L. Pangaro¹;

¹Uniformed Services University of the Health Sciences, Bethesda, MD; ²Walter Reed Army Medical Center, Washington, DC

PURPOSE: To determine if faculty development seminars can improve the amount and quality of feedback in medical student ambulatory teaching encounters without sacrificing efficiency.

METHODS: Faculty members participated in a pre-post study of a faculty development program consisting of three 90-minute interactive seminars teaching evaluation, feedback, and One-Minute Preceptor microskills. Audiocassettes were collected of ambulatory teaching encounters with 3rd year medical students before and after the intervention. The audiocassettes were transcribed and coded by individuals blind to the identity of the teachers and learners. Transcripts were coded using the Teacher Learner Interactive Assessment System, a qualitative tool designed to comprehensively code all utterances into mutually exclusive categories. Ten percent of audiocassettes were double coded to assess inter-rater agreement. Written surveys after each encounter assessed learner, teacher, and patient satisfaction. Learner and teacher perceptions of the amount and quality of several aspects of the encounter, including feedback, were also evaluated.

RESULTS: Nine teachers and 64 third year medical students participated, providing 45 encounters before and 48 encounters after the seminars. 17,859 utterances were coded. Coders achieved a high degree of agreement (Spearman's rho > 0.8). In the baseline encounters, 17% of teacher utterances were some form of feedback, predominantly (92%) minimal feedback statements such as "right" or "I agree". Only 8% of feedback utterances were specific and none were interactive. Most (91%) of the feedback was positive. After the faculty development workshops, the amount and quality of feedback increased; teachers were more likely to provide feedback (OR 1.21; 95% CI 1.07–1.36) and that feedback was nearly twice as likely (OR 2.08; 95% CI 1.45–2.99) to be specific. Subjectively, teachers reported a higher likelihood of allowing students to present their own plans ($p=0.004$) and of providing specific feedback to students ($p=0.004$) after the intervention. Students also reported greater ability to present their own plans ($p=0.02$) and rated the encounters as more successful ($p=0.003$). Patient satisfaction was unchanged. The total time spent teaching during the ambulatory encounters was 14.2 ± 5.3 minutes before and 15.6 ± 6.7 minutes after the seminars, a non-significant difference ($p=0.28$). **CONCLUSION:** Faculty development seminars focused on delivery of feedback can significantly improve the amount and quality of feedback delivered in the ambulatory setting without increasing the time required to teach.

CLINICAL EXPERIENCES AND TRAINING IN END-OF-LIFE CARE DURING RESIDENCY. C.E. Schwartz¹; ¹Montefiore Medical Center/Albert Einstein College of Medicine, Bronx, NY

PURPOSE: Over the past 2 decades, medicine and society have become increasingly concerned with palliative and end-of-life care. In order to train residents effectively, the first step is to assess the current adequacy of their clinical exposure to dying patients and the quality of their faculty supervision and training.

METHODS: Graduating third year residents in Internal Medicine, Family Medicine, and Social Pediatrics at Montefiore Medical Center/Albert Einstein College of Medicine were surveyed in June, 2000 about end-of-life (EOL) clinical experience and education during their residencies. Results are presented as univariate descriptive statistics and using Pearsonchi-square tests to assess the statistical significance of differences in proportions between groups on categorical variables. The level of significance for all tests was 0.05. All analyses were performed using SAS statistical software, version 6.11 (SAS Institute, Cary, NC, 1995).

RESULTS: A total of 55 of the 61 graduating third year residents were surveyed. Over their three years of training, residents reported caring for a median of two patients who died in outpatient primary care, and 10 patients who died during inpatient rotations. In addition, they reported caring for a median of only 3 patients who they felt might have had a life expectancy of 6–12 months. They reported median "little/poor" EOL teaching from faculty, and "little/poor" support from resident colleagues in their EOL clinical care. Only 6% rated oncology rotations as "relevant or helpful in learning about EOL care;" only 10% rated intensive care rotations as helpful; and only 27% rated geriatrics rotations as educationally beneficial.

CONCLUSION: These results suggest that residents are meaningfully and intensively involved in the care of remarkably few terminally ill patients, and when they are, these teaching opportunities are too often being squandered. They suggest that we must improve EOL education in acute inpatient settings, capitalizing on the opportunities to teach residents around the dying patients that they currently care for. They further suggest that we must increase residents' experience in EOL patient care in community settings, targeting primary care clinics, home care, nursing home and hospice services.

NUMERACY SKILLS OF MEDICAL STUDENTS. S.L. Sheridan¹, M.P. Pignone¹; ¹University of North Carolina at Chapel Hill, Chapel Hill, NC

PURPOSE: Although the ability to work with and interpret numbers is vital to patient care, little is known about the numeracy skills of health care professionals. To facilitate mastery of numeracy in medicine, we sought to (1) assess the numeracy skills of first-year medical students, and (2) determine how different risk presentations affect student ability to compare and calculate treatment benefits.

METHODS: We surveyed 62 first-year medical students who were attending a seminar on risk communication. Students were asked to (1) answer three questions assessing their ability to handle numbers in a non-health setting (non-health numeracy; i.e. Imagine that we flip a fair coin 1000 times-what is your best guess about how many times the coin would come up heads?), (2) state which of two drug treatments for a hypothetical disease Y provided more benefit, and (3) calculate the effect of drug treatment for a patient with a given baseline risk of disease. Risk information was presented to each student in one of four randomly allocated risk formats-relative risk reduction (RRR), absolute risk reduction (ARR), number needed to treat (NNT), or a combination of all three of these risk presentation formats (COMBO). Students' abilities to compare and calculate drug treatment benefits were stratified according to the number of correct answers they gave to the non-health numeracy questions: 0 or 1 correct=minimal numeracy, 2 correct=moderate numeracy, 3 correct=advanced numeracy.

RESULTS: Fourteen (23%) students incorrectly answered one or more non-health numeracy questions. Lower non-health numeracy skills correlated with lower health numeracy skills. While more than 90% of students with advanced ($n=48$) or moderate ($n=11$) non-health numeracy skills correctly stated which of two treatments provided more benefit, only 33% of students with minimal ($n=3$) non-health numeracy skills correctly did so ($p=0.03$). Similarly,

71% of students with advanced non-health numeracy skills, but only 36% with moderate and 0% with minimal non-health numeracy skills correctly calculated the effect of drug treatment on a given baseline risk of disease ($p < 0.01$). Regardless of the risk presentation format, students were able to correctly state which of two treatments provided more benefit (range 81–93%). When asked to calculate the effect of drug treatment on a given baseline risk of disease, however, students had significantly more difficulty when risk was presented as NNT (25% answered correctly, compared to 75% for all other presentations; $p=0.01$).

CONCLUSION: First-year medical students with lower non-health numeracy skills demonstrated less skill in estimating both relative and exact treatment benefits. For all students, information presented as NNT was interpreted less successfully than information presented in other risk formats. If these findings persist through training, educators may need to de-emphasize NNT relative to other formats that communicate the same information more successfully.

HOW WELL DO SECOND-YEAR MEDICAL STUDENTS OSCE SCORES PREDICT USMLE SCORES? S.R. Simon¹, K. Volkan¹, C. Hamann², C. Duffey¹, S. Fletcher¹; ¹Harvard Medical School, Boston, MA; ²Massachusetts General Hospital, Boston, MA

PURPOSE: The Objective Structured Clinical Examination (OSCE) is a widely used method of evaluating medical students' performance and competency, but its predictive validity early in medical school is not known. We therefore undertook a study to correlate OSCE scores in the Harvard Medical School second-year physical diagnosis course with scores from the United States Medical Licensing Examination (USMLE) Step 1.

METHODS: Our 16-station OSCE used nine different evaluation formats alone or in combination: question and answer, preceptor role play, standardized patients, actual patients, structural models, 35-mm slides, audiotape, videotape, and CD-ROM. Correct answers were recorded by faculty preceptors. Total OSCE scores and station scores were calculated as follows. Each task or question counted one point, and the percentage of tasks performed or questions answered correctly for each station was designated the station score. The sum of station scores produced a total OSCE score for each student. We linked OSCE scores from 1998, 1999 and 2000 with demographic information, Medical College Admission Test (MCAT) scores, and USMLE Step 1 scores. We conducted correlation analyses and developed a linear regression model with USMLE Step 1 score as the dependent variable and the following covariates: gender, ethnicity, MCAT subscores, and OSCE station scores.

RESULTS: Total OSCE scores and MCAT scores were available for 390 students. Of these, USMLE Step 1 scores were available for 355 students (91%). Average total OSCE score was 59 ± 6 . Average Step 1 score was 234 ± 17 . Correlation coefficient for the total OSCE score with USMLE Step 1 score was 0.41 ($p < 0.0001$). In the linear regression model ($R^2 = 0.52$), 5 of the 16 station scores were significant ($P < 0.05$) predictors of USMLE Step 1 score, as were MCAT biological sciences and physical sciences sub-scores. The 5 predictive stations (alcoholism and abdominal exam, arthritis, calf pain, lung exam, and skin exam) were weighted toward identification of abnormalities (28% of all items on the 5 predictive stations, as compared with 17% of all 16 OSCE stations) and differential diagnosis (32% versus 19%).

CONCLUSION: An OSCE in physical diagnosis taken by second-year medical students is correlated with scores on the USMLE Step 1 exam, with OSCE skills that foreshadow the clinical clerkships (identification of abnormality and development of differential diagnoses) most predictive of USMLE scores. This correlation suggests predictive validity of the OSCE and supports its use in identifying students who need remedial education. This OSCE provides an additional measure to predict students' success on subsequent national board exams.

IMPLEMENTING WEB-BASED CURRICULUM AND EVALUATION COMPONENTS IN A COMMUNITY-BASED MEDICINE CLERKSHIP. D.J. Solomon¹, G. Ferencik¹; ¹Michigan State University, East Lansing, MI

PURPOSE: Community-based experiences can help prepare medical students for addressing disparities in health care. Clerkship management and instruction can be difficult with multiple community-based teaching sites. To help address this challenge we have been developing Web-based components for our internal medicine clerkship. To date these components include forms evaluating the clerkship and individual preceptors, an interactive patient logbook with real-time faculty access to completed logs, clerkship calendar and, on a pilot basis, the ACP/ASIM clinical problem solving cases. This study addressed 1) if students were able to conveniently access these components from all of our six community-based campuses, each with multiple hospitals and clinics, 2) students' acceptance of these web-based components, 3) the efficiency gains of Web-based data collection and clerkship management.

METHODS: We appended questions concerning the ease of accessing the Web-based components and students' preference for this format on the clerkship feedback form. Students completing the ACP/ASIM cases filled out a Web-based survey. Efficiency gains were based on comparison with the former paper-based evaluation system.

RESULTS: To date we received feedback from 87 students who completed the third year basic and fourth year advanced internal medicine clerkship. 91% of the students indicated they had convenient access to the web-based components. 77% of the students preferred Web-based to paper evaluation forms while 60% preferred Web-based to paper logbooks. All eight students completing the ACP/ASIM cases found them useful or extremely useful and only one encountered any technical difficulty.

CONCLUSION: Our students appeared to have little difficulty accessing the web-based components of the clerkship even though they were in a variety of community-based settings with no special provisions for Internet access. Students clearly preferred the Web-based to paper evaluation forms. While on the whole they preferred Web-based logbooks, they were less enthusiastic. The concerns they expressed to a large extent reflected design problems with the new system. Many of these have been resolved. The Web-based feedback forms and logbook dramatically streamlined the process of collating the data and producing feedback reports. These now take less than an hour to prepare and avoid the time consuming and error prone task of transcribing data from paper forms. Pilot results evaluating the use of the ACP/ASIM cases in the clerkship curriculum are encouraging and suggest these cases could provide a valuable adjunct learning experience. For the next academic year we plan to develop two Web-based

instructional modules and a more comprehensive clerkship management system that provides additional real-time feedback to students and preceptors.

COMPARING CARDIAC AUSCULTATORY SKILLS USING REAL PATIENTS AND CD-ROM HEART SOUNDS. J. Solomonides¹, D.S. Hatem¹, M. Belanger²; ¹University of Massachusetts Memorial Medical Center, Worcester, MA; ²University of Massachusetts Medical School

PURPOSE: Recognition of cardiac auscultatory events has been shown to be deficient. Whether the assessment method influences documented skills acquisition has not been determined. Our purpose was to compare second year medical students' abilities to recognize cardiac physical examination findings in patients with known murmurs with recognition of abnormal heart sounds produced by a CD-ROM.

METHODS: Ninety-one second year medical students were evaluated on cardiac auscultation skills at the end of a Physical Diagnosis course. Students examined 1 Real Patient (RP) with a known murmur and listened to a CD-ROM (CD), to 2 murmurs and 2 extra sounds, then recorded their findings. Findings assessed included Heart Sounds (HS); 5 items for RP and CD) and Murmur (M); 5 items for RP, 4 items for CD) scores. RP murmurs included a systolic ejection murmur (SEM; n=44 students) and aortic stenosis with aortic insufficiency (AS/AI; n=47). Gold standard for the RPs was a recent exam by their cardiologist. The Criley CD-ROM was used to assess HS with abnormalities including S3 and S4 sounds (5 items; n=91), and murmurs with abnormalities including mitral regurgitation (4 items; n=91) and aortic stenosis (4 items; n=91). Mean scores were compared using paired sample T-tests.

RESULTS: Students' recognition of cardiac findings differed significantly with the two methods of evaluation. Overall mean scores for the CD-ROM cardiac findings (HS and M scores) were 77.5 while those for the RPs were 59.3 ($p < .001$). Mean HS scores were similar between groups (CD 81.7, RP 78.7; $p=0.1$) while M scores accounted for the differences in student performance for the two evaluation methods (CD 72.0, RP 37.7; $p < .001$). These mean score differences were similar when comparing single murmurs on CD-ROM to the RP with a single murmur (CD 68.5, RP 37.7; $p < .001$) and to the RP with two murmurs (CD 68.5, RP 38.1; $p < .001$).

CONCLUSION: Medical students scored significantly higher when cardiac auscultation skills were assessed using a commercially produced CD-ROM compared to their skills in examining real patients. Differences may be due to examination technique, patient factors (body habitus), competing sounds (breath sounds), or patient complexity (2 murmurs v 1 murmur). The lack of difference in HS scores may be due to the small numbers of abnormal findings resulting in lack of discriminatory capacity as opposed to a true skills difference. The use of a CD-ROM alone to assess cardiac findings may overestimate student skills. This reinforces the need to focus our cardiac auscultatory skills teaching to enhance student examination skills acquisition.

THE IMPACT OF A MISSION BASED FACULTY DEVELOPMENT PROGRAM. M. Sostok¹, L. Coberly¹, N. Walker¹, G. Rouan¹; University of Cincinnati, Cincinnati, OH

PURPOSE: We designed a faculty development program with the mission of achieving the following goals: (1) increase the number of student contacts with community based faculty, (2) promote an active learning environment with community based faculty and (3) increase the number of independent patient contacts by students in community teaching sites. Our current study looks at the impact of this mission based program.

METHODS: We developed a survey evaluating six key components of an Active Learning Environment (ALE) (table 1). Third year students rated their preceptors on each ALE component at the end of the 4-week internal medicine ambulatory clerkship using Likert-type scales to generate an ALE score (min.=6, max.=30). Faculty development programs were designed to improve preceptors' ALE teaching skills. Follow-up letters were sent to each participant reviewing program highlights and providing them with the ALE ratings they received from students for 6 months prior to their workshop. These were compared to preceptors' ratings received for 6 months after their faculty development participation. Student's t-test was used to compare differences in pre- and post-workshop mean survey results.

RESULTS: To date 14 community preceptors have participated in three faculty development programs. Table 2 results show a trend towards higher post-workshop ALE scores. Their was a significant increase in the post-workshop independent patient contacts reported by students.

CONCLUSION: A mission-based faculty development program can have a significant impact on the learning environment of community teaching sites in an ambulatory clerkship.

Components of an Active Learning Environment (Table 1)

Independent pt. contact	Develop an assessment
Propose a plan	Discuss visit with patient
Document visit in chart	Self-directed learning

Faculty Development Group's mean ALE results (Table 2)

Indicator	Pre-workshop	Post-Work (p)
Student contacts	1.6	2.1 (p=0.15)
ALE Score	19.1	23.3 (p=0.08)
Independent pt. contact	11	27 (p=0.002)

DIAGNOSTIC CHARACTERISTICS OF CARDIAC AUSCULTATION BY CLINICAL TRAINEES IN A REAL-WORLD SETTING. J. Stahl¹, A.N. Babu²; ¹Massachusetts General Hospital, Boston, MA; ²St. Louis VAMC, St. Louis, MO

PURPOSE: A limited number of studies have evaluated the skills of medical students (MES) and internal medicine residents (IMR) in cardiac auscultation. Subjects have been tested in a

classroom environment largely using simulated or recorded sounds. Typically, only one lesion was presented at a time. Overall diagnostic accuracy has not been found superior for IMR compared to MES. Our study examines the performance of MES and IMR in cardiac auscultation in a real-world inpatient setting.

METHODS: Study subjects were a self-selected sample of MES and IMR at North Shore University Hospital, Manhasset NY between 9/4/2 to 9/5/3. Participants were invited to submit clinical cardiac exams they performed on patients who were subsequently to undergo transthoracic echocardiography with color Doppler (TTE) for appropriate clinical indications. Subjects submitted their findings directly into a computerized database or via a paper clinical exam form. The submitted exam was included only if it had been performed and recorded prior to the performance of the TTE. All TTE were performed in standard fashion by the cardiology department which had no prior knowledge of which patients were participating. **RESULTS:** 53 blinded cardiac auscultations (37 IMR/16 MES) were compared to findings on TTE. Sensitivity (Sn) and specificity (Sp) could only be calculated for aortic stenosis (AS), insufficiency (AI) and mitral regurgitation (MR) as there were insufficient events for assessing other valvular lesions. The crude Sn/Sp for the whole group was AS: .67/.81, AI: .15/.9, MR: .48/.42. The Sn/Sp by subgroup was IMR(AS:.67/.89, AI:.13/.93, MR:.48/.38) v MES(AS:.67/.58, AI:.28/.55, MR:.5/5) The skill in discriminating amongst these murmurs was examined by Receiver Operating Characteristic (ROC) curve analysis. The ROC areas were IMR/MES — AS: .71/.38, AI: .31/.2, MR: .23/.25. The most common errors were MR False Positive (MRFP), MR False Negative (MRFN), AI False Negative (AIFN) and AS False Positive (ASFP). When a single murmur was present the strongest positive correlation between errors was ASFP:MRFN ($r^2=.3$). When multiple murmurs were present, it was ASFN:AIFN ($r^2=.28$) and AIFP:MRFN ($r^2=.22$).

CONCLUSION: In a clinical setting, IMR perform better diagnostically than MES. Both groups were more accurate for systolic than diastolic murmurs. The most common errors were mistaking AS and MR for each other and mistiming murmurs, especially when multiple lesions co-existed. To enhance diagnostic accuracy, training in the cardiac exam should focus on correct temporal orientation in the cardiac cycle and the contextual clinical findings associated with a murmur.

A PROBLEM WITH REPEATED STANDARDIZED PATIENT EXAMINATIONS? G.M. Talente¹, S.A. Haist¹, S.M. Quinlivan¹, J.F. Wilson¹; ¹University of Kentucky, Lexington, KY

PURPOSE: Standardized Patient (SP) examinations are being used more frequently for the evaluation of residents and medical students' clinical and interpersonal skills. There is little information available on the effect of repeated testing on performance. We analyzed this effect on our residents' performance on a SP exam.

METHODS: We have recently developed a standardized patient exam to replace the traditional ABIM Clinical Evaluation Exercise administered by faculty. This test was given to the PGY1, PGY3 and PGY4 residents in Internal Medicine and Medicine/Pediatrics at the University of Kentucky. The exam consisted of an assessment of history, physical examination and verbal and non-verbal communication skills scored by the SP. The SP scored a 225-item exam checklist as yes/no and rated non-verbal communication skills on a 7-point Likert scale with anchors at the extremes. The number of previous SP tests for each resident was recorded. The relationship between performance and number of SP experiences was evaluated by correlation analysis.

RESULTS: 50/52 residents participated. The number of previous tests ranged from 0 – 50 with a median of 4, by self-report. The number of previous SP tests was associated with better physical exam performance ($r = 0.442$, $p=0.001$). Previous testing was associated with a lower rating of non-verbal interpersonal traits ($r = -0.415$, $p=0.002$). Specific traits including: friendly tone of voice ($p=0.013$), interested voice ($p=0.004$), frequency of smiling ($p=0.05$), eye contact ($p=0.033$), nodding ($p=0.017$), willingness to listen ($p=0.028$), sympathy and warmth ($p=0.029$), ease to talk with ($p=0.009$), openness to questions ($p=0.05$), open body posture ($p=0.012$), and relaxed hand position ($p=0.02$) were all scored lower for residents with more SP test experience.

CONCLUSION: The association of improved objective performance and diminished interpersonal skills may be the result of repeated testing where trainees learn to perform for improved objective performance and develop testing behaviors that negatively effect their interactions with the standardized patient.

SETTING PERFORMANCE STANDARDS FOR A STANDARDIZED PATIENT FORMAT CLINICAL EVALUATION EXERCISE FOR INTERNAL MEDICINE RESIDENTS. G.M. Talente¹, S.A. Haist¹, J.F. Wilson¹; ¹University of Kentucky, Lexington, KY

PURPOSE: Standardized Patient (SP) examinations are being developed as an alternative means of assessing clinical competence among trainees at all levels. In order to utilize the information obtained in such a test reasonable standards for acceptable and unacceptable performance must be determined. We recently implemented a SP format of the American Board of Internal Medicine's required Clinical Evaluation Exercise (CEX). Standards were determined using the Modified Angoff method, a criterion based method of standardization using multiple expert raters.

METHODS: For the CEX, residents were instructed to perform a problem-focused history, comprehensive physical exam (PE), and to write a detailed assessment and plan (AP) for one of four cases. Checklists (yes/no) were completed by the SP regarding history, PE and humanism. Faculty developed keys for the AP for each case. Seven faculty in General Internal Medicine acted as expert raters. They were given the materials for each of the four cases and asked to determine standards separating four levels of performance, unsatisfactory, marginal, satisfactory and superior. They were provided with performance data and were given an opportunity for discussion. They then provided final performance cutoffs. The overall standard used was the average of the seven raters' adjusted standards.

RESULTS: 50/52 Residents in Internal Medicine and Medicine/Pediatrics participated in the SP format CEX. Performance on the history component was: 54% marginal, 46% satisfactory. For PE content: 6% unsatisfactory, 24% marginal, 54% satisfactory and 16% superior. For Humanism: 10% marginal, 34% satisfactory, and 56% superior. For clinical judgement: 18% unsatisfactory, 46% marginal, and 36% satisfactory. Of the 51 individual performance cutoff decisions made by the seven raters the variation between raters decreased in 42 of 51 instances (p

= .0001) from the initial to the adjusted standards. Overall performance was judged unsatisfactory for 15 residents (7 interns).

CONCLUSION: Use of the Modified Angoff technique for standardization decreased the variation between raters' standards. Greater than 30% of the residents were judged as marginal or unsatisfactory in history taking, PE skills, and clinical judgement (AP), a number much higher than expected and much higher than determined by the traditional CEX. Further study will be required to explain this finding in order to confirm the validity of these standards.

INVOLVING RESIDENTS IN DEVELOPING AND IMPLEMENTING A CLINICAL GUIDELINE. I.G. Tape¹, R.A. Walker¹, S.W. Shummur¹, J.R. Bessmer¹, M.S. Jackson², D.V. O'dell¹; ¹University of Nebraska, Omaha, NE; ²University of Kansas, Wichita, KS

PURPOSE: The success of clinical practice guidelines is enhanced by involving practicing physicians in guideline development, yet resident physicians are rarely included in such efforts. We hypothesized that residents could develop and implement a successful practice guideline as a way of gaining quality improvement experience. We studied their guideline development process and evaluated the effect of their guideline on patient care.

METHODS: We asked a group of nine resident volunteers to develop a clinical guideline on a topic of their choice. The hospital funded the effort, extended after-hours availability to guideline-related diagnostic services and provided a nurse coordinator. We observed the process of guideline development, tracked guideline utilization, and measured physician satisfaction with it after six months of use. We also compared admission rate, length of stay and resource costs for a cohort of patients before and after guideline implementation.

RESULTS: The residents selected acute chest pain evaluation and developed an evidence-based guideline. They focused on risk stratification and rapid diagnostic testing. They streamlined documentation by developing evaluation templates that substituted for written notes and orders. When applied to 319 patients 30 and older presenting with acute chest pain, 49% were classified as "possibly cardiac" of which 41% were considered "high suspicion" and 59% "low suspicion." An acute cardiac diagnosis was made in 12% overall (24% of possibly cardiac, 46% of high suspicion and 9% of low suspicion). This cohort of 319 patients was compared to 150 chest pain patients evaluated before the guideline was implemented. Median length of stay and cost of care both decreased by 34%. Admission rate was unchanged. Physician satisfaction with the guideline was uniformly positive six months after its implementation. Physicians rated 24-hour availability of a nurse-coordinator and after-hours functional testing particularly highly. Guideline utilization remains high more than two years after its implementation.

CONCLUSION: Resident physicians were the key participants in developing and implementing a successful clinical guideline that enjoys ongoing use. By experiencing the success of their own efforts, residents learned important lessons about quality improvement. Such experiences are an important addition to resident education and to the quality of care at academic health care centers.

PRIMARY CARE PHYSICIANS INTEREST IN SPECIAL FEATURES OF ELECTRONIC PUBLICATION. D. Torre¹, S.M. Wright¹, R.F. Wilson¹, M. Diener-west², E.B. Bass¹; ¹ Johns Hopkins University School of Medicine, Baltimore, MD; ² Johns Hopkins University School of Hygiene and Public Health, Baltimore, MD

PURPOSE: Primary care physicians (PCPs) can access electronic versions of an increasing number of medical journals. Academic PCPs are particularly important users of journals because of their roles in education, research and patient care. Our objective was to assess academic PCPs' interest in using special features that could accompany electronic publication of journal articles and whether interests differ between general internists (GIMs) and family practitioners (FPs).

METHODS: We surveyed 175 GIMs and 175 FPs randomly selected from the Society of General Internal Medicine and the Society of Teachers of Family Medicine — two societies that have a strong commitment to education and research in primary care.

RESULTS: The overall response rate from the first mailing of the survey was 48%. Computer proficiency was rated as "good" by 38% of GIMs and 30% of FPs; 55% of GIMs and 57% of FPs reported using on-line journals. The primary work activity was patient care for 43% of GIMs and 45% of FPs, teaching for 31% of GIMs and 49% of FPs, and research for 35% of GIMs and 5% of FPs. GIMs were more likely than FPs to regularly read original articles (56% vs. 35%) and editorials (48% vs. 25%) but not reviews (50% vs. 51%). GIMs were more likely than FPs to report high interest in having an electronic link to an article's references (62% vs. 46% for original articles [$p < .05$], 54% vs. 43% for reviews [$p < .05$], and 48% vs. 34% for editorials [$p < .05$]) or having an electronic link to appendices and supportive materials (44% vs. 19% for original articles [$p < .05$], 33% vs. 14% for reviews [$p < .05$]), but less likely to report high interest in having an electronic link to health-related web sites (29% vs. 44% for original articles [$p < .05$], 29% vs. 38% for reviews [$p < .05$]), electronic communication with authors (25% vs. 31% for original articles [$p < .05$], 25% vs. 28% for reviews [$p < .05$], 24% vs. 56% for editorials [$p < .05$]), or electronic communication with other readers (31% vs. 50% for original articles [$p < .05$], 28% vs. 40% for reviews [$p < .05$], 13% vs. 21% for editorials [$p < .05$]). The majority of both groups reported moderate to high interest in receiving an electronic copy of the table of contents before each journal issue is released.

CONCLUSION: Academic GIMs and FPs have considerable interest in potential features of electronic medical journals, but they differ in their level of interest in specific features. Journals that target these important groups should consider these views when deciding on specific electronic features they want to offer.

WEB-BASED RESOURCES IN UNDERGRADUATE AMBULATORY EDUCATION — READY FOR PRIME TIME? A. Tulsky¹, N. Tannery¹, D. Lescisin¹, D. Elnicki¹; ¹University of Pittsburgh, Pittsburgh, PA

PURPOSE: Using community sites for medical student ambulatory training decentralizes their education by moving students away from the traditional source of learning resources — the medical school library. At the University of Pittsburgh School of Medicine (UPSOM) we developed a web site to provide online resources specifically oriented to learning ambulatory

medicine. We wanted to know what predicted student comfort in its use and overall satisfaction with the site.

METHODS: The Community/Ambulatory Medicine Clerkship (CAMC) is a 12 week ambulatory clerkship providing 3 week community-based experiences in Internal Medicine, Pediatrics, Family Practice and a selective. UPSOM librarians developed a web site with the curriculum online and links to relevant full text articles, textbooks, Medline and to specific patient and physician oriented web sites. Students completed a self-administered survey on the web site at the end of their CAMC clerkship during the 1999 academic year. We asked students if they owned a computer with Web access, how many of their clinical sites had Web access, their level of comfort finding and using Web medical sources before and after the clerkship and the overall value of the site to their learning. A forward stepwise linear regression model was built to predict comfort and satisfaction in the use of the Web site as a learning resource.

RESULTS: We surveyed 147 students (98% response rate) of whom 64% owned their own computer. On a scale of 1 to 5 (5 = extremely valuable), 88% of students rated the value of the Web site > 3. While 76% were extremely or somewhat comfortable finding and using Web resources before the clerkship, 90% ($p < .0001$) reported the same at the end. The site was accessed on a weekly or greater basis by 58% of the students. Although 13% of the students experienced no clinical sites with Web access, 61% had 1 or 2 sites with access and 26% had 3 or 4 sites with Web access. In a stepwise linear regression model, comfort in using Web medical resources at the end of the rotation was best explained by comfort in using the Web before the rotation, owning a computer and the number of clinical sites that had Web access ($R^2 = .48$). Overall satisfaction with the Web site was explained by the time of year the clerkship was taken, owning a computer and the number of sites with Web access ($R^2 = .067$).

CONCLUSION: Students found value in an ambulatory clerkship Web site as an information resource. It could be made more useful to them by assuring better access to a computer at their clinical sites and providing a personal computer. While comfort in using a Web site as a resource improved by the end of the clerkship, this can be enhanced by training prior to the beginning of the clerkship and again assuring better computer access. Given the relatively low frequency of use however, it cannot yet suffice as the sole curricular resource.

CAN PALLIATIVE CARE EDUCATION IMPROVE THE OPIOID PRESCRIBING PRACTICES OF MEDICAL RESIDENTS? W.A. Ury¹, M. Rahn¹, V. Tolentino¹, M. Pignotti¹, D.P. Sulmasy¹; ¹Saint Vincent's Hospital, New York Medical College, New York, NY

PURPOSE: To assess whether a case-based palliative care curriculum improved the opioid prescribing practices of medical residents.

METHODS: The curriculum consisted of 10 one-hour case-based seminars, including two devoted to pain management. We reviewed consecutive billing and pharmacy records of patients of medical residents ($n=733$) and a comparison group of patients of neurology and rehabilitative medicine residents ($n=273$) who received an opioid during two 8-month periods before (1/1/97–4/30/97) and after (1/1/99–4/30/99) the implementation of the curriculum. The data abstraction protocol was validated by standard chart review of a random subsample of 50 charts (concordance = 89.9%). Three outcomes were measured: 1) % opioid orders for meperidine in non-gastrointestinal patients (excluded because of a longstanding belief that meperidine should be used in biliary disease); 2) % opioid orders accompanied by a bowel regimen (to prevent constipation); and 3) % opioid orders accompanied by adjuvant nonsteroidal anti-inflammatory drugs (to achieve the "additive effect").

RESULTS: We had a power of 80% to detect a 10% difference at $\alpha = .05$, two-tailed. Percent change before/after for each outcome for each group was assessed by t-testing. Logistic regression models were developed for each of the outcome measures in order to control for independent variables that could be confounders and to assess the magnitude of effects. The models used controlled for age and race. Significance is noted by * if $p < .05$.

CONCLUSION: These data suggest that a palliative care curriculum can improve the opioid prescribing practices of medical residents, over and above a secular trend among other house officers. Larger samples and further research are needed to understand these findings and how palliative care education can improve patient care.

	Experimental Group			Control Group		
	Pre-Post % Change	Logistic Regression Model Odds Ratio	95% Confidence Interval	Pre-Post % Change	Logistic Regression Model Odds Ratio	95% Confidence Interval
Medication Received						
Meperidine	-4.4 *	0.55	0.32 to 0.96 *	+0.8	1.08	0.51 to 2.30
Bowel Regimen	+8.5 *	1.48	1.07 to 2.03 *	+4.4	1.21	0.74 to 2.00
Nonsteroidal	+5.3 *	1.53	1.01 to 2.32 *	+8.4 *	2.27	1.10 to 4.67

AN ASSESSMENT OF RESIDENTS ABILITIES TO DETECT AND CARE FOR VICTIMS OF DOMESTIC VIOLENCE. N. Varjavand¹, D.H. Novack¹, D.G. Cohen¹; ¹MCP Hahnemann University School of Medicine, Philadelphia, PA

PURPOSE: Domestic Violence (DV) is the number one cause of injury to women in the United States. Victims of DV often present to their doctors complaining of vague or chronic symptoms. Delayed and incorrect diagnosis can lead to needless suffering, be potentially life threatening, and also be costly to society as physicians spend significant health care resources to work-up presenting symptoms. Because of increased public awareness of DV, and reports of underdetection by practicing physicians, many have sought to improve DV education at the medical student and residency levels. Have these educational interventions improved residents'

abilities? In the current study, we examined the abilities of internal medicine residents to identify and care for a victim of domestic abuse.

METHODS: Seventy-one internal medicine residents in four programs participated in a ten-station standardized patient-based Clinical Skills Assessment. One standardized patient (SP) portrayed a woman, complaining of headaches, who was a victim of domestic abuse. The SP assessed residents' elicitation of information and counseling interventions on a fifteen-item checklist. After the encounter, residents documented the patient's problems, etiologies, work-up and treatment recommendations.

RESULTS: Forty (56%) residents correctly diagnosed domestic violence and discussed the diagnosis with the patient. Eighteen residents (25%) asked about immediate safety concerns. Twenty-three (32%) asked about concomitant child abuse. Forty-one (58%) did not refer the patient for DV counseling. Forty-eight (68%) made one or more incorrect recommendations. Some of these recommendations could be dangerous: recommending marital therapy (n=19, 27%), prescribing potentially addictive medicines (n=9, 12%), or suggesting that the patient bring her abusive partner in to the doctor (n=23, 32%). To work-up the patient's headache, 36 (52%) ordered unnecessary tests which would have cost \$32,500.

CONCLUSION: In our sample there were many deficiencies in the residents' abilities to diagnose and manage DV. The majority of residents provided inappropriate care that in some cases would be dangerous. In addition, the residents' ordering of unnecessary and costly tests is a drain to already-limited health care resources. There appears to be a need for more intensive training in DV at the undergraduate and residency levels.

OUTCOMES OF AN EVIDENCE BASED MEDICINE COURSE FOR PRIMARY CARE INTERNAL MEDICINE RESIDENTS. K.A. Vom Eigen¹; ¹University of Connecticut School of Medicine, Hartford, CT

PURPOSE: Improvement of knowledge and skills in Evidence Based Medicine (EBM) is an important goal of residency training in Internal Medicine, though many programs do not have the resources to provide intensive EBM curricula. I evaluated the outcomes of a simple EBM course taught by a single instructor as part of a 3-month firm rotation for second year Primary Care Internal Medicine residents.

METHODS: Groups of 4 to 5 residents participate in 10 to 12 weekly 1-hour sessions during the rotation. After 2 didactic sessions to review basic biostatistics, clinical epidemiology, and other core EBM topics, the sessions follow a journal club format. One resident searches for, selects, and distributes an article relevant to the clinical question chosen for that session. They present a summary and initial evaluation of the article, which is followed by participatory discussion facilitated by the instructor. The Users' Guides to the Medical Literature series from JAMA is used to guide topic selection and as supplementary reading. Group discussions focus on the Users' Guides core concepts and evaluative criteria. The residents also do a more intensive final presentation in a similar format. Changes in EBM knowledge and skills were evaluated with a 38-item test given before and after the course, with blinding of participants and the instructor to the results. Participants also self-evaluated changes in their knowledge, skills, and practice on an anonymous evaluation form.

RESULTS: From September 1998 through December 2000, 39 residents took the course. Of 36 who completed both tests, 33 (92%) improved their scores. The mean score improved from 64.4% to 74.5% correct responses ($p < .000001$, paired t-test), a 15.7% relative increase, or an average of 3.8 more correct items. Of the 37 who completed the evaluation form, 78% said participation in the course had improved their evaluative skills (>3 on a 5-point agreement scale). 73% felt more confident in evaluating medical literature, and 78% said the course had changed the way they read and evaluate articles. 49% said the course had significantly improved their clinical knowledge and 43% said it had changed their patient care. Only 30% said it had increased their chances of participating in research in the future.

CONCLUSION: Participants in this course improved their performance on a test of EBM knowledge and skills. Most participants said their skills and confidence had improved as a result of the course, and that they had changed the way they evaluate medical literature. Fewer felt the course had changed their clinical knowledge or practice, and most said it had not increased their interest in participating in research. Similar results might be achieved in other settings with this course format.

COMPARING CLINICAL TEACHING SKILLS OF FACULTY IN THE INPATIENT AND OUTPATIENT SETTINGS USING STUDENTS RATINGS FROM VALIDATED INSTRUMENTS. TR. Vu¹, D.K. Litzelman²; ¹Washington VAMC, Washington, D.C.; ²Indiana University, Indianapolis, Indiana

PURPOSE: To compare medical students' mean ratings of faculty's clinical teaching in the outpatient (OP) and inpatient (IP) settings using internally validated instruments, and to evaluate whether the amount of time a teacher spends in both the OP and IP settings and years of teaching experience predict teaching performance ratings for discrete elements of clinical teaching.

METHODS: Following IP and OP internal medicine rotations, third-year medical students completed corresponding validated instruments to evaluate their IP and OP teachers. The IP instrument reflects the seven categories of the Stanford Faculty Development Program's (SFDP) educational framework for analyzing clinical teaching and employs five-point Likert scales (1=least to 5=most effective), while the OP instrument reflects three of the SFDP's educational constructs. Between 1996 and 1999, 1193 evaluation forms were collected on 102 faculty who all had at least five student-evaluators. Repeated measures analysis of variance (ANOVA) compared the means of corresponding categories between IP and OP teaching. Pairwise differences within each teaching setting were examined by the Tukey-Kramer method. Multiple regression analyses assessed the relationship between teaching experience and time spent on the IP wards and OP clinics and performance ratings on discrete elements of clinical teaching.

RESULTS: Analysis of variance revealed that "Communication of Goals" (CG) ratings for IP teaching (mean=3.70; 95% CI=3.63-3.76) were significantly higher ($F=241$; $p<.001$) than for OP teaching (mean=3.50; 3.35-3.65). No significant difference was found between IP and OP teaching for the "Learning Climate" (LC) categories (IP mean=4.44; OP mean=4.48) or for the

"Self-Directed Learning" (SDL) categories (IP mean=4.31; OP mean=4.31). Pairwise comparisons revealed that the three categories within each teaching setting were all significantly different from one another ($p<.04$). Regression analysis revealed that time spent teaching in the IP setting was a significant negative predictor ($p<.001$) in the LC and CG ratings for IP teaching, while total years of teaching experience was a significant positive predictor ($p<.001$) in the SDL ratings for IP teaching. For the OP teaching ratings, time spent teaching in the OP setting was a significant negative predictor ($p<.002$) for CG and SDL, while LC ratings were unaffected by any of these variables.

CONCLUSION: Our results suggest the need for focused faculty development programs to enhance CG skills-particularly for teachers who spend more clinical time in either the IP or OP setting, SDL skills for less experienced IP teachers and teachers who spend more clinical time in the OP settings, and LC skills for faculty with more inpatient clinical time.

INTERNAL MEDICINE TRAINEES PERCEPTIONS OF CROSS-CULTURAL MEDICINE. S. Wahi-Gururaj¹, S. Franco², J. Orlander³; ¹Section of General Internal Medicine, Boston University Medical Center/ENRM Veterans Hospital, Boston, MA; ²Boston University Medical Center, Boston, MA; ³VA Boston Healthcare System/BUSM, Boston, MA

PURPOSE: To understand Internal Medicine trainees' attitudes towards cross-cultural medicine and its impact on clinical care for purposes of developing curricula.

METHODS: Internal Medicine residents at one urban, multi-center residency program were asked to complete a 71-item questionnaire. The survey was self-administered at conferences, on wards, and by e-mail. It was piloted on faculty, fellows, and chief residents for clarity of the questions and content validity. We solicited residents' observations of the impact of cross-cultural issues on clinical care. Attitudes and self-perceived knowledge and skills were assessed with 6-point Likert scales.

RESULTS: 71 of 133 residents (53%) responded. There were 35 PGY-1, 16 PGY-2, and 20 PGY-3. The mean age was 28 years, 52% were women, 60% white, 39% were fluent in a language other than English, and 49% had non-U.S. born parents. Residents reported experiencing a situation where a patient's care was negatively impacted due to sexual orientation (28%), gender (29%), ethnicity (38%), race (44%), socioeconomic status (56%), or a language barrier (90%). 76% of residents responded that cultural issues are "very"/"extremely" important to patient care, but only 50% thought training in cross-cultural medicine could substantially improve health delivery. Residents viewed the following as major barriers to integrating cross-cultural issues in residency training: time constraints (60%), lack of experts (30%), lack of interest among learners (19%), lack of interest among faculty (16%), perception that these issues are already adequately addressed (12%), belief that these issues might offend some people (12%). Most residents rated themselves as being overall "somewhat" to "moderately" knowledgeable (69%) and skilled (81%) about cultural issues and medicine. A significant association was found between PGY level and perception of overall knowledge ($p<0.04$). Interestingly, 16-18% of PGY-1 and 2 residents felt very knowledgeable about cultural issues and medicine, while no PGY-3 did.

CONCLUSION: Successful development of curricula in cross-cultural medicine requires an understanding of trainees' attitudes and observations. Residents regard cultural issues as important to patient care and many report observing a direct impact. Despite diverse backgrounds, they do not consider themselves to be highly knowledgeable or skilled in this area. Based on these observations, a need for cross-cultural curricula exists.

THE AVAILABILITY AND IMPACT OF MENTORSHIP ON JUNIOR FACULTY DEVELOPMENT. J.M. Watanabe¹, L.D. Chew¹, D.S. Buchwald¹, D.S. Lessler¹; ¹University of Washington, Seattle, WA

PURPOSE: Mentors are regarded as important for the academic success of junior faculty, but the availability and impact of mentors on the careers of junior faculty has not been well studied. The purpose of this study was to evaluate junior faculty perceptions of senior faculty mentors at one large academic medical center.

METHODS: We surveyed all 154 physicians in the clinician-educator (CE) and clinician-scientist (CS) tracks from the Departments of Medicine, Surgery and Obstetrics/Gynecology who held the rank of Instructor or Assistant Professor at the University of Washington. We collected demographic information and asked respondents to describe their academic productivity and rate the adequacy of their mentors.

RESULTS: Among 122 (79%) junior faculty who responded the mean age was 38.7 years (SD=4.3); 56 (46%) were women; 52 (43%) were in the CE track; and the mean years in rank was 2.5 (SD=1.5). CEs reported authoring 2.6 (SD=3.7) book chapters and 4.5 (SD=8.1) original manuscripts whereas the means for CSs were 3.5 (SD=12.1) and 10.9 (SD=12.6), respectively. CEs and CSs had a mean of 1.1 (SD=1.8) and 3.1 (SD=2.6) funded grants, respectively. When asked if they had access to a senior faculty member in their department whom they trusted and who supported them in achieving their career goals, similar percentages of CEs and CSs answered affirmatively (75% and 74%, respectively). On the other hand, when asked if they had adequate mentoring, 36% of all respondents "strongly agreed" or "agreed," (mentored faculty) versus 64% whom "neither agreed nor disagreed," "disagreed" or "strongly disagreed" (non-mentored faculty). In bivariate analyses, mentored faculty were more likely to be men (44% vs. 27%, $p=0.049$), have completed fellowship training (41% vs. 21%, $p=0.048$) and to be CSs (51% vs. 15.4%, $p<0.0001$). 93% of mentored respondents "agreed" or "strongly agreed" that they would achieve their short-term career goals vs. 65% of the non-mentored faculty ($p=0.0006$). Gender (OR=3.6, $p=0.02$) and academic track (OR=9.9, $p=0.0002$) were independently associated with junior faculty perception of adequate mentoring in logistic regression adjusting for age, years at current rank, department affiliation and fellowship training. Mentored faculty reported more publications than non-mentored faculty. This difference did not reach statistical significance in linear regression ($p=0.1$) adjusting for all independent variables entered in the logistic multivariate analysis.

CONCLUSION: Over half of the junior faculty in this study did not feel that they were being adequately mentored. Women and CEs were more likely to perceive not being adequately mentored. Strategies to improve mentoring need to be developed and evaluated.

QUALITY OF PAP SMEARS BY INTERNISTS: EXPLORING THE INFLUENCE OF GENDER, STATUS AND PRACTICE SITE. R.S. Watkins¹, D. Case¹, D.E. Bonds¹; ¹Wake Forest University School of Medicine, Winston-Salem, NC

PURPOSE: Little is known about the quality of Pap smears obtained by internists. The presence of endocervical cells increases the quality of pap smears and has been previously evaluated in family physicians and ob-gyn. We examined the influence of gender, status (faculty vs. resident) and practice site (university vs. community) on rates of satisfactory smears as defined by the presence of endocervical cells.

METHODS: Retrospective review of Pap smears evaluated by one cytology department over a 2 year period that were performed by internal medicine faculty and residents from 1 academic department at three clinic sites: two university-based clinics (one staffed by faculty and one staffed by residents) and one community-based clinic (staffed by both faculty and residents). Vaginal smears were excluded.

RESULTS: 1086 smears were evaluated. 56% of smears were obtained at the faculty university clinic (FC), 18% at the resident university clinic (RC), 26% at the community clinic (CC). FC patients were older than RC or CC (mean age 56,40,42). 86 Internists were included: 65% males, 17% faculty (60% FC), 83% residents (52% CC). 75% of smears were satisfactory, 19% satisfactory but limited by no endocervical cells, 5.6% limited for other reasons, 0.18% unsatisfactory. Males and female internists had similar satisfactory rates (78% vs. 71%). Physicians at the university clinics had higher satisfactory rates (83%) compared to the community clinic (53%). University faculty had higher satisfactory rates than community faculty (83% vs. 43%). University residents had satisfactory rates similar to university faculty (83%) but higher rates than community residents (56%). Community residents had higher satisfactory rates than community faculty (56% vs. 43%). Using the generalized estimating equations approach to control for volume of smears done by each physician we found that younger patient age ($p=0.02$), faculty status ($p=0.03$), and community clinic site ($p=0.0001$) were all associated with lower satisfactory rates. The odds of a satisfactory reading were 1.13 for each additional decade of age, 2.04 for residents vs. faculty, and 6.22 for FC vs. CC, 3.49 for RC vs. CC, and 1.78 for FC vs. RC.

CONCLUSION: Satisfactory rates among this sample of Internists were 11–18% lower than published reports for family physicians and ob-gyn. Rates of satisfactory smears did not differ by gender. We did find higher satisfactory rates in older patients, residents and at university-based sites. These results suggest that academic institutions should place more emphasis on teaching Pap smear sampling technique to internists.

CHANGES IN INTERNAL MEDICINE PROGRAM DIRECTORS RECOMMENDATIONS AND PERCEPTIONS REGARDING GRADUATES MASTERY OF PROCEDURAL SKILLS. R.S. Wigton¹, L.L. Blank², D.R. Nickol¹, F.D. Duffy²; ¹University of Nebraska, Omaha, NE; ²American Board of Internal Medicine, Philadelphia, PA

PURPOSE: To assess the effect of changes in medical care practices and new residency requirements on procedural skills training, we compared responses of internal medicine (IM) program directors in a new procedural skills survey with responses to a similar survey in 1987 — before the implementation of related requirements by the American Board of Internal Medicine (ABIM) and Residency Review Committee (RRC).

METHODS: Mailed and web-based surveys were conducted of all 395 internal medicine residency programs regarding 45 procedural skills. We asked directors which methods of procedural skills training they used, which procedures residency graduates should master, and the amount of training needed to attain and maintain competency in each procedure. This paper reports analysis of responses of 269 (68%) program directors or their designees.

RESULTS: As in the 1987 survey, the proportion of program directors who said all residents should master these procedures differed from the proportion who said that all residents do master them. For 21 of the 45 procedures in the survey, at least 30% more respondents said residents should master the procedure than said all their residents do master the procedure. This gap is larger in the new survey. For the 35 procedures that appeared in both surveys the new survey found 12% fewer reporting that all residents mastered those procedures. For example, although more than 90% of respondents in both surveys thought thoracentesis should be mastered by all residents, 79% in the 1987 survey vs. 58% in the new survey thought it was mastered by all. In addition, estimates of the amount of training needed to master the procedures were lower in 31 of 35 procedures and were 30% lower overall. Some procedures were less strongly recommended, as might be expected from changes in practice or new regulations (e.g. bone marrow aspiration, gram stain). For 4 of 6 procedures required by both the ABIM and RRC, fewer than 60% of directors said that all of their graduates master them.

CONCLUSION: Compared with 1987, similar numbers of IM program directors consider procedural skills' training necessary, but fewer think that their residents are mastering the skills. Program directors' estimates of the amount of training needed are 30% lower. Although residents appear to be getting less training in procedural skills, these skills remain important for internists to learn. Given the implications for patient care, this apparent mismatch between expectations and actual practice requires careful review and intervention by medical educators and policy makers alike.

RESIDENT PHYSICIAN ATTITUDES TOWARD MANAGED CARE. B.C. Williams¹, T. Bailey¹, J.G. Frohna¹, K.J. Sheets¹, J.O. Woollicroft¹; ¹University of Michigan, Ann Arbor, MI

PURPOSE: Managed care curricula are being developed and implemented in graduate health professions education programs throughout the country. The purpose of our study was to measure the attitudes of resident physicians in three departments at a large academic medical center during 1999 and 2000. Specifically, we wished to compare the attitudes of incoming interns and graduating seniors, since differences between these groups could indicate effects of graduate medical education on physicians' attitudes towards managed care.

METHODS: A 35-item questionnaire was administered to all first- and third-year residents in Family Medicine, Internal Medicine, and Pediatrics at a large midwest academic medical

center through two cross-sectional surveys in 1999 and 2000 ($n=320$). Topics included the effects of managed care on physician autonomy, costs, and quality of care (7 items); practice guidelines (3 items); capitation (3 items); Medicaid managed care (4 items); factors necessary for success under managed care (7 items); and expected practice payor mix (4 items), geographic setting (6 items), and practice type (1 item). All items were 4- or 5-point Likert-type scales.

RESULTS: Over 60% of residents agreed that managed care has restrictive effects on physician autonomy, compensation, referrals, and test ordering. Residents were relatively neutral regarding managed care's effects on quality and costs of care. Residents recognized (>60% of residents rated as important) several "non-traditional" areas as important for success under managed care, including patient outcomes, assessing the needs of populations, and team building. First- and third-year residents' attitudes towards managed care were largely similar, though first-year residents were more positive about managed care's effectiveness in controlling costs (64% vs. 46% agreed, $p<.04$), and more likely to expect to care for underserved (79% vs. 66%, $p<.005$) and Medicaid patients (98% vs. 88%, $p<.04$).

CONCLUSION: Resident physicians are concerned about managed care's threat to professional autonomy, are relatively neutral regarding other aspects of managed care such as quality of care or cost control. Differences between first- and third-year residents were consistent with greater idealism among interns than seniors.

Residency curricula should directly address issues of autonomy under managed care and provide accurate information on the effects of managed care on quality and costs. Recognition of the importance of several "new" physician competencies suggests a positive effect of recent changes in medical school and residency curricula. Future research should be directed towards whether differences in idealism between interns and seniors are due to the effects of training, or secular trends in attitudes towards managed care and practice intentions, or both.

ISSUES OF MEDICAL STUDENT PROFESSIONALISM: RESULTS OF A NATIONAL SURVEY. R.Y. Wong¹, M. Battistone², P.A. Hemmer³, M. Elnicki⁴, M.M. Udden⁵, T.A. Albritton⁶; ¹Loma Linda University School of Medicine, Loma Linda, CA; ²University of Utah School of Medicine, Salt Lake City, Utah; ³Uniformed Services University of the Health Sciences, Bethesda, Maryland; ⁴University of Pittsburgh School of Medicine, Pittsburgh, PA; ⁵Rice University, Houston, Texas; ⁶Medical College of Georgia, North Augusta, SC

PURPOSE: The fostering, nurturing and evaluation of professionalism in medical students have attracted great attention in recent years. The Clerkship Directors in Internal Medicine (CDIM) recognizes its role in this area and in its 2000 annual survey to clerkship directors (CDs) included a section of questions pertaining to student professionalism.

METHODS: The CDIM Evaluation and Research Committee developed the 2000 annual survey. This questionnaire was sent to all US medical schools in May and the results tabulated at the national CDIM office. Questions asked 1) whether formalized courses on student professionalism existed at the school, 2) whether white coat ceremonies were held, 3) what percentage of third year students had been counseled by the CD concerning professionalism issues, 4) what percentage of students received unsatisfactory clerkship grades because of these deficiencies in professionalism and 5) whether separate forms or mechanisms existed to document and deal specifically with issues of misconduct.

RESULTS: 74% (92 of 124) of US medical schools completed the survey. 25% of respondents stated that a formalized course for professionalism existed at their institution. 84% of schools hold a white coat ceremony or similar event. During the past year CDs counseled almost 4% (range 0–15%) of third year students and 1% of all students received an unsatisfactory clerkship grade because of these deficiencies. Only 32% of medical schools reported a separate form or mechanism to document breaches of professionalism.

CONCLUSION: The results of this national survey of Internal Medicine CDs confirm the increasing interest in professionalism in medical education. The vast majority of institutions have established white coat ceremonies to foster the values of the profession. CDs are counseling students for unprofessional behavior and for a small minority this has significantly impacted the grade they received. However, formalized courses and effective policies to evaluate and document student misconduct still need broader implementation.

UNDERSTANDING THE EXCELLENT ATTENDING-PHYSICIAN ROLE MODEL: INSIGHT FROM THE PROS. S.M. Wright¹, J.A. Carrese¹; ¹Johns Hopkins University, Baltimore, MD

PURPOSE: Role modeling is an integral component of medical education and an important factor in shaping the values, attitudes, behavior, and ethic of medical trainees. This study sought to examine and understand how highly respected physician role models think about role modeling.

METHODS: In-depth semi-structured 30-minute interviews were conducted by the primary investigator in the offices of 29 of the 30 most highly regarded role models, as judged by the house officers, at two large teaching hospitals in Baltimore. Interview transcripts were independently coded by two readers and compared for agreement. Content analysis identified several major categories of themes, which were examined and conceptually organized.

RESULTS: The analyses revealed that role models have identifiable characteristics. Subcategories under the domain of personal qualities include a commitment to excellence and seeking continual improvement, integrity, a positive outlook, leadership, and being interpersonally skilled. Under the domain of teaching, the subcategories were establishing rapport with learners, being committed to their growth, and developing specific teaching philosophies and approaches. Subjects thought there was some overlap between teaching and role modeling, but role modeling was felt to be more implicit, more intimate, and more encompassing. In the clinical domain, a recurrent theme was that being a strong clinician was necessary but not sufficient for being considered a role model for medical learners. The informants identified barriers to effective role modeling and they included being over-extended, being passive, and being impatient, inflexible and overly opinionated. Although any given role model might embody a range of talents, subjects believed it was valuable for each medical learner to have multiple role models.

CONCLUSION: Highly regarded role models shared their opinions about the critical components of role modeling in medicine and the barriers that can make it difficult. The

identification of personal qualities and features of teaching and clinical work associated with effective role modeling by physicians advances our understanding in this area.

CAN SPECIALISTS AND GENERALISTS TEACH CLINICAL SKILLS TO FIRST AND SECOND YEAR MEDICAL STUDENTS WITH EQUAL EFFECTIVENESS? L.J. Zakowski¹, C.S. Seibert¹, S. Van Eyck¹, S. Dotti¹, M. Albanese¹, S.E. Skochelak¹; ¹University of Wisconsin-Madison, Madison, WI

PURPOSE: Due to the dwindling numbers of generalists available to teach interviewing and physical exam skills to small groups of medical students in the first and second year, all clinical departments at the University of Wisconsin Medical School agreed to supply faculty for small group teaching in the Patient, Doctor, and Society Course (PDS.) Much concern remained, however, as to whether specialist and generalists could teach pre-clinical students with equal effectiveness.

METHODS: Specialists and generalists were randomly assigned to the third semester of PDS to teach groups of four students. All leaders were provided with the same verbal and written orientation materials. The specialists from nine clinical specialties were assigned to lead 18 groups of students and the generalists from general internal medicine and family medicine were assigned to lead 16 groups.

To determine from the students' point of view, whether specialists and generalists teach with equal effectiveness, students were asked to evaluate their small group leaders on a seven point Likert scale on nine items that included: enthusiasm for teaching; fosters discussion; prepared/knowledgeable; availability; constructive feedback; timely return of work; good role model; treats students with respect; and an overall rating of the group leader. Scores for leaders were compared using a chi square test.

In addition, a questionnaire was distributed to the leaders to determine whether specialists and generalists were equally confident in teaching basic history and examination skills. The questionnaire asked leaders to evaluate their confidence in their ability to teach the abdomen, neurology, cardiovascular, pulmonary exams, and in providing feedback to students in the complete history and physical exam skills. Confidence ratings for leaders were compared using a chi square test.

To determine if there might be differences in specialist and generalists' teaching effectiveness, student scores on an objective, structured, clinical examination (OSCE) were compared. Finally, we compared small group grades assigned by specialist and generalist leaders to their students. **RESULTS:** Responses from 77% of the students indicated that specialists and generalists teach basic examination skills with equal effectiveness. Responses from 71% of the leaders indicated that they are equally confident in teaching basic examination skills.

There were no differences in the generalists and specialists assignment of grades to their students. Most importantly, there were no differences in the scores on the OSCE between students taught by generalists and specialists.

CONCLUSION: Medical schools can expand their pool of physicians teaching basic skills by utilizing specialists without decreasing teaching effectiveness. With changes in the organization, financing and delivery of health care, this strategy could help schools preserve their academic mission in the face of mounting pressure to increase clinical revenue.

PATIENT-CENTERED RESEARCH

PHYSICIANS CAN BE TRAINED IN 3-HOURS TO ASSESS LV FUNCTION USING POINT-OF-CARE ECHOCARDIOGRAPHY. J.H. Alexander¹, E.D. Peterson², T.M. Harding², A.Y. Chen³, C.H. Cabell³, D. Adams², J. Kisslo²; Duke University, Durham, NC; Duke University Medical Center, Durham, NC; Duke Clinical Research Institute, Durham, NC

PURPOSE: Knowledge of left ventricular (LV) systolic function is important prognostically and therapeutically in the care of patients with suspected cardiovascular disease. Novel point-of-care echo machines are becoming available and may provide a rapid and cost-effective method for accurately assessing LV function. It is unknown, however, whether physicians without prior echo experience can be trained to use echo to assess LV systolic function.

METHODS: Physicians, without prior echo experience, completed a 3-hour training course including didactic and hands-on training in echocardiography. Patients scheduled for standard transthoracic echocardiography as part of their clinical care were eligible for enrollment. Enrolled patients had a point-of-care echocardiogram performed and interpreted for LV ejection fraction (EF) by a trained physician within 24 hours of their standard echo. LV EF was classified as 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, or 55%. Point-of-care echos were performed using the new Agilent "OptiGo" cardiac ultrasound device. Each standard echo was read twice. All echocardiographic interpretations were performed blinded to one another. Agreement was defined as a difference in LV EF of 15%. Agreement rates between point-of-care echo and standard echo were determined and compared to standard echo interobserver agreement rates. **RESULTS:** Twenty-five physicians enrolled 510 patients. The mean age of the patients was 59.4 years and 52.8% were male. The indications for echocardiographic assessment was evaluate ejection fraction in 70.4% and evaluate murmur in 11.1%. A total of 57.9% of patients had an LV EF of > 55%, 14.2% had an LV EF of 45–50%, 9.2% had an LV EF of 35–40%, 7.2% had an LV EF of 25–30%, and 11.5% had an LV EF of <15–20%. Rates of agreement for evaluation of LV function are shown in the table.

CONCLUSION: Physicians without prior echo experience can be trained in 3-hours to assess LV function with a novel point-of-care echo device. Agreement between point-of-care echo and standard echo is similar to standard echo interobserver agreement. Wider availability of point-of-care echocardiography could improve medical care.

Agreement Rates

	Point-of-care Echo	Standard Echo
Agreement rate between	83.9%	89.1%
Standard Echo and:		

A MODEST SERUM CREATININE ELEVATION IS INDEPENDENTLY ASSOCIATED WITH POSTOPERATIVE MORBIDITY AND MORTALITY. M.M. O'Brien¹, R. Gonzales², L. Shroyer¹, G.K. Grunwald², J. Daley³, W. Henderson⁴, S. Khuri³, R.J. Anderson²; ¹Denver Veterans Administration, Denver, CO; ²University of Colorado Health Sciences Center, Denver, CO; ³Boston Healthcare System, West Roxbury, MA; ⁴Hines Veterans Administration, Chicago, IL

PURPOSE: The impact of modest renal insufficiency on operative outcome remains unclear. The purpose of this study was to determine the effect of an elevated preoperative serum creatinine concentration on outcome after general surgery.

METHODS: Pre-operative variables (n=46) and postoperative morbidities and mortality were analyzed in 49,081 patients undergoing general surgery from 11/1/96 through 9/30/98 at 100 Veterans Affairs Medical Centers.

RESULTS: Unadjusted 30-day mortality and several cardiac, respiratory, infectious, neurologic and hemorrhagic morbidities were significantly (p < .001) greater in patients with a serum creatinine > 1.5 mg/dl. Multivariable analyses found an adjusted odds ratio for mortality to be 1.44 (95% confidence intervals 1.22 to 1.71) for a preoperative serum creatinine of 1.5–3.0 mg/dl and 1.93 (95% confidence intervals 1.51 to 2.46) for a preoperative serum creatinine > 3.0 mg/dl, compared to a preoperative serum creatinine < 1.5 mg/dl. Three methods (multiple linear regression, recursive partitioning and Cramer V) yielded a threshold serum creatinine of approximately 1.5 mg/dl for significant excess mortality. Multivariable analyses also found that elevated preoperative serum creatinine was independently and significantly associated with postoperative respiratory (prolonged mechanical ventilation and re-intubation), cardiac (cardiac arrest), infectious (sepsis) and hemorrhagic morbidities. The approximate threshold serum creatinine concentration for these complications was 1.5 mg/dl for respiratory complications, 2.0 mg/dl for sepsis and 3.0 mg/dl for cardiac arrest and hemorrhage, with an approximate threshold of 1.5 mg/dl for overall morbidity.

CONCLUSION: A preoperative serum creatinine greater than 1.5 mg/dl is a readily available marker that is independently associated with morbidity and mortality after general surgery. Prospective interventional studies designed to reduce morbidity and mortality are indicated in this high risk population.

TRUST IN HEALTH CARE AND USE OF CANCER SCREENING AMONG LOW INCOME WOMEN. K. Armstrong¹, G. Fitzgerald¹, J.S. Schwartz¹; ¹University of Pennsylvania, Philadelphia, PA

PURPOSE: Despite considerable media attention, the role of mistrust of the health care system in socioeconomic health disparities is currently not known. The purpose of this study was to describe trust in health care among low income women and the impact of trust on use of cancer screening.

METHODS: Cross-sectional survey of women between 50 and 75 years old enrolled in a Medicaid MCO in Philadelphia. To reach women with and without telephones, both telephone interviews and mailed questionnaires were used. Using focus group discussions and literature review, three components of trust were identified and measured: (1) trust in personal health care provider (HCP); (2) trust in HCPs in general; and (3) beliefs about unethical research practices, including that experiments happen on patients without their knowledge and the Tuskegee syphilis study could happen again. Use of screening mammography, clinical breast exams and PAP smears were measured by self-report.

RESULTS: 399 women completed the survey (response rate 65%). 64% were African American, 14% Caucasian, 13% Hispanic; mean age was 61.9 years; 46% had not completed high school; 53% had an annual household income <\$10,000; 97% had a regular HCP. 63% thought that experiments happen on patients without their knowing it and 73% of the 90 women who had heard of the Tuskegee study thought it could happen again today. 67% rated their trust in their own HCP as complete and 32% rated their trust in HCPs in general as complete. Lower trust in personal HCP and lower trust in HCPs in general were associated with a personal experience where doctors were not honest (RR 3.92, 95%CI 2.0–7.7, RR 2.81, 95%CI 1.8–4.3 respectively) and knowing someone who had an experience where doctors were not honest (RR 2.48, 95%CI 1.3–4.5 and RR 2.04, 95%CI 1.4–3.0 respectively), but not with ethnicity, education, age or physician characteristics. African-American women were more likely to believe experiments happen without patient knowledge and that Tuskegee could happen again (RR 2.28, 95%CI 1.3–4.1, RR 3.55, 95%CI 1.0–12 respectively). After multivariate adjustment, trust in personal HCP was associated with adherence to screening mammography, clinical breast exams and PAP smears (OR 2.30, 95%CI 1.2–4.7, OR 3.55, 95%CI 1.7–7.3, OR 2.27, 95%CI 1.1–4.6 respectively). Trust in HCPs in general or beliefs about unethical research were not associated with use of cancer screening.

CONCLUSION: Although only a minority of low income women report low levels of trust in their personal health care provider, lack of trust is strongly associated with lack of adherence to recommended cancer screening. Mistrust of health care providers in general and beliefs about unethical medical research practices are more common, but have less effect on use of preventive health services. Increasing trust of primary care physicians among low income populations may offer a novel approach to addressing socioeconomic health disparities.

DEFINING THE UNDERSERVED: SCREENING MAMMOGRAPHY IN PHILADELPHIA. K. Armstrong¹; ¹University of Pennsylvania, Philadelphia, PA

PURPOSE: Traditional efforts to address underservedness have focused on low income, minority populations. Preliminary evidence suggests these characteristics may no longer serve as

universally useful markers of medical underservedness. The purpose of this study was to determine how well race/ethnicity and income identify women who do not receive screening mammography in the City of Philadelphia.

METHODS: Telephone survey of adult residents of the City of Philadelphia. Households were sampled using random digit dialing stratified by service area. Interviews were conducted in English or Spanish between June and September 1998. Women ≥ 50 yr were asked "About how long has it been since you last had a mammogram?" and women who had not had a mammogram in past year were asked the most important reason they had not had a mammogram.

RESULTS: Of 6,329 eligible households, 4,114 individuals (65%) agreed to participate. 828 respondents were women between 50 and 75 years of age. Mean age was 61.9 yrs; 46% were Caucasian, 48% African-American and 4% Hispanic; 26% had not completed high school. Mean annual household income was \$19,000 (range <\$8,000 to >\$100,000). 75% had undergone screening mammography in the prior year, 9% one to two years previously, 11% more than two years ago and 5% had never had a mammogram. In the overall sample, mammography in the prior year was associated with African-American ethnicity (RR 1.19, 95% CI 1.1–1.3), but not with age, income or education. However, there were significant interactions between ethnicity and income and ethnicity and age. Income was associated with mammography among Caucasian women only (RR 1.28, 95% CI 1.0–1.7 for above vs. below federal poverty level) and age was associated with mammography among African-American women only (RR 1.34, 95% CI 1.0–1.8 for over 60 vs. 60 and under). Among women who had not had a mammogram in the prior year, African-American women were less likely to cite lack of motivation as their reason for not undergoing screening (RR 0.70, 95% CI 0.48–0.99) and more likely to cite system barriers (e.g. insurance, transportation) than Caucasian women.

CONCLUSION: In the city of Philadelphia, low income, minority women no longer represent the underserved for screening mammography. Furthermore, the characteristics of women underusing screening mammography differ between African-American and Caucasian women. These data suggest that the sociodemographic characteristics of the medically underserved may depend upon type of medical care, geographic location and even ethnic group. Generalist efforts to reduce medical underservedness should be based on accurate assessments of the communities they serve.

HIV THERAPY ADHERENCE AND RESISTANCE IN DRUG USERS. J. Amsten¹, P. Demas¹, H. Farzadegan², M. Gourevitch¹, K. Hertogs³, E. Schoenbaum¹; ¹Montefiore Medical Center, Bronx, NY; ²Johns Hopkins University, Baltimore, Maryland; ³Virco, Mechelen, Belgium

PURPOSE: Failure of HIV suppression may occur because of poor adherence, drug resistance, or suboptimal drug potency. Our purpose was to describe the relationship between both self-reported (SR) and electronically monitored (MEMS) adherence (ADH) and HIV viral load (VL), and to characterize drug resistance patterns in subjects who failed to sustain virologic suppression on potent HAART regimens.

METHODS: Subjects on HAART were recruited from a longitudinal study of HIV-infected drug users. During a median of 166 days, VL and ADH with all antiretrovirals were assessed by SR and MEMS at each of 6 monthly visits. ADH (% doses taken/doses prescribed) was assessed for both 1 day and 1 week preceding each visit, and mean ADH was calculated for the entire study period. We also studied drug resistance in 11 patients with viral rebound (VL >1000 copies/ml for >2 consecutive mos. after previously maintaining VL <400 copies/ml for >2 consecutive mos.).

RESULTS: 501 SR estimates and 417 MEMS estimates were obtained from 85 subjects. Mean 1 week SR ADH correlated with mean 1 day SR ADH (76% v. 77%; $r=0.84$, $p=0.0001$), as did mean 1 week and 1 day MEMS ADH (55% v. 57%; $r=0.86$, $p=0.0001$). Though SR ADH was greater than MEMS ADH, both were correlated with VL ($r=0.6$ for 1 week MEMS, $r=0.5$ for 1 day MEMS, $r=0.5$ for 1 week SR, $r=0.4$ for 1 day SR). The following factors were associated with poor ADH: active heroin use (64% for active v. 80% for former use, $p=.02$), active cocaine use (38% for active v. 60% for former use, $p=.009$), active alcohol use (60% for subjects who drank several days/week v. 80% for subjects who drank less, $p=.002$), side effects (73% for subjects with >2 side effects v. 83% for subjects with <2, $p=.08$), and female gender (44% for women v. 60% for men, $p=.04$). Among the 11 patients in whom we analyzed ADH, resistance and rebound, we found that median ADH during viral suppression was 54% (range 0–83%), and median ADH after rebound was 8% (range 0–36%). Only 2/11 developed high-level phenotypic resistance, and 4/11 developed significant mutations in either RT or protease; M184 in 3/11 on 3TC, K103N in 2/2 on NNRTIs, and various IDV mutations in 1/11 on PIs. In 2/4 patients, M184 mutations were present without corresponding phenotypic resistance.

CONCLUSION: High correlation between 1 day and 1 week estimates suggest that ADH does not change in the day before a visit. While SR overestimates true ADH, repeated quantitative SR estimates correlate with VL. Our data further show that SR ADH >90% is necessary to maintain virologic suppression, and that active drug/alcohol use and medication side effects lower adherence. Our analysis of drug resistance suggests that significant RT or protease mutations may be present as early indicators of phenotypic resistance. However, due to lack of sufficient selective pressure, high level phenotypic resistance is uncommon after viral rebound when ADH is poor.

PREDICTORS OF FAILURE TO RESCUE FROM POSTOPERATIVE PNEUMONIA IN PATIENTS UNDERGOING MAJOR NONCARDIAC SURGERY. A.M. Arozullah¹, W. Henderson², S. Khuri³, J. Daley⁴; ¹University of Illinois at Chicago, Chicago, IL; ²Cooperative Studies Program Coordinating Center, Hines, Illinois; ³VA Boston Healthcare System, West Roxbury, MA; ⁴Harvard University, Boston, MA

PURPOSE: To determine the patient-specific and operation-specific predictors of mortality subsequent to developing postoperative pneumonia (POP).

METHODS: We studied 401,480 cases undergoing major noncardiac surgery that were enrolled in the National VA Surgical Quality Improvement Program from September 1, 1996 through August 31, 1999. We excluded 15,649 cases with either preoperative pneumonia,

ventilator dependence, do not resuscitate status, or comatose status leaving 385,831 cases for analysis. POP was defined using the Centers for Disease Control and Prevention (CDC) definition of nasocomial pneumonia. Failure to rescue was defined as POP followed by mortality within 30 days of the index operation. Patient-specific and operation-specific variables were selected as potential predictors based on separate, previously validated, models predicting POP and 30-day postoperative mortality. A base logistic regression model including all potential predictors was developed with failure to rescue following POP as the dependent variable. Those variables that were not significant predictors ($p > 0.05$) were sequentially deleted from the base model. The deleted variables were reintroduced at various stages of model development to reassess their contribution to the model. The final model included 25 significant predictors ($p < 0.05$).

RESULTS: There were 6759 cases (1.8% of total cases analyzed) that developed POP. Among these POP cases, there were 1256 deaths or cases with failure to rescue. Patient-specific predictors of failure to rescue included age > 60 years, dependent functional status, ASA class, history of alcohol use, recent weight loss, disseminated cancer, chronic steroid use, chronic obstructive pulmonary disease (COPD), stroke, impaired sensorium, central nervous system tumor, ascites, and bleeding disorder. Preoperative laboratory values that were significant predictors of failure to rescue included albumin level < 3.5 g/dl, blood urea nitrogen > 30 mg/dl, sodium < 135 mmol/L, and increased bilirubin and serum glutamic oxaloacetic transaminase (SGOT) levels. The operation-specific predictors of failure to rescue included type of surgery (thoracic, abdominal aortic aneurysm repair, upper abdominal, and neurosurgery), emergency surgery, longer operation time, and higher number of red blood cell units given intraoperatively. The c-statistic for the final model including all significant patient-specific and operation-specific predictors was 0.887.

CONCLUSION: Patient-specific predictors of failure to rescue following POP relate to general health status (age, functional status, ASA class, recent weight loss), immune status (alcohol use, disseminated cancer, chronic steroid use), neurological status (stroke, impaired sensorium, central nervous system tumor), pulmonary status (COPD), and liver function (ascites, bilirubin and SGOT level). Operation-specific predictors relate to type and location of surgery, operative time, and blood transfused during surgery. These predictors may be useful in predicting the risk of failure to rescue following POP for subjects undergoing major noncardiac surgery. Studies to evaluate interventions such as preoperative nutritional supplementation, aggressive pulmonary rehabilitation, and intensive perioperative respiratory care for patients at high risk for failure to rescue should be pursued in the future.

PREDICTORS OF INPATIENT MORTALITY AMONG HIV PATIENTS WITH COMMUNITY-ACQUIRED PNEUMONIA. A.M. Arozullah¹, P.R. Yarnold², M. Deloriaknoll², S. Ali³, C.L. Bennett²; ¹University of Illinois at Chicago/VA Chicago Healthcare System, Chicago, IL; ²Northwestern University, Chicago, Illinois; ³VA Chicago Healthcare System-Lakeside Division, Chicago, IL

PURPOSE: Pulmonary infections, including bacterial pneumonia, cause significant morbidity and mortality for patients infected with HIV. The purpose of this project was to determine the predictors of inpatient mortality among HIV patients with community-acquired pneumonia admitted in the early highly active antiretroviral therapy era, between 1995–1997.

METHODS: Trained abstractors performed retrospective chart reviews of hospitalized HIV patients diagnosed or suspected of having community-acquired pneumonia, who were admitted between 1995–1997 at 86 hospitals in 7 geographic areas. We used hierarchically optimal classification tree analysis to develop a pre-admission severity of illness staging system for predicting inpatient mortality.

RESULTS: Among HIV patients with confirmed or suspected community-acquired pneumonia ($n=1112$), the inpatient mortality rate was 9.5%. Multivariate modeling indicated a 5-category staging based on four predictors (Table 1). Bootstrap validity analysis (50% resample) supported the stability of this model.

CONCLUSION: Our staging system identifies HIV patients with low, moderate, and high risk of inpatient mortality from community-acquired pneumonia and may be useful for case-mix adjustment while exploring variations in community-acquired pneumonia mortality rates among hospitals and across cities caring for HIV patients.

Table 1 Staging System for Predicting Inpatient Mortality

Stage	N(%)	Respiratory Rate	Albumin	Creatinine	CD4	Mortality Rate
1	456 (41.0)	≤ 24.5 /minute	> 2.95 g/dl	-----	---	0.9%
2	126 (11.3)	> 24.5 /minute	-----	≤ 1.25 mg/dl	≥ 50	3.2%
3	274 (24.7)	≤ 24.5 /minute	≤ 2.95 g/dl	-----	---	10.2%
4	97 (8.7)	> 24.5 /minute	-----	≤ 1.25 mg/dl	< 50	14.4%
5	159 (14.3)	> 24.5 /minute	-----	> 1.25 mg/dl	---	35.2%

CHARACTERIZATION OF HERBAL WEIGHT LOSS PRODUCTS SOLD OVER THE INTERNET. B. Ashar¹, K. Getz¹; ¹Johns Hopkins University, Baltimore, MD

PURPOSE: The potential risks of herbal supplement use is of growing concern. Reports of serious side effects and drug interactions have prompted many physicians to incorporate questions of supplement use into their medical history. Despite these efforts, clinicians are often faced with inadequate information due to the recent boom of combination herbal products known only by their trade names. Supplements marketed to consumers as weight loss remedies are one such group of products that frequently contain a multitude of "natural" ingredients. This study was designed to characterize the ingredients found in a number of popular weight loss products in order to assist physicians in preventing potential adverse effects.

METHODS: A systematic search of the World Wide Web was conducted to identify twenty herbal weight loss products for sale on the internet. Three search engines were used (HotBot,

Google and Excite) and the search term "herbal weight loss." A site abstraction form was developed to collect data on the ingredients contained within each product. Descriptive statistics were used to characterize the most common ingredients found among the twenty combination products.

RESULTS: Nine ingredients were identified in six or more of the products. These included: MaHuang, Ginger, Guarana, Siberian Ginseng, White Willow Bark, Chromium, Garcinia Cambogia, Gotu Kola, and Zinc. The number and frequency of each of these ingredients among the herbal weight loss products characterized are included in the table below.

CONCLUSION: Despite little evidence of efficacy and the potential for significant toxicity, ingredients such as MaHuang, Guarana, and Chromium are found in a majority of weight loss products. Physicians need to take this into consideration when faced with a patient who discloses the use of an unfamiliar over-the-counter weight loss supplement.

Ingredients Contained in Herbal Weight Loss Supplements

Ingredient	Number of Products	Ingredient	Number of Products
Chromium	15 (75%)	White Willow Bark	8 (40%)
Ma Huang (ephedra)	14 (70%)	Gotu Kola	8 (40%)
Guarana	12 (60%)	Zinc	6 (30%)
Siberian Ginseng	11 (55%)	Garcinia Cambogia	6 (30%)
Ginger	9 (45%)		

COST EFFECTIVENESS OF SCREENING FOR HYPERCOAGULABLE STATES IN PATIENTS WITH DEEP VEIN THROMBOSIS. A.D. Auerbach¹, G. Sanders², J. Hambleton¹; ¹University of California, San Francisco, San Francisco, California; ²Stanford University, Stanford, California

PURPOSE: Deep venous thrombosis (DVT) is common, accounting for nearly 100,000 hospital admissions in 1997. Numerous hypercoagulable states (HCS) pose an increased risk for recurrent thrombosis, with recent research suggesting HCS are common. However, little data exist to aid physicians trying to balance the relationship between relative risk for recurrent thrombosis, prevalence of HCS, characteristics of screening tests, and benefits and risks of therapy when considering screening for HCS.

METHODS: We developed a population-based Markov model comparing usual care (initial therapy for deep venous thrombosis followed by surveillance for recurrent events) to a strategy that included screening for HCS (with affected patients being placed on lifetime anticoagulation) for patients suffering an idiopathic DVT. Using this model, we explored the relationship between relative risk (RR) for thrombosis due to HCS, prevalence of HCS, and the characteristics of screening tests with benefits and risks of therapy to determine clinical situations where screening may be cost-effective.

RESULTS: In our base case analysis (RR=4, prevalence=20%), screening was more costly (\$1,700 higher than usual care) and less effective (quality adjusted life expectancy (QALY) 17.63 years in screened group vs. 17.99 in usual care). The marginal cost-effectiveness (MCE) of screening became cost effective at \$50,000 per QALY only if the complication rate (CR) from therapy was reduced 90% or more (<0.2%/year for major complications, <0.6%/year for minor complications). When we examined the relationship between RR and prevalence assuming baseline CR, the MCE of screening was cost effective for illnesses with RR > 20 and prevalence > 10%. This range of prevalence and RR fits published data describing homozygous Factor V Leiden mutation. For prevalence between 5 and 10%, RR of thrombosis would need to exceed 50 for screening to become cost-effective, and for prevalence < 5%, RR would need to exceed 100 for screening to become cost-effective. Increasing the specificity and sensitivity of the test panel to 100%, or varying age at presentation, costs of the test panel, or utilities for thrombotic events and bleeding complications did not significantly affect our findings.

CONCLUSION: Screening for HCS in patients with an idiopathic deep venous thrombosis is cost-effective only when HCS are common and associated with a high relative risk of recurrent thrombosis, or if complications of treatment are reduced significantly. Current evidence suggests few defects meet this profile. Wider screening may be justified if treatments that result in fewer bleeding complications can be found.

PROCESSES AND OUTCOMES OF END-OF-LIFE CARE IN A VOLUNTARY HOSPITALIST MODEL. A.D. Auerbach¹, S.Z. Pantilat¹, R.M. Wachter¹, L. Goldman¹; ¹University of California, San Francisco, San Francisco, California

PURPOSE: Because most Americans die in the hospital, the growth of hospitalist services may significantly impact the care of dying patients. However, few data exist comparing end-of-life care in the hospitalist model with that provided in the traditional model of care.

METHODS: We performed a retrospective chart review of 148 patients who died while under the care of community-based physicians (CP) (74 patients) or hospitalists (H) (74 patients) at a community-based teaching hospital. Patients of CP and H were of similar age (74 vs. 72 yrs, p=0.4); and race (62% white vs. 50%, p=0.14). Although more patients of CP were admitted with a primary diagnosis of cancer (24% vs. 7%, p<0.001), fewer patients of CP were admitted with Class IV cardiopulmonary symptoms (8% vs. 27%, p=0.001) or AIDS (1% vs. 8%, p=0.04).

RESULTS: At admission, more patients of CP had documented requests to limit care (such as DNR/DNI or comfort care measures), although this was not statistically significant (47% vs. 35%, p=0.14); at the time of death almost identical percentages of had documented requests for limitations in care (71% vs. 72%, p=0.89). This change may have been related to a higher frequency of family meetings among hospitalists' patients (63% of CP vs. 91% of H, p<0.01). Hospitalists were also more likely to be documented as present at the time of the family meeting (51% of CP vs. 91% of H, p<0.001). For patients who requested limitation in care, patients of CP were more likely to have pain, anxiety, or dyspnea in the 48 hours prior to death (75% of CP vs. 57% of H, p=0.047). For each symptom, similar proportions of patients in

each group were documented to be in pain (33% of CP vs. 22% of H, p=0.17) or have anxiety (20% of CP vs. 10% of H, p=0.12) in the 48 hours prior to death. However, despite worse NYHA class, patients of hospitalists were less likely to have dyspnea (62% of CP vs. 40% of H, p<0.01).

CONCLUSION: Our findings suggest that care by a hospitalist does not negatively impact coordination of end-of-life care, and may improve patient symptoms at the time of death. Future studies should confirm these findings and seek ways to continue to improve end-of-life care.

THE HYPERVENTILATION TEST: A NEW DIAGNOSTIC MANEUVER FOR WHITE COAT EFFECT IN HYPERTENSIVE PATIENTS. F. Augustovski¹, M. Deprati¹, C. Calvo¹, G. Waisman²; ¹Unidad de Medicina Familiar y Preventiva, Hospital Italiano, Buenos Aires, Argentina; ²Unidad de Hipertensión, Servicio de Clínica Médica, Buenos Aires, Argentina

PURPOSE: White coat effect is common, increases with age, and can lead to overtreatment of hypertension (HBP) and excessive resource use. Hyperventilation produces a transient decrease in blood pressure (BP). Our objective was to evaluate the hyperventilation maneuver as a diagnostic test of the white coat effect in HBP.

METHODS: We included adult uncontrolled HBP patients (>140/90) visiting their primary care physicians. Basal BP was measured as the average of 2 readings. Then, the hyperventilation test was done -deep breaths during 30 seconds- and the BP response was recorded immediately after the maneuver. 24-hour ambulatory BP monitoring was used as the gold standard. White coat effect was defined as a >15mmHg decrease between office and ambulatory systolic blood pressure (SBP). The hyperventilation test operating characteristics were evaluated.

RESULTS: 67 patients were included for the analysis (mean age 59.7 [SD 10], 41% females). The prevalence of white coat effect was 56%. Comparing patients with white coat effect vs. those without it, the hyperventilation test produced a mean SBP decrease of 19.3 and 10.6 mmHg (p>0.001) and a mean DBP decrease of 6.1 and 5.4 mmHg (pNS) respectively. Mild dizziness was reported by 5% of the patients. The area under ROC curve of SBP decrease was 0.74 (CI95% 0.62-0.86), and that for DBP was 0.53 (CI95% 0.38-0.67).

CONCLUSION: The hyperventilation test can be a useful tool to detect white coat effect in HBP. It is done at the office in about two minutes and it has no significant adverse effects. It may have important implications in avoiding overtreatment, its adverse consequences, and unnecessary resource utilization -mainly ambulatory BP monitoring.

Hyperventilation Test: Operating characteristics (SBP decrease)

Cutoff Value, in mmHg	Sensitivity	Specificity	LR+	LR-
> 5	92.1	24.1	1.2	0.3
> 10	84.2	37.9	1.4	0.4
> 15	63.2	65.5	1.8	0.5
> 20	55.3	82.8	3.2	0.5
> 25	31.6	96.5	9.2	0.7

UTILITY OF THE COLLAPSING PULSE IN SCREENING FOR AORTIC INSUFFICIENCY. A.N. Babu¹, S.R. Bitar¹, S.M. Kymes¹, C. Isenberg¹; ¹St. Louis VAMC, St. Louis, MO

PURPOSE: Aortic insufficiency (AI) is a common valvular disorder, particularly in the elderly. Its detection has implications for hemodynamic management, endocarditis prophylaxis and diagnosis of systemic disorders. Multiple reports suggest that auscultation is the most accurate physical examination technique for detecting AI. Uniformly, studies have demonstrated a high specificity (Sp) (over 90%) but low sensitivity (Sn)(0-75%) for auscultation. Sp does not vary significantly by class of examiner or lesion severity unlike Sn. For mild lesions, Sn has ranged from 36-61% for cardiologists to 0-9% for non-cardiologists. The "collapsing" pulse (CP) has long been regarded a useful sign of AI. This study is the first systematic assessment of its diagnostic utility and operating characteristics.

METHODS: 145 patients (95% male, mean age 63) were prospectively recruited when presenting for transthoracic echocardiography (TTE) for any indication. A designated internist blinded to TTE findings and indication checked all patients for the presence of a CP by palpation of the right radial artery before and after elevating the patient's wrist per standardized protocol. The procedure took about 15 seconds. A clear increase in pulse amplitude with wrist elevation was considered a positive, and if so, was further graded from 1 (mild) to 3 (severe). The examiner was not allowed to take a history or perform any other diagnostic maneuver. TTE were subsequently reviewed by a single blinded cardiologist for AI by accepted color and continuous Doppler criteria. AI if present was graded from 1+ (mild) to 3+ (severe).

RESULTS: 34 patients had 1+ and 8 patients 2-3+ AI. Sn, Sp, and negative/positive predictive values (NPV/PPV) with 95% confidence intervals (95% CI) for the CP are tabled. CP amplitude did not correlate with AI severity.

CONCLUSION: In older males, the CP has good Sn and NPV for AI and outperforms auscultation. For ruling out mild AI in particular, the CP is notably superior to auscultation. It also has excellent NPV for excluding significant AI. The CP test is quick, simple and independent of examiner class for successful execution. Given the high Sp but low Sn of auscultation, addition of the CP test when screening for AI can be highly beneficial.

CP attributes in the diagnosis of any, mild (1+) or significant (2-3+) AI

AI	N	Sn%	95% CI	Sp%	95% CI	NPV%	95% CI	PPV%	95% CI
Any	145	86	75-97	16	9-23	73	54-92	29	21-37
1+	137	85	73-97	16	9-23	76	58-94	25	17-33
2-3+	111	88	66-100	16	9-21	95	89-100	6	4-8

BLOOD PRESSURE CHANGES IN AORTIC INSUFFICIENCY. A.N. Babu¹, S.R. Bitar¹, S.M. Kymes¹, C. Isenberg¹; ¹St. Louis VAMC, St. Louis, MO

PURPOSE: Aortic insufficiency (AI) is a common valvular disorder, most prevalent in older adults. Its detection has implications for hemodynamic management, endocarditis prophylaxis and diagnosis of systemic disorders like Marfan's syndrome. Multiple studies have shown that physical examination techniques have poor sensitivity in diagnosing AI, particularly mild AI. There is abundant literature stating that blood pressure (BP) changes are useful in the diagnosis of AI. Specifically, AI is felt to be associated with a wide pulse pressure (PP) and low diastolic blood pressure (DBP). However, there is little supporting data. Most of the limited studies done in this area have suffered from spectrum bias, as patients known to have hemodynamically significant AI were selected for study. This investigation was conducted to determine the utility of BP changes in diagnosing and quantifying AI in a general medical population.

METHODS: Study design: Prospective, blinded assessment of 61 patients, predominantly older males, presenting for transthoracic echocardiography (TTE) for any indication. Procedure: Patients had their BP measured in the right upper extremity while seated, using a mercury sphygmomanometer by the designated investigator. A standardized protocol was followed. The initial appearance of Korotkoff sounds signified the SBP, and DBP was the BP at which the sounds disappeared. The examiner was blinded to echo findings and indication and was not allowed to take a history or perform any other diagnostic maneuver. The TTE was subsequently reviewed by a single blinded cardiologist for the presence or absence of AI by accepted echo Doppler criteria. AI, if present, was graded from 1+ (mild) to 3+ (severe).

RESULTS: Out of 61 patients studied, 16 patients were found to have AI. 1+ AI was present in 10 patients and 2–3+ AI in 6. Mean DBP was 78mmHg for normals, 76 for all AI, 79 for 1+ AI and 69 for 2–3+ AI. Mean PP was 62mmHg for normals, 61 for all AI, 53 for 1+ AI and 73 for 2–3+ AI.

CONCLUSION: There was no significant difference in DBP or PP between patients with or without AI. Thus, BP changes cannot be used to screen for the presence of AI. Further, BP changes have no apparent utility in the diagnosis of mild AI, which forms the large majority of cases in a general medical population. There was a statistically insignificant trend towards lower DBP and wider PP in patients with significant AI, which would be consistent with findings from prior studies. The generalizability of our findings to females and the young is uncertain. Future literature and teaching on AI should highlight the limitations of BP in its diagnosis.

IMPROVEMENTS IN ACCESS TO CARE FOR HIV AND AIDS IN A STATEWIDE MEDICAID MANAGED CARE SYSTEM. J.E. Bailey¹, D.L. Van Brunt¹, S. Raffanti², W. Long³; ¹University of Tennessee, Memphis, Memphis, TN; ²Vanderbilt University, Nashville, TN; ³Tennessee Department of Health, Nashville, TN

PURPOSE: Medicaid managed care systems can impact access to health care for persons with chronic illnesses. This retrospective longitudinal study sought to determine if the initiation of a statewide Medicaid managed care system (TennCare) resulted in increased access to health care for persons with HIV and AIDS in Tennessee.

METHODS: The study population consisted of all persons living in Tennessee with HIV and AIDS from 1992 to 1997. The State Health Department HIV/AIDS Reporting System database was linked with Medicaid/TennCare enrollment files to identify the percentage of the state's entire HIV/AIDS population enrolled in Medicaid/TennCare and their eligibility status and demographic characteristics. Percentages of persons enrolled in the program were calculated annually by dividing the number enrolled by the absolute number of persons with HIV and AIDS living in the state during that year.

RESULTS: The absolute number of persons living with HIV or AIDS served by Tennessee's Medicaid system more than doubled from 1993 to 1997 following the initiation of TennCare. The percentage of persons living in Tennessee with HIV and AIDS enrolled in the Medicaid/TennCare program 1 or more day per year increased from 1993 to 1997 for HIV (41% to 43%) and AIDS (49% to 54%). The percentage of persons enrolled in the program greater than 320 days increased for HIV (28% to 34%) and AIDS (32% to 44%). The largest percentage of persons added to the program were eligible on the basis of being uninsured or uninsurable. TennCare enrollees were similar in demographic characteristics to the HIV/AIDS population previously covered by Medicaid in that the majority were male and urban dwelling but an increased percentage was of black race.

CONCLUSION: This study documents substantial increases in the number of persons with HIV and AIDS covered through a statewide Medicaid managed care program initiative. Increases in the percentages of persons with HIV and AIDS covered by the system were more modest due, in part, to large increases in the number of Tennessean's living with HIV and AIDS during the study period. Increases in Medicaid insurance coverage were greatest for the AIDS population; a group that has traditionally been more likely to experience difficulties maintaining insurance coverage. This study demonstrates that Tennessee's broad expansion of eligibility for persons with HIV and AIDS resulted in improved access to care for these vulnerable populations.

QUALITY OF CARE FOR HIV AND AIDS IN TENNESSEES STATEWIDE MEDICAID MANAGED CARE SYSTEM. J.E. Bailey¹, D.L. Van Brunt¹, G. Somes¹, C. Dorko¹, M. Appleton¹, P. Jenkins¹; ¹University of Tennessee, Memphis, TN

PURPOSE: Medicaid managed care systems provide insurance coverage for many persons living with HIV and AIDS but the impact of these systems on quality of care for these vulnerable populations remains unknown. This longitudinal study sought to assess the impact of a statewide Medicaid managed care system (TennCare) on quality of care for HIV and AIDS.

METHODS: This retrospective longitudinal descriptive analysis used repeated measures obtained over time on the entire population meeting study criteria in each year of the study. The State Health Department HIV/AIDS Reporting System database was linked with Medicaid/TennCare encounter files and vital records to identify all persons with HIV or AIDS in Tennessee enrolled in the Medicaid (1992 – 1993) or TennCare program (1994 – 1997) for at least one year. For each reporting year, participants were required to have a prior HIV diagnosis

date, to be alive at the year's beginning, and to meet age and eligibility criteria. The main outcome measures included measures of health care utilization, incidence of opportunistic infections and mortality.

RESULTS: Average number of hospitalizations decreased from 1992 to 1997 for HIV (0.72 to 0.37) and AIDS (1.27 to 0.52). Average number of emergency visits decreased from 1992 to 1997 for HIV (1.70 to 1.12) and AIDS (1.65 to 1.02). Outpatient visits decreased slightly for HIV (5.94 to 5.00) and AIDS (8.37 to 7.35) and average number of antiretroviral medications filled increased for HIV (1.27 to 2.45) and AIDS (1.31 to 3.34). 1992 – 1997. The percentage of persons diagnosed with pneumocystis carinii pneumonia decreased from 1992 to 1997 for AIDS (10% to 6%) and those with community acquired pneumonia decreased for HIV (14% to 9%) and AIDS (27% to 12%). The annual incidence of AIDS in the HIV population decreased from 33% to 10%. Annual mortality decreased for HIV (3% to 1%) and AIDS (16% to 3%).

CONCLUSION: This study documents marked improvements in health care utilization patterns, morbidity and mortality in a statewide Medicaid managed care system. Although these findings are due in part to major gains in medication therapy during the study period, this study suggests that patients served by TennCare are benefiting from these recent advances in HIV therapy and AIDS treatment.

FUNCTIONAL HEALTH LITERACY AND THE RISK OF HOSPITAL ADMISSION AMONG MEDICARE MANAGED CARE ENROLLEES. D. Baker¹, J. Gazmararian², M. Williams³, T. Scott², R. Parker³, D. Green², J. Ren², J. Peel²; ¹Case Western Reserve University, MetroHealth Medical Center, Cleveland, OH; ²USQA Center for Health Care Research, Atlanta, GA; ³Emory University, Atlanta, GA

PURPOSE: National studies indicate that 44% of elderly Americans have readings skills that are inadequate to perform routine activities required to function in society. This study sought to determine whether inadequate functional health literacy is an independent risk factor for hospital admission among the elderly.

METHODS: New Medicare managed care enrollees in Cleveland, Houston, Tampa, and South Florida were contacted by telephone and asked to participate. Individuals who were institutionalized, unable to speak English or Spanish fluently, blind, or who had severe cognitive impairment were excluded. Those who consented underwent a 1-hour in-home interview between April–May 1997 to determine demographics, income, health behaviors, chronic diseases, and self-reported health (SF12 physical and mental health summary scales). Reading ability was measured using the Short Test of Functional Health Literacy in Adults, and patients were classified as having adequate, marginal, or inadequate literacy. Hospital admissions through April 30, 1999 were determined using administrative data. Cox proportional hazards models (crude and risk-adjusted using all covariates listed above) were used to compare the time to first admission according to literacy. Interaction terms between literacy and all other covariates were also examined.

RESULTS: 7,471 enrollees were contacted by phone; 3,390 refused to participate, 737 were ineligible, and 184 had incomplete interviews, leaving 3260 for analysis. Participants and non-participants had similar age, gender, and SES. A total of 29.5% of participants were hospitalized during follow-up. The crude relative risk of hospitalization was higher for individuals with inadequate functional health literacy (N=800; 1.43, 95% CI 1.24 – 1.65) and marginal literacy (N=366; 1.33, 95% CI 1.09 – 1.61) compared to those with adequate literacy (N=2094). After adjusting for demographics, socioeconomic status, health behaviors, the number of chronic diseases, and self-reported physical and mental health at the time of the baseline interview, the adjusted RR of hospital admission was 1.29 (95% CI 1.07 – 1.55) for individuals with inadequate literacy and 1.21 (95% CI 0.97 – 1.50) for those with marginal literacy. The relationship between literacy and hospitalization varied substantially according to baseline self-reported physical health, with the effect of literacy being higher for those in better health (p < .001 for interaction term).

CONCLUSION: Inadequate functional health literacy is an independent risk factor for hospital admission among elderly managed care enrollees. This increased risk appears confined to individuals with above average physical functioning, perhaps because those in poor health receive more assistance from others with their care.

RACE/ETHNICITY, LANGUAGE, SOCIOECONOMIC STATUS AND HEALTH CARE USE FOR MEDICARE MANAGED CARE ENROLLEES. D. Baker¹, J. Gazmararian², M. Williams³, T. Scott², R. Parker², D. Green², J. Ren², J. Peel²; ¹Case Western Reserve University, MetroHealth Medical Center, Cleveland, OH; ²USQA Center for Health Care Research, Emory University, Atlanta, GA

PURPOSE: To determine if there are differences in physician visits according to race, ethnicity, language, literacy, and income among Medicare managed care enrollees.

METHODS: New Medicare managed care enrollees in Cleveland, Houston, Tampa, and South Florida were contacted by telephone and asked to participate. Individuals who were institutionalized, unable to speak English or Spanish fluently, blind, or who had severe cognitive impairment were excluded. Those who consented underwent a 1-hour in-home interview between April–May 1997 to determine demographics, income, literacy (Short Test of Functional Health Literacy in Adults) health behaviors, chronic diseases, and self-reported health (SF12 physical and mental health summary scales). Outpatient physician visits (office and emergency department) during the first year following enrollment were determined using administrative data. Time to first visit was analyzed using Cox proportional hazards models (crude and risk-adjusted using all covariates listed above). χ^2 tests and logistic regression were used to determine participant characteristics associated with not having any physician visit in the first year. T-tests and linear regression was used to analyze differences in the number of physician visits in the first year among those with at least 1 visit. A p value of .05 was considered significant.

RESULTS: 7,471 enrollees were contacted by telephone; 3,390 refused to participate, 737 were ineligible, and 184 had incomplete interviews, leaving 3260 for analysis. Participants and non-participants had similar age, gender, and socioeconomic status (based on ZIP code). Among study participants, those with income < \$ 10,000/yr had a longer time until their first visit compared to

those with incomes of \$15–24,999/yr (hazard ratio 0.84, 95% CI 0.74–0.95), but there were no differences by race/ethnicity, language, or literacy. A total of 8.7% of participants had no physician visit during the first year; the proportion of people with no visit was similar regardless of race/ethnicity, language, literacy, or income. The mean number of visits among those with at least 1 visit was 14.3 (SD + 14.5). The number of visits was strongly related to the number of chronic diseases, physical and mental health, impairments in activities of daily living, health behaviors, and study site. After adjusting for these covariates, the mean visits was 15.6 for participants with income < \$10,000/yr and 17.4 for those with higher incomes ($\Delta=1.81$, 95% CI 0.72–2.89). There was no difference in the number of visits for patients with low literacy vs. those with adequate literacy and for African Americans or Latinos vs. whites.

CONCLUSION: Income, but not literacy, race, ethnicity, or language is associated with having fewer physician visits among new Medicare managed care enrollees.

HIV PREVENTION SERVICES IN CORRECTIONAL DRUG TREATMENT PROGRAMS: DO THEY CHANGE RISK BEHAVIORS? R. Ballard Lubelczyk¹, P.D. Friedmann¹, S.C. Lemon², M.D. Stein¹, D.R. Gerstein³, ¹Brown University, Providence, RI; ²Brown University School of Medicine, Providence, RI; ³National Opinion Research Center, Washington, DC

PURPOSE: In the United States, there are over 2.0 million individuals in prison or jail. HIV prevention education is a part of substance abuse treatment at many correctional facilities, but the effectiveness of these programs has not been evaluated. The goals of our study were to evaluate who receives HIV prevention services in correctional substance abuse treatment programs, and to determine the impact of these services on short term risk behaviors.

METHODS: This study was a secondary analysis of data from the National Treatment Improvement Evaluation Study (NTIES), conducted during 1993–1995. A total of 1,223 HIV-free adult inmates, enrolled in nine correctional substance abuse treatment programs were evaluated. A composite index modeled after the validated Risk Assessment Battery (RAB) measured HIV risk behavior at treatment entry and at twelve-month follow-up. We employed multivariable analysis of covariance to assess the influence of receiving HIV prevention services, controlling for potential confounders, on HIV risk behavior at twelve-month follow-up. Because of assumed differences between those who were in and those who were out of custody during the follow-up period, all analyses were stratified by this variable.

RESULTS: Overall, 77.1% of the sample received HIV prevention services while in treatment. Among the out of custody group, Hispanics, persons of younger age and those in outpatient substance abuse treatment programs were significantly less likely to receive HIV prevention services. Among the continuously incarcerated group, women, persons of younger age and those in outpatient substance abuse treatment programs were least likely to receive these services. Both the in custody and out of custody groups exhibited HIV risk behavior reductions after twelve months of follow-up. However, controlling for baseline HIV risk behavior and other possible confounders, receiving HIV prevention services was associated with less risk behavior only among the out of custody group ($\beta=-.14$, $P=.04$). No association was observed among the group that was continuously incarcerated throughout the follow-up period.

CONCLUSION: This analysis suggests that HIV prevention services are beneficial in reducing risk behavior among inmates whose discharge is expected in the near future. The benefit of administering these services in individuals with longer sentences earlier in the incarceration period is unclear.

TIME ORIENTATION AND LIFESTYLE CHANGES IN ANGIOGRAPHY PATIENTS. E. Jean-pierre¹, J.M. Barnhart², ¹Ferkhof Graduate School of Psychology, Bronx, NY; ²Albert Einstein College of Medicine, Bronx, NY

PURPOSE: Lifestyle changes, such as smoking cessation, can play a pivotal role in outcomes following the diagnosis and treatment of coronary heart disease (CHD). However, many patients with CHD are not motivated to make lifestyle changes even after undergoing coronary revascularization. Prior studies have shown that time orientation is a major motivational determinant of an individual's behavior and more specifically as it relates to healthy lifestyle choices. Therefore, we sought to examine the association of future orientation with readiness to make healthy lifestyle changes in a diverse group of patients at high risk of CHD.

METHODS: Forty patients scheduled for elective coronary angiography were identified in the fall of 2000. High risk patients were those with one or more of the following: diabetes, hypertension, or hyperlipidemia. Following informed consent, a baseline 20-minute structured bedside interview was conducted, which collected information on the following: demographics (e.g. age, race, and sex), high-risk behaviors (e.g. smoking) using the interview format of the Computerized Lifestyle Assessment (CLA), and future orientation using Carstensen and Lang's Future Orientation Scale. This latter scale measures the extent that one perceives there is sufficient time to achieve his/her goals. A follow-up interview was conducted 3 months from baseline to assess readiness to change as measured by the CLA.

RESULTS: Follow-up data were completed on 33 subjects (83%). Their mean age was 58 years (S.D.=10.4), 64% were women, 42% were Hispanic, 27% were White, 21% were Black, and 10% were Others. Most patients (79%) did not exhibit any movement in readiness to change their lifestyle or habits as measured by the CLA subscale. Overall, 15% of patients had movement toward adopting a healthier lifestyle following angiography, such as improving their eating habits and exercising; 6% regressed. Patients who were future oriented ($n=16$) had the most progress in movement toward making healthy changes in exercise (25%), diet (10%), and smoking (10%), compared to 17%, 23%, and 6% respectively of patients who were present oriented. However, 13% of future oriented subjects had an overall decrease in their readiness to change score following angiography, compared to none (0%) of the 17 patients who were present oriented.

CONCLUSION: These data suggest that patients at high-risk of CHD, as well as those who are future oriented, may be slow to make lifestyle changes following angiography. These findings also indicate that there is a need for more research to ascertain patients' perception of their personal coronary risks. Primary care and other health providers can use such novel information to implement individually relevant educational programs to address motivational factors related to health behavior.

A SHORTENED INSTRUMENT FOR LITERACY SCREENING. P.F. Bass¹, J.F. Wilson¹, C.H. Griffith¹, ¹University of Kentucky, Lexington, Kentucky

PURPOSE: Approximately 25% of Americans are functionally illiterate. Low literacy has been associated with poorer physical health, psychological health, and higher health care costs. While there are several well-validated screening instruments for literacy, their length makes them less useful in a busy practice setting. This study was conducted to validate a shortened version of the Rapid Estimate of Adult Literacy in Medicine (REALM), the REALM-R.

METHODS: The original 66 item REALM was administered to 50 patients in the Internal Medicine Clinic (IMC) at the University of Kentucky. Using item analysis and pilot data, eight of the original 66 items were selected to create a new instrument, the REALM-R. The eight items chosen were: osteoporosis, allergic, jaundice, anemia, fatigue, directed, colitis, and constipation. The REALM-R has face validity in that the chosen words are health related and are commonly used in general medical practice. Criterion validity was based on the correlation between the raw scores on the REALM-R, the REALM, and the WRAT. The REALM-R and the Wide Range Achievement Test-Revised (WRAT), a reliable and well validated literacy screen, were then administered to 157 patients in the IMC.

RESULTS: Patients ranged in age from 17 to 93, were 85% Caucasian, and completed high school on average. Scores on the REALM-R ranged from 0 to 8 with a mean and standard deviation (S.D.) of 6.8 and 2.1. The reliability of the REALM-R was .91 using Cronbach's alpha. Scores on the WRAT ranged from 14 to 56 with a mean and S.D. of 45.2 and 7.5. The part whole correlation between the REALM-R and the REALM was 0.72. The correlation between the REALM-R and the WRAT was 0.78. A score of 6 on the eight item REALM-R identified 26 of 30 persons scoring more than 1 S.D. below the mean on the WRAT, and only identified one person scoring above the mean as having a potential literacy problem.

CONCLUSION: The REALM-R is reliable and substantially correlated with both the REALM and the WRAT while being tested in a new population. As a result of the improved efficiency with this screening instrument, the shortened instrument is advantageous in busy practice settings. While not providing the user with grade equivalent scores, it quickly and efficiently identifies persons with potential literacy problems which can adversely affect their care.

RELATIONSHIP BETWEEN SOCIOCULTURAL ISSUES AND ACCESS TO CARE IN A PUBLIC HOSPITAL URGENT CARE CLINIC. H. Batal¹, S. Prock¹, S. Majeres¹, L. Lasater¹, R. Lundgren², J. Adams¹, P. Mehler¹, ¹Denver Health Medical Center, Denver, CO; ²Colorado Prevention Center, Denver, CO

PURPOSE: To describe access to care issues in an ethnically diverse population of patients seeking care at a public hospital urgent care clinic.

METHODS: 20% of patients presenting to the Denver Health Medical Center urgent care clinic (UCC) between June 15 and August 11, 2000 were randomly approached to participate in a pre-visit interview, offered in both English and Spanish. Patients were asked standardized questions regarding physical and mental health, their reasons for accessing care in the UCC, barriers to health care, history of prior Emergency Department (ED) and UCC usage, reasons for delaying health care, chronic medical conditions, and prior receipt of preventive health services. Study patients' charts and admission records were abstracted for the number of prior visits and documentation of receipt of preventive health services at our institution, ethnicity, age, gender, and insurance status. Data were analyzed using multiple logistic regression.

RESULTS: There were 1006 patients surveyed with a refusal rate of 20.3%. The study patients did not differ from the total population who accessed care during this time period in regards to age, insurance status, ethnicity, or gender. After controlling for several factors, including age, gender, and the presence of chronic medical conditions, Hispanic patients were more likely than white patients to report having a regular source of health care ($p=0.008$) or a regular physician ($p=0.018$). However, Spanish speaking patients were less likely to note a regular physician ($p<0.0001$) than English speaking patients. Black patients were more likely than white patients to note a regular source of care ($p=0.04$) but not a regular physician ($p=0.8108$). Hispanic patients were 1.7 times more likely to have visited the ED or UCC in the past year than our white patients ($p=0.006$). However, Spanish speaking patients had less use of these services ($p=0.01$) than English speaking patients.

CONCLUSION: Sociocultural issues such as race and English fluency likely contribute to utilization of resources such as EDs and UCCs. For example, in our population minorities tended to utilize EDs and UCCs more often than white patients, but were also more likely to report that they had a regular source of care. Given the racial disparities in health outcomes there is an imperative to address this problem with additional studies. Simply attempting to redirect patients who inappropriately utilize EDs and UCCs to their regular source of care is unlikely to succeed unless language and cultural issues are addressed.

DOES AGGRESSIVE CARE FOLLOWING ACUTE MYOCARDIAL INFARCTION REDUCE MORTALITY? ANALYSIS WITH INSTRUMENTAL VARIABLES TO COMPARE EFFECTIVENESS IN CANADIAN AND UNITED STATES PATIENT POPULATIONS. C.A. Beck¹, J. Penrod¹, T.W. Gyorkos¹, S.H. Shapiro², L. Pilote¹, ¹Division of Clinical Epidemiology, Montreal General Hospital Research Institute, Montreal, Quebec; ²McGill University, Montreal

PURPOSE: In the United States, where an aggressive approach to care following acute myocardial infarction (AMI) is more commonly adopted in comparison to a conservative approach, the aggressive approach may be associated with small incremental (marginal) mortality benefits. We conducted this study to evaluate the effectiveness of aggressive care following AMI in Canada. We hypothesized that the marginal benefits should be larger in Canada, as the country is operating on a lower margin because the approach to care is more conservative overall.

METHODS: We conducted a retrospective cohort study using administrative data of acute care hospital admissions and in- and out-patient services for all patients who sustained a first AMI in

Quebec in 1988 (n=8674). We used differential distances to hospitals offering aggressive care as instrumental variables to control for unobserved case-mix variation when measuring mortality up to 4 years after first AMI.

RESULTS: Of the 4422 subjects who were > 64 years old, 11% received cardiac catheterization within 90 days after admission. In a previous study that applied similar methodology to the 1987 United States (US) Medicare population of first AMI patients, 30% of subjects received catheterization within 90 days. As in the US study, we found that subjects living relatively close to hospitals offering aggressive care were more likely to receive aggressive care (26% of "close" versus 19% of "far" subjects received cardiac catheterization within 90 days; 95% CI: 5% to 9%). Unlike the US study, we found no differences in mortality across the "close" versus "far" differential distance groups (unadjusted differences at 1 year: 1%; 95% CI: -1% to 3%). This absence of association was found in elderly (> 64 years) and younger age groups. Adjusted results also showed no differences between subjects receiving aggressive versus conservative care (at 1, 2 and 4 years: 4%, 2%, -4%; 95% CI: -11% to 20%, -15% to 18%, -26% to 8%, respectively).

CONCLUSION: Contrary to our hypothesis but consistent with results from numerous randomized trials and observational studies, the aggressive approach to post-AMI care does not appear to be associated with marginal mortality benefits even in Canada, where the approach to post-AMI care is conservative overall.

MUTUAL INFORMATION AS A SUMMARY INDEX OF DIAGNOSTIC TEST PERFORMANCE. W.A. Benish¹; ¹Case Western Reserve University, Cleveland, OH

PURPOSE: The area under the receiver operating characteristic curve [AUC] is a popular but suboptimal summary index of diagnostic test performance. An alternative test performance summary statistic, mutual information, is proposed.

METHODS: Mutual information is a mathematical function that quantifies the amount of information that one random variable provides about another random variable. If patient diagnosis [Dx] and test result [T] are viewed as random variables, then their mutual information [I(Dx;T)] provides a measure of how much the test result can tell us about the diagnosis (and vice versa). Mutual information is a central concept in information theory. It is defined as the difference between the entropy [H] of one random variable minus the entropy of that variable conditional upon the value of another random variable. Hence, $I(Dx;T) = H(Dx) - H(Dx|T)$. $I(Dx;T)$ is expressed in terms of bits of information. If there is complete uncertainty prior to testing about which of two diagnoses is correct, a test that establishes the diagnosis provides one bit of information.

RESULTS: $I(Dx;T)$ has advantages over the AUC. Unlike the AUC, the adequacy of $I(Dx;T)$ is not dependent upon "well-behaved" distributions of test results. Moreover, $I(Dx;T)$ can be calculated when multiple diagnoses are under consideration. Furthermore, calculation of $I(Dx;T)$ does not require that test results be dichotomized into "normal" and "abnormal" categories. Although clinicians may prefer results that are reported as "normal" or "abnormal," this usually requires the specification of an artificial diagnostic criterion. Diagnostic information is typically lost in this simplification process. One application of the mutual information function is the calculation of the amount of information lost as test results are dichotomized. Consider, for example, a population of individuals in which the probability of disease is 0.5. Assume that the "healthy" and "diseased" individuals in this population are normally and identically distributed with respect to some diagnostic marker, except that the means of the two populations are separated by one standard deviation. In this case, $I(Dx;T) = 0.16$ bits. If, however, the results are dichotomized using the optimal cutoff (the midpoint between the means of the two populations) then $I(Dx;T)$ is reduced to 0.11 bits.

CONCLUSION: Information theory provides a framework for understanding diagnostic testing. This perspective suggests that mutual information, rather than the AUC, provides the appropriate summary index of diagnostic test performance.

DOES THIS WOMAN HAVE AN ACUTE UNCOMPLICATED URINARY TRACT INFECTION: A SYSTEMATIC AND QUANTITATIVE ANALYSIS. S. Bent¹, B. Nallamothu², S.K. Saint²; ¹University of California, San Francisco, San Francisco, CA; ²University of Michigan, Ann Arbor, MI

PURPOSE: Acute uncomplicated urinary tract infection (UTI) is one of the most common complaints of young women seeking urgent medical care. Although the clinical encounter typically involves taking a history and performing a physical examination, the diagnostic accuracy of the clinical assessment remains uncertain. We, therefore, sought to rigorously and critically review the diagnostic accuracy of the clinical examination in diagnosing UTI in women.

METHODS: We performed a quantitative and systematic review of the literature using the following data sources: 1) MEDLINE (1966 - May 2000), 2) manual review of bibliographies, 3) contact with experts in the field. Studies were included if they contained original data on the accuracy or precision of the history and/or physical examination in diagnosing acute UTI in women. Studies were excluded if they evaluated infants, children, pregnant women, or patients with complicated UTI, or if they presented insufficient data to calculate likelihood ratios for specific signs and symptoms. Two authors independently abstracted all pertinent data.

RESULTS: Eleven studies met inclusion criteria. Overall, the prevalence of UTI was high (50%) in women who presented with one or more symptoms. Five symptoms (dysuria, frequency, hematuria, back pain, vaginal discharge), one sign (costovertebral tenderness) and one risk factor (diaphragm use) were found to be useful in the evaluation of patients with suspected UTI. The diagnostic power of individual tests was relatively weak, with likelihood ratios ranging from 0.34 to 3.09. The most powerful item from the history and physical examination was a history of vaginal discharge, which significantly decreased the probability of UTI (LR+ = 0.34).

CONCLUSION: The history and physical examination alone cannot lower the post-test probability of UTI below the treatment threshold. Patients with suggestive symptoms and/or a positive urinalysis should receive empiric antibiotic therapy. Patients with a negative urinalysis

should generally have treatment guided by the results of a urine culture. Patients with vaginal discharge or with a negative urinalysis but positive symptoms should have a pelvic examination since the probability of other infections is high.

THE EFFECT OF SERUM CARDIAC TROPONIN TESTING ON HEALTH RESOURCE USE FOR PATIENTS WITH SYMPTOMS SUGGESTIVE OF ACUTE CARDIAC ISCHEMIA: A RANDOMIZED CONTROLLED TRIAL. M. Berkowitz¹, S.E. Kimmel²; ¹Philadelphia VAMC and the University of Pennsylvania, Philadelphia, PA; ²University of Pennsylvania, Philadelphia, PA

PURPOSE: Serum cardiac troponin testing has become standard of care for detecting myocardial injury in patients with acute coronary syndromes, but its utility over CK testing alone in general populations is open to question. We studied the effect of troponin I testing on health resource use for unselected patients with clinical presentations suggestive but not diagnostic of acute cardiac ischemia.

METHODS: Patients who underwent cardiac enzyme testing in the outpatient settings (clinics and ER) of an urban VA hospital were eligible for inclusion. Those with non-diagnostic ECGs and no arrhythmias were randomized to CK and troponin I testing (CKTn) or CK testing alone (CK). The study had 80% power to detect a 15% difference in cardiac catheterization among patients hospitalized at their index visit. Secondary outcomes included hospitalization rates, lengths-of-stay, and use of cardiac medications, functional studies, and revascularization procedures. Statistical comparisons were performed using unpaired t and rank sum tests for parametric and nonparametric continuous data, and odds ratios and chi-square tests for proportions.

RESULTS: Three hundred ninety two of 775 patients who underwent cardiac marker testing were enrolled (51% [47,54]). CKTn patients were more likely to be current or former smokers (83 vs 74%, p=0.03) but the two groups were otherwise similar in age (mean 60.7 years), sex (95% male), race (65% black), cardiac risk factors, and history of coronary artery disease and procedures. The overall incidence of unstable angina and myocardial infarction was 12% [9,16] and similar in the two groups. There was no difference in cardiac catheterization rates (OR for CKTn patients 1.3 [0.74,2.4], p=.34). However, more CKTn than CK patients were discharged from their index visit without hospitalization (OR 1.85 [1.08, 3.18], p=0.02). There was no difference in total, ICU, or hospital lengths-of-stay (3.8, 2.1, and 1.6 days), in the odds of patients undergoing functional studies (1.0 [0.61,1.6]) or revascularization procedures (OR 2.1 [0.87,5.0]) while hospitalized, or of receiving beta-blockers (.99 [0.64,1.5]) or aspirin (.85 [0.52,1.4]) at discharge (all p's > .10).

CONCLUSION: Cardiac troponin I testing had no effect over CK testing alone on health resource use in patients hospitalized for their symptoms but resulted in significantly more discharges from outpatient settings. If studies confirm that discharged patients had no adverse outcomes, a policy of linked CK and troponin testing may offer a cost-effective strategy for managing outpatients with clinical presentations suggestive of acute cardiac ischemia.

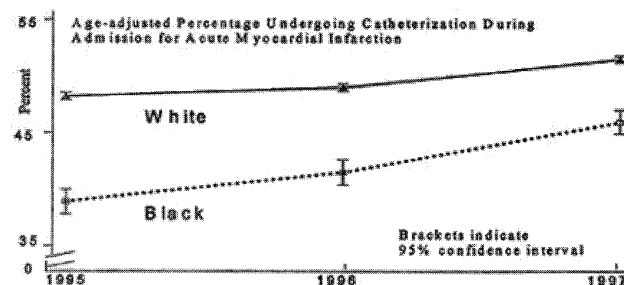
RACIAL DISPARITIES IN CATHETERIZATION FOR ACUTE MYOCARDIAL INFARCTION IN THE UNITED STATES, 1995-1997. A. Bertoni¹, F. Brancati¹; ¹Johns Hopkins School of Medicine, Baltimore, MD

PURPOSE: Racial disparities in procedure use for acute myocardial infarction (AMI) have been well documented in selected populations in the 1980s and early 90's, but little is known about recent trends in disparities in the general population.

METHODS: We conducted a series of cross-sectional analyses of data from the Nationwide Inpatient Sample (which includes race and demographic, diagnostic and procedure data on all discharges from over 900 representative hospitals in 18 states) for 1995 through 1997 to compare rates of catheterization performed prior to discharge for acute myocardial infarction (ICD9 code 410) in whites vs. blacks.

RESULTS: We identified 298,496 discharges among whites and 23,496 discharges among blacks with AMI as the first listed diagnosis during 1995-97. Age-adjusted catheterization rates by race are presented in the figure. From 1995 to 1997, age-adjusted peri-AMI catheterization rates increased more sharply in blacks (39% to 46%) than in whites (48% to 52%; p<0.01). After adjustment for age, gender, comorbidity, region, and rural locale using Poisson regression, the incidence rate ratio (IRR) for peri-AMI catheterization in blacks compared to whites in 1995 was 0.84 (95% confidence interval (CI) 0.78-0.90). During the next two years, the gap progressively narrowed, such that in 1997 the adjusted IRR was 0.90 (95% CI 0.83-0.96). Similar results were obtained when we restricted the analysis to discharges from hospitals that performed catheterizations.

CONCLUSION: These recent nationwide data suggest that, while blacks remain less likely to undergo catheterization during a hospitalization for AMI than whites, this racial gap appears to be narrowing.



ARE WOMEN VETERANS WORRYING ABOUT HEART DISEASE? M.S. Biswas¹, P.S. Calhoun², H.B. Bosworth³, L.A. Bastian⁴; ¹Department of Veterans Affairs, Durham, NC; ²Duke University, Durham, NC; ³Durham VAMC, Durham, NC; ⁴Durham VAMC/Duke University, Durham, NC

PURPOSE: Women veterans, like their male counterparts, are at increased risk for heart disease because they are more likely to be obese and smoke cigarettes than the general population. In order to develop clinical interventions that will improve risk behaviors among women veterans, we assessed their awareness of heart disease. The purpose of this study was to determine whether women veterans are worried about heart disease and to specify whether women with risk factors for coronary artery disease (CAD) are more likely to worry about heart disease compared to women without risk factors.

METHODS: Cross-sectional survey of women veterans receiving primary care at the Durham VAMC. We measured demographics, CAD risk factors, other medical conditions, and worry about heart disease. Statistical analysis was performed using the chi-square test and logistic regression analysis using worry about heart disease as the outcome variable.

RESULTS: Overall, 409 women veterans completed the questionnaire (response rate 71%). We excluded women with heart disease (n=64) and incomplete data (n=17). Our analyses are based on 328 women, mean age 49 years (range 36–90 years). Overall, only 42% worried about heart disease. In bivariate analyses, factors associated with worrying about heart disease were younger age ($p=0.04$), single marital-status ($p=0.009$), elevated cholesterol ($p=0.002$), obesity ($p=0.03$), and premature family history of CAD ($p=0.001$). Of those with family history, elevated cholesterol, or obesity, however, only 50% worried about heart disease. In logistic regression analyses, age, marital status, elevated cholesterol, obesity, and family history, were independently associated with worry about heart disease ($p<0.05$). No other risk factors for CAD assessed in this study, including diabetes, smoking, hypertension, or post-menopausal status, were significantly associated with worry about heart disease. In fact, there were trends towards not being worried about heart disease despite having such risk factors as smoking (54% of smokers were not worried), hypertension (52% with hypertension were not worried), and post-menopausal status (61% of post-menopausal women were not worried). While women with increasing numbers of risk factors were more likely to worry about heart disease ($p<0.001$), it was not until the women had four or more risk factors that more than 50% of them worried about heart disease.

CONCLUSION: Despite evidence in the literature indicating that the strongest risk factors for CAD in women are diabetes, smoking, and hypertension, women veterans with these risk factors were not worried about heart disease. Of the traditional risk factors examined, only elevated cholesterol, obesity, and family history were significantly associated with worry about heart disease. The majority of women with multiple risk factors remained unworried. The current study strongly suggests that women who should be concerned about heart disease are not. Clinical interventions should be developed that emphasize risk behavior modification to reduce the risk of first coronary events in women veterans.

OVERRULE OF THE RESUSCITATION DECISIONS IN JAPAN. S. Bito¹, Y. Simoda¹, S. Matsumura¹, Y. Saito¹, M. Aoki¹; ¹National Tokyo Medical Center, Tokyo, Japan

PURPOSE: Little is known about how resuscitation decisions are made and the predictors of receiving resuscitation in Japan. We described the use of resuscitation and of Do Not Resuscitate (DNR) orders for older patients dying in a Japanese hospital and evaluated patient characteristics associated with such decisions.

METHODS: We reviewed the medical records of all patients >70 years who died in a teaching hospital in Japan over 12 months. We identified resuscitations (including timing, treatments and whether the primary attending physician was present), orders concerning resuscitation and demographic and clinical patient characteristics. Cardiopulmonary resuscitation (CPR) was defined as any emergent intubation or cardiac massage or both.

RESULTS: Of the 221 patients who died, 210 medical records were accessible. Seventy-three percent had DNR decisions and 20% percent of the patients who had a DNR decision received resuscitation; 49 of 56 patients without a DNR order decision received resuscitation. None of the 79 patients who received resuscitation survived to leave the hospital. Only 10% of those received resuscitation while with DNR decisions had both emergent intubation and cardiac massage. Compared to those whose DNR orders were respected, patients for whom physicians overruled a DNR decision to perform resuscitation were more likely to have non-malignant disease (OR = 3.9; 95% C.I. 1.5–10.2) and to have had low conscious level on admission (OR = 3.1; 95% C.I. 1.1–8.7). Patient's sex and age, timing of the resuscitation and whether their physician was present at the time of the resuscitation were not associated with overriding DNR decisions.

CONCLUSION: It is common for DNR decisions to be overruled in Japan. This is an affront to the Japanese patient that should be remedied. Policies are needed to ensure that patient preferences are translated into care.

ARE BLACKS ALWAYS LESS WILLING THAN WHITES TO DONATE ORGANS? L. Boulware¹, L.E. Ratner¹, J. Sosa¹, L.A. Cooper¹, T.A. Laveist¹, N.R. Powe¹; ¹Johns Hopkins Medical Institutions, Baltimore, MD

PURPOSE: Blacks are less willing to donate cadaveric organs, the most common type of transplantation, but whether the same race differences exist for other donations is unknown. We compared patterns of race differences in willingness to donate cadaveric organs with those of becoming a living kidney donor and donating blood.

METHODS: We conducted a cross-sectional study in Maryland via standardized telephone questionnaire using random digit dialing to identify households. Persons age 18–75 years were randomly selected within households. Subjects were considered willing to become cadaveric donors if they had designated themselves as an organ donor on their driver's license, living-related donors if they were willing to donate a kidney to a sibling, and blood donors if they had previously donated blood. Using logistic regression we analyzed the independent effects of race on willingness to donate adjusting for demographic, clinical and attitudinal characteristics.

RESULTS: Of 385 respondents (84% of randomized households), 19% were White men (WM), 30% White women (WW), 12% Black men (BM), and 31% Black women (BW). BM and BW were more likely than WM and WW to have < high school education ($p<0.01$), to have annual income <\$40,000 ($p<0.01$), to have dependents ($p<0.01$), to be unemployed ($p=0.01$), to consider spirituality important ($p<0.01$), to believe hospitals "want to know more about personal affairs or business than they really need to know" ($p<0.01$), and more likely to believe "hospitals have done harmful experiments on patients without their knowledge" ($p<0.01$). BM were least willing to become cadaveric donors (17%) compared to BW (34%), WW (59%), and WM (65%) ($p<0.01$). However, WM were least willing to become living related donors (60%) compared to BW (61%), BM (72%), and WW (75%) ($p=0.05$). BW were least willing to donate blood (40%), compared to WW (58%), BM (65%), and WM (86%) ($p<0.01$). Despite adjustment for differences in respondent characteristics, BM were still much less likely to become cadaveric organ donors [odds ratio (95% CI)=[0.2 (0.1–0.7)], equally as likely to donate blood [0.8(0.3–2.5)], and almost 3 times more likely to become living-related donors [2.8 (0.9–8.7)] than WM. It is unclear whether BW were less likely to be cadaveric organ donors [0.6 (0.2–1.5)], but they were much less likely to donate blood [0.1 (0.04–0.3)], and equally as likely to be willing to be a living related donor [1.3 (0.5–2.7)] as WM.

CONCLUSION: Race differences in willingness to donate are not consistent across all types of donation, refuting a commonly held belief that Blacks are always less willing to donate. Understanding why Blacks donate differently for non-cadaveric situations may augment efforts to recruit minority organ donors.

ETHNIC VARIATIONS IN PRE-OPERATIVE DEPRESSION AMONG PATIENTS UNDERGOING CORONARY ANGIOPLASTY. C. Boutin-Foster¹, C. Young¹, J. Peterson¹, M.E. Charlson¹; ¹Cornell University Medical College, New York, NY

PURPOSE: Ethnic minorities with coronary artery disease have worse outcomes compared to Caucasians. Depression is known to be a major risk factor for poor outcomes among patients with coronary artery disease. However, among ethnic minorities with coronary artery disease, little is known about the prevalence of depression. The purpose of this study was to determine whether ethnic variations in pre-operative depression exist between Caucasians and ethnic minorities undergoing angioplasty.

METHODS: Subjects in the study were patients who had recently undergone coronary angioplasty and were eligible for enrollment in a randomized trial to reduce cardiac risk factors. Baseline evaluation included an assessment of demographic characteristics, cardiac history, and severity of illness. Depression was assessed using the Center for Epidemiologic Studies Depression Scale (CES-D). A score of 16 or greater indicated the presence of depression.

RESULTS: Of 542 patients enrolled, 33% had evidence of pre-operative depression. Their mean age was 61(+12), 34% were female, 14% were African-American and 16% were Latino-American. Patients who were depressed were more likely to be younger in age ($P=0.03$). A greater proportion of women were depressed in comparison to men ($P=.003$). Mean CES-D scores were significantly higher among ethnic minorities compared to Caucasians ($P=.002$). The highest CES-D scores were observed among Latino-Americans. In general, African-Americans and Latino-Americans were 2 times more likely than Caucasians to have evidence of depression (95% CI 1.3–3.2).

CONCLUSION: Among patients undergoing coronary angioplasty, minorities were 2 times more likely to have evidence of depression at baseline. This finding is clinically important since baseline depression among patients with coronary artery disease is a poor prognostic indicator. Future studies are needed to determine whether baseline depression among ethnic minority patients with coronary artery disease is associated with poor outcomes.

Rates of Depression by Ethnicity ($P=.001$)

	Depressed (%)	Not Depressed (%)
Caucasian	30%	70%
African-American	39%	61%
Latino-American	58%	42%

DOES THE SOURCE OF MEDICAL INFORMATION AFFECT PERCEIVED HEALTH RISKS? R.S. Braithwaite¹, J.M. Fortin², B.E. Bond¹, N.F. Col²; ¹Tufts University/New England Medical Center, Boston, MA; ²Harvard Medical School, Chestnut Hill, MA

PURPOSE: Patients often receive medical information from the media and internet, but little is known about how information sources affect risk perceptions. Our objective was to investigate whether the source of medical information affected perceived risks for common diseases.

METHODS: Women treated for breast cancer at an urban hospital were recruited through flyers and an oncology social worker. Participants completed a health survey about their sources of health information (doctor, television (TV), or internet) and their perceived lifetime risk of heart disease, hip fracture, and recurrent breast cancer. Risk estimates among patients receiving vs not receiving information from each source were compared using Wilcoxon Rank tests.

RESULTS: The mean age of the 51 participants was 52 years, 16% were non-Caucasian, and the median educational level was 2 years of college. Patients receiving medical information from either TV or the internet had higher perceived risks for both heart disease and hip fracture than those not receiving information from those sources. Patients receiving information from TV perceived breast cancer recurrence as being more likely even though fewer among them had late stage or node-positive disease. Patients who received information from their doctors did not have consistently higher or lower perceived risk estimations compared to patients not receiving information from their doctors.

CONCLUSION: Patients' perception of risk may be affected by the source of health information, with those who rely on the TV or internet having higher risk perceptions than those who rely on other sources.

Perceived Lifetime Disease Risk by Use (+) or Nonuse (–) of Information Source (*p < 0.05)

	+ TV (n=37)	–TV (n=14)	+Web (n=12)	–Web (n=12)	+Doctor (n=35)	–Doctor (n=16)
Heart Disease	15%*	6%*	24%	9%	11%	18%
Hip Fracture	5%*	1%*	8%	3%	5%	2%
Breast Cancer	19%	10%	13%	17%	15%	17%

A META-ANALYSIS OF VIRAL LOAD DETECTABILITY WITH HIGHLY-ACTIVE RETROVIRAL THERAPY (HAART). R.S. Braithwaite¹, M.P. Wolf¹, P.R. Skolnik², J. Lau², J.B. Wong², ¹Tufts University/New England Medical Center, Boston, MA; ²Tufts University, Boston, Massachusetts

PURPOSE: HAART effectively reduces HIV RNA to undetectable levels, but its durability and degree of suppression vary. Our aim was to combine and quantify published HIV viral load detectability over time for HAART.

METHODS: A Medline search from 1966–Dec 2000 for RCTs and clinical cohorts of HIV patients treated with HAART but with no prior exposure to protease inhibitors identified 203 studies. Selection criteria and subgroup analyses were pre-specified. We abstracted data regarding patient characteristics, HAART, CD4, viral load and survival over time, and used the DerSimonian and Laird random effects model to combine data

RESULTS: Our selection criteria yielded 75 studies: 53 with undetectable viral load data involving 12,104 patients. For intent-to-treat HAART trials (21 studies, n=6,187), the table summarizes the proportion of patients with viral load <500 over time, distinguishing between studies permitting 1 or >1 HAART regimen. One HAART is inferior to >1 HAART because of lack of compliance or efficacy for any one particular regimen. Among patients compliant with HAART (6 on-treatment trials, n=508), 86±4% had viral load <500 at 6 months and 80±4% at 12 months. Improvement in viral assays affected the definition of undetectable viral load. In 4 HAART trials (n=508), 9±8% of patients were <500 but ≥50 at 6 months, and 14±6% were <500 but ≥50 at 12 months.

CONCLUSION: HAART effectively suppresses viral load <500 for up to 30 months, but many patients require at least one change in HAART. Improvements in viral assays have lowered the threshold for viral load detection and reduced the proportion of HAART patients with undetectable viral load.

Intent to Treat by # of HAART Regimens Permitted

	6 mo	12 mo	18 mo	24 mo	30 mo
1 HAART	61±2%	49±2%	38±3%	28±3%	23±5%
>1 HAART	56±2%	55±2%	63±2%	57±4%	51±6%

ALCOHOL CONSUMPTION AND PROBLEM DRINKING IN VA SURGICAL PATIENTS. R.C. Brooks¹, J. Conigliaro¹, J.T. King¹, J. Chang¹, N.J. Brucker¹; ¹University of Pittsburgh, Pittsburgh, PA

PURPOSE: Alcohol abuse is a common but frequently unrecognized problem in surgical patients, with important implications for Medical consultants. Studies have shown that 10–50 % of hospitalized patients on surgical services have alcohol problems, depending on the subspecialty service and the detection methods used. We screened a large cohort of VA surgical patients to determine the prevalence of problem drinking in a combined inpatient and outpatient population as the initial step in a study of alcohol consumption and surgical outcomes.

METHODS: All patients at the VA Pittsburgh Healthcare System admitted to surgical services or scheduled for elective surgery requiring at least an overnight hospital stay were considered candidates for screening. Patients were asked to complete a confidential screen for alcohol problems using the Alcohol Use Disorders Identification Test (AUDIT) and quantity-frequency measures. Non-cirrhotic patients who met criteria for alcohol problems (AUDIT score 8 or weekly consumption >15 drinks in men or >11 drinks in women) and control patients who did not meet these criteria were invited to participate in a descriptive cohort study. Controls were matched by surgical service and admission status (inpatient versus outpatient at screening, planned overnight versus longer admissions for elective procedures) in a 2:1 control:study ratio. Participants received a baseline preoperative assessment including demographic information and the Alcohol Timeline Follow-Back to measure alcohol consumption. Descriptive statistics were reported as mean standard deviation. Current drinking and drinking over the last 6 months were analyzed using a non-parametric test for trend.

RESULTS: Among 1624 patients presenting for surgical admission or preoperative assessment over one year, 1067 (65.7%) were screened for alcohol problems, of which 99 (9.3%) screened positive. The mean AUDIT score for screen-positive patients was 12.0 ± 9.6 and the mean number of drinks per week was 33.2 ± 37.2, compared with 1.1 ± 2.1 and 2.0 ± 8.5, respectively, for screen-negative patients. Forty-seven (47.5%) screen-positive patients agreed to further assessment and follow-up, as did 93 matched controls. Mean AUDIT and alcohol consumption scores were not statistically different between enrolled and non-enrolled study patients (p=0.3). Among enrolled problem drinkers, there was a trend toward less alcohol consumption in a typical drinking day as surgery approached, from an average of 9.2 ± 12.5 drinks/day over 6 months prior to their assessment to 5.4 ± 4.4 drinks/day over the previous 1 month (p=0.13 for trend). Binge drinking episodes (>6 drinks in one day) showed a similar reduction, with the average problem drinker reporting 5.6 ± 8.7 episodes/month at 3–6 months prior to assessment but only 3.4 ± 6.7 episodes/month over the previous 1 month (p=0.18 for trend). The problem drinkers were most commonly found on the Orthopedic (29.8%) and General surgery (25.5%) services.

CONCLUSION: The prevalence of problem drinking in our large cohort of veterans undergoing surgery was lower than that found in prior studies of surgical inpatients but similar

to that seen in published studies of medical outpatients. The inclusion of outpatients and patients undergoing elective procedures may shift the drinking profile of surgical patients towards that of the general population. Intensive screening of surgical patients for problem drinking by Medical consultants may be most effective on selected subspecialty services. Alcohol consumption levels among problem drinkers tended to decrease as the date of surgery approached, suggesting that the perioperative period may provide a good opportunity for brief interventions.

ALPHA BLOCKERS AND HYPERTENSION — HOW MANY ARE AT RISK FOR SUBOPTIMAL THERAPY? C.L. Bryson¹; ¹Puget Sound VA, Seattle, Washington

PURPOSE: This study defines the number of people at risk for suboptimal therapy with hypertension who are only on alpha blockers. Results of randomized controlled trials have provided substantial evidence that antihypertensives differ in their ability to mitigate outcomes of CHF, CVA, MI and renal failure in the hypertensive population despite nearly equivalent blood pressure control. This is of particular interest because of the early termination of one arm of the ALLHAT study in which the alpha-1 blocker doxazosin was associated with a doubling of the rate of incident heart failure. Doxazosin is less effective for preventing the outcome of CHF than chlorthalidone in hypertensive patients. This result was unexpected, since doxazosin has favorable effects on other surrogate markers including serum lipids. Alpha-1 blockers are an effective treatment for prostatism, and fifty percent of men have histologic evidence of BPH by 60 years of age. It is difficult to quantitate the exposure of alpha-1 blockers for hypertension in elderly men on pharmacy data alone because information about the patient's comorbid illnesses is necessary to interpret the indication for the agent. This is true not only for alpha-1 blockers but also for other antihypertensives.

METHODS: The National Ambulatory Medical Care Survey (NAMCS) is a national survey of physician visits that records not only physician diagnoses but also patient therapy. It was a sampling study that was carefully performed and allows extrapolation to the national visits during the study times. The NAMCS data was analyzed with STATA. The provided weights were utilized to expand the results to projected national averages. Summary measures were computed for variables of interest while maintaining acceptable relative standard errors.

RESULTS: This presentation or poster will present the results of the analysis of the 1998 NAMCS for antihypertensive use, with an accent on alpha blocker usage as a single drug. Approximately 3.5 to 4% of the hypertensive population appears to be treated with only an alpha blocker, which is between 840,000 to 1 million patients. Interestingly, about half of these are women. Only 15% of hypertensives were on diuretics. Time trends and usage by indication may also be presented from prior NAMCS data. A correlation with VA data will be available as well by the time of presentation as a result of an ongoing project.

CONCLUSION: An estimation of the use of different antihypertensive agents correlated with indication is important, particularly in light of recent clinical trials. An estimate of the use of alpha-1 blockers as single agents prior to changes in recommendations will help enumerate the size of the population at risk for suboptimal pharmacotherapy with regard to cardiovascular endpoints. It will also provide context for the importance of findings from clinical trials of antihypertensive agents.

CAN WE DIAGNOSE ACUTE SINUSITIS OVER THE TELEPHONE?: A COMPARISON OF THREE METHODS IN PATIENTS CALLING PRIMARY CARE PHYSICIANS WITH UPPER RESPIRATORY ILLNESS. D. Budnitz¹, R. Kessler¹; ¹University of Pennsylvania, Philadelphia, PA

PURPOSE: Physicians commonly manage patients with acute upper respiratory illness (URI) solely through telephone contact. Management often includes diagnosing acute sinusitis and prescribing antibiotics over the telephone although little data has been published on the accuracy or effectiveness of such telephone triage and treatment. The purpose of this study was to determine the correlation of three approaches to the diagnosis of acute sinusitis in patients calling their physician with URI complaints.

METHODS: Between March 15 and July 15, 2000, sequential patients calling an internal medicine practice complaining of URI were requested to participate in a structured symptom survey. The diagnosis of acute sinusitis was determined in a blinded fashion by applying three diagnostic approaches modified for telephone use: (1) criteria set forth by the Task Force on Rhinosinusitis of the American Academy of Otolaryngology-Head and Neck Surgery; (2) a clinical prediction rule published in the JAMA Rational Clinical Examination Series; and (3) clinical cues found empirically to be used by clinicians to differentiate sinusitis from viral URIs. Agreement between diagnostic approaches was determined using the Kappa statistic.

RESULTS: Of 91 eligible patients, 61 (67%) agreed to participate in the study. Mean age was 42 years (range 12–78), 43 (70%) were women, and mean duration of symptoms was 4.7 days (range 1–23). Forty-four patients (72%), 6 patients (10%), and 11 patients (18%) were diagnosed with sinusitis by each diagnostic approach respectively. Agreement between diagnostic approaches ranged from kappa=0.08 (approach 1 compared to approach 2) to kappa=0.39 (approach 2 compared to approach 3).

CONCLUSION: Three published approaches to diagnosing sinusitis modified for telephone use exhibited poor to marginal agreement. More research is needed to determine a valid tool for telephone diagnosis of sinusitis before the diagnosis and treatment of acute sinusitis solely through telephone contact can be recommended.

DOUBLE JEOPARDY: THE IMPACT OF RACE/ETHNICITY AND HEALTH INSURANCE. H. Burstin¹, R. Weinick², N. Lurie³; ¹Agency for Healthcare Research and Quality, Rockville, MD; ²Center for Primary Care Research, Rockville, MD; ³Department of Health and Human Services, Washington, DC

PURPOSE: Efforts to reduce health care disparities have focused on the impact of race/ethnicity, as well as lack of health insurance. The purpose of this study was to examine how these factors interrelate in the acquisition of health care services.

METHODS: Using the 1996 Medical Expenditure Panel Survey (MEPS), a national household cross-sectional survey of health care utilization, we examined the differential impact of non-white race combined with lack of health insurance. Outcome measures for the last year included usual source of care (USC), doctor's office visit (MD), emergency room visit (ER), and use of prescription medications (RX). Insurance was categorized as uninsured, public only, and any private insurance. Disparities by insurance status were examined for African-Americans (AAs) and Latinos.

RESULTS: The national survey included 11,826 Whites, 2,627 AAs, and 4,393 Latinos. In all insurance categories, Latinos were less likely than whites to have a USC. Among the uninsured, 45.2% of Latinos had a USC, compared to 65.1% of whites and 61.2% of AAs. Among the privately insured, AAs were significantly less likely to have USC than whites. For all types of insurance, whites were significantly more likely than AAs or Latinos to have an MD visit. Overall, whites were not more likely to have ER visits compared with AAs and Latinos, though publicly insured whites (24.1%) were more likely to seek ER care than AA's (16.8%) and Latinos (17.7%). In all insurance categories, whites were more likely than AAs or Latinos to have any RXs. Among the uninsured, whites were more likely to get RXs (50.8%), compared to Latinos (33.8% RX) and AAs (38.4% RX).

CONCLUSION: We found that uninsured patients of color suffered disproportionate access barriers, even when compared to uninsured whites. This "double jeopardy" effect was most striking among uninsured Latinos. These data suggest the need for strategies to reduce disparities among racial/ethnic minorities that go significantly beyond health insurance coverage.

ECONOMIC COSTS OF PERCUTANEOUS ENDOSCOPIC GASTROSTOMY AMONG A COHORT OF OLDER ADULTS. C. Callahan¹, N. Buchanan¹, T. Stump¹, R. Nisi¹; ¹Indiana University Center for Aging Research, Regenstrief Institute for Health Care, Indianapolis, IN

PURPOSE: We have recently reported high mortality rates and low rates of improvement in nutrition and health-related quality of life among a prospective cohort of older adults receiving percutaneous endoscopic gastrostomy (PEG). The purpose of the current study is to describe the direct charges associated with PEG tube feeding over one year.

METHODS: All patients aged > 60 receiving PEG over a 24-month period in a defined community were eligible for this study. Consent to collect data from medical records was obtained on 105 (70%) of 150 eligible patients. We obtained charge data from the two area health systems in the defined community where the PEG procedures occurred. Patients were interviewed every 2 months for 1 year to determine the use of outpatient services, the volume of PEG feeding, and the rate of complications related to PEG.

RESULTS: The mean patient age was 79 and the most common indications for PEG were stroke, dementia, and cancer. Total hospital charges for this cohort were \$1.8 million over one year or sim;\$17,000 per patient. Censoring patients at death or one year post-PEG, the mean number of days of PEG tube feeding was 167 (range 4–365). The mean volume of feeding per day was 1375 cc which represented sim;1560 kcal/day. Direct charges per patient for providing PEG feeding for a 3, 6, & 12 month period are shown below.

Item	3 months	6 months
Initial PEG operative procedure	\$2200	\$2200
Videosophogram (swallowing study)	\$ 475	\$ 475
Enteral formula	\$1318	\$2635
Replacement of PEG tube	\$ 0	\$1200
MD charges for minor complications	\$ 65	\$ 131
Hospital charges for major complications	\$2180	\$2531
Total direct charges related to PEG	\$6238	\$9172
Average daily cost	\$ 67.80	\$ 50.12

CONCLUSION: Direct charges associated with PEG tube feeding over one year are estimated at ~\$13,000. Overall charges are significantly increased by the charges for PEG tube replacement and complications. Daily charges are modest but highly dependent on the duration of PEG tube feeding.

THE INFLUENCE OF DIABETES MELLITUS ON MORTALITY AND MORBIDITY AFTER CORONARY ARTERY BYPASS GRAFT SURGERY. J.L. Carson¹, P.S. Scholz¹, A. Chen², E. Peterson², J. Gold³, S. Schneider¹; ¹Robert Wood Johnson Medical School, New Brunswick, NJ; ²Duke University, Durham, NC; ³Albert Einstein College of Medicine, New York, NY

PURPOSE: The impact of diabetes on short-term mortality and morbidity in patients undergoing coronary artery bypass surgery is unclear. The aims of our analysis were to determine if diabetes is independently associated with 30-day mortality, morbidity, and to characterize the causes of death.

METHODS: We performed a retrospective cohort study of consecutive patients undergoing coronary artery bypass surgery from the United States and Canada that are included in the 1997 Society of Thoracic Surgeons national database. We included patients undergoing both first and re-operations. We excluded patients who underwent simultaneous valvular heart surgery, patients with diabetes treated with diet alone or in whom the treatment was unknown, and those with missing information on the presence of diabetes. The primary outcome was 30-day mortality. Secondary outcomes were in-hospital morbidity, infections, and composite outcomes of mortality or morbidity and mortality or infection.

RESULTS: There were 146,786 patients undergoing coronary artery bypass surgery; 41,663 patients with diabetes mellitus and 105,123 without diabetes mellitus. The prevalence of diabetes treated with medications or insulin was 28.4%. The 30-day mortality was 3.74% in patients with diabetes mellitus and 2.71% in those without diabetes mellitus; the unadjusted odds ratio was 1.40 (95% confidence interval, 1.31–1.49). After adjusting for other baseline risk

factors, the overall adjusted odds ratio for diabetics was 1.23 (95% confidence interval, 1.15–1.32). Patients treated with oral hypoglycemic medications had adjusted odds ratio 1.13; 95% confidence interval, 1.04–1.23, whereas, those on insulin had an adjusted odds ratio 1.39; 95% confidence interval, 1.27–1.52. Morbidity, infections, and the composite outcomes occurred more commonly in diabetic patients and were associated with an adjusted risk about 35% higher in diabetics than non-diabetics, particularly among diabetic patients treated with insulin (adjusted risk between 1.50–1.61). Compared to patients without diabetes, infections were the cause of death more often in patients with diabetes treated with insulin ($p = 0.033$) and neurologic causes of death occurred more frequently in diabetes patients overall ($p = 0.01$) and in diabetes patients treated with oral medications ($p < 0.005$).

CONCLUSION: Diabetes mellitus is an important risk factor for mortality and morbidity among those undergoing bypass surgery. Further research is needed to identify the mechanism(s) responsible for and methods to improve the deleterious outcomes.

DIFFERENCES BETWEEN PATIENTS REFERRED TO HOSPICE FROM ACADEMIC VS. NON-ACADEMIC SETTINGS. D. Casarett¹; ¹University of Pennsylvania, Philadelphia, PA

PURPOSE: Hospice is designed to provide end of life care to patients who are terminally ill. Increasingly, hospices are developing affiliations with academic medical centers. However, little is known about hospice patients referred from academic medical centers, and how their needs may differ from those of the general population of hospice patients. Therefore, the purpose of this study was to identify differences between patients referred to an urban, non-profit hospice from academic vs. non-academic medical centers.

METHODS: In this retrospective cohort study, 1,691 patients admitted to an inpatient and outpatient hospice program were identified between January of 1997 and January of 1999, and data were gathered until discharge or death.

RESULTS: Patients referred from academic medical centers were younger, had higher incomes, and were less likely to have Medicare or Medicaid. Patients referred from academic medical centers were less likely to have a Do Not Resuscitate order or a living will, and had more medical and nursing needs. Survival analysis revealed no difference in length of stay between patients referred from academic and non-academic medical centers.

CONCLUSION: Patients referred to hospice from academic medical centers have greater needs for nursing and medical care than do patients referred from non-academic medical centers. These results have important implications for hospice planning and policy. If they are replicated in other hospice settings, they suggest that hospices that establish ties with academic medical centers may be caring for patients with more symptoms, and more needs for care. Such differences should be considered in drawing up plans for affiliations. They might also be considered on a more global scale in assessing reimbursements to hospices by Medicare and other payers.

IN-HOSPITAL RESUSCITATION: RELATIONSHIP OF DIAGNOSTIC CATEGORIES TO RHYTHM AND SURVIVAL. K.D. Chandra¹, D.C. Parish¹, F.C. Dane¹, M.D. Durham¹, J.A. Hudson¹; ¹Mercer University School of Medicine, Department of Internal Medicine, Macon, GA

PURPOSE: To determine the relationship of diagnostic categories to cardiac rhythm and survival of in-hospital resuscitation. To date, limited data are available.

METHODS: Retrospective, cohort study of in-hospital first arrests in 3,167 admissions from 01/01/87 through 12/31/96 at one hospital. Diagnostic related groups (DRG) were combined into ten clinically meaningful categories: cardiovascular surgery, pulmonary, non-cardiac surgery, miscellaneous, cardiovascular, metabolic/endocrine, gastrointestinal, infections, heme/ onc, and hypertension/stroke.

RESULTS: There is significant association between diagnostic category and rhythm severity, $X^2(36)=297.43$, $p < .0001$. PEA is the most prevalent rhythm in all categories except pulmonary, where organized was more frequent. Survival varied significantly across categories. Patients admitted for cardiac surgery had the highest survival rate; hypertension/stroke had the lowest survival, $X^2(9)=182.55$, $p < .0001$. (See Table.)

CONCLUSION: Significant association exists between diagnostic categories and rhythms. Survival and rhythm frequency differed among categories. In-hospital resuscitation studies should include diagnostic categories to allow comparisons among studies and institutions.

*** % of Survival/N**

	SVT/VT	ORG	VF	PEA	ASYS	TOTAL
CVS	70/20	57/91	66/111	23/157	11/36	43/415
PULM	100/10	70/132	11/18	9/89	5/22	42/271
SURG	62/16	55/206	33/51	23/237	19/87	35/597
MISC	61/13	59/116	20/39	11/137	13/54	29/359
CV	55/38	41/118	34/143	8/246	15/80	24/625
M/E	100/3	50/24	16/19	6/46	8/39	18/131
GI	100/1	34/29	9/11	15/53	0/30	16/124
ID	50/8	37/70	10/30	5/115	3/71	14/294
H/O	0/2	38/29	14/7	8/51	0/32	13/121
HTN/S	33/9	14/70	11/28	12/82	5/41	12/230
%/N *	62/120	50/885	35/457	14/1213	10/492	28/3167

USE OF ANTIBIOTICS AND RISK OF DEEP VEIN THROMBOSIS IN A POPULATION WITHOUT KNOWN RISK FACTORS. J. Chang¹, J. Luchsinger¹; ¹Columbia University, NY, NY

PURPOSE: Venous thrombosis is a major health concern with at least 201,000 first lifetime cases reported each year in the United States. Although numerous risk factors have been

identified, many thrombotic events occur in patients without these predisposing factors. There are conflicting reports of a link between Chlamydia pneumoniae and deep venous thrombosis (DVT). Our objective is to ascertain whether the use of antibiotics, particularly those effective against C. pneumoniae, is associated with a lower rate of DVT.

METHODS: We analyzed a claims-based dataset comprised of 170,011 enrollees of both sexes, aged 46 and over, who did not have risk factors for DVT and who had at least 2 years of continuous full coverage for pharmaceutical and health services. DVT (ICD-9 codes 451.1, 451.11, 451.19, 451.2, 451.81, 453.8, 453.9) and antibiotic use (national drug codes) were defined through insurance claims. We examined the association between use of macrolides, quinolones, tetracyclines, penicillins, cephalosporins, and trimethoprim/sulfamethoxazoles (TMP/SMZs) and DVT with Cox regression, stratified by birth, beginning and end of observation, and total claims. Time to event was defined as time from beginning of observation to time of DVT. Antibiotic use was treated as a time-dependent covariate.

RESULTS: There were 2236 cases of DVT (1.31%) during 4.6 mean years of follow up. The average age of the sample was 67 with 51% being female. Claims for antibiotics included 15.6% macrolides, 10.0% quinolones, 4.1% tetracyclines, 25.8% penicillins, 18.9% cephalosporins, and 1.7% TMP/SMZs. The hazard ratios of DVT for the antibiotic groups were the following: macrolides 1.15 95% CI (0.93–1.42), quinolones 1.43 95% CI (1.12–1.82), tetracyclines 1.01 95% CI (0.67–1.54), penicillins 1.01 95% CI (0.84–1.22), cephalosporins 1.61 95% CI (1.33–1.95), and TMP/SMZs 1.39 95% CI (0.75–2.59). Further analysis of individual antibiotics within each of the groups failed to show a protective effect.

CONCLUSION: Our analysis failed to show a protective association between the use of antichlamydial antibiotics and DVT.

BALANCING THE RISKS AND BENEFITS OF ANTICOAGULATION IN PATIENTS WITH ATRIAL FIBRILLATION. J. Chau¹, H.N. Lee², L. Sirie³. ¹Northern Ontario Family Medicine Program, Sudbury, Ontario; ²McMaster University and Group Health Centre, Sault Ste Marie, Ontario; ³Group Health Centre, Sault Ste Marie, ON

PURPOSE: Anticoagulation has been proven to prevent stroke in patients with atrial fibrillation. However, this intervention is generally under-utilized, especially in the elderly. It has been suggested that addressing physician and patient concerns of the relative risks of stroke or bleeding by the use of various published indices might help correct this problem. However, these indices have not been compared to one another or validated in the elderly. The purpose of this study is to stratify a community-based population of seniors according to their stroke risk and compare it to their bleeding risk according to published indices to evaluate their utility in clinical practice.

METHODS: All senior (age >65) patients with chronic or paroxysmal atrial fibrillation enrolled in the anticoagulation clinic of a Canadian community health service organization were enrolled and audited. Data relevant to the risk indices was collected from the clinic's computerized medical record. The stroke risk stratification indices were the Atrial Fibrillation Investigators (AFI) scale (Arch Intern Med 1994; 154:1449–57) and the Stroke Prevention in Atrial Fibrillation (SPAF) Investigators scale (Chest 1998; 114:579S–589S). The Outpatient Bleeding Risk Index (OBRI, Am J Med 105;91–9) was used to estimate bleeding risk.

RESULTS: Of the 171 patients with atrial fibrillation enrolled in the anticoagulation clinic, 18 were younger than 65 years, and 8 had concomitant valvular disease leaving 145 patients for analysis. Data collection was complete. The mean age was 77 years, and women comprised 55% of the cohort. According to the AFI criteria, 38% of the population had an annual stroke risk of <4.3% while 50% of the population had an annual stroke risk < 3.6% by SPAF (p < .05). There was agreement between the AFI and SPAF in estimating the patients with the highest stroke risk of 8% annually (43% and 50%, respectively). The OBRI estimated 91% of all patients to have a bleeding risk of 12% over two years while 9% of the cohort had the highest estimated bleeding risk of 53% over 48 months. In all but the highest stroke risk groups, the OBRI-estimated bleeding risk exceeded the expected benefits of anticoagulation.

CONCLUSION: In this population of elderly patients with atrial fibrillation, the use of published risk stratification indices has limited clinical utility, mainly because of the difficulty in balancing the relative risks in the group with low and intermediate stroke risk. In particular, the validated OBRI does not seem applicable to the stable outpatient elderly population, and its use might actually deter the appropriate anticoagulation of patients.

SOCIAL AND WORK DISABILITY IN PRIMARY CARE PATIENTS WITH DEPRESSION AND BACK PAIN: EVALUATION OF THE WHO DISABILITY ASSESSMENT SCHEDULE. L. Chwastiak¹, M. VonKorff², S. Hauge². ¹University of Washington, Seattle, WA; ²Center for Health Studies, Group Health Cooperative, Seattle, WA

PURPOSE: The prevention of disability and the preservation of work and social functioning are clinical and policy priorities for the health care of patients with chronic illness. Most health status instruments used in outcomes research to measure the burden of illness do not account for work and social functioning, and often confuse the concepts of symptoms and functioning. Current instruments also are not responsive to change over time or with treatment. This study examines the utility of a new general disability instrument, the WHO Disability Assessment Schedule (WHODAS II) in health services research. The specific aims are to determine whether the WHO DAS II has different profiles in two disease states, whether it is responsive to change following treatment; and to compare the effect size to those of other disability measures.

METHODS: We surveyed adult patients making a primary care visit for depression (n=73) or back pain (n=76) at clinics in the Group Health Cooperative of Puget Sound (WA). Telephone interviews were conducted at baseline (within 2 weeks of the visit), and after 3 months of treatment for the condition. The interview included the WHODAS II, SF-36, Work Limitations Questionnaire (WLQ), Patient Health Questionnaire (PHQ), Sheehan Disability Instrument (depression); and the Roland Pain Scale and the Graded Chronic Pain Scale (back pain).

RESULTS: At baseline, most of the subscales of the WHO DAS II (5 of 6) and the SF-36 (5 of 8) were statistically significantly different in the two disorders. The subscale that best distinguished the groups was the WHO DAS II physical mobility subscale. The WHODAS II

summary score was highly correlated with the PHQ in the depression sample (Pearson r=.694, p<.0001), and with the Roland score in the back pain sample (r=.716, p<.0001). In both depression and back pain, the work subscale of the WHODAS II was most highly correlated with WLQ time management subscale (rs=.70 and .68). After 3 months, 92% of subjects completed the follow-up interview. Responsiveness to change of the instruments will be compared by two methods: effect size, and regression models that evaluate the relationship between the WHO DAS II and the other measures.

CONCLUSION: In this study, the WHO DAS II performed well in discriminating the disability associated with two different disease states. In both depression and back pain, it distinguished physical functioning from interpersonal functioning. Its work subscale is highly correlated with longer work-specific instruments. Preliminary analyses of the follow-up data reveal that the instrument can detect change following primary care treatment interventions.

BODY TEMPERATURE, BLOOD PRESSURE, VISCOSITY AND PULSE RELATIONS: A PILOT STUDY. Y. Cinar¹. ¹University of Yeditepe, Haydarpasa Numune Education and Research Hospital, Istanbul, Uskudar

PURPOSE: Temperature effect on blood viscosity is reported and related hemodynamic changes like blood pressure, flow rate and diameter of vessels are calculated by hydrodynamic principle. The aim of this study is search of the effect of body temperature on viscosity, pressure and pulse in vivo.

METHODS: Healthy cases (n=17) was selected by simple random sampling method. Cases are kept in supine position for two hours. Case body temperatures are increased by electrical blanket in 60 minutes and then decreased by air condition in 60 minutes. Pulse, blood viscosity, temperature of the calf, biceps and the axilla and blood pressure of the upper (UE) and lower (LE) extremities were monitored. Wilcoxon Rank test was used for statistical evaluation.

RESULTS: Differences between cooling (CC) and heating (HC) conditions were as follows: Temperature (°C) of axilla (4.18±6.76%; Z=-3.518, P=0.000), UE temperature (20.31±7.84%; Z=-3.575, P=0.000), LE temperature (26.20±4.61%; Z=-3.623, P=0.000), blood viscosity [(Second), -3.25±2.8%; Z=-3.517, P=0.000], plasma viscosity (-1.84±1.46%; Z=-3.466, P=0.001), pulse/min (11.26±14.71%; Z=-2.440, P=0.015), systolic UE pressure [(mm/hg) -1.49 ±13.98%; Z=-0.579, P=0.562], diastolic UE pressure (-5.04 ± 16.67%; Z=-1.239, P=0.215), systolic LE blood pressure (-5.59 ± 15.24%; Z=-1.713, P=0.087), diastolic LE pressure (-11.84 ±13.88%; Z=-2.858, P=0.004). Correlation are as follows; erythrocyte sedimentation rate (ESR) and UE temperature (r=0.540, p=0.025), axilla temperature and pulse (r=0.577, p=0.015), LE temperature and pulse in HC (r=0.505, p=0.039), systolic pressure and pulse in CC (r=0.98, p=0.042), UE diastolic pressure and pulse in CC (r=0.623, p=0.008), diastolic pressure and pulse in CC (r=0.568, p=0.017).

CONCLUSION: Results showed that temperature related change in blood viscosity, pressure and pulse are important.

THE MECHANISM OF COLD ANGINA. Y. Cinar¹. ¹University of Yeditepe, Haydarpasa Numune Education and Research Hospital, Istanbul, USKUDAR

PURPOSE: Cold angina is explained with increased coronary "vascular resistance". The temperature of the lower extremities in cold weather and the temperature of blood during hibernation is approximately 22°C. It is suggested that when blood temperature decreases from 36.6°C to 22°C, blood viscosity increases 26.13%. The aim of this study is determination of the effect of the increased blood viscosity on circulatory system.

METHODS: Changes in circulatory parameters due to increased viscosity can be calculated with Poiseuille's hydrodynamics equation. In the equation, blood flow rate in m3/sec is $Q = \pi r^4 / 8 \eta L (F1 - F2)$ and peripheral resistance of a vessel is $R = (F1 - F2) / Q$, where r is the radius, L is the length of vessel, η is blood viscosity, $F1$ is the initial and $F2$ is the final pressure at the two ends of a vessel.

RESULTS: According to these equations, flow rate decreases 20.72 % and peripheral resistance increases 20.72 % when viscosity increases 20.72 % due to cold environment. This state can be compensated by increased cardiac work as 20.72 % increase in blood pressure or 5.9 % increase in the vessel radius.

CONCLUSION: The term "peripheral resistance" is more correct for the explanation of cold angina than vascular resistance, because peripheral resistance can cover blood viscosity (resistance against blood flow) and vascular resistance (change in diameter, length and shape of vessel).

These calculations show that decreased temperature and increased viscosity of the blood in the skin and lower extremities in cold weather results in increased peripheral resistance and circulatory work. Due to sclerotic vessel has no vasodilatation capacity, Yschemia risk is high for the coronary atherosclerotic patient. Unfortunately, similar circulatory changes caused by the cooling of the pulmonary blood in cold weather are unknown.

The information above shows that the mechanism of cold angina gains an additional explanation with increased blood viscosity and peripheral resistance due to decreased blood temperature, for the first time. This method is important for further evaluation of the dynamics of the circulatory system.

VALIDATION OF PERIOPERATIVE CARDIAC RISK INDICES. S.L. Cohn¹, A. Haridas¹, S. Hegde¹. ¹SUNY Downstate, Brooklyn, NY

PURPOSE: Cardiac complications (CC) are major causes of perioperative morbidity. Various risk indices and algorithms for predicting cardiac risk have been published. The purpose of this study was to evaluate performance of these tools.

METHODS: We retrospectively reviewed records of 90 patients undergoing 104 major intra-abdominal or vascular procedures in a large, urban teaching hospital. Outcome measures included the following CC - (major) death, MI, ischemia, CHF, VT; (minor) other hemodynamically significant arrhythmias, hypotension requiring pressors. We examined performance of the ASA, Goldman, Detsky, Eagle, Larsen, and Lee indices, and the ACC and ACP guidelines.

RESULTS: CC occurred in 13 patients (14.4%) including 8 patients who experienced 11 major complications. There was a trend toward more CC with age > 60, anesthesia duration > 3 hours, creatinine > 2.4, and vascular and emergency surgery. The ASA scale and revised cardiac index(Lee) performed better than the original Goldman and Detsky indices. The ACC and ACP algorithms were best at identifying low risk patients, but some of this effect was due to the results of non-invasive tests. (Table 1) Exercise capacity, Eagle criteria, and Larsen's index were not predictive of CC.

CONCLUSION: Although our group of patients had relatively few major complications, we were able to validate the predictive ability of various indices to stratify perioperative risk; however, statistical significance of the differences could not be achieved due to the small numbers. The goals of risk indices are to identify low risk patients requiring no further workup and to select other patients potentially benefiting from additional testing or interventions. Other management strategies, including perioperative beta-blockers, will be needed to reduce complications in the intermediate and high risk groups, and in emergency surgery patients.

Table 1: Cardiac Complications (All/Major) vs Risk Indices

	ASA	ASA	GOL	GOL	GOL	DET	DET	ACC	ACC	ACC	ACP	ACP	ACP	LEE	LEE	LEE
%	1-2	3-4	1	2	3	0-10	15+	Lo	Int	Hi	Lo	Int	Hi	1	2	3
All	9.8	18.4	10.2	17.1	33.3	13.8	23.5	14.3	15.0	17.1	11.7	14.3	19.4	11.3	16.0	50.0
Maj	5.1	10.5	5.4	14.3	16.7	4.6	17.6	0.0	10.0	11.4	1.7	0.0	16.7	5.8	13.8	33.3

PERIOPERATIVE CARDIAC RISK GUIDELINES: AMERICAN COLLEGE OF CARDIOLOGY (ACC) OR AMERICAN COLLEGE OF PHYSICIANS (ACP)? S.L. Cohn¹, S. Hegde¹, A. Haridas¹, ¹SUNY Downstate, Brooklyn, NY

PURPOSE: The ACC and ACP each published guidelines for perioperative cardiovascular evaluation for noncardiac surgery. Both were evidence-based, but the ACC also used expert consensus opinion in the absence of strong evidence. Because of this difference, the guidelines may present conflicting opinions. The purpose of this study was to compare their recommendations with respect to non-invasive testing (NIT) and evaluate risk stratified clinical outcomes.

METHODS: We retrospectively reviewed records of 90 patients (pts) undergoing 104 major intra-abdominal or vascular procedures in a large, urban teaching hospital. Patients were classified into low, intermediate, or high (including emergency) risk groups based on the algorithms. We then compared the ACC and ACP recommendations for NIT and subsequent clinical outcomes for each group. Major complications were defined as cardiac death, MI, ischemia, CHF, and VT.

RESULTS: Eleven major complications occurred in 8 pts, 3 of whom had MI's (1 cardiac death). Of the 104 procedures, 30 were emergent (1 MI). Additionally, 9 and 5 pts respectively were classified in a high risk group by ACC and ACP criteria. The remaining pts were divided into test or no test subgroups (Table 1). The ACC recommended NIT for 11 of these low-intermediate risk pts, 2 of whom had postop MI's, and no test for the other 54 pts, none of whom had major complications. The ACP recommended NIT for 6 of 14 intermediate risk pts, 1 of whom had an MI, and no test for the 55 low risk pts, 1 of whom had an MI. ACP guidelines recommended no test in 7 of the 11 pts (1 had an MI) in whom the ACC suggested testing. The guidelines agreed otherwise.

CONCLUSION: Both ACC and ACP algorithms separated low risk patients from intermediate-high risk ones. Because of differing concepts regarding exercise capacity and NIT in non-vascular surgery, the ACP classified more patients as low risk and recommended fewer NIT's; however, the clinical outcomes appeared comparable (despite the small numbers). Physician judgement and personal preference for the guidelines will determine the approach regarding NIT for an individual patient undergoing elective surgery. Future studies will need to clarify the best approach for high risk/emergency patients.

Table 1 ACC vs ACP: Risk Groups & Major Complications

	Low	Complic	Int	Complic	High	Complic
ACC	45	1	20	3	39	4
ACP	55	2	14	2	35	4

SCREENING FOR PERIPHERAL ARTERIAL DISEASE IN AN ETHNICALLY DIVERSE BILINGUAL POPULATION. T. Collins¹, W. Foster¹, M.E. Suarez-almazor¹, N.J. Petersen¹; ¹Baylor College of Medicine, Houston, TX

PURPOSE: The purpose of this pilot study was to assess the feasibility of screening Non-Hispanic Whites, African Americans, and Hispanics, both English and Spanish speaking, for peripheral arterial disease.

METHODS: We screened patients who were ≥50 years of age for peripheral arterial disease (PAD) within three primary care clinics. PAD was diagnosed by measuring the ankle-arm index (AAI) which is a ratio of the systolic blood pressure in the ankle to the systolic blood pressure in the arm. An AAI of < 0.9 defined disease. In addition, patients completed an assessment of symptoms related to PAD, the San Diego Claudication Questionnaire (SDCQ). Patients were screened from the Houston VA Medical Center and the Harris County Hospital District, both of which were hospital-based clinics. Additional questionnaires were used to determine the patient's medical history (questionnaire designed by the team), walking impairment (Walking Impairment Questionnaire), and overall health status (SF-36). Each questionnaire was translated into Spanish by a team of bilingual investigators (the SF-36 was translated by the Health Assessment Lab, part of the Medical Outcomes Study). The translation process involved two

forward and two backward translations. The research assistant who enrolled patients into the study was bilingual, fluent in both English and Spanish.

RESULTS: We screened 48 patients (18 Non-Hispanic Whites, 22 African Americans, 7 Hispanics, 3 of whom were Spanish speaking, and 1 mixed race). The mixed race patient was excluded from the analysis. The mean ages of the patients who were screened and did not agree to enrollment vs. the patients who were screened and did agree to enrollment were 61.05 (±7.62), and 62.5 (±8.48), respectively, P = 0.54. Among all patients screened, 15 (83.3%) of Non-Hispanic Whites, 7 (30.4%) of African Americans, and 5 (71.4%) of Hispanics (3 Spanish speaking) agreed to enrollment. There was no difference by gender in the percent of patients who agreed to enrollment (females 60%; males 61.3% P = 0.93). Among the 20 patients screened who refused enrollment, 75% were African American (66.7% male), 15% were Non-Hispanic White (100% male), and 10% were Hispanic (50% male). Two patients were defined as having PAD based on an AAI of < 0.9 (1 Non-Hispanic White male and 1 African American female). Patients were able to complete all questionnaires.

CONCLUSION: We were able to perform a pilot study to determine the feasibility of screening for PAD in an ethnically diverse population, including Spanish speaking patients. Among all three ethnic groups, African American patients, particularly males, were least likely to participate in our screening study. Further research is needed to determine ways to increase research participation by African Americans. We are presently enrolling all three ethnic groups in a larger study to determine the epidemiology of PAD by race within primary care clinics. As the demographics of the U.S. shift, future research should include multiple ethnic groups, including non-English speaking patients.

RACE/ETHNICITY AS A RISK FACTOR FOR LOWER EXTREMITY NON-TRAUMATIC AMPUTATION IN PERIPHERAL ARTERIAL DISEASE. T. Collins¹, M.L. Johnson¹, J. Daley², W. Henderson³, S. Khuri²; ¹Baylor College of Medicine, Houston, TX; ²Harvard University, Boston, MA; ³Hines VA Medical Center, Hines, IL

PURPOSE: To determine if race/ethnicity was independently associated with an increased risk for non-traumatic lower extremity amputation versus lower extremity bypass revascularization among patients with peripheral arterial disease (PAD).

METHODS: Data were analyzed from the National VA Surgical Quality Improvement Program (NSQIP) and from the Veterans Affairs Patient Treatment File (PTF). Data from the NSQIP (Phase I, 10/1/91 – 12/31/93 and Phase II, 8/1/94 – 9/30/95) were used to determine patients who underwent a lower extremity amputation vs. lower extremity bypass revascularization and to obtain information on important preoperative clinical variables, including race, that are likely associated with the outcome of amputation in patients with PAD. The PTF was used to verify that the amputations were associated with a diagnosis of PAD. Race/ethnicity was defined as non-Hispanic White, African American, or Hispanic. Variables that were univariately associated (p < 0.05) with the outcome of amputation were then placed into a multivariate logistic regression model to determine independent predictors for the dependent variable, lower extremity amputation vs. lower extremity bypass revascularization.

RESULTS: We identified 3,085 lower extremity amputations and 8,409 lower extremity bypass operations. Among all cases included, there were 416 Hispanics (3.6%), 2,337 African Americans (20.3%), and 8,741 non-Hispanic Whites (76.1%). Among the amputation cases, 4.9% were African American, 29% were African American, and 66.2% were non-Hispanic White. Approximately 11% of patients who underwent an amputation were 50 – 59 years of age, 37% of the patients who underwent an amputation were ages 60 – 69 years, and 48% were > 70 years. Over 99% of all cases were male patients. Risk factors associated with the outcome of lower extremity amputation vs. bypass revascularization were age > 70 years, diabetes mellitus, dialysis, wound infection, rest pain/gangrene, renal insufficiency/failure, history of congestive heart failure, functional status, previous lower extremity bypass revascularization, functional status, emergency surgery, Hispanic ethnicity, and African American race. Among the 18 variables within the model, Hispanic ethnicity and African American race were associated with a greater risk of amputation than a history of diabetes requiring oral medication or rest pain/gangrene (Hispanic ethnicity 1.4, 95% Confidence Interval (CI) 1.1, 1.8; African American race 1.8, 95% CI 1.6, 2.0; diabetes requiring oral medication 1.3, 95% CI 1.1, 1.5; rest pain/gangrene 1.2, 95% CI 1.0, 1.3). The final model had a c-statistic of 0.82.

CONCLUSION: After adjusting for preexisting illnesses such as diabetes, cardiovascular disease, functional status, and other risk factors, both Hispanic ethnicity and African American race were independent risk factors for lower extremity amputation in patients with PAD. Although the burden of certain atherosclerotic risk factors is higher in minority patients, the impact of this burden does not account for the increased risk of the outcome of lower extremity amputation in these two populations. As we do not have information on the severity of disease, race/ethnicity may be a marker for the severity of PAD. Further research is needed to better understand the reason(s) why race/ethnicity is independently associated with poor outcomes in PAD.

COMPARISON OF PATIENT AND PROVIDER PERCEPTIONS OF BACK PAIN SEVERITY, LIKELIHOOD OF IMPROVEMENT, AND TREATMENT PREFERENCES. M.T. Connelly¹, A. Hrbeck², D. Post², R.B. Davis², C. Canning¹, R. Phillips², D. Cherkin³, D.M. Eisenberg², ¹Harvard Medical School and Harvard Pilgrim Health Care, Boston, MA; ²Harvard Medical School; ³Group Health Cooperative, Seattle, WA

PURPOSE: Whether patients and their providers make comparable assessments of the severity of pain, the likelihood of improvement and optimal treatment strategies when patients present with an acute episode of low back pain is not known. We sought to evaluate the level of concordance on these dimensions between patients and providers at the time that individuals with acute low back pain (ALBP) enrolled in a randomized controlled trial (RCT) of usual care versus choice of complementary and alternative medical (CAM) therapy.

METHODS: Baseline data obtained by face to face interview at RCT enrollment. Eligible subjects were adult members of a multi-specialty medical practice who reported experiencing uncomplicated ALBP for less than 3 weeks with no co-morbid explanation for symptoms. Enrollees were randomized to either usual care or to a choice of chiropractic, massage,

acupuncture or usual care. Enrollees' medical providers completed a written survey at the time they referred a patient to the study.

RESULTS: Of the 80 enrolled in the study to date, the mean age was 42, 51% were women, and 58% were white. Sixty-seven of the enrollees' providers (84%) responded to a questionnaire about the enrollees' ALBP. The mean patient-reported severity score for back pain on a scale of 0–10, with 10 representing the worst pain ever, was 8.0 (s.d. 1.6, range 4–10); providers reported a mean score of 6.3 (s.d. 2.1, range 1–9). Providers substantially underestimated patients' pain ($t=5.48$, $p<.001$), with 24% of the enrollees describing their pain as 10/10 and none of the providers rating pain higher than 9/10. Providers and patients had similar expectations for recovery. On a 0–10 scale, with 10 representing complete improvement in 6 weeks and 0 representing no improvement, patients reported a mean score of 8.4 (s.d. 1.8, range 4–10) and providers a score of 8.1 (s.d. 2.5, range 1–10), $t=.84$, $p=.40$. The majority of patients (55%) expressed a preference for massage, 31% for chiropractic, 12% for acupuncture, and 1% for usual care. Thirty-two percent of the providers accurately anticipated the type of therapy the patient said s/he would choose, 18% thought that the modality the patient preferred was the best choice for that patient. Only 7% believed the therapy preferred by the patient was the best treatment for ALBP in general. Almost half (44%) of the providers stated that they did not know what the best treatment for ALBP was in general.

CONCLUSION: These findings suggest that providers of patients enrolling in an RCT of usual care versus a choice of CAM therapies for ALBP do not appreciate the severity of patients' perceived pain or the beliefs and expectations patients have regarding specific treatment options.

PATIENT SATISFACTION WITH THEIR PRIMARY CARE PHYSICIAN PREDICTS RECOVERY FROM MAJOR DEPRESSION. N.C. Connolly¹, B.H. Hanusa¹, H.C. Schulberg², B.L. Rollman¹; ¹University of Pittsburgh, Pittsburgh, PA; ²Cornell University School of Medicine

PURPOSE: Major depression (MD) is prevalent among patients who present for primary care services and is responsible for an excess burden of disability. Identifying patient characteristics associated with recovery can allow us to better target treatments for those patients less likely to recover under their primary care physician's (PCP's) usual care.

METHODS: We examined data collected as part of a clinical trial testing the effectiveness of disseminating depression treatment guidelines via electronic medical records system (EMR). Patients presenting to a university-based clinic were screened for MD using the PRIME-MD. Eligible patients also had a 17-item Hamilton Rating Scale for Depression (HRS-D) ≥ 12 . Patient satisfaction with their PCP was assessed at baseline using a 5-point Likert scale and later dichotomized to "very satisfied" or "not very satisfied" for our analysis. The HRS-D was repeated at 6 months. Recovery was defined as a HRS-D ≤ 7 . Information about depression treatment, such as pharmacotherapy, number of visits, and referral to a mental health specialist was abstracted from the EMR.

RESULTS: Between 4/96 and 12/98, 211 depressed patients met all protocol eligibility criteria and completed a baseline interview. Of the 193(91%) patients who completed 6 month interviews, their baseline mean HRS-D was 20, 71% were women, 24% were African American and 119 (62%) reported being "very satisfied" with their PCP. No differences in HRS-D or satisfaction were noted by gender, race or medical comorbidity. Although "very satisfied" patients had significantly lower baseline HRS-D, the difference of only one point was not considered clinically significant. At 6 months 43(22%) of the patients recovered. "Very satisfied" patients were more likely to recover at 6 months than those who were "not very satisfied" (31% vs 8%, OR 5.1, 95% CI=2.0–12.8). This association persisted after controlling for baseline HRS-D scores, medical co-morbidity and process measures for depression care (adjusted OR 5.3, 95% CI=2.0–13.8). A baseline report of "very satisfied" was not significantly associated with subsequent number of PCP visits, medications prescribed, or mental health referrals.

CONCLUSION: Patients who are "very satisfied" with their PCP are more likely to recover from a MD episode. Patient recovery has been found to be related to physicians engendering confidence in their patient's recovery. It is tempting to speculate that PCPs who stimulate greater optimism in their patients, improve satisfaction and subsequent recovery from depression. Assessment of patient satisfaction early in the therapeutic relationship may identify patients at risk for poor recovery from MD.

SMOKING CESSATION IN THE HOMELESS: THERE IS A WILL, BUT IS THERE A WAY? R.L. Cook¹, S.E. Connor², M.I. Herbert³, S.M. Neal¹, J. Williams⁴; ¹School of Medicine, University of Pittsburgh, Pittsburgh, PA; ²School of Pharmacy, Program for Healthcare to Underserved Populations, University of Pittsburgh, Pittsburgh, PA; ³Program for Healthcare to Underserved Populations, University of Pittsburgh, Pittsburgh, PA; ⁴Health Care for the Homeless, Pittsburgh, PA

PURPOSE: Smoking is a significant issue among the homeless, but little is known about their willingness to quit or their interest in assistance with quitting. We therefore sought to determine what proportion of homeless persons smoked, whether they were ready to stop smoking, and whether they had preferences for smoking cessation treatments.

METHODS: We anonymously surveyed homeless adults at nine sites in Pittsburgh, including emergency shelters, drop-in centers, transitional housing, a community health center, and residential substance abuse treatment programs. All persons at each site were approached during 2-hour blocks; 97% of those present completed the survey. Measures included demographic characteristics, living situation, smoking behavior, readiness to change smoking behavior (based on a stage-of-change model), self-efficacy to stop smoking (rated from 1 to 10), nicotine dependence (Fagerstrom scale), and preferred method of smoking cessation.

RESULTS: 273 persons completed the survey, of whom 87% met criteria for homelessness. The sample was middle aged (mean age 42 years), male (81%), and minority (63%). Of the 68% who were current smokers, 87% had interest in quitting, and 39% indicated readiness to quit smoking within the next six months. Compared to those not ready to quit, persons were significantly ($p < .05$) more likely to be ready to quit if they had high self-efficacy (47% vs. 21%), knew persons who would support their smoking cessation efforts (47% vs. 13%), or smoked within 30 minutes of awakening (40% vs. 31%). Readiness to quit was not associated

with gender, ethnicity, substance abuse, nicotine dependence, or type of shelter. Participants self-efficacy to quit smoking was significantly higher with assistance, compared to their self-efficacy without assistance (6.3 vs. 4.8, $p=0.000$). Treatments preferred by persons ready to quit included nicotine replacement (33%), counseling (15%), bupropion (Zyban) (7.5%), and nicotine in combination with other treatments (12%); one-third (33%) indicated "nothing" or "quitting cold-turkey" would help.

CONCLUSION: The majority of homeless persons smoke, many are ready to stop smoking, and most believe they will be more successful with treatments such as nicotine replacement. There is an urgent need to develop and implement smoking cessation programs for the homeless.

DISTRUST, RACE AND RESEARCH. G. Corbie-Smith¹, D. St. George², S. Thomas³; ¹University of North Carolina at Chapel Hill, Chapel Hill, NC; ²University of North Carolina at Greensboro, Greensboro, NC; ³University of Pittsburgh, Pittsburgh, PA

PURPOSE: Investigators have voiced concerns that distrust of research and the medical community impedes successful recruitment of African Americans and other minority groups into clinical research. Our objective was to examine possible differences in distrust by race in a national sample of respondents and to determine to what extent other sociodemographic factors explain any racial differences in distrust.

METHODS: We used data from a national telephone survey that examined attitudes and beliefs of Americans toward participation in clinical research. A distrust index, scores ranging from 0–most trusting to 7–most distrusting, was created using the sum of the responses to a seven-item measure of trust. The dependent variable in these analyses, high levels of distrust, was categorized as 5 on the distrust index. Of 1000 survey respondents, there were 527 Black and 382 White respondents eligible for this analysis ($n=909$).

RESULTS: Black respondents were more likely than White respondents not to trust that "their doctor would fully explain research participation" (41.7% vs. 23.4%, $p < .001$) and less likely to believe that they could ask their doctor questions (15.2 vs. 7.6%, $p < .001$). Black respondents were also more likely to disagree that "their doctor would not ask them to participate in research if the doctor felt there was harm" (37.2% vs. 19.7%, $p < .001$) and more likely to state that they felt their doctors sometimes exposed them to unnecessary risks (45.5% vs. 34.8%, $p < .001$). Blacks were more likely to believe that "someone like them" would be used as a guinea pig without their consent (79.2 vs. 51.9, $p < .001$) and that their physician had given them treatment as part of an experiment without their permission (24.5% vs. 8.3%, $p < .01$). Black respondents has a significantly higher mean index score (3.1) than Whites (1.8) ($p < .001$). Nearly 30% of blacks compared to 9% of whites had scores 5 ($p < .001$). In unadjusted analyses, Blacks had more than 4 times the odds of whites of having a distrust index score 5. After controlling for other sociodemographic variables in the logistic regression model, race remained strongly associated with a higher distrust score (OR=3.6; 95% CI=2.4,5.4).

CONCLUSION: We found important differences by race in distrust. Even after controlling for markers of social class, African Americans were less trusting. Investigators trying to engage African Americans in research must focus on developing interpersonal trust with community members by actively engaging the members in all aspects of research design development and dissemination of findings.

ADDRESSING HEALTH DISPARITIES: DO WE HAVE THE DATA? G. Corbie-Smith¹, D. St. George², D.F. Ransohoff¹; ¹University of North Carolina at Chapel Hill, Chapel Hill, NC; ²University of North Carolina at Greensboro, Greensboro, NC

PURPOSE: The NIH has recognized the importance of racial parity in clinical research so findings from clinical investigation can benefit the broadest populations. We conducted this study to determine the impact of federal mandates enacted in 1993/1994 on recruitment and reporting of clinical trials (RCTs) published in peer reviewed journals in diseases where significant disparities in health exist.

METHODS: We performed a Medline search covering the period January 1989 to Oct 2000 to identify RCTs published in the New England Journal of Medicine, JAMA and Annals of Internal Medicine. We included clinical trials conducted in patients in the following areas: cardiovascular disease, diabetes mellitus, HIV/AIDS, and cancer. For each RCT we collected data on the number and percent of white and minorities (i.e., African Americans, Hispanic, Asian/Pacific Islander, and Native Americans) reported in the study samples. In addition to simple descriptive statistics, we used ANOVA to assess reporting of race before during and after 1993 – 1995.

RESULTS: A total of 253 RCTs met our inclusion criteria (29 diabetes, 84 cardiovascular and 50 cancer and 90 HIV/AIDS). Forty percent of RCTs did not report the race/ethnicity of the study sample. Of the RCTs that reported race ($n=150$), 29% of the study samples were minorities. There was no statistical difference in the percent of minorities in RCTs published before 1993, between 1993 and 1995, and after 1995 (27%, 26% and 33%, $p=0.3$). In addition there was no difference in the percent minorities when stratified by disease.

CONCLUSION: In diseases where there are significant disparities by race in health outcomes, a large number of RCTs did not report the race/ethnicity of the study sample. In addition, there has been no increase in the proportion of minorities in study samples before, during or after enactment of federal initiatives to increase the proportion of minorities in RCTs. For clinicians to have the highest level of evidence to effectively care for minority patients and reduce disparities in health outcomes, RCTs must consistently report race/ethnicity and the proportion of minorities included in study samples needs to increase.

PODIATRIC PROBLEMS ARE ASSOCIATED WITH WORSE HEALTH STATUS. C.K. Crews¹, K.O. Vu¹, A.J. Davidson¹, L.A. Crane², P.S. Mehler¹, J.F. Steiner¹; ¹University of Colorado Health Sciences Center, Denver, CO; ²Department of Preventive Medicine and Biometrics, University of Colorado Health Sciences Center

PURPOSE: Persons with severe mental illness suffer from high rates of medical disorders and increased difficulty with self-care. We interviewed a sample of persons with severe mental illness to determine the prevalence of self-reported podiatric impairments and their effect on the participants' health status.

METHODS: 309 psychiatric outpatients underwent a personal interview assessing medical conditions and health status with the Medical Outcomes Study Short Form-36 (SF-36) survey. We assessed psychiatric severity with the Brief Psychiatric Rating Scale (BPRS) which rates psychiatric severity from 0 (none) to 6 (extremely severe) in 18 symptom constructs. Podiatric health was assessed using 9 items from the National Health Interview Survey (NHIS) and an additional item addressing foot pain.

RESULTS: 57% of our sample were diagnosed with schizophrenia and the mean BPRS score was 22 reflecting moderate severity of psychiatric illness at the time of the interview. 70% of participants reported at least one of 8 medical problems. 82% of patients surveyed reported at least one podiatric problem and 40% reported at least 3 problems. The most common problems were foot pain (48%), nail disorders (35%), and corns/calluses (28%). The prevalence rates of podiatric disorders in our cohort were 4–10 times higher than those reported by the general population in the 1990 NHIS. For example, 5.2% of the general population reported nail disorders, 5.0% reported foot infections, and 5.7% reported corns/calluses.

In bivariate analysis, the total number of podiatric problems reported by the participants was inversely related to their scores in all eight domains and both summary scores of the SF-36 (all $p < 0.001$). After controlling for sociodemographic factors, psychiatric illness, and medical conditions, the total number of podiatric limitations remained significantly associated with lower patients ratings in 4 of the 8 SF-36 domains: bodily pain ($p < 0.001$), role limits-emotional ($p = 0.002$), social functioning ($p = 0.03$), and general health ($p = 0.003$) and both summary domains, physical component ($p = 0.04$) and mental component ($p = 0.03$).

CONCLUSION: Persons with severe and persistent mental illness have markedly elevated rates of podiatric problems when compared to the general population. These problems are associated with worsened self-perceived health status. Addressing podiatric health may be a successful way to improve the overall health of this population.

DIFFERENCES IN HEALTHCARE EXPENDITURES AMONG PATIENTS TREATED FOR DEPRESSION WITH OR WITHOUT ANXIETY. W.H. Crown¹, E. Berndt¹, S. Finkelstein¹, D. Ling¹, G.J. Wan²; ¹The MEDSTAT Group, Inc., Ann Arbor, MI; ²Wyeth-Ayerst Pharmaceuticals, St. Davids, PA

PURPOSE: To compare healthcare expenditures of treatment with serotonin norepinephrine reuptake inhibitors (SNRIs) and selective serotonin reuptake inhibitors (SSRIs) in depressed patients with or without anxiety.

METHODS: Using administrative claims from the MEDSTAT MarketScan database, we identified patients with a new episode of depression enrolled in a participating health plan during the years 1994–1998. Patients included for analysis had both a diagnosis of depression (ICD-9-CM criteria) and a prescription for either an SNRI (venlafaxine [$n = 290$], venlafaxine XR [$n = 63$]) or an SSRI (fluoxetine [$n = 2854$], paroxetine [$n = 1772$], sertraline [$n = 2580$], fluvoxamine [$n = 124$]) antidepressant.

RESULTS: The SNRIs ($n = 353$) had lower inpatient non-mental health costs (\$206 vs \$472; $P = 0.02$) and lower antidepressant medication costs (\$302 vs \$338; $P = 0.01$) compared with the SSRIs ($n = 7330$). In particular, venlafaxine ($n = 290$) costs were lower than fluoxetine ($n = 2854$) costs (\$281 vs \$395; $P < 0.05$). Among patients with depression and anxiety, SNRIs ($n = 219$) had lower inpatient non-mental health costs (\$273 vs \$635; $P = 0.04$) and lower antidepressant medication costs (\$304 vs \$350; $P = 0.01$) than SSRIs ($n = 4351$). Among patients with depression without anxiety, SNRIs ($n = 134$) had lower inpatient non-mental health costs (\$96 vs \$234; $P = 0.05$) and lower inpatient mental health costs among users of inpatient mental health services (\$2,301 vs \$4,847; $P = 0.02$) relative to SSRIs ($n = 2979$).

CONCLUSION: Patients receiving SNRIs appear to have lower healthcare expenditures in some areas compared with SSRIs among depressed patients with or without anxiety. Further research is needed to determine whether patients treated with SNRIs can be shown to accrue more favorable economic benefits relative to patients treated with SSRIs over time.

A DESCRIPTIVE STUDY OF HUMAN IMMUNODEFICIENCY VIRUS (HIV) POSITIVE SINGLE ROOM OCCUPANCY (SRO) HOTEL RESIDENTS IN THE SOUTH BRONX. C. Cunningham¹, S. Shapiro¹, K. Berg¹, G. Sacajiu¹, G. Paccione¹; ¹Albert Einstein College of Medicine, Bronx, NY

PURPOSE: HIV continues to be a destructive problem in the urban poor. The SRO hotel population, a unique subset of the urban poor, is inadequately described in the literature. Our purpose is to describe the level of comprehensive health care, perceptions of health care, and utilization of health care services in the SRO hotel population served by both the Department of Acquired Immunodeficiency Syndrome (AIDS) Services Income Support (DASIS), a division of New York City's Department of Welfare, and CitiWide Harm Reduction, a community-based organization in the South Bronx.

METHODS: Following a pilot survey, we conducted an extensive community-based survey of SRO hotel residents in the South Bronx from 8/99 to 4/00. We went door-to-door in all 10 DASIS- and CitiWide-served SRO hotels administering anonymous surveys to residents. The 34-item questionnaire included demographic information, health care utilization patterns, status of HIV disease and treatment, perceptions of quality of and access to health care, and drug use.

RESULTS: Of the 190 hotel residents approached, 97% completed the survey. Of these residents, 91% reported being HIV-positive, 63% male, 59% Black, 34% Latino, and 62% active drug users. The average age is 41 years (range 20–74 years), and 56% are high school graduates. The median length of stay in a single hotel is 4 months (range 1 day–32 years), and 96% have Medicaid. Of the 167 HIV-positive residents, 67% report having primary care providers. Of these individuals, 71% report receiving care in an infectious disease (ID) or HIV-specialty clinic. However, only 22% of 18 eligible residents report taking *Mycobacterium avium* complex (MAC) prophylaxis, 65% of 57 eligible residents report taking *Pneumocystis carinii* pneumonia (PCP) prophylaxis, and 44% of 88 eligible residents report taking antiretroviral medications. In the previous 6 months HIV-positive residents report a mean of 1.5 Emergency Department visits, and in the previous year they report a mean of 1.7 hospitalizations. Quality of care and access to care are perceived to be less than "good" by 44% and 36% of HIV-positive residents, respectively.

CONCLUSION: Among South Bronx HIV-positive SRO hotel residents, a largely substance-using population with Medicaid, two thirds report having primary care providers. Yet a significant proportion of these individuals are not receiving medically recommended HIV therapy. Approximately half are dissatisfied with their quality of care, and one third with their access to care. In light of these findings, steps are currently being undertaken to address this deficiency in health care among HIV-positive individuals living in SRO hotels.

USE OF ELECTRONIC MEDICAL RECORDS: COMMUNICATION PATTERNS IN OUTPATIENT ENCOUNTERS. R.H. Curry¹, G. Makoul¹; ¹Northwestern University, Chicago, IL

PURPOSE: Although use of electronic medical record (EMR) systems is increasing, the effects on actual communication between patients and physicians have not been studied extensively. We sought to assess physician-patient communication patterns associated with use of an EMR system in an outpatient setting and provide an empirical foundation for larger studies.

METHODS: We conducted a cross-sectional, case-control study involving analysis of videotaped physician-patient encounters, surveys of physicians and patients, and medical record reviews. The setting was an academic general internal medicine practice. We studied three physicians who used an EMR system (EMR physicians) and three who used solely a paper record (control physicians). A total of 204 patient visits were included in the analysis (mean = 34 for each physician). Makoul's SEGUE instrument was utilized for content analysis of whether physicians accomplished communication tasks during encounters. A separate, qualitative analysis by the authors sought evidence of general patterns of EMR and paper chart use.

RESULTS: Compared to the control group, EMR physicians adopted a more active role in clarifying information and providing patient education. Overall, there were no statistically significant differences between the physician groups in mean duration of encounters, number of treatment options offered, or number of lab tests ordered. The qualitative analysis showed that physicians in both groups tended to direct their attention to the patient record during the initial portion of the encounter. The physical orientation of the computer keyboard/monitor and physicians' facility in typing were important determinants of attentiveness to the patient throughout the EMR encounters.

CONCLUSION: EMR physicians' communication behaviors may reflect styles established before they began using an EMR system. Coupling education on patient-centered communication when physicians are trained on the use of EMR systems could enhance the effectiveness of providers using this tool.

DO PATIENTS PERCEPTIONS OF METASTATIC BREAST CANCER INFLUENCE DECISIONS CONCERNING BREAST CANCER SCREENING? S. Cykert¹, N.W. Phifer¹, C.J. Hansen²; ¹University of North Carolina at Chapel Hill, Greensboro, NC; ²Moses Cone Memorial Hospital, Greensboro, NC

PURPOSE: It is unclear why many women who are advised by their physicians to obtain breast cancer screening (BCS) choose not to comply. Financial concerns and fear of the procedure are examples of factors that might deter women from recommended screening. Another possible explanation for avoidance of BCS might be that some women are not sufficiently concerned about the consequences of metastatic breast cancer to submit to screening. We, therefore, measured women's perceptions of metastatic breast cancer and compared them to self-reported compliance with BCS.

METHODS: A trained research assistant interviewed 106 women aged 50 to 75 years using a structured questionnaire that assessed patients' attitudes toward mammography and their compliance with BCS. Interviews were conducted in North Carolina and Florida. Patients preferences for the health state "life with metastatic breast cancer" were measured using a standard gamble approach. Standard gamble results were converted to health utility scores (HUS) for which 1.0 = normal health and 0 = death. Descriptive statistics were compiled. Bivariate analysis was performed to identify possible associations with recent noncompliance with BCS. Demographic data, attitudes toward screening, and HUS were used as independent variables in this analysis. Due to the exploratory nature of this study and small sample size, a p -value of < 0.1 was used as the cutoff for statistical significance.

RESULTS: Of the 106 women interviewed, 83% were high school graduates or above, 35% were African-American, and 89% had some form of health insurance. The mean age was 60 yrs. Two thirds of participants reported themselves to be at least in good health. The mean HUS for life with metastatic breast cancer was 0.39 with a median score of 0.25. Thirty-two of the one hundred-six women interviewed had not received BCS during the 2 years preceding this survey. Over 90% agreed that mammograms could save their life and that the actual x-ray would not lead to breast cancer or needless breast surgery. Bivariate associations with BCS non-compliance included worse self-reported health, education less than high school, African-American race, and high HUS (0.7 or above) for life with metastatic breast cancer.

CONCLUSION: Poverty, African-American race, and lack of education have long been cited as sentinel characteristics for lower BCS rates. The above data suggest that optimistic perceptions of metastatic cancer might also contribute to these lower rates. If indeed, higher individual HUS, can be confirmed as an independent predictor of BCS non-compliance, then a standard gamble approach could be used to identify a group of patients that could benefit from a targeted educational intervention that aggressively portrays consequences of metastatic breast disease.

RACIAL DISPARITY IN LUNG CANCER SURGERY: DO PATIENTS' PERCEPTIONS OF PROGRESSIVE LUNG CANCER EXPLAIN THE DIFFERENCE? S. Cykert¹, N.W. Phifer¹, C.J. Hansen²; ¹University of North Carolina at Chapel Hill, Greensboro, NC; ²Moses Cone Memorial Hospital, Greensboro, NC

PURPOSE: A recent analysis of secondary data revealed marked differences in lung surgery rates between African-Americans and whites who suffered from early stage, non-small cell, lung cancer. The disparity in treatment according to race led to an 8% reduction in 5 year survival for African-American patients. The authors were unable to sort out whether patient or physician factors explained the difference in surgical decision making. Because of this uncertainty, we decided to

measure patient perceptions of progressive lung cancer and evaluate whether more optimistic perceptions of this disease could reasonably explain differences in surgical management.

METHODS: We recruited a stratified, random sample of 160 individuals from church groups, health fairs, and medical clinics in urban areas of North Carolina and Florida. A trained research assistant administered an oral questionnaire to obtain demographic information and assess patient preferences using a standard gamble approach. Standard gamble results were converted to health utility scores (HUS) for which 1.0 = normal health and 0 = death. HUS for progressive lung cancer were analyzed according to race. We then constructed a decision model that evaluated lung cancer surgery vs. no surgery for stages I and II non-small cell cancer. The model accounted for racial differences in HUS for progressive disease. The model was intentionally biased against surgery by overestimating surgical risk and survival with progressive lung cancer. Sensitivity analysis was performed.

RESULTS: The average age of the 160 patients interviewed was 59 yrs. Sixty-three percent were women, 42% were educated beyond high school, 41% had incomes < \$20,000, 15% had no health insurance, and 41% were African-American. The mean HUS for progressive lung cancer among African-Americans surveyed was 0.32 compared to 0.21 among whites. The decision model when applied to a 60 yr-old African-American male demonstrated a quality adjusted survival of 3.38 life years (QALY's) for the surgical therapy group compared to 0.48 QALY's for the "no surgery" group. In the sensitivity analysis, if patient age was adjusted to 80 yrs., surgery still maintained a large advantage (1.58 QALY's for surgical treatment vs. 0.48 QALY's for "no surgery"). Even if surgical cure rate was reduced to 10% in the model, the advantage in QALY's remained for surgical patients (1.2 surgery vs. 0.48 "no surgery").

CONCLUSION: Although African-Americans hold a much more optimistic view of progressive lung cancer than whites, the difference does not explain discrepancies in lung surgery decisions even when decision models are significantly biased against surgical intervention. Other issues, such as physicians' misinterpretation of patients' surgical risk and patients' mistrust or misunderstanding of physicians, need to be assessed in order to resolve discrepancies in lung cancer treatment and survival.

INDIVIDUALIZED TREATMENT FOR ALCOHOL WITHDRAWAL. J. Daepfen¹, P. Gache¹, U. Landry¹, E. Sekera¹, V. Schweizer¹, S. Gloor¹, B. Yersin¹, ¹Alcohol Treatment Center, Lausanne

PURPOSE: To assess the effect of an individualized treatment regimen on the quantity and duration of benzodiazepine prescribed for alcohol withdrawal.

METHODS: A randomized double-blind controlled trial including 117 consecutive patients admitted in an alcohol treatment program was conducted. Patients were randomized into two groups: i) 56 patients were treated with oxazepam only in response to the development of signs of alcohol withdrawal (symptom-triggered), and ii) 61 were treated with oxazepam every 6 hours with additional doses as needed (fixed-schedule). The administration of additional oxazepam was determined using the CIWA-Ar, a validated measure of the severity of alcohol withdrawal. **RESULTS:** 40% of the patients in the symptom-triggered group were treated with oxazepam, compared to 100% in the fixed-schedule group ($p < .001$). Mean oxazepam administered in the symptom-triggered group was 40 mg compared to 230 mg in the fixed-schedule group ($p < .001$). Mean duration of oxazepam treatment was 20 hours in the symptom-triggered group compared to 63 hours in the fixed-schedule group ($p < .001$). There were no differences in the incidence of complications and in the measures of comfort between the two groups.

CONCLUSION: A symptom-triggered pharmacological treatment decreases the quantity and duration of benzodiazepine prescribed in alcohol withdrawal, with similar safety and comfort.

MEASUREMENT AND VALIDATION OF SELF-REPORTED SYMPTOMS AMONG PEOPLE WITH HIV. N. Dave¹, A.W. Wu², M. Diener-West², S. Sorensen³, D. Revicki³, ¹University of Pittsburgh Medical Center, Pittsburgh, Pennsylvania; ²Johns Hopkins University, Baltimore, MD; ³MEDTAP International

PURPOSE: Symptoms are an important component of the health status of people with HIV/AIDS. Since both HIV/AIDS and its treatments commonly cause adverse somatic effects, symptom assessment is necessary for delivering and evaluating care for people with HIV. This study compared three methods to assess patient reported symptoms (presence, frequency, and bother) and to validate report of individual symptoms using longer patient-reported scales and clinical parameters in people with HIV disease.

METHODS: Prospective cohort study. A questionnaire completed by 160 individuals on two occasions, four months apart. The questionnaire included 39 HIV-related symptoms, the Medical Outcomes Study-HIV Health Survey (MOS-HIV), and Health Transition items. Correlation coefficients were calculated to assess relationships between summary scores for the three symptom methods with health-related quality of life domains, CD4 counts, and HIV disease stage. Analyses were stratified by HIV disease stage to determine if correlations differed with respect to clinical stage.

RESULTS: Sixty-seven percent of subjects were male, 63.5% were African-American, and 26% had less than a high school education. The percentage of participants with AIDS, symptomatic HIV, and asymptomatic HIV were 46.2%, 28.5%, and 25.3%, respectively with mean CD4 counts of 194, 386, and 400, respectively. The average number of symptoms reported was 15.2 (SD=8.4). There were statistically significant correlations between CD4 count and symptom frequency, symptom bother, and symptom presence. As hypothesized, correlations with the MOS-HIV Quality of Life domain were moderate and statistically significant whereas there were no significant correlations with the Quality of Life Transition score. Examination of individual symptoms revealed that fatigue and decreased memory were significantly correlated with both the Health Transition and MOS-HIV scores. Presence, frequency, and bother scores for trouble falling asleep were correlated with Sleep scale scores. Presence and frequency of fever correlated significantly with CD4 count. Bodily pain had statistically significant correlations with the Pain domain of the MOS-HIV. Hot/cold spells were not significantly correlated with CD4 count.

CONCLUSION: The three symptom methods and their scores were strongly intercorrelated. Participants with more symptoms reported greater frequency and bother from symptoms.

Several individual symptoms correlated strongly and statistically significantly with longer measures of the same concept, like the MOS-HIV Health Survey. Overall, no one method is clearly superior; different individual symptoms were better measured using different methods. Studies are needed to develop a measure of symptoms commonly experienced by HIV-infected people.

DO AFRICAN-AMERICANS OVERESTIMATE THEIR RISK OF BREAST CANCER MORE THAN CAUCASIAN WOMEN? S. Davids¹, M. Schapira¹, L. Eparvier¹, A.B. Nattinger¹, ¹Medical College of Wisconsin, Milwaukee, WI

PURPOSE: Women have been found to overestimate breast cancer risk. Perceptions are more accurate in higher educated and more numerate women. The objectives of this study are to evaluate the association between African-American (AA) race and breast cancer risk perceptions. **METHODS:** We assessed 5-year and Lifetime Risk Perceptions of AA and Caucasian women recruited from a primary care clinic. Actual risk was calculated using the Gail Model. Error in estimation (EE) was defined as perceived risk minus calculated risk. The following categories were defined. Under-estimators had an error of < -10%, Accurate Estimators had an error of > -10% but < 10%, Moderate Over-estimators had an error of > 10% and < 30%, High Over-estimators had an error of > 30% and < 50%, Extreme Over-estimators had an error of > 50%. Nonparametric statistics were used to compare EE between groups.

RESULTS: There were 254 subjects. Sixty-eight percent (68%) were Caucasian, 30% AA, and 2% other. AA were younger (mean of 55 yrs vs. 58 years, $p=0.03$), of lower income (84% vs. 35% of Caucasians < \$20,000/year, $p<0.0001$), had lower Rapid Estimate of Adult Literacy in Medicine (REALM) scores (mean 53 vs. 65, $p<0.0001$), and less likely to answer a set of three numeracy questions correctly (11% vs. 55%, $p<0.0001$), compared to Caucasians. The calculated mean lifetime and five-year risk of AA women was lower than Caucasian women (lifetime risk: 5.5% vs. 9.7%, $p<0.0001$, five-year risk: 0.89% vs. 1.8%, $p<0.0001$). In univariate analyses, race, age, and reading level were not associated with accurate risk perception. A trend of increased accuracy of lifetime and five-year risk with higher numeracy was found ($p=0.08$). The table shows error in estimation stratified by race (* p -values > 0.05 using a chi-square test).

CONCLUSION: In a primary care population, AA women had similar levels of accurate breast cancer risk estimation as Caucasian women, despite lower education and numeracy levels. Both populations could benefit from education to improve the accuracy of risk estimation.

Estimation Error

	Lifetime Risk: AA*	Lifetime Risk: Caucasian*	5-Yr Risk: AA*	5 Yr Risk: Caucasian*
Under-estimators	1.3%	1.2%	0%	0%
Accurate Estimators	22.7%	26.6%	45.3%	42.2%
Moderate Over-estimators	24%	29.5%	26.7%	24.3%
High Over-estimators	28%	26.6%	14.7%	22.6%
Extreme Over-estimators	24%	16.2%	13.3%	11%

DECIDING ON ANTICOAGULATING THE OLDEST OLD WITH ATRIAL FIBRILLATION. N.A. Desbiens¹, ¹University of Tennessee College of Medicine, Chattanooga Unit, Chattanooga, TN

PURPOSE: Deciding on anticoagulating the oldest-old (OO) with atrial fibrillation (AF) with warfarin is an increasingly common problem for general internists. Age is an important risk factor not only for AF, but also for stroke in patients with AF. In addition, other risk factors for stroke in AF, such as hypertension (HBP) and diabetes (DM), also increase with age. Though anticoagulation (AC) has been shown to be effective in preventing strokes in some patients with AF, these studies cover only short periods and include few OO. Also, contraindications and side effects of AC are greater in the OO, the fastest growing segment of the population.

METHODS: A cost-effectiveness analysis was developed using 3 Markov states: Life without stroke, life with stroke residuals and death. Quality-adjusted life expectancy (QALE) was calculated for each state. Baseline probabilities and rates were obtained from systematic reviews and age-adjusted. Age-specific death rates were calculated from the 1997 US Life Tables. Rates of stroke increased with age and with number of risk factors. Hemorrhagic strokes (HS) were explicitly modeled to study tradeoffs between decreasing ischemic strokes and increasing HS with AC. Costs of hospital care for each of type of stroke and other major bleeding were obtained from the Healthcare Cost and Utilization Project 1997 Nationwide Inpatient Sample. Other costs were estimated from published data. Medicaid national average nursing home costs were used. Costs and effects were discounted and a societal perspective was used.

RESULTS: In patients with AF and no other risk factors, AC leads to worse outcomes at any age. In patients with AF and previous histories of DM, HBP and stroke or transient ischemic attacks (stroke/TIA), AC leads to greater QALE and decreases costs, but the effect is seen primarily in patients younger than 85. Both the absolute QALE gained from AC and the percent increase in remaining QALE decrease with age—the latter from a 23% increase at 65 to 4% at 100 years of age. The outcomes for patients with one or two risk factors are intermediary to those seen for patients with 0 and 3 risk factors, though the risks of AC outweigh the benefits in the OO. Sensitivity analyses indicate that the risk of HS and death from HS stroke with age are influential variables in affecting QALE.

CONCLUSION: Though a close call, in the OO at the highest risks of stroke (HBP, DM, previous stroke/TIA) the benefits of AC outweigh the risks. This is not so for those at less risk, but the results of the decision analysis are based on extrapolation of data on the risks and outcomes in the OO. More direct evidence and better markers of risk are needed in the OO with AF before a categorical recommendation about AC can be made.

MODERATE ALCOHOL USE IN ELDERLY WOMEN IS ASSOCIATED WITH LESS DEPRESSION AND A STRONGER SOCIAL NETWORK. S. Dellefesen¹, J. Rizzo², H. Nelson³, ¹Portland VA Medical Center, Portland, OR; ²Kaiser Permanente Center for Health Research, Portland, OR; ³Oregon Health Sciences University, Portland, OR

PURPOSE: Clinicians often associate alcohol use with negative personal factors, such as depression and social isolation. However, there are limited data describing these potential associations, especially for elderly women.

METHODS: To determine whether elderly women's alcohol use is related to depression and social withdrawal, we recently evaluated Visit 6 (1997–1998) data collected from 1,585 women (mean age 81.5 yrs) in Portland, OR who were enrolled in the Study of Osteoporotic Fractures. Based on self-report, women were categorized into three subgroups — current, past, and never drinkers. A drinker was defined as someone who drank moderately (< 2 drinks/day). Depression was evaluated by using the Geriatric Depression Scale (low score indicating less depression) and Social Networks were defined by using the Lubben Social Network Scale (higher score indicating worse social network).

RESULTS: When compared to past drinkers, current drinkers are less likely to carry a previous diagnosis of depression, score low on the depression scale, and take antidepressants. Current users are more similar to never users for these measures.

Depression Measure	Drinkers		
	Current (n=554)	Past (n=701) n(%)	Never (n=330)
Previous depression diagnosis*	33(27)	54(44)	37(30)
Diagnostic depression score*	45(24)	102(54)	45(22)
Antidepressant medication use*	6(14)	26(60)	11(26)
Social Network Measure	Current	Past	Never
Low social Network score*	122(31)	188(48)	81(21)

*P < 0.018 (Chi-Squared)

CONCLUSION: Women who currently drink a moderate amount of alcohol are less likely to be depressed and more likely to have a stronger social network than women who are past or never users. Questions regarding possible confounders and why past users quit using alcohol may be answered after analyzing a recent questionnaire describing this cohort's pattern of alcohol use.

WOMEN'S CLINICAL PREVENTIVE SERVICES IN THE SAFETY NET: WHO IS MISSING OUT? A. Diamant¹, S. Asch¹, W. Ford¹, L. Gelberg¹, ¹UCLA - Department of Medicine, Los Angeles, CA

PURPOSE: Public health departments have traditionally provided episodic safety net services to the uninsured. Increasingly, these providers have taken up the role of providing primary care clinical preventive services as well, particularly to racial/ethnic minority populations. To assess the association of race/ethnicity and other patient and system characteristics with receipt of preventive health care services by low-income women, we studied patients cared for through the Los Angeles-County-Department of Health Services(LAC-DHS) Primary Care Network.

METHODS: We sampled patients receiving primary care services at 50 facilities to represent all patients in care throughout the LAC-DHS Primary Care Network. 1,288 adult females were interviewed for a response rate of 80%. The sample was racially and ethnically diverse-63% Latina/Hispanic, 22% white, 13% African American, 1% Asian/Pacific Islander and 1% Mixed race/Other. We measured receipt of the following tests and services: Pap smear within 3 years, mammogram within 2 years among women age 50 years and older, flu shot among women at risk, and cardiovascular disease testing and counseling among women at risk. We used bivariate and multivariate logistic regression analyses to assess the association of patient characteristics (age, race/ethnicity, education, income, immigration status, health insurance status, health status) and system characteristics (type of facility-comprehensive health center, personal health center, hospital outpatient clinic and public/private partnership clinic) with receipt of preventive health care services by low-income women within a publicly funded health care system.

RESULTS: Eighty-eight percent of women had undergone cervical cancer screening within 3 years and 82% of women age 50 years and older had undergone breast cancer screening within 2 years, with the highest rates of screening among Latina patients. Among women at increased risk for morbidity and mortality associated with influenza 41 % had received a flu shot within one year. For women at increased risk for cardiovascular disease 91 % had undergone cholesterol screening within 3 years, 82% had been counseled about the importance of exercise, 86% about nutrition and healthy eating and 46% about tobacco use. In multivariate regression analyses Latinas remained the most likely to have undergone cervical and breast cancer screening and among women at risk for influenza African Americans were the most likely to have received a flu shot. **CONCLUSION:** For low-income women who are able to gain access to a large urban publicly funded health care system for their primary care needs their level of receipt of preventive health care is comparable or better than that of women in other healthcare settings, although shortcomings remain.

THE EFFECT OF INSURANCE STATUS AND ACCESS TO CARE ON PUBLIC HEALTH PATIENTS' USE OF EMERGENCY DEPARTMENT SERVICES. A. Diamant¹, R. Hays¹, E. Fielder¹, W. Ford¹, L. Gelberg¹, ¹UCLA-Department of Medicine, Los Angeles, CA

PURPOSE: Studies have demonstrated that people seek emergency department services for a variety of reasons, including not being able to otherwise access needed health care. To assess the use of emergency department services(emergent and non-emergent) by low-income patients with a regular source of care we interviewed a sample of patients who received their medical care through a large public health system.

METHODS: We performed a stratified cross sectional study utilizing probability sampling techniques and survey methods. We sampled patients receiving primary care services through the Los Angeles County Department of Health Services Primary Care Network. 1819 adults were successfully interviewed for a response rate of 80%. The sample was racially, and ethnically diverse. We used bivariate and multivariate logistic regression analyses to assess the association of patient characteristics including insurance status and use of primary care services with visits (1 or more) to the emergency department within the preceding 12 months.

RESULTS: Twenty-eight percent of the adults in this sample had made one or more visits to an Emergency Department for medical treatment within the preceding 12 months. Almost half of those who had used emergency department services reported that they felt their medical problem was an emergency (49%), but a substantial number of patients sought emergency care for financial reasons(10%) and ease of accessibility to needed medical care (12%). Multivariate analyses revealed that patient characteristics associated with a greater likelihood of having used emergency department services included being white, poor health status, having Medicaid rather than being uninsured, and having delayed seeking needed medical care.

CONCLUSION: Patients within a publicly funded medical system have high rates of emergency department use. Type of health insurance and whether people had delayed receiving needed medical care during the preceding year were associated with use of emergency department services. Thus, access and barriers to needed medical care are related to use of emergency department services for emergent and non-emergent medical care.

TREATMENT OF RHINOSINUSITIS WITH OR WITHOUT INTRANASAL STEROIDS: THE CAFFS TRIAL. R.J. Dolor¹, D.L. Witsell¹, A.S. Hellkamp¹, D.L. Simel¹, ¹Duke University, Durham, NC

PURPOSE: Conflicting study results regarding the role of intranasal steroids in patients with rhinosinusitis create a dilemma for clinicians. Using a double-blind, randomized, placebo-controlled methodology, we examined whether the addition of an intranasal steroid to conventional antibiotic therapy speeds the recovery of patients with recurrent acute rhinosinusitis.

METHODS: Ninety-five (95) patients, ages 18 or older, presenting with acute rhinosinusitis and a previous history of recurrent sinusitis or chronic rhinitis were enrolled from 22 primary care and otolaryngology practices nationwide. Either plain film sinus radiograph or nasal endoscopy supported the diagnosis of sinusitis for all patients. Patients with a history of chronic bacterial sinusitis, previous sinus surgery, and recent intranasal steroid or antibiotic use were excluded. Subjects were randomized to receive either 2 puffs of fluticasone propionate (n = 47) or saline nasal spray (n = 48) in each nostril once daily for 21 days. All patients received 2 puffs of oxymetazoline hydrochloride in each nostril twice daily for 3 days initially and cefuroxime axetil 250 mg twice daily for 10 days. Patients recorded their daily symptom status in a diary during the 3-week treatment phase. Telephone follow-up was conducted at 10, 21, and 56 days after enrollment; 88 patients (92.6%) completed all telephone follow-up. The primary endpoint was time to clinical success over the 8-week follow-up period (defined a priori as cured or much improved).

RESULTS: Patients using the regimen which included fluticasone propionate achieved higher rates of clinical success, 93% versus 74% in the placebo group (p = 0.009 by Chi-square test). The relative benefit increase was 26% with the inclusion of fluticasone to the regimen and the absolute benefit increase was 19%; the number needed to treat to achieve one additional treatment success was 6. Fluticasone-treated patients improved significantly more rapidly (median days to clinical success was 9.5 and 6.0 days, respectively; p = 0.012 by log rank test). Other covariates, including specialty vs. primary care study site, age, number of comorbidities, gender, race, recent upper respiratory infection, or history of allergy, were tested using a Cox proportional hazards model, but only the inclusion of fluticasone to antibiotic and decongestant therapy significantly influenced the clinical outcomes.

CONCLUSION: The addition of a 3-week course of fluticasone propionate to oxymetazoline (3 days) and antimicrobial therapy with cefuroxime (10 days) improves clinical success rates and accelerates recovery of patients with a history of either chronic rhinitis or recurrent sinusitis who present for treatment of acute rhinosinusitis.

COLONIZATION AND CATHETER TYPE IN ALLOGENIC AND AUTOLOGOUS BONE MARROW TRANSPLANT RECIPIENTS. L. Dubin¹, F. Momin¹, ¹Oakwood Healthcare System, Dearborn, MI

PURPOSE: Since bone marrow transplantation has become an accepted procedure in cancer therapy, more patients have been treated with this technique. The occurrence of infection in these immunocompromised patients can become a life-threatening event, and has led to closer examination of all aspects of the intravascular access devices for chemotherapy and bone marrow delivery.

METHODS: 81 patients from the Oakwood Hospital Bone Marrow Transplant (BMT) Program were assessed. These patients have undergone either autologous, allogenic or autologous stem cell (ASCT) transplants during the last 2.5 years. The type of the intravascular access device, the type of transplant, the presence of graft vs. host disease (GVHD), and neutropenia were compared.

RESULTS: The overall incidence of infectious complications was 26/81 cases. Out of 46 patients who underwent autologous BMT, 14 had an infection (11 out of 36 Neostar catheters and 3 out of 10 Hickman). In the 21 patients who received allogenic BMT, 8 had an infection (1 out of 4 Neostar catheters and 7 of 17 Hickman). There was not a single episode of Gram negative infection. In the ASCT group, 4 of 14 patients were infected (all with the Neostar type catheter). Five patients died; a contributing factor in two was infection in the presence of a Hickman catheter. Fifty-six patients had an optimal and 17 a suboptimal duration of their intravascular access device. The rest continue to undergo treatment.

CONCLUSION: In our study, for patients undergoing BMT (allogenic, autologous or ASCT), the type of intravascular access device (Neostar or Hickman) appeared to make no difference in the rate of infectious complications. The presence of neutropenia, GVHD, and the duration of successful vascular access are predisposing factors affecting the incidence of infection and catheter colonization.

RESPECTFUL SOLUTIONS TO UNMET NEED FOR DEPRESSION TREATMENT IN PRIMARY CARE: A NOVEL APPLICATION OF MARKET RESEARCH TECHNIQUES. M. Dwight-Johnson¹, I.T. Lagomasino², A. Fink³, J. Unutzer⁴, N. Duan⁵; ¹University of Southern California, Los Angeles, CA; ²Charles R. Drew University, Los Angeles, CA; ³UCLA School of Public Health, Los Angeles, CA; ⁴UCLA Neuropsychiatric Institute, Los Angeles, CA; ⁵UCLA Center for Community Health, Los Angeles, CA

PURPOSE: Disadvantaged populations are often the last to benefit from innovation, including quality improvement (QI) programs for depression. Public sector clinics may not have the resources to implement and maintain QI programs and QI programs designed for private patients may not address needs, treatment preferences, and resources available to depressed, indigent patients. Careful needs assessment can help formulate objectives of QI for such populations, but assessment may be limited by the fact that such populations have little prior exposure to appropriate treatments. We present the development of a unique approach that demonstrates respect for indigent patients and their providers by employing market research strategies to rigorously assess their preferences for and barriers to depression care, as the first step to developing a tailored QI program.

METHODS: We conducted literature reviews of existing primary care depression interventions and reviewed methods for understanding patient treatment preferences, including the use of market research methods for assessing consumer preferences. Key informant interviews were conducted with 4 principal investigators of depression treatment interventions.

RESULTS: The reviews indicated that few existing QI studies for depression explicitly incorporate assessment of and respect for patient treatment preferences in intervention design. Further, valid methods for assessing patient and provider preferences for and barriers to depression care have not been systematically applied to treatment interventions for the poor. Market research techniques including conjoint analysis surveys have been validated and used to design appealing and acceptable consumer products and services. Because market research methods, such as conjoint analysis, have not been commonly used in developing health services interventions, especially in populations with little treatment experience, we have developed an iterative process for assessing and validating patient and provider preferences and barriers to care. The method includes 4 phases: 1.) patient and provider focus groups, 2.) a market research survey (using conjoint analysis) 3.) a pilot depression intervention in which patient preferences, barriers to care, and opportunities for treatment are identified and results from step 2 are validated, and 4.) indepth qualitative interviews with patients and providers. For all phases of the design depressed patients from the general internal medicine clinic are identified using waiting room screening.

Focus groups were held with depressed patients to inform the design of the conjoint analysis survey. Of 86 patients in the general internal medicine clinic screened for focus groups, 26 (36%) had current depressive disorders, 23 patients were eligible to participate in focus groups, and 15 (65% of those eligible) did so. The mean age of focus group participants was 52 (SD=8.9); 14 of 15 were female; 73% had less than 6 years of education; and all were Spanish speaking. Only one participant reported prior treatment for depression. Patients preferred counseling as the first line treatment for depression, but would consider antidepressants if recommended by a caregiver. A majority preferred group over individual counseling. Patients preferred to receive treatment at the primary care clinic, but from a mental health specialist rather than a primary care provider. Patients reported that their lack of knowledge regarding depression and its treatment and their fear of treatment costs were the main reasons that they had not sought care.

In the second phase, 175 depressed patients are recruited to complete a conjoint analysis survey. In this technique, key attributes of a proposed product and possible variations (called "levels") of the attributes are defined. Hypothetical products with variations of the defined attributes are rated or ranked by patients according to their preferences. Conjoint analysis then determines the relative importance consumers attach to each of the attributes and the utilities they attach to each variation or level of the attributes. We assess the validity of conjoint analysis for determining preferences for and barriers to depression care by comparing survey results with actual treatment choices and barriers to care following a limited depression intervention. Finally, indepth qualitative interviews developed with a medical anthropologist are used to complement quantitative data gathered during the pilot intervention.

CONCLUSION: Previous QI programs for depression in primary care have generally not included systematic assessment of patient and provider preferences for and barriers to care in their design. Rather than making assumptions about the needs of providers and consumers in disadvantaged settings, researchers should rigorously assess the preferences of those most affected by a planned QI intervention. A rational and respectful approach for assessing patient and provider preferences using market research methods is proposed for developing and adapting health care QI interventions for disadvantaged populations.

USING ANTI-XA ACTIVITY TO MONITOR ENOXAPARIN IN ELDERLY PATIENTS AND THOSE WITH DECREASED RENAL FUNCTION. E.J. Dzielak¹, R. Getts¹, N. Chiappetta²; ¹Moses Taylor Hospital, Scranton, PA; ²Scranton Temple Residency Program, Scranton, PA

PURPOSE: To use anti-Xa activity to monitor enoxaparin dosing in the following patient types: greater than 65 years old, with a creatinine clearance less than 30ml/min and in morbid obesity. To develop a weight adjusted dosing method for patients over 65 years old and for patients with a creatinine clearance less than 30ml/min.

METHODS: All patients were dosed by their physician using 1mg/kg sub-cutaneous every 12 hours. Peak anti-Xa activity was obtained 3-5 hours after the third dose, trough activity was obtained just prior to the dose. Anti-Xa activity was measured using Dioagnostica Strago instrumentation. aPTT was obtained simultaneously.

RESULTS: Thirty-eight patients were involved in the evaluation. Thirty-six peak levels and 17 trough levels were obtained. Thirteen patients greater than 65 years old were able to be evaluated. All values are mean: age 77, Scr 1.29mg/dl, CrCl 42ml/min, peak anti-Xa 1.3IU/ml. aPTT was prolonged in the vast majority of patients who had anti-Xa activity > 1.0IU/ml. One patient experienced a major hemorrhage.

CONCLUSION: Doses of 1mg/kg of enoxaparin in elderly patients, especially females, seems to produce higher than expected peak anti-Xa activity. This can result from a decrease in renal

function and an altered volume of distribution. As a follow-up to this evaluation we developed a weight-adjusted dosing method for enoxaparin in our elderly patients. We realize anti-Xa activity is only one of several pharmacologic properties of enoxaparin.

Patients > 65 years old

	Mean
Age	77
Scr (mg/dl)	1.28
CrCl (ml/min)	42
Actual Body Wt. (kg)	84
Adjusted Body Wt. (kg)	67
Dose (mg)	82
Peak anti-Xa activity	1.3IU/ml
Female/Male	10/3

DOES PHYSICIAN RECOMMENDATION OF EXERCISE CAUSE CHANGE IN PHYSICAL ACTIVITY? D. Edelman¹, M.K. Olsen¹, T.K. Dudley¹, A.C. Harris¹, E.Z. Oddone²; ¹Durham VA Medical Center, Durham, NC; ²Duke University, Durham, NC

PURPOSE: Physical activity protects against and treats many chronic medical illnesses and is often prescribed by physicians. Our purpose was to determine the predictors of change in physical activity and in particular to determine if physician recommendation to exercise led to a change in physical activity.

METHODS: Setting: A tertiary care Veterans Affairs Medical Center. Patients: 1253 outpatients age 45-65 enrolling in a prospective diabetes screening study. Measures: Demographics, height, weight, and comorbidity (Kaplan-Feinstein scale) were collected at a baseline patient interview. At 1 year we asked patients if their physicians had recommended exercise at any time in the past year. Quality of Life was measured by Medical Outcomes Study Short Form 36 (SF-36). Physical activity was measured at baseline and a year later by the Framingham Physical Activity Index (FPAI), a well-validated instrument which collects data for both leisure-time and overall (including job-related) physical activity. Analysis: Because FPAI scores and change scores were not normally distributed and linear regression models were heavily influenced by outliers, change in FPAI from baseline was dichotomized (as > 0 or <= 0) and logistic regression was performed to identify independent predictors of change in FPAI. Baseline FPAI score was added to the model to account for regression to the mean, which was significant in our sample.

RESULTS: Patients' mean age was 54.6 and 6% were female. Mean SF-36 Physical Functioning (PF) was 55.2. 45% of patients increased their overall FPAI, and 28% of patients increased their leisure-time FPAI. In a multivariable model including age, race, gender, percent of ideal body weight, comorbidity, and baseline FPAI, the only independent predictor of change in overall FPAI was PF (for each 10 points of PF, OR=1.07, 95% CI 1.02, 1.12). No demographic or disease factors tested predicted a change in overall FPAI. Physician recommendation for exercise during the year of follow-up was also not associated with change in overall FPAI over that year (OR=1.1, 95% CI 0.82, 1.5). The predictors of change in leisure-time physical activity were PF (for each 10 points of PF, OR=1.15, 95% CI 1.09, 1.21) and female gender (OR 1.8, 95% CI 1.03, 3.2). Again, physician recommendation for exercise was not associated with change in leisure-time physical activity (OR=1.2, 95% CI 0.89, 1.6).

CONCLUSION: Predictors of change in overall and leisure-time physical activity include baseline physical functioning (for both outcomes) and female gender (for leisure-time activity). Physician recommendation for exercise leads to changes neither in overall physical activity nor in leisure-time physical activity.

DISPARITIES IN INFLUENZA AND PNEUMOCOCCAL VACCINE COVERAGE AMONG ADULTS WITH DIABETES, 1998. L.E. Egede¹, D. Zheng¹; ¹Medical University of South Carolina, Charleston, SC

PURPOSE: Persons with diabetes mellitus (DM) are at increased risk for complications and deaths from influenza and pneumonia. The influenza vaccine (IV) and the pneumococcal vaccine (PV) are safe and effective preventive measures. Data from the National Health Interview Survey (NHIS) was analyzed to identify disparities in IV and PV coverage among adults with diabetes in 1998.

METHODS: NHIS is a national household survey of non-military and non-institutionalized adults in the U.S. We analyzed data on 2247 adults with DM excluding a diagnosis during pregnancy. IV and PV coverage was defined as percent of adults that reported receiving the IV in the preceding 12 months and PV in the preceding 5 years. Statistical analyses and population estimates were done with SUDAAN to account for the complex sampling design of the NHIS. **RESULTS:** 49% reported receiving IV and 31% reported receiving PV in 1998. Among IV recipients, adults > 65 years were more likely to receive IV than adults < 65 years (66% Vs 38%, p<0.0001). Whites were more likely to receive IV than Blacks (53% Vs 36%, p<0.0001) but Whites Vs other minorities (53% Vs 43%, p=0.06) and Blacks Vs other minorities (36% Vs 43%, p=0.24) were not different. Primary care visit (PCV) in the past year increased IV coverage (53% Vs 31%, p<0.0001). Multiple logistic regression was used to adjust for age, income, gender, marital status, census region, education, and primary care visit. Whites were 1.6 times more likely to get IV than non-whites (OR=1.63, CI 1.26-2.11). Adults > 65 years were 3 times more likely to get IV than adults < 65 years (OR=3.00, CI 2.42-3.74) and adults with PCV were 2.4 times more likely to get IV than those without visits (OR=2.42, CI 1.84-3.18). Similar results were found among PV recipients. Adults > 65 Vs < 65 (49% Vs 19%, p<0.0001). Whites Vs Blacks (34% Vs 20%, p<0.0001), whites Vs others (34% Vs 26%, p=0.11), and blacks Vs others (20% Vs 26%, p=0.18). Adults with PCV Vs no PCV (33% Vs 22%, p=0.0003). After multiple logistic regression, Whites (OR=1.65, CI 1.26-2.15), Adults > 65 years (OR=4.26, CI 3.34-5.44) and adults with PCV (OR=1.68, CI 1.20-2.36) were more likely to receive PV.

CONCLUSION: National coverage for IV and PV among adults with DM is less than optimal. IV and PV coverage is highest among adults >65 years, whites, and persons with a primary care visit within the past year. Blacks, other minorities, and adults <65 years are at high risk for not receiving both vaccines.

THE ASSOCIATION BETWEEN SMOKING AND DRINKING IN A PRIMARY CARE POPULATION. D. Einstadter¹, T. Parran¹, R. McCormick¹, N. Dawson¹; ¹Center for Health Care Research and Policy at MetroHealth Medical Center, Case Western Reserve University, and the VA Medical Center, Cleveland, OH

PURPOSE: The association between alcohol and tobacco use among heavy drinkers is well known. Whether this association is present across the spectrum of alcohol use and whether recent societal changes in substance use have affected this relationship is not clear. We examined the relationship between alcohol and tobacco use in an urban primary care population.

METHODS: We utilized a computerized interactive voice response system during 1199-3100 to administer a 44 question general health survey to patients enrolled in the General Medicine Clinic at the MetroHealth Medical Center in Cleveland, OH, a large urban teaching hospital. Survey questions included: demographics, work status, educational attainment, health ratings, social support, and current alcohol and tobacco use. Alcohol use was assessed by AUDIT and CAGE questionnaires and by reported drinks per day. Alcohol use behavior was classified by a combination of AUDIT and CAGE scores: Lifetime Abstainers(LA), Low Risk Social (LRS), Recovering Problem (RP), or Current Hazardous/Harmful (HH) drinkers. Smokers were defined by a 'yes' response to: 'Do you currently smoke?' The proportion of patients smoking was examined across the spectrum of AUDIT and CAGE scores, across drinks per day categories, and alcohol use behavior categories. The association between smoking and alcohol use was assessed using chi-square and logistic regression.

RESULTS: 4761 patients completed the survey. The average age was 51 years, 71 % were women, 62% white, 23% uninsured; 50% reported current alcohol use and 34% were smokers. Overall, drinkers were more likely to smoke than non-drinkers (40% vs. 27.5%, $p < 0.001$). The proportion of smokers increased significantly with increasing AUDIT (27% for AUDIT=0 vs. 61 % for AUDIT >=8), CAGE (28% for CAGE=0 vs. 65% for CAGE=4), and drinks per day (36% for <1 drink per day to 70% for 7+ drinks per day). After adjustment for age, race, gender, insurance type, education, and marital status, LA's were significantly less likely to smoke compared to LRS drinkers (RR 0.5, 95% CI 0.5 - 0.7). In contrast, both RP (RR 1.37, 95% CI 1.2 - 1.5) and HH drinkers (RR 1.76, 95% CI 1.6 - 1.9) were significantly more likely to smoke.

CONCLUSION: Smoking and drinking are closely associated across the entire spectrum of drinking behaviors. Increased smoking rates in RP drinkers suggests that these individuals have an especially difficult time quitting, and may require extra effort to achieve successful smoking cessation.

THE USE OF COMPUTERIZED REMINDERS TO IMPROVE ROUTINE TESTING AND SCREENING RECEIPT AMONG PATIENTS WITH DIABETES. J. Elston Lafata¹; ¹Henry Ford Health System, Detroit, MI

PURPOSE: Although mailed reminders have been widely used for prevention among the general population, few studies have evaluated their effectiveness among the chronically ill. With increased computer technology, many physician practices are able to track routine service receipt and implement automated reminder systems. Using existing computerized technology within a multi-specialty group practice, we implemented and evaluated the effectiveness of mailed reminders for improving routine monitoring and screening among patients with diabetes.

METHODS: Automated visit and prescription claims data were used to identify adult HMO patients with diabetes aligned to one of 195 primary care physicians (PCPs). 107 PCPs agreed to have their patients with diabetes randomized to receive the reminder (N=1633) or usual care (N=1659). The reminder included a letter from the individual's PCP with tailored testing recommendations, a self-care handbook, and a preventive care checklist. Using data from automated clinical and administrative databases, logistic regression models were fit to evaluate the effect of reminder receipt on fasting lipid profile and glycated hemoglobin testing, dilated retinal exam receipt, and visit frequency during 6 and 12 months following randomization. To control for non-independence among patients aligned to the same PCP, generalized estimating equation approaches were used.

RESULTS: Reminder and usual care recipients did not differ in sociodemographic, clinical or prior testing characteristics. In the 6 months following randomization, reminder recipients were more likely to receive a retinal exam (OR=1.29, $p < 0.01$), and a visit (OR=1.28, $p < 0.00$), and tended to be more likely to receive a glycated hemoglobin test (OR=1.12, $p=0.09$). In the 12 months following randomization, reminder recipients were more likely to receive a glycated hemoglobin test (OR=1.21, $p=0.02$), a retinal exam (OR=1.24, $p < 0.01$), and a visit (OR=1.25, $p=0.03$), and tended to be more likely to receive a lipid test (OR=1.14, $p=0.08$).

CONCLUSION: We found significant improvements in the use of routine monitoring and screening among patients with diabetes receiving a computerized mailed reminder compared to usual care.

FREQUENCY IN PHYSICIAN COMMUNICATIONS WITH CANCER PATIENTS NEAR END OF LIFE. C.P. Escalante¹, C.G. Martin¹, U. Alexis¹, J. Geraci¹; ¹University of Texas M. D. Anderson Cancer Center, Houston, TX

PURPOSE: Dyspnea, often a terminal symptom in patients with advanced cancer (ca), is the fourth most common complaint of patients (pts) presenting to the Emergency Center (EC) at the UT MD Anderson Cancer Center. Our previous work determined progressive disease, poor performance status, abnormal triage pulse and respiration and lung ca were predictive of death within 30 days. We report our findings of frequency of physician end of life (EOL) communications with 143 ca pts presenting to our EC with dyspnea who died within 6 months.

METHODS: Demographic and clinical information was abstracted from the medical record. Univariate analyses of clinical factors that relate to whether pts received SCC and how long before death the SCC was provided were conducted using chi-square and t tests. Logistic and linear regression were used to test interrelationships among the variables predicting SCC.

RESULTS: The median age was 58 years (range, 23 - 92), and 45% were female. Underlying ca were 22% breast, 38% lung and 40% other. The majority, 74%, had metastatic disease at presentation, 83% had progressive disease, and 58% had a 3-4 (poor) Zubrod performance score. Of 143 pts (all dead within 6 months), 119 (83%) had documented EOL/SCC. 89% of breast and other ca pts had SCC with their physician compared to 74% of lung ca pts ($p = 0.0227$). The median number of days to death of SCC for breast and other ca pts was 15 and for lung ca pts 34 days ($p = 0.049$). Pts whose death was predicted within 30 days were no more likely to receive SCCs than pts with longer expected survival.

CONCLUSION: EOL communications in advanced ca pts at high risk of death within 30 days is not optimum. Earlier SCC would benefit both patients and their families. Rates should be closer to 100% and timing should usually be greater than 30 days. Variation in SCC by diagnosis could be functions of type of pt and physician preferences as well as the nature of the disease itself.

FREQUENCY OF SCC BY CA DIAGNOSIS

SCC	BREAST & OTHER CA (N)	LUNG CA (N)	TOTAL (N)
NO	10	14	24
YES	79	40	119
TOTAL (N)	89	54	143

VALIDATION OF CASE-MIX MEASURES DERIVED FROM SELF-REPORTS OF DIAGNOSES AND HEALTH. V.S. Fan¹, D.H. Au¹, P.J. Heagerty², R.A. Deyo¹, M.B. McDonnell³, S.D. Fihn¹; ¹University of Washington, Seattle, Washington; ²Fred Hutchinson Cancer Research Center, Seattle, Washington; ³A Puget Sound Health Care System, Seattle, Washington

PURPOSE: Self-reports of chronic diseases and general measures of health status predict use of health services but it is not established how well they predict mortality or hospitalization. We developed the Seattle Index of Co-morbidity (SIC) using self-reported chronic illnesses and tested its ability to predict mortality and hospitalizations among general outpatients.

METHODS: Using data from the Ambulatory Care Quality Improvement Project (ACQUIP), we conducted a prospective cohort study of patients from GIM clinics at 7 VA medical centers. Our primary outcomes were all-cause mortality and first hospitalization. Of the 34,103 subjects who were eligible for ACQUIP, 12,388 subjects returned both a health inventory and the SF-36 at entry to the study and were eligible for this analysis. The health inventory asked whether a provider had told patients if they had any of 25 common chronic illnesses. We used Cox-modeling to estimate the hazard ratio for mortality and for first hospitalization. Because patients below 50 years of age violated the proportional hazards assumptions, the analysis was restricted to the 10,947 patients over 50 years of age. These patients were followed for a mean of 722.5 (plus or minus 84.3) days.

RESULTS: Using a derivation set of 5,469 patients, the SIC was constructed using age, smoking status and 7 of 25 self-reported medical conditions that were univariately associated with increased mortality. The SIC was predictive of both mortality and hospitalizations when tested in a validation set of 5,478 patients. Two-year Kaplan Meier estimated mortality for different scores was: 0 to 3 points — 1.3%, 4 to 5 — 3.4%, 6 to 7 — 7.6%, 8 or more — 15.2% ($p < 0.00005$ for difference in the survival curves). A separate model was constructed in which only age and the PCS and MCS scores of the SF-36 were entered to predict mortality. Both the SF-36 component scores and the SIC had comparable ROC curves (AUC, 0.70). However, combining the SIC and SF-36 modestly improved discrimination for all cause mortality (AUC=0.74).

CONCLUSION: A new outpatient comorbidity score developed using an simple, self-administered, health inventory was predictive of 2-year mortality and hospitalizations in the primary care setting. Adding the PCS and MCS scales from the SF-36 modestly, though significantly, increased the ability of the model to predict both mortality and hospitalizations.

MULTIPLE IMPUTATION VERSUS DATA ENHANCEMENT FOR DEALING WITH MISSING DATA IN OBSERVATIONAL HEALTH CARE OUTCOME ANALYSES. P.D. Faris¹, W.A. Ghali¹, R.F. Brant¹, C.M. Norris², P.D. Galbraith¹, M.L. Knudtson¹; ¹University of Calgary, Calgary, Alberta; ²University of Alberta, Edmonton, Alberta, Canada

PURPOSE: The problem of missing data is frequently encountered in observational studies of health care delivery. The purpose of this study was to compare two strategies for handling missing data. These strategies were multiple imputation and a method of enhancing a clinical database through merging with administrative data.

METHODS: The missing data strategies were applied to a clinical database containing information collected from 6065 cardiac catheterization patients in 1995 in the province of Alberta, Canada. Observations were missing for several important clinical variables. For example, 27.3 % of patients had missing observations for ejection fraction, and 24.6 % had missing observations for congestive heart failure. Three multiple imputation methods were employed: multiple imputation by chained equations (MICE; van Buuren and Oudshoorn, 2000), transformations/imputations using canonical covariates (transcan; Harrell, 2000), and a data augmentation algorithm utilizing a multivariate normal model (norm; Schafer, 2000). Multiple imputation methods were compared with a strategy of enhancing the clinical database by merging it with diagnoses derived from administrative ICD-9 codes in hospital discharge data and day procedure data (Norris et al., 2000). Logistic regression models predicting death at one year were based on the different missing data strategies applied to the 1995 data and were evaluated using measures of discrimination and goodness of fit. The strategies were further evaluated by examining how well the logistic regression models predicted outcomes in data collected from patients in 1996.

RESULTS: The different methods produced similar logistic regression coefficients. The C-statistics for the logistic regressions were 0.825, 0.819, 0.815 and 0.802 for the MICE, transcan, norm and data enhancement methods respectively. Decreases in deviance from the null model were 469.4 (MICE), 432.0 (transcan), 412.3 (enhanced data) and 412.2 (norm). When the logistic regression models were applied to 1996 data, the C-statistics were 0.806, 0.803, 0.803, and 0.800 for the MICE, transcan, enhanced data, and norm models respectively. Decreases in deviance from the null model were 236.0, 229.5, 226.5, and 225.5 for the MICE, transcan, norm, and enhanced data models.

CONCLUSION: The performance of the data enhancement and multiple imputation strategies, although not identical, was generally similar. Multiple imputation methods require considerable statistical expertise, while data enhancement requires availability of data resources. Researchers should therefore base their choice of methods on their team's expertise and data resources, rather than on performance considerations alone.

IRON SUPPLEMENTATION FOR FATIGUE IN NON-ANEMIC WOMEN : A DOUBLE-BLIND RANDOMIZED PLACEBO-CONTROLLED TRIAL. B. Favrat¹, F. Verdon¹, C. Fallab¹, C. Bonard¹, M. Graff¹, A. Michaud¹, T. Bishoff¹, M. Devevey¹, J. Studer¹, L. Herzig¹, A. Pecoud¹; ¹University Outpatient Clinic and Department of General Medicine, Lausanne, Switzerland

PURPOSE: To document a subjective response to iron therapy in non-anemic women complaining of fatigue.

METHODS: The design is a double-blind, placebo-controlled prospective study. Subjects were recruited in an academic primary care center and in 8 general practices. The inclusion criteria were women between 18 and 55 years old, complaining of unexplained fatigue after ruling out somatic disorders. Patients with the chronic fatigue syndrome or evidence of depression were excluded; 131 participants were randomly assigned either to oral ferrous sulfate (80 mg once daily) or placebo for 4 weeks. After one month, women (n=69) with a low serum ferritin concentration (below or equal 20 ug/l) received iron for two additional months in an open design. The follow-up for all patients was 3 months. The effect of iron treatment was assessed by questionnaires (a 10-point Likert scale of fatigue and a self-report visual analogic scales of fatigue, depression and anxiety). Adherence to treatment was verified by an electronic device recording the date and time of every opening of the pill container. Intention-to-treat and per-protocol analysis were done.

RESULTS: Of the 148 enrolled subjects (76 were randomized in the iron group and 72 in the placebo group), 131 (89 %) completed the study (69 in the iron group and 62 in the placebo group). Mean age, serum ferritin concentration, level of fatigue, depression and anxiety did not differ significantly at baseline. Compliance and drop-out rates were similar in both groups. Univariate and multiple-linear regression analysis showed that the level of fatigue after one month were significantly improved for the group who received iron in comparison with the placebo group (p=0.0049). This improvement was independent of the depression or anxiety scores. Interestingly, among the women in the iron group, the best predictor of response (in a regression analysis) was the amount of iron consumed and not the serum ferritin at baseline (p=0.007). After the first month, the patients who received iron because of a low level of ferritin were greatly improved at 3 months in comparison with no treatment at all (p=0.0005).

CONCLUSION: Iron supplementation even in the absence of anemia may be beneficial to women complaining of unexplained fatigue.

PRESCRIPTION DRUG COVERAGE AND CARDIOVASCULAR DRUG USE AMONG ELDERLY MEDICARE BENEFICIARIES WITH CORONARY HEART DISEASE. A.D. Federman¹, A.S. Adams², D. Ross-degnan², J.Z. Ayanian¹; ¹Brigham and Women's Hospital, Boston, MA; ²Harvard Medical School, Boston, MA

PURPOSE: Despite the recent intense focus on prescription drug coverage for the elderly, few studies have examined drug coverage and utilization of essential medications while adjusting for confounding factors. We used multivariate regression analyses to study drug coverage and cardiovascular drug use and drug expenditures among Medicare beneficiaries with coronary heart disease (CHD).

METHODS: From the 1997 Medicare Current Beneficiaries Survey, we included patients >65 years with CHD or past myocardial infarction and excluded those with <12 months Medicare Part B, end stage renal disease, Veteran's Administration drug coverage, or nursing home residence. Insurance groups were: Medicare fee-for-service only (FFS), private supplemental coverage with no drug benefit, and drug coverage from Medicaid, other public (e.g. state pharmaceutical assistance programs), self-purchase (Medigap), HMO, and employer-sponsored (ES) sources. The association of insurance group and any use of beta-blockers (BB), long-acting nitrates, anti-anginal calcium channel blockers (CCB) and statins was examined in weighted multivariate logistic regression models, adjusting for demographic and health characteristics and the complex sample design. The percent of household income spent out-of-pocket on drugs was analyzed with linear regression. Because economic access to private drug coverage could bias results, we analyzed subgroups with restricted access: household income <150% of poverty and public insurance only (FFS, other public, Medicaid).

RESULTS: Among 1908 (15%) beneficiaries who met entry criteria, the mean age was 76 years. 66% of FFS had household income below 150% of poverty. Prevalence of drug use varied widely with insurance status: BB, 20.7–36.1%; nitrate, 20.4–39.5%; CCB, 10.6–26.8%; statin, 4.1–27.3%. Compared to ES, FFS had lower odds for use of 3 drug classes: nitrate, OR 0.63 (95% CI 0.39–1.00); CCB, OR 0.47 (95% CI 0.23–0.96); and statin, OR 0.16 (95% CI 0.05–0.49). In low-income and publicly-insured subgroup analyses, odds ratios for statin use remained lower for FFS (adjusted p < 0.05). For all drug expenditures, FFS patients spent 8.3% of household income out-of-pocket compared with 1.7% for ES (adjusted p < 0.001).

CONCLUSION: Elderly Medicare beneficiaries with CHD who lack drug coverage have less use of essential medications and a larger medication-associated financial burden than those with ES drug coverage. A Medicare prescription drug benefit that targets poor and near-poor elderly may improve their access to important drugs for CHD and reduce financial burden related to drug expenditures.

DEPRESSIVE SYMPTOMS AND MYOCARDIAL INFARCTION OR ISCHEMIA IN OLDER WOMEN: A PROSPECTIVE STUDY. M.D. Feldman¹, F. Harris¹, D.C. Bauer¹; ¹University of California, San Francisco, San Francisco, CA

PURPOSE: Some previous studies have found that depressive symptoms predict coronary artery disease. This association has been less strong and not as well studied in women, especially older women. Our aim was to analyze prospective data from a cohort of older women to determine if depressive symptoms predict myocardial ischemia or infarction.

METHODS: The Fracture Intervention Trial (FIT) was a multicenter randomized control trial of alendronate vs. placebo among 6459 women with low bone mass 55-80 years of age. At baseline, each participant was examined and completed an extensive questionnaire that included the 16 item Geriatric Depression Scale (GDS). The 140 women with a previous myocardial infarction (MI) were excluded from this analysis. Over a mean follow-up of 4.25 years, 77 women had fatal or non-fatal MIs, and an additional 102 had ischemic events/unstable angina. Medical records, discharge summaries, and death certificates were collected from each event, and were blindly adjudicated by 3 cardiologists. Relative hazard models, adjusted for potential confounders, were used to model the association between GDS scores and documented cardiovascular events. Results are reported as relative risk and 95% confidence intervals.

RESULTS: Mean GDS (±SD) was 1.5±2.0, and 168 (5%) reported 5 or more depressive symptoms. In age-adjusted analyses, a GDS score of 5 or more was associated with a trend towards increased MI and acute myocardial ischemia. After adjustment for age, self-reported health, BMI, smoking, alcohol consumption, exercise, hypertension, stroke, diabetes, and use of estrogen, however, there was no longer an association between depressive symptoms and subsequent ischemia or MI (Table 1).

CONCLUSION: In this cohort of older women, there was no significant association between depressive symptoms and subsequent myocardial ischemia and/or infarction when the data was adjusted for potential confounders.

Relative Risk (95% CI) of Cardiac Events in Women with GDS score of 5 or greater.

	Age Adjusted	Multiply Adjusted
Myocardial Infarction	1.37 (0.59, 3.17)	1.00 (0.41, 2.45)
MI or Acute Ischemia	1.59 (0.95, 2.67)	1.13 (0.65, 1.96)

SOCIAL INTEGRATION INTO COMMUNITY: EFFECT ON HEALTH CARE SEEKING BEHAVIORS. K. Felix-Aaron¹, R. Weinick², D.M. Levine³, H. Burstin¹; ¹Agency for Healthcare Research and Quality, Rockville, MD; ²Center for Primary Care Research; ³Johns Hopkins University, Baltimore, MD

PURPOSE: While poverty and minority status have been associated with poor health and health care outcomes, research suggests that social networks may improve health status. However, the relationship between the degree of social integration into a community and health care seeking behaviors is unclear. The objective of the study was to evaluate the relationship between social integration and positive health care seeking behaviors in a low-income African-American community.

METHODS: This secondary cross-sectional analysis used data from a door-to-door survey of 2,196 residents in a low-income, urban, and African-American community. The measures of social integration included 1) frequency of church attendance and 2) duration of residence in that community. Health care seeking measures were 1) Pap smear, 2) mammogram and, 3) dental visit all within 2 years, and 4) blood pressure measurement within 1 year, 5) having a regular source of care, and 6) not seeking care when needed. In multivariate analyses, we controlled for socioeconomic factors (age, gender, education, employment, marital status, and health insurance) and number of chronic diseases. Interactions were examined between measures of social integration and other covariates.

RESULTS: The study population had a mean age of 44 years, 37% male, 63% completed high school, 21% married, 74% had health insurance, and 25% had ≥2 chronic diseases. Eight hundred and eighteen (37%) went to church at least once a month and 1116 (51%) resided in that community for 10 or more years. In multivariate analyses, regular church attendance increased dental visits (OR 1.62, 95% CI 1.3–2.0) and blood pressure measurements (OR 1.76, 95% CI 1.3–2.4). Significant interactions were found between regular church attendance and insurance, number of chronic illness, and age. Regular attendance significantly increased the rate of pap smears only for the uninsured (OR 2.3, 95% CI 1.3–4.2) and women with ≥2 diseases (OR 1.8, 95% CI 1.1–3.0). Similarly, regular church attendance improved the rate of persons who did not seek care when needed only for persons ≥65 years (OR 0.52, 95% CI 0.3–0.9). Duration of community residence was associated with 3 of the six outcomes evaluated. Longer duration of community residence significantly increased the likelihood of having a dental visit (OR 1.7, 95% CI 1.2–2.5) and regular source of care (OR 1.5, 95% CI 1.0–2.1) but only for the uninsured. Interestingly, longer duration of community residence improved the rate of persons who did not seek care when needed only for those persons with insurance (OR 0.74, 95% CI 0.59–0.93).

CONCLUSION: In a low-income, urban, and African-American, community, regular church attendance and longer duration community residence are important correlates of positive health care seeking behaviors, especially for the most vulnerable, the uninsured and elderly.

SOCIAL INTEGRATION INTO COMMUNITY: EFFECT ON HEALTH CARE SEEKING BEHAVIORS. K. Felix-Aaron¹, R. Weinick², D.M. Levine³, H. Burstin¹; ¹Agency for Healthcare Research and Quality, Rockville, MD; ²Center for Primary Care Research; ³Johns Hopkins University, Baltimore, MD

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COST-EFFECTIVENESS OF AIRLINE DEFIBRILLATORS: IS PEACE OF MIND MORE IMPORTANT THAN SAVING LIVES? P.M. Cram¹, S. Vijan¹, A.M. Fendrick¹; ¹University of Michigan, Ann Arbor, MI

PURPOSE: Airline passengers are particularly vulnerable to the effects of cardiac arrest due to a lack of access to emergency medical services. To offset this relative isolation, airlines are installing automated external defibrillators (AEDs) on commercial aircraft. Our objective was to measure the cost-effectiveness of airline AED programs and estimate their value to the flying public.

METHODS: A decision analytic model was constructed to estimate the clinical and economic effects of airline AEDs. Inputs were obtained from published data and the FAA. Utility estimates were derived from cardiac arrest survivors. The medical event rate was .0004 per flight in the base case; 15% of medical events were cardiac arrest. Of these cardiac arrests, 48% were ventricular fibrillation or tachycardia. Cardiac arrest survival with an AED on-board was 40%, compared to 8% without AED access. The AED cost (including training) was \$2.51 per flight. Sensitivity analyses evaluated changes in AED cost, probability of cardiac arrest (and arrest survival), and the likelihood of airplane diversion. Since AEDs may provide utility gains through "peace of mind" for passengers not experiencing a medical event, the impact of this added passenger confidence was also evaluated.

RESULTS: AEDs on commercial aircraft cost an incremental \$5.16 per flight (approximately \$.05 per passenger). AED deployment resulted in an estimated additional \$162,000 per QALY gained (16 quality-adjusted minutes per flight). Sensitivity analysis of event probabilities and cost inputs did not substantially change the results. However, the cost effectiveness of AEDs was significantly enhanced by the inclusion of utility gain experienced by passengers from increased peace of mind. While the magnitude of this benefit is unknown, an incremental increase of .003 in utility over the flight duration would reduce the incremental cost-effectiveness of AEDs to less than \$50,000 per QALY gained.

CONCLUSION: Our model estimated that when the benefits of on-board AEDs are limited to patients experiencing a medical event, the incremental cost-effectiveness is inferior to most recommended medical interventions. However, if passengers gain utility from knowing an AED is on the aircraft, then these incremental expenditures may be justified. Utility gains from "peace of mind" may have significant implications in determining the value of health care interventions. Thus, further research should be conducted into this potentially important area.

UTILIZATION PATTERNS OF ACE INHIBITORS AND ACE INHIBITOR-ALTERNATIVES IN A CHF OUTPATIENT POPULATION. B. Fenster¹; ¹Stanford University, Stanford, CA

PURPOSE: Angiotensin Converting Enzyme (ACE) inhibitors have been well established as first-line therapy in a wide spectrum of congestive heart failure (CHF) patients (1,2). ACE inhibition has been proven to ameliorate symptoms, decrease hospitalizations, and decrease mortality. In patients in whom ACE inhibitors cannot be used due to contraindications, intolerances, or allergies, angiotensin receptor blockers (ARB's) or isosorbide dinitrate and hydralazine, particularly in combination, have been shown to provide similar therapeutic benefits (3,4). Although the use of ACE inhibitors in treatment of CHF has become standard of care, previous studies have demonstrated wide variations in utilization patterns. Furthermore, few studies have directly assessed ACE inhibitor utilization among CHF outpatients (5,6). Finally, little is known about the usage of ACE inhibitor-alternatives in the setting of ACE inhibitor intolerance. The purpose of this study is to determine the utilization patterns of ACE inhibitors relative to established ACE inhibitor-alternatives in an outpatient CHF population.

METHODS: The Palo Alto Medical Foundation (PAMF) is a large multi-specialty clinic located in Palo Alto, California. The PAMF CHF Registry consists of a database listing demographic and clinical data for 229 PAMF outpatients who currently carry the diagnosis of CHF. A retrospective electronic medical record audit of the PAMF CHF Registry population

was performed to assess current utilization and dose administration of ACE inhibitors. Patients not receiving ACE inhibitors were assessed for ACE inhibitor-alternative usage, specifically ARB's, hydralazine, or isosorbide dinitrate. Reasons for ACE intolerance, contraindications, or allergies were also recorded.

RESULTS: Review of all 229 patients in the PAMF CHF registry revealed that 87% of patients were receiving with either ACE inhibitors or established ACE inhibitor-alternative therapy. Sixty seven percent of all patients were receiving ACE inhibitor therapy, and 68% of these patients were treated with doses considered to be within the target range. An additional 20% of the population was being treated with either ARB's or hydralazine and/or isosorbide dinitrate. Many patients (13%) who were treated with ACE inhibitors experienced allergies or intolerances.

CONCLUSION: The overwhelming majority of PAMF CHF outpatients are receiving either ACE inhibitor or ACE inhibitor-alternative therapy. In addition, the majority of patients receiving ACE inhibitors are receiving doses within target range. Although current practice patterns are better than predicted, a small but significant population exists that may benefit from ACE inhibitor initiation and/or more aggressive titration of existing ACE inhibitors. However, such efforts may be limited by a significant number of ACE inhibitor intolerances, allergies, and contraindications. 1. SOLVD Investigators. Effect of enalapril on mortality and the development of heart failure in asymptomatic patients with reduced left ventricular ejection fractions. *N Engl J Med* 1992; 327:685–691. 2. CONSENSUS Trial Study Group. Effects of enalapril on mortality in severe congestive heart failure: results of Cooperative North Scandinavian Enalapril Survival Study (CONSENSUS). *N Engl J Med* 1987; 316: 1429–1435. 3. Pitt, B, et al. Angiotensin II receptor antagonists in heart failure: Rationale for design of the Evaluation of Losartan in the Elderly (ELETE) trial. *Cardiovasc Drug Ther* 1995;9:693–700. 4. Cohn, JN, et al. Effect of vasodilator therapy on mortality in chronic congestive heart failure: results of a Veterans Administration Cooperative Study (V-eHeft I). *N Engl J Med* 1986;314: 1547–1552. 5. The Large State Peer Review Organization Consortium. Heart failure treatment with angiotensin converting enzyme inhibitors in hospitalized Medicare patients in 10 large states. *Arch Int Med* 1997;157:1103–1108. 6. Sueta CA, et al. Analysis of the degree of undertreatment of hyperlipidemia and congestive heart failure secondary to coronary artery disease. *Am J Cardiol* 1999;83:1303–1307.

EVIDENCE FOR TREATMENT OF ALCOHOL PROBLEMS IN PRIMARY CARE? D.A. Fiellin¹, M.C. Reid¹, P.G. O'Connor¹; ¹Yale University, New Haven, CT

PURPOSE: Patients with alcohol problems are common in primary care and there may be advantages to treating these patients in this setting. However, there are few systematic reviews that focus on the efficacy of treatment for alcohol problems in primary care. Our goal was to evaluate the literature on treatment of alcohol problems in primary care according to alcohol consumption outcomes and methodological standards.

METHODS: We searched the published literature for years 1966–1999 to identify studies on treatment of alcohol problems in primary care. We included studies from English language peer reviewed journals that were (1) performed in a primary care setting; (2) had as the main focus the treatment of alcohol problems; (3) had two or more treatment arms; and (4) reported alcohol consumption outcomes. Two reviewers appraised all articles for methodological content and results according to pre-specified coding criteria.

RESULTS: We identified eleven studies on treatment of alcohol problems in primary care. Seven described treatments for at risk drinking, while four described treatment for alcohol abuse and dependence. The studies were performed in a variety of primary care settings and all reported on the efficacy of brief interventions (BI). There was marked variability in these brief interventions with 4 (36%) using a single session, 6 (55%) using a drinking diary and 5 (45%) discussing a treatment plan. BI were effective in reducing alcohol consumption compared to control treatments in only three of the eleven studies (27%). This finding was only demonstrated in studies of at risk drinking. The studies inconsistently adhered to methodological standards with only 3 (27%) using a single step recruitment procedure and reporting the participation rate, none (0%) providing a full description of patient demographics and medical/psychiatric comorbidity, 1 (9%) assessing prognostic susceptibility, 10 (91%) performing randomization, 1 (9%) assessing ancillary treatments, 0 (0%) assessing discrimination between the treatments, and 5 (45%) providing follow-up on all enrolled subjects.

CONCLUSION: This systematic review, restricted to studies performed in primary care found that, in contrast to expert recommendations, there is contradictory evidence on the efficacy of brief interventions for at risk drinkers in primary care. Based upon evidence from the medical literature, patients in primary care with alcohol abuse or dependence are unlikely to decrease their alcohol consumption as a result of brief interventions alone. Finally, studies of brief interventions in primary care inconsistently adhere to methodological standards.

DO BELIEFS AND ATTITUDES AFFECT UNDERUTILIZATION OF TOTAL KNEE REPLACEMENT AMONG AFRICAN-AMERICANS WITH OSTEOARTHRITIS? M.K. Figaro¹, P. Williams-Russo¹; ¹Cornell University, New York, NY

PURPOSE: Total Knee Replacement (TKR) improves mobility, decreases pain, and improves quality of life for patients with severe knee osteoarthritis (OA). Despite the higher prevalence and higher reported disability of knee OA among Blacks, whites are over 1.5 times more likely to have TKR than Blacks. This study examined beliefs of Harlem Blacks with OA regarding TKR, including its efficacy and safety.

METHODS: Community-dwelling older adults were recruited from churches, senior centers or other social networks in Harlem. In-person interviews elicited familiarity with others in the community who had TKR, perceived knowledge of TKR, referral to specialists, concerns and beliefs regarding surgery. We also assessed demographic data and severity of OA using the Western Ontario and McMaster Osteoarthritis Index (WOMAC).

RESULTS: Women made up 84% of the sample of 65 patients. The mean age was 73, with a mean duration of symptoms of 9 years; 61% had at least a high school education and 45% had an income $< \$10,000$ in 1999; 99% of subjects had Medicare insurance. Subjects had a mean WOMAC score of 56 +18, indicating high levels of OA disability; 45% had been referred to specialist. Table 1 summarizes major factors and concerns.

CONCLUSION: The most important of subjects' concern were regarding post-surgical pain, low expectations of the eventual efficacy of surgery, and concern over the cost of TKR. In addition, despite the high WOMAC disability score of the group, less than 50% of patients were referred to a specialist. Efforts to address the current disparity in use of TKR should attend to these potentially modifiable factors.

Table 1.

Description of Variable	Percent of Subjects
Familiarity with others post TKR	84%
Perceived improvement of others post TKR	71%
Perceived knowledge TKR	22% (very knowledgeable)
Concern regarding post-surgical pain	43% (very or extremely concerned)
Expectation of improvement post TKR	36% (very or extremely likely)
Cost of surgery	44% (very or extremely important)

ROUTINE REPORTING OF HEALTH STATUS DATA DOES NOT IMPROVE GENERAL HEALTH OR SATISFACTION. S.D. Fihn¹, M. McDonnell², P. Diehr¹, D.H. Au¹; ¹University of Washington, Seattle, WA; ²VA Puget Sound Health Care System, Seattle, WA

PURPOSE: In previous studies, short-term feedback of general measures of health status to health care providers has not improved patients' overall health. We hypothesized that sustained, individualized reporting of general and condition-specific health measures in a sophisticated format would enhance health and satisfaction.

METHODS: We conducted a randomized effectiveness trial in the General Internal Medicine clinics at 7 Veteran Affairs facilities that had discrete firms. All patients who were assigned a primary provider and who had an appointment in the prior year were eligible. Randomization, intervention and analysis were by firm. Respondents to a baseline health inventory were regularly mailed the SF-36 and, as relevant, validated questionnaires about 6 chronic conditions: coronary disease, COPD, depression, diabetes, alcohol use and hypertension. From the SF-36, we computed physical and mental component scales (PCS and MCS). We also sent surveys about satisfaction with humanistic and organizational aspects of care and retrieved clinical data from computerized records. The information collected was presented to primary care providers of eligible patients at every visit for 2 years in graphical format (e.g., blood pressure, HbA1C, SF-36 scale scores and anginal frequency) plus interpretations (e.g., "Patient unable to climb stairs due to shortness of breath") and "tips" derived from national guidelines. Clinicians also received summary reports comparing their patient panels with local and national norms and highlighting patients with potentially serious problems. In addition, we conducted training on quality improvement and interpretation of health status measures.

RESULTS: Of 34,103 patients initially eligible, 22,223 returned baseline health inventories. Of these, 15,346 returned an SF-36 and condition-specific questionnaires. Over 300,000 questionnaires (comprising >3 million pages) were mailed to patients, and over 44,000 individualized feedback reports were distributed. After adjustment for baseline values and deaths, mean changes in intervention and control firms for the PCS (-7.28 [sd 1.15] vs. -6.63 [sd 1.04]) and MCS (-3.63 [sd 1.43] vs. -3.83 [sd 1.71]) over 2 years were not significantly different in paired analyses ($p=.34$ and $p=.70$). There were also no significant differences in satisfaction in the humanistic or organizational domains between intervention and control firms. Similar results were found using several analytic strategies including restriction to patients who completed all forms.

CONCLUSION: Routine collection and feedback of measures of general and condition-specific health and satisfaction did not improve outcomes. It is likely that such data must be linked to specific management suggestions if they are to influence patients' outcomes.

DEPRESSION IS ASSOCIATED WITH HIGHER LEVELS OF C REACTIVE PROTEIN IN MEN IN THE THIRD NHANE SURVEY. D.E. Ford¹, T.P. Erlinger¹; ¹Johns Hopkins University, Baltimore, MD

PURPOSE: Depression is associated with poorer survival after myocardial infarction and in several studies appears to be an independent risk factor for developing incident coronary artery disease (CAD). The biological mechanism explaining the increased risk is not certain. We wanted to determine if C-reactive protein (CRP), a marker of increased risk for CAD, is associated with major depression.

METHODS: Our analysis is based on data from the third National Health and Nutrition Examination survey conducted between 1988-1994. The Diagnostic Interview Survey (DIS) was administered to 8451 individuals between 17-39 years of age to assess current and past episodes of major depression. The analysis is based on 7091 individuals (3849 females and 3242 males) with complete data on CRP and other important covariates. Analysis was stratified by gender because of the influence of estrogens on CRP levels. Logistic models were used with detectable CRP as the dependent variable. Unadjusted and adjusted estimates for current (last year) and past major depression compared to never lifetime depression were calculated. Estimates are adjusted for sampling design and nonresponse.

RESULTS: Men with current major depression (OR=3.29 95% CI 1.56, 6.92) and past major depression (OR=2.05 95% CI 0.84, 5.04) had elevated levels of detectable CRP after adjustment for age, African-American race, BMI, total cholesterol, log triglycerides, diabetes, systolic blood pressure, tobacco smoking (self-reported current or past or cotinine level above 10 ppm), and alcohol intake compared to never depression. A similar trend was found for single episode of depression (OR=1.13 95% CI 0.26, 4.90) and multiple episodes of depression (OR=3.87 95% CI 1.67, 8.96) compared to never depression after adjustment. There was a significant linear trend between number of lifetime depressive symptoms and elevated CRP ($p < 0.001$). The relationship between depression and elevated CRP was present for smokers, nonsmokers, and those with low or high total cholesterol. For women, there were no associations between any of the measures of depression and elevated CRP. Adjusting for

current use of estrogens or ever use of oral contraceptive agents and excluding those who were pregnant did not substantially change the results.

CONCLUSION: In young men, current major depression is associated with elevated CRP levels that appear to return toward normal when the depression remits. CRP may be one mechanism by which depression leads to increased coronary heart disease in men. The same associations were not present in women but we may not have been able to adequately account for their hormonal status. This cross-sectional data needs to be confirmed with prospective studies.

STRESS AND THE URGE TO DRINK. E. Fouquereau¹, A. Fernandez¹, E. Mullet², P.C. Sorum³; ¹Universite Francois-Rabelais, Tours, Indre-et-Loire; ²Ecole Pratique des Hautes Etudes, Plaisance-du-Touch, Haute-Garonne; ³Albany Medical Center, Schenectady, NY

PURPOSE: Understanding why people drink alcohol is important for the health and safety of individuals and the public. The aim of this study was to examine from a cognitive point of view the hypothesized link between drinking and stress.

METHODS: 25 scenarios were constructed by combining two items, either two life-change events or a social situation and an emotional state. In the initial three experiments, 159 male and 43 female alcoholics and 157 male and 93 female non-alcoholics in France judged the degree to which these scenarios were stressful and subsequently the degree to which they stimulated an urge to drink. In the final experiment, 126 of the male alcoholics were studied at the beginning and end of an in-patient alcohol rehabilitation program.

RESULTS: The alcoholics and non-alcoholics, regardless of gender, assigned similar stress values to the scenarios and used the same cognitive rules for combining the stress associated with two items (disjunctive rules for two life-change events and additive ones for a personal emotion combined with a social situation). They differed, however, in how they judged the urge to drink. The non-alcoholics reported little stimulus to drink from any combination of items, whereas the alcoholics not only perceived the individual items as stimulating an urge to drink, but also used the same cognitive rule in judging the combined urge to drink of two items as they used in judging the combined stress. After completing rehabilitation, the alcoholics judged the combinations of life-change events as stimulating less stress and less urge to drink; nevertheless, they continued to use a disjunctive combination rule.

CONCLUSION: Stress and drinking are linked at a fundamental cognitive level among alcoholics, though not among non-alcoholics. Alcoholics should be helped to recognize this link, to reduce their feelings of stress, and to respond to stress in ways other than drinking.

ASSESSING SOCIAL DISPARITIES IN HEALTH PLANS PERFORMANCE ON QUALITY MEASURES: ADVISING SMOKERS TO QUIT. A.M. Fremont¹, M.M. Shah², S.L. Wickstrom², C.E. Bird¹, T.S. Rector², E.A. Moglynn¹; ¹RAND, Santa Monica, CA; ²Center for Health Care Policy and Evaluation, Minneapolis, MN

PURPOSE: Disparities in care by gender, race/ethnicity, and socioeconomic status are well documented. Yet, widely used systems of monitoring quality such as the NCQA's HEDIS measures do not require plans to report results separately by patient sociodemographic group. This AHRQ sponsored project is investigating the need for and feasibility of plans reporting selected HEDIS measures stratified by key social groupings. Our initial analysis examines plans' performance on advising smokers to quit, stratified by social group.

METHODS: We obtained HEDIS 2000 data for 3,663 commercially-insured and 2,824 Medicaid-insured members enrolled in 7 UnitedHealthcare plans located in 3 regions. As specified by the NCQA, the Consumer Assessment of Health Plans Survey (CAHPS 2.0) was used to collect information on whether the enrollee was a current or recent smoker (denominator) and advised to quit smoking (numerator). We obtained information on patients' sociodemographic characteristics, visits, and health from the survey. Response rates were 55% and 38% for commercial and Medicaid patients, respectively. We analyzed rates of advising smokers to quit separately for commercially insured and Medicaid patients as is customary with HEDIS measures. For comparisons between patients in different sociodemographic groups, we pooled data from the different plans. We used Chi-Square tests to compare the proportion of smokers advised to quit by gender (female vs. male), race/ethnicity (nonHispanic White vs. (nonWhites and Hispanics)), and education (high school or less vs. some college or more). Pending multivariate analyses will assess independent effects of these factors adjusting for age, number of visits, and self-reported health.

RESULTS: Of the respondents, 693 commercially insured and 973 Medicaid patients were smokers. The average rate of advising smokers to quit was the same for commercial (66%) and Medicaid (66%) patients. Among commercially insured patients, men were less likely to be advised to quit than women (61% vs. 70%, $P=.01$) and a lower proportion of less educated patients were advised to quit than among more educated patients (70% vs. 64%, $P=.10$), however, the latter difference was not significant. In contrast, among Medicaid patients, nonHispanic Whites were more likely than other patients to be advised to quit (69% vs. 61%, $P=.01$). The proportion of commercially insured patients advised to quit was also higher for nonHispanic Whites than for others (67% vs. 60%), but the difference was not significant ($P=.32$).

CONCLUSION: Preliminary results suggest important disparities in care received by patients in different sociodemographic groups on one HEDIS performance measure: advising smokers to quit. However, adjustment for likely confounders such as the number of visits, larger sample sizes, and evaluation of other HEDIS measures is needed before these results can be meaningfully interpreted.

ACCULTURATION AND CIGARETTE SMOKING IN URBAN CHINESE AMERICANS. S.S. Fu¹, G.X. Ma², P.T. Siu³, X.M. Tu¹, J.P. Metlay¹; ¹University of Pennsylvania, Philadelphia, PA; ²Temple University, Philadelphia, PA; ³Thomas Jefferson University, Philadelphia, PA

PURPOSE: Less acculturated Chinese Americans are more likely to experience cultural and language barriers. The purpose of this study was to assess the relationship between acculturation and cigarette smoking in Chinese Americans.

METHODS: We conducted a cross-sectional survey of Chinese American adults attending four pediatric, medical, or dental practices located in Philadelphia's Chinatown. A self-administered questionnaire translated into Chinese or English was given to all consecutive Chinese American adults who had scheduled visits or were accompanying a child on selected days during November 2000 through January 2001. To measure acculturation, we adapted a reliable and valid 13-item acculturation scale that was developed for Southeast Asians. The acculturation scale includes a subscale to assess English proficiency. Items to assess smoking status, stages of change, knowledge and attitudes regarding smoking were adapted from the literature. The instrument was independently translated into Chinese, back translated, and pilot tested among experts as well as in community settings for content validity and appropriate cultural context. Univariate analyses examined the relationship between smoking status and English proficiency. Scores on the English proficiency subscale were categorized into quartiles. Multiple logistic regression analysis controlled for age, sex, study site, education and income.

RESULTS: To date 394 patients have completed the questionnaire. The response rate was 91% of eligible respondents. Respondents mean age was 44.5 years, 41% were male, 56% had not completed high school and 28% had no health insurance. Among Chinese males, 58% reported having ever smoked; 26% were current smokers. Among Chinese females, only 6% reported having ever smoked; 2% were current smokers. Among Chinese males, rates of ever smoking were 72% in the lowest quartile of English proficiency vs. 41% in the highest quartile ($p = .01$ for trend). Rates of current smoking were 34% in the lowest quartile and 18% in the highest quartile ($p = .01$ for trend). In multivariate analysis, increased English proficiency was significantly associated with decreased rates of ever smoking (adjusted OR .68 per increasing quartile, 95% CI .47, .98). Although not statistically significant, increased English proficiency was associated with decreased rates of current smoking (adjusted OR .65, 95% CI .34, 1.21).

CONCLUSION: Lower English proficiency is significantly associated with increased cigarette smoking in Chinese American males. Although not significant, current smokers tended to show lower English proficiency than former smokers. Acculturation level should be considered when developing culturally appropriate interventions to decrease cigarette smoking in this vulnerable population.

MANAGEMENT OF HYPERCHOLESTEROLEMIA AMONG OLDER DIABETIC PATIENTS. C. Furman¹, C.S. Ritchie¹, C. Mundt¹, J. Burke², W. Mc Kinney¹; ¹University of Louisville, KY; ²Louisville Veterans Affairs Medical Center, Louisville, KY

PURPOSE: Diabetes mellitus and hypercholesterolemia are common diseases in the aging population. Diabetes increases cardiovascular risk twofold or more, compared with the non-diabetic population. Therefore, it is important to address the management of hypercholesterolemia in the older diabetic patient. We sought to understand how hypercholesterolemia was managed among a large primary care population of older diabetics.

METHODS: Subjects who had diabetes mellitus based on physician diagnosis, diagnostic criteria and the use of diabetic medications were identified using the Louisville Department of Veterans Affairs Medical Center Decentralized Hospital Computer Program (DHCP). We evaluated treatment management of diabetic patients aged 55–64 years old, 65–74 years old, and 74 years old and greater over a three-year period from 10/1/96 until 9/30/99. Quality of care standards were the Adult Treatment Panel (ATP) II guidelines of the National Cholesterol Education Program (NCEP) and the Department of Veterans Affairs guidelines for The Pharmacological Management of Hyperlipidemia. We defined diabetes as a person taking hypoglycemic medications (oral or insulin). Definitions for hypercholesterolemia were based on NCEP guidelines.

RESULTS: The study population consisted of 1394 diabetics with a mean age of 69.2 (± 7.2 SD), 98.4% male, and 1.6% female. The old-old diabetic population ($n=360$) comprised 13% of the Louisville VAMC patient population 75 years old and older in 1997 (equivalent to the national rate of diabetes in persons aged 75 and older (13.2%)). The mean cholesterol was 200 mg/dL (± 42.8 SD) and the mean LDL was 120 mg/dL (± 34.3 SD). The oldest-old were least likely to have their cholesterol checked ($p < .001$). They were also least likely to be placed on cholesterol-lowering medications when their LDL Cholesterol was greater than or equal to 160 mg/dL. However, in 1998 and 1999, the oldest-old were more likely to be treated with medication than the middle old.

CONCLUSION: The oldest old are less likely to have their cholesterol checked, but once it is checked, they are more likely than the middle old to be treated. Since diabetes and cardiovascular disease are common, cardiovascular disease is more common in the elderly, and morbidity and mortality from cardiovascular disease is great, physicians need to attempt to reduce morbidity and mortality by adhering to the practice guidelines for cholesterol management — even in the oldest diabetics. Practice patterns demonstrated in this study suggest increasing use of cholesterol-reducing medications in older adults.

VALIDATION OF CLINICAL PREDICTION RULES FOR STROKE: RESULTS FROM THE NATIONAL REGISTRY OF ATRIAL FIBRILLATION (NRAF) PROJECT. B.F. Gage¹, A.D. Waterman¹, W.D. Shannon², M. Boechler³, M.W. Rich², M.J. Radford⁴; ¹Washington University in St. Louis, St. Louis, MO; ²Washington University, Saint Louis, Missouri; ³Innovative Emergency Management, Inc., Baton Rouge, LA; ⁴University of Connecticut, Farmington, CT

PURPOSE: Patients who have atrial fibrillation (AF) have an increased risk of stroke, but their absolute rate of stroke depends on age and comorbid conditions. An accurate, objective scheme to estimate the risk of stroke in the AF population would allow physicians and patients to choose antithrombotic therapy more judiciously.

METHODS: We combined two existing classification schemes into a new stroke-risk scheme, the CHADS₂ index, and then validated all three classification schemes. We formed CHADS₂ by assigning 1 point for the presence of each of the following conditions — congestive heart failure, hypertension, age greater than 75 years, diabetes mellitus — and 2 points for a history of stroke or transient ischemic attack. To test the three classification schemes, we collaborated with Peer Review Organizations representing seven states to assemble a National Registry of AF

(NRAF) consisting of Medicare beneficiaries who have non-rheumatic AF. The NRAF dataset contained health outcomes based on Medicare part A claims data.

RESULTS: Of 1733 NRAF cohort members who had not been prescribed warfarin at the time of hospital discharge, 94 were readmitted for an ischemic stroke (stroke rate, 4.4 per 100 patient-years). As indicated by a c-statistic significantly greater than 0.5, the two existing classification schemes predicted stroke better than chance: 0.78 for the scheme developed by the Atrial Fibrillation Investigators (AFI) and 0.82 for the Stroke Prevention in Atrial Fibrillation (SPAF) III scheme. However, the new CHADS₂ index performed better with a c-statistic of 0.87. The stroke rate per 100 patient-years without antithrombotic therapy increased by a factor of 1.5 (95% CI, 1.3–1.7) for each 1-point rise in the CHADS₂ score (Table).

CONCLUSION: The two existing classification schemes and especially a new stroke-risk index, CHADS₂, can quantify the risk of stroke for patients who have AF. Use of the CHADS₂ index should aid in the selection of antithrombotic therapy for patients who have non-rheumatic AF.

Risk of Stroke in Medicare Beneficiaries Stratified by CHADS₂ Score

CHADS ₂ Score	No. of Patients	No. of Strokes	NRAF Crude Stroke Rate/100 pt-yrs	NRAF Adjusted Stroke Rate(95% CI)/100 pt-yrs
0	120	2	1.2	1.9 (1.2 – 3.0)
1	463	17	2.8	2.8 (2.0 – 3.8)
2	523	23	3.6	4.0 (3.1 – 5.1)
3	337	25	6.4	5.9 (4.6 – 7.3)
4	220	19	8.0	8.5 (6.3 – 11.1)
5	65	6	7.7	12.5 (8.2 – 17.5)
6	5	2	44.0	18.2 (10.5 – 27.4)

MEDICATION ERRORS AND ADVERSE DRUG EVENTS IN THE AMBULATORY SETTING. T.K. Gandhi¹, S.N. Weingart², D.S. Seger¹, A. Seger¹, J.S. Borus¹, L. Burdick¹, L.L. Leape¹, D.W. Bates¹; ¹Brigham and Women's Hospital, Boston, MA; ²CareGroup Healthcare System, Boston, MA

PURPOSE: Medication errors (MEs) and adverse drug events (ADEs) are common in the hospital setting. However, relatively little is known about the frequency of MEs and ADEs in outpatients. Therefore, we prospectively examined the frequency, type, severity, and preventability of MEs and ADEs in the ambulatory setting.

METHODS: We studied 2 academic hospital practices and 2 community based clinics in the Boston area, and collected copies of prescriptions written by 24 primary care providers from Sept 1999 to March 2000 (6 weeks per clinic). Prescription copies were reviewed by a pharmacist to screen for MEs and potential ADEs. In addition, patients who received prescriptions were telephoned 2 weeks after their visit to ask about problems with their medications, with a response rate of 59%. Patient-reported problems were presented to 2 MD reviewers, who classified them as ADEs or not, and determined severity and preventability. Events were considered preventable if they persisted for an unnecessarily prolonged period (even if they could not have been completely prevented).

RESULTS: Of 1173 prescriptions screened during this time period, 202 (17%) were rule violations (orders that violate strict standards but are generally understood and generate no additional work), 44 (4%) were MEs, 59 (5%) were potential ADEs. The most frequent rule violations were missing route, and the most frequent MEs were errors in route or dose. The most frequent potential ADEs were errors in frequency and dose. Of 661 patients surveyed, 178 (27%) reported a total of 206 ADEs. Of these, 29 (14%) were serious, 74 (36%) were preventable, and 13 (6%) were both. The most frequent types of ADEs and preventable ADEs were CNS-related (35%, 36%), gastrointestinal (21%, 9%), and cardiovascular (16%, 18%). The most frequent medications involved in preventable ADEs were antidepressants (22%), ACE-inhibitors (14%), beta-blockers (10%), and calcium-channel blockers (10%). Of preventable ADEs ($n=74$), physicians were primarily responsible in 63% and patients in 33%. The main types of physician error were failure to act on results of monitoring or tests (50%) and inappropriate drug choice (24%). The main type of patient error was failure to inform their physician of problems (100%).

CONCLUSION: We conclude that errors and ADEs are common in the outpatient setting. A large fraction of ADEs were preventable. Improved monitoring for ADEs, including better patient-physician communication about ADEs, appears to be critical to ADE prevention in the ambulatory setting.

COMPUTER MULTIMEDIA TO EDUCATE UNDERSERVED DIABETIC POPULATIONS ON ANNUAL EYE EXAMINATIONS: A PILOT STUDY. B.S. Garber¹, I.G. Brodsky², A. Lima¹, M. Girotti¹, K. Lawless¹, L. Smolin¹, K. Honore¹, D. Henry¹, A.R. Eiser¹; ¹University of Illinois at Chicago, Chicago, IL; ²University of Illinois at Chicago, Chicago, IL

PURPOSE: There are significant health disparities among various diabetic populations, particularly with respect to ethnicity, race, socioeconomic status, and educational background. Unfortunately, educational brochures are often generalized, and do not address the different needs and barriers that impair the self-management required for care of this chronic disease. Few materials are available that cater to individuals of different ethnic backgrounds, especially those with low literacy or education backgrounds. This study evaluated a new multimedia software application for diabetic populations with these needs in mind. The target population of subjects included diabetic patients from an urban, minority community not accustomed to computer usage.

METHODS: 22 diabetic African-American patients were consecutively interviewed and digitally videotaped in an urban general medicine clinic. These individuals provided testimonials regarding the barriers, fears, benefits, and myths surrounding the ophthalmologic eye examination including pupil dilation. Specific issues, such as faith, lack of

insurance, and misconceptions about eye exams were addressed. The video clips were edited and integrated into a multimedia application that included simple activities to engage the user over 15–20 minutes. The application was available near the waiting room of a clinic for patients to view when waiting for an appointment, or afterwards before leaving. The program uses a touch screen, so previous technological experience was not required. A separate sample of 26 patients viewed the multimedia program, and completed a brief questionnaire.

RESULTS: 11 male and 15 female diabetic patients completed the multimedia program and survey, with 17 (65%) African-American, and 18 (69%) having elementary or high school education. 12 of 26 (46%) have never used a computer before, and 18 (69%) described the computer as “easy” or “very easy” to use. Preferences for medium of learning about diabetes were greatest for computers (11), followed by doctors and nurses (8), television (5), and brochures and articles (4).

CONCLUSION: A multimedia patient education program that addressed barriers related to literacy and culture was found to be acceptable by a convenience sample. Additional telephone surveys will provide data regarding knowledge and attitudes toward obtaining annual eye examinations, as well as actual compliance with visits. Given the growing interest in on-line consumer health information, the availability of similar applications for underserved populations may help to bridge the “digital divide” (or, the gap between different communities in terms of access to technology).

GIVEN A CHOICE DO PATIENTS SELECT A PHYSICIAN OF THEIR OWN GENDER AND RACE? B. Gerbert¹, S. Berg-Smith¹, M. Mancuso¹, R. Brand¹, J. Wofsy¹, S. McPhee¹, D. Null¹; ¹Univ. of California, San Francisco, CA

PURPOSE: While gender selection of physicians has been studied extensively, mostly by survey, selection of physicians by race has rarely been addressed. We used an experimental design and an actor-portrayed video doctor to investigate patients’ preferences for physician gender and race.

METHODS: Participants were asked which one of six video doctors (one male and female African American, Latino, and Caucasian) they would pick to be their physician after viewing each deliver a 45-second health prevention message. The videotapes were professionally produced and great care was taken to keep all variables constant, including the age, attractiveness, and clothing of the video doctors; the video setting; and the delivery of the prevention message (script, non-verbal behaviors, facial expressions, and interpersonal style—e.g., warm, empathic, collaborative). Each group of 20–22 participants viewed one of 18 configurations of the presentation order of the six video doctors to control for any order effect.

RESULTS: 400 participants were recruited in a shopping mall in the San Francisco Bay Area. The sample was diverse in gender (39% male, 61% female); race (29% Caucasian, 8% African American, 26% Latino, 30% Asian, and 7% Other); and age (18 years–87 years). When asked who they would choose to be their physician, both male and female participants were more likely to choose a female (82%) as their physician rather than a male (18%). Among female participants, 88% chose a physician of their own gender, while only 29% of males chose a male physician. When asked who they would choose to be their physician, 23% of participants chose an African-American physician, 32% chose a Latino physician, and 44% chose a Caucasian physician. Fifty-two percent of African-American participants chose an African-American physician; 44% of Latino participants chose a Latino physician; and 56% of Caucasian participants chose a Caucasian physician. Among Asian participants, 20% chose an African-American physician, 37% a Latino physician, and 43% a Caucasian physician.

CONCLUSION: The vast majority of participants in our study chose a female video doctor as their physician, regardless of their own gender. Although approximately half of African-American, Latino, and Caucasian participants chose their own race, the other half were willing to choose a video doctor of a different race. Our data suggests that when selecting a physician, patients may strongly associate women, regardless of a standardized patient-centered delivery, with those qualities they want in their physician. Future studies may want to use video doctor technology to further investigate patients’ preferences in selecting their physician, in particular the effect of physician race on patient choice.

FOOD CHOICES, PHYSICAL ACTIVITY, AND SELF-ESTEEM IN ADDRESSING ADOLESCENT OBESITY. E.J. Gertner¹; ¹Lehigh Valley Hospital, Allentown, PA

PURPOSE: Adolescent obesity continues as public health epidemic. Data from the National Health and Nutrition Examination Surveys (NHANES I, II, III) reveal that the percentage of youth above the 85th percentile for body mass index (BMI) increased to approximately 22% by the late 1980’s. Further, since the beginning of these surveys, the prevalence of adolescent obesity has increased by 39%, with Hispanic, Native American, and African American children more affected than other populations. A 1997 survey conducted by the Allentown (PA) Health Bureau revealed a 31% obesity rate among adolescent students in the Allentown School District (ASD), where 57% of students are non-white and 62% are eligible for free or reduced lunches. Both decreased physical activity and poor eating behaviors contribute to obesity. General internists have a responsibility but limited opportunities to understand and influence teen nutrition, especially in minority populations.

METHODS: To better understand students’ attitudes toward nutrition and physical activity, as well as barriers to improving their choices, we conducted focus groups among representative students to determine their food preferences, and dietary and activity patterns. We developed a 57-question survey that we utilized in six focus group sessions with high school students in an alternative program within the ASD. Sessions were audiotaped, and responses to questions categorized into four areas: Food choices, Physical Activity, Self-Esteem/Identity, and Knowledge.

RESULTS: In general, students expressed a willingness to eat healthier foods, if provided a variety including fresh fruits and vegetables, as well as culturally acceptable products. Adolescents desired daily gym class with a variety of activities, although admitted that much of their free time was spent engaged in sedentary activities. Groups also expressed indifference, although many had basic knowledge of the consequences of obesity.

CONCLUSION: Providing greater choices and opportunities for food and physical activity can promote healthier behaviors among adolescents, impacting upon the prevalence of obesity in this population. In particular, minority students may have different needs than others. General internists can have a large impact on adolescent obesity and the subsequent development of related diseases by working together with community leaders and other health care professionals in advocating policies to provide adolescents with these opportunities.

THE IMPORTANCE OF DETAILED CLINICAL DATA FOR EVALUATING SEX DIFFERENCES IN ACCESS TO CARDIAC REVASCLARIZATION. W.A. Ghali¹, P.D. Faris¹, P.D. Galbraith¹, C.M. Norris², M.J. Curtis¹, L.D. Saunders², V. Dzavik², L.B. Mitchell¹, M.L. Knudtson¹; ¹University of Calgary, Calgary, Alberta; ²University of Alberta, Edmonton, Alberta

PURPOSE: While some studies have suggested that there may be sex differences in access to cardiac procedures, others have found no evidence of ‘gender bias’ in cardiac care. Possible explanations for these inconsistent results include the focus, in some studies, on non-representative patient populations, and the occasional use of data sources that lack clinical detail. We used a clinically-detailed cardiac database to study crude and adjusted rates of cardiac revascularization for males and females in a population-based cohort of patients undergoing cardiac catheterization.

METHODS: The Alberta Provincial Project for Outcome Assessment in Coronary Heart Disease (APPROACH) is an inception cohort of all patients undergoing cardiac catheterization in the province of Alberta, Canada. We studied 21,816 patients undergoing catheterization in calendar years 1995 through 1998, and determined rates of percutaneous coronary intervention (PCI) and/or coronary artery bypass grafting (CABG) for males and females in the year following cardiac catheterization. Cox proportional hazards models were then used in a two-step analysis to determine adjusted relative risks for revascularization and adjusted procedure rates. An initial ‘partial adjustment’; controlled for variables such as age and comorbidities that are routinely available in most databases, including administrative databases. A subsequent ‘full adjustment’ additionally controlled for extent of coronary artery disease and ejection fraction, variables that are only available in detailed clinical databases.

RESULTS: For the endpoint of any revascularization procedure (PCI or CABG), the unadjusted relative risk for revascularization for females relative to males was 0.67 (0.65–0.71). The relative risk increased to 0.69 (0.66–0.72) with partial adjustment, and to 0.98 (0.94–1.03) with full adjustment, indicating equivalent access to revascularization for males and females. For PCI, the corresponding relative risks were 0.77 (0.73–0.82), 0.84 (0.80–0.89), and 1.02 (0.96–1.08). For CABG, the corresponding relative risks were 0.54 (0.51–0.58), 0.51 (0.48–0.55), and 0.93 (0.87–1.01). The fully-adjusted proportions of patients having undergone any revascularization procedure (PCI or CABG) at one year after catheterization were 51.5% for males and 51.1% for females (p=0.49).

CONCLUSION: In Alberta, Canada, clinical variables fully explain sex differences in access to revascularization, and misleading conclusions would arise without full adjustment for male-female differences in extent of coronary artery disease and ejection fraction. Extreme caution is needed in interpreting reports on access to care that use sparsely-detailed clinical data sources.

ABORIGINAL ETHNICITY AND ACCESS TO CARDIAC PROCEDURES. W.A. Ghali¹, P.D. Faris¹, P.D. Galbraith¹, C.M. Norris², M.L. Knudtson¹; ¹University of Calgary, Calgary, Alberta; ²University of Alberta, Edmonton, Alberta

PURPOSE: Anecdotal reports suggest that individuals of aboriginal ethnicity may be less likely to undergo cardiac procedures than those of non-aboriginal ethnicity. We studied crude and adjusted rates of cardiac catheterization, and subsequent coronary revascularization, for individuals of aboriginal ethnicity living on native reservations.

METHODS: Data from APPROACH, the Alberta Provincial Project for Outcome Assessment in Coronary Heart Disease, were used to identify all individuals undergoing cardiac catheterization in Alberta, Canada between January 1, 1995, and December 31, 1998. Cases were linked via postal codes to Canadian census data to determine population rates of cardiac catheterization by census-defined geographic regions. Individuals were flagged as being of aboriginal ethnicity when they resided in regions where greater than 95% of 1996 census respondents indicated that they were of aboriginal ethnicity. Such regions generally represent native reservations. Using the Alberta-wide population as a reference, we employed indirect standardization to adjust rates of cardiac catheterization to sequentially control for 1) age and sex, 2) income quintile, and 3) urban vs. rural residence. We also evaluated survival and the occurrence of revascularization procedures after catheterization (i.e., coronary artery bypass grafting [CABG] or percutaneous coronary intervention [PCI]).

RESULTS: We studied a total of 21,520 unique Alberta residents undergoing cardiac catheterization in the period studied. Of these, 98 (0.46%) met the study’s criterion for defining aboriginal ethnicity based on postal code of residence. Using the Alberta-wide catheterization rate as a reference, we would have expected 204 unique individuals from these postal codes to undergo catheterization. The resulting unadjusted observed-to-expected (O/E) ratio of 0.48 (0.34–0.62) indicates that individuals of aboriginal ethnicity living on reservations are 52% less likely to undergo cardiac catheterization than are individuals living in other regions. After adjustment for age and sex, the O/E ratio increased to 0.70 (0.54–0.86). Additionally controlling for income quintile increased the O/E ratio to 0.74 (0.57–0.91), and adjustment for urban vs. rural residence further increased the O/E ratio to 0.93 (0.74–1.11). The latter ratio indicates only a statistically non-significant 7% lower likelihood of undergoing catheterization for individuals of aboriginal ethnicity. Subsequent to cardiac catheterization, individuals of aboriginal ethnicity (as defined above) were as likely to undergo CABG or PCI, and as likely to be alive at one year, as were others not meeting the study definition for aboriginal ethnicity.

CONCLUSION: Individuals of aboriginal ethnicity living on native reservations were less likely to undergo cardiac catheterization than were others living in the province of Alberta, Canada. However, analyses controlling for age, sex, income, and location of residence suggest that it is those factors, and not aboriginal ethnicity per se, that explain the lower observed rates of cardiac catheterization.

DIABETES PREVALENCE BY ACCULTURATION STATUS AMONG MEXICAN AMERICANS. R. Gianani¹, T.D. Mackenzie¹; ¹University of Colorado Health Sciences Center, Denver, CO

PURPOSE: The prevalence of type II diabetes is much higher among Mexican Americans than among non-Hispanic Whites. Although much of this difference is thought to be genetic, environmental factors such as diet and exercise may also play a role. One way to try to separate genetic from environmental factors is to use acculturation data in immigrant populations. Prior studies on the effects of acculturation on the prevalence of diabetes in Mexican Americans have reported conflicting results. A study of Mexican Americans in San Antonio, Texas showed a decrease in diabetes rate with increasing acculturation. This decrease was hypothesized to be secondary to improved diet and increased levels of exercise in the more acculturated group. The Hispanic Health and Nutrition Examination Survey (HHANES) studied Mexican Americans in the Southwestern United States and found no effect of acculturation on rate of diabetes in Mexican Americans.

METHODS: All patients seen in the adult clinic of a large community health center in Denver, Colorado over a fifteen month period were included in our analysis. For each patient seen, diabetes was considered present if the ICD9 code for diabetes was billed as the primary diagnosis on at least one visit during the 15 months. Level of acculturation was assessed by using language preference. Hispanic patients who spoke primarily Spanish were considered less acculturated than those who spoke English. Demographic and language preference data were obtained from registration records.

RESULTS: Approximately 5,300 patients were included in our analysis. Of these, 1,500 patients were non-Hispanic, 2,600 were English-speaking Hispanics, and 1,200 were Hispanic who spoke primarily Spanish. By our current diagnostic criteria, overall diabetes prevalence was 14%. Hispanics had a higher prevalence of diabetes than non-Hispanics at all age groups and this difference was more pronounced with increasing age. In the 55–64 yo age group, 31% of Hispanics vs. 20% of non-Hispanics had diabetes. There was no significant difference in diabetes prevalence between primarily Spanish-speaking and English-speaking Hispanics.

CONCLUSION: We found no significant effect of acculturation on diabetes prevalence in this clinic population of Mexican Americans. Our findings also support previous reports of increased diabetes prevalence among Mexican Americans when compared to non-Hispanics, and a magnification of this effect in the higher age groups.

UNDERESTIMATION OF DISABILITY AMONG COMMUNITY-LIVING OLDER PERSONS. T. Gill¹, C. Williams¹; ¹Yale University School of Medicine, New Haven, CT

PURPOSE: When determining the incidence of disability, long assessment intervals may be problematic because they do not account for the possibility of recovery not for losses to follow-up due to deaths. In this study, we tested two related hypotheses: (1) longitudinal studies with long assessment intervals underestimate the incidence of disability; and (2) these underestimates increase as the length of the assessment interval increases.

METHODS: The study population included the 754 members of the Precipitating Events Project, an ongoing longitudinal study of nondisabled, community-living persons, aged 70 years or older. After a comprehensive, home-based assessment, participants were categorized into three groups according to their risk for disability (low, intermediate, and high) and were subsequently followed for up to two years with monthly telephone interviews (98% completion rate) to ascertain the presence of disability in four key activities of daily living (ADLs). We compared the rate of ADL disability obtained from a single follow-up assessment with that obtained from monthly assessments for intervals of 6, 12, 18, and 24 months.

RESULTS: For any ADL and for each of the four individual ADLs (walking, bathing, dressing, and transferring), the rates of disability at 12 months were at least twice as high for the monthly assessments as for the single follow-up assessment. For any ADL disability, for example, the corresponding rates were 0.22 and 0.10. Although the overall rates were lower, the results for persistent disability, defined as a new disability that was present for at least two consecutive months, were similar. As the length of the assessment interval increased, the "true" rate of disability, as determined by the monthly assessments, was increasingly underestimated by a single follow-up assessment. For example, the rates of any ADL disability for the monthly and single follow-up assessments were 0.28 and 0.09 at 18 months and 0.37 and 0.11 at 24 months, respectively. While these underestimates of disability were due almost exclusively to high recovery rates for low-risk participants, they were due increasingly to losses to follow-up from deaths for high-risk participants.

CONCLUSION: The incidence of disability is substantially underestimated by longitudinal studies with long assessment intervals. These underestimates could lead policy-makers to inaccurately estimate the health care needs of community-living older persons.

GIVING HOMELESS YOUTH INCREASED ACCESS TO MEDICAL CARE IN A SHELTER IMPROVED HEALTH OUTCOMES. C. Gilroy¹, C. Kwiatkowski¹, R. White¹; ¹HealthONE Presbyterian St. Luke's/UCHSC, Denver, CO

PURPOSE: Homelessness among youth has increased nationwide over the last decade, and recent counts in Colorado estimate 200–300 adolescents sleep on the street every night in Denver. While adolescents in general face many serious health risks, these youth face worse and more chronic health problems, psychiatric and behavioral problems, alcohol and substance

abuse, sexually transmitted diseases (STD) and teen pregnancies. School-based clinics have been reported to improve health among adolescents through increased access. We hypothesized that increased access to medical care within the shelter environment would improve health outcomes in this under served population.

METHODS: Homeless youth were surveyed in 1992 and 1998 at the only Denver-area shelter dedicated to youth under 21. The first survey, in 1992, occurred when medical services were provided in the shelter 8 hours per week. At the time of the second survey in 1998, medical services had increased to 36 hours per week. The surveys were of structured interview form based on the Adolescent Health Survey instrument. Outcomes assessed included access to health care, health satisfaction, STD's and pregnancies.

RESULTS: The two study samples differed somewhat with regard to demographics. In 1998, there were more males (62%) than in 1992 (55%). There were also substantially more African-Americans (16% vs. 7%). The proportion of African Americans among shelter users was twice the proportion of African Americans in the community. 'Good' or 'Excellent'; self-reported health increased from 65% to 74% between 1992 and 1998. Endorsement of frequent difficulty accessing care decreased from 31% to 14% between 1992 and 1998. Agreement with the statement, "the last time I needed healthcare I took care of it myself" decreased from 24% in 1992 to 8% in 1998. The number of self-reported STD's decreased from 26% in 1992 to 17% in 1998, compared with the 2.4% reported in Colorado high school students in 1996. This decrease was seen at the same time a Centers for Disease Control-sponsored STD screening project was working to reach homeless youth, and detection rates may actually have increased. The number of teen pregnancies decreased from 14% in 1992 to 6% in 1998.

CONCLUSION: Among homeless youth, increased access to medical care through increased availability of services in the shelter environment was associated with improvements in perceived health, access to health care, and reductions in some common health problems. Continued expansion of services within the shelter could have a profound impact on problems such as mental health and substance abuse.

HOMELESS YOUTH ARE AT INCREASED RISK FOR BRAIN INJURY AND PSYCHIATRIC SYMPTOMS. C. Gilroy¹; ¹HealthONE Presbyterian St. Luke's Hospital, Denver, CO

PURPOSE: Concussion has been associated with disturbance of physical well-being, thought process, and emotions, that may last 6 months or longer after the time of injury. Homeless adolescents are at higher risk for brain injury than adolescents from safer environments due to increased behaviors leading to injury. Impulsivity and affective liability are part of the DSM-IV definition of Postconcussional Disorder. These attributes have also been noted to be more common in homeless youth. We hypothesized an association between previous head injury and impulsivity and behavioral dyscontrol, as possible symptoms of postconcussional disorder.

METHODS: Homeless adolescents accessing services at a shelter were surveyed about health histories, including the statement "Have you ever been hit on the head so hard you lost consciousness". In the same survey a psychologic symptom battery, the SCL-90, was administered. The incidence of head injury was then compared with psychiatric symptom indices in the highest quartile. An odds ratio for the severity of psychologic symptoms in those with history of concussion was then calculated.

RESULTS: Of the 97 youth interviewed, 38 endorsed a history of grade III concussion, generating a prevalence of brain injury which is 40% higher than that reported in one study of high school football players. Odds ratios for psychiatric symptom score in the 75h percentile or greater among those with history of grade III concussion were as follows:

Symptom	Odds Ratio (95% CI)
Hostility	3.5(1.5 – 8.4)
Phobia	21(0.8 – 5.2)
Impulsivity	2.0(0.6 – 7.2)
Anxiety	1.5(0.6 – 5.4)
Obsessive/Compulsive	1.5(0.6 – 3.7)
Depression	1.3(0.5 – 3.4)

CONCLUSION: The symptoms of hostility, impulsivity, anxiety, and depression are included in the DSM-IV definition of Postconcussional Disorder. Impulsivity, anxiety, and depression were modestly, but not significantly, increased in those with history of grade III concussion. A significant association between hostility and concussion was found. As with other studies of brain injury in adolescents, the direction of the relationship between hostility and brain injury is unclear. However, the effect brain injury may have on the skills necessary to exit street life may be serious, and should be further investigated.

ECHINACEA: NO TRUTH IN LABELING? C. Gilroy¹; ¹HealthONE Presbyterian St. Luke's Hospital and University of Colorado Health Sciences Center, Denver, CO

PURPOSE: Echinacea is one of the most popular dietary supplements with over \$10 million in annual sales in the U.S. However, since dietary supplements are not regulated by the FDA in the same manner as other over-the-counter products, there is concern as to whether contents are consistent with the labeling. Three species of Echinacea are sold commercially, augustifolia (EA), pallida (EPA), and purpurea (EPU). Each species has a different biochemical footprint, analogous to different contents. Clinical trials have reported decreases in cold symptoms using EPU, decreases in cold symptoms and duration using EPA, and no effect on cold symptoms using EA. It follows that labeling of species content on Echinacea supplements is important consumer information. We hypothesized that species content in Echinacea preparations may not be consistent with labeling.

METHODS: Ten single-herb liquid-only preparations of Echinacea from 5 Denver area stores were purchased and sent to a blinded independent laboratory. The preparations were analyzed

by thin layer chromatography (TLC) for fingerprints unique to each species of Echinacea. Of the 10 samples sent, 3 were not analyzed due to difficulties with glycerol in the preparation. RESULTS: The following table shows the labeled and actual species contained in each Echinacea sample.

Sample	Labeled	Analysis	Difference
1	EA, EPU	EA	-1
2	EA	EA, EPA	+1
3	EPU	EA, EPU	+1
4	EPU	EPU	0
5	"Echinacea"	EPU	0
6	EA, EPU	EPA, EPU	+1/-1
7	EA	EA	0

CONCLUSION: In this modest sample, 57% of the samples did not contain the species they were labeled with. In 2 of the samples, the difference between content and labeling may have meant the difference between an 'active' or 'inactive' supplement. While many consumers object to federal regulation of dietary supplements, this difference between labeling and content emphasizes the need for better regulation.

WILL INSURED MINNESOTANS GIVE UP BENEFIT COVERAGE TO INCLUDE THE UNINSURED? S.D. Gould¹, A.K. Biddle², E. Benavides³, M. Danis⁴; ¹University of Michigan, Ann Arbor, Michigan; ²University of North Carolina, Chapel Hill, North Carolina; ³Independent Consultant, Minneapolis, MN; ⁴National Institutes of Health, Bethesda, Maryland

PURPOSE: Describe the willingness of insured Minnesotans to tradeoff their own health benefits to cover the uninsured.

METHODS: Thirty-one groups of laypeople (n=322) recruited through employers and community groups participated in CHAT "Choosing Healthplans All Together," (c) a simulation exercise in which participants choose whether and how extensively to cover health services in a hypothetical health plan. Choices are constrained by limited resources (\$5000/yr household insurance premium); participants allocate their resources amongst a variety of health services. We describe individual and group decisions whether to contribute 2% of their premium to cover uninsured children in Minnesota, or 4% of their premium to cover both uninsured children and adults. Bivariate associations between categorical variables used chi-square tests.

RESULTS: Twenty-nine groups and 282 individuals had complete data.

PARTICIPANT CHARACTERISTICS		HEALTH AND INSURANCE	
Mean age (+/- SD)	42.4(11.6)	Employer provides insurance	82.2%
Female	66.7%	Exc/Very good health status	77.6%
Caucasian	93.6%	Hospitalized last yr (household)	9.4%
Income < \$35,000	18.4%	Chronic illness (household)	28.0%

In initial selections before group discussion, 31.9% of individual participants used 4% of their allotted premium to provide all children with insurance; another 22.0% used an additional 4% to cover uninsured adults as well. In selections after group discussion, 62.4 percent of individuals chose to insure children only; 2.1% chose to insure both adults and children. Only higher levels of education were associated with decisions to include coverage for the uninsured ($p < .05$). In group decisions, all 29 groups chose to insure children; 22 of these chose to insure both adults and children. The uninsured category had the most individual-group disagreement of any health services category under consideration. Despite this disagreement, and others, 252/282 (89.2%) said they were willing to abide by the health plan developed by their group.

CONCLUSION: In the context of tradeoffs with their own health insurance benefits, all 29 groups of Minnesotans chose to contribute a portion of their premium to insure all children and most (22/29) chose also to insure uninsured adults. Although one-third of individuals differed by not selecting the uninsured category in their post-game selections, the vast majority of individuals were willing to abide by their groups' decisions.

EMPOWERING HOMELESS SUBSTANCE USERS: EVALUATION OF A WORK STABILIZATION PROGRAM. A.J. Gordon¹, J. Gibson², P.J. Freyder³, T.P. O'toole⁴; ¹The Program for Health Care to Underserved Populations, University of Pittsburgh, Pittsburgh, PA; ²Center for Research on Health Care, University of Pittsburgh, Pittsburgh, PA; ³The Public Inebriate Program, The Salvation Army of Pittsburgh, Pittsburgh, PA; ⁴The Johns Hopkins University, Baltimore, MD

PURPOSE: Homeless people with drug or alcohol addictions often lack continuity of health and social care. Structured, collaborative vocation assistance programs may provide means to improve the health and independence of this distressed population. We evaluated a unique community and academic partnered work stabilization program for substance using homeless individuals as an innovative means of improving health, housing, and social outcomes of this vulnerable population.

METHODS: We identified adult, homeless, vocationally-able, substance using people in drop-in clinics and substance abuse treatment settings. Consenting persons were enrolled in a program consisting of employment in a 90-day work component, financial counseling and monitored checking accounts, case management, and medical care. A structured interviews with clients at enrollment, after the work component, and six months after completion of the work component assessed for work stability, sobriety and clean time, social support and functional status, housing stabilization, and health service utilization.

RESULTS: Of the 182 persons interviewed, 120 (66%) eventually enrolled into the program. Enrollees in the work component were middle aged (mean age = 39), minority race (92%), and male sex (64%) and abusing primarily alcohol (31%), crack/cocaine (21%), heroin (5%) alone or poly-substance abuse (44%). They lived in emergency shelters (23%), doubled-up (49%), and bridge-housing (25%) arrangements. Self identified co-morbid medical and psychiatric illness were common and a third were on at least one prescribed medication. Fifty-seven (48% of enrolled) completed the work component of the program. Of these, 95% were subsequently employed at completion of the work component. During the work component, clients were abstinent (40%), improved their housing status (47%), attended medical appointments (87%), reported career skill improvement (69%), and arrest free (84%). Six months after the completion of the program, over 33% continued to be employed full-time, 39% had further improved their housing, and 36% remained abstinent of substance use. A vast majority reported having medical (90%) and psychiatric (93%) service needs met.

CONCLUSION: A work stabilization program, supported by a collaborative network of community and academic partners, decreased substance use, criminal behavior, and health care utilization for a drug and alcohol abusing homeless population. Community and academic partnerships can be a powerful impetus to promote health and social improvement for this distressed population.

DIAGNOSTIC VALUE OF ADENOSINE DEAMINASE (ADA) IN TUBERCULOUS PLEURAL EFFUSIONS: A META-ANALYSIS. M. Goto¹; ¹Kyoto University, Kyoto, Kyoto, Japan

PURPOSE: To summarize the test characteristics of ADA in diagnosis of pleural tuberculosis, we undertook a meta-analysis using a SROC curve.

METHODS: A search of the English-language medical literature published between 1966 and 1999 retrieved 98 reports. Studies were included if (1) the absolute numbers of true-positive (TP), false-negative (FN), true-negative (TN), and false-positive (FP) observations were available or derivable from the data presented; (2) the gold standards were described explicitly; and (3) the positivity criteria for ADA was reported. We constructed a summary receiver operating characteristic curve (SROC curve) out of extracted data, and estimated the test characteristics.

RESULTS: Forty-five reports met the inclusion criteria. Pathological study of pleural biopsy, microbiological examination of pleural fluid, clinical course and so on were used as gold standards. The summary measure of test characteristics derived from the SROC curve was 92.8% for sensitivity and specificity, respectively.

CONCLUSION: The test performance of ADA in tuberculous pleural effusion is reasonably good, and ADA level measurement in pleural effusion would be a useful diagnostic tool for diagnosing of tuberculous pleuritis.

THE OUTCOME OF PHYSICAL SYMPTOMS WITH TREATMENT OF DEPRESSION. I. Greco¹, K. Kroenke²; ¹St. Vincent Hospital, Indianapolis, IN; ²Regenstrief Institute and Indiana University, Indianapolis, IN

PURPOSE: The majority of depressed patients initially present in primary care with physical symptoms rather than psychological complaints. We examined the outcome of physical symptoms with antidepressant treatment of depression during a 9-month period.

METHODS: This clinical trial enrolled 573 depressed patients cared for by 40 physicians in two primary care research networks. Patients were randomized to receive one of three selective serotonin reuptake inhibitor (SSRI) antidepressants: paroxetine, fluoxetine, or sertraline. Data was collected from participants via computer-administered telephone interviews both at baseline and after 1, 3, 6, and 9 months of treatment. This data included validated measures for depression, anxiety, and 14 common physical symptoms (SCL-20 and PRIME-MD), health-related quality of life (SF-36), work functioning (Work Limitation Questionnaire), and Medical Outcomes Study measures for social functioning, positive well-being, sleep, concentration, and sexual functioning. The prevalence of each physical symptom was assessed at baseline and all follow-up intervals and symptom severity graded from 0 (none) to 2 (bothered a lot). Stepwise linear regression models were used to determine the independent effects of physical symptoms and depression on HRQoL and other domains at baseline and over 9 months of treatment. All models were adjusted for demographics and anxiety scores.

RESULTS: Most of the individual symptoms had a baseline prevalence in depressed patients of 50% or greater. Moreover, the prevalence of symptoms that were graded as "bothered a lot" exceeded 10% for 11 of the 14 symptoms. Physical symptom prevalence and severity dropped substantially during the first 4 weeks of antidepressant therapy with minimal improvement thereafter. While also showing the greatest improvement during the first month, depression and all other outcomes continued to improve over the 9-month trial. Physical symptoms had the strongest association with physical functioning (11%–20% of the variance), pain (23%), and overall health perceptions (19%), whereas depression had the greatest association with mental (32%–49%), social (11%–36%), and work functioning (9%–37%).

CONCLUSION: Many physical symptoms are present in at least half of depressed patients presenting in primary care. The predominant benefits of antidepressant therapy occur during the first 4 weeks of treatment for physical symptoms, while depression and other HRQoL domains continue to improve for some months thereafter. Physical symptoms and depression have differential effects on HRQoL domains.

PREVALENCE OF ARTHRITIS IN A MANAGED CARE POPULATION. D.C. Green¹, J.K. Rao², L.A. Schild¹; ¹USQA Center for Health Care Research, Atlanta, GA; ²Durham VAMC, Durham, NC

PURPOSE: Arthritis is a prevalent chronic condition and a leading cause of disability among Americans. While national and state-based surveys provide population estimates of arthritis, little is known about its prevalence in managed care populations. This analysis describes the prevalence and characteristics of persons with self-reported arthritis in a national managed care organization.

METHODS: We surveyed a stratified random probability sample representative of managed care members aged 45 and older in 2 health plan locations (n=2168). Respondents reporting doctor-diagnosed arthritis or chronic joint symptoms (pain, stiffness or swelling of a joint within the past year that is present on at least half the days of a typical month) were considered to have arthritis. Weighted descriptive analyses were conducted to estimate the age- and gender-specific arthritis prevalence, and to describe the impact of arthritis (i.e., health status, activity limitations). We also describe treatments respondents reported using for arthritis and the type of physician who provided the arthritis diagnoses.

RESULTS: The prevalence of arthritis ranged from 32.0% for the 45–54 age group to 36.8% for the ≥75 age group. Women were more likely to report arthritis than men (38.3% vs. 29.9%). Compared to those without arthritis, respondents with arthritis were more than twice as likely to report fair or poor health (24.9% vs. 11.1%). Of those with arthritis, 39.9% indicated activity limitations related to their joint symptoms. Respondents reported using prescription medicines (43.9%), over-the-counter pain medications (57.2%), complementary therapies (i.e., herbs, glucosamine) (32.8%), physical or occupational therapy (20.2%), and exercise (i.e., walking, swimming) (70.5%) to treat their arthritis symptoms. Just over half of those reporting physician-diagnosed arthritis received their diagnosis from a primary care physician.

CONCLUSION: One-third of these managed care members have self-reported arthritis and 40% report arthritis-related activity limitations. The prevalence and impact of arthritis on health-related quality of life suggest an opportunity to explore interventions, including education on self-help activities, that may complement care provided by clinicians.

KNOWLEDGE, BELIEFS, AND BEHAVIORS AMONG MULTI-CULTURALLY DIVERSE COLLEGE WOMEN CONCERNING THE PREVENTION OF OSTEOPOROSIS. J.P. Grimes¹, J.A. Orzano¹, F. Frech², R. Russo², A. Petrova¹, C. Delnevo². ¹UMDNJ-Robert Wood Johnson Medical School Department of Medicine and Family Medicine, New Brunswick, NJ; ²UMDNJ-School of Public Health

PURPOSE: Few researchers have evaluated what young women know about osteoporosis, their beliefs about the disease, and to what extent they practice preventive behaviors. None have explored these issues in a multi-culturally diverse population of young women. The purpose of this exploratory study was to assess knowledge, beliefs and behaviors about osteoporosis prevention in this target population.

METHODS: We performed a cross-sectional study of multi-culturally diverse college women attending classes over a two week period in 2000. In this convenient sample (N=50), we collected data using a survey instrument of 20 questions designed to measure knowledge, beliefs and behaviors about osteoporosis prevention. Incorporating the self-reported data, we characterized the respondents by demographic information, medical conditions and family history; calculated Body Mass Index, general consumption of calcium, their level of alcohol and nicotine use and degree of physical activity; and tabulated a "Belief-about-Disease" Score to capture their perception of osteoporosis in relation to other medical conditions.

RESULTS: Nearly a third of the respondents were non-White and 14% were of Asian origin. Respondents were able to identify the risk factors of low calcium intake(96%), lack of exercise(86%), and menstrual irregularities(68%). However, other risk factors such as White race(20%), Asian race(21%), and excessive exercise (42%) were less likely to be identified. Ten percent had a medical condition putting them at risk. There was no relationship between correctly identifying lack of exercise and low calcium intake as risk factors and getting adequate osteoprotective exercise (> 360 minutes/month)(p=0.53) and dietary calcium(> 1200mg/daily) (p=0.85). Osteoporosis (3.31) (on Likert scale from 1–5, 1 indicates not at all; 5, extremely) was believed to be significantly less serious than heart disease(3.60), breast cancer(3.66), AIDS(3.79), and Alzheimer's disease(3.54)* (*p<0.05). There were no associations between ethnicity and respondents' responses, knowledge and beliefs.

CONCLUSIONS: Many college women, regardless of race, lacked knowledge about the relationship between osteoporosis and low calcium intake, use of tobacco, alcohol, physical inactivity, and excessive exercise. Increasing levels of osteoporosis knowledge was not associated with young women's beliefs and behavior. Our data suggest a need to educate young women. Comprehensive osteoporosis educational programs during college should be evaluated as an intervention to improve osteoporosis prevention.

THE COST-EFFECTIVENESS OF AUTOMATED EXTERNAL DEFIBRILLATORS ON COMMERCIAL AIRCRAFT. P.W. Groeneveld¹, J.L. Kwong¹, Y. Liu¹, A.J. Rodriguez¹, M.P. Jones², G.D. Sanders¹, A.M. Garber³. ¹Stanford University, Stanford, CA; ²Front Line Strategic Management Consulting, Foster City, CA; ³Stanford University and VA Palo Alto Health Care System, Palo Alto, CA

PURPOSE: Recently published studies suggest that automated external defibrillators (AEDs) on commercial aircraft may save the lives of passengers who have out-of-hospital cardiac arrest (OHCA). However, AED equipment and training costs are high, while cardiac arrests onboard aircraft remain rare. We conducted a cost-effectiveness analysis to explore the economic and health impact of AED deployment in the U.S. passenger air fleet.

METHODS: We developed a decision analytic model using a societal perspective for costs and benefits. Estimates of the incidence of OHCA onboard aircraft, effectiveness of AEDs in resuscitation, subsequent hospital survival rate, and annual mortality after surviving OHCA were derived from the medical literature. Equipment, maintenance, and flight attendant training costs were obtained from AED manufacturers, the Federal Aviation Administration, and the Air Transportation Association. Hospitalization and downstream medical costs were abstracted from published cost analyses. Published estimates of health-related utility after cardiac arrest were used to express effectiveness in terms of quality-adjusted life years (QALYs) gained. We compared four strategies: 1) AEDs on all aircraft with basic life support (BLS) training for flight attendants; 2) AEDs on wide-body aircraft only; 3) AEDs on no aircraft—but attendants trained in BLS; and 4) no aircraft AEDs and no BLS training.

RESULTS: Placing AEDs on all passenger aircraft with concurrent BLS training would cost \$49,800 per QALY gained compared to no AEDs and no BLS training. BLS training without AEDs was dominated by a combination of strategies #1 and #4. Deploying AEDs exclusively on wide-body aircraft would be cost-effective only if total training costs could be limited to less than 49% of the cost of training all attendants. Sensitivity analyses indicated that the ability of AEDs to improve survival was the most important influence on cost-effectiveness. AEDs must demonstrate an incremental survival rate of at least 14% to have a cost-effectiveness ratio less than \$50,000/QALY. A Monte Carlo analysis, which varied all assumptions simultaneously, indicated that the cost/QALY value for strategy #1 would be less than \$50,000 at a probability of 34%, while the probability of the cost-effectiveness being greater than \$100,000/QALY was only 0.1%.

CONCLUSION: The use of AEDs on passenger aircraft is similar in cost-effectiveness to many widely accepted medical interventions and health policy regulations. Sensitivity analysis suggests this result is robust, even when underlying assumptions are varied widely. This study implies that requiring the placement of AEDs on U.S. commercial aircraft would be a cost-effective use of resources.

ACTIVE CLINICAL ISSUES ON HOSPITAL DISCHARGE ARE ASSOCIATED WITH WORSE CLINICAL AND FUNCTIONAL OUTCOMES IN PATIENTS WITH HIP FRACTURE. E.A. Halm¹, J. Wang¹, A.L. Siu¹. ¹Dept of Health Policy, Mount Sinai School of Medicine, NY, NY

PURPOSE: Pressures to shorten length of stay increase the chance that patients (Pts) may be discharged 'sicker and quicker.' Pts hospitalized with hip fracture are very frail and maybe at particularly high risk. We sought to measure prevalence of active clinical issues (ACIs) on discharge (DC) in Pts hospitalized with hip fracture and their associated clinical and functional outcomes.

METHODS: Information on vital signs, eating status, mentation, mobility, incontinence, and wound status, and other active problems on DC, and in the 24 and 48 hours prior to DC was collected on 559 Pts hospitalized with hip fracture as part of a 4 hospital prospective cohort study. Deaths, readmission, and functional mobility within 60 days of DC were ascertained by telephone interviews and querying the NY State hospitalization database. Mobility was measured with the validated Functional Independence Measure-Locomotion scale (FIM). Logistic and linear regression assessed associations between the number of ACIs on DC and post-DC outcomes. ACIs on DC were defined as any of the following in the 24 hrs prior to DC (Temp > 100, 100 > HR > 60, SBP < 90, DBP < 60, RR > 24, O2 sat < 90%, altered mentation, inability to eat, incontinence, and not mobilized beyond a chair (all different from baseline), and other active medical problems [dyspnea, chest pain, arrhythmias, CHF, or wound infection]). We used a validated risk adjustment model to control for covariates known to influence clinical and functional hip fracture outcomes.

RESULTS: Pts mean age was 81 yrs, 82% were female, 22% had dementia, and 12% came from a nursing home. The mean length of stay was 8.5±6.2 days. Overall, 74% of Pts had ≥1 ACI on DC (29% had 1, 28% had 2, and 17% had ≥3 ACIs), most commonly not mobilized beyond a chair (37%), new urinary incontinence (34%), Temp > 100 (11%), abnormal mentation (5%), DBP < 60 (5%), and new bowel incontinence (3%). Within 60 days of DC, 3.8% of Pts died, 18.8% were readmitted, and 20.2% died or were readmitted (major events). The mean FIM locomotion score at Day 60 was 6.1±4.0 (scale from 2–14; higher is better). The greater the number of ACIs on discharge, the higher the risk of post-DC death, readmission, and major events and worse functional mobility [FIM score] (p < .001 for all). The odds of major events increased 40% for every additional ACI on DC (OR=1.4; 95% CI 1.1–1.8). The number of ACIs on DC remained a significant predictor of all post-DC outcomes even after controlling for age, sex, comorbidities, initial APACHE score, and pre-fracture nursing home residence, dementia, home health assistance, and FIM score. Similar results were obtained for outcomes at 30 and 90 days as well as for analyses that considered only the last set of vital signs along with the other ACIs or ACIs in the 48 hours prior to DC.

CONCLUSION: The greater the number of ACIs on DC, the worse the post-DC clinical and functional outcomes even after controlling for other important prognostic factors. Clinicians should factor this information into deciding appropriateness for DC and the type and intensity of post-DC care and medical observation.

DEMOGRAPHIC CHARACTERISTICS OF PEOPLE WITH EXTREMELY ELEVATED HIGH DENSITY LIPOPROTEIN. A. Hamad¹, L. Ward¹, M. Zaghmout¹, J. Singh¹, H. Mahmoud¹, M. Salameh¹. ¹Nassau University Medical Center, Eastmeadow, NY

PURPOSE: Elevated HDL is considered a negative risk factor for ischemic heart disease and HDL level > 75mg/dl has been found to be associated with the "longevity syndrome". We wanted to study people with extreme HDL elevation as no studies have been done to evaluate them before.

METHODS: We identified all persons presented to our hospital with HDL level > 95mg/dl in the last 2 years. All subjects were contacted by phone.

RESULTS: 102 subjects were contacted (83 females and 19 males). The majority were African Americans (52 subjects) (13 males and 39 females). The mean age was 59.3 ± 1.5 years. The mean cholesterol was 234 ± 5 mg/dl with a mean HDL of 108 ± 2 mg/dl, triglyceride of 85 ± 4 mg/dl, and LDL of 112 ± 5 mg/dl. The total cholesterol to HDL ratio was 2.4 ± 0.2. Mean Body Mass Index (BMI) was 25.8 ± 0.6 kg/m². (table)

Only 15% were younger than 50 years old. We found an unusually high prevalence of hypertension in our subjects (53%) especially in the African American population (71%). Seven patients have coronary artery disease (CAD) but all had other cardiac risk factors.

CONCLUSION: Our study showed a predominance of African Americans and postmenopausal females. This population had normal TG, LDL, and BMI, and a high prevalence of hypertension and smoking. Half exercised or reported following a special diet. Although elevated HDL is associated with longevity syndrome, it may not be protective against CAD in people with multiple other risk factors.

comorbid conditions in our population with extreme HDL elevation

	Patients Number (%)
CAD	7 (6.8%)
Stroke	8 (7.8%)
diabetes mellitus	16 (15.5%)
hypertension	52 (51.8%)
age > 60 years	49 (48.6%)
chronic renal failure	8 (7.8%)
obesity (>30kg/m2)	16 (15.5%)
postmenopausal	70 (69.4%)

DECISION-MAKING AND OUTCOMES FOR ELDERLY PATIENTS WITH SEVERE, SYMPTOMATIC AORTIC STENOSIS. M.B. Hamel¹, P.S. Douglas², J.Y. Wei¹, R.S. Phillips¹; Beth Israel Deaconess Medical Center, Boston, MA; University of Wisconsin School of Medicine, Madison, WI

PURPOSE: Patients with severe, symptomatic aortic stenosis (AS) are at high risk for mortality. Aortic valve replacement (AVR) can prolong survival and alleviate symptoms, but may have substantial morbidity in older patients. Decisions about whether to pursue major cardiac surgery for elderly patients are often difficult.

METHODS: We studied decision-making and outcomes for patients age 60 and older who had severe aortic stenosis (mean valve gradient >50mmHg or valve area <0.8cm by echocardiogram) with associated symptoms. Patients were identified from databases containing results of inpatient and outpatient echocardiograms performed at two urban teaching hospitals in 1996 or 1997. We reviewed hospital records as well as office notes from patients' primary care doctors, cardiologists and cardiothoracic surgeons to describe decision-making and outcomes.

RESULTS: For the 94 patients studied, 62% were female, the mean (sd) age was 82 (8), and the mean (sd) number of comorbid illnesses was 1.5 (1.2). Baseline functional status information was available for 81 patients; 27% of patients needed assistance with at least one basic activity of daily living (ADL=walking, transferring, dressing, bathing, toileting, feeding). By 60 days after the date of the echocardiogram, a decision NOT to have AVR was documented for 21 patients. Of these 21 patients, 10 were not offered cardiac catheterization to further evaluate their AS, 9 were advised not to have or were not offered AVR by a cardiologist or cardiothoracic surgeon, and 2 declined further evaluation for AVR. For the remaining patients a decision was made to proceed with AVR or no decision had been documented by 60 days. At the time of the chart review (2-4 years after echocardiogram), 37 patients (39%) had AVR and 13 (14%) had palliative aortic valvuloplasty. Compared with patients who did not have AVR, patients who had AVR were younger (78 vs 84, $p < .0001$), more likely to be male (46% vs 26%, $p=.05$), had fewer comorbid illnesses (mean number comorbidities 1.1 vs 1.7, $p=.04$), and were less likely to need assistance with ADLs (mean number of ADL dependencies 1.1 vs 1.7, $p=.04$). In a multivariable logistic regression model including age, sex, number of comorbid illnesses, and ADL dependency, these differences persisted, but only the effect of age was statistically significant at $p < 0.05$. Of the 33 patients who had AVR at one of the study hospitals, 19 also had coronary artery bypass surgery, 2 also had mitral valve repair or replacement, 2 died postoperatively, and 1 suffered a stroke.

CONCLUSION: In this cohort of elderly patients with severe, symptomatic AS, the majority did not have AVR surgery. Patients who were younger and who had better baseline health status were more likely to have AVR. Few patients who were selected for AVR died postoperatively or suffered a major surgical complication.

INADEQUATE FOLLOW-UP OF BREAST ABNORMALITIES IN LOW-INCOME WOMEN. J.H. Han¹, J.A. Anapolsky¹, S. Reich¹, K.M. Freund¹; Boston University, Boston, MA

PURPOSE: African American women experience higher breast cancer mortality than Caucasian women despite increasing rates of screening mammography. One postulated factor in these worse outcomes is a lower rate of follow-up for diagnostic testing once a breast abnormality is detected. We examined factors associated with inadequate follow-up in a referral group of predominantly low-income minority women.

METHODS: The study population consisted of women referred to a Breast Center at an urban medical center from January to June 2000. Demographic information was collected via medical charts, registration files completed at the time of appointment scheduling, and consultation/referral forms. Adequate follow-up was defined as patient arrival to the scheduled Breast Center appointment and inadequate follow-up was defined as failure to arrive on the designated appointment day. We analyzed factors associated with inadequate follow-up such as patient age, race, median household income by zip code, referral source, and insurance type.

RESULTS: Of 390 women referred to the Breast Center during the study period, 51% were African American, 31% Caucasian, 14% Hispanic, and 4% Asian. 67% had public or no health insurance. Reasons for referral included abnormal clinical breast exam (42%), abnormal screening mammogram (10%), follow-up for prior breast abnormality (22%), or other breast symptoms (25%). 51% were referred from neighborhood health centers, 41% from hospital-based offices, and 8% from private practice. During the study period, 36% had inadequate follow-up for their breast abnormality. Factors associated with inadequate follow-up included patient age, insurance type, and referral source. 44% of women under age 40 had inadequate follow-up, compared to 30% for women age 40-64 and 28% for those older than age 65 ($p < 0.01$). 38% of women with public or no health insurance had inadequate follow-up while 23% with private insurance had inadequate follow-up ($p < 0.01$). 40% of women with a neighborhood health center referral source had inadequate follow-up vs. 30% of patients with a hospital-based office referral ($p=0.06$). Race, median household income by zip code of residence, and reason for referral were not associated with inadequate follow-up.

CONCLUSION: In this referral group of predominantly low-income minority women, 36% did not have adequate follow-up for a breast abnormality. Younger age, lack of private health insurance, and neighborhood health center referral were associated with inadequate follow-up.

These factors may contribute to the observed disparities in breast cancer outcomes of African American women.

CLINICAL PREDICTION RULE FOR MAINTENANCE WARFARIN DOSE. R. Hanratty¹, L.E. Feinberg¹, K. Vu¹, R. Gonzales¹; ¹University of Colorado Health Sciences Center, Denver, CO

PURPOSE: Warfarin is a commonly prescribed oral anticoagulant with a narrow therapeutic window. Current recommendations are to initiate warfarin with a 5 mg daily dose, and make adjustments thereafter. Because approximately 40% of individuals have significantly lower or higher maintenance requirements, a better approach to predicting maintenance dose could reduce adverse events and be more cost-effective.

METHODS: A retrospective chart review of patients enrolled in a university-affiliated anticoagulation clinic was performed to develop a clinical prediction rule for maintenance warfarin dose for patients with a goal International Normalized Ratio (INR) of 2.0 to 3.0. Maintenance dose was defined as being greater than one month out from initiation of warfarin therapy with at least two consecutive measurements one week apart within the target range. Potential predictor variables included: age, sex, weight, race, indication for anticoagulation, individual medications and total number of medications. A combined approach using recursive partitioning and multivariate logistic regression was used to develop and evaluate potential prediction rules. Two different rules were tested: one looking at low maintenance requirements (less than or equal to 3 mg per day), the other looking at high maintenance requirements (greater than or equal to 7 mg per day).

RESULTS: 131 patients meeting entry criteria were identified with an average age of 68 years (range 27 to 93 years), 52% male sex, 70% Caucasian race, 75% anticoagulated for either atrial fibrillation (38%) or deep venous thrombosis/pulmonary embolus (37%), average weight of 184 pounds (range 100 to 350 pounds) and average number of chronic medications (in addition to warfarin) of 6 (range 0-29). The simultaneous presence of 3 of the following variables predicted low maintenance requirements with 80% specificity (95% CI 73-87%), 67% sensitivity (95% CI 59-75%) and a positive predicted value of 92% (95% CI 87-97%): age greater than 50, weight less than 200 pounds and being on 5 or more chronic medications. Although there were factors (age less than 50 years, and weight greater than 200 pounds) that on univariate analysis were associated with high maintenance requirements, a statistically meaningful rule could not be developed.

CONCLUSION: A prediction rule that identifies those with the lowest warfarin maintenance requirements has been developed which could potentially aid in determining how to initiate warfarin therapy. This rule needs to be validated and refined using different patient populations.

RURAL LESBIAN WOMEN: RISK FACTORS AND SCREENING BEHAVIOR RELEVANT TO BREAST AND CERVICAL CANCER. D. Harper¹, E. Konings¹, K. Masuda¹, L. Titus-Ernstoff¹; ¹Dartmouth Medical School, Lebanon, New Hampshire

PURPOSE: Our objectives were three-fold: To conduct a pilot study of risk factors and screening behaviors relevant to breast and cervical cancer in rural lesbian women, to assess women's willingness to participate in a study of these factors, and to assess the feasibility of contacting women through regional lesbian organizations.

METHODS: Using a brief self-administered questionnaire, we obtained information concerning major risk factors and screening behaviors relevant to breast and cervical cancer. The questionnaire, targeted to rural lesbian who were at least 40 years of age, was distributed at lesbian community events in rural New Hampshire and Vermont between June and August 1997. A separate survey, assessing willingness to participate in a future study, was distributed (between October and December 1998) by regional lesbian organizations to their membership. **RESULTS:** The first questionnaire was completed by 105 women. Of these, 82% reported at least one physician visit during the previous year. With regard to breast cancer risk factors and screening, 66% were nulliparous, 66% had at least one clinical breast exam during the previous year, 43% had had a mammogram, and 40% conducted breast self-exams. With regard to cervical cancer risk factors and screening, 80% had a history of heterosexual activity, and 62% had undergone a Pap smear during the previous year. Over the past 3 years, 21% had an abnormal pap smear. The later survey assessing willingness to participate in breast and cervical cancer research was mailed to 250 women; 47 surveys were undeliverable due to unknown address. Of the remaining 203 (62%), 108 women (53%) expressed willingness to participate.

CONCLUSION: This pilot study suggests that breast cancer risk may be higher among lesbian women than non-lesbian women, and that lesbian women are at risk of cervical cancer despite of their current sexual practices. Our preliminary data also suggest that rural lesbian women are medically underserved with regard to breast and cervical cancer screening. The mailed survey also supports the feasibility of recruiting large numbers of rural lesbian women for health research.

PREVALENCE AND TREATMENT OF MENOPAUSAL SYMPTOMS AMONG BREAST CANCER SURVIVORS. P.F. Harris¹, P.L. Remington¹, A. Trentham-Dietz¹, C. Allen¹, P.A. Newcomb¹; ¹University of Wisconsin-Madison, Madison, WI

BACKGROUND: Women diagnosed with breast cancer often experience early menopause secondary to treatment effects, yet physicians may be reluctant to prescribe hormone replacement therapy (HRT) in order to reduce their risk of cancer recurrence.

The objective then is to assess the burden of menopausal symptoms, HRT use, and alternative treatments in recent breast cancer survivors compared to age-matched controls.

METHODS: This is a population-based case-control study using breast cancer survivors and age-matched controls. Wisconsin women 18-74 years old with a new diagnosis of breast cancer 8-11 months prior to interview (n=110) and control subjects randomly selected from population lists (n=73). A standardized telephone questionnaire was administered to elicit information on menopausal symptoms, estrogen and alternative (prescription medications, vitamins, herbal preparations, soy products, acupuncture, chiropractic) therapies used to alleviate symptoms.

Multivariate logistic regression was used to obtain odds ratios between cases and controls. Main outcomes were symptoms of menopause, use of estrogen, and use of alternative therapies.

RESULTS: Breast cancer survivors were 5.4 (95% CI 2.9–10.4) times more likely to experience symptoms, 25 (95% CI 8.3–100) times less likely to use estrogen, and 7.4 (95% CI 2.5–21.9) times more likely to use alternatives than controls. Soy, vitamin E, and herbal remedies were the most common alternative therapies reported by participants; use was greater in cases compared to controls. Among cases, tamoxifen users (n=62) reported a higher prevalence of symptoms and a higher prevalence of alternative treatments. Most soy users reported increasing soy products specifically to reduce the chances of a diagnosis of recurrent breast cancer.

CONCLUSION: This is the first population-based survey of menopausal symptoms and treatments that compares breast cancer cases with disease-free controls. Prevalence of menopausal symptoms and use of alternative menopausal therapies were higher in breast cancer survivors than in controls; the rate of estrogen use was much lower. The increased use of soy products in this population has not been previously documented.

Reported Use of Nonestrogen Treatments: WI Breast Cancer Cases, 1998–1999

Nonestrogen Preparations	Controls n=73	Cases n=110	Odds ratio	p value
All	5	29	7.4	0.003
Herbals	7	8	1.4	0.26
Vitamin E	3	12	3.3	0.08
Clonidine	0	3	—	—
Vitamin B6	0	4	—	—
SOY:				
For any reason	3	12	3.7	0.08
For symptoms	1	8	7.2	0.12
Cancer prevention	2	7	3.1	0.24

INCREASING SCREENING MAMMOGRAPHY AMONG RETICENT MEDICARE BENEFICIARIES: A PERSONALIZED LETTER INTERVENTION. R.V. Harrison¹, N.K. Janz¹, R.A. Wolfe¹, P.J. Tedeschi¹, M.E. Chernew¹, J.K. Stross¹, L.F. McMahon¹; ¹University of Michigan, Ann Arbor, MI

PURPOSE: (1) Develop standard methods to identify the mammography status of Medicare beneficiaries. (2) Test the effect of a personalized letter directed to women who have not had a recent mammogram — reticent beneficiaries. (3) Assess the cost-effectiveness of the intervention.

METHODS: HCFA data for Medicare beneficiaries were used to identify women living in Michigan continuously from 1993–98, age ≥65 in 1993, with no obvious comorbidity affecting screening, and no mammogram for ≥5 years (1993–97). A randomized design included paired intervention and control women matched on residential zipcode and race. The study sample of 2,458 women had 1,229 pairs of either African American (AA) or non-AA women and either urban or rural women. The intervention used principles from the Health Belief Model in a personally addressed letter from the Medical Director of Michigan Medicare noting the individual's lack of use of the mammography screening benefit, with additional breast cancer related information enclosed. Letters were sent in Nov. 1997 with Medicare mammography claims followed through 1998.

RESULTS: All women were age ≥70, with a mean of 79 in both control and intervention groups. Overall, 5.2% of controls and 8.1% of the intervention group subsequently had mammograms, + 2.9%, OR 1.6 (p < .005). The findings were similar by urban/rural and AA/non-AA subgroups. Rates and the effect were higher among the women age 70–79: 6.5% of control and 10.6% of intervention, + 4.1%, OR 1.6 (p < .02). Projecting to the Medicare population in Michigan, a total cost of \$117,000 to \$278,000 for a state wide intervention would produce 3,700 to 5,000 mammograms at \$32 to \$54 per additional mammogram in this reticent group.

CONCLUSION: This approach for measurement and intervention is feasible, effective, and can be directly implemented in other states and nationally. Targeting a reticent group of older women, this intervention demonstrated a significant improvement in subsequent mammography. This targeted approach is likely to be more cost effective than “blanket” community-based approaches with significantly higher overall cost but the same target group of reticent older women. Future research can address variations to increase the communication's effectiveness (e.g., multiple communications). The approach to measurement and intervention can be tried for other preventive services in the Medicare or other defined populations.

PATIENT KNOWLEDGE OF OUTPATIENT MEDICATION INDICATIONS. H.L. Heiman¹, T.K. Gandhi¹, S.N. Weingart¹, J. Borus¹, E. Burdick¹, A. Seger¹, D.S. Seger¹, L.L. Leape¹, D.W. Bates¹; ¹Brigham and Women's Hospital, Boston, MA; ²CareGroup Health Systems, Boston, MA

PURPOSE: Studies show that patient education about medications can improve adherence. However, few data are available about how often patients do not know or misunderstand the reasons for their medications. We studied a group of general medicine outpatients to assess patient understanding of their medications.

METHODS: Patients seen in four Boston general medicine practices from Sept 1999 to March 2000 (6 weeks per clinic) and who received at least one prescription from a study physician (6 MDs per clinic) were eligible for the study. Approximately two weeks after the index visit, patients were telephoned (response rate 59%) and asked to retrieve all of their pill bottles, read the name of each medication, and state their reason for taking it. Answers were coded “know” or “don't know,” and the reasons were recorded verbatim. The accuracy of “know” responses was coded by a physician-reviewer on a 5-point scale.

RESULTS: Of 598 patients, 58 (9.7%) did not know or gave definitely inaccurate indications for at least one medication. The drugs most often associated with inaccurate indications were blood pressure and lipid-lowering agents. An example of an inaccurate response was “atorvastatin is for my blood pressure.” Education and younger age were directly related to knowledge (see Table). There was a trend toward association of non-white race and male gender with poorer medication knowledge, but they were not statistically significant. Patients on more than one medication were far less likely to know the indication of all medications (p < .001).

CONCLUSION: In this relatively well-educated population, knowledge about drug indications was high. However, older and less educated patients, as well as patients on multiple medications, were much less likely to know why they were taking all of their medications. Such patients may benefit from intensive education programs to enhance understanding of their medications, which could improve adherence.

Predictors of Medication Knowledge

	Do Not Know All Med Indications (n=58)	Know All Med Indications (n=598)	p value
Mean age	70	50	0.001
% Not finished high school	17%	3%	0.001
% Non-white	28%	18%	0.1
% Female	58%	67%	0.14

IMPROVING ADHERENCE TO GUIDELINES: IMPACT OF AN OPINION LEADER QUALITY IMPROVEMENT INTERVENTION. G.R. Heudebert¹, E.S. Berner¹, S. Baker¹, E. Funkhouser¹, C.A. Fargason¹, N.W. Weissman¹, C.I. Kiefe¹; ¹University of Alabama at Birmingham, Birmingham, AL

PURPOSE: Adherence to evidence-based practice guidelines is frequently suboptimal. We wished to determine whether an opinion leader quality improvement project would increase adherence to the Unstable Angina (UA) AHCPR guidelines compared to a traditional HCFA quality improvement model.

METHODS: We designed a three-armed randomized controlled trial of 22 acute care hospitals in one state. The intervention arms were: 1) hospital-specific data-feedback of performance on quality indicators combined with a physician opinion leader-driven improvement intervention (OL); 2) hospital-specific data-driven feedback with traditional HCFA quality improvement model efforts (HCFA); and 3) no intervention (NI). We convened a national panel of experts to translate selected elements from the AHCPR UA guidelines into quality measures. We selected the following indicators: 1) ASA within 24 hours of admission, 2) ASA at discharge, 3) heparin during hospitalization, 4) beta-blockers at discharge, and 5) EKG within 20 minutes of presentation. For chart abstraction, we identified potential cases of UA using a stratified random sampling scheme based on ICD-9 codes from Medicare claims data. A computerized algorithm was developed to confirm UA cases from the abstracted data. Centrally trained abstractors reviewed the complete medical records. We used GLM to adjust for patients-nested within hospitals.

RESULTS: Charts were abstracted for 2,516 patients; their average age was 72 ± 11 years, of whom 45% were male and 85% were white. Average baseline performance for the indicators was 71.6 (ASA in 24hrs), 69.1 (ASA at discharge), 40.2 (heparin), 55.8 (beta-blockers), and 49.4 (EKG). All indicators, except EKG within 20 minutes, showed greater improvement in the OL group. Results did not change in the adjusted analysis. The table reports percent change in adherence to indicators from baseline to remeasurement.

CONCLUSION: No OL hospitals showed deterioration for any indicator, whereas in the other two groups performance decreased somewhat on three indicators in each group. There was statistically significant improvement for OL hospitals in ASA within 24 hours. OLs appear to contribute some added improvement over traditional HCFA models.

	ASA 24hr [*]	ASA discharge	Heparin	Beta-blockers	EKG 20'
OL	12.05%	0.17%	9.9%	1.93%	0.82%
HCFA	-4.66%	-1.46%	6.4%	-1.34%	4.14%
NI	-0.74%	-4.28%	4.1%	-6.37%	5.95%

*: p < 0.05 (Breslow-Day heterogeneity test)

SATISFACTION WITH TREATMENT DECISIONS FOR CLINICALLY LOCALIZED PROSTATE CANCER. RESULTS FROM THE PROSTATE CANCER OUTCOMES STUDY. R.M. Hoffman¹, W.C. Hunt², S.N. Stone², F.D. Gilliland³; New Mexico VA Health Care System, Albuquerque, NM; New Mexico Tumor Registry, Albuquerque, NM; University of Southern California, Los Angeles, California

PURPOSE: Men with clinically localized prostate cancer have several treatment options including conservative management (C), radiation therapy (RT), or radical prostatectomy (RP). We used data from the population-based Prostate Cancer Outcomes Study (PCOS) to evaluate patient satisfaction with these treatments.

METHODS: The PCOS evaluated 3,830 incident cases of prostate cancer reported in 6 regional Surveillance, Epidemiology, and End Results (SEER) programs between October 1, 1994 and October 31, 1995. The current analysis was restricted to the 2,387 subjects with clinically localized prostate cancer who completed a 24-month post-diagnosis follow-up questionnaire. Weighted multivariate logistic regression analysis was used to determine factors associated with treatment satisfaction. Independent variables included demographics, socioeconomic status, comorbidities, tumor characteristics, additional cancer treatments,

perception of being free of cancer, scales measuring self-reported bowel, urinary, and sexual function, and perception of problems with these three functions.

RESULTS: Treatment selections were 431 C, 583 RT, and 1,373 RP. The cohort was 74% non-Hispanic white, 13% African American, and 13% Hispanic; 41% were 65 years and younger. Perception of being free of cancer was reported by 17% (C), 87% (RT), and 94% (RP), even though 16% (RT) and 14% (RP) required additional androgen deprivation therapy. Moderate/big sexual problems were reported by 33% (C), 44% (RT), 53% (RP); moderate/big urinary problems were reported by 4% (C), 3% (RT), 11% (RP); and moderate/big bowel problems were reported by 6% (C), 7% (RT), 3% (RP). Men undergoing radiation therapy were more likely to be delighted/pleased with their treatment choice (70%) than those undergoing either conservative management (55%) or surgery (59%), $P < 0.0001$. The following factors were independently associated with treatment satisfaction for conservative management: receiving androgen deprivation therapy (OR = 2.0; 95% CI 1.2, 3.3) and being free of cancer (OR = 2.5; 1.0, 6.1); for radiation: not receiving androgen deprivation therapy (OR = 2.4; 1.4, 4.4) and having moderate/big sexual problems (OR = 0.46; 0.25, 0.84); and for radical prostatectomy: being free of cancer (OR = 3.2; 1.6, 6.3), and having moderate/big bowel (OR = 0.21; 0.07, 0.60); urinary (OR = 0.24; 0.15, 0.39), or sexual problems (OR = 0.37; 0.25, 0.55).

CONCLUSION: The majority of men with localized prostate cancer were very satisfied with their treatment decision. Sexual problems were common with all treatments but significantly decreased satisfaction for only RT and RP. RP most frequently led to urinary problems and this significantly decreased satisfaction. Androgen deprivation therapy increased satisfaction for C but decreased satisfaction for RT. For all treatments, the perception of being free from cancer was highly associated with satisfaction.

PACIFIC ISLANDER CANCER CONTROL NETWORK. *FA Hubbell¹, S. Mishra², P.H. Luce³*, ¹University of California, Irvine, Irvine, CA; ²University of California, Irvine, Irvine, CA; ³National Office of Samoan Affairs, Carson, CA

PURPOSE: This abstract describes the structure, objectives, and achievements to date of the Pacific Islander Cancer Control Network (PICCN), one of 18 Special Populations Networks (SPNs) established by the National Cancer Institute (NCI) to address the cancer control needs of America's medically underserved populations.

METHODS: The PICCN focuses on three major Pacific Islander groups: American Samoans, Chamorros/Guamanians, and Tongan Americans. The project is a collaborative effort between the University of California, Irvine (UCI), the National Office of Samoan Affairs (NOSA) and six community-based organizations (CBOs). The CBOs are located in the western United States and the Territories of American Samoa and Guam-areas in which the large majority of these Pacific Islanders reside. A Steering Committee with membership from the CBOs, UCI, NOSA, NCI, and voluntary groups concerned with cancer control coordinate the network activities. The project objectives include enhancing cancer awareness, increasing minority participation in clinical trials, enlarging the number of minority investigators, and encouraging research among Pacific Islanders.

RESULTS: The PICCN is in its first year of operation. Process evaluation has documented the successful implementation of the following activities 1) organizing the coordinating structure, 2) assessing existing cancer education materials aimed at Pacific Islanders, 3) conducting cancer education programs through the CBOs, 4) establishing relationships between the CBOs and their local cancer centers, 5) informing CBOs about existing cancer center studies in which Pacific Islander could participate, 6) identifying potential candidates for research training, and 7) developing pilot projects to assess the cancer control needs of Pacific Islanders.

CONCLUSION: The Pacific Islander Cancer Control Network is off to a successful start in assessing and addressing the cancer control needs of Pacific Islanders. These efforts will contribute to the National goal of eliminating health disparities between racial/ethnic groups by the year 2010.

COUNSELING OVERWEIGHT AND OBESE PATIENTS IN PRIMARY CARE: WHAT DO PHYSICIANS SAY? HOW DO PATIENTS BEHAVE? *J. Humair¹, L. Seematter², J. Cornuz²*; ¹Geneva University Hospital, Geneva, GE; ²Lausanne University Hospital, Lausanne, Switzerland

PURPOSE: Effective counseling of overweight and obese patients should include behavioral strategies facilitating weight loss and a healthy diet. Data about their use in primary care are scarce. This study aims to assess frequency, content and impact of weight and diet counseling provided by primary care physicians to such patients.

METHODS: We conducted an observational study on clinical prevention in 2 university-based primary care clinics staffed by 35 residents. We included 893 consecutive patients of whom 39% (44%) were overweight or obese (body mass index > 25 kg/m²). We interviewed patients after the index visit to assess counseling performed by residents during the three last visits. At 1 year, we mailed a questionnaire to assess patients' self-reported behavior change and weight loss (60% responders). We compared outcomes according to the quality of physician counseling (4–6 vs. 1–3 strategies used).

RESULTS: Physicians infrequently advised overweight and obese patients about weight and diet: informing of health risks (40%); recommending to lose weight (40%); assessing motivation to lose weight (34%). They rarely used strategies facilitating behavior change: planning specific dietary changes (24%), setting a target weight (14%) and written material (14%). At 1 year, patients who received better counseling were more likely to have set a target weight (57% vs. 43%), changed their diet (50% vs. 38%), visited a dietician (33% vs. 8%), and lost weight (49% vs. 32%) (all $p < 0.01$).

CONCLUSION: Primary care physicians provided sub-optimal weight and diet counseling to overweight and obese patients and rarely applied behavioral strategies. Patients who received interventions of higher quality reported behavior change and weight loss more frequently. Clinical trials should test the efficacy of brief weight and diet counseling delivered by physicians to overweight and obese patients.

UPPER EXTREMITY MUSCULOSKELETAL DISORDERS ASSOCIATED WITH ADVERSE COMPUTING HABITS AMONG COLLEGE STUDENTS. *N. Hupert¹, A.H. Fossel², M. Cortes³, J.N. Katz²*; ¹Weill Medical College of Cornell University, New York, NY; ²Brigham and Women's Hospital, Boston, MA; ³Harvard School of Public Health, Boston, MA

PURPOSE: We recently found that $> 50\%$ of graduating seniors at a large private university reported upper extremity (UE) pain with computer use. Upper extremity musculoskeletal disorders (UEMDs) are a leading cause of work disability in the United States. We surveyed college students to determine risk factors for UE pain and functional disability related to UEMDs.

METHODS: We surveyed a convenience sample of 250 students at a 425-undergraduate dormitory of a major university. 97% of students live on campus and are randomly assigned to a dormitory. 204 (82%) responded, but 11 were subsequently deemed ineligible, for a final response rate of 81% (193/239) representing 45% (193/425) of the source population. We used previously validated, self-reported measures of computer work habits, upper extremity pain, functional limitation, and mental health (Mental Health Inventory (MHI-5)). We calculated descriptive statistics using the Chi-square test and used generalized linear models to determine the independent effect of adverse computing habits on UEMD symptoms and functional impairment.

RESULTS: Our sample included 46% women and 38% racial/ethnic minorities. 29% reported never having UE pain related to computer use, 30% had experienced pain in the past, 36% had pain in the past and current pain (in preceding 2 weeks), and 5% only had current pain. We defined "intensive computer users" as students who typically computed for > 4 h/day or spent ≥ 4 h computing without getting out of the chair at least once a month. 66% were intensive computer users. Women were more intensive users ($p = 0.05$), reported more functional impairment due to UEMDs ($p = 0.013$), and had more upper arm ($p = 0.07$), shoulder ($p = 0.001$), and neck pain ($p < 0.0001$) than men. Worse mental health scores were associated with increased reported UE pain ($p = 0.01$) and functional impairment ($p < 0.0001$). Controlling for sex, mental health, and overall health, intensive computer use independently predicted higher UEMD symptom (Type III SS $p = 0.03$) and pain severity scores ($p = 0.01$).

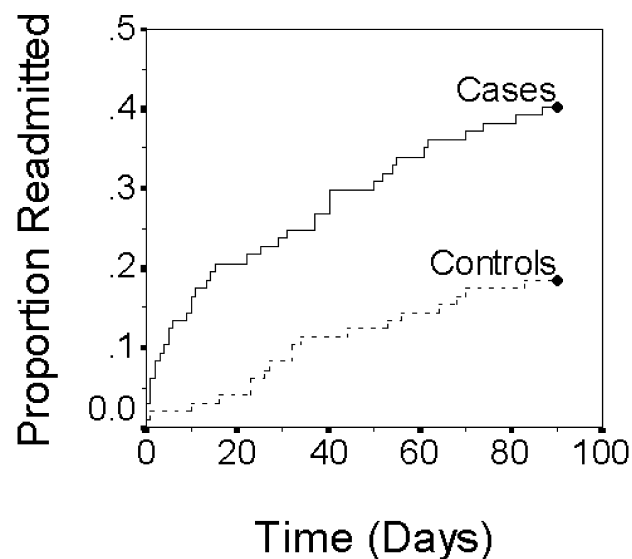
CONCLUSION: Most college students in this sample engage in intensive computer use and have UE pain with computing. We found a significant independent association between intensive use and UEMD symptomatology. Our results suggest that programs to foster healthy computing skills may prevent long-term disability from UEMDs as students enter the work force.

WHAT HAPPENS TO PATIENTS WHO LEAVE AGAINST MEDICAL ADVICE? *S.W. Hwang¹, J. Li¹*; ¹St. Michael's Hospital, Toronto, ON

PURPOSE: To examine readmission rates among general medical patients who leave against medical advice (AMA) and to identify predictors of readmission.

METHODS: We prospectively identified patients leaving AMA from the general medical service of St. Michael's Hospital, a community-based teaching hospital in downtown Toronto. Each case was paired with a routinely discharged control patient, matched for age, sex, and diagnostic group. Readmissions within 90 days were identified at St. Michael's Hospital and 4 other hospitals in a 3 mile radius. We used stepwise regression to obtain cubic spline functions that were best-fitted to the Kaplan-Meier curves for cases and controls. For case patients, predictors of readmission were examined using logistic regression models.

RESULTS: 97 case patients left AMA over an 18 month period. Case patients were more likely to readmitted than controls ($p = 0.0006$ by log-rank test). Cubic spline regression models demonstrated that case patients were readmitted at an accelerated rate for the first 15 days, followed by a 75 day period in which their readmission rate was not significantly different from controls. Readmission was not associated with age, sex, length of stay before leaving AMA, homelessness, alcohol abuse, drug abuse, or major mental illness.



CONCLUSION: The increased risk of readmission among patients leaving AMA from a general medical service is limited to the first 15 days after departure. Readmission cannot be predicted on the basis of simple demographic or clinical variables.

QUALITY OF ANTICOAGULATION WITH UNFRACTIONATED HEPARIN (UFH) AND WARFARIN IN THE TREATMENT OF ACUTE THROMBOSIS. E.M. Hylek¹, S. Regan¹, L.E. Henault¹, M. Gardner¹, D.E. Singer¹, M.J. Barry¹; ¹Massachusetts General Hospital, Boston, MA

PURPOSE: The goals of anticoagulation (AC) in the treatment of acute thrombotic disease are to minimize clot propagation and decrease the risk of recurrence. Current guidelines recommend a minimum of 4 days of heparin administered concurrently with warfarin and continuation of heparin until 2 consecutive INRs exceed 2.0. We sought to define the quality of acute AC therapy among hospitalized patients being treated for acute venous and arterial thrombosis at our academic medical center.

METHODS: Hospitalized patients receiving UFH and warfarin were identified from July 1998 to December 1998. Patients with at least 2 PTT measurements >40 seconds with at least one warfarin bill were screened. Three hundred and thirty-five patients who received UFH and warfarin for treatment of acute venous or arterial thrombosis were identified. Detailed review was completed on 316 (94%). Five patients with known heparin induced thrombocytopenia were excluded. Analyses involving warfarin and days of UFH and warfarin overlap were limited to patients continuing on and completing their UFH course prior to discharge.

RESULTS: Of the 311 patients who initiated UFH therapy, 134 (43%) were being treated for venous thromboembolism (VTE), 122 (39%) for embolic stroke, and 55 (18%) for peripheral arterial thrombosis. The mean age was 64.9 years (range 18–97) and 52% were female. Following initiation of UFH, the first PTT measured was ≥ 55 seconds for 57% of patients. The proportion of all PTT tests on patients being treated with UFH that were within the therapeutic range (55–85 seconds) was 43% overall, 44% for VTE and 46% for stroke. Thirty-five percent were <55 seconds.

Of the 311 patients who initiated UFH, 251 completed their heparin course prior to discharge. The period of overlap with warfarin was ≤ 3 days for 28% and ≥ 6 days for 38%. UFH was discontinued prior to the first INR exceeding 2.0 in 37% and less than 16% of patients fully met the guideline criteria.

CONCLUSION: Acute management of thrombotic disease remains suboptimal in real-world practice. Few patients actually meet the currently recommended AC guidelines for conversion of UFH to warfarin. Strategies are needed to improve the overall quality of anticoagulant care including expanding the use of LMWH.

POTENTIAL AVOIDABLE DAYS AMONG STABLE HOSPITALIZED PATIENTS UNDERGOING ANTICOAGULATION FOR ACUTE THROMBOTIC DISEASE. E.M. Hylek¹, S. Regan¹, M. Gardner¹, L.E. Henault¹, D.E. Singer², M.J. Barry¹; ¹Massachusetts General Hospital, Boston, MA; ²Partners Health Care, Boston, MA

PURPOSE: Low molecular weight heparin (LMWH) has been shown to be effective in the treatment of acute thrombotic disease. Its stable dose response and longer plasma half life make it attractive for treatment in the outpatient setting. We sought to identify the potential LOS savings among a hospitalized cohort of patients being treated with unfractionated heparin (UFH) and warfarin for acute venous and arterial thrombosis at one academic medical center.

METHODS: Hospitalized patients receiving UFH and warfarin were identified from July 1998 to December 1998. Patients with at least 2 PTT measurements >40 seconds with at least one warfarin bill were screened. Three hundred and thirty-five patients were identified who received UFH and warfarin for treatment of acute venous or arterial thrombosis. Clinical information was complete on 316 (94%). Five patients with known heparin induced thrombocytopenia were excluded. Experienced nurse case managers conducted detailed chart review using Medicare's Appropriateness Evaluation Protocol Criteria (AEP) to determine those days of hospitalization attributable solely to the overlap of UFH with warfarin while awaiting therapeutic INR. The AEP criteria are based on severity of illness and intensity of service measures. All patients had to be medically ready for discharge home.

RESULTS: Of the 311 patients identified, 134 (43%) were being treated for venous thromboembolism, 122 (39%) for embolic stroke, and 55 (18%) for peripheral arterial thrombosis. One hundred eighty-three (59%) patients were discharged directly to home. One hundred sixty-two (89%) of the 183 patients met eligibility criteria for LMWH. Six patients were deemed ineligible because of renal insufficiency, 4 were nonadherent with therapy, 4 developed heparin induced thrombocytopenia, 3 sustained bleeding complications, and 4 for other reasons. Of the 162 patients eligible for LMWH, 52% were female and the mean age was 61 years. Forty-four percent of these patients were being treated for venous thromboembolism, 39% for embolic stroke, and 17% for peripheral arterial thrombosis. Based on case manager chart review applying Medicare's AEP criteria, 290 days of acute hospitalization could have been averted among 121 patients by transition to LMWH. Among all of those eligible for early discharge on LMWH, this is an average of 1.8 days per patient.

CONCLUSION: Treatment of acute thromboembolism in the hospital setting is an intensive use of hospital days. Among selected patients, transition to LMWH and home care has the potential to substantially reduce medical care costs.

OUTCOME CONCERNS MEDIATE RACIAL DIFFERENCES IN ATTITUDES TOWARD JOINT REPLACEMENT. S.A. Ibrahim¹, C.J. Burant², L.A. Siminoff², K. Kwok³; ¹Louis Stokes VAMC/CWRU, Cleveland, OH; ²Cleveland; ³University of Pittsburgh, Pittsburgh, PA

PURPOSE: Whites are 7 times more likely than African-Americans (AA) to undergo joint replacement (JR). The reasons behind this profound racial disparity are not known. We examined factors that may influence attitudes of African-Americans and whites toward joint replacement.

METHODS: 596 AA (46%) and white (54%) patients, aged >50 with chronic knee/hip pain (potential future candidates for joint JR) were surveyed to assess attitudes, awareness/understanding of JR, as well as post-surgery concerns. We used multivariate logistic regression models to control for SES, age, severity of disease/functional status and depression. **RESULTS:** AAs and whites were comparable with respect to age (66 ± 10 vs. 66 ± 9 , $P = 0.60$), severity of disease/functional status (WOMAC index mean score 46 ± 17 vs 45 ± 17 , $P =$

0.32); disease burden (Charlson score 2.3 ± 2 vs 2.5 ± 2 , $P = 0.29$), and geriatric depression scores (4.4 ± 3.3 vs. 4.9 ± 3.8 , $P = 0.08$). However, AAs were more likely to report lower annual household income (less than \$10,000, 40% vs. 21%, $P = 0.000$), less employment (9% vs. 15%, $P = 0.03$), and lower educational levels (less than high school, 42.9% vs 29.4%, $P = 0.002$). AAs were less likely than whites to have ever heard of JR or to have family or friends who underwent knee or hip replacement. They were also less likely to have a good understanding of what JR involves. Adjusted ORs were 0.65 (0.40 – 1.00); 0.32 (0.22 – 0.40); 0.55 (0.38 – 0.70), respectively. AA to white adjusted OR for the attitude question was 0.54(0.32 – 0.91). Adjusted ORs for AA to white responses to post-surgical pain and difficulty walking responses were 2.61 (95% CI 1.74 to 3.89) and 2.76 (95% CI 1.83 to 4.16), respectively. Post-surgery pain and difficulty walking responses were correlated (negatively) with attitudes ($r = -0.15$, $P = 0.000$ and $r = -0.12$, $P = 0.004$, respectively) at the bivariate level. When included in the final model, the relationship between ethnicity and attitudes toward joint replacement was no longer significant; adjusted OR 0.66 (95% CI 0.38 – 1.14). Responses on familiarity with JR did not mediate the relationship between ethnicity and attitudes toward JR.

CONCLUSION: Differences in attitudes toward JR between AA and white patients is mediated by post-surgical concerns about pain and disability. If confirmed, our findings suggest an opportunity for generalists to address patient-level factors that contribute to racial disparity in the utilization of joint replacement.

NO RACIAL AND GENDER DIFFERENCES IN 18-MONTH MORTALITY FROM DIASTOLIC HEART FAILURE. S.A. Ibrahim¹, C.J. Burant², K. Kwok³; ¹Louis Stokes VAMC/CWRU, Cleveland, OH; ²Cleveland, OH; ³University of Pittsburgh, Pittsburgh, PA

PURPOSE: Although racial and gender differences in mortality have been reported for systolic heart failure, little is known regarding diastolic heart failure prognosis; even though diastolic heart failure is the most common type of heart failure in the US.

METHODS: Our sample consisted of 1058 patients 65 years of age or older who were admitted to 30 hospitals in Northeastern Ohio with a principal diagnosis of heart failure and had documented diastolic dysfunction by Echo. Diastolic heart failure in this cohort was defined as "having a principal diagnosis of heart failure and a LVEF of 50% by Echo." Mortality data for all patients were obtained from Ohio MEDPRO files for Medicare beneficiaries. Logistic regression was used to compare 18-month mortality by ethnicity and by gender, adjusting for age, gender and comorbidities.

RESULTS: Of the 1058 patients with documented diastolic heart failure (13% AA and 87% white), AAs and whites were comparable with respect to history of angina, stroke, being on dialysis, alcohol use, and the proportion of males. AAs were more likely to have hypertension (50% vs 36%; $P = 0.001$), diabetes (46% vs 29%; $P = 0.000$), history of tobacco use (27% vs 18%; $P = 0.011$), and high serum creatinine (1.99 ± 2 vs 1.50 ± 1 ; $P = 0.003$); they were also younger (76 ± 7 vs 79 ± 8 ; $P = 0.000$). Whites were more likely to have a history of ischemic heart disease (48% vs 32%; $P = 0.000$), metastatic cancer (3% vs 0%; $P = 0.034$), DNR status on record (14% vs 7%; $P = 0.013$), and atrial fibrillation (24% vs 14%; $P = 0.002$). The AA to white adjusted OR for 18-month mortality was 1.03 (0.66–1.59). For men vs women (30% vs 70%), the above-mentioned comorbidities were comparable, except women were more likely to have DNR status (16% vs 7.3%; $P = 0.000$) and to be older (79.5 ± 8 vs 77 ± 7 ; $P = 0.000$). Males were more likely to have a history of tobacco use (30% vs 14%; $P = 0.000$), alcohol use (36% vs 15%; $P = 0.000$), and higher serum creatinine level (1.7 ± 1.2 vs 1.4 ± 1.1 ; $P = 0.001$). Men to women adjusted OR for 18-month mortality 1.06 (0.76–1.46).

CONCLUSION: In this cohort of elderly patients admitted with diastolic heart failure, there were no ethnic or gender differences in 18-month mortality.

ALTERNATIVE MEDICINE USE AMONG MEDICAL OUTPATIENTS. M. Herman¹, L. Inouye¹; ¹Naval Medical Center Portsmouth, Portsmouth, VA

PURPOSE: The use of complementary and/or alternative medicine (CAM) is increasing in popularity among patients in the United States. Hundreds of herbal products and homeopathic remedies are readily available to the consumer, but most of these have not been proven safe or effective. Potential side effects and interactions with traditional medications are often unknown. Studies suggest that between 30% and 50% of the adult population in industrialized nations use some form of CAM to prevent or treat a variety of health-related problems. The purpose of our study was to evaluate CAM use by Internal Medicine. The analysis would assess the proportion of patients in the clinic who are using alternative medicines, provide a description of the most common alternative medicines being used, and evaluate characteristics associated with alternative medicine use.

METHODS: A convenience survey was completed by patients of the Internal Medicine clinic at Naval Medical Center, Portsmouth (NMCP) between 6DEC99 and 7JAN00. Information was gathered on the patients gender, age, race, medical problems and medications, educational background, military status, herbal and non-herbal usage to include confidence of efficacy and safety, and the source of information for herbal and non-herbal products. Chi-squared analysis was used to compare CAM use differences in gender and educational groups.

RESULTS: There were 212 surveys completed. Of those responding, 15% were active duty or a dependent of an active duty member, and 60% were female. The mean age of respondents was 53 years. Complementary therapies had been used in 60% of the patients. The most common herbal products were garlic, ginseng, ginkgo biloba, and St. Johns wort. The most common non-herbal products were vitamins, chromium, glucosamine, and chondroitin sulfate. Sixty-five percent of females used CAM compared with 53% males, although this difference was not statistically significant ($p = 0.125$). The average age of CAM users was 53 years. Thirty percent of CAM users had a college degree, 15% were active duty, 35% were retired, and 50% were dependent. Sixty percent of CAM users had been using an alternative medicine for at least one year. Use of herbal CAM was associated with a higher educational background, although the difference was not statistically significant ($p = 0.132$). However, vitamin use was significantly correlated with a higher educational background. In those subjects with less education than a college degree, only 25% used vitamins. Among college graduates 54% used vitamins

($p < 0.0001$). Seventy-one percent of respondents felt that complementary therapy was safe and only 17% were very confident of its efficacy.

CONCLUSION: CAM use among Internal Medicine patients at NMCP is higher than reported use in studies of civilian populations, which may reflect a difference in patient profile. Of interest is that a significant proportion of people using alternative medicines were not confident of its safety and only a few were very confident of its efficacy. This information may correlate with compliance issues pertaining to physician prescribed medications. While the social profile may reflect those most likely to use CAM, our data demonstrates widespread use of alternative medicine. This information has important implications for health care providers, who must consider safety, efficacy and possible interactions with standard therapies. Further analysis will include the results from over 400 respondents.

IMPACT OF UNDIAGNOSED MENTAL DISORDERS IN PRIMARY CARE. M.L. Passamonti¹, J.L. Jackson², ¹WRAMC, Washington, DC; ²Uniformed Services University of the Health Sciences, Bethesda, Maryland

PURPOSE: To assess the frequency and impact of undiagnosed mental disorders over time. **METHODS:** In 1995, 500 primary care patients presenting to a walk-in clinic with physical complaints were screened for mental disorders with the PRIME-MD. Five years later, patient surveys assessed symptom outcome, mental disorders (PRIME-MD), functional status (MOS SF-6) and whether they had been diagnosed or treated for a mental disorder. Vital status was assessed with the Social Security Index.

RESULTS: Forty patients (8%) died. Among the remainder, 65% completed surveys, with follow up losses equal between patients with and without a mental disorders. The prevalence of mental disorders declined from 30% at baseline to 12%, though patients with a disorder at baseline were more likely to still have a disorder at 5 years (RR: 3.5, 95% CI: 2.0–6.0). Only 32% of patients with a mental disorder at enrollment were diagnosed during the 5 years of follow up. The rate of diagnosis varied by disorder: 52% of patients with major depression, all patients with panic disorder and 66% of patients with generalized anxiety at baseline were diagnosed. For subthreshold disorders, diagnoses rates were lower: 32% for depression and 5% for anxiety. Patients with unrecognized depression (RR: 5.0, 95% CI: 1.2–21.1) or anxiety (RR: 4.8, 95% CI: 1.1–23.5) were more likely to still have their disorder 5 years later, compared to patients whose disorders were recognized. While most patients with minor depression or subthreshold anxiety (73–83%) had their disorder at only one time point and no patient with subthreshold anxiety progressed to generalized anxiety or to panic disorder, 18% of those with minor depression progressed to major depression by 5 years. Patients whose mental disorder went undiagnosed over 5 years were more likely to have worsening of their initial symptom (RR: 3.6, 95% CI: 1.6–8.4), greater stress (RR: 2.3, 95% CI: 1.9–2.8), greater worry that their symptom represented a serious illness (RR: 2.9, 95% CI: 1.7–4.9), a greater number of “other” bothersome physical symptoms ($p < 0.0001$), and worse functional status ($p < 0.0001$). Outcomes among patients whose mental disorders were diagnosed during the intervening 5 years, were the same as among those without mental disorders.

CONCLUSION: Patients with unrecognized major mental disorders were more likely to still be struggling with their disorder 5 years later, had less resolution of their presenting physical symptom and experienced worse health-related outcomes.

SYMPTOMS IN PRIMARY CARE, OUTCOMES AT FIVE YEARS. J.L. Jackson¹, M.L. Passamonti², ¹Uniformed Services University of the Health Sciences, Bethesda, Maryland; ²WRAMC, Washington, DC

PURPOSE: To assess the outcome of physical symptoms over 5 years of follow up.

METHODS: In 1995, 500 adults presenting with a physical complaint completed surveys that assessed symptom characteristics, functional status (MOS SF6) and mental disorders (PRIME-MD). Follow up surveys assessed symptom outcome, functional status and satisfaction at 2 weeks, 3 months and 5 years.

RESULTS: Patients averaged 43 years in age (18–92), were 52% female, 45% African American and 49% white. Patients presented with a myriad of complaints that we collapsed to 13 categories. Musculoskeletal symptoms were the most common category, present in a third and half had some sort pain complaint. Twenty-one percent had experienced their symptom less than 3 days, 40% less than a week, 68% less than a month; 46% had seen a previous doctor for the same problem. Most patients experienced improvement, 70% at 2 weeks, 79% at 3 months and 75% at 5 years. There was no relationship between type of symptom and likelihood of improvement. However, there was good correlation between improvement in symptoms between each of the three time points. (Spearman's rho: 0.40–0.43). Among patients whose symptom had improved by 3 months, most (83%) were still improved at 5 years. Conversely among those reporting no improvement, 52% were still not improved at 5 years. There was no relationship between likelihood of symptom improvement and age, sex, or baseline functional status. Patients without symptom improvement had worse functional status ($p < 0.001$), more illness worry ($p < 0.001$), a greater number of “other” bothersome symptoms ($p = 0.003$) and were more likely to have an active mental disorder at 5 years (RR: 1.3, 95% CI: 1.02–1.7). Independent predictors of improvement included improvement at 3 months, symptom less than 3 days in duration at presentation and no patient worry that the illness could be potentially serious at presentation.

CONCLUSION: Patients with symptoms present for more than 3 days at presentation, who are worried their symptom may be serious or who have not improved by 3 months are likely to still have problems with their symptom at 5 years.

Predictors of Improvement

	OR (95% CI)
Improved at 3 months	3.8 (2–7.3)
Duration < 3 days at presentation	2.8 (1.1–7.6)
Worried at presentation	0.48 (0.25–0.95)

ALTERNATIVE HEALTH PRACTICES AMONG PATIENTS WITH PHYSICAL SYMPTOMS.

J.L. Jackson¹, M.L. Passamonti², ¹Uniformed Services University of the Health Sciences, Bethesda, Maryland; ²WRAMC, Washington, DC

PURPOSE: To assess the prevalence and impact of alternative health care practices among patients presenting with physical symptoms.

METHODS: In 1995, 500 primary care patients presenting to a walk-in clinic with physical complaints completed surveys that assessed functional status and mental disorders (PRIME-MD). Five years later, patient surveys assessed symptom outcome, mental disorders (PRIME-MD), functional status (MOS SF-6), satisfaction, and the use of alternative health practices for their specific problem as well as the impact of such treatment.

RESULTS: Forty patients died during the interval. Among the remaining 460, 65% completed surveys. Seventy-five (25%) reported trying an alternative health practice for their problem. Among those using an alternative practice, a variety of methods were reported: spiritual healing 26%, diet supplements 25%; minerals 17%; chiropractic 17%; herbs 13%, copper 10%, acupuncture 8%, allopathy 8%. Four percent reported the alternative health method to be very helpful for their problem, 22% somewhat helpful, 43% neither helpful nor harmful and 31% reported it to have made the problem worse. Patients using alternative health practices tended to be older (59 vs 54, $p = 0.02$), and were more likely to be college educated (RR: 1.7, 95% CI: 1.3–2.2). There were no differences in stress, functional status, presence of a mental disorder, serious illness worry and other demographic characteristics. Patients using alternative health care practices were less likely to be satisfied with the care they had received from traditional health practices (RR: 0.48, 95% CI: 0.32–0.72) and were less likely to have experienced resolution of their symptom (RR: 0.39, 95% CI: 0.24–0.64).

CONCLUSION: Alternative health care practices are common among primary care patients. Such patients, surprisingly, may be older and better educated. They may choose alternative methods because they are less satisfied with the care received from traditional practitioners or because they have failed to improve. A third of patients reported that these methods made their problems worse.

THE IMPACT OF NATIONAL LOW BACK PAIN GUIDELINES ON CLINICAL PRACTICE.

J.L. Jackson¹, R. Browning², ¹Uniformed Services University of the Health Sciences, Bethesda, Maryland; ²NNMC, Bethesda, MD

PURPOSE: The 1994 AHRQ Low Back Pain Clinical Guidelines recommended treatment with tylenol in patients less than 55 years of age with no historical or physical examination findings suggestive of more serious causes. Nonsteroidal use was discouraged. Muscle relaxants, narcotics and routine radiographs were deemed inappropriate. Our purpose was to assess the impact of these guidelines on clinical practice.

METHODS: Patients greater than 17 but less than 55 years in age, being seen in primary care settings (FP, IM, GP) with low back pain, without an inflammatory or secondary diagnosis to explain their back pain were abstracted from the National Ambulatory Medical Care Survey. Data from 1991–1993 (prior to release of the 1994 AHRQ), were compared to data from 1995–1997.

RESULTS: A total of 11,854,653 encounters met criteria and were included in this analysis. The mean age of patients was 37 years, 57% were male and 34% of the visits were injury related. Prior to the guidelines, 0.33% of patients with back pain were treated with acetaminophen, 39% with a nonsteroidal, 27% with muscle relaxants and 4% with narcotics. Twenty-seven percent were referred for physiotherapy and 16% had an Xray associated with the visit. After the guideline release in 1994, acetaminophen use increased to 3%, muscle relaxant use declined to 17% and referrals to physiotherapy declined to 22%. Nonsteroidal (40%), muscle relaxants (20%), narcotic use (6%) and the rate that Xray's were obtained (19%) were unchanged.

CONCLUSION: The AHRQ guidelines increased acetaminophen use, but only to 3%. Physicians were less likely to prescribe a muscle relaxant and were less likely to refer for physical therapy. There was no impact of the guidelines on clinician use of nonsteroidals, narcotics or radiographic evaluation.

ALCOHOL CONSUMPTION AND MORTALITY IN MEN WITH PRE-EXISTING CEREBROVASCULAR DISEASE.

V. Jackson¹, H. Sesso^{1,2}, J.E. Buring^{1,2}, J. Gaziano², ¹Massachusetts Veterans Epidemiology Research and Information Center, Boston, MA; ²Brigham and Women's Hospital, Boston, MA

PURPOSE: Clinicians are often reluctant to recommend alcohol consumption among those with a history of stroke. We therefore sought to examine the relationship between alcohol intake and the risk of total and cardiovascular mortality in men with a previous history of stroke.

METHODS: The study population consisted of the 104,353 physicians who comprised the enrollment cohort of the Physicians' Health Study. We focused our analyses on the 1,320 men who indicated a history of stroke at baseline and provided self-reported data on alcohol consumption. Cox proportional hazards was used to model the association between alcohol and mortality after adjustment for major lifestyle and clinical risk factors. Alcohol consumption was classified into four categories, with those reporting rarely/never used as the referent group.

RESULTS: During a mean follow-up of 4.6 years, 369 men died, of whom 267 died from cardiovascular disease. Multivariate relative risks of total mortality were decreased in men with a history of stroke who consumed < 1 drink per week, 1–6 drinks per week, and at least 1 drink per day, at 0.88(0.60,1.28), 0.64(0.48,0.85) and 0.71(0.54,0.94) respectively (p , linear trend 0.028). Multivariate relative risks of cardiovascular mortality in men with a history of stroke were also decreased in men who consumed < 1 drink per week, 1–6 drinks per week, and at least 1 drink per day, at 0.89(0.58,1.36), 0.56(0.40,0.79), and 0.64(0.46,0.88) respectively (p , linear trend 0.008). Adjustment for lifestyle and other clinical risk factors did not significantly change the risk estimates for total or cardiovascular mortality as compared with age-adjusted models. Because only fifty-five men died as a result of stroke, we were underpowered to examine this association, but the risk estimates were nonsignificantly lower among men consuming at least 1 drink per week.

CONCLUSION: These data suggest that alcohol may reduce the risks of total and cardiovascular mortality in men with a history of stroke. More data are needed to confirm or refute these results.

THE ASSOCIATION BETWEEN SEXUAL FREQUENCY AND PROSTATE AND GENERAL HEALTH IN MEN. S.J. Jacobsen¹, D.J. Jacobson¹, R.L. Thiemann¹, C.J. Girman², T. Rhodes², R.O. Roberts¹, M.M. Lieber¹; ¹Mayo Clinic, Rochester, MN; ²Merck Research Laboratories, West Point, PA

PURPOSE: Historically, it has been suggested that the symptoms associated with benign prostatic hyperplasia can be aggravated by infrequent sex. We evaluated this by assessing the cross-sectional association between frequency of ejaculation and lower urinary tract symptom severity and general self-perceived health among participants in the Olmsted County Study of Urinary Symptoms and Health Status among Men.

METHODS: In 1989 and 1990, 2,115 Caucasian men between the ages of 40 and 79 years were recruited from a random sample of Olmsted County residents (55% response rate). These men completed a self-administered questionnaire that assessed lower urinary tract symptom severity with questions similar to those in the American Urological Association Symptom Index (AUASI) and completed a similar questionnaire biennially thereafter through 1998. In 1994, a question was added to assess the frequency of ejaculation during the previous month. This was answered by 81 percent of participants in that round.

RESULTS: Overall, men who reported ejaculating at least once a week were less likely to have moderate-severe (AUASI > 7) symptoms than men reporting no ejaculations (Odds ratio=0.62, 95% Confidence Interval=0.51, 0.75). There was a similar association with health-related quality of life (Poor-good vs. very good-excellent, OR=0.45, 95% CI=0.37, 0.60). The association with symptom severity did not exist within age decade, however. The odds ratios for ejaculating at least once a week were 0.93, 0.79, 1.21 and 1.12 for men in their forties, fifties, sixties and seventies, respectively. Thus, after adjusting for age, the odds ratio was 0.95 (0.75, 1.21). The odds ratio for health-related quality of life, however, remained virtually unchanged (OR=0.47, 95% CI=0.37, 0.60).

CONCLUSION: While there has been speculation about the adverse effects of abstinence on prostate health, these cross-sectional data suggest that frequency of ejaculation has no effect on lower urinary tract symptoms; the apparent protective association appears to be an artifact caused by the confounding effects of age. This is not the case, however, for the cross-sectional association with quality of life.

REASONS FOR AND PREDICTORS OF RE-HOSPITALIZATION FOR PATIENTS WITH COMMUNITY-ACQUIRED PNEUMONIA. H. Jasti¹, E. Mortensen², M.J. Fine², W.N. Kapoor²; ¹University of Pittsburgh, Pittsburgh, PA; ²UPMC, Pittsburgh, PA

PURPOSE: Re-hospitalization after in-patient treatment for community-acquired pneumonia (CAP) is common. The aims of this study were to examine reasons and identify predictors of re-hospitalization (rehosp).

METHODS: This analysis was performed as part of the randomized, controlled trial, "Dissemination of Guidelines for Pneumonia Length of Stay," implemented in 7 hospitals in Pennsylvania from February 1998 to March 1999. The trial assessed the effect of a guideline intervention to reduce the length of stay (LOS) for patients hospitalized with CAP. Inclusion criteria included a clinical diagnosis of pneumonia and a chest radiograph with a new pulmonary infiltrate. Severity of illness was quantified using the Pneumonia Severity Index (PSI), a predictor of 30-day mortality in patients with CAP. For this study, 2 physicians independently reviewed the records of patients re-hospitalized within 30 days and afterwards reached a consensus on reasons for rehosp. Demographic, clinical, laboratory, and radiological data were used to identify reasons for rehosp. Categorical and continuous variables were assessed using the Chi-square test and the Student's t-test, respectively. Any variable significant at the p < 0.05 level was entered into a multiple logistic regression model.

RESULTS: Of the 608 patients (283 in the intervention arm and 325 in the control arm), 69 patients (11%) were re-hospitalized within 30 days. There was no significant difference in re-admission rates between the 2 groups. The mean time to rehosp was 8.49 (+/- 0.96) days. The reasons for rehosp were CAP-related (n=14), combination of CAP and comorbid condition (n=3), or comorbid condition alone (n=52). The kappa statistic, assessing inter-rater reliability, was 0.784. The major comorbid conditions requiring rehosp were cardiovascular (n=19), neurologic (n=6), pulmonary (n=6), gastrointestinal (n=5), genitourinary (n=5), and others (n=11). Significant predictors of rehosp included age (p < 0.001), unemployment status (p=0.04), congestive heart failure (p < 0.001), coronary heart disease (CAD) (p < 0.001), ventricular arrhythmias (p=0.014), chronic obstructive pulmonary disease (COPD) (p < 0.001), chronic oxygen treatment (p=0.001), and PSI (p=0.05). Significant predictors in the multiple logistic regression model included CAD (odds ratio 3.2 with a 95% confidence interval of 1.8-5.5), COPD (2.5, 1.5-4.2), and unemployment status (2.4, 1.1-5.3).

CONCLUSION: This study describes the incidence and reasons for re-hospitalization in patients with CAP and identifies subsets of patients that could be targeted for interventions to decrease re-admissions after CAP.

LIVE DISCHARGE FROM HOSPICE CARE. JB JEANS, KS KUTNER, DE NOWEL, C KASSNER, S MEYER. UNIVERSITY OF COLORADO HEALTH SCIENCES CENTER, DENVER, CO. J.B. Jeans¹; ¹University of Colorado Health Sciences Center, Denver, CO

PURPOSE: The value of hospice lies in its ability to provide continuous quality care until the time of death. However, some patients are discharged from hospice, usually because they no longer meet hospice criteria or choose more aggressive care. We sought to determine differences between patients who are discharged vs. those who die in the hospice setting.

METHODS: We identified 48 patients who were discharged from a local hospice organization between the years 1996 and 2000. We then selected 47 control cases, matched on primary diagnosis and month of admission, who died while in the same hospice organization. Data were collected on these patients through retrospective chart review.

RESULTS: Of the 95 patients, 66% were female, 87% were Caucasian, 37% were married and 74% had Medicare as the primary payor source. The mean age of admission was 70 years, mean length of stay 50 days, and mean Karnofsky status on admission 30%. The most common primary diagnosis was cancer (38%), followed by neurologic disease (14%) and dementia (14%). At the time of admission, 43% received care at home, 40% in assisted living or a nursing home (AL/NH), and 17% in the hospice facility. At the time of discharge/death, 33% were at home, 41% in AL/NH, and 26% in the hospice facility. The reason for discharge was stabilization of condition in 54%, revocation of hospice benefit in order to pursue more aggressive treatment in 13%, and transfer to different hospice in 11%. Discharge disposition was to a different hospice in 11%, to AL/NH in 48%, and to home in 23%. The discharged patients vs. deaths did not differ in age, sex, race, payor source, site of hospice care, admission weight, level of cognition, severity of pain, level of mobility at admission, number of ADL's performed at admission, or Karnofsky status at admission. Having a primary caregiver was also not significant. Compared to patients who died, those discharged had a longer length of stay (mean 62 and 38 days, p=0.05), were less likely to be married (25% and 50%), and less likely to be bedbound at discharge (23% and 76%, p=0.000).

CONCLUSION: Patients discharged alive from hospice, compared to case controls matched on diagnosis and date of admission who died in hospice, had similar demographics and level of functioning at the time of admission. The presence of a primary caregiver did not alter outcome and being married was associated with death. We speculate that hospice provides additional social contact or improved medical care that prolongs the life of some patients.

PATIENTS EXPECTATIONS AMONG NATIVES AND IMMIGRANTS: ARE THEY DIFFERENT AND HOW DO PHYSICIANS PERCEIVE THEM? N. Junod¹, B. Favrat¹, F. Secretan¹, M. Vannotti¹, A. Pecoud¹; ¹University Outpatient Clinic of Lausanne, Switzerland

PURPOSE: Recognizing patients' expectations is considered as an important objective for primary care physicians. However, few studies have investigated the influence on such matter of race and/or ethnicity, specially among immigrants. The aim of the study was to investigate whether patients' expectations among immigrants and natives are different and if physicians can perceive them.

METHODS: The design is a two month pre-consultation patient survey and post-consultation physician survey (matched pairs) in an academic primary care center. Subjects were natives (Swiss French-speaking people) and immigrants (ex-Yugoslav, European, French-speaking African and other) consulting without appointment. The main outcome measures were patients' expectations (14-item questionnaire: receiving a prescription, a sickness certification, reassurance, seeing a specialist, etc...), patients' perception of their health status, physicians' perception of their patients' expectations (14-item questionnaire) and agreement between patient and physician. Questionnaires were available in 3 languages (French, Serbo-Croat and Albanian).

RESULTS: 343 patient and 333 physician questionnaires were analyzed. Patients were separated into 5 groups: Swiss (n=167), European (n=50), Ex-Yugoslav (n=37), African (n=62) and other (n=27). Immigrants (n=176) do not consult more quickly than native patients and initiate a treatment at home as often (n=ns). However, more immigrants perceive their health as bad (22.7% vs 5.4% p < 0.01) and ask their doctor for reassurance (84.3% vs 62.7% p < 0.01). More Ex-Yugoslav patients expect to be referred to specialists (60% vs 30% p < 0.01) and more African patients wish to receive medication than Swiss patients (83% vs 59% p < 0.01). These results are confirmed by a stepwise logistic regression including independent variables such as age, sex, civil status and education. Finally, we found a poor agreement between patients' expectations and their perception by physicians, regardless of patients' origin (k < 0.36).

CONCLUSION: Our study suggests that patients' expectations may differ according to their origin and that physicians certainly have a poor perception of their patients' expectations in an outpatient emergency setting, regardless of patients' origin. However, further investigation is needed to analyze whether the physicians' lack of knowledge of their patients' expectations influences the latter's satisfaction and the quality of medical care.

URINARY TRACT INFECTIONS IN POSTMENOPAUSAL WOMEN: EFFECT OF HORMONE TREATMENT AND RISK FACTORS. A.M. Kanaya¹, J.S. Brown¹, E. Vittinghoff¹, S.K. Agarwal², S.B. Hulley¹, B. Foxman³; ¹University of California, San Francisco, San Francisco, CA; ²Cedars-Sinai Medical Center, Los Angeles, CA; ³University of Michigan, Ann Arbor, MI

PURPOSE: While risk factors for urinary tract infection (UTI) among young women are well described, risk factors in postmenopausal women are poorly understood. Previous studies have shown that vaginal administration of hormones decreases UTI risk in postmenopausal women with recurrent UTIs, but evidence supporting a similar effect of oral hormones is mixed. We assessed the effects of oral hormone treatment in a randomized trial and evaluated potential risk factors for UTI frequency among post-menopausal women.

METHODS: 2,763 postmenopausal women were enrolled in the Heart and Estrogen/progestin Replacement Study (HERS), a randomized, double-blinded, controlled trial to evaluate hormone treatment for the prevention of coronary heart disease events in women with established coronary heart disease. Participants were randomly assigned to daily 0.625 mg of conjugated estrogens plus 2.5 mg of medroxyprogesterone acetate or placebo and followed for a mean of 4.1 years. We analyzed results for 1,318 women randomized to hormone therapy and 1,336 women randomized to placebo who had complete data. Participant demographic information, reproductive history, sexual activity, vaginal symptoms, urinary incontinence symptoms, overall health status and chronic medical illnesses including diabetes were assessed at baseline and each annual visit. Number of UTIs diagnosed by a physician in the previous year was assessed by self-report at each annual visit. A proportional-odds model for repeated ordinal measures was used to test the treatment effect at all annual visits together, and risk factors for UTI were assessed using generalized estimating equation logistic models.

RESULTS: UTI frequency tended to be a little higher in the group randomized to hormone treatment, although the difference was not statistically significant [Odds Ratio (OR) 1.2; 95% confidence interval 1.0-1.4]. Among the 116 women who reported multiple UTIs in the

previous year at baseline, there was no difference in the odds of UTI for the oral hormone vs. placebo group [OR 1.1 (0.7–1.9)]. The average incidence of UTI at each annual visit was 12.3%. Among women with at least one UTI during a reporting period, 32.2% had two or more UTIs in the previous year. In multivariate analysis, risk factors significantly associated with UTI included diabetes requiring medications [OR 1.6 (1.3–2.0)], urinary urge incontinence symptoms [OR 1.5 (1.3–1.7)], frequent vaginal itching or dryness [OR 1.4 (1.1–1.7)], history of child-birth [OR 1.4 (1.0–1.9)], and self-reported poor health [OR 1.4 (1.2–1.6)]. UTI in the previous year was strongly associated with incident UTI [single UTI: OR 6.9 (5.8–8.1); multiple UTI: OR 18.2 (14.0–23.5)]. After controlling for diabetes status, fasting blood glucose level was not associated with risk of UTI. Sexual intercourse was not a risk factor for UTI.

CONCLUSION: Oral hormone therapy did not reduce the frequency of urinary tract infections. Potentially modifiable risk factors to decrease risk of urinary tract infection in postmenopausal women are different from those for younger women, and include diabetes, vaginal symptoms, and urge incontinence.

THYROID DYSFUNCTION IN AN ELDERLY BIRACIAL POPULATION: THE HEALTH, AGING, AND BODY COMPOSITION STUDY. A.M. Kanaya¹, F. Harris¹, S. Volpato², E.J. Perez-stable¹, T. Harris², D.C. Bauer¹; ¹UCSF, San Francisco, San Francisco, CA; ²National Institute on Aging, NIH, Bethesda, MD

PURPOSE: Thyroid dysfunction increases with age, is more common among women, and among whites is associated with lipid changes. There is limited information on racial differences in thyroid dysfunction and the association with cholesterol abnormalities.

METHODS: Health ABC is a cohort of well-functioning whites and blacks, ages 70–79, recruited for a population-based study in 1997. We analyzed thyrotropin (TSH) and serum cholesterol test results for 2,742 participants performed at the second annual visit. Free thyroxine (FT4) levels were performed on those with abnormal TSH results. Medical history and physical measurements were gathered at baseline and the second annual visit. The relationship between TSH and total cholesterol was analyzed with multivariable linear regression adjusting for multiple covariates.

RESULTS: 267 (9.7%) participants were taking thyroid hormone medication and 223 (83.5%) of these men and women were biochemically euthyroid. Among those not taking thyroid hormone, 2,364 (95.5%) participants were euthyroid. Subclinical hypothyroidism was the most prevalent disorder among all race/gender groups; white women had a significantly higher prevalence of subclinical hypothyroidism than black women (4.5% vs. 1.6%, $p=0.004$). Mean cholesterol levels were highest for those who were biochemically hypothyroid and lowest among those who were hyperthyroid. After adjustment for age, gender, race, body-mass index, smoking, alcohol use, diabetes, hormone-replacement therapy and thyroid hormone use, a TSH > 7.0 uIU/ml was associated with a 18 mg/dl increase in cholesterol and a TSH < 0.1 uIU/ml, with a 21 mg/dl decrease and did not differ between black and white participants.

CONCLUSION: Healthy community-dwelling elderly blacks have a lower prevalence of thyroid dysfunction compared to whites. The association between increased TSH and increased cholesterol is similar in blacks and whites.

Thyroid status of those not taking thyroid hormone therapy (n = 2,475): (%)

	Euthyroid	Hypo-thyroid	Hyper-thyroid	Subclinical Hypo.	Subclinical Hyper.
	TSH > 0.1, TSH < 7.0	TSH > 7.0, FT4 < 0.8	TSH < 0.1, FT4 > 1.8	TSH > 7.0, normal FT4	TSH < 0.1, normal FT4
White Men	95.5	0.6	0.1	3.4	0.4
Black Men	97.1	0.4	0	2.0	0.4
White Women	93.3	1.1	0.5	4.5	0.6
Black Women	96.8	0.7	0.2	1.6	0.7

URINARY TRACT INFECTIONS IN POSTMENOPAUSAL WOMEN: EFFECT OF HORMONE TREATMENT AND RISK FACTORS. A.M. Kanaya¹, J. Brown¹, E. Vittinghoff¹, S. Agarwal², S. Hulley¹, B. Foxman³; ¹University of California, San Francisco, San Francisco, CA; ²Cedars-Sinai Medical Center, Los Angeles, CA; ³University of Michigan, Ann Arbor, MI

PURPOSE: Previous studies have shown that vaginal administration of hormones decreases urinary tract infection (UTI) risk in post-menopausal women with recurrent UTIs, but evidence supporting a similar effect of oral hormones is mixed. We assessed the effects of oral hormone treatment in a randomized trial and evaluated potential risk factors for UTI frequency among post-menopausal women.

METHODS: 2,763 postmenopausal women were enrolled in the Heart and Estrogen/progestin Replacement Study (HERS), a randomized, blinded, controlled trial to evaluate hormone treatment for the prevention of coronary events in women with established heart disease. Participants were randomly assigned to daily 0.625 mg of conjugated estrogens plus 2.5 mg of medroxyprogesterone acetate or placebo and followed for a mean of 4.1 years. We analyzed results for 1,318 women randomized to hormone therapy and 1,336 women randomized to placebo who had complete data. Participant demographic and clinical information was assessed at baseline and each annual visit. Number of UTIs diagnosed by a physician in the previous year was assessed by self-report at each annual visit. A proportional-odds model was used to test the treatment effect and risk factors for UTI were assessed using generalized estimating equation logistic models.

RESULTS: UTI frequency tended to be a little higher in the group randomized to hormone treatment, though not statistically significant (Odds Ratio (OR) 1.2; 95% CI 1.0–1.4). Among the 116 women who reported multiple UTIs in the previous year at baseline, there was no difference in the odds of UTI for the oral hormone vs. placebo group [OR 1.1 (0.7–1.9)]. The average incidence of UTI at each annual visit was 12.3%. Among women with at least one UTI during a reporting period, 32.2% had two or more UTIs in the previous year. In multivariate

analysis, risk factors significantly associated with UTI included diabetes requiring medications [OR 1.6 (1.3–2.0)], urinary urge incontinence symptoms [OR 1.5 (1.3–1.7)], frequent vaginal itching or dryness [OR 1.4 (1.1–1.7)], history of child-birth [OR 1.4 (1.0–1.9)], and self-reported poor health [OR 1.4 (1.2–1.6)]. UTI in the previous year was strongly associated with incident UTI [single UTI: OR 6.9 (5.8–8.1); multiple UTI: OR 18.2 (14.0–23.5)]. Sexual intercourse was not a risk factor for UTI.

CONCLUSION: Oral hormone therapy did not reduce the frequency of urinary tract infections. Potentially modifiable risk factors to decrease risk of urinary tract infection in postmenopausal women are different from those for younger women, and include diabetes, vaginal symptoms, and urge incontinence.

COMPARISON OF SINGLE HIGH DOSE ORAL PROPAFENONE VERSUS IBUTILIDE THERAPY IN THE ACUTE CONVERSION OF NEW ONSET ATRIAL FIBRILLATION/FLUTTER. J. Kassotis¹, E. Festic¹, N. Mirhoseni¹; ¹New York Methodist, Brooklyn, NY

PURPOSE: Atrial Fibrillation (AF) is the most common sustained arrhythmia, and is the cause of significant morbidity. Risk factors which increase the incidence of AF are: increasing age, ischemic heart disease, valvular heart disease, cardiomyopathy, congestive heart failure and chronic lung disease. AF may also occur in a normal heart. Ibutilide, a class III antiarrhythmic agent, has emerged as an effective intravenous agent for treatment of AF to normal sinus rhythm (NSR). However it is associated with a 4% chance of ventricular arrhythmias. In addition, Ibutilide remains a very expensive medication often requiring a second dose. Oral Propafenone, a class IC agent, has been shown to be an effective treatment for recent-onset AF in patients without structural heart disease. It is an inexpensive medication usually administered as an oral medication in the dose range of 150 to 300 mg. Recently it was shown that a single dose of 600 mg of Propafenone, was successful compared to control, in converting recent-onset AF to NSR. Although the effectiveness of both oral Propafenone and Ibutilide, individually, in converting recent-onset AF to NSR has been well established, the comparative studies of these agents are lacking. The aim of this study is to evaluate in a randomized fashion the efficacy of Intravenous Ibutilide versus a single 600 mg dose of oral Propafenone in treatment of recent-onset AF.

METHODS: All eligible patients with recent-onset AF/atrial Flutter (AFL) were identified and placed in the telemetry unit. Recent-onset AF was defined as arrhythmia lasting less than two weeks duration. The onset of AF was established based on symptoms such as palpitation, dizziness, chest pain or electrocardiogram (ECG) finding. If timing of onset of AF/AFL was unclear or greater than 48 hours, before attempting cardioversion, patients underwent a transesophageal echocardiogram (TEE). All patients were started on heparin and their prothrombin time (PTT) was maintained at 1.5–2.0 times control. If patients had a rapid ventricular rate of greater than 100 bpm, AV nodal blocking agents were used to control the ventricular response. Exclusion criteria included: age less than 18 year old; hemodynamically unstable patient; previous myocardial infarction; known coronary artery disease; unstable angina; NYHA functional class greater than 1 heart failure; ventricular rate less than 60 bpm; hypokalemia (less than 3.5 mEq/L); hyperkalemia (greater than 5.5 mEq/L); untreated thyroid disease; recent GI bleeding; history of adverse drug reaction; contraindication to heparin; second or third degree atrioventricular block; bifascicular block; Complete bundle branch block; sick sinus syndrome; other antiarrhythmic therapy used within 8 hours prior to the enrollment in this study. Patients were monitored, TSH was checked and all electrolytes including potassium and magnesium were supplemented prior to drug administration. After it was determined that AF/AFL has been stable with a documented duration of greater or equal to one hour, eligible patients were randomly assigned to receive a single 600 mg dose of oral Propafenone or an intravenous infusion of 1 mg of Ibutilide over 10 minutes. A second infusion of Ibutilide was given 10 minutes after completion of the first infusion if conversion did not occur. Serial ECG's were performed following drug administration. A 12-lead ECG was performed immediately upon cardioversion to NSR. Cardioversion was identified as a stable sinus rhythm lasting for at least one hour. In case of persistent AF/AFL, patients underwent electrical cardioversion after the 24-hour observation period.

RESULTS: We identified a total of 100 patients with recent-onset AF/AFL. Of the 100 patients 37 were eligible for enrollment in this study. 19 patients (51%) converted to NSR spontaneously. 18 patients were randomized to either receive Propafenone(8) or Ibutilide(10). 6 of 8 patients converted to NSR following Propafenone(75% conversion rate) with a mean time to conversion of 4.5 hours. 7 of 10 patients converted to NSR following Ibutilide (70% conversion rate). No significant difference in conversion rate between drug arms. All patients except for one required two doses of Ibutilide for conversion. Mean time for conversion to NSR with Ibutilide was 2 hours. No significant ventricular tachyarrhythmias were noted.

CONCLUSION: Propafenone is a safe, inexpensive and well-tolerated agent for the acute conversion of recent-onset AF/AFL to NSR, with the same efficacy as Ibutilide. In select patients(e.g. patients with no structural heart disease) Propafenone may be safe to use as an outpatient basis.

PATTERNS OF USE AND ATTITUDES TOWARDS ELECTRONIC COMMUNICATION BETWEEN PATIENTS AND THEIR PHYSICIANS. S.J. Katz¹, C.A. Moyer¹, K. Dobias¹, D. Cox¹, D.T. Stern¹; ¹University of Michigan Medical Center, Ann Arbor, MI

PURPOSE: Patients are increasingly using email to communicate with their physicians. However, both providers and patients have reported concern about the impact of electronic patient-provider communication on the doctor-patient relationship and on utilization of clinic resources. In the context of a randomized controlled trial of both email and web-based patient-provider communication that we are performing at selected primary care sites at the University of Michigan, we examined current use and attitudes of patients and their physicians towards the use of email in medical practice.

METHODS: We performed a survey of primary care physicians (N=29 faculty and 97 residents, response rate=93.3%) in two large primary care clinics and a consecutive sample of patients visiting these physicians over a two week period in 1999 (N=476, response rate=80.0%).

RESULTS: 52.1% of patients reported having personal email accounts and 33.8% used email at least daily but only 10% of email users ever emailed their doctor. 63.6% were willing to use

email to communicate with their doctor. These patients were more likely than those unwilling to use email to be younger (mean age 47.1 vs 55.4), have higher incomes (>\$75,000, 32.2% vs 17.0%) and to report better health (self assessed 10pt scale, 7.2 vs 6.5, All $p < .01$). Patients and physicians reported a number of barriers to use of email: 46.2% of patients who were email users "would rather speak to a real person" and 31.5% thought email "would take too long" but only 11.1% were worried about privacy at work or home. 47.4% of physicians were afraid they would be overwhelmed with patient emails, 30.2% did not feel they had enough time to check patient emails, and 44.8% felt patient email would add rather than substitute for other tasks. More physicians than patients thought that email could improve the patient-physician relationship (67.0% vs 54.3% of patient email users, $p < .05$) but physicians were more concerned than patients over security issues (43.2% vs 33.4% of patient email users, $p < .05$).

CONCLUSION: Patients and their physicians are increasingly connected to email but few patients are connecting to their doctors. Our results suggest that patients perceive different barriers to use of email than their physicians. Patients may be more concerned about whether email will substitute for phone calls or visits versus concerns about privacy. Physicians appear to be optimistic about the role of email in improving the patient-physician relationship, but have concerns about privacy and workload. Research is needed to address these concerns about the efficiency, effectiveness and appropriate use of electronic patient-physician communication.

EVALUATION OF A COMPREHENSIVE IN-HOME CASE MANAGEMENT INTERVENTION IN IMPROVING STROKE SURVIVORS ABILITY TO RECOGNIZE AND PROPERLY RESPOND TO STROKE SYMPTOMS. G. Kefalas¹, S. Hazelett¹, K. Hua², K.R. Allen¹, G.C. Wickstrom³, ¹Summa Health System, Akron, OH; ²Northeastern Ohio Universities College of Medicine, Rootstown, OH; ³Summa Health System, Akron, OH

PURPOSE: Lack of knowledge of stroke signs and symptoms is prevalent, even among stroke survivors. This lack of knowledge can translate into a delay in seeking treatment at the onset of stroke symptoms. Such delays limit the effectiveness of current treatments for stroke. It is assumed that the education stroke survivors receive on an acute stroke unit is sufficient to ensure that they will recognize and properly respond to stroke symptoms after discharge. This assumption has not been tested.

METHODS: This study examined stroke survivors' knowledge of stroke symptoms and their anticipated response to these symptoms at the time of discharge from an acute stroke unit and after either a 3 month post-discharge comprehensive team management intervention or usual post-discharge care. This was a secondary analysis of data from a randomized controlled pilot study measuring the overall impact of the post-discharge intervention and included 96 patients discharged from an acute stroke unit. The investigator-generated "stroke knowledge test" assessed 1) how patients would respond to specific stroke symptoms (i.e., call 911, call their Dr, call a friend, take care of themselves), 2) whether they could correctly label varied symptoms as stroke-related, and 3) if they would call 911 if they knew they were having a stroke.

RESULTS: Overall, the percentage of correct responses declined over three months in the control group but increased in the intervention group. At three months 91% of intervention patients and 85% of controls recognized numbness in an extremity as a stroke symptom, and, of these, significantly more intervention patients ($p = 0.02$) reportedly would have called 911 if they experienced such numbness. Likewise, 100% of intervention and 97% of control patients recognized sudden speech difficulties as a stroke symptom at three months, but significantly more intervention patients would have called 911 if they experienced these speech difficulties ($p = 0.002$). Lastly, 94% of intervention patients and 88% of controls recognized sudden onset of dizziness as a stroke symptom at three months, but significantly more intervention patients ($p = 0.005$) would have called 911 if they experienced such dizziness. When asked at three months what they would do first if they knew they were having a stroke, 86% of intervention patients and 74% of controls said they would call 911.

CONCLUSION: This study showed that stroke education incorporated in a comprehensive post-discharge intervention improves post-discharge knowledge of stroke symptoms compared to education on the stroke unit alone, and, in as much as self-reported anticipated behavior correlates with actual performance, results in more patients calling 911 at the onset of stroke symptoms. These results suggest that in-hospital education alone may not be as effective as that combined with education delivered in an outpatient setting. They also suggest that the emphasis of educational efforts should be on the appropriate response to stroke symptoms, not just on stroke symptom recognition.

THE COST-EFFECTIVENESS OF NEW THERAPIES FOR ACUTE CARDIAC ISCHEMIA DEPENDS ON BASELINE RISK: THE EXAMPLE OF EPTIFIBATIDE. D.M. Kent¹, R. Ruthazer¹, H.P. Selker¹, ¹New England Medical Center, Tufts University School of Medicine, Boston, MA

PURPOSE: In the PURSUIT trial, eptifibatide was found to reduce the risk of acute myocardial infarction (AMI) and mortality (composite) from 15.7% to 14.2% (relative risk reduction [RRR] = 9.6 percent) in patients presenting with non-ST elevation acute cardiac ischemia (ACI). A published cost-effectiveness analysis, based only on the more favorable United States (US) results of the trial, found that the cost-effectiveness ratio for eptifibatide for this indication was \$16,491 per year of life saved. We sought to estimate the cost-effectiveness of eptifibatide based on the results of the entire trial and the expected risk of AMI/death in a community based-sample of patients.

METHODS: Using logistic regression equations to predict patient-specific risks of AMI and death and assuming a constant RRR of 9.6 percent for AMI and death, we estimated the cost-effectiveness of routine use of eptifibatide across a population of 2,780 consecutive patients with non-ST elevation ACI who were admitted to hospital, but were not treated with eptifibatide.

RESULTS: Predictions were obtained on 2,766 (99.5 %) of patients who met inclusion criteria. Of these patients, 2.0 percent died at 30 days and 12.2 percent had the composite outcome of AMI or death. Given these overall risks and a constant RRR, the average cost per year of life

saved in our sample was estimated to be \$41,832 per year of life saved. Moreover, adjusting each patient's risk of AMI and death using individual patient characteristics, only 6 percent of patients were at sufficiently high-risk to warrant therapy under a threshold of \$20,000 per year of life saved. More than half of admitted patients were at such low risk for a bad outcome, even without therapy, that the marginal cost-effectiveness in these patients was greater than \$50,000 per year of life saved. According to our analysis, one in twenty admitted patients were at insufficient risk to warrant therapy even under a threshold of \$100,000.

CONCLUSION: Given a constant RRR, use of eptifibatide is economically attractive in high-risk patients, but less attractive in low risk patients. Strategies to risk stratify patients to optimize clinical and economic outcomes should be considered.

THE IMPACT OF PROCEDURE-RELATED DELAY AND MORTALITY RISK ON THE BENEFITS OF PRIMARY ANGIOPLASTY: A METAREGRESSION. D.M. Kent¹, J. Lau¹, H.P. Selker¹, ¹New England Medical Center, Tufts University School of Medicine, Boston, MA

PURPOSE: It has been demonstrated that primary angioplasty (PTCA) in high-PTCA-volume hospitals is more effective in reducing mortality in reperfusion-eligible patients with acute myocardial infarction (AMI) than thrombolytic therapy. However, the benefits of PTCA disappear completely at low-volume hospitals. These findings support the strategy of bypassing low-volume community hospitals to deliver reperfusion-eligible patients with AMI to high-volume cardiac centers. However, bypassing community hospitals, or transferring patients from their emergency departments, causes delays that may nullify the expected benefits of PTCA. Moreover, immediate thrombolytic therapy leads to excellent outcomes in most patients. Thus, it is not clear which patients might benefit from PTCA, in the face of additional transport-related delay.

METHODS: We used the results of ten published trials and meta-regression techniques to assess the relationship between the treatment benefit of PTCA (in terms of the reduction in mortality rate in a trial) and procedure-related time delay (i.e. the difference between the median "door-to-needle" time with thrombolytic therapy and the median "door-to-balloon" time with PTCA in a trial). The magnitude and statistical significance of the relationship was estimated using weighted least squared regression, weighting each trial by the inverse of the variance of its log odds ratio. We also assessed the relationship between treatment benefit and mortality risk (using the mortality rate in the control group treated with thrombolytic therapy as a measure of baseline risk).

RESULTS: A statistically significant trend demonstrates that the treatment effect of PTCA decreases across trials as procedure-related delay increases ($p = 0.014$). The regression line crosses the x-axis at 48 minutes, suggesting that, where the median procedure-related delay is greater than about three quarters of an hour, PTCA may be no more effective than thrombolytic therapy and may be harmful, even at the high volume centers represented in these trials. Additionally, all trials in which the baseline mortality risk of the patients was moderate to high (reflected by a mortality rate in the thrombolytic treatment arm of greater than 5 percent), showed treatment benefit, while four out of five trials with low mortality risks demonstrated no benefit for PTCA, and potential harm.

CONCLUSION: Our meta-regression suggests that high-risk patient populations get substantially greater benefit from PTCA than low risk patients with AMI. It also suggests that the benefits of PTCA, on average, are nullified where the procedure-related delay is about 45 minutes or more. How this "time interval to mortality equivalence" varies with a patient's mortality risk, and with the time elapsed from symptom onset, requires modeling on individual patient data.

THE INFLUENCE OF HEALTH STATUS ON SATISFACTION AMONG PATIENTS WITH DIABETES. E. Kerr¹, D. Smith¹, R. Hayward¹, S. Krein¹, S. Kaplan², ¹VA Center for Practice Management and Outcomes Research and University of Michigan, Ann Arbor, MI; ²Primary Care Outcomes Research Institute, Tufts University School of Medicine, Boston, MA

PURPOSE: Studies suggest that health status influences patient satisfaction, but little work has examined the influence of general physical functioning versus disease severity health status measures on satisfaction. Using data from a survey of diabetic patients, our objectives were to examine 1) the association between health status and patient satisfaction using two different health status measures and 2) whether the associations differed with different dimensions of satisfaction.

METHODS: We surveyed 2000 patients receiving diabetes care across four Veterans Integrated Service Networks in fiscal years 1998 and 1999. Diabetes severity was measured using diabetes-related components of the Total Illness Burden Index (TIBI), a measure of diabetes complications and co-morbidities. Physical health status was measured by the Physical Function Index (PFI10) from the Short Form 36. Using satisfaction with overall quality as the dependent variable, we constructed two separate multiple linear regression models that examined the association between physical health status (PFI10 or TIBI) and overall satisfaction, controlling for gender, race, income, education, age, number of primary care visits, and presence of mental health diagnosis. We re-ran these models using a scale of satisfaction with patient-provider communication as the dependent variable.

RESULTS: 70% of eligible patients responded to the survey (n=1314). These patients reported low levels of physical function (PFI10 mean(SD) = 46.5(30.2)), and high levels of diabetes severity (TIBI mean(SD) = 40.5(18.8)). Overall, these patients were moderately satisfied, with 65% percent reporting excellent or very good quality of care. Lower levels of physical functioning were associated with less satisfaction ($p < .01$), but generic physical health status explained less than 1% of the variation in satisfaction (model r-squared =.03). In contrast, diabetes-related severity and co-morbidities were more strongly associated with lower levels of satisfaction ($p < .001$), with the TIBI explaining 4% of satisfaction variation (model r-squared =.05). Results were similar when we examined satisfaction with patient-provider communication.

CONCLUSION: In this diabetes population, diabetes-related disease severity and co-morbidities explained a greater portion of the variance in satisfaction than did a measure of physical functioning. Adjusting only for general physical function in studies comparing

satisfaction among individual providers or organizations may not be sufficient if patients of these providers have similar physical functioning but differing degrees of disease severity.

DO ADDICTED HOMELESS PERSONS RELAPSE SOONER? S. Kertesz¹, P. Friedmann², N. Horton¹, M. Winter³; ¹Boston Medical Ctr & Boston Univ School of Medicine, Brookline, MA; ²Rhode Island Hospital & Brown University, Providence, RI; ³Boston University School of Public Health, Boston, MA

PURPOSE: Although attention to substance abuse is considered important in the care of the homeless, published data do not indicate whether homelessness carries a worse prognosis following detoxification. We studied the association between housing status and substance abuse relapse following detoxification (detox).

METHODS: Subjects underwent inpatient detox for alcohol, cocaine or heroin from 6/97 to 3/99. Baseline interviews assessed substance use, homelessness during the 6 months prior to detox, and demographic, health and psychosocial variables. Six months later, subjects reported recurrent substance use (relapse) and post-detox treatment history. Survival analysis (proportional hazards regression) was used to assess the association between homelessness, other clinically relevant predictors, and time to relapse following detox. We specified a single interaction term, homelessness with Inpatient Sobriety Stabilization Program (ISSP) exposure. ISSP's are voluntary, short-term transitional sobriety-oriented facilities, often called "holding programs."

RESULTS: Of 254 subjects (54% of 470) available at 6 months, 72% reported relapse, occurring a median of 31 days after detox. Forty-nine percent of subjects had reported 1 homeless night before detox. Follow-up rates did not significantly differ between homeless and nonhomeless subjects. There was no difference in time to relapse for subjects stratified by housing status or by varying durations of homelessness. Controlling for demographic and psychosocial variables, subjects with full-time employment were at significantly lower risk of relapse ISSP- (Hazard Ratio=0.71, 95% CI .50-.98). There was no main effect for homelessness ($p=.19$) but there was a significant interaction between homelessness and ISSP exposure ($p=.04$). The Table shows the proportion relapsed at 6 months, broken down by housing status and ISSP exposure, and helps to demonstrate the interaction. Homeless persons who used ISSP's had the lowest relapse risk.

6-month Relapse Rates, by Housing & ISSP exposure status

	Housed	Homeless
ISSP-	80%	87%
ISSP+	74%	62%

CONCLUSION: Homelessness did not predict more rapid relapse after detoxification. This counter-intuitive finding may be explained in part by homeless persons' use of Inpatient Sobriety Stabilization Programs. These data suggest that ancillary services for a vulnerable population were associated with a desirable clinical outcome. This experience may inform the development of new care initiatives, estimated at \$1 billion, designed to help the homeless successfully transition off the street.

AGE AND RACE RELATED DIFFERENCES IN HIV DISEASE EXPRESSION: THE HCSUS AND VACS EXPERIENCE. A. Kilbourne¹, D.S. Zingmond², A.C. Justice¹, N.S. Wenger², M. Rodriguez-barradas³, L. Rabeneck³, D. Taub⁴, S.A. Bozzette⁵; ¹University of Pittsburgh, Pittsburgh, PA; ²UCLA, Los Angeles, CA; ³Baylor College of Medicine, Houston, TX; ⁴National Institute of Aging, NIH, Baltimore, MD; ⁵University of California, San Diego, San Diego, CA

PURPOSE: Previous work in the HIV Cost and Services Utilization Study (HCSUS) demonstrated poorer survival among older, nonwhite patients. We examine differences in disease expression (HIV symptoms, HIV comorbidities, CD4 count), by age and race and compare these results from HCSUS with a more recent cohort of HIV+ veterans in care: Veterans with HIV/AIDS Cohort Study (VACS).

METHODS: HCSUS is a national probability sample of 2864 HIV+ adults receiving care in the U.S. interviewed 1/96-2/97. VACS is a multi-site VA cohort of 881 HIV+ adults assembled between 6/99-7/00. Patients in both studies completed a survey concerning symptoms and HIV comorbidities. In separate analyses of VACS and HCSUS data, we compared the prevalence of 8 symptoms and 18 CDC-defined HIV comorbidities common to both studies, among older (>=50 years) nonwhites, younger nonwhites, older whites; compared to younger whites (reference group). Illness severity was assessed using CD4 count/mm3.

RESULTS: HCSUS compared to VACS participants were more likely white and younger (respectively 49% vs. 33% white- $P<0.001$; 90% vs. 56% <50 years- $P<0.001$). Few differences were apparent in either cohort when symptom and comorbidity results were compared by age and race alone; however differences were apparent when stratified by age/race groups. Symptom counts were higher among younger whites compared to older nonwhites (HCSUS: 3.1 vs. 2.5- $P=0.07$; VACS: 4.4 vs. 3.5- $P<0.001$). HIV comorbidity counts were higher for younger whites compared to older nonwhites (HCSUS: 1.9 vs. 1.4, $P<0.05$; VACS: 1.2 vs. 0.8, $P<0.01$). In VACS, no significant differences were apparent in current lab-based CD4 count. In HCSUS, younger whites had higher self-reported CD4 counts (374) compared to younger nonwhites (297; $P<0.01$), older whites (295; $P<0.05$), and older nonwhites (238; $P<0.01$).

CONCLUSION: Similarities in results between HCSUS and VACS are striking given different data collection methods and populations. Older nonwhites reported fewer HIV-related symptoms and comorbidities than younger whites. However, in HCSUS, older nonwhites appeared more severely ill than younger whites, suggesting that HIV symptoms and comorbidities are underreported. Better understanding of differences in HIV expression by age and race may decrease disparities in HIV diagnosis and management for older minorities.

PREVALENCE OF GENERAL MEDICAL COMORBIDITY AMONG HIV+ VETERANS: IS THERE A ROLE FOR GENERAL MEDICINE IN HIV CARE? A. Kilbourne¹, A.C. Justice¹, S. Weissman², L. Rabeneck³, M. Rodriguez-Barradas³; ¹University of Pittsburgh, Pittsburgh, PA; ²VA Cleveland Medical Center, Cleveland, OH; ³Baylor College of Medicine, Houston, TX

PURPOSE: There is an ongoing debate on whether HIV+ patients are more optimally managed by generalists or specialists. The profile of comorbidity among HIV+ patients has evolved. General medical comorbidities such as hepatitis and diabetes are becoming increasingly prevalent. No large cohort studies have reported general medical comorbidities among HIV+ patients. We examined the prevalence of HIV comorbidities and general medical comorbidities based on a recent cohort of HIV+ individuals in care within the VA system: the Veterans with HIV/AIDS Cohort Study (VACS).

METHODS: VACS is a multi-site cohort of 881 HIV+ adults from Cleveland, Manhattan, and Houston VA ID clinics assembled between 6/99-7/00, representing 85% of the HIV+ patients seen at these clinics during this interval. Providers were queried regarding their patients' CDC-defined HIV comorbidities (19 total) and general medical comorbidities (13 total- based on the Duke and Charlson comorbidity scales). CD4 count was obtained from VA computerized laboratory data. We identified CD4 count lab results nearest to the date of study enrollment for all patients, 71% of patients had CD4 laboratory data on or before the survey.

RESULTS: The mean age of VACS participants was 49 years, 67% were nonwhite, and 27% had a CD4 count of <200/mm3. The mean number of HIV comorbidities experienced was 1.1, yet the mean number of general medical comorbidities was 1.4 ($p<0.001$). HIV+ patients with a CD4 count of <200/mm3 had a higher number of HIV comorbidities (respectively 1.7 vs. 0.8, $P<0.001$), yet there was no significant difference in general medical comorbidity count (respectively 1.4, 1.4, $P=0.68$). The most common HIV comorbidities reported by providers were esophageal candidiasis (21%), peripheral neuropathy (16%), shingles (16%), bacterial pneumonia (13%), and PCP (12%). The most common general medical comorbidities included hepatitis (51%), hypertension (24%), hyperlipidemia (17%), diabetes (11%), and pulmonary disease (11%). Over a third (39%) of patients experienced >2 general medical comorbidities.

CONCLUSION: General medical comorbidities may constitute a higher disease burden for HIV+ Veterans than HIV comorbidities. As the HIV epidemic enters its 3rd decade, HIV clinicians and researchers are faced with an increasing prevalence of general medical comorbidities experienced by their HIV patients. Our results suggest a substantial role for general medical management in HIV care.

DOES DEPOT MEDROXYPROGESTERONE CONTRACEPTION INCREASE DIABETES RISK IN NAVAJO WOMEN? C. Kim¹, K. Seider², E.A. Begier², Y.S. Kwok¹; ¹University of Washington, Seattle, WA; ²Crownpoint Healthcare Facility, Crownpoint, NM

PURPOSE: In Navajo women, a population at high-risk for type 2 diabetes mellitus, depot medroxyprogesterone (DMPA) is a common contraceptive due to its ease of administration. However, recent evidence suggests that DMPA contraception may lead to weight gain and independently decrease insulin sensitivity. The objective of this study was to determine if DMPA was associated with development of diabetes in Navajo women.

METHODS: We conducted a clinic-based case-control study. Eligible subjects were Navajo women aged 18-50 years who had seen a healthcare provider at a Navajo Area Indian Health Service clinic at least once in 1998. Diabetic cases ($n=284$) and non-diabetic controls ($n=570$) were matched by age. Medical records were reviewed to determine contraception use before the diagnosis date of diabetes in each case and its matched controls.

RESULTS: DMPA users were more likely to develop diabetes than patients who had used combination estrogen-progesterone oral contraception only (OR 3.8, 95% CI [1.8-7.9]). The greater odds persisted after adjustment for body mass index (BMI) (OR 3.6, 95% CI [1.6-7.9]). When DMPA use of 3 months vs. 4-11 months vs. 12 or more months was compared, risk of diabetes was associated with longer use ($p=0.02$). Longer DMPA use was also associated with greater BMI; each additional month of use was associated with a 0.12 kg/m2 greater BMI (95% CI 0.001-0.24).

CONCLUSION: DMPA contraception was associated with greater risk of diabetes compared to combination oral contraception, and risk was greater with longer use and persisted after adjustment for body mass index. Additional research is needed to confirm these results and elucidate the mechanism, but DMPA may have this previously unrecognized side effect that could influence choice of contraceptive method, especially in women at high risk for diabetes mellitus.

ORAL CONTRACEPTIVE USE AND RISK OF DIABETES IN YOUNG ADULT WOMEN. C. Kim¹, D.S. Siscovick², S. Sidney³, C.E. Lewis⁴, C.I. Kiefe⁴, T.D. Koepsell²; ¹University of Washington, Seattle, WA; ²University of Washington, Seattle, WA; ³Kaiser Permanente, Oakland, CA; ⁴University of Alabama, Birmingham, AL

PURPOSE: Diabetes mellitus affects an increasingly younger population. Although over 10 million women in the United States use oral contraceptives (OCs), the association between diabetes and current OC use in young adult women is unclear. We studied the associations between 1)current OC use and 2)glucose levels, insulin levels, and diabetes in young adult women.

METHODS: Female participants ($n=2,787$) in the Coronary Artery Risk Development in Young Adults study (CARDIA), a multi-center, prospective observational study of African-Americans and whites, were aged 18-30 years at enrollment. OC use, fasting glucose, fasting insulin levels, and diabetes diagnosis were assessed at study examination years 0, 7, and 10. Current OC use was defined as OC use at each time of examination; non-use (the reference group) was defined as combined past OC use and never use at each examination. Using generalized estimating equations, we analyzed the cross-sectional associations between 1)current OC use and 2)fasting glucose, fasting insulin, and presence of diabetes.

RESULTS: Current OC users differed from non-users on several factors. Current users were younger, had a lower mean body mass index (BMI), lower smoking prevalence, lower parity, and higher mean total cholesterol at all exams. By year 7, current users were more highly educated and by year 10 current users were more likely to be white and were more physically active than non-users. In unadjusted analyses, current use was associated with lower fasting glucose levels [-4.9 mg/dl, 95% CI (-5.7 , -4.1)], lower fasting insulin levels [-1.1 uU/ml, 95% CI (-1.6 , -0.5)], and a non-significant reduction in the odds of diabetes [OR 0.7, 95% CI (0.4, 1.3)] compared to non-use. After adjustment for the covariates noted above, current OC use was associated with lower fasting glucose levels [-2.4 mg/dl, 95% CI (-3.4 , -1.4)] and lower risk of diabetes [OR 0.6, 95% CI (0.4, 1.0)] than non-use, but not with insulin levels [0.2 uU/ml, 95% CI (-0.6 , 1.0)].

CONCLUSION: Current OC use is associated with lower glucose levels and less diabetes in a population of African-Americans and white women than non-use.

EARLY SPECIALIST EVALUATION OF CHRONIC RENAL DISEASE: WHO GETS IT AND DOES IT MAKE A DIFFERENCE? K. Kinchen¹, N.E. Fink¹, J. Sadler², R. Brookmeyer¹, N.R. Powe¹; ¹Johns Hopkins University, Baltimore, MD; ²University of Maryland, Baltimore, MD

PURPOSE: End-stage renal disease (ESRD) affects almost 300,000 people and has an annual mortality of about 20%. Pre-ESRD care involves managing the complications of chronic renal failure and establishing permanent vascular access. Prior research suggests that patients with ESRD often are not referred to nephrologists in a timely manner. We performed this study to determine the clinical and sociodemographic factors associated with late evaluation (LE) and to assess whether LE is associated with worse survival?

METHODS: We conducted a national concurrent prospective cohort study of 427 new-onset, hypertension or diabetes-related ESRD patients enrolled between 1995 and 1998. We categorized patients based on the time from first evaluation by a nephrologist to the initiation of dialysis: LE (<4 months), middle evaluation (ME) (4–12 months) and early evaluation (EE) (>1 year). Multinomial logistic regression was used to determine the factors associated with LE and ME compared to EE patients. We used Kaplan-Meier survival estimates and Cox proportional hazards models to examine the independent association between LE and mortality. Mean follow-up is 2.3 years.

RESULTS: LE patients have significantly lower baseline hematocrits than EE patients (27.8 vs. 29.1 respectively), lower use of pre-ESRD erythropoietin (15.1% vs. 24.6%), and are less likely to have permanent vascular access established 1.5 months before hemodialysis (13.3% vs. 43.6%). After adjusting for age, race, sex, lack of health insurance, education, and marital status, the odds of LE are 1.47 (95% CI 1.15–1.88) for higher comorbidity/disease severity as measured by the Index of Coexistent Disease (ICED). Compared to white males, black males have a 5 times higher odds (OR 4.97 [95% CI 2.25–10.98]) and black females a 2.4 times higher odds (OR 2.42 [1.15–5.12]) of LE. When compared to EE patients, the RR of death for LE patients is 1.6 (1.01–2.53) adjusted for age, race, sex, dialysis modality, education, pre-ESRD employment, years smoking, and ICED score. The RR decreases to 1.48 (.89–2.44) after including baseline serum albumin, hematocrit, and creatinine in the model.

CONCLUSION: Late evaluation by a nephrologist was associated with black race and higher ICED scores. When compared to EE, LE appears to be associated with worse survival. While the RR is attenuated when baseline serum albumin, creatinine, and hematocrit are controlled for, it is important to consider that late evaluation itself may contribute to lower albumin and hematocrit through poorer nutritional guidance and less use of pre-ESRD erythropoietin

FINDING EXECUTIVE DYSFUNCTION: COMPARISON OF THREE MENTAL STATUS TESTS. B.P. Kinosian¹, A. Sevrin¹, R. Berlin¹, A. Nelson¹, J. Handa¹; ¹University of Pennsylvania, Philadelphia, PA

PURPOSE: Standardized mental status examinations are widely used to identify cognitive impairment in frail, older patients. However, they differ in the domains they assess, particularly executive function. We compared the Short Portable Mental Status Questionnaire (SPMSQ), the standard instrument used in the Program for All-Inclusive Care of the Elderly and many state geriatric assessment programs, with the Mini-Mental Status Examination (MMSE), and the Modified MMSE (3MSE). We hypothesized that the 3MSE would identify more frail elders as cognitively impaired than either the SPMSQ or the MMSE.

METHODS: Population: Urban PACE site, with 75 frail elderly PACE participants assessed on entry. Instruments: SPMSQ was administered by the program social worker, the MMSE and 3MSE were administered by the primary care provider, and a psychiatrist blind to the other tests administered a second MMSE and applied DSM IV criteria for dementia. Test Evaluation: Standard test cutoffs were applied (SPMSQ >2 errors, 3 if <9th grade education; MMSE <24, 3MSE <78); test sensitivity and specificity were calculated compared to clinical diagnosis. MMSE was linearly converted to the 3MSE scale, for comparison of test calibration. Test performance was compared in prior specified subgroups of special interest (e.g. MMSE >24).

RESULTS: Average age was 84 $\pm$ 9 yrs, 72% female, 33% African American, average 10.3 years of education, with 3.2 activity of daily living (ADL) impairments (87% dependent in 2+ ADLs). Mean SPMSQ: 3.2 $\pm$ 2.3, MMSE: 23 $\pm$ 6.1, mean 3MSE: 62.9 $\pm$ 19. Prevalence of dementia was 61% (46/75). Test sensitivity/specificity were: .73/.83 (SPMSQ), .67/.89 (MMSE), and .88/.96 (3MSE). Among the 15 patients with dementia and an MMSE >24, 11 (73%) were identified by 3MSE, 3 by SPMSQ. Of the 4 patients with dementia and a 3MSE >78, 0 were identified by MMSE and 2 by SPMSQ. If the MMSE is transformed into a 3MSE equivalent (MMSE*(100/30)), 18% of participants had a 3MSE score 10 or more points below the transformed MMSE score (equivalent to a 3-point change on MMSE); 29% had a 3MSE score 7 or more points less than the transformed MMSE (2-point change equivalent). The 14 participants with discordant 3MSE/MMSE results exhibited executive deficits on the 3MSE not assessed on the MMSE.

CONCLUSION: We conclude that the 3MSE is a more sensitive test for cognitive impairment than the MMSE and SPMSQ, while both the 3MSE and MMSE are more specific than the SPMSQ. In those cases where the 3MSE added incremental value to the MMSE, items testing executive function were responsible for the added testing information.

BARRIERS TO HIV CARE FOR LATINOS IN MINNESOTA. J. Kirsch¹, M. Ruiz², A. Zhu³; ¹Hennepin County Medical Center, Minneapolis, MN; ²West Side Community Health Services, St. Paul, MN; ³University of Minnesota, Minneapolis, MN

PURPOSE: Across all ethnic groups, individuals are diagnosed with HIV and fail to follow-up for medical care. The Latino population has a disproportionately high HIV infection rate in Minnesota (MN) and is the most uninsured ethnic group. As it grows, the rate of HIV infection is likely to rise accordingly. Latinos are diagnosed later in the course of the disease and have significantly worse clinical outcomes. We believe that Latinos experience unique barriers. However, the impact of HIV on the Latino population in MN has not previously been studied. Knowing why people delay seeking HIV care post-diagnosis could help identify people at high risk for appointment adherence failure. Knowledge of factors associated with attrition and poor adherence could help to identify high-risk individuals and to direct outreach efforts.

METHODS: Between 11/98 and 3/99, we interviewed 60 self-identified Hispanic/Latinos, 33.3% of the known HIV positive Latino population living in MN. Participants were identified through clinics, social service agencies and advertisements. The survey was 94 questions with an open-ended discussion. Medical records were reviewed to assess use of medical facilities and record clinical measures.

RESULTS: All respondents completed the interview. Most interviews (63.3%) were done in Spanish. Mean age at diagnosis was 32.2 and 75% were men. Once diagnosed, 30% of subjects delayed > 1 year before seeking medical attention. Factors associated with this delay, using multivariate analysis, included intravenous drug use (IVDU), lack of insurance, fear of disclosure, and feeling well ($p=0.0001$). Consistent with ethnographic studies, 55.6% of subjects born in Puerto Rico (PR) had IVDU as a risk factor. Difficulty in finding Spanish-speaking MD's was a barrier for 31.7% of subjects. Other factors included immigration, social stigma of HIV, and finances. Using multivariate analysis, risk factors for missing appointments included younger age, homosexuality, and having been tested due to reasons other than self-referral ($p=0.001$). Subjects listed forgetting, feeling well, transportation, language barrier, and the social stigma of HIV as reasons for missing appointments. While diagnosed later in the course of the disease, those born outside of US/PR had significantly better appointment adherence ($p=0.02$).

CONCLUSION: Dealing with a patient's decision to not receive medical care is challenging and complex. Identifying high-risk individuals, such as those with IVDU, may help direct efforts at these people. However, while IVDU is rarely a problem for the rapidly growing non-US/PR born population, language barriers and immigration are obstacles. Efforts should also include reassuring patients about confidentiality, addressing immigration issues and educating about the risks of asymptomatic HIV infection. Special attention needs to be paid to homosexual, young and non-self referring patients to increase appointment adherence. Individualized methods, such as reminder systems, should be used to increase adherence. Barriers to care need to be assessed at the time of post-test counseling, and efforts tied to social services.

DELAYED HIV TESTING FOR LATINOS IN MINNESOTA. J. Kirsch¹, M. Ruiz², A. Zhu³; ¹Hennepin County Medical Center, Minneapolis, MN; ²West Side Community Health Services, St. Paul, MN; ³University of Minnesota, Minneapolis, MN

PURPOSE: HIV has become a disease of minorities. The Latino population in Minnesota (MN) is the most uninsured ethnic group and as it grows, the rate of HIV infection will rise correspondingly. Latinos are diagnosed later in the course of their disease and have significantly worse clinical outcomes. It is our belief that HIV positive Latinos in MN are being underserved, and this problem had not been studied. This study was intended to help understand the barriers Latinos encounter causing them to test late in the course of HIV disease. With the long latency between infection and clinical manifestations, early testing may decrease morbidity, mortality and the spread of infection. Knowing why people delay seeking HIV testing could help identify people at high risk. We believe that gathering information directly from affected individuals is relevant and may be generalized to a metropolitan Latino population.

METHODS: Between 11/98 and 3/99, we interviewed and reviewed medical records of 60 self-identified Hispanic/Latinos, 33.3% of the known HIV positive Latino population living in MN. The survey was 94 questions with an open-ended discussion. Medical records were reviewed to assess use of medical facilities and record clinical measures.

RESULTS: Participants were identified through clinics, social service agencies and advertisements. All respondents completed the interview. Most interviews (63.3%) were done in Spanish. Of interviewees, 75% were men and the mean age at diagnosis was 32.2. Initial mean CD4 counts following diagnosis are lower for Latinos than whites (204 vs. 288) in MN (MN Dept. Health). Within the study population, being born outside of the US or PR was associated with late presentation (228 vs. 392, $p=0.01$). Reasons people were initially tested included illness (45%), screening (26.7%) and self-referral (25%). Using multivariate analysis, factors associated with late testing included male sex, advanced age, born outside US/PR, heterosexual and higher educational level ($p=0.0006$). Birthplace was the most significant delay factor. Immigration was a key barrier to interacting with the medical system, regardless of documentation status.

CONCLUSION: With the long latency between HIV infection and clinical disease, early detection is particularly important. In this metropolitan population of Latinos, diagnosis was relatively late in the course of the disease, as measured by decreased CD4 counts, and consistent with the finding that 75% of diagnoses were not by self-referral. The most rapidly growing Latino population, those born outside US/PR, tested significantly later in the course of the disease. Being of advanced age and heterosexual may cause further delays in

that many people still consider HIV to be an infection of young gay men and thus do not perceive self-risk. This population may need to be targeted for increased testing. Interestingly, risky behaviors decreased significantly following diagnosis, supporting the need for widespread access to HIV testing and general support for access to medical care for Latinos in MN.

DISPARITY BETWEEN PATIENTS AND PHYSICIANS IN THE IMPORTANCE OF FAITH ON MEDICAL DECISION MAKING. S. Knittig¹, P. Nietert¹, J. Zoller¹, G. Silvestri¹; ¹Medical University of South Carolina, Daniel Island, SC

PURPOSE: Advanced lung cancer is a disease with little hope of long term survival. The two treatment options for patients are chemotherapy or supportive care. Several factors may influence the decision to undergo treatment. One poorly understood factor is the influence of a patient's faith on how they make medical decisions. We compared the effect of faith on treatment decisions among doctors, patients (PTs) and their caregivers (CGs).

METHODS: 100 PTs with newly diagnosed advanced (stage IIB) or metastatic (stage IV) lung cancer, their CGs, and 257 medical oncologists were interviewed separately using a standardized questionnaire. Participants were asked to rank, from 1 to 7, the factors that influenced their treatment decisions: (1)cancer doctor's recommendation, (2)faith in God, (3)ability of treatment to cure disease, (4)side effects, (5)family doctor's recommendation, (6)spouse's recommendation, (7)children's recommendation. Rankings were compared between the 3 groups of participants using restricted simultaneous ordered logistic regression models. Comparisons between those PTs who ranked faith as a high priority (1 or 2), and those PTs who ranked faith as a low priority (3–7), were made using t-tests and chi-square tests.

RESULTS: All 3 groups ranked the oncologist's recommendation as the most important factor. However, PTs and their CGs ranked faith in God second, while physicians placed it last ($p < 0.001$). The ability of the treatment to cure disease and side effects were important to all groups. PTs and CGs ranked their family doctor, spouse or child's recommendation as less important. Physicians were generally in agreement with PTs and CGs with the exception of physicians feeling the spouse's input was more important than did the PTs or CGs. PTs who placed a high priority on faith in God ($n=53$) were more likely to be African American (40% vs. 19%; $p<0.05$). Irrespective of race, PTs who placed a high priority on faith in God were less likely to have graduated high school than those who ranked faith lower (47% vs. 72%; $p<0.001$). There was a trend for PTs who ranked faith high to leave all decisions regarding treatment to their oncologist.

CONCLUSION: PTs and CGs are in complete agreement on the factors that are important in deciding on treatment options for advanced lung cancer but differ substantially from doctors. All 3 agree that the single most important factor is the oncologist's recommendation. However, faith in God is an extremely important factor for PTs and CGs but not physicians. PTs who ranked faith high are disproportionately African American and less likely to have graduated high school.

Medical decision making incorporates tangible probabilities and utilities, (i.e. morbidity, mortality, side effects and costs of differing treatment options), into decision models that aid in a patient's treatment choice. This study suggests the previous model of decision making may be oversimplified. To our knowledge this is the first study to demonstrate that, for some, faith is an extremely important factor in medical decision making, more so than even the efficacy of treatment. If faith plays an important role in how some PTs decide treatment, and physicians are unaware of or do not account for it, the decision making process may be unsatisfactory to all involved. Future studies should clarify how faith impacts on individual decisions regarding treatment.

ADVANCE CARE PLANNING: A COMPARISON OF TWO METHODS. R. Kolarik¹, R.M. Arnold¹, G. Fischer¹; ¹University of Pittsburgh, Pittsburgh, PA

PURPOSE: Different advance directive (AD) forms have been developed, some focusing on values, others on patient preferences for specific interventions. Our goal was to compare patient responses to value-based and intervention-based modes of advance care planning.

METHODS: Outpatients age 55 or older who did not have an AD and who scored less than 9 on the Short Blessed Test for dementia were randomly assigned to complete either Emanuel's Medical Directive (EMD) or Pearlman's values history (PVH) form. Length of time to complete the form and questions patients asked while completing the form were documented. Patients were asked to take the form home and to speak with others about it. Patients were called 2–4 weeks later and asked 30 questions about the form. Questions focused on their assessment of the form's ability to reflect their goals for care, to educate them about end of life care, to reduce surrogate burden, to enhance patient control, and to improve communication. Patients were asked to mail back the completed AD.

RESULTS: Of the 246 patients approached, 137 refused, 56 already had an AD, and 3 screened out due to dementia. 50 patients were enrolled and 19 in each group completed the telephone survey. The mean age was 65 years, 66% were female, and 68% reported being in good to excellent health. There were no significant differences in age, gender, income, marital or health status between the EMD and PVH groups. Both groups took a mean of 22 min. to complete the form. Patients asked a similar number of questions about the forms (2.3 EMD, 1.9 PVH $p=.45$). There was also no difference in the percentage of individuals who had conversations with others about the AD (58% EMD, 52% PVH $p=.71$) or the proportion returning the forms by mail (36% EMD, 56% PVH $p=.15$). All PVH forms designated a surrogate, whereas 82% of EMD forms did so ($p=.036$). More patients (45%) found the PVH form to be too general to make specific treatment decisions than the EMD (25%) ($p=.02$). All of the EMD respondents and 18/19 of PVH agreed that the forms were a good first step in planning care at the end of life. Both forms were rated equally favorably (totally or mostly agree) in their ability to reflect goals for care (89% EMD vs 84% PVH $p=.59$), to educate (68% EMD vs 79% PVH $p=.67$), to reduce burden (83% EMD, 63% PVH $p=.38$) to enhance control (79% EMD, 58% PVH $p=.38$) and to improve communication (68% EMD, 47% PVH $p=.16$).

CONCLUSION: Both forms were rated favorably as tools for advance care planning. A values history was slightly more likely to result in the designation of a surrogate than a standard directive. Patients may find different types of forms equally useful for communicating their goals at the end of life.

PATIENTS KNOWLEDGE ABOUT THEIR MEDICATION IS DISTURBED BY CHANGES MADE DURING A HOSPITAL STAY. M.P. Kossovsky¹, P. Micheli¹, M. Louis Simonet¹, F.P. Sarasin¹, P. Chopard¹, T.V. Perneger¹, J. Gaspoz¹; ¹Department of Internal Medicine and Groupe de Recherche et d'Analyse en Systèmes et Soins Hospitaliers (GRASSH), Geneva, Switzerland

PURPOSE: To determine the magnitude of the changes made to patients' usual medication during a hospital stay and their impact on patients' knowledge about their treatment.

METHODS: Settings: The department of Internal medicine of the Geneva University Hospitals, Geneva, Switzerland. Methods: During 3 consecutive months, data were collected about patients' medication upon admission and at discharge. A week following discharge, patients were called at home and were asked to list their medications, to precise the indication of each drug, and to say whether they had received an information about them during their hospital stay. Only medications prescribed for a chronic condition were taken into account.

RESULTS: 367 patients were included into the study. During the hospital stay, 470 (26.2%) among the 1795 drugs present upon admission were stopped. Among the 2055 drugs prescribed at discharge, 730 (32.5%) had been introduced during the stay. The mean number of drugs per patient increased by 0.7 during the stay (4.9 upon admission and 5.6 at discharge). According to the patients, information was given to them for only 15% of the drugs prescribed at discharge; their knowledge of the indications was significantly worse for the drugs introduced during the hospital stay than for the older ones (odds-ratio 0.69; 95% CI: 0.53 – 0.89). In multivariate analysis, patients' knowledge of the indication of a drug was negatively associated with a new prescription during the hospital stay (odds-ratio 0.60; 95% CI: 0.45 – 0.82), with patients' age (odds-ratio 0.76 for each additional decade; 95% CI: 0.66 – 0.88) and with length of stay (odds-ratio 0.97 for each additional day; 95% CI: 0.95 – 1.00); it was positively associated with the fact that information was provided during the stay (odds-ratio 6.90; 95% CI: 3.0 – 16.0).

CONCLUSION: Patients' usual medication may be profoundly modified during a hospital stay on the wards of a department of internal medicine. In addition, information about the drugs prescribed at discharge is rarely given, even though it significantly improves patients' knowledge of their treatment. During a hospitalization, patients' usual medication should be modified only when necessary, and patients should be carefully informed about changes in their drug regimen.

A STRUCTURED PATIENT-CENTERED INTERVIEW ON HOSPITAL DISCHARGE SIGNIFICANTLY IMPROVES PATIENTS KNOWLEDGE ABOUT THEIR MEDICATION.

M.P. Kossovsky¹, M. Louis Simonet², F. Sarasin², P. Chopard³, T.V. Perneger⁴, J. Gaspoz³; ¹Department of Internal Medicine and Groupe de Recherche et d'Analyse en Systèmes et Soins Hospitaliers, Geneva, Geneva; ²Department of Internal Medicine and Groupe de Recherche et d'Analyse en Systèmes et Soins Hospitaliers (GRASSH), Geneva, Geneva; ³Quality of Care Unit and Department of Internal Medicine and Groupe de Recherche et d'Analyse en Systèmes et Soins Hospitaliers (GRASSH), Geneva, Geneva; ⁴Quality of Care Unit and Groupe de Recherche et d'Analyse en Systèmes et Soins Hospitaliers (GRASSH), Geneva, Geneva

PURPOSE: To determine if a structured patient-centered discharge interview, with presentation and explanation of a treatment card, can improve patients' knowledge about their medication.

METHODS: Design: Prospective controlled study.

Settings: Two clinics of general internal medicine of the Geneva University Hospitals: the intervention clinic and the control one.

Population: 809 patients discharged home.

Measurements: Patients' knowledge about their medication, assessed by a telephone interview a week after discharge.

Intervention: Before discharge, every patient leaving the intervention clinic had a structured interview with the physician in charge of him. During it, questions about the hospital stay and concerning medications were encouraged. A treatment card was handed over to the patient, discussed with him, and then given to him. Information on the card included the name of all drugs, their indications and dosages, the time and way to take them, the precautions to observe, as well as main possible side effects.

RESULTS: Results: Once returned home, patients of the intervention clinic had a significantly better knowledge of drug indications (odds-ratio 2.71; 95% CI: 1.37 – 5.40), of precautions to observe (odds-ratio 1.50; 95% CI: 1.03 – 2.19) and of possible side effects (odds-ratio 2.28; 95% CI: 1.52 – 3.42) than those of the control clinic. Adjustment for age, gender, education level or comorbidity index did not modify these results.

CONCLUSION: A structured patient-centered interview on hospital discharge, enhanced by a medication card that is explained and given to the patients, significantly increases their knowledge about their medication. These results may have positive therapeutic effects, since compliance is known to be improved by better medication knowledge.

REDUCTION IN ALCOHOL CONSUMPTION IMPROVES QUALITY OF LIFE AND DECREASES ALCOHOL-RELATED CONSEQUENCES INOUTPATIENT PROBLEM DRINKERS. K.L. Kraemer¹, S.A. Maisto², J. Conigliaro¹, M. McNeil¹, M. Kelley¹;

¹University of Pittsburgh, School of Medicine, Pittsburgh, PA; ²Syracuse University, Syracuse, NY

PURPOSE: To determine whether sustained decreases in alcohol consumption are associated with improved quality of life and fewer adverse consequences in problem-drinking outpatients.

METHODS: We analyzed data from 301 subjects enrolled in the Early Lifestyle Modification study, a randomized controlled trial of 2 types of brief intervention for problem drinking. A total of 13,439 adult patients in 12 primary care clinics were screened for alcohol problems using the Alcohol Use Disorders Identification Test and quantity-frequency questions. Enrolled subjects completed baseline, 6-month, and 12-month assessments, which included the RAND 36-item Health Survey (SF-36) and the 15-item Short Inventory of Problems (SIP), a measure of alcohol-related consequences. Alcohol consumption was assessed at each time point by the Time Line Follow Back method.

RESULTS: 213 subjects (mean age 46 years, 79% white, 71% male, 62% hazardous alcohol use, and 38% alcohol abuse /dependence) had complete SIP and SF-36 data. Alcohol consumption decreased in all treatment arms. The 77 subjects (36%) who sustained a 30% decrease in drinks/month from baseline through 12 months had significant improvement in the SF-36 Mental Component Summary score ($P = 0.037$) and Physical Component Summary score ($P = 0.058$) and had fewer alcohol-related consequences ($P < 0.001$) when compared to those who did not sustain a decrease. In addition, the 34 subjects who sustained a decrease to non-hazardous levels (defined as < 16 drinks/week for men and < 12 drinks/week for women) through 12 months had significant improvement in the SF-36 Physical Component Summary score ($P = 0.002$) and had fewer alcohol-related consequences ($P = 0.047$) when compared to those who continued a hazardous level of use.

CONCLUSION: Outpatient problem drinkers who sustain a reduction in alcohol consumption have improved health-related quality of life and fewer alcohol-related adverse consequences. Benefits are seen across a wide spectrum of alcohol problems. These findings provide additional motivation for primary care physicians to identify patients with alcohol problems and to initiate intervention.

MOTOR VEHICLE ACCIDENTS AMONG NEW JERSEY HOUSESTAFF. T. Hossain¹, D. Kuo¹; ¹Morristown Memorial Hospital/Atlantic Health System, Morristown, NJ

PURPOSE: Residency training is notorious for rigorous work schedules and demanding work load. Typical on-call schedules require residents to care for hospitalized patients for 36 hours continuously. Residents are likely to experience more sleep deprivation. Previous studies have reported that sleep deprivation is directly correlated with an increased rate of motor vehicle accidents, and motor vehicle accidents are a major cause for mortality in the United States. None of these studies have looked at housestaff from various training programs. We were interested in finding out the rate of motor vehicle accidents among residents training in various specialties in New Jersey.

METHODS: We sent out a 16 item questionnaire to a randomly selected group of residents who were training in New Jersey in internal medicine, pediatrics, general surgery, and family medicine 2000–2001. The questionnaire comprised questions about sleep and overnight call responsibilities, and the circumstances surrounding motor vehicle accidents, sleeping while driving, and traffic violations.

RESULTS: Of the 819 surveys to residents working in New Jersey — 199 internal medicine, 218 family medicine, 183 general surgery, and 216 pediatrics — we received 283 completed surveys. Our overall response rate was 34.6%. We found that over half of the residents who responded had ever fallen asleep while driving, and 21.9% of respondents were involved in an automobile accident during residency. Over half of the respondents had received a moving violation, most often for driving fast and driving to or from clinic or the office. There was no correlation between frequency of call and hours of sleep to incidence of motor vehicle accidents and receiving a citation; however, there was a higher incidence of falling asleep while driving in patients who were on call every third night and slept fewer than four hours a night ($p = 0.001$ for both). If they did have an accident, it was most likely on a post-call day returning from the hospital. 89.9% of the residents got four or fewer hours of sleep on a typical on-call night.

CONCLUSION: Residents-in-training are at high risk for motor vehicle accidents and falling asleep while driving when they are post-call and have less sleep. Mechanisms for reducing resident fatigue post-call such as a night-float system may reduce these incidences.

FACTORS ASSOCIATED WITH REPEATED EMERGENCY DEPARTMENT USE AMONG HOMELESS AND MARGINALLY HOUSED PERSONS. M. Kushel¹, S. Perry¹, D. Bangsberg¹, R. Clark¹, A.R. Moss¹; ¹University of California, San Francisco, CA

PURPOSE: Homeless persons commonly seek care in emergency departments (ED). Factors associated with high utilization of the ED are not known. This study was conducted to identify factors associated with frequent use of the ED among homeless and marginally housed persons in San Francisco (SF).

METHODS: We surveyed homeless and marginally housed persons in SF in 1996 and 1997. We interviewed and HIV tested all consenting adults at major homeless shelters and food lines and a random sample of residents of low income hotels. Consistent with prior research, subjects with 4 or more ED visits in the prior year were classified as being a high utilizer. In a multivariate model, we examined factors associated with high utilization in six domains: 1) demographic variables (age, sex, ethnicity, veteran status, receipt of public benefits); 2) mental health and substance abuse problems; 3) health status (self-reported health status, HIV status, comorbid illness, need for ongoing medical treatment); 4) crime/ victimization (arrested, victim of assault in past year); 5) housing history (annual days street/ shelter) and 6) health insurance. **RESULTS:** 2578 subjects (67% of those approached) completed the survey. The median age was 43 years (range 15–77). 50% had an alcohol problem, 62% had a lifetime history of drug use and 22% had a history of a psychiatric hospitalization. 29% had been arrested and 57% assaulted in the prior year. Subjects reported spending an average of 88 street/shelter nights in the prior year. 35% reported their health as being fair/poor. 9% were HIV infected. 52% were uninsured. 40% of the subjects reported an ED visit in the prior year: 19% had 1; 14% 2–3 and 8% had four or more visits. High utilizers, despite being only 8% of the sample, accounted for 55% of the overall ED use. In a multivariate model, younger age (5 yr decrease OR 1.1), street/ shelter days (100% increase OR 1.8), mental health (OR 1.5) or alcohol problem (OR 1.6), skin popping (OR 1.8), poorer self reported health (OR 1.3) co-morbid illness (OR 1.8) or an ongoing treatment need (OR 3.6), being arrested in the past year (OR 1.7) and being the victim of an assault (OR 2.1) were significantly associated ($p < 0.05$) with being a high utilizer. Neither HIV status nor insurance was associated.

CONCLUSION: Among homeless and marginally housed persons, a small group account for the majority of the use of the ED. Chronic health problems (medical, psychiatric and substance abuse) and street/shelter factors (days homeless, crime and victimization) are risk factors for being a high utilizer. These factors should be considered when designing interventions to reduce use of the ED in this population.

QUALITY OF LIFE PERSISTS AT THE END OF LIFE. J.S. Kutner¹, D.E. Nowels¹, C. Kassner¹, L.L. Bryant¹; ¹University of Colorado Health Sciences Center, Denver, CO

PURPOSE: Maximizing quality of life (QOL) at the end of life (EOL) is a central goal of hospice and palliative care. The purpose of this study was to describe QOL among patients receiving hospice care and to determine whether QOL at the EOL is associated with particular patient characteristics, such as diagnosis or functional status.

METHODS: Cross-sectional study conducted in the Population-based Palliative Care Research Network (PoPCRN) among English-speaking adults who consented and had a Short Portable Mental Status Questionnaire (SPMSQ) adjusted score < 5 . Data were collected via an interviewer-administered questionnaire. QOL was measured by the McGill QOL Questionnaire (MQOL). Descriptive statistics, t-tests, and ANOVA (SPSS version 10) were used to describe the population and determine associations between the variables of interest.

RESULTS: 66/82 (80%) individuals from 14 PoPCRN sites consented to and were able to participate. 56% were female, 89% were white, 39% were cared for at home, and 53% had a cancer diagnosis. The median age was 76 years, median Karnofsky score 50%, and median time between hospice admit and interview was 49 days. MQOL responses (range 0 – 10; 0=bad, 10=good) indicate that these patients had few physical symptoms, but the ones they had were problematic (single troublesome symptom score=4.4). The MQOL subscale scores indicate that these patients were less troubled by psychosocial and existential issues than by physical symptoms: Total MQOL score=7.1, Physical well-being score=6.0, Physical symptom overall score=5.9, Psychological symptom score=7.1, Existential well-being score=7.4, Support score=8.6. There were no significant associations between age, marital status, gender or length of hospice care and any of the QOL scores. A Karnofsky score $< 50\%$ was associated with worse Existential well-being (6.9 vs. 7.9, $p=0.017$) while a cancer diagnosis was associated with greater Existential well-being (7.8 vs. 6.9, $p=0.047$). Being cared for at home and a cancer diagnosis were both associated with a greater sense of Support (9.2 vs. 8.2, $p=0.005$; 9.0 vs. 8.1, $p=0.042$).

CONCLUSION: Among this older, terminally ill population receiving hospice care, whose functional level was poor and for whom physical symptoms were troublesome, quality of life persisted. Quality of life was particularly preserved among patients with cancer, those who were cared for at home and those who had better functional status.

MEDICAL THERAPY OR CORONARY ARTERY BYPASS GRAFT SURGERY FOR CHRONIC STABLE ANGINA: AN UPDATE USING DECISION ANALYSIS. Y. Kwok¹, P.A. Heidenreich², C. Kim¹; ¹University of Washington, Seattle, WA; ²Stanford University, Palo Alto, CA

PURPOSE: Randomized trials comparing medical and surgical therapies for treatment of chronic stable angina were completed in the early 1980s. New therapies developed since then have decreased mortality and myocardial infarction from coronary artery disease. Using decision analysis, we simulate a trial comparing coronary artery bypass graft surgery versus medical therapy incorporating contemporary recommendations for treatment.

METHODS: We constructed a Markov decision analysis model to simulate the outcome of a trial lasting 5 and 10 years comparing medical treatment to bypass surgery for the treatment of stable chronic angina. We obtained baseline data from a meta-analysis of trials comparing bypass surgery to medical therapy. Risk reduction from contemporary therapies was obtained from randomized trials and meta-analyses. Medical therapies included in the model were: aspirin, cholesterol lowering with statins, beta-blockers post myocardial infarction and in congestive

heart failure, and angiotensin converting enzyme inhibitors in congestive heart failure. Medical treatments were applied to both medical and surgical patients. The left internal mammary artery graft was added to the model for surgical patients.

RESULTS: All subgroups experienced modest gains in survival with current therapies. At five years, 90.4% of the medical group were alive (a gain of 6.2%) compared with 93.8% of the surgical group (a gain of 4%). At ten years, 82.4% of the medical group were alive (gain of 12.9%) compared with 87.5% of the surgical group (a gain of 13.9%). Subgroup analysis of patients with triple vessel disease yielded similar results. For patients with low ejection fraction, 85.4% medical group patients and 92.2% of surgical patients were alive at 5 years. Sensitivity analyses did not substantially affect the conclusions.

CONCLUSION: Both patients treated with initial surgery and those treated initially with medical therapy have experienced gains with advances in the treatment of chronic stable angina. The gains are proportional in both groups such that the fundamental conclusions of the bypass trials are unchanged.

TEMPORAL CHANGES IN INTENSITY OF ANTIRETROVIRAL THERAPY FOR DRUG USERS: IMPACT OF OUTPATIENT CARE PATTERNS. C. Laine¹, D. Zhang², W.W. Hauck³, B.J. Turner². ¹Jefferson Medical College, Philadelphia, PA; ²University of Pennsylvania, Philadelphia, PA; ³Thomas Jefferson University, Philadelphia, PA

PURPOSE: More intensive combination ART has important clinical benefits, but drug users' access to this treatment has been limited. We hypothesized that drug users' outpatient care patterns would be associated with the likelihood of receiving more intensive ART over time.

METHODS: We studied 8,987 drug users in New York State (NYS) Medicaid for > 10 months in each year of 1996–97 and/or 1997–98 and prescribed ART for > 6 months in 1996 or 1997. From pharmacy data, we categorized the type of prescribed ART for > 50% of treatment time in each year into 4 intensity levels: none; 1 ART; 2+ ART but not highly active ART (HAART); or HAART including a protease inhibitor or non-nucleoside reverse transcriptase inhibitor. Changes in ART from the base year to the next were classified as: suboptimal (ie, continuing or increasing to 1 ART or changing from a more intense to a less intense regimen); acceptable (ie, continuing or increasing to 2+ non-HAART drugs); or optimal (ie, continuing or increasing to HAART). Indicators were also created for visits (N=0, 1–3, 4+) to a provider with a NYS contract to offer HIV-focused care in exchange for higher Medicaid payment. Each patient's outpatient care pattern in the base year was categorized as: regular medical care alone (> 35% of visits to one provider); regular substance abuse treatment alone (6+ months with one provider); both; or neither. We estimated ordinal logistic regression models of ART pattern corrected for clustering of observations.

RESULTS: Temporal changes in ART were: suboptimal (44%), acceptable (20%), and optimal (36%). Adjusted odds ratios (AOR) of greater intensity of ART were higher for persons with HIV-focused care (AOR 2.6 [CI 2.4, 2.8] for 4+ visits and 1.6 [CI 1.4, 1.7] for 1–3 visits vs. none). In a second model that included an interaction between HIV-focused visits and outpatient care pattern, AORs were higher for persons with both regular substance abuse and regular medical care compared to those with neither type of care at all levels of HIV-focused visits, but the AOR for both types of care was highest when subjects had no HIV focused care (AOR 1.67 [CI 1.39, 1.27]).

CONCLUSION: Drug users appear to be prescribed more intensive ART over time if they have visits to providers who offer HIV-focused care and/or regular outpatient substance abuse and medical care. Both regular medical care and regular substance abuse care were particularly beneficial for persons with no HIV-focused visits.

DEMENTIA AND PSYCHIATRIC ILLNESS AS PREDICTORS OF NURSING HOME ADMISSION: RESULTS FROM A NATIONALLY REPRESENTATIVE SAMPLE OF U.S. ELDERLY. K. Langa¹, M. Kabeto¹, A.R. Herzog¹, N. Foster¹, W. Straus², A.M. Fendrick¹; ¹University of Michigan, Ann Arbor, MI; ²Merck, West Point, PA

PURPOSE: Dementia and psychiatric illness have been shown to be predictors of nursing home admission (NHA), but most prior studies have used non-representative patient samples rather than population-based samples. In addition, it is unclear whether dementia and psychiatric illness remain as independent predictors of NHA after adjusting for co-existing chronic health conditions. We sought to determine the independent effect of dementia and history of psychiatric illness on NHA using a population-based sample of individuals age 70 or older.

METHODS: We used data from the first three waves ('93, '95, '98) of the Asset and Health Dynamics Study, a longitudinal survey of a nationally representative cohort of U.S. elderly born in 1923 or earlier (N=7443). Dementia was classified as absent, mild/moderate, or severe at study entry (1993). Baseline history of "emotional, nervous, or psychiatric problems" was self-reported. A Cox proportional hazard model was used to determine the independent effect of dementia and history of psychiatric illness on NHA during the 5 years of follow-up after adjusting for age, race, gender, net worth, social support, disability in ADLs and IADLs, and other chronic health conditions.

RESULTS: Of the 7443 respondents, 1799 (24%) had died after 5 years of follow-up, and an additional 190 (3%) were excluded from the analysis due to missing data. Of the remaining 5454 individuals, 387 (7%) entered a nursing home. The Cox proportional hazard model showed that NHA was 2.1 times ($p < .01$) more likely for those with mild/moderate dementia (compared to those with no dementia), 3.2 times ($p < .01$) more likely for those with severe dementia, and 1.4 ($p < .05$) times more likely for those with a history of psychiatric illness (compared to those without a psychiatric history), after adjusting for all other covariates. No other chronic condition (including stroke, diabetes, heart disease, and cancer) was independently associated with a statistically significant increase in NHA risk.

CONCLUSION: In a large nationally representative sample of community dwelling elderly, the presence and severity of dementia, and history of psychiatric illness were important independent predictors of NHA, even after adjusting for multiple co-morbid conditions and ADL/IADL disabilities. Further study should assess if the onset of dementia in those with prior psychiatric illness may lead to even greater risk of NHA, and whether effective

treatment of cognitive impairment and psychiatric symptoms among the elderly can reduce the risk of NHA.

RACIAL DIFFERENCES IN USE OF UNCONVENTIONAL MEDICAL THERAPY. B.B. Lee¹, J. Speckman², M.A. Moskowitz², A. Ash²; ¹Bedford VAMC and Boston University School of Medicine, Boston, MA; ²Boston University School of Medicine, Boston, MA

PURPOSE: Recent studies have found substantial use of unconventional medical therapies in the United States population, with less use among minorities, especially African-Americans. We examined the factors associated with racial differences in the use of these unconventional therapies.

METHODS: We analyzed data from the first three rounds of the 1996 Medical Expenditure Panel Survey (MEPS), a national subsample of the National Health Interview Survey. We examined self-reported use of conventional and unconventional medical treatment among 13,259 adults over age 18. Unconventional treatment was defined through the MEPS questionnaire as alternative or complementary treatment (e.g. acupuncture, herbal remedies, homeopathy). We further included chiropractic care in our definition. We examined the relationship between unconventional care use and race, dividing the sample into Non-Hispanic Whites (NHW), Non-Hispanic Blacks (NHB), Hispanics, and "Other" minorities (American Indians, Aleutians/Eskimos, and Asian/Pacific Islanders). We controlled for confounding by education, geographic region, poverty level, gender, age, urban locale, and insurance status. We used weighted logistic regression to adjust for survey design factors. We report associations for NHBs vs. NHWs via odds ratios (ORs) and 95% confidence intervals (CIs).

RESULTS: Of those sampled, 6420 individuals sought some form of medical care (e.g. hospital admissions, emergency department visits, outpatient clinic visits, or unconventional practitioner visits), of which 1251 (19%) sought unconventional therapy. In bivariate analysis, NHBs were nearly half as likely to use unconventional therapy compared to NHWs (OR=0.46, CI 0.35–0.61). Hispanics were less likely to use unconventional therapy, although this relationship was marginally significant (OR=0.81, CI 0.64–1.01). "Other" minorities were more likely to use unconventional therapy (OR=1.4, CI 1.02–2.02). Poverty status, education, and insurance were significantly related to unconventional care use. In multivariate analyses controlling for these confounders, however, being a NHB was still associated with less unconventional therapy use (OR=0.57, CI 0.43–0.76). This difference persisted within all strata of poverty, education, and insurance.

CONCLUSION: We found significantly less use of unconventional therapy among NHBs compared to NHWs regardless of insurance status, poverty level, or education. Further qualitative research may be required to identify additional non-socioeconomic factors responsible for this difference.

THE VALIDATION OF AN OSTEOPOROSIS RISK ASSESSMENT INSTRUMENT IN A COMMUNITY SETTING. H.N. Lee¹, C. Wilson², M. Vargha³, M. Hemy⁴, P. Ciaschini⁴; ¹McMaster University, Sault Ste Marie, Ontario; ²Northern Ontario Family Medicine Program, Sudbury, Ontario; ³Algoma University College, Sault Ste Marie, Ontario; ⁴Group Health Centre, Sault Ste Marie, Ontario

PURPOSE: Although osteoporosis-related complications are preventable, there is no evidence-based consensus on how to best identify which "non-high risk" women should have testing for low bone mineral density (BMD) by dual-energy x-ray absorptiometry (DXA). Recently, the Osteoporosis Risk Assessment Instrument (ORAI), a simple three-point scale using age, weight and current estrogen use, reports accuracy in predicting women with low BMD (T score < -2.0) in the femoral neck or lumbar spine (CMAJ 2000; 162(9):1289–94). The present study aims to validate the utility of the ORAI in a community, primary care setting.

METHODS: There is only one DXA machine in Sault Ste Marie, a Northern Ontario city serving a population of 105,000. Data on the age, weight and current estrogen status of 800 consecutive women presenting for initial DXA scanning was prospectively gathered. Records were excluded if patients were at "high-risk" of osteoporosis (steroid use, menopause < 40 years) or currently taking non-hormonal antiresorptive therapy. The main outcome measure was low bone mineral density (T score < -2.0) at either the femoral neck or the lumbar spine.

RESULTS: Of 800 women presenting for initial DXA scanning over a 6 months in 2000, 14 were excluded because of steroid use, early menopause (onset < 40 years) or use of non-hormonal antiresorptive agents. The prevalence of low BMD (T score < -2.0) was 24.2% (95% confidence interval [CI], 21.3–26.1%). The 3-item ORAI had a sensitivity of 85.2% (95% CI 80.1%–90.3%) and a specificity of 63.9% (95% CI 66.1%–66.5%) for selecting women with low BMD (T score < -2.0). The Likelihood Ratio of a positive ORAI (LR+) is 5.0 and LR(–) is 0.23. In this community population the positive predictive value of the ORAI is 42.9% with a negative predictive value of 93.1%. Hypothetically, the use of the ORAI might have reduced 44.9% of screening DXA tests in this primary care setting.

CONCLUSION: The ORAI accurately identifies the majority of women who have low bone mineral density in the community, and could be a useful clinical prediction guide to target appropriate screening interventions in the primary care setting.

EFFECT OF TRANSLATION METHOD ON PATIENT SATISFACTION AT AN URBAN WALK-IN CLINIC. L. Lee¹, J. Kutner¹, H. Bata²; ¹University of Colorado Health Sciences Center, Denver, CO; ²Denver Health Medical Center, Denver, CO

PURPOSE: Patients with limited English proficiency who are seen by language discordant providers may communicate through a variety of interpretation methods. We studied the effect of translation method on satisfaction among Spanish-speaking patients at a hospital-based walk-in clinic serving primarily indigent patients.

METHODS: All Spanish-speaking patients and a random selection of 10% of English-speaking patients presenting to the walk-in clinic at Denver Health Medical Center between July and October 2000 were approached for study entry. Participants completed a self-administered post-

visit questionnaire assessing demographics, general health, and satisfaction with clinic visit. "Language concordant" patients were defined as Spanish-speaking patients seen by Spanish-speaking providers and English-speaking patients. These patients were compared with "language discordant" patients who used AT&T interpreters, family, or other interpreters for translation. Patients were considered "satisfied" if they rated their satisfaction as very good or excellent. Evaluation of satisfaction was by multiple logistic regression, controlling for age, sex, race, language spoken, education, and health.

RESULTS: Of the 536 study participants, 67% were Spanish-speaking. Among Spanish-speaking patients, 42% were seen by language concordant providers. For patients who did not see language concordant providers, translation was through AT&T telephone interpreters (19%), family members (23%), and other interpreters (16%). Language concordant and language discordant participants did not differ in age, sex, race, insurance status, or education level. Language concordant patients were less likely to be Hispanic (59% vs 95%, $p=0.001$) or describe their health as poor or fair (29% vs 51%, $p=0.001$). Language concordant patients and patients using AT&T telephone interpreters reported identical overall visit satisfaction (77%), while patients using family or other interpreters were significantly less satisfied (54% and 49%, $p<0.01$ and $p=0.007$, respectively). Compared to language concordant patients, patients who had family members translate were less satisfied with their provider in regards to listening to health concerns (62% vs 85%, $p=0.003$) and discussing sensitive issues (60% vs 76%, $p=0.02$). Patients who used other interpreters were less satisfied with provider skills (60% vs 83%, $p=0.02$), manners (71% vs 88%, $p=0.02$), listening (54% vs 85%, $p=0.002$), explanations (57% vs 84%, $p=0.02$), answers (57% vs 84%, $p=0.05$), and support (63% vs 84%, $p=0.02$).

CONCLUSION: Patients utilizing less formal translation methods, such as family members and ad hoc interpreters, are less satisfied with care than those seeing language concordant providers or using AT&T telephone interpretation.

ARE WOMENS PERCEPTIONS OF MAMMOGRAPHY AFFECTED BY HOW THE INFORMATION IS FRAMED? C.L. Lewis¹, M.P. Pignone¹, S.M. Downs¹, R.P. Harris¹, D.F. Ransohoff¹, M.S. O'malley¹, L.S. Kinsinger¹, S.L. Sheridan¹; ¹University of North Carolina, Chapel Hill, NC

PURPOSE: To determine if framing information about mammography affects women's perceptions of the potential benefits and harms of mammography.

METHODS: Women ages 35–50 without a history of breast cancer were recruited from the waiting room of a university internal medicine practice. Following a baseline questionnaire about breast cancer risk factors, and benefits/harms of mammography, 179 women were randomized to one of three videos. For women ages 40–49 who undergo 10 years of mammography, the videos described the number of lives extended(1/1000), the number of false positives expected(300/1000), and the number of women expected to worry after a false positive mammogram screening(100/1000). Information was numerically equivalent in each video but presented in a positive frame(1 woman would have her life extended) a negative frame(999 women would not have their lives extended) or a balanced frame(positive and negative information given). Participants completed a follow-up questionnaire immediately after viewing the video.

RESULTS: Participants were predominately insured(80%), white(60%) and high school graduates(63%). At baseline, 3% endorsed the correct response(1 per 1000 women screened for 10 years would live longer because they had mammograms) and 76% thought 100 or 500 women would live longer. After the video, 53% endorsed the correct response with no differences by video ($p=.21$). For the number of false positives expected, 15% gave the correct response at baseline, but at followup the correct responses were given by more women in the negative(65%) and balanced(65%) frames than in the positive frame(40% $P<.01$). The results were parallel for the number expected to worry after a false positive (62%, 56%, and 32% for negative, balanced and positive frames respectively). A majority of respondents in all three groups thought the benefits of mammography were more important than the harms before(75–80%) and after seeing the the video(77–81%).

CONCLUSION: Women's baseline knowledge of the benefits of mammography was overly optimistic. All three videos improved women's knowledge of the benefits/harms of mammography. Despite this improvement in knowledge and exposure to varying frames of information, women's perceptions of the benefits/harms did not change.

DOES NUMERACY INFLUENCE WOMENS ABILITY TO LEARN ABOUT THE POTENTIAL HARMS AND BENEFITS OF MAMMOGRAPHY? C.L. Lewis¹, M.P. Pignone¹, S.M. Downs¹, R.P. Harris¹, D.F. Ransohoff¹, M.S. O'malley¹, L.S. Kinsinger¹, S.L. Sheridan¹; ¹University of North Carolina, Chapel Hill, NC

PURPOSE: To determine if "numeracy" affects women's ability to learn information about the potential benefits and harms of mammography.

METHODS: In the waiting room of a university internal medicine practice we recruited 179 women ages 35–50 without a history of breast cancer and randomized them to one of three videos. For women ages 40–49 who undergo 10 years of mammography, the videos described the number of lives extended(1/1000), the number of false positives expected(300/1000), and the number of women expected to worry after a false positive mammogram screening(100/1000). Participants completed a questionnaire immediately after viewing the video. Information presented in each video was numerically equivalent, but to obtain the correct answers, subtraction was required for at least one question in each video. Numerate women were defined as being able to correctly note that 40% was equivalent to a 4 out of 10 chance.

RESULTS: Participants were predominately insured(80%), white(60%), high school graduates(63%), and over half were numerate(65%). The table shows the percentage of correct responses for numerate and non-numerate women when the answers were given in the video compared to when subtraction was required to get the correct answers.

CONCLUSION: Most numerate women were able to learn numeric information from the video but had difficulty manipulating this information if they were not given the correct response in the video. Non-numerate women were unable to learn numeric information even when given the correct response. Although women in their forties are supposed to be informed of the benefits/harms of mammography to make decisions about screening, many women may have difficulty understanding this information if it is relayed numerically.

% Correct for Numerate and Non-numerate Women

	Answers Given Numerator	Answers Given Non-num	Subtraction Required Numerator	Subtraction Required Non-num
Live longer	77	26	51	32
False +	81	32	46	26
Worry	71	36	37	22

HEALTH CARE UTILIZATION AMONG CHINESE AMERICAN ADULTS. D. Liao¹, C.E. Blixen¹, J. Hammel¹, X. Jin¹; ¹Cleveland Clinic Foundation, Cleveland, OH

PURPOSE: It is estimated that by 2010, there will be more than 17 million Asian/Pacific Islanders in this country. Although the rapid growth of this population will certainly result in more Asian/Pacific Islanders being seen by U.S. health care providers, little is known about the health care practices of this fastest growing minority populations. In addition, the term "Asian/Pacific Islanders" is misleading in that this title refers to a great number of cultural groups. As a first step to understanding the health care needs and preferences of such a diverse population, we examined the health care practices of a sample of Chinese Americans.

METHODS: 1.Study Design - A cross-sectional survey. 2.Setting - A large Midwestern city. 3.Sample - 75 community residing Chinese Americans. 4.Measures - Demographic data and health care practices. 5.Analytic Procedures - Responses were summarized as mean and standard deviation, or as percentages. Associations between socioeconomic variables and utilization variables were measured using the test for a non-zero Spearman correlation between the crosstabulated variables.

RESULTS: The sample consisted of a primarily well educated group of 30–50 year old (75%) professional men (37%) and women (63%). Almost all of the sample had been born in China (93%), claimed Mandarin Chinese as their native language (88%), and had resided in the United States an average of 8.9 (SD=7) years. Although most of the respondents had health insurance (93%) and more than half (69%) had a primary care provider, only 37% stated that they would have a routine check-up with this provider if they "felt fine." However 54% of the females in our sample reported having annual pap screening and breast exam by their physicians. Over half of the sample (63%) reported problems with understanding medical terms. Less educated respondents were more likely to be hesitant to see the doctor due to language difficulties ($p=0.001$) and more likely to use western medicine for a cold ($p=0.023$). Respondents with lower incomes were also more likely to be hesitant to see the doctor due to language difficulties ($p=0.036$) but respondents with higher incomes were more more likely to have a primary care physician ($p<0.001$).

CONCLUSION: Although our sample was well educated and had adequate access to health care services, the inability to understand medical terminology may be a prime contributor to the under-utilization of health care services. A more diverse and larger sample is needed to further assess the utilization patterns from all socioeconomic levels. In addition, health literacy levels of Chinese Americans need to be assessed in order to provide culturally competent health care to this population.

CONTRASTING VIEWS OF PHYSICIANS AND PATIENTS REGARDING CLINICAL E-MAIL COMMUNICATION AT AN ACADEMIC MEDICAL CENTER. C. Lin¹, L. Ware², G.A. Albertson², E.M. Cyran², L. Schilling², J.F. Steiner², R.J. Anderson²; ¹University of Colorado, Denver, CO; ²UCHSC, Denver, CO

PURPOSE: We wished to study the differences in views and motivations between physicians and patients regarding e-mail communication.

METHODS: Mailed survey of 1100 practicing physicians at the University of Colorado and 1000 patients enrolled in a managed health plan at UC.

RESULTS: We received 546 responses (49.6%) to two physician (MD) mailings, and 409 responses (40.9%) to a patient (PT) mailing. 97% of surveyed MD's have access to email, while 89% of surveyed PT's have access. 35% of MD's currently communicate with PT's via e-mail, while only 18% of PT's e-mail their MD. In contrast, 44% of all MD's would be interested in e-mailing PT's in the future, while 91% of all PT's desired e-mail access to their MD in the future. Patients were more likely to view e-mail in a positive light. Both MD's and PT's felt that e-mail eliminated "phone tag" (MD: 79%, PT: 90%, $p=0.001$), was faster (MD: 71%, PT: 81%, $p=0.001$), more convenient (MD: 54%, PT: 94%, $p=0.001$), more efficient (MD: 56%, PT: 71%, $p=0.001$), and allowed printable documentation (MD: 76%, PT: 77%, $p=NS$). More PT's than MD's felt that e-mail saved the patient money (MD: 24%, PT: 34%, $p=0.001$). Patients were less likely to cite risks of e-mail communication including, possibility of e-mail message being lost (MD 67%, PT: 65%, $p=NS$), lack of confidentiality (MD: 75%, PT: 47%, $p=0.001$), impersonal (MD:49%, PT:34%, $p=0.001$), unable to discuss complex issues (MD:77%, PT: 55%, $p=0.001$), possibility of misunderstanding (MD: 73%, PT: 27%, $p=0.001$), not timely (MD: 38%, PT: 34%, $p=NS$), lack of access to a computer (MD: 39%, PT: 4%, $p=0.001$), and difficult to use system (MD: 10%, PT: 2%, $p=0.001$).

CONCLUSION: About 1/3 of MD's and 1/5 of PT's currently use e-mail to communicate with each other. 44% of MD's and 91% of PT's would like to do so. Patients have more positive views about clinical e-mail usage than physicians.

ANTIBIOTIC TREATMENT OF SORE THROAT BY PRIMARY CARE PHYSICIANS, A NATIONAL SURVEY 1989–1998. J. Linder¹, R. Stafford¹; ¹Massachusetts General Hospital, Boston, MA

PURPOSE: Group A β -hemolytic streptococci (GABHS) are cultured in 10–25% of adults with sore throat. GABHS is the main cause of sore throat requiring antibiotic therapy. The Infectious Disease Society of America recommends either penicillin or erythromycin as first-line agents to prevent complications and reduce symptoms. Our goals were: 1) define the rate and type of antibiotics given to patients with sore throat by community primary care physicians over a ten-year period, and 2) determine predictors of antibiotic prescription and use of non-recommended antibiotics for sore throat.

METHODS: The National Ambulatory Medical Care Survey (NAMCS) collects data from office-based physician practices in the United States including patient demographics, patients' reasons for visit, physician diagnoses, and medications prescribed. Using NAMCS from 1989 to 1998, we analyzed 2,141 adult primary care visits with a chief complaint of sore throat. We estimated national antibiotic prescription rates over time and determined both the most frequent diagnoses made and the most frequent antibiotics prescribed. We evaluated significant predictors of antibiotic prescription and non-recommended treatment using multivariate logistic regression.

RESULTS: An estimated 6.8 million adult patients each year made acute visits for sore throat to office-based, primary care physicians between 1989 and 1998. The most common diagnoses were acute pharyngitis (42%), acute upper respiratory tract infection (21%), acute tonsillitis (8%), and streptococcal infection (6%). Antibiotics were prescribed to 73% (95% CI 70–76%) of patients. The most common antibiotics prescribed were aminopenicillins (23% of visits), cephalosporins (14%), penicillin (14%), and erythromycin (10%). Over ten-years the proportion of patients receiving any antibiotic decreased while the proportion of patients receiving non-recommended treatment increased (Table). In multivariate logistic regression modeling, predictors of antibiotic prescription were younger patient age (OR 1.17 per decade, CI 1.08–1.27), general practice (OR 1.48, CI 1.12–1.95), and residence in the South (OR 1.40, CI 1.02–1.94). Predictors of non-recommended antibiotic use were study year (OR 1.26 for each grouping in the table, CI 1.11–1.42), residence in the South (OR 1.38, CI 1.06–1.81), and HMO membership (OR 0.75, CI 0.58–0.98).

Study Years	1989–1992	1993–1995	1996–1998	p for trend
Any antibiotic (%)	75	72	70	0.01
Non-recommended antibiotic (%)	45	53	55	0.0001

CONCLUSION: Three-quarters of adults with sore throat presenting to community primary care physicians are treated with antibiotics. This greatly exceeds the prevalence of GABHS in adults. Overall use of antibiotics has declined over a ten-year period, but the use of non-recommended, more expensive, broad-spectrum antibiotics is increasing.

GENDER DIFFERENCES IN THE READINESS TO ADOPT FLEXIBLE SIGMOIDOSCOPY AS A COLORECTAL CANCER SCREENING TEST. B. Ling¹, J. Trauth¹, R. Schoen¹, M. Hayran¹, J. Weissfeld¹; ¹University of Pittsburgh, Pittsburgh, PA

PURPOSE: Population-based surveys indicate that women are less likely than men to have ever had a flexible sigmoidoscopy (FSG) for colorectal cancer screening. This study's objective was to compare age-appropriate women to men in their readiness to adopt FSG as a screening behavior by using the Transtheoretical Model.

METHODS: A telephone survey of randomly selected individuals aged 50–79 years from two Pennsylvania communities was conducted. Participants were asked about their past and current behavior as well as future intent toward FSG. With the use of the Transtheoretical Model, these responses were then integrated to assess one's readiness in adopting FSG as a screening behavior. According to the model, one must pass through a series of stages before a behavior is achieved and maintained. For this study, participants were categorized into the following stages: Unaware (never heard of test), Precontemplation (never had nor intending to have test), Contemplation (never had test but intending to), Relapse (had test, but not current with a nationally recommended guideline of screening FSG every 5 years), Relapse Risk (current with testing, but not intending to have test again), and Compliant (current with testing and intending to have test again). To test for differences in the distribution of these stages between genders, chi square test was used. A subanalysis was also performed to test the a priori hypothesis that, in a population aware of FSG but who had never had the test previously, women compared to men would show less intent in having it done in the future. Odds ratio was computed.

RESULTS: There were 414 respondents (59% response rate) considered average-risk for colorectal cancer of which 239 (58%) were women. From this cohort, women (W) and men (M) were distributed into the following stages respectively: Unaware (W49%, M49%), Precontemplation (W25.9%, M14.3%), Contemplation (W8.8%, M13.1%), Relapse (W6.3%, M6.9%), Relapse Risk (W6.7%, M10.3%), and Compliant (W3.3%, M7.4%). There is a statistically significant difference ($p=.02$) in the distribution between genders with the largest difference being more women in the Precontemplation stage. In the subanalysis of those aware of FSG but had never had the test previously, women were less intent than men on having it done in the future (OR 0.37, 95% CI 0.17–0.78).

CONCLUSION: The awareness level of FSG was low for both women and men. Among those aware of the test, women were less ready to adopt FSG as a colorectal cancer screening test. Future work should assess barriers and facilitators for advancing women beyond the Precontemplation stage. Furthermore, efforts to evaluate and address awareness of FSG in the community setting should be performed.

THE USE OF HORMONE REPLACEMENT THERAPY BY BREAST CANCER SURVIVORS. B. Littenberg¹, R.G. Pinckney¹, B.M. Geller¹, M. Coahran¹, P.M. Vacek¹; ¹University of Vermont, Burlington, VT

PURPOSE: Controversy surrounds the use of estrogen and other hormone replacement therapy (HRT) by breast cancer survivors. Although strong epidemiologic evidence (such as controlled trials) implicating HRT as a cause of breast cancer recurrence is lacking, many believe it predisposes to recurrence. The American Society of Health-Systems Pharmacists Drug Information, for example, describes estrogen as “contraindicated” in known breast cancer (except for occasional use in suppressing metastatic disease). Nonetheless, some survivors report using HRT. We were concerned that HRT use in this setting may be the result of medical error and sought to discover its frequency.

METHODS: We examined two data sources. The Vermont Mammography Registry (VMR) includes records of 114,943 women who have had mammograms since 1996. Each woman completed a questionnaire including her personal history of breast cancer and her current use of HRT at the time of each mammogram. The Third National Health and Nutrition Examination Survey (NHANES) database is a national probability sample of 39,695 persons who were interviewed in their homes by trained staff. Data include personal history of breast cancer and a complete medication history including visual inspection of prescription bottles by the interviewer.

RESULTS: In the VMR, 4,962 women reported a personal history of breast cancer. 394 of these survivors (7.9%) reported current use of HRT at one or more mammogram examinations. NHANES included 157 women with a personal history of breast cancer. Eleven of these survivors (7.0%) were currently using HRT.

CONCLUSION: A surprisingly large fraction of breast cancer survivors use HRT. The concordance between two estimates obtained using very different methods argues against random or data error. Other possible explanations include: 1) the patient opts to accept the putative risks of recurrence in an informed manner, 2) the prescriber does not know that HRT is contraindicated in breast cancer, 3) the prescriber rejects the theory of hormone induced recurrence, 4) the prescriber does not know the patient is a breast cancer survivor, or 5) the patient self-prescribes. We believe that HRT use in breast cancer survivors is a sentinel event that may represent medical error.

IMPUTATION OF SF-12 HEALTH SCORES FOR RESPONDENTS WITH PARTIALLY MISSING DATA. H. Liu¹, R.D. Hays¹, J. Adams², W. Chen¹, D. Tisnado¹, C.M. Mangione¹, C.L. Damberg³, K.L. Kahn¹; ¹UCLA, Los Angeles, CA; ²RAND, Santa Monica, CA; ³Pacific Business Group on Health, San Francisco, CA

PURPOSE: Background: The standard SF-12 scoring algorithm requires answers to all 12 items to estimate physical (PCS) and mental (MCS) component summary measures. The loss of observations due to missing items can reduce analytical power and possibly lead to bias, especially when the SF-12 is the primary outcome measure. Accurate imputation of SF-12 PCS and MCS scores in this situation is critical.

METHODS: part of the Physician Value Check Survey. The 13,438 patients include 2287 chronic and 11,151 non-chronic patients based upon self-report of chronic diseases. In stage 1, multi-pattern regressions were performed to impute PCS and MCS using SF-12 items only (simple model), and then using SF-12 items plus patients demographics and comorbidities (enhanced model) in an attempt to reduce bias from the simple model. Distribution and variation of the imputed SF-12 values were evaluated. Based on confidence bands of the imputed SF-12 values, a cut point of number of missing SF-12 items was selected for using the simple or enhanced model to impute SF-12 PCS and MCS. In stage 2, patients with no missing SF-12 items who also had demographic and comorbidity data were randomly classified into 11 subgroups. Permutations of missing patterns (missing 1 to 11 out of the total 12 items) on SF-12 items were generated to simulate missingness (missing 1 to 11 SF-12 items) for these patients. The imputed PCS and MCS were compared to the observed values for these patients separately by the identified cut point. The analysis was performed separately for chronic and non-chronic subgroups.

RESULTS: Results: In stage 1, the standard errors of means of imputed summary scores increased almost monotonically by the number of missing SF-12 items, ranging from 0.11 to 4.55 for PCS and 0.09 to 2.18 for MCS in the non-chronic subgroup, and from 0.28 to 5.84 for PCS and 0.24 to 5.62 for MCS in the chronic subgroup. A significantly consistent increase of the width of confidence bands (about 50% in standard error) occurred between 5 and 6 items missing. Of the 13,438 patients, 10,766 (8949 non-chronic and 1817 chronic) patients had non-missing data for all SF-12 items as well as demographic and comorbidities. Product-moment correlations between the imputed and the observed scores were large, ranging from 0.73 to 0.97. The correlations from imputation with the enhanced model were consistently higher than the correlations from the simple model. The increments of correlations by the enhanced model were statistically significant for patients with more than 6 missing SF-12 items (7%, $P<0.05$).

CONCLUSION: Conclusion: For patients with partially missing SF-12 items, direct imputation of summary SF-12 physical and mental scores is efficient. Adding patients' demographics and comorbidity information to the imputation model for PCS and MCS can improve the imputed value and reduce bias. The enhancement of adding patients' demographics and comorbidities to the imputation models for PCS and MCS is necessary and important for patients with 6 or more SF-12 items missing.

LONG TERM OUTCOMES AFTER LAPAROSCOPIC SURGERY FOR GASTROESOPHAGEAL REFLUX. J.Y. Liu¹, S. Woloshin¹, W.S. Laycock², L.M. Schwartz¹; ¹VA Outcomes Group, White River Junction, VT; ²Dartmouth-Hitchcock Medical Center, Lebanon, NH

PURPOSE: Increasingly, patients are undergoing minimally invasive surgery to treat gastroesophageal reflux disease. While short term data suggest this surgery can cure reflux,

long term outcomes are unknown. We sought to assess the long term results and quality of life after laparoscopic surgery.

METHODS: We surveyed adult patients one to five years after laparoscopic surgery for gastroesophageal reflux at an academic tertiary care center. To date, 151/248 (61%) have responded to our mailed survey.

RESULTS: The mean age for respondents was 52 years; 54% were male. The median time since surgery was 2.5 years. Ninety percent felt their surgery was working well. Overall, 28% had some typical reflux symptoms (e.g. heartburn or regurgitation) but only 5% reported either of these symptoms 'most' or 'all of the time.' Sixty-three percent reported other upper gastrointestinal symptoms (e.g. bloating and dysphagia) which may be related to either surgery or underlying disease: 23% had one of these other symptoms 'most' or 'all of the time.' About half of the respondents reported that they take some action to manage their symptoms: 4% take frequent over the counter medications, 13% take daily prescription antireflux medications, 34% make lifestyle changes (e.g. eating smaller meals), and 35% avoid certain foods (e.g. carbonated beverages).

CONCLUSION: Many people continue to have upper gastrointestinal symptoms which require action after surgery. However, the vast majority of patients feel their surgical treatment is working well.

A NEW MEASURE OF QUALITY OF LIFE WITH GASTROESOPHAGEAL REFLUX. J.Y. Liu¹, L.M. Schwartz¹, W.S. Laycock², S. Woloshin¹; ¹VA Outcomes Group, White River Junction, VT; ²Dartmouth-Hitchcock Medical Center, Lebanon, NH

PURPOSE: The goal of therapy for patients with gastroesophageal reflux disease (GERD) is to improve quality of life. Existing quality of life measures for GERD focus on symptoms, failing to account for the actions taken to manage symptoms (e.g. chronic medications, lifestyle or dietary changes). We sought to validate a quality of life measure that assesses both symptoms and actions taken to manage GERD.

METHODS: Written survey administered to 140 adult patients in clinic with severe GERD who were seeing a gastroenterologist or surgeon for their treatment and 41 unselected patients in the primary care clinic. The unselected patients completed a mailed retest survey two weeks later and a subset who reported stable symptoms were evaluated for test-retest reliability.

RESULTS: The patients with severe GERD were younger than those in the unselected group (mean age 42 years vs. 65 years). Symptoms and management burdens were substantially greater for patients with severe GERD than unselected patients supporting construct validity. Notably, many primary care patients reported reflux symptoms and took actions to manage them.

	Severe GERD (n=140)	Unselected patients (n=41)
Symptoms in the past 2 weeks:		
Had heartburn or sour taste	97%	60%
Had these symptom 'most' or 'all of the time'	65%	15%
Feels 'unhappy' or 'terrible' with symptoms	85%	21%
Actions taken		
Take daily prescription antireflux medication	88%	17%
Change lifestyle (e.g. elevating head of bed)	86%	32%
Avoid specific foods (e.g. coffee, spicy foods)	82%	34%
Feels 'unhappy' or 'terrible' with management	79%	13%

We found good test-retest reliability for our questions about symptoms and management actions (26 patients, Pearson's correlation coefficients averaged 0.51 for 33 correlations).

CONCLUSION: This new measure appears to be a valid and reliable way to assess reflux symptoms and the actions taken to manage them. Since many patients with severe GERD take substantial actions to manage reflux, measurement of symptoms alone underestimates the impact of the disease.

ANTIOXIDANT VITAMIN INTAKE AND RISK OF ALZHEIMERS DISEASE. J.A. Luchsinger¹, M. Tang¹, S. Shea¹, R. Mayeux¹; ¹Columbia University, New York, NY

PURPOSE: Oxidative stress has been hypothesized to have a role in the pathogenesis of Alzheimer's disease (AD). Our objective was to determine whether dietary intake of antioxidant vitamins is associated with a lower risk of AD.

METHODS: 980 elderly subjects free of dementia at baseline were followed for a mean of 4 years. Information on the supplement and dietary intake of vitamins A, C, and E were obtained with semi-quantitative food frequency questionnaires administered at baseline. Dietary intake of vitamins was adjusted for total caloric intake. Multivariate analyses with Cox regression were conducted with quartiles of each vitamin as the exposure of interest and incident AD as the outcome, adjusted for age, education, gender, APOE-4 status, ethnicity, and smoking. Trend tests were calculated with Cox regression adjusting for all covariates.

RESULTS: There were 260 incident cases of AD in 77,931 person-years of follow-up. The mean age was 75 years and 67% of the subjects were women. Subjects reporting use of vitamin supplements tended to have lower adjusted risks of AD, but the results were not statistically significant. Calorie-adjusted dietary intake of vitamins A, C, and E alone was not associated with a lower risk of AD. The table shows the adjusted hazard ratios (95% CI) of AD for quartiles of total vitamin intake (dietary + supplement).

CONCLUSION: Subjects in the highest quartile of intake of total vitamin C had a decreased risk of AD. Randomized trials of vitamin C in AD are needed before treatment recommendations can be made.

HR (95%CI) of AD per quartile of vitamin intake

	Vitamin A	Vitamin C	Vitamin E
First	1.00	1.00	1.00
Second	0.82 (0.56–1.19)	0.70 (0.49–1.01)	0.77 (0.53–1.11)
Third	0.92 (0.64–1.33)	0.83 (0.58–1.19)	0.89 (0.62–1.29)
Fourth	0.69(0.47–1.03)	0.66 (0.45–0.95)	0.74 (0.50–1.08)
p for trend	0.13	0.06	0.22

HOW VALUABLE IS TALKING TO OUR PATIENTS? A CLOSER LOOK AT TAKING THE HISTORY. J.W. Peabody¹, J. Luck²; ¹Veterans Affairs San Francisco Healthcare System, San Francisco, CA; ²Veterans Affairs Greater Los Angeles Healthcare System, Los Angeles, CA

PURPOSE: Most clinicians rely on the history to make an accurate diagnosis and prescribe appropriate treatment. However, as increasing administrative and financial pressures shorten the outpatient clinical visit, the opportunity to take a careful history is becoming more abbreviated. This study prospectively evaluated whether accurate history taking, under precisely controlled conditions, actually leads to better outpatient clinical practice.

METHODS: We prospectively collected data in the primary care general internal medicine clinics at two VA-university teaching hospitals. Greater than 95% of residents and attendings consented to participate—and 20 were randomly selected as subjects. Eight unannounced standardized patients (SPs) were introduced into the clinic of each selected provider presenting four common conditions: COPD, Low Back Pain, diabetes mellitus and coronary artery disease (i.e., 2 SPs per condition). Thus, every physician saw an identical set of cases. All visits were new patient appointments. Visits were scored by the SP using a closed-ended checklist of quality criteria; other criteria were assessed by abstracting the medical record. Scoring was based on explicit criteria derived from national guidelines and expert panels. Separate scores were generated for History Taking, the Physical Examination, Testing (laboratory and imaging studies), and Diagnosis plus Treatment. Scores were expressed as a percentage of criteria that were correct. Multiple linear regression models were used to predict if better history taking led to more accurate diagnosis and treatment plans, more correct testing, and appropriate physical examinations. The models controlled for case, site, and level of training.

RESULTS: Physicians who scored higher on history taking (that is, took more accurate histories) made more accurate diagnoses and more appropriate treatment decisions (p = 0.006). By contrast, more accurate physical examinations and diagnostic test ordering did not predict better diagnosis and treatment (p > 0.05). The quality of diagnosis and treatment varied between the two sites (p < 0.001) but not by level of training or case. Taking an accurate history did predict the appropriateness of the physical examination (p = 0.004), but it did not predict testing (p > 0.05).

CONCLUSION: Under controlled conditions, where case mix was specified by experimental design, our results show that accurate history taking predicted when residents and attendings were more likely to make the correct diagnosis and prescribe the best treatment. While accurate history taking also guided the physical examination, the examination itself did not lead to subsequently higher process scores. The site effects were also strong. These findings demonstrate that successful completion of the history effectively predicts of diagnostic and treatment accuracy. Emphasis on development of history taking skills in residency programs, and adequate visits time allocations in practice settings, should contribute to higher quality clinical care.

CHARACTERISTICS OF PATIENTS CHOOSING A PATIENT-DIRECTED, TELEPHONE-BASED, INTERACTIVE VOICE RESPONSE INTERVENTION TO SUPPORT CHANGE IN AT-RISK ALCOHOL BEHAVIOR. W. Mackay¹, S. Wolfe¹, D. Einstadter², N. Pelletier³, C. Pickens³, C. Thomas³, N. Dawson³; ¹MetroHealth Medical Center, Cleveland, OH; ²Case Western Reserve University, Cleveland, OH; ³Visiting Nurse Association, Cleveland, OH

PURPOSE: Telephone-based self-help systems have been suggested as a method to assist problem drinkers reduce alcohol use. There is little information however regarding characteristics of patients who are willing (or not) to participate in this form of intervention. We examined differences between at-risk patients who did and did not select to enroll in an Interactive Voice Response (IVR) intervention to reduce drinking.

METHODS: During 1/99–3/00, patients seen in the General Medicine Clinic at the MetroHealth Medical Center in Cleveland, OH, a large urban teaching hospital, were administered a health screening which included demographics, SES, general health, social support, smoking, and alcohol consumption and history including the Alcohol Use Disorders Identification Test (AUDIT), CAGE, family CAGE, self-recognition of a problem, and interest in change. Patients who screened as hazardous (AUDIT=8–10) or harmful (AUDIT=11) drinkers were contacted by a community nurse, counseled about their drinking and offered an IVR intervention which included: a closed voice-mail system for contacting the PC physician or community nurse, for identifying referral resources, and for listening to health and substance abuse related topics and messages. We compared characteristics of patients willing to enroll and those who declined.

RESULTS: Of 409 eligible patients, 298 (73%) were contacted and 157 (53%) agreed to enroll in the IVR intervention. Enrollment was similar for men (83/170, 49%) and women (74/128, 58%), p=0.15. Mean age was similar for male enrollees (50 yrs) and decliners (51 yrs), but female enrollees (43 yrs) were significantly younger than decliners (49 yrs) p<0.005. Enrollees had higher alcohol abuse risk measures: the median AUDIT was 16 for enrollees vs. 11 for decliners (p < .001); harmful drinkers enrolled more frequently 125/208 (60%) than hazardous drinkers 32/90 (36%), p<0.001; personal CAGE was higher among male enrollees (p < 0.005), and both personal and family CAGE values were higher among female enrollees (p < 0.001). Also, enrollees were more likely to be "interested in making changes" in their drinking at screening (p < 0.001). In a multivariate logistic regression analysis, CAGE2 (OR=2.07), "interest in change" (OR=1.97), and family CAGE2 (interacting with gender) were significantly (p < 0.05) associated with enrollment; age and education level were not independently significant.

CONCLUSION: Alcohol risk level, readiness for change, and gender appear to be significant factors in a patient's willingness to enroll in an IVR intervention to reduce high-risk alcohol

behavior. Age and education did not appear to influence enrollment. Understanding factors influencing enrollment is needed to improve the attractiveness and impact of such an intervention, or to inform the decision whether to employ it at all.

PATIENT AND PHYSICIAN DISCUSSION OF SPIRITUALITY: SHOULD IT BE A ROUTINE PART OF PRACTICE? C.D. Maclean¹, L. Schultz², B.E. Susi³, N.W. Phifer⁴, F. Respect Group⁴; ¹University of Vermont, Essex Junction, VT; ²Emory University, Atlanta, GA; ³Carolinas Healthcare System, Charlotte, NC; ⁴University of North Carolina at Chapel Hill, Chapel Hill, NC

PURPOSE: Previous studies have reported that patients and physicians value the spiritual aspects of health and would welcome discussion on the topic. Our objective was to determine specific patient and physician preferences for addressing spirituality and to explore whether patients are willing to forego time spent discussing medical problems to discuss spiritual issues. **METHODS:** The RESPECT study is a multi-center structured survey of patient and physician attitudes regarding religion/spirituality in the medical encounter. A trained research assistant administered the patient survey. This questionnaire included demographic items, functional status (SF-36), a spiritual assessment (Spiritual Well Being Scale), and statements measuring patients' preferences regarding physician involvement in their religious/spiritual life. The physician survey was administered to residents and attending physicians by mail. Bivariate analyses were done to examine associations between patient characteristics and the questions that measured patients' desires for spiritual involvement in specific medical settings.

RESULTS: Preliminary data were based on the first 300 patient and 446 physician surveys. Patients' ages ranged from 20 to 77 years. 44% percent were female, 46% were African-American, 51% Caucasian, and 63% belonged to various Protestant denominations. 60% were internists and 24% family physicians. 67% of patients and 84% of MDs agreed that doctors should be aware of their patients' religious and spiritual beliefs. About a third of patients and MDs felt that doctors should ask their patients about these beliefs. 35% of patients and 22% of MDs felt doctors should play no role in the religious/spiritual life of their patients. 16% of patients felt their doctors were not qualified to address spiritual issues. Despite high interest, only 12% of patients had actually raised religious issues with their doctor, and only 9% of patients were willing to discuss these issues in a medical encounter if it meant spending less time on medical problems. Patient preferences for a spiritual or religious dialogue are associated with higher levels of religiosity and spirituality, but not with age, race, and functional status.

CONCLUSION: A majority of patients and physicians feel that discussion of religion and spirituality in the medical encounter are appropriate, but few patients have actually raised religious issues with their doctor. Since very few patients are willing to forego time spent on medical issues for these discussions, it may not be reasonable to incorporate religious dialogue as part of the daily practice routine.

ACCESSING CARE IN A PUBLIC HOSPITAL URGENT CARE CLINIC — A MARKER OF INADEQUATE RECEIPT OF PREVENTIVE HEALTH MEASURES? S. Majeres¹, H. Batal¹, S. Prock¹, L. Lasater¹, J. Adams¹, R. Lundgren², P. Mehler¹; ¹Denver Health Medical Center, Denver, CO; ²Colorado Prevention Center, Denver, CO

PURPOSE: To describe the barriers to receipt of preventive care services in patients seeking care in a public hospital urgent care clinic.

METHODS: 20% of patients presenting to the Denver Health Medical Center urgent care clinic (UCC) between June 15 and August 11, 2000 were randomly approached to participate in a pre-visit interview in either English or Spanish. Patients were asked standardized questions regarding their health, reasons for accessing care in the UCC, barriers to care, chronic medical conditions, and prior receipt of preventive health services. Study patients' charts and admission records were abstracted for the documentation of receipt of preventive health services at our institution, ethnicity, age, gender, and insurance status. Data were analyzed using chi-square tests of association.

RESULTS: There were 1006 patients surveyed with a refusal rate of 20.3%. The study patients did not differ from the total patient population who accessed care during this time period in regards to age, insurance status, ethnicity, or gender. Only 37.1% of patients identified a regular source of care other than an ED or UCC. Patients who noted a regular source of care were significantly more likely to report having had a flu shot in the past twelve months (34.9% vs. 21.3%; $p < 0.0001$) or having had a cholesterol test in the past five years (46.8% vs. 27.3%; $p < 0.0001$). Chart review at our institution confirmed these results in regard to cholesterol testing in the past three years (24.9% vs. 7.6%; $p < 0.0001$). In women, those who reported a regular source of care were significantly more likely to have had a documentable Pap smear in the past three years ($p < 0.0001$). However, even among those with a regular source of care, only 53.9% had a documented Pap smear in the past three years. This trend towards greater receipt of preventive care services among those patients who identified a regular source of care was also seen in those women who necessitated mammography. Among women over 50, 47.5% of those with a regular source of care had a documented mammogram in the past two years, compared to 36.4% of those without a regular source of care ($p = 0.37$).

CONCLUSION: Patients who present to an urgent care clinic frequently do not have a regular source of health care. Those without a regular source of health care are less likely to receive needed preventive care services. This suggests a mechanism to provide preventive care measures to patients accessing care in urgent care clinics, namely to offer preventive care services within the urgent care system to serve this at risk population.

HOW DO SCREENING QUESTIONNAIRES AFFECT COMMUNICATION ABOUT RISK? A. Rubin¹, G.T. Makoul¹, R.H. Curry¹; ¹Northwestern University, Chicago, IL

PURPOSE: Since risky behavior involving poor diet, inactivity, tobacco, alcohol, unsafe sex, motor vehicle accidents, and guns are major causes of morbidity and mortality in the United States, we sought to determine the extent to which these topics were addressed and to identify obstacles to communication during primary care visits at two sites: Chicago IL and Burlington VT.

METHODS: As part of a larger study of physician-patient communication and decision making (~250 visits per site), we indexed, transcribed, and coded all discussions regarding health risks during initial visits. We reviewed charts to see if risks were noted in the record and, in Vermont, reviewed screening questionnaires to record which risks were flagged.

RESULTS: Despite similarities in practice demographics and physicians' attribution of value to health risk discussion, the physicians who used a screening questionnaire (VT) asked fewer questions about health risks and recorded fewer notes about health risks in medical records.

CONCLUSION: The use of a screening questionnaire for health risks does not ensure that risks are either discussed or recorded. Patients should be asked about health risks even if a screening tool is used.

OUTCOMES OF FALLS IN A FRAIL ELDERLY COMMUNITY. J. Aaron¹, M. Gavinsky², J. Prochnow², B. Kean², J. Gulla², M. Malone¹; ¹University of Wisconsin Medical School (Milwaukee Clinical Campus), Milwaukee, WI; ²Geriatrics Institute, Milwaukee, WI

PURPOSE: Falls in the elderly pose a real threat to the objective of keeping persons independent and in their own homes. The purpose of this study is to describe the outcomes of falls in a community dwelling frail elderly population at two Program of All Inclusive Care for the Elderly (PACE) program sites in Milwaukee, WI.

METHODS: A retrospective chart review was performed on all elderly adults who fell ($n=90$) between 1/1/2000 and 4/30/2000 from a total program population of 343.

RESULTS: 179 falls occurred among 90 participants during the study period resulting in a rate of falls of 1.56 per person per year. Twenty-nine (32.2%) of the participants fell once, 44 (48.8%) fell twice and 17 (18.8%) fell three or more times. Seventy-seven (43%) of the 179 falls resulted in soft tissue injuries. Four (2.2%) of the 90 fallers sustained fractures. Five (2.8%) of the 90 fallers were hospitalized immediately after the fall. Twelve of the 90 fallers had a change in their living location as a result of their falls. Six of these people remained in their new group home/nursing home at least 8 months after the fall. Six of the 90 fallers died during the 4 month course of the study.

CONCLUSION: In this program that is responsible for the health care needs of a frail elderly population, falls resulted in an increased level of housing needs and of supervision in addition to the morbidity and mortality that has previously been described.

DEFINING ALCOHOL DISORDERS IN ELDERLY PRIMARY CARE PATIENTS: DOES DSM-IV WORK? D.L. Mansell¹, P. Sekar¹; ¹University of Alabama at Birmingham, Birmingham, AL

PURPOSE: Elderly patients with alcohol disorders differ from typical adult patients with alcohol abuse and dependence. As people age, tolerance to alcohol decreases, but other symptoms of alcohol abuse/dependence may still occur. We report the frequency of DSM-IV criteria for alcohol abuse and dependence in elderly primary care patients.

METHODS: Elderly primary care patients (age 65 and older) participated in a study to evaluate screening methods for alcohol disorders. Participants were completed the SCID, a structured clinical interview for DSM-IV to determine whether criteria for alcohol abuse and dependence were present.

RESULTS: 331/399 study participants completed the SCID. Mean age was 72.5; 39% were male, 43% were black and 39% were white. 74 (21 women) of the participants met one or more DSM-IV criteria for alcohol abuse and dependence. The mean number of criteria answered yes for the 74 was 2.6 for men and 2.5 for women.

CONCLUSION: Unlike younger adults with alcohol disorders, only a minority of elderly men and women report DSM-IV tolerance and withdrawal criteria for alcohol dependence. However, elderly primary care patients still have clinically important ill effects from alcohol. Classification systems for alcohol disorders in the elderly should reflect age specific characteristics of the population.

Percent of Men and Women with DSM-IV criteria

	Men (N=53)	Women (N=21)
work/school affected	9.4%	0.0%
drink in dangerous situation	47.2%	19.1%
legal trouble from drinking	26.4%	26.4%
problems with family/friends	15.1%	19.1%
drink more than intended	49.1%	57.2%
desire or efforts to cut down	62.3%	76.2%
spend a lot of time drinking/recovering	17.0%	23.8%
social, occupational activities curtailed	7.6%	4.8%
drank despite physical or psychological problems	7.6%	7.6%
tolerance to alcohol	11.3%	19.1%
withdrawal symptoms	9.4%	4.8%

ARE POSTMENOPAUSAL WOMEN READY TO ACCEPT TREATMENT FOR OSTEOPOROSIS AFTER HIP FRACTURE? K.F. Mauck¹, M. Cuddihy¹, G. Pond¹, V.S. Pankratz¹, L.J. Melton¹, R.T. Trousdale¹; ¹Mayo Clinic, Rochester, MN

PURPOSE: Although treatment for osteoporosis can reduce future fracture risk and improve overall function, most women do not receive medical management for osteoporosis after sustaining an osteoporotic fracture. This study investigated postmenopausal women's willingness to accept osteoporosis therapy after an acute proximal femur (hip) fracture.

METHODS: A consecutive series of postmenopausal women (age > 50) admitted to a tertiary care hospital with an acute proximal femur fracture was identified. Sociodemographic information was gathered on all patients. Patients were invited to participate if the fracture

resulted from mild to moderate trauma, no previous diagnosis of dementia or bone disease was noted, and an adequate score on a Mini Mental Status Exam was achieved. Patients were then surveyed on readiness to accept osteoporosis treatment, personal knowledge of osteoporosis, and medical history. A modified form of the Weinstein Precaution Adoption Process Model (PAPM) was used to measure stage of readiness to accept osteoporosis treatment. According to this model, at any point in time, a person will be in one of seven stages with respect to adoption of a specific behavior change: Stage 1 corresponds to being completely unaware of the health problem, while Stage 7 corresponds to taking and maintaining treatment for a health problem for at least 6 months. The Ailinger Facts on Osteoporosis Quiz (FOQ) was used to measure osteoporosis knowledge.

RESULTS: Between May and August 2000, 70 women with a mean age of 85 years were treated for an acute proximal femur fracture at our institution. All patients were reviewed and of these, only 29 were eligible for the study, with 21 agreeing to participate (mean age 82 years). We excluded 41 patients, the majority of which had dementia (61%) or delirium (20%). The majority of women (12, 57%) were in PAPM Stage 2—corresponding to no serious thoughts about personally taking treatment for osteoporosis. The FOQ was completed by 11 patients, who achieved a mean score of only 53%. Only 3 patients (14%) voluntarily asked for more information about osteoporosis during our interview.

CONCLUSION: This study demonstrates that women who sustain a proximal femur fracture with minimal trauma, a primary manifestation of osteoporosis, are not aware of their disease. Their readiness to accept a therapeutic intervention was also low, confirming the lack of awareness. This presents a challenge to physicians caring for osteoporotic patients, who are in need of both education and intervention.

ACCESS TO HOSPICE SERVICES AMONG PATIENTS WITH LUNG CANCER. E.P. McCarthy¹, R.B. Burns¹, R.B. Davis¹, R.S. Phillips¹; ¹Beth Israel Deaconess Medical Center, Harvard Medical School, Boston, MA

PURPOSE: Dying patients are often referred to hospice too late to benefit optimally from palliative care. Barriers to earlier hospice care are poorly understood. Therefore, we used the Linked Medicare-Tumor Registry Database to identify factors associated with late enrollment into hospice among patients with lung cancer.

METHODS: We studied Medicare beneficiaries diagnosed with first primary lung cancer at age 66 or older in the Surveillance, Epidemiology, and End Results (SEER) Program and who died on or between Jan 1, 1988 and Dec 31, 1994. We linked diagnostic information to Medicare hospice claims. Data available included patient demographics, staging, and date of diagnosis. We performed Cox regression analyses to identify factors independently associated with time to hospice entry. Models were stratified on tumor registry. We measured time to hospice enrollment from the date of diagnosis. Patients who died without receiving hospice care were censored. Adjusted hazard ratios (aHR) <1.0 signify later enrollment to hospice.

RESULTS: Of the 36087 lung cancer patients, the majority were male (64%) and white (88%); 8% were black and 18% resided in a rural area. The mean age at diagnosis was 74 years and 62% were married. Most patients were Stage III (24%) or Stage IV (33%) at the time of diagnosis, and two-thirds died within one year. Only 18% of patients received hospice care prior to death. Among hospice patients, 25% entered within 3 months of diagnosis while 57% entered hospice care more than 6 months after their diagnosis. The median length of stay in hospice was 25 days.

After adjustment, factors associated with late enrollment in hospice care include being female [aHR=93, 95% CI (.88-.98)]; being black [.89, (.81-.98)]; and residing in a rural area [.81, (.74-.88)]. Younger patients enrolled in hospice later than older patients. We also observed a significant trend over time. Since the implementation of the Medicare hospice benefit in 1982, the time from diagnosis to hospice care has decreased over time. For example, patients diagnosed between 1986 and 1989 enrolled in hospice significantly later in the course of their illness than patients diagnosed between 1990 and 1993 [.53, (.50-.56)].

CONCLUSION: Access to hospice care has improved dramatically over time. However, women, African-Americans, younger patients and those residing in rural communities continue to experience barriers to earlier hospice care.

EMERGENCY DEPT USE FOR PATIENTS WITH PRIMARY CARE OPTIONS. J.M. McCauley¹, M.W. Jenckes¹, A. Channugam¹, M.L. McCarthy¹; ¹Johns Hopkins Medical Institutions, Baltimore, MD

PURPOSE: Lack of primary care access is reported to be the major reason for the 40-65% of Emergency Departments (ED) visits reported to be nonurgent. Nonurgent use is especially high for Medicaid patients. Our 18-site primary care group has 24/7-nurse triage, night/weekend Urgi-care, and same day primary care appointments. ED use remains high among our capitated Medicaid (CMA) population (N=29,000). To design further performance improvement initiatives, we piloted a survey of our patients presenting to the inner city, academic, ED seeing the largest volume of our patients. Our objectives were to determine if our patients: 1) could identify their primary care physician (PCP) 2) knew where to call when acutely ill 3) perceived differences in care provided by PCP vs. ED physicians.

METHODS: From 1 June to 14 August 2000 we surveyed all adult patients presenting to the ED identified as belonging to our primary care practices. Our survey had 20 variables on access and 10 ratings of care. We calculated frequencies and significance tested with chi square.

RESULTS: Of 68 patients identified, 88% completed the survey. Respondents were 75% female, 84% AA, 77% 18-54 years old, 79% CMA. Of these, 71% knew about the night/weekend triage phone line. 98% stated they had a regular primary care clinician (PCP); 81% were able to recall their PCP's name. When asked, "When you are sick and need care right away, how quickly do they see you?" 70% answered the same day or the next day. Regarding ED usage, 49% of patients called triage before coming to the ED; those with CMA insurance were less likely to call (42% vs 75%, p=.04). Of those that called, 96% reported they were told to go to the ED ("callers"). ED nurses triaged 59% of "callers" and 44% of non "callers" to Level 1 or 2 (out of 4; 1 is high). Eighty per cent of respondents rated "How well doctor listens to you," "Doctors respect for you," and "Doctors ability to figure out what is the matter" as excellent or very good. There were no significant differences in ratings between PCPs and ED physicians.

CONCLUSION: Our small pilot survey indicates that patient knowledge of, and access to, acute primary care was very good. However a large number of patients with non-urgent conditions still didn't call before presenting to the ED. Almost all that did call were sent to the ED, even though 41% were then judged to be of lower sickness acuity by the ED triage nurses. These results suggest that our CMA and other patients need more education and/or incentives to access primary care alternatives to ED use. Our results also suggest that our triage nurses could benefit with further training on appropriate ED referral.

COMMON LEG SYMPTOMS IN PERIPHERAL ARTERIAL DISEASE AND THEIR IMPLICATIONS FOR LOWER EXTREMITY FUNCTIONING. M.M. McDermott¹, P. Greenland¹, K. Liu¹, J. Guralnik², M. Criqui³, N. Dolan¹, W. Pearce¹, J. Schneider¹, L. Sharma¹, D. Gibson¹, G.J. Martin¹; ¹Northwestern University Medical School, Chicago, IL; ²National Institute on Aging, Bethesda, MD; ³University of California, San Diego, CA

PURPOSE: We identified clinical characteristics and functional limitations associated with all types of leg symptoms commonly present in lower extremity peripheral arterial disease (PAD). We hypothesized that physically active PAD persons with no exertional leg symptoms would have better leg functioning than other leg pain groups.

METHODS: Study participants were 454 men and women with PAD identified from three medical centers in Chicago. Participants were categorized into one of six mutually exclusive leg symptom groups: Group 1: No exertional leg pain/active (walked > 6 blocks in the last week) (13%); Group 2: No exertional leg pain/inactive (<= 6 blocks walked last week) (6%); Group 3: Atypical exertional leg pain and walk through leg pain (8%); Group 4: Intermittent claudication (37%); Group 5: Atypical exertional leg pain/stop walking with leg pain (17%); Group 6: Leg pain on exertion and rest (19%). Functional measures included six-minute walk, four meter walking velocity, time for five chair rises, and standing balance. All statistical analyses were adjusted for multiple comparisons using Bonferroni method.

RESULTS: Group 1 and Group 3 had significantly less severe PAD, as measured by the ankle brachial index, than the other groups. Compared to all other groups, Group 6 had the highest prevalence of lower extremity arthritis, disc disease, diabetes, and depression. In general, functional performance was poorest in Group 6 and best in Groups 1 and 3. Compared to Group 1, Group 6 had slower walking velocity (0.75 m/sec vs. 0.89 m/sec, p<0.05), achieved shorter distance in the six-minute walk (972 feet vs. 1201 feet, p<0.05), had slower time for rising from a seated position five times (14.2 sec vs. 11.3 sec, p<0.05), and was less able to hold the tandem stand position for ten seconds (32% vs. 51%, p<0.05). Compared to Group 1, Group 4 was more likely to stop during the six-minute walk (38% vs. 16%, p<0.05) and walked fewer blocks in the last week (35.7 vs. 53.0, p<0.05). Group 2 had slower walking velocity than Group 1 (1.02 vs. 1.22 meters/sec p<0.01).

CONCLUSION: Comorbid disease may contribute to the nature of leg symptoms reported by PAD patients. Clinicians can use the leg symptom categories defined above to gauge the degree of functional impairment associated with PAD.

THE NATURAL HISTORY OF OBESITY: WEIGHT CHANGE IN A LARGE US LONGITUDINAL SURVEY. K.M. McQueen¹, J.M. Garrett¹; ¹University of North Carolina at Chapel Hill, Chapel Hill, NC

PURPOSE: Understanding the natural history of weight gain on a population basis may be a critical step towards developing effective clinical or public health interventions. While the marked rise in overweight and obesity prevalence in this country has been well documented by the serial cross sectional data of the US National Health and Nutrition Examination Survey (NHANES), longitudinal description of the trend is lacking. This study assesses the development of body mass of a cohort of young US adults and identifies predictors including age, gender and ethnic group that may have implications for targeted intervention.

METHODS: The National Longitudinal Survey of Youth (NLSY79) is a nationally representative sample of over 12,000 persons aged 14-22 years at baseline, with oversampling of minority ethnic groups. Body mass index (BMI = kg/m²) was calculated from self-reported height and weight data at 12 sample points over 17 years. 'Overweight' and 'obese' were defined as BMI > 25 and > 30, respectively. Mean BMI was calculated by age group, gender and ethnicity. In this study, unadjusted data are used to describe trends in BMI, and to calculate incidence of overweight and obesity using survival analysis.

RESULTS: BMI trends were strongly related to age, race and gender. Women's BMI increased steadily with age, with average BMI reaching "overweight" status by age 26 in Black women, age 29 in Hispanic women and age 35 in all others. The average BMI of Black women was in the obese range before age 40. Ethnic differences in weight prevalence were evident by age 17. Survival analysis quantified the incidence rate of obesity as 2.5 times faster for Black women as for Non-Hispanic/Non-Black women (CI 2.2 - 2.8) and 2.1 times faster for Hispanic women versus those who were Non-Hispanic/Non-Black (CI 1.8 - 2.4). Male average BMI showed less dramatic age trends and a different ethnicity pattern: Hispanic men consistently had the highest mean BMI values. Hispanic men developed obesity 2.0 times faster (CI 1.7 - 2.3) and Black men developed obesity 1.8 times faster than Non-Black/Non-Hispanic men (CI 1.6 - 2.0).

CONCLUSION: This large longitudinal survey shows marked differences in the rate of weight accumulation for different ethnic groups, evident at a very young age. This pattern persisted through young adulthood with excess weight, on average, increasing gradually with age. To alter the trajectory of obesity in this country, interventions should target young adults, and focus on those of minority ethnic groups.

COUNSELING FOR DEPRESSION IN STAFF-MODEL MANAGED CARE ORGANIZATIONS. L.S. Meredith¹, M. Rowe¹, P. Nutting, Md, Msp², L.V. Rubenstein, Md, Msp¹; ¹RAND, Santa Monica, CA; ²Center for Research Strategies, Denver, CO

PURPOSE: Manualized short-term psychotherapies such as Cognitive Behavioral Therapy (CBT) have been shown to be as effective as antidepressant medications for treating patients with major depression. The success of these methods may be related in part to the use of specific counseling techniques. These analyses assess the extent to which primary care clinicians and

mental health specialists use CBT techniques to treat primary care patients with major depression.

METHODS: Cross-sectional descriptive evaluation of counseling received by 567 patients meeting criteria for 1-year major depressive disorder based on a structured diagnostic interview. The 567 enrolled participants are from a consecutive sample of patients attending one of nine participating primary care practices in California (3 based in a single academically-affiliated VA medical center and 6 in independent medical centers from 1 region of Kaiser Permanente). Participants completed an interviewer-administered survey of health perceptions and utilization patterns. Patients reported the frequency of visits to medical and mental health providers and the extent to which their providers used four different CBT techniques. These included helping the patient reduce negative thinking, encouraging enjoyable activities, assisting with problem-solving, and helping the patient feel better about their life as it is.

RESULTS: Patients reported an average of 3.2 visits (median=1.0, range 0–144, 61.3% with no visits) to medical providers in which they discussed their personal or emotional problems and 4.5 (median=0, range 0–96, 60.7% with no visits) visits to mental health specialists. Of those patients with at least one visit to a medical provider during the past six months, between 42.1% and 57.6% per therapeutic technique reported that their medical providers had used it. The range was 78.9% to 84.9% per therapeutic technique for patients visiting mental health specialists. The most prevalent therapeutic technique was encouragement of more enjoyable activities, 59.3% and 85.5% of patients reported receiving counseling that included at least two of the four therapeutic techniques from medical and mental health providers, respectively.

CONCLUSION: Although the mental health specialists used therapeutic counseling techniques more often than did the primary care clinicians in these practices, primary care clinicians used them more often than we expected given the competing demands of generalist practice. This optimistic result suggests that further evaluation of the efficacy and effectiveness of primary care clinician therapeutic counseling in improving outcomes is warranted.

PATIENT CHARACTERISTICS ASSOCIATED WITH ALTERNATIVE MEDICINE USE AMONG VETERANS ATTENDING OUTPATIENT VA GENERAL INTERNAL MEDICINE CLINICS. M.A. Micek¹, K.A. Bradley¹, C.H. Braddock¹, P. Diehr¹, M. McDonnell¹, S.D. Fihn¹; ¹University of Washington, Seattle, WA

PURPOSE: Little is known about the predictors of alternative medicine use in the veteran population. This study investigates whether veterans' sociodemographic characteristics, self-reported health status, health-related beliefs, and satisfaction with their VA health care providers, are related to their use of alternative medicine.

METHODS: Participants included patients attending 7 VA general internal medicine clinics participating in the VA Ambulatory Care Quality Improvement Project (ACQUIP), a multicenter randomized controlled trial to test the effectiveness of an information feedback system on improving health status and satisfaction outcomes. This analysis is based on a random subsample of 1034 patients who also received the ACQUIP Health Beliefs Scale. This scale assessed patients' attitudes towards conventional medicine and adherence to health-conscious lifestyles. Patients' satisfaction with their provider was measured with the 12-item humanistic component of the Seattle Outpatient Satisfaction Questionnaire. Self-reported health status was measured with the mental, physical, and pain subscales of the SF-36. Additional data included patients' age, gender, ethnicity, level of education, and income. Alternative medicine use was determined with a questionnaire assessing the use of the following: biofeedback/relaxation, chiropractic, acupuncture, massage, herbs, homeopathy, naturopathy, spiritual healing, folk remedies, and other. Multiple logistic regression techniques were used to evaluate the association between alternative medicine use within the past year (yes or no), sociodemographic characteristics, health beliefs, and patient satisfaction.

RESULTS: 495 patients returned the Health Beliefs Scale (48% response) and were included in this analysis. 124 patients (25%) reported the use of at least one alternative medicine therapy in the past year, with the most common modalities including herbs (13.5%), chiropractic (7%), and massage (6%). In multivariate analyses adjusted for sociodemographic characteristics and self-reported health status, the following were associated with alternative medicine use: more education (OR 1.5; 95% CI 1.3–1.9); better physical health (OR 1.044; 95% CI 1.009–1.081); and more pain (OR 1.022; 95% CI 1.005–1.039). After adjustment for sociodemographic factors and subjective health status, alternative medicine use was associated with the following health-related attitudes and behaviors: believing more in the value of non-medical treatments (OR 1.4; 95% CI 1.1–1.9); adhering to health-conscious lifestyles (OR 1.7; 95% CI 1.3–2.2); and actively seeking health information from non-medical sources (OR 1.6; 95% CI 1.3–2.1). In adjusted analyses, there were no significant associations between alternative medicine use and attitudes towards conventional medicine or patient satisfaction.

CONCLUSION: A quarter of VA general internal medicine patients in this study reported the use of alternative medicine in the past year. Patients who used alternative medicine had more education, and reported better physical health but more pain. Alternative medicine users also believed more in the value of non-medical treatments, reported adherence to health-conscious lifestyles, and reported obtaining more health information from non-medical sources. Similar to previous studies, negative attitudes towards conventional medicine and lower levels of patients' satisfaction with their VA providers did not appear to influence use of alternative medical modalities.

TO ERR IS HUMAN: THE ELDERLY AT RISK. E. Michota¹, C.E. Blixen¹, K. Shermock¹, M. Secic¹; ¹The Cleveland Clinic Foundation, Cleveland, OH

PURPOSE: Geriatric patients represent 40% of all hospital discharges and over half of all inpatient days of care. Despite explicit criteria identifying inappropriate medications for the elderly, patients aged 70 and older routinely receive drugs that increase their risk for acute confusional states or drug-induced delirium. The purpose of this study was to identify the number and kinds of inappropriate medications dispensed to hospitalized elders and calculate the medical error rate for a 12-month period.

METHODS: 1. Study Design — A 12-month retrospective study. 2. Setting — A 942-bed tertiary care academic teaching hospital in the Midwest. 3. Sample — Patients 70 years of age

and older who were hospitalized during a 12-month period in 1999 and 2000. 4. Measures — Number and kinds of inappropriate medications dispensed; medical error rate. 5. Analytic Procedures — Using established criteria, potentially inappropriate medications were identified in the hospital formulary and assessed by the central pharmacy database for frequency of use. The hospital admissions database was used to identify the total number of elders hospitalized in the previous year. The two databases were merged, sorting by patient age, non-intensive care unit location, and our defined inappropriate medications. A medical error rate was calculated by dividing the total of inappropriate medications dispensed by 100 patient-days. Patient days were calculated by length-of-stay data also obtained from the hospital admissions database. The study was repeated in the subsequent 6-months to assess for secular trends and the stability of the error rate.

RESULTS: A total of 11,797 doses of inappropriate medications were dispensed to 15,011 patients aged 70 and over during a 12-month period. The most common of these prescribed medications were meperidine (3,652 doses), amitriptyline (1,736 doses), and diazepam (1,644 doses). The medical error rate was 0.12 per day, or 12 errors per 100 patient days. Findings for the subsequent 6 months were similar.

CONCLUSION: The results of our study are troubling though not surprising. Academic teaching hospitals often require all patient orders to be written by resident physicians, with variable levels of senior supervision. Educational interventions targeted at internal medicine residents have the potential to reduce the medical error rate in elderly hospitalized patients.

HORMONE REPLACEMENT THERAPY AND RISK OF VENOUS THROMBOEMBOLISM: A META-ANALYSIS. J. Miller¹, B. Chan¹, H. Nelson¹; ¹Evidence-based Practice Center, Oregon Health Sciences University and Portland VA Medical Center, Portland, OR

PURPOSE: Although postmenopausal hormone replacement therapy (HRT) is widely used, it poses important health risks. This study assesses the risk of venous thromboembolism (VTE) with the use of postmenopausal HRT and selective estrogen receptor modulators (SERMs). This work was part of a comprehensive assessment of hormone replacement conducted for the third U.S. Preventive Services Task Force.

METHODS: We identified English-language studies in the MEDLINE and HealthSTAR databases from January 1966 to June 2000, and the Cochrane Controlled Trials Register. References of key articles were also reviewed for related studies. Studies were included if they enrolled postmenopausal women and reported deep venous thrombosis (DVT), pulmonary embolism (PE), or VTE as either part of a predetermined outcome or as a reportable adverse event related to HRT. Another Medline search from January 1991 to June 2000 identified studies addressing SERMs and VTE risk. Ten HRT studies (two randomized controlled trials, seven case-control studies, and one cohort study) and five SERM randomized controlled trials met inclusion criteria. Two investigators abstracted data on participants, intervention, VTE rates, and confounders. We used the Bayesian data analysis framework for the meta-analysis of HRT studies. The SERMs were not included in the meta-analysis.

RESULTS: HRT was associated with an increased risk of VTE in current users (RR=2.34; 95% CI, 1.83–2.94). Five studies showed a significant increase in VTE with HRT. The highest VTE risk was reported with the first year of URT use (OR =2.9 to 6.7). Two of the SERM randomized controlled trials reported an increased risk of VTE (RR= 1.91 to 3.1).

CONCLUSION: Postmenopausal HRT is associated with an increased risk of VTE. SERMs may be similarly associated with VTE risk. These findings are pertinent to clinicians and patients considering HRT.

BREAST CANCER SCREENING IN AMERICAN SAMOAN WOMEN. S. Mishra¹, F. Hubbell¹, P. Luce²; ¹University of California, Irvine, CA; ²National Office of Samoan Affairs, Carson, CA

PURPOSE: Little is known about breast cancer screening practices or predictors of age-specific screening for Samoan women. We determine population-based estimates of utilization of clinical breast exam (CBE) and mammography by Samoan women and examine predictors of recent age-specific utilization of screening exams.

METHODS: Through systematic, random sampling procedures, we identified and interviewed 720 adult (≥30 years) Samoan women residing in American Samoa, Hawaii, and Los Angeles. Multivariate logistic regressions were performed to determine independent predictors for recent age-specific screening.

RESULTS: Only 55.6% of women (≥30 years) had ever had a CBE and 32.9% of women (≥40 years) had ever had a mammogram. Furthermore, only 24.4% and 22.4% of Samoan women (≥40 years) residing in Hawaii and Los Angeles, respectively, had an age-specific mammogram within the prior year. Independent predictors of age-specific CBE screening included age, education, health insurance, ambulatory visit, and being a resident of Hawaii or Los Angeles; and those for mammography included ambulatory visit and awareness of screening guidelines.

CONCLUSION: Population-based estimates of age-specific breast cancer screening among Samoan women are lower than nation objectives and those reported for other minorities. Targeted efforts that address doctor patient communication on preventative behavior, improved access to health care services (especially in American Samoa), and focused educational awareness programs are needed to improve the dismal screening rates observed in this indigenous population.

IS POSTMENOPAUSAL HORMONE REPLACEMENT THERAPY USE ASSOCIATED WITH LESS COGNITIVE IMPAIRMENT? J.L. Mitchell¹, K.J. Cruickshanks¹, B.E. Klein¹, D.M. Nondahl¹; ¹University of Wisconsin-Madison, Madison, WI

PURPOSE: Controversy exists on whether postmenopausal hormone replacement therapy (HRT) can lessen or avert cognitive impairment that may lead to dementia. The purpose of this study is to examine the association of HRT use and impaired cognition.

METHODS: Postmenopausal women (n=1403) participating in the population-based Beaver Dam Eye Study and the Epidemiology of Hearing Loss Study in Beaver Dam, Wisconsin were included in these analyses. Data were from the 10-year follow-up for the eye study and the

concurrent 5-year follow-up for the hearing study (1998–2000). HRT use was assessed by questionnaire and confirmed by participants' presentation of prescribed medications at the eye examination. Cognitive impairment is defined as less than 24 points, or 80%, on the mini-mental status exam, or doctor-diagnosed Alzheimer's disease. Odds ratios were obtained from logistic regression models.

RESULTS: The average age of the women was 69 years (range 53 to 97). Twenty-five percent of participants were current HRT users. The average age of current users was 65 years; the average age of non-users was 72 years. More current users completed high school than non-users, 91% and 78%, respectively. In preliminary analyses, the overall prevalence of cognitive impairment was 6%. HRT users were less likely to be cognitively impaired than non-users (5.7% vs 0.5%, $p=0.0003$). After adjusting for age, education, and depression (as measured by Short Form-36 mental health subscale), the odds of being cognitively impaired were not statistically significantly different for HRT users versus non-users (OR=0.40; 95% CI=0.14–1.15, $p=0.09$).

CONCLUSION: These preliminary analyses suggest that postmenopausal HRT use may be associated with a slighter lower odds of having impaired cognition. However, this association may be marking selection bias, as the relationship is no longer statistically significant after adjusting for the confounding effects of age, education, and depression. Longitudinal studies, adequately controlling for these confounders, are needed to determine the effect of HRT use on cognition.

PATIENTS ARE LESS SATISFIED WITH CARE BY RESIDENTS IN CONTINUITY CLINICS. S. Mitrovich¹, K. Pituch¹, D. Rich², G. Freed¹, D. Oliansky¹; ¹University of Michigan, Ann Arbor, MI; ²University of Michigan Hospitals, Ann Arbor, MI

PURPOSE: Assessing patient satisfaction with care received is important to measure as it is a primary goal of providers and clinic practices. Studies done, to date, have been limited in that most measure patient satisfaction with experiences related to medical student education and have been retrospective in design. We designed a prospective study to find the differences in patient satisfaction with outpatient care provided by continuity clinic resident physicians compared to satisfaction with care provided directly by faculty physicians in the same practice. **METHODS:** During one week in June 2000, 1107 adult and pediatric clinic patients completed a patient satisfaction questionnaire following their visit, during which the patient was either seen by the faculty member or by the resident physician year 1 through 4. The data was collected from 5 primary care clinics; pediatrics, internal medicine and a combined medicine/ pediatric clinic. Four clinics were in community practices and one clinic was an academic hospital based practice. The questionnaire comprised of 26 questions addressing service characteristics of the office staff, provider and facility. Ten questions assessed satisfaction of care provided by the doctor. Differences in results for faculty vs. resident patients were analyzed for the entire set of collected responses, as well as for each clinic.

RESULTS: 1772 total patients were seen during the study week. Sixty-three percent of the questionnaires were completed (1107); 824 from faculty patients and 283 from visits with the resident. 42% of the faculty patients and 24% of the resident patients saw the doctor they wanted to see on the day of the visit. 67% of the faculty patients and 60% of the residents patients "strongly agreed" that the resident answered questions in a way the patient could understand. 52% of faculty patients and 43% of the resident patients indicated that the time spent with the physician was "excellent". 62% of the faculty patients rated thoroughness of the exam as excellent versus 50% of the resident visits. 47% of the faculty visits and 36% of the resident visits rated the care received overall as "excellent". All differences reached statistical significance with $p<0.05$. A similar distribution of small differences, all indicating better satisfaction with care by faculty occurred in all 5 participating practices.

CONCLUSION: Patients were more satisfied with care received by faculty as compared with residents. Difficulty for the residents patients to schedule an appointment with their physician of choice is an important barrier to improving patient satisfaction.

ACCESS TO, USE OF, AND UNMET NEED FOR HEALTH CARE SERVICES BY HIV+ HISPANICS IN THE UNITED STATES. L. Morales¹, W. Cunningham¹, R. Andersen², F. Galvin³, T. Nakazono², M. Shapiro¹; ¹University of California, Los Angeles and RAND Health, Los Angeles and Santa Monica, CA; ²University of California, Los Angeles, Los Angeles, CA; ³Charles R. Drew University, Los Angeles, CA

PURPOSE: Hispanics comprise a large and growing proportion of HIV-infected population. Overall access to and use of HIV care is lower for Hispanics than for whites, but little is known about how these vary within the diverse Hispanic population in the US. We examined access to, use of, and unmet need for care among HIV+ Hispanics according to level of acculturation, language in which subjects chose to complete surveys, citizenship status, gender, education and insurance status among the 415 Hispanic patients (15% of all subjects) in the HIV Costs and Services Utilization Study (HCSUS).

METHODS: HCSUS is a nationally representative study of HIV-infected persons in care. Weights were used to adjust for sampling, multiplicity, and non-response. Standard errors were corrected for the multistage sampling design. Use of care was measured in four dimensions: ambulatory care, emergency care, hospital care and mental health services. Overall access was measured with a six-item scale ($\alpha = 0.72$) asking about urgent care, emergency care, hospital care, clinic location, costliness of services, and access to specialists. Unmet need for care was measured by five items asking about foregone medical care due to: inability to get time off work, being too sick to seek care, lack of transportation, needing to care for someone else, or insufficient money to pay for care. All results reported are significant at a $p<0.05$ level.

RESULTS: In separate multivariate analyses controlling for CD4 count, Hispanics who were US citizens reported greater use of emergency care (Odds Ratio [OR]=1.92), hospital care (OR=2.59) and mental health services (OR=1.92) than non-citizens. Hispanics who were more acculturated (OR=1.76), who completed an English-language survey (OR=2.43), who were US citizens (OR=1.85), who were more educated (OR=1.93) and who were insured (OR=3.54) reported greater access to care. Women reported less use of ambulatory care (OR=0.66) and mental health services (OR=0.63) and worse overall access (OR=0.63) than men. Women were

more likely to report unmet need for medical care (OR=2.17) than men (due to any reason of the five reasons asked about).

CONCLUSION: Significant variation exists among HIV+ Hispanics in access to and use of care by language preference, acculturation, citizenship status and gender. Interventions to improve access for Hispanics should prioritize those Hispanics in greatest need.

PREDICTORS OF LONG-TERM MORTALITY FOR PATIENTS WITH COMMUNITY-ACQUIRED PNEUMONIA. E. Mortensen¹, W.N. Kapoor¹, J. Chang¹, M.J. Fine¹; ¹University of Pittsburgh, Pittsburgh, PA

PURPOSE: Community-acquired pneumonia (CAP) is the leading infectious disease cause of death in the United States. Although risk factors for short-term mortality are widely studied, there is very little data on the risk factors for long-term mortality after surviving CAP. The aim of this study is to examine the predictors of long-term mortality after initial CAP.

METHODS: This analysis was a follow-up of the 4 United States sites of the Pneumonia Patient Outcomes Research Team (PORT) prospective, multicenter cohort study. Baseline demographic and clinical data, including comorbid illnesses, laboratory and radiology results, were obtained for all patients. Long-term mortality was determined using the National Death Index. Patients were enrolled between October 1991 and March 1994, and mortality information was obtained up to December 1998. Charlson's comorbidity scale was used to assign a comorbidity score for preexisting comorbid conditions and age. All patients who died within 90-days of enrollment were excluded from the analysis in order to study the long-term mortality of those surviving after the acute illness. Statistical modeling included the use of Kaplan-Meier survival plots and the log rank test for categorical variables, and univariate Cox proportional hazard modeling for continuous variables. Any variable that was statistically significant ($p \leq 0.05$) was then entered into a step-wise forward multivariate Cox proportional hazard model.

RESULTS: Of the 1555 subjects enrolled at the 4 U.S. centers, 1419 subjects survived past 90-days after enrollment. Of those patients 472 (33.3%) died by December 1998. During the first year after enrollment 29.2% of patients died, in the second year 19.1%, third year 16.1%, fourth year 10.0% and $\geq$ fifth year 25.4%. Significant baseline predictors of long-term mortality were being DNR (hazard ratio 2.1 with 95% confidence interval of 1.5–2.9), poor nutritional status (1.9, 1.3–2.8), pleural effusion (1.4, 1.1–1.8), steroid use (1.6, 1.3–2.1), nursing home residence (10.1, 3.2–32.1), Charlson's score of 1 or 2 (2.5, 1.6–4.0), Charlson's score 3 or 4 (5.8, 3.9–8.9), and a Charlson's score ≥ 5 (14.6, 10–21.5). Significant protective factors against mortality included $\geq$ college education (0.6, 0.5–0.8), female gender (0.7, 0.6–0.8), and the symptom of fever (0.7, 0.6–0.9).

CONCLUSION: This study demonstrates that there is a significant long-term mortality burden after CAP, and that age, education, gender, comorbidity, nutritional status and nursing home status are important predictors of long-term mortality after CAP.

TEA CONSUMPTION AND SURVIVAL AFTER ACUTE MYOCARDIAL INFARCTION. K.J. Mukamal¹, M.A. Mittleman²; ¹Beth Israel Deaconess Medical Center, Boston, Massachusetts; ²Harvard School of Public Health, Boston, MA

PURPOSE: Although studies of tea consumption and mortality among healthy adults report conflicting results, the effect of tea consumption on patients with acute myocardial infarction is unknown.

METHODS: In a prospective cohort study, we followed 1935 patients (601 women and 1334 men) hospitalized with a confirmed acute myocardial infarction between 1989 and 1994 at 45 community hospitals and tertiary care centers in the United States, as part of the Determinants of Myocardial Infarction Onset Study. During hospitalization, participants reported their usual intake of caffeinated tea during the year prior to infarction to trained interviewers. We excluded 35 patients with missing information on tea intake. We analyzed survival through 1995 using Cox proportional hazards regression, controlling for age, sex, body-mass index, previous myocardial infarction, diabetes, hypertension, education, income, current and former smoking, alcohol use, regular exertion, acute infarct complications, previous medication use, and receipt of thrombolytic therapy.

RESULTS: Of the 1900 eligible patients, 1019 consumed no tea, 615 consumed less than 2 cups daily (moderate use), and 266 consumed 2 or more cups daily (heavy use). Heavy tea drinkers were older, less obese, and more likely to be female than non-drinkers, but their sociodemographic and clinical characteristics were otherwise similar. During a median follow-up of 3.8 years, a total of 313 patients (16%) died. Compared to non-drinkers, total mortality was lower among moderate tea drinkers (hazard ratio 0.78; 95% confidence interval (CI), 0.60–1.01) and heavy tea drinkers (hazard ratio 0.65; 95% CI, 0.44–0.97; p for trend = 0.01). When modeled continuously, the hazard ratio for each additional daily cup of tea was 0.89 (95% CI, 0.78–1.01). The effect of tea was similar in men and women, in younger and older patients, and for total and cardiovascular mortality. Controlling for coffee intake did not change the results.

CONCLUSION: Self-reported tea consumption during the year prior to acute myocardial infarction is associated with improved survival following infarction.

FACTORS ASSOCIATED WITH ADVERSE OUTCOMES AFTER DISCHARGE FROM HOME CARE. R.L. Murkofsky¹, R.B. Davis¹, R. Phillips¹; ¹Division of General Medicine and Primary Care, Beth Israel Deaconess Medical Center, Harvard Medical School, Boston, MA

PURPOSE: With recent changes in the financing of health care, home care patients may be discharged from home care prematurely, placing them at risk for adverse outcomes. Little is known about what happens to patients after home care discharge. We examined the frequency of adverse outcomes within 3 months of home care discharge, and identified factors associated with their occurrence.

METHODS: Patients admitted to a single home care agency between April and September 2000 were invited to participate in the study. We collected data by chart review and telephone interview. We followed patients for 3 months after home care discharge and collected information on 5 adverse outcomes: death, readmission to home care, nursing home placement,

emergency room use, and hospitalization. As potential correlates of adverse outcomes, we analyzed patient demographics, reason for home care, dependency in activities of daily living (ADLs) and instrumental activities of daily living (IADLs), cognitive function, depression, number of medications, caregiver support, and living alone. We developed a logistic regression model for adverse outcomes within 3 months of home care discharge.

RESULTS: Of 217 eligible patients, 180 (83%) agreed to participate in interviews. To date, 155 have been discharged from home care (85% remained home, 13% were hospitalized, and 2% were transferred to hospice). We obtained 3-month outcomes on 108 (84%) of the 129 patients who remained at home after home care discharge. Of these patients, 60% were female, 78% were white, 74% were ≥ 65 years of age (mean age 74 years), and 72% had Medicare or Medicare managed care as their primary insurance. The median home care length of stay was 23 days. At the time of home care discharge, 57% had ≥ 1 ADL dependency; 95% had ≥ 1 IADL dependency; 37% lived alone; and 16% had no caregiver. The most common diagnoses were congestive heart failure (25%), chronic pulmonary disease (28%), diabetes (28%) and cancer (30%). Within 3 months of home care discharge, 34 patients (32%) had one or more adverse outcomes: 5% died, 4% were readmitted to home care, 26% had an emergency room visit, and 15% were hospitalized. Patients with ≥ 1 ADL dependency were more likely to have an adverse outcome compared to those with no ADL dependency (42% vs. 17%, $p = 0.007$). In a multivariable model including patient age and sex, factors associated with adverse outcomes within 3 months of home care discharge were ≥ 1 ADL dependency (Adjusted Odds Ratio 3.9 [95% CI: 1.3, 11.5]) and a history of cancer (4.7 [1.7, 12.9]).

CONCLUSION: Home care patients are at substantial risk for adverse outcomes within 3 months of home care discharge. Those with functional dependence or cancer are at highest risk. Further research is necessary to determine whether additional home care services would prevent adverse outcomes in these high risk patients.

ENHANCED QT DISPERSION OCCURS IN PATIENTS WITH REVERSIBLE PERFUSION DEFECTS DURING PHARMACOLOGIC STRESS TESTING. M. Murdaza¹, M. Singh¹, V. Dimitrov¹, B. Kanna¹, T. Saleem¹, A. Soni¹; ¹Lincoln Medical Center, Bronx, NY

PURPOSE: To determine if Enhanced QTc Dispersion occurs in ischemia during pharmacologic stress testing. This may serve as a non-invasive method to detect ischemia in patients suspected of having CAD.

BACKGROUND: QTdispersion (Qtd), defined as the interlead variability in QT intervals on a 12 lead surface electrocardiogram (EKG) reflects regional inhomogeneity of ventricular repolarisation. Increased Qtd has been reported in patients with ischemic heart disease (IHD) during exercise and after myocardial infarction (MI). Pharmacological coronary vasodilatation is known to alter regional myocardial blood flow. However, the effect of redistribution of coronary blood flow during pharmacological coronary vasodilatation on myocardial function is unclear. We hypothesized that increased disparity of myocardial blood flow can lead to non-homogeneous regional ventricular repolarisation.

METHODS: We examined the effect of pharmacologic coronary vasodilatation on Qtd in 81 subjects who received dipyridamole during myocardial stress imaging study (imaging). Patients with history of MI in the past 2 weeks, electrolyte imbalance, cardiac arrhythmia, long QT syndrome, those on pro-arrhythmic agents were excluded. Routine EKGs were performed at rest (0 Mts) and peak stress (7–9 Mts after dipyridamole infusion). All EKGs were interfaced to a personal computer equipped with a digitizing tablet consisting of a magnifier used to enhance the accuracy of measurements. QT intervals were measured in all 12 leads and corrected for heart rate (QTc) using Bazett's formula. Intra & interpersonal variations were minimized by repeated blinded measurements. Patients were divided into three groups based on clinical probability and the results of the myocardial perfusion scan. 25 subjects with high pre-test probability for IHD and high risk reversible defects on imaging (reversible group) were compared with 33 other subjects with low clinical probability and normal imaging study ("normal" group). 12 patients with low risk fixed defects on imaging and variable probability ("fixed" group) were also studied. Mean Qtd values at rest and during pharmacologic stress testing were compared in each group. Statistical analysis was done using t test. Results are expressed in mean $\pm$ SD.

RESULTS:

Imaging results (n)	QTcd at rest	QTcd at peak stress	p value
Reversible defects (25)	0.0575 $\pm$ 0.02	0.0754 $\pm$ 0.03	0.018
Normal imaging (33)	0.0614 $\pm$ 0.02	0.0638 $\pm$ 0.02	ns
Fixed defects (12)	0.0633 $\pm$ 0.02	0.0619 $\pm$ 0.02	ns

(*n = no. of patients; Qtd = corrected QT dispersion; ns = not significant)

CONCLUSION: Enhanced QT dispersion occurs in patients who have high risk reversible perfusion defects on myocardial perfusion imaging studies during pharmacological vasodilatation. This finding suggests that QT dispersion may be a marker of increased inter-regional variation of myocardial perfusion among patients with IHD. Further studies are needed to validate this finding.

ETHNIC AND SOCIOECONOMIC DETERMINANTS OF TREATMENT, ADHERENCE TO TREATMENT AND CONTROL OF HYPERTENSION IN ADULTS. THE THIRD NATIONAL HEALTH AND NUTRITION EXAMINATION SURVEY, 1988–1994. S. Natarajan¹, S.R. Lipsitz¹, K.M. Magruder¹, D. Moge¹; ¹Medical University of South Carolina, Charleston, SC

PURPOSE: BACKGROUND: Prior studies have shown striking ethnic and socioeconomic differences in the prevalence of hypertension. While it is known that blood pressure (BP) is poorly controlled in the US population, studies have not evaluated the ethnic and socioeconomic determinants of Hypertension (HTN) treatment, adherence to treatment and

control of hypertension. AIM: The goals of the study were to evaluate ethnic and socioeconomic determinants of 1) treatment of hypertension, 2) adherence to treatment recommendations and 3) control of hypertension.

METHODS: The study is a cross-sectional analysis of the Third National Health and Nutritional Examination Survey, 1988–1994, a stratified, multistage sample of the US civilian non-institutionalized population. The study involved detailed home interviews, a careful physical exam, laboratory and other measurements. The study sample comprised 5,063 subjects with HTN. Ethnicity indicators were non-Hispanic Whites (NHW), non-Hispanic Blacks (NHB) and Mexican-Americans (MA). Socioeconomic indicators were years of education and the poverty-income ratio (PIR). The outcome measures were treatment (self-report), adherence (self-report) and control of HTN (SBP < 140 and DBP < 90). Treatment measures were medication prescriptions, counseling to restrict salt or sodium, Counseling to lose weight, or use of other methods like exercise, restrict alcohol intake, and to change diet. Adherence measures were self-report of taking medications, report of salt or sodium restriction, attempts to lose weight, and report of other approaches like exercise, alcohol restriction, and to change diet. BP control was measured by taking the mean of 6 BP measurements taken on 2 different days. Multivariate regression analyses, while accounting for the complex sampling design by using the appropriate sampling weights, adjusted for age, sex, access to health care, obesity, and urbanization. Significant results ($p < .05$) are reported as means or as odds ratios (OR) with the referent category being NHW, PIR less than or equal to 1 or < 12 years education.

RESULTS: There were no racial or socioeconomic differences in likelihood of prescription of medicines or weight reduction counseling. NHB were more likely to be counseled for salt restriction (OR 1.4). Subjects with > 12 years education were more likely (OR 1.5) to be counseled to exercise, restrict alcohol intake, and to change diet. In subjects who were counseled, NHB (OR 1.6) and subjects with > 12 years education (OR 1.6) were more likely to report salt restriction. NHB were also more likely to attempt to lose weight (OR 1.6), and to use other methods (OR 2) like exercise, alcohol restriction, and to change diet. Even among subjects with HTN who are aware of their HTN and report taking medications, mean systolic BP was higher in NHB (152 mm Hg) and MA (150 mm Hg) compared to NHW (141 mm Hg). There were no socioeconomic differences in mean BP. Among patients who reported adherence to medications, NHB (OR 1.4) and MA (OR 1.6) were more likely to have uncontrolled HTN (BP $> 140/90$ mm Hg).

CONCLUSION: Though self-reported treatment and self-reported adherence to treatment are similar among ethnic and socioeconomic categories, there are substantial ethnic differences in the actual blood pressure level, particularly in NHB and MA. While it is reassuring to note the similar odds of medication prescription among the different ethnic and socioeconomic categories, it is important to understand that initiating treatment, though crucial, is not sufficient. Future efforts should emphasize better counseling and in escalating intensity of treatment to achieve adequate control, particularly in ethnic minorities.

DISPARITIES IN THE DIAGNOSIS AND TREATMENT OF HIGH SERUM CHOLESTEROL BY RACE AND ETHNICITY. DATA FROM NHANES III. K. Nelson¹, K. Norris², C. Mangione¹; ¹UCLA Division of GIM/HSR, Los Angeles, CA; ²King Drew/ UCLA School of Medicine, Los Angeles, CA

PURPOSE: To examine racial and ethnic differences in the diagnosis and treatment of high cholesterol.

METHODS: We analyzed data from 7679 white, 4467 African American, and 4113 Mexican American adults over age 25 who completed the household questionnaire and laboratory exam in NHANES III. All respondents were asked: "Have you ever had your cholesterol checked?" If respondents answered yes, they were asked: "Have you ever been told by a doctor that your cholesterol level was high?" and "Because of your high cholesterol, have you ever been told by a doctor to take prescribed medication?" If a prescription medication was taken, the interviewer asked to see the medication container. Bivariate and multivariate analyses were used to examine the impact of race and ethnicity on cholesterol screening and on taking a cholesterol lowering medication. Unweighted counts and weighted percentages and odds ratios are presented. Odds ratios are adjusted to account for age, gender, income and, education, insurance status, comorbid illness and regular source of health care.

RESULTS: African Americans and Mexican Americans were significantly less likely than whites to report ever having their blood cholesterol checked (Table 1). Of those who were told to take medication, African Americans and Mexican Americans were significantly less likely to be taking a cholesterol-lowering agent (Table 1). The mean total serum cholesterol (from the laboratory exam) of those who reported being told their cholesterol was high was 234 mg/dL as compared to a mean of 198 mg/dL of those who were not told their cholesterol was high ($p < .001$).

Table 1

	Cholesterol Checked n=8996 % OR	Cholesterol was high n=3174 %	Told to take Medication n=761 %	On medication n=302 % OR
White	66 1.0	35	21	49 1.0
African Am	50 0.7	29	22	29 0.3
Mexican Am	37 0.7	32	23	28 0.5
	$p < .001$	$p < .001$	ns	$p < .05$

CONCLUSION: African Americans and Mexican Americans are less likely to report serum cholesterol screening than whites. Even when identified as requiring medication, African Americans and Mexican Americans were less likely than whites to be on cholesterol lowering agents. These disparities in primary prevention are likely to lead to an increase in the risk of coronary artery disease in these populations.

DO RECRUITMENT PROCEDURES AND INFORMED CONSENT PROVISIONS AFFECT PARTICIPATION RATES IN QUALITY IMPROVEMENT STUDIES? DATA FROM THE IMPROVING CHRONIC ILLNESS CARE EVALUATION. K. Nelson¹, R. Garcia¹, J. Brown¹, C. Mangione¹, E. Keeler¹, S. Cretin²; ¹Division of GIM/HSR, UCLA School of Medicine, Los Angeles, CA; ²RAND Health, Santa Monica, CA

PURPOSE: To describe the outcome of the Institutional Review Board (IRB) process from a multi-site QI evaluation and to examine participation rates as a function of recruitment and informed consent procedures.

METHODS: We analyzed participation rates from a patient survey and medical record abstraction from 1504 individuals with either congestive heart failure or diabetes treated at 13 clinical sites in the United States that had implemented an evidence based QI intervention. We used bivariate analyses to assess the effect of recruitment procedures and informed consent requirements on participation rates.

RESULTS: Recruitment procedures mandating patient permission to be contacted about the study varied by clinical site (Table 1). Among the sites that required advanced permission, only 58% (584/1012) of potentially eligible participants granted permission to be contacted about the study. Participation rates varied significantly by the type of recruitment and informed consent procedures required (Table 2).

Table 1. Consent procedures required by clinical sites in ICICE (n=13), # Sites requiring

		Medical Record Abstraction	
		No Written consent	Written consent
Telephone	No Advance consent	5	
Survey	Advance consent to contact	5	1
	Written consent to contact		2

Table 2. Participation Rates by recruitment and informed consent requirements (n=13)

		Medical Record Abstraction	
		No Written Consent	Written Consent
Telephone	No Advance consent	86%	
Survey	Advance consent to contact	40%	64%
	Written consent		30%
		P < .05	

CONCLUSION: We found substantial variation in provisions for subject recruitment and informed consent. These differences significantly impacted participation rates, and possibly the generalizability of this multi-site QI evaluation study.

RACIAL AND ETHNIC VARIATION IN RESPONSE RATE TO MAILED VERSUS TELEPHONE SURVEYS. K. Nelson¹, A.M. Geiger², F. Nevarez², C.M. Mangione¹; ¹University of California, Los Angeles, CA; ²Southern California Permanente Medical Group, Pasadena, CA

PURPOSE: To describe the demographic characteristics of participants who completed a mailed survey as compared to those initial non-responders who completed the same survey over the telephone.

METHODS: To better understand barriers to care for abnormal cervical cytology, we performed a cohort study of all women with an abnormal pap smear who received care at Kaiser Permanente Los Angeles Medical Center between July 1998 and October 1999. We administered a mailed survey with telephone follow-up for non-responders between April and August 2000. Potential participants were mailed an introductory letter and survey in both English and Spanish with a self-addressed stamped return envelope. If no response was obtained within three weeks, a second letter and survey were mailed. If no response was received within three weeks, a trained bi-lingual interviewer called potential participants and offered to complete the survey over the telephone. The survey asked questions regarding patient satisfaction, health beliefs, cancer knowledge, and socio-demographics characteristics. We used bivariate and multivariate analyses to describe the population characteristics of respondents who replied by mail as compared to those who completed the survey by telephone.

RESULTS: Of the 1049 potentially eligible participants, 733 women completed the survey for an overall response rate of 70%. Thirty one percent (n=226) of the surveys were received after the first mailing, 25% (n=185) were returned after the second mailing, and 44% (n=322) were completed over the telephone. Sixty seven percent of those who completed the survey in Spanish were obtained by telephone, compared to only 37% of the surveys completed in English. (p < .001) In multivariate analysis, Latinas who completed the survey in Spanish and African American women were significantly more likely to complete the telephone survey than their white counterparts. Telephone respondents were younger, had lower household incomes, and were significantly less satisfied with their health care than those who responded to the mailed survey. Women who completed the telephone survey had more misconceptions about cancer and were more likely to report no knowledge of having an abnormal pap smear.

CONCLUSION: Our results suggest that women from under-represented minorities were significantly more likely to participate in our research study if they are offered the survey by

telephone. If confirmed in other study populations, this finding has important implications for the design of studies that examine racial and ethnic disparities in care.

DELAYS IN CARE FOR ABNORMAL CERVICAL CYTOLOGY: THE EFFECT OF RACE/ETHNICITY AND HEALTH BELIEFS. K. Nelson¹, A.M. Geiger², F. Nevarez², C.M. Mangione¹; ¹University of California, Los Angeles, Los Angeles, California; ²Southern California Permanente Medical Group, Pasadena, CA

PURPOSE: To determine if race/ethnicity, health beliefs, and cancer knowledge influence delays in care for abnormal Pap smears.

METHODS: We conducted a cohort study of all women with an abnormal pap smear who received care at Kaiser Permanente Los Angeles Medical Center between July 1998 and October 1999. We administered a mailed survey with telephone follow-up for non-responders between April 2000 and August 2000. Women were eligible for the study if they were over the age of 18. A delay in care was defined as failing the first scheduled clinic visit to follow-up on an abnormal pap smear and/or requiring multiple attempts, including certified letters, to contact. We used multi-trait scaling to construct four scales: satisfaction with care, fatalism, misconceptions about cancer, and cervical cancer risk factor knowledge. We used bivariate and multivariate analyses to determine if these factors and race/ethnicity were significantly associated with delays in care.

RESULTS: Our response rate was 70% (n=733). Our sample was 51% Hispanic, 23% white, 13% African American and 12% Asian. Twenty three percent of participants completed the survey in Spanish. Fifteen percent of the sample had less than a high school education and 40% had annual incomes of less than \$30,000. Seventy two percent of the pap smear abnormalities were ASCUS (atypical squamous cells of undetermined significance), while the remaining abnormalities were low grade (24%) or high grade (4%) intraepithelial lesions. 15% of the sample had a delay in care for follow up on their abnormal pap smear. In multivariate analyses, women who delayed care were more likely to agree that: "A woman can tell if she has cervical cancer without going to the doctor" and that: "Women need to get a pap smear only with abnormal bleeding." Women who delayed care were also more likely to endorse fatalistic beliefs such as "There is very little a person cancer do to reduce their risk of cancer" and "I would rather not know if I had cancer". There were no significant associations between delays in care with race or ethnicity, education, income, marital status, country of origin, language preference, type of pap smear abnormality, satisfaction with care, or knowledge of cervical cancer risk factors. Barriers to care such as cost, transportation, childcare, and making time for doctors appointment were not associated with delays in care.

CONCLUSION: Delays in care were not associated with race and ethnicity in this insured managed care population. However, fatalistic health beliefs and misconceptions about cancer were significantly associated with delays in care.

A COST-EFFECTIVENESS ANALYSIS OF DIAGNOSTIC TESTING AND TREATMENT OF ADULTS WITH PHARYNGITIS. J. Neuner¹, R. Phillips², M. Aronson³, M.B. Hamel²; ¹Medical College of Wisconsin, Milwaukee, WI; ²Beth Israel Deaconess Medical Center, Boston, MA; ³Harvard University, Boston, MA

PURPOSE: Pharyngitis is the second most common presenting complaint in the primary care office. Since rheumatic fever has become uncommon in the United States while rapid streptococcal antigen test technology has improved and concerns over drug side effects have grown, we examined cost-effective diagnosis and treatment of patients with suspected GABHS pharyngitis.

METHODS: We constructed a decision analysis model from a societal perspective to examine the short-term cost-effectiveness of five strategies for the management of adult patients with pharyngitis: 1) Observation without testing or treatment, 2) Empiric treatment with penicillin, 3) Throat culture using a two-plate selective culture technique 4) Optical immunoassay (OIA) followed by culture to confirm negative OIA tests, or 5) OIA alone. We obtained data on test characteristics and event probabilities from published studies. We estimated utilities from a published patient survey which used a time-tradeoff technique to measure perceptions of outcomes (mild penicillin reaction, severe penicillin rxn, acute rheumatic fever) expressed in pharyngitis day equivalents. Based on published studies in which patients assigned a utility of 0.95 to other common symptoms such as diarrhea and dyspepsia, we estimated pharyngitis to be associated with a utility of 0.95. We estimated costs from our hospital's financial systems, prior publications, and Medicare reimbursement rates. We converted all costs to year 2000 dollars. In sensitivity analyses, we varied values for probabilities and costs across ranges that included all published values or across ranges of 50–200% of our baseline estimates.

RESULTS: At a baseline prevalence of GABHS pharyngitis of 9.7%, a culture alone strategy was both most effective and least expensive, resulting in an average of .2618 quality-adjusted life days (QALDs) lost and an average cost of \$6.19 per patient. For our base-case analysis, average costs in dollars, effectiveness in lost QALDs, and incremental cost-effectiveness were as below (see table). Our results were sensitive to the prevalence of GABHS, so that OIA followed by culture was most effective at a prevalence greater than 17%. Observation was least expensive at a prevalence less than 3% and empiric treatment least expensive at a prevalence greater than 75%. As the probability of allergic reactions was varied throughout a possible range from 0 to 5%, OIA with culture, culture, and observation were all optimal strategies. Our results did not change with variations in costs of diagnosis or treatment.

CONCLUSION: We found that observation, culture, and two rapid antigen test strategies for diagnostic testing and treatment of suspected GABHS pharyngitis in adults were all very similar in terms of both effectiveness and costs, although culture was least expensive and most effective at our institution's GABHS prevalence of 9.7%. We did not find empiric treatment to be most effective or least expensive at any prevalence of GABHS seen in adult populations.

Costs and effectiveness of management strategies for streptococcal pharyngitis (base-case)

	Average Cost	Average Effectiveness (Lost QALDs)	Incremental C-E
Culture	\$6.19	.2618	—
Observation	\$10.20	.2758	Dominated
OIA alone	\$11.13	.2674	Dominated
Empiric therapy	\$12.09	.4015	Dominated
OIA + culture	\$13.75	.2656	Dominated

HOW VALID ARE PAP SMEAR AND MAMMOGRAM SELF-REPORTS IN A MULTI-ETHNIC POPULATION? T.T. Nguyen¹, S.J. McPhee¹, C.P. Somkin², P. Vo³, B. Nguyen³, S.J. Shema³; ¹Division of GIM, UCSF, San Francisco, CA; ²Kaiser Permanente Division of Research, Oakland, CA; ³Northern California Cancer Center, Union City, CA

PURPOSE: Self-reports are commonly used to assess screening behaviors. Prior, studies have evaluated the accuracy of such reports, but few have compared their validity across different ethnicities and health care delivery systems.

METHODS: In 1999, we compared self-reports of Pap smears and mammograms to medical records among African-American, Latina, Chinese, Filipina, and White women in low-income census tracts of Alameda County, CA. Using random digit dialing telephone interviews in 4 languages, we collected self-reports of the date and site of the most recent Paps and mammograms. We considered a self-report valid if the test was documented at the site within 12 months of the reported date.

RESULTS: Among 1465 women, 1390 (95%) reported ever having had a Pap smear and 1274 (89%), a mammogram. We examined self-reports of Pap smears for 448 women in 3 locations and mammograms for 846 women in 4 facilities. Table 1 shows that accuracy varied significantly by ethnicity for mammogram and Pap self-reports. Mammogram self-reports were significantly less accurate for women seen at the county hospital (48%) compared to community hospitals (77%) and HMO (76%) ($p=0.001$). Accuracy of Pap self-reports was also significantly lower at county clinics (49%) and health center (56%) compared to HMO (72%) ($p=0.001$).

CONCLUSION: Self-reports of breast and cervical cancer screening frequently could not be validated. Accuracy of such reports varied by ethnicity and site of care. Researchers relying on self-reports need to be aware of such differences.

Table 1. Accuracy of Self-Reports by Ethnicity

	Pap n/N	%Pap valid	Mam n/N	%Mam valid
Afr-Am	102/162	63	236/335	70
Latina	85/132	64	132/189	70
Chinese	27/41	66	50/78	64
Filipina	26/39	67	43/58	74
White	62/74	84 ($p=0.026$)	159/186	85 ($p=0.001$)
TOTAL	302/448	67	620/846	73

LOW HEALTH LITERACY SKILLS IN EMERGENCY DEPARTMENT USERS. G.M. O'Brien¹, M. Lopez-perez², M.J. Shapiro³; ¹Brown University, Providence, RI; ²Brown University School of Medicine, Providence, RI; ³Rhode Island Hospital, Providence, RI

PURPOSE: Many persons use the emergency department (ED) as their main source of health care. Persons with low health literacy skills have more hospital visits, longer stays, and use more hospital resources than others. We hypothesized that low health literacy may be a determinant in the use of the ED as a main source of health care. The purpose of this study was to assess health literacy rates in these persons compared to persons who have a regular physician.

METHODS: We performed an observational study of 96 persons presenting to the ED who identified the ED as their regular source of care or had no doctor (NODOC), and a comparison group of a convenience sample of 100 persons presenting to the ED who stated they had a regular physician (DOC). The study was conducted from 6/00 to 8/00 in an busy urban ED. The test of functional health literacy in adults (TOFHLA) was administered in English or Spanish to all subjects, and was the main outcome measure, categorized as adequate or low for analysis.

RESULTS: The NODOC subjects had more males (66 vs.25%) $p=.0001$, fewer whites (38.5 vs.53%) and more blacks (19 vs.7%) $p=.054$, lower mean age (30 vs.34) $p=.02$, were more uninsured (46 vs.14%) $p<.0001$, fewer had chronic disease (25 vs.41%) $p=.02$, lower education level (<12 yrs) (61.5 vs.43%) $p=.01$, and more subjects with low literacy (36 vs.22%), $p=.017$; when compared to the DOC group. By multiple logistic regression, those in the NODOC group were more likely to be <40 yrs OR 2.5, (1.12–5.48) $p=.03$, male OR 4.26 (2.06–8.77) $p<.0001$, have no insurance vs private OR 5.13 (2.22–11.8) $p<.0001$, and showed a trend for having lower health literacy OR 1.78 (.83–3.8) $p=.13$.

CONCLUSION: In addition to being male, <40 years old and uninsured, low health literacy seems to be a factor in patients using the ED care or not having a doctor. Larger studies need to be done to validate this trend.

PRIMARY CARE DELIVERY AND LOW-INCOME WOMENS ADHERENCE TO CANCER SCREENING. A.S. O'malley¹, C.B. Forrest², J. Mandelblatt³; ¹Georgetown University, Washington, DC; ²Johns Hopkins University, Baltimore, Maryland; ³Georgetown University Medical Center, Washington, DC

PURPOSE: Background: African-American and low-income women have lower rates of cancer screening and higher rates of late-stage disease.

OBJECTIVE: To examine the effects of health insurance, primary care, and health maintenance organizations on adherence to regular breast, cervical and colorectal cancer screening.

METHODS: Design: Random-digit-dial telephone survey of a population-based household sample.

SETTING: Washington, D.C. census tracts with at least 30% of households below 200% of federal poverty line.

Participants: 1205 women over age 40, 82% of whom were African-American.

Main Outcome Measures: Adherence was defined as receipt of the last two screening tests within recommended intervals for age.

RESULTS: The survey response rate was 85%. Overall, 75% of respondents were adherent to regular Pap smears, 66% to clinical breast exams, 65% to mammography, and 29% to fecal occult blood test recommendations. Continuity with a single primary care practitioner, comprehensive service delivery, and higher patient satisfaction with the relationships with primary care practitioners were consistently associated with higher adherence across the four screening tests, after considering other factors. Coordination of care also predicted screening adherence for women age 65 and over, but not for the younger women. Compared with counterparts in non-HMO plans, women enrolled in health maintenance organizations were also more likely to be adherent to regular screening (e.g. OR Pap 1.89, 95%CI 1.11–3.17; OR Clinical breast exam 2.04, 95%CI 1.21–3.44; OR Mammogram 1.95, 95%CI 1.15–3.31; OR fecal occult blood test 1.70, 95%CI 1.01–2.83). Of note, controlling for primary care variables, the presence of insurance was not a significant correlate of screening adherence.

CONCLUSION: Healthcare services organized to promote better primary care delivery—especially continuity, and enrollment in health maintenance organizations are stronger determinants of whether low-income women are adherent to regular cancer screening than the presence of health insurance. Efforts to eliminate cancer disparities could include incentives for primary care delivery systems to improve continuity of care, comprehensiveness of services, and patient-provider relationships.

PRIMARY CARE PERFORMANCE & LOW-INCOME WOMENS PATIENT-PHYSICIAN RELATIONSHIPS. A. O'malley¹, C.B. Forrest²; ¹Georgetown University, Washington, DC; ²Johns Hopkins University, Baltimore, Maryland

PURPOSE: The physician-patient relationship is highly valued by consumers; and, it is especially crucial to the most vulnerable members of our society. This study assessed whether women in ambulatory care delivery systems with heavier emphasis on primary care characteristics have stronger patient-physician relationships in terms of trust, compassion and communication.

METHODS: A population-based telephone survey of 1205 women from a combination of random-digit-dial and targeted listed household samples from lower income census tracts of Washington, D.C. was conducted from January–March, 2000. Valid and reliable measures of specific aspects of primary care and of the physician-patient relationship were used.

RESULTS: The survey response rate was 85%. Socioeconomic characteristics of the respondents reflected success in reaching the population sought. Four attributes of primary care (continuity, organizational accessibility, comprehensiveness of services, and coordination of specialty care) were important positive correlates of all aspects of the patient-physician relationship (trust, compassion and comprehension), regardless of insurance, demographics, socioeconomic or health status. For example, women with the highest level of comprehensiveness of services at their usual source of care, were 11 times as likely to trust their physician ($p<.01$) and 6 times as likely to find their physicians compassionate and communicative ($p<.01$), compared to those with the lowest level of comprehensiveness. Higher organizational accessibility was strongly associated with greater trust, compassion and communication aspects of the physician-patient relationship. (OR 3.2, OR 7.4 and OR 6.9 respectively, $p<.01$ for each).

CONCLUSION: Women in ambulatory care systems that most strongly exhibit the features of primary care report stronger physician-patient relationships than women whose usual sources of ambulatory care lack those primary care features. Ambulatory care systems which are organized to be more accessible to consumers, which permit patients to see the same clinician for their visits, which provide more comprehensive services, and which allow clinicians to coordinate specialty services are more likely to foster strong relationships between their patients and physicians, and to have patients who are more satisfied with those relationships.

THE IMPACT OF PSYCHOLOGICAL FACTORS ON MOTIVATION FOR CHANGE IN A SCREENING POPULATION. J. Flynn¹, P.G. O'malley¹, D.L. Jones¹, A.J. Taylor¹; ¹Walter Reed Army Medical Center, Washington, DC

PURPOSE: Motivation for change is important to optimize healthy changes in modifiable risk factors for disease. We studied the relationship between multiple psychological factors and motivation for change.

METHODS: Prospective study of a consecutive sample of 1,000 consenting, U.S. Army personnel (39 to 45 years old). Participants were assessed for depression, anxiety, somatization, stress, hostility, and functional status using validated psychometric tools. Motivation for change (based on the transtheoretical model) was assessed using a ladder score (0–10).

RESULTS: The cohort was 82% male, 73% white, and the mean age was 42. DSM-IV disorders were present in 12.7%. Women had lower motivation for change ($p<0.03$), lower mental health functional status, and higher anxiety, depression, stress, and somatization scores ($p<0.001$ for all). Smokers also had lower overall motivation for change ($p<0.001$), functional status ($p=0.05$), and higher hostility ($p<0.001$) and depression scores ($p<0.01$). In multivariate models (controlling for age, gender, education, and smoking), all psychological variables except hostility were independently negatively associated with motivation for change.

CONCLUSION: Depression, anxiety, somatization, and stress are negatively associated with motivation for change. This suggests that screening for psychiatric disorders and stress may be a useful strategy in people who are less ready to change behaviors.

Correlation of Psychological Scores with Motivation for Change

MOTIVATION FOR CHANGE	ANX	DEPR	HOST	STRESS	SOMAT	SF-36
Overall	-.13**	-.21**	-.08*	-.18**	-.20**	+.26**
Fat Intake	-.07	-.08*	-.06	-.08*	-.10**	+.10*
Fr/Veg Intake	-.14*	-.13**	-.11**	-.16**	-.12*	+.18**
Manage Stress	-.33**	-.31**	-.15**	-.32**	-.25**	+.38**
Tob(Smokers only)	-.23	-.11	-.12	-.03	-.03	+.12
Phys Activity	-.19*	-.22**	-.13**	-.22**	-.20**	+.28**

* $P < .05$; ** $P < .001$

RACE, PATIENT PREFERENCES AND USE OF CAROTID ENDARTERECTOMY. E. Oddone¹, R. Horner², K. Stechuchak², L. McIntyre²; ¹Duke University, Durham, NC; ²VAMC, Durham, NC

PURPOSE: Stroke is the third leading cause of death and the leading cause of disability in the United States. Carotid endarterectomy (CE) has been proven to reduce the risk of stroke for certain patients. While black patients are at higher risk for stroke than whites, prior research had shown that they are only one-third to one-fourth as likely as white patients to receive CE. This study was designed to determine the importance of patient preferences in predicting use of carotid angiography (CA) and CE in a racially stratified sample of patients at risk for stroke and potentially eligible for CE.

METHODS: Prospective, racially stratified, cohort study of patients undergoing evaluation for carotid artery stenosis at five VAMC's (Atlanta, Durham, Pittsburgh, Richmond, and St. Louis). Patient interviews were conducted at the time of the index carotid ultrasound and each patient was followed for 6 months to determine clinical events and receipt of CA or CE. We assessed demographic characteristics, clinical status (asymptomatic, transient ischemic attack-TIA, or stroke), severity of illness, health related quality of life (SF-12), previous surgical history, operative risk, and trust in physician. We also measured patients' aversion to CE, using a standard gamble technique. The main outcome was receipt of CA and CE within six months of the carotid ultrasound.

RESULTS: Between 1997 and 1999 we enrolled 719 patients (625 white and 94 black) who had at least one carotid artery with a 50% stenosis. Black and white patients were similar in terms of age, gender, education level, and social support. More black than white patients received their ultrasound for a completed stroke (35% vs. 13%) and fewer black patients were classified as asymptomatic or as having had a TIA ($p < 0.001$). Health related quality of life scores, trust in physician, and medical comorbidity scores were similar for black and white patients. Based on responses to the standard gamble, black patients expressed higher aversion to CE (30% in highest aversion category) than did white patients (16% in highest aversion category), $P=0.01$. However, there were no racial differences in receipt of either CA or CE in the 6-month follow-up period. During follow-up, 30% of white patients and 24% of black patients received either CA or CE ($p=0.27$). In adjusted analyses, the patients' clinical status (i.e., asymptomatic vs. TIA vs. stroke) was associated with receipt of CA or CE and not the patients' race.

CONCLUSION: Contrary to prior research, results from this study show that race was not associated with receipt of CA or CE for patients with $>50\%$ stenosis and who receive their care in an equal access health care system (VA). These findings are robust in that we controlled for symptom status and all patients had anatomic information concerning the degree of carotid artery stenosis. Clinical status was the primary determinant of the care received.

DECISION ANALYSIS: HOW STRONG IS THE EVIDENCE? J. Oh¹, D. Conner², B. Flansbaum², W. Rifkin², D. Rose³; ¹East Meadow, New York; ²NUMC, East Meadow, NY; ³LJMC, New Hyde Park, NY

PURPOSE: Clinical decision analyses (CDA) can be very useful in determining an optimal medical strategy among competing alternatives. One weakness of CDAs is their reliance on evidence from other studies to estimate clinical outcomes and costs. Since 1995, it has been recommended that CDAs should report explicitly the search strategy and relative strength of the studies used, similar to the recommendations for other integrative studies, such as overviews. The purpose of this study is to determine how frequently the search, article selection and appraisal of the evidence used in CDAs are reported.

METHODS: We performed a Medline search on CDA and decision analysis based Cost-effectiveness studies published on 5 major medical journals (Annals of Internal Medicine, BMJ, JAMA, Lancet, and New England Medical Journal) between January 1997 and October 2000. We excluded studies based on previously published decision analyses. The articles were reviewed by 2 investigators who were blinded to author and journal. We extracted information necessary to determine if the search strategies and databases searched were reported, if reference lists were reviewed for additional studies, if a systematic effort was made to find relevant unpublished studies, and if an a priori selection criteria or any validity grading system was used.

RESULTS: We identified 257 articles through our Medline search and 36 met our eligibility criteria. The search strategy, including databases searched, was reported in 9 (25%) of the articles. Reviewing the reference list for additional studies was reported by 5 (14%) of the articles. Though no article reported a systematic effort to locate unpublished studies, 3 articles did report using data from unpublished sources. Two studies (6%) reported using a selection criteria and one study (3%) a grading system for the quality of the evidence.

CONCLUSION: Our study shows that only 25% of CDAs published in major medical journals reported their process for selecting and evaluating the data they used. This is very concerning in view of the importance these items play when readers must appraise the validity of a CDA. Although it is possible that most studies were very rigorous in their selection of data, the lack of a clear report in the methods section makes it difficult for readers to identify stronger CDAs from ones that are less evidence-based. Our limitations include the fact that some of these data elements can only be obtained from databases like SEER, but all of these studies included probabilities that were obtained from the medical literature. We believe that the addition of

evidence selection and appraisal in CDAs can strengthen these studies and give readers more confidence to evaluate their results.

SMOKING AMONG PATIENTS IN ADDICTION TREATMENT: A MISSED OPPORTUNITY. Y. Olsen¹, D.P. Alford¹, N.J. Horton², R. Saitz¹; ¹Boston University School of Medicine, Boston, MA; ²Boston University School of Public Health, Boston, MA

PURPOSE: Smoking is a major cause of preventable death among patients at methadone maintenance (MM) clinics. Although most MM patients smoke cigarettes, and all must attend required weekly counseling for drug addiction, little is known about how well methadone clinics address nicotine dependence. We surveyed MM patients to test the hypotheses that (1) many are interested in receiving help to quit smoking and (2) despite weekly clinic visits, few have received nicotine dependence counseling or treatment.

METHODS: We conducted a cross-sectional survey of MM patients at three large urban methadone clinics. Subjects had smoked at least 1 cigarette in the last month, at least 100 cigarettes in their lifetime, and had met with their assigned MM counselor at least once. The survey assessed demographics, readiness to quit smoking, and assistance with smoking cessation from MM counselors using questions with Likert-type responses.

RESULTS: Of 575 MM patients in the clinics, 425 (74%) were eligible smokers; 100% completed the survey. Fifty percent were female; the mean age was 42; 64% were white, 22% African-American, and 13% Hispanic/Latino. Just over half (55%) reported no illicit drug use in the past 30 days. A majority of subjects were heavy smokers: 56% smoked at least one pack per day for a mean of 26 years, with 64% starting smoking by age 16. Study subjects had frequent contact with their counselors: the mean number of sessions attended over a 2-month period was 7.7 (SD 3.8); the median number of months with the same counselor was 12. We found a high level of interest in quitting: 81% expressed the desire to quit, and 71% had plans to quit within the next six months. Sixty-nine percent agreed at least somewhat that counseling may aid people with nicotine addiction, and 73% that smoking cessation should be discussed in counseling sessions. However, 56% reported that their counselors had never asked about smoking. Over two-thirds reported that counselors never or rarely 1) discussed the adverse health effects of nicotine (68%), 2) advised them to quit (68%), nor 3) discussed smoking cessation options (72%).

CONCLUSION: While all MM patients receive addiction counseling and many report interest in receiving help to quit smoking, primary care physicians should not assume that MM clinics are providing adequate cessation services. To combat a leading cause of mortality in this vulnerable population, MM clinics should routinely offer treatment for nicotine dependence. In this motivated population, every patient encounter may be an opportunity for change. Supported by Grant IRG-72-001-26-IRG from the American Cancer Society.

PREOPERATIVE CARDIAC RISK ASSESSMENT PRIOR TO AMBULATORY SURGERY. A.A. Onitilo¹, A.O. Olatosi², J. Iqbal³, L.G. Capps³; ¹Harlem Hospital, New York, New York; ²Harlem Hospital Center, New York, NY; ³Harlem Hospital Center, NY, NY

BACKGROUND: Various guidelines for the assessment of perioperative cardiac risk beginning nearly 40 years ago. All stress clinical evaluation and most studies have shown no additional benefit of additional testing except in high-risk situations. Both patient factors and the type of surgery contribute to risk. Most ambulatory procedures are in the lowest risk category.

OBJECTIVE: To evaluate the risk of cardiac complication in patients who underwent ambulatory procedures.

DESIGN: Prospective cohort study. Setting: An urban teaching hospital in New York City with predominantly black population. Patients: 281 patients referred for medical consultation in the ambulatory surgery unit. Method: We collected data on risk factors for coronary artery disease and type of surgery to be performed during the preoperative evaluation. All patients had CBC, routine chemistry, coagulation studies, electrocardiogram, and chest x-ray according to the protocol followed in the ambulatory surgery unit. A chart review conducted at least a month after the surgery was done to assess the incidence of postoperative cardiovascular complications.

RESULTS: Average age of patients studied was 57, with a range of 28-86 years. There were 108 men and 173 women. 30% had an abnormal electrocardiogram, 61% had hypertension, 30% had diabetes, 37% patients had at least one other risk factor for coronary disease, and 53% had two or more cardiac risk factors for surgery. The type of surgery included a variety of ambulatory procedure. The most common were eye 21%, breast 15%, cholecystectomies/laparotomies 10%, hernias 7%, and ENT 5%.

Overall outcome There were no patients in this cohort group who had postoperative cardiac complications.

CONCLUSION: The incidence of cardiac complications after low-risk ambulatory procedures is low, even in the patients with risk factors. Additional testing is very unlikely to add to the ability to predict risk in this group of patients.

Table 1 Risk Factors

	Abnormal EKG	Hypertension	Diabetes	Other CAD risk Factors
Percent	30	61	30	37

ARE USERS OF SILDENAFIL WILLING TO CHANGE INSURANCE PROVIDERS TO OBTAIN COVERAGE FOR THE MEDICINE? N.M. Paige¹, R.D. Hays², M.S. Litwin², J. Rajfer², M.F. Shapiro²; ¹West Los Angeles VAMC/UCLA School of Medicine, Los Angeles, CA; ²UCLA School of Medicine, Los Angeles, CA

PURPOSE: Pharmacy benefits may influence one's choice of a health insurance plan. This study explores the willingness of users of sildenafil (Viagra) to change insurance providers in order to obtain coverage for the medication.

METHODS: We conducted a cross-sectional survey in a university hospital urology and internal medicine clinic and university-affiliated community primary care clinics. Letters were sent to eligible patients by their primary care provider or urologist inviting them to participate in the study.

RESULTS: One hundred and twenty four patients (52.5% of the eligible sample) completed and returned a survey. Mean age was 60.5 years (range 36–80). 53% were married; 95% were white; 62% had a 4-year college degree; mean household income was \$75,000. 30% had insurance coverage through Medicare plus a supplemental plan; 20% by a HMO; and 42% by a FFS or PPO plan. 32% of respondents reported that their current health insurance paid for at least four doses of sildenafil per month. The remainder did not have coverage or did not know the number of doses covered. When respondents were asked to assume that their current health insurance provider did not cover sildenafil, 45% reported that they would possibly or definitely consider switching their health plan to obtain coverage for the medication. Nearly two-thirds of those willing to switch believed that sildenafil should be covered by health insurance even if it increases insurance premiums. Respondents who would consider switching were younger (55.8 years vs. 64.0 years, $p < 0.01$); used more doses in the last 30 days (6.5 vs. 4.4, $p = 0.05$); had current coverage for 4 or more doses per month (65% vs. 35%, $p < 0.01$); reported greater improvement (change) in their mental health summary score with use of the medication (3.4 vs. 0.2, $p = 0.01$); and had greater improvement (change) in their overall sexual satisfaction score (3.6 vs. 2.0, $p = 0.02$). Moreover, respondents who would most likely consider changing insurance providers were those who had current coverage with an FFS/PPO plan (56%) or HMO plan (50%); 30% of those who had current coverage by Medicare and a supplemental plan would consider changing.

CONCLUSION: Whether or not a health insurance provider covers sildenafil may influence a person's decision to switch insurance providers. Additional studies are needed to further investigate this potentially important finding.

FACTORS ASSOCIATED WITH READMISSION AND LEAVING HOSPITAL AGAINST MEDICAL ADVICE AMONG HIV-POSITIVE PATIENTS. A PALEPU, AH ANIS, H SUN, D GUH, A Palepu¹, AH Anis¹, H Sun¹, D Guh¹, D Skippen¹, J Russell¹, P Phillips¹, MV O'Shaughnessy¹, ¹University of British Columbia, Vancouver, British Columbia

PURPOSE: A specialized 18-bed HIV/AIDS ward was instituted in February 1997 at St. Paul's Hospital in Vancouver, BC which coincided with the explosive outbreak of HIV among injection drug users (IDUs) in this city. The purpose of this study was to assess the differential health service utilization pattern including readmission rates among HIV-positive IDUs and non-IDUs as well as to examine the factors associated with leaving hospital against medical advice (AMA).

METHODS: We reviewed the discharge abstracts of all patients admitted to St. Paul's Hospital with HIV/AIDS between April, 1997 and August, 1999. The medical record department has reliably coded illicit injection drug use diagnoses since 1995 and we used this designation to define IDUs and non-IDUs. Leaving hospital AMA was determined from physician notes. Readmission was defined as being hospitalized within 30 days of discharge from the first hospitalization. Related readmissions were determined by a physician who reviewed the discharge abstracts.

RESULTS: There were 1355 HIV-positive patients who incurred 3289 admissions during the 28-month study period. The median age was 33 (IQR 33–44) and 21% were women. 46% had AIDS-defining illnesses and 632 (47%) were IDUs. 49% had > 1 admission, 268 (8%) were homeless and 551 (17%) left hospital AMA. 191 patients were readmitted within 30 days and 134 (10%) were related to the index hospitalization. Most readmissions occurred within the first week after discharge (median 6 days, IQR 2–15). There were no differences between IDUs and non-IDUs in their readmission rates (11% vs 9%, $p = 0.22$) or between patients who were initially admitted to the HIV/AIDS ward versus the medical ward (both 12%). The length of stay was similar among patients who were and were not readmitted within 30 days. The three most frequent diagnoses were: Bacterial pneumonia; PCP; Cellulitis and Abscess. Leaving AMA from the index hospitalization was associated with 30-day readmission ($p = 0.001$). Factors associated with leaving AMA included: IDUs vs. non-IDUs (21% vs. 6%, $p = 0.001$); medical vs. AIDS ward (20% vs. 12%, $p = 0.003$); absence vs. presence of AIDS-defining illness (15% vs. 9%, $p = 0.002$); homeless vs. not (22% vs 12%, $p = 0.004$).

CONCLUSION: HIV-positive patients were often readmitted to hospital with a diagnosis related to their index hospitalization and this did not differ by IDU status or hospital ward. Patients more likely to leave AMA were IDUs, on the medical ward, did not have AIDS and were homeless. Better linkages to care providers in the community to ensure monitoring and completion of therapy may reduce hospital readmission rates. In addition, the reasons why some HIV-positive patients leave AMA need to be explored and strategies developed to address this problem.

FACTORS OTHER THAN THE DUKE CRITERIA ASSOCIATED WITH INFECTIVE ENDOCARDITIS AMONG FEBRILE INJECTION DRUG USERS. A Palepu¹, S Cheung¹, R Wood¹, C Thompson¹, ¹University of British Columbia, Vancouver, British Columbia

PURPOSE: The Duke criteria have incorporated echocardiography results in the diagnosis of infective endocarditis (IE). St. Paul's Hospital is an inner-city hospital that serves a large proportion of injection drug users (IDUs) at risk of IE. The purpose of this study was to validate the Duke criteria in a high-risk population and to determine if other clinical factors available on admission predicted the presence of IE.

METHODS: We collected clinical data of consecutive febrile IDUs who were hospitalized at our hospital over a 15 month period. Included patients were referred to Internal Medicine, recently injected illicit drugs, febrile and the housestaff had a clinical suspicion of IE and ordered an echocardiogram. The housestaff collected history and physical examination variables on a standardized form and one of the investigators (SC) collected follow-up data. Each patient's hospital course was followed until discharge or death. Stepwise logistic regression was used to determine independent predictors of IE exclusive of the Duke criteria.

RESULTS: 100 febrile IDUs were enrolled and half were female and 58 were HIV-positive. 22 met Duke criteria for IE. IDUs with IE more often had chief complaints of myalgia, fever, chills

and weakness than those without IE. They also had a longer length of illness prior to presenting to hospital (10.8 d vs. 6.9 d), a past history of IE (45.5% vs. 25.6%), evidence of systemic emboli (40.9% vs. 7.7%), multiple opacities on chest radiograph (45.5% vs. 20.5%), and fewer had an obvious source of infection (50% vs. 78.2%). Most IDUs with IE had positive blood cultures with typical organisms (68.2% vs. 23.1%), and all met major echocardiography criteria for IE. The following factors were independently associated with IE (aOR, 95% CI): past history of IE (5.3, 1.4–19.5); no obvious source of infection (5.4, 1.6–17.5); and multiple opacities on chest radiograph (5.2, 1.5–17.8).

CONCLUSION: Most of the IDUs with IE met 2 major Duke criteria for IE. The distribution of minor Duke criteria between IDUs with and without IE was similar except for the high prevalence of vascular phenomenon in those with IE. Finally, we found 3 clinical factors that should heighten the suspicion of IE amongst febrile IDUs: past history of IE, no obvious source of infection and multiple opacities on chest radiograph. Such patients should be kept in hospital until the results of their blood cultures are available.

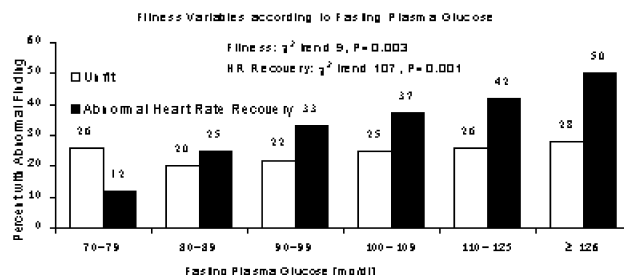
FASTING PLASMA GLUCOSE, PHYSICAL FITNESS, AND HEART RATE RECOVERY: ASSOCIATIONS AND PROGNOSTIC IMPORTANCE. C Panzer¹, M.S. Lauer¹, E. Blackstone¹, B. Hoogwerf¹, ¹Cleveland Clinic Foundation, Cleveland, OH

PURPOSE: Diabetes mellitus is associated with abnormal autonomic function. Impaired glucose tolerance and diabetes mellitus are related to increased mortality. Abnormal heart rate recovery (aHRR), a measure of autonomic dysfunction, is also known to be associated with increased mortality. Whether impaired fasting glucose (IFG) is associated with aHRR has not been characterized.

METHODS: 5190 healthy adults without medically treated diabetes (mean age 45, 39 % women) enrolled in the Lipid Research Clinics Prevalence study underwent exercise testing. Low physical fitness was defined according to the lowest quartile of peak METs. HRR was defined as the change from peak heart rate to that measured after 2 minutes of recovery, an abnormal value was < 42 BPM.

RESULTS: 504 subjects (10%) had IFG and 131 (3%) were diabetics. An aHRR was found in 1699 (33%) adults; 1196 (23%) were unfit. Increasing levels of fasting plasma glucose (FPG) were strongly associated with aHRR (see figure), even at glucose levels < 110 mg/dl. The association between increasing FPG and poor physical fitness was weaker. FPG remained an independent predictor of aHRR after adjustment for standard risk factors and resting heart rate (RHR) (adjusted OR for increase of glucose of 500/ glucose by 1, 1.17, 95% CI 1.05–1.30, chi-square = 9, $P = 0.003$). There was a strong interaction between FPG and RHR for prediction of aHRR. The association between FPG and aHRR was very strong with RHR 80 BPM (P for interaction = 0.0008); no association was present with RHR < 80 BPM.

CONCLUSION: 1. FPG is strongly and independently associated with aHRR, even at non-diabetic levels. 2. The association between FPG and aHRR is substantially affected by resting heart rate.



USE OF EMAIL REMINDERS TO IMPROVE INFLUENZA VACCINATION RATES IN THE ELDERLY. MA Parkula¹, J.J. Mentel¹, ¹Mayo Clinic Jacksonville, Jacksonville, FL

PURPOSE: In the United States, influenza causes an average of 20,000 deaths per year; > 90% of these deaths are among persons aged greater than or equal to 65 years. Based on the results of the 1996 Medicare Current Beneficiary Survey, the 2 most common reasons for not receiving the influenza vaccine were a lack of knowledge that the vaccine was needed and not thinking of or missing the vaccination. We evaluated the effectiveness of a single timely email reminder which highlighted the reasons for receiving the influenza vaccine, the date and time of convenient walk-in vaccination clinics, and that the participant's personal physician recommended the vaccine.

METHODS: From July until September of 1999, 3249 consecutive patients in the Primary Care Internal Medicine Clinic at the Mayo Clinic in Jacksonville, Florida were surveyed concerning their email capability. Of those patients over the age of 65 who responded, 337 had email capability and an interest in communicating with their practitioner. Two hundred twenty two agreed to participate in the study. Based on the results of the aforementioned survey, 86% of the 222 participants had self-reported that they had received an influenza vaccination in 1998. (Previous studies have validated the reliability of self-reported influenza vaccination rates.) A single email reminder was sent to all participants on November 11, 1999 reminding them of the need and location for influenza vaccinations. From February to April of 2000, participants were contacted by phone to verify their vaccination status for 1999.

RESULTS: The self reported vaccination rate for 1999 increased to 91%. Analysis using the McNemar test for paired samples reveals the increase in vaccination rate from 1998 to 1999 to be statistically significant with a p-value of less than 0.003.

CONCLUSION: A single timely email reminder is an effective means of improving influenza vaccination rates in elderly patients with email capability. Increases in patients' email capability should provide greater opportunities for the use of this technology.

PREVALENCE OF ELECTRONIC COMMUNICATING (EMAIL) CAPABILITY IN A PRIMARY CARE CLINIC. M.A. Parkulo¹, J.J. Mentel¹; ¹Mayo Clinic Jacksonville, Jacksonville, FL

PURPOSE: Email is rapidly becoming a major avenue for communication in America. Patient-practitioner email communication could potentially have a dramatic impact on the practice of medicine especially chronic disease management in the primary care setting. A systematic survey assessing the email capability and interests of patients in a primary care clinic has never been reported.

METHODS: We surveyed consecutive patients presenting for care in the Primary Care Internal Medicine Clinic at the Mayo Clinic in Jacksonville, Florida from 7/1/99 to 9/30/99. The survey included demographic questions as well as questions about Internet and email access, how frequently email was used, respondents desire to communicate by email with their practitioner, and history of chronic diseases.

RESULTS: Three thousand two hundred forty nine patients were offered the survey. Two thousand twenty two respondents completed the survey yielding a 62% response rate. The age of the respondents was as follows: 3.2% <35 years old, 12.3% 35–49 years old, 19.2% 50–64 years old, 44.8% 65–79 years old, and 20.2% greater than or equal to 80 years old. The respondents were predominately white (91%)

CONCLUSION: There is a high prevalence of email capability in our population even though the population has a substantial percentage of elderly patients. Of those who have email capability over two thirds expressed an interest in communicating with their provider via email. Many of these patients have chronic illnesses which could provide an opportunity for further research about managing these chronic illnesses electronically.

CONTROLLED DRUG PRESCRIBING IN INTERNAL MEDICINE: FAR TO GO BEFORE WE SLEEP. T.V. Parran¹, W. Mackay¹; ¹Case Western Reserve University, Cleveland, OH

PURPOSE: Most prescribing of controlled drugs in the American Healthcare System is done by Internists. When practicing physicians are asked about the areas of medical practice where they feel the least prepared by their formal medical education and the most uncomfortable in clinical practice, they frequently cite scenarios around controlled drug prescribing. These include acute / chronic / and malignant pain management, management of anxiety vs. depression, management of insomnia, identification and management of addictive disorders, and opioid and benzodiazepine pharmacology. Pilot data exist to suggest that the prescribing of controlled drugs in primary care clinic settings bear little if any resemblance to current practice recommendations. As part of a controlled clinical trial of alcohol screening and interventions in a primary care population, we performed a chart review in the Resident and Attending Primary Care Medical Clinic of a large urban teaching hospital. In an effort to assess controlled drug prescribing practices, part of the chart review included assessing whether there was addiction screening information in the patient record when controlled drugs were prescribed.

METHODS: A total of eight hundred charts were randomly chosen from the three Internal Medicine Clinic resident firms as well as from the Attending clinic. Charts were categorized based upon 1) whether controlled drugs were prescribed at any time in the 12 months prior to the audit, 2) the type of controlled drug prescribed, 3) whether there was any evidence of an alcohol or drug abuse history documented in the chart, and 4) the result of that history if documented. The charts were abstracted by trained research assistants using a data template and supervised by the Project Director and the Project Administrator.

RESULTS: A total of 768 charts were completely reviewed, of which 135 indicated some prescribing of controlled drugs (17.6%). The percentage of patients receiving controlled drug prescriptions was not different based upon the gender of the patient (18% of men and 17% of women), and the type of controlled drugs showed no gender differences. 45% of women and 35% of men had no evidence of a substance use history having been taken in their charts, and 3% of women and 13% of men had charts that documented hazardous or harmful use of alcohol or drugs. Of the 70 male patients prescribed controlled drugs, they represented 17.6% of those with no substance use assessment, 18.8% of those with a low risk substance use assessment, and 17% of those with active substance use problems documented in their charts. For the 65 women patients prescribed controlled drugs, they represented 18.5% of those with no substance use assessment, 14.7% of those assessed as low risk, and 33.3% of those with chart documentation of active addictive problems. There were no differences between the controlled drug prescribing patterns of attendings versus residents.

CONCLUSION: The rate of prescribing controlled drugs and the appropriateness of prescribing decisions is becoming more and more closely monitored, but by regulatory agencies as well as advocacy groups. Little is known about the controlled drug prescribing practices of practicing physicians, but they self-report it as an area of great unease. Although guidelines are not well established in this area of practice, there is general agreement that ambulatory controlled drug prescribing is relatively contraindicated in the face of active or past substance abuse on the part of the patient. This pilot data contains concerning information to suggest that a lack of any documentation of substance use history is common place when prescribing controlled drugs in primary care. Worse, the percentage of patients receiving controlled drug prescriptions who have a documented active substance abuse problem, is as high as those who receive a prescription without any history taken, and at least as high as those who both have an appropriate history AND whose history indicates that they are a low risk population. Ominously, attending prescribing practice seems identical to house staff practice patterns.

PREDICTORS OF RESPONSE TO INFLIXIMAB IN PATIENTS WITH CROHN'S DISEASE. M.A. Parsi¹, J.P. Achkar¹, S. Richardson², J. Katz², B. Lashner¹, A. Brzezinski¹; ¹Cleveland Clinic Foundation, Cleveland, OH; ²Case Western Reserve University School of Medicine, Cleveland, OH

PURPOSE: Infliximab is an antibody to tumor necrosis factor-alpha. This drug has in several studies been shown to be effective in treating patients with inflammatory and fistulous Crohn's disease (CD) refractory to conventional therapy. However, treatment with infliximab is expensive and has associated toxicities. In addition, the mean duration of response to

infliximab is approximately 12 weeks which necessitates re-dosing at regular intervals. Therefore identifying predictors of rate of response to infliximab could be of great benefit in selection of patients for this treatment. Furthermore, identifying factors that are associated with a prolonged duration of response could allow for modification of these factors to achieve a longer duration of response in treated patients. The purpose of this study was to identify factors predictive of rate and duration of response to infliximab in patients with CD. **METHODS:** 100 patients with refractory CD (59 with inflammatory CD and 41 with fistulous CD) and at least 3 month of follow-up following infliximab infusion were evaluated (mean follow up = 9 months; range = 3–17 months). Clinical-response, duration of response, smoking history, gender, race, duration of disease, concurrent immunosuppressive or steroid use, age at diagnosis and age at infusion were analyzed.

RESULTS: Rate of response: 67% of patients responded to infliximab. Rates of response were significantly higher in patients with fistulous disease (80%) compared to those with inflammatory disease (58%); $p < 0.02$. 77% of non-smokers responded compared to 49% of smokers ($p < 0.004$). The favorable response in non-smokers applied mostly to patients with inflammatory disease ($p < 0.001$). Concurrent use of immunosuppressive medication was also associated with higher rates of response in inflammatory disease ($p < 0.007$). Duration of response: There was a statistically significant longer duration of response among non-smokers compared to smokers both for inflammatory and fistulous disease. Among patients with inflammatory Crohn's disease, 87% on concurrent immunosuppressive medication had a duration of response longer than 2 months compared to only 45% not on any immunosuppressants ($p < 0.04$). Use of immunosuppressives was not associated with longer duration of response in patients with fistulous disease.

CONCLUSION: Rate of Response: Non-smoking and concurrent immunosuppressive use are associated with higher rates of response to infliximab in patients with inflammatory Crohn's disease. Duration of response: Smoking decreases duration of response to infliximab in both types of Crohn's disease. Concurrent immunosuppressive use increases duration of response to infliximab in patients with inflammatory disease.

BODY IMAGE ASSESSMENT: COMPARISON OF FIGURE RATING SCALES AMONG URBAN BLACK WOMEN. M.R. Patt¹, L.R. Yane¹, A. Lane², C. Finney¹, D.M. Becker¹; ¹Johns Hopkins University, Baltimore, Maryland; ²Ohio State University, Columbus, Ohio

PURPOSE: The social norm for a larger body image may support the propensity for overweight and obesity, which has a prevalence of 66%, in black women. The goal of this study was to determine a psychometrically stable and socially acceptable Figure Rating Scale (FRS) for assessing body image in black women.

METHODS: The study sample ($n = 50$) was selected sequentially from black women (mean age 52.3+/- 10 years) who were being screened for a larger randomized trial of nutrition and exercise education in urban black churches (Project Joy). Three standard published FRS were compared with a new culturally specific scale, the Reese FRS, developed from digitized photographs of black women and modified according to recommendations from focus groups. All four FRS consisted of nine ordered body images increasing in size from very thin to obese. Distributions of respondent's selections on all scales were examined relative to anthropometric measures, including body mass index, BMI (weight in kg/m²) and categories of obesity and overweight (National Expert Panel on Overweight and Obesity Guidelines). Cultural identity, based on questions adopted from an African American Acculturation scale, and its relationship to FRS preference was also assessed.

RESULTS: Body weight distribution for US black women over 40 years matched that found in NHANES III. All four FRS performed similarly and correlated significantly with BMI, $r = -0.70$ to -0.75 , $p < 0.0001$. The percentage of obese women who identified with one of the three largest images in the FRS was 22% on three scales and 38% on one; this is markedly below 56% of women who were obese based on the national guidelines. Overall, 44% of the women preferred the new Reese FRS; among those women with cultural identity scores in the upper quintile, 72% preferred the new FRS.

CONCLUSION: Overall, for all FRS, there is considerable overlap among images selected for all weight categories. There is a strong preference for the culturally specific Reese scale, especially among women with strong cultural identity scores. This new FRS appears to be more socially acceptable. The failure of a large percentage of obese black women to identify themselves as obese needs further investigation.

IMPLEMENTING PROACTIVE CARE MANAGEMENT FOR DEPRESSION. M.L. Pearson¹, J. Unutzer², K. Minnium¹, L.V. Rubenstein³; ¹RAND, Santa Monica, CA; ²University of California, Los Angeles, Los Angeles, California; ³Sepulveda VA, Sepulveda, CA

PURPOSE: Evidence supports the effectiveness of collaborative care for depression in primary care, incorporating clinician education, proactive care management, and collaboration with mental health specialists. Little is known, however, about the influence of proactive care management on processes and outcomes of care for depression. The purpose of these analyses is to assess adherence to specific aspects of the nurse care management protocol, and evaluate the effects of adherence on quality of care and intermediate outcomes.

METHODS: We used a randomized encouragement design to study the effects of assisting 6 managed care organizations to implement collaborative care for depression in 30 experimental and 16 control practices. We identified a total of 920 depressed patients and referred them to their clinical practice site's collaborative care program. Access to the program was through a nurse care manager. Care managers recorded all patient contacts on pre-structured forms. Patients completed self-administered surveys at baseline and every six months. Our evaluation uses bi-variable and multi-variable regression analyses.

RESULTS: Overall, 73% of patients completed initial assessment visits. There was considerable variation among sites in adherence to the care management protocols, e.g., assessment of patient treatment preferences at the initial visit ranged from 18% to 100% at different sites (73% of patients assessed overall).

In bivariate regression analyses, greater performance of care management processes was strongly associated with higher quality of depression care, including the degree to which patients

received and adhered to appropriate anti-depressant and psychotherapy regimens. Performance also related to improved levels of patient knowledge, probable depression, and active coping. When nursing assessment, proactive follow-up, and other self-management support processes were included together in regression analyses, the number of follow-up contacts had the greatest effect on patient outcomes ($p < 0.001$).

CONCLUSION: Practices assisted in implementing collaborative care models for depression achieved varying levels of adherence to the care management design. Of all the processes studied, the amount of proactive follow-up had the strongest effect on care and outcomes.

PRIMARY CARE PHYSICIAN DEPARTURE: EFFECTS ON HEALTH CARE QUALITY. A.G. Pereira¹, S.D. Pearson¹; ¹Harvard Medical School/Harvard Pilgrim Health Care, Boston, MA

PURPOSE: Discontinuity of care is an important health policy issue, but few data exist to assess its effects on health care quality or outcomes. This study evaluated measures of health care quality in patients whose primary care providers (PCPs) left their medical practice.

METHODS: This study was performed in a large, multi-site multispecialty group practice associated with a single insurer. We used a controlled pre-post design to compare measures of health care quality received by patients whose PCPs left the practice (LEAVEmDs) with the measures of health care quality in patients whose PCPs did not leave the practice (STAYmDs). For all patients, we compared rates of preventive care, and in patients with hypertension, we compared blood pressure control. The practice had a standardized process for reassigning patients of departing PCPs. Patients were eligible for analysis if their insurance and care within the practice was uninterrupted over a four-year study period. LEAVEmDs were matched to eligible STAYmDs by age, sex and practice site.

RESULTS: During the study period, nine PCPs, caring for 3,931 patients, voluntarily left the group practice. These LEAVEmDs were matched to 16 STAYmDs, caring for 8,009 patients. Mean age of LEAVEmD patients and STAYmD patients was the same: 46 (± 13); 76% of the LEAVEmD patients were women, as compared to 80% of the STAYmD patients ($p < 0.01$). During the two baseline years prior to LEAVEmD departures, there were no differences in quality of care measures between the two groups of patients. In the two years following PCP departure there was no difference in rates of Pap smears among women aged 25–65 (88% vs. 88%, $p = 0.85$), or in mammography in women 50–65 (90% vs. 93%, $p = 0.09$). Rates of fecal occult blood testing for colon cancer in all patients aged 50–65 were lower among LEAVEmD patients (51% vs. 56%, $p = 0.02$). There was no difference in the proportion of patients with hypertension who experienced a rise of $\geq 10\%$ in mean systolic and/or diastolic blood pressure (6.0% vs. 7.0%, $p = 0.53$).

CONCLUSION: Overall, PCP departure from this group practice was not associated with substantial decreases in rates of several important preventive screening measures for their patients who continued to receive care within the practice. Among patients with hypertension, we found no difference in rates of blood pressure control. Further research is needed to address two questions: first, whether these findings are consistent with the experiences of patients in a variety of practice settings, and second, whether PCP discontinuity may adversely affect subgroups of patients with other chronic conditions.

PERFORMANCE OF HELICAL COMPUTED TOMOGRAPHY IN UNSELECTED OUTPATIENTS WITH SUSPECTED PULMONARY EMBOLISM. A.F. Perrier¹, N. Howarth², D. Didier², P. Unger², P. De Moerloose³, D. Slosman², A. Junod², H. Bounameaux²; ¹University of Geneva, Geneva, GE; ²Geneva University Hospital, Geneva, GE; ³Geneva, GE

PURPOSE: Helical computed tomography (CT) is commonly used to diagnose pulmonary embolism (PE), although its operating characteristics have been insufficiently evaluated. Therefore, we aimed to assess the performance of helical CT.

METHODS: Two-hundred and ninety-nine consecutive patients admitted to the emergency ward for clinically suspected PE, with a D-dimer level above 500 $\mu\text{g/L}$ (ELISA assay) were included. The diagnosis of PE was established by a validated algorithm including clinical assessment, lower limb compression ultrasonography, lung scan and pulmonary angiography. The CT scans were read by radiologists blind to all clinical data 3 months after image acquisition.

RESULTS: Pulmonary embolism was present in 118 patients (39%) in the study population. In 12 patients (4%), the CT scan was inconclusive, of whom two had a PE. Among patients with a conclusive CT, sensitivity of CT was 70% (95%CI 62 to 78), specificity was 91% (95%CI 86 to 95), positive predictive value was 84% (95%CI 76 to 91) and negative predictive value was 82% (95%CI 76 to 87). Sensitivity of a strategy adding lower limb ultrasound to helical CT increased to 79% (95%CI 71 to 86), and associating a lung scan in patients with normal ultrasound, helical CT being performed in patients with a nondiagnostic lung scan, would have a sensitivity of 95% (95%CI 89 to 98) and a specificity of 93% (95%CI 88 to 96). Interobserver agreement was high (kappa coefficients 0.82 to 0.90).

CONCLUSION: Helical CT should not be used as a single test in suspected PE, but it could replace angiography in combined strategies including ultrasound and lung scan.

DIAGNOSTIC STRATEGIES FOR SUSPECTED PULMONARY EMBOLISM INCLUDING HELICAL CT SCAN: A COST-EFFECTIVENESS ANALYSIS. A.F. Perrier¹, F. Sarasin¹, N. Howarth¹, H. Bounameaux¹; ¹University of Geneva, Geneva, Switzerland

PURPOSE: Helical computed tomography (CT) is increasingly used in the diagnosis of suspected pulmonary embolism, alone or in combination with other diagnostic tests, such as lung scan, lower limb ultrasound (US) or plasma D-dimer measurement (DD). Therefore, we assessed the cost-effectiveness of including helical CT in the diagnosis of suspected pulmonary embolism.

METHODS: We performed a formal cost-effectiveness analysis by a decision model. Probabilities for test characteristics and clinical outcomes were obtained from the literature. Cost estimates were derived from our hospital's database. We compared 1) CT as a single test, and 2) inclusion of CT in a sequential strategy resting on clinical assessment, DD, US, lung scan and pulmonary angiography, either as a substitute for angiography or for lung scan. These

strategies were compared with a reference strategy in which all patients with a nondiagnostic lung scan underwent an angiogram. Outcome measures were costs per patient, 3-month quality-adjusted survival, and incremental costs per quality-adjusted life years (QALYs) gained. **RESULTS:** Helical CT as a single test is associated with a 1% higher mortality than the reference strategy, and higher costs (\$3,439 per QALY compared to \$3,202 per QALY for the reference strategy). Replacing angiography by CT in the sequential strategy, was the most cost-effective strategy (\$2,447 per QALY gained). Replacing lung scan by CT was also cost-effective (\$2,700 per QALY gained) provided an angiogram was performed in patients with a high clinical probability and a negative CT. Omitting DD measurement from the strategies increased costs but did not change effectiveness. The results were stable over the entire range of values tested in sensitivity analysis.

CONCLUSION: Helical CT as a single test in suspected pulmonary embolism is not cost-effective. However, CT may be cost-effective when included in combined diagnostic strategies.

TRUST OF PHYSICIANS AND SATISFACTION AMONG MINORITY PATIENTS WITH ISCHEMIC HEART DISEASE. L.A. Petersen¹, T.C. Collins², N.R. Kressin³; ¹VAMC, Houston, Texas; ²Baylor College of Medicine; ³Boston University, Bedford, Massachusetts

PURPOSE: Studies show that African-American (AAs) are less satisfied with their health care. Since AAs have historically experienced bias within the health care system, and recent work suggests bias on the part of physicians toward minority patients, the goal of this study was to assess racial differences in patients' trust and satisfaction with their physicians regarding decision-making for coronary disease treatment.

METHODS: We prospectively enrolled 431 males with nuclear imaging studies graded as positive for cardiac ischemia (71.6% self-reported white, 28.4% self-reported AA) at 5 VA hospitals (Houston, Atlanta, St. Louis, Pittsburgh, Durham). We collected survey data using the published Trust in Physician scale, Seattle Angina Questionnaire, and new items developed from focus groups.

RESULTS: There was no difference in mean age of the groups. AAs were less likely than whites to be "completely satisfied" that everything possible is being done for their cardiac symptoms (44.4% vs 51.4%, respectively; $P = 0.05$). There was a trend toward more AAs reporting dissatisfaction with explanations provided by the physician (14.8% vs 8.0%; $P = 0.12$). Though all patients had positive imaging studies, equal percentages (33.4% vs 33.8%; $P = 0.98$) thought they "did not have heart disease". There were no racial differences in the percentage reporting that their heart condition was "not at all serious" (3.2% vs 3.1%; $P = 0.97$). There were no racial differences in the percentage of patients who preferred to leave all treatment decisions to their doctors, who agreed with statements that doctors knew the patient's medical and personal situations, or who reported that doctors were respectful or were concerned about them. Despite racial differences in overall satisfaction with medical care, there were no differences in the percentage who agreed that they "trust the doctor's judgments about medical care" (85.3% vs 88.7%; $P = 0.17$) or who stated that the doctor was "well qualified to manage medical problems" (83.0% vs 87.5%; $P = 0.79$).

CONCLUSION: Though we found racial differences in satisfaction with care and with physician-patient communication, there were no racial differences in patients' trust of their physicians. Further work should assess whether there are racial differences in the construct of trust not captured with current measurement scales as well as develop strategies to improve physician-patient communication and satisfaction with care.

FIRST EXPERIENCE WITH THE ORAL DIRECT THROMBIN INHIBITOR H 376/95 COMPARED WITH WARFARIN IN PATIENTS WITH NON-VALVULAR ATRIAL FIBRILLATION (NVAF). P. Petersen¹; ¹University Hospital of Copenhagen, Hvidovre, Copenhagen

PURPOSE: H 376/95 is a novel, oral direct thrombin inhibitor that shows predictable pharmacokinetics and has not shown clinically significant food or drug interactions. Anticoagulants such as warfarin are used to lower the risk of stroke in patients with non-valvular atrial fibrillation (NVAF). In the SPORTIF II study, the tolerability and safety of three doses of H 376/95 were compared with warfarin in NVAF patients.

METHODS: This was a randomized, parallel-group study of NVAF patients who had at least one additional risk factor for stroke. The primary outcome was the number of adverse events, e.g., bleeding and thromboembolic events. The duration of treatment was 12 weeks, during which three groups received H 376/95 (n=187) 20, 40 or 60 mg bid, given double blind. In a fourth group, warfarin (n=67) was managed and monitored according to normal routines, aiming for an INR of 2.0–3.0.

RESULTS: A total of 257 patients were randomized to treatment and 254 patients received study drug. The median age was 70 years (range: 39–95). In addition to NVAF, all patients had one additional risk factor for stroke, 75% had two additional risk factors, and 42% had three or more additional risk factors. The number of minor bleeds was comparable and low in all four groups. There was only one major bleed (vaginal) observed in a patient receiving warfarin and none in the H 376/95 groups. Out of 67 patients in the warfarin group, two patients had transient ischaemic attacks. Of the 187 patients in the H 376/95 groups, one patient had a transient ischaemic attack and one patient had an ischaemic stroke; both patients were in the H 376/95 60-mg bid group.

CONCLUSION: Fixed doses of H 376/95 up to 60 mg bid were well tolerated, without the need for dosage adjustment or coagulation monitoring, during a 3-month treatment period in NVAF patients with medium-to-high risk for stroke and systemic embolism.

INADEQUATE FOLLOW-UP FOR ABNORMAL PAP SMEARS IN AN URBAN POPULATION. N.B. Petersen¹, J.A. Anapolsky¹, S. Reich¹, S. Franco¹, K.M. Freund¹; ¹Boston University, Boston, MA

PURPOSE: African-American women have a higher mortality rate from cervical cancer than Caucasian women despite similar rates of screening. One possible explanation for this difference

in mortality is a discrepancy in the rate of follow-up of abnormal Pap smears among African-American women. The objective of this study is to determine the demographic factors associated with a delay to follow-up of abnormal pap smears.

METHODS: Eligible subjects were a sample of women aged 18 years and older with an abnormal cervical cytology report between February 1999–April 2000 screened at an academic medical center or one of three neighborhood health centers. We obtained information on subjects from a registration database and a pathology database. We excluded subjects who did not have a race category specified (17%). We defined adequate follow-up conservatively, as cytology or pathology within 4 months for dysplasia (low or high grade squamous intraepithelial lesion or carcinoma in-situ) and within 7 months for atypia (atypical squamous or glandular cells of unknown significance). We analyzed differences in follow-up rates by race, age, insurance status, source of care (academic medical center or neighborhood health center), and type of pap abnormality (dysplasia or atypia).

RESULTS: Of the 345 subjects, 57% were African American, 16% were Caucasian, and 26% were from other racial/ethnic groups. 49% of the subjects were under the age of 30. 62% were screened at the medical center and 38% at the neighborhood health centers. Only 18% of the subjects had private insurance. Overall, only 53% of the subjects had follow-up for an abnormal pap smear either by additional cytology or pathology within 4 months for dysplasia (50%) or 7 months for atypia (56%). 55% of African-American women, 55% of Caucasian women, and 48% of women from other racial/ethnic groups received follow-up (not significant). The follow-up rate of 45% for subjects less than 30 was significantly lower than that of subjects aged 30 or older (61%, p -value 0.003). No differences in follow-up rates were seen according to the subject's race, insurance status, source of care, or type of pap abnormality.

CONCLUSION: In a low income, predominantly minority population, the rate of follow-up for abnormal pap smears was low, only 53%. This may explain the higher mortality rate seen in African-American women. Younger women were less likely to receive adequate follow-up. Race, source of referral, insurance status, and type of pap abnormality were not associated with follow-up rates in this cohort.

RACIAL DIFFERENCES IN PATIENTS ATTITUDES ABOUT PHYSICIANS RELIGIOUS BELIEFS. N. Phifer¹, C.D. Maclean², L.J. Schultz³, B.E. Susi⁴, F. Et Al⁵; ¹Moses Cone Memorial Hospital, Greensboro, NC; ²University of Vermont, Burlington, VT; ³Emory University, Norcross, GA; ⁴Carolinas Healthcare System, Charlotte, NC; ⁵Moses Cone, Greensboro, NC

PURPOSE: The cultural competence of a physician likely plays a role in medical decision making when caring for African American patients. It is also likely that physicians may not comprehend the importance of religion in the medical care of African American patients. We wanted to see how strongly African American patients felt, compared to other patients, about the religious beliefs and behaviors of their physicians.

METHODS: We surveyed patients at 7 medical centers in 4 states (NC,GA,FL,VT). A trained research assistant administered a 112-item survey that included questions pertaining to demographics, health status, functional status (SF-36), and spirituality assessment (SWB scale). Patients were also asked to express their preferences regarding physician involvement in their religious/spiritual life. Descriptive statistics were tabulated. Associations between patient characteristics and patients' preferences for specific physician spiritual behaviors were assessed using bivariate and multivariate techniques.

RESULTS: Two hundred-ninety nine patients were surveyed. Patients' ages ranged from 19 to 86 years. Fifty six percent were male, 46% were African American, 51% white, and 63% belonged to Protestant denominations. Fifty three percent had annual incomes of \$20,000 or less. Fifty nine percent of African American patients (AA) felt it was important that their physician have strong spiritual beliefs compared with 40% of whites (W) ($p < .001$); 45% of AA vs 32% of W would want a different physician if their doctor did not believe in God or a higher power ($p < .001$). Forty five percent of AA felt their health would improve if their physician prayed for them vs 32% of W ($p = .03$). Fifteen percent of AA were willing to give up time spent on medical problems to discuss spiritual issues with their physicians while only 4% of W agreed with this viewpoint ($p = .001$). In bivariate analysis, significant associations with patient desires for more physician religious involvement in medical encounters ($p < .05$) were AA race, older age, education less than college, and higher score on the SWB scale. In logistic regression, only AA race remained significantly associated with willingness to give up time spent on medical care in exchange for spiritual discussion (OR 3.8, 95%CI 1.3–11). AA race (OR 1.9, 95%CI 1.1–3.4) and higher SWB score (OR 1.8, 95%CI 1.3–2.4) were associated with patients' desire for physicians to have strong religious beliefs. Patients who would change physicians if theirs did not believe in God/higher power were more likely to be AA (OR 2.1, 95%CI 1.1–3.9), have higher SWB score (OR 1.5, 95%CI 1.0–2.1) and education less than college (OR 1.9, 95%CI 1.1–3.4). AA race (OR 2.0, 95%CI 1.1–3.5) and being married (OR 2.1, 95%CI 1.2–3.7) were associated with the belief that physician prayer would improve patients' health. Age, sex and physical function had no significant associations.

CONCLUSION: Physician religious beliefs and behaviors are more important to AA patients than to white patients. Physicians should be aware that religious discussions might be important for effective communication with AA patients in making medical decisions.

PATIENT PREFERENCES FOR PHYSICIAN INVOLVEMENT IN THEIR RELIGIOUS/SPIRITUAL LIFE: WOULD PHYSICIANS RESPOND TO THESE PREFERENCES? N. Phifer¹, C.D. Maclean², L.J. Schultz³, B.E. Susi⁴, F. Et Al⁵; ¹Moses Cone Memorial Hospital, Greensboro, NC; ²University of Vermont, Burlington, VT; ³Emory University, Norcross, GA; ⁴Carolinas Healthcare System, Charlotte, NC; ⁵Moses Cone, Greensboro, NC

PURPOSE: It has previously been reported that many patients would like their physicians to engage in prayer with them as part of their medical care. We studied the relationship of care settings to this preference and physicians' willingness to respond to patients' religious/spiritual needs.

METHODS: We surveyed patients and physicians at 7 medical centers in 4 states (NC,GA,FL,VT). For the patient questionnaire, a trained research assistant administered a 112-item survey that included questions pertaining to demographics, health status, functional status (SF-36), and spirituality assessment (SWB scale). Patients were also asked to express their preferences regarding physician involvement in their religious/spiritual life. The physician questionnaire was administered to residents and primary care practitioners through the mail. Descriptive statistics were tabulated.

RESULTS: Two hundred ninety-nine patients and 444 physicians were surveyed. Patients' ages ranged from 19 to 86 years. Fifty six percent were male, 46% were African-American, 51% white, and 63% belonged to Protestant denominations. Fifty three percent had annual incomes of \$20,000 or less. Of the physicians, 45% were practicing physicians and 55% were residents/fellows. Seventy three percent were internists and 21% family practitioners. Forty five percent of physicians were of Protestant denominations. Twenty percent of patients wanted their physicians to pray with them during a routine office visit, but only 5% of physicians wanted to pray with their patients under the same circumstances. However, 55% of physicians said they would pray with their patients during a routine office visit if the patients requested. If patients were hospitalized and near death, 52% wanted their physicians to pray with them compared to 26% of physicians who felt this appropriate. This number increases to 77% of physicians if a patient requests such prayer behavior.

CONCLUSION: Patients' preferences for prayer in medical settings increases in settings associated with more severe illness. Physicians' willingness to engage in prayer with patients rises in a parallel manner but does not match the level of patients' desires unless prayer by a patient is requested. These data suggest that patients should communicate their feelings about prayer to their doctor if they truly desire their physician's participation.

SCREENING FOR ALCOHOL PROBLEMS IN HIV-INFECTED PRIMARY CARE PATIENTS. S.J. Phillips¹, K.A. Freedberg², E.T. Traphagen¹, N.J. Horton³, J.H. Samet¹; ¹Boston University, Boston, MA; ²Massachusetts General Hospital, Boston, MA; ³Harvard University, Boston, MA

PURPOSE: Alcohol is known to affect HIV risk behaviors and adherence to antiretroviral medications, but its prevalence among HIV-infected persons and effective screening approaches are less well described. Our objective was to determine the prevalence of a history of alcohol problems in patients entering primary care for HIV infection and to assess the positive predictive value of the CAGE questionnaire for alcohol abuse or dependence.

METHODS: Between 7/97 and 10/00, HIV-infected patients presenting to a multidisciplinary clinic for evaluation and linkage to primary care were assessed for alcohol problems using the standard threshold of 2 or more positive responses to the CAGE questionnaire. The predictive value of the CAGE was evaluated by administering the CIDI-SAM, an interview for DSM-IV lifetime diagnoses of alcohol abuse and dependence, to a sample of those with a positive CAGE screening test.

RESULTS: Among the 715 patients who spoke English or Spanish, mean age was 38.6 years; 70% were male; 50% were Black, 25% Latino, 23% White, 2% other. Primary HIV risk factors were injection drug use 47%, heterosexual 37%, and men who have sex with men 16%. Most (673/715, 94%) were evaluated for alcohol problems: 41% (279/673) had a positive CAGE screening test (2+); 7% (47/673) had 0 or 1+ CAGE scores, but were deemed to have had an alcohol problem by physician clinical judgment. Of the 326 patients with reports of alcohol problems, 116 underwent a diagnostic interview: 80% (93/116) met DSM-IV criteria for diagnosis of lifetime alcohol dependence, 14% (16/116) for lifetime alcohol abuse, and 6% (7/116) did not meet criteria for either diagnosis. In the group of patients with 2 or more positive responses to the CAGE questionnaire ($n = 102$), 95% (97/102) met DSM-IV criteria for alcohol dependence or abuse. In the group not identified by the CAGE but by clinical judgment ($n = 14$), 86% (12/14) met the same criteria.

CONCLUSION: Nearly half of patients presenting for HIV care in an urban clinic had a history of past or present alcohol problems, and the positive predictive value of the CAGE questionnaire was 95% for lifetime diagnoses of alcohol abuse or dependence. Given the prevalence of alcohol problems in this population, the potential impact of such problems on treatment, and the high predictive value of this simple screening tool, alcohol screening should be routinely implemented for all patients initiating HIV medical care.

THE IMPACT OF LITERACY ON THE EFFECTIVENESS OF A COMPREHENSIVE DIABETES MANAGEMENT PROGRAM. M. Pignone¹, R. Rothman¹, R. Malone¹, B. Bryant¹, B. Dennis¹, D. Dewalt¹; ¹Departments of Medicine and Pharmacy, UNC-Chapel Hill, Chapel Hill, NC

PURPOSE: Previous research suggests that diabetic patients with low literacy have lower disease-specific knowledge. We sought to examine the prevalence of low literacy among high-risk diabetic patients and determine if literacy level affects the change in knowledge or average hemoglobin A1C (HBAIC) after a comprehensive intervention to improve diabetes care.

METHODS: Using a well-validated literacy instrument, the Rapid Estimate of Adult Literacy in Medicine (REALM), we screened 91 diabetic patients who were referred for enrollment in a comprehensive, pharmacist-based diabetes care program in a university internal medicine clinic. The intervention involved direct verbal teaching, simple diagrams, and phone-based follow-up to help patients better manage their blood sugar levels. We obtained diabetes knowledge scores and HBAIC values at entry and after 3–4 month follow-up. We then analyzed the effect of literacy level on change in knowledge and HBAIC from baseline to follow-up using non-parametric Wilcoxon rank sum tests.

RESULTS: Mean age of subjects was 58 (range 27–87); 63% were female, 65% were African-American, and 38% completed high school. Average duration of disease was 9 years (range 0–35). Low literacy, defined as a reading level below 9th grade (REALM < 61) was present in 78% of patients (32% grade 3 or below, 22% grade 4–6, 24% grade 7 or 8). The effect of literacy on the response to the intervention is shown in the Table.

	Literacy Level	Base-line	4 mo. F/U	Pre-Post Difference (95%CI)	Net Difference (95%)
Mean Knowledge	Higher*	43%	73%	+30% (21%, +39%)	+8% (-, +17%)
	Lower**	29%	51%	+22% (18%+26%)	
Mean HbA1C	Higher*	10.5%	8.4%	-2.1% (-3.5%, -0.7%)	-1.1% (-2.5%, +0.4%)
	Lower**	9.6%	8.5%	-1.1% (-1.6%, -0.5%)	

*Higher literacy is ≥ 9 th level (n=19); **Lower literacy is <9th grade (n=72)

CONCLUSION: Low literacy is extremely common among high-risk adult patients with diabetes. Our intervention improved knowledge and outcomes for patients with high and low literacy, but the effect was greater for patients with higher literacy. To prevent disparities in health outcomes, specific interventions to assist low literacy patients should be developed.

UNDIAGNOSED PRIMARY AND CHRONIC HIV INFECTION: PREVALENCE AND PREDICTORS IN AN URGENT CARE SETTING. J.M. Pincus¹, J. Vidrih¹, S.S. Crosby¹, E. Losina², E.R. King¹, K.A. Freedberg³; ¹Boston Medical Center, Boston, MA; ²Boston University School of Public Health, Boston, MA; ³Massachusetts General Hospital, Boston, MA

PURPOSE: To determine the prevalence and predictors of primary and chronic HIV infection in patients presenting with viral symptoms to an urban urgent care clinic (UCC).

METHODS: We prospectively evaluated patients with any viral symptoms (temp >100°F, headache, fatigue, sore throat, myalgias, night sweats, rash, oral ulcers, or diarrhea) and any risk for HIV infection (sex or drug use) in the prior two months. Primary infection was defined as ELISA negative and RNA positive; chronic infection was defined as ELISA/Western Blot positive and RNA positive.

RESULTS: 1010 patients who presented to the UCC with viral symptoms were screened. Of 413 interested and eligible patients, 375 (91%) enrolled and had antibody and RNA testing. HIV infection was diagnosed in 9 (2.4%, 95%CI: 1.1%, 4.5%). Primary infection was diagnosed in 4 (1.1%, 95%CI: 0.3%, 2.7%) and chronic infection in 5 (1.3%, 95%CI: 0.4%, 3.0%). Fever was associated with a higher prevalence of primary HIV infection (7.0%, OR = 24.8, p < 0.0001). Symptoms associated with a higher prevalence of chronic HIV were oral ulcers (5.7%, OR = 6.8, p = 0.018) and diarrhea (4.2%, OR = 12.3, p = 0.0047). The four patients with chronic HIV and available CD4 counts all had very advanced disease (CD4 < 50/uL).

CONCLUSION: Undiagnosed primary and advanced chronic HIV infection are prevalent in patients presenting with viral illness to an urban urgent care clinic. Testing for both should be recommended for these patients.

USING CORONARY ARTERY CALCIUM TO GUIDE CHOLESTEROL-LOWERING THERAPY: A COST-EFFECTIVENESS ANALYSIS. M.J. Pletcher¹, J.A. Tice¹; ¹University of California, San Francisco, San Francisco, CA

PURPOSE: Coronary artery calcium (CAC), a marker of coronary artery disease (CAD), may be detected and quantified using electron beam computed tomography (EBCT). Proposals have been made to use the CAC score to guide cholesterol-lowering therapy, but the costs and benefits of this strategy have not been quantified. We performed a cost-effectiveness analysis to determine whether measurement of CAC with EBCT is a cost-effective method of targeting patients for cholesterol-lowering drug therapy.

METHODS: A Markov model was constructed to estimate the incremental cost-effectiveness, over 10 years, of five different strategies: "Treat none," "Treat all," or test with EBCT and treat if the CAC score is greater than 0 ("Test and treat>0"), 100 ("Test and treat>100"), or 400 ("Test and treat>400"). Overall rates of CAD events were based on Framingham risk equations. Distributions of CAC scores given risk factors were calculated from the literature. Summary relative risks for CAD events in each of four CAC score strata (0, 1–100, 101–400, >400) were calculated with a random-effects model using data obtained from meta-analysis of articles identified through a systematic search of the literature. Therapy with HMG-CoA Reductase Inhibitors (STATINS) was assumed to reduce CAD event rate by 31%. Cost data were derived from literature review. Two base-case analyses were performed. Case 1 was a 50 year old man with low density lipoprotein cholesterol (LDL) of 175, high-density lipoprotein cholesterol level (HDL) of 45, and no other CAD risk factors. In Case 1, STATINS would not be recommended according to the National Cholesterol Education Program II (NCEP II) guidelines. Case 2 was the same 50 year old man with high cholesterol, but also Stage I hypertension, for whom STATINS would be recommended by NCEP II.

RESULTS: Meta-analysis: Four studies met inclusion criteria. Relative risks (with 95% confidence intervals) for CAD events were 1.0 (reference), 3.9 (1.7–8.8), 7.9 (2.5–24.4), and 9.9 (2.8–34.6), given CAC scores of 0, 1–100, 101–400, or >400, respectively. Cost-effectiveness analysis: In each case, the "Test and treat" strategies were more cost-effective than the "Treat all" strategy. This result held true while the cost of EBCT was less than \$2400 (base case \$400/scan), and the cost of STATINS was more than \$150/year (base case \$849.84/year). It was also insensitive to reasonable variation of other parameters, including the cost of CAD events, overall risk of events, relative risk given CAC score stratum, relative risk reduction with STATINS, and distribution of CAC scores. The "Test and treat >400" strategy failed by extended dominance in both cases. In Case 1, the incremental cost-effectiveness ratios (C/E's) of all "Test and treat" strategies, though lower than "Treat all," were greater than 300,000 \$/life-year compared with "Treat none" (the NCEP II recommendation). Given a

reasonable C/E threshold of 50,000 \$/life-year, this result was also insensitive to variation of parameters. In Case 2, the C/E of "Treat all" (the NCEP II recommendation), in comparison to the "Test and treat >0" strategy, was 1,600,000 \$/life-year. The per-year cost of STATINS would have to be less than \$160/year, or the cost of EBCT greater than \$2400, in order to make the NCEP II standard ("Treat all") cost-effective or dominant in comparison to the optimal "Test and treat" strategy. Note that the NCEP II standard ("Treat all") in comparison with "Treat none" also did not meet reasonable C/E standards, consistent with a recent cost-effectiveness analysis of STATINS for primary prevention of CAD (Ann Intern Med 2000;132:769–779).

CONCLUSION: Screening of patients with EBCT who would not otherwise qualify for cholesterol-lowering therapy, based on NCEP II guidelines, does not meet reasonable cost-effectiveness guidelines. On the other hand, using EBCT to identify patients at very low risk for CAD events among patients who would otherwise be treated with STATINS may be a cost-effective strategy in selected patient populations.

FREE MAMMOGRAMS DO NOT IMPROVE UTILIZATION OF SCREENING MAMMOGRAPHY. S. Rusch¹, J. Poasi², S. Goel³, J. Aldag⁴; ¹OSF St Francis Hospital and Medical Center, UICOM-P, Peoria, IL; ²OSF St Francis Hospital and Medical Center, Peoria, Illinois; ³Holy Family Medical Center, Des Plaines, IL; ⁴Univ. of Illinois College of Medicine at Peoria (UICOM-P), Peoria, IL

PURPOSE: Breast cancer is the leading cause of cancer and the second most common cause of cancer death among women in the United States. Research on the utilization of screening mammography for women has consistently shown that women are not taking advantage of recommended screening services. Most of the research on breast cancer screening has utilized retrospective cross-sectional survey designs, convenience samples and has not controlled for important confounding variables. These studies have limited themselves to describing characteristics of patients who do not get screened or listing patient-reported reasons for underutilization. Studies have implicated cost, insurance status and socioeconomic status as strong predictors of who will follow through with screening.

METHODS: Randomized, unblinded, controlled interventional design. Consecutive clinic attendees who were >40 years of age and eligible for a screening mammogram were enrolled in the study (N = 334). Patients were randomized within resident patient panels to be offered free screening mammograms or to be offered mammograms without mention of cost. Residents used a standardized script to offer all eligible patients a mammogram. Those who agreed were given appointments prior to leaving the clinic. The hypothesis was that those patients who were assigned to receive a free mammogram would be more likely to follow through with their mammogram appointment.

RESULTS: Baseline characteristics between the control and intervention group were similar. Overall compliance rates were similar for both groups. Fifty four percent of the control group and 63% of the intervention group completed mammograms. This difference was not significant (chi square = 2.88; p = .237). Black women were 23% less likely than their non-black counterparts to complete the mammogram (p = .069). There was no association between completion of mammogram and age, education, family history of breast cancer or type of insurance.

CONCLUSION: Offering free screening mammograms and eliminating many of the known barriers to screening was not sufficient in this population to encourage compliance with screening mammography.

HEALTH LITERACY AND RATES OF HOSPITAL READMISSION, AND EMERGENCY ROOM VISITS IN AN INNER-CITY HOSPITALIZED POPULATION. J.R. Polanco¹, M.J. Fagan¹; ¹Brown University, Providence, RI

PURPOSE: Low health literacy, a hidden obstacle to adequate health care, has significant impact on health services utilization. We set out to determine the association between health literacy of medical inpatients and subsequent emergency room visits, and hospital readmissions.

METHODS: The study is a prospective cohort study with 3 months of follow-up, and took place in the medical service of an urban, teaching, hospital in Providence, Rhode Island. Patients admitted to the medical service at Rhode Island Hospital were eligible to participate in the study. A total of 293 hospitalized patients were approached over a seven week period for participation in the study, of which 161 (55%) completed the instrument. Patients who agreed to participate completed the short version of the Test of Functional Health Literacy in Adults (TOFHLA), which was administered by a bilingual research assistant.

RESULTS: Of the participants, 54% were female, and the mean age was 59 years old. 126 (78%) of the participants scored in the adequate health literacy range, which was defined as a score of 23–36 (maximum score = 36). 35 (22%) patients scoring in the marginal health literacy range (17–22), and those scoring in the inadequate range (0–16) were combined into one group as low health literacy. Three months after discharge from the hospital, the hospital computer records were reviewed for the main outcomes—number of emergency room visits and hospital readmissions. There were 22 emergency room visits in the 126 patients with adequate literacy and 15 emergency room visits in the 35 patients with low health literacy (mean number of emergency room visits 0.17 (95%CI 0.082–0.267) vs. 0.43 (95%CI 0.092–0.765), p = 0.04). Patients with low health literacy had 2.5 times more visits to the emergency room than patients with adequate literacy. There were 52 hospital readmissions in the 126 patients with adequate literacy and 25 readmissions in the 35 patients with low health literacy (mean number of readmissions 0.413 (95%CI 0.237–0.589) vs. 0.714 (95%CI 0.346–1.083), p = 0.12).

CONCLUSION: In our study of a medical inpatient population, patients with low health literacy had an increased number of emergency room visits within three months of discharge when compared to patients with adequate health literacy. Future studies should be focused on interventions aimed at patients with low health literacy to improve health services utilization.

ACCESS TO CARE ISSUES IN A PUBLIC HOSPITAL URGENT CARE CLINIC. S. Prock¹, H. Batal¹, S. Majeres¹, L. Lasater¹, R. Lundgren², J. Adams¹, P. Mehler¹; ¹Denver Health Medical Center, Denver, CO; ²Colorado Prevention Center, Denver, CO

PURPOSE: To describe access to care issues in a population of patients seeking care at a public hospital urgent care clinic.

METHODS: 20% of patients presenting to the Denver Health Medical Center urgent care clinic (UCC) between June 15 and August 11, 2000 were randomly approached to participate in a pre-visit interview, offered in both English and Spanish. Patients were asked standardized questions regarding physical and mental health, their reasons for accessing care in the UCC, barriers to health care, history of prior Emergency Department (ED) and UCC usage, reasons for delaying health care, chronic medical conditions, and prior receipt of preventive health services. Study patients' charts and admission records were abstracted for the number of prior visits and documentation of receipt of preventive health services at our institution, ethnicity, age, gender, and insurance status. Data were analyzed using multiple logistic regression.

RESULTS: There were 1006 patients surveyed with a refusal rate of 20.3%. The study patients did not differ from the total population who accessed care during this time period in regards to age, insurance status, ethnicity, or gender. A regular source of care, other than an ED or UCC, was identified by only 37.1% of patients. Those patients with insurance were 1.7 times more likely to note a regular source of health care ($p=0.0046$) than those without insurance or those on the state indigent care discount program. Hispanic ($p=0.0075$) and black ($p=0.04$) patients were more likely than white patients to note a regular source of health care. Older patients ($p<0.0001$) and female patients ($p<0.0001$) were also more likely to report a regular source of health care. Patients accessing care in the UCC were likely to have delayed accessing care, 71% had been sick for more than two days, with 49% noting that their current medical problem had been present for a week or longer. Those patients without insurance or on the state indigent care program were more likely to report a delay in seeking care ($p=0.0279$). Among those who delayed care for more than two days, 26.4% reported that a lack of insurance contributed to their delay.

CONCLUSION: Patients presenting to a public hospital urgent care clinic often delay care and often do not have a regular source of health care. Having a regular source of care was associated with age, gender, ethnicity, and insurance status. Patients without insurance were more likely to report delaying health care.

ASSOCIATION BETWEEN PHYSICAL ILLNESS AND SUICIDE AMONG THE ELDERLY. H. Quan¹, J.E. Arboleda-florez², G.H. Fick¹, H.L. Stuart², E.J. Love¹; ¹University of Calgary, Calgary, Alberta; ²Queen's University, Kingston, Ontario

PURPOSE: Only a few small studies have explored the association between various physical illnesses and suicide in the elderly and they have produced inconsistent results. Thus, we undertook this larger study to more definitively assess the association between elderly suicide and physical illness.

METHODS: This case-control study included all suicides (920 cases) and motor vehicle accident deaths (1,050 controls) for 1984–95 in the Province of Alberta, Canada among Alberta residents aged 55 years and older. We reviewed Medical Examiner records and extracted sociodemographic information. Then, deterministic linkage was used to link subjects to the Alberta provincial health care registry to determine personal identifiers. Those identified in the registry (1766, 90%) were linked with hospital discharge and physician claims data. We extracted coded diagnoses for each subject in the two years prior to the date of suicide or accident. The accuracy of the administrative data diagnoses was assessed through patient chart review in a sub-set of the study population, revealing kappa values ranging from 0.70 for prostatic disorder to 1.0 for cancer.

RESULTS: Compared to the motor vehicle accident victims, the elderly who committed suicide were more likely to be men (78.6% vs. 62.2%), aged 55–64 years old (50.6% vs. 37.5%), unmarried (46.4% vs. 41.0%), white (96.4% vs. 91.7%), residents of high median income areas (43.7% vs. 26.9%) or urban areas (58.4% vs. 40.5%), and to have a history of depression (37.0% vs. 6.2%) or other psychiatric illnesses (57.1% vs. 26.0%). The two groups were similar in the proportion of unemployed (80.9% vs. 82.8% currently). When controlling for sociodemographic characteristics and psychiatric disorders, the elderly who committed suicide were more likely to have had cancer (odds ratio [OR]: 1.73, 95% confidence interval [CI]: 1.16–2.58) and prostatic disorders excluding prostate cancer (OR: 1.70, CI: 1.16–2.49) than were motor vehicle accident victims. Among those who were married, chronic pulmonary disease was more frequent among the elderly who committed suicide than among motor vehicle accident victims (OR: 1.86, CI: 1.22–2.83). Ischemic heart disease, cerebrovascular disease, peptic ulcer disease, and diabetes mellitus were not independently associated with elderly suicide.

CONCLUSION: Cancer, chronic pulmonary disease, and prostatic disorders appear to be associated with suicide among the elderly. Physicians and other clinicians should consider assessment and monitoring of patients' suicidal tendencies when such conditions are present.

HOMOCYSTEINE STATUS AND ITS MODULATORS IN PATIENTS WITH CLINICAL AND SUBCLINICAL THYROID DISEASE. P. Rao¹, V. Shukla¹, F. Khan¹, R. Carmel¹, A. Malik¹; ¹New York Methodist Hospital, Brooklyn, NY

PURPOSE: Hypothyroidism (HYPO) is associated with increased atherosclerosis in autopsy studies. Higher plasma homocysteine (Hcy) levels, even within the normal range, are an independent risk factor for cardiovascular disease. A few studies have shown that patients with HYPO have increased Hcy levels. Some have suggested that this may be due to the slightly increased creatinine (Cr) level in HYPO but other factors affecting Hcy status were not examined. Therefore, we assessed Hcy, vitamin and Cr levels together in patients with HYPO. Because little comparative information exists for patients with hyperthyroidism (HYPER) and none for patients with subclinical hypothyroidism (SC-HYPO), these conditions were also studied.

METHODS: Hcy, thyroid stimulating hormone (TSH), free thyroxine (FT4), vitamin B12 (B12), folate and Cr levels were measured in 11 (2 men, 9 women), 10 (2 men, 8 women), and 19 (5 men, 14 women) patients with HYPO, SC-HYPO and HYPER respectively.

RESULTS: Mean ($\pm$ SD) Hcy level was higher in HYPO than in HYPER (9.6 ± 3.0 vs 8.4 ± 2.7 μ mol/l) but not significantly so ($p=.26$) and only 1 patient had an abnormal Hcy (normal in women = $4.4-12.1$ μ mol/l). Hcy in SC-HYPO (10.3 ± 2.2 μ mol/l) was also higher than in HYPER ($p=.055$). Plasma Hcy did not correlate with either TSH or FT4 levels. Although Cr levels were higher in HYPO than in the other groups ($p=.0003$), the Cr level did not correlate with Hcy. On the other hand, folate levels were significantly lower in HYPO than in HYPER (9.2 ± 6.0 vs 17.6 ± 11.2 μ g/l, $p=.013$); folate levels were also lower in SC-HYPO (8.9 ± 3.2 μ g/l; $p=.005$). The folate levels correlated inversely with Hcy levels in the entire population ($p=.001$) and within the HYPO group ($p=.06$). B12 levels showed no significant patterns in thyroid disease.

CONCLUSION: Our data show that Hcy levels are higher in HYPO than in HYPER but not significantly so. While Cr levels were significantly higher in HYPO than in HYPER, our data suggest that Cr levels are not the major determinant of Hcy changes in thyroid disease. Instead, folate status appears to play an important role in thyroid disease that was not appreciated because vitamin and renal status had not been examined together until now. Finally, our data raise the possibility that early changes in Hcy levels similar to those in HYPO may appear in SC-HYPO. If confirmed by our therapeutic trials, these data may provide an additional reason for intervention in SC-HYPO to mitigate atherosclerotic risk.

SEX DIFFERENCES IN CARDIAC CATHETERIZATION AND CORONARY REVASCULARIZATION: THE ROLE OF PHYSICIAN SEX. S.S. Rathore¹, J. Chen¹, Y. Wang¹, M.J. Radford¹, H.M. Krumholz¹; ¹Yale University School of Medicine, New Haven, Connecticut

PURPOSE: Female patients are less likely to undergo cardiac procedures after a myocardial infarction (MI) than male patients. We examined the use of cardiac catheterization and coronary revascularization post-MI to determine whether sex disparities in procedure use were associated with physician sex or were more pronounced when a patient and physician were of different sexes.

METHODS: We evaluated data from the Cooperative Cardiovascular Project, a sample of Medicare beneficiaries hospitalized for MI in 1994 and 1995. Patients age 65 years and older with a confirmed MI who presented directly to the hospital with no prior history of revascularization ($n=111,319$) were linked with physician data provided by the American Medical Association and evaluated for the use of cardiac catheterization (CATH) and coronary revascularization (REVASC, by PTCA or CABG) within 60 days of admission for MI. Separate multivariable logistic regression analyses were employed to ascertain the influence of patient sex and physician sex for CATH and REVASC use adjusting for patient sociodemographic characteristics, illness severity, physician factors (specialty, age, race, practice type), and hospital characteristics. A patient sex/physician sex interaction term was incorporated in multivariable analysis in order to test whether differences in procedure use were greater when a patient and physician were of different sexes.

RESULTS: Female patients were less likely to undergo CATH (39.4% vs. 52.1%, $p=0.001$) and REVASC (25.8% vs. 36.7%, $p=0.001$) than male patients, while patients treated by male physicians were more likely to undergo CATH (45.9% vs. 39.5%, $p=0.001$) and REVASC (31.4% vs. 25.8%, $p=0.001$) than patients treated by female physicians. Female patients remained less likely to undergo CATH (odds ratio [OR]: 0.82, 95% confidence interval [CI] 0.80, 0.85) and REVASC (OR 0.80, 95% CI 0.77, 0.82) in multivariable analysis, regardless of the treating physician's sex. In contrast, patients treated by male physicians were more likely to undergo CATH (OR 1.15, 95% CI 1.07, 1.23) and REVASC (OR 1.14, 95% CI 1.06, 1.22) than those treated by female physicians, regardless of patient sex, in adjusted analysis. Differences in CATH ($p=0.66$) and REVASC ($p=0.32$) use were not greater when a patient and physician were of different sexes.

CONCLUSION: Though male patients and patients treated by male physicians were more likely to undergo cardiac catheterization and coronary revascularization post-MI, differences in procedure use were not greater when a patient and physician were of different sexes, suggesting that factors other than sex bias account for sex differences in cardiac procedure use.

WHAT IS MOST IMPORTANT TO PATIENTS? FOCUSING THE CONTENT OF AN AMBULATORY PATIENT SATISFACTION QUESTIONNAIRE. K.L. Ravenell¹, D.A. Asch², A.C. Aguirre¹, K.M. Fosnocht¹, C. Henry¹, J.J. Murphy², J.A. Shea¹; ¹University of Pennsylvania, Philadelphia, PA; ²University of Pennsylvania, Philadelphia VAMC, Philadelphia, PA

PURPOSE: Patient satisfaction instruments are frequently used to measure the quality of care provided by physicians and practices. Because these instruments are used for diverse purposes, however, they often become long and difficult for patients to complete, especially those from vulnerable populations. We attempted to reduce the 63-item Veterans Administration (VA) Ambulatory Customer Care Satisfaction Survey into a concise instrument focused on patients' concerns.

METHODS: Patients were recruited sequentially at two VA Medical Center clinics and two medical clinics at a nearby University. Patients were first asked an open-ended question, "What is important to you regarding your outpatient clinic visits?" They were instructed to provide comments based only on their ambulatory care, not inpatient or specialty care. Comments with similar themes were grouped into categories. For the second part of the task, they were shown a list of 15 randomly chosen items from the VA instrument and asked to rate each on a 5-point scale from "not at all important" to "extremely important." A range of content and complexity was represented within each set of 15 items.

RESULTS: A total of 496 patients participated. Most were middle-aged (mean age = 50.5, $sd=17.8$), male (67.1%), and African-American (59.9%); 38.7% had a high school education or less. A total of 879 comments were provided about their outpatient clinic visits. The majority of patients gave at least one (80.6%) or two comments (53.8%). The five topics volunteered most frequently were the wait time for appointments (31%), doctor-patient interactions (14%), getting good service and care (11%), overall speediness (11%), and friendliness/courtesy by all staff (10%). However, when patients rated a list of existing items, the average rating for every item was above 4.0 (i.e., between "very important" and "extremely important").

CONCLUSION: Reducing the number of items in patient satisfaction questionnaires is complex. Patients are willing to specify what is important to them in their ambulatory care, however, most patients volunteer only one or two issues, and these issues vary across patients. When shown a partial list of items from an existing instrument, ceiling effects limit the patients' ability to reveal the relative value of potentially competing items. The disparity in results with the two methods suggests researchers need to be aware of how existing instruments were generated, think about the ability and willingness of patients to discriminate among multiple items with similar themes, and find ways to capture what is important to patients without overburdening them.

COST-EFFECTIVENESS OF CANCER SCREENING IN THE ELDERLY. J.S. Rich¹, J.D. Birkmeyer¹; VA Medical Center, White River Junction, VT

PURPOSE: Although the benefits of continuing to screen for cancer in the elderly are unknown, the elderly are screened. We developed a Markov model to estimate the cost-effectiveness of continuing to screen for cancer beyond 70 years of age.

METHODS: For each of three cancers (breast, cervical and colon), we estimated the marginal cost-effectiveness of continuing to screen average risk elderly men and women who had previously undergone regular screening. All-cause and cancer-specific mortality data were obtained from the National Center for Health Statistics and the Surveillance Epidemiology and End Results survey. Assuming that the benefits of screening seen in younger patients extend to the elderly, we used a reduction in cancer-specific mortality of 27% for biennial mammography and 33% for annual fecal occult blood testing (FOBT) to model screening benefits. We assumed a 70% reduction in cervical cancer mortality to model benefits with triennial screening Pap smears. Three costs were included: cost of the screening test, cost to evaluate an abnormal, and the cost of cancer care. We assumed that there were no harms with screening.

RESULTS: The table below shows the incremental gains in life expectancy and cost-effectiveness of continuing to screen the elderly for cancer. The values for the 70–79 year old interval are compared to stopping screening at age 69; the values for the 80 years and over interval are compared to stopping screening at age 79.

	Gain in life expectancy, days	Cost per life year saved (rounded to nearest 1,000)
Pap smears		
70–79	0.65	157,000
80+	0.04	733,000
Mammography		
70–79	4.67	50,000
80+	0.29	226,000
Fecal occult blood		
70–79	5.59	50,000
80+	0.37	115,000

CONCLUSION: Under assumptions that are very favorable to screening, continuing to screen for cancer beyond 80 years of age does not appear to be cost-effective. Although screening for breast and colon cancer in 70–79 year olds may be cost-effective, the gains in life expectancy are small and may be outweighed by potential harms.

AT HOME WITH HEART FAILURE & TRADE: EFFECT OF STANDARDIZED TELEPHONIC CASE MANAGEMENT FOR HEART FAILURE. B. Riegel¹, B. Carlson¹, Z. Kopp², B. Lepetir², A. Unger³; ¹Sharp HealthCare, San Diego, CA; ²Pfizer, Inc., New York, NY; ³Science Applications International Corporation, Reston, VA

PURPOSE: Case management that promotes heart failure (HF) self-care is thought to decrease the need for hospitalization but few randomized, controlled clinical trials have tested the approach. Since much of the effectiveness of case management depends on the unique abilities of the provider, decision-support software from Pfizer, Inc., "At Home with Heart Failure™", was used to standardize care.

METHODS: A prospective randomized controlled clinical trial was conducted to evaluate the effectiveness of software-supported telephonic case management in decreasing acute care resource use. 281 physicians from two hospitals in So. California were matched on specialty and practice size and randomized to intervention or usual care. Patients who were cognitively intact and spoke English or Spanish were identified during a HF hospital admission; 358 patients of the randomized physicians were included. Mean age was 72 years, 51% female, 56% unmarried, 72% class III or IV. The intervention group (n=130) was telephoned within 5 days after hospital discharge and thereafter at a frequency guided by the software based on patient symptoms, knowledge, and needs. On average, patients received 17 calls (median=14) from the case manager over the 6-month intervention period. Printed educational materials were mailed. Physicians were notified of patient progress in writing and telephoned as needed. Care for patients in the usual care group (n=228) was not standardized. T-tests were used to test the hypothesis of equal mean acute care resource use in the two groups.

RESULTS: Resource use was consistently lower in the intervention group: 6-month all-cause hospitalization rates were 27% lower (.62 vs. .86, p=0.03), HF hospitalization rates were 48% lower (.22 vs. .41, p=0.005), HF hospital days were 46% lower (1.1 vs. 2.1, p=0.04). Multiple readmissions were 43% lower (.13 vs. .23, p=.025).

CONCLUSION: Standardized case management using telephonic decision-support software provided in the early months after a HF admission can augment care and significantly reduce acute care resource use.

DISCLOSURE OF MEDICAL ERRORS: PRIMARY CARE PHYSICIANS DO BEST. J. Rifkin¹, A. Robinson¹, D. Topp², C. Gilroy¹, J. Pickard¹, L. Winslow¹, J. Saari¹, H. Shapiro², R.J. Anderson¹; ¹HealthONE Presbyterian/St. Luke's Medical Center and University of Colorado Health Sciences Center, Denver, CO; ²HealthONE Presbyterian/St. Luke's Medical Center, Denver, CO

PURPOSE: In 1999, the Institute of Medicine reported that medical errors kill between 44,000 and 98,000 patients each year. There is little information on physician and public attitudes regarding disclosure of medical errors.

METHODS: One thousand Colorado physicians were randomly selected to receive a mail survey describing three medical error scenarios (one minor, one moderate and one serious). The same three scenarios were presented to five hundred Colorado residents via telephone survey. For each scenario, the physicians and the public were asked if the error should be disclosed to the patient. The moderate error scenario was slightly modified from one previously tested and reported by M. Hingorani et al (*BMJ* 1999;318:640–1).

RESULTS: Response rates were 57% for physicians and 82% for the public. The public was less

PERCENT WHO WOULD NOT DISCLOSE:	Public	Physician	P
Would not disclose at least one error	20.6%	13.5%	<0.01
Would not disclose minor error	8.4%	8.5%	<0.91
Would not disclose moderate error	12.2%	6.1%	<0.001
Would not disclose serious error	5.6%	3.4%	<0.45

likely to desire disclosure of all three errors compared to physicians (p < 0.01) especially for the moderate error (p < 0.001) and the serious error (p < 0.045).

Retired physicians (vs. active physicians, p < 0.05) and the over 65 public (vs. public under 65, p < 0.004) would not disclose the error for at least one of the scenarios. Primary care physicians (Internal Medicine, Family Medicine, Pediatrics, and General Practice) were more likely to disclose all three errors (p < 0.04) than non-primary care physicians. Physician response was not influenced by gender, years in practice, practice location or previous malpractice litigation. Desire for disclosure of the moderate error was roughly comparable for Colorado residents (88%) and United Kingdom respondents (92%). Gender, race, education, and income did not influence public response.

CONCLUSION: Overall, 87% of Colorado physicians would disclose all medical errors to patients. The Colorado public was less inclined to disclose errors than physicians. Retired physicians and senior citizens were less likely to disclose one or more errors. Primary care physicians relative to all other physicians were most likely to disclose all three errors.

OBESITY AND ACTIVITY SELF-PERCEPTIONS AND BARRIERS AMONG OLDER, LOW-INCOME HIGH FUNCTIONING WOMEN. C.S. Ritchie¹, B.A. Stetson¹, K. Adams¹, E. Rucker¹; ¹University of Louisville, Louisville, KY

PURPOSE: Obesity is increasingly recognized as a highly prevalent condition among older women and is associated with multiple comorbidities. Physical activity is an important component of weight reduction and attempts to improve function. We sought to identify perceptions regarding physical activity among older, low-income high functioning women.

METHODS: We performed prompted survey completion with a group of 38 older, low-income women who volunteered as senior companions and caretakers for homebound older adults. BMI classifications are described using WHO criteria.

RESULTS: Mean age: 73.81 (range 63–93). Seventy percent of subjects were African American; 30% Caucasian. Nearly 65 % were overweight or obese (2.9% underweight; 32.4% normal weight; 8.8% overweight; 41.2 %obese; 14.7% morbidly obese). Of the non-obese women, 85.7% were satisfied with their activity level; compared with only 26.3% of obese women (p<.01). Despite their mobility and involvement in volunteer activities, participants cited numerous barriers to health-promoting physical activity. In contrast to common assumptions regarding social and environmental obstacles to exercise, such as family caregiving responsibilities and weather; this sample indicated that their primary barriers were cost, experiencing pain, illness or injury, safety concerns and self-consciousness. Obese women were more likely to report barriers associated with illness and injury (p<.05) and self-consciousness (p<.01). Obese women also indicated a greater dislike of solitary, "lifestyle" exercise (p<.05). Despite a high prevalence of obesity in this group, only half of the participants reported being encouraged by their physicians to engage in physical activity.

CONCLUSION: Physical limitations and obesity are significant problems even among this relatively high functioning group of older low-income women. Perceived barriers including underlying illness and self-consciousness need to be addressed when considering interventions.

A CONTROLLED TRIAL OF COLLABORATIVE MEDICATION EDUCATION AND PHARMACEUTICAL CARE. M.S. Roberts¹, K. Chalka¹, J. Chang¹, C. Amy¹, W.N. Kapoor¹; ¹University of Pittsburgh, Pittsburgh, PA

PURPOSE: To assess whether a multidisciplinary intervention designed to improve patient medication education and evaluate specific medication regimens can improve patient knowledge concerning medication use, increase compliance with medication use post-discharge, and decrease medication errors.

METHODS: A multidisciplinary intervention was designed that included 1) a folder with medication information sheets placed at the bedside, 2) nurse-directed teaching about each medication with each administration throughout the hospital stay, 3) automatically triggered specialized teaching plans for anticoagulation, diabetes, and inhaled medications, 4) protocol-triggered review of medications by a pharmacist to decrease administration complexity and identify potential medication-related problems (interactions, dosing problems, ADRs). The intervention was studied in 4 nursing units in a tertiary teaching hospital utilizing a non-

repeated pre-post/intervention-control design. The intervention was applied to all patients admitted to the intervention unit, and was not instituted on the control unit. Random samples of patients from both the control and intervention units before and after the implementation of the intervention were selected for participation in the evaluation component of the project. Evaluation consisted of baseline data collection and a structured survey that occurred 14 days post-discharge to assess knowledge and compliance. Multivariate logistic regression was used to test for significance.

RESULTS: 2397 patients were admitted during the study period (1264 on the control unit, 1133 on the intervention). 1246 randomly selected patients (602 control, 644 intervention) were screened for participation in the evaluation component of whom 611 would be responsible for their own medications post-discharge and were eligible. 302 patients agreed to participate and 239 (155 control, 124 intervention) completed the 14-day follow-up. Improvement in post-discharge assessment of knowledge regarding specific medications was significantly higher in the intervention group (7.4% for control vs 28.0% intervention, $p=0.0129$), as was patient satisfaction with medication education. Protocol-initiated pharmacy consults were completed on 43 patients, of which 18 resulted in improvements in therapy.

CONCLUSION: This pilot trial demonstrates that a multi-disciplinary intervention composed of nurse-directed education given at the time of medication administration linked with protocol-based consults to pharmacists and other health care providers can significantly increase patient knowledge regarding medication and satisfaction with medication education. Larger sample sizes are required to assess impact on medication error reduction, readmission and health status.

QUALITY OF HEALTH CARE AND THE PROBLEM OF MEDICAL ERRORS IN THE U.S.: PHYSICIANS AND THE PUBLIC DISAGREE. A.R. Robinson¹, J. Rifkin¹, D. Topp¹, C. Gilroy¹, J. Pickard¹, L.C. Winslow¹, J. Saari¹, H. Shapiro¹, R.J. Anderson¹; ¹HealthONE Presbyterian/St. Luke's Hospital and University of Colorado Health Sciences Center, Denver, CO

PURPOSE: The 1999 Institute of Medicine (IOM) report on medical errors has focused attention on the quality of health care in the U.S., yet little information exists on physician or public opinion regarding conclusions from that report.

METHODS: We randomly surveyed 1000 Colorado physicians using a mail questionnaire and asked them to rate their agreement with several statements from the IOM report. A telephone survey of 500 Colorado residents asked respondents to rate their agreement with the same statements. Chi-square tests were used to compare proportions of physicians and the public who agreed with the statements and to assess for demographic differences by response.

RESULTS: Response rates were 57% for physicians and 82% for the public. The following table shows percentages of physicians and the public who agreed with each statement:

Statement	Physician%	Public%	P
"Quality of health care in the U.S. is a significant problem."	33.2	67.6	<0.001
"Our health care system does not match the safety record attained in other industries."	24.7	36.1	<0.001
"Error reduction in medical practice should be a national priority."	69.8	86.8	<0.001
"We need a national agency to provide leadership and research in reducing medical errors."	22.9	59.8	<0.001
"We should implement a system of mandatory reporting of serious medical errors."	52.7	90.4	<0.001

Physicians 65 and older were more likely to feel that quality of health care is significant problem ($p<0.01$), but there were no differences in physician responses by gender, specialty, practice type, or by whether they had ever had a malpractice suit filed against them. Of the public respondents, females, those over 65, and those with lower incomes were more likely to believe quality of health care is a problem ($p<0.01$). Public responses did not differ by race or level of education.

CONCLUSION: There is a marked disparity in public vs. physicians' perceptions of the quality and safety of health care in the U.S., as well as in the need for a national agenda directed towards reporting and reduction of medical errors. Physicians should recognize that a majority of the public believes that the quality of health care is a significant problem and that mandatory reporting of medical errors is widely supported.

RACE, QUALITY OF DEPRESSION CARE, AND RECOVERY FROM MAJOR DEPRESSION IN A PRIMARY CARE SETTING. B.L. Rollman¹, B.H. Hanusa¹, H.C. Schulberg²; ¹University of Pittsburgh, Pittsburgh, PA; ²Cornell University School of Medicine

PURPOSE: Studies have found that African-Americans (AA) tend to receive poorer quality care and experience poorer clinical outcomes than Caucasians (C) across a variety of medical conditions. However, little is known about racial differences in care and recovery from major depression (MD) in primary care settings.

METHODS: We examined data collected as part of a clinical trial testing the effectiveness of disseminating a depression treatment guideline via an electronic medical record system (EMR). Patients presenting for routine care by a board-certified primary care physician (PCP) at a university-based clinic were screened for MD using the PRIME-MD and their PCPs were subsequently informed of this finding. Study patients had a Hamilton Rating Scale for Depression (HRS-D) score ≥ 12 and were not in treatment with a mental health specialist (MHS). The HRS-D was repeated at 6 months to ascertain recovery (HRS-D ≤ 7). Data about depression treatment such as number of visits and discussions of depression, counseling, pharmacotherapy, and referral to a MHS, was abstracted from the EMR. Non AA/C patients

were excluded from our analyses (3%) given the focus of this report on comparing differences between AAs and Cs.

RESULTS: Between 4/96 and 12/98, 204 depressed patients (25% AA) met all protocol eligibility criteria and completed a baseline interview. They were cared for by 15 PCPs (100% C; 53% male; median age 39 years). At 6 months, 187 (92%) had follow-up information available including 47 (24%) AAs and 140 (73%) Cs. AA and C patients were similar at baseline on level of depressive severity (mean: HRS-D AA=21 vs. C=20), age (mean: 44 vs. 43), gender (female: 64% vs. 73%), employment (fulltime: 45% vs. 58%), medical comorbidity (2+ conditions: 57% vs. 59%), and history of depression (43% vs. 49%). AAs had a mean of 3.1 contacts with their PCP including 1.4 contacts discussing depression while Cs had 2.9 and 1.3 contacts, respectively over a 6-month period following study entry. PCPs documented counseling their AA patients for MD for 15%, recommending pharmacotherapy for 57%, and recommending a MHS referral for 28%. These rates were comparable for C patients (15%, 54%, and 35%, respectively). Overall, a similar proportion of AA and C patients recovered from their MD episode at 6-months (21% vs. 24%).

CONCLUSION: Although inadequate caregiving and poor treatment outcomes are commonly found among patients with MD who are cared for by PCPs, we were unable to identify any disparities in care or recovery rates for MD among the AA and C patients enrolled in our clinical trial. Future studies should confirm our results and examine methods for improving the delivery of guideline-based treatments for patients of all races experiencing MD.

MORTALITY ASSOCIATED WITH HEPATITIS C VIRUS (HCV) IN NEW YORK CITY 1992-1997. G. Sacaiju¹, E. Weis², A. Fine³; ¹Albert Einstein College of Medicine, Bronx, NY; ²Columbia University School of Public Health, New York, NY; ³NYC DOH, New York, NY

PURPOSE: Hepatitis C viral infection represents a substantial clinical and public health burden. In New York City (NYC) with its diverse ethnic population and large proportion of immigrants, characterizing the subgroup-specific burden of HCV is important to target HCV diagnosis, treatment and prevention efforts. Our objective was to describe the clinical and epidemiologic features of HCV infection in NYC, and to assess the sub-population specific burden of HCV mortality.

METHODS: We used data collected by the New York City Department of Health (NYCDOH) Vital Statistics office to identify all persons who died with a diagnosis of HCV coded as a primary or contributing cause of death between 1992 and 1997. Demographic data collected by this office was analyzed. To obtain information regarding cause of death, clinical characteristics, and risk factors for HCV infection, we performed medical record reviews on the subset who died during 1996-1997.

RESULTS: Between 1992 and 1997, 1002 persons met the case definition for HCV-associated death. Eight hundred and ninety were identified as NYC residents. Median age at death was 55 (3-97); 56% were male; 39% white non-Hispanic, 32% Hispanic, 22% black non-Hispanic, 4% Asian-Pacific Islander, 3% other and unknown. Forty eight percent were foreign born. Three hundred and nine (81%) of the 1996-1997 medical records requested were received. Of these, 26 could not be reviewed because of insufficient information. Of the 283 medical records reviewed 156 (55%) had a risk factor for HCV infection reported. Of these, injection drug use accounted for 73% and blood product transfusion accounted for 26%. We calculated a median years of potential life lost (YPLL) of 12 for the entire study population. During 1997 the proportion of deaths attributable to HCV in comparison to other causes of death was 0.9% among Hispanics, 0.5% among Asian among black and 0.2% among white.

CONCLUSION: Although HCV causes death among all segments of the NYC population, it has disproportionately burdened Hispanics and young adults. While additional research on incidence, prevalence, and distribution of risk factors is necessary, these data suggest that primary and secondary prevention efforts against HCV infection be directed at the most vulnerable populations.

COLORECTAL CANCER SCREENING STRATEGY AMONG DIFFERENT SPECIALTY PHYSICIANS. R. Sadurski¹, A.C. Mainous¹; ¹Medical University of South Carolina, Charleston, SC

PURPOSE: Different colorectal cancer (CRC) screening strategies have been shown to reduce CRC mortality. Fecal occult blood testing (FOBT) done yearly, flexible sigmoidoscopy (flex-sig) every 5 years with FOBT yearly and colonoscopy every 10 years are recommended for screening asymptomatic, average CRC risk individuals by renowned physician organizations. The purpose of this study was to examine preferences of CRC screening strategies among physicians of different specialties and their perception of patients barrier to screening.

METHODS: We conducted a web page, anonymous survey among physician in one university hospital and one VA hospital. The survey consisted of 3 questions: first asked about preferred and alternative choices of CRC screening that physician would recommend for their patients, second about determinant of recommending alternative strategies and third about perceived barriers that keep patients from having colonoscopic procedure if such is recommended.

RESULTS: Fifty six physicians responded: 19 family practitioners (FP), 15 gastroenterologists (GI) and 22 internists (IM). Figure 1 shows proportions of physicians' "optimal choice" according to different specialties. Family medicine favored flex-sig + FOBT strategy, gastroenterologists-colonoscopy and internists any of these two. Only about 13% of all physician favoured FOBT strategy. Most physicians indicated patient preference (78%), clinical status (48%) and cost to the patient (41%) as determinants of using strategy other than their preferred. Factors that would, in the opinion of physicians, keep patients from undergoing colonoscopy were: apprehension of pain (77%) and embarrassment (39%) as well as patient financial cost (55%).

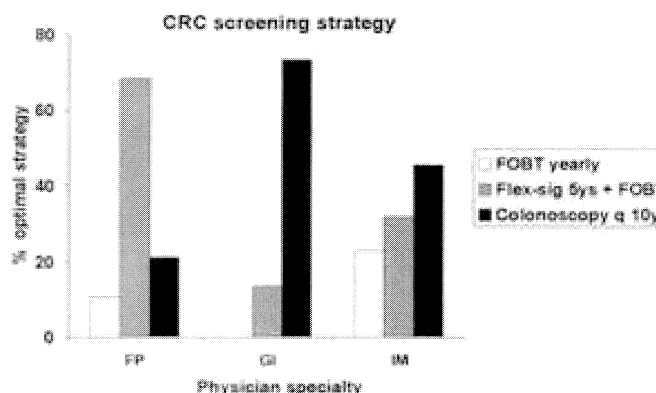


Figure 1.

CONCLUSION: Physicians of different specialties differ according to their perception of optimal strategy for colorectal cancer screening. Family practitioners prefer Flex sig+FOBT, gastroenterologists-colonoscopy and internists both strategies similarly.

EXPLAINING LOW PATIENT SATISFACTION RATINGS AMONG ASIAN AMERICANS. S. Saha¹, D.H. Hickam¹; ¹Portland VAMC/Oregon Health Sciences University, Portland, OR

PURPOSE: To determine whether low patient satisfaction ratings among Asian Americans as compared with other ethnic groups are attributable to worse experiences with health care, higher expectations, or different survey response tendencies.

METHODS: We analyzed data from a national random-digit-dial telephone survey conducted in 1994. We measured respondents' satisfaction with their regular physicians, health care services, health insurance, and life in general, using 4-point Likert scales (4 indicating highest rating). We hypothesized that if low patient satisfaction ratings among Asians were attributable to response tendencies, we should observe a similar pattern of responses among Asians across domains related and unrelated to health care. We also asked if respondents had ever changed physicians due to dissatisfaction.

RESULTS: Respondents included 627 Asian, 1005 black, 1000 Hispanic, and 1114 white adults. Asian respondents were mainly Chinese (33%), Vietnamese (32%), and Korean (32%). Asians reported lower satisfaction with their physicians and health care services than other ethnic groups, even after adjusting for demographic and health-related variables. Asians also reported lower satisfaction with their health insurance and with life in general (Table; $p < .001$ for all comparisons of Asians vs. non-Asians). Asians' responses to all satisfaction measures were similar and were more normally distributed than non-Asians' responses, which were more skewed toward the highest ratings. Asians were less likely than non-Asians to report having changed physicians due to dissatisfaction (24% vs. 31%, $p = .001$).

CONCLUSION: Asians recorded lower satisfaction ratings not only for physicians and health care services but also for life in general and, despite lower reported satisfaction, were less likely than non-Asians to have changed physicians due to dissatisfaction. These findings suggest that lower satisfaction ratings among Asians are due to different response tendencies rather than worse experiences or higher expectations of health care. Further research is needed to explain this phenomenon and assess its potential impact on performance measurement for physicians and health care organizations providing care for large numbers of Asian Americans.

	Satisfaction with physician	Satisfaction with health care	Satisfaction with health insurance	Satisfaction with life
Asians (mean +/- SD)	3.20 +/- .68	3.20 +/- .66	3.11 +/- .77	3.27 +/- .69
Non-Asians (mean +/- SD)	3.47 +/- .67	3.37 +/- .79	3.29 +/- .82	3.40 +/- .74

WHICH ANTIMICROBIAL CENTRAL VENOUS CATHETER SHOULD WE USE? K.D. Marciano¹, D.L. Veenstra¹, B.A. Lipsky², S. Saint³; ¹University of Washington, Seattle, WA; ²University of Washington, Seattle, Washington; ³University of Michigan, Ann Arbor, MI

PURPOSE: Vascular catheter-related blood stream infections are costly and associated with substantial morbidity and mortality. Central venous catheters coated with antibacterial agents have been shown to be superior to non-coated catheters. Trial results suggest, however, that central venous catheters impregnated with minocycline/rifampin, although more expensive, are clinically superior to catheters impregnated with chlorhexidine/silver sulfadiazine. It remains unclear if minocycline/rifampin catheters are cost-effective for all high-risk patients or only those requiring longer-term catheterization.

METHODS: We developed a series of decision analytic models using patient-level clinical trial data to determine if minocycline/rifampin catheters are cost-effective for patients requiring various durations of catheterization. We calculated incremental cost-effectiveness ratios for

patients catheterized for durations ranging from 1 to 25 days. We simulated the use of 10,000 catheters and calculated the proportion of times that the catheters were cost-effective or cost-saving at each duration of catheterization. The perspective was that of the healthcare payer; the time horizon was the period of hospitalization.

RESULTS: The data were too sparse to estimate cost-effectiveness for patients catheterized less than 8 days. The probability that minocycline/rifampin catheters were cost-effective or cost-saving compared to chlorhexidine/silver sulfadiazine catheters in patients expected to be catheterized at least 8 days was 91%. The probability minocycline/rifampin catheters were cost-effective or cost saving in patients expected to be catheterized 13 days or longer was over 95%. These findings remained relatively consistent during one-way and multi-way sensitivity analyses.

CONCLUSION: The probability that minocycline/rifampin catheters are cost-effective or cost-saving increases relative to the expected duration of use. There is insufficient evidence to suggest that minocycline/rifampin catheters are clinically or economically preferable to chlorhexidine/silver sulfadiazine catheters in patients catheterized less than 8 days. However, minocycline/rifampin catheters are likely to be cost-effective for patients expected to be catheterized for periods longer than 1 week, and are clinically and economically preferable for patients expected to be catheterized for 2 or more weeks.

FOUR-YEAR PROSPECTIVE EVALUATION OF COMMUNITY-ACQUIRED BACTEREMIA. S. Saint¹, R. Lark¹, C. Chenoweth¹, J. Zemencuk², B.A. Lipsky³, J.J. Plorde³; ¹University of Michigan, Ann Arbor, MI; ²Ann Arbor VA Medical Center, Ann Arbor, MI; ³University of Washington, Seattle, Washington

PURPOSE: Over 250,000 patients in the United States develop bacteremia or fungemia each year, with an associated mortality of 10 to 20% per episode. Though many recent studies of nosocomial bacteremia have been reported, there are limited data focusing on community-acquired bacteremia. Given the morbidity, mortality, and economic consequences of community-acquired bacteremia, we decided to: (1) describe the epidemiology and microbiology of community-acquired bacteremia; (2) determine the crude mortality associated with such infections; and (3) identify independent predictors of mortality.

METHODS: This prospective study was conducted at the Seattle Division of the Veterans Affairs Puget Sound Healthcare System from January 1, 1994 to December 31, 1997. All patients with clinically significant community-acquired bacteremia or fungemia were evaluated. Data were collected on demographics, co-morbid conditions, clinical parameters, microorganisms, source of infection, and patient outcome.

RESULTS: During the study period, 387 bacteremic episodes occurred in 334 patients. *Staphylococcus aureus* (18%), *Escherichia coli* (15%), and coagulase-negative staphylococci (12%) were the most commonly isolated organisms. The most frequent sources were the urinary tract, intravascular catheters, and pneumonia. Overall, almost one-third of bacteremia cases were directly related to indwelling catheters: either intravascular (20%) or urinary (10%). Approximately 14% of patients died. Patient characteristics independently associated with increased mortality included shock (OR 3.7, $p = 0.02$), renal failure (OR 4.0, $p = 0.003$), and a "Do Not Attempt Resuscitation" order (OR 21.7, $p < 0.001$). The risk of death was also higher in those whose source was pneumonia (OR 6.3, $p = 0.03$) or an intra-abdominal site (OR 10.7, $p = 0.02$), or if multiple sources were identified (OR 13.4, $p = 0.003$). The presence of fever (temperature $> 38.0^{\circ}\text{C}$) was associated with a decreased risk of death (OR 0.4; $p = 0.005$).

CONCLUSION: Vascular and urinary catheters were implicated in a substantial proportion of infections, emphasizing the need for appropriate and judicious use of such devices. We will likely observe an increase in the incidence of such device-related infections as healthcare expands into the outpatient setting. Fortunately, many of these infections are potentially avoidable via established infection control practices used to prevent hospital-acquired infections. Thus, strategies that have been useful in preventing nosocomial device-related bacteremia could prove successful if adapted into the outpatient setting. If this is borne out, community-acquired bacteremia may increasingly become viewed as a preventable disease.

UNRECOGNIZED ALCOHOL AND MEDICATION INTERACTIONS. B. Saitz¹, N.J. Horton¹, M.A. Moskowitz¹, J.H. Samet¹; ¹Boston University Schools of Medicine and Public Health, Boston, MA

PURPOSE: Despite known interactions, the occurrence of inappropriate medication and alcohol use has not been well described. We sought to assess 1) the prevalence of potential alcohol and medication interactions in hazardous and harmful drinkers in a primary care setting, 2) the association between this medication use and alcohol consumption, and 3) whether users of medications known to interact with alcohol are counseled by physicians regarding drinking.

METHODS: We interviewed current hazardous drinkers (at least one drink in the past month, and at least one positive response to the CAGE alcoholism screening questionnaire or > 14 drinks per week/ > 4 drinks per occasion for men, 7 and 3 respectively for women) just prior to a visit with a primary care physician to determine the self-reported current (30 day) prevalence of use of medications that can interact with alcohol. Alcohol consumption was assessed by trained interviewers using a validated calendar method. Immediately after the physician visit, patients reported whether there had been a discussion about alcohol with the physician during the visit that day.

RESULTS: The 312 subjects were 63% male, mean age was 44, and 63% had a high school education. Subjects saw one of 41 physicians. Most (78%) reported current use of medication that can interact with alcohol: 59% used non-steroidal anti-inflammatory drugs, 44% acetaminophen, 16% antihistamines, 10% each narcotics, antidepressants, and medication for sleep, 6% anxiolytics, and 3% blood thinners (i.e. warfarin). Users of these medications drank more drinks per drinking day (median 4.0 vs. 3.5, mean 5.8 vs 4.6, $p = 0.04$) and binge drank more often (median 2 vs. 1 binges in the past month, mean 6 vs. 5, $p = 0.03$) than non-users. There was no discussion about alcohol use with the physician for 37% of users and 40% of non-users of medications that can interact with alcohol ($p = 0.68$).

CONCLUSION: Most hazardous and harmful drinkers in a primary care setting were using prescription and nonprescription medications that can interact with alcohol. Users of these medications drank more than non-users. Many had no discussion about alcohol with their

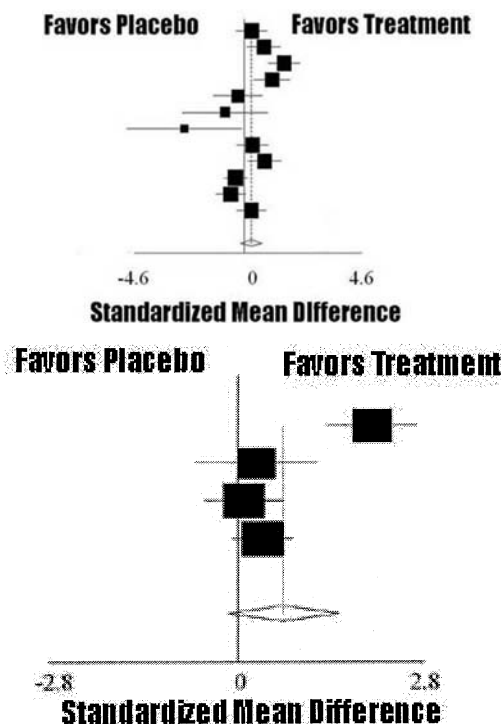
physicians. While the extent and impact of alcohol-medication interactions require further study, these results suggest that potentially dangerous alcohol and medication interactions are common and unrecognized by primary care physicians.

ARE ANTIDEPRESSANTS EFFECTIVE THERAPY FOR LOW BACK PAIN? A META-ANALYSIS. S. Salerno¹, R. Browning², J. Jackson¹; ¹Uniformed Services University of the Health Sciences, Bethesda, MD; ²National Naval Medical Center, Bethesda, MD

PURPOSE: To determine the efficacy of antidepressants in the treatment of back pain.

METHODS: We searched MEDLINE (1966–2000), PsycLit, Cinhal, Embase, Aidsline, Healthstar, Cancerlit, the Cochrane library (clinical trials registry and the Database of Systematic Reviews), Micromedex and FEDRIP as well as references of reviewed articles. Inclusion criteria included English-language, randomized, placebo-controlled trials of antidepressant medication among adults with back pain. Data were abstracted independently by two reviewers. Two continuous outcomes, back pain severity and ability to perform activities of daily living, were measured. Study quality was assessed using the methods of Jadad and data were synthesized using a random effects model.

RESULTS: Ten trials were included, the majority of which studied tricyclic antidepressants (8/10 studies) in the setting of chronic back pain (7/10 studies). Patients treated with antidepressant medication were not more likely to improve than those treated with placebo either in pain severity (Standardized Mean Difference: (SMD) 0.14, 95% CI: –0.27, 0.55) or in activities of daily living (SMD: 0.77, 95% CI: –0.07, 1.61). Patients receiving antidepressants experienced greater side effects (24% vs. 16%, $p=0.03$) than those receiving placebo.



CONCLUSION: Antidepressants are no more effective than placebo in the management of back pain. Further randomized controlled trials of antidepressants are needed before they can be routinely prescribed for treatment of back pain.

IMPACT OF PATIENTS HEALTH LITERACY ON ORAL ANTICOAGULATION. C. Sam¹, M.V. Williams¹, N.K. Wenger¹; ¹Emory University, Atlanta, GA

PURPOSE: To determine (1) the relationship of health literacy to patient's knowledge of information considered important to safely taking warfarin, and (2) the relationship of patients' health literacy to an objective measure of warfarin compliance.

METHODS: We conducted a prospective study of patients attending an anticoagulation clinic at a urban public hospital. We measured literacy using the Short Test Of Functional Health Literacy in Adults (S-TOFHLA). Patients' knowledge was assessed using 22 questions based on key elements in educational material and deemed important based on a review of the literature such as the dose of medication and recognizing bleeding complications. We assessed compliance by the proportion of INR measurements in therapeutic range among patients with at least 5 INR measurements.

RESULTS: A total of 237 patients were enrolled in the study. Only 22% of patients in our sample had adequate health literacy, 8% had marginal literacy, and 46% had inadequate literacy. Mean knowledge scores (95%CI) were highly correlated with patients' health literacy: 17.9 (17.2, 18.6), 16.2 (15.2, 17.2), and 14.0 (13.3, 14.8), respectively. There was no difference of percent in-range INRs across the three literacy groups.

CONCLUSION: Inadequate literacy is common among patients in our anticoagulation clinic and correlates with poorer knowledge of important facts regarding warfarin safety. However, we did not find a relationship between patients' health literacy and proportion of time their INR was in range. Additional research on outcomes such as bleeding and recurrent thrombosis is ongoing.

THE ASSOCIATION BETWEEN EXPECTATIONS REGARDING AGING AND HEALTH-CARE SEEKING BELIEFS FOR AGE-ASSOCIATED CONDITIONS. C.A. Sarkisian¹, R.D. Hays¹, C.M. Mangione¹; ¹UCLA Department of Medicine, Los Angeles, California

PURPOSE: Older adults vary greatly in their expectations regarding aging, with some expecting to maintain high function, and others expecting to experience functional decline. Whether having low expectations regarding aging causes older adults to seek less health care for modifiable age-associated conditions is unknown. We set out to determine whether older adults with low expectations regarding aging are less likely to think that they should seek health care for age-associated conditions associated with functional decline.

METHODS: We surveyed by mail a random sample of 588 English-speaking patients aged 65 and older cared for by 20 UCLA network physicians. We measured expectations regarding aging using a recently developed and validated 38-item Expectations Regarding Aging (ERA) Survey, which includes 10 domains of expectations such as physical health, cognitive function, and mental health. Additionally, to identify beliefs regarding care-seeking, we asked participants if they felt it was "very", "somewhat", or "not at all" important to seek health care for 13 age-associated conditions such as falling, urinary incontinence and pain. Participants also reported on their chronic medical conditions, ability to perform activities of daily living, level of depressive symptoms, religiosity, and generic physical and mental health status. We then constructed a multiple linear regression model to assess the independent relationship between ERA Survey scores and beliefs regarding care-seeking, adjusting for potential confounders.

RESULTS: Surveys were returned by 72% ($n=429$) of those surveyed; 54% were women; the mean age was 76 years; 80% were Caucasian, 9% Latino, 6% African American; 59% reported >2 medical conditions, and 21% reported disability. After controlling for all covariates including age, comorbidity, activities of daily living, religiosity and generic physical and mental health status, having lower expectations regarding aging was associated significantly with lower beliefs regarding care-seeking for age-associated conditions ($p=.01$).

CONCLUSION: In this community-based sample of older adults, having lower expectations regarding aging was independently associated with placing less importance on seeking health care for age-associated, potentially modifiable conditions. These results suggest that older adults with low expectations regarding aging may not receive health care for modifiable conditions, and thus may experience preventable functional decline.

PREDICTORS OF DRUG ADHERENCE IN AN INDIGENT POPULATION. J.M. Schechter¹, J.D. Voss¹; ¹University of Virginia, Charlottesville, VA

PURPOSE: Prescribed drug adherence (DA) is often suboptimal regardless of socioeconomic status, but disadvantaged populations face greater adherence barriers. We evaluated the association of DA for diabetes mellitus (DM), hypertension (HBP) and hypercholesterolemia (HC) with demographic and other factors in a medically indigent population to develop a predictive model of DA.

METHODS: For 1984 indigent patients, adherence was modeled using multivariable linear regression with an alpha of 0.01 for statistical significance. DA was defined as either mean DA for all drugs taken by each patient or minimum DA based on the lowest adherence drug using prescription refill data from a closed pharmacy system. Additional data were drawn from a clinical data repository.

RESULTS: Based on mean DA, 31% of patients took <80% of prescribed doses. Based on minimum DA, 52% of patients took <80% of prescribed doses for at least one drug. Increasing age, ethnicity (white), and greater medication supply per prescription were independently associated with higher mean and minimum DA. Number of drugs taken had a positive mean but negative minimum DA association. Gender, number of primary care visits, dosage schedule, and copayment were not independently associated with DA. The model explained only 6% of the variance in mean DA.

CONCLUSION: In an indigent population with HBP, DM or HC, DA was associated with ethnicity, age, and quantity of drug supplied. However these factors explained only a small amount of the substantial DA variability. To target DA efforts, better predictors must be developed to identify patients most in need of adherence intervention. Prescription refill claims data could serve this purpose.

Drug Adherence Parameters

Variable	Model: mean(SD)	Mean DA par est	p value	Min DA par est	p value
Age	57(13)	0.18	<0.0001	0.22	<0.0001
Race (W)	60%	4.10	<0.0001	5.60	<0.0001
# Drugs	3.0(2.0)	0.76	0.003	-3.00	<0.0001
Day supply	38.4(18)	0.19	<0.0001	0.27	<0.0001

CONGESTIVE HEART FAILURE PATIENTS EMERGENCY VISITS: EPIDEMIOLOGY AND CASE MIX ISSUES. G. Schiff¹, E. Abadilla¹, J. Wang¹, G. Shah¹, D. Goldberg¹, R. McNutt¹; ¹Cook County Hospital, Chicago, IL

PURPOSE: Because of its high prevalence and cost, preventing emergency departments (ED) visits and hospitalizations for congestive heart failure (CHF) has become the focus of extensive "disease management" activities. As part of a national chronic disease improvement collaborative, we sought to better understand and improve the care for patients with CHF. Because virtually all admissions in our public hospital system are the result of an emergency department visit, and our finding that approximately one in three ED visits of patients in our CHF cohort resulted in a hospital admission (unpublished data), we sought to better characterize the frequency and patterns of ED visits in this patient population.

METHODS: We assembled two cohorts of patients with the diagnosis of congestive heart failure – 1) an outpatient cohort attending one of four outpatient sessions at a public hospital general medical clinic over a three month period (7/99–10/99) who had CHF listed on their

outpatient chart problem list (N=470), and 2) inpatients hospitalized with an admitting diagnosis of CHF over a 15 month period (5/99–9/99; N=1297). All emergency room visits from 7/99–11/00 were electronically recorded, along with primary and secondary diagnoses, and then matched with our two CHF registries to query for ER visits for registry patients.

RESULTS: For the outpatient cohort there were a total of 525 ED visits for the 470 CHF outpatients during the 17 month study period (=111.7 visits per 100 patients overall, annualized rate of 78.8 visits/100 patients/yr). The inpatient CHF cohort of 1297 patients made 2,626 ED visits (=202.5 visits per 100 patients overall, annualized rate 142.9 visits/100 patients/yr). 203 patients (43.2%) of the outpatient cohort had one or more visit to the ED (1 visit-93 (19.8%), 2 visits-231 (17.8%), 3 visits-139 (10.7%) 4 or more visits-208 (16.0%). For the Ed visits, CHF was listed as a primary or secondary Ed diagnosis in 866 (33.0%) of the inpatient cohorts' visits, and only 99 (18.9%) of the outpatient cohort patients' ED visits.

CONCLUSION: Patients with a diagnosis of congestive heart failure have high rates of emergency department encounters. CHF registry patients identified from an inpatient admission, have twice the rate of ED visits as patients identified in a continuity general medical clinic, suggestion that comparative studies of patients and disease management programs outcomes are likely to be biased by the method/site of patient identification. IN addition, many Ed visits appear to be not directly related to CHF, suggesting broader interventions may be required if reducing ED visits is an expectation of CHF chronic care improvement.

SHOULD WE BE SCREENING FOR FUNCTIONAL HEALTH LITERACY PROBLEMS AMONG PATIENTS WITH DIABETES? D. Schillinger¹, J. Piette², C. Daher¹, H. Liu¹, A.B. Bindman¹; ¹University of California, San Francisco, CA; ²Center for Health Care Evaluation, Palo Alto, CA

PURPOSE: Low levels of functional health literacy (FHL) are common among public hospital patients with chronic conditions. Identifying patients with low FHL may improve care at the risk of disclosing a stigmatizing issue. We evaluated the acceptability and utility of FHL screening among type 2 diabetes patients (DM2).

METHODS: This study was a randomized controlled trial of FHL screening. We measured FHL for all patients with the short Test of Functional Health Literacy in Adults. We enrolled 155 patients who had low FHL (scores <23), DM2, spoke English or Spanish, and were cared for by one of 61 primary care physicians (PCPs) at a public hospital. PCPs in the intervention group were alerted prior to the visit when the patient had low FHL; control PCPs were not. We surveyed patients regarding the acceptability of FHL screening. We surveyed PCPs regarding their satisfaction with the visit, communication techniques, and self-rated effectiveness. We also asked intervention PCPs to report their prior estimates of the patient's FHL and the extent to which knowledge of the patient's FHL would impact future diabetes care. Post-visit survey data were available for both patients (141) and physicians (152). Post-visit differences between intervention and control patients and physicians were examined, controlling for the clustering of patients by physicians.

RESULTS: Nearly all patients (95%) felt that FHL screening was useful; of these, 99% felt it was important to share this information with their PCP. After adjusting for clustering, PCPs who received FHL feedback felt less trained and confident in caring for their patient than control physicians (p=.02), were more likely to involve patient's family or friend in decision-making (p=.03), or use pictures and diagrams to promote understanding (p=.08). However, intervention and control PCPs had similar self-rated effectiveness (p=0.66). Intervention PCPs accurately estimated the FHL of patients with inadequate FHL only 30% of the time. Intervention PCPs believed that FHL screening was clinically useful (63% of visits), would result in improved medication adherence (63%), and would improve future DM2 care (58%).

CONCLUSION: Patients with DM2 and low FHL found FHL screening acceptable. PCPs, despite working in a setting with high prevalence of FHL problems, frequently overestimated patients' FHL. FHL feedback was associated with greater feelings of PCP inadequacy and use of recommended communication strategies, but no differences in physician self-rated effectiveness. Most PCPs believed FHL screening was clinically useful. Further research should explore the impact of FHL screening combined with provider communication training on patient-centered measures and clinical outcomes.

VEGETARIAN DIETS AMONG ADULTS IN THE UNITED STATES. C. Sciamanna¹, M. Flynn¹; ¹Brown University, Providence, RI

PURPOSE: Our goal was to identify the prevalence of self-reported vegetarian diets in the United States.

METHODS: We examined differences between 270 adult vegetarians and 810 randomly selected adult controls from the 1994–1996 Continuing Survey of Food Intakes of Individuals. **RESULTS:** Only 2.2% of the sample reported eating a vegetarian diet was 43% (116) of self-reported vegetarians reported eating some meat (beef, pork, poultry or fish) during the 24-hour food recall surveys. Overall, reports of vegetarian diet were more likely in women (67.8% of vegetarians v. 49.0% of controls; p<.001) and “other” races (12.6% of vegetarians vs. 7.7% of controls; p<.05). Medical history was strongly associated with reporting a vegetarian diet. Among those who were vegetarian, 22.3%, 16.3% and 6.7% reported a history of high cholesterol, heart disease and stroke compared to 16.6%, 9.2% and 1.7% of controls, respectively (all comparisons significant at p<.01). Self-report of other risk behaviors was also significantly associated with reporting a vegetarian diet; 16.7% and 59.5% of vegetarians reported current smoking and current vigorous exercise at least two times per week compared to 23.2% and 50.9% of controls, respectively. No differences were seen between vegetarian and non-vegetarian subjects with respect to age, body mass index, household income, education level, region of country and self-reported health status.

CONCLUSION: In conclusion, self-report of a vegetarian diet is uncommon among U.S. adults and is strongly associated with gender, chronic medical conditions, and the self-report of other health-promoting behaviors.

EPIDEMIOLOGY OF DIET COUNSELING BY PHYSICIANS. C. Sciamanna¹; ¹The Miriam Hospital and Brown University, Providence, RI

PURPOSE: We analyzed a representative sample of 19,058 visits from the 1998 National Ambulatory Medical Care Survey (NAMCS) to examine the frequency that physicians reported counseling patients about their diet.

METHODS: Patients with diabetes, coronary heart disease (CHD), hyperlipidemia, obesity, and hypertension were identified using a combination of visit codes, diagnoses and current medications from forms completed by physicians after patient encounters. Independent determinants of diet counseling were evaluated using multiple logistic regression.

RESULTS: Overall, diet counseling was performed during 12.3% of all visits and was more common among patients who were middle-aged (45–64 years) (13.4% vs. 11.8% for other age groups; p<.05) nonwhite (14.2% vs. 12.3%; p<.01), presenting for routine chronic care or a routine physical vs. acute care (14.7% and 18.5%, respectively: vs. 9.7%; p<.01), for patients with diabetes (33.8%), obesity (55.8%), hyperlipidemia (36.2%), and hypertension (36.6%) (p<.01 for all comparisons), if the physician was a primary care provider (22.0%), cardiologist (27.0%) or obstetrician/gynecologist (16.0%) vs. other providers (5.8%, p<.001), and if the patient was also counseled about physical activity (67.5% vs. 11.5%; p<.001). These effects persisted after multiple logistic regression analyses controlling for potential confounders. Gender, coronary heart disease history, region of the country and insurance type were not associated with diet counseling.

CONCLUSION: In conclusion, health care providers primarily use diet counseling as a treatment for diet-related illnesses, rather than as primary prevention, where it was noted that fewer than 1 in 5 patients were counseled about their diet during routine physicals.

TESTING STRATEGIES FOR DETECTION OF A LUPUS ANTICOAGULANT: A DECISION ANALYSIS. J.B. Segal¹, M. Petri¹, H.P. Lehmann¹, T.S. Kickler¹; ¹Johns Hopkins University School of Medicine, Baltimore, MD

PURPOSE: Clinicians often screen patients with a history of thrombosis or a prolonged activated partial thromboplastin time (aPTT) for the presence of a lupus anticoagulant (LA). This testing is typically with a sensitive activated partial thromboplastin time (aPTT) and dilute Russell viper venom time (dRVVT), followed by mixing studies and confirmatory tests. With decision analysis, we investigated the optimal testing strategy for detecting LA in three clinical settings.

METHODS: A decision-tree was constructed with 12 strategies, using a combination of aPTT and dRVVT with confirmatory tests, tissue thromboplastin time (TTI), platelet neutralizing procedures, and mixing studies. Probabilities of adverse events, utilities of these events, and costs were obtained from a literature review. Sensitivity and specificity of each strategy was calculated by testing 90 healthy people and 77 patients, with true positives defined as per the algorithm recommended by the International Society on Thrombosis and Haemostasis.

RESULTS: For healthy people with a prolonged aPTT, the optimal strategy is not to test for LA and assume that no patient has LA. For patients with systemic lupus erythematosus (SLE), it is cost saving to test with TTI alone if the cost of the test is less than \$13 (base case - \$9), or if the prevalence of LA in the population is >33% (base case-35%). For patients with past thrombosis or fetal loss, it is cost saving to avoid testing and assume no patient has LA, or to use TTI alone if the expected prevalence of LA is >33% (base case-9%).

CONCLUSION: It is optimal in healthy people and in those with a low likelihood of LA to avoid testing and to assume that these patients do not have LA. However, for patients with SLE, or for patients with past thromboses and a high likelihood of LA, TTI is a cost saving test. These results suggest that the current strategy for detecting LA needs to be modified.

PREVALENCE OF THYROID DISEASE IN CANCER PATIENTS. D. Shani¹; ¹Lenox Hill Hospital, NY, NY

PURPOSE: symptoms of hyperthyroidism and hypothyroidism include weight changes, fatigue which can be confused with symptoms of cancer. very few data exist about the prevalence of thyroid disease in cancer patients. This was a prospective study testing thyroid functions in patients with various diagnosis of cancer in order to assess the prevalence of thyroid disease in those patients.

METHODS: cross sectional study was conducted at the Comprehensive cancer center of Saint Vincents Hospital in NYC. Patients were recruited to have thyroid function done at the same time other blood work was done. Later data was collected from the patients chart regarding their type of cancer, the stage of their disease, and type of therapy they were receiving (chemotherapy, radiotherapy, hormonal therapy, etc) Thyroid function were performed by the hospital laboratory using standard kits. TSH normal range 0.49–4.67 Free T4 normal range 0.17–1.85

RESULTS: Total of 158 cases analyzed to date. 44 cases of breast cancer, 25 cases of gastrointestinal cancer, 7 cases of lung cancer, 60 cases of hematological malignancies, and 22 cases of other types of cancer (ovarian, prostate, ks, etc). The mean age of patient was 57 y/o (range 28–86). There were 85 females and 73 males in the study. 17 patients had abnormally low TSH suggesting subclinical hyperthyroidism and 1 patient had high T4 suggesting Thyrotoxicosis. 6 patients had elevated TSH and 3 had low T4 suggesting hypothyroidism. In total 26 cases (16%) of abnormal thyroid function in the population studied.

CONCLUSION: The prevalence of thyroid disease in cancer patients is significant, and since the symptoms mimic those of cancer, both internists and oncologists caring for the patients should be aware of that.

COMMUNICATION ABOUT SIDE EFFECTS OF PRESCRIPTION DRUGS: DO PHYSICIANS UNDERSTAND PATIENTS EXPECTATIONS? M. Shama¹, M.C. Mosier², D.K. Ziegler², D.R. Calkins³; ¹University of Missouri at Kansas City, Overland Park, KS; ²Kansas University Medical Center, KC, KS; ³Harvard University, Boston, MA

PURPOSE: Writing a prescription for a new drug is a common event in medical practice. The expectations of patients for information about the side effects of prescription drugs and the understanding of those expectations by physicians has not been well studied.

METHODS: We surveyed 2,500 adult patients seen in the ambulatory clinics of a university hospital. We asked patients about their desire for information regarding the side effects of prescription drugs. We asked 190 physicians practicing in the same clinics about their perceptions of patients' expectations for this information. We also collected information on age, gender and level of education for patients and age, gender and specialty for physicians. We found significant differences in responses of patients and physicians.

RESULTS: Please see the table for results.

CONCLUSION: Physicians underestimate patients' desire for information about drug side effects. Communication on this topic can be improved by a better understanding of patients' expectations.

Results

	Survey Item	Patient Response	Physician Response	- P Value
1.	Patients want to be told about all side effects, no matter how rare.	76%	24%	< .001
2.	Patients want to be told about any serious side effect.	83%	66%	< .001
3.	Physicians should give the same detailed information to all patients	68%	23%	< .001
4.	There are occasions when physicians are justified in withholding information	27%	50%	< .001

COMMUNICATING TREATMENT BENEFIT INFORMATION TO PATIENTS: NUMBER NEEDED TO TREAT (OR TRICK?)

S.L. Sheridan¹, M.P. Pignone¹, C.L. Lewis¹; ¹University of North Carolina at Chapel Hill, Chapel Hill, NC

PURPOSE: Researchers have suggested that patients may understand treatment benefits better when they are presented as numbers needed to treat (NNT) rather than as absolute or relative risk reductions. We sought to determine if NNT helps patients interpret treatment benefits better than absolute risk reduction (ARR), relative risk reduction (RRR) or a combination of all three of these risk reduction presentations (COMBO).

METHODS: We surveyed 357 men and women, ages 50 to 80, who presented for care at a university internal medicine clinic. After answering three questions assessing their ability to handle numbers (numeration skill), subjects were asked to (1) state which of two drug treatments for a hypothetical disease Y provided more benefit, and (2) calculate the effect of drug treatment on a patient with a given baseline risk of disease. Risk information was presented to each subject in one of four randomly allocated risk formats—NNT (100 versus 250 people just like you would need to be treated for 5 years for a benefit against disease Y to be seen in one of you), ARR (treatment reduces the chance that you will develop disease Y by 10 versus 4 out of 1000 over the next 5 years), RRR (treatment reduces the chance that you will develop disease Y by 25% versus 10% over the next 5 years), or COMBO.

RESULTS: Subject ability to interpret treatment benefits varied significantly with the risk presentation format they received. Patients correctly interpreted the treatment benefit most often when it was presented in the RRR format: when asked to state which of two treatments provided more benefit, subjects who received treatment benefit information in the RRR format responded correctly 60% of the time, whereas subjects receiving the COMBO, ARR, and NNT presentation formats responded correctly only 43%, 42%, and 30% of the time, respectively ($p < 0.01$). Most patients were unable to calculate the exact effect of drug treatment on a patient with a given baseline risk of disease, although subjects receiving the RRR and ARR formats responded correctly slightly more often (21% and 17%, respectively, compared to 7% for COMBO and 6% for NNT, $p < 0.01$). Higher numeracy skills were associated with increased ability to interpret treatment benefits: 88% of subjects who answered all three numeracy questions correctly stated which of two treatments provided more benefit versus only 63% of subjects who answered two questions correctly and 35% of subjects who answered one or no questions correctly ($p < 0.01$). This association persisted when subjects were asked to calculate the effect of drug treatment on a patient with a given baseline risk of disease.

CONCLUSION: Patients are best able to interpret the benefits of treatment when they are presented in a RRR format. NNT is often misinterpreted by patients and should not be used alone to communicate risk to patients.

PHYSICAL AND MENTAL IMPAIRMENT AMONG VETERANS WITH RECENT DEPRESSIVE SYMPTOMS

S.E. Sherman¹, E. Yano², D. Mittman², J.T. Braslow³, L.V. Rubenstein⁴; ¹University of California, Los Angeles, Sepulveda, CA; ²Department of Veterans Affairs, Sepulveda, CA; ³UCLA; ⁴RAND, Sepulveda, CA

PURPOSE: Depression is very common, particularly in primary care. While depression in otherwise healthy young patients is rather straightforward, it is less clear how to manage depressed patients with multiple medical problems. We assessed the level of physical and mental health impairment in primary care patients who had recently had depressive symptoms.

METHODS: We identified two samples of veterans from the Primary Care Clinic at the VA Sepulveda Ambulatory Care Center who had depressive symptoms during a 7-month period: 290 patients whose primary care encounter form had a depression-related diagnosis (PC group) and 130 patients seen in primary care by Consultation L group). These patients all were sent a letter from their attending physician, inviting them to participate in a study of assisted referral back to primary care for patients who were still depressed. 19 PC group patients (7%) and 23 C&L group patients (18%) declined to participate and over half never responded (65% PC, 52% C&L). Respondents (82 PC, 39 C&L) were contacted 6 months after their initial diagnosis of depressive symptoms. The interviewer-administered initial survey included the Veterans Short Form-12 (SF-12V) and three questions on where they received health care. Respondents

not currently seeing a mental health specialist were asked to complete a more detailed survey, which included measures of depression, anxiety, alcohol use and sociodemographics.

RESULTS: Completed interviews were available for 86 patients (53 PC, 33 C&L). 14 patients who agreed to participate could not be reached and 20 interviews were still pending. The average Mental Component Score for respondents was 36.4. While 40% had no mental health impairment, 17% had impaired mental health (MCS 31–40) and 43% had severely impaired mental health (MCS 30 or less). The average Physical Component Score for respondents was 37.3. The breakdown was similar to that for mental health: 38% had no physical health impairment, 24% had impaired physical health (PCS 31–40), and 37% had severely impaired physical health (PCS 30 or less). 20% of respondents were both physically and mentally impaired and another 14% were severely impaired physically and mentally. Of those with current mental health impairment, the prevalence of physical health impairment was similar between the group seeing psychiatry and those not currently seeing psychiatry.

CONCLUSION: Six months after having depressive symptoms identified, most veterans still have profoundly impaired mental health. Of note, the degree of physical health impairment is equally profound. This suggests that for many depressed patients in primary care, treatment models need to account for severe disability both physically and mentally.

HEALTH STATUS AND UTILIZATION OF VETERANS WHO SMOKE

S.E. Sherman¹, E. Yano², A.B. Lanto², M. Wang², M.L. Lee², B.S. Mittman², L.V. Rubenstein³; ¹University of California, Los Angeles, Sepulveda, CA; ²Department of Veterans Affairs, Sepulveda, CA; ³RAND, Sepulveda, CA

PURPOSE: Smokers tend to be sicker and have more health care utilization than non-smokers. We assessed whether or not this same relationship holds among veterans, who themselves tend to be sicker and have more health care utilization than the general public.

METHODS: As part of the baseline patient survey for a multi-site trial of evidence-based quality improvement for smoking cessation, we interviewed veterans at 18 sites in the Southwestern U.S. The survey, conducted by computer-assisted telephone interview among patients with at least 3 primary care visits in the prior 18 months, covered smoking history, health habits, health status and demographics. Questions were adapted from previously validated sources, including the California Tobacco Survey, Medical Outcomes Study, CES-D (depression), AUDIT (alcohol abuse), and others. Outpatient utilization data for FY 1999 were extracted from the VA Outpatient File at Austin and grouped by type of visit. Inpatient utilization data were obtained from the Patient Treatment File at Austin. Comparisons were done using chi-squared tests and ANOVA. For this preliminary analysis, we also analyzed the utilization data splitting subjects into those under 65 ($n=3745$) and those 65 or older ($n=3961$).

RESULTS: Of the 7,706 subjects interviewed, current smokers were more likely than former smokers or those who never smoked to be younger, less active, and divorced (all $p < 0.001$). Current smokers were more likely than the other two groups to have severe mental health impairment (21% vs. 12% vs. 14%, $p < 0.001$) and severe physical health impairment (55% vs. 48% vs. 46%, $p=0.026$). Current smokers were less likely than former smokers or those who never smoked to report being in excellent or very good health (16% vs. 22% vs. 26%, $p < 0.001$). Among current smokers, approximately half reported they never drink alcohol, but 18% met AUDIT criteria for alcohol abuse. Forty-three percent of current smokers met CES-D criteria for depression at the time of the interview. Current smokers, former smokers, and those who never smoked did not differ in the total number of outpatient visits or for emergency room or general medicine visits. Current smokers did average more mental health and social work visits and fewer medical subspecialty visits than former smokers or those who never smoked ($p < 0.001$). The rate and length of acute care hospitalizations did not differ among the three groups, but smokers had more nursing home admissions. When we analyzed the data separately for subjects over 65 and under 65, the utilization differences tended to occur predominantly in the half of the sample under age 65.

CONCLUSION: While smokers had higher outpatient mental health use, they did not have higher utilization for most other categories. They tended to have worse functional and health status than former smokers and those who never smoked.

GENDER DIFFERENCES IN HEALTH CARE UTILIZATION AND COSTS FOR MEDICARE BENEFICIARIES WITH CANCER

L.R. Shugart¹, C. Bird¹, J. Lynn¹; ¹RAND, Santa Monica, CA

PURPOSE: Medical care at the end of life is quite costly; over one quarter of Medicare payments support care of beneficiaries who will die within a year. The aggregate costs and utilization patterns of men and women facing death may not be the same due to differences in their age, preferences, and cultural stereotypes. We analyzed Medicare services utilization and payments for men and women dying of cancer to determine whether differences occur within 5-year age strata. We selected cancer because of its clear diagnosis, generally accepted standards of care and limited treatment options for which we do not expect gender differences.

METHODS: We analyzed 1993 through 1998 Medicare claims and eligibility data for an overall 0.1% random sample (a 2% sample of an initial 5% sample) of beneficiaries who: were age 65 and over with a diagnosis of cancer but without a diagnosis of end stage renal disease, died in the target time interval and had one year of continuous data before death. Beneficiaries were identified as having a cancer diagnosis if cancer accounted for the plurality of physician spending in the year of death. Age strata were based on the beneficiary's age at death.

RESULTS: Across all age strata, women were more likely to use skilled nursing facility care than men (35% vs. 27%, $p < .10$) and hospice care (42% vs. 37%, $p < .10$). With the exception of 80–84 year-olds, Medicare total reimbursements for hospitalizations were higher for women than for men (\$18,531 vs. \$17,166, $p < .05$, overall). The beneficiary's responsibility for Medicare costs (e.g., deductibles and co-payments) was higher for men aged 65–69 (\$5,188 vs. \$4,863, $p < .10$) and for women in the 70–74 age stratum (\$5,163 vs. \$4,403, $p < .05$) and in the 75–79 age stratum (\$4,501 vs. \$4,298, $p < .10$).

CONCLUSION: Observed gender differences in aggregate spending and utilization are attenuated by age stratification. The persisting differences in use of skilled nursing facilities and hospice and higher hospital costs represent both differences in care and coverage under

Medicare. These findings require more study, such as analyses stratified by other primary diseases, type of cancer, and type of service. Our data support the contention that social setting, cultural setting, and patient preferences may combine to create modest but persistent differences in treatment patterns by gender.

Selected Characteristics of Medicare Beneficiaries by Gender

	Men	Women	Signif.
Total Deaths	1885	2373	
% Cancer Deaths	23.4	18.1	p=.16
% w/Cancer Hospitalized	86.6	90.0	p=.28
Avg \$ in LYOL	\$26,883	\$29,326	p=.04

CERVICAL CANCER SCREENING IN THE ELDERLY: IS THE BURDEN OF PREVENTION GREATER THAN THE BURDEN OF DISEASE? B.E. Sirovich¹, D.J. Gottlieb², E.S. Fisher²;

¹VA Outcomes Group, Department of Veterans' Affairs Hospital, White River Junction, VT; ²Center for the Evaluative Clinical Sciences, Dartmouth Medical School, Hanover, NH

PURPOSE: Although cervical cancer deaths account for fewer than one percent of cancer deaths among women aged 65 and older, most elderly women report continuing to undergo periodic Pap smear screening. We sought to describe the burden of downstream testing following Pap smears in elderly women who undergo screening.

METHODS: Using three years of Medicare Part B 5% Files (1995–1997), we identified women over age 65 who underwent Pap smear screening during a 16-month period in 1996 and 1997. In order to exclude women undergoing follow-up Pap smears or surveillance smears (women with a previous procedure or diagnosis suggestive of cervical dysplasia or malignancy), we required a one-year observation period without such testing prior to the screening Pap smear for each woman. We measured downstream events including subsequent Pap smears, colposcopies, and other diagnostic and therapeutic surgical procedures (e.g. conization) during the 8 months following the screening Pap smear.

RESULTS: In 1996, almost two and a half million female Medicare beneficiaries over age 65 underwent Pap smear screening. For every 100,000 women screened, 3020 had at least one additional Pap smear, 669 underwent colposcopy, and 183 had other surgical procedures during the eight-month period following the screening Pap smear. Rates of downstream interventions were similar for women aged 66–75 and 76–85, and declined only modestly in women age 86 and older. Overall, for every 100,000 women screened, 3497 experienced downstream interventions within 8 months of the initial Pap smear. For comparison, according to population-based surveillance data (SEER,1996), 16 women per 100,000 aged 65 and older are diagnosed with cervical cancer each year, while nine are expected to die from the disease.

CONCLUSION: Relative to the small number of expected cases of cervical cancer, women aged 65 and older continue to be screened in large numbers and undergo substantial follow-up testing in response to abnormal Pap smear results. Whether this testing reduces the subsequent risk of death from cervical cancer is unknown.

THE FREQUENCY OF PAP SMEAR SCREENING IN THE UNITED STATES. B.E. Sirovich¹, H.G. Welch¹;

¹VA Outcomes Group, Department of Veterans' Affairs Hospital, White River Junction, VT

PURPOSE: Most US professional medical societies and the national health systems of all other industrialized nations recommend that most women need not undergo Pap smear screening annually. There are no data, however, regarding the frequency at which women actually undergo screening. We sought to describe the frequency of cervical cancer screening in the United States.

METHODS: We used the 1998 Behavioral Risk Factor Surveillance System of the Centers for Disease Control and Prevention (CDC), a cross-sectional population-based telephone survey conducted annually on a random sample of civilian non-institutionalized adults. We focused on female respondents aged 18 and older (n=86,715). For women who reported ever having had a Pap smear, we estimated screening intervals based on the timing of a woman's most recent Pap smear. We assumed that each woman either was not being regularly screened, or was being screened at one of three discrete screening intervals (every year, every 2 years, or every 3 years). **RESULTS:** The vast majority (93%) of American women report having had at least one Pap smear in their lifetime. Among women who have not had a hysterectomy and who have been screened at least once, 90% report having had a Pap smear within the past three years. Based on the reported time since the last Pap smear, we estimate that 59% of women undergo Pap smear screening annually, 18% have a two-year screening interval, 13% are screened every three years, and 10% are not being screened regularly. Even the very elderly report regular screening — an estimated 34% percent of women aged 75–84 and 18% of women aged 85 and older undergo annual Pap smear screening. Extrapolating these data to the country as a whole, more than 45 million American women undergo Pap smear screening annually.

CONCLUSION: The majority of American women are being screened for cervical cancer too frequently. Lengthening the screening interval would not only reduce the number of pelvic exams, but would also reduce follow-up testing for abnormal smears and the volume of specimens that cytotechnologists are required to read.

STATINS AND THE INCIDENCE OF ABNORMAL TRANSAMINASE AND CREATINE PHOSPHOKINASE. C.C. Smith¹, L.I. Bernstein¹, R.B. Davis¹, R.H. Shmerling¹;

¹Beth Israel Deaconess Medical Center, Harvard Medical School, Boston, MA

PURPOSE: Recommendations for monitoring transaminase (alanine and aspartate aminotransferase, ALT and AST) and creatine phosphokinase (CPK) in patients taking HMG-CoA reductase inhibitors (statins) vary and are not based on data from clinical practice.

We performed a study to determine the utility of routine screening of transaminase and CPK values in patients taking statin medications.

METHODS: We performed a retrospective on-line medical record review of the primary care practice at Beth Israel Deaconess Medical Center. A computerized search identified all patients at our institution with a statin on their medication list, as well as their ALT, AST and CPK values for 1998. Patients were separated into four categories based on which laboratory data had been measured during the year: CPK only, transaminase only, both CPK and transaminase or neither. From each of these four categories, 45 patient records were randomly selected for review to verify that they were followed at this primary care practice and were on a statin. The random samples were used to identify the rate of physician monitoring of transaminase and CPK values within this practice. We reviewed the on-line medical records of all patients within this practice on statins who had significantly abnormal test results (defined in prior trials of statin therapy as greater than 3x the upper limit of normal for transaminases and greater than 10x the upper limit of normal for CPK) to determine the relationship to statin therapy and outcome.

RESULTS: 4556 patients at our institution had a statin on their medication list. Of the 180 random charts reviewed, 26% (corresponding to 1189 of the total cohort) were receiving care from this primary care practice and were taking a statin. Among the patients whose charts were reviewed, 53% were female with a mean age of 61 years. These data also demonstrated that physicians within our practice monitored serum transaminase values (defined as measurement of either AST or ALT during 1998, or within six months of starting a statin) in 84% of patients; physicians monitored CPK values in 54% of patients. 29% of the patients within this practice were on a dose greater than the recommended starting dosage. Of all the patients in this practice who had been monitored, 12 (1.2%) had a significant elevation of AST or ALT and 6 (0.9%) had a significant elevation of CPK. However, after a detailed chart review, none of these abnormalities were attributable to statins.

CONCLUSION: In this study of statin use in a primary care practice, no cases of significantly abnormal transaminase or CPK values attributable to statins were discovered through routine monitoring. This questions the necessity of routine measurement of transaminase and CPK in all patients taking statin medications.

COST-EFFECTIVENESS OF VARICELLA VACCINATION TO PREVENT HERPES ZOSTER: AN EXPLORATORY ANALYSIS. K.J. Smith¹, M.S. Roberts²;

¹Mercy Hospital of Pittsburgh, Pittsburgh, PA; ²Center for Research on Health Care, University of Pittsburgh School of Medicine, Pittsburgh, PA

PURPOSE: Varicella vaccination for persons 60 years of age and older for prevention of herpes zoster is being studied in randomized, controlled trials, however it is unclear what the cost-effectiveness of this intervention might be.

METHODS: We developed a Markov simulation model to estimate the incremental cost-effectiveness of vaccination vs. no vaccination in 60-year-olds. The Markov cycle length was one month with a lifetime time horizon. We took a societal perspective, discounting costs and benefits at 3%/yr and using quality of life utility values to calculate costs per quality adjusted life year (QALY) gained. Vaccinated patients had immunologic boosting for 56 months (median) and had no risk of zoster during that time. Thereafter, vaccinated patients had half the usual zoster and postherpetic neuralgia (PHN) incidence rates (or post-boost effect=50%). Age-specific zoster incidence was that of Rochester, Minn. Half the zoster patients received antiviral therapy; 20–40% of all zoster patients developed PHN, depending on their age. Multiple sensitivity analyses were performed.

RESULTS: In the baseline analysis, vaccination gained 0.0004 QALY compared to no vaccination at a cost of \$51.13 or \$143,000/QALY gained. Vaccination is least expensive at age 66, costing \$129,000/QALY gained. In 60-year-olds, costs/QALY gained is greater than \$50,000 with wide variation of: immunologic boosting duration, post-boost effect, PHN probability or duration, herpes zoster duration or utility, or antiviral therapy or hospitalization parameters. Vaccinating 60-year-olds costs \$103,000 or \$61,500 per QALY when lifetime zoster risk (baseline 11.0%) increased to 1 in 7 (14.3%) or 1 in 5 (20%) respectively. If lifetime zoster risk is 20%, costs/QALY gained are less than \$50,000 if post-boost effect is <38%, median PHN duration is >114 days (baseline 90), average PHN utility is <0.76 (where 0=death and 1=perfect health), or vaccination cost <\$54.21 (baseline \$61.96, including \$5 administration cost).

CONCLUSION: Varicella vaccination of 60-year-olds is expensive compared to many accepted medical interventions in a model biased toward its use. Results are most sensitive to the lifetime risk of zoster, which varies significantly in different populations.

ASSESSING RISK FOR EARLY RELAPSE TO SMOKING WITH THE PRIME-MD. S.S. Smith¹, M.C. Fiore¹, T.B. Baker¹;

¹University of Wisconsin Medical School, Madison, WI

PURPOSE: The PRIME-MD (Primary Care Evaluation of Mental Disorders; Spitzer et al., 1994) is a valid and reliable assessment tool designed to enable rapid determination of current or past psychiatric disorders in primary care patients. The current analyses were conducted to assess whether or not smokers with current psychiatric diagnoses, as determined with the PRIME-MD, are at higher risk for early relapse to smoking after quitting.

METHODS: The PRIME-MD was administered as part of the study screening procedures in an NIH-funded smoking cessation clinical trial that enrolled 677 smokers. Brief individual smoking cessation counseling was provided prior to quitting and on the quit day; nicotine patches were dispensed on the quit day. The outcome measure in the current analyses was biochemically-verified smoking status at one week post-cessation. PRIME-MD diagnoses included major depressive disorder (MDD), panic disorder (PD), generalized anxiety disorder (GAD), and probable alcohol abuse/dependence.

RESULTS: A total of 182 smokers (27%) were diagnosed with at least one of the four disorders and 30 smokers had two or more diagnoses. 63% of the participants without any current diagnoses were abstinent at one week compared to 20% of participants with MDD (n=133), 5% with GAD (n=20), and 0% with PD (n=3). No statistically significant increase in risk of relapse was observed for 62 participants with alcohol abuse/dependence (50% abstinence rate). Only 7% of participants with two or more disorders were abstinent at one week post-quit.

CONCLUSION: Smokers with current mood and anxiety disorders appear to be at significant risk for early relapse to smoking. The PRIME-MD can provide rapid assessment of these psychiatric diagnoses that are commonly seen in the primary care setting. Identification of smokers with significant depression and anxiety can aid in the development of smoking cessation treatment plans (e.g., including an anti-depressant such as bupropion SR for depressed patients) and can focus attention on the importance of aggressive relapse prevention in these individuals.

PREVENTIVE HEALTH SERVICE USE AMONG A RUSSIAN IMMIGRANT POPULATION.

S. Smoger¹, A. Furr¹, W.P. McKinney¹, C. Hornung¹, C.J. Mundt¹, J.J. Miller¹; ¹University of Louisville, Louisville, KY

PURPOSE: To compare the use of disease screening and preventive services among an immigrant ethnic population with published National and state reports.

METHODS: Adult immigrants from the former Soviet Union relocated into the Jefferson County, KY, area were surveyed by phone. A market research firm hired, trained, and monitored Russian language interpreters who administered the survey to adults with a valid phone number, using a list provided by the resettlement agency. We report comparisons with the 1999 Behavioral Risk Factor and Surveillance System (BRFSS) data (www.cdc.gov).

RESULTS: Of 426 available adults, completed interviews were obtained in 56% (n=241), with mean age 54.2 yrs (18–94), 54% women, and subjects had resided in the county a mean of 7.5 yrs (1–29). 91% had some type of health care coverage 91% (n=220), comparable to reports for BRFSS. Self-rated health was rated poor/fair by 60% (n=241), compared with responses on the 1999 BRFSS of 13.1% National and 21.6% Kentucky. Preventive measures shown in Table [P values for comparison with the study group].

CONCLUSION: Despite having health insurance, these immigrants report high levels of poor/fair health. Compared to the BRFSS data, there is underutilization of immunizations and cancer screening. Outreach efforts are needed to educate this population and target specific preventive services.

	Study Group %, n=241	1999 National BRFSS %	1999 Kentucky BRFSS %
Hypertension	29.5	23.9*	27.5*
Diabetes	8.8	5.6*	6.4*
PneumoVax ever	10.4	18.4**	16.7**
Flu shot ever	24.1	31.3***	30.9***
Stool test,40+	12.0	31.1**	30.8**
Sigmoid ever,40+	23.4	33.7**	28.2***
Dentist last yr	68.9	68.1*	63.0*
Mammogram ever,40+	68.7	86.2**	85.6**
Breast exam ever	62.2	89.0**	88.6**
Pap ever	50.7	95.1**	92.9**
*P=NS	**P=0.001	***P=0.01	****P=0.05**

THE EFFECT OF THE PREVENTIVE MEDICAL MESSAGE ON INTENTION TO CHANGE

BEHAVIOR. P.C. Sorum¹, S. Frileux², E. Mullet³, M. Munoz Sastre²; ¹Albany Medical Center, Albany, NY; ²Université du Mirail, Toulouse; ³Ecole Pratique des Hautes Etudes

PURPOSE: To elucidate the impact of the individual and combined components of a physician's preventive "medical message" on people's intention to adopt more healthy behaviors. The specific risk used in this study was that of developing manifestations of coronary artery disease.

METHODS: 86 young people (ages 20–30) and 64 older people (ages 60–80) in France indicated their degree of intention to adopt a healthy behavior in response to a series of 64 hypothetical messages from their physician about their risk of developing a cardiac event. The messages were all possible combinations of 4 components: the type of cardiac event (angina or a heart attack), the probability of developing it (5, 10, 15, or 20%), the time frame (within 5, 10, 15, or 20 years), and the degree of the patient's control over developing it ("given your family history, it is probable that you could not do much to reduce this risk" or "the reduction of this risk depends entirely on your behavior"). They were asked about adopting 1 of 3 behaviors: a "medical treatment," a strict dietary regimen, or regular exercise. The results were examined graphically and by ANOVA.

RESULTS: All components had a significant effect on the intention to change behavior, but these effects varied according to age. The young people were influenced most by the degree of controllability, the probability, and the nature of the cardiac event; the older people by the time horizon. Even without a promise of controllability, participants were inclined to adopt protective behavior. An increase in probability had less added effect as the probability got larger; nonetheless, the level of intent to change behavior associated with a large probability over a long time (e.g., 20% over 20 years) was higher than that associated with a smaller probability over a short time frame (e.g., 5% over 5 years). The type of protective action had no effect on intention. **CONCLUSION:** The preventive medical message should include the type of event, the probability of developing it, the time frame, and the degree to which the patient can prevent it. The physician should stress the time frame in talking with older people and the other factors with young people.

NATIONAL PATTERNS OF DEPRESSION TREATMENT, 1987 TO 1998. R.S. Stafford¹, E. Macdonald¹, S. Finkelstein²; ¹Massachusetts General Hospital, Boston, MA; ²Massachusetts Institute of Technology, Cambridge, MA

PURPOSE: Physicians' approach to the treatment of depression has changed drastically in the last 15 years with the availability of selective serotonin reuptake inhibitors (SSRIs) in 1988. We

sought to closely examine trends in antidepressant use, as well as assess broader changes in treatment patterns.

METHODS: Using data available from the National Disease and Therapeutic Index (NDTI), a physician survey conducted by IMS Health, we examined trends in antidepressant prescribing from 1987–98. NDTI provides nationally representative diagnostic and prescribing information on patients treated by U.S. office-based physicians. We selected visits by patients reported to have depression. Annual sample sizes varied from 3,901 visits in 1987 to 6,639 in 1998. We examined annual prescribing data to measure: 1) the frequency of visits where depression was noted and 2) the likelihood of specific medications being reported as treatment. **RESULTS:** The NDTI data show a dramatic shift in the treatment of depression between 1987 and 1998. In 1987, tricyclic antidepressants (TCAs) were the predominant drug class prescribed to patients with depression (47% of patients). Among individual antidepressants, the most common were amitriptyline (14%), trazadone (12%), doxepin (8%), and desipramine (6%). In 1989, a year after its introduction, fluoxetine was prescribed to 21% of patients with depression. Despite the introduction and growing use of other SSRIs in the 1990s, fluoxetine has maintained a constant share of patients at around 25%. The introduction of sertraline in 1992, paroxetine in 1993 and more recent SSRIs led aggregate SSRI use to grow to 38% in 1992, 56% in 1994, 60% in 1996 and 75% in 1998. In 1998, fluoxetine (24%), sertraline (21%), paroxetine (17%) and bupropion (6%) were the leading antidepressants. TCAs accounted for only 8% of depression patients in 1998. The prescribing of non-TCA, non-SSRI antidepressants decreased from 13% (chiefly trazadone) in 1987 to 10% (chiefly bupropion) in 1998. The use of benzodiazepines in depression declined from 11% of patients in 1987 to 3% in 1998. The rate of patients with reported depression not receiving an antidepressant decreased from 30% in 1987 to 9% in 1998. The estimated national number of physician visits by patients with depression increased from 14.4 million visits in 1987 to 22.5 million in 1998.

CONCLUSION: The increasing therapeutic dominance of SSRIs has been pivotal to changes in depression treatment. Perhaps because of their effectiveness and side-effect profile, increasing SSRI use also may have contributed to the declining use of benzodiazepines, increased aggregate antidepressant treatment rates, and the increasing frequency of physician visits where depression was reported.

NATIONAL TRENDS IN RECOMMENDED CARDIAC MEDICATIONS. R. Stafford¹, C. Chaisson¹; ¹Massachusetts General Hospital, Boston, MA

PURPOSE: Previous studies suggest that recommended cardiac medications are underutilized. We evaluated recent national patterns of medication use in the ambulatory care setting for warfarin use in atrial fibrillation (AF), beta blocker (BB) and aspirin (ASA) use in coronary artery disease (CAD), and ACE inhibitor use (ACEI, including angiotensin receptor blockers) in congestive heart failure (CHF).

METHODS: We used the 1989–1998 National Ambulatory Medical Care Surveys to identify nationally representative samples consisting of patients with AF, CAD and CHF who lacked specific contraindications to therapy. For AF, 1,370 office visits were identified; for CAD, 8,773 visits (for BB) and 8,808 visits (for ASA); and for CHF, 3,310 visits. We examined time trends in the proportion of visits reporting the selected medications, weighted to reflect national patterns. Logistic regression was used to evaluate the independent predictors of cardiac medication use in 1995–1998, including patient and physician characteristics.

RESULTS: In patients with AF, warfarin use increased steadily from 13% of visits in 1989 to 38% in 1994 to 50% in 1998 (see Table). In CAD patients, BB use increased from 16% in 1989 to 19% in 1994 to 33% in 1998. ASA use in CAD was 11% in 1989 increasing to 22% in 1992 where it plateaued, reaching 25% in 1998. In CHF patients, ACEI use increased slowly from 22% in 1989 to 28% in 1994 to 32% in 1998. Patterns of medication use for 1995–98 varied by patient and physician characteristics. Women were less likely to be taking warfarin for AF (OR=0.35 P=0.06) or ASA for CAD (OR=0.79 P=0.04). Nonwhite CAD patients were less likely to take BB (OR=0.68 P=0.03) or ASA (OR=0.69 P=0.05) than were whites. The oldest and youngest patients tended to be less likely to receive medications than those aged 60–69; this was most dramatic for BB in CAD in those 80+ (OR=0.57 P<0.001). Patients with CAD visiting cardiologists were more likely to receive recommended drugs.

CONCLUSION: Recommended cardiac medication use increased from 1989 to 1998, but some increases have not continued into the late 1990's and have not been uniform across subpopulations. Further adoption of these effective therapies could result in additional benefit for patients with cardiac conditions.

Trends in Percentage of Visits with Recommended Cardiac Medications

	1989	1992	1994	1996	1998	P trend
AF/Warf	13	29	38	34	50	0.07
CAD/BB	16	22	19	25	38	<0.01
CAD/ASA	11	22	24	25	25	<0.01
CHF/ACEI	22	26	28	29	32	0.65

CHRONIC DISEASE MEDICATION USE IN MANAGED CARE AND INDEMNITY HEALTH

PLANS. R. Stafford¹, H. Davidson², H. Miracle-mcmahill³, S. Crawford³, D. Blumenthal¹, S. Davidson⁴; ¹Massachusetts General Hospital, Boston, MA; ²Beth Israel Deaconess Hospital, Boston, MA; ³New England Research Institute, Watertown, MA; ⁴Boston University, Boston, MA

PURPOSE: Systems of health care reimbursement may influence patterns of chronic disease management. We used claims data to evaluate concerns that financial incentives in managed care may deter physicians from the use of expensive and newer medications.

METHODS: We analyzed claims data on 175,000 members of three managed care and indemnity plans in a mid-sized northeastern city. For 1997, adult patients with diabetes mellitus (DM, N=26,915), congestive heart failure (CHF, N=8,231) and asthma (N=10,150) were

identified by ICD-9 codes. Our outcome measure was the pattern of medication use for these conditions. In comparing indemnity and managed care, we used logistic regression to adjust for case-mix: patient age, gender, number of physician visits, Diagnostic Care Group (DxCG) risk score, and a series of DxCG co-morbidity indicators.

RESULTS: With few exceptions, the use of chronic disease medications was more likely among managed care patients. For patients with DM, managed care plans had greater use of sulfonylureas than indemnity (42% vs. 37%), metformin (26% vs. 17%) and troglitazone (9% vs. 6%). In multivariate analysis to adjust for case-mix, these differences remained significant ($p < 0.001$). Insulin use was no different in managed care and indemnity plans. For CHF patients, managed care patients had greater use of loop diuretics (44% vs. 39%), ACE inhibitors/ARBs (50% vs. 39%), and beta-blockers (25% vs. 16%); differences that were significant in multivariate analysis ($p < 0.001$, except loop diuretics [$p=0.003$]). No differences in digoxin use were observed. For asthma patients, managed care had greater use of inhaled corticosteroids (33% vs. 28%), systemic corticosteroids (18% vs. 15%), short-acting beta-agonists (41% vs. 31%), long-acting beta agonists (10% vs. 8%), and leukotriene modifiers (5.3% vs. 3.9%); differences that were significant in multivariate analysis ($p < 0.001$, except for long-acting beta agonists [$p=0.03$] and leukotriene modifiers [$p=0.04$]). No differences were found for cromolyn or methylxanthine use.

CONCLUSION: Chronic disease patients in managed care plans are more likely to receive a broad range of medications. Three of 4 exceptions to this pattern are older medications whose use has been partly supplanted by newer therapies. The general pattern of greater medication use in managed care could reflect patient selection, reduced out-of-pocket cost barriers, or a strategy to avoid other, more costly services. Managed care plans do not appear to withhold expensive, newer medications.

BREAST SURGERY FOR BENIGN DISEASE: CONSERVATIVE PRACTICE OR MISSING EVIDENCE? C.A. Steiner¹, C. Case², M. Johantgen³, ¹Agency for Healthcare Research and Quality, Rockville, MD; ²Georgetown University, D.C.; ³University of Maryland, Baltimore, MD

PURPOSE: The emphasis on screening and minimally invasive breast techniques has led to an increase in detection of both benign and malignant breast disease. Optimal management for benign breast disease remains uncertain, and population-based studies of surgical interventions are lacking. This study uses population-based administrative data to examine use of excisional breast procedures across 5 states and 7 years, comparing benign to malignant breast diagnoses.

METHODS: Data are from the Healthcare Cost and Utilization Project, a federal-state-industry partnership in administrative data. Ambulatory surgery(AS) and inpatient(IP) data are used from CO, CT, MD, NJ and NY for 1990–1996. All women with a lumpectomy(ICD-9 85.21) or subtotal mastectomy(85.22, 85.23) were extracted. Lumpectomy(LUMP) and subtotal mastectomy(STMAS) were further classified as either benign disease or cancer using diagnoses codes. Overall IP and AS age-adjusted rates per 100,000 women for each state, year and procedure were calculated using direct standardization.

RESULTS: The rates of breast conserving surgery(LUMP+STMAS)for breast cancer increased across all five states, from between 33–71/100,000 women in 1990 to between 70–140/100,000 women in 1996. The rate of LUMP is 3 to 4 fold higher for benign disease as compared to that for cancer in every state. For instance, in 1996, the rate of LUMP in NY for cancer was 91/100,000 women, while for benign disease, 271/100,000. However, in contrast to cancer, the rate of LUMP for benign disease is relatively stable across all of the states and years. The overall rate of STMAS is one-tenth that of LUMP in all the states. Similar to LUMP, the rate of STMAS for benign disease was stable across time. However, there was a great deal of variation among the states in use of STMAS for benign breast disease. The rate varied from a negligible rate in CO, to as high as 50/100,000 women in CT in 1996. In addition, for CT and NJ there were equal rates of STMAS for benign and cancer diagnoses(50 and 27/100,000 respectively), while the STMAS rate for cancer was higher in the other 3 states.

CONCLUSION: This is the first population-based study comparing interventions for benign breast disease to cancer. This study demonstrates unexpectedly high rates of surgical interventions (LUMP and STMAS) for benign breast disease. These rates may be explained by women and physician fear of missed cancer diagnoses, lack of best practices, upcoding of procedures, or AS data not capturing cancer at discharge for short stays. However, the latter would indicate a lack of two-stage decision-making process for women.

NOT TAKING MEDICATIONS BECAUSE OF COST: RISK FACTORS AND THE IMPACT OF PRESCRIPTION COVERAGE IN OLDER AMERICANS M.A. Steinman¹, L.P. Sands², K.E. Covinsky², ¹University of California, San Francisco and San Francisco VA Medical Center, San Francisco, CA; ²University of California, San Francisco, San Francisco, CA

PURPOSE: Anecdotal reports have highlighted the stories of patients who skip doses or otherwise avoid using their medications because they cannot afford the expense. However, little is known about which elderly patients without prescription coverage are at highest risk of not taking their medications because of cost, and how prescription coverage modifies this risk.

METHODS: We performed a cross-sectional study of 4896 subjects in the 1995–1996 wave of the Survey of Asset and Health Dynamics Among the Oldest Old (AHEAD), a population-based survey of Americans age 70 years and older. Subjects were asked the extent of their prescription coverage, and whether they had taken less medicine than prescribed for them because of cost over the prior 2 years. We used bivariate and multivariate analyses to identify risk factors for restricting one's use of medications because of cost in subjects who lacked prescription coverage. Among subjects with these risk factors, we then examined the effect of prescription coverage on rates of medication restriction.

RESULTS: Of 4896 seniors who regularly used prescription medications, 39% had no prescription coverage, 44% had partial coverage, and 17% had full coverage. Not taking medications because of cost was reported by 8%, 3%, and 2% of patients with no, partial, and full prescription coverage, respectively ($p < .01$ for trend). Among subjects with no prescription coverage, the strongest independent predictors of foregoing medications because

of cost were ethnicity (prevalence of restriction 21% for minority subjects, 6% for white subjects, $P < .01$), income (16% for annual income $< \$10,000$, 8% for \$10,000 to \$19,999, 4% for $\geq \$20,000$, $P < .01$), and out-of-pocket prescription drug costs (13% for monthly cost $> \$100$, 7% for \$20 to \$100, 3% for $< \$20$, $P < .01$). Prescription coverage markedly reduced the rate of medication restriction in these high-risk groups. For example, rates of medication restriction in minority and low-income subjects were 21% and 16% among those with no coverage, 8% and 8% among those with partial coverage, and 4% and 2% among those with full coverage, respectively ($p < .001$). Almost half (43%) of minority patients with low income, high drug costs, and no prescription coverage reported restricting their use of medications, compared with 12% of patients with these three risk factors and partial prescription coverage.

CONCLUSION: Restricting one's use of medications because of cost is common in vulnerable groups of seniors who lack prescription coverage. Among these high-risk groups, prescription coverage markedly reduces the rate of medication restriction.

ERRORS IN ISOLATION: RECORDED VITAL SIGNS FOR PATIENTS WITH COMMUNICABLE INFECTIONS. H.T. Stelfox¹, D. Redelmeier², ¹Harvard University, Cambridge, MA; ²University of Toronto, Toronto

PURPOSE: Hospital contact isolation policies are designed to prevent the nosocomial transmission of infectious diseases, but may inadvertently promote the neglect of isolated patients. We tested whether the frequency and quality of recorded vital signs differed between isolated and non-isolated inpatients.

METHODS: We identified consecutive adults admitted to a large Canadian teaching hospital between January 1, 1999 and January 1, 2000 who were placed in contact isolation during their hospital stay ($n=81$). Controls were selected by identifying the two patients who occupied each isolated patient's hospital bed immediately before and after their admission ($n=162$). Vital signs recorded in the medical records were compared for the two groups using *t* tests. Adjustments for age, gender, Charlson Comorbidity Score, living status (nursing home vs. home) and admitting service (medicine vs. surgery) were done using propensity score matching.

RESULTS: We found no difference in the frequency of daily vital signs ordered for isolated and non-isolated patients (2.8 vitals/day vs. 3.2 vitals/day, $p > 0.20$). However, isolated patients had a significantly higher percentage of their vital signs incompletely recorded (missing at least one of heart rate, blood pressure, respiratory rate or temperature) as compared to non-isolated patients (17.3% vs. 12.7%, $p=0.03$). Both groups had surprisingly high percentages of their respiratory rates recorded as twenty (53.2% vs. 50.9%, $p > 0.20$).

CONCLUSION: Patients placed in contact isolation appear to have their vital signs ordered as frequently though not recorded as completely as non-isolated patients. Differences in the recording of vital signs may reflect underlying differences in the quality of care received by isolated patients and warrant further investigation.

PATIENT AND PHYSICIAN PERCEPTIONS REGARDING THE EFFECT OF PRAYER AND SPIRITUALITY ON HEALTH - ARE WE CONNECTING? B.E. Susi¹, D. Bynum², L.J. Schultz³, E. The Respect Group², ¹Carolinas Healthcare System, Charlotte, NC; ²University of North Carolina at Chapel Hill, Chapel Hill, North Carolina; ³Emory University, Norcross, GA

OBJECTIVE: Several studies have previously reported that patients want their physicians to address spiritual issues as part of their health care. It has also been reported that patients feel their individual spiritual faith can help them recover from illness. Surveys of physicians have shown that they often do not routinely address these issues with patients. We sought to examine patient and physician perceptions regarding inclusion of religiosity and spirituality in the medical encounter.

METHODS: We surveyed primary care physicians and patients in primary care clinics at 7 sites in 4 states (NC, FL, GA, VT). For the patient survey, a trained research assistant verbally administered a 112 item survey that included questions pertaining to demographics, health status, utilization of health care, physical function (SF-36), and a religious/spiritual assessment (Spiritual Well Being, or SWB) scale. Physician surveys were mailed to primary care faculty, residents, and recent graduates at the same 7 sites. Bivariate analyses were done to examine associations between patient/physician characteristics and questions regarding the impact of prayer on health outcomes.

RESULTS: 299 patients and 444 physicians were surveyed. Patients' ages ranged from 19–86 years. Fifty-six percent were male, 46% were African-American, 51% Caucasian, and 63% were Protestant. Fifty-three percent of the patients surveyed had annual incomes of \$20,000 or less. Of the physicians, 45% were in practice, and 55% were residents/fellows. Forty-five percent were Protestant. Fifty-three percent work in an academic setting. Of the patients, 30% felt that praying with their doctor would improve their own health. Thirty-nine percent of patients vs. 19% of physicians agreed that their doctor praying for them would improve their health. In addition, 42% felt that praying for their doctor would improve their own health, while 32% of physicians felt that patient prayer for them would help them provide better health care. Nearly half of physicians felt that patients who prayed for themselves or had strong religious/spiritual beliefs would have improved health outcomes. Fifty-three percent of patients as compared to 19% of physicians agreed that faith alone can cure disease. Bivariate analyses in the patient sample showed that the perception that prayer improves health outcomes is associated with higher attendance of organized worship services, hospitalization within the past year, higher score on the SWB scale, and lower total household income. In the physician surveys, bivariate analyses associated the equivalent perceptions with attendance of organized worship services and higher SWB score.

CONCLUSION: A significant percentage of patients perceive that prayer (whether it is done by their doctor, with their doctor, or for their doctor) is linked to improvements in health. Although there are a significant number of physicians who perceive that prayer is linked to improved health outcomes, there is still a large disparity in the prevalence of physicians compared to patients who share this perception. These data suggest that a more reliable

quantification of the benefit of spiritual behavior in medical encounters might be needed to bridge this gap.

FACTORS ASSOCIATED WITH HOSPITALIZATION AMONG INJECTION DRUG USERS WITH SOFT TISSUE INFECTIONS. T. Takahashi¹, J.O. Merrill², K. Bradley³, ¹VA Puget Sound, University of Washington, Seattle, WA; ²University of Washington, Seattle, WA; ³VA Puget Sound Health Care System, University of Washington, Seattle, WA

PURPOSE: The prevalence of soft tissue infections among injection drug users (IDUs) is estimated between 11%–32% with 40% of those presenting for care requiring hospitalization. Risk factors for hospitalization from these infections are not known. This study sought to identify factors associated with increased health care utilization.

METHODS: This was a cross-sectional study of all IDUs seeking initial care for soft tissue infections at an urban, public hospital emergency department (ED) from November, 1999 through April, 2000. Demographics, infection prodrome, and clinical measures were extracted from medical records. Systemic infection was defined as temperature $\geq 38^{\circ}\text{C}$ or WBC $\geq 15,000$ per milliliter, prolonged hospitalization as ≥ 2 days, and delay in seeking care as ≥ 5 days of symptoms before the ED visit.

RESULTS: Two hundred forty-two patients sought initial care for IDU-related soft tissue infections. Most patients were male (64%), Caucasian (69.4%), unemployed (78%), and uninsured (52%). Soft tissue infections included abscesses (74.4%), cellulitides (31%), and infected ulcers (5%) and were mainly located on the lower arm (49.6%), deltoid (14.1%), leg (22.7%), and buttock (19.8%). Ninety-seven (40.1%) patients delayed seeking care. Systemic signs of infection were identified in 118 (48.8%) patients. Of the 97 (40.1%) patients hospitalized, 70 (72.2%) stayed 2 or more days. Ethnicity, gender, age, employment status, and homelessness were not associated with having systemic infection, with hospitalization or with prolonged hospitalization.

Compared to those with other types of infection, patients with abscesses were much less likely (OR 0.13, 95% CI 0.02–0.73) to be hospitalized while those with cellulitides were 3.2 times more likely (95% CI 1.47–6.79) to be hospitalized and 3.8 times more likely (95% CI 1.70–8.57) to be hospitalized ≥ 2 days.

Among patients who delayed seeking care, those with systemic signs of infection were 5.36 times (95% CI 1.67–17.19) more likely to have a prolonged hospitalization compared to those without systemic signs of infection. Prolonged hospitalization was not associated with systemic infection among those who did not delay seeking care.

CONCLUSION: Among IDUs seeking ED care for soft tissue infections, systemic signs of infection after delay in seeking care, and cellulitides, were each associated with increased use of health services. Gender, ethnicity, employment status, insurance status, and homelessness did not alter the risk of hospitalization, of prolonged hospitalization, or of having systemic signs of infection.

POSTMENOPAUSAL HORMONE REPLACEMENT THERAPY AND STROKE: A SYSTEMATIC EVIDENCE REVIEW AND META-ANALYSIS. L. Takano¹, B. Chan¹;

¹Evidence-based Practice Center, Oregon Health Sciences University and Portland VA Medical Center, Portland, OR

PURPOSE: Stroke is the third leading cause of death and a major cause of disability among postmenopausal women in developed countries. Although postmenopausal hormone replacement therapy (HRT) is one of the most widely prescribed drugs, it is associated with increased rates of thromboembolic events and therefore may be important etiologically in stroke. We conducted a systematic evidence review and meta-analysis for the third US Preventive Services Task Force (USPSTF) to investigate the relationship between HRT and stroke.

METHODS: We searched the MEDLINE database from 1992 to 2000 for all published English language studies reporting the association between HRT and stroke. In addition, reference lists of key articles were reviewed for related studies, including those predating the search. Thirty-three observational studies met inclusion criteria and were reviewed; however, only those studies considered good or fair quality based on criteria developed by the USPSTF were included in the detailed review and meta-analyses. We identified no randomized controlled trials. We used the Bayesian data analysis framework for the meta-analysis.

RESULTS: The pooled relative risk of stroke mortality in women who had ever used HRT was 0.83 (95% CI 0.64–1.05). Stroke incidence was significantly increased among ever users, with a pooled relative risk of 1.15 (1.03–1.29). On subanalyses, the risk of thromboembolic stroke was significantly elevated among women who had ever used HRT (RR 1.30 [1.10–1.58]); however, not subarachnoid hemorrhage (RR 0.93 [0.69–1.25]) or hemorrhagic stroke (RR 0.71 [0.25–1.29]).

CONCLUSION: Studies evaluating the association between URT and stroke mortality suggest no effect. Our meta-analysis suggests an increase risk of the incidence of total stroke, largely due to thromboembolic stroke, among women with exposure to HRT. These results are consistent with preliminary findings from the Women's Health Initiative. The results are limited by the observational nature of the data and randomized controlled trials will be the most valid way of clarifying the association between stroke and HRT.

CERVICAL CANCER SCREENING AMONG CHINESE AMERICAN WOMEN. V.M. Taylor¹, J. Jackson², S. Tu³, Y. Yasui¹, S.M. Schwartz¹, A. Kuniyuki¹, E. Acorda¹, T. Hislop⁴, ¹Fred Hutchinson Cancer Research Center, Seattle, WA; ²Harborview Medical Center, Seattle, WA; ³University of Washington, Seattle, WA; ⁴British Columbia Cancer Agency, Vancouver, BC

PURPOSE: One of the Year 2010 goals for reducing racial/ethnic health disparities specifically addresses cancer screening. Chinese women in North America have high rates of invasive cervical cancer and low levels of Papanicolaou (Pap) testing use. This study examined cervical cancer screening barriers and facilitators among Chinese American women.

METHODS: A community-based, in-person survey of Chinese women was conducted in Seattle, Washington during 1999. The total estimated response rate was 64% and the cooperation rate was 72%. Four hundred and thirty-two women in the 20–79 age-group were

included in this analysis. The main outcome measures were a history of at least one previous Pap smear and Pap testing within the last two years.

RESULTS: Nineteen percent of the respondents had never received cervical cancer screening and 36% percent had not been screened in the previous two years. Eight characteristics were independently associated with a history of at least one Pap smear: being married, thinking Pap testing is necessary for sexually inactive women, having concerns about embarrassment or cancer being discovered, having received a physician or family recommendation, having obtained family planning services in North America, and having a female provider. The following characteristics were independently associated with recent screening: thinking Pap testing is necessary for sexually inactive women, having concern about embarrassment, having received a physician recommendation, having obtained obstetric services in North America, and having a female provider.

CONCLUSION: Pap testing levels among the study respondents were well below the National Cancer Institute's Year 2000 goals. The findings suggest that cancer control interventions for ethnic Chinese women are more likely to be effective if they address multiple barriers and facilitators. Results also indicate that efforts to increase Pap testing rates among Chinese should target the health care providers who serve Asian American communities.

HEALTH RELATED QUALITY OF LIFE IN SYMPTOMATIC MENOPAUSAL WOMEN: SF-36 SCORES FOR PARTICIPANTS IN A CLINICAL TRIAL OF PHYTOESTROGENS. J.A. Tice¹, S.R. Cummings¹, T. Blackwell², B. Ettinger³, ¹University of California, San Francisco, San Francisco, California; ²University of California, San Francisco, San Francisco, CA; ³Kaiser Permanente Division of Research, Oakland, California

PURPOSE: Women experiencing menopausal symptoms are increasingly choosing alternative therapies to improve their quality of life. This is the first clinical trial evaluating the quality of life effects of two dietary supplements derived from red clover.

METHODS: We have randomly assigned 252 women with at least 3 months amenorrhea who experience ≥ 35 hot flashes per week to 3 months of daily Promensil(r) (80 mg total isoflavones), Rimostil(r) (50 mg total isoflavones), or an identical placebo. At randomization and close-out visits, participants health related quality of life is evaluated using the Medical Outcomes Study 36 Item Short Form (SF-36). Non-parametric statistical tests were used as most of the SF-36 scale scores were not normally distributed.

RESULTS: Participants' mean (SD) age is 52.3 (3.1) years and they average 3.3 (4.5) years since menopause. Ten percent of the participants are African American and 84% are Caucasian. Forty-two percent had completed college and all but 2 participants had completed high school. At baseline, women in the study scored higher than age and gender equivalent normative data on the Physical Component Summary (PCS mean 51.2, median 52.9) and the Mental Component Summary (MCS mean 51.9, median 54.8). Among the 8 domains of the SF-36, the participants scored lower than population norms on only the bodily pain scale. The weekly hot flash count decreased 36% for the 232 women who have completed the 12 week protocol. The PCS was unchanged over 12 weeks, while the MCS increased 6%. Change in hot flash count was not correlated with change in any of the eight SF-36 domains or with changes in the summary components.

CONCLUSION: Since the blind has not been broken, it is not possible to definitively assess the effects of these dietary supplements on health related quality of life. The lack of association between the change in hot flash count and the SF-36 scales suggests that the SF-36 may not be a sensitive measure of menopausal quality of life.

SYMPTOMS, QUALITY OF LIFE, AND SATISFACTION WITH CARE AMONG PATIENTS WITH CHRONIC HEART FAILURE. W.M. Tierney¹, U. Subramanian¹, L. Plue¹, G. Eckert¹, F. Smith¹, H. Temkit¹, M. Weinberger¹, S. Fihn², ¹Roudebush VA Medical Center, Indiana University, Indianapolis, IN; ²VA Puget Sound Health Care System and University of Washington, Seattle, WA

PURPOSE: Chronic heart failure (CHF) is costly, morbid, and common. CHF care is often suboptimal because appropriate drug regimens are complex and should be modified according to patients' symptoms. Standard measures of CHF symptoms and health status would be a valuable tool in providing CHF care.

METHODS: To assess a measure of CHF symptoms and compare it with CHF-specific and generic health status and satisfaction with care, we studied 497 veterans (80% of those eligible) who had an outpatient diagnosis of CHF, objective left ventricular systolic dysfunction, and being actively treated for CHF by their primary care physicians (PCPs). We administered the Kansas City Cardiomyopathy Questionnaire (KCCQ) (to assess CHF-specific symptoms) and the SF-36 (to obtain generic summary measures: Physical Composite Score (PCS) and Mental Composite Score (MCS)). Symptoms were used to assign New York Heart Association (NYHA) functional class. We also assessed patients' satisfaction with their PCP and the most recent prior outpatient PCP visit using locally validated measures; higher scores denote better status or greater satisfaction.

RESULTS: Patients' mean age was 69 years; 98% were men and 86% Caucasian. Patients' had moderate to severe limitations (means: NYHA=2.6, PCS=34, MCS=54). As shown in the table, KCCQ symptoms and NYHA class were highly correlated with CHF-specific physical limitations, functional status, and PCS ($p < .0001$ for each but less so with MCS ($p < .005$). KCCQ physical limitations and functional status scores and the PCS were highly correlated, but less with the MCS ($p < .005$ for each). There was no correlation between satisfaction with the PCP and any symptom or health status measure. Satisfaction with the most recent primary care visit was correlated modestly with PCS and MCS ($p < .0001$ for each) but less strongly with symptoms ($p < .05$).

CONCLUSION: Measures of CHF-specific and generic health status are highly correlated with each other and with severity of CHF symptoms, but they tap different aspects of patient status. But there was little impact on satisfaction with their PCPs or primary care among these veterans with CHF. These health status measures can provide important data to assess and improve the care of patients with CHF.

	symp	nyha	p-lim	f-stat	pcs	mcs
nyha	-.56					
p-lim	.68	-.79				
f-stat	.90	-.75	.87			
pcs	.52	-.43	.38	.51		
mcs	.32	-.15	.13	.26	.06	
p-sat	.07	-.04	.04	.06	.08	.07
v-sat	.10	-.02	-.01	.05	.19	.20

INTEREST IN BREAST CANCER CHEMOPREVENTION AMONG OLDER WOMEN. J. Tjia¹, R.A. Silliman²; ¹University of Pennsylvania, Philadelphia, PA; ²Boston University School of Medicine, Boston, MA

PURPOSE: To determine the level of interest in taking tamoxifen for the primary prevention of breast cancer among women over age 50.

METHODS: We conducted a cross-sectional, one-time structured interview of women over 50 years old in outpatient geriatrics and women's health clinics at an inner-city academic medical center. Data on age, race, socioeconomic factors, comorbidities using the Charlston comorbidity index, and objective/subjective risk of breast cancer and interest in taking a breast cancer chemopreventive agent were obtained.

RESULTS: From the forty-eight (48) participants enrolled in this pilot study, 31 (65%) were black, 17 (35%) white, Hispanic or other, 35 (73%) were > 65 years old. A substantial proportion, 37.5%, of older women were interested in taking preventive medication for breast cancer. Interest in taking a cancer preventive agent appeared to be higher among white women (59% vs 39%), women who have had a breast biopsy (71% vs 35%), and those who believe their risk of cancer warrants taking an agent (93% vs 24%). Of women receiving Medicare, those with additional drug coverage were more willing to take a preventive agent than those without drug coverage (61% vs 38%). Interestingly, those women who paid the highest percentage of their monthly income for medications were also the most interested in taking a breast cancer preventive agent (87.5% vs 36%). No difference was observed by educational level, income, family history, number of comorbidities, objective breast cancer risk, or personal acquaintance with a person with breast cancer.

CONCLUSION: A substantial number of women in our pilot study were interested in taking a chemopreventive agent for the primary prevention of breast cancer. Interest appeared to be associated with race, subjective cancer risk, breast biopsy history, and prescription drug coverage. Given our limited numbers, further study is warranted to test the robustness of these associations.

PREVENTIVE MEDICATION USE AMONG OLDER WOMEN. J. Tjia¹, R.A. Silliman²; ¹University of Pennsylvania, Philadelphia, PA; ²Boston University School of Medicine, Boston, MA

PURPOSE: To determine the level, pattern, and factors associated with preventive medication use among women over age 50.

METHODS: We conducted a cross-sectional, one-time structured interview of women over 50 years old in outpatient geriatrics and women's health clinics at an inner-city academic medical center. Data on preventive medication use, age, race, socioeconomic factors, insurance status, and comorbidities using the Charlston comorbidity index were obtained.

RESULTS: From the forty-eight (48) participants enrolled in this pilot study, 31 (65%) were black, 17 (35%) white, Hispanic or other, 35 (73%) were > 65 years old. All (100%) women in this study took at least one preventive agent every day (defined as a multivitamin, calcium, hormone replacement therapy, aspirin, cholesterol lowering drug or antihypertensive (to prevent myocardial infarction or stroke)). Over 55% took at least two preventive medications a day. The frequency of medications used were: antihypertensive (69%), aspirin (50%), multivitamin (50%), calcium (35%), cholesterol lowering agent (31%), and hormone replacement therapy (15%). Women with Medicare alone versus those with Medicare plus additional drug coverage were less likely to take: 1) a cholesterol lowering agent (23% vs 46%); 2) an aspirin (38% vs 61%); and 3) an antihypertensive (69% vs 84%). Lower income women were less likely to take calcium (25% vs 53%) or a multivitamin (43% vs 65%). Older women were less likely to take a cholesterol lowering agent than younger women (0% vs. 38%), even though the same older women were much more likely to be taking an antihypertensive agent (90% vs 62%). Women cited that physician recommendation was the most important factor in deciding to take a preventive medication. Conversely, women cited that cost and insurance coverage were the least important factors.

CONCLUSION: There is a high level of interest in taking preventive medications among older women. However, low-cost, proven therapies such as aspirin and calcium are underutilized. Ability to pay appears to drive decisions about medication use despite the fact that women cited cost/insurance coverage as the least important decision-making factor. Women whose Medicare lacks supplemental drug coverage were less likely to take both expensive drugs (cholesterol lowering agents) and inexpensive drugs (aspirin). Lower income women were less likely to take over the counter preparations such as calcium and multivitamins. In addition, the pattern of use of some medications appeared to be inconsistent; the very elderly women in our study, at risk for coronary artery disease as evidenced by their high rate of antihypertensive use, were not taking cholesterol lowering agents. Additional study is needed to test the robustness of our findings and to help understand how women make decisions about their medication use so that inequalities due to age or ability to pay may be addressed.

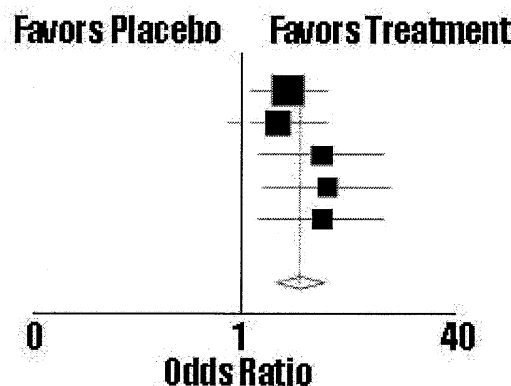
TREATMENT OF FIBROMYALGIA WITH CYCLOBENZAPRINE (FLEXERIL): A META-ANALYSIS. J. Toffen¹, J.L. Jackson², P.G. O'malley¹; ¹WRAMC, Washington, DC; ²Uniformed Services University of the Health Sciences, Bethesda, Maryland

PURPOSE: To assess the effectiveness of cyclobenzaprine in fibromyalgia.

METHODS: Randomized, placebo-controlled trials of cyclobenzaprine in fibromyalgia were obtained from searching MEDLINE, EMBASE, PSYCLIT, unpublished literature and

bibliographies. We performed independent duplicate review of each study for both inclusion and data extraction. Global improvement was abstracted as a dichotomous variable. Effect on pain, fatigue and sleep were abstracted as continuous variables at 4 time points (2, 4, 8 and 12 weeks). Analysis was done using a random effects model. Quality was assessed using the methods of Jadad.

RESULTS: Six randomized, placebo-controlled trials were identified of which all were appropriate for some data extraction. Overall the quality of the studies was fair (mean score: 4.7), range 0–8. The odds ratio for improvement with therapy was 3.7 (95% CI: 2.2–6.3). The pooled risk difference for these studies was 0.28 (95% CI: 0.16–0.40), which calculates to 3.6 (95% CI 2.5 to 6.25) individuals needing treatment for one patient to experience symptom improvement. Patients experienced improvement in pain at 2 weeks (Standardized Mean Difference: 0.44, 95% CI: 0.18–0.88) but none at 4, 8 or 12 weeks. There was no significant effect on patient fatigue at any time point. Patients experienced a statistically significant improvement in sleep at all time points with SMD's ranging from 0.35–0.49. Eighty-five percent of patients given cyclobenzaprine reported at least one side effect.



CONCLUSION: While patients were nearly 4 times as likely to improve if given cyclobenzaprine, the effect was modest, with no effect on fatigue, and improvement in pain only at two weeks. Patients treated with cyclobenzaprine did report modest improvement in sleep at all time points as well as more side effects.

PREGNANCY AND FAMILY PLANNING SERVICES AMONG INCARCERATED WOMEN. L. Tong¹, J.G. Clarke²; ¹Brown University, Department of General Internal Medicine, Providence, RI; ²Brown University, Providence, RI

PURPOSE: Among incarcerated women, pregnancies are often unplanned and high-risk due to high rates of psychiatric disease and illicit drug use. Obtaining birth control can be costly and challenging for this underserved population given its lack of medical insurance and poor access to health care. Offering women an intramuscular injection of progesterone or a three-month supply of oral contraceptive pills prior to their release from prison could prevent or delay potentially high-risk and unwanted pregnancies. In addition to pregnancy prevention, birth control provision would also introduce control and stability into the lives of this disenfranchised population. The aims of this study are to obtain pilot data to determine 1) the number of pregnancies among incarcerated women in Rhode Island and 2) the number of pregnancies occurring within 12 weeks of a prior incarceration.

METHODS: Charts of all patients with positive (+) serum beta-human chorionic gonadotropin (b-HCG) values from January 1998 to January 2000 were reviewed. Women who were post-partum or who had a recent termination of pregnancy were excluded from the study. Chi-square and t-tests were used where applicable.

RESULTS: A total of 129 women and 140 pregnancies were included in the study. Of the 140 pregnancies, seventy-seven (55%) pregnancies occurred in women with prior incarcerations. Of these 77 pregnancies, 38 (27%) occurred within 12 weeks of a prior release. Women who conceived within 12 weeks of their prior release had a higher number of prior incarcerations (5.5 vs. 3.7, $p = 0.0043$) and a longer mean length of incarceration during their pregnancy (61.4 days vs. 26.7 days, $p = 0.0016$). Women who were incarcerated for greater than 30 days during their pregnancy had a higher number of incarcerations (5.3 vs. 3.7, $p = 0.0043$) and a longer stay during the prior incarceration (58.9 days vs. 21.4 days, $p = 0.0082$).

CONCLUSION: Twenty-seven percent of conceptions among incarcerated women occurred within 3 months of a prior release, suggesting that initiating birth control prior to release from prison could prevent a significant number of pregnancies. Women with potentially preventable pregnancies had multiple prior incarcerations and longer lengths of stay, thus providing multiple opportunities for medical intervention and counseling. Family planning services may also provide cost savings as women with potentially preventable pregnancies have longer lengths of stay while pregnant, thus requiring more financial support for prenatal care. The establishment of a family planning program for incarcerated women would not only provide an important service to a medically underserved population, but it may also be a cost-effective measure for the Department of Corrections.

IS THE MORTALITY ADVANTAGE OF PHYSICIANS DUE TO LOWER RATES OF SMOKING? D.M. Torre¹, N. Wang¹, L.A. Meoni¹, M.J. Klag¹, D.E. Ford¹; ¹Johns Hopkins University, Baltimore, MD

PURPOSE: Physicians in the U.S. have been reported to have lower mortality than the general population. Smoking is strongly related to most health outcomes. However the contribution of smoking behavior to mortality benefits in this group has not been investigated.

METHODS: We compared the mortality experience of a cohort of 1,213 white male medical students, graduating classes 1948–64, to the U.S. white male population using age-race-sex and calendar time specific death rates from U.S. vital statistics. We calculated all cause and specific cause mortality compared to the general population, stratifying by smoking status at baseline and lifetime smoking. Mortality through 1998 was assessed by annual questionnaires, contacting family members and co-workers, scanning obituaries and National Death Index searches. Vital status was known for > 99% of the cohort.

RESULTS: The average age was 26, average follow-up 40 years and mean age at death 58 years. During 48,589 person-years of follow-up, 195 deaths occurred. When smoking from baseline to 1993 was considered, all cause mortality in men compared to the general population was 74% lower for lifelong nonsmokers (SMR 0.26, 95% CI 0.19–0.36), intermediate for those who quit smoking during follow-up (SMR 0.38, 95% CI 0.31–0.47) and high for lifelong smokers (SMR 1.33, 95% CI 0.92–1.86). Mortality from cardiovascular disease was 86% lower than expected for lifelong non-smokers (SMR 0.14, 95% CI 0.06–0.26) and 46% lower than expected for lifelong smokers (SMR 0.54, 95% CI 0.20–1.18). Risk of dying from lung cancer was 93% lower than expected for those who never smoked (SMR 0.07, 95% CI 0.00–0.37), greater than the general population for lifelong smokers (SMR 1.92, 95% CI 0.62–4.49), and intermediate for intermittent smokers (SMR 0.13, 95% CI 0.01–0.48). The risk of dying from suicide compared to the general population was higher among baseline smokers (SMR 2.25, 95% CI 0.16–3.92) and lifelong smokers (SMR 8.60, 95% CI 3.14–18.72).

CONCLUSION: These results demonstrate the substantial health benefits associated with low rates of smoking among physicians, compared to the general population, and the lack of this advantage for physicians who smoked.

GENDER DIFFERENCES IN PHARMACY-BASED ADHERENCE TO COMBINATION ANTIRETROVIRAL (ARV) THERAPY AND EFFECT OF HEALTH CARE DELIVERY. B.J. Turner¹, C. Laine², J. Rothman³, W.W. Hauck², ¹University of Pennsylvania, Philadelphia, PA; ²Thomas Jefferson University, Philadelphia, PA; ³New York State Department of Health, Albany, NY

PURPOSE: In a recent national study, women reported poorer adherence to ART than men but socioeconomic differences may explain this finding. We used a pharmacy-based measure to assess adherence in HIV-infected drug users enrolled in New York State (NYS) Medicaid, a population where men and women have similar socioeconomic backgrounds.

METHODS: We analyzed longitudinal claims files for 9,557 non-pregnant HIV+ drug users enrolled in Medicaid >10 months in each year of 1997–98. In persons prescribed ARVs for >6 months and on 2+ ARVs for >2 months in each year, adherence was defined as >90% days covered by 2+ ARVs from the start of combination therapy through the last prescribed ARV(s) in each year, with inpatient days judged adherent. Demographics and clinical conditions (i.e., HIV, substance abuse, and general medical) and health care factors were identified from eligibility/claims files in 1997. Outpatient care in 1997 was categorized as: regular medical care (>35% of visits to one provider); regular substance abuse treatment (6+ months with one provider); both; or neither. We created separate indicators for any care in 1997 from: a provider offering HIV-focused care under NYS Medicaid contract; an infectious diseases (ID) specialist; and a psychiatrist. Two logistic regression models were estimated for adherence to 2+ ARVs in 1997 and, to reduce possible confounding due to concurrent assessment of medical care patterns, adherence to 2+ ARVs in 1998.

RESULTS: Of 9,557 persons on Medicaid in 1997–8, 2+ ARVs were prescribed for 4,299 in 1997 and 4,589 in 1998. In 1997, 29.4% of 1,564 women and 37.6% of 2,735 men were >90% adherent while, in 1998, 34.6% of 1,730 women and 40.3% of 2,186 men were adherent. In both models, the adjusted odds of adherence were lower for women than men (adjusted odds ratio (AOR) 0.67 [CI 0.58, 0.77] in 1997), and (AOR 0.81 [CI 0.71, 0.92] in 1998). Of health care factors, only psychiatric care was associated with adherence in both years (AOR 1.19 [CI 1.01, 1.39] in 1997 and AOR 1.23 [CI 1.04, 1.46] in 1998) while ID care was associated with adherence in 1998 (AOR 1.25 [CI 1.05, 1.49]).

CONCLUSION: Women appear less likely to adhere to ARVs than men even in this socioeconomically homogeneous cohort of indigent drug users. Psychiatric care and to a lesser extent ID care may promote adherence to combination ARV treatment in drug users.

QUALITY OF LIFE IN SYNCOPAL PATIENTS; A FIRST REPORT. N. Van Dijk¹, N. Colman², M.A. Sprangers³, W. Wieling², M. Linzer⁴, ¹Academisch Medisch Centrum Amsterdam, Department of Medicine, Amsterdam; ²Academic Medical Centre, Amsterdam, Department of Medicine, Amsterdam, NH; ³Academic Medical Centre, Amsterdam, Department of Medical Psychology, Amsterdam, NH; ⁴University of Wisconsin-Madison, Madison, WI

PURPOSE: Loss of consciousness is a potential physically and socially incapacitating condition. Uncertainty about the diagnosis and fear of an episode taking place in a crowded or dangerous situation can influence social functioning. Few and only small studies on quality of life of patients with loss of consciousness have been performed.

METHODS: As part of the Dutch Fainting Assessment Trial (FAST) we asked consecutive patients presenting to all departments of the AMC with loss of consciousness to participate in this study. All patients were asked to fill in the SF-36, a generic quality of life questionnaire, and the Syncope Dysfunction Scale (SDS), a previously validated disease-specific questionnaire which assesses specific areas of impairment due to syncope and fear or worry about syncopal spells.

RESULTS: Of the first 144 patients 125 (87%) were included in analysis of quality of life. Disease specific impairment was moderate, with acknowledged impairment in 32% of listed activities (e.g. driving, physical activities and ability to work). Fear and worry due to syncope was also moderately high at 40 (SD 29) (possible range 0–100, 0 = no fear or worry, 100 = terrified/all I do is worry).

CONCLUSION: Quality of life in patients with episodes of syncope is seriously affected. Knowledge of problems and fears which these patients encounter are vital knowledge for physicians that can improve patient-doctor understanding and guide (practical) medical advice to this group of patients.

Results SF-36 (scores 0–100; 100 = perfect health)

	Mean (SD)	Mean norm pop (SD)	P-value
Physical functioning	67 (29)	83 (22)	< .001
Role physical	45 (44)	76 (36)	< .001
Bodily pain	66 (28)	75 (23)	< .001
General health	54 (23)	71 (21)	< .001
Vitality	49 (24)	69 (19)	< .001
Social functioning	68 (28)	84 (22)	< .001
Role emotional	66 (42)	82 (33)	< .001
Mental Health	67 (22)	77 (17)	< .001

SERVICE UTILIZATION DIFFERS FOR AFRICAN-AMERICANS, LATINOS, AND WHITES WITH DIABETES. R.B. Vargas¹, R.B. Davis², E. McCarthy³, L.I. Iezzoni², ¹Beth Israel Deaconess Medical Center, Boston, MA; ²Harvard University, Boston, MA; ³Beth Israel Deaconess Medical Center, Boston, Massachusetts

PURPOSE: African-Americans and Latinos are more likely to have diabetes and experience worse outcomes from diabetes than Whites. We compared rates of outpatient office visits, emergency ward (EW) visits, and hospitalizations for persons with diabetes by race and ethnicity. We studied Medicaid recipients in New Jersey, thus controlling roughly for income and financial access to care.

METHODS: From the 1994 New Jersey State Medicaid Research Files, we identified persons with diabetes by ICD-9-CM diagnosis code or by prescriptions for sulfonylureas or insulin. The final database excluded patients who were: < 18 or > 64 years old; enrolled < 3 months on Medicaid; in Medicaid managed care or Medicare; or pregnant. We compared Black, Hispanic, and White patients, eliminating persons with other or unknown race/ethnicity. We identified visits using CPT evaluation and management procedure codes for outpatient office visits and for EW services, verifying each by place of service documentation. Unique hospitalizations were identified based on date of admission. We used a Poisson regression multivariate model to examine the effect of race on service use, adjusting for age, sex, reason for Medicaid eligibility, selected co-morbidities, and duration of enrollment.

RESULTS: Of 11,841 patients, 44% were Black, 15% were Hispanic, and 41% were White; 73% were women; mean age was 47 years; and 66% had claims for sulfonylureas or insulin. Of Blacks 51% had no outpatient visits, compared to 47% of Whites and 42% of Hispanics; 35% of Blacks had at least one hospitalization, as did 31% of Whites and 19% of Hispanics. For outpatient visits, compared to Whites, Blacks had an adjusted relative risk (aRR) of 0.84 (95% CI, 0.79–0.89), and Hispanics had an aRR of 1.00 (0.92–1.08). For EW visits, compared to Whites, Blacks had an aRR of 0.95 (CI, 0.89–1.02), and Hispanics had an aRR of 0.68 (CI, 0.61–0.77). For hospital admissions, compared to Whites, Blacks had an aRR of 1.21 (CI, 1.14–1.28), and Hispanics had an aRR of 0.91 (CI, 0.82–1.01). Additionally aRRs for all three services differed significantly between Blacks and Hispanics.

CONCLUSION: Among persons with diabetes, Blacks had significantly lower rates of outpatient visits and higher hospitalization rates than Whites despite similar incomes and identical insurance benefits. Hispanics had similar outpatient visits and admissions rates, but significantly fewer EW visits than Whites. These results suggest that factors other than income and insurance, such as availability of primary care services and physician-patient factors, contribute to differences in utilization by race and ethnicity.

DIABETES: CHARACTERISTICS OF THE SOUTH ASIAN POPULATION IN AN ACADEMIC COMMUNITY-BASED PRIMARY CARE SETTING. A. Vidyarthi¹, H. Zeytinoglu², M. Brunt², ¹Cambridge Hospital, Cambridge, Massachusetts; ²Cambridge Hospital, Cambridge, MA

PURPOSE: Populations of South Asian descent in India, the West Indies, and England have a higher prevalence of Type 2 diabetes mellitus than their Europid counterparts. Observational data suggest that this population also demonstrates poorer disease management and higher rates of clinical complications, in particular, diabetic nephropathy. We sought to confirm and further elucidate these findings in the U.S. South Asian immigrant population that utilizes care at the Cambridge Health Alliance (CHA), the local primary care network and two city hospitals serving Cambridge and Somerville Massachusetts.

METHODS: We conducted a retrospective review of 937 South Asian patients and 49,206 Europid patients (>age 18). We defined South Asian as patients who state on registration forms that they speak Bengali, Gujarati, Hindi, Punjabi, Nepalese or Urdu in the home. We defined diabetic as any patient with a billing diagnosis of diabetes in two or more visits over the last two years. Europid controls are those who meet the same criteria for a diabetes diagnosis and state that they speak English in the home. We controlled for socioeconomic class by including only patients who use CHA Free Care or a Medicaid variant as their primary insurance. We compared mean HbA1c, mean LDL cholesterol, and prevalence of microalbuminuria between the two populations. We will continue the analysis with the comparison of BP and ophthalmologic changes, and will report statistical tests of significance for all variables.

RESULTS: Sixty-eight of 937 (7.26%) South Asian patients were diabetic, compared with 1,698 of 49,206 (3.45%) Europid patients (RR: 2.10). Mean HbA1c (8.71% vs. 8.47%), mean LDL-C (124.17 mg/dl vs. 116.44 mg/dl) and prevalence of microalbuminuria (21% vs. 13%) were all higher in the South Asian patients than in the Europid patients

CONCLUSION: The South Asian population is at higher risk of type 2 diabetes mellitus and its complications, and may exhibit poorer metabolic control as compared to the Europid population. Our findings warrant increased research into this population to determine the factors that influence observed differences, and to elaborate culturally appropriate interventions for this high-risk population.

RACIAL DISPARITIES IN DISABILITY IN TYPE 2 DIABETES. S. Vijan¹, K. Langa², N. Stuart³, R.A. Hayward²; ¹University of Michigan, Ann Arbor, MI; ²University of Michigan, Ann Arbor, Michigan; ³Ann Arbor VA HSR&D, Ann Arbor, MI

PURPOSE: The complications of type 2 diabetes result in substantial morbidity and mortality. Racial disparities in clinical outcomes of diabetes care are well documented; however, racial variations in disability rates due to diabetes have not been studied. We sought to evaluate the impact of higher complication rates in minorities on health status and rates of disability.

METHODS: We used data from the 1992 wave of the Health and Retirement Study (HRS), a national household sample of people age 50 or older in the United States, to provide estimates of prevalence of disability in patients with type 2 diabetes. Logistic regression models were constructed to evaluate predictors of disability. Self-reported disability was the dependent variable. Independent variables included race and other sociodemographic characteristics and the presence or absence of common diabetes complications including stroke, heart failure, coronary artery disease, visual loss, and renal failure. Costs of disability were calculated by applying the median reported income in the HRS to the lost work time associated with disability. Results were extrapolated to the US population using survey weights.

RESULTS: In bivariate analyses, we found that African-Americans with diabetes were substantially more likely to be disabled than non-Hispanic whites with diabetes (unadjusted OR = 2.38; 95% CI 1.79, 3.16), as were Hispanic-Americans (OR = 1.54, 95% CI 1.05, 2.26). However, after adjustment for age, education, and gender, only African-Americans were at higher risk of disability (adjusted OR = 2.09; 95% CI 1.78, 2.45). These effects persisted when adjusting for the microvascular and macrovascular complications of diabetes (OR 2.74, 95% CI 1.89, 4.00). There were no significant interaction effects between race and other demographic or clinical variables. In the US population, the total lost income since the onset of disability for African-Americans with type 2 diabetes is \$26 billion, which represents an incremental lost income of \$12 billion compared to non-Hispanic whites with diabetes.

CONCLUSION: African-Americans with diabetes have substantially elevated risks of disability and loss of income compared to non-Hispanic whites. However, the reasons for this disability remain unclear, and do not appear to be solely related to the increased risk of microvascular and cardiovascular complications. More research is needed to better define the causes and effects of increased rates of disability in minorities with type 2 diabetes.

RATES, CAUSES, AND COSTS OF DISABILITY IN PATIENTS WITH TYPE 2 DIABETES. S. Vijan¹, K. Langa², N. Stuart³, R.A. Hayward²; ¹University of Michigan, Ann Arbor, MI; ²University of Michigan, Ann Arbor, Michigan; ³Ann Arbor VA HSR&D, Ann Arbor, MI

PURPOSE: The complications of type 2 diabetes, such as cardiovascular disease, stroke, visual loss, and renal failure, result in high rates of morbidity and mortality. Disability from these complications is substantial and leads to dramatic losses in income and productivity. We sought to quantify the rates, causes, and costs of disability in patients with type 2 diabetes in the United States, which have not been well delineated.

METHODS: We used data from the 1992 wave of the Health and Retirement Study (HRS), a national household sample of people 50 and older in the United States, to estimate risks of disability in patients with type 2 diabetes. Self-reported disability was the primary outcome variable; duration of disability was a secondary outcome. Predictors of disability in the population of patients with type 2 diabetes (n=1330) were evaluated using logistic regression models. Independent variables included sociodemographic characteristics, health status, and presence or absence of common diabetes complications including stroke, heart failure, coronary artery disease, visual loss, and renal failure. Costs of disability were calculated by applying the median reported income in the HRS to the lost work time associated with disability; these were estimated for the entire US population using survey weights.

RESULTS: 23.1% of patients with diabetes were disabled, compared to 7.7% of subjects without diabetes (unadjusted OR = 3.62; 95% CI 3.14–4.18). Although more patients with diabetes were disabled, they were not disabled for longer than patients without diabetes (2550 days versus 2847 days; p=.1663). In multivariate analyses, the strongest predictors of disability in patients with diabetes were the presence of poor visual acuity (OR = 3.48; 95% CI 1.23, 9.81), congestive heart failure (OR = 3.47; 95% CI 1.69–7.14), coronary artery disease (OR = 2.41; 95% CI 1.26, 4.64), or renal failure (OR = 2.41, 95% CI 1.25, 4.62); stroke was not a significant independent predictor of disability. After adjustment for these outcomes, diabetes was still an independent predictor of disability (adjusted OR=2.29; 95% CI 2.61, 3.55). Based upon a median reported income of \$26,000 in this sample, the lost income in the current United States cohort of patients with type 2 diabetes is approximately \$14 billion annually; the total incremental lost income attributable to diabetes is \$9.3 billion annually. To date, the total lost income over the lifetime of the current US diabetes population is nearly \$100 billion.

CONCLUSION: Type 2 Diabetes is associated with a substantially increased risk of disability, and diabetes-related disability leads to staggering losses in income and productivity. Some, but not all of this increased risk is explained by their rates of microvascular and cardiovascular diabetes complications. Both health care and societal costs should be considered when setting health care policy.

THE ASSOCIATION BETWEEN HEARING IMPAIRMENT AND DEPRESSION IN OLDER PERSONS WITH DIABETES. J. Virk¹, A.F. Brown¹, P.R. Gutierrez¹, C.M. Mangione¹; ¹UCLA, Los Angeles, CA

PURPOSE: Persons with diabetes and older persons are at high risk for sensory impairment and depression. This study evaluates the relationship between hearing loss and depression among Medicare beneficiaries with diabetes cared for in fee-for-service (FFS) and managed care (MC). **METHODS:** We conducted secondary analysis of data obtained in two studies of quality of care for older persons. Telephone interviews and clinical examinations were performed on subjects identified through random sampling of persons enrolled in FFS Medicare and one large Medicare managed care plan in Los Angeles County. Participants completed a 4-item Geriatric Depression Scale (GDS-4) and a survey that asked about socio-demographic and clinical characteristics and health status. Hearing was tested with a hand-held audioscope during the study's physical examination. Hearing impairment was defined as better ear threshold at 40dB at

the 1kHz or 2kHz frequencies. Logistic regression models were constructed to evaluate the influence of hearing impairment on depressive symptoms. All models were adjusted for patient socio-demographic and clinical characteristics, health status, and system of care.

RESULTS: There were 484 study participants (response rate 68%). Mean age was 75±6 years; 47% were women; 52% were white, 20% Hispanic, 18% African-American, 6% Asian/Pacific Islander, and 5% Other/Multiracial; 48% had annual income under \$20,000 per year; 9% received Medicaid; 24% had not graduated from high school; and 24% were enrolled in FFS. Depressive symptoms were reported by 25%. Hearing aids were worn by 9%, 17% had hearing loss at 1 kHz, and 23% demonstrated hearing loss at 2 kHz. In the multivariate analysis, hearing impairment was significantly associated with depression (OR, 1.8; P=0.05). Although there was an association between depression and hearing impairment at a frequency of 2kHz, it was not statistically significant (OR=0.67; p=.09). Other characteristics that were significantly associated with depression were higher comorbidity score (OR=1.2; p=.005); worse physical well-being (OR=1.04; p=.009); low income (OR=1.8; p=.02); and male gender (OR=1.6; p=.05).

CONCLUSION: Hearing impairment is significantly associated with symptoms of depression among older persons with diabetes, a group that is already at high risk for sensory impairment due to visual loss and diabetic neuropathy, and points to a potentially correctable cause of depression.

THE EFFECT OF REFERRAL ON SENSITIVITY AND SPECIFICITY. J.M. Wagner¹, S. Straus², F. McAlister³, B. Foster⁴, D. Sackett⁵; ¹University of Texas Southwestern Medical Center at Dallas, Dallas, TX; ²University of Ontario; ³University of Alberta; ⁴University of Texas Southwestern; ⁵Retired

BACKGROUND: Although traditional teaching has insisted the sensitivity and specificity of a diagnostic test should not change as the pre-test probability of disease changes, several authors have suggested this may not be true in practice as patients are referred from a setting of undifferentiated problems with lower pre-test probabilities in primary care to tentative diagnoses with higher pre-test probabilities in secondary and tertiary care. The purpose of this paper is to demonstrate how and understand why sensitivity and specificity change as a patient population is divided by the physician's decision whether to refer.

METHODS: Using MEDLINE, a collection of published studies with various inclusion criteria was assembled(216). Studies were included when a) the study provided enough information about the studied population to determine its position along the referral algorithm b) the article provided history or physical examination component data that could be combined with data from other articles, and c) the article provided enough information to construct 2 × 2 tables to calculate sensitivity and specificity for those components of the clinical examination. The studies were divided into two groups, depending on whether the patient population was studied before or after referral for surgery. Group A consists of those studies evaluating all patients in whom appendicitis was considered as a diagnosis, regardless of whether they were referred for surgery. Group B consists of those studies evaluating all patients who had been referred to the operating suite for suspected appendicitis.

RESULTS: In the metanalysis of the two-by-two table data, there were several trends observed as the population studied changed from group A to group B. Sensitivity tended to rise and specificity tended to fall across most of the exam components reported. These changes were found to be statistically significant for the specificity of migration and rebound pain. When all components of the examination were combined, the fall in specificity was statistically significant for the aggregate data.

CONCLUSION: This study provides the first evidence that sensitivity and specificity change in populations of patients evaluated at different positions along the referral spectrum. These changes in the operating characteristics of the finding on clinical examination are important to recognize. These changes have implications for teaching the clinical examination as well as interpreting their presence or absence in varying patient populations. For example, a finding that is felt to be non-specific in tertiary care settings may be quite specific in primary care settings.

Changes in Sensitivity and Specificity

	Measurement	Group A	Group B	p value
Combined	Sensitivity	58	67	0.0826
Combined	Specificity	64	52	0.0105

RISK OF HIP FRACTURE IN DISABLED COMMUNITY-LIVING OLDER ADULTS. L.C. Walter¹, L. Lui¹, C. Eng², K.E. Covinsky¹; ¹University of California, San Francisco, CA; ²On Lok Senior Services and University of California, San Francisco, CA

PURPOSE: To determine the rate of hip fracture and the risk factors associated with hip fractures in disabled elderly persons who choose to live in the community rather than a nursing home.

METHODS: We assessed potential risk factors in 5,187 persons who enrolled 1/90 – 12/97 in 12 nationwide sites of the Program of All-Inclusive Care for the Elderly (PACE), which provides comprehensive care to nursing-home-eligible men and women living in the community (mean age = 79, 71% women, 49% white, 47% with dementia). Functional status on enrollment was assessed by each site's nursing staff and included degree of dependence in walking and 5 Activities of Daily Living (ADL): bathing, dressing, toileting, transferring and eating. Cognitive status was assessed using a 10-item mental status questionnaire (SPMSQ). Demographics and comorbidities were also recorded on enrollment. The main outcome measure was new hip fracture that resulted in hospital admission. Average follow-up was 2 years.

RESULTS: A total of 238 hip fractures (4.6%) occurred during follow-up. The rate of hip fracture was 2.3% per person-year. Four independent predictors of hip fracture were identified using Cox proportional hazard analysis: age ≥75 years (adjusted hazard ratio (HR) = 2.0, 95% CI 1.4–2.8); white ethnicity (HR= 2.1, 95% CI 1.6–2.8); ability to transfer independently from bed to chair (HR= 3.0, 95% CI 1.2–7.2); and SPMSQ errors ≥5 (HR= 1.6, 95% CI 1.3–2.1). Several risk factors, including gender and history of stroke, were not independently associated with hip fracture after adjustment for the above risk factors. The incidence of hip fracture

ranged from 0.5% per person-year among persons with 0–1 independent risk factors to 4.7% per person-year among those with all four risk factors.

CONCLUSION: This cohort of disabled community-living older adults showed a higher rate of hip fractures (2.3% per person-year) than studies of community-dwelling white women over age 80 (1.4% per person-year) but lower than that reported for nursing home cohorts (4.4% per person-year). Older age, white ethnicity, ability to transfer independently and cognitive impairment were independent predictors of hip fracture in this population. Persons with these risk factors should particularly be targeted for preventive interventions.

DEPRESSION: A SIGNIFICANT PREDICTOR OF NON-COMPLIANCE WITH APPOINTMENTS BY CLINIC PATIENTS. L. Wedderburn¹, P. Metha², A.B. Leong², V. Chandrakar², L. Tank²; ¹New York Methodist Hospital, Brooklyn, NY; ²New York Methodist Hospital, Brooklyn, NY

PURPOSE: Missed clinic visits interfere with good medical care and have adverse effects on clinic efficiency. Explanations for patient non-compliance in keeping appointments generally emphasize practical physical issues such as finance, transportation and socioeconomic ones. We evaluated other factors, as well, in a community hospital's medical clinic in order to define ways to improve compliance with appointments.

METHODS: A 10-item questionnaire covering demographic, medical conditions, socioeconomic and support conditions, missed visits, and reasons for missed visits, were filled out by 24 clinic visitors. All charts were separately analyzed for kept and missed appointments in the previous year. 72 patients were compliant (<20% appointments missed) and 36 were noncompliant (>30% appointment missed; range=30.7–85.7% median=42.3%).

RESULTS: Contrary to expectations non-compliant and compliant patients did not differ by medical insurance, availability of social support, transportation, education, age, or ethnicity. The sole significant difference in medical conditions was the presence of depression. The odds ratio for depression was 3.18 (95% confidence interval(1.7–8.7); p=.016) times greater among non-compliers, 39% of whom admitted to depression, than compliers. The non-compliers were aware of their noncompliance. The most common reason cited for missed appointments were forgetting (n=13), inadequate money for transportation (n=12), personal problems (n=9), and too long a wait (n=9). However, these did not differ significantly from the reasons cited by the others. **CONCLUSION:** The most striking, and we believe important, finding in our survey was the significant association of depression with noncompliance far outstripped all other characteristics, including financial and social ones. This observation suggests a very important problem, as well as a potential solution to the problem of noncompliance. We conclude that it is essential to address the possibility of depression actively in such patients as both a therapeutic goal in itself, and a potential means to improve compliance with medical care.

THE RELATIONSHIP BETWEEN INTENTIONAL WEIGHT LOSS AND ALL-CAUSE MORTALITY AMONG MEN. C.C. Wee¹, R.B. Davis¹, R.S. Phillips¹; ¹Div. of General Medicine, Beth Israel Deaconess Medical Center, Harvard Medical School, Boston, MA

PURPOSE: Although intentional weight(wt) loss is recommended to many patients, whether it improves survival, particularly among men, is unclear. We examined the association between intentional wt loss and mortality in a national sample of men.

METHODS: We used data from the 1989 National Health Interview Survey where 20,131 adults (91% response rate) were asked about sociodemographic and basic health information including self-reported health and smoking status, health care utilization, height, current and maximum(max) wt, and whether they were trying to lose wt. Date of death was obtained by linking to the National Death Index through 12/31/95. We determined wt loss by subtracting wt in 1989 from max wt. We classified wt loss as intentional if respondents reported that they were trying to lose wt in 1989; otherwise, it was considered of unclear intention. We used Cox models to examine if intentional wt loss was associated with time to death. We limited our analyses to men only to avoid misclassification and confounding due to pregnancy. We weighted all percentages to reflect population estimates and adjusted standard errors to account for the complex sampling. **RESULTS:** Among 8395 eligible men, 604 had died by the end of follow-up. Among men who lost wt intentionally, those who lost 5–10 lbs had the lowest unadjusted mortality (2.3%) when compared to those who lost up to 5 lbs (5.2%), p=.004. After adjusting for age, max. BMI, race, education, smoking, health status, number of doctor visits, hospitalized days, and days in bed, and intent to lose wt, intentional wt loss was associated with improved survival among men who lost 5–10 lbs (see Table). The benefits of weight loss were most consistent among men with BMIs of 25–29.9 kg/m². **CONCLUSIONS:** Modest intentional weight loss appears to be associated with mortality benefit among men. The benefit of greater weight loss is less clear.

Hazard Ratios(HR) Associated with Intentional Weight Loss

	% of men	HR (95% CI), all	HR (95% CI), BMI 25–29.9
Lost up to 5 lbs	35.2	1.00	1.00
Lost > 5–10 lbs	6.5	0.50 (0.25,0.98)	0.46 (0.19,1.14)
Lost > 10–20 lbs	10.0	1.04 (0.55,2.00)	0.75 (0.31,1.85)
Lost > 20–30 lbs	5.4	1.01 (0.53,1.93)	0.65 (0.25,1.70)
Lost > 30–40 lbs	2.9	0.63 (0.26,1.57)	1.12 (0.27,4.54)
Lost > 40 lbs	2.8	1.16 (0.53,2.54)	1.63 (0.50,5.32)

AN OBSERVATION OF THE CLINICAL EFFECT OF GLYCOPROTEIN ULINASTATIN ON ACUTE PANCREATITIS. Z. Weikang¹; ¹Union Hospital, Wuhan, Hubei

PURPOSE: Glycoprotein ulinastatin is a newly discovered proteinase inhibitor extracted from human urine that can restrict the activity of pancreatin. This study was designed to evaluate its clinical effect on patients suffering from acute pancreatitis.

METHODS: 257 patients were divided into two groups. The patients in the experimental group (132 patients) received ulinastatin (300,000U/day for two weeks). The patients in the control group (125 patients) were treated with routine protocol (anisodamine 20mg/d). Both groups received similar supportive treatments. Observations of clinical symptoms and signs were made on daily basis and measurements of serum amylase, hepatic/renal function, and TNF- α were performed at the 3-day interval for three weeks.

RESULTS: The results showed that the clinical symptoms and signs such as fever, abdominal distention and pain, belch, and constipation of the experimental improved faster than those of the control group. Analyses of serum amylase, glutamic pyruvic transaminase (GPT), creatinine, and TNF- α showed that all these indexes were significantly lower in the treatment group than those of the control group (p<0.01) at the 6th, 9th and 12th day, even though no such difference was observed at the 3rd day.

CONCLUSION: Acute pancreatitis patients treated with ulinastatin recovered significantly faster than those patients treated by routing protocol. Thus, ulinastatin is an effective medicine for treating patients with acute pancreatitis.

A RANDOMIZED TRIAL OF PHARMACEUTICAL CARE. M. Weinberger¹, M. Murray², D. Marrero³, N. Brewer², M. Lykens³, L. Harris³, R. Seshadri³, H. Caffrey³, J. Newell⁴, J. Collins⁴, C. McDonald^{2,3}, W. Tierney¹; ¹Regenstrief Institute, Roudebush VAMC, Indiana University, Indianapolis, IN; ²Regenstrief Institute, Indianapolis, IN; ³Indiana University, Indianapolis, IN; ⁴CVS/Pharmacy, Woonsocket, RI

PURPOSE: Pharmacists may improve patient outcomes by engaging in pharmaceutical care (PC) activities (e.g., monitoring symptoms, counseling about medications, helping resolve drug-related problems). We conducted a randomized trial in 36 community drug stores to evaluate the effectiveness of PC for patients with asthma or chronic obstructive pulmonary disease (COPD).

METHODS: 36 drug stores were divided into 12 clusters of 3 stores matched on key criteria. Within each cluster, stores were randomized to PC or to 1 of 2 control groups. Pharmacists in the PC group were provided with: recent patient-specific data [e.g., peak expiratory flow rates (PEFR), acute exacerbations resulting in hospitalization or emergency department visit] displayed on a computer in their stores; materials to facilitate implementation of PC; and training on the PC program. Customers of these stores were eligible if they: filled a prescription for a breathing medication during the past 3 months; reported having asthma or COPD; were 18 years or older; filled $\geq 70\%$ of their prescriptions in a single study store; were able to be interviewed; and resided in the community. Primary outcomes (PEFR, health-related quality of life, acute exacerbations) were assessed at 6 and 12 months. We present 6-month interim data; differences at the p<0.05 level are considered significant.

RESULTS: Asthma subjects (N=660) had a mean age of 45 years, 80% were Caucasian, and 82% female. COPD subjects (N=453) had a mean age of 62 years, 86% were Caucasian, and 66% female. At the 6-month interim analysis, the PC group did not have significantly better health-related quality of life or PEFR than either control group. The proportion of patients in the PC group with an acute exacerbation (11.4%) was not significantly different from either control group (9.4%, 9.1%). Neither was their a difference in months until the first acute exacerbation between the PC group (2.9 $\pm$ 1.8) and either control group (3.1 $\pm$ 1.6 and 2.3 $\pm$ 1.7).

CONCLUSION: Our interim (6-month) analyses found that an intensive PC program in community drug stores had no significant effect on our primary outcomes. We are currently examining long-term (12-month) outcomes, secondary outcomes (e.g., patient satisfaction), and dose-response effects.

THE IMPACT OF DOUBLE COUNTING AND TRANSFER BIAS ON ESTIMATED RATES AND OUTCOMES OF ACUTE MYOCARDIAL INFARCTION. J.M. Westfall¹; ¹University of Colorado Health Sciences Center, Denver, CO

PURPOSE: Background: Ischemic heart disease is the leading cause of death in the United States. Recent studies report inconsistent findings on the changes in the incidence of hospitalizations for ischemic heart disease. These reports have relied primarily on hospital discharge data. Preliminary data suggest that a significant percentage of patients suffering acute myocardial infarction in rural communities are transferred to urban centers for care. Patients transferred to a second hospital may be counted twice for one episode of ischemic heart disease. The purpose of this study is to describe the impact of double counting and transfer bias on the estimation of incidence rates and outcomes of ischemic heart disease, specifically acute myocardial infarction, in the U.S.

METHODS: Methods: Analysis of state hospital discharge data from Kansas, Colorado, Nebraska, Arizona, New Jersey, Michigan, Pennsylvania, and Illinois for the years 1995–1997 for patients reported to have suffered ischemic heart disease (ICD9 codes 410–414, 786.5). We developed a matching algorithm for hospital discharges to determine patients counted twice for one episode of ischemic heart disease. We validate the matching algorithm with a sensitivity analysis and chart audit.

RESULTS: We found that double count rates range from 10–15% for all states and have increased over the past 3 years. Moderate sized rural counties had the highest estimated double count rates at 15–20% with a few counties having estimated double count rates as high as 35–50%. Older patients and females were less likely to be double counted (p < .05).

CONCLUSION: Conclusions: Double counting patients has resulted in a significant overestimation in the incidence rate for hospitalization for acute myocardial infarction. Correcting for this double counting reveals a significantly lower incidence rate and a higher in-hospital mortality rate for acute myocardial infarction. Transferred patients differ significantly from non-transferred patients, introducing significant bias into myocardial infarction outcome studies. Double counting and transfer bias should be considered when conducting and interpreting research on ischemic heart disease, particularly in rural regions.

OUTCOMES AFTER HOSPITALIZATION FOR PULMONARY EMBOLISM COMPARED TO HOSPITALIZATION FOR VENOUS THROMBOSIS. R.H. White¹, S. Murin¹; ¹University of California, Davis, Sacramento, California

PURPOSE: There is increasing evidence that pulmonary embolism (PE) and deep vein thrombosis (DVT) may not simply be different manifestations of a single disease process, venous thromboembolism (VTE).

METHODS: Linked California hospital discharge records of 71,250 patients hospitalized between Jan 1, 1991 and June 30, 1996 for either PE or DVT were analyzed. For each record, the first hospitalization with a principal diagnosis of DVT (N= 51,233) or PE (N = 20,017) was considered the index hospitalization. Demographic and clinical characteristics (ICD-9 codes) were compared, as were the 6-month incidences of re-hospitalization for recurrent PE or DVT, bleeding and death. Surgery defined as operation within 61 days.

RESULTS:

Cohort	Age65	AA	Latino	CAD	COPD	Cancer	Surg
DVT	55.9%	7.5%	9.1%	15.7%	12.6%	17.4%	15%
PE	58.5%	9.4%	6.6%	22.8%	19.35%	15%	19%

All differences $p < 0.001$ except: bleed — 6 month follow-up $p = 0.05$; death — 6 months $p = 0.52$, NS. In multivariate logistic modeling, odds ratios (ORs) for recurrent VTE in 6 months: All differences significant with $p < 0.001$. Each race compared to others combined.

Cohort	Recurrent DVT — 6 mo	Recurrent PE — 6 mo	Hosp bleed	Bleed — 6 mo	Hosp Death	Death — 6 mo
DVT	5.5%	0.9%	1.39%	2.2%	1.1%	9.4%
PE	1.7%	3.8%	2.6%	2.0%	7.3%	19%

All differences $p < 0.001$ except: bleed — 6 month follow-up $p = 0.05$; death — 6 months $p = 0.52$, NS.

In multivariate logistic modeling, odds ratios (ORs) for recurrent VTE in 6 months:

Characteristic	Outcome = Recurrent PE	= Recurrent DVT
Age (65 vs. <65)	0.7 (0.67–0.76)	0.7 (0.67–0.75)
Sex (male vs. female)	1.0 (0.9–1.1)	1.0 (0.9–1.1)
VTE type (PE vs. DVT)	4.4 (3.9–4.9)	0.3 (0.26–0.33)
Cancer (vs. no cancer)	1.3 (1.1–1.5)	1.5 (1.4–1.6)
IVC filter within 6 mo.	1.0 (0.6–1.8)	1.7 (1.2–2.3)
Prior VTE (vs. none)	1.4 (1.1–1.8)	1.4 (1.2–1.6)

CONCLUSION: Patients with PE (versus DVT) were less likely to be Latino, more likely to be African-American, and more likely to have cardiopulmonary disease or recent surgery. Recurrent VTE manifested as PE was strongly associated with an initial diagnosis of PE, and conversely, recurrent DVT was strongly associated with an initial diagnosis of DVT. Coupled with recent data indicating that the prevalence of factor V-Leiden among patients with PE is significantly lower compared to patients with DVT, our findings suggest that there are undefined genetic or acquired factors that effect the phenotypic expression of VTE.

A POPULATION-BASED COMPARISON OF THE 3-MONTH INCIDENCE OF THROMBOEMBOLISM AFTER MAJOR ELECTIVE/URGENT SURGERIES. R.H. White¹, P. Romano¹, H. Zhou¹; ¹University of California, Davis, Sacramento, CA

PURPOSE: Background: Although the incidence of venous thromboembolic (VTE) complications after specific kinds of surgeries has been intensely studied, few studies have compared the 3-month incidence of VTE after different surgeries.

METHODS: We analyzed the California linked discharge records of 2,159,747 patients hospitalized between Jan 1, 1991 and Oct 1, 1996 who underwent one of 75 selected surgical procedures on the first or second hospital day, and determined the 3 month incidence of rehospitalization for VTE. Data were stratified by age (> 65 and < 65) and malignancy (present within 6 months or absent). A multivariate model was developed to compare the 3 month incidence of VTE using inguinal hernia repair (ICD-9-CM=53.0) as the referent surgery, adjusting for age, sex, presence of malignancy, Charlson Index > 1 , race (Asian vs. non-Asian), and prior VTE (< 6 mo.).

RESULTS: The table below shows the VTE incidence in a analysis restricted to age > 65 yrs, no cancer, $> 10,000$ procedures.

In a logistic model that included all surgeries, parameters significantly associated with VTE included age > 65 (OR=1.5, CI 1.4–1.6), malignancy (OR=2.2, CI 2.0–2.2), Asian ethnicity (OR=0.4, CI 0.3–0.5) and prior VTE (OR=4.7, CI 4.3–5.1). Surgeries with the highest adjusted ORs for VTE were: hip replacement (OR=6.4), open brain biopsy (OR=6.3), pericardiectomy (OR=4.8), cystectomy (OR=4.3), partial hip replacement (OR=4.2), gastroenterostomy (OR=4.1), craniotomy (OR=4.1), fixation of hip fracture (OR=3.4), excision lower limb artery (OR=3.0).

CONCLUSION: Even with widespread use of thromboprophylaxis, major orthopaedic surgery on the lower extremity had the highest risk of VTE. Major vascular surgery, neurosurgery, and procedures performed on patients with malignancy or unique risk factors were also associated with higher risk of VTE.

SURGERY - high frequency	VTE - %	SURGERY - low frequency	VTE - %
Total Hip Arthroplasty	3.3	Cystocele/Rectocele	0.35
Resection & Graft of Aorta	2.9	Transurethral Prostate	0.42
Partial Hip Replacement	2.8	Lap Cholecystectomy	0.49
Hip - ORIF	2.6	Carotid Endarterectomy	0.50
Total Knee Replacement	2.6	Pacemaker Insertion	0.51
Peripheral Vascular Bypass	2.4	Urinary Incontinence	0.55
Partial/total Colectomy	2.4	1-vessel PTCA	0.83
Coronary Bypass (1–4 vessel)	2.3	Total Cholecystectomy	1.09

DO BETTER PHYSICIAN-PATIENT (MD-PT) RELATIONSHIPS PRODUCE BETTER ADHERENCE (ADH) TO ANTIRETROVIRAL THERAPY (ART) IN HIV DISEASE? J. Schneider¹, I.B. Wilson², S. Greenfield¹, S.H. Kaplan¹; ¹Tufts University, Boston, MA; ²New England Medical Center, Boston, MA

PURPOSE: It is often asserted that better MD-PT relationships result in higher rates of ADH with recommended therapies. The goal of this research was to assess how measures of MD-PT relationships are related to ADH with ART in patients with HIV.

METHODS: Experienced HIV providers (> 50 patients) in Boston, and their patients, were sampled. To measure ADH, we formed a scale ($\alpha = 0.72$) from 4 items asking patients about frequency of recent changes from their prescribed ART regimen. To measure the quality of MD-PT relationships we used 6 published scales (general communication, HIV-specific information, participatory decision making, overall satisfaction, willingness to recommend MD, and MD trust; $\alpha > 0.70$ for all) and one new 3-item scale, ADH dialogue ($\alpha = 0.92$). For ADH dialogue, patients rated their MD at understanding and solving problems with ART regimens. Covariates included sociodemographic variables and health status (HCSUS physical (PCS) and mental component (MCS) summary scores). Separate multiple linear regression models were estimated for each of the 7 measures, adjusting for significant covariates.

RESULTS: Of 29 eligible providers, 22 (78%) participated. Of 774 participating patients, 620 (83%) returned questionnaires. Mean patient age was 42 years, 17% were female, 71% were white, and 66% were gay. In bivariate tests, general communication ($r = 0.12$, $p = 0.003$), ADH dialogue ($r = 0.19$, $p < 0.001$), overall satisfaction ($r = 0.10$, $p = 0.01$), willingness to recommend ($r = 0.12$, $p = 0.006$) and MD trust ($r = 0.10$, $p = 0.01$) were significantly associated with ADH. MCS ($r = 0.226$, $p < 0.001$), PCS ($r = 0.160$, $p < 0.001$), and age ($r = 0.16$, $p = 0.0003$) were significantly associated with ADH. Blacks reported worse ADH than whites ($p < 0.05$). In multivariable models, after controlling for MCS, age, and race, ADH dialogue was the only variable significantly associated with ADH ($R^2 = 0.12$, $p = 0.001$).

CONCLUSION: Only ADH dialogue was independently associated with adherence. Optimizing patients' ADH probably requires detailed and specific information gathering and problem solving directed at specific dosing times and/or specific medications. These data do not appear to be an automatic byproduct of good general MD-PT communication.

BARRIERS TO FREE DENTAL CARE IN FRAIL URBAN ELDERS. K. Wilson¹, C. Ritchie¹, M. Weinrich¹; ¹University of Louisville, Louisville, KY

PURPOSE: The Surgeon General's Report on Oral Health highlights the inadequate oral health care provided to the vulnerable population of homebound older adults. Cost of dental care and transportation challenges serve as major barriers to dental care in this population. We sought to examine the impact of removing both cost and transportation barriers on dental care utilization in urban homebound older adults.

METHODS: 225 older adults participating in a home and community based waiver program and/or a home-delivered meals program were invited to receive free nutritional and oral health assessments. Individuals with both poor oral health and nutritional needs were offered free dental care including free transportation. A dental hygienist and nurse coordinator made appointments on behalf of patients, provided appointment reminders and arranged transportation.

RESULTS: Mean age of participants was 81.6. 81% were female, 53% were African American. Of 159 subjects who completed both nutritional and oral health assessments, one-third were eligible for free dental intervention and transportation. Of those offered free dental care, 80% agreed to receive care. Of those who agreed to receive care, 14% did not keep any dental appointments, 19% kept some of their dental appointments but did not complete their dental intervention, 67% fully completed their dental interventions. The only significant barrier to completing the recommended dental intervention was depression score ($p = .03$). Even when functional status and Mini Mental State Examination scores were adjusted in the model, depression remained an independent predictor for not completing the recommended dental care visits.

CONCLUSION: When cost and transportation difficulties are removed as barriers to dental care, it is possible to complete needed dental interventions in a significant number homebound adults. Public policy changes would be needed to provide an infrastructure to facilitate this process. Depression remains an independent risk factor for incomplete interventions.

SELF-CARE INTERVENTIONS TARGETING OLDER, AFRICAN AMERICAN OR LATINO ADULTS WITH DIABETES: A SYSTEMATIC REVIEW. R.L. Wintz¹, C.A. Sarkisian¹, R. Shanman², C.M. Mangione¹; ¹UCLA Department of Medicine, Los Angeles, California; ²RAND Health Corporation, Santa Monica, California

PURPOSE: Diabetes disproportionately affects older African Americans and Latinos. During the development of a self-care intervention targeting older African Americans and Latinos, we set out to identify and examine previously published self-care interventions among older adults, African Americans or Latinos with diabetes.

METHODS: We searched 6 databases (Medline, HealthSTAR, EMBASE, PsycINFO, Ageline, Sociological Abstracts) for potentially eligible English language intervention studies

from 1985–2000, using a reproducible strategy. Eligible interventions were limited to those: 1) aimed at changing knowledge, beliefs, or behavior among adults with diabetes; 2) targeting either older (>55 years), African American (AA) or Latino adults; 3) measuring the outcomes glycemic control, symptoms, or quality of life (QOL). Obviously ineligible titles were eliminated; potentially eligible abstracts were independently reviewed by 2 investigators, and eliminated when ineligible. Full articles of remaining abstracts were reviewed using a standardized abstraction tool by 2 investigators who rated the quality of the investigations based on the criteria used by the U.S. Preventive Health Task Force.

RESULTS: Of 3175 titles identified, we reviewed over 1700 abstracts, and 200 potentially eligible papers. Thirteen papers met inclusion criteria: 5 among older white adults, 2 among middle-aged AAs, 4 among middle-aged Latinos, 1 among older AAs, and 1 among older Latinos. Of these, 8 (62%) were fully randomized controlled trials (RCTs). Studies enrolled between 7 and 596 subjects; there were no large multicenter trials. In 8 RCTs, glycosylated hemoglobin improved; however, no study described follow-up beyond 1 year. QOL was rarely measured. Among older adults, interventions that involved sustained personal contact, for example in ongoing support groups, improved glycemic control. Among AAs and Latinos, personalized culturally-tailored interventions improved glycemic control.

CONCLUSION: Despite the public health burden of diabetes upon older adults, AAs, and Latinos, only 13 self-care intervention studies were identified that attempted to improve glycemic control or QOL in these groups. The RCTs that have succeeded have incorporated either sustained social support or culturally-tailored interventions. Given the disproportionate prevalence of diabetes among older AAs and Latinos, future large randomized controlled trials should build upon these findings in order to determine which components of self-care interventions are most effective.

THE EFFECT OF THE SEVERITY OF DISEASE ON THE PERFORMANCE OF DIAGNOSTIC TESTS. J. Wisnivesky¹, T.G. McGinn¹, A. Siu¹; ¹Mount Sinai School of Medicine, New York, NY

PURPOSE: To avoid erroneous results in the estimation of the sensitivity and specificity of diagnostic tests, it has been recommended that a broad spectrum of disease be represented in the study population when evaluating the efficacy of a test. However, the actual effect of severity of disease on these indexes has not been evaluated on a clinically relevant population. The objective of this study was to assess the effect of the severity of disease on the diagnostic performance of a positive PPD and upper lobe disease on chest radiograph (defined as infiltrates and/or consolidation above the third rib posteriorly) for the diagnosis of pulmonary tuberculosis (TB).

METHODS: Seventy five patients with positive sputum culture for TB and 75 subjects who were considered at risk of TB but whose sputum tested negative for TB were identified from the hospital's TB registry. Patients were divided into two groups, severe disease and non-severe disease according to the presence or absence of specific criteria that are known to be associated with worse clinical outcomes. Patients with TB were classified as having severe disease if the sputum smear tested positive for acid-fast bacilli. Patients without TB (most of which had pneumonia) were classified as having severe disease if they were older than 65 years of age, had a temperature greater than 40°C, or had pleural effusion on chest radiograph. The sensitivity and specificity of a positive PPD and upper lobe disease were determined independently among the group of patients with severe and non-severe disease.

RESULTS: The sensitivity of upper lobe disease for the diagnosis of TB was 41.6% and 21.6% among those with severe and non-severe disease respectively ($P=0.03$). The specificity of this radiologic finding increased from 77.2% among patients with severe disease to 94.7% among those with non-severe disease ($P=0.05$). Conversely, the sensitivity of PPD was lower among those with severe disease (22.2%) compared to those with nonsevere disease (42.0%, $P=0.05$). The specificity of PPD was not statistically significantly different among this two groups (100.0% vs 89.0%, severe vs non severe disease respectively; $P=0.10$).

CONCLUSION: The spectrum of disease significantly modified the sensitivity and specificity of a positive PPD and upper lobe disease on chest radiograph for the diagnosis of TB. Interestingly, the effect of the severity of disease was not in the same direction in the two tests evaluated. This potential source of error in measuring test performance should be considered when translating the results of research studies to the clinical practice.

ASSOCIATIONS BETWEEN PATIENT CHARACTERISTICS AND AUDIT. S. Wolfe¹, N. Dawson¹, C. Thomas¹, D. Einstadter¹; ¹Case Western Reserve University at MetroHealth Medical Center, Cleveland, OH

PURPOSE: The Alcohol Use Disorders Identification Test (AUDM) has widely been reported as the best alcohol use screening tool to use in primary care, yet little is known about the associations of characteristics of patients and AUDIT score. We used a health appraisal telephone interview including the AUDIT, which produces scores 0–7 (safe), 8–10 (hazardous), 11–40 (harmful), to screen primary care patients. The purpose was to examine the relationship of responses to the other interview questions with AUDIT scores. The direction and strength of the presumed associations was stated a priori.

METHODS: Patients ($n=4967$) of an urban, teaching hospital's primary care clinic were telephoned for the interview. Patients were 71% female, mean age 51. Patients were administered a health screening interview that includes the AUDIT, CAGE, family CAGE, and questions about demographics, SES, general health, cigarettes smoking, and social support. We used the Tobit model, censoring for AUDIT=0, to analyze the relationship of other interview questions to AUDIT scores. The Tobit model is a 2-stageregression technique appropriate for a bimodal distribution of the dependent variable.

RESULTS: Variables (and beta weights) significantly ($p<0.05$) positively associated with AUDIT scores are: CAGE score (1.3), reporting that most friends have alcohol problems (3.5), number of cigarettes smoked per day (.08), frequency of emotional problems interfering with social activities (0.52), reporting ever having had an alcohol problem (1.26), poor physical health (0.41), frequency of participating in spiritual activities (.63), having a friend willing to help (.76). Variables (and beta weights) negatively ($p<0.05$) associated: age in decades (–0.9), being female

(–2), being employed, retired, homemaker or student (–0.89) compared to being unemployed, having friends who provide encouragement when faced with a difficult situation (–.76), and frequency of physical problems interfering with social activities (–.26). Family CAGE, race, marital status, education level, having a close friend, losing a friend or relative, frequency of contact with those closest to you, self-report number of visits to a doctor in the past year were not significantly associated with AUDIT score.

CONCLUSION: The associations presented do not necessarily suggest causality. As expected, being male, younger, having friends who are problem drinkers, and higher CAGE scores had the highest association with AUDIT scores. Counter to our a priori predictions, frequency of participating in spiritual activities and having a friend willing to help were associated with higher AUDIT scores. Interestingly, frequency of emotional problems is positively associated with AUDIT scores while frequency of physical problems is inversely associated. Also unexpected was the lack of association between Family CAGE and AUDIT score. Further investigation may help explain these unexpected findings.

RACIAL DIFFERENCES IN PREFERENCES FOR SPECIALIST CARE. M.D. Wong¹, S.M. Asch², R.M. Andersen¹, R.D. Hays¹, M.F. Shapiro¹; ¹UCLA, Los Angeles, CA; ²VAMC, Los Angeles, CA

PURPOSE: Having difficulty seeing a specialist is a common reason for dissatisfaction with care and for health plan disenrollment. We evaluated correlates of patients' expectations for care by specialists versus primary care providers (PCPs).

METHODS: Between 11/99 and 2/00, we surveyed patients in the waiting rooms of 3 academic, general internal medicine practices in California and New York. Patients were asked about their preference for first seeing their PCP or a specialist for 3 clinical scenarios and the health problem for which they sought medical care on the day on which they were surveyed. To identify the independent predictors of preferring to see a specialist, we combined patients' responses to the 3 scenarios and health problem and performed a multiple logistic regression, adjusted for clustering at the patient level using the Huber-White method. We also performed analyses separately for each of the 3 scenarios and the presenting health problem.

RESULTS: The proportion of patients who preferred to see a specialist first for the 3 scenarios was 13% for chest pain, 15% for knee pain, 11% for a rash, and was 8% for the presenting health problem. In the multivariable analysis, blacks were less likely than whites to prefer a specialist ($OR=0.45$, $p<0.0001$). Compared to those with fee-for-service insurance, those with VA coverage and Medicaid insurance were less likely to prefer a specialist ($OR=0.55$, $p=0.04$ and $OR=0.42$, $p=0.01$, respectively). Those who preferred a specialist had less trust in their PCP ($OR=1.6$, $p<0.0001$), had less confidence in their PCP's ability to diagnose and treat their problem ($OR=2.6$, $p<0.0001$), were more certain about their diagnosis ($OR=1.4$, $p<0.0001$), and perceived the health problem as being more serious ($OR=1.2$, $p=0.02$). Non-significant predictors included age, gender, employment, education, income, managed care insurance, health status and several attitudes including perceived susceptibility to illness, entitlement and assertiveness. These results were similar across the 3 scenarios and the presenting health problem.

CONCLUSION: Controlling for other sociodemographic characteristics, health beliefs and attitudes, blacks were much less likely to prefer care from a specialist. Though blacks have generally had worse access to care, lower quality of medical care, and worse health outcomes, their satisfaction with medical care has not consistently been lower than that of whites. Our results support the hypothesis that differences in expectations for care might explain this incongruity and raise the question about whether differences in expectations might contribute to disparities in health care quality and outcomes.

WITHHOLDING PROTEASE INHIBITORS FROM NON-ADHERENT PATIENTS WITH HIV INFECTION AND RACIAL DISPARITIES IN TREATMENT. M.D. Wong¹, N.S. Wenger¹, R.M. Andersen¹, M.F. Shapiro¹, P.D. Cleary², I. Wilson³, B.E. Landon², H. Liu¹, S.A. Bozzette⁴, W.E. Cunningham¹; ¹UCLA, Los Angeles, CA; ²Harvard University, Boston, MA; ³New England Medical Center, Boston, MA; ⁴UCSD, San Diego, CA

PURPOSE: Most providers say that they withhold protease inhibitors (PIs) from HIV-infected patients whom they believe will be non-adherent. We hypothesized that this prescribing attitude might influence patients' access to PIs and sought to determine whether this prescribing attitude explained why African Americans tend to receive PIs later than whites.

METHODS: In the HIV Cost and Services Utilization Study (HCSUS), we prospectively studied a nationally representative sample of HIV-infected patients ($n=2864$) and their providers in the U.S. We examined data for those with completed provider and patient surveys (398 providers taking care of 1892 patients (66% of original patient cohort)). We asked providers whether they prescribe PIs to patients whom they believe will be non-adherent. Patients were followed over a 2-year period and were asked at baseline and both follow-up surveys whether and when they began using PIs. Using parametric survival models, we determined the impact of providers' PI prescribing attitude on time to first PI use and on racial differences in time to first PI use.

RESULTS: Contrary to our hypothesis, having a provider who considers adherence before prescribing PIs (More Restrictive) predicted earlier PI use, adjusting for provider characteristics only (e.g. HIV knowledge and type of specialty) ($p=0.01$). However, providers' PI prescribing attitude was not associated with time to PI use after additionally adjusting for patient CD4 count, symptoms, age, gender, HIV risk factor, education, income, insurance and perceived access to care. After adjusting for patient and provider characteristics, the median time to first PI use was 74.2 days later for African Americans than for whites ($p=0.001$). This racial difference in time to PI use was similar for an individual with a More or Less Restrictive provider (adjusted racial difference in time to PI use was 74.7 and 73.3 days respectively, $p>0.20$).

CONCLUSION: Providers' PI prescribing attitude was not associated with patients' access to PIs. African Americans received PIs later than whites, but providers' PI prescribing attitude did not explain this racial disparity. Future studies will need to examine whether other patient or provider characteristics better explain why African Americans receive PIs later than whites.

WHICH DISEASES CONTRIBUTE MOST TO THE DISPARITY IN MORTALITY BETWEEN BLACK AND WHITE ADULTS? M.D. Wong¹, M.F. Shapiro¹, S.L. Ettner¹, ¹UCLA, Los Angeles, CA

PURPOSE: Though blacks have greater mortality rates than whites, it is not known which diseases contribute most to this disparity. We examined mortality rates as well as years of potential life lost (YPLL), which is a more sensitive measure of loss of life due to premature death.

METHODS: To examine nationally representative samples of adults in the U.S., we used data from the 1986, 1988 and 1990 National Health Interview Surveys (NHIS) for participants greater than age 25 with corresponding National Death Index data on each cohort through 1995. YPLL was calculated as the difference between the age at death and the maximum years that a person could have lived, which we set at age 85. Mortality rates and YPLL due to all and specific causes of death were calculated for blacks and whites and standardized to the age, sex, and educational distribution of the total U.S. population. We then calculated the percent contribution of racial differences in YPLL from specific causes of death to racial differences in YPLL from all-cause mortality.

RESULTS: Adjusted for age and sex only, YPLL due to all-cause mortality was 0.62 years/person greater for blacks than for whites, averaging across the 3 NHIS cohorts. However, this disparity disappeared after adjusting for education in addition to age and sex (black-white difference was -0.08 years/person). Age and sex-adjusted YPLL was greater among blacks compared to whites for all of the specific causes of death that we examined and for all 3 NHIS cohorts. The contribution of specific causes of death to the racial disparity in YPLL was 25.6% for infections, 14.0% for cancer, 6.1% for diabetes, 4.6% for stroke, 4.0% for ischemic heart disease, 3.3% for renal disease, 3.1% for lung disease, and 4.0% for accidents, homicides, and suicides combined. An analysis of mortality rates revealed similar results, with the exception that cancer contributed most to the racial disparities in mortality (22.6%) followed by infectious diseases (14.6%). For deaths due to infections, the black-white mortality rate difference was more pronounced in those younger than 60 years old compared to those 60 years or older (6.0 vs. 0.6 deaths/10,000 person-years, respectively). In contrast, the racial disparity for deaths due to cancer was much greater in those age 60 years than in those age <60 years (black-white mortality difference=20.2 vs. 3.5 deaths/10,000 person-years, respectively).

CONCLUSION: YPLL was greater among blacks than whites, though differences in socioeconomic status (using education as a proxy) appear to account for this disparity. Mortality rates and YPLL yielded slightly different results because YPLL accounts for the age at death and racial disparities in death due to infections were more pronounced at younger ages as compared to deaths from cancer. Future efforts to reduce the racial disparity in mortality may have the greatest impact if focused on infectious diseases and cancer, which we found contributed the most to the difference in YPLL and mortality rates between blacks and whites.

CHARACTERISTICS OF PATIENTS WITH ISCHEMIC HEART DISEASE WHO FOUND MEDICATION USE DIFFICULT. L. Woodard¹, N.R. Kressin², L.A. Petersen³, ¹Baylor College of Medicine, Houston, TX; ²Center for Health Quality, Outcomes & Economic Research, Bedford, Massachusetts; ³VAMC, Houston, TX

OBJECTIVES: Multiple studies indicate that patient compliance with cardiovascular medications is poor. One possible explanation is that patients perceive medication use as difficult. The goal of this study was to assess characteristics of patients who found medication use difficult, including the physician-patient relationship, health beliefs, and socioeconomic factors. **METHODS:** We prospectively surveyed 273 White and African American patients who underwent nuclear imaging studies graded as positive for cardiac ischemia at 5 VA hospitals (Houston, Atlanta, St. Louis, Pittsburgh, and Durham). Of these, 33 (12.1%) patients rated medication use as difficult. Difficulty was assessed by asking patients "How bothersome is it for you to take your pills for chest pain, chest tightness, or angina as prescribed?"

RESULTS: Patients who found medication use difficult were less likely than those who did not find it difficult to agree that they were satisfied with their current treatment (59.4% vs. 77.0% respectively, $P=0.03$) and that their doctors used words they could understand (68.8% vs. 88.3%, $P=0.003$). Among patients who found medication use difficult, there were also trends towards having less trust in their doctor's judgement about their medical care (78.8% vs. 87.5%, $P=0.17$) and feeling that they were less able to get to know their doctor (59.4% vs. 70.6%, $P=0.20$). There were no differences between patients who found medication use difficult and those who did not in the perception that their doctors were well qualified to manage their medical problems (84.4% vs. 86.1%, $P=0.80$), were experienced (66.7% vs. 74.4%, $P=0.35$), or that they were seen by doctors in training (71.9% vs. 60.8%, $P=0.22$). There were also no differences in health beliefs or socioeconomic factors, including percentage of African Americans (18.2% vs. 23.8%, $P=0.48$), mean age (59.4 years [± 10.3] vs. 61.6 years [± 10.1], $P=0.24$), mean education (11.5 years [± 3.0] vs. 12.0 years [± 2.9], $P=0.31$), percentage of married subjects (45.5% vs. 39.2%, $P=0.49$), unemployment (78.8% vs. 82.1%, $P=0.65$), and low-income status (71.0% vs. 65.6%, $P=0.57$).

CONCLUSION: We found that there are differences between patients who find medication use difficult and those who do not in patient satisfaction, attitudes toward physicians, and physician-patient relationships. However, there were no differences in socioeconomic factors, health beliefs, or perceived financial barriers to medication use. This study was limited by lack of information regarding the complexity and side effects of the medication regimen. Further studies should assess interventions aimed at increasing medication compliance by improving the physician-patient relationship.

PATIENT, PROVIDER AND SYSTEM BARRIERS TO DIABETES CARE IN A HIGH-RISK PRIMARY CARE SETTING. K.F. Woods¹, M.K. Hargreaves¹, U. Nwachuku¹, D.G. Schlundt², L. Mcneil¹, T. Adeola¹, U. Nwokoro¹, E.S. Butts¹, ¹Meharry Medical College, Nashville, TN; ²Vanderbilt University, Nashville, TN

PURPOSE: Type 2 diabetes mellitus (DM) is a costly and debilitating disease that disproportionately affects African Americans (AAs). In Nashville/Davidson County, TN, the

economic impact of DM is substantial and it is the 7th leading cause of death. In Davidson County, age-adjusted mortality rates for DM have increased 40.6% ($>2 \times$ the national standard) over the past decade and are 185.1% higher for AAs than for whites.

METHODS: To identify barriers to care, a personal diabetes questionnaire was administered to 84 patients (pts) enrolled in an intensive DM management program at a primary care clinic of a major teaching hospital [82% AA; 64% female; 91% MCD/MCR, 2% uninsured; age: 55.4 yr (SD 13.8); BMI: 33.5 kg/m² (SD 8.0)]. A 12-mo. retrospective analysis was performed on 100 charts of diabetic pts from the same clinic [77% AA; 67% female; 80% MCD/MCR, 4% uninsured; age 58.8 yr (SD 13.8); BMI: 33.7 kg/m² (SD 10.4)]. In addition, 35/50 faculty and residents that provide DM care in the clinic completed physician surveys. Variables were coded as pt, provider or system barriers if 30% or > of respondents endorsed an item as a problem some or all of the time. Provider and system barriers were also identified from chart review when 30% or > of data was missing.

RESULTS: Thirty-five (35) variables and 63 barriers to DM care were identified. Pt barriers (25.3%) included adherence with medication, glucose monitoring, diet and exercise. Non-compliance with appointments was affected by transportation, childcare, and work-related issues. Costs of care, pt literacy, lack of family support and other pt priorities were also cited as pt barriers. Provider barriers (45.2%) included knowledge of/agreement with ADA guidelines, reimbursement issues, prescribing uncertainty, and provider perceptions of pts with DM. System barriers (33.3%) included lack of insurance, DM care costs, reimbursement problems, lack of DM educators/nutritionists, poor record-keeping, difficulty obtaining lab data and provider communication issues. Chart review documented: HbA1c - 67%, UA - 52%, urine microalbumin - 20%, total cholesterol - 23.5%, LDL - 11.8%, foot exam - 25%, fundoscopic exam - 19%, influenza and pneumonia vaccine - 18% and 7%, respectively.

CONCLUSION: We found that provider and system barriers outweighed patient barriers to DM care in this high-risk population. In addition to pt empowerment, DM improvement programs must address provider and system obstacles that impede optimal DM management. Further clarification of specific barriers and their impact on diabetes care is needed, particularly in AA, underserved and/or high-risk populations.

IS PATIENT TRUST OF PHYSICIANS ASSOCIATED WITH THE LEVELS OF TRUST IN GENERAL? S. Yamashiro¹, ¹Saga Medical School, Saga, Japan

PURPOSE: To investigate the correlation between patient trust of physicians and the levels of trust in general or other factors at two sites: A university hospital outpatient (general internal medicine) clinic and a private primary-care clinic.

METHODS: The study of patient trust was conducted in November and December of 1999. Adult patients who had been cared for by the same physicians for more than 3 months and who had at least 4 visits during this period, were selected for this study. The patient trust of physician ratings were obtained by self-administered questionnaires, the Trust in Physician Scale (modified by the Stanford Trust Study Group). In addition to patient trust, we obtained a General Trust Scale (Yamagishi, Japan), patients' demographic data, the lengths of the patient-physician relationships, the reasons for the patients' choice of physicians. 275 patients answered the questionnaire to rate 11 different physicians during their visits. Multiple regression analysis was used for this study.

RESULTS: The Trust in Physician scale (5 Likert scales: 1 = totally disagree to 5 = totally agree) were transformed to a 0 to 100 scale, and it showed high internal consistency (Cronbach's alpha = .82) for Japanese patients. Mean scores of male and female were 86.3 (SD: 10.3) and 85.9 (SD: 10.5) respectively. The Trust in Physician scores were strongly associated with the General Trust scores ($p = .017$), the reasons for the patients' choices ($p = .037$) and the satisfaction with care ($p < .001$) after adjusted by patient characteristics. However, the Trust in Physician scores were not statistically associated with gender, age, education levels and the lengths of the relationships.

CONCLUSION: The patient trust, a measure of the health care seeking behaviors, was associated with the societal trust in general in our study. In addition, the patient trust was a useful concept to assess the relationship between patient and physician, especially the satisfaction with care.

THE RELATIONSHIP BETWEEN OBESITY AND HEALTH-RELATED QUALITY OF LIFE. W.S. Yancy¹, M.K. Olsen¹, H.B. Bosworth¹, E.C. Westman², D. Edelman¹, ¹Durham VAMC, Durham, NC; ²Duke University, Durham, NC

PURPOSE: Obesity causes substantial morbidity and negatively impacts health-related quality of life (HRQOL). The purpose of this study was to examine the relationship between obesity and HRQOL in a population with high morbidity and low HRQOL.

METHODS: DESIGN: A cross-sectional analysis using the Medical Outcomes Study Short-Form 36 (SF-36) questionnaire and body weight and height measurements. SETTING: Durham Veterans' Affairs Medical Center outpatient clinic. PARTICIPANTS: 1253 outpatients who volunteered to participate in a one-day screening session. ANALYSIS: We used multiple linear regression to examine the relationship of body mass index (BMI) with each of the eight SF-36 subscales while adjusting for age, race, gender, comorbid illnesses (Kaplan-Feinstein comorbidity index), depression (CES-D depression scale) and exercise history. BMI was divided into 6 categories using the World Health Organization classification system. Normal range (18.5–24.9 kg/m²) was the referent category in the analyses.

RESULTS: Participants had a mean age of 54.6 years (SD=5.6); 69% were white, 29% were black and 94% were male. The distribution of BMI was: <18.5 kg/m² (n=16, 1.3%), 18.5–24.9 kg/m² (n=254, 20%), 25–29.9 kg/m² (n=531, 42%), 30–34.9 kg/m² (n=308, 25%), 35–39.9 kg/m² (n=112, 8.9%) and ≥ 40 kg/m² (n=32, 2.6%). Mean SF-36 subscale scores for the sample were lower than U.S. norms by a mean of -18.7 points (range: -32.9 on Role-Physical to -3.5 on Mental Health). After adjusting for covariates, individuals with BMI ≥ 40 kg/m² had significantly lower scores compared with normal weight individuals on the Role-Physical ($\beta = -18.8$, $p=0.008$), Vitality ($\beta = -12.5$, $p=0.001$) and Social Functioning ($\beta = -9.3$, $p=0.017$) subscales (β represents the deviation in mean score from the referent BMI group). On the Physical Functioning subscale, lower than average scores were seen in individuals with BMI of 35–39.9 kg/m² ($\beta = -5.7$, $p=0.054$) and ≥ 40 kg/m² ($\beta = -18.4$, $p=0.001$). On the Bodily Pain

subscale, lower than average scores were seen in individuals with BMI of 25–29.9 kg/m² ($\beta = -4.8$, $p = 0.008$), 30–34.9 kg/m² ($\beta = -5.2$, $p = 0.011$), 35–39.9 kg/m² ($\beta = -7.2$, $p = 0.008$) and > 40 kg/m² ($\beta = -11.3$, $p = 0.012$). BMI had no significant relationship with the General Health, Role-Emotional and Mental Health subscales.

CONCLUSION: In a population with high morbidity and low HRQOL, class 3 obesity (BMI ≥ 40 kg/m²) was associated with decreased scores on 5 of 8 SF-36 subscales while BMI ≥ 35 kg/m² was associated with decreased physical functioning and BMI ≥ 25 kg/m² was associated with increased bodily pain. These relationships persisted even after controlling for severity of comorbid illness and other covariates.

CLINICIAN/ORGANIZATION-CENTERED RESEARCH

GENDER DISPARITIES AFFECTING DEMAND FOR ON-CALL SERVICES. S.A. Bookire¹, D.W. Bates¹, H. Sandige², T.H. Lee¹; ¹Brigham and Women's Hospital and Partners Health Care, Boston, MA; ²Partners Health Care, Boston, MA

PURPOSE: Studies have shown that female physicians earn less, are less likely to hold high level academic positions, publish less frequently, and are less likely to have professional mentors than male physicians. Female physicians generally have more female patients than their male colleagues, since patients generally prefer same-sex physicians. Female patients tend to visit their physicians more frequently and ask more questions of their physicians. Therefore female primary care physicians may also experience more demands from within their practices than male physicians do. We performed a study asking: (1) Do female patients call (or are called about) more frequently than male patients call? (2) Do the patients of female physicians call (or are called about) more often than patients of male physicians? (3) What factors other than gender of physicians and patients may contribute to patient calling rate?

METHODS: We studied the hospital-based primary care practice of the Brigham and Women's Hospital in Boston, Massachusetts. Physicians on call recorded a standardized log for thirteen consecutive weekends. The time each physician was signed out was tracked to estimate missed calls. We determined, from our computerized physician records, physician panel sizes, proportion female, the number of weekly clinical sessions, and physician demographics. We adjusted each primary care physician's calls for potential missed calls by dividing by percentage of time signed out. Using Chi Square analysis, patient call rates were evaluated based on overall patient gender mix, and physician gender was evaluated based on physician panel size. Physician age, years since training, sessions per week and panel size were assessed for their possible effect using logistic regression analysis. Statistical analyses were performed using SAS version 6.12.

RESULTS: Among the 38 physicians in this hospital-based primary care practice, 61% were female. They cared for 31,986 patients who were 72% female. Female physicians had a higher proportion of female patients (86% vs. 42%), larger mean panel sizes (958 vs. 666), and practiced more sessions per week (3.9 vs. 2.8) than their male counterparts. Of the 652 calls during the study, female patients called more frequently than male patients ($p < .001$), and female patients generated more total calls than male patients ($p < .001$). Also, female physicians had significantly more calls and callers than male physicians did ($p < .001$). Physician age, specialty, and years in practice did not affect calling rate.

CONCLUSION: Compared to male physicians, female physicians experience more demand for on-call service. In addition to slower advancement, factors intrinsic to their practice may contribute to higher reported job stress among female physicians. Physician gender may be an important factor when considering severity adjustments to determine reimbursement schedules.

"WE'RE JINXED!" - ARE RESIDENT'S FEARS FOUNDED? A.C. Ahn¹, B. Nallamothu², S.K. Saint²; ¹Massachusetts General Hospital, Jamaica Plain, MA; ²University of Michigan, Ann Arbor, MI

PURPOSE: Many house officers believe that there is bad karma bestowed on them when supportive remarks are expressed during call days. Remarks such as "Hope you have a great call!" or "Hope you have no more admissions!" are interpreted as ill-fortuned greetings that may do more harm than good. The purpose of this study is to determine whether this fear of being "jinxed" is based on objective findings or mere psychological superstition.

METHODS: We performed a randomized controlled trial using senior medical residents at 2 university-affiliated hospitals. On the morning of a call day, each resident was randomized by selecting either a Jinx ("You will have a great call day!") or No Jinx message (i.e. blank form) from a collection of unlabeled envelopes. Primary outcomes measured included the number of total admissions and hours of sleep for each resident, and the subjective level of difficulty of the resident's call day (on a scale from 1 to 5 with 1 being easiest and 5 the most difficult). Some residents were randomized more than once during a call month. Student *t*-tests assessed for differences in outcomes between the Jinx and No Jinx groups, with robust variance estimates used to adjust for the effects of clustering at the resident level.

RESULTS: Thirty-three Jinxes and 36 No Jinxes were performed on 30 residents. No statistically significant differences were seen between the 2 groups in important baseline characteristics such as *a priori* level of superstition or admitting service (i.e., general medical service, cardiology service). Relative to those in the No Jinx group, residents who were jinxed had significantly *fewer* admissions (5.1 versus 6.2; $P = 0.021$), *more* hours of sleep (3.8 versus 2.9; $P = 0.034$), and a *lower* subjective level of difficulty (2.9 versus 2.5; $P = 0.040$) during their call day. **CONCLUSION:** There is no evidence that offering encouraging remarks to residents on a call day "jinxes" them. Therefore, such remarks may be given safely to residents without fear of causing more admissions or less sleep, or worsening a call day's level of difficulty.

GAMING THE SYSTEM - PUBLIC PREFERENCES THAT DOCTORS PLAY? G. Alexander¹, R. Werner¹, A. Fagerlin², P. Ubel²; ¹University of Pennsylvania, Philadelphia, PA; ²Ann Arbor Veterans Affairs Medical Center, Ann Arbor, MI

PURPOSE: Efforts to contain rising costs have lead third party payers to restrict access to some healthcare services. Previous research suggests that the "hassle" of the appeals process influences physicians' sanctioning of deception to obtain medically necessary services. However, little is known about whether the public supports such deception, and if so, whether this support is sensitive to the burden of the appeals process.

METHODS: A clinical vignette was administered to 700 jurors and to a randomly selected national sample of 1617 physicians. Vignettes differed by condition severity (severe angina vs. moderate low-back pain), likelihood of successful appeal (95% vs. 50%), and length of appeals process (5-10 minutes vs. 1 hour). Participants were asked whether in response to a restriction on medically necessary healthcare the doctor in the vignette should (1) appeal the restriction, (2) accept the restriction, or (3) misrepresent the facts in order to obtain the desired service.

RESULTS: Overall, the public was more likely than physicians to sanction deception (26% vs. 11% respectively, $p < .001$), and less likely to support accepting (4% vs. 12%, $p < .001$) or appealing (70% vs. 77%, $p < .001$) the restriction. Sanctioning of deception was greater among the public than physicians regardless of the vignette used. The public's support of misrepresenting (vs. appealing) was greater with a lower likelihood of successful appeal (Odds ratio [OR] 1.67, 95% confidence intervals [CI] 1.18, 2.37) but was not significantly related to the length of appeals process (OR 1.10, 95% CI .78, 1.55). Among physicians, the likelihood of sanctioning deception was significantly greater with both a lower likelihood of successful appeal and a longer length of the appeals process.

CONCLUSION: Sanctioning of physician deception is even greater among the general public than among physicians, and unlike physicians, the public's support of deception is not sensitive to the length of the appeals process. Patients' preferences regarding deception of insurance companies may further pressure physicians who struggle to balance the tension between patient advocacy and honesty.

EVALUATING THE IMPACT OF ADMISSION DAY ON COST AND QUALITY OF CARE. N. Armistead¹, M. Davidson¹, K. Evans¹, G. Hobbs¹, W. Shockcor¹, M. Elnicki¹; Dept of Medicine, West Virginia University, Morgantown, WV

PURPOSE: Previous studies have demonstrated increased cost and length of stay (LOS) associated with weekends. The factors contributing to this impact are not clear. We hypothesized that patients admitted to our institution on Fridays have increased cost and LOS compared with patients admitted on Mondays. We also investigated specific factors that could be contributing to the increased cost.

METHODS: The study population included all General Medicine admissions from 1/1/96 through 12/31/99, who were hospitalized for 7 days or less. We compared admissions on Fridays versus Mondays with regards to the following parameters: Direct cost, LOS, Labs, Radiology, Admission status of "Observation" (OBS) versus "Inpatient" and for the OBS patients, the hours spent in OBS, Discharge destination and LOS for patients needing Skilled Nursing placement (SNF) or home health (HH) services and Mortality.

RESULTS: Patient distribution was similar on all days of the week with respect to age, gender, race, payer mix, case-mix index (CMI) and distribution of DRGs. The key findings are listed in the table.

Group (n)	Mean Direct Cost $\pm$ SD*	LOS $\pm$ SD*	Labs*	Diag Rad†	% Admitted to OBS*	Mean Hrs in "OBS"†
Fri (951)	1696 $\pm$ 1194	3.1 $\pm$ 1.9	29	1.9	24	29
Mon (1176)	1554 $\pm$ 1133	2.6 $\pm$ 1.6	26	1.7	30	27

* $p < 0.01$

† $p < 0.05$

There was no difference in the two groups in the percentage of patients discharged to Skilled Nursing facilities or to home with home health. However, for the patients needing SNF placement there was increased LOS for Friday admissions (mean LOS: Fri 4.7, Mon 3.7; $p = 0.03$). Similar trend was present for patients needing HH but did not reach statistical significance (mean LOS: Fri 4.4, Mon 4.0, $p = 0.20$). There was no difference in mortality (% Expired: Fri 1.22%, Mon 1.88%; $p = 0.45$). Analysis of the yearly trend revealed decreasing cost and LOS through the four years, but the Mon-Fri difference was maintained.

CONCLUSION: Patients admitted on Fridays have higher costs and LOS, despite similar case mix. They have more labs and diagnostic radiology testing done. They are admitted under "Observation" status less frequently and spend more time in observation. They wait longer for placement. Despite recent successes in decreasing cost and LOS, weekends continue to play a significant role in the cost of healthcare. Further reduction in cost will require attention to the factors we have identified including reduction in diagnostic testing and more efficient discharge planning over the weekend.

HOW WELL DO POSTOPERATIVE MORTALITY AND PULMONARY COMPLICATION RATES CORRELATE AT THE HOSPITAL LEVEL? A.M. Arozullah¹, W. Henderson², S. Khuri³, J. Daley⁴; ¹University of Illinois at Chicago/VA Chicago Healthcare System, Chicago, IL; ²Hines VA Hospital, Hines, IL; ³VA Boston Healthcare System, West Roxbury, MA; ⁴Harvard University, Boston, MA

PURPOSE: Reporting other outcome-based measures in addition to traditional mortality rates may improve the measurement of the quality of surgical care. The main objective of

this study was to determine the correlation between risk-adjusted hospital rates and rankings for postoperative mortality, postoperative pneumonia (POP), and respiratory failure (RF).

METHODS: Cases were selected from those who underwent major noncardiac surgery at 44 VA hospitals participating in Phase I and II of the National VA Surgical Quality Improvement Program. Postoperative mortality was defined as death within 30 days postoperatively. POP was defined as a positive sputum culture with antibiotic treatment or an infiltrate on chest x-ray diagnosed as pneumonia or pneumonitis following surgery. RF was defined as mechanical ventilation greater than 48 hours postoperatively and/or reintubation and mechanical ventilation subsequent to postoperative extubation. Separate logistic regression models predicting mortality, POP, and RF were developed using Phase I cases (10/91–12/93) and validated using Phase II cases (1/94–8/95). These models estimated the probability of mortality, POP, and RF for each case and the sum within each hospital represented the 'expected' number of deaths or complications. Hospital mortality, POP and RF observed to expected (O/E) ratios were calculated. Hospitals were ranked by O/E ratio resulting in six hospital outcome measures: mortality O/E ratio, mortality rank, POP O/E ratio, POP rank, RF O/E ratio, and RF rank. O/E ratio correlations were assessed using Pearson correlation coefficients. Rank correlations were assessed using Spearman rank-correlation coefficients.

RESULTS: POP vs. RF vs. Mortality O/E ratio: There was significant correlation between POP and RF O/E ratios in Phase I ($r=0.33$, $p=0.03$) and Phase II ($r=0.52$, $p<0.01$). There was no significant correlation between POP and mortality O/E ratios in Phase I ($r=0.11$, $p=0.49$), but there was significant correlation in Phase II ($r=0.37$, $p=0.01$). There was no significant correlation between RF and hospital mortality O/E ratios in Phase I ($r=0.29$, $p=0.054$) or Phase II ($r=0.21$, $p=0.17$). RF vs. POP vs. Mortality rank: There was a significant correlation between RF and POP rank in Phase I ($r=0.36$, $p=.02$) and Phase II ($r=0.54$, $p<0.01$). There was also significant correlation between RF and mortality rank in Phase I ($r=0.32$, $p=0.03$) and Phase II ($r=0.33$, $p=0.03$). There was no correlation between POP and mortality rank in Phase I ($r=-0.02$, $p=0.90$), but there was a significant correlation in Phase II ($r=0.30$, $p=0.04$). In comparing Phase I O/E ratios and rankings with their Phase II counterparts, mortality O/E ratios ($r=0.39$, $p<0.01$), mortality rankings ($r=0.47$, $p<0.01$), POP O/E ratios ($r=0.77$, $p<0.01$), POP rankings ($r=0.74$, $p<0.01$), RF O/E ratios ($r=0.57$, $p<0.01$), and RF rankings ($r=0.63$, $p<0.01$) were all highly correlated across Phases.

CONCLUSION: There is significant correlation between risk-adjusted hospital O/E ratios and hospital rankings of POP and RF. There is significant correlation between hospital mortality ranking and RF ranking, but no correlation between mortality and RF O/E ratios. There is inconsistent correlation between hospital mortality and POP O/E ratios and rankings. Reporting complementary measures, such as risk-adjusted hospital POP and RF rates and rankings, along with mortality rates may improve the detection of high quality and low quality hospitals by measuring dimensions of surgical care not captured by mortality rates alone.

THE IMPACT OF FINANCIAL PRESSURES ON PHYSICIAN PARTICIPATION IN SUPPLEMENTAL INCOME-GENERATING ACTIVITIES. B. Ashar¹, R. Miller¹, K. Getz¹, G. Prokopowicz¹, N. Powe¹, ¹Johns Hopkins University, Baltimore, MD

PURPOSE: Due to a number of factors, many clinicians have experienced a decline in their income over the past few years. To compensate for this decline, some physicians have chosen to supplement their income through means not associated with their primary practice of medicine (e.g. drug trials, speaking engagements, malpractice case review, product sales). This is the first study to qualify and quantify the extent of participation in such supplemental income-generating activities (SIGAs).

METHODS: We conducted a cross-sectional mailed survey of 1000 Maryland members of the American College of Physicians to determine the frequency of participation in various SIGAs. Members were sampled sequentially with a random start. The survey consisted of 24 items designed to collect information on demographics, practice patterns, income variables, and participation in nine different SIGAs. We used Chi-square analysis to determine whether relationships existed between personal and professional characteristics and involvement in SIGAs.

RESULTS: After the first two mailings, our response rate was 40%. Sixty-three percent of those surveyed reported engaging in at least one SIGA. The three most common activities noted by these physicians are shown in the table below. Statistical analysis suggested a significant association between participation in SIGAs and male gender, married status, age between 41 and 50, having dependents, dissatisfaction with income, urban or suburban practice setting, and medical subspecialization.

CONCLUSION: Physician participation in income-generating activities outside of their primary medical practices is widespread, perhaps reflecting an increase in financial pressures caused by decreasing reimbursements and managed care limitations. The impact of time diverted from the primary practice of medicine on continuity of care, as well as the ethical issues surrounding the enrollment of patients in clinical trials, the sale of medical and/or non-medical products from the office, and physician relationships with the pharmaceutical industry needs to be further examined.

Common Supplemental Income-generating Activities

	Physician Participation
Pharmaceutical company sponsored lectures	27%
Legal review/expert witness testimony	24%
Pharmaceutical company sponsored clinical trials	22%

INTEROBSERVER AGREEMENT IN THE ECHOCARDIOGRAPHIC DIAGNOSIS OF AORTIC INSUFFICIENCY AND STENOSIS. A.N. Babu¹, S.R. Bitar¹, S.M. Kymes¹, ¹St. Louis VAMC, St. Louis, MO

PURPOSE: Transthoracic echocardiograms (TTE) are frequently ordered by generalists to assess a heart murmur or screen for valvular heart disease. Little is known about interobserver agreement (IOA) in diagnosing aortic insufficiency (AI) or stenosis (AS) by TTE. The limited studies done have suffered from one or more of the following: 1) verification bias, 2) small sample size, 3) limited generalizability as observers were given a specific interpretation protocol and 4) possible Hawthorne effect. Despite these features fostering IOA, some studies have reported significant lack of IOA, particularly for AI. Our submission avoids the above deficiencies and appears to be the largest analysis of IOA in the diagnosis of AI and AS by TTE. **METHODS:** 143 patients undergoing TTE for any indication were prospectively enrolled in a blinded study assessing components of the physical exam for diagnosing AI. The TTE, incorporating color and continuous flow Doppler analysis, were performed by an experienced technician and officially read by one of three staff cardiologists. These studies were also interpreted by a blinded study cardiologist (SRB) for the presence of AI or AS. All observers were experienced in TTE interpretation and none (including SRB) were aware of each other's findings or the planned IOA analysis. No interpretation protocol was specified. SRB's findings were compared to the official readings. AI/AS was graded 0 (absent) to 3 (severe). IOA was assessed by the kappa statistic (K) using SPSS v. 9.0.

RESULTS: Intraobserver agreement for SRB evaluated on 8 TTE was 100%. IOA findings for presence/absence of AI/AS for the remaining 135 TTE are tabled. IOA for AI and AS severity was within 1 grade for 100% of all observer pairs.

CONCLUSION: There was good-excellent IOA in diagnosing AS and good IOA for AI. For the group as a whole, IOA was better for AS than AI. IOA for AI/AS severity was within 1 grade for 100% of observer pairs. Our analysis supports the practice of generalists utilizing TTE to reliably establish the presence, absence or severity of AI and AS.

K values for IOA in diagnosing AI/AS by TTE, comparing SRB to observer A/B/C

Observer	N	AI	AS
A	64	.52	.58
B	56	.77	.79
C	15	.59	1.00

CHANGES IN HMO PHYSICIAN BURNOUT: 1991 TO 1999. K.H. Bachman¹, D.K. Freeborn², ¹Northwest Permanente, Clackamas, OR; ²Center for Health Research, Portland, OR

PURPOSE: 1) To investigate change in self-reported burnout in HMO physicians in 1991 to 1999; 2) To identify attitudinal and demographic correlates of burnout; and 3) To identify the changes in attitudinal and demographic correlates of burnout between 1991 and 1999.

METHODS: Data came from similar mail surveys performed in 1991 and 1999 of all physicians practicing at a large group model HMO. Response rates were 85% and 77% respectively. Data were not linked by respondent. The single item burnout measure used in this survey has previously been validated in this and other settings.

RESULTS: Self-reported burnout increased in prevalence from 19% of the 439 respondents in 1991 to 41% of the 482 respondents in 1999 ($p=.001$). Between 1991 and 1999, the number of female and part time physicians increased as did the average age and organizational tenure. Burnout was higher in physicians in the middle range of tenure (6–15 years) and in 1999 among physicians working part-time but was not associated with physician specialty, age, or gender. Burnout was associated with workload factors such as stress from long hours, inadequate time for patient visits, and the perception that the visit rate was too high to do a good job ($p=.001$). Negative attitudes regarding relationships with colleagues, dissatisfaction or stressful relations with patients, and negative attitudes about organizational control of practice were also associated with burnout ($p=.001$).

From 1991 to 1999, the number of physicians reporting great stress from long hours increased from 15 to 41%, physicians reporting great stress from inadequate time for patient visits increased from 37 to 48%, and the number of physicians reporting that the visit rate was too high to do a good job increased from 31 to 54% ($p=.001$ for all). The quality of professional relations was perceived to be worsening by 20% of physicians in 1991 but by 40% in 1999 ($p=.001$). Although satisfaction with patient relations and continuity of care remained stable, physicians experienced higher levels of stress from demanding patients in 1999 than in 1991 ($p=.004$). Measures of perceived organizational control of practice were unchanged.

CONCLUSION: Key correlates of the increased burnout rates noted in 1999 appear to be stress from workload factors, stress from demanding patients, and perceptions of declining quality of professional relations. Further research needs to elucidate organizational and structural changes that can prevent burnout by modifying perceptions of physician workload and improving relationships with colleagues and patients.

DO SATISFIED PHYSICIANS MAKE FOR SATISFIED PATIENTS? D.R. Barnett¹, P.F. Bass¹, J.F. Wilson¹, C.H. Griffith¹, ¹University of Kentucky, Lexington, Kentucky

PURPOSE: Little research has been done to evaluate patient and physician satisfaction within the same visit. We were especially interested in how continuity of care, the order in which patients are seen, and whether dissatisfaction with the previous patient would influence the satisfaction of the physician and patient within a clinic visit.

METHODS: Physician and patient satisfaction was assessed in our internal medicine resident continuity clinic. Trained interviewers administered a patient satisfaction questionnaire upon completion of the patient's clinic visit. The patient questionnaire was comprised of eight 10-point Likert-type items. Cronbach's alpha for the patient questionnaire was 0.97. Physician satisfaction was obtained by written self-administered questionnaires. The questionnaire was

comprised of three 5-point Likert-type items. Cronbach's alpha for the physician questionnaire was 0.88. Other items assessed by both questionnaires included contextual aspects of the clinic, diagnoses of patients, continuity, and demographic data. 416 matched physician and patient satisfaction questionnaires were obtained.

RESULTS: Most clinic encounters were satisfying to physicians and patients. The mean physician satisfaction score was 4.01, SD .92 (5-point scale) and the mean patient satisfaction score was 9.46, SD 1.13 (10-point scale). Continuity of care was associated with patient satisfaction. Patient satisfaction was 9.65 if they had seen the physician previously; whereas, it was 9.24 if they had not seen the physician previously ($t=3.61$, $p=.0004$). However, physician satisfaction was not influenced by previously seeing the patient. Physician satisfaction for an encounter was greatest if the patient was the first of the clinic day compared to all others (4.15 vs. 3.96, $t=3.05$, $p=.002$). Similarly, patient satisfaction was greatest if they were the first patient of the clinic day (9.69 vs. 9.39, $t=3.13$, $p=.001$). Physician satisfaction with the immediately preceding patient encounter also appears to influence satisfaction within a clinic visit. Physician satisfaction with a clinic visit is lower if the physician was dissatisfied with the patient from the immediately preceding clinic visit ($p=.001$). However, physician satisfaction with the preceding clinic visit does not influence patient satisfaction.

CONCLUSION: The majority of clinic visits are satisfying to physicians and patients. However, clinic continuity, the order in which patients are seen and even the previous patient can influence satisfaction within a clinic visit. These insights can guide the development of educational strategies to improve the patient-physician interaction, which in turn should enhance the satisfaction of physicians and patients with clinical encounters.

IS THERE A JULY PHENOMENON? THE EFFECT OF JULY ADMISSION ON INTENSIVE CARE MORTALITY AND LENGTH OF STAY. *W.A. Barry*¹, G.E. Rosenthal¹; ¹University of Iowa Hospitals and Clinics and Iowa City VAMC, Iowa City, IA

PURPOSE: It has been hypothesized that the inexperience of new housestaff early in the academic year (i.e., July Phenomenon) may lead to worse patient outcomes. However, little empirical data exist. The purpose of this study was to determine relationships between intensive care unit (ICU) admission during the first three months of the academic year and in-hospital mortality and length of stay (LOS) in teaching and non-teaching hospitals.

METHODS: We conducted a retrospective cohort study of 156,136 consecutive admissions (mean age 63.0 years, 51.7% male) to 38 ICUs in 28 hospitals in a large mid-western metropolitan area during 1991–1997. Hospitals included 5 major ($n=48,853$), 6 minor ($n=29,995$), and 17 non-teaching ($n=77,288$) facilities. Admission severity of illness was measured using the APACHE III methodology, which considers, age, comorbidity, admission source, diagnosis, and abnormalities during the first 24 hours of admission in 17 physiologic variables.

RESULTS: Overall unadjusted in-hospital mortality was 12.2%, and was similar during July–September and October–June in major (14.2% vs. 14.5%; $p=.37$) and minor (11.5% vs. 11.5%; $p=.97$) teaching hospitals, but differed in non-teaching hospitals (10.3% vs. 11.3%; $p<.001$). Adjusting for severity of illness using logistic regression, the odds of death was similar ($p>.1$) in July–September admissions, relative to October–June admissions, in major (OR 0.96; 95% CI, 0.89–1.03), minor (OR 1.06; 95% CI, 0.96–1.19), and non- (OR 0.95; 95% CI, 0.88–1.01) teaching hospitals. Results were similar when admissions in July, August, and September were examined separately, as shown in the table below (data reflect OR (95% CI)).

Teaching Status	July	August	September
Major	0.92 (0.81–1.04)	0.88 (0.78–1.00)	1.08 (0.96–1.21)
Minor	1.07 (0.90–1.27)	1.02 (0.86–1.21)	1.10 (0.93–1.30)
Non-teaching	0.97 (0.87–1.08)	0.92 (0.83–1.03)	0.95 (0.85–1.06)

Likewise, the relative LOS of ICU admissions in July–September was nearly identical to admissions in October–June in major teaching hospitals (1.00; 95% CI, 0.98–1.02). This was also true in minor teaching and non-teaching hospitals.

CONCLUSION: We found no evidence to support the existence of a July phenomenon in ICU patients. Patients admitted to teaching hospital ICUs during July, August, and September, have similar adjusted mortality and LOS. These findings suggest that residency programs may compensate for the inexperience of new housestaff during the first 3 months of the academic year.

DOES DATE OF ADMISSION DURING THE MONTH AFFECT INTENSIVE CARE OUTCOMES? RESULTS OF A REGIONAL ANALYSIS. *W.A. Barry*¹, G.E. Rosenthal¹; ¹University of Iowa Hospitals and Clinics and Iowa City VAMC, Iowa City, IA

PURPOSE: Discontinuity of care in teaching hospitals may occur when teams rotate on and off service. The purpose of this study was to determine relationships between the timing of admission (during the month) and in-hospital mortality and length of stay (LOS) in patients admitted to intensive care units (ICUs).

METHODS: We conducted a retrospective cohort study of 156,136 consecutive admissions (mean age 63.0 years, 51.7% male) to 38 ICUs in 28 hospitals in a large mid-western metropolitan area during 1991–1997. Hospitals included 5 major ($n=48,853$), 6 minor ($n=29,995$), and 17 non-teaching ($n=77,288$) facilities. Data were collected from patients' medical records. Admission severity of illness was measured using the APACHE III methodology. Multivariable analyses compared mortality for admissions on Days 1–5, 6–10, 11–15, 16–20, 21–25, and 26–31 of each month, adjusting for severity of illness and month and weekday of admission.

RESULTS: In-hospital mortality was higher in patients admitted on Days 1–5, 6–10, and 26–31, compared to admissions on Days 11–25 in major teaching hospitals (14.7%, 14.7%, 14.5%, and 10.9%, respectively; $p<.01$). Adjusting for severity of illness and other factors using logistic

regression, the odds of death was higher in patients admitted to major teaching hospitals ICUs on Days 1–10 or 26–31, relative to patients admitted on Days 11–25 (OR 1.11; 95% CI, 1.04–1.18; $p=.01$). Adjusted odds of death were not higher during these periods in minor teaching (OR 1.03; 95% CI, 0.94–1.13; $p=.59$) or non-teaching (OR 0.95; 95% CI, 0.90–1.00; $p=.07$) hospitals. Results were similar when admissions on Days 1–5, 6–10, and 26–31 were examined separately, as shown in the table below (data reflect OR (95% CI)).

Teaching Status	Days 1–5	Days 6–10	Days 26–31
Major	1.12 (1.04–1.20)	1.10 (1.00–1.21)	1.11 (1.04–1.18)
Minor	1.05 (0.95–1.16)	1.01 (0.88–1.17)	1.03 (0.94–1.13)
Non-teaching	0.96 (0.90–1.03)	0.95 (0.87–1.03)	0.95 (0.90–1.00)

In linear regression analyses, LOS was nearly identical in major, minor, and non-teaching hospitals (relative LOS 1.01, 1.00, and 1.00 respectively).

CONCLUSION: Patients admitted to ICUs in major teaching hospitals early or late in a month have a modestly higher risk of death, even after adjusting for admission severity of illness. A similar pattern was not observed in minor teaching and non-teaching hospitals. These findings may reflect the discontinuity of care that typically occurs when housestaff and attending teams rotate on and off service.

PATIENT LOYALTY, SOCIETAL RESPONSIBILITY AND PHYSICIAN CAREER SATISFACTION. *M.C. Beach*¹, L. Meredith², K.B. Wells², D.E. Ford¹; ¹Johns Hopkins University, Baltimore, MD; ²RAND, Santa Monica, CA

PURPOSE: To explore the extent to which physicians agree that their main responsibility is to individual patients rather than to society, to understand which physician and practice characteristics may influence this attitude, and to examine how this attitude affects physicians' career satisfaction.

METHODS: We surveyed 500 primary care providers from 80 outpatient clinics in 11 managed care organizations (MCOs) and received 414 completed surveys (response rate 83%). We used t-tests, chi-squared tests and logistic regression to compare the socio-demographic and practice characteristics of physicians who strongly agreed versus all other responses with the statement "The practitioners main responsibility is to each individual patient rather than to society" (the individual-patient ethic). We then examined the relationship between the individual-patient ethic and career satisfaction using chi-squared tests and logistic regression with career satisfaction as the dependent variable.

RESULTS: 28% of physicians strongly agreed, 42% somewhat agreed, 13% were neutral and 17% somewhat or strongly disagreed that their main responsibility was to each individual patient rather than society. Physicians with the individual-patient ethic were older (43% physicians >50 years vs 21% physicians <35 years, $p=.009$), spent more hours/week in direct patient care (36 vs 32 hours, $p=.048$), and tended to practice in network rather than staff model MCOs (33% vs 24%, $p=.077$). There was no difference in gender, race, specialty, or geographic region between physicians with or without the individual-patient ethic. In multivariate analyses, physician age remained the only significant predictor of the individual-patient ethic after controlling for all other physician and practice characteristics. When examining the relationship of the individual-patient ethic to career satisfaction, physicians with the individual-patient ethic were more satisfied with their ability to provide good quality care (38% vs 19% very satisfied, $p=.000$) and more satisfied with their ability to serve enrollee needs (28% vs 12% very satisfied, $p=.000$) but were not significantly more satisfied with their overall careers (20% vs 14%, ns). After controlling for physician age, gender, specialty, number hours/week in direct care, type of MCO and geographic region, physicians with the individual-patient ethic were more likely to be very satisfied with their ability to provide good quality patient care (AOR 3.16, 95% CI 1.88–7.75), to serve enrollee needs (AOR 3.81, 95% CI 1.87–7.75) and more likely to be very satisfied with their overall career (AOR 2.13, 95% CI 1.04–4.39).

CONCLUSION: Physicians who strongly agree that their main responsibility is to each individual patient rather than society are older and more satisfied with their careers. The extent to which this attitude impacts patient care is unknown.

QUALITY OF PRIMARY CARE DELIVERY IN GENERAL AND WOMENS HEALTH CLINICS. *B.A. Bean-mayberry*¹, C.H. Chang¹, M.A. Mcneil¹, J.C. Whittle¹, P.M. Hayes¹, S.H. Scholle¹; ¹University of Pittsburgh, Pittsburgh, PA

PURPOSE: Fragmentation of health services may hamper women's ability to obtain comprehensive, coordinated healthcare. The Veterans Administration (VA) has encouraged its medical facilities to create specialized women's health clinics (WHC) to meet these needs. This study compares patient perceptions of the quality of primary care delivery in WHC versus traditional VA general medicine or primary care clinics (PCC).

METHODS: We mailed an anonymous survey to women veterans from the 10 VA facilities in the Mid-Atlantic region. For the 8 facilities with a WHC code, we randomly sampled 170 veterans who used the WHC, and 80 who used a PCC from March 1, 1999, to March 1, 2000. At the other 2 sites, we selected 250 women who visited a PCC. The initial mailing was followed at one week intervals by a reminder postcard then a repeat survey. Patient perceptions of primary care were measured using the original 19-item Components of Primary Care Index (CPCI) by Flocke (1997). The 4 subscale categories include: patient preference for provider; interpersonal communication; accumulated knowledge; and coordination of care. Due to the skewed distribution of ratings, we used stepwise ordinal logistic regression to evaluate the impact of WHCs.

RESULTS: Patients in WHC rated their care significantly higher on all four CPCI subscales. Compared to patients using a PCC, the odds of a perfect score in a WHC was 1.51 for patient preference, 1.20 for communication, 1.58 for accumulated knowledge, and 1.60 for

coordination ($p < 0.01$ for all subscales). Additionally, other provider characteristics significantly influenced patient ratings. The presence of a regular VA provider was associated with higher ratings on all four subscales (OR 2.53, 1.53, 2.44, and 3.26 respectively, $p < 0.001$). Among the women who had a regular provider, those who used a female practitioner also rated their care significantly higher on all subscales (OR 1.67, 1.80, 2.16, and 1.83, $p < 0.001$).

CONCLUSION: Female veterans using specialized women's clinics perceive a higher quality of primary care delivery even after controlling for demographics, health status, and provider characteristics. Further details about model of care, type of provider, and health outcomes for women veterans need to be examined. Since WHC appear to differ from PCC on some factors, one may infer that selection bias accounts for the differences in patient ratings between clinics. Further analyses will account for selection bias through Heckman models or other methods.

GIVING CREDIT FOR PEER REVIEW: REVIEWER OPINIONS ABOUT MANUSCRIPT-SPECIFIC ACKNOWLEDGEMENT. M. Berkowitz¹, F. Davidoff¹; ¹University of Pennsylvania, Philadelphia, PA

PURPOSE: Peer reviewers make important contributions to many published journal manuscripts but receive limited credit for their work. We assessed peer reviewers' opinions of a proposal to acknowledge reviewer contributions to published manuscripts by publishing their names with those manuscripts.

METHODS: Reviewers for *Annals of Internal Medicine* were asked what they thought generally about published manuscript-specific acknowledgement, if they would want it personally, if it would be recognized at their institutions, and if they would want it if it were institutionally recognized. Respondents answered on a 5-point scale with 1 representing unfavorable and 5 favorable responses. We hypothesized that junior and clinical more than senior and research reviewers would desire published manuscript-specific acknowledgement and compared these groups using rank sum and t tests.

RESULTS: Seventy-eight reviewers responded (80%). Reviewers were an average of 22.5 years out of training, and spent equivalent mean times of 30% in clinical and research activities. 66%[55,76] were against manuscript-specific acknowledgement (mean score with s.d 2.7±1.3), 68%[57, 79] said they would not want it personally (2.6±1.4), 84%[73,91] thought it unlikely that published acknowledgement would be recognized by their institutions (2.2±1.3), but 51%[39,63] said they would want it if it were (3.3±1.3). There were no differences in responses based on years from training or clinical or research focus, but the difference in wishes for acknowledgement without and with institutional recognition was significant ($p = .004$).

CONCLUSION: A majority of peer reviewers at various career levels and in clinical and research tracks did not favor publishing manuscript-specific acknowledgements, in part because of perceptions of a lack of institutional recognition for the effort. Close to half favored the idea if it were linked to institutional recognition, however, suggesting divided support for broadening acknowledgement and recognition of peer review activities.

WAITING FOR CORONARY REVASCULARIZATION: COMPARING DUTCH PRACTICE WITH A EUROPEAN EXPERT PANELS PRIORITY SCHEME. S.J. Bernstein¹, P. Lazaro², K. Fitch², M.D. Aguilar², H. Rigter³, J.P. Kahan⁴; ¹University of Michigan, Ann Arbor, Michigan; ²Instituto de Salud Carlos III, Madrid, Spain; ³Trimbos Institute, Utrecht; ⁴RAND Europe, Leiden, The Netherlands

PURPOSE: Waiting lists for coronary revascularization are frequently managed without any explicit queuing criteria. Patients may thus not receive priority based on the severity of their clinical symptoms. This is of particular concern in Europe. We therefore convened a European expert panel to develop criteria for maximum acceptable waiting times for coronary revascularization. We assessed the applicability of this criteria and compared Dutch patients' waiting times with the panel's recommendations.

METHODS: A panel of 13 cardiothoracic surgeons and cardiologists from Spain, the Netherlands, Sweden, Switzerland and the United Kingdom was convened to assess the appropriateness of, and priority for, a set of hypothetical scenarios for coronary revascularization. They rated the appropriateness of these scenarios using a modified delphi process and then assigned a maximum waiting time, on a scale of 7 time frames, for 200 indications that were not judged inappropriate. We then measured the waiting time for coronary revascularization (i.e. the number of days between when a recommendation was made that a patient should undergo revascularization and the time the procedure was performed) for 1713 chronic stable angina patients who were treated at one of 10 hospitals in the Netherlands. We also collected data on how the patient's clinical data was presented at the meeting where the recommendation was made for revascularization: (1) direct presentations occurred when the referring cardiologist attended the meeting; and (2) indirect presentations occurred when the patient's clinical data was provided by telephone, letter or facsimile. We assessed the proportion of patients who underwent revascularization within the maximum recommended time.

RESULTS: There was significant variation in the maximum recommended waiting time among the panelists (mean maximum recommended waiting time 96 days; standard deviation 85 days). Angioplasty patients waited fewer days than bypass patients (48 vs. 83 days, $p < 0.001$). Thirty-seven percent of patients waited longer than the maximum recommended waiting time. The majority of patients with excess waiting times were referred for bypass surgery rather than angioplasty (e.g., 83% vs. 17%, respectively). Patients whose cases were discussed during a direct presentation waited fewer days than those patients whose cases were presented indirectly (61 vs. 74 days).

CONCLUSION: One-third of patients referred for coronary revascularization waited for periods longer than those recommended by a multinational panel. Two contributing factors were the type of revascularization procedure the patient was referred for and how the patient's

case was presented, factors not considered by the panel as they felt a patient's waiting time should be determined by clinical symptoms.

PRIORITIZING THE CORONARY REVASCULARIZATION QUEUE: DOES POTENTIAL THREAT TO ABILITY TO WORK ALTER EUROPEAN SPECIALISTS DECISIONS? S.J. Bernstein¹, P. Lazaro², K. Fitch², M.D. Aguilar², J.P. Kahan³; ¹University of Michigan, Ann Arbor, Michigan; ²Instituto de Salud Carlos III, Madrid, Spain; ³RAND Europe, Leiden, The Netherlands

PURPOSE: Waiting lists are usually considered from the perspective of the physician or health care system. However, patients may receive preferential access to care based on socioeconomic status or occupation. Canadian physicians have been reported to place considerable weight on a patient's age and work status in determining queue placement. We sought to determine the effects of social factors on how European cardiovascular specialists place hypothetical patients in the queue for coronary revascularization.

METHODS: A 13 member panel of cardiothoracic surgeons and cardiologists from 5 European countries (the Netherlands, Spain, Sweden, Switzerland and the United Kingdom) was convened to assess the appropriateness of, and priority for, a set of hypothetical scenarios for coronary revascularization. They rated the appropriateness of these scenarios using a modified delphi process and then assigned a maximum waiting time, on a scale of 7 time frames, for all non-inappropriate indications. They also assessed the impact of social factors on a subset of 45 scenarios in which the patient was considered to have mild-moderate angina, moderate left ventricular function, and had a low-moderate operative risk. The scenarios varied in the patients ability to work, live independently or care for dependents which was categorized as being: (1) not threatened, but more difficult; (2) threatened, but not immediately; and (3) immediately threatened. The time frame was converted to a linear scale and multiple regression was performed on the 540 individual panelist's ratings to assess the impact of each factor on waiting time.

RESULTS: Twelve of the 13 panelists responded. Large shifts in ($p < 0.001$) occurred in ratings of waiting time, with the order of priority being those whose ability to work, live independently or care for dependents was immediately threatened first, those whose activities were threatened but not immediately second, and those whose activities were not threatened last. The overall mean shift in maximum recommended waiting time due to a threat to ability to work, live independently or care for dependents (37 days longer wait for those who were not threatened vs. those immediately threatened) was less than the mean shift due to extent of coronary disease (e.g., 93 days longer wait for those with 1 or 2 vessel non-proximal left anterior descending artery disease compared to those with left main disease) or stress test results (62 days longer for those with a negative compared to very positive stress test).

CONCLUSION: Cardiovascular specialists may place considerable weight on the threat of cardiovascular disease on a patient's ability to perform their usual social activities. However, the impact of this factor varies according to clinical presentation. The impact of these factors should be assessed in actual practice.

COST-EFFECTIVENESS OF HEPATITIS A AND B VACCINATION IN CHRONIC HEPATITIS C PATIENTS. A.M. Borzecki¹, I. Davis¹, K. Peltekian¹; ¹Dalhousie University, Halifax, Nova Scotia

PURPOSE: In the US, hepatitis C virus (HCV) infects approximately 4 million people and kills over 8000 annually. With pre-existing HCV infection, the morbidity and mortality of acute hepatitis A (HAV) and hepatitis B virus (HBV) are markedly increased. Accordingly, the Centers for Disease Control (CDC) has recommended vaccination of all chronic HCV patients against HAV and HBV. In Nova Scotia, following the CDC's recommendation, the Department of Health recently implemented a universal HAV and HBV vaccination program for HCV infected patients. However, the optimal vaccination strategy is unknown. This study examines the cost-effectiveness of different vaccination strategies for HAV and HBV in HCV patients.

METHODS: We designed a decision tree to evaluate the cost-effectiveness (CE) of three vaccination strategies: (1) universal - vaccinate all HCV patients against HAV and HBV vaccination regardless of initial HAV or HBV serologic status; (2) screening - screen all HCV patients for HAV and HBV and vaccinate for either if negative; (3) combined - screen all patients and vaccinate with an initial dose at the first visit. If serology is negative patients complete the vaccination course. Pre-vaccination seroprevalence was based on data from our HCV Clinic which is a referral center for the province of Nova Scotia. Vaccination compliance and vaccine efficacy were derived from the literature. Direct medical costs including medical services and vaccination costs were obtained from the 1999 Nova Scotia Department of Health fee schedule. We measured cost per patient immunized and marginal CE ratios based on incremental immunity obtained.

RESULTS: The mean costs per patient were \$171, \$132, and \$162 for the universal, screening and combined strategies respectively. The CE analysis revealed slightly different results. At a baseline compliance of 80% the combined strategy dominated. The respective CE ratios per additional percent immunity obtained were \$2.57, \$2.48 and \$2.44. The model was sensitive to the compliance rate, pre-vaccination seroprevalence and costs of screening and vaccines. At compliances below 72% the universal strategy dominated. Above 83% the screening strategy dominated. When the HAV prevalence was below 29% or HBV below 33%, the universal strategy was preferred. Above 48% and 51% respectively, the screening strategy dominated. Between these values, the combined strategy was favored. In our HCV patients the seroprevalence of HAV was 43% and HBV was 45%. At higher screening costs the universal strategy dominated and at higher vaccination costs the screening strategy was preferred.

CONCLUSION: The appropriate vaccination strategy in HCV patients is determined by local HAV and HBV seroprevalence, vaccination compliance rates and costs. In our setting the combined strategy is preferred and we recommend that the Nova Scotia Department of Health adopt this as their approach.

PRIMARY CARE ALCOHOL-RELATED DISCUSSIONS WITH AT-RISK DRINKERS. K.A. Bradley¹, A.J. Epler², K.R. Bush², J.L. Sporleder², C.W. Dunn¹, N.E. Cochran³, C.H. Braddock III¹, D.R. Kivlahan¹, M.B. McDonnell¹, S.D. Fihn¹; ¹University of Washington, Seattle, WA; ²VA Puget Sound Health Care System, Seattle, WA; ³Dartmouth College, White River Junction, VT

PURPOSE: No published study has directly evaluated the nature of discussions that primary care providers have with patients regarding alcohol use. The objective of this study was to determine the content and style of discussions that primary care providers have with at-risk drinkers during scheduled appointments in clinics with and without a provider prompting intervention.

METHODS: Male patients and their providers in two firms of a Veterans Affairs (VA) General Internal Medicine Clinic were invited to participate in an audio-tape study of patient-provider communication. Patients were eligible if they indicated on a mailed questionnaire that they were willing to consider having their clinic appointments taped, if they reported past-year at-risk drinking, and if their provider consented. Patients and providers were blinded to the focus of this sub-study on alcohol. At-risk drinking was defined as a score of ≥ 1 point on a previously validated augmented CAGE questionnaire, or report of drinking 5 or more drinks on an occasion in the past year. All utterances from alcohol-related discussions were coded into Motivational Interviewing Skills Code categories. At each scheduled primary care visit, providers in the Intervention firm received tailored ACQUIP Drinking Practices Reports summarizing participating patients' responses to mailed alcohol-screening questionnaires. Providers in both firms were offered alcohol-related continuing medical education (one 45-minute conference). Permutation T tests were used to compare the mean number of discussions per provider in intervention and control clinics.

RESULTS: Of 68 taped visits of at-risk drinkers with their providers, 39 (57.4%) included any discussion of alcohol, 17/41 (41.5%) in the Control firm and 22/27 (81.5%) in the Intervention firm ($p=0.026$). Providers contributed 58% of utterances during alcohol-related discussions, including primarily questions (24%), information giving (23%) and facilitation (34%). Advice, reflective listening, and supportive or affirming statements were less common, 5%, 4%, and 5% of utterances respectively. Only 21% of all taped visits included alcohol-related advice, 33% and 12% in the Intervention and Control firms, respectively ($p=0.64$).

CONCLUSION: Alcohol-related primary care discussions mainly involved assessment and information giving. A minority of taped at-risk drinkers in this study received explicit alcohol-related advice. Prompting providers about at-risk drinking was associated with increased frequency of alcohol-related discussions.

THE TREATMENT OF ALCOHOL AND DRUG RELATED DIAGNOSIS IN TEACHING HOSPITAL: HOSPITAL RESOURCE USE AND EFFICIENCY. J. Bramble¹, H. Sakowski¹, D. Esterbrooks¹, E. Rich¹; ¹Creighton University, Omaha, NE

PURPOSE: Teaching hospitals, due to their size, expertise, and location may find themselves in a position to treat alcohol and drug related diagnoses more often than their community counterparts. Previous studies have shown the existence, in some cases, of a volume-to-outcome relationship. Thus, though past research has shown that overall teaching hospitals may be less efficient, they may experience efficiencies in uniquely qualified or niche areas. This study examines the relationship of hospital resource use and the treatment of alcohol and drug diagnoses. Specifically the study tests the hypothesis that teaching hospitals will have greater efficiency than other community hospitals in the treatment of alcohol or drug related diagnoses. This study also examines the hypothesis that hospitals with more experience in treating alcohol and drug related diagnoses would experience greater efficiencies than hospitals with less experience.

METHODS: This is a retrospective cross-sectional study that examines the association of hospital resource use and the treatment of alcohol and drug related diagnoses. The population for this study comes from the 1996 Healthcare Costs and Utilization Project's National Inpatient Sample. The final sample consisted of patients with the appropriate alcohol and drug related diagnosis ($n=9,528$ patients). Data from the American Hospital Association, the Area Resource File, InterStudy, and the Health Care Financing Administration supplemented the patient level data from NIS. Teaching hospitals were identified using the Council of Teaching Hospital designation. Hospital experience was measured by the number alcohol and drug related admissions relative to the hospital's total admissions. Hospital resource use was measured by taking both the log of length of stay and total hospital charges. Least square regression was used to analyze hospital resource use while controlling for hospital, patient, and market characteristics.

RESULTS: The average age of the sample was 39 with 68% of the patients being male and 66% of them white and 23% black. Twenty percent of the admissions were to teaching hospitals. Teaching hospitals had significantly lower charges ($p=0.029$) than non-teaching hospitals and shorter lengths of stay ($p=0.042$) than their community hospital counterparts. When examining the hospitals relative experience, hospitals with greater experience had significantly lower total charges ($p<0.001$); however, contrary to our hypothesis they had significantly longer lengths of stay ($p<0.001$).

CONCLUSION: Though previous studies have shown teaching hospitals, as compared to non-teaching hospitals, are less efficient overall, the results of this study show that they are able to experience certain efficiencies in niche areas. Specifically, teaching hospitals are more efficient at treating alcohol and drug related diagnoses. As teaching hospitals continually find themselves competing for managed care contracts, finding and documenting niche areas in which they excel becomes increasingly important.

QUALITY OF CARE AND PATTERNS OF SERVICE USE FOR MEDICARE BENEFICIARIES WITH DIABETES IN FEE-FOR-SERVICE AND MANAGED CARE. A.E. Brown¹, J. Adams², M.F. Shapiro¹, C.M. Mangione¹; ¹UCLA, Los Angeles, CA; ²RAND, Santa Monica, CA

PURPOSE: To compare quality of care and patterns of primary care provider (PCP) and specialty service use for Medicare beneficiaries with diabetes enrolled in fee-for-service (FFS) settings and network model managed care (MC).

METHODS: Telephone interviews and clinical examinations were performed on a random sample of subjects with diabetes cared for in FFS and managed Medicare. Managed care participants were from one of 17 provider groups that contract with a large health plan in Los Angeles County. The main dependent variables were 4 self-reported process indicators (HbA_{1c} and cholesterol measurement, dilated eye exam, and foot exam in the prior year), and 2 measured outcomes (pulse pressure and HbA_{1c} level). Per visit adherence to quality indicators for each subject was evaluated for process and outcome indicators separately using the ratio of number of indicators met to the number of PCP visits in the past year. Multivariate regression models were used to evaluate the influence of FFS vs MC on each quality indicator and per visit adherence, adjusted for visits to an endocrinologist or diabetes educator and clinical and sociodemographic characteristics.

RESULTS: Interviews were completed by 724 persons (response rate 67%). Mean age was 75 ± 7 years; 51% women; 51% white, 20% Latino, 16% African American, and 8% Asian Pacific Islander; 47% earned under \$20,000 per year; 7% received Medicaid; and 25% had not graduated from high school. Mean number of visits to a provider was 6.8 ± 4.6 in FFS and 5.6 ± 3.9 in MC ($p=.0004$). More FFS (33%) than MC (16%) subjects saw an endocrinologist ($p<.0001$). However, more MC (26%) than FFS (19%) study participants visited a diabetes educator ($p=.04$). Adjusted rates of adherence to the 6 quality indicators were similar in FFS and MC. However, enrollment in FFS was associated with lower per visit adherence for the composite process of care indicators ($p=.004$). In adjusted models, visits to an endocrinologist and/or diabetes educator resulted in higher rates of adherence for 3 of the quality indicators (see table). **CONCLUSION:** In this cohort of older persons with diabetes, although care under FFS was more resource intensive, we observed similar levels of adherence to quality indicators in FFS and MC settings. In both settings, coordination with endocrinologists and/or diabetes educators appeared to improve process of care.

Influence of Provider Type (vs. PCP Only) on Quality of Care - Odds Ratio (p)

	Chol check	Dilated Eye Exam	Foot Exam
Endocrine only	2.7 (.02)	NS	2.0 (.02)
DM Educator only	2.0 (.04)	NS	1.8 (.02)
Endo and Educ	2.9 (.02)	3.2 (.009)	4.9 (.0001)

DOMESTIC VIOLENCE VICTIMS DENY ABUSE TO THEIR PHYSICIANS. R. Buranosky¹, B. Hanusa¹, M. Mcneil¹, K. Dowd¹, S. Scholle¹; ¹University of Pittsburgh, Pittsburgh, PA

PURPOSE: Routine screening for domestic violence (DV) is increasingly recommended in outpatient healthcare settings. However, little is known about the screening process. The purpose of this study was to document the screening process for women counseled for DV in a primary care setting.

METHODS: We identified pregnant women who were counseled for DV between 8/98–6/99 at a university obstetrics clinic. The DV protocol at this clinic, similar to national guidelines, recommends that women be asked about DV at every visit. Identified women are offered counseling using a specific form (usually performed by a social worker). Data were collected from the DV counseling form and from medical records for the previous 6 months. We assessed how the women were screened, who had counseled the women, and what types of abuse led to the counseling. We compared demographic and health characteristics of counseled women to a control group of pregnant women, matched on race, age and type of visit, who were not counseled for abuse.

RESULTS: Of 63 pregnant women counseled for abuse, 53 had complete medical records available for review. Fourteen (26%) of the women were identified as victims at their first clinic visit. Thirty-three (62%) of women denied abuse in at least one previous screening attempt. Eighteen (67%) of the 27 women who were asked about abuse by their physician on the same day that they received counseling denied abuse to their physician. Of the 48 women (91%) referred by an MD/RN to social work for counseling, only 25 (52%) were referred for DV. Twenty-three (48%) were referred for non-DV reasons with the social worker subsequently identifying the abuse. Women referred for DV were more likely to be seen within a month of referral than women referred for other reasons (72% vs. 39%, $p=0.01$). The majority of abuse reported was physical (96%) and recent (83% with event within past month); 37% had been experienced abuse for more than 2 years. In the 40 cases we were able to match to controls, women counseled for abuse were more likely to be depressed (51% vs. 29%, $p<0.05$), to be currently using drugs (20% vs. 6%, $p<0.1$), to have one or more children (69% vs. 47%, $p<0.05$), and to have a current unplanned pregnancy (91% vs. 74%, $p<0.1$).

CONCLUSION: Victims of domestic violence often deny their abuse. Frequent screening for abuse may be needed, and early referral of high-risk patients to social work may increase detection rates. Future research should consider the system effects, such as the roles of various healthcare personnel and context of the visits, on the effectiveness of detecting DV in a primary care practice.

PHYSICIAN ATTITUDES CONCERNING INFLUENCE OF RELIGION AND SPIRITUALITY ON HEALTH. D. Bynum¹, M. Monroe², A. Klioze³, A. Klioze³, M. Franco⁴; ¹University of North Carolina Faculty Development Program, Chapel Hill, North Carolina; ²Carolinas Medical Center, Charlotte, NC; ³University of Florida, Gainesville, Florida; ⁴East Carolina University, Greenville, North Carolina

PURPOSE: Reports have linked intercessional prayer and religiosity with better health outcomes though the strength of these results has been questioned. We sought to clarify physician's beliefs regarding the effect of religion and prayer on health outcomes.

METHODS: THE RESPECT study is a multi-center study of patient and physician attitudes and preferences regarding religion and spirituality in the medical encounter. Physician surveys were administered to residents, recent graduates, and practicing physicians at four centers in North Carolina, one in Florida, and one in Vermont. Survey items included standard

demographic questions, Likert scales addressing attitudes and preferences, and the Spiritual Well Being (SWB) Scale.

RESULTS: Four hundred forty-four physicians responded, of which 74% were general internists and 24% family practitioners. Nineteen percent agreed that faith alone can cure disease; 32% felt that patient prayers for physicians improve health outcomes while 19% agreed that physician prayers for patients could achieve such improvements. Over half (55%) of physicians felt it appropriate to encourage religious and spiritual beliefs for the medical benefit of the patient and 45% felt that patients with strong beliefs would have better health outcomes. Bivariate analysis revealed that physicians with a higher score on the SWB scale and more frequent church attendance were more likely to hold beliefs that health outcomes are improved with patient and physician prayer and that faith alone can cure disease. There were no associations between physician sex, age, race, or spirituality training and these attitudes.

CONCLUSION: A large number of physicians believe religious and spiritual behaviors by patients will improve health outcomes. A smaller but significant number believe that physician prayer will improve patient outcomes. These findings suggest that there is a need for more rigorous examination of the connection between health outcomes and spirituality as many physicians seem willing to incorporate aspects of spiritual behavior into their practices.

WOMENS HEALTH CENTERS: PREDICTORS OF DEVELOPMENT. C. Caffrey¹, B. Simon², M. Wang², E. Yano³; ¹VA Greater Los Angeles (GLA) Healthcare System, Sepulveda, CA; ²VA GLA, Sepulveda, CA; ³VA HSR&D Center of Excellence, Sepulveda, CA

PURPOSE: Addressing gender disparities in care for women is a priority in the Department of Veterans Affairs (VA), where typically >90% of patients are men. Separate women's health centers (WHC's) have grown in the private sector as a means to deliver more gender-sensitive services to women, although the majority have focused on reproductive services (Weisman et al 1995). Little is known about the impetus to develop a separate WHC, especially within the VA, where issues of access to care, privacy, and quality of care for women veterans have often been of concern. We examined organizational, provider, and patient-level determinants of developing a separate WHC for delivery of gender-specific and primary care services.

METHODS: We used data from the 1999 VHA Primary Care Practices Survey, a national survey of senior VA primary care leaders at all VA health care facilities with >4,000 unique patients and >20,000 outpatient visits during fiscal year 1998 (response rate 93%, n = 219). Analysis of facility characteristics associated with a self-reported WHC for delivering primary care was performed using bivariate analysis and independent predictors were determined using logistic regression.

RESULTS: Nationally, 62% of facilities reported having a separate WHC to provide primary care for women. WHC's tended to reside in urban locations ($p < 0.05$) and in more complex facilities ($p < 0.05$). Facilities with WHC's had more female patients ($p < 0.001$) with more visits ($p < 0.001$), but comparable visits-per-patient (7.1 vs. 7.4). Facilities with team-based primary care that integrated providers from multiple disciplines were more likely to have developed a WHC ($p < 0.05$). WHC's were more likely where primary care leadership was distinct from subspecialty care ($p < 0.001$) and a separate budget existed for primary care ($p < 0.05$). In addition, WHC's were more common in academic facilities (69% vs. 56%, $p = 0.09$) with higher numbers of internal medicine houseofficers (43.5 vs. 26.9, $p < .05$) and longer outpatient block rotations ($p < .05$). Having separate primary care leadership, and higher women veteran case-load remained significant predictors of WHC development after adjusting for other variables.

CONCLUSION: Separate WHC's were present in larger, complex, more intensively academic medical centers with more established and authoritatively independent primary care programs. As a cross-sectional study, it is unclear whether WHC development occurred as a result of or resulted in a higher female patient volume. The implications of separate service delivery for women with respect to clinical quality, and patient satisfaction are still unclear and somewhat controversial; assessing WHC performance is a crucial next step.

THE IMPACT OF TESTING FOR MUTATIONS IN BRCA1 AND BRCA2 ON LIFE ON LIFE EXPECTANCY AND QUALITY-ADJUSTED LIFE EXPECTANCY. M.H. Cameron¹, T.B. Newman¹; ¹University of California, San Francisco, San Francisco, CA

PURPOSE: To estimate the benefit of testing women for BRCA1 and BRCA2 mutations, and to determine threshold risk and utility values for testing to be beneficial.

METHODS: We used DATA 3.5 software to build a Markov model that estimates the effect of testing for BRCA1 and BRCA2 mutations on survival and quality-adjusted survival. The model includes two decision nodes, the first for whether to test and the second for whether to have prophylactic mastectomy and oophorectomy if a mutation is found. The base case values used were: age 40 years; prior probability of mutation 2.5% (literature value for Ashkenazi Jewish women); lifetime risk of breast of 56% with and 11% without a mutation; lifetime risk of ovarian cancer of 16% with and 1% without a mutation; reduction in breast cancer risk with prophylactic mastectomy 90%; reduction in ovarian cancer risk with prophylactic oophorectomy 50%; five-year survival for breast cancer 85% and for ovarian cancer 50%; disutility of prophylactic mastectomy 5% and disutility of prophylactic oophorectomy 2%; and a discount rate of 3% per year. Age-specific death rates from other causes were obtained from vital statistics.

RESULTS: Using the base case values, testing for BRCA1 and BRCA2 genetic mutations prolongs average survival by 12 days and average quality-adjusted survival by 1.1 days. Testing results in increased quality adjusted survival if oophorectomy reduces the risk of ovarian cancer by at least 30%, mastectomy reduces the risk of breast cancer by at least 80%, the risk of perioperative death is less than 10%, the lifetime risk of breast cancer with a mutation is greater than 46% and the disutility of mastectomy and oophorectomy are less than 20%. Testing provides benefit at all levels of risk of ovarian cancer with a mutation and at all probabilities of having a mutation.

CONCLUSION: Under a wide range of baseline assumptions, testing for BRCA1 and BRCA2 mutations can increase women's average survival and quality-adjusted average survival. The benefits of this test depend on the individual's disutility for undergoing prophylactic oophorectomy and bilateral mastectomy, the risk of breast cancer associated with BRCA1 and BRCA2 mutations and the efficacy of prophylactic procedures for reducing breast and ovarian cancer risk. However, under some assumptions, the benefit of testing becomes very small, suggesting low cost efficacy.

PROBLEM DRINKING IN PRIMARY CARE PRACTICES: ECONOMIC IMPLICATIONS. R.D. Cebul¹, D. Einstadter², N.V. Dawson², S. Wolfe³; ¹Case Western Reserve University, Hunting Valley, OH; ²Case Western Reserve University, Cleveland, Ohio; ³Center for Health Care Research & Policy, Cleveland, Ohio

PURPOSE: Recent studies have reported paradoxically lower costs for problem drinkers (PDs) than for non-PDs in primary care practices (PCPs). We used validated screening instruments and objective utilization measures to examine the prevalence and economic implications of drinking in a large urban PCP.

METHODS: We used the CAGE and AUDIT questionnaires to screen continuity patients between 1/99–3/00. PD was classified using established cutpoints (CAGE > 1; AUDIT > 7). Using NIH-NIAAA-identified ICD-9 codes, patient encounters in the year before screening were categorized as Alcohol-Caused (AC), -Highly Associated (AHA), or -Nonspecifically Associated (ANSA). For patients with completed surveys, we compared utilization among PDs and non-PDs using rate ratios (RR), including adjustments for age, sex, and race in Poisson regression analyses.

RESULTS: Among 4,680 patients with completed surveys, there were a total of 48,367 encounters, including 21.9% (10,552) for alcohol-related conditions (AC, 0.7%; AHA, 5.6%; ANSA, 15.6%). CAGE identified more patients as PDs than did AUDIT (19.8% vs. 10.9%) and was more strongly associated with utilization. After adjusting for age, sex, and race, CAGE-identified PD was associated with higher utilization for AC conditions (RR 17.1, 95% CI 12.8–23.0), AHA conditions (RR 1.45; 1.33–1.58), all alcohol-related conditions combined (RR 1.20; 1.14–1.25), and for all encounters combined (RR 1.07; 1.04–1.09). As compared to their use of Outpatient services, PDs were especially more likely than non-PDs to use high cost Emergency Department and Inpatient services.

CONCLUSION: Using CAGE results and objective encounter data from continuity patients in a large group practice, problem drinking was common and associated with higher overall utilization as well as use of higher cost sites for specific alcohol-related conditions. These findings support initiatives to screen and intervene for problem drinking in primary care practices.

INFORMED CONSENT AND PROSTATE CANCER SCREENING: THE IMPORTANCE OF FACTS MEN OUGHT TO KNOW ABOUT PROSTATE SPECIFIC ANTIGEN BY PHYSICIAN SPECIALTY. E. Chan¹, S. Vernon², F. O'donnell¹, M. Haynes²; ¹University of Texas - Houston Health Science Center, Dept of Internal Medicine, Houston, TX; ²Center for Health Promotion and Prevention Research, School of Public Health, University of Texas-Houston Health Science Center, Houston, TX

PURPOSE: The purpose of this study was to determine whether there are differences in ratings of importance among internists, family physicians, and urologists for 17 key facts about prostate cancer screening with prostate specific antigen (PSA). Because screening for prostate cancer with PSA is controversial, several professional organizations recommend informing men about the risks and benefits of PSA screening. In a previous study, a Delphi panel of experts in prostate cancer and focus groups of couples with screened and unscreened men identified 17 key facts about PSA they believe asymptomatic men ought to know.

METHODS: A survey containing 17 key facts about PSA was mailed to a nation-wide random sample of internists, family physicians, and urologists (n=768, response rate = 58.6%) listed in the 2000 Directory of Board Certified Medical Specialists. Respondents were asked to rate how important it is for asymptomatic men to know each of the 17 facts about PSA screening using a 5-point Likert scale ranging from "not at all important" to "extremely important." Because internists and family physicians had similar responses, they were combined as non-urologists and compared to urologists. We used logistic regression to compare differences between urologists and non-urologists and adjusted for the effects of age and gender.

RESULTS: Urologists and non-urologists differed in rating how important it is for men to know 9 of the 17 facts. Urologists considered it extremely or very important for men to know that it is unclear whether regular PSA screening will reduce prostate cancer mortality, that done together the PSA and digital rectal exam (DRE) can screen for prostate cancer, and that PSA and DRE are most appropriate for men whose life expectancy is at least 10 years. Non-urologists considered it extremely or very important for men to know that PSA screening is controversial, that there are risks and benefits to PSA screening, that prostate cancer may grow slowly without symptoms, that it is unclear whether treatment of early, localized prostate cancer is helpful, that there are side effects to treatment, and that a man over age 70 is less likely to die from prostate cancer even though he has a higher risk of having it.

CONCLUSION: Urologists and non-urologists differ in rating how important it is for men to know facts about PSA screening. Despite professional recommendations for informing men about the risks and benefits of PSA screening, physician disclosure may differ depending on physician specialty.

QUALITY OF DIABETES CARE AT AN ACADEMIC HEALTH CENTER: RESIDENT CONTINUITY CLINIC AND A FACULTY PRACTICE IN GENERAL INTERNAL MEDICINE- IS THERE A DIFFERENCE? S. Chheda¹, G. Griffing¹, D. Geisler¹; ¹Saint Louis University, St. Louis, MO

PURPOSE: Many academic health centers have a primary care practice site that contains both a resident continuity clinic practice supervised by faculty and a faculty- only clinical practice. Few studies have been conducted to compare the quality of care provided by residents supervised by faculty versus faculty alone in the ambulatory setting. We conducted a pilot study to compare the quality of ambulatory diabetes care provided by residents in a continuity clinic to that provided by faculty at the same site.

METHODS: We conducted a retrospective study of a convenience sample of the medical records of patients with the diagnosis of diabetes mellitus (ICD-9 250.xx) seen between October 1998 and October 1999. A total of 139 faculty patient charts and 141 resident patient charts were evaluated. Patients were excluded if less than one year of care had been provided or if patients were also followed through the endocrinology clinic. Descriptive data, adherence to the American Diabetes Association (ADA) Clinical Practice Recommendations for exams and tests as well as blood pressures, HbA1C levels and lipid profiles were assessed in 94 faculty patients

and 96 resident patients. Statistical analysis was done using the students' t-test and the chi-square test.

RESULTS: Faculty and resident patients were similar with respect to sex, race, weight, years of diagnosis of diabetes mellitus and type of diabetes therapy. However, resident patients were slightly older than faculty patients were (mean age 63 vs. 57, $p=0.008$). Residents and faculty were no different in frequency of adherence to the ADA Clinical Practice Guidelines for visits per year, HbA1C testing, lipid profile testing, and foot exam. Faculty were significantly more adherent to urine evaluation (44% vs. 26%, $p=0.04$) and referral for dilated eye exam (54% vs. 39%, $p=0.02$). There were no differences between residents and faculty patients with respect to mean HbA1C levels, mean control of blood pressure, use of ACE inhibitors and use of lipid lowering therapy. Resident patients were more likely than faculty patients to have LDL levels controlled to less than 100 (42% vs. 25%, $p=0.03$).

CONCLUSION: Residents and faculty provide primary care for similar groups of diabetic patients at this academic health center. Adherence to ADA Clinical Practice Recommendations was similar between faculty and residents for many, but not all areas. Outcomes such as HbA1C levels and blood pressure control were similar for both groups, though more resident patients had better LDL cholesterol control. Though resident continuity practices are faculty supervised and often in the same setting as faculty practices it cannot be assumed that quality outcomes are the same for patients of these two groups. Interventions to improve rates of adherence to specific recommendations should be geared towards both faculty and residents at this academic health center.

BARRIERS TO HEALTH CARE ACCESS FOR INDIGENT MEN. S. Connelly¹, K.B. Desalvo¹, ¹Tulane University, New Orleans, LA

PURPOSE: The vast majority of patients receiving care in our indigent Internal Medicine continuity clinics are female. We hypothesized that their male relatives were either receiving their health care outside of the system, or were experiencing special barriers to access. The purpose of this survey was to identify if and where male relatives of our patients are receiving health care and to identify the potential barriers to accessing health care by men.

METHODS: The authors conducted a structured interview on a convenience sample of 92 patients from an urban internal medicine clinic. Information on employment status, health care coverage, place of medical services, and potential barriers to access was obtained for 465 of their relatives living in the community. Responses were analyzed with chi square and multivariate analysis for relationships where appropriate using SPSS software.

RESULTS: Patients who attend the Internal Medicine Clinic when compared to their relatives were less likely to have health care coverage 39% vs. 63% ($p<.001$) and were less likely to be working than their relatives 36% vs. 61% ($p<.001$). Only 22 (6.5%) of the male relatives went to the VA for care. Among relatives, those with Full/Part-time work status were significantly more likely not to receive any primary care, 52% vs. 71% ($p<.001$). Lack of health care coverage was significantly associated with decreased primary care, as well, only 18.5% of relatives without health coverage received primary care compared to 76.5% of those with coverage ($p<.001$). Younger generation for females and males were significantly less likely to receive primary care compared to older generation males and females (females $p<.001$ and males $p<.006$). Male relatives are significantly less likely to access primary care compared to females, 57% vs. 70% ($p<.015$) and are less likely to access the Charity system compared to females 20% vs. 34% ($p<.014$). Male relatives are more likely to obtain the majority of their health care in the ER and Walk-in Clinic compared with female relatives 20% vs. 8% ($p<.001$). There was no difference between work status, age distribution, and health care coverage between male and female relatives.

CONCLUSION: The barriers to health care access in both males and females were Full/Part time-work status, younger generation, and lack of health care coverage. Our patients' male relatives are less likely to have a regular source of primary care compared to female relatives. Offering evening and weekend hours in our indigent continuity clinics may possibly reduce this barrier. Further studies are needed to better assess this inequality of primary care access between males and females.

APPROPRIATENESS OF DISCHARGE AND TIME TO RETURN VISIT IN A CARDIOLOGY CLINIC. S. Connelly¹, K.B. Desalvo¹, A. Dalal¹, ¹Tulane University, New Orleans, LA

PURPOSE: Decreasing waiting times to next available appointment involves evaluation of the physician appropriateness and effectiveness of practice in the clinic system. Some studies have shown that patients receive "better" care from a sub-specialist than those followed by a generalist. The appropriateness and variability of this care, however, has not been well scrutinized. The purpose of this study, was to identify if patients enrolled in the cardiology clinic belong there based on a set of criteria agreed between internal medicine and cardiology. Also, to assess the variation among cardiology fellow's discharge rates and revisit interval assignment.

METHODS: Data was extracted from 194 cardiology clinic notes between 1/14/00–2/27/00. Providers were cardiology fellows practicing at an urban VA. The main outcomes of interest were appropriate discharge from the cardiology clinic (per criteria established by the authors) and assigned revisit interval. Predictors of interest included provider data and patient level information such as a severity of illness score and ejection fraction.

RESULTS: Overall, discharge rate was 24/194 (12.4%). Of the 161 remaining in the clinic, 74 (46%) could have been discharged to their primary provider by the agreed criteria. In logistic regression, disease severity was correlated with discharge from clinic ($r=0.69$). The more stable patients were more likely to be discharged from clinic. Overall, average clinic revisit interval was 3.8 months. There was a large amount of variability among providers. Individual provider average follow up time ranged from 2.2–7.7 months. Individual discharge rates ranged from 0–26%. Individual clinic appropriateness ranged from 0–90%. There was no correlation of revisit intervals with severity of illness score ($r=.021$, $p=0.794$) or with ejection fraction ($r=.013$, $p=0.91$). At best, patient severity of disease accounted for only 4% of the variance. Individual providers saw patients back at a consistent length of time regardless of the needs of the individual patients for instance, provider #10 saw every patient back in 6 months and provider #31 saw every patient back in 3 months.

CONCLUSION: There needs to be established criteria which patients must meet to remain in the clinic that will be utilized by all providers in the clinic to reduce variability in care. There needs to be improved consistency and decreased variability in return to clinic times to improve access in the cardiology clinic setting. Return visit time should be based on severity and stability of patient's illness and not on individual provider preference.

BLOOD PRESSURE CONTROL IN THE DIABETIC POPULATION. L.E. Connors¹, M.R. Ezer¹, T.L. Uchida¹, J.M. Fiskio¹, J.S. Borus², D.W. Bates¹, D.G. Fairchild¹, ¹Brigham & Women's Hospital, Boston, MA; ²Harvard Medical School, Boston, MA

PURPOSE: In 1997, the Joint National Committee on Prevention, Detection, Evaluation, and Treatment of High Blood Pressure (JNC VI) recommended that the goal blood pressure (BP) for diabetic patients be less than 130/85. Our clinical observations however suggested that clinician awareness of these guidelines and diabetic BP control were inferior to that of the general hypertensive population.

METHODS: To assess the recognition of hypertension and BP control among diabetic patients, we conducted a chart review of 322 randomly selected diabetic patients of whom 233 (72.4%) carried a diagnosis of hypertension. BP measurements were included only from routine office visits with the patient's primary care provider. For those patients with known hypertension, patients were considered to be uncontrolled if the average of the three most recent BP readings resulted in a systolic and/or diastolic BP greater than or equal to ($\geq$) 130/85. For those diabetic patients without a diagnosis of hypertension, the six most recent BP readings within the last two years were averaged.

RESULTS: Of the 233 diabetic patients with known hypertension, only 53 (23%) were controlled (BP <130/85). Of the 89 diabetics without a diagnosis of hypertension, 32 (36%) met the JNC VI criteria for hypertension (BP $\geq$ 130/85). Within this group, 26 (29.2%) had average BPs which fell between 130/85 and 140/90. In comparison, a chart review of 100 randomly selected patients with hypertension showed a 45% rate of BP control (<140/90) in these same practices ($p<0.001$).

CONCLUSION: The rate of blood pressure control among our diabetic population was low and poorer than for hypertensives as a whole in our ambulatory practices, despite the fact that diabetics benefit more from therapy. Furthermore, a third of diabetics had undiagnosed hypertension and a significant portion of these patients had marginally elevated BPs (130/85 $\geq$ BP <140/90). These data suggest that physician awareness of the JNC VI guidelines for diabetics may be lacking, and that this may be contributing to the relatively low rate of BP control among our diabetic patients. We are currently studying a computer-based physician reminder system to see if this can enhance diabetic BP control.

SMOKING STATUS AND LIKELIHOOD OF UNDERGOING CORONARY REVASCLARIZATION AFTER CARDIAC CATHETERIZATION. J. Cornuz¹, P.D. Faris², W.A. Ghali², M.L. Knudtson², ¹Lausanne University, Lausanne; ²University of Calgary, Calgary, Alberta

PURPOSE: Smoking is viewed by many as a personal lifestyle choice and as a result, smokers are often judged to be responsible for the health consequences of their behavior. Such a stigmatizing approach might incite health care providers to be less enthusiastic about offering cardiac procedures such as percutaneous transluminal coronary angioplasty (PTCA) or coronary artery bypass grafting (CABG) to smokers. We aimed to assess whether smokers were less likely to undergo such revascularization procedures after undergoing cardiac catheterization.

METHODS: We evaluated data from APPROACH, the Alberta Provincial Project for Outcome Assessment in Coronary Heart disease, an ongoing prospective cohort study of all patients undergoing cardiac catheterization in the province of Alberta, Canada. Smoking (defined as smoking at least one cigarette a day) was considered to be present if mentioned in patient's hospital record or reported by the patient at time of catheterization. The occurrence of CABG or PTCA after cardiac catheterization is captured in the APPROACH database. Cox proportional hazards models were used to determine the adjusted likelihood of undergoing revascularization procedures for smokers vs. non-smokers, while controlling for other factors that might influence likelihood of revascularization, i.e., age, sex, comorbidities (e.g., diabetes), previous invasive cardiac procedures, thrombolytic therapy, indication for catheterization (eg, recent myocardial infarction vs. angina), left ventricular ejection fraction, and extent of coronary artery disease.

RESULTS: From January 1995 to December 1997, 20406 patients underwent cardiac catheterization in Alberta. Their mean age was 62.5 and 30% were female. 25% were current smokers and 37% were former smokers. When compared with non-smokers, the hazard ratio for undergoing any revascularization procedure (PTCA/CABG) after catheterization was 0.98 (95% CI 0.93–1.03) for current smokers and 0.99 (0.94–1.04) for former smokers. The hazard ratio for undergoing PTCA was 0.92 (0.86–0.98) for current smokers and 0.98 (0.92–1.04) for former smokers. For CABG, the corresponding hazard ratios were 1.10 (1.00–1.20) and 1.01 (0.94–1.08) for current and former smokers, respectively.

CONCLUSION: Despite potential for discrimination by smoking status in revascularization treatment decisions, current and former smokers undergoing cardiac catheterization in Alberta were as likely as non-smokers to undergo revascularization procedures. While reassuring, these results do not preclude the possibility of biased clinical decision-making prior to referral for cardiac catheterization or variations in other geographic regions.

A SURVEY OF CRITICAL PATHWAY USE AND EVALUATION PRACTICES AMONG HOSPITALS WITHIN TWO QUALITY CONSORTIA. J. Dares¹, E.B. Bass², P.J. Pronovost¹, ¹Johns Hopkins University, Baltimore, MD

PURPOSE: The drive towards improving efficiency and quality has led to the rapid adoption of critical pathways as a utilization management tool in U.S. health care institutions. Hospitals have devoted substantial resources to critical pathways but it is unknown whether hospitals routinely evaluate the clinical or economic effects of pathways. We sought to determine how different types of hospitals evaluate the critical pathways they use.

METHODS: We performed a cross-sectional survey of hospital administrators of two consortia of hospitals that have made a commitment to improving quality of care. Our main outcome measures were percent of hospitals tracking utilization and effects of pathways.

RESULTS: Fifty-five percent of the hospitals surveyed responded including 10 academic medical centers (AMC), 11 community teaching hospitals (CTH), and 11 community hospitals (CH). The median number of adult critical pathways currently used by AMCs, CTHs, and CHs, respectively, were 24.5, 18, and 2.5. All AMCs and CTHs used pathways, but only 6 community hospitals used pathways. The most common pathways included community-acquired pneumonia, total hip and knee replacements, and stroke. Of hospitals with pathways, actual use of the pathways by patients was reportedly measured by 60% of AMCs, 73% of CTHs, and 50% of CHs. Of hospitals that measured pathway use, the mean percentage of patient admissions eligible for pathways was 38%; the mean percentage of patient admissions placed on pathways was 32%. Among hospitals with pathways, the evaluation of pathways included monitoring of: length of stay in 100% of AMCs, 82% of CTHs, and 66% of CHs; total hospital costs in 90% of AMCs, 73% of CTHs, and 66% of CHs; in-hospital mortality in 80% of AMCs, 55% of CTHs, and 50% of CHs; infectious complications in 80% of AMCs, 45% of CTHs, and 33% of CHs; procedure complications in 60% of AMCs, 64% of CTHs, and 33% of CHs; adverse drug events in 30% of AMCs, 18% of CTHs, and 0% of CHs; readmissions rate in 60% of AMCs, 45% of CTHs, and 33% of CHs.

CONCLUSION: Although AMCs, CTHs, and CHs all are likely to evaluate economic outcomes of critical pathways, they (and especially CHs) are less likely to evaluate actual utilization and clinical outcomes of pathways, thereby making it difficult to monitor the overall effectiveness of pathways.

PHYSICIANS RESPONSES TO INFLUENZA VACCINE DELAYS, 2000–01. M. Davis¹, S. McMahon², J. Santoli², B. Schwartz², S. Clark¹; ¹University of Michigan, Ann Arbor, MI; ²Centers for Disease Control and Prevention, Atlanta, GA

PURPOSE: Influenza vaccine is the primary method for preventing influenza. Delay in influenza vaccine distribution was expected for the 2000–01 influenza season. We conducted a cross-sectional study to characterize practice adjustments and expectations of physicians regarding influenza vaccine delay.

METHODS: We sent a two-page mail survey to a national random sample of 1606 internists (IM) and family physicians (FP). Participants answered questions regarding their usual influenza vaccine administration period, their practices' capacity for identifying high-risk patients, and modifications in their clinical practice for the 2000–01 influenza season. The first mailing was conducted in September, 2000; non-respondents received another mailing in October, with a \$2 incentive. Tests of significance were computed using likelihood ratio chi-square.

RESULTS: Response rate was 60% after two mailings and did not differ significantly by respondent specialty. Of 952 respondents currently practicing, 756 (79%) typically administered influenza vaccine; family physicians were more likely than internists to administer influenza vaccine (82% vs. 76%; $p < 0.05$). Among physicians administering vaccine, 356 (47%) self-identified as IM and the remainder as FP. At the time of survey completion, 13% of respondents had received their full order of vaccine; another 23% reported receipt of a partial order. Whereas 37% of physicians who had already received their full order had modified their practices this influenza season, 70% of physicians who had received either partial or no shipment reported practice changes ($p < 0.001$). The most commonly reported practice modification was to target high-risk patients for receipt of available doses. However, only one-quarter of physicians (28% IM vs. 24% FP; NS) had experience with mail or telephone reminder systems to contact high-risk patients. Of concern, 43% of providers were hesitant about administering vaccine after the onset of influenza season in their practice area.

CONCLUSION: Facing delays in influenza vaccine distribution, many physicians reported an intention to vaccinate high-risk patients first. However, few physicians had experience using reminder systems, and nearly half of physicians were hesitant about administering vaccine after influenza season has begun. These findings raise concerns about the capacity for physicians to meet domestic vaccination goals during this influenza season and in the future, and the possibility of preventable morbidity and mortality attributable to influenza as a result.

FALSIFICATION OF THE MEDICAL RECORD: ETHICAL IMPLICATIONS FROM A STANDARDIZED PATIENT COMPARISON. I.R. Dresselhaus¹, J. Luck², J.W. Peabody³; ¹VA San Diego Healthcare System, San Diego, CA; ²School of Public Health, University of California, Los Angeles, ; RAND, San Francisco, CA

PURPOSE: Payers and institutions use chart abstraction to measure physician performance, despite its underestimation of the quality of care due to recording bias. We wondered if the medical record might also overestimate the quality of care through false, and potentially unethical, documentation by providers. To determine this, we compare the quality of care as documented in the medical record with the reports of actor patients.

METHODS: Twenty physicians in the primary care clinics of two Veterans Affairs Medical Centers were randomly selected among consenting residents and faculty (97% agreed to participate). Data were collected from standardized (actor) patients, who served as the gold standard and presented undetected as patients to physician subjects, and from the medical record generated from these visits. Quality criteria were developed from national guidelines and a modified Delphi technique for four common medical conditions. These were then recorded by a standardized patient or abstracted from the medical record. Physician subjects completed 160 evaluations of standardized patients (8 cases $\times$ 20 physicians). We determined the proportion of criteria reported in the medical record but not by the standardized patient (false positives). We also determined the distribution of false positives according to domain (history, physical exam, diagnosis, treatment), study site, physician subjects, and actor patients. False positive rates at the two study sites were compared by t-test.

RESULTS: Compared to the gold standard of standardized patients, false positives were identified in the medical record for 6.4% of measured items overall. False positives were higher for physical examination (13.5%) and diagnosis (14.6%) than for history (3.8%) and treatment (3.4%). The difference in false positive rates between site 1 (7.1%) and site 2 (5.7%) was not statistically different ($p = 0.34$). The proportion of false positives for individual physician subjects ranged from 2.2% to 13.0% and for actor patients from 1.4% to 11.6%.

CONCLUSION: These results suggest that chart abstraction may overestimate the quality of care due to false positives. The clustering of false positives in the domains of physical examination and diagnosis suggests that these are not incidental occurrences or under-reporting by actor patients. Though false positives in the physical examination could result from careless documentation by physician subjects, they may indicate intentional misrepresentation of the process of care, perhaps to up-code a visit or save time by adding an exam element not performed. Such fabrication would violate ethical standards essential to the integrity of clinical practice, potentially putting patients at risk by including misinformation in the medical record. By contrast, documentation of diagnosis in the medical record but not by the actor patient would represent an important lapse in communication of essential information. Improved evaluation methods are needed to detect such irregularities as well as guidelines to determine appropriate actions when potentially unethical conduct is identified in studies of quality.

CAREER TUMULT: INTERNISTS EXPECTATIONS IN A CHANGING PROFESSIONAL ENVIRONMENT. K.A. Edwards¹, D.L. Kasman², E.B. Larson³, E.C. Caga-Anan³; ¹University of Washington School of Medicine, Seattle, WA; ²VA Puget Sound Health Care System; ³University of Washington, Seattle, WA

PURPOSE: For many physicians, the environment in which they trained differs from the current practice environment of managed care, declining revenues, patients on the Internet, and insurance paperwork. Studies of physician career satisfaction have suggested that unmet expectations may play a role in declining satisfaction, but this area remains largely unexplored. We designed a survey to examine career expectations and satisfaction, as well as evaluating trends in satisfaction over time.

METHODS: We mailed surveys to half the ACP-ASIM membership in Oregon ($n = 550$) in Fall 2000. Survey categories were developed from previous literature on career satisfaction and physician self-care. We asked respondents to rate 16 features as "not important, important, or very important" to their current view of their career. The features included: having an adequate income, being valued by others, intellectual challenge, control of professional life, helping others, collegial relationships, being part of a team, doctor-patient relationships, working on the cutting edge of science, providing quality health care, managing own business, and others. We then asked whether expectations about these features were "not met, matched, or exceeded" in their career currently.

RESULTS: We had a 75% response rate ($n = 412$). We removed surveys returned from retired physicians and residents, analyzing 345 (67% male, 33% female). The highest number of physicians ranked "providing a high standard of care" as a "very important" feature of their career (81%), with 93% stating that this expectation was either matched or exceeded in their current work. "Physician-patient relationships" and "helping others" were also ranked as very important (68% and 63% respectively) and were matched or exceeded (84% and 94% respectively). 98% of respondents reported "autonomy/control over professional life" as an important or very important to their career, yet a high proportion of unmet expectations appeared in the following areas: "having control over my professional life" (52% unmet), and "having autonomy in current position" (38% unmet). There was also high rates of unmet expectations in "making as much money as anticipated" (42% unmet) and "work is fun" (33% unmet). 71% report being "somewhat-very satisfied" with their current practice situation, while 83% selected these satisfaction levels for their situations 10 years ago. 61% would go to medical school again.

CONCLUSION: Overall career satisfaction appears to be declining with high unmet expectations regarding professional autonomy and control while, in this sample, internists reported that expectations regarding quality of care were generally met. Improved understanding of expectations and practice changes should help educators and professional groups in their efforts to educate and promote professional development.

PRACTICE-LEVEL VARIANCE IN AMBULATORY QUALITY OF CARE. E.F. Ellerbeck¹, K.K. Engelman¹, M.C. Mosier¹; ¹Department of Preventive Medicine, Kansas City, Kansas

PURPOSE: Although shortcomings in the quality of ambulatory care have been well documented, there is little information available about how physician's practices vary in quality of care. This study was designed to compare quality in the best ambulatory practices in the state of Kansas (benchmark practices) with other practices in the state.

METHODS: We identified all Medicare patients continuously enrolled in Medicare Part B in 1998 and 1999. We utilized Medicare claims to measure quality of care based on whether or not beneficiaries had received an influenza immunization in 1999, a pneumococcal immunization between 1991 and 1999, whether females had a mammogram in 1998 or 1999, and whether persons with diabetes had an eye examination or a lipid profile in 1998 or 1999 or a hemoglobin A1C in 1999. We linked patients to primary care providers based on office visit claims and merged providers into practices based on common provider numbers. We restricted our analyses to 141 practices with at least 50 patients in the denominator for each quality indicator. We identified benchmarks as the 90th percentile in performance on each indicator and compared benchmark performance with median performance and performance at the 10th percentile.

RESULTS: We observed large discrepancies in performance among practices on all quality indicators (see table). Performance on certain quality measures was highly correlated (i.e. practices performing well on one measure were more likely to perform well on other measures). **CONCLUSION:** There is substantial variation among physician's practices in their performance on commonly utilized quality markers. Further efforts are needed to describe the characteristics of practices that are performing at the highest levels.

Percent of patients in different offices receiving the indicated service

Indicator	Benchmark	50th Percentile	10th Percentile
Influenza	64	50	29
Pneumococcal	55	43	30
Mammograms	80	68	51
Eye examinations	88	81	69
HgbA1c	96	88	61
Lipid profile	80	57	36

A NEW DECISION AID IMPROVES PHYSICIANS TRIAGE DECISIONS FOR PATIENTS WITH SUSPECTED CARDIAC ISCHEMIA. B.M. Reilly¹, A. Evans¹, K. Das¹, J. Schaidler¹, J.E. Calvin¹; Cook County Hospital, Chicago, IL

PURPOSE: Previous attempts to improve the triage of patients with suspected acute cardiac ischemia in the Emergency Department (ED) have been disappointing. We developed and prospectively evaluated a decision aid based on a validated clinical prediction rule to determine if physicians' triage behavior would change and whether any changes would be beneficial.

METHODS: We constructed a decision aid from Goldman and colleagues' published clinical prediction rule, which predicted major cardiac complications (N Engl J Med 1996). After achieving over 80% compliance with use of the decision aid in our ED, we prospectively collected data on 1011 consecutive patients admitted from the ED with suspected cardiac ischemia during late 1999. Clinical data collected in the ED allowed us to adjust for possible differences in case mix when comparing decisions and outcomes before and after the intervention. We compared triage decisions (CCU vs. inpatient telemetry vs. observation unit or ward) and their safety and efficiency with similar patient cohorts studied prospectively in our institution during 1997 (n=207) and 1998-99 (n=1033). Triage safety was defined as the proportion of all patients who had major cardiac complications within 3 days who were admitted to either the CCU or inpatient telemetry unit. Triage efficiency was defined as the proportion of all patients who did not have major cardiac complications who were not admitted to either the CCU or the telemetry unit.

RESULTS: Among the 1011 patients, 1008 were eligible, with clinical follow-up on over 98%. Major complications occurred within 3 days for 3.5% (35/1008). A greater proportion of patients were admitted to an observation unit in 1999 compared to 1997: 35% vs. 21%; $P < 0.001$. In 1999, a smaller proportion of admissions to the inpatient telemetry unit were very low risk patients compared to 1998-99: 41% vs. 52%; $P < 0.001$. Triage safety was slightly higher in 1999 compared to 1997: 94% (33/35) vs. 89% (8/9); $P = 0.3$. Triage efficiency was much greater in 1999: 36% vs. 21%; difference: 15 percentage points (95% CI: 8% to 21%); $P < 0.001$.

CONCLUSION: A new decision aid changed physicians' triage decisions and improved triage efficiency without compromising triage safety.

THE GOOD NEWS ABOUT GIVING BAD NEWS TO PATIENTS. N.J. Farber¹, S.Y. Urban², V.U. Collier³, J.L. Weiner⁴, R. Polite⁵, E.G. Boyer⁶, E. Davis⁷; Christiana Care Health System, Wilmington, DE; New York University, New York, NY; Christiana Care Health System, Newark, DE; Drexel University, Philadelphia, Pennsylvania; St. Joseph's University, Philadelphia, PA

PURPOSE: Physicians are often required to give bad news to patients. While recommendations about how to do so have been published in the literature, there is little data available on how physicians actually do inform patients of life-threatening diagnoses. We therefore surveyed internists about how they inform patients about bad news.

METHODS: A survey instrument asked how frequently physicians perform different activities in giving bad news to patients, based on a four point Likert scale. Categories included previously recommended activities involving emotional support of the patient (11 items) and those involving the proper setting (9 items). The average amount of time spent with patients while giving bad news was assessed. The impact of demographic variables on the number of emotional support items, setting and time items, and number of minutes spent informing patients were analyzed via ANOVA. All significant variables for each category were entered into a multiple linear regression model.

RESULTS: Of the 961 surveys which were received by subjects, 461 (48%) were completed and returned. A majority of physicians (52-93%) always or frequently performed 10 of the 11 recommended emotional support items and 6 (60-92%) of the 9 proper setting items while giving bad news to patients. The average time spent in giving bad news was 27 minutes. Although training in giving bad news had a significant impact on the emotional support items provided to patients ($p < 0.05$), only 25% of respondents had any previous training in this area. A number of demographic factors were also associated with an increased number of emotional support items provided to patients, including being single ($p < 0.05$), being female ($p < 0.005$), and having personally had a life-threatening illness ($p < 0.05$).

CONCLUSION: By their own report, internists generally inform patients of bad news in an appropriate fashion. Some deficiencies do exist, but training in giving bad news can improve this important type of communication. Educational opportunities about how to give bad news should be provided to practicing physicians; and curricula designed for medical students and residents.

BIAS IN ESTIMATES OF CONFIDENCE INTERVALS FOR HEALTH OUTCOME REPORT CARDS. P.D. Faris¹, W.A. Ghali¹, R.F. Brant¹; University of Calgary, Calgary, Alberta

PURPOSE: In health outcome report cards comparing providers for binary outcomes such as mortality, a commonly used method of profiling providers is to use risk factor data from patients treated by the providers to construct logistic regression models. Such models are then used to obtain the expected number (E) of outcomes for each provider and the ratio of observed (O) to expected outcomes (O/E ratio) is used as a risk-adjusted measure of provider performance. To account for chance variation and to determine "outlier status", confidence intervals (CI's) derived from standard deviations are placed around the O/E ratios. Here we compare two

methods for calculating standard deviations (SD's) and CI's. The typically used method is compared with a more complex method based on the propagation of errors (PE).

METHODS: Typically, when calculating the variances and SD's of O/E ratios, only O is treated as a random variable, and variability in E is ignored. A variance estimate treating both O and E as random variables was derived using the propagation of errors (PE) method. The resulting PE-SD estimates were compared with typical SD estimates using a data set (N=50,357) previously employed to profile CABG providers in Canada. O/E ratios were used to profile three levels of hospital patient volume (low, medium, high), 8 provinces, and 23 hospitals. For each O/E ratio, the ratio of the typical SD to the PE-SD was used to evaluate the relative sizes of the SD estimates. In addition, computer simulations based on a hypothetical data set with three providers were used to evaluate the SD estimates. Measures of performance included empirical coverage and bias.

RESULTS: For the CABG data profiles, the ratios of the typical SD's to the PE-SD's were 1.04, 1.65, and 1.23 for the low, medium and high volume hospitals, respectively. For 7 of 8 provinces, the ratios of the SD's were only marginally larger than 1. However, for the province treating the largest proportion of patients (52%) the ratio was 1.40. For the 23 hospitals, the largest SD ratio was 1.09. The computer simulations confirmed that the typical SD's were less accurate than the PE-SD estimates, and the coverage probabilities indicated that confidence intervals based on the typical SD's are too wide.

CONCLUSION: The typically used SD estimates and CI's were always larger than the PE-SD estimates that accounted for variability in E. The bias in variance and SD was greatest when one or more providers treated a large proportion of the patients. When only a few providers are being compared, or when some providers treat a large proportion of the patients, the use of typical SD's may lead to incorrect conclusions regarding the outlier status of providers.

MEASURES OF PHYSICIAN LANGUAGE COMPETENCE THAT PREDICT CARE OF SPANISH SPEAKING PATIENTS. A. Fernandez¹, A.E. Rosenthal¹, E.J. Perez-stable¹, A.B. Bindman¹, H. Liu¹, K. Vranizan¹, K. Grumbach¹; ¹University of California, San Francisco, San Francisco, California

PURPOSE: Language concordance between physicians and patients is associated with greater patient satisfaction. However, it is not known what level of physician Spanish language competence is important to Spanish speaking patients. We used four measures to test the association between physician Spanish ability and the probability of these physicians having Spanish speaking patients in their practice.

METHODS: Primary care physicians for Spanish and English speaking diabetics from eleven public health clinics completed a questionnaire containing four questions about Spanish language competence: 1) self-rated fluency 2) use of interpreters 3) confidence in conducting an H&P 4) confidence in discussing complex topics. We used chi-squared to test the association between measures of physicians' Spanish competence and the proportion of Spanish speaking patients in their practice.

RESULTS: 105 MDs caring for 365 Spanish and 1023 English speaking diabetics participated. 17 (16%) MDs rated their Spanish fluency as excellent and they cared for 144 (40%) of monolingual Spanish speaking patients. Conversely 27 (26%) of MDs rated their fluency as poor or none and they cared for 26 (7%) of Spanish speaking patients. MDs whose self-rated Spanish was excellent, good, or fair were much more likely to treat Spanish speaking patients than those whose Spanish was self-rated as poor or none (OR 3.5, CI 2.3 - 5.3). Similarly, MDs who never, rarely, or occasionally use an interpreter were more likely to treat Spanish speaking patients than those who usually, or always use an interpreter (OR 3.8, CI 2.6 - 5.4). The two additional questions assessing clinician confidence in conducting a Spanish patient history or carrying out a complex discussion were less predictive of treating Spanish speaking patients.

CONCLUSION: While factors other than patient preference may affect the distribution of patients, our study demonstrates that a brief physician questionnaire can discriminate degrees of language competence that appear meaningful to patients in selecting their physician.

HIGH USE OF IN-PATIENT SERVICES AT A PUBLIC HOSPITAL IS ASSOCIATED WITH ALCOHOL AND SUBSTANCE ABUSE. A. Fernandez¹, M. Ehman¹, H. Liu¹, D. Schillinger¹; ¹University of California, San Francisco, CA

PURPOSE: Prior research has shown that old age and chronic conditions such as congestive heart failure are associated with multiple admissions of patients to the medical wards of community hospitals. Risk factors associated with multiple admissions to public hospital wards are unknown, though poor access to primary care is known to be associated with preventable hospitalizations. We studied the demographic, medical, and utilization patterns of high utilizing patients of an urban public hospital to determine risk factors for multiple admissions.

METHODS: Using an administrative data set, we obtained data on patient demographics, in- and out-patient utilization, and in-patient and out-patient billing diagnosis on all patients discharged from a public hospital medicine/cardiology service between 7/97-6/98. Patients with 3 or more discharges in a one-year interval were defined as high utilizers (HU). We used chi square to test differences between high utilizers and non-high utilizers (NHU) in utilization of emergency room and primary care services in the year following the index discharge, in diagnosis and in demographic characteristics.

RESULTS: 3562 patients had 5573 discharges in the study period. 465 patients (13%) were HU who accounted for 35% of discharges, and 36% of hospital bed-days. Compared to NHU, HU were slightly more likely to be male (71% vs. 66% $p = 0.02$) and more likely to be African American (46% vs. 30% $p = 0.001$). They were no different in age. HU had more visits to primary care clinics (mean 5.72 vs. 2.5 $p = 0.01$), to the emergency department (mean 4.38 vs. 2.5 $p = 0.01$) and to any network clinical site (median 24 visits vs. 5 visits $p = 0.001$). HU had more congestive heart failure (21% vs. 12% $p = 0.01$), AIDS (30% vs. 14% $p = 0.01$) and COPD (21% vs. 13% $p = 0.02$). HU were much more likely to abuse alcohol (47% vs. 25% $p = 0.01$) or drugs (58% vs. 30% $p = 0.01$) or either alcohol or drugs (71% vs. 42% $p = 0.01$).

CONCLUSION: High users of urban medical wards in a clinical system with good access to primary care have high rates of use of primary care services. Our data suggests that interventions to reduce multiple hospitalizations in urban public hospitals need to integrate substance abuse services to traditional disease management programs.

SYSTEMATIC REVIEW: MULTIDISCIPLINARY DISCHARGE TRANSITION PROGRAMS DECREASE HOSPITAL READMISSION FOR HEART FAILURE PATIENTS. V. Flintoft¹, H. Lee², F. Sridhar³, D. Lee¹; ¹Institute for Clinical Evaluative Sciences, Toronto, Ontario; ²McMaster University, Sault Ste Marie, Ontario; ³McMaster University, Hamilton, Ontario

PURPOSE: Heart Failure (HF) is a leading cause of hospitalization and re-admission in most hospital systems. Multidisciplinary "Discharge Transition" programs aimed specifically at the education and close follow-up of HF inpatients have been evaluated in a number of relatively small randomized controlled trials (RCTs). This is the first meta-analysis to evaluate the effectiveness of peri-discharge, multidisciplinary HF patient management programs.

METHODS: Electronic database searches were conducted on MEDLINE, HealthSTAR and EMBASE. Reference lists of identified articles and experts' opinions were also reviewed. All potentially relevant articles were obtained. Study selection criteria were: (1) RCTs of adult inpatients hospitalized for HF enrolled at the peri-discharge transition period; (2) HF-specific patient education intervention coupled with a post-discharge follow up assessment; (3) Primary outcome of unplanned all-cause readmission secondary outcomes of mortality, compliance and quality of life. Study inclusion and quality assessment using a modified Jadad Scale were independently assessed by all four authors. Agreement was rated by a weighted Kappa and final decision agreed upon by consensus.

RESULTS: A total of 529 citation titles were identified: 199 from MEDLINE; 148 from HealthSTAR; 162 from EMBASE and 20 from personal files, reference lists, or communication with experts. Of these, 94 were deemed potentially eligible for pre-test selection. Two research trials in progress were identified and interim data obtained from the principal investigators. The four reviewers selected 10 papers and two studies "in progress" for inclusion in the overview. The Kappa agreement statistic for multiple reviewers was 0.73 (SE 0.09). One study was subsequently excluded because it represented an earlier publication of the same trial, and another study was excluded because of substantial heterogeneity in methods, patient population and results. The results of two other studies were not transferable, leaving 8 RCTs for meta-analysis for the primary outcome. The pooled risk ratio for hospital readmission was 0.755, 95% CI(0.680-0.839), $p < 0.001$. There was a non-significant trend towards decreased mortality, pooled risk ratio 0.859 (95% CI, 0.691-1.067), $p = 0.17$. There was insufficient data to evaluate intervention effects on quality of life or compliance.

CONCLUSION: This systematic review suggests that specific HF intervention programs targeted at the discharge transition period significantly decreases hospital readmissions.

ETHICS CONSULTATION IN U.S. HOSPITALS. E. Fox¹, S. Myers², R. Pearlman³; ¹National Center for Ethics, Veterans Health Administration, Washington, DC; ²RAND, Arlington, VA; ³VA Puget Sound Health Care System, Seattle, WA

PURPOSE: This descriptive study fills previous gaps in knowledge about the prevalence, structure, processes, and practitioners of ethics consultation in U.S. hospitals.

METHODS: We surveyed 600 randomly selected hospitals (12% of all general hospitals) in 1999-2000 to characterize ethics consultation services in U.S. hospitals. We asked: What is the prevalence of ethics consultation? Who performs ethics consultation and what are their backgrounds and training? How do ethics consultation services function? Descriptive statistics were employed. Chi-square, ANOVA, and multivariate analyses were used to compare responses across hospitals.

RESULTS: Ethics consultation services existed in approximately 82% of all U.S. general hospitals. In a majority of hospitals, consultations are most often performed by a small team (68%) as opposed to a full committee (23%) or an individual consultant (9%). On average each ethics consultation service provided 7 consultations during the preceding year; an average of 4 individuals performed each consultation. Physicians and patients/surrogates were generally notified prior to the consultation, but their permission was usually not required. Data gathering always involved discussions with clinical staff and patients/families in 63% and 41% of services, respectively. Medical records were always examined in 70% of services; patients were always examined in 30%. Individuals who performed consultation usually either learned to do consultations independently without formal, supervised training (47%) or were trained through direct supervision (36%). Less than 5% of individuals performing consultation completed a fellowship or graduate degree program in bioethics. Formal evaluative activities occurred in less than one third of the services.

CONCLUSION: Ethics consultation services are quite prevalent in general U.S. hospitals. The small team model is more common than other models (full ethics committees and individual consultant) that have been characterized in the literature. Our data suggest that nationally over 45,000 ethics consultations are performed each year and that over 35,000 individuals perform these consultations. Despite this level of activity, many individuals performing consultations lack formal supervised training. Moreover, these services are usually not formally evaluated.

DEPRESSION MANAGEMENT BY VA AND NON-VA MEDICAL PHYSICIANS. S.M. Frayne¹, K.M. Freund¹, S.J. Smith¹, K.M. Skinner¹, A. Ash¹, M.A. Moskowitz¹; ¹Boston University, Boston, MA

PURPOSE: Depression typically presents in a medical rather than mental health setting. VA and non-VA settings may differ in physician characteristics, training opportunities, and organizational factors. We sought to assess VA and non-VA medical physicians' recognition and treatment of depression.

METHODS: We selected a random sample of VA and non-VA primary care physicians and medical specialists practicing in the Northeastern United States. Physicians viewed a 5 minute, professionally produced video of a patient with somatic symptoms, who described 5 symptoms meeting clinical criteria for major depression. After viewing the video, physicians answered interviewer-administered questions about differential diagnosis and management recommendations. Because of the experimental design, every physician reviewed identical clinical data, permitting direct comparisons of physicians' diagnostic reasoning and management approaches. Physicians also viewed two "control" scenarios (a patient with chest pain and a patient with polymyalgia rheumatica), further masking the key study purpose.

RESULTS: 81 VA and 129 non-VA physicians participated. Regarding diagnosis, 85% of VA versus 95% of non-VA physicians listed depression as a possible diagnosis ($p = .02$), and 47% versus 62% assigned depression a probability of greater than 50% ($p = .04$). Regarding management recommendations, 15% of VA physicians versus 2% of non-VA physicians said they would recommend a mental health referral ($p = .001$), and 9% versus 13% would recommend an antidepressant ($p = NS$). 35% of VA physicians versus 53% of non-VA physicians would see the patient portrayed in the videotape for a return visit within 2 weeks ($p = .01$).

CONCLUSION: Non-VA physicians were more likely to recognize depression, although many VA physicians also considered it. Neither group followed Agency for Healthcare Research and Quality guidelines for follow-up and for initiation of pharmacologic or psychotherapy, although non-VA physicians were more likely to see the patient back within 2 weeks and VA physicians were more likely to refer to mental health. Differences in management by VA versus non-VA physicians may point to systems issues underlying non-adherence to depression management guidelines.

DO HOSPITALISTS FOLLOW EFFICIENCY GUIDELINES BETTER THAN NON-HOSPITALISTS? S. Freer¹, J. Cotter¹, C. Pugh¹, W. Smith¹; ¹Virginia Commonwealth University, Richmond, VA

PURPOSE: The hospitalist model of inpatient care on general medicine wards offers the potential for a more efficient use of hospital resources, but limited formal evaluation has been conducted. The ability to effectively move patients out of the hospital on day of discharge is an important indicator of the improved use of hospital resources. This is the first study we know of to examine if hospitalists, when compared to non-hospitalists, could more efficiently discharge patients. Physicians have control over the initiation but not the completion of this process.

METHODS: We studied discharge orders written by 11am and completion of discharge by noon as a measure for evaluating efficiency. These efficiency guidelines were established by the institution. We evaluated compliance with these guidelines through a retrospective analysis of clinical and utilization data on 3,040 patients admitted to general medicine wards of the Medical College of Virginia Hospital at Virginia Commonwealth University. We analyzed the number and percentage of discharge orders written before 11am and the number of patients discharged by noon by hospitalist and non-hospitalist physicians using chi-square to compare proportions.

RESULTS: In FY00 the percentage of discharge orders written before 11am for the hospitalist teams (44.3%) was greater than that for the non-hospitalist teams (28.2%; $p < 0.01$). Very low percentages of patients were discharged by noon from either general medicine service. **CONCLUSION:** Hospitalist physicians performed significantly better than did non-hospitalist physicians on entry of discharge orders by 11am, a process measure of efficiency over which they have control. There was no difference between the groups on the completion of discharge, a process measure of efficiency over which physicians have less control.

HIGHER PHARMACEUTICAL COSTS DO NOT CORRELATE WITH DIABETES CONTROL. S.L. Fultz¹, C.B. Good¹, M.E. Kelley¹, M.J. Fine¹; ¹VA Pittsburgh Health Care System, University of Pittsburgh, Pittsburgh, PA

PURPOSE: To examine the relationship between pharmaceutical costs and glycemic control in the outpatient management of DM.

METHODS: Pharmacy records for all outpatient diabetes prescriptions during 1997 were obtained at four VA hospitals. Primary care providers (PCP) were defined as writing > 50% of prescriptions, and coded as staff attendings (AT), medicine house officers (HO), nurse practitioners (NP), endocrinologists (EN), or no PCP. Patients were grouped based on whether they received oral agents only, insulin only or both. ANOVA was performed to evaluate medication costs, glucose monitoring supply costs, total costs and glycemic control by site, provider type and treatment classification, controlling for demographics. Supply costs were calculated only across patients receiving supplies; but all patients were included in total monthly costs. Chart reviews were done on a subset from 2 sites to evaluate for differences in DM severity and comorbid conditions.

RESULTS: 4544 patients were identified with complete information for cost analyses (Site A=1957, Site B=1194, Site C=611, Site D=782); 3690 (81%) of these had laboratory data available for glycemic control analysis. For total monthly costs, there was a significant site effect with $D(\$31.12) > A(\$15.85) > B(\$12.18) > C(\$8.44)$; provider type also showed a significant effect with $EN > AT = HO = NP = \text{no PCP}$. For monthly medication costs, $D(\$12.01) > A(\$6.62) = B(\$7.15) = C(\$6.16)$; provider type analysis showed $EN > \text{no PCP} > AT = HO = NP$. Monthly supply costs showed $B(\$41.17) > D(\$33.75) > C(\$22.63) > A(\$14.36)$; provider type effects showed $EN > AT = \text{no PCP} > HO = NP$. Across all of the cost analyses, patients treated with oral agents and insulin > insulin > oral agents. For glycemic control, as measured by last HgbA1C, $A(8.26) > B(7.37) = C(7.40) = D(7.18)$; treatment groups showed oral agents and insulin > insulin > oral agents. When controlling for the difference from the upper limit of normal at each site, $A(1.75) > D(1.32) > B(1.15) > C(0.64)$; treatment group again showed a difference: oral agents and insulin > insulin > oral agents. There were no provider type effects for glycemic control. Chart review data showed no difference in diabetes severity or the presence of comorbid conditions across site or provider types.

CONCLUSION: As the site with the lowest monthly cost had the best glycemic control, this data demonstrates that higher pharmaceutical costs are not associated with improved glycemic control. This suggests that efforts to reduce costs by standardizing treatment can be accomplished without losing glycemic control.

CLINICAL TRIAL OF STANDARD WARFARIN MANAGEMENT VERSUS A TELEPHONE-BASED ANTICOAGULATION SERVICE. B.E. Gage¹, A.D. Waterman², G.A. Banet², P.E. Milligan²; ¹Washington University in St. Louis, St. Louis, MO; ²Washington University School of Medicine, St. Louis, MO

PURPOSE: To compare the rate of adverse events in patients who have their warfarin managed by either their physician or by a multidisciplinary, telephone-based anticoagulation service (ACS).

METHODS: In December 1998 we randomized 4 of 8 outpatient health centers that participated in a local HMO to have access to an ACS. The multi-disciplinary, telephone-based ACS evaluated the appropriateness of warfarin therapy, dosed the warfarin therapy, educated the patients, and monitored their international normalized ratios (INRs). At the control sites, physicians and their staff performed these tasks. We assessed the rate of hospitalization for a hemorrhagic or thrombotic adverse event by analyzing hospital claims submitted to the HMO. Using two-sided tests of the Poisson approximation, we compared the relative rates of adverse events between control and ACS sites using an intent-to-treat approach.

RESULTS: Of the approximately 1100 patients taking warfarin at the 8 centers, approximately half (N = 503) were eligible and enrolled in the ACS. The remaining patients had their warfarin monitored and dosed by their physician. Using hospital claims data through July 30, 2000, the relative risk for an adverse event was 0.71 among patients who had access to the ACS, corresponding to a significant ($p < 0.05$) relative risk reduction of 29% compared with patients whose warfarin was managed by their physician.

CONCLUSION: The telephone-based ACS was associated with a significantly decreased rate of hospitalization for warfarin-related adverse events. This finding suggests that multi-disciplinary, disease-state management programs that empower nurses and pharmacists can improve medical care.

DOES A HALF-DAY WORKSHOP IN DOCTOR-PATIENT COMMUNICATION IMPROVE PATIENT SATISFACTION WITH THE DOCTOR-PATIENT INTERACTION? T.H. Gallagher¹, J. Munro¹; ¹Washington University in St. Louis, St. Louis, MO

PURPOSE: Physicians' communication skills are strongly associated with patient satisfaction. However, it is not known whether training physicians in doctor-patient communication improves patient satisfaction with the doctor-patient interaction.

METHODS: 42 community-based primary care physicians participated in a randomized trial testing a half-day doctor-patient communication workshop's impact on patient satisfaction. The communication workshop was developed by the Bayer Institute for Health Care Communication. Patient satisfaction was measured via a telephone interview with 15 of each physicians' patients before and 15 different patients after the workshop. 26 physicians were randomly selected to attend the workshop, with the remaining 22 serving as a control group.

RESULTS: Baseline patient satisfaction with their doctor was quite high. The 1,270 patients interviewed gave their physician a median rating of 9 on a scale of 0 (worst doctor possible) to 10 (best doctor possible). 47% of patients rated their doctor as a "10." Physicians also scored highly using a validated 10-item scale to designed assess communication skills and humanistic qualities; the median score was 46 of a possible 50 points, with 35% of patients rating their doctor as "excellent" on all 10 items. Patients' ratings of physicians' communication skills were a powerful independent predictor of overall patient satisfaction ($p < .001$). However, analysis of variance showed that attending the workshop had no significant effect on patients' rating of their doctor's communication and humanistic skills ($p = 0.32$) or patients' overall rating of their doctor ($p = 0.33$). Physicians with baseline satisfaction ratings in the lowest quartile were no more likely to show improvement in their ratings than were physicians with higher baseline satisfaction ratings. **CONCLUSION:** Patients' assessments of physicians' communication skills are an important component of overall satisfaction. However, a half-day workshop in communication skills has little impact on patients' satisfaction with the doctor-patient interaction. A more in-depth and sustained intervention may be necessary if communication training is to improve patient satisfaction.

IMPACT OF BASIC COMPUTERIZED PRESCRIBING ON OUTPATIENT MEDICATION ERRORS AND ADVERSE DRUG EVENTS. T.K. Gandhi¹, S.N. Weingart², A. Seger¹, D.S. Seger¹, J.S. Borus¹, E. Burdick¹, L.L. Leape¹, D.W. Bates¹; ¹Brigham and Women's Hospital, Boston, MA; ²CareGroup Healthcare System, Boston, MA

PURPOSE: In the inpatient setting, physician order entry has been shown to significantly reduce serious medication errors (MEs). However little is known about the impact of computerized prescribing systems in the ambulatory setting on MEs and adverse drug events (ADEs). We compared the frequency of these events in outpatient clinics using handwritten versus computerized prescribing.

METHODS: We prospectively studied 2 sites with handwritten prescribing and 2 with basic electronic prescribing in the Boston area. The computerized sites had printed prescriptions and required fields, but no defaults and optional or non-existent checks for allergies and drug interactions. We collected copies of prescriptions written by 24 primary care providers from Sept 1999 to March 2000 (6 weeks per clinic). Prescription copies were reviewed by a pharmacist to screen for MEs and potential ADEs. In addition, patients who received prescriptions were telephoned 2 weeks after their visit to ask about problems with their medications (response rate 59%), which were then classified as ADEs or not by 2 MD reviewers.

RESULTS: Of 1173 prescriptions screened during this time period, 202 (17%) were rule violations (orders that violate strict standards but are generally understood and generate no additional work), 44 (4%) were MEs, and 59 (5%) were potential ADEs. Sites with computerized prescribing were significantly less likely to have rule violations ($p < .02$) and MEs ($p < .01$), but potential ADE rates were not significantly different. Of 661 patients surveyed, 178 (27%) reported a total of 206 ADEs, of which 74 (36%) were preventable. There was no significant difference between computerized and non-computerized sites in ADE rates and preventable ADE rates (37%, 35%). Of preventable ADEs (n = 74), 45 (62%) occurred in the ordering stage, and 26 (36%) in the patient administering stage. The main types of physician error were failure to act on results of monitoring or tests (50%) and inappropriate drug choice (24%). Improved computer ordering checks would have prevented only 20% of physician errors. **CONCLUSION:** Errors in the drug process were common in the outpatient setting. However, while basic computerized prescribing systems were associated with lower rates of rule violations and MEs, serious error rates (potential and preventable ADEs) were similar. Monitoring for ADEs was unexpectedly important. Prescribing systems with advanced decision-support, including monitoring and communication features as well as allergy/interaction checking, may be required to substantially reduce the frequency of serious errors.



PHYSICIANS SCREENING AND INTERVENTION BEHAVIORS FOR DOMESTIC VIOLENCE COMPARED TO HIV/STD RISK, ALCOHOL ABUSE, AND TOBACCO USE.

B. Gerber¹, J. Tang¹, S. Gansky¹, R. Carlton¹, S. Mcphee¹; ¹Univ. of California, San Francisco, CA

PURPOSE: Despite mandates for physicians to screen for and intervene with domestic violence as they should do for HIV/STD risk, alcohol abuse, and smoking, studies show that physicians face many barriers to doing so. We wanted to know how asking and intervening with domestic violence compares to asking and intervening with patients in these other three health-risk areas.

METHODS: In November 2000, we mailed a questionnaire on physicians' screening and intervention behaviors for domestic violence, HIV/STD risk, alcohol abuse, and tobacco use to a national random sample of 1200 physicians (internal medicine and family practice). Our response rate to date, prior to our final mailing, is 49%. Chi-square tests with 3 degrees of freedom (p) and Bonferroni-Holm adjusted pairwise comparisons among topics (p') were used to test results.

RESULTS: Sixty-six percent of our physician sample reported that intervening when domestic violence is identified is an essential part of their role as a physician. In contrast, 79% reported that intervening with patients identified for HIV/STD risk is an essential part of their role; 82% with alcohol abuse; and 85% with tobacco use (Chi-square=13.8, $p=.003$; $p^* < .032$ for domestic violence vs. tobacco and vs. alcohol). Regarding screening, only 10% of physicians reported that they always ask new patients about domestic violence, compared to 22% for HIV/STD risk, 72% for alcohol use, and 84% for tobacco use (Chi-square=188, $p < .001$; $p^* < .035$ for all 6 pairwise comparisons). When asked about their knowledge of and confidence in screening in these areas, only 19% strongly agreed that they knew how to assess patients for the risk of domestic violence, whereas 47%, 59%, and 70% strongly agreed that they knew how to assess patients for HIV/STD risk, alcohol abuse, and tobacco use respectively (Chi-square=68.3, $p < .001$; $p^* < .001$ for all 3 pairwise comparisons with domestic violence and HIV/STD vs. tobacco). When asked if they would rather refer domestic violence victims to outside resources than provide counseling themselves, 79% agreed or strongly agreed that they would. This compares with 42%, 51%, and 23% who agreed or strongly agreed that they would rather refer patients identified with HIV/STD, alcohol, and tobacco risks respectively (Chi-square=74.7, $p < .001$; $p^* < .006$ for all pairwise comparisons except HIV/STD vs. alcohol).

CONCLUSION: These data suggest that, despite physicians' view that screening for domestic violence is part of their role, physicians are not prepared or able to overcome barriers to ask their patients about abuse. It is even more difficult for physicians to screen their patients for the health risk of domestic violence than to screen for the other health risks, including the risk for HIV/STD, which is also considered a sensitive and stigmatized topic. One solution may be simplifying the physicians' role and augmenting patient care with other experts and assessment tools.

THE COMPUTERIZED PATIENT RECORD SYSTEM: A TRADE-OFF BETWEEN PHYSICIAN TIME AND PATIENT BENEFIT. P.A. Glassman¹, P. Belperio², B. Simon², K. Lim¹, J. Sayers²; ¹VA Greater Los Angeles, Los Angeles, CA; ²VA Greater Los Angeles, LA, CA

PURPOSE: Improving patient safety is a critical concern. As part of a larger evaluation study designed to reduce adverse drug events, we conducted a baseline survey on clinicians' attitudes towards and knowledge of the Computerized Patient Record System (CPRS) and associated drug interaction alerts.

METHODS: We surveyed 319 clinicians at 12 sites within an integrated VA Healthcare System located in Southern California. Clinicians included attending-level physicians, nurse practitioners and physician assistants. The questionnaire, developed and tested locally, assessed extent of and comfort with CPRS. Questions assessed how CPRS in general, and clinician order entry and drug interaction alerts in particular, have affected clinician efficiency and patient care. Additionally, we assessed general knowledge about 21 common drug-drug and drug-disease interactions. This abstract reports the results of the first 63 (20%) respondents in this abstract. The survey period is expected to end in January 2001, with an anticipated response rate of 65% to 70%.

RESULTS: In this preliminary analysis respondents were 71% male, 87% full-time employees and 72% Internists. Participants averaged 48 years of age, practiced 2.5 days per week in outpatient clinics and wrote 53 prescriptions per week. A majority of clinicians (range: 70% to 86%) preferred using CPRS to conventional written methods for entering patient notes, requesting consults, ordering radiological procedures, ordering laboratory tests, and prescribing medications. Overall, clinicians reported that CPRS improved the safety and quality of patient care but reduced clinician efficiency. For example, 81% of clinicians felt that drug interaction alerts increased the potential for prescribing safely but 70% perceived that clinician order entry increased the time required to write prescriptions. Also, while 71% felt that clinician order entry reduced errors in ordering laboratory tests, 68% reported that doing so required extra time. Clinicians recognized a median of 53% (range: 17% to 89%) of 21 common drug interactions. Eighty five percent (85%) of clinicians reported that they would have felt more confident about their answers had they had drug alerts to identify these interactions. In practice, however, respondents reported that they would be more likely to change a patient's medication based on personal interaction with a pharmacist (58%) rather than depend on a CPRS drug interaction alert (4%). Thirty-eight percent were equally likely to change medications based on an alert from either source. Clinicians reported the greatest barriers to effective use of drug alerts included non-relevant alerts (72%), system slowdowns and shutdowns (65%), and lack of time to review alerts (57%).

CONCLUSION: Early analysis of an on-going survey suggests that clinicians perceive that CPRS improves the quality and safety of patient care but decreases their efficiency. Clinicians favor the CPRS drug alert system as a means to improve their recognition of drug interactions, an issue that needs to be further addressed based on our data, but note several implementation problems that impede effective utility.

CLINICIAN ACCEPTANCE OF A GUIDELINE IMPLEMENTATION. M.K. Goldstein¹, N. Hastings², M.R. Bergen², M. Vance¹, B.B. Hoffman¹ ¹VA Palo Alto and Stanford University, Palo Alto, CA; ²Stanford University, Stanford, CA

PURPOSE: Clinical practice guidelines (CPGs) are being implemented in many large health care systems; yet, little is known about clinician reaction. We evaluated clinician response to a hypertension (HTN) guideline implementation at a large, geographically-diverse VA medical center.

METHODS: In the context of a facility-wide implementation of 10 CPGs, the HTN guideline was implemented with 36 attending physicians and nurse practitioners in a randomized trial comparing a general intervention (educational components and a list of the patient's antihypertensive drugs delivered to clinicians at each primary care clinic visit) with an individualized intervention (the general intervention plus patient-specific recommendations about drug therapy delivered to clinicians during primary care clinic visits). For the 4500 hypertensive patients of these study clinicians meeting study criteria, the individualized intervention improved guideline concordance of drug therapy (*Med Dec Mkg* 2000;20:488). At the end of the study period, two physicians, previously unaffiliated with the study, conducted structured interviews with 32 (89%) study clinicians.

RESULTS: 31 (97%) clinicians were aware of CPG implementation at the facility and 28 (88%) were specifically aware of the HTN CPG. 29 (91%) and 28 (88%) clinicians, respectively, reported that they use CPGs and that CPGs help with patient care. 16 of 18 (89%) clinicians in the individualized group agreed with the recommendations, but only 3 (17%) reported that seeing the recommendations affected their management of patients. Barriers to following the recommendations included patient reluctance to change medication; concern about need for additional clinic visits and laboratory monitoring of recommended medications; previously demonstrated patient intolerance of the guideline drugs. Time constraints were mentioned frequently.

CONCLUSION: Clinicians had a high level of awareness of the CPG implementation and a generally positive attitude toward CPGs. In the context of the randomized trial evidence of the impact of the guideline implementation, the small percent of clinicians who reported that the recommendations affected their management of patients suggests that clinicians may not be aware of the impact of guideline implementation.

USUAL CARE ADHERENCE COUNSELING PRACTICES AMONG PHYSICIANS, PHARMACISTS AND CASE MANAGERS CARING FOR PATIENTS WITH HIV IN NORTH CAROLINA (NC). C. Golin¹, S. Smith², S. Reif³; ¹University of North Carolina at Chapel Hill, Chapel Hill, North Carolina; ²UNC School of Pharmacy, Chapel Hill, NC; ³Duke University

PURPOSE: Patient adherence to antiretroviral therapy (ART) is critical to effective treatment of HIV and may depend partly on health professionals' practices. However, little is known about providers' knowledge, practices, and barriers in promoting ART adherence. We sought to assess adherence counseling practices among the physicians (MD), pharmacists (PH), and case managers (CM) caring for HIV+ patients in NC and to identify barriers they faced in facilitating adherence.

METHODS: In February 2000, using NC AIDS Drug Assistance Program records and commercial prescription records, we identified and surveyed, by mail, the 1,301 MDs, PHs, and CMs caring for HIV+ patients in NC. After 4 mailings, 440 (77%) of the PHs, 94 (85%) of the CMs, and 380 (63%) of the MDs responded. Among the responding MDs, only the 190 reporting having prescribed a protease inhibitor in the last year were included in the analytic sample.

RESULTS: 31% of the MDs were general internists, 24% family practitioners, and 22% infectious disease specialists; half cared for <10 HIV+ patients. 66% of PHs worked in chain

pharmacies, 28% in independents and only 2% in hospitals; half cared for <4 HIV+ patients. On average, CMs cared for 34 HIV+ patients. Regarding the average number of minutes counseling a patient on a new 3-drug ART regimen, PHs spent 7, MDs 13 and CMs 21. MDs seeing a higher volume of HIV+ patients spent no more time counseling but were more likely to: advise patients of side effects (SE), tailor the regimen, explain intake requirements, and provide a pill box ($p < .01$ for each). Over 90% of MDs explained dosing, asked for patients' questions and discussed drug resistance most or all of the time; <60% discussed SE management, handling missed doses, ways to remember doses, or planning dose times. Over 70% of PHs explained dosing instructions and asked for questions but <50% discussed SE management, handling missed doses, drug interactions or storage requirements. The most common CM adherence counseling behaviors were: praising adherent clients (84%), discussing nonadherence repercussions (74%), asking about clients' treatment concerns (74%) side effects (65%), and if medications were taken on time (71%). The vast majority of MDs, PHs, and CMs reported strong interest in doing adherence counseling and had positive attitudes toward ART, ART counseling, and HIV+ patients. The primary barriers reported were lack of space, time, and reimbursement for counseling. MDs viewed CMs' doing adherence counseling as acceptable but were dissatisfied with HIV CM availability. Both MDs and CMs indicated a high interest in improving collaborations with each other in the care of HIV+ patients.

CONCLUSION: Innovative practice arrangements between all three provider groups may be needed and feasible to facilitate adherence counseling and address the multifaceted problem of antiretroviral adherence.

IMPACT OF AN EDUCATIONAL PHYSICIAN PROFILING AND PRACTICE GUIDELINE INTERVENTION ON ANTIBIOTIC TREATMENT OF ACUTE BRONCHITIS BY NETWORK-MODEL HMO PHYSICIANS. R. Gonzales¹, J. Maselli¹, B. McCafferty², K. Gershman³, K.F. Baum⁴, L. Latts⁴, R. Faraci²; ¹University of Colorado Health Sciences Center, Denver, CO; ²Colorado Medical Society, Denver, CO; ³Colorado Department of Public Health and Environment, Denver, CO; ⁴Colorado Clinical Guidelines Collaborative, Denver, CO

PURPOSE: Recent antibiotic use is an important risk factor for spread and infection with antibiotic-resistant *S. pneumoniae*. Effective strategies to improve ambulatory antibiotic dispensing for acute respiratory illnesses are needed.

METHODS: Time-series analysis. Practice guidelines for sore throat, nasal congestion and cough illness developed by the Colorado Clinical Guidelines Collaborative were mailed to all primary care physicians belonging to the Colorado Medical Society in November 1999. "Profiled physicians" (having at least 10 office visit claims for adults with acute bronchitis in 1998) ($n = 352$) also received antibiotic prescribing profiles based on aggregated office visit and pharmacy claims from 7 HMOs participating in the Colorado Medical Society Joint Data Project; and "non-profiled" physicians ($n = 514$) received summary data based on the entire cohort of patient visits. The office visit was the primary unit of analysis. Multivariate mixed effects models included patient age, physician specialty, HMO and time segments (baseline period = 1/98–10/99; study period = 11/99–2/00) as fixed effects, and unique physicians as random effects. Units of time were measured in months.

RESULTS: There were 12,993 and 2,899 adult office visits for acute bronchitis in the baseline and study periods, respectively. The majority of patients were 18–44 years old and treated by family physicians. Prescription rates for family physicians were about 5% greater than for internists ($p = 0.006$), and were positively associated with increased patient age. Adjusted monthly antibiotic prescription rates for acute bronchitis were stable during the baseline period for both physician groups (profiled: 65% to 63%, $p = 0.10$; non-profiled: 64% to 63%, $p = 0.78$), and declined significantly during the study period among profiled physicians (63% to 54%; $p = 0.0005$), but not among non-profiled physicians. The rate of decline in antibiotic prescription rates during the study period did not differ by patient age ($p = 0.09$) or physician specialty ($p = 0.87$).

CONCLUSION: This study demonstrates that antibiotic treatment of adults diagnosed with acute bronchitis by physicians in private practice can be reduced using a combination of educational physician profiling and practice guideline dissemination.

TRENDS IN ANTIMICROBIAL TREATMENT OF ACUTE RESPIRATORY TRACT INFECTIONS BY PRIMARY CARE PHYSICIANS, 1994–1998. R. Gonzales¹, J. Maselli¹; ¹University of Colorado Health Sciences Center, Denver, CO

PURPOSE: Rising rates of *S. pneumoniae* resistant to penicillin and other antibiotics have been publicized in the US since 1994, and have led to numerous calls to limit excess antibiotic use in ambulatory practice. This study evaluates whether changes in antibiotic treatment of acute respiratory infections (ARIs) between 1994 and 1998 have occurred, and whether these changes are associated with specific patient and physician characteristics.

METHODS: The National Ambulatory Medical Care Survey data files for years 1994–98 were obtained from the National Center for Health Statistics web site. Office visits with a principal diagnosis of otitis media, sinusitis, pharyngitis, bronchitis and upper respiratory tract infections (URIs)/common cold were identified, and further limited to visits to a general or family practitioner (GFP), internist (IM) or pediatrician (PED). Antimicrobial treatment was assigned to the visit only if it was entered as the principal medication entry (out of 6 possible). Multivariate logistic regression analysis was used to evaluate independent associations between antibiotic treatment for ARIs and time (year); patient age, race and diagnosis; and physician specialty and practice location. To determine if a change in antibiotic prescription rates during this period were associated with one of the other variables, interaction terms were utilized.

RESULTS: Overall, annual antibiotic prescription rates for ARIs declined between 1994 and 1998 (65%, 64%, 60%, 62% and 55%, respectively; $p = 0.0001$). This equates to 7 million fewer antibiotic prescriptions for ARIs dispensed by primary care physicians in 1998 compared to 1995 (the peak year). Independent of specific diagnosis and time, antibiotic treatment was less likely for patients age >45 years (vs. children age <5 years) (OR = 0.62, 95% CI = 0.51–0.76), and of non-white race (OR = 0.75, 95% CI 0.65–0.88). Because of a significant interaction between time and physician specialty, we stratified further analyses by specialty. Between 1994 and 1998, time was associated with a decrease in adjusted annual antibiotic prescription rates among PED and IM, but not among GFP (relative rate change = –8.9%; $p = 0.0001$; –9.3%;

$p=0.043$; and -4.5% ; $p=0.17$, respectively). Antibiotic prescription rates for specific diagnoses declined variably across specialties: sinusitis: no declines for any specialty; otitis media: PED; pharyngitis: PED; bronchitis: GFP; URIs/colds: IM decline $>$ PED $>$ GFP.

CONCLUSION: Primary care physicians are heeding the call to limit antibiotic prescribing in ambulatory practice, although there remains much room for improvement. Changes in prescribing need to be correlated with trends in antibiotic-resistance rates.

CORRELATION BETWEEN ANTIBIOTIC PRESCRIBING FOR COLDS AND UPPER RESPIRATORY TRACT INFECTIONS BY PRIMARY CARE PHYSICIANS AND PUBLIC AND PROFESSIONAL MEDIA ATTENTION. K. White¹, L.K. Traditi¹, J. Maselli², B. Gonzales¹; ¹University of Colorado Health Sciences Center, Denver, CO

PURPOSE: Antibiotic treatment of the common cold and upper respiratory tract infections (URIs) by US primary care physicians declined between 1994 and 1998. This study evaluates the amount of public and professional media exposure provided to the topics of antibiotic-resistance and overuse of antibiotics, and the correlation between media exposure and changes in antibiotic treatment of colds and URIs during this period.

METHODS: Treatment related to primary care physician (general and family practitioner, internist or pediatrician) office visits with a principal diagnosis of the common cold or URI were analyzed using the National Ambulatory Medical Care Survey (years 1994–98). Key word searches relating to excess antibiotic use and antibiotic resistance were performed on all databases, and final tallies confirmed by manual review. Lexis-Nexis was used to identify US news stories from large-circulation newspapers corresponding to the northeast (NY Times), midwest (Chicago Sun-Times), south (Atlanta Journal-Constitution) and west (LA Times), and one national newspaper (USA Today). The Vanderbilt Television News Archive was used to identify evening news stories on ABC, CBS, NBC, and CNN. Articles from medical journals represented in the Abridged Index Medicus were identified using MEDLINE. "Sentinel articles" were defined as receiving evening network news coverage by at least 1 network within 3 days of publication. Segmented time series analysis (logistic regression) was used to compare the change in 60-day (the smallest time unit providing adequate sample size) antibiotic prescription rates between time periods corresponding to increases in media exposure.

RESULTS: From 1994–98, there were 207 newspaper stories, 31 major network evening news stories, and 65 journal articles identified. The number of newspaper stories reported in 1994–1996 was stable, but more than doubled in 1997, and remained high in 1998. Evening news stories doubled in 1996 and 1997 compared to the previous years, but reverted to baseline levels in 1998. In contrast, journal articles have increased linearly between 1994 and 1998. There were 6 sentinel publications during this period: June 94, January 95, August 95, January 96, September 97 and March 98. Inspection of 60-day antibiotic prescription rates for colds and URIs reveals only 2 major trends: 1) relatively stable rates between 1994 and 1996 ($p=0.73$); and 2) declining prescription rates from 1996 through 1998 (parameter estimate = -0.036 ; $p=0.06$).

CONCLUSION: The change in antibiotic prescribing for colds and URIs that occurred in 1996 appears to be more strongly correlated with an increase in public media exposure than with medical journal publications. However, one can not rule out an impact of other forms of physician education (eg. local or national conferences) on prescribing behavior.

IS THERE BENEFIT USING GLUCOSE TEST STRIPS IN DIABETES? C.B. Good¹, S.L. Fultz², M.E. Kelley¹, M.J. Fine¹; ¹VA Pittsburgh Healthcare System, University of Pittsburgh, Pittsburgh, PA

PURPOSE: Home glucose monitoring is a common practice in diabetes (DM) treatment, but adds significantly to costs of care. Several studies have questioned the benefit of glucose monitoring in patients with type II DM. We examined the impact of glucose test strips on glycemic control in outpatient DM patients at four VA facilities.

METHODS: All patients receiving outpatient DM medications during 1997 were evaluated at four VA hospitals in Pennsylvania. Administrative data identified demographic variables (age, sex, race, marital status), glycosylated hemoglobin (HbA1C), all medications and DM supplies, and type of clinician (resident, nurse practitioner, staff internist, endocrinologist, and other). We examined differences among patients who received glucose test strips (strip+) and those that did not (strip-). Simple descriptive statistics was performed comparing strip+ and strip- patients, and analysis of variance was done, including main effects (medication type [oral agents, insulin, and both oral agents and insulin], site of care, and physician type), interaction terms, and demographic variables. Chart review of a subset of patients examined Charlson comorbidity scores, and presence of diabetes complications (neuropathy, retinopathy, and nephropathy).

RESULTS: There were 2912 strip+ patients, and 2221 strip- patients. Strips added an average of \$22.56 each month to cost of diabetes care. Average HbA1C for strip+ patients was 7.92; for strip- patients, 7.34. There were significant differences for use of strips in patients for site (ranging from 26% to 73% strip+), race (favoring white race), and marital status (favoring married). Differences in HbA1C persisted after adjusting for type of medication, site, and demographic covariates. There was no difference in Charlson comorbidity scores between groups in the substudy, but there were more diabetic complications in the strip+ group (15% more had at least one diabetic complication).

CONCLUSION: Glucose control was poorer in patients who received test strips. Differences persisted after controlling for important confounders. Thus, our study does not suggest benefit from home glucose monitoring in terms of improved glycemic control, although presence of more DM complications in the strip+ group suggests that strips may be used more often in difficult to control patients. Because home glucose monitoring is costly, further study should evaluate the cost-effectiveness and clinical benefit of its use.

RUN-IN PERIODS IN RANDOMIZED TRIALS AND COMMERCIAL SUPPORT. R.L. Goodman¹, A. Biviano¹, R. Press¹; ¹Columbia University, New York, NY

PURPOSE: The ultimate objective of the randomized clinical trial is the application of the results of the trial to the care of individual patients. Run-in periods in clinical trials are often

used to exclude subjects who are likely to be non-adherent with therapy or to experience side effects. While this approach may increase the chances of finding a treatment effect by eliminating non-compliers and subjects likely to experience adverse events, it diminishes the generalizability of the results of the trial, as patients are less likely to be as adherent and more likely to experience side effects. This is especially so if measures such as the "Number Needed to Treat" (NNT) are used, as the "true" NNT is likely to be higher than that obtained in the trial if an appreciable number of patients are excluded during the run-in period. As run-in periods would seem to increase the chance of finding a positive result, we hypothesized that there might be an association between the use of run-in periods in the design of randomized trials and the presence of commercial sponsorship, particularly by the pharmaceutical industry.

METHODS: We searched all randomized trials published in The New England Journal of Medicine, The Journal of the American Medical Association, The Lancet, and The BMJ, from January 1, 1998 through December 31, 2000. We included trials that had at least one arm consisting of either PO, dietary, intranasal, or subcutaneous therapy administered by the study subjects, of at least 4 weeks duration. We excluded studies of treatments not administered by subjects (e.g., intravenous or intramuscular treatments) and studies of treatment of acute events (e.g., myocardial infarction or COPD exacerbations), in which setting a run-in period would not be feasible. However, studies of treatments following acute events that did not require acute treatment were included. Trials of treatments during pregnancy were also excluded.

For the purpose of our study, we defined a run-in period as follows: Any period preceding randomization during which time subjects took active treatment, placebo, or a dietary intervention to which they might subsequently be randomized. Pre-randomization periods used to assess subjects' symptoms were not considered run-in periods, even if the authors of the studies designated them as such. This is because these screening or "baseline" periods serve a very different function from the above defined run-in periods. Drug "wash-out" periods were also not considered run-in periods in our analysis, for similar reasons.

Sponsorship was determined based on stated funding source at the conclusion of the article. We considered a study to be industry sponsored if at least one of the funding sources was from a commercial entity (e.g., a pharmaceutical company). We did not consider the provision of medication or placebo to constitute commercial support.

All studies were evaluated separately by two reviewers, and determination of the presence or absence, as well as type of run-in period was made while blinded to the source of funding. Likewise, ascertainment of funding source was made blinded to presence or absence of a run in period. Any disagreement regarding presence or type of run-in period was resolved by consensus among the three authors, while blinded to the funding source.

RESULTS: There were 214 eligible trials. Of these, 138, or 65%, were industry sponsored. Run-in periods as defined above were found in 39 of the trials. Twenty-eight of these were placebo run-in, 3 were placebo and diet, 2 were treatment, 2 were diet alone, 2 were treatment and placebo, and in 2 reports the type of run-in was not stated. In only 12, or 31%, of the 39 trials was the number of subjects excluded because of non-compliance or adverse events included in the result. Of the 39 trials with run-in periods, 36, or 92%, were commercially funded. The odds that a trial with a run-in period was commercially funded was 12:1, compared to 1.4:1 for trials without run-in periods (OR=8.6).

CONCLUSION: Run-in periods in clinical trials appear to serve a commercial, rather than a scientific, purpose. As their presence limits the generalizability of the results of these trials, we question their use in the design of randomized clinical trials done for scientific purposes.

EFFECTIVENESS OF CARDIOVASCULAR RISK REDUCTION IN PRIMARY CARE OF TYPE 2 DIABETES. R.W. Grant¹, D.E. Singer¹, E. Cagliero¹, D.M. Nathan¹, P. Murphy-sheehy¹, J.B. Meigs¹; ¹Massachusetts General Hospital, Boston, MA

PURPOSE: Cardiovascular disease (CVD) is the leading cause of mortality in type 2 diabetes (DM2). Despite demonstrated efficacy of medical therapy in clinical trials, effectiveness of CVD risk reduction in actual practice may be sub-optimal. Our goals were to: 1) determine patterns of hyperglycemia (HG), hypertension (HTN) and hyperlipidemia (HL) management among patients with DM2, and 2) assess whether effective HG management was associated with effective HTN and HL management.

METHODS: 601 confirmed DM2 patients attending our outpatient clinics between 3/96 – 8/97 were randomly selected; chart abstraction data were linked to laboratory testing results. We defined 3 components of effective CVD risk reduction: 1) risk marker testing for HbA1c, systolic blood pressure (SBP), and LDL cholesterol, 2) initiation of medical therapy if above goal, and 3) use of high dose therapy if above goal. To assess the relationship between HG, HTN, and HL management, we compared risk marker levels, testing frequency, and medication dose intensity, using linear and logistic regression.

RESULTS: Cohort characteristics included: mean age 64.7, 58.4% male, median DM2 duration 6 years, with 73% HTN and 50% HL prevalence. Sub-optimal management effectiveness was greatest for HL compared to HG or HTN. Fewer patients were tested for LDL than HbA1c (Fisher's exact test, $p<0.0001$), and among tested patients, significantly fewer were at LDL goal than HbA1c goal ($p=0.045$) [Table]. Levels of HbA1c, SBP, and LDL were not significantly correlated after adjusting for BMI. Achieving goal HbA1c did not predict meeting either SBP or LDL goals. There were no significant correlations between HG medication dose intensity (oral HG agent or insulin U/Kg dose) and either HMG CoA reductase (statin) dose or number of HTN medications.

Risk Factor (goal)	Tested within 18 months	Percentage of Tested Patients:		
		Above Goal	On Rx if Above Goal	On High Dose Rx if > Goal (definition of high dose)
HbA1c ($<7.5\%$)	92%	53%	95%	65% (on maximum oral agent dose or on insulin)
SEP (<130 mm Hg)	99%	58%	78%	30% (>1 HTN medication)
LDL (<130 mg%)	66%	60%	36%	7% (on high dose statin)

CONCLUSION: Management of HL and HTN was suboptimal compared to HG. In particular, HL was least effectively managed despite proven efficacy and convenience of statin therapy. Effective management of HG did not correlate with effective HL or HTN management. In DM2, more effective CVD risk reduction requires increased emphasis on reducing HL and HTN.

FACTORS INFLUENCING PHYSICIAN USE OF ANTICOAGULATION IN ATRIAL FIBRILLATION: A NATIONAL SURVEY. C.P. Gross¹, E.W. Vogel², A.J. Dhond¹, C.B. Marple³, R. Edwards⁴, O. Hauch³, E.A. Demers², M. Ezekowitz²; ¹Yale University School of Medicine, New Haven, CT; ²MCP Hahnemann School of Medicine, Philadelphia, PA; ³Astra Zeneca LP, Wayne, PA; ⁴Health Services Consulting Corp., Santa Fe, NM

PURPOSE: To explore factors associated with physicians' decisions to use warfarin in patients with non-valvular atrial fibrillation (NVAF).

METHODS: We mailed a self-administered questionnaire to a random sample of general internists, selected from the AMA Masterfile. The instrument included questions on physicians' demographic characteristics, prior experiences and beliefs about the natural history and treatment outcomes for patients with NVAF. Case scenarios described patients with varying degrees of risk of thromboembolism or hemorrhage. Respondents who were in the upper quartile of warfarin use, as defined by the proportion of cases in which they recommended warfarin, were "high-users". Results were summarized as proportions; chi-square was used for comparisons with a significance level of 0.05.

RESULTS: Surveys were returned by 120 of 427 eligible respondents. The mean age was 46 years (79% male; 38% academic). The median perceived risk for TIA and ischemic stroke (CVA) in a NVAF patient who was 76 year old (otherwise healthy, not receiving therapy) were 10% (Interquartile Range - (IQR): 5 - 20%) and 7.5% (IQR: 5 - 16%) per year, respectively. For NVAF patients on warfarin, the median risk estimates for TIA and ischemic stroke were 2% for TIA (IQR: 1 - 5%) and 2% for CVA (IQR: 1 - 5%); the median annual risk estimate of intracranial hemorrhage (ICH) for these patients was 3% (IQR: 1 - 5%). Respondents indicated they would recommend warfarin for a median of 9 of the 14 cases (IQR: 7 - 11 cases). Physician age, gender, perceived risk of ischemic stroke and estimated stroke risk reduction attributable to warfarin were not associated with warfarin use. However, respondents whose estimated risk of warfarin-associated ICH was in the highest quartile were significantly less likely to be high-users than those who provided lower estimates (6.9% vs. 30.8%; $p=0.01$). Use of warfarin was also inversely related to the belief that an ICH was a "worse outcome" than an ischemic stroke for a NVAF patient or a "more regrettable" outcome for their physician.

CONCLUSION: There is substantial variability in beliefs about the natural history of NVAF and the risks and benefits of warfarin. Warfarin use was impacted more by estimated risk of intracranial hemorrhage and by the perceived severity and "regretability" of this complication than by the potential benefits.

THE PATIENT RECRUITMENT PROCESS IN PUBLISHED RANDOMIZED CONTROLLED TRIALS: ARE TRIAL RESULTS RELEVANT TO OUR PATIENTS? C. Gross¹, A. Heiat¹, R. Mallory¹, H. Krumholz¹; ¹Yale University School of Medicine, New Haven, CT

PURPOSE: A common criticism of clinical trials is the enrolled patients have passed through a highly selective enrollment "filter" and may not represent patients in the general population. However, little is known about the recruitment and evaluation of potential subjects in published trials. We therefore performed a review of published randomized controlled trials (RCT's) to assess the rigor with which enrollment experience was reported, and to analyze the available data.

METHODS: We selected all RCT's that were published in the Annals of Internal Medicine, The New England Journal of Medicine, Lancet, or JAMA between April 1, 1999 and March 31, 2000. From each article, we abstracted the number of patients that were screened by the investigators to determine if they were eligible, the number who were eligible, and the number who were enrolled. Results were summarized as proportions; chi-square tests were used to compare groups with a significance level of 0.05.

RESULTS: A total of 172 RCT's were reviewed; the median number of participants was 260 (range 18 to 54,654). Of these, 90 (52%) papers included an estimate of the number of patients that were screened by the investigators for eligibility. The number of patients who were eligible for participation was reported in 74 (43%) studies. Multicenter trials were significantly less likely to report the number of eligible patients (33.9%) than single center trials (58.7%; $p=0.002$). Only 54 (31.4%) papers included information about both the number of patients screened and the number eligible. In these studies, the median proportion of screened patients who were eligible for participation was 58.8% (range 1.4% - 100%); and the median proportion of eligible patients who enrolled was 91.8% (range 35.1 - 100%).

CONCLUSION: Many RCT's published in major medical journals provided incomplete information about their patient recruitment experience. In the papers that did include this information, there was marked variability in the proportion of patients who were eligible and the proportion that enrolled. Clinician-scientists should be encouraged to collect and report this information in order to allow readers to consider the external validity of their work.

IS DISEASE-SPECIFIC MORTALITY A VALID ENDPOINT FOR CANCER SCREENING? D.A. Haggstrom¹, H.G. Welch², W. Black¹; ¹Dartmouth-Hitchcock Medical Center, Lebanon, NH; ²Dartmouth College, White River Junct, VT

PURPOSE: Disease-specific mortality (DSM) is the standard outcome measure used in randomized trials of cancer screening. However, the determination of cause of death is often problematic. Because there is little ambiguity about when death itself occurs, we reviewed the published randomized trials using all-cause mortality (ACM), in addition to DSM.

METHODS: We derived ACM for the major randomized trials of cancer screening: mammography (n=8), fecal occult blood testing (FOBT) (n=3), and chest radiography (n=3) from the available data. For each study, we compared the relative risk of death in the screened group versus the control group (RR) based on DSM to the RR based on ACM. For each study,

we considered the RR's to be discrepant when they were not both less than or greater than 1.0. To provide some perspective on the disease burden, we also calculated the ratio of DSM to ACM in the control groups for each cancer type.

RESULTS: The RR's were discrepant in 5 of 14 trials. See table for details of RR and disease burden for each screening test. In addition, in one of the trials in which the RR's were both less than 1.0 (Edinburgh mammography trial), the observed difference in ACM was far greater than could be expected on the basis of decreased breast cancer mortality (5 times larger than the observed difference in DSM).

CONCLUSION: ACM should be considered in randomized trials of cancer screening because it helps to identify threats to validity (such as the randomization flaws in the Edinburgh trial) and because it provides needed perspective on the magnitude of the benefit of screening. RR's based on ACM and DSM are often discrepant, raising questions about the validity of DSM and the potential for unrecognized harm from screening.

DSM & ACM Statistics

	Mammography	FOBT	CXR
RR DSM	.64 - 1.36	.67 - .86	1.11 - 1.36
RR ACM	.800 - 1.06	.986 - 1.00	.964 - 1.16
DSM-ACM discrepancy	2 of 8	2 of 3	1 of 3
DSM/ACM	7.7%	10.8%	10.1%

IS VOLUME RELATED TO OUTCOMES IN HEALTH CARE? A SYSTEMATIC REVIEW AND METHODOLOGIC CRITIQUE OF THE LITERATURE. E.A. Halm¹, C. Lee², M.R. Chassin¹; ¹Dept of Health Policy, Mount Sinai School of Medicine, New York, NY; ²Dept of Surgery, New York-Presbyterian Medical Center, New York, NY

PURPOSE: Many studies report that higher volume is associated with better health outcomes for certain procedures and clinical conditions; others do not. Purchaser and advocacy groups have recommended that patients be selectively referred to high volume hospitals and physicians. The wisdom of this strategy depends on the strength, magnitude, and meaning of the association between volume and outcome. We systematically reviewed this literature focusing on the methodological rigor and results of these studies.

METHODS: MEDLINE was searched from 1980 to 2000 for English-language, population-based studies examining the independent relationship between hospital or physician volume and clinical outcomes using numerous MESH term and key word algorithms. Bibliographies were reviewed to identify other articles. Experts were contacted about missing or unpublished studies. For each study, we calculated the absolute difference in mortality rate between the highest and lowest volume strata reported and the number needed to treat (NNT) to prevent one death attributable to low volume. Two investigators independently abstracted each article using a standard form.

RESULTS: Of 257 studies reviewed, 128 met inclusion criteria covering 27 procedures and clinical conditions. The methodological rigor of most studies was modest. Few studies used clinical data for risk-adjustment or examined hospital and physician volume effects simultaneously. Overall, 70% of all hospital volume and 74% of physician volume studies reported a significant association between higher volume and better outcomes. The strongest associations were for surgery on pancreatic cancer, esophageal cancer, abdominal aneurysm, and pediatric cardiac problems, and treatment of AIDS (median, 8 to 14 excess deaths per 100 cases attributed to low volume; NNT, 7 to 12). Although statistically significant, the magnitude of the volume-outcome relationships for CABG, PTCA, MI, carotid endarterectomy, other cancer surgery, and orthopedic procedures was much smaller (median, 0.2 to 3 deaths per 100; NNT, 35 to 500).

CONCLUSION: High volume is associated with better outcomes across a wide range of procedures and conditions, but the magnitude of the association varies greatly. The clinical and policy significance of this finding is complicated by many methodological shortcomings. Differences in processes of care between high and low volume providers may explain much of the relationship between volume and outcome.

IMPROVING HEADACHE CARE: IMPACT OF A DISEASE MANAGEMENT PROGRAM. L.H. Harpole¹, A.L. Bernstein², G.P. Samsa¹, D.B. Matchar¹; ¹Duke University Medical Center, Durham, NC; ²Kaiser Permanente, Santa Rosa, CA

PURPOSE: Chronic headache is a cause of significant morbidity, increased health care costs, and lost productivity. Despite substantial evidence about effective therapies, considerable barriers to providing optimal care exist: failure to establish the correct headache diagnosis, unfamiliarity with treatment options, and inadequate patient education. The purpose of this study was to design, implement and evaluate an evidence-based headache management program (HMP) that provided standardized evaluation, education and treatment to patients referred with chronic headache.

METHODS: We convened an expert panel to review current headache guidelines and to define the necessary components of a HMP. A HMP was then established at a staff model health care maintenance organization (HMO). Patient eligibility included: age >18, migraine disability assessment questionnaire (MIDAS) score >5, not seeing neurologist/headache specialist for headache care, and enrollment in the HMO for at least six months. The initial HMP visit included patient assessment, education and treatment planning. Baseline data collection included the MIDAS, SF-36, headache worry and pain (validated 10 point scales), patient satisfaction with prior headache care, and emergency room visit rates. Follow-up data were collected at 3 and 6 months.

RESULTS: Between March and December 1999, 54 of 155 referred patients were eligible and enrolled in the HMP. All patients completed the initial comprehensive visit; 75% attended one or more follow-up appointments. At baseline, enrolled patients (mean age 40, 80% female) were severely disabled by their headaches, dissatisfied with their headache care, and expressed

considerable difficulty managing their headaches. Follow-up data at 3 months demonstrated significant improvement in headache-related disability and pain: MIDAS score (40 to 25, $p < 0.0001$), days with headache over preceding 3 month period (34 to 20 days, $p < 0.0001$), and average headache pain (7.2 to 5.8, $p < 0.005$), with results constant at 6 months. At six months, patients experienced significant improvement in multiple SF-36 subscales, including role-physical, pain, vitality, social functioning, role-emotional and mental health (all $p < 0.006$), as well as improvement in the mental health summary scale score ($p < 0.002$). Patients became less worried about their headaches (6.8 to 5.4, $p < 0.005$) and a greater proportion were satisfied with their headache care (47% to 88%, $p < 0.0001$). Significantly less patients reported problems with their headache management (91% to 46%, $p < 0.0001$). Visit rates to the emergency room for headache care declined (0.4 to 0.2, $p < 0.04$).

CONCLUSION: Patients with chronic headache who enrolled in a HMP experienced significant improvement in headache related disability, pain, functional health status, and patient satisfaction, and utilized the emergency room for headache care at a lower rate. These results should be validated in a randomized controlled trial to assess the potential role for HMPs in managing problematic headache patients.

THE INFLUENCE OF DISCOUNT DRUG PRICING ON THE COST-EFFECTIVENESS OF CHOLESTEROL LOWERING FOR PRIMARY PREVENTION OF CORONARY ARTERY DISEASE. M. Hard¹, D.H. Hickam², S. Saha³; ¹Portland VA Medical Center, Portland, OR; ²None Given, Portland, OR; ³Oregon Health Sciences University, Portland, OR

BACKGROUND: Recent large clinical trials of healthy persons with elevated cholesterol levels have shown that cholesterol lowering with HMG Co-A reductase inhibitors (statins) modestly reduces the incidence of coronary artery disease (CAD). However, because of the high cost of statins, their use in the primary prevention setting has been controversial.

PURPOSE: To examine the impact of discount drug pricing on the results of published cost-effectiveness analyses (CEAs) of statins for primary prevention of CAD, for the purpose of developing a drug utilization policy for a single health care institution.

METHODS: We searched MEDLINE to identify all studies that provided cost and effectiveness data for primary prevention of CAD using statins. The articles were evaluated for quality using criteria published by the Panel on Cost Effectiveness in Health and Medicine. We determined the proportion of total costs within cost-effectiveness models that was attributable to direct costs for drugs. We then determined the costs of statins at our own institution and used these to recalculate cost-effectiveness for all patients in different clinical subgroups.

RESULTS: The MEDLINE search produced 7 CEAs that provided sufficient data on costs. Only one of these (Ann Intern Med 2000; 132(10):769) met all important quality criteria. In that study, 90% of total costs were attributable to drug costs alone, and the cost-effectiveness ratios were $> \$50,000/\text{QALY}$ for all 240 sub-groups analyzed. The statin dosage range used in that study is achieved at our institution by using half tablets of currently available preparations, and the actual VA costs for statins were 14% of those drug costs assumed in this study. When the VA costs were applied to this analysis, the cost-effectiveness of using statins for primary prevention was $< \$50,000/\text{QALY}$ for the majority of patient subgroups.

CONCLUSION: In at least one large health care system, statins are available at acquisition costs considerably lower than assumed in published CEAs. These acquisition costs permit statin use for primary prevention of CAD to have an acceptable cost-effectiveness ratio. If these drug prices were widely available to consumers, this important category of medication would have great potential for improving the nation's health at reasonable cost. Mechanisms for creating a uniform pricing structure for these widely used medications should be explored.

MEDICAID MANAGED CARE: ARE ACADEMIC MEDICAL CENTERS PENALIZED BY ATTRACTING HIGH-COST PATIENTS? M. Heisler¹, S.M. Demmoner¹, J.E. Billi¹, R.A. Hayward¹; ¹University of Michigan, Ann Arbor, MI

PURPOSE: Academic medical centers (AMCs) have a long history of providing medical care for ill, publicly insured patients on a primary care and referral basis. Under capitation, however, attracting and caring for a disproportionate number of chronically ill patients may result in serious financial liabilities. This is an especially salient issue with the growth of Medicaid Managed Care (MMC). This study's purpose is to determine whether there are case-mix and health utilization disparities between Medicaid enrollees within a managed care organization (MCO) in Michigan who selected providers affiliated with a major AMC and enrollees who selected community providers.

METHODS: A retrospective cohort study using cost estimates obtained from claims data and the Medicaid fee schedule. The cohort included all capitated Medicaid enrollees in the study MCO from January 1, 1997 through October 31, 1999. The prevalence of 25 high-cost chronic medical conditions were identified from the utilization records. T-tests and linear regressions were used to assess differences in total health care cost estimates per member for Medicaid enrollees at AMC primary care sites vs. other community sites, including analyses stratified for the two main Medicaid programs: Temporary Assistance for Needy Families (TANF) and Aid to Blind and Disabled Program (ABAD).

RESULTS: Enrollees with AMC providers had a much higher cumulative prevalence of the 25 high-cost chronic medical conditions (95.6 per 1000 enrollees vs. 65.6 per 1000; $p < .001$). Estimated total costs were higher for Medicaid enrollees with AMC PCPs than for enrollees with other PCPs. Average annual estimated costs per member (excluding pharmacy costs) at AMC sites were \$358 higher than those at the other sites (\$1277 vs. \$919, $p < .001$). Controlling for age, gender and Medicaid program did not substantially affect these cost estimate differences between sites. Almost 70% of the observed cost differences occurred in the 20% of enrollees who were in the ABAD program. The average estimated costs per ABAD member were \$1,245 higher per member per year at the AMC sites, averaging a total estimated cost disparity of \$974,000 a year for all ABAD patients. Most of this cost difference was due to costs of inpatient hospitalizations, but separate analyses of outpatient pharmacy costs also showed significantly higher costs for ABAD enrollees with University PCPs ($p < .001$). Comparisons of the estimated costs with 1998 capitation payments to the AMC showed that over the three year study period the University sites received about \$2.8 million less than they would have received under

Medicaid's Fee-For-Service (FFS) payment system and that most of this was accounted for by the ABAD program comprised of less than 1,000 enrollees.

CONCLUSION: This study found both significant case-mix and cost disparities suggesting that AMC primary care sites experienced substantial adverse selection. Unless approaches to account for adverse selection are put in place, this phenomenon could jeopardize Medicaid recipients' ongoing access to needed medical care. Current efforts to implement risk adjustment strategies are thus encouraging and deserve careful evaluation.

VARIATIONS IN HEALTHCARE USE AND PATIENT COMPLEXITY IN RESIDENTS' CONTINUITY CLINIC PANELS. D. Helmer¹, H. Nelson², C.M. Ashton¹; ¹Baylor College of Medicine, Houston, Texas; ²Houston Center for Quality of Care and Utilization Studies, Houston, TX

PURPOSE: There has been no report of the variation in clinical characteristics among patient panels seen by internal medicine residents in their continuity-clinics in the Veterans Health Affairs (VHA) system. This study describes the variability among resident continuity-clinic patient panels using various demographic, utilization, cost, diagnosis, and clinical complexity measures.

METHODS: Forty internal medicine resident panels were identified from a single VHA site as of June 30, 2000. The analysis period was July 1, 1999, to June 30, 2000. Patient identifiers were generated from the local administrative computer system and linked to a summary database used to profile primary care staff providers. The following variables were summarized for each resident panel and for the residents' patients as a group: the five most common diagnoses, age, race, gender, bed days of care (BDOC), number of encounters, laboratory cost, prescription cost, and clinical complexity as measured by Starfield 6 category and number of Ambulatory Diagnostic Groups (ADG's).

RESULTS: There were 2,143 patients assigned to the 40 resident panels after excluding duplicated patients and those with missing records. The five most common diagnoses in resident continuity clinic patients in general were benign essential hypertension (ICD9=401.1), coronary atherosclerosis (414.0), diabetes mellitus type II (250.0), hypercholesterolemia (272.0), and chronic airway obstruction (496). These diagnoses represented the top five diagnoses in 20% of resident panels. An additional 60% of resident panels had four of these diagnoses represented in their top five diagnoses. The mean age of the patients was 62.1 years, racial distribution was 32.1% black, 49.5% white, and 6.1% Hispanic and 98.3% of patients were male. The clinical complexity of patients did not vary significantly among resident panels by mean number of ADG's (mean 6.42, ANOVA, $F(39, 2086) = 1.14$, $p = 0.2528$) or Starfield 6 category (Chi-square test, $p = 0.36$). The presence of significant variation among the panels by oneway ANOVA of log transformed values was detected in number of encounters ($F(39, 2086) = 2.07$, $p = 0.0001$), lab costs ($F(39, 2086)$, $p = 0.0004$), and prescription costs ($F(39, 2086)$, $p = 0.0212$), but not in BDOC ($F(39, 2086)$, $p = 0.3193$).

CONCLUSION: There is no significant variation in clinical complexity among resident continuity-clinic panels as measured by number of ADG's or Starfield 6 category. However, there is significant variation among panels by utilization measures such as number of encounters, lab costs and prescription costs. Further investigation into the sources of this variation is recommended, as practice-pattern variation may have implications for resident training and patient care.

COMMUNICATING ABOUT LOW BACK PAIN: WHY ARE PATIENTS OF CHIROPRACTORS MORE SATISFIED? R. Hertzman-Miller¹, H. Morgenstern², E. Hurwitz³, P. Harber¹, T. Belin², G. Kominski², F. Yu², A. Adams³; ¹UCLA Medical Center, Los Angeles, CA; ²UCLA School of Public Health, Los Angeles, CA; ³Southern California University of Health Sciences, Whittier, CA

PURPOSE: Patients who see chiropractors (DCs) for their low back pain are more satisfied than patients who see medical doctors (MDs). Qualitative studies suggest that DCs give patients more information and more control over their treatment. Among primary care patients, information and control generally lead to greater satisfaction. We estimate the effects of provider type, explanation of treatment, and self-care advice on satisfaction in a randomized trial of DC vs. MD treatment for low back pain.

METHODS: 681 adult low back pain patients were randomly assigned to treatment by either a DC or a MD and were surveyed at baseline, 2, and 4 weeks; follow-up was 99.7%. We used ordinary least squares linear regression to estimate the effects of provider type, explanation, self-care advice, and other factors on satisfaction with provider. Explanation of treatment was reported as "yes" or "no." Self-care advice was the number of items of advice (0-10) the patient reported receiving. Satisfaction was measured on a 10-item scale (5 = low to 50 = high).

RESULTS: The mean satisfaction score was 36.1 for patients in the DC group and 30.6 in the MD group; the crude difference was 5.5 (95% confidence interval [CI] = 4.5, 6.5). Adjustment for sociodemographic and baseline illness characteristics alone did not substantially change these estimates. The mean number of items of self-care advice was 2.3 in the DC group and 1.6 in the MD group ($p < 0.001$). Sixty-one percent of patients in the DC group vs. 16% in the MD group received an explanation of treatment ($p < 0.001$). In a regression model that included provider type, explanation, self-care advice, improvement in disability, prior experience with and confidence in treatment, baseline sociodemographic and illness characteristics, and interaction terms, the DC-MD difference in satisfaction was 4.1 (95% CI = 2.6, 5.5) among patients who received no self-care advice and 2.5 (95% CI = 2.0, 3.0) among patients who received 2 items of self-care advice. On average, satisfaction increased 1.2 points (95% CI = 0.8, 1.6) per item of self-care advice for patients in the DC group and 1.9 (95% CI = 1.5, 2.3) per item for patients in the MD group. In both groups, explanation of treatment increased satisfaction by 3.3 points (95% CI = 2.4, 4.3). Improvement in disability, confidence in treatment, and visit duration all increased satisfaction but did not affect the DC-MD difference in satisfaction.

CONCLUSION: The greater satisfaction with low back pain care among patients of DCs is partly explained by explanation of treatment and self-care advice. These communication skills may account for much, though not all, of DCs' success in treating low back pain. To treat patients with low back pain better, MDs should consider improving their own communication skills.

QUALITY OF PHARMACOLOGIC CARE FOR VULNERABLE ELDERLY. T. Higashi¹, J. Avorn², E. Knight², D.H. Solomon¹, P.G. Shekelle³, N.S. Wenger¹; ¹University of California, Los Angeles, Los Angeles, CA; ²Harvard Medical School; ³RAND, Los Angeles, CA

PURPOSE: Drug therapy is a cornerstone of medical treatment for older persons, but has substantial risks. We evaluated the quality of pharmacologic care for community-dwelling vulnerable elders.

METHODS: The Assessing Care of Vulnerable Elders project developed quality indicators (QIs) of medication treatment for community-based persons > 65 years at increased risk of functional decline and death. Based on systematic review of literature, an expert panel rated 12 process measures as valid QIs of pharmacologic care that focused on avoidance of inappropriate medications, patient education, and monitoring response and toxicity. These were applied to a sample of inpatient and outpatient medical records of vulnerable elders to examine the quality of care they receive.

RESULTS: Of 3206 elders randomly selected from two managed care plans, 2278 (71%) were interviewed, 475 were identified as "vulnerable". For 400 who consented to participation and had abstractable medical records, 444 new medications were started for chronic conditions during a 13-month period; 259 patients were prescribed at least one new drug.

Ninety-seven percent of medications had a specific indication evident from the chart, but only 66% of medications were monitored for a therapeutic response within 6 months of initiation. Laboratory testing for medication follow-up was poor with only 9 of 23 newly prescribed ACE inhibitors and 7 of 23 new diuretics followed-up with a potassium and creatinine check within one month; 79% of patients taking diuretics had an annual electrolyte check.

Inappropriate medication prescription was uncommon: chlorpromazine, meperidine, barbiturates and strong anticholinergics were inappropriately prescribed in only 8 cases. Only 18% of new medication prescriptions were accompanied by any chart documentation of patient education.

CONCLUSION: Although vulnerable elders are at risk of adverse reactions to medications, monitoring for response and toxicity is lax. Documentation of medication education is rare, but may have occurred without recording.

DO PHYSICIANS EXAMINE PATIENTS UNDER CONTACT PRECAUTIONS LESS FREQUENTLY? L. Higgins¹, S. Saint², B. Nallamothu², C. Chenoweth²; ¹Hunter Holmes McGuire Veteran Affairs Medical Center, Richmond, VA; ²University of Michigan, Ann Arbor, MI

PURPOSE: Patients infected with multi-drug resistant bacteria are increasingly common in U.S. Hospitals. Standard of care dictates that these patients be placed under contact precautions, which requires hospital personnel to gown and glove prior to patient examination. While it is unclear whether placing these patients under contact precautions reduces the spread of multi-drug resistant bacteria in the hospital, contact precautions may actually impede the ability to examine patients due to the time and effort required to fully gown and glove. We therefore sought to assess whether hospitalized patients placed under contact precautions are examined less often by senior medical residents and attending physicians during morning rounds compared to those not under contact precautions.

METHODS: We conducted a prospective cohort study on inpatient medical services at the University of Michigan Hospitals and Ann Arbor Veteran Affairs Medical Center. We directly observed senior medical residents and attending physicians during morning rounds and recorded the contact precaution status of the patient and whether they were examined by either physician. We defined "examined" as physician examination of at least one organ system. Chi-square tests were used to test for significant differences in examination rates between the 2 groups.

RESULTS: Of a total of 136 patients, 31(23%) were under contact precautions. Senior medical residents examined 87% of patients not under contact precautions versus 84% of those under contact precautions (relative risk [RR], 0.96; 95% confidence interval [CI], 0.81 – 1.14; p=0.65). In comparison, attending physicians examined 72% of patients not under contact precautions versus 37% of those under contact precautions (RR, 0.51; 95% CI, 0.31 – 0.82; p=.0001).

CONCLUSION: Attending physicians are half as likely to examine patients under contact precautions when compared to patients not under contact precautions. Because patients under contact precautions are often the most ill in the hospital, the use of contact precautions may be potentially harmful by decreasing the likelihood that an attending physician will actually examine a patient during morning rounds.

EXPERIENCES OF PHYSICIANS WHO FREQUENTLY USE E-MAIL WITH PATIENTS. I. Houston¹, D.Z. Sands², D.E. Ford¹, M. Patt¹, B. Nash³; ¹Johns Hopkins University, Baltimore, MD; ²Beth Israel Deaconess Medical Center, Boston, MA; ³Physicians Online (formerly), Scarsdale, NY

PURPOSE: Although patient demand is high, physicians have not widely adopted use of e-mail with patients. Our purpose was to explore the experiences of physicians frequently using e-mail with their patients, as a window to the future.

METHODS: To identify the rare "vanguard" physicians who frequently use e-mail with their patients, we recruited 11,859 physicians using Physicians' Online, an Internet professional information portal, to participate in a survey. To identify frequent users of e-mail, we asked "In a typical day, please estimate the average number of e-mails you and your immediate staff receive from patients who are currently in your care." Additional questions assessed common clinical topics in e-mail, physician's adherence to published guidelines from the AMA and American Medical Informatics Association, and physician satisfaction, measured by "Would you recommend to a colleague use of e-mail with patients."

RESULTS: Among the 1,325 physicians responding (including individuals from all 50 states), we identified 204 frequent users who received one or more e-mails from patients daily. The mean age was 49 years, 82% were male, and 35% were primary care providers. Participants were from a variety of specialties including adult medical subspecialties(9%), pediatrics(11%), surgery(11%), and psychiatry(7%). Common clinical topics, received frequently or sometimes by over 40% of physicians, include: non-urgent new symptoms, questions about lab results, and questions about information on the Internet. Although infrequent, 7% of our sample did report frequently or sometimes receiving e-mails about urgent issues such as chest pain. The most

common practice adherent with published guidelines was including the e-mail in the medical record (38%), but they rarely incorporated other guideline issues such as educating patients about appropriate topics (10%).

Despite daily use of e-mail with patients, 25% of the 204 were not satisfied. Physicians who were not satisfied more frequently (80%) reported "patient request" as the most important reason for using e-mail, compared with those who were satisfied (39%). (p<0.01). Compared with dissatisfied physicians, physicians who were satisfied with e-mailing patients more frequently noted a decrease in their amount of telephone medicine (29% versus 2%; p<0.001), and were more likely to use the emails to educate their patients (25% versus 14% p<0.01).

CONCLUSION: Experiences of these "vanguard" physicians were mixed. Although patient demand is high, physician satisfaction of e-mail with patients may also be important to sustained use of e-mail with patients. Our data suggest that increased dissemination/adherence of published guidelines and increased perception of the efficiency of e-mail communication may improve acceptance.

CONTINUITY AND SATISFACTION IN MANAGED CARE. J. Hsu¹, J. Schmitt¹, B.H. Fireman¹, J.V. Selby¹; ¹Kaiser Permanente Division of Research, Oakland, CA

PURPOSE: Continuity of care is a highly desirable characteristic that may be threatened by recent changes in the health care system. We sought to assess patients' perceptions of continuity of care and the relationship between continuity and satisfaction in a managed care organization.

METHODS: Within a randomized controlled trial of a patient-provider matching intervention in a group model health maintenance organization (HMO), we evaluated whether subjects reported usually seeing their own primary care provider (PCP), and the association of this perception with satisfaction and trust. Between October 1999 and September 2000, 2501 (64%) subjects completed a mailed, self-administered questionnaire, one year after receiving a new PCP.

RESULTS: Among patients who had seen their PCP at least once during the study period, the majority (76%) reported usually seeing their own PCP during their visits. These subjects tended to be older (58 v. 52 yo, p<0.0001), white (75% v. 69%, p<0.02), and had more visits during the past year to their PCP (3.6 v. 3.2 visits, p<0.04) than subjects who reported usually seeing another provider, i.e. lacking continuity of care. Moreover, compared to subjects without continuity, subjects perceiving that they had continuity reported greater satisfaction with their PCP and the health system. For example, subjects reporting that they usually saw their PCP had significantly higher levels of satisfaction with their PCP (72% v. 38% reporting excellent or very good overall satisfaction, p<0.0001). These subjects were more likely to recommend their PCP to others (82% v. 52%, p<0.0001), reported greater trust in their PCP (p<0.0001), and in the health system (p<0.0001). In addition, these subjects also were less likely to perceive that the PCP created barriers to access, such as with specialist referrals (p<0.0001).

CONCLUSION: In short, the majority of subjects in this study perceived that they usually saw their own PCP during visits. Furthermore, the perception of continuity of care with one's PCP was associated strongly to satisfaction and trust in the PCP, and also in the health system. Further research is needed to assess the direction of this association, as well as to identify potential areas for organizational improvements.

PHYSICIAN ATTITUDES AND PRACTICE PATTERNS IN PROSTATE CANCER SCREENING. L. Hu¹, A. Geller¹, M.A. Moskowitz², A.M. Harrington¹, K.M. Freund¹; ¹Boston University, Boston, MA; ²Boston Medical Center, Boston, MA

PURPOSE: Screening for prostate cancer remains controversial in the medical community. In the setting of conflicting screening guidelines, this study aims to understand the decision-making process of primary care physicians when counseling patients on prostate cancer screening.

METHODS: We surveyed 398 primary care physicians in Massachusetts, over-sampling providers who care for African-American men. The survey consisted of 52 questions, divided to assess physicians' knowledge, attitudes and practice patterns surrounding prostate cancer screening. The survey incorporated one of four versions of a clinical vignette, varying patient age (55 vs. 79 years) and race (black vs. white) as a way of unobtrusively identifying patient factors that physicians use in their decision making. Knowledge of prostate cancer was scored on a 100 point scale. Attitude questions were scored on a 100 point scale, with 0 being complete disagreement and 100 being total agreement.

RESULTS: The response rate was 82%. Forty-eight percent of all physicians reported that they would recommend prostate cancer screening for the patient in the vignette. Physicians were more likely to recommend prostate cancer screening when the vignette patient was black as opposed to white (52% vs. 41%, p=.04) and younger as opposed to older (64% vs. 28%, p=.01). Physicians who recommended screening the vignette patient scored lower on the knowledge questions than physicians who did not (73.8% in screeners vs. 77.9% in non-screeners, p=.002). Physicians who recommended screening agreed more strongly than non-screeners with the statement that "screening reduces mortality" (68 in screeners vs. 40 in non-screeners, p<.001). Physicians who did not recommend screening agreed more strongly than screeners with the statements "screening may result in unnecessary treatment" (76 vs. 63, p<.001) and "screening for prostate cancer results in more harm than good" (53 vs. 39, p<.001). **CONCLUSION:** Physicians incorporate patients' age and race when making decisions on prostate screening recommendations. Physicians' level of prostate cancer knowledge correlates inversely with screening rate. An increase in knowledge leads to a decrease in screening. Physicians who chose not to recommend screening for their patients appeared to do so out of the concern that prostate cancer screening does not necessarily decrease mortality and may actually cause more harm than good to their patients.

THE COSTS OF CARE OF TYPE 2 DIABETES MELLITUS PROVIDED IN A DIABETES CLINIC VERSUS A GENERAL MEDICINE CLINIC. E. Huang¹, S. Gleason¹, R. Gaudette¹, S. Gazelle¹, E. Cagliero¹, P. Murphy-sheehy¹, D.M. Nathan¹, D.E. Singer², J.B. Meigs²; ¹Massachusetts General Hospital, Boston, MA; ²Partners Health Care, Boston, MA

PURPOSE: The organization of preventive services for patients with chronic diseases may have implications for costs of care. At our academic medical center, patients with type 2 diabetes

mellitus receive care in a diabetes-specific clinic or in a general internal medicine clinic. The goal of this study was to characterize the impact of clinic membership on the costs of care.

METHODS: Type 2 diabetes patients attending the Diabetes Clinic (DC) or the Adult Medicine Clinic (AMC) between 3/96 and 8/97 were identified by ICD-9 diagnoses. Charts were randomly selected and reviewed (N=601). Outpatient costs for the period of chart review were based on the average costs of outpatient medications, visits, referrals, and tests. Inpatient costs were determined for the period 8/97 to 8/99. We analyzed the impact of clinic membership on 1) outpatient costs, 2) the probability of hospitalization, 3) the average costs of a hospitalization, 4) the average cumulative costs of all hospitalizations per patient. We analyzed crude outcome differences using Wilcoxon rank-sum or chi-square tests. To assess the independent effect of clinic membership, we conducted multivariable regression analyses, adjusting for disease severity and clinic case-mix.

RESULTS: The patients in the DC were younger (63 years of age vs. 65 year of age, $p < 0.01$) than AMC patients but had a longer mean duration of diabetes (12 years vs. 6 years, $p < 0.01$) and more baseline microvascular complications (65% vs. 44%, $p < 0.01$). DC patients received more intensive outpatient care (including HbA1c testing, eye and foot exams, and insulin therapy). Accordingly, in unadjusted analysis of outpatient costs, DC patients received more expensive care than AMC patients (\$5156 vs \$4555 per 1.4 years, $p = 0.02$). On crude analysis of 2-year hospitalization data, DC and AMC patients had no statistically significant differences in diabetes-related admissions (27% vs. 24%), mean costs of hospitalizations (\$10,393 vs \$9,617), and total cost of hospitalizations (\$21,053 vs. \$14,933). In multivariable models, we continued to find no significant difference between DC and AMC patients' hospitalization outcomes, although membership in the DC tended to be associated with a lower likelihood of hospital admission (OR 0.37, 95%CI 0.29–4.61), lower average hospital costs (13.5% lower, $p = 0.32$), and lower cumulative hospital costs (13.4% lower, $p = 0.33$).

CONCLUSION: DC patients had more severe diabetes and more costly outpatient care than AMC patients. However, we found no significant difference in the likelihood of hospitalization, average cost of hospitalization, and average cost of total hospitalization between DC and AMC patients. Further study is needed to establish the difference in overall clinic costs. Diabetes-specific clinics may have features which are cost saving and may be incorporated into primary care generalist settings.

OVERCOMING LANGUAGE BARRIERS TO ACCESS TO MEDICAL CARE: THE COSTS AND BENEFITS OF INTERPRETER SERVICES. E. Jacobs¹, J. Suaya², E. Stone³, D. Shepard²; ¹Cook County Hospital, Chicago, IL; ²Brandeis University, Waltham, MA; ³Division of Medical Assistance, Boston, MA

PURPOSE: Many US residents face language barriers that reduce access to medical care. Despite a growing focus on this problem many health care organizations provide inadequate interpreter services. A principal reason is the concern that uncertain benefits do not justify the costs of adequate services. The objective of this study was to assess the impact of an interpreter service program on the utilization and cost of health care services at a staff model HMO.

METHODS: We conducted a 2-year cohort study of continuously enrolled adult members of a staff model HMO where new comprehensive interpreter services for Spanish and Portuguese-speaking ambulatory patients were implemented in year 2 of the study. Two groups were studied: an interpreter service group (ISG, n=380) consisting of members who used the new interpreter services and a comparison group (CG, n=4119) consisting of a 10% random sample of all other members who received ambulatory care in year 2. We abstracted demographic information and utilization of primary health care services (preventive services and outpatient services) and hospital-based services (ED visits and hospitalizations) from the HMO's administrative database. We calculated the unit cost per interpretation based on the cost of the services (salaries, fringe benefits, supervision, and overhead) and reported volume of use of services. We calculated induced costs of interpreter services by multiplying the change in utilization for each health care service by its 1997 fee-for-service reimbursement rate from the Massachusetts' Division of Medical Assistance.

RESULTS: Utilization of primary health care increased in both groups after implementation of interpreter services. The changes (yr2-yr1) in utilization of preventive services ($p < 0.05$), utilization of office visits ($p < 0.01$), prescriptions filled ($p < 0.01$) and prescriptions written ($p < 0.01$) were significantly greater in the ISG compared to the CG. Utilization of hospital-based services remained the same for both groups, except for a reduction in ED use by the ISG. The change in rate of ED use (yr2-yr1) was not significant when compared to the CG. The net cost of providing interpreter services in 1997 prices was \$79/per interpretation $\times$ 2.95 visits per person/per year in the ISG. The net induced cost of the increased care delivered to the ISG compared to the CG was \$22 per person per year. The overall cost impact was an increase of \$256 per person per year in the ISG or \$2.41 per HMO member per year.

CONCLUSION: We found that professional interpreter services improve the delivery of health care to limited English proficient patients, increasing the delivery of preventive and all primary care and possibly reducing the use of ED services. The net first-year cost was \$256 per person using the interpreter services or \$2.41 per HMO member per year.

RELATIONSHIPS BETWEEN PATIENT SATISFACTION AND SEVERITY-ADJUSTED HOSPITAL MORTALITY. C.K. Jaipaul¹, G.E. Rosenthal¹; ¹University of Iowa & Iowa City VAMC, Iowa City, IA

PURPOSE: Patient satisfaction is increasingly being used to benchmark hospital performance. However, relationships between patient satisfaction and other performance measures are poorly studied. The goal of this study was to examine the relationship between patient satisfaction and severity-adjusted mortality rates.

METHODS: The current study used data from a regional initiative to measure performance in 31 hospitals in Northeast Ohio during 1991–1995. Satisfaction was assessed using the Patient Judgement System, a 41 item survey mailed to patients after discharge. Responses were obtained from 40,344 medical patients (response rate 48%). We utilized 5 scales (physician care, nursing care, coordination of care, information provided, and discharge instructions) and 1 single item assessment (overall quality). For each hospital, mean scores were determined, adjusting for self-

reported health status, age, gender, education and other demographic factors. Severity-adjusted mortality was determined for patients with 6 medical diagnoses. For each hospital, observed mortality was compared to predicted mortality derived from validated disease-specific multivariable models that were based on data abstracted from medical records. Comparisons of observed and predicted mortality rates were summarized using the z-statistic.

RESULTS: Mean satisfaction scores for the 5 scales and 1 single-item indicator ranged from 68.0 (information provided) to 72.6 (physician care). Z-statistics for mortality rates ranged from -7.6 to 4.6. In analyses including data for all 5 years (1991–95), patient satisfaction scores, at a hospital level, were inversely correlated with mortality for each of the 6 measures of satisfaction. Correlations were of moderate strength, ranging from -0.31 to -0.43. Results were generally consistent in analyses of individual years. The table below shows correlations based on data for all 5 years (top row) and range of correlations for individual years (bottom row).

Physician Care	Nursing Care	Coordination of Care	Information Provided	Discharge Instructions	Overall Quality of Care
-03.1	-0.35	-0.41	-0.43	-0.39	-0.43
(p=.09)	(p=.06)	(p=.02)	(p=.02)	(p=.03)	(p=.02)
-0.12 to -0.48	-0.16 to -0.44	-0.18 to -0.48	-0.16 to -0.57	-0.20 to -0.57	-0.23 to -0.52

CONCLUSION: Hospitals with higher patient satisfaction scores tended to have lower severity-adjusted mortality rates. Correlations were moderate in magnitude and consistent across individual years. These findings support the validity of patient satisfaction as a measure of hospital performance.

ARE OLDER PATIENTS MORE SATISFIED WITH THEIR CARE THAN YOUNGER PATIENTS? C.K. Jaipaul¹, G.E. Rosenthal¹; ¹University of Iowa & Iowa City VAMC, Iowa City, IA

PURPOSE: Previous studies have found that patient satisfaction increases with age. It is hypothesized that this may reflect lower expectations of older patients who are more accustomed to being ill. The goal of this study was to re-examine the relationship between age and satisfaction and account for differences in self-reported health.

METHODS: We conducted a retrospective cohort study of patients from 31 hospitals in a large metropolitan region in 1990–95. Satisfaction was assessed with the Patient Judgement System, a 41 item survey mailed to patients after discharge. Analyses were based on responses from 64,900 patients (response rate 52%) and examined 5 multi-item scales (physician care, nursing care, coordination of care, information provided, and discharge instructions) and 1 single-item measure of quality. Scores ranged from 0 to 100 (poor to excellent) for each measure. Analyses compared scores in 5 age groups (18–35, 36–50, 51–65, 66–80, and >80 years).

RESULTS: Satisfaction scores for all 6 satisfaction measures initially increased with age and then declined. For patients rating their health as poor or fair, scores peaked at age 65. For those rating their health as good to excellent, scores peaked at age 80. Satisfaction was also higher in patients with better self-reported health. Declines in satisfaction in older patients were less pronounced in patients with better health. The table below displays mean satisfaction with physician care according to age and health status. Results for the other 5 measures were similar.

		Age					
		N	18–35	36–50	51–65	66–80	>80
Health Status	All patients	64,900	6,159	11,182	16,985	25,059	5,515
	Poor	4,485	70.3	74.9	77.0	75.5	71.3
	Fair	17,435	64.2	67.1	70.1	66.3	64.3
	Good	22,757	64.0	70.5	72.9	70.7	68.4
	Very good	14,568	67.7	73.4	75.8	76.2	73.3
		Excellent	77.1	82.3	85.9	87.9	85.0

CONCLUSION: Satisfaction exhibits a complex relationship with age, with scores increasing until age 65 to 80 and then declining. This relationship was consistent across individual satisfaction scales and levels of health status. The results suggest that age and health status are important modifiers of satisfaction and should be taken into account when interpreting patient satisfaction data.

PHYSICIANS RECOMMEND INTERVENING WITH PATIENTS — NOT PHYSICIANS — TO BETTER IMPLEMENT HYPERTENSION GUIDELINES. R.L. Jayes¹, M. Mintz¹, M.A. Corcoran¹; ¹George Washington University, Washington, DC

PURPOSE: The Joint National Committee (JNC) 6 hypertension (HTN) practice guidelines are well accepted by physicians. However, it is widely recognized that they are not implemented optimally and a significant reduction in morbidity and mortality could be achieved with better HTN care. We convened focus groups of practicing physicians from a range of practice settings to determine what barriers they see to more effective HTN care and to suggest effective interventions.

METHODS: We selected 2 focus groups of 8 physicians, Group 1 from small office practices, and Group 2 from a staff-model HMO. Group 1 was selected by faxed invitations to 200 primary care physicians in a local medical society directory. Group 2 was selected by e-mailed invitation to 80 primary care physicians at one staff-model HMO. We selected 8 respondents

from each group who met these criteria: > half-time practice for the last 2 years, > 40% minority patients, and providing primary care of HTN. The 2-hour sessions with 2–4 observers were audiotaped. A facilitator elicited comments and discussion on JNC6 recommendations, perceived barriers, efficacy of traditional guideline interventions, and recommended solutions for future HTN care. One investigator analyzed tapes for emerging themes and concepts. The resulting summaries were distributed to the research team to discuss alternative interpretations. All focus group observers did a final reading of the summary to confirm the validity of the themes and conclusions.

RESULTS: Similar themes emerged in the two groups. Physicians felt the JNC6 recommendations were realistic but that physician behavior was not the problem with implementation. Physicians outlined the barriers to optimal control of HTN: poor office systems for tracking care, inability to address lifestyle modification in office visits, impracticality of self-monitoring, the disincentive of co-pays, lack of generic samples for cost-effective prescribing. There was no interest in traditional physician directed interventions such as seminars, academic detailing, incentives for chart review to change therapy, pocket or wall versions of the guidelines, or practice profiling. Physicians suggested making monitoring more practical, providing generic samples, and utilizing group appointments, pharmacies, the internet, and ethnically sophisticated mass-media campaigns for patient education.

CONCLUSION: Physicians agreed with the goals and recommendations of the JNC6 HTN guidelines but felt that existing efforts to improve guideline compliance by changing their behavior are bothersome and ineffective. They suggest intervening with the patient to address barriers outside the control of their practices. Future research should examine patients' perceptions of the acceptability and efficacy of these interventions.

DIFFERENCE IN DEMOGRAPHICS, CLINICAL CHARACTERISTICS, QUALITY OF CARE AND OUTCOMES BETWEEN TRANSFERRED AND NON-TRANSFERRED PATIENTS WITH ACUTE MI.

J.W. Jones¹, J.J. Allison¹, J.M. Westfall², C.I. Kiefe¹, P. Sekar¹, N.W. Weissman¹, R.M. Centor¹; ¹University of Alabama at Birmingham, Birmingham, AL; ²University of Colorado Health Sciences Center, Denver, CO

PURPOSE: Each year thousands of patients admitted with acute myocardial infarction (AMI) are transferred during their hospital course with profound implications for quality and outcomes of care. Objective: To evaluate the demographic and clinical characteristics of transferred AMI patients and the association of transfer status with quality of care and mortality.

METHODS: We analyzed the Cooperative Cardiovascular Project (CCP) dataset for 203313 Medicare patients from 4777 hospitals nationally admitted with confirmed AMI between February 1994 and July 1995. Demographic and clinical characteristics of transferred patients were compared with non-transferred patients. Main Outcome measures: Appropriate use of beta blocker, aspirin and ACE-inhibitor; 30 day mortality.

RESULTS: 28.0 % of AMI patients are transferred to another facility at some point during their hospital stay. 24.5% of transfers occurred on the day of admission. Patients were more likely to be transferred out from a smaller, rural, non-teaching facility and in to a larger, urban, teaching facility. Transferred patients were on average 4 years younger. A smaller percentage of African American patients and women are transferred relative to patients remaining in the admitting facility. (4.8% vs. 7.2% and 41.2% vs. 48.9%) Transferred AMI patients tend to have fewer comorbid conditions and are less critically ill as measured by Charlson co-morbidity index, Krumholz predicted mortality and APACHE scores. Patients who were transferred were more likely to receive appropriate therapy with aspirin, and beta blockers. 30 day mortality was lower for transferred patients even after adjustment for severity of illness and co-morbid conditions and after excluding patients who died on the day of admission. (3835) (P>.001 for all comparisons).

CONCLUSION: In this study of elderly AMI patients, older age, female gender, African American race, greater comorbidity and more severe illness were negatively associated with interhospital transfer. Transfer of care was associated with better quality of care based on two out of three indicators and improved mortality.

Table 1: Differences Between Transferred and non-Transferred AMI Patients

	Transferred	Non-transferred
N	57053 (28.0%)	146260 (72%)
Average Age in years	71.8	75.8
female	41.2%	48.9%
African-American	4.8%	7.2%
Charlson score	0.56	0.72
Krumholz predicted mortality	.1400	.1922
APACHE score	8.03	9.8
Eligible patients who received aspirin	88.8%	84.9%
Eligible patients who received beta-blocker	57.2%	50.2%
Eligible patients who received ACE-inhibitor	37.3%	45.4%
Observed 30 day mortality	10.5% (5966)	18.3% (26015)
30 day mortality: Observed/predicted	.75	.98

LENGTH OF STAY AS A SOURCE OF BIAS IN COMPARING PERFORMANCE OF VA AND PRIVATE SECTOR HOSPITALS: LESSONS FROM A REGIONAL EVALUATION OF INTENSIVE CARE OUTCOMES.

P.J. Kaboli¹, M.J. Barnett², S.M. Fuehrer³, G.E. Rosenthal¹

^{a2}; ¹University of Iowa, Iowa City, IA; ²Iowa City VAMC, Iowa City, IA; ³Cleveland VAMC, Cleveland, OH

PURPOSE: Compare intensive care unit (ICU) mortality and length of stay (LOS) in a VA hospital and private sector hospitals and examine the impact of hospital utilization on mortality comparisons.

METHODS: This retrospective cohort study evaluated consecutive ICU admissions to a VA hospital (n=1,142) and 27 private sector hospitals (n=51,249) serving the same health care market in 1994–95. Mortality and ICU LOS were adjusted for severity of illness using APACHE III, a validated method that considers age, comorbidity, ICU admission diagnosis, admission source, and abnormalities in 17 physiologic variables during the first 24 hours of ICU admission. We used two multivariable statistical methods to estimate the risk of death in VA patients, relative to private sector patients. 1) Logistic regression provided the odds of in-hospital death in VA patients. 2) Cox proportional hazards regression was used to account for potential differences in LOS and in the timing of death by censoring patients at the time of discharge.

RESULTS: Unadjusted in-hospital mortality was higher in VA patients (14.5% vs. 12.0%; p=.01), as was hospital (28.3 vs. 11.3 days; p<.001) and ICU (4.3 vs. 3.9 days, p<.001) LOS. Using logistic regression to adjust for admission severity of illness, the odds of death was similar in VA patients, relative to private sector patients (OR 1.16, 95% C.I. 0.93–1.44; p=.18). However, a higher proportion of VA deaths occurred after 21 hospital days (33% vs. 13%; p<.001). Using proportional hazards regression, the risk of death was actually lower in VA patients (hazard ratio, 0.70, 95% CI, 0.59–0.82; p<.001). This result was consistent in stratified analyses of medical (hazard ratio, 0.80, p=.02) and surgical (hazard ratio, 0.46, p<.001) patients. After adjusting for severity using linear regression, differences in ICU LOS were no longer significant (p=.19).

CONCLUSION: Severity-adjusted mortality in ICU patients was lower in a VA hospital than in private sector hospitals in the same health care market, based on proportional hazards regression. This finding differed from logistic regression analysis, in which mortality was similar. This difference suggests that comparisons of hospital mortality between systems with different hospital utilization patterns may be biased if LOS is not considered. Moreover, if generalizable to other markets, our findings suggest that ICU outcomes are at least similar in VA hospitals compared to private sector hospitals.

PATIENT AND ORGANIZATIONAL CORRELATES OF SCREENING FOR COLORECTAL CANCER.

K.L. Kahn¹, H. Liu², J. Adams³, M. Spar⁴, D.M. Carlisle¹, W. Chen¹, R.D. Hays², C.M. Mangione¹, D. Tisnado¹, C.L. Damberg¹; ¹University of California, Los Angeles, Los Angeles, CA; ²RAND Corporation, Santa Monica, CA; ³West Los Angeles VA, Los Angeles, CA; ⁴Pacific Business Group on Health, San Francisco, CA

BACKGROUND: The evidence to support colorectal cancer (CRC) screening has become definitive though it is well known that most adults do not receive the screening that is recommended.

METHODS: We merged patient self-report data from 1998 for respondents >=50 years (n=20,199) with surveys of medical directors of 53 medical organizations [29 medical groups (MG) and 24 Independent Practice Associations (IPAs)] from 3 west coast states to evaluate rates of CRC screening and patient and organizational predictors of those rates. We used logistic regression adjusted for intra-organization clustering of patients within medical organizations to estimate CRC screening rates as a function of patient demographic and comorbid characteristics, receipt of care within a MG or IPA, receipt of care within an organization that uses their corporate structure to implement guidelines, tracking, or feedback of CRC screening results, and the unique identity of each of the medical organizations.

RESULTS: Overall, screening rates were modest [45% for sigmoidoscopy or colonoscopy during the last 5 years (SIG), 43% for fecal occult blood tests by the patient at home (FOBT), 64% for either SIG or FOBT, and 24% for both]. Screening rates varied by demographic characteristics. After adjustment for comorbidity, use of any CRC screen was significantly lower for females (OR 0.69***), Asians (OR 0.73***), Hispanics (OR 0.86*), and those from other non-white race (OR 0.76*); screening rates were higher for African Americans (OR 1.3**). Respondents receiving care in IPAs reported less screening than those in MGs (OR 0.83*) after adjustment for patient characteristics as well as use of guidelines, tracking and feedback pertinent to CRC screening, and clustering of patients within organizations. In addition to the amount of variation in CRC screening explained by patient characteristics alone, receipt of care in a MG or IPA and corporate policy for use of guidelines, tracking or feedback for CRC screening explained 8% of the unique variation in CRC screening rates, while the unique identity of the 53 medical organizations explained 51% of the unique variation.

CONCLUSION: Although there is no clinical reason demographic characteristics should influence rates of CRC screening, they do. In addition to demographics, the corporate structure of the medical organization is associated with rates of patient receipt of CRC screening such that screening occurs more frequently within MGs than IPAs. Even after adjustment for patient characteristics, the use of guidelines, feedback, and tracking of guideline performance, and whether the patient received care in a MG or IPA, most of the variation in screening is explained by the unique identity of their medical organization. *p<.03, **p<.002, ***p<.001.

PHYSICIAN-ASSISTED SUICIDE: PROSPECTS FOR CONSENSUS AMONG CONNECTICUT MEMBERS OF THE AMERICAN COLLEGE OF PHYSICIANS—AMERICAN SOCIETY OF INTERNAL MEDICINE (ACP-ASIM).

L. Kaldjian¹, J. Jekel², J. Bernene³, T. Duffy⁴; ¹Yale University, New Haven, CT, Iowa City, IA; ²Griffin Hospital, Derby, CT; ³ACP-ASIM, New Britain, CT; ⁴Yale University, New Haven, CT

PURPOSE: State medical associations may choose to participate in public debates surrounding physician-assisted suicide (PAS). However, physicians lack consensus on the ethics of PAS and its legalization. We examined physician attitudes about PAS and the prospects for consensus regarding its legalization.

METHODS: Anonymous questionnaires were mailed to all (1456) members of the Connecticut Chapter, ACP-ASIM. The survey instrument measured attitudes toward PAS; consensus on health policy issues generally; participation in public discussions about PAS; prospects for consensus on its legalization; demographic data including age, gender, experience caring for terminally ill patients, religious affiliation, and religious service attendance.

RESULTS: We received 677 completed questionnaires (response rate, 47%). 32% of physicians believed PAS is sometimes appropriate and 28% believed it should be legalized. Respondents who did not agree with legalization were more likely to be concerned about possible abuses (odds ratio [OR], 4.3 [95% CI, 2.9 to 6.2]) and risks to vulnerable populations (OR, 8.0 [CI, 5.1 to 12]), and to believe that PAS violates the physician's role as healer (OR, 29 [CI, 14 to 59]); they were also more likely to have cared for greater numbers of terminally ill patients ($p=0.02$), to report certain religious affiliations ($p<0.001$), and to attend religious services more frequently ($p<0.001$). 87% of physicians believed that they should strive for consensus on health policy issues generally, and 84% believed that Connecticut ACP-ASIM should participate in discussions regarding PAS in Connecticut.

CONCLUSION: Connecticut ACP-ASIM members' heterogeneous attitudes about PAS are associated with differences in professional experience and judgment and religious background. Nevertheless, these physicians favor organizational participation in public discussions regarding PAS and efforts to build consensus on health policy issues generally.

THE LANGUAGE DIVIDE: USE OF INTERPRETERS IN OUTPATIENT PRACTICE. L.S. Karliner¹, E.J. Perez-Stable¹; ¹University of California, San Francisco, San Francisco, CA

PURPOSE: The proportion of foreign-born residents in the United States is at its highest point in over 100 years. Provision of interpreter services for non-English speaking patients is now a federal requirement and general internists will need to use interpreters to care for these patients. We surveyed clinicians in three academic generalist practices to describe how interpreters are being used.

METHODS: We surveyed all of the primary care internists and nurse practitioners ($n=194$) in three academic outpatient settings in San Francisco regarding their most recent patient encounter which involved an interpreter. Questionnaires were self-administered and took about 10 minutes to complete.

RESULTS: 158 questionnaires were completed (81% response rate). Among the respondents, 53% were women, 64% White, 66% residents, 78% spoke and understood a language other than English, and 60% had some prior training regarding working with interpreters. Reflection of the most recent patient encounter requiring an interpreter showed that 67% used professional interpreters, 49% felt that not enough time was allotted for the appointment, 90% felt satisfied that they had accomplished their own goals during the encounter, and 83% felt satisfied that the patient had accomplished his/her goals. Most respondents (78%) were very satisfied or satisfied with the medical care they provided, 85% felt satisfied with their ability to diagnose a disease and treat a disease, but only 45% were satisfied with their ability to empower the patient with knowledge about their disease, treatment or medication. Even though 71% felt they were able to make a personal connection with their patient, only 33% felt they had learned about another culture as a result of the encounter. Clinicians had trouble eliciting exact symptoms (70%), explaining treatments (57%), and eliciting treatment preferences (51%). Almost half (45%) felt that too much time was spent on translation. A majority of clinicians perceived that lack of knowledge of the patient's model of disease and lack of knowledge of the patient's culture hindered their ability to provide quality medical care. However, only 18% felt they were unable to establish trust or rapport.

CONCLUSION: Although most clinicians feel good about their encounters with patients in which they utilize an interpreter, many of them have received no training regarding that use, and their ability to elicit information, convey information, understand the patient in his/her cultural context, and empower the patient with regard to his/her own healthcare appears to be compromised.

APPROPRIATENESS OF FOLLOW-UP CARE IN PATIENTS WITH POSSIBLE ACUTE CARDIAC ISCHEMIA. D. Katz¹, T. Aufderheide², M. Bogner¹, P. Rahko¹, J. Beshansky³, H. Selker³; ¹University of Wisconsin, Madison, WI; ²Medical College of Wisconsin, Milwaukee, WI; ³Tufts-New England Medical Center, Boston, MA

PURPOSE: The AHRQ Unstable Angina (UA) guideline recommends early follow-up and diagnostic evaluation of patients with symptoms suggestive of an acute coronary syndrome (ACS) who are at intermediate or high likelihood of coronary artery disease (CAD). The purpose of this study is to determine the appropriateness of follow-up care for emergency department (ED) pts who are managed as outpatients, based on guideline recommendations.

METHODS: We conducted a prospective cohort study of 374 adults who presented to two University EDs with chest pain or other symptoms suggestive of ACL. Pts' ED evaluation included 12-lead ECG and pertinent history. An AHRQ guideline-based algorithm was used to stratify pts into 4 risk groups: 1) very low risk of CAD (VLR), or intermediate-high likelihood of CAD, with 3 subgroups based on the short-term risk of adverse outcomes, low (LR), intermediate (IR), and high risk (HR). Adverse events and follow-up care (clinic visits, cardiac testing, ED visits, hospitalization) were assessed by 30-day interview and chart review. Physician reviewers assigning final diagnosis were blinded to guideline risk group assignment.

RESULTS: The proportion of pts with confirmed ACI in VLR, LR, IR, and HR guideline groups was 3, 13, 17, and 23%, respectively. There were no MIs in the VLR group (0/108 vs. 9/266 in the remaining risk groups, $p=0.05$). Of pts in the LR, IR, and HR groups, 47% were triaged home. Of these 124 pts, 65% received outpatient follow-up by a clinician, but only 7% received cardiac work-up (non-invasive testing or coronary angiography) during follow-up; 2% were confirmed to have ACI (0% AMI), and 6% of these pts required ED re-evaluation or hospitalization for heart-related symptoms during 30-d follow-up.

CONCLUSION: Nearly half of ED pts at intermediate-high likelihood of CAD are currently managed as outpatients. While the risk of missed ACI in these pts is low, our data demonstrate that a substantial fraction fail to receive recommended clinical follow-up or cardiac work-up. For the AHRQ guideline to be maximally effective as a triage aid, structural changes to facilitate the timely and adequate outpatient follow-up of ED pts at increased risk of ACI are needed, particularly in EDs without rapid non-invasive testing protocols.

EFFECTIVENESS OF A NURSE-BASED STRATEGY FOR IMPLEMENTING THE AHRQ SMOKING CESSATION GUIDELINE IN PRIMARY CARE. D. Katz¹, D. Muehlenbruch¹, N. Zillmer¹, J. Martin-Mckay¹, M. Fiore¹, T. Baker¹; ¹University of Wisconsin, Madison, WI

PURPOSE: To determine whether a multi-modality strategy which emphasizes implementation of the AHRQ Smoking Cessation Clinical Practice Guideline by clinic nurses results in improved smoking cessation outcomes in primary care.

METHODS: We conducted a controlled trial of a guideline-derived intervention, which included 3 components: 1) training of clinic nurses and medical assistants to use a modified vital signs stamp in identifying and counseling smokers at each visit, 2) feedback to clinic staff on performance of AHRQ guideline-recommended activities, and 3) proactive telephone counseling by a smoking cessation counselor (2 sessions) and free nicotine replacement therapy (NRT) for all pts who indicated willingness to make a quit attempt within 30 d of their clinic visit. After a period of baseline data collection (8/99–1/00), the intervention was implemented at 1 pilot family practice (FP) clinic (2/00–3/00); patterns of usual care were observed concurrently at 5 control FP clinics. We obtained exit interviews of consecutive adult smokers who presented for routine, non-emergency care. Abstinence (no cigarettes over prior 7 d) and quit attempts were determined by telephone interview during 6 mo follow-up. Odd ratios (OR) were computed for each outcome.

RESULTS: There were no significant differences in demographic or smoking variables between baseline and intervention pts, 87 and 85% of whom completed 6 month follow-up. Concordance with guideline recommendations at the clinic visit and quit status are shown below ($p<0.05$). For comparison, the 6 mo quit rate for pts at control clinics during baseline and intervention periods was 12 and 10%, resp. 37% of intervention group pts received telephone counseling plus NRT.

CONCLUSION: Effective reduction of tobacco use requires redesigning health care systems to increase the identification of smokers and the delivery of smoking cessation advice in a time-efficient manner. The 9% absolute difference in 6 mo quit rates associated with this nurse-based, primary care intervention is comparable to that observed in clinical trials of NRT.

Smoking Cessation Outcomes at Intervention Clinic, %

	Baseline (N=108)	Intervention (N=116)	OR (95% CI)
Ask about tobacco use*	86	97	6.2 (1.9–21)
Ask if willing to quit*	25	88	23 (11–48)
Advise pt to quit*	35	70	4.2 (2.4–7.4)
2-month quit rate*	5	22	5.4 (2.1–14.3)
6-month quit rate	12	21	1.9 (0.9–3.9)

COMMUNITY-WIDE INCIDENCE AND OUTCOMES OF PNEUMONIA IN PATIENTS HOSPITALIZED FOR ACUTE STROKE. I.L. Katzan¹, R.D. Cebul², S.S. Husak², N.V. Dawson², D.W. Baker²; ¹Cleveland Clinic Foundation, Cleveland, OH; ²Case Western Reserve University, Cleveland, OH

PURPOSE: Pneumonia is an important and often avoidable complication of acute stroke. Our primary objectives were to determine: (a) the incidence of pneumonia in patients hospitalized for acute stroke in Greater Cleveland over a 7-year period; and (b) the effect of pneumonia on mortality up to 30 days after admission.

METHODS: Subjects were 14,293 patients >64 years old admitted for ischemic (ICD-9-CM 434, 436) or hemorrhagic (ICD-9-CM 431, 997.02) stroke to one of 27 hospitals in Greater Cleveland between 1991 and 1997. Patient level data were obtained through the Cleveland Health Quality Choice (CHQC) program; mortality was obtained by linking CHQC data with MEDPAR files. Pneumonia was identified from secondary ICD-9-CM codes. Cox regression analysis was used to estimate the effect of pneumonia on mortality up to 30 days from admission after adjusting for severity of illness using a model of 30-day predicted mortality from admission data (c -statistic = 0.78). Patients dying or having a "Do Not Resuscitate" order within 3 days of hospitalization were removed a priori from the regression analysis to focus on those for whom avoiding pneumonia was most pertinent.

RESULTS: Pneumonia occurred in 6.9% ($n=985$) of all patients hospitalized for acute stroke. There was a significantly higher incidence in patients: with hemorrhagic stroke (11.5% vs 6.3%, $p<0.001$), men (9.0% vs 5.4%, $p<0.001$), from nursing homes (11.4 vs 6.4%, $p=0.001$), aged >74 (7.2% vs 6.3%, $p=0.032$), and with greater severity of illness on admission (predicted 30-day mortality 24.8% vs 13.2%, $p<0.001$). Average length of stay for patients with pneumonia was 16.2±0.44 days compared to 7.5±0.05 days in patients without pneumonia ($p<0.001$); fewer patients with pneumonia were discharged to home (12.0% vs 41.4%, $p=0.001$). Crude 30-day mortality rates were 37.4% for patients with pneumonia and 13.0% for those without ($p<0.001$). There was a steady increase in the adjusted hazard ratio for death among stroke patients with pneumonia throughout the 30-day period; adjusting for admission severity, the hazard ratio for death within 30 days was 4.31 (95% C.I. 3.55–5.23).

CONCLUSION: In this largest community-wide study of stroke complications, pneumonia conferred significantly higher severity-adjusted hazard for mortality within 30 days of admission and was associated with dramatically longer lengths of hospital stay and reduced chances of discharge to home. Measures to reduce the incidence of pneumonia following stroke are warranted to reduce morbidity, mortality, and cost.

BREAST CANCER SCREENING: THE ROLE OF PHYSICIAN PERSONAL CHARACTERISTICS AND PROFESSIONAL EXPERIENCE. L. Kelly¹, W.P. Moran², R. Velez¹, G. Russell¹, D. Case¹, E.D. Paskett¹; ¹Wake Forest University, Winston-Salem, NC

PURPOSE: Many studies have found that physician characteristics, such as age, gender, practice type, professional experience, and personal experience, are associated with physician attitudes concerning breast cancer screening (BRCA). We examined the relationship between these variables and a favorable attitude towards BRCA.

METHODS: As a part of the Forsyth County Cancer Screening (FoCaS) project, we surveyed physicians from community health centers on their attitudes about BRCAS. Independent variables included age, gender, type of medical practice, professional experience with breast cancer (BRCA), and personal experience with BRCA. The dependent variable, a sum score of attitudes towards BRCAS, was calculated such that a higher score indicated a more favorable attitude towards screening. Univariate and multivariate analyses with backwards-stepwise regression assessed the relationship between the dependent and independent variables.

RESULTS: The response rate for the survey was 82% (N=140 responses); 120 physicians provided complete data for analysis. Of these physicians, 62% were male, 60% were under 30 years old, 12% worked in a private office, 42% worked in a health center, 46% worked in an outpatient setting, 38% had some personal experience with BRCA, and 68% had some professional experience with BRCA. In both univariate and (multivariate) models professional experience $p=0.006$ (0.03) was associated with a more favorable attitude concerning BRCAS, but age $p=0.10$ (0.54), gender $p=0.54$ (0.59), practice type $p=0.23$ (0.62), and personal experience 0.54 (0.37) were not.

CONCLUSION: Prior professional experience of the physicians, but not demographic or personal variables, was significantly related to a favorable attitude towards BRCAS. Our data differs from previous studies, which found that BRCAS may be influenced by physician characteristics like age, gender, practice type, and personal experience. Increasing awareness through the education of physicians has been a focus of many efforts to increase the incidence of screening for BRCA. If professional experience is associated with a favorable attitude towards BRCAS, then medical trainees should be exposed to environments where BRCAS is a high priority in an effort to increase BRCAS in women.

COLLATERAL DUTIES: A HUGE UNAPPRECIATED BURDEN FOR CLINICIAN-EDUCATORS. S. Kick¹, D. Tanaka¹; ¹University of Colorado Health Sciences Center, Division of General Internal Medicine, Denver, CO

PURPOSE: Physicians spend a great deal of time on patient-care, educational, administrative and research activities outside of the examination room. Most often, these activities are not compensated monetarily or through productivity standards such as relative value units (RVU's). The purpose of this study was to utilize a novel method to document physician time spent in a number of these collateral activities.

METHODS: The Titrax[®] software was developed as a tool for keeping time of various activities on a palm pilot. Permission for the use of this program for research was granted by the developers. The program allows for the creation of different categories. At a touch of the stylus, time is kept for the activity chosen. Several categories were chosen in the areas of patient/colleague communication, administration, teaching and research. Clinical faculty in the division of General Internal Medicine were asked to use the palm pilot in one-week increments. The clinicians were given instruction in the use of the program and were asked about acceptability of the program following the week's use. Data are presented as descriptive summaries of the time spent in the various activities. Total time was indexed to the degree of clinical effort.

RESULTS: Three physicians have used the program to date. All found it acceptable with minimal problems of use. Each of these three physicians used the program for two weeks. The total time spent in these collateral activities ranged from 12.3 hours to 23.2 hours per week. When indexed to the number of patients seen, and hours actually spent in clinic the clinicians spent 20.3 to 24.1 minutes per patient seen, and 45 to 66 minutes per hour spent in patient care, specifically on collateral duties occurring outside of the examination room.

CONCLUSION: Clinician-educators spend a substantial amount of time in activities other than face-to-face patient care. Depending on how one defines the number of hours in a work week, clinicians are spending between 30 and 60% of their time in these activities which are neither compensated nor accounted for in traditional means of measuring physician productivity.

A CONCORDANCE BASED METHOD TO EVALUATE GUIDELINE IMPLEMENTATION. L. Kleinman¹, B. Feldman¹, P. Stillman¹, M. Potylycki¹, E. Karoly¹, M. Young¹; ¹Lehigh Valley Hospital, Allentown, PA

PURPOSE: We demonstrate a novel method to analyze systematically implementation of a clinical guideline. The guideline asked attending physicians to classify the indications for cardiac telemetry into one of three categories that segregated the risk of a life threatening arrhythmia according to expert judgement and the medical literature.

METHODS: We developed a concordance based analysis of guideline implementation, using five key indicators of performance: 1. Use of the guideline by the physicians (proportion of patients whom the physicians classify at all); 2. Impact on key clinical or resource outcomes targeted by the guideline (hours of telemetry per patient before and after implementation); and 3 - 5, various aspects of concordance: a. accuracy of guideline application by the doctors (doctors' classification compared to retrospective expert classification); b. accuracy of translation of the doctors' orders into practice (actual length of telemetry vs. the guideline recommendation for the doctor's classification); and, c. the match between the service provided and the patient's need (actual length of telemetry versus the length recommended, when classified retrospectively by an expert reviewer). We collapsed adjoining categories (low and moderate risk for one set of analyses, high and moderate risk for another) to calculate kappa, sensitivity (Se), specificity (Sp), and predictive values positive (PVP) and negative (PVN), considering, for example, the doctors' classification as if it were a diagnostic test for the expert's assessment, that is: 'how well do the doctors identify patients in the highest (or lowest) risk category.' We studied 184 patients admitted to a single telemetry unit of a large academic community hospital Jul - Aug, 1999 and 236 patients admitted one year later, after guideline implementation. 127/236 were sampled for detail review.

RESULTS: Telemetry use dropped from 60 hours / admission to 47 hours, $p<.05$. Kappa varied from 0.10 (poor agreement) for length of telemetry vs. guideline recommendations for low risk patients (indicating persistent overuse of telemetry) to 0.54 (good agreement) between the doctors' and the expert's classifications for patients at highest risk. Agreement between the doctors' and expert's classifications are shown in the table.

CONCLUSION: We present a simple method for analyzing the performance of guidelines in practice. Variations in the levels of concordance (range of kappa .10-.54) demonstrate the importance of multifaceted measurement to describe guideline performance. This approach answers key substantive questions about the safety and effectiveness of a guideline as implemented, and identifies opportunities to improve.

CONCORDANCE OF DOCTORS' AND EXPERT'S CLASSIFICATIONS

Category classified:	Se	Sp	PVP	PVN	kappa
Highest risk	0.86	0.89	0.46	0.98	0.54
Lowest risk	0.61	0.84	0.61	0.84	0.45

EVALUATION OF THE EFFECT OF MOBILE COMPUTERS ON THE ACCEPTANCE OF A COMPUTERIZED PATIENT RECORD SYSTEM. J.M. Kramer¹, A.M. Fendrick¹, T.J. Laing¹, S. Rhodes²; ¹University of Michigan, Ann Arbor, MI; ²Ann Arbor VA Medical Center, Ann Arbor, MI

PURPOSE: Computerized order entry systems have the capability to decrease order errors and reduce complications due to adverse events. Their use has been limited because the necessary hardware is often distant from the point of care. The improved convenience of a mobile ordering system may increase caregiver satisfaction with electronic systems. We aimed to evaluate the use and attitudes regarding the Computerized Patient Record System (CPRS) as affected by the availability of a wireless mobile computing device.

METHODS: All residents were trained on the Veterans Administration's CPRS and then exclusively used the CPRS on desktop computers for 14 days. Subsequently, one-half of the inpatient internal medicine teams were randomly provided wireless mobile computers for 14 days. Mobile computer use was voluntary and desktop-based computers were equally available for all user groups. The CPRS application was identical on all computing devices. We recorded the number of orders written for each device and collected residents subjective opinions of CPRS in both groups before and after the introduction of the mobile computers.

RESULTS: The study populations were similar with 16 participants in each arm. Mobile computer users reported that they entered an average of 50% (36-62, 95%CI) of orders using the mobile devices. On a 5 point Likert scale with 1 or 2 being favorable, 81% responded favorably that the computer helped them complete their work sooner and 50% responded that it improved patient care during rounds. Correlation between technical problems reported and the percent of orders written on the mobile device was 0.02, (-0.07 to 0.11, p-value of 0.64). Mobile computer users required the use of the paper chart in addition to CPRS for 17% of patients (4.4-29.4, 95%CI). Most commonly cited missing elements that were needed included current notes types that were not online (33%), notes predating the system (20%), EKGs (16%), and vital signs (12%). On the same Likert scale, 75% agreed that they could give up the traditional paper chart in favor of the mobile computer.

CONCLUSION: Overall, internal medicine residents responded positively to the implementation of the mobile CPRS system. Based on surveys and computerized order logs, all residents voluntarily used the devices to enter nearly half of orders over available desktop computers. Reported difficulties with the technologies did not appear to significantly affect the mobile computer users ordering pattern, suggesting that the technical problems were limited and the technology was robust. Finally, feedback from this study was sufficiently positive to justify implementation to all internal medicine teams.

THE POSITIVE IMPACT OF HOSPITAL-BASED CLINICIAN EDUCATORS ON LENGTH OF STAY AND RESIDENT EDUCATION. M.E. Kulaga¹, J.P. Cleary¹, S.P. O'mahony¹, E. Mazur¹; ¹Norwalk Hospital, Norwalk, CT

PURPOSE: Caring for indigent patients and educating medical residents are important challenges for teaching hospitals that we addressed by hiring two inpatient academic clinician educators. After one year, we assessed their impact on patient length of stay as well as resident education and service from the viewpoint of both medical residents and community physicians.

METHODS: Two clinician educators were hired to provide inpatient attending medical care and resident education, including 1:1 intern mentoring. Length of stay data was calculated for Medicaid and self-pay patients based on DRG groups where >50% of patients were admitted to the clinician educator service and was compared to data on the same patient mix from the preceding year. Categorical medicine residents (N=36) were given an anonymous survey assessing the clinician educators' impact on resident education and service, including resource utilization and resident roles. Community internists (N=150 on staff, 100 who actively use hospital) also received an anonymous survey assessing their perceptions of hospitalist medicine and the clinician educators' impact on housestaff behavior.

RESULTS: Length of stay: LOS was reduced from 5.3 to 4.4 days comparing the last 6 months of the clinician educators' first year with the last six months of the previous year. Reductions in LOS >1 day were found with GI, pulmonary, infectious disease, endocrine and neurology admissions and 1/3 of a day for cardiology admissions.

Resident survey: Return rate was >90% with all residents noting improvement in the quality of attending rounds, bedside teaching and the overall inpatient experience. Increased use of evidence based medicine resources was noted by 90%. Residents' roles as teachers and team leaders were increased or unchanged for over two-thirds and none felt they had been reduced to unacceptable levels. The new system was preferred by 100%. Attending survey: Many of the attending internist on staff responded (N=60, 40% of all on staff, 60% of active hospital users). Approximately 60% of responders noted improvement in residents' clinical reasoning skills and leadership roles. None felt that their personal roles as teachers were reduced to unacceptable.

CONCLUSION: Hiring clinician educators to serve as inpatient teaching attendings is effective in reducing length of stay and improving resident education.

PERSONAL, ORGANIZATIONAL, AND MARKET LEVEL INFLUENCES ON PHYSICIANS PRACTICE PATTERNS: RESULTS OF A NATIONAL SURVEY OF PRIMARY CARE PHYSICIANS. B.E. Landon¹, J. Reschovsky², M. Reed², D. Blumenthal¹; ¹Harvard University, Boston, MA; ²Center for Studying Health Systems Change, Washington, DC

PURPOSE: To determine how clinical decisions are associated with individual characteristics, practice setting and organizational characteristics, attributes of the patient population under care, and the market environment.

METHODS: We surveyed a cross-sectional, nationally representative sample of primary care physicians who provide direct patient care at least 20 hours per week. We examined the proportion of physicians who would order a referral, diagnostic test, or treatment for 5 clinical scenarios thought to be representative of discretionary medical decisions.

RESULTS: We received responses from 4,825 primary care physicians who cared for adult patients (Response Rate 65%). The distribution of results for each of the 5 clinical scenarios demonstrates significant variability both within and between physicians. We saw no evidence of a consistent practice style across the vignettes (e.g. "aggressive" or "conservative"). The organizational setting of practice was the most consistent predictor of behavior across all of the clinical scenarios, with the exception of back pain, which was minimally related to any of the environmental factors. When compared to physicians in solo practice, physicians in all other practice settings were less likely to order a test or referral or pursue treatment. Practice involvement with managed care and measures of financial influences and administrative strategies associated with managed care were minimally and inconsistently associated with reported physician behaviors.

CONCLUSION: The ability of managed care to improve the quality and consistency of care while also controlling the costs of care depends on its ability to influence medical decisions. Our findings generally demonstrate that managed care has a weak influence on discretionary medical decisions and that the influence of managed care pales in comparison to personal and practice setting influences.

EVOLVING DISSATISFACTION AMONG PRIMARY CARE PHYSICIANS. B.E. Landon¹, R. Aseltine², J.A. Shaul¹, Y. Miller³, B.A. Auerbach³, P.D. Cleary¹; ¹Harvard University, Boston, MA; ²Boston University, Boston, MA; ³The Massachusetts Medical Society, Waltham, MA

PURPOSE: To examine trends in physician career satisfaction, particularly within the managed care environment.

METHODS: We administered cross sectional surveys to primary care physicians in Massachusetts in 1996 and 1999. We examined physicians' ratings of overall satisfaction with their current practice situation and their current experience with managed care.

RESULTS: 1336 primary care physicians responded to the 1996 survey (56% response rate) and 966 physicians responded to the 1999 survey (42%). Fourteen percent of the respondents were family physicians, 52% were general internists, and 27% were general practitioners. About one-quarter of the respondents were in solo practice and 38% were in single specialty primary care groups. In 1996, 19.8% of physicians were somewhat or very dissatisfied with their current practice situation as compared to 33.4% in 1999, an increase of over 50% ($p < .001$). Overall dissatisfaction with managed care increased from 28.3% to 62.2% ($p < .001$). In multivariable models, external influences on practice were important predictors of overall dissatisfaction. Physicians whose choice of hospitals was restricted (O.R. 2.23, 95% C.I. 1.30, 3.78) and those who reported that managed care plans influenced their practice "a lot" (O.R. 1.85, 95% C.I. 1.10, 3.11) were more likely to be dissatisfied. The adequacy of reimbursement was an important predictor of overall satisfaction and satisfaction with managed care and the ways in which physicians experienced the financial incentives were associated with managed care dissatisfaction.

CONCLUSION: Primary care physicians in Massachusetts are increasingly dissatisfied with their practice and with managed care. Continued erosion of physician satisfaction could have significant implications for quality of care and the quality of the work force that is attracted to medicine as a career.

NEW DRUGS: WHEN CAN CLINICIANS KNOW THEY ARE SAFE? K. Lasser¹, P.D. Allen¹, S.J. Woolhandler¹, D.U. Himmelstein¹, D. McCormick¹, D.H. Bor¹; ¹The Cambridge Hospital/Harvard Medical School, Cambridge, MA

PURPOSE: To determine the median time interval between the introduction of a new prescription drug onto the market and the identification of serious adverse drug reactions, defined by the appearance of a new black box warning in the Physician's desk reference (PDR) or the withdrawal of the drug from the market for safety reasons.

METHODS: We identified all drugs that had a black box warning in the 2000 PDR, and examined earlier PDRs to determine when the black box warning first appeared. For those drugs that did not have a black box warning in the PDR in which they first appeared, we measured the amount of time that elapsed between the introduction date and the publication year of the first PDR in which a black box warning appeared. We also measured the time that elapsed from drug introduction to withdrawal for drugs withdrawn for safety reasons from 1975 through 2000. We excluded drugs that were introduced before 1975, over the counter drugs, biologics, and topical agents. We also analyzed the content of the black box warnings and the reasons for drug withdrawal.

RESULTS: 558 new chemical entities were approved from 1975–1999. Of these, 10.6% ($n=59$) obtained a new black box warning or were withdrawn from the market for safety reasons. There were 81 changes in the PDR over the study period, representing either the addition of one or more black box warnings per drug or the withdrawal of a drug. 50% of all changes to the PDR occurred within 4 years following drug introduction. Seventeen drugs were withdrawn from the market between the years 1975 and 2000; half of these drugs were withdrawn within 1.5 years following their introduction. The most common warnings/reasons for withdrawal were: liver toxicity (18%), hematologic toxicity (14%), cardiotoxicity (12%), and risk in pregnancy (12%). **CONCLUSION:** Since 1975, one in ten new drugs was found to have serious adverse effects, half of which were documented in the PDR within 4 years after introduction. Clinicians should prescribe new drugs cautiously, especially when there is a similarly efficacious agent that has

been in longer use, and should avoid use of new drugs in pregnant women. If a new drug is prescribed, clinicians should observe patients for liver, hematologic, and cardiac toxicity. Improved reporting of adverse drug reactions might decrease the time until such reactions are discovered.

THE INFLUENCE OF PHARMACEUTICAL SAMPLES ON THE PRESCRIBING PRACTICE OF CLINICIANS IN AN ACADEMIC CLINIC. A.S. Lee¹, B.A. Mincey¹, M.A. Parkulo¹; ¹Mayo Clinic, Jacksonville, FL

PURPOSE: Clinician antibiotic prescribing practices have come under increasing scrutiny due to concerns about resistant organisms. A potential influence on these behaviors is the availability of pharmaceutical samples. A prospective blinded study was designed to evaluate this influence in an academic clinic.

METHODS: The antibiotics in the drug closet of our teaching clinic were manipulated over 20 consecutive weeks. The following changes were made at 4 week intervals: 1) removal of all antibiotics, 2) removal of macrolides, 3) removal of quinolones, 4) removal of cephalosporins, 5) removal of amoxicillin/clavulanate and nitrofurantoin. Upon completion of the study, a chart review was conducted to identify acute infectious illnesses for which an antibiotic was likely to be prescribed. Datapoints collected include: clinical diagnosis, antibiotic prescribed, post-graduate year (if a resident), date of service, and patient's gender and age. Fishers Exact 2-sided tests were performed to determine whether the presence of antibiotics in the closet influenced their frequency of use.

RESULTS: 465 encounters fulfilled our inclusion criteria. 77% of these encounters led to an antibiotic prescription. Of these, 73% had an antibiotic prescribed that could be found in the drug closet. Defining the relative risk as the risk of being prescribed a specific antibiotic if it was available in the closet versus if it were not, the relative risk ratios (RR) with their associated 95% confidence intervals (CI) for each class of antibiotics are listed in the following table.

CONCLUSION: The prescribing practices of clinicians in our academic clinic do not appear to be directly influenced by the availability of antibiotic samples in a drug closet. Although a direct relationship between the drug closet and the prescribing practices of clinicians was not observed, it is notable that 73% of all prescriptions were for antibiotics that could be found in the drug closet. This still suggests that pharmaceutical marketing exerts an influence on clinician antibiotic prescribing practices.

Relative Risks

	RR (95%CI)	2-sided P-value
Macrolides	0.98 [0.78–1.22]	0.91
Quinolones	1.27 [0.85–1.91]	0.30
Cephalosporins	0.28 [0.09–0.84]	0.03
Amoxicillin/Clavulanate, Nitrofurantoin	0.88 [0.24–3.23]	1.00

ACADEMIC PHYSICIANS VIEW CLINICAL E-MAIL WITH PATIENTS MORE POSITIVELY THAN THOSE IN PRIVATE PRACTICE. C. Lin¹, L. Ware², G.A. Albertson², E.M. Cyran², L. Schilling², J.F. Steiner², R.J. Anderson²; ¹University of Colorado, Denver, CO; ²CU, Denver, CO

PURPOSE: We studied the differences between physicians in academic and private practice settings regarding e-mail communication with patients.

METHODS: Mailed survey of 1100 practicing MD's at University of Colorado (CU MD) and 1000 randomly selected private practice MD's in Colorado (CO MD).

RESULTS: We received 574 responses (52%) to the University survey, and 501 responses (50%) to the Colorado survey. Among respondents, more CO MD's were in primary care (CO: 40%, CU: 14%, $p=0.001$), and more spent greater than 50% in direct patient care (CO: 84%, CU: 55%, $p=0.001$). Age and sex were comparable. CU MD's were more likely to have access to e-mail (CU: 95%, CO: 90%, $p=0.001$), to use it with colleagues (CU: 93%, CO: 61%, $p=0.001$), to use it for curbside consultation (CU: 66%, CO: 38%, $p=0.001$), and to use it with patients (CU: 35%, CO: 16%, $p=0.001$). Both CU MD's and CO MD's were aware of federal legislation on patient privacy standards (CU: 42%, CO: 37%, $p=NS$), but this awareness did not affect their likelihood of e-mailing patients. CU MD's perceptions of the benefits of e-mailing patients were shared by CO MD's: faster (CU: 72%, CO: 57%, $p=0.001$), no phone tag (CU: 80%, CO: 76%, $p=NS$), more efficient (CU: 56%, CO: 43%, $p=0.001$), and printable documentation (CU: 76%, CO: 72%, $p=0.001$). CU MD's perceptions of risk of e-mailing patients was shared by CO MD's: message would not be seen (CU: 67%, CO: 78%, $p=0.001$), message is not confidential (CU: 75%, CO: 81%, $p=0.026$), cannot convey complex message (CU: 77%, CO: 84%, $p=0.006$), patient over use (CU: 75%, CO: 79%, $p=NS$). Among CU and CO MD's, no subgroups were more likely to e-mail patients.

CONCLUSION: Compared to private practice physicians, academic physicians at one institution have greater access to e-mail, use it more often with colleagues and patients, perceive greater benefit and less risk in using e-mail with patients. Surprisingly, we found no consistent subgroups of academic or private practice physicians more likely to use e-mail to communicate with patients.

PRECONCEPTION CARE BY PRIMARY CARE PHYSICIANS. L.E. Lincoln¹, P.S. Heckerling²; ¹University of Illinois at Chicago, Chicago, IL; ²Virginia Interment College, Lincolnwood, IL

PURPOSE: Preconception care and counseling should be integrated into the health maintenance of all female patients of childbearing age. Healthy People 2000 includes preconception care as an objective for all women who can become pregnant who present for continuing care in primary care settings. Because 50% of pregnancies are unplanned and 50% of pregnant women present to their pregnancy provider after the critical period of organogenesis, it

is important that primary care physicians be aware of the safety of drugs in pregnancy given to nonpregnant women, as well as the effect of common medical problems in pregnancy. For the same reasons, it is also important that counseling and prevention of risk factors known to affect pregnancy outcome occur prior to conception. The purpose of this study was to assess the knowledge and practice of preconception care by primary care attending physicians.

METHODS: All primary care attending physicians in general internal medicine and family practice at our institution were given a written survey regarding their knowledge and practice of preconception care. Knowledge questions assessed were 1) safety of medication use in pregnancy 2) risk associated with diabetes in pregnancy, and 3) the rate of unplanned pregnancies and estimated gestational age at presentation to a pregnancy provider. Practice questions included counseling of the nonpregnant patient regarding 1) the hazards of smoking, alcohol and illicit drug use 2) folic acid supplementation, 3) rubella screening, and 4) the effect of chronic medical disorders were pregnancy to occur. Surveys were mailed. Responses were kept anonymous and a number was used to identify participants.

RESULTS: Of 25 surveys mailed, 21 were returned. Analysis was done of the returned surveys. 80% of responding attending primary care physicians were aware that 50% of women present to their pregnancy provider after organogenesis and that similar rates of pregnancies are unplanned. 45% counseled women with substance abuse of potential effects on pregnancy, 40% recommended folic acid use, and 10% screened for rubella. 55% counseled women with chronic disease of deleterious effects on pregnancy. Safety of medications in pregnancy were correctly identified by 35% for pain relievers, 45% for antibiotics, 25% for asthma medications, and 10% for antihypertensives. Only 45% of providers were aware of the risk associated with diabetes in the first trimester. No statistical difference using Chi Square analysis was found when the results were computed by the sex of the provider or by primary care specialty, possibly due to the small sample size.

CONCLUSION: Despite the knowledge that women often do not plan pregnancy and do not present to a pregnancy provider early enough to prevent the toxic effect of medications, chronic medical disorders, or personal behaviors, the majority of primary care providers were not knowledgeable regarding preconception care and did not incorporate it into the continuing care of women of childbearing age.

DOES THE QUALITY OF CARE FOR PATIENTS WITH HEART FAILURE DIFFER BETWEEN HOSPITALISTS AND NON-HOSPITALISTS? P. Lindenauer¹, R. Chehabeddine², J. Fitzgerald³, P. Pekow², E.M. Benjamin¹; ¹Tufts University, Springfield, MA; ²University of Massachusetts at Amherst, Amherst, MA; ³Baystate Medical Center, Springfield, MA

PURPOSE: The quality of care provided to patients hospitalized for heart failure has been shown to vary by physician, hospital and region. Hospitalists are physicians whose primary professional focus is the general medical care of hospitalized patients. While hospitalists appear to reduce costs and length of stay, their impact on quality of care is less certain. We sought to compare the quality of care provided to patients with heart failure by hospitalists and non-hospitalist physicians.

METHODS: We conducted a retrospective medical record review at a 550 bed community-based teaching hospital in Massachusetts that is home to 3 hospitalist groups as well as numerous small group practices that provide care for their own hospitalized patients. We identified all patients cared for by hospitalists and by non-hospitalist general internists who were discharged with an ICD-9-CM principal diagnosis of heart failure between 4/1/99 and 3/30/00. We evaluated quality of care by measuring adherence to a set of process measures derived from the JCAHO's "Core Measures Set" and HCFA's Health Care Quality Improvement Initiative. The study was powered to have an 80% chance of detecting a 15% difference in ACEI or ARB use in patients with EF < 40%.

RESULTS: 280 patients accounting for 326 heart failure admissions were included in the analysis. 20 hospitalists cared for 137 (42%) cases while 65 non-hospitalist, general internists, cared for the other 189 (58%) cases. Patients cared for by hospitalists were younger, more frequently male, and were more likely to receive housestaff coverage. 129 (94.2%) of 137 hospitalist cases had left ventricular ejection fraction documented or measured compared to 165 (87.3%) of 189 (p=.04) non-hospitalist cases. In cohorts of ideal candidates, performance of hospitalists and non-hospitalists were similar for ACEI or ARB prescribed to patients with EF < 40% (96.9% vs 95.9% p=1.00), and for warfarin prescribed for patients with AF (60% vs 54.5% p=.64). Performance of hospitalists and non-hospitalists were similar for counseling surrounding smoking cessation (18.2% vs 11.4% p=.51), medication use (98.4% vs 98.3% p=1.00), crisis management (65.1% vs 71.1% p=.26), diet (93.0% vs 89.4% p=.28), and weight monitoring (24.0% vs 26.7% p=.60). After adjusting for the confounding effects of age, gender, and housestaff coverage using multivariable modeling the observed difference in the rate of left ventricular assessment was no longer significant.

CONCLUSION: Quality of care for heart failure at our institution was generally high and we were unable to detect significant differences between the practices of hospitalists and non-hospitalist general internists.

PREDICTING AND PREVENTING PHYSICIAN BURNOUT: RESULTS FROM THE US AND THE NETHERLANDS. M. Linzer¹, M. Visser², F. Oort², E. Smets², J. McMurray¹, J. De Haes²; ¹University of Wisconsin, Madison, WI; ²University of Amsterdam, Netherlands

PURPOSE: Physician burnout, which includes physical and emotional exhaustion, has only recently been recognized as a significant problem. This is the first study to identify and compare burnout predictors in two countries, the US and the Netherlands (NL).

METHODS: Data were collected in two separate surveys of physician job satisfaction, stress and burnout between 1996 and 1998 in the US (adjusted response rate 52%) and the NL (adjusted response rate 63%). A single item assessing increasing degrees of burnout was used in the US; the NL survey used the Maslach Burnout Inventory. For this analysis, we created two datasets of comparable physicians (n = 1749 US, n = 1349 NL), operationalized a series of research variables, and performed structural equation modeling to develop a predictive burnout model. A root mean square error of approximation (RMSEA) of <.05 indicates close fit of the model to the data.

RESULTS: In the US, 22% of physicians acknowledged burnout. In the NL, 20% felt burned out a few times per month or more. Background predictors in the model (age, sex, children, practice type and work hours) do not impact on burnout but rather affect key mediator variables (work control, work-home interference and home support). These mediators impact on job stress and satisfaction. Stress, satisfaction and the mediator work-home interference directly affect burnout. This model has excellent fit (RMSEA .035 US and .037 NL) and predicts 50% of the burnout variance in both countries.

CONCLUSION: Physician burnout is prevalent, and predictors of burnout may be similar across Western nations. This robust model, with excellent fit to the data in two countries, suggests that burnout is directly related to job stress and satisfaction, and can be prevented by preserving work control and modulating work-home interference.

IS THE VA MEETING ITS MANDATE TO CARE FOR LOW INCOME AND UNINSURED VETERANS? J.A. Long¹, D.A. Asch¹; ¹University of Pennsylvania, Philadelphia, PA

PURPOSE: Part of the VA mandate is to care for poor and uninsured veterans, however, there is little data on how well the VA is serving these vulnerable populations. Therefore, the objective of this study was to determine the rates of services received among veterans by income level and insurance status.

METHODS: Using the 1992 National Survey on Veterans, we looked at the rates of outpatient and preventive care as well as rates of unmet medical needs and hospitalizations among 10,476 non-institutionalized veterans over the age of 25.

RESULTS: In logistic regression, after adjusting for age, sex, race/ethnicity, marital status, education, general health, number of chronic conditions, ADL status, disability preventing work, and service connected disability, low income and uninsured veterans, received less outpatient and preventive care and had more unmet medical needs and hospitalizations. These findings held true regardless of where veterans sought care. The table below shows the adjusted odds of receiving preventive care by income level and insurance status for veterans who received some care at a VA medical center and veterans who received no care at a VA medical center.

CONCLUSION: In a system with a mandate to care for the poor and uninsured, income and health insurance remain important determinants of who receives health services. These results suggest that the VA health system may not be adequately meeting its mandate to care for these vulnerable veterans.

Adjusted Odds Ratios for Receiving Preventive Care By Income and Insurance Status

	VA Users	Non-VA User
Income		
\$40,000+	1.0	1.0
\$20,001 – \$40,000	0.65 (0.46 – 0.93)	0.83 (0.70 – 0.97)
\$10,001 – \$20,000	0.56 (0.40 – 0.79)	0.67 (0.54 – 0.82)
<\$10,000	0.57 (0.33 – 0.68)	0.53 (0.41 – 0.68)
Health Insurance Coverage		
Yes	1.0	1.0
No	0.82 (0.78 – 0.98)	0.67 (0.50 – 0.90)

CONTINUING MEDICAL EDUCATION PROGRAMS VIA TELEMEDICINE. A. Lopez¹, S. Lazarus², G. Barker¹, E. Krupinski¹, R. Weinstein¹; ¹Arizona Telemedicine Program, University of Arizona, Tucson, AZ; ²University of Arizona, Tucson, AZ

PURPOSE: To assess cost of CME telemedicine programming and satisfaction of the participants with the session.

METHODS: A costing model was developed to determine the annual per-physician expense of obtaining CME credits by traditional means as compared to the cost of providing these services via the network. Each participant completed a satisfaction survey after attending the program.

RESULTS: The Arizona Telemedicine Program (ATP) began interactive educational programming to rural communities in Arizona in 1998. Continuing Medical Education (CME) credit is provided for conferences with existing accreditation. 2718 individual sign-ins have been recorded over 269 sessions. 5089.35 credit hours have been awarded since the program's inception. 39 sites have participated. They are located an average 4 hour travel time from ATP. On average, 10 clinicians were present during each session. Overall attendance was 38% Internal Medicine Grand Rounds (IM), 15% special broadcasts (SB), 12% Cardiology Series (CS), 12% Psychiatry Grand Rounds (PsGR), 11% satellite broadcasts (SGR), 5% Pediatric Grand Rounds (PGR), and 5% Public Health Grand Rounds (PHGR). The costing model demonstrates that providing educational services over the network are cost effective when as few as 2 physicians per site attends a session. The satisfaction survey revealed: interested in topic (96%), learned new information (96%), teaching technique conducive to learning (94%), objectives met (94%), clinically relevant (84%), very good to excellent handouts (84%), no perceived product bias (82%), very good to excellent audiovisuals (78%), very good to excellent discussion (77%).

CONCLUSION: Telemedicine provides high quality, clinically relevant and cost-effective medical updates that increase rural providers' access to educational programs.

DETERMINANTS OF UNREIMBURSED HOSPICE CARE: FINDINGS OF A NATIONAL SURVEY. K.A. Lorenz¹, K.E. Rosenfeld², S.M. Asch³, S.L. Ettner⁴; ¹VA Greater Los Angeles Healthcare System, Los Angeles, CA; ²VAGLHS, Los Angeles, CA; ³RAND, Los Angeles, CA; ⁴UCLA, Los Angeles, CA

PURPOSE: Reimbursement rates for hospice care have recently received national policy attention; however, many deaths occur among persons without insurance coverage for hospice services. We therefore sought to understand patient and agency characteristics associated with unreimbursed hospice care.

METHODS: The 1998 National Home and Hospice Care Survey (NHHCS) conducted by the National Center for Health Statistics employed the sampling frame of the 1990 National Provider Inventory. The NHHCS surveys home care and hospice agencies and includes a chart review of up to 6 randomly selected discharges from each agency's census. We conducted bivariate comparisons of unreimbursed care with patient characteristics age (categorized as $\geq$ or $<$ 65 years), gender, ethnicity, diagnosis, and agency characteristics profit status, urban / rural location, census region, and integration. Using logistic regression weighted to account for the complex sampling strategy we evaluated the independent association of patient and agency characteristics to the receipt of unreimbursed hospice.

RESULTS: In bivariable analyses, patients receiving unreimbursed hospice care were younger (48% vs 21% $<$ 65 years of age, $p < 0.001$), more likely to be nonwhite (11% vs. 4%, $p < 0.01$), have a cancer diagnosis (76% vs. 64%, $p < 0.01$), and receive care from a hospice agency that was rural (55% vs. 35%, $p < 0.001$), nonprofit (87% vs. 73%, $p < 0.001$), and freestanding (68% vs. 37%, $p < 0.001$). In multivariable analysis, age less than 65 (OR=4.55, $p < 0.001$), cancer diagnosis (OR=2.8, $p < 0.05$), and nonwhite, non-African American race were associated with greater odds of receiving unreimbursed care (OR=10.8, $p < 0.001$), as were receiving care from a freestanding hospice agency (OR=6.5, $p < 0.001$). Hospice care delivered in the in the Midwest (OR=10.5, $p < 0.001$), South (OR=5.5, $p < 0.001$), and West (OR=11.5, $p < 0.001$) were more likely to be unreimbursed than that delivered in the Northeast.

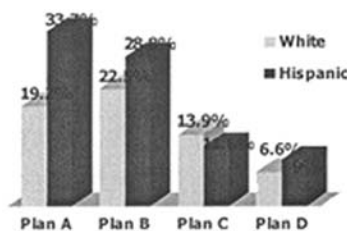
CONCLUSION: Patient demographic factors (age, race) associated with receipt of free hospice care may reflect patient insurance status. The integration of hospices with other healthcare agencies should be evaluated for its impact on the delivery of unreimbursed services. The lack of association between profit status and free care should be assessed in light of the total community benefit provided by for profit hospices.

RACIAL AND ETHNIC DIFFERENCES IN EXPERIENCES WITH MANAGED CARE: INSIGHTS FROM THE CONSUMER ASSESSMENT OF HEALTH PLANS (CAHPS). N. Lurie¹, A. Bierman¹, J. Sangi¹, C. Zhan¹; ¹US DHHS and AHRQ, Washington, DC

PURPOSE: Eliminating health disparities requires equalizing access to and quality of care for different racial/ethnic (r/e) groups. Because understanding r/e differences in patient experiences with care can help identify opportunities for intervention to achieve this goal, we assessed racial/ethnic differences in access to and satisfaction with health care and health plan among Medicare managed care enrollees and insured commercial population using a standardized instrument (CAHPS).

METHODS: We analyzed 1998 CAHPS data, a random sample of enrollees in each plan, for 142,586 people in 313 Medicare + Choice (M+C) plans and 138,896 enrollees in 245 plans participating in the Federal Employees Health Benefit Plan (FEHBP). We examined differences in CAHPS global ratings, composite scores (e.g., getting the care you need when you need it, being treated with courtesy and respect) and individual items (e.g., having a regular doctor, visit in the last year) between white's (W), Blacks (B), Hispanics (H) and Asians (A), adjusting for age, education, income, race, time in plan and health status using multivariable regression techniques and accounting for clustering within plans.

RESULTS: The largest r/e differences were seen in measures of access to care. The % with no regular doctor (7%W, 11%B, 13%H, 18%A, for M+C; 14%W, 19%B, 22%H, 30%A, for FEHBP; $p < 0.01$, adjusted) and the % who did not report a visit in the last year (11%W, 14%B, 16%H, 19%A for FEHBP, 21%W, 23%B, 24%H, 32%A for M+C; $p < .01$, adjusted) varied significantly by r/e. H-W and B-W differences within plans for % without a regular doctor ranged from 3% to 14.5%, and .4% to 6.9% ($p < .01$). For example, see figure below. Large between plan differences were also seen in ratings such as ease of access to specialists. Results for global ratings of quality and composite scores for satisfaction were mixed. In general, As had the lowest CAHPS ratings and composite scores. Percent reporting no regular doctor (Hispanics compared to whites)



CONCLUSION: Significant r/e differences exist in experiences of care for insured patients within and across health plans. Variations in the magnitude of differences in access to care plans suggest modifiable factors that health plans could target to eliminate well-documented racial and ethnic disparities in care.

END-OF-LIFE CARE: WHAT DETERMINES PHYSICIAN CONFIDENCE? J.W. Magnani¹, T.A. Raffin¹; ¹Stanford University, Stanford, California

PURPOSE: Competent medical care at the end-of-life (EOL) involves clinically managing patient's physical symptoms, acute and chronic pain, and psychological and emotional distress. Little research has examined physicians' perceptions of confidence to provide EOL care. We therefore sought to investigate physician confidence in the EOL setting and to identify demographic, clinical, and subjective factors correlated with confidence. We hypothesized that self-perceptions of confidence would differ by physician specialty and practice setting.

METHODS: This study utilized a questionnaire in which primary care physicians, cardiologists, and oncologists assessed their confidence in 4 domains of EOL and palliative care by Likert scale: communication; psychosocial distress; acute/chronic pain management; and symptom management. We specifically examined data on: demographics and practice setting (academic vs. community); clinical experience in EOL care; emotional responses to patient

deaths; disagreements over goals of care; support following patient deaths; and role perceptions in the EOL setting (emotional and spiritual counseling). 133 of 171 (78%) distributed questionnaires were returned (for primary care physicians, cardiologists and oncologists: $n = 44$, 46, and 43, respectively).

RESULTS: In univariate analysis (F-test), oncologists described significantly greater confidence in EOL communication ($p < .01$) and pain management ($p < .01$) domains. Multivariate regression models were conducted independently in the 4 confidence domains of EOL care. Emotional counseling in the setting of EOL care was significantly associated ($p < .05$) with confidence in all 4 domains. EOL experience in the last 12 months was significantly associated ($p < .005$) with confidence in the communication, pain and symptom management domains. Grieving for patients' deaths was associated with confidence in the communication ($p < .005$) and psychosocial symptom ($p < .05$) domains. Disagreements with patients, colleagues, or families over goals of care were negatively associated with confidence in all four domains ($p < .005$).

CONCLUSION: Oncologists were found to have greater confidence in EOL communication and pain management, which may be due to increased experience in these areas. EOL experience in the last 12 months, interest in providing emotional counseling to dying patients, and grieving for patients' deaths were associated with increased confidence in care of patients at the EOL. Disagreements over goals of care were associated with decreased confidence. The above factors should be considered when developing interventions for training physicians in EOL care.

DOES PARTICIPATING IN A POSITIVE CLINICAL TRIAL INFLUENCE SUBSEQUENT MEDICAL PRACTICE? S.R. Majumdar¹, W. Chang¹, P.W. Armstrong¹; ¹University of Alberta, Edmonton, AB

PURPOSE: Those with the earliest awareness of new evidence should be the investigators who generated that evidence. We hypothesized that participating in the positive SAVE trial would lead to increased adoption of ACE-inhibitors (ACE-Is) by trialists when caring for subsequent AMI patients. We also determined if participating in MDPIT, a trial of null-effect, would lead to abandonment of calcium channel-blockers.

METHODS: We performed a cross-sectional analysis of data collected for 25,886 North American AMI patients enrolled in GUSTO-1 from 659 hospitals (22 SAVE sites, 9 MDPIT sites). One-third of patients were enrolled after SAVE was published. The primary outcome measure was use of an ACE-I at discharge. We used logistic regression analyses to examine the association between prior trial participation and the use ACE-Is at discharge, adjusting for covariates at the level of the site and the patient.

RESULTS: Physicians at sites that had participated in SAVE were no more likely to use ACE-Is at discharge than those at non-SAVE sites (16% versus 15%, $p = 0.15$; adjusted odds 1.08, 95%CI 0.93 - 1.26). Although use of ACE-Is increased somewhat after SAVE was published (from 14% before to 18% after, $p < 0.001$) there was no difference in uptake between SAVE sites (3% increase) and non-SAVE sites (4% increase; $p > 0.5$ for difference) over time. Physicians who participated in either MDPIT or SAVE were less likely to use CCBs (21% versus 26% at other GUSTO-1 sites, $p < 0.01$) and more likely to use beta-blockers (61% versus 54%, $p < 0.01$).

CONCLUSION: Participating in a positive trial had little impact on subsequent adoption of that drug. If those physicians who generate evidence are slow to translate it into practice, it is unlikely that passive forms of dissemination can improve the quality of care. To accelerate adoption of new evidence further research should be directed at understanding those factors other than knowledge and awareness that influence clinical practice.

PHYSICIAN ATTITUDES VERSUS ACTIONS ON RELIGION AND SPIRITUALITY IN PATIENT CARE: A STRIKING DISCORDANCE. R. Mansfield¹, W.P. Moran¹, B. Kendrick¹; ¹Wake Forest University, Winston-Salem, NC

PURPOSE: There has been an increasing amount of research in the area of religion and spirituality and its effect on patient care and health outcomes, but how this should be incorporated into physicians' practice has been an area of debate. We evaluated the extent to which general internal medicine residents and faculty acknowledge a personal belief in spirituality and religion; and to what degree internists feel it is appropriate to incorporate religion and spirituality into health care.

METHODS: A twelve-item survey of the general internal medicine residents and faculty at Wake Forest University.

RESULTS: 87/112 physicians responded (response rate 77% of residents and 82% of faculty), and an impressive majority (90%) of respondents professed a belief in God or other higher power. 82% considered themselves to be religious or spiritual. An overwhelming 92% felt that patients' beliefs can influence their health. 71% agreed that it would be helpful to inquire about patients' religious/spiritual concerns relating to their health. However, less than half (47%) of respondents felt there is a role for direct physician involvement in meeting patients' religious or spiritual needs, and even less (43%) had actually been directly involved in patients' spiritual care.

CONCLUSION: A vast majority of respondents profess their own personal religious and spiritual beliefs, and feel that religion and spirituality can have a positive effect on health outcomes. Yet only a minority feels that doctors should actively pursue any action in this area, and even fewer have actually done this. This is a striking discordance between physician attitude and action. Given the growing amount of evidence that suggests religion and spiritual factors can affect health outcomes, efforts should be made to determine what barriers lie between these attitudes and actions, and improve physician education on how to approach these important patient-centered issues.

EFFECT OF LOW HEDIS SCORES ON PUBLIC DISCLOSURE OF SUBSEQUENT HEDIS SCORES BY HMOs. D. McCormick¹, D.U. Himmelstein¹, S. Woolhandler¹, D.H. Bor¹, S.M. Wolfe²; ¹The Cambridge Hospital / Harvard Medical School, Cambridge, MA; ²Public Citizen Health Research Group, Washington, DC

PURPOSE: HMOs' financial incentives to restrict medical care have generated public concern about the quality of care they provide. Public disclosure of quality of care data might improve public accountability, enhance informed consumer decision making and promote quality

improvement. Yet, public disclosure of this data is voluntary. We sought to determine the association between HMO quality of care and willingness to publicly disclose quality of care scores.

METHODS: We analyzed data from the National Committee for Quality Assurance's annual Quality Compass databases, including the Health Plan Employer Data and Information Set (HEDIS), a standardized set of quality of care measures (e.g. % of women > 50 who received biannual mammography). We determined how frequently HMOs that publicly disclosed HEDIS scores in 1997 or 1998 withdrew from public disclosure the subsequent year. We also assessed whether withdrawal from public disclosure was associated with being in the lowest (compared with the highest) tertile of plans ranked by HEDIS scores for 9 clinical quality measures, after adjustment for the method of data collection (administrative vs. medical record review).

RESULTS: Of the 329 HMOs which publicly disclosed HEDIS scores in 1997, 161 (48%) withdrew from public disclosure in 1998. Of the 292 HMOs which publicly disclosed in 1998 (including newly disclosing plans), 67 (23%) withdrew from public disclosure in 1999. Quality varied greatly between plans with absolute differences in HEDIS scores between lowest and highest tertile plans ranging from 14.6 to 42.3 percentage points. When HEDIS measures were analyzed individually, HMOs ranked in the lowest (compared with the highest) tertile in 1997 were significantly more likely to withdraw from public disclosure in 1998 for 7 of the 9 measures (RR=1.6 – 2.7, $p<0.05$ for all). In 1999, HMOs ranking in the lowest tertile in 1998 for 8 of the 9 measures were significantly more likely to withdraw from public disclosure (RR=1.9 – 7.0, $p<0.05$ for all). When examined by average rank across all 9 measures, plans in the lowest tertile were significantly more likely to withdraw from public disclosure in 1998 (OR=4.0; 95% CI, 2.2 – 7.3) and 1999 (OR=6.1; 95% CI, 2.4 – 15.8).

CONCLUSION: HMO's that receive low scores are more likely to stop disclosing their quality data in the subsequent year compared with higher scoring plans. Efforts to improve public accountability and quality of care through HEDIS or similar programs must be based on mandatory reporting of quality data.

DISPARITIES IN HEALTH CARE FOR PERIPHERAL ARTERIAL DISEASE PATIENTS: A NATIONAL PHYSICIAN SURVEY. M. McDermott¹, E. Hahn², D. Cella², K. Hanley², L. Odom², J. Ockene³, A. Hirsch⁴, W. Pearce¹, M. Lipsky¹, M. Criqui⁵, P. Greenland¹; ¹Northwestern Univ. Medical School, Chicago, IL; ²Evanston Northwestern Hospital, Evanston, IL; ³University of Massachusetts, Worcester, MA; ⁴University of Minnesota, Minneapolis, MN; ⁵Univ. of California, San Diego, CA

PURPOSE: Lower extremity peripheral arterial disease (PAD) is associated with a high rate of coronary heart disease (CHD). This study assessed physicians' knowledge and behavior regarding CHD risk factor interventions for PAD patients.

METHODS: The American Medical Association's national database was used to randomly select 5,221 physicians, including general internists (GI), family physicians (FP), cardiologists (CD), and vascular surgeons (VS). GI, FP, and CD were randomized to receive one of three surveys describing a typical 55–65 year old patient. One survey described a PAD patient, one survey described a patient with coronary artery disease (CAD), and one survey described a patient with no atherosclerosis (no athero). All VS received the PAD survey.

RESULTS: 51% of contacted physicians responded, of whom 73% were eligible. Of these, 79% completed the survey. Selected results are shown in the table.

Almost always prescribed...	CAD (n=422)	PAD (n=4662)	No athero. (n=386)	p value
Anti-platelet therapy	75.2%	59.3%	25.6%	0.001
Aerobic exercise	61.0%	53.3%	49.2%	0.003
Mean LDL level at which chol lowering drugs prescribed (mgs/dl)	121.6%	136.8%	50.1%	<0.001%
It is extremely important to prescribe...				
Anti-platelet therapy	62.2%	33.0%	21.5%	0.001
Aerobic exercise	43.8%	31.7%	21.5%	0.001
Chol. Lowering drug for high LDL	75.8%	48.2%	51.0%	0.001

For PAD patients, CD prescribed anti-platelet therapy (74.3%) more often than other physicians (41.2%–52.6%, $p=0.001$). CD prescribed cholesterol-lowering drugs more intensively than other physicians (123.8 mgs/dl vs. 138.2–150.4 mgs/dl, $p<0.001$). CD and VS prescribed exercise at a higher rate than other physicians (60% vs. 42%, $p=0.001$).

CONCLUSION: Reported CHD risk factor interventions are less intensive for PAD than for CAD patients. Insufficient knowledge regarding the importance of CHD risk factor interventions in PAD may contribute to suboptimal care in PAD.

PART-TIME MEDICAL PRACTICE IN THE US AND NETHERLANDS: CROSS-NATIONAL COMPARISONS. J.E. McMuray¹, M. Linzer¹, P. Heiligers², J. Douglas¹, R. Shugerman³; ¹University of Wisconsin-Madison, Madison, WI; ²Nivel, Utrecht, Netherlands; ³U. Washington-Seattle, Seattle, WA

PURPOSE: Part-time (PT) medical practice is likely to increase as greater numbers of women enter the medical work force. Using data from the Netherlands (NL), where PT is common, and the US, we analyzed 1) physician job satisfaction, 2) work life issues for PT and Full-time (FT) physicians, and 3) work force implications for the US.

METHODS: In the US, we compared personal and practice characteristics and career satisfaction of PT and FT physicians from a random, stratified sample of 5704 MDs (2326 respondents; adjusted response rate=52%). In the Netherlands, 3026 specialists and general practice physicians were surveyed regarding details of their practices and attitudes towards PT work (n=1976; adjusted response rate=66%). Data were analyzed separately for "within

country" findings; cross-national comparisons were performed when comparable data were available.

RESULTS: PT was more common in Dutch physicians (32% NL vs 12% US; $p<.001$). Dutch women were more likely to practice PT than US women (74% NL vs 20% US; $p<.001$), especially in primary care (86% NL vs 23% US; $p<.001$). Even in Dutch Internal Medicine subspecialties, large numbers of female MDs worked PT (45% NL vs 20% US; $p<.01$). PT in the Netherlands typically persisted until children were aged 18. Fifty five per cent of Dutch physicians expressed a positive attitude towards PT work. In the US PT physicians were significantly more satisfied than FT MDs with patient care issues, administration, and personal time ($p<.001$). PT US MDs reported higher global job satisfaction, significantly less stress, and a trend towards decreased burnout (18% PT vs 24% FT; $p<.09$). Should US PT practice assume proportions similar to NL, a 10% decrease in the primary care workforce could be expected.

CONCLUSION: The positive aspects of PT practice in the US and NL, the substantially higher prevalence of PT practice among women in NL, and the rapid rise in numbers of US women MDs suggest that PT practice will sharply increase in the US over the next decade. This could have major implications for the US physician workforce and for medical practice organization.

ARE PRINTED PROMOTIONAL MATERIALS FROM PHARMACEUTICAL SALES REPRESENTATIVES IN ARGENTINA EVIDENCE-BASED? R. Mejia¹, A. Avalos¹, E.J. Perez-Stable²; ¹Programa de Medicina Interna General, Hospital de Clinicas, Universidad de Buenos Aires, Buenos Aires, Capital Federal; ²University of California, San Francisco, CA

PURPOSE: Pharmaceutical sales representatives (drug reps) frequently visit 70–90% of physicians in their daily clinical practice. Many physicians consider their printed promotional materials to be an accurate source of clinical information. We evaluated a sample of the printed promotional materials distributed to physicians by drug reps to determine whether the data contained in the printed promotional materials are correct and supported by references accessible in Argentina.

METHODS: A consecutive sample of all printed promotional materials distributed by drug reps in the general internal medicine program based at a major teaching hospital of the University of Buenos Aires (Hospital de Clínicas) was collected between March 15 and April 15, 2000. Reprints and monographs were excluded. Clinical information was reviewed by two general internists and compared to information in Goodman's 9th edition textbook of pharmacology, in the pharmacology section of the UpToDate version 8.2, or in both. References cited were reviewed for correct citation and accessibility in any of the four major medical libraries in Buenos Aires.

RESULTS: Of the 64 pieces of printed promotional materials collected, 30 were randomly selected and evaluated. In 21 (70%) of the thirty promotional printed materials evaluated the therapeutic effect promoted in the advertisement appeared in the references used. Only 18 (60%) had statements supported by cited references. From a total of 131 references cited in printed promotional materials, 60 (46%) were incorrectly listed according to the International Committee of Medical Journal Editors. These references were also inaccessible. Of the 71 references correctly cited, 49 (69%) were not available in any of the four major medical libraries in Buenos Aires and 8 (11%) were available in only two of the libraries. Twenty-two references were read, and in only 12 (54%) of these, did the objective of the research study occur with the statement of the printed promotional material. Adverse drug reactions, warnings about drug interactions and drug contraindications were absent from all printed promotional materials.

CONCLUSION: The printed promotional materials distributed by pharmaceutical sales representatives in Buenos Aires are biased and provide misinformation more often than not. We recommend that practicing physicians should routinely disregard printed promotional materials as a source of clinical information.

DECREASED LENGTH OF STAY, COSTS AND MORTALITY IN A RANDOMIZED TRIAL OF ACADEMIC HOSPITALISTS. D.Q. Meltzer¹, M.N. Shah¹, J. Morrison¹, L. Jin¹, W. Levinson¹; ¹University of Chicago, Chicago, IL

PURPOSE: Hospitalist physicians who specialize in inpatient care are rapidly increasing in number but there is limited evidence from randomized trials concerning their effects on resource utilization and outcomes or how they achieve their effects. This project aims to determine the effect of hospitalists on resource use and outcomes on a general medicine service in an academic medical center and the mechanism for their effects.

METHODS: A longitudinal trial from July 1997–June 1999 with all patients admitted every fourth day assigned to teams led by hospitalist physicians (HPs) who care for inpatients 6 months per year versus teams led by non-hospitalist physicians (NHPs) who care for inpatients 1 or 2 months per year. Resource utilization was measured by length of stay and costs. Patient outcomes were measured by 30-, 60-, and 365-day mortality rates, readmission rates, reported physical function, and patient satisfaction.

RESULTS: Of 611 admissions to the general medicine service, 24.8% were to HPs and 75.2% to NHPs. Patients cared for by HPs and NHPs did not differ in age, race, gender, diagnosis mix, Charlson index, or payer mix. Average length of stay for the general medicine service was 4.7 days and average cost was \$8517. In multiple regression analysis controlling for diagnosis with DRG weight and for comorbidity with Charlson index, HPs did not have different length of stay or costs than NHPs in year 1, but HPs had 0.5 day lower length of stay ($p<0.01$) and \$740 lower costs ($p<0.01$) in year 2. There were no differences in mortality in year 1, but in year 2 HPs had lower mortality at 30 days (4.2% vs. 6.0% for NHP, $p<0.04$), and 60 days (8.8% vs. 6.8%, $p<0.07$). There were no statistically significant differences between HPs and NHPs in 365-day mortality rate, readmission rate, physical function, or overall patient satisfaction, but the trends favored HPs in all these measures. In analyses controlling for diagnosis-specific fixed effects, month of admission, total volume of patients seen by the physician to date, and total volume of patients with the same diagnosis seen by the physician to date, the effect of hospitalists on length of stay and costs, but not mortality, was explained by diagnosis-specific clinician volume to date. **CONCLUSION:** Over time, HPs decreased length of stay and costs and mortality at 30 and 60 days compared to NHPs. HPs' greater disease-specific experience treating patients with

common diagnoses partially explains their lower resource use, but does not appear to explain their reductions in mortality. Experience treating patients with a given illness may be an important determinant of resource utilization, even for common conditions. The use of HPs and other approaches that increase physician experience with specific clinical conditions may reduce resource use while maintaining or improving outcomes.

HOW DO PHYSICIANS BALANCE INDIVIDUAL AND SOCIAL CONCERNS IN THEIR USE OF ANTIBIOTICS? J.P. Metlay¹, J.A. Shea², D.A. Asch¹; ¹VA Medical Center, Philadelphia, PA; ²University of Pennsylvania, Philadelphia, PA

PURPOSE: Rising antimicrobial drug resistance has led to heightened awareness of the unique societal risks of antimicrobial drug use. Many public health and professional organizations have emphasized the importance of considering these risks in influencing the choice of antimicrobial drugs. This study assessed the degree to which societal vs. patient concerns influence antibiotic prescribing decisions in the treatment of patients with community-acquired pneumonia (CAP).

METHODS: We conducted a national survey of generalists (internists and family practitioners) and infectious disease (ID) specialists. Specific survey components included (1) preference rankings of reasons for selecting drugs in the treatment of patients with CAP, (2) knowledge and attitude questions regarding antimicrobial drug prescribing, and (3) measurement of thresholds for switching from hypothetical older drugs with increasing resistance to newer drugs with much less current resistance. Physicians were sampled from the AMA Masterfile and surveyed by mail with two follow-up mailings. Responses were calculated for the overall sample as well as stratified by clinical specialty.

RESULTS: 833 out of 1582 physicians (53%) completed the survey. Response rates did not differ across specialty, gender, or US region. Overall, physicians ranked efficacy, severity of illness, and prior experience as the top reasons for selecting antimicrobial drugs, and this ranking was consistent across specialty. Concern over contributing to rising antimicrobial resistance was ranked low by both ID specialists and generalists. Although 88% agreed that "antibiotic resistance is a major health problem" and 89% agreed that "over-prescribing antibiotics is a major cause of antibiotic resistance," only 54% agreed that "before prescribing an antibiotic, I weigh the potential benefit to the patient against the potential harm to society." In a hypothetical scenario for the treatment of an outpatient with CAP due to *S. pneumoniae*, 55% of generalists and 67% of ID specialists were unwilling to accept a 20% or greater level of resistance with an older drug before switching to a newer drug without resistance ($p=.01$). Physicians reported a higher threshold for switching to a newer antibiotic if they agreed that before prescribing an antibiotic, they weigh the potential benefits to patients against the potential harms to society.

CONCLUSION: Many physicians report that societal issues created by antimicrobial drug resistance have at most a small impact on their prescribing decisions. These attitudes were generally consistent across different response tasks and between generalists and ID specialists. Since efforts to reduce antibiotic resistance focus on increasing provider knowledge about the societal risks of these drugs, these results suggest that such efforts alone will have limited success.

EVALUATION OF TWO PRIMARY CARE INTERVENTIONS TO IMPROVE SMOKING CESSATION RATES. C. Milch¹, J. Edmunson², J. Beshansky¹, M.H. Eckman³, H.P. Selker⁴;

¹New England Medical Center, Boston, MA; ²Blue Cross Blue Shield of Massachusetts, Boston, MA; ³University of Cincinnati Medical Center, Cincinnati, OH; ⁴Tufts University, Boston, MA

PURPOSE: Primary care physicians have been the focus of efforts to improve the quality of preventive health care such as smoking cessation counseling. Many primary care practices have not implemented systematic protocols to identify patients who smoke or to encourage clinicians to provide smoking cessation counseling because such interventions are thought to be too time and effort intensive. With collaboration from Blue Cross Blue Shield of Massachusetts, we designed a project to assess the relative effectiveness, as compared with no intervention (usual care), of two brief primary care interventions on patient smoking cessation rates. We hypothesized that improved documentation would increase smoking cessation among patients.

METHODS: We performed the study at the adult primary care practice of a large urban tertiary care center. We chose interventions that could be used routinely in a busy primary care practice, and which differed in the level of patient and physician involvement. Each intervention was implemented on a geographically separate "team" consisting of 5-6 attending physicians, 2 nurse practitioners, and 3-5 residents. Three teams served as controls (usual care). The "minimal" intervention consisted of a smoking status "vital-sign" stamp, which identified smokers and documented patient smoking status. The "enhanced" intervention consisted of a simple, one page, 5-question form which identified smokers, assessed level of cessation readiness and motivation, and provided smoking cessation counseling prompts for clinicians. Outcomes were collected 8-10 months after implementation of the interventions. Patients who had identified themselves as current smokers at the beginning of the study period and who had seen a clinician at least once during the study period were contacted and their smoking status assessed by self-report. Two individuals unaware of study design or purpose performed structured chart reviews to assess clinician documentation of patient smoking status and cessation counseling.

RESULTS: Four hundred seven charts were reviewed for documentation assessment. Smoking status was documented at 84% of the visits on the minimal intervention team, 91% on the enhanced intervention team, and 49% on the control teams ($p\text{-value}<0.0005$). Cessation counseling was documented at 36% of the visits on the minimal intervention team, 47% on the enhanced intervention team and 30% on the control teams ($p\text{-value}=0.015$). The improvement in smoking status documentation was mainly noted 1) for routine and non-cardiac/pulmonary urgent care visits compared with new patient or cardiac/pulmonary urgent care visits, and 2) among nurse practitioners and a patient's primary care physician compared with residents or urgent care physicians. The trends were similar for cessation

counseling documentation. Two hundred forty-five patients completed the smoking status assessment at the end of the study period: 58% were female, the mean age was 46 years, and 40% were evenly divided between the minimal and enhanced intervention teams. There were no significant differences between groups in mean length of follow-up, or median number of visits or cigarettes smoked. A total of 33 (13.5%) patients had stopped smoking by the end of the follow-up period: 4% in the minimal intervention team, 30% in the enhanced intervention team, and 11% in the control teams ($p\text{-value}<0.0005$). The proportion of patients reporting increased awareness of the health risks associated with smoking or improved motivation to stop smoking was higher among the intervention (56 and 67%) than control teams (40 and 59%).

CONCLUSION: An enhanced intervention that uses a short questionnaire to identify patients who smoke and assess their level of cessation readiness may significantly improve smoking cessation rates compared with minimal or no intervention. Both enhanced and minimal interventions can improve physician documentation of patient smoking status and cessation counseling.

PHYSICIANS INTENT TO REMAIN IN MEDICAL PRACTICE. R.G. Miller¹, B.H. Ashar¹, K.J. Getz¹, G.P. Prokopowicz¹, N.R. Powe¹; ¹Johns Hopkins University School of Medicine, Baltimore, MD

PURPOSE: The rate of new applications to medical school in the United States has been declining over the past three years, possibly because of less interest in medicine as a career among young persons. It has been suggested, however, that even established physicians are leaving medicine for a host of reasons (e.g., financial issues, time factors). Our study attempts to qualify and describe this issue in relation to financial pressures of physicians.

METHODS: We conducted a state-wide, randomized survey of 1000 Maryland chapter members of the American College of Physicians. Surveys were mailed, with a follow-up reminder letter and survey sent at one month to non-responders. The 24-item survey collected data on physician demographics, satisfaction with income, activities to generate income outside of medical practice, and plans to remain in medicine in 5 years.

RESULTS: After initial mailings, the response rate was 40%. Mean age was 47.7 years, and 76.2% were male. Excluding the 11.9% of physicians who planned to retire, only 193 (62.1%) intended to remain in their current practice or position in 5 years. 11.6% felt they would still be in medicine, but in a different practice or setting, and another 6.5% expected to be in medical industry or management. A surprising 16.4% planned to be in a profession outside of medicine or were unsure about remaining in the medical field. In univariate analysis, factors associated with a desire to change positions were female gender, unmarried status, no dependents, generalist (vs. specialist), decrease in primary medical income over 5 years, and low level of satisfaction with primary medical income. If starting over, 25% of all responders would not have chosen medicine again as a career.

CONCLUSION: In our survey of Maryland physicians, over 35% planned to be in a different practice or position, many not related to medicine. This documents further the growing dissatisfaction with medicine among physicians, much of which is related to financial pressures. The impact of these findings on continuity and quality of patient care deserves further study.

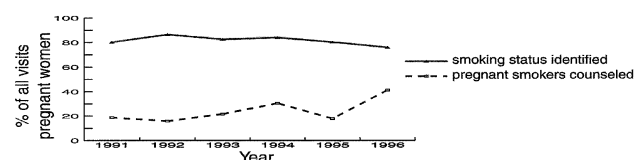
COUNSELING OF PREGNANT SMOKERS BY U.S. PHYSICIANS. S. Moran¹, A.N.

Thordike², K. Armstrong³, N.A. Rigotti³; ¹University of Pennsylvania, Philadelphia, PA; ²Partners Health Care, Boston, MA; ³Harvard Medical School, Boston, MA

PURPOSE: The effects of maternal smoking on infant mortality and morbidity are well recognized. Brief physician advice has been shown to increase quit rates among pregnant smokers and to improve pregnancy outcomes. Frequent prenatal visits offer repeated opportunities for smoking cessation counseling. We evaluated how frequently physicians identify the smoking status of pregnant patients and how frequently physicians counsel pregnant smokers.

METHODS: The National Ambulatory Medical Care Survey is an ongoing national survey of U.S. office based physicians. During the years 1991 through 1996, physicians answered questions both about patient smoking status and whether or not tobacco counseling was performed. We analyzed data from these six years to determine the frequency with which physicians identified the smoking status of pregnant patients and the frequency with which physicians counseled pregnant smokers. We compared smoking status identification and counseling rates for different specialties.

RESULTS: 9807 physicians recorded data on 5622 visits by pregnant women. The Figure presents the frequency of identification of smoking status among pregnant patients and tobacco counseling provided to pregnant smokers. Overall, smoking status was identified at 81.5% of visits by pregnant women. Smokers were identified at 563 visits and physicians provided smoking cessation counseling at 22.5% of visits by pregnant smokers. Pregnant smokers were counseled at 23.5% of visits to obstetrician/gynecologists, compared to 18.1% of visits to family practitioners and 11.0% of visits to general practitioners.



CONCLUSION: Physicians frequently identified smoking status of pregnant patients, but provided smoking cessation counseling to pregnant smokers only 22.5% of the time. These data suggest that physicians are missing an important opportunity to counsel pregnant smokers about smoking cessation. Interventions are needed to facilitate physician counseling of pregnant smokers.

COMPARING PHYSICIAN SATISFACTION WITH TWO ORDER ENTRY SYSTEMS. H. Murff¹, J. Kannry², D.W. Bates³; ¹Brigham and Women's Hospital, Boston, MA; ²Mount Sinai-NYU Health System, New York, NY

PURPOSE: Many are advocating widespread implementation of physician order entry. However, while physician order entry can improve patient safety and reduce hospital costs, it is not yet widely used. A wide variety of physician order entry systems currently exists making it important to determine which systems physicians are likely to accept. While studies have evaluated physician satisfaction with institution-specific order entry systems, few published studies have assessed commercial products, even though nationally these products will represent the majority of systems that will be implemented.

METHODS: We distributed the Questionnaire for User-Interface Satisfaction (QUIS) to Internal Medicine physician housestaff who use both a commercially available order entry system and the Veterans Affairs Computerized Patient Record System (CPRS). The QUIS is a validated questionnaire that has been used to measure user satisfaction with the human-computer interface with electronic medical records. It is a twenty-seven-item survey and questions are scored on a 0 (lowest) to 9 (highest) scale.

RESULTS: 90 and 84 surveys were returned giving a response rate of 63% and 64% for the commercially available product and the CPRS surveys respectively. Overall, residents were dissatisfied with the commercial system giving it an overall mean score of 3.67 (95% confidence intervals 3.37,3.97). In contrast, the Veterans Affairs CPRS had a mean score of 7.21 (95% confidence interval 7.00, 7.43) indicating the residents were more satisfied with the CPRS (p -value <0.0001). Overall satisfaction was strongly correlated ($\rho=0.71$) in both surveys with the ability to perform tasks in a "straight-forward" manner. Overall satisfaction was also correlated with terminology consistency ($\rho=0.60$, $\rho=0.71$) and screen sequence ($\rho=0.61$, $\rho=0.67$) for the commercial product and the CPRS respectively.

CONCLUSION: Satisfaction with the user interface for the two order entry systems differed dramatically between the two systems, suggesting that all physician order entry systems are not equally accepted. Satisfaction was highly correlated with the ability to perform given tasks in a direct and consistent manner. These data support prior studies that have found correlations between user satisfaction and efficiency with order entry systems. If maximal benefit is to be gained from the implementation of such systems further studies are needed to investigate what specific order entry features physicians find most useful and least disruptive to daily workflow.

WHAT ARE CALIFORNIA PHYSICIANS DOING TO MAINTAIN THE FINANCIAL VIABILITY OF THEIR PRACTICES? S. Mutha¹, K. Grumbach¹, E.H. O'Neil¹, C. Dower¹; ¹University of California, San Francisco, San Francisco, CA

PURPOSE: Nearly half of all California physicians still practice in solo or self-employed group practices. The recent and well-publicized demise of independent practice associations (IPA) and practice management organizations in the state has underscored the tenuous financial status of these practices. As part of a larger study about the state of physician practice in California, we interviewed community-based physicians and asked them to describe various aspects of their practice lives. We describe here the variety of approaches that physicians are taking to enhance or maintain the viability of their practices.

METHODS: We interviewed fifteen physicians throughout California who spend > 50% time providing care in office-based practices. Participants were identified using the snowball sampling method. Each individual was interviewed by telephone by a study physician. The average age of participants was 41 years (31–55), 43% were specialists, 36% were female and 30% practiced in community health center or public clinics. Only 13% expressed any dissatisfaction with their practices. Transcribed interviews were thematically analyzed using ATLAS 4.1 software.

RESULTS: Physicians in solo or small group practices were more apt to discuss financial issues related to their practices than those affiliated with larger groups or with health maintenance organizations. Specific methods for improving the financial viability of their practices included improving "customer service" for patients and referring physicians, limiting contracts with low paying insurers and or patients when feasible, lengthening work hours, shortening duration of office visits, offering supplemental patient services for out-of-pocket pay, streamlining office processes and staffing, optimizing billing and using physician extenders. A few physicians described colleagues whose response was to leave practice for administrative positions or early retirement.

CONCLUSION: Physicians are experiencing financial pressures that in California have been amplified by the financial insolvency of IPAs and other practice organizations. The predominant response to improve the financial viability of their practices has been more efficient and effective utilization of existing resources (including personnel). Less common approaches include providing supplemental services that patients are willing to pay for out-of-pocket and leaving clinical practice. These experiences, to some degree, mimic those experienced by other professions such as dentistry. The frequency with which the approaches are used and their implications for access, patient outcomes and physician workforce requires further study.

VIRGINIA'S FREE CLINICS: VALUABLE ASSETS FOR UNINSURED CARE. M.M. Nadkarni¹, M. Cruise², J.T. Philbrick¹; ¹University of Virginia, Charlottesville, VA; ²Virginia Association of Free Clinics, Lynchburg, VA

PURPOSE: Free clinics have earned a place among the safety net providers for the uninsured. Virginia has the greatest number of free clinics of any state. We evaluated the type and dollar value of medical services provided in 1999.

METHODS: Survey completed by the directors of all 32 clinics registered in 1999 with the Virginia Association of Free Clinics. Dollar value of services provided was estimated to be the "usual and customary" charges for the region of each clinic; prescription value was estimated to be \$45, the average retail price of a prescription for Virginia in 1999. Operating expenses were expenditures for clinic administration, fund raising, and direct services, excluding capital expenses.

RESULTS: See table.

CONCLUSION: Free clinics in Virginia effectively leveraged \$6.8 million operating expenses into \$24.5 million worth of medical services. The average expense per visit, including prescriptions and health care provider services, compares favorably to private practice office benchmarks. Although many patients were served, they represented only the tip of the iceberg in Virginia in 1999, comprising only 4% of the approximately one million uninsured. While free clinics provide an outlet for patients who might not receive care otherwise, physicians and policy makers must continue to work for a more comprehensive solution.

Survey Results — 32 Clinics

	Total All Clinics	Clinic Median	Range
Total Patients	37,715	896	45–5,247
Total Visits	107,896	2,454	90–16,427
Primary Care	82,521	1,864	81–14,228
Specialty Visits	16,986	356	0–3,773
Dental Visits	8,479	175	0–1,922
Total Visit Value	\$7,785,857	\$162,448	\$7,085–1,061,025
Prescriptions (RX)	282,590	6,602	112–37,431
Total RX Value	\$12,716,550	\$297,090	\$5,040–1,684,395
Donated Lab Value	\$4,014,626	\$100,616	\$7,669–606,561
Total Services	\$24,517,034	\$643,099	\$13,295–2,774,508
Operating Expense	\$6,843,569	\$136,800	\$2,176–816,033
Expenses/Visit	\$63.43	\$66.01	\$10.41–160.40

MEDICALLY UNDERSERVED PATIENTS: DO THEY GET WHAT IS COMING TO THEM? M. Nadkarni¹, D. Hoffman¹; ³University of Virginia, Charlottesville, VA

PURPOSE: To ascertain how many patients in a medically underserved population receiving continuity care at a University Clinic actually qualify for additional health and social assistance benefit programs but are not yet enrolled.

METHODS: A random sample of 200 medically indigent patients at University Medical Associates, a combined resident-faculty practice with 9000 active patients which provides care to a primarily underserved population in central Virginia, was selected for analysis of healthcare benefits they were eligible for but were not currently receiving. Computer records of demographic and insurance status were reviewed and compared to relevant qualification criteria for Medicare, Medicaid, Medicaid QMB, General Assistance, Food Stamps, or Virginia's Children's Health Insurance Program (CMSIP) Programs.

RESULTS: 74 patients had no billable insurance, 19 were on Medicaid, 26 had private insurance and 63 had Medicare B benefits. (insurance status of 18 patients was questionable or unclassifiable).

OF 74 UNINSURED PATIENTS:

- 40 (53.7%) definitely were eligible for at least 1 other benefit
- 9 (12.6%) were definitely ineligible for any other benefits
- 25 (33.7%) needed further assessment but could potentially qualify for other benefits

OF 63 MEDICARE PATIENTS

- 25 (39.7%) definitely were eligible for at least 1 other benefit
- 38 (60.3%) were ineligible for other benefits

CONCLUSION: Many patients without health insurance are actually eligible for assistance programs and potentially could receive some form of healthcare insurance. A significant number of patients in underserved populations who are already covered by Medicare may still qualify for additional benefits but have not accessed them. Aggressive attempts to overcome barriers for enrollment in important assistance programs could substantially increase the welfare of many patients in indigent or uninsured populations. Interventions to increase enrollment of eligible patients and their families should be undertaken on national, state, and county level. Additionally, individual clinics and practices can help increase enrollment with relatively simple procedural measures.

BENEFIT ELIGIBILITY

	Eligible	Ineligible	Possibly Eligible
UNINSURED	40 (54.6%)	9 (12.6%)	25(32.8%)
MEDICARE	25 (39.7%)	38 (60.3%)	

SURGEON VOLUME OF EARLY STAGE BREAST CANCER CASES. A. Nattinger¹, R. Hoffmann¹, R. Sparapani¹; ¹Medical College of Wisconsin, Milwaukee, WI

PURPOSE: The percentage of U.S. women with early stage breast cancer who receive initial treatment in accordance with national guidelines has been declining. We are interested in whether physician volume of cases is related to processes or outcomes of care, and therefore determined for a population-based cohort the distribution of cases cared for by each surgeon operating on one or more cohort patients.

METHODS: We used data from a population-based SEER-Medicare linked data base to study women aged 65 or older, who were diagnosed with a stage I or II breast cancer in 1993–95, and underwent either breast-conserving surgery (BCS) or mastectomy. We determined the operating surgeons from the Medicare part B claims, and eliminated surgeons with an office zip code outside the SEER states.

RESULTS: Over the three study years, 15,574 women were operated on by 3,265 different physicians. The median number of cases performed per physician per year was 1.0 with 1st/3rd quartiles of 0.0/4.0. Over three years, 35.3% of physician-years were zero-case years. We

categorized physicians based on 3-year volume of operations, and determined the average percentage of the patients undergoing BCS (as opposed to mastectomy) for each volume category. As seen in the table, the higher volume surgeons had greater use of BCS ($p < 0.0001$). **CONCLUSION:** The volume of potentially curative breast cancer operations per surgeon per year was quite low in this population-based cohort, even when considering the fact that women aged 65 and older account for only about 45% of all breast cancer diagnoses. Volume of cases was directly related to use of BCS, which raises the question of whether volume may be associated with other outcomes of care.

Volume vs. BCS Use

3-year Volume of Operations	Physicians (N)	BCS (%)
1–3	1586	38%
4–8	659	50%
9–15	425	57%
> 15	600	63%

RACE, PROVIDER RECOMMENDATIONS AND USE OF CAROTID ENDARTERECTOMY. E. Oddone¹, R. Horner², K. Stechuchak², L. McIntyre²; ¹Duke University, Durham, NC; ²VAMC, Durham, NC

PURPOSE: The degree to which provider recommendations influence racial disparities in receipt of invasive procedures has been understudied. Prior research has shown that black patients are only one-third to one-fourth as likely as white patients to receive carotid endarterectomy (CE). This study was designed to determine if a patient's race influenced provider beliefs and recommendations for referral to either carotid angiography (CA), CE, or subspecialty care.

METHODS: Providers at five VAMC's (Atlanta, Durham, Pittsburgh, Richmond, and St. Louis), who evaluated patients after an index carotid ultrasound, were surveyed at the time of their patients' ultrasound. Patients were comprised of a prospective, racially stratified, cohort who were undergoing evaluation for carotid artery stenosis. For study providers, we assessed beliefs about their patient's operative risk, anticipated efficacy of CE, and their final recommendation regarding CA and CE, or referral. For all patients, we determined receipt of either of these procedures within 6-months of the ultrasound.

RESULTS: Between 1997 and 1999 we enrolled 719 patients (625 white and 94 black) who had at least one carotid artery with greater than 50% stenosis. Black and white patients were similar in terms of age, gender, education level, social support, and trust in their physicians. More black than white patients received their ultrasound for a completed stroke (35% vs. 13%), and fewer black patients were classified as asymptomatic or as having had a TIA ($p < 0.001$). Provider surveys were completed for 634 of the 719 enrolled patients (88%). Providers (N=183) included general internists (44%), PAs or NPs (21%), other medical specialists (15%), vascular surgeons (11%), and neurologists (9%). Most providers were white (74%), 5% were black. On average, providers felt that black patients were at greater risk for complications from CA (mean 5% vs. 3%, $P=0.05$), and would have a greater operative risk during CE (46% high-risk category vs. 34%, $P=0.03$) than were white patients. There was no difference in providers' recommending black or white patients for CA, CE or referral to subspecialty care (54% vs. 48%, $P=0.34$). In adjusted analyses accounting of baseline clinical status and other factors, race was not found to be an independent predictor for any provider recommendation. During the six-month follow-up for patients whose providers recommended CA, CE or referral, 53% of whites received either CA or CE compared to 46% for blacks ($p=0.50$).

CONCLUSION: We failed to find evidence of racial bias in provider recommendation after directly questioning providers of patients who were potentially eligible for CA or CE. Race appears to have little influence and clinical status dominates in referral or receipt of these procedures for patients in this equal access healthcare system.

WHEN DO GUIDELINES GO OUT-OF-DATE? A SURVIVAL ANALYSIS. E. Ortiz¹, S. Rhodes², S. Morton², M.P. Eccles³, J.M. Grimshaw⁴, S. Woolf⁵, P.G. Shekelle²; ¹VA San Diego Healthcare System and University of California, San Diego, CA; ²Southern California Evidence-Based Practice Center, Santa Monica, CA; ³University of Newcastle upon Tyne, United Kingdom; ⁴University of Aberdeen, Scotland; ⁵Virginia Commonwealth University, Richmond, VA

PURPOSE: Considerable effort and resources are expended on the development and dissemination of guidelines. Guidelines must be based on the best available evidence if they are to be useful to clinicians. Yet information about the rate at which guidelines become outdated is not known. We undertook a systematic assessment of the AHCPR's clinical practice guidelines to evaluate their current validity and then used this information to develop estimates of the rate at which guidelines become outdated.

METHODS: We developed a conceptual model for assessing the validity of existing guidelines using literature searches and expert guidance. Two physicians with expertise in literature retrieval and evaluation conducted limited literature searches for new evidence that could affect the validity of the guideline statements. We also searched for practice guidelines that had been published after the release of the AHCPR guidelines. Guideline statements were sent to original guideline panel members and other content experts, who were asked to evaluate the current validity of each statement and include evidence to support their assessment. The evidence collected was then used to determine the validity of the guidelines, which were classified as "still valid" or "needs updating." Finally, we conducted a "survival analysis" to estimate the rate at which guidelines become outdated using a Kaplan-Meier survival curve and the Weibull parametric model.

RESULTS: We reviewed 17 guidelines and surveyed 170 experts, 121 (71%) of whom returned their surveys. Our literature searches identified 6,994 titles, which were all reviewed.

We reviewed the abstracts of 610 of these articles and then narrowed our selection down to 173 full-text articles. We also reviewed 159 guidelines. There were 13 (76%) guidelines in need of an update, 3 (18%) guidelines that were still valid, and 1 (6%) guideline for which we could not make a determination due to insufficient information. The guidelines had an estimated "shelf-life" of 3 years (90% of the guidelines were still valid) to 5.9 years (upper bounds of the 95% CI).

CONCLUSION: Our study provides the first empirical evidence of the rate at which guidelines go out-of-date. More than 75% of the AHCPR guidelines are in need of updating. We recommend re-evaluation at a minimum every three years with modifications in this timeline based on whether advances in the field are rapidly or slowly evolving. Future guidelines could facilitate the updating process by including an estimate of the type of evidence necessary to prompt revisions.

POLICIES FOR HUMAN IMMUNODEFICIENCY VIRUS SCREENING IN DEPARTMENT OF VETERANS AFFAIRS HOSPITALS. D.K. Owens¹, G. Sanders², V. Sundaram¹, L.C. Lazzaroni², E.M. Yano³, P. Tempio¹, B. Phillips⁴, V. McWhorter⁴, L.R. Douglass¹, S.A. Bozzette For The VA Queri HIV⁴; ¹VA Health Care System, Palo Alto, CA; ²Stanford University, Stanford, CA; ³Department of Veterans Affairs, Sepulveda, CA; ⁴VA San Diego Healthcare System and the UC San Diego Dept of Medicine, San Diego, CA

PURPOSE: The Department of Veterans Affairs (VA) cares for 18,000 patients with human immunodeficiency virus (HIV) infection, making it the largest provider of HIV care in the U.S. Early identification of HIV infection is critical if patients are to receive the maximum benefit from therapy. As part of the VA HIV Quality Enhancement Research Initiative, we surveyed VA facilities to determine whether they had formal policies for identification of HIV infection.

METHODS: We used a paper-based questionnaire to survey the lead clinician for HIV care at 139 VA facilities with 20 or more HIV patients. The questionnaire addressed screening and testing policies for HIV infection in distinct patient groups (primary care, mental health, etc). Facilities that did not respond were re-contacted. We used logistic regression to determine whether facility size (rated on a 1 to 4 scale with 1 being largest and more complicated centers), complexity (weighted score from 1 to 100), HIV prevalence, or location in an urban area predicted the presence of screening policies.

RESULTS: Of the 139 facilities surveyed, 115 responded (83%). Forty-six facilities (40%) had written policies for screening for chronic HIV infection in at least one patient group; 44 (38%) facilities had no policy for any group. Only 14 facilities (12%) had a policy for detection of acute HIV infection in any group. Over 90% of facilities had written policies on informed consent for testing, and 80% of facilities had policies on pre- and post-test counseling. In a multivariate analysis, neither facility size, complexity, location, nor HIV prevalence significantly predicted the presence of screening policies.

CONCLUSION: Although the vast majority of VA hospitals have policies about how HIV testing should be done, less than half of VA facilities have policies about who should be screened. Only a few VA centers have policies regarding detection of acute HIV infection. Development of such policies may facilitate appropriate screening for HIV infection in the VA.

AFTER DISCHARGE EFFECTS OF A HOSPITALIST SERVICE. H. Palmer¹, M. Delamata¹, C. McBride¹, T. Dunsworth¹, K. Evans¹, G. Hobbs¹; ¹West Virginia University, Morgantown, WV

PURPOSE: Critics state that some of the cost reductions demonstrated by hospitalist services may be explained by "cost shifting" to the outpatient setting. Timely and appropriate patient follow up is also of concern. The purpose of our study was 2 fold: 1)determine if a significant portion of the cost savings realized by our hospitalist service could be attributed to "cost shifting" and 2)determine if follow up issues were addressed as an outpatient.

METHODS: During the 1998–99 academic year, WVU implemented a hospitalist service (H) which was compared with the two other medicine services, one staffed by General Internists (GIM) and the other by subspecialists (SS). Inpatient cost (COST) and length of stay (LOS) were the main outcomes. 956 charts (321 for H, 257 for GIM, 378 for SS) were retrospectively analyzed in a blinded fashion to determine the following for each group: 1) percentage of patients discharged with workups needing to be performed (% W/TJ) 2) percentage of patients with tests or procedures scheduled as outpatient (% TEST) 3) average costs of test/procedures specified at discharge (F/U COST)(elective surgeries excluded). Of patients who had follow up within our system, 25 charts in each group were reviewed to determine follow up rates (% F/U).

RESULTS: See table. Among the 3 groups, there were no significant differences in the percentages of patients discharged with test procedures or workups pending ($p > 0.6$). Patients who had longer LOS or who were followed in our system tended to have more outpatient test and workups scheduled ($p < 0.01$). In terms of cost specified at discharge, there were no significant differences ($p = 0.09$). There were no significant differences in percentage of issues followed up, however the numbers were small ($p = 0.2$).

CONCLUSION: "Cost shifting" does not appear to explain the cost benefits realized within our hospitalist system. Most of the issues specified at discharge tend to be followed up in the outpatient setting.

Service (N)	COST \$SD	LOS (SD)	%TEST	F/U COST\$ (SD)	%W/U	%F/U
H (829)	4290 (6512)	4.4 (4.0)	38.9	135 (517)	5.9	79
GIM (716)	4850 (7027)	5.2* (5.2)	35.0	61 (280)	5.5	72
SS (874)	6066* (7550)	6.0* (5.5)	37.0	108 (382)	5.6	92

* $p < .01$ compared with H

DID MANAGED CARE SAVE US MONEY? S. Pati¹, S. Shea², O. Carrasquillo²; ¹Columbia University, Pediatrics and Medicine, New York, NY; ²Columbia University, Medicine, New York, NY

PURPOSE: Health insurance offered by managed care organizations (MCOs) has lower premiums than fee-for-service (FFS) plans. Thus, it is widely assumed that health care expenditures for patients enrolled in MCOs are lower than for patients in FFS yet few studies have examined this issue.

METHODS: To compare per capita health expenditures among people enrolled in MCOs versus those in FFS we analyzed data from the 8207 adults age ≤ 65 sampled in the 1996 Medical Expenditure Panel Survey (MEPS). This nationally representative survey calculates expenditure data based on information provided by patients, health care providers, and hospitals. The design of MEPS allows comparison of expenditures for patients in MCOs and FFS plans. Total expenditures included all sources of payment (i.e. out-of-pocket, third party payers, and other sources). MCO enrollees were those in health maintenance organizations or plans requiring a gatekeeper. We excluded adults with public health insurance. We used linear regression to obtain expenditure data (log transformation) adjusted for age, race/ethnicity, gender, education, poverty status, functional status, and self-perceived health status.

RESULTS: 55% of privately insured adults in MEPS were enrolled in MCO plans versus 45% in FFS. Adults in FFS were more likely to be non-Hispanic white (NHW) ($p < 0.05$). Older adults, females, NHWs, and those in fair/poor health had greater per capita health expenditures ($p < 0.05$). Mean overall health expenditures in 1996 were \$1803 (median \$563) per adult in MCOs vs. \$1820 (median \$478) in FFS plans ($p > 0.05$ [NS]). Mean inpatient expenditures were \$460 per adult in MCOs vs. \$473 in FFS and mean ambulatory expenditures were \$466 (median \$140) per adult in MCOs vs. \$448 (median \$99) in FFS [NS]. Analysis by source of payment showed that per capita expenditures by private insurers tended to be greater for MCO enrollees \$1376 (median \$323) than FFS \$1271 (median \$166) [NS]. After multivariate adjustment for demographic variables, expenditures for adults in MCOs were significantly greater than in FFS ($p < 0.01$, model $R^2 = .12$).

CONCLUSION: In 1996, annual per capita expenditures in managed care plans were not lower than in fee-for-service plans. Despite similar expenditures, MCOs appeared to charge lower premiums than FFS plans, perhaps to gain market share.

EXPANDING THE LIKELIHOOD FUNCTION IN CLINICAL DECISION MAKING. J.C. Peirce¹, R.M. Hoffman²; ¹Good Samaritan Reg Med Cntr, Phoenix, AZ; ²VAMC, Albuquerque, NM

PURPOSE: Likelihood functions are necessary for clinical decisions where several hypotheses are considered in a differential diagnosis with fixed case findings in contrast to probabilities where the disease is fixed and findings are variable. The likelihood ratio (LR) is defined as: posterior odds / clinical finding $\div$ prior odds, or $\ln LR = \ln$ posterior odds $-\ln$ prior odds.

When the posterior odds given the clinical finding equals the prior odds then the value for the $\ln LR$ of the finding is zero, leading Edwards to define $\ln LR$ as the *informational value of the finding in support of the hypothesis*. LRs define the slopes of a trapezoidal receiver operating characteristic (ROC) curve that always lie under the asymptotic maximum likelihood estimation (MLE) of the ROC curve. Using a single cutpoint, the trapezoidal area under the ROC curve (AUROC) usually is significantly less than the MLE AUROC. The purpose of this study is to determine the utility of creating greater than two strata of likelihood ratios when continuous or ordinal data exist so that the trapezoidal AUROC approximates the MLE AUROC.

METHODS: We reanalyzed two published reports: (1) a prospective study looking at uptake of fluorine 18-deoxyglucose (FDG) by positron emission tomography (PET) in solitary pulmonary nodules - found indeterminate on chest radiography or CT evaluation, and (2) a nested case-control study in a large cohort of men looking at prostate specific antigen in diagnosing aggressive and non-aggressive prostate cancer. The methods of Peirce and Cornell were used to determine optimum stratum-specific likelihood ratios and the trapezoidal AUROC, and the methods of Dorfman and Alf were used to determine the MLE AUROC.

RESULTS: In a 9 center study, Lowe et al evaluated 89 patients with solitary pulmonary nodules using FDG uptake by PET. They reported a 92% sensitivity and 90% specificity using a cutpoint of 2.6 standard uptake value (SUV) for FDG. Calculating a MLE AUROC (s.e.) of 0.969 (0.018), we found a trapezoidal AUROC of 0.907, 3.44 standard errors (se's) below the MLE AUROC. We developed four distinct strata, one strongly suggestive of a pathognomonic stratum (FDG-SUV > 5.0) and another weakly suggestive of an absolute negative stratum (FDG-SUV < 1.0) with a trapezoidal AUROC of 0.944; 1.39 standard errors below the MLE AUROC. Going from 2 to 4 strata changed the odds of reaching a correct answer from 9.7 to 16.8. Gann et al reported 183 cases of aggressive prostate cancer and 549 controls as part of the Physicians Health Study. At the usual cutpoint of 4.00 ng/mL the trapezoidal AUROC was 0.757, 3.63 se's less than MLE AUROC (s.e.) of 0.826 (0.019). Adding a stratum above 10ng/mL increased the trapezoidal AUROC to 0.767 and the upper LR from 6.12 to 21.8, and adding a stratum below 2.0 ng/mL decreased the lowest LR from 0.43 to 0.29 and made the trapezoidal AUROC equivalent to MLE AUROC. Going from 2 to 4 strata increased the odds of reaching a correct answer from 3.29 to 4.75.

CONCLUSION: Important information is obtained that otherwise would be overlooked in tests with continuous or ordinal variables by developing multiple strata, each with their own LRs so that the trapezoidal AUROC approximates the MLE AUROC.

ACCESS TO CLINICALLY NECESSARY ANGIOGRAPHY IN VA COMPARED WITH MEDICARE: A NATIONAL COMPARISON OF PROCESS AND 3-YEAR SURVIVAL. L.A. Petersen¹, S. Normand², L. Leape³, M. Volja², B.J. McNeil²; ¹Houston VAMC, Houston, Texas; ²Harvard University, Boston, Massachusetts; ³Harvard School of Public Health, Boston, MA

PURPOSE: To compare access to clinically necessary angiography in VA with traditional fee-for-service care under Medicare financing, we used clinical data collected by chart review from two cohorts discharged with acute myocardial infarction (AMI).

METHODS: The first cohort was a stratified national random sample of male veterans age ≥ 65 discharged from VA hospitals with the primary diagnosis of AMI (ICD-9-CM-410). The second cohort contained all males age ≥ 65 discharged with AMI from the states of CA, FL, MA, NY, OH, PA, and TX under Medicare financing. Data on comorbid conditions and severity of AMI were collected on both samples using the Cooperative Cardiovascular Project data collection instrument. We studied 1,631 veterans in 81 VA hospitals and 19,114 Medicare patients in 1,530 non-VA hospitals who were eligible for angiography more than 12 hours after admission but prior to discharge. We compared use of angiography, revascularization procedures, and 1- and 3-year mortality among all patients and those rated clinically Necessary for angiography using established criteria.

RESULTS: Among all patients eligible for angiography 12 hours or more after admission, 48.9% of Medicare and 38.3% of VA patients underwent angiography during the index admission ($p = 0.005$). Of those who underwent angiography during the index admission, 33.9% of Medicare and 25.3% of VA patients underwent percutaneous transluminal coronary angioplasty (PTCA). For coronary artery bypass graft (CABG) surgery, the rates were 27.1% of Medicare and 15.4% of VA. For any revascularization the rates were 59.0% of Medicare and 39.4% of VA (all $P < 0.01$). According to modified RAND criteria, 28.4% of Medicare patients and 43.4% of VA patients met criteria for Necessary angiography prior to discharge. Medicare patients who were rated Necessary for angiography were more likely than the VA patients who were rated as Necessary to undergo angiography (60.5% vs. 47.3%; $P = 0.001$). The rates of revascularization procedures for these patients during the index admission, conditional upon undergoing angiography, were 33.1% of Medicare vs. 29.5% of VA for PTCA ($p = 0.58$), for CABG, 32.3% of Medicare and 19.1% of VA, and for any revascularization the rates were 62.5% Medicare and 46.6% VA (both $P < 0.001$). Despite these significant differences in procedure rates, there was no significant difference between the two groups in 1-year mortality (19.5% of Medicare vs. 19.1% of VA Necessary patients; $P = 0.63$). Using pairs of matched Necessary VA and Medicare patients to control for illness burden, there was no significant difference in use of angiography (McNemar $P = 0.52$). As in the unadjusted analyses, we also observed no significant difference in 1-year mortality (McNemar $P = 0.20$) or 3-year mortality (McNemar $P = 0.59$). Our findings did not change when we substituted ACC/AHA guideline clinical criteria for RAND ratings to judge clinical need for angiography.

CONCLUSION: These findings suggest that practice pattern differences in use of angiography are being driven more by patient characteristics than by organizational differences between VA and non-VA care. Reassuringly, such differences in practice are not accompanied by long-term mortality differences at up to 3 years after accounting for dissimilarities in population risk between the two cohorts.

PHYSICIAN RESPONSES TO LIFE-THREATENING DRUG-DRUG INTERACTION ALERTS. J.E. Peterson¹, G.J. Kuperman², C. Shek², D.W. Bates¹; ¹Brigham and Women's Hospital, Boston, MA; ²Partners Health Care, Boston, MA

PURPOSE: Drug-drug interactions have potential for harm, yet physicians are often unaware of them. While alerts about life-threatening drug-drug interactions are integrated into many medical information systems, few data are available regarding whether these warnings affect prescribing behavior or prevent patient injury.

METHODS: We examined all consecutive cases of alerts warning of life-threatening drug-drug interactions during a six month period at a 700-bed academic tertiary care hospital. The alerts examined were the 7 with the most severe consequences from a database of known drug-drug interactions. Data were automatically collected each time a physician entered an order that triggered an interaction alert. The hospital information system logged the drug pair involved in the interaction, whether the alert was overridden, and any subsequent orders to discontinue one of the drugs in the pair. Adverse events attributable to the drug interaction were ascertained by chart review.

RESULTS: Over 6 months, there were 168 life-threatening drug-drug interaction alerts (see table). After 65(43%) of the alerts, physicians immediately discontinued one of the medications. For the 85(57%) alerts where both medications were given despite the alert, there were 2 adverse drug events, both involving the drug cisapride, and neither resulted in subsequent disability.

CONCLUSION: Physicians overrode alerts about life-threatening drug-drug interactions more than half the time. When they did so, few adverse drug events occurred. These data suggest that the severity level of some drug-drug interactions may be set too high. For the small number of drug pairs that should never be given jointly, stronger safety measures should be put in place.

	Ratio (%) of Alerts Overridden
Cisapride-phenothiazines	25/65(38%)
Cisapride-azole antifungals	6/9(67%)
Gemfibrozil-HMG CoA inhibitor	46/64(72%)
Allopurinol-azathioprine	7/10 (70%)
Dextroamphetamine-selegiline	2/2 (100%)
Meperidine-ritonavir	0/4 (0%)
Pergolide-phenothiazines	0/2 (0%)

FACULTY EXPERIENCE WITH RACIAL BIAS IN ACADEMIC MEDICINE. N.B. Peterson¹, R.H. Friedman¹, S. Franco¹, P.L. Carr², A.S. Ash¹, M.A. Moskowitz¹; ¹Boston University, Boston, MA; ²Massachusetts General Hospital, Boston, MA

PURPOSE: To determine the prevalence of self-reported experiences of racial bias among medical school faculty and to determine the factors associated with experiencing bias.

METHODS: We mailed surveys in 1995 to a stratified sample of 3332 medical faculty at 24 randomly selected U.S. medical schools. The sample was stratified by sex, year of graduation from medical school, and medical school department and supplemented to include all faculty

who were minorities, primary care physicians, or female associate and full professors. 60% of eligible faculty responded. We divided faculty into 3 racial categories as defined by the AAMC: underrepresented minority (URM) - all Blacks, Native Americans/Alaskan natives, Cuban and Puerto Rican Hispanics], non-underrepresented minority [(NURM) - all other Hispanics and Asians], and majority faculty. We defined bias as a "yes" response to, "In your professional career, have you personally encountered racial/ethnic discrimination (unfair or injurious distinction or treatment) by a superior or colleague?". Bivariate and logistic regression analyses were performed to determine factors associated with experiencing racial bias.

RESULTS: 1833 medical faculty were analyzed; 10.1% URM, 7.7% NURM, and 82.2% majority. For URM, NURM, and majority faculty, mean ages were 44.1, 44.3, and 46.5 years and percent male were 59.5%, 39.7%, and 51.2%, respectively. In unadjusted analyses, faculty were more likely to perceive racial bias if they were minority faculty [URM (48.1%) and NURM (26.2%) vs. majority (6.9%)], were born in countries other than the U.S., had a primary language other than English, or were in medical or surgical sub-specialties. After adjustment for age, sex, being a member of a surgical specialty department, and primary language, URM and NURM faculty were significantly more likely to perceive racial bias [(OR 13.0, 95% CI (9.0, 18.7)), (OR 4.6, 95% CI (2.9, 7.3)), respectively]. Also significantly associated with perception of racial bias were age [per year, OR 1.024, 95% CI (1.01, 1.04)] and being a member of a surgical specialty department [OR 1.5, 95% CI (1.02, 2.17)], but not marital status, country of birth, departments other than surgical specialties, and rank.

CONCLUSION: Minority faculty, both URM and NURM, perceive significant racial bias in their professional careers. The reasons for this need to be better understood so that improvements in the academic environment for faculty, staff, and trainees can be pursued by medical schools and teaching hospitals.

IS HEALTH PLAN QUALITY PERFORMANCE ASSOCIATED WITH WITHDRAWAL FROM MEDICARE MANAGED CARE? H. Pham¹, K.D. Frick¹, M. Diener-West¹, H.R. Rubin¹, N.R. Powe¹; ¹Johns Hopkins University School of Medicine, Baltimore, MD

PURPOSE: Withdrawals of managed care plans from Medicare have affected over 700,000 people. Some plans claim that providing higher quality care raises costs and lowers profits, and that plans withdraw because they cannot sustain high quality care under current payment levels. We performed a study to assess whether higher quality performance by Medicare managed care plans was associated with withdrawal.

METHODS: Taking each county where a contract was active as a unit of analysis, this was a cohort study of Medicare managed care plans active in 2310 contract-county combinations in 1997 and followed for three years. We measured quality using fourteen indicators from the Health Employer Data and Information Set which we collapsed into four summary measures. We used a separate Cox proportional hazards regression for each indicator and each summary measure to assess the association of high quality care with time-to-withdrawal from Medicare while adjusting for market, organizational, environmental, and demographic factors.

RESULTS: Of 2310 managed care contract-county combinations, 877 (38%) withdrew. The withdrawal rate among plans with high scores on a summary clinical quality measure was one-fifth that for low scorers (4.2% v. 20.5%). For the summary ambulatory care access measure, the corresponding ratio was two-fifths (12.8% v. 32.0%). Lower payment rates were associated with higher withdrawal risk, contrary to plan claims, with higher clinical and ambulatory care access quality performance. In separate multivariable analyses, both high summary clinical performance (HR 0.18, 95%CI 0.08-0.42), and high summary ambulatory care access performance (HR 0.53, 95%CI 0.27-1.07) were independently associated with lower withdrawal risk. There was no independent association between either physician qualification or primary care provider stability measures and withdrawal.

CONCLUSION: Managed care plans that continue to provide care to Medicare beneficiaries have higher average performance on clinical and ambulatory care access measures than plans that withdrew.

PREDICTORS OF PRIMARY CARE PHYSICIAN CHANGE IN A MANAGED CARE SETTING. A. Pichon-Riviere¹, A. Rubinstein²; ¹Programa de Efectividad Clínica, Capital Federal, Buenos Aires; ²Medicina Familiar - Hospital Italiano, Capital Federal, Buenos Aires

PURPOSE: Continuity and longitudinality of care are strong determinants of utilization of services and health outcomes in a managed care setting. In this study we analyzed different patient's characteristics, and attempted to identify the predictors of change of primary care physician (PCP) in this population.

METHODS: The study took place in the HMO of a large teaching hospital. PCPs have a defined panel of patients who can change their PCP without restrictions. Data were collected on all consecutive patients 16 years of age or older enrolled in the HMO from May 1997 to June 1999 and personally registered with one PCP. Patients were followed until December 1999. Our outcome of interest was time (in days) to PCP change after patient enrollment. Patients who did not change their PCP during the study period contributed person-time until they were censored at the end of follow-up or until they disenrolled from the HMO. Using univariate Cox proportional hazards modeling, we identified significant predictors of time to PCP change. We selected the best multivariate model by fitting a Cox model using forward stepwise selection.

RESULTS: 9,839 patients fulfilled the study criteria. The mean age was forty-five years (SD 20.3), sixty three percent (63.39%) were females. From the seventy-four PCPs included, nineteen (25.7%) were females, forty-three were family physicians (58.1%) and thirty-one (41.9%) general internist. Twenty-eight (37.8%) of them had more than ten years of practice and thirteen (17.6%) were residents. The median follow-up was 16 months and a total of 1002 patients (10.18%) changed their PCP at least once during the study period. This represented a hazard of 7.46 changes per 100 person-year. Fifty percent of the changes were observed during the first seven months of enrollment. The hazard of change increased during the first three months and then declined continuously during the remaining follow-up period. Nine predictors remained significant in the multivariate Cox model: high utilizer of physician visits (HR 1.95), high use of emergency room (HR 1.27), previous change (HR 1.76), psychosocial problems (HR 1.52), functional disorders (HR 1.38), minor chronic

conditions (HR 1.44), mayor chronic conditions (HR 1.29), female physician (HR 1.31) and family physician PCP (HR 0.79).

CONCLUSION: The hazard of 7.46 changes per 100 person-year seems to be high although it is difficult to evaluate given that this measure is seldom reported. Most of the changes occurred during the first months of enrollment. Patients with a higher use of medical services and those with psychosocial problems were at higher risk of change. Interestingly, the training status of the PCP did not influence the risk of change.

ARE ERRORS DISCUSSED IN MORBIDITY AND MORTALITY CONFERENCES? E. Pierluissi¹, M. Fischer², A.R. Campbell³, C.S. Landefeld³; ¹San Francisco VAMC, San Francisco, CA; ²Stanford University, Palo Alto, CA; UCSF, San Francisco, CA

PURPOSE: Morbidity and Mortality Conference (M&M) is the the only conference in residency education where adverse events (AE) and errors are discussed with a goal to improve patient care, but it is not known whether M&M is used this way. Therefore, we determined how AE and errors are discussed in medical and surgical M&Ms.

METHODS: Trained physician reviewers used a structured implicit review to rate case presentations at medical and surgical M&Ms in four teaching hospitals. For each case presentation, the reviewer determined whether an AE occurred, whether the AE was due to an error, and whether the error was attributable to an individual, team, system, or some combination of these. The reviewer then determined whether the error was discussed in the conference and, if so, how it was discussed and attributed (individual, team, system). Interrater agreement was moderate to good (kappa for AEs was 0.56 and 0.73 for errors).

RESULTS: The reviewers rated 54 cases in 35 medical M&Ms and 135 cases in 46 surgical M&Ms. Adverse events were present less often in medical than surgical cases 35% (19/54) vs. 79% (107/135; p<0.002). AE were due to errors in 20% (11/54) of medicine and 46% (62/135) of surgery cases (p<0.002). In medical M&M, only 54% (6/11) of these errors were discussed at all, whereas, virtually all errors in surgical cases (95%, 59/62) were discussed (p<0.002). The rater-identified errors were discussed as errors 36% (4/11) of the time in medicine conferences compared to 85% (53/62) of the time in surgery conferences (p<0.002). In medical M&Ms, only 6/11 cases with errors had enough information to attribute the error to an individual, team, or system. However, only 2 attributions were actually made in the conference. In surgical M&Ms, discussants attributed error to an individual in 58% of cases, to the team in 46%, and to the system in 22%. The physician reviewers attributed error to the individual and system as often but attributed error to the team more often (73% vs. 46%; p<0.01).

CONCLUSION: Medicine MM conferences rarely discuss errors. Surgery MM conference frequently discuss adverse events and errors, however, conference discussion de-emphasizes the role of the team in assigning responsibility for an error. There are significant opportunities for improving the way adverse events and errors are discussed in MM conferences.

DELAY TO BIOPSY AFTER A POSITIVE MAMMOGRAM. R.G. Pinckney¹, V.M. Pam¹, M. Coahran¹, G.M. Berta¹, B. Littenberg¹; ¹University of Vermont, Burlington, VT

PURPOSE: Delay from detection to diagnosis has been associated with more advanced stage at diagnosis and decreased survival in women with breast cancer. We sought to determine the delay to biopsy after mammographic detection of suspicious lesions at a statewide level.

METHODS: The Vermont Mammography Registry (VMR) collects information on all mammograms and breast pathology obtained in Vermont. Mammograms performed on Vermont women between 1/1/1994 and 8/31/1999 with Breast Imaging Reporting and Data System (BI-RADS) category 4 assessments (suspicious for malignancy) or category 5 assessments (highly suggestive of malignancy) were considered positive and in need of biopsy. These mammograms were matched to a subsequent biopsy obtained by fine needle aspirate, core biopsy, or surgery. We calculated the proportion of patients who had biopsies performed within 30 days, had biopsies within 1 to 12 months, and not biopsied within 12 months after the positive mammogram.

RESULTS: 5,713 mammograms met eligibility criteria. 57.8% of mammograms were followed by biopsy within 1 month, 18.5% within 1 month to 12 months, and 23.8% did not have a biopsy within one year. There was a significant variation (p<.001) in time to biopsy by facility with a range of 33.5% to 77.5% of patients biopsied within one month and 9.3% to 60.0% not having had a biopsy within one year.

CONCLUSION: Though a majority of women received an expedient biopsy after a positive mammogram, the large proportion of women having a delayed biopsy or apparent lack of biopsy is of concern.

DELAY TO BIOPSY AFTER A POSITIVE MAMMOGRAM. R.G. Pinckney¹, B.M. Geller², M. Coahran², P. Vacek², B. Littenberg¹; ¹University of Vermont, Burlington, VT

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FAILURE TO FOLLOW MAMMOGRAPHERS RECOMMENDATIONS ON MARGINALLY ABNORMAL MAMMOGRAMS: DETERMINATION OF ASSOCIATED FACTORS. E.G. Poon¹, E.S. Burdick¹, J.S. Haas², T.A. Brennan³, D.W. Bates¹, ¹Brigham and Women's Hospital, Boston, MA; ²University of California, San Francisco, San Francisco, CA; ³Partners Healthcare, Boston, MA

PURPOSE: The short-term follow-up of marginally abnormal mammograms presents a challenge to both physicians and patients. A variety of systems and individual factors may impact on the adherence to mammographers' recommendations. We sought to assess the extent of this problem and to examine the factors associated with the provision of appropriate follow-up care. **METHODS:** Among a sample of 579 women with abnormal screening mammograms or breast complaints, we analyzed the clinical data of 126 women who had marginally abnormal mammograms for which the mammographers recommended 6-month repeat follow-up mammograms. All women were contacted for detailed telephone surveys at the time of their index mammogram, and 7 to 8 months later. The medical records of these women were reviewed to determine the breast care they received before and during the follow-up period. Women who received follow-up mammograms as recommended, saw breast specialists, or underwent breast biopsies within the 7–8 month follow-up period were considered to have received appropriate follow-up care. We then studied the differences between women who did and did not receive appropriate follow-up care.

RESULTS: While 81 women (64.3%) received appropriate follow-up care for marginally abnormal mammograms, 45 women (35.7%) did not. Comparing these groups, women who reported being told about the follow-up plan (O.R. 3.25, CI=1.48 to 7.13), women who scheduled their own mammograms (O.R. 2.29, CI=1.04 to 5.02), and women who saw more physicians during the follow-up period (0.50 MD's vs. 0.18 MD's, $p=0.03$) were more likely to receive appropriate follow-up care. There was a trend indicating that women whose physicians documented discussions about findings on the mammograms (O.R. 1.95, $p=0.10$), and women who had managed care health insurance (O.R. 2.65, $p=0.06$) were more likely to receive appropriate follow-up care. Logistical regression analysis demonstrated that being told about the follow-up plan was the only factor independently associated with receiving appropriate follow-up care. (adjusted O.R. 3.31)

CONCLUSION: A substantial portion of women who had marginally abnormal mammograms did not receive appropriate follow-up care. Patients who feel informed about the follow-up plan are more likely to receive appropriate follow-up care.

QUALITY OF CARE FOR UNINSURED PATIENTS WITH DIABETES IN A RURAL AREA. D. Porterfield¹, L. Kinsinger¹, ¹North Carolina Division of Public Health, Raleigh, NC

PURPOSE: Quality of care for uninsured patients is not well described, and may vary by practice setting. Three-quarters of uninsured patients with a usual source of care report a traditional private physicians' office as their source of care. We compared quality of care for uninsured patients with diabetes in private primary care physicians' offices and community/migrant health centers (C/MHCs).

METHODS: We conducted a cross-sectional chart review in a convenience sample of eight physician offices and three C/MHC sites in the same seven-county area in rural North Carolina. Billing systems generated lists of self-pay patients with diabetes. Chart abstraction ($n=142$; 61 from private physicians' offices and 81 from C/MHCs) yielded data on process and intermediate outcome measures of diabetes care, which were derived from the Diabetes Quality Improvement Project. Data analysis accounted for clustering by individual physician.

RESULTS: Charts of patients in C/MHCs demonstrated higher rates on 5 of 6 process measures of quality of care, including measurement of HbA1C (98% vs. 75%; $p<0.001$), cholesterol (82% vs. 51%; $p<0.001$), and urine protein (90% vs. 25%; $p<0.001$). No differences were seen in intermediate outcomes of glucose, blood pressure, or lipid control.

CONCLUSION: In our sample, uninsured patients with diabetes in CIMHCs had higher quality of care as suggested by higher rates of processes of care. Further work is required to understand specific barriers to care for the uninsured, and resources may be needed to assist private practitioners in caring for uninsured patients with diabetes.

ASSESSING ACCURACY OF DIAGNOSIS TYPE INDICATORS FOR FLAGGING COMPLICATIONS IN CANADIAN ADMINISTRATIVE DATA. H. Quan¹, G. Parsons², W.A. Ghali¹, ¹University of Calgary, Calgary, Alberta; ²Foothills Hospital, Calgary, Alberta

PURPOSE: Canadian administrative hospital discharge data contain a diagnosis type indicator for each coded diagnosis that allows researchers to distinguish complications from pre-existing diagnoses. Given that the validity of diagnosis type indicators is unknown, we conducted a detailed chart review to evaluate the accuracy of diagnosis type indicators for flagging complications.

METHODS: We obtained administrative hospital discharge data for 1200 randomly-selected adult inpatient separations in Calgary, Alberta occurring between April 1, 1996 and March 31, 1997. Each discharge record contains up to 16 diagnoses, 16 corresponding diagnosis type indicators (value of '2' = complication), and up to 10 procedures. A clinically trained chart reviewer examined the corresponding medical charts for evidence of diagnoses and complications. A complication was defined as a new diagnosis arising after the start of hospitalization. We defined 12 common conditions using the ICD-9-CM coding system: stroke, cystitis/pyelonephritis, atrial fibrillation, respiratory failure, post-hemorrhage anemia, pneumonia, bowel obstruction, congestive heart failure, acute renal failure, myocardial infarction, atelectasis, and pleural effusion. We then selected records with presence of a condition in both administrative data and chart review data. Finally, we determined the extent to which the diagnosis type indicator in administrative data agreed with the chart reviewer's assessment (criterion standard) of whether a diagnosis was a complication or not.

RESULTS: Observed agreement for flagging complications using diagnosis type indicators was generally high between the data sources (agreement ranging from 47 to 97%, and agreement >80% for 8 of the 12 conditions studied). However, kappa varied greatly across conditions studied (range: -0.06–0.87) and was low at <0.20 for 6 conditions. Specificity was generally high (range: 68–100%) and was >90% for 8 conditions, suggesting that pre-existing conditions were usually appropriately coded as such in the administrative data. In contrast, sensitivity was lower (range: 0–83%), with sensitivity <50% for 7 conditions, indicating a tendency for complications to often be miscoded as baseline comorbidities.

CONCLUSION: The validity of diagnosis type indicators in Canadian administrative discharge data appears to be sub-optimal for some types of complications. This is likely to be of greatest concern in studies that rely on diagnosis type indicators to define complications as outcomes.

USE OF PROCEDURES DRIVES PHARMACOLOGIC THERAPY IN ACUTE MYOCARDIAL INFARCTION. C. Requeiro¹, A. Hart¹, N. Gill¹, R. Shannon¹, ¹Allegheny General Hospital, Pittsburgh, PA

PURPOSE: Recent data suggest that there are substantial variations in the treatment of acute myocardial infarction (AMI) based on age, race, gender and socioeconomic status. Specifically, the use of proven pharmacologic interventions and lifestyle modification counseling have been shown to vary by patient characteristics. We evaluated the use of post-AMI therapies and counseling in patients with AMI treated in an academic referral center with special attention to those receiving medical management alone for AMI.

METHODS: We reviewed the records of a convenience sample of 384 patients with AMI from 1997–2000. Our outcomes of interest included use of post-AMI therapies such as aspirin (ASA), beta-blockers (BB), angiotensin converting enzyme inhibitors (ACEI) and other anti-coagulants (AC). We collected variables including use of primary angioplasty (PA), defined as a cardiac procedure within twenty-four hours of admission, demographics, comorbidities, do-not-resuscitate (DNR) status, physician characteristics, and severity of illness using the ATLAS system. Univariate analyses were performed on the variables of interest, and subsequent logistic regression modeling incorporated all variables significant on univariate analyses.

RESULTS: Unadjusted analyses of pharmacologic therapies revealed that severely ill patients were less likely to receive any post-AMI therapies (all $p<.05$). Older patients were significantly less likely to receive both ASA and BB post-AMI ($p<.05$), but were not less likely to receive ACEI or AC. Race and gender were not associated with differences in any medication use. Conversely, patients who had undergone PA were significantly more likely to receive all medications ($p<.05$). Unadjusted analyses of tobacco and cholesterol counseling, defined as any chart reference to tobacco use in current smokers and defined as any chart reference to serum cholesterol in all AMI patients, was significantly more likely to occur if one underwent PA, and significantly less likely to occur as severity of illness increased ($p<.05$). After adjustment for potential confounders, patients who underwent PA were significantly more likely to receive proven post-AMI therapies and counseling ($p<.01$), and this increased likelihood persisted across increasing levels of severity of illness.

CONCLUSION: Our results indicate significant differences in the use of post-AMI pharmacologic therapies and counseling correlated with the use of primary angioplasty for AMI, despite evidence of proven benefit in AMI in patients undergoing medical management alone. These differences are not explained by severity of illness and suggest that standard therapies may be withheld from those who may benefit most.

A NEW TRIAGE DECISION AID FOR SUSPECTED ACUTE CARDIAC ISCHEMIA: POTENTIAL VERSUS ACTUAL IMPROVEMENT IN PHYSICIANS DECISIONS. B.M. Reilly¹, A. Evans¹, K. Das¹, J. Schaidler¹, ¹Cook County Hospital, Chicago, IL

PURPOSE: Analyze differences between the potential impact and the actual impact of a decision aid (DA) for triage of ED patients with suspected acute cardiac ischemia.

METHODS: The DA was based on Goldman and colleagues' previously published prediction rule (NEJM 1996). We compared triage by the DA and by physicians (MDs) in simulated cases ($n=2867$) and in actual ED patients ($n=1215$). In the simulated cases, patients had no comorbidities and physicians had unlimited access to beds. Outcome measures included: triage safety, the proportion of all patients who suffer major cardiac complications triaged from the ED to inpatient monitored beds (CCU or telemetry); triage efficiency, the proportion of all patients who do not suffer major cardiac complications not triaged to inpatient monitored beds; and, the proportion of all patients triaged to inpatient monitored beds (% Mon.)

RESULTS: Potential triage efficiency (DA alone) was superior to triage efficiency of unaided MDs (MD, no DA) both in simulated cases (efficiency 0.41 vs. 0.38; $P<0.001$) and in actual patients (efficiency 0.49 vs. 0.24; $P<0.001$). In actual patients, triage was more efficient when physicians used the DA (MD, use DA) than when they did not use the DA (MD, no DA) (efficiency 0.38 vs. 0.24; $P<0.001$) but less than potential efficiency (0.49 for the DA alone; $P<0.001$). Triage safety of the DA was superior in simulated cases (safety 0.96 vs. 0.85 in MD, no DA; $P<0.001$) with a trend toward improved safety in actual patients (safety MD, use DA 0.97 vs. MD, no DA 0.85; $P=0.11$). Fewer patients were triaged to monitored beds (% Mon.) when the DA was used, both in simulated cases ($p=0.04$) and in actual patients ($p<0.001$).

Triage by:	Simulated Cases			Actual Patients		
	Safety	Efficiency	%Mon	Safety	Efficiency	%Mon
DA alone	0.96	0.41	0.62	0.89	0.49	0.53
MD, no DA	0.85	0.38	0.64	0.85	0.24	0.76
MD, use DA				0.97	0.38	0.63

CONCLUSION: Use of the DA significantly improves physicians' triage efficiency but the gap remains large between potential and actual triage efficiency when using the DA in actual patients. The effects of patient comorbidity, bed availability and other factors on triage safety and efficiency merit further study.

COMPARING HOSPITALISTS AND COMMUNITY BASED PRIMARY CARE PHYSICIANS CARE OF PNEUMONIA PATIENTS. W.D. Rifkin¹, D.S. Conner¹, A. Silver², A.M. Eichorn²; ¹Nassau University Medical Center, East Meadow, NY; ²NS-LIJ Health System, New Hyde Park, NY

PURPOSE: Hospitalists have been reported to be more efficient than community-based primary care physicians (PCPs) in caring for hospitalized patients, but how they achieve these efficiencies is not known. In order to identify specific practices that lead to efficient patient care, we investigated the medical care given by hospitalists and PCPs to patients with community-acquired pneumonia.

METHODS: We reviewed the charts of patients hospitalized with pneumonia (ICD-9 codes 481-486) at a tertiary care center during 1998 and 1999. We excluded patients with HIV, lung cancer, active mycobacterium tuberculosis, mechanical ventilation, prior hospitalization within 7 days or length of stay greater than 14 days. We collected data necessary to determine severity of disease (Pneumonia Severity Index), clinical outcomes, and process of care. Stability criteria were defined as temperature less than 101 F, respiratory rate less than 24, systolic blood pressure >100, baseline mental status and tolerating food.

RESULTS: There were 455 patients in the study: 270 cared for by PCPs and 185 by hospitalists. The patients cared for by the PCPs were on average five years older than those cared for by hospitalists. The gender distributions of the patient groups were equivalent. The PCPs had more patients with risk class 5 pneumonias (22% vs. 13%). Length of stay, adjusted for high risk class, insurance status, mortality, residence in a skilled nursing facility and age, was 5.6 days for hospitalists and 6.5 days for PCPs. A difference in door-to-needle time of 1.7 hours favoring the PCPs did not contribute to LOS in this study. Mean time to stability of illness was 3.2 days for hospitalists and 3.3 days for PCPs (not significant). Time from stability to switch to oral antibiotics averaged 1.6 days for hospitalists and 2.3 days for PCPs ($p=0.003$). Inpatient mortality was 2.2% for hospitalists, 8.9% for PCPs. Hospitalists requested pulmonary consults for 11% of their patients and PCPs for 16% of their patients. At discharge, 14% of hospitalist patients and only 7% of PCPs patients had at least 1 unstable parameter. Only 6 patients had 2 unstable parameters, 2 (1.8%) cared for by hospitalists and 4 (1.5%) cared for by PCPs. The most common unstable parameter was SBP <100 mmHg.

CONCLUSION: Although patients cared for by hospitalists had almost a full day shorter length of stay and were switched to oral antibiotics more quickly once stable, they were discharged twice as often with an unstable clinical parameter. It is unclear whether this represents an organizational need to discharge patients quickly or whether the hospitalists interpret stability differently. Future research should focus on patient outcomes and physicians' practice to optimize patient length of stay.

SURVEY OF PHYSICIANS PRACTICE REGARDING ORAL HEALTH CARE. C.S. Ritchie¹, C.H. Drisko¹, L.J. Goldsmith¹; ¹University of Louisville, Louisville, KY

PURPOSE: Recent research has highlighted the connection between oral health and systemic disease. Oral symptoms can indicate general medical problems and systemic disease; treatment can lead to oral complications. We surveyed physicians in 2 major metropolitan areas to determine if oral exams are conducted, if information regarding the relationship between oral health and systemic disease is disseminated by physicians, and to detect consulting and referral activity between physicians and oral health practitioners.

METHODS: We obtained mailing lists of physicians in the Louisville, Kentucky major metropolitan area and the Cincinnati, Ohio major metropolitan area from the Indiana Health Professions Bureau, the Kentucky Medical Association, and the State Medical Board of Ohio. Of 6482 surveys mailed to physicians, 642 surveys were received back. Results were entered into an ACCESS database and analyzed with SPSS (Statistical Package for the Social Sciences).

RESULTS: Although the response rate was low, the responders to the survey represented all medical specialties. In particular, 139 responders were physicians in adult primary care. We found that primary care physicians differed from other specialties in number of oral exams given ($p<.00005$), with primary care physicians tending to give more oral exams. Of primary care physicians, 69% routinely gave information regarding the connection between oral health and smoking, compared to 33.7% of other specialists ($p<.0005$). Less than 20% of primary care physicians gave information regarding the relationship between oral health and pregnancy, preterm delivery, and menopause. Of primary care physicians, 44% consulted with a dental professional regarding sub-acute bacterial endocarditis prophylaxis, compared to 19% of other specialists ($p<.0005$). Primary care specialists and other specialists did not differ in reporting reading "some" of the literature regarding the association of oral health and systemic disease (74% overall, $p=.08$).

CONCLUSION: Although primary care physicians are more likely to give oral exams, they provide inadequate education regarding the role of oral health in pregnancy, preterm delivery, and menopause. The low response rate to the survey may indicate a lack of interest and emphasis on the relationship of oral health to systemic disease.

PHYSICIANS OPINIONS ON DIRECT-TO-CONSUMER PHARMACEUTICAL ADVERTISING ARE LARGELY NEGATIVE. A. Robinson¹, J. Rifkin¹, D. Topp¹, C. Gilroy¹, J. Pickard¹, L. Winslow¹, J. Saari¹, H. Shapiro¹, R. Anderson¹; ¹HealthONE Presbyterian/ St. Luke's Hospital and UCHSC, Denver, CO

PURPOSE: Direct-to-consumer (DTC) advertising of prescription drugs has increased markedly in the U.S. While pharmaceutical companies tout the benefits of these ads in improving public knowledge, increasing competition, and decreasing the cost of medications, little is known about how physicians perceive these ads or how they affect the physician-patient relationship.

METHODS: We randomly surveyed 1,000 Colorado physicians using a mail questionnaire and asked them to rate how often DTC advertising affects specific aspects of their relationship with patients and their practice, as well as how often these ads are complete and accurate. They were also asked to rate their agreement with statements regarding the effect of DTC ads on the cost and consumption of medications. Responses were then examined for differences based on the age, gender, specialty, number of years in practice, and practice type of the respondents.

RESULTS: Response rate was 44%. A majority of respondents felt that DTC advertising rarely or never makes patients better informed (53%) or that they do not adequately inform patients about either other treatment options (94%) or cost (96%). Sixty-two percent felt that ads increase the amount of time spent with patients, 72% that they change patients' expectations of them, and 71% that they increase dissatisfaction for patients with restricted drug formularies. Overall, 63% of physicians felt that these ads increase drug consumption, 91% felt they increase cost, and 74% felt that better regulation of DTC advertisements is needed. Primary care physicians were more likely than non-primary care physicians to feel that DTC ads increase the time spent with patients (67% vs. 52%, $p=0.002$), that they change patients' expectations of them (78% vs. 64%, $p=0.001$), and that they increase overall drug consumption (68% vs. 56%, $p=0.01$). HMO physicians were more likely than non-HMO physicians to feel that ads increase time spent with patients (90% vs. 58%, $p=0.001$). No significant differences were found based on age, gender, years in practice, or practice site.

CONCLUSION: A majority of physicians reported that DTC advertising adversely affects aspects of the doctor-patient relationship by changing patients' expectations, by adversely increasing the time spent with patients, and by increasing dissatisfaction among patients with restricted formularies. Physicians' opinions regarding the value of these advertisements is largely negative, and most agree that better regulation is warranted.

AN OBSERVATIONAL STUDY OF OUTPATIENT CHEMOTHERAPY SAFETY. J.M. Rothschild¹, S. Bartel², E.A. Flynn³, K.N. Barker², T.K. Gandhi¹, L.L. Leape⁴, D.W. Bates¹; ¹Brigham and Women's Hospital, Boston, MA; ²Dana Farber Cancer Institute, Boston, MA; ³Auburn University, Auburn, AL; ⁴Harvard School of Public Health, Boston, MA

PURPOSE: Medication errors (MEs) in hospitals are frequent, often preventable and may result in adverse drug events (ADEs). However, few data are available concerning outpatient settings. Because even highly toxic chemotherapy is increasingly delivered to outpatients, we sought to examine the type and frequency of MEs and ADEs in this setting.

METHODS: We performed a prospective 3-month study in 3 outpatient oncology clinics; 2 adult (hematologic malignancy and solid tumor) and 1 pediatric. Researchers were trained in the observation technique of detecting MEs. Observations were conducted during pharmacy preparation and nursing administration. This technique limits observations to only some of the prescribed medications patients receive during a visit. We defined MEs as any deviation from the prescribed written orders, standard base solutions, or accepted techniques. Each observed medication preparation, administration, or omission was considered an "opportunity for error" (OE).

RESULTS: We observed 1008 medications associated with 1053 OEs. Overall there were 27 MEs with an error rate of 2.6% per OE. Errors were less frequent in pediatrics (1.8%) as compared to adult hematologic and solid tumor clinics (2.6% and 3.6%, respectively). When analyzed by observation phases, the error rate per OE for preparation and administration were 1.9% and 3.3%, respectively. The most common types of MEs were dosing errors (eg. acetaminophen 650 mg instead of 325 mg) and incorrect infusion rates (eg. docetaxel given over 30" rather than 60"). Among the MEs, we observed no ADEs. Five non-preventable ADEs were observed including infusion site reactions. Potential ADEs (intercepted before reaching the patient) were more common during preparation ($n=20$, 3.7%) than administration ($n=3$, 0.6%).

CONCLUSION: These data suggest that the observed error rate was lower than in acute care settings, but several still occurred. While no ME resulted in harm, some had the potential to do so. These findings will help us develop, implement and test systems changes designed to result in safer medication processes.

MEDICATION-RELATED MALPRACTICE CLAIMS ANALYSIS: CAUSES, PREVENTABILITY AND COSTS. J.M. Rothschild¹, T.K. Gandhi¹, R. Kaushal¹, F.A. Federico², P.S. Dasse², J.S. Borus¹, D.W. Bates¹; ¹Brigham and Women's Hospital, Boston, MA; ²Risk Management Foundation, Cambridge, MA

PURPOSE: While they represent only the "tip of the iceberg", malpractice claims provide a rich resource of adverse drug events (ADEs). However, such claims may or may not involve actual injury, or be preventable. We studied medication-related malpractice claims in order to understand their frequency and causes, elucidate error prevention opportunities, and determine claims related costs.

METHODS: We conducted a retrospective review of medication-related malpractice claims between 1990 and 1999 from the Harvard Risk Management Foundation (RMF). Claim abstracts were screened for medication-related allegations. Two physicians independently judged the abstracts as to the presence of an ADE and their preventability. Agreement was compared using the kappa statistic. Charts were reviewed to detect human factors and /or systems-related deficiencies and infer opportunities for prevention interventions. Costs were calculated from RMF records.

RESULTS: Among 2090 claims against RMF insureds, 218 involved medication use (10.4%), including 113 judged due to ADEs (5.4%). Agreement between investigators was excellent (92%, $k=.94$). Two-thirds of ADEs were judged preventable. The most common drug classes were anticoagulants (10%), anesthetics (10%) and antibiotics (9%). ADEs occurred in outpatients (57%) more frequently than inpatients (43%). ADEs resulted in life-threatening or fatal injuries in 20% of cases. MEs were often associated with multiple causes. Latent error causes included poor team and interdisciplinary communication (49% and 29%), complicated system designs (21%) and faulty ergonomic conditions (14%). Human performance errors were knowledge based (60%), skilled-based (47%) and rule-based (20%). ME prevention strategies included improvements in standardization of processes (89%), training (86%), and preparation (39%). Disposition of closed claims ($n=83$) were: 58%- settled; 40%- dismissed, denied, or defense verdict; 2%- plaintiff verdict. Closed claims' incurred costs (mean) for inpatient and outpatient ADEs were \$409,000 and \$72,000, respectively. Preventable ADEs cost a total of \$15.5 million.

CONCLUSION: Medication-related claims data reveal that ADEs were severe, costly and often preventable. Claims data can provide another source of information for development of ADE prevention strategies, and the costs associated with claims can be used to help justify investing in prevention.

EFFECTS OF PRIMARY CARE CLINICIAN DIALOGUE WITH PATIENTS ABOUT DEPRESSION ON DEPRESSION TREATMENT. L.V. Rubenstein¹, M. Rowe², L.S. Meredith³, S. Hickey², N.P. Gordon⁴, ¹VA Greater Los Angeles, RAND and UCLA, Sepulveda, CA; ²RAND, Santa Monica, CA; ³RAND, Santa Monica, CA; ⁴Kaiser Permanente Division of Research, Oakland, CA

PURPOSE: While most experts acknowledge the critical role of primary care clinicians (PCCs) in the diagnosis and treatment of depression, PCC success in this area is problematic, and direct patient access to mental health care has been proposed as a solution. The purpose of these analyses is to learn how often PCCs in two large healthcare organizations that allow direct patient access to mental health services discuss depression diagnosis or treatment with depressed outpatients, and the relationship between PCC discussion and treatment rates.

METHODS: We screened 19,793 consecutive patients attending one of nine primary care practices in one of two staff model managed care organizations in California: three practices based in a single academically-affiliated VA medical center and six in separate Kaiser Permanente (KP) medical centers. Of the 597 who met full criteria for current major depression by diagnostic interview and met other study inclusions criteria, 567 completed a 40-minute telephone-administered survey assessing their mental health care over the preceding six months. Measures include PCC discussion of depression (the PCC asked about depression or gave the patient a treatment for depression or emotional problems at the last visit); recent depression treatment (the patient took an antidepressant or had at least 4 mental health specialty visits in the past 6 months); and high or low depression symptoms by CESD score. Significance is by Chi-Square.

RESULTS: Of 567 patients with major depression, 414 (73%) had been recognized as depressed (64% HMO, 76% VA) as indicated either by PCC discussion of depression (333 or 59%) or by recent depression treatment in the absence of PCC discussion (81 or 14%). A total of 290 of the 567 (51%) reported recent depression treatment; treatment rates were significantly higher among the 333 patients who had discussed depression with their PCC than among the 233 who had not (63% vs 35%, $p < .001$). Excluding patients reporting recent treatment, PCCs who discussed depression were more likely to assess suicidality, mania, and alcoholism. Among the 567 patients, those whose PCC discussed depression were more likely to be treated with an appropriate antidepressant (35% vs. 18%, $p < .001$), and to visit mental health (51% vs 33%, $p < .001$). Results comparing treatment for patients with PCC discussion and those without remained similar and significant when analyzed separately for patients with high or low CESD scores.

CONCLUSION: In two organizations with direct primary care patient access to mental health specialty, discussion with PCCs remains an important determinant of depression treatment quality. These findings support the importance of PCC recognition of depression, and the need for better integration of mental and physical health care.

ANTIBIOTIC PRESCRIPTION FOR UPPER RESPIRATORY TRACT INFECTIONS: INTERNISTS DO BETTER! O.T. Rutschmann¹, D. Tomatore², R. Lopez², E. Costich², N. Nadkarni², M.E. Domino², ¹Duke University, Durham, NC; ²University of North Carolina, Chapel Hill, NC

PURPOSE: Using the 1992 National Ambulatory Medical Care Survey (NAMCS) dataset, Gonzales et al. (JAMA 1997, 278: 901) showed that overuse of antibiotics for upper respiratory tract infection / common cold / bronchitis (URTI) is widespread and suggested that strategies should be broadly implemented to change prescribing behavior of physicians. We aimed to explore if antibiotic prescription practice has changed 6 years later, using the NAMCS 1998 dataset.

METHODS: NAMCS 1998 is a national probability survey of visits (n=23,339) to office-based physicians (n=1,226). Data on antibiotic prescriptions were abstracted from the dataset and predictors of antibiotic use for URTI were included in a regression logistic model. Predictor factors included age, sex, race, ethnicity, geographic characteristics, type of insurance, physician type, type of practice, duration of visit, and if a X-ray was performed or not.

RESULTS: In 1998, there were 20.6 million office visits for URTI resulting in more than 10 million antibiotic prescriptions (=19.9% of all antibiotic prescriptions for adults in 1998). 49% of patients diagnosed with an URTI were treated with an antibiotic. In a multivariate logistic model including all predictor variables, the only parameter statistically significantly associated with lower antibiotic prescriptions was treatment by an internal medicine specialist (OR: 0.56, 95% CI 0.33–0.93, referral category = general/family medicine). Two other parameters were associated with a higher probability of receiving antibiotics: practicing in a solo practice (OR: 2.21, 95% CI 1.38–3.57, referral category = not a solo practice), practice located in the Midwest (OR: 2.12, 95% CI 1.08–4.24, referral category = Northeast). All other parameters were not associated with antibiotic prescription rates.

CONCLUSION: Despite of current recommendations, misuse and overuse of antibiotics continue to be widespread for the treatment of URTI. Internists seem to be more aware of the danger associated with this overuse. Other physician groups should be targeted as well as physicians practicing in solo practice.

PHYSICIAN ETHNICITY, PROFESSIONAL SATISFACTION, AND WORK-RELATED STRESS: RESULTS FROM THE PHYSICIAN WORKLIFE STUDY. M.M. Glymour¹, S. Saha², M.D. Schwartz³, J. Bigby¹, ¹SGIM Career Satisfaction Study Group; Brigham and Women's Hospital, Boston, MA; ²Portland VAMC/OHSU, Portland, OR; ³New York University, New York, NY

PURPOSE: There are limited data about minority physicians' satisfaction with their professional lives. Previous studies suggest that among academic physicians, minorities may be less satisfied than whites. In this study, we describe, by race and ethnicity, satisfaction and job stress among a national sample of physicians.

METHODS: We analyzed data from 2,326 respondents to a career satisfaction survey of physicians drawn from the AMA Physician Masterfile. Scales (from 1–5) measuring overall job satisfaction, career satisfaction, and work-related stress were constructed from Likert-response items and tested using factor and reliability analyses. We examined the association between

physician ethnicity and each of these scales. We also examined how the proportion of patients from ethnic backgrounds other than the physician's own affected satisfaction and job stress.

RESULTS: Respondents included 56 black, 136 Hispanic, 409 Asian, and 1,651 white physicians. Black and Hispanic physicians, as compared with others, cared for higher proportions of minority, non-English speaking and uninsured patients. Scores for job and career satisfaction and stress by ethnicity are displayed in the table. Higher scores indicate higher satisfaction or stress, and p-values refer to comparisons of the highest and lowest scores. After adjusting for physician age, gender, practice setting, specialty, and patient panel characteristics (proportion of patients who were female, elderly, non-English speaking, medically complex, psychosocially complex, or substance-abusing), Hispanic physicians had higher job and career satisfaction than all other groups. The proportion of physicians' patients from ethnic groups other than their own was associated with lower job satisfaction and higher stress ($p < .001$). In adjusted analyses, these findings persisted but were most consistent among white and black physicians.

CONCLUSION: Hispanic physicians were more satisfied with their jobs and careers than physicians from other ethnic groups. For white and black physicians, caring for ethnically dissimilar patients was associated with lower job satisfaction and higher stress. Future study should investigate how physician cultural factors and cross-cultural relationships with patients affect physician worklife and professional satisfaction.

	Black (SD)	Hisp (SD)	Asian (SD)	White (SD)	p-value
Job sat	3.53 (.81)	3.82 (.77)	3.63 (.75)	3.73 (.82)	.02
Career sat	3.63 (.86)	3.92 (.82)	3.69 (.79)	3.73 (.82)	.03
Stress	2.43 (.63)	2.30 (.61)	2.41 (.63)	2.32 (.70)	.05

PROVIDING PHYSICIANS WITH PATIENT-SPECIFIC INFORMATION INCREASES THE LIKELIHOOD OF ALCOHOL COUNSELING AND DECREASES PATIENT DRINKING. B. Saltz¹, N.J. Horton¹, L.M. Sullivan¹, M.A. Moskowitz², J.H. Samet¹, ¹Boston University Schools of Medicine and Public Health, Boston, MA

PURPOSE: Screening and intervention for alcohol problems are underutilized. We tested whether providing primary care physicians (PCPs) with patients' alcohol screening results and individualized recommendations increases the likelihood of alcohol counseling by the PCP and decreases patient drinking.

METHODS: PCPs were randomized to an intervention or control group, stratified by level of training (resident or faculty). During office visits, intervention PCPs received patients' alcohol screening results (CAGE and consumption), a readiness to change measure, and action recommendations based on problem severity; control physicians received no such information. Eligible patients visiting these PCPs were current hazardous or harmful drinkers. PCP counseling was assessed by patient report immediately after the visit and was categorized as 1) specific counseling (regarding safe drinking, to quit, or cut down), or referral (to alcohol treatment or a mutual help group), 2) any advice regarding drinking and 3) the occurrence of any alcohol-related discussion. Patient drinking was assessed using a validated calendar method 6 months later.

RESULTS: Of 41 PCPs, 20 were in the intervention group and 48% were residents. PCPs were similar by randomized group. The 312 patients were 64% male, 56% black, 19% white, 16% Latino, 63% high school graduates, 40% unemployed. Mean age was 44. Patients drank a mean of 6 drinks per drinking day, and 36% were in Precontemplation, and 32% the Contemplation stage of readiness to change. For faculty PCPs, mean counseling rates were higher in the intervention group (mean specific counseling rates 51% vs 39%, $p = 0.08$, any advice 58% vs 40%, $p = 0.03$, discussion 72% vs 48%, $p = 0.01$). Mean counseling rates for residents did not differ significantly by randomized group. Hierarchical models considering patient level data confirmed that the impact of the intervention on counseling differed by resident/faculty status ($p = 0.03$ for interaction). Stratified hierarchical models confirmed the findings. At six months, 76% completed follow-up. Patients who saw intervention physicians drank fewer drinks per drinking day (mean difference adjusted for baseline drinking: 2.4 drinks, $p = 0.05$).

CONCLUSION: A simple systems intervention, providing physicians with screening results and recommendations about patients with hazardous or harmful drinking, increased faculty PCP counseling and decreased alcohol consumption at six months.

ARE PATIENTS AND PRACTICE PATTERNS OF INTERNISTS AND FAMILY PHYSICIANS DIFFERENT IN THE AGE OF MANAGED CARE? S. Salerno¹, J. Jackson¹, ¹Uniformed Services University of the Health Sciences, Bethesda, MD

PURPOSE: To identify differences in patient characteristics and practice patterns between internists and family physicians in the ambulatory setting.

METHODS: 108,098,716 encounters in the 1995–1998 National Ambulatory Medical Care Survey were compared for differences in patient age, encounter time, patient diagnoses, and acuity during visits to internists and family physicians. Laboratory tests, radiologic procedures, medications, and preventive services provided at each visit were compared between the two specialties examined after adjusting for age and gender.

RESULTS: Patients under age 18 were seen in 4.7% of internist visits compared with 18.9% of family physician encounters ($p < 0.001$). Adult patients seen by internists were older (56.2 vs. 50.1 years, $p < 0.001$) and had longer visit lengths (17.4 vs. 15.8 minutes, $p < 0.001$) than patients seen by family physicians. Patients were more likely to see an internist as a result of a referral (OR 1.428, 95% CI 1.427–1.429) than a family physician. Internists saw patients with diabetes, COPD, diabetes mellitus, coronary artery disease, arthritis, and hypertension more frequently than family physicians ($p < 0.001$ for all) and were more likely (OR 1.308, 95% CI 1.307–1.309) to be seeing the patient for follow up of a chronic medical problems. Internists were more likely to perform mental status (OR 1.459, 95% CI 1.457–1.460), rectal (OR 1.237, 95% CI 1.236–1.238) and breast (OR 1.296, 95% CI 1.295–1.297) exams and to test for hypertension, hyperlipidemia, and order a PSA or mammogram than family physicians ($p < 0.001$ for all).

Family physicians were more likely (OR 1.254, 95% CI 1.253–1.255) to see patients for acute problems unrelated to chronic medical conditions. Family physicians performed more pelvic exams (OR 1.447, 95% CI 1.446–1.448) and Papanicolaou smears (OR 1.391, 95% CI 1.390–1.392), and more frequently provided counseling on smoking cessation, mental health, injury prevention, family planning and sexually transmitted diseases ($p < 0.001$ for all). Family physicians were more likely to prescribe medication at patient visits (OR 1.229, 95% CI 1.228–1.230) than internists, but internists more frequently ordered advanced radiologic tests such as CT, MRI, and ultrasound ($p < 0.001$ for all) even after adjustment for age, gender, and number of ICD-9 diagnoses.

CONCLUSION: Significant differences exist between ambulatory practice patterns of primary care physicians independent of patient age and gender.

LAB-PHARMACY LINKS: LEGAL LABELING LANGUAGE OVERLOAD. G. Schiff¹, G. Shah¹, H. Lubin¹, D. Klass¹, D. Bates²; ¹Cook County Hospital, Chicago, IL; ²Brigham and Women's Hospital, Boston, MA

PURPOSE: Lab and pharmacy are intimately related. However their departments, functions and information systems are often disconnected. The extent of these clinical interrelationships, and expectations upon prescribing physicians to remember critical information (warnings, toxicity, monitoring) is vast and taxes the memory of busy clinicians. A number of drugs have recently been withdrawn because of operational failure in meeting such expectations (eg troglitazone-hepatic toxicity monitoring). To better understand the extent of such information, and because the Physicians Desk Reference (PDR) is the most widely used drug information source and represents legally-mandated prescribing labeling for drugs marketed in the U.S, we analyzed the content of lab-pharmacy interactions and warnings for frequently prescribed and recently approved drugs.

METHODS: We developed a taxonomy of critical lab-pharmacy interaction categories, and reviewed all solid oral medications for the most frequently prescribed drugs (N=40), plus new molecular entities (NME's) marketed in 1998–99 (N=37). We conservatively counted multiple warnings as a single test whenever possible (eg thrombocytopenia, anemia, leukopenia = CBC). **RESULTS:**

Lab-Drug Category Lab Results:	Top 40 Drugs Total Warnings	New Drugs (N=37) # Drugs w/ Warnings	Total Warnings	# Drugs w/ Warning
Contraindicating Drug	13	10	20	16
Indicating for Drug	8	8	6	6
Dose Adjustments	40	31	32	25
Indicating Toxicity	149	37	147	34
Baseline Monitoring	15	11	13	8
Follow-up Monitoring	19	11	19	12
Interfered w/ by drug	5	3	1	1

CONCLUSION: Officially labeled lab-pharmacy interactions/warnings are frequent—totaling 249 (6.2/drug for commonly prescribed drugs, and 238 (6.4/drug) for NME's)—a number that exceeds the unaided prescribers' abilities to remember and monitor. Automated screening and reminders will need to be programmed into routine prescribing processes to prevent errors related to such interactions. A noteworthy discrepancy exists between number of toxicities and recommended monitoring, and relatively few drugs (especially NME's) appear to have lab interference evaluated, suggesting a need for additional research to ensure safe prescribing and error-free lab test interpretation.

RACIAL DISPARITIES IN INFLUENZA VACCINATION: DOES MANAGED CARE NARROW THE GAP BETWEEN AFRICAN AMERICANS AND WHITES? E.C. Schneider¹, P.D. Cleary¹, A.M. Zaslavsky¹, A.M. Epstein¹; ¹Harvard University, Boston, MA

PURPOSE: To assess racial disparities in influenza vaccination among Medicare beneficiaries and evaluate whether or not the magnitude of racial disparities is smaller for those enrolled in managed care than among those with fee-for-service insurance.

METHODS: We analyzed 13,674 responses to the 1996 Medicare Current Beneficiary Survey of African American and white Medicare beneficiaries with managed care and fee-for-service insurance. We compared sociodemographic characteristics and health attitudes among the groups, the proportion receiving influenza vaccination, and differences in racial disparities in influenza vaccination between those with managed care and fee-for-service insurance. Using a propensity model based on sociodemographics, comorbid illnesses, attitudes toward health care, and their interactions with race, we adjusted for differences in the two insurance groups and compared the adjusted racial disparities.

RESULTS: Eight percent were African American. Eleven percent were enrolled in managed care. On average, African Americans and fee-for-service enrollees were older, had worse health status, had a higher prevalence of diabetes, and were more likely to avoid doctor visits. Overall, 65.8% of Medicare beneficiaries received influenza vaccination. Whites were much more likely than African Americans to receive influenza vaccination (67.7% vs. 45.1%, absolute disparity = +21.6%; 95% C.I. 18.2% to 25.0%). Managed care enrollees were more likely to receive influenza vaccination than those with fee-for-service insurance (71.2% vs. 65.4%, difference = +5.8%; 95% C.I. 3.3% to 8.3%); however racial disparities in influenza vaccination were nearly identical among beneficiaries both with managed care and with fee-for-service insurance. These disparities were not significantly different after adjustment for propensity to enroll in managed care.

CONCLUSION: While managed care is associated with higher rates of influenza vaccination, we found no evidence that managed care reduces racial disparities in influenza vaccination. Our results suggest that health plans are not yet adequately addressing racial disparities in the use of preventive services.

SECONDARY PREVENTION OF CORONARY ARTERY DISEASE AT A LARGE TEACHING HOSPITAL. J.L. Schnipper¹, R.S. Stafford¹; ¹Massachusetts General Hospital, Boston, MA

PURPOSE: To determine the extent to which outpatients with coronary artery disease (CAD) receive adequate secondary prevention measures, and to identify the factors that predict variations in the provision of these services.

METHODS: Cross-sectional study of 3920 patients with coronary disease (ICD-9 codes 410.0–414.99) who were seen in one of ten outpatient practices affiliated with Massachusetts General Hospital (MGH) during fiscal year 1997. Demographic and laboratory information were collected on all patients, and a medical record review was performed on a subset of patients to confirm CAD and to determine blood pressures, medical conditions, and medications during fiscal years 1997 and 1998 (N = 302).

RESULTS: Our study showed evidence of variable short-comings among different CAD prevention strategies. 17% of patients never received a low density lipoprotein (LDL) cholesterol test during the study period, and only 19% of patients had an LDL level ≤ 100 mg/dL. Among those with an LDL ≥ 130 mg/dL, 41% received no lipid-lowering medication and only 5% were on a standard maximum dose of an HMG CoA reductase inhibitor (statin). The average of the last two blood pressure readings was less than 130/85 mm Hg in 42% of patients, and 53% of patients met JNC VI guidelines for blood pressure control. Among hypertensive patients, 94% were on some type of anti-hypertensive medication. 75% of all patients were prescribed a beta-blocker during the time period, and 86% were prescribed aspirin or another antithrombotic medication.

In multivariable models adjusted for several patient and physician characteristics, having an LDL < 130 mg/dL was less likely for patients in the oldest age quartile (odds ratio (OR) 0.59, 95% confidence interval 0.49–0.72), female patients (OR 0.60 [0.51–0.69]), those without insurance coverage of medications (OR 0.54 [0.37–0.79]), and those whose type of insurance indicated low socioeconomic status (OR 0.85 [0.70–1.01]). Older patients and women were less likely to have a blood pressure $< 140/90$ mm Hg. Younger age was of borderline significance in predicting beta-blocker use.

CONCLUSION: At outpatient practices affiliated with a major academic medical center, there is still much room for improvement in the secondary prevention of CAD. Deficiencies were greatest in the area of lipid management. Lipid management also was the most susceptible to practice variation on the basis of socioeconomic status. Older patients and women also were less likely to receive adequate secondary prevention. Interventions such as physician education, computerized reminder systems, and population management may be effective in correcting some of these shortcomings.

FOR-PROFIT HOSPITALS AND DRG "UPCODING". E. Silverman¹, J. Skinner²; ¹Dartmouth Hitchcock Medical Center, Lebanon, NH; ²Dartmouth College, Hanover, NH

PURPOSE: One large for-profit hospital chain has admitted to inflating the severity of discharge diagnoses ("upcoding") for economic gain. It is unknown how widespread this practice is, or whether it is endemic to for-profit hospitals. We examined the association between for-profit hospital ownership and upcoding for Medicare discharges for pneumonia, the most widely recognized condition under which upcoding seems to occur.

METHODS: To analyze time trends in coding patterns, we used a 20 percent national sample of Medicare inpatient claims for 1989 through 1997 with the hospital as the unit of analysis. Hospital ownership was defined as for-profit, not-for-profit, or government based on the American Hospital Association's Annual Survey of Hospitals for each year. The relevant diagnosis related groups (DRGs) were for simple pneumonia with and without complications (DRGs 89 and 90), and for complex pneumonia with and without complications (DRGs 79 and 80). The dollar reimbursement for complex pneumonia is approximately 50 percent higher than for simple pneumonia. For each year and ownership group we calculated an upcoding ratio defined as the percentage of discharges coded for complex pneumonia out of the total discharges for simple and complex pneumonia combined.

RESULTS: Crude upcoding ratios for pneumonia were highest among for-profit hospitals for each year between 1989 and 1997. In addition, between 1989 and 1996, the upcoding ratio increased from 0.20 to 0.30 in not-for-profit hospitals, 0.21 to 0.30 in government hospitals, and from 0.28 to 0.51 in for-profit hospitals. Between 1996 and 1997 the ratio fell slightly for each group, which corresponds to a period of great publicity and legal charges of fraud. Average thirty-day mortality rates for the DRGs under consideration were similar across ownership categories, and in fact declined over the study period, suggesting that differences in the percentage of pneumonia discharges coded as complex was not related to case-mix.

CONCLUSION: For-profit hospital ownership was associated with higher upcoding ratios among Medicare discharges for pneumonia than in not-for-profit hospitals.

PHYSICIAN ATTITUDES TOWARD HOMOSEXUALITY AND HIV INFECTION: SAN DIEGO COUNTY PHYSICIAN SURVEY. D. Smith¹, W. Mathews¹; ¹University of California, San Diego, CA

PURPOSE: Physicians' attitudes are important because of the possible impact they may have on the structure and quality of patient care [1]. In 1982 Mathews et al [2], surveyed San Diego County Medical Society's physicians about their attitudes toward homosexuality. They found significant differences in prevalence of homophobic attitudes by gender, year of medical school graduation, specialty and practice setting [2]. Our objective was to assess physicians' attitudes toward homosexuality and toward persons with HIV infection and to compare the current prevalence of homophobia to that measured seventeen years previously.

METHODS: An anonymous, self-administered, 17-item survey was mailed to all 4,385 members of the SDCMS and 1,271 UCSD physicians. The survey included 8 likert-scaled items measuring general attitudes toward homosexuality (Cronbach's alpha = 0.88). Additional items assessed attitudes toward entry to medical school and consultative referral patterns, conditional on sexual orientation and HIV status of hypothetical referents.

RESULTS: 736 physicians responded (13% response rate). Seven percent self identified as homosexual or bisexual. Response rates did not vary by gender, but ranged by specialty from 24% for internists to 5% for psychiatrists. Levels of homophobia declined markedly since the 1982 survey. Only 3% would not admit a highly qualified homosexual applicant to medical school (30% in 1982). Nine percent would discontinue referrals to a gay pediatrician (46% in 1982). Forty two percent would not admit a "highly qualified but asymptomatic applicant with excellent response to antiretroviral therapy to medical school." Sixty six percent would discontinue referral to a general surgeon known to be HIV-infected. In multiple logistic regression analyses controlling for sex and medical school affiliation, significant ($p < .05$) independent predictors of being in the highest 10% on an HIV-phobia scale were year of graduation from medical school and degree of homophobia (model ROC=0.77).

CONCLUSION: Although limited by a low response rate, this survey suggests substantial reduction in homophobia since 1982. However, physician attitudes toward homosexuals and year of graduation from medical school appear to be significant predictors of attitudes toward persons with HIV infection.

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SCREENING FOR HIGH UTILIZING SOMATIZING PATIENTS USING A PREDICTION RULE DERIVED FROM THE MANAGEMENT INFORMATION SYSTEM OF AN HMO. R. Smith¹, J. Gardiner¹, S. Armatti¹, M. Johnson¹, J. Lyles¹, C. Given¹, C. Lein¹, B. Given¹, J. Goddeeris¹; ¹Michigan State University, East Lansing, MI

PURPOSE: Somatization is a common costly problem with great morbidity, but there has been no effective screening method to identify these patients and target them for treatment. We tested an hypothesis that we could identify high utilizing somatizing patients from a management information system (MIS) by total number of visits and what we termed "somatization potential," the percentage of visits for which ICD-9 primary diagnosis codes represented disorders in the musculoskeletal, nervous, or gastrointestinal systems or ill-defined complaints.

METHODS: We identified high utilizing patients from the MIS of a large staff model HMO as those having 6 or more visits during the year studied (65th percentile). A physician rater then reviewed the medical records of these patients and identified somatizing and nonsomatizing patients. In two-thirds of the population (the derivation set), we used logistic regression to refine our hypothesis and identify predictors of somatization available from the MIS: demographic data, all medical encounters, and primary diagnoses made by usual care physicians [ICD-9 codes]. We then tested our prediction model in the remaining one-third of the population (the validation set) to validate its utility.

RESULTS: The derivation set contained the following significant correlates of somatization: gender, total number of visits, and percent of visits with somatization potential. The c-statistic was .90. In the validation set, the explanatory power was less with a c-statistic of .78. A predicted probability of .04 identified almost all somatizers, while a predicted probability of .40 identified about half of all somatizers but produced few false positives.

CONCLUSION: We have developed and validated a prediction model from the MIS that helps to distinguish chronic somatizing patients from other high utilizing patients. Our method requires corroboration but carries the promise of providing health plan directors with an inexpensive, simple approach for identifying the common somatizing patient and, in turn, targeting them for treatment. The screener does not require clinicians' time.

MINOR ACUTE ILLNESS: A COMMON BUT IGNORED PROBLEM IN HIGH UTILIZING PATIENTS - A PRELIMINARY RESEARCH REPORT. R. Smith¹, J. Gardiner¹, J. Lyles¹, M. Johnson¹, K. Rost¹, Z. Luo¹, J. Goddeeris¹, C. Lein¹, C. Given¹, F. Dwamena¹, C. Collins¹, L. Vanegeren¹; ¹Michigan State University, East Lansing, MI

PURPOSE: High utilizing patients with medically unexplained symptoms often are believed to be somatizing patients, defined here as no organic disease explanation for symptoms of at least 6 months duration. High utilizing patients with unexplained symptoms of shorter duration have not been studied and there is no effective treatment for this group, which we call "minor acute illness."

METHODS: Our aim was to determine how many high utilizing patients with medically unexplained symptoms met our chart-rating criteria for somatization and/or minor acute illness and what the stability of these diagnoses was over time. We performed chart review at baseline; one and two years later, we re-rated the charts of patients initially rated as having somatization as well as a 15% sample of those with minor acute illness. Subjects were a random sample of high utilizing staff model HMO patients (6 or more visits/year) from 21-55 years old, identified from the management information system (MIS). Chart review designations as organic disease, somatization, or minor acute illness were the measurements made.

RESULTS: Among 883 high utilizing patients, 35% had organic diseases, 14% had somatization, and 51% had minor acute illness as their primary problems. No patients with initial minor acute diagnoses were reclassified as somatization one and two years later and all but 2 patients had minor acute illness in one or both follow-up years. Among somatizers, 54% were re-classified as minor acute illness the following year and, of these, 50% remained as minor acute illness in the second follow-up year. 70% of initial somatizers and 45% of initial minor acute patients continued to be high utilizers in one or both follow-up years. Persistent high utilization in both follow-up years occurred in 17% of minor acute patients and 33% of somatizing patients.

CONCLUSION: Minor acute illness appears more common among high utilizing patients than somatization and organic diseases combined. It has not previously been studied. Diagnoses of somatization were unstable over two years follow-up while minor acute diagnoses were stable, supporting the latter as a valid entity. Utilization did not remain high in either group. We recommend further study of minor acute illness to determine its relationship to somatization, to

determine if it is a valid entity, and to better understand the broad spectrum of patients with medically unexplained symptoms.

AVOIDANCE OF REGRET AS A MOTIVE FOR ORDERING PROSTATIC SPECIFIC ANTIGEN TESTS. P.C. Scrum¹, E. Mullet², G. Chasseigne³, S. Bonnin-scaon², J. Shim⁴, J. Cogneau⁵; ¹Albany Medical Center, Schenectady, NY; ²Ecole Pratique des Hautes Etudes; ³Université François-Rabelais, Tours; ⁴University at Albany, Albany, NY; ⁵MG France, St Avertin

PURPOSE: When making management decisions, physicians try to avoid the regret that would be provoked by unwanted consequences of their actions or inactions. We explored the strength and determinants of such anticipated regret in a study of decisions to order prostatic specific antigen (PSA) tests for hypothetical patients.

METHODS: 32 U.S. and 33 French primary care physicians looked at 32 patient scenarios and estimated the probability each patient had prostate cancer and the likelihood the participant would order a PSA. They were then given 12 scenarios and were told to suppose they had performed routine exams on these patients, had not ordered PSAs, and now several years later found advanced prostate cancer in these patients. They were asked to indicate how much regret they would feel that they had not ordered PSAs. The 12 hypothetical patients differed according to age (55, 65, or 75), the request to be screened for prostate cancer (no or yes), and the shape of the prostate on the earlier rectal exam (no irregularity or some irregularity). The assessments of regret were analyzed by ANOVA (with significance set at $p < .02$). For each group of physicians, a correlation was calculated between the mean rating for regret and the mean likelihood of ordering a PSA.

RESULTS: The regret scores were correlated with the likelihood of ordering PSAs for both the French ($r = .64$, $p < .005$) and the Americans ($r = .42$, $p < .02$). The level of regret was higher when the patient had requested a PSA, when the shape had some irregularity, and when the patient was younger. When the patient had requested a PSA, the shape of the prostate had less effect than when the patient had not requested it. The U.S. physicians indicated a higher level of regret than did the French. A patient request had a greater impact on U.S. than on French physicians. Increasing patient age reduced regret more among the French than among the Americans.

CONCLUSION: Anticipated regret over a failure to have diagnosed an aggressive prostate cancer is correlated with a policy of ordering of PSAs. The extent and determinants of this regret appear to be culturally sensitive.

HOW OFTEN DO MEDICAL ORGANIZATIONS HAVE CORPORATE POLICIES PERTINENT TO CLINICAL STRUCTURE? M. Spar¹, D.M. Carlisle², C. Damberg³, H.H. Liu², W. Chen², C.M. Mangione², K.L. Kahn²; ¹UCLA/PricewaterhouseCoopers, LLP, Los Angeles, CA; ²University of California, Los Angeles, Los Angeles, CA; ³Pacific Business Group on Health, San Francisco, CA

PURPOSE: Several studies have suggested that a central medical organizational structure can influence processes and outcomes, for example with audit and feedback, economic incentives, and implementation of guidelines. One form of organizational structure is clinical structure, the application of organizational policies which, in aggregate, allow medical organizations to improve the efficiency of care and outcomes delivered to cohorts of patients. We used literature review supplemented by clinical and administrative input to design a model of clinical structure which was tested in a 45 minute telephone survey by physician interviewers of medical directors of 56 medical organizations.

METHODS: Responses were obtained from the directors of 53 medical organizations (response rate 95%) providing data from 29 medical groups (MGs) and 24 independent practice associations (IPAs) from 3 west coast states. The survey queried directors about whether their organization had in place corporate-wide policies to standardize clinical operations across the offices and providers that define their corporate entity. The survey asked about each of 30 dimensions of clinical corporate structure distributed across 183 individual survey items. We considered an organization to have corporate policy (CP) if the respondent indicated CP for at least one item associated with a dimension of clinical structure.

RESULTS: There was substantial variation across organizations in the extent to which the directors reported corporate policies. On average, organizations had CP for 49% of the 183 items across 30 dimensions of corporate structure. The clinical corporate policies most frequently cited by respondents pertained to the use of clinical guidelines (94%); specifications regarding the exact content of the ambulatory medical record (92%), and whether or not prospective review was required prior to specialty consultation (91%). Organizations infrequently cited having clinical corporate policies for assigning new patients to providers according to clinical need (19%), for office staff counseling patients following MD visits regarding next steps (21%), and use of multidisciplinary team meetings (25%). Of the 30 dimensions of clinical structure studied, CP was more prevalent in MGs than IPAs ($p < .05$) for 17 (57%) dimensions. For example, 66% of MGs and 33% of IPAs reported CP regarding written educational materials being available to patients ($p < .05$).

CONCLUSION: While some medical organizations have developed sophisticated systems to organize the delivery of clinical care to cohorts of patients using corporate administrative policies, most have not. For most dimensions for which the literature suggests care could be improved with organizational support, most organizations, especially IPAs, do not report corporate policies in place.

INCONSISTENCY IN THE DIAGNOSIS OF CHRONIC HEART FAILURE. U. Subramanian^{1,2}, M. Weinberger^{1,2,3}, W. Tierney^{1,2,3}; ¹Roudebush VAMC Indiana University School of Medicine Indiana Regenstrief Institute for Health Care, Indianapolis, IN

PURPOSE: CHF is a highly prevalent, morbid and costly condition. Agency for Healthcare Research and Quality (AHRQ) evidence-based guidelines recommend treating patients for CHF based on ejection fraction from echocardiogram (echo). Unfortunately, physician compliance with these guidelines is sub-optimal. One possible reason for noncompliance is if physicians do

not accept a diagnosis of CHF based upon ejection fraction alone. Thus, it is important for both clinical and research purposes to estimate the agreement between quantitative measurements and the cardiologists' qualitative interpretation reported on echo

METHODS: All patients at a university-affiliated VAMC who had an echo between 12/96 and 12/99 were identified by electronic medical records (N=5,505). One record per patient was identified and for each echo, we abstracted both quantitative (fractional shortening and fractional area change) and qualitative (cardiologists' interpretation) data. We excluded reports with missing quantitative (N=912) or qualitative (N=85) data. Abnormal numeric data was defined as fractional shortening less than 0.18 or fractional area change less than 0.35. Abnormal textual report was defined as left ventricular systolic dysfunction or generalized wall motion abnormality. We then calculated agreement from a 2x2 table.

RESULTS: Of 4,508 patients with echos in our study, 1237 had an abnormal echo by quantitative and/or qualitative data (Table). Of these 1237 patients, an abnormality was identified by both strategies in only 548 (44%) of patients. Discrepancies rates between qualitative and quantitative reports were similar. The cardiologists' interpretation of the echo was normal when the numeric data were abnormal 41% (386/937) and the cardiologist's interpretation was abnormal with normal numeric data 35% (303/851).

Qualitative (Textual) Interpretation

	ABNORMAL	NORMAL	TOTAL
Qualitative	548	386	937
Data	303	3256	3571
	TOTAL	851	3642
			4508

CONCLUSION: There is significant discrepancy between qualitative and quantitative strategies to diagnose CHF, which may explain non-compliance with AHRQ recommendations. Physicians who use either standard alone, risk undertreating or overtreating patients which may lead to morbidity. We conclude that CHF guidelines must develop a clinically acceptable way of identifying patients with CHF in order to insure that appropriate care is delivered and that patient outcomes are maximized.

GEOGRAPHIC VARIATION IN HEALTHCARE UTILIZATION AND OUTCOMES IN PATIENTS WITH ACUTE MYOCARDIAL INFARCTION. U. Subramanian¹, W. Tierney¹, G. Eckert¹, G. L'italien¹, P. Lapuerta¹, M. Weinberger¹; ¹Roudebush VA Medical Center, Indiana University, Indianapolis, IN

PURPOSE: Studies examining some of the clinical and non-clinical determinants of outcomes of acute myocardial in the VA suggest that age, race, co-morbid illnesses, and availability of cardiac surgical services all play a role. However, few data are available on the regional variation in health care utilization and outcomes during and after acute myocardial infarction. The purpose of our study was to examine regional variation in utilization and outcomes during the acute and chronic care of patients with acute myocardial infarction (AMI). Examining these differences will lead to a better understanding of potentially modifiable variables and processes of care that influence patient outcomes

METHODS: Using national VA databases, we identified all veterans who were hospitalized for AMI at any VA Medical center between 10/90 to 10/97. Demographic, inpatient, outpatient, mortality and readmission data were extracted from the VA's administrative databases for the four regions: Northeast, South, Midwest and West. Multivariable Cox proportional hazards models, controlled for comorbidity, were used to assess the effect of independent variables on time to death and readmission.

RESULTS: We identified 67,889 patients with AMI who were 98% male at mean age of 65 years. There were no significant regional differences in patients' demographic data. Patients in the Northeast had a greater prevalence of comorbid conditions and longer lengths of stay during the index AMI hospitalization (14 days in the NE Vs 10 days in the W). There was no variation in the rates of cardiac procedures (PTCA and/or CABG) among the regions during the index admission. Region of the country was an independent predictor of mortality. Subsequent adjusted all-cause mortality after discharge from the index hospitalization was lower in the Northeast (OR=.910 CI .86 - .95) and the West (OR=0.871 CI .87 - .95). Patients in the Northeast and West had greater cardiology or primary care follow up at 60 days and 1 year post discharge than patients in the South. Outpatient follow up appeared to account for a substantial amount of the regional variation in all-cause mortality.

Any F/U (% or pts)	NE	S	NW	N
60 days	45	40	39	52
1 year	71	67	66	76

CONCLUSION: There was substantial geographic variation in clinical care and outcomes among veterans with AMI. Outpatient follow-up was highly variable and associated with mortality. Further studies are needed to determine the most effective strategies for improving outcomes after AMI.

PATIENTS RATINGS OF QUALITY AND SATISFACTION WITH END-OF-LIFE CARE. D.P. Sulmasy¹, J.M. McIlvane²; ¹St Vincents Hosp & New York Medical College, New York, NY; ²St Vincents Hosp, New York, NY

PURPOSE: A lack of established measures, among other factors, has limited our knowledge about patients' ratings of quality and their satisfaction with end-of-life care.

METHODS: We recently described a new, valid, and reliable instrument called QUEST, which has 4 scales measuring patients' perceptions of quality and their satisfaction with care by physicians (MDs) and nurses (RNs). We administered QUEST and other standardized instruments to 84 medical inpatients (45 terminally ill and 39 comparison) at 2 hospitals. We used correlations, t-tests, and ANOVA to test associations between sociodemographic/clinical factors and QUEST scores.

RESULTS: Mean patient age was 60, APACHE-III physiology score 16.6; 71% were white, 60% men, 54% Catholic, 28% Medicaid/uninsured, 38% had cancer or HIV, 41% moderate/severe pain, 19% depressed, and 58% private (PVT). Age, APACHE, race, gender, religion, insurance, and diagnosis were unassociated with quality or satisfaction. Quality and satisfaction scores for MDs were lower among "service" (SERV) than PVT patients and lower among terminal than non-terminal patients. Quality and satisfaction scores for RNs were lower among SERV patients and depressed patients. P-values for ANOVA models for QUEST were: MD Quality=.01, MD Satisfaction (adjusted for pain)=.007, RN Quality=.16, RN Satisfaction=.002.

CONCLUSION: Terminally ill SERV patients rated MD quality and satisfaction lower than PVT patients and lower than SERV patients not terminally ill. RN satisfaction scores were high only among PVT patients who were not depressed. Understanding these apparent differences in end-of-life care will require further investigation.

ANOVA Models of Quality and Satisfaction QUEST Scores: Physicians

	Mean MD Quality	Mean MD Satisfaction
PVT/non-Terminal	50.2	53.6
PVT/Terminal	51.4	44.4
SERV/non-Terminal	52.1	45.6
SERV/Terminal	32.1	33.9

ANOVA Models of Quality and Satisfaction QUEST Scores: Nurses

	Mean RN Quality	Mean RN Satisfaction
PVT/non-Depressed	50.1	53.3
PVT/Depressed	38.7	29.2
SERV/non-Depressed	37.9	37.4
SERV/Depressed	37.1	31.6

FACTORS ASSOCIATED WITH THE TIME NURSES SPEND AT THE BEDSIDES OF DYING INPATIENTS. D.P. Sulmasy¹, J.M. McIlvane²; ¹St Vincents Hosp & New York Medical College, New York, NY; ²St Vincents Hosp, New York, NY

PURPOSE: At the end of life, when cure is no longer possible, perhaps the most important medical intervention that can be offered is time. However, almost nothing is known about the time health professionals spend with dying inpatients.

METHODS: We asked day-shift nurses to use a diary to record how much time they spent in various activities for 47 medical inpatients at 2 teaching hospitals (25 terminally ill patients with Do Not Resuscitate (DNR) orders and 22 comparison patients). To minimize Hawthorne effects, nurses were told to record multiple categories of activities and were only informed that this was a "time-motion study for seriously ill patients." This method has been validated against video-recordings of actual time spent in the room (r=.70, p<.0001). Non-English speaking patients, those whose clinical condition prohibited interview, and those with a reduced set minimal status (MMS) score indicating dementia were excluded. Charts were reviewed and patients were interviewed using a battery of standardized instruments. Because nursing time was not normally distributed, we used log-transformed time as our dependent variable. We then used t-tests, correlations, and ANOVA to test associations between sociodemographic/clinical data and bedside nursing time.

RESULTS: The mean patient age was 59, and the mean APACHE-III physiology score 16.3; 64% were white, 19% African-American, 11% Latino, and 6% Asian; 60% were men, 39% Protestant, 41% had cancer or HIV, and 23% were depressed. The mean amount of time nurses spent with patients per day shift was 27 minutes. In bivariate analyses, age (p=.23), APACHE-III physiology score (p=.78), gender (p=.81), religion (p=.27), diagnosis (p=.99), and DNR status (p=.26) were unassociated with nursing bedside time. Nurses spent more time at the bedsides of patients with lower reduced-set MMS scores (indicating worse mental function (r=.26, p=.08)), more time with white patients compared with non-white minorities (31 minutes vs. 22 minutes, p=.08), and more time with depressed compared with non-depressed patients (43 minutes vs. 23 minutes, p=.02). In an ANOVA model, lower reduced-set MMS scores (p=.02), white race (p=.02), depression (p=.15), and a race-depression interaction term (.045) were associated with more nursing bedside time.

CONCLUSION: In this population, nurses spent more time with patients who had worse mental function (even after excluding those who were frankly demented), and more time with white patients who were depressed compared with both non-depressed white patients and minority patients regardless of depression score. DNR orders were not associated with less nursing contact. While some of these results appear to suggest unexpected differences in the time nurses give to patients, firm conclusions will require further study.

CHALLENGING BUT SUPPORTIVE ENVIRONMENT FOR GENERAL INTERNAL MEDICINE AT A JAPANESE UNIVERSITY HOSPITAL. Y. Takeda Yamashita¹, K. Yamamoto¹, Y. Jin¹, H. Sakugawa¹, M. Masuda¹, Y. Komine¹, Y. Yoshii¹, C. Ogura¹; ¹Ryuky University Hospital, Nakagami, Okinawa

PURPOSE: General internal medicine (GIM) as an academic discipline was recently introduced to medical schools in Japan. Residency programs are still quite young and fully trained primary care physicians are in shortage. In May 1999, the Ryuky University Hospital started a general medicine clinic to provide better care for its patients. However, since there

were no staff trained in GIM, specialists from the Department of Medicine and Department of Psychiatry were assigned to the clinic. We conducted two surveys to assess the attitudes toward and perceptions of GIM practice among specialists at different academic disciplines in the hospital.

METHODS: First a survey was carried out using a questionnaire that was distributed to all physicians who work at the Ryukyu University Hospital. We also conducted a 2nd survey which focused on the experience at the GIM clinic for specialty physicians who had ever practiced at the clinic.

RESULTS: Overall response rates were 66.3% (n=236) and 75% (n=48), respectively. The clinic performance over the first year was evaluated and rated from -3 (extremely poor) to +3 (excellent). Respondents whose specialty was not in internal medicine perceived the GIM clinic more positively (average +1.0) compared to internists with and without GIM clinic experience (average -1.9 and -1.6, respectively), which reached statistical significance ($p < 0.001$). Among residents, fellows and faculties, there was no statistical difference regarding attitudes toward the competence of primary care physicians, encouragement and positive regard for GIM. Eighty percent of internists who had practiced at the GIM clinic agreed that their clinical competencies as a specialist were not sufficient to be a primary care physician. 92% of them disagreed with the hospital's continued operation of the GIM clinic under current administration, however, if GIM trained physicians were stationed at the clinic, the same number of internists agreed that the clinic would play an important role at their hospital. Psychiatrists were more willing to cooperate with GIM clinic compared to other specialties.

CONCLUSION: Historically, the culture of academic health centers had not been hospitable toward primary care. However, our survey demonstrated that the chilly climate has been changing. Only very few physicians perceive primary care tasks as not requiring a high level of expertise. In many schools, increased attention to education for primary care practice is evident in Japan. A more generalist workforce with competence and expertise in primary care is essential.

A SURVIVAL ADVANTAGE FOR AFRICAN AMERICAN PATIENTS IN A COMMUNITY BASED PROGRAM OF ALL-INCLUSIVE CARE FOR FRAIL ELDERLY. E.J. Tan¹, L. Lui¹, C. Eng¹, K.E. Covinsky¹; ¹University of California, San Francisco, San Francisco, CA

PURPOSE: Previous work has demonstrated that African American elders experience higher functional and cognitive disability than White elders. The Program of All-Inclusive Care for the Elderly (PACE) provides comprehensive community-based long-term care services for nursing home eligible elders who remain in the community. Once enrolled patients have equal access to all needed medical and long-term care services at no additional cost. We examined the impact of enrollment in the PACE system on mortality differences between African American and White patients.

METHODS: 2002 White patients and 859 African American patients enrolled at 12 nationwide PACE sites from 1990-1996. All patients were followed from enrollment through death, or 12/31/97. On enrollment, measures of demographic characteristics, functional status, and comorbid medical diagnoses were obtained by social workers, nurses, and physicians. We used the method of Kaplan-Meier to describe mortality rates in African American and White patients. We used the log-rank test and Cox-proportional hazards models for bivariate and multivariate comparisons of mortality rates for African American patients vs. White patients.

RESULTS: On enrollment into PACE, African American patients were younger than White patients (77.4 v 79.6, $P < .01$). However, African American patients had worse functional status (mean score on 10 point ADL scale, 6.5 v 7.2, $p < .01$) and were more likely to be diagnosed with dementia (56% v. 45% $p < .01$). 74% of both African American and White patients were women. A survival advantage for African Americans emerged after 9 months of enrollment in PACE. Rates of mortality at 1, 2, and 3 years after enrollment were 13%, 24%, and 33% for African American patients and 16%, 29%, 39% for White patients (HR for African American patients = 0.76; 95% CI 0.66 - 0.88). After adjustment for PACE site, age, gender, education, caregiver support, ADL function, presence of dementia and other comorbid conditions, African American patients continued to have a lower rate of mortality than White patients (HR = 0.79; 95% CI 0.66 - 0.95).

CONCLUSION: In a system providing equal access to comprehensive medical and long-term care services to frail elders, African American patients have substantially lower rates of mortality than White patients after adjustment for age, comorbidity, and ADL function on enrollment.

POTENTIALLY AVOIDABLE HOSPITALIZATION DAYS: A PROSPECTIVE ANALYSIS. A. Torn¹, A. Pichon-Riviere², C. Romero¹; ¹Sanatorio de la Trinidad, Capital Federal, Buenos Aires; ²Programa de Efectividad Clínica - Argentina, Capital Federal, Buenos Aires

PURPOSE: Analyze the distribution and causes of the potentially avoidable additional hospitalization days (PAAD) in an institution that has a hospitalist program.

METHODS: From 06/99 to 01/00 the surgical and medical admissions of adult patients to a private hospital in Buenos Aires, Argentina were followed prospectively to detect PAAD. The additional days were defined using the Millman and Robertson's Optimal Recovery Guidelines. The case management nurse assigned a cause from a set of 16 possibilities to each additional day and categorized the admissions in 5 severity levels.

RESULTS: There were 1812 medical and surgical admissions of adult patients; 626 (34.55%) were medical and 1186 (65.45%) surgical from which 27.4% were emergency and 72.6% programmed procedures. The mean age was 56.3 years with a 25% of the patients older than 74 years; 58.5% were women. The average length of stay (ALOS) was 4.1 days with a median of 2.1; 5.8 for the medical admissions and 3.2 for the surgical ones. The difference between the programmed and emergency operations was significant: 2.6 vs. 4.9. There was also a significant increase of the ALOS with categories of age and severity. The inpatient mortality was 3.5%, 93.4% were discharged and 3% were transferred to a rehabilitation center or to home care. A total of 723 PAAD, representing a 9.64% of the hospitalization days, were generated by 600 admissions. The physicians were responsible of

668 PAAD (92.40%). Inappropriate admission (potentially ambulatory procedures), was the cause of 392 (54.22%) PAAD, 181 (46.17%) were due to gynecological procedures, 73 (18.62%) to general surgical and 62 (15.82%) to traumatological ones. Prolongation of hospital stay was the cause of 331 (45.78%) PAAD divided in 55 days (16.62%) attributed to the health care system and 276 (83.38%) to the physicians. Of the latter 106 (38.40%) were responsibility of traumatology and 82 (29.71%) of gynecology. A logistic regression model adjusted for severity, age, sex, type of admission and specialty was analyzed. The following specialties were positive predictors of PAAD compared to general surgery: gynecology (OR 7.84), traumatology (OR 2.75) and neurosurgery (OR 4.58). Emergency surgery (OR 0.61) was a negative predictor of PAAD and age (OR 1.01) was a positive one.

CONCLUSION: Even though this study was conducted in an institution where a hospitalist program is in place, there are still a significant number of potentially avoidable additional days. They are concentrated in programmed surgical procedures mainly gynecological and orthopedic, and in the lower levels of severity.

MANAGED CARE AND PRIMARY CARE PHYSICIANS PERCEPTION OF PATIENT ACCESS TO HIGH QUALITY MENTAL HEALTH SERVICES. B.W. Van Voorhees¹, N. Wang¹, D.E. Ford¹; Johns Hopkins University, Baltimore, MD

PURPOSE: Primary care physicians (PCPs) are often required to act as gatekeepers and or patient advocates for mental health services in an extremely complex administrative environment that includes managed care organizations with and without mental health carve-outs. This analysis addresses two aspects of this environment: (1) number of managed care contracts per practice (2) percentage of patients for whom the physician serves as managed care gatekeeper.

METHODS: Data from the 1996-97 Physician Community Tracking Study performed by the Center for Studying Health System Change were analyzed. This cross-sectional survey consisted of telephone interviews of 12,385 physicians (5,583 primary care physicians caring for adults) and had a response rate of 65 percent. Access to high quality mental health services was defined as PCPs indicating that they always or almost always could obtain high quality mental health services for their patients.

RESULTS: Perceived availability of high quality services was much lower for outpatient mental health than for outpatient medical specialists (30% vs. 81 %, p -value < 0.001). Perceived access to high quality outpatient mental health services declined with increasing numbers of managed care contracts: zero (44%, Odd Ratio 2.3, CI 1.8, 3.1), one (37%, OR=1.8), two to ten (27%, OR=1.1), eleven to twenty-five (29%, OR=1.2) and more than twenty five (25%, OR=1.0). These findings persisted after adjustment for patient demographics (Medicare percentage, Medicaid percentage), physician demographics (age, gender, foreign vs. US medical graduate and overall career satisfaction) and practice characteristics (type of practice and concern about financial penalties for practice decisions) using multiple logistic regression analysis. An adjusted OR of 1.9 (p -value of < 0.001) was observed for zero contracts compared to twenty- five or more. A U-shaped distribution in perceived access was observed for the percent of patients in the practice for whom physician served as gatekeeper: three percent or fewer patients in gatekeeper status (42%, OR 2.6), more than ninety percent of patients (43%, OR 2.6) and fifty-two to sixty percent of patients were in gate-keeping status (22%, OR 1.0). These findings persisted after adjustment using the same model described above (p -value < 0.001). The same patterns were observed for inpatient psychiatric services. The same relationships between access to high quality services and number of managed care contracts and gatekeeper percentage were not observed for medical services.

CONCLUSION: The mental health referral process may be uniquely vulnerable to the administrative burdens imposed by the current managed care environment. Paradoxically, while large numbers of managed care plans with a variety of options are thought to increase choice, they may actually decrease access to high quality mental health services.

INDIGENT PATIENTS AND PRESCRIPTION DRUG ADHERENCE: A PHYSICIAN SURVEY. J.D. Voss¹, J.M. Schectman¹, M.M. Nadkarni¹; ¹University of Virginia, Charlottesville, VA

PURPOSE: For many chronic diseases, effective drug therapy is now available. However, patients often fail to adhere to prescribed regimens with suboptimal clinical results. We surveyed resident and faculty physicians (MD) of a university-based internal medicine clinic serving an indigent population to determine MD attitudes, beliefs and behaviors with respect to patient drug adherence (DA).

METHODS: One hundred two MDs completed a 14 question survey asking them to define 'good' adherence, estimate the proportion of their patients who achieve 'good' patient DA, and to respond to a series of statements regarding DA. Two mailings produced an 86% response rate. At least 20 responses were obtained from each of PGY1,2,3 and attending MD levels.

RESULTS: Seventy nine percent of MDs defined 'good' adherence as patients taking at least 80% of prescribed doses. However, MDs estimated that only 66+/-15% (mean/SD) of their patients achieved 'good' adherence. In the prior 6 months, 50% of MDs discussed DA with at least 80% of patients, and 91% of MDs reported discussing DA with at least 50% of patients. Despite these discussions, only 32% of MDs felt that at least half of their patients had a good understanding of the importance of drug therapy to avert adverse outcomes. Almost all (97%) MDs felt that DA assessment and intervention was an important component of the office visit and 100% felt that prescription refill feedback would help them target patients for adherence intervention. There was no association between year of training and DA responses.

CONCLUSION: In this indigent care setting, MDs indicated that they frequently discussed DA with patients but felt that patients lacked a good understanding of the role of their medication and that many were not adherent to their prescribed regimens. All MDs felt that specific DA feedback based on prescription refill data would assist them to identify and manage non-adherent patients. Since automated pharmacy claims systems cover the majority of the US population, such feedback is feasible and worthy of future study in both indigent and more affluent populations.

PATIENT AND PROVIDER REPORTED ADHERENCE: A CLINICALLY USEFUL APPROACH TO MEASURING ADHERENCE. J.H. Wagner¹, A.C. Justice¹, G. Sinclair², S. Weissman³, ¹VA Pittsburgh Healthcare System, Pittsburgh, PA; ²University of Texas Southwestern Medical Center at Dallas, Dallas, Texas; ³Yale University School of Medicine, New Haven, CT

PURPOSE: Adherence to antiretroviral therapy is important to prevent the development of resistance to treatment. There is however, no gold standard for the assessment of adherence. If we are to improve adherence we must first be able to clinically identify nonadherent individuals. Because there is no gold standard measure of adherence, this can best be achieved with a clinical measure or combination of measures that demonstrate the strongest and most consistent expected associations with viral load, active alcohol and drug use, depression, and homelessness. **METHODS:** VACS is a multi-site cohort of 881 HIV+ adults from Cleveland, Manhattan, and Houston VA HIV clinics assembled between 6/99-7/00, representing 85% of the patients seen at these clinics during this interval. Patients filled out a survey that included questions regarding adherence to their HIV medication, alcohol and drug use, depression measured using the CESD-10, and homelessness (past four weeks and lifetime). Providers were also queried about their patients' adherence to HIV therapy, suspected drug and alcohol use, and psychiatric comorbidities. HIV-1 RNA viral load was obtained from VA computerized laboratory data and was analyzed as a continuous measure (log10).

RESULTS: Both patient and provider reported adherence were strongly related to decreasing viral load ($p < .01$). Current alcohol consumption ($p < .01$) and patient and provider reported drug use ($p < .01$) were strongly related to poorer adherence. Poorer adherence was also associated with patient reports of binge drinking ($>=6$ drinks on one occasion) ($p < .01$), drug use more than 2-4 times per month ($p < .01$), failing to do what was normally expected of them ($p < .01$), and finding that they were unable to stop taking drugs in the past year ($p < .01$). Provider reports of patient depression were only weakly associated with poorer adherence but patient completed CESD scores demonstrated a strong relationship with poorer adherence ($p < .01$). The association between patients reported poorer adherence and homelessness were strong ($p < .01$). After controlling for the significant association between patient reported adherence and the outcomes of interest using multivariate analyses, there was a clear and consistent independent association between provider reported adherence and viral load ($p < .001$), present alcohol use ($p < .001$), present drug use ($p < .001$), and depression ($p < .01$). **CONCLUSION:** Both provider and patient reported adherence demonstrated expected and independent associations with viral load, alcohol consumption, illicit drug use, depression, and homelessness. Patient and provider report together may provide a foundation for a clinically useful standardized measure of adherence.

ACADEMIC GENERAL INTERNAL MEDICINE, FAMILY MEDICINE, AND GENERAL PEDIATRICS: WHOS MOST SATISFIED? S. Wahi-Gurra¹, R. Friedman², J. Speckman², A. Ash², P. Carr³, M. Moskowitz²; ¹Section of General Internal Medicine, Boston University Medical Center/ENRM Veterans Hospital, Boston, MA; ²General Internal Medicine, Boston University Medical Center, Boston, MA; ³Department of Medicine, Massachusetts General Hospital, Boston, MA

PURPOSE: Little is known about the careers of academic generalist physicians. We compared career satisfaction and achievement for academic faculty in Family Medicine (FM), General Internal Medicine (GIM), and General Pediatrics (GP).

METHODS: All full-time primary care physicians from 24 randomly selected U.S. medical schools were sent a 177-item self-administered survey in 1995. We constructed an overall job satisfaction scale based on faculty evaluation of: work setting, potential to achieve goals, overall professional satisfaction, and professional expectations (Cronbach's alpha=0.87). In addition, faculty assessed their satisfaction with rate of academic progress, the extent they felt welcomed at their institutions, and how likely they were to leave academic medicine; and also reported their academic rank and salary. We categorized physicians by dominant role: clinician, researcher, educator, administrator, or "hybrid", based on self-reported percentages of time spent in various professional activities. Analyses were adjusted for age, gender, race, seniority, hours worked per week, institution, and status as a clinician or researcher, or primary care type when applicable. **RESULTS:** The 500 physician responders consisted of 142 FM, 207 GIM, 151 GP (overall response rate 62%). For all groups, the mean ages ranged from 41-45 and most were white (80-89%). In multivariable analyses, FM physicians reported higher job satisfaction ($p=0.01$) and satisfaction with their rate of academic progress ($p=0.21$). GIM and GP faculty were less likely to achieve the rank of full professor [odds ratio 0.3 ($p < 0.02$) for both] and more likely to leave academic medicine ($p=0.01$). Mean salary for FM (\$120,000) was significantly higher than for GIM (\$102,000) and GP (\$97,000), $p < 0.001$. All three groups reported similar scores (mean 4.1 on 5-point Likert scale) on the extent they feel welcome at their respective institutions ($p=0.95$). Adjusted analyses examining faculty role revealed that clinicians were less satisfied with their jobs ($p=0.01$) and were more likely to leave academic medicine ($p=0.001$).

CONCLUSION: FM physicians had higher career satisfaction and salaries and were more likely to achieve senior rank than other generalist faculty. Physicians with primarily clinical responsibilities were less satisfied. Interestingly, FM, GIM, and GP faculty all felt welcome at their institutions. Understanding the sources and differing levels of dissatisfaction is important in recruiting and retaining academic primary care faculty.

PHYSICIAN WILLINGNESS TO PRESCRIBE MIFEPRISTONE. L. Waldman¹, E. Losina¹, S. Franco¹, A. Ash¹, M. Moskowitz²; ¹Boston University School of Medicine, Boston, MA

PURPOSE: Mifepristone (RU-486) is an anti-progesterone agent that has just become available for early medical abortion in the US. We examined willingness to provide mifepristone for physicians likely to face the issue of medical abortion: Primary care physicians (PCPs) in Internal Medicine (IM) and Family Medicine (FM), as well as Obstetrician/Gynecologists (OBs). We also compared PCPs to OBs on factors related to willingness to provide Mifepristone.

METHODS: A 29-item survey was mailed to random samples of Massachusetts IMs, FMs, and OBs. Topics included demographics, practice patterns of reproductive health, and attitudes regarding mifepristone. Willingness to provide mifepristone was collapsed from a 5-point scale

to a dichotomous variable: 'willing' and 'not willing'. Bivariate analysis was conducted, using Chi-square analysis to test for associations. We examined physician willingness to prescribe mifepristone and factors associated with willingness.

RESULTS: Of 600 physicians surveyed, 512 met eligibility requirements, of whom 300 responded (50%). Most physicians surveyed (86%) were aware of mifepristone and its use in reproductive services. PCPs were almost as willing to provide mifepristone as OBs (53% vs. 57%, $p=0.12$). Of PCPs who had never performed a surgical abortion, almost half (47%) were willing to offer mifepristone. PCPs with prior experience prescribing emergency hormonal contraception (EC), at least once, were more willing to prescribe mifepristone than PCPs who had never prescribed EC (58% vs. 24%, $p=0.001$). Many PCPs (58%) were dissuaded by fear of complications, while fewer OBs (23%) were put off by this factor ($p=0.001$). Most physicians (67%) indicated that patient request for mifepristone would influence them to prescribe it.

CONCLUSION: Given high physician willingness to provide mifepristone, US abortion services may now expand to include PCPs who have not formerly performed surgical abortion. PCPs more likely to prescribe mifepristone may be those who have comfort with EC. If PCPs are to offer mifepristone, they may need training to address fears of complications. As physicians consider whether to become mifepristone providers, patient requests for this service may play a large role.

PRESCRIPTION OF BETA-ADRENERGIC ANTAGONISTS AMONG OLDER ADULTS HOSPITALIZED FOR ACUTE MYOCARDIAL INFARCTION, AS ASSESSED THROUGH ELECTRONIC MEDICAL RECORDS M. Weiner¹, T. Stump¹, J. Lewis², C. Callahan¹; ¹Indiana University Center for Aging Research, Regenstrief Institute for Health Care, Indianapolis, IN; ²Health Care Excel, Louisville, KY

PURPOSE: Despite mounting evidence that beta-adrenergic antagonists (beta-blockers) are beneficial immediately following acute myocardial infarction (AMI), a median of only 64% of Medicare beneficiaries nationwide receive beta-blocker within 24 hours of AMI. Measuring use of beta-blockers in AMI usually requires expensive chart reviews. Using a comprehensive electronic medical records system instead of chart review, we sought to measure use of beta-blockers in AMI as well as differences in charges between those who received beta-blocker and those who did not.

METHODS: We examined data from all 15,610 Medicare-covered hospitalizations during 1995-99 for 7,251 patients in an urban academic medical center, evaluating for International Classification of Diseases codes for AMI. Restricting study to patients 65 years of age or older, we collated and merged into this all available prescription drug data pertaining to these patients, separating inpatient from outpatient drugs based on date. We then calculated rates of beta-blocker use in AMI and differences in estimates of total hospital charges and length of stay depending on whether beta-blocker were prescribed.

RESULTS: Of 9,191 hospitalizations among 4,490 older adults, 385 (4%) contained a diagnostic code for AMI. Beta-blockers were prescribed in 252 (65%) of the encounters, and 85% of prescriptions occurred on the date of admission or one day later. Overall, beta-blocker prescriptions occurred on one of these days in 55% of hospitalizations for AMI. Metoprolol was the beta-blocker prescribed most often (77% of beta-blocker use in AMI). Of those receiving any beta-blocker, 3% had a diagnostic code for obstructive lung disease, compared to 5% among those not receiving beta-blockers ($p = 0.37$). Those receiving beta-blockers had mean length of stay 1.5 days longer ($p < 0.05$) but mean charges not significantly different from those not receiving beta-blockers.

CONCLUSION: A comprehensive, electronic medical records system can be used instead of resource-intensive chart review to measure use of beta-blockers after AMI. Uncertainty about the precise relationship between the two methods should prompt, as a next investigative step, a more direct comparison of chart review to the use of electronic records to measure indicators of quality of healthcare.

ANTIHYPERTENSIVE MIGRATION: HOW FREQUENTLY DO MEDICATION CLASS SWITCHES OCCUR? M.G. Weiner¹, A. Cohen¹; ¹University of Pennsylvania, Philadelphia, PA

PURPOSE: Despite the proven survival benefit of beta blockers and diuretics, research has shown that these agents are underutilized in practice. Contributing to this finding is the perception that other agents may work better or have fewer side effects. Under the assumption that a medication class switch represents a physician's belief that the medication was of little benefit, or was not well-tolerated by the patient, this study compares the degree to which each of the major classes of antihypertensive medications is discontinued in favor of another class of antihypertensive.

METHODS: We examined pharmacy data from a population of Medicare patients enrolled in a managed care organization (MCO) between December 1996 and October 2000. An overall interval of antihypertensive medication usage was determined for each patient. The subset of this time interval in which the patient received a beta blocker (BB), calcium channel blocker (CCB), ACE inhibitor (ACEI), or diuretic was determined. A therapy class switch was defined as prescribing a new class of antihypertensive medication more than one month after the end of the original antihypertensive class interval ended. In this manner, the addition of a second antihypertensive would not be considered a class switch, unless the first medication was stopped permanently, and the second medicine continued for more than a month.

RESULTS: Of the 9798 patients who filled at least 3 antihypertensive prescriptions over at least 90 days, 4260 received a BB, 4225 received a CCB, 4533 received an ACEI and 3480 received a diuretic (sum exceeds total since patients could have been on more than one class). Kaplan Meyer survival analysis indicates that after 1000 days, a higher percentage of patients continue to take BB (68.5%) than ACEI (63.8%), CCB (61.8%) or diuretics (53.4%). The length of time until 25% of patients had switched their medication class was longest for the group taking BB (760 days), then ACEI (589 days), CCB (568 days) and diuretics (387 days). This trend favoring BB is also observed when limiting the analysis to classes of antihypertensives that were initiated in the observation period.

CONCLUSION: The results of the study suggest that, contrary to common perceptions, among a Medicare population, beta blockers are not more commonly discontinued than the

more modern, and expensive antihypertensive agents. While the clinical factors that influence a physician's decision to start a particular medication class are not known, this potential bias is mitigated by the common tenets of good clinical practice. These tenets dictate that when choosing to prescribe any class of drug, the physician believes it would be safe, effective and well tolerated. The study results suggest that, regardless of the initial indications for choosing a particular therapeutic class of antihypertensive, or the reason for eventually changing the class (non-compliance, patient complaints, ineffectiveness, expense, etc.), BB seem less likely than ACEI and CCBs to be switched, while diuretics are more likely than the other classes to be discontinued.

QUALITY OF END-OF-LIFE CARE FOR VULNERABLE ELDER. N.S. Wenger¹, K.E. Rosenfeld², C. Maclean¹, D.H. Solomon¹, C.P. Roth³, D. Saliba³, R. Louie³, R.T. Young¹, P.G. Shekelle³. ¹University of California, Los Angeles, Dept of Medicine, Los Angeles, CA; ²none given, Santa Monica, CA; ³RAND, Los Angeles, CA

PURPOSE: Attention to issues of autonomy, surrogates and treatment preferences are important for vulnerable elders, who are at risk of decisional incapacity and death. We investigated these elements of end-of-life care among a cohort of vulnerable elders.

METHODS: The Assessing Care of Vulnerable Elders project developed and tested quality indicators (QIs) for community-based persons' 65 years at increased risk of death or functional decline. Based on a systematic literature review, an expert panel rated 14 QIs related to documentation about surrogates and preferences, and translation of preferences into care as valid process measures for end-of-life care. We measured these QIs by reviewing 13 months of inpatient and outpatient charts of patients in 2 managed health plans.

RESULTS: Of 3206 seniors randomly selected from the plans, 2,278 (71%) were interviewed and 475 were identified as "vulnerable". Of 400 (84%) who consented to study participation and had abstractable medical records, 376 were eligible for at least one QI. Fourteen elders (4%) had any documentation (formal or informal) about surrogate decision makers in their outpatient charts. Charts for 103 hospitalizations yielded 8 advance directives, 23 noted a surrogate or an attempt to find one, and 72 (70%) contained no documentation. When hospitalized with altered mental status, 5 of 20 elders had any documentation about a surrogate within the first 3 hospital days and 1 of 6 elders admitted to an ICU had any documentation about preferences within 3 days. Seventy percent of orders to limit care in the hospital considered patient participation in decision-making and 2 of 2 ventilated patients had documentation about preferences.

CONCLUSION: Despite much recent attention to end-of-life care, surrogates and preferences are not elicited prospectively from vulnerable elders. Improvement in the quality of end-of-life care is needed, particularly for persons at risk of life-threatening events.

VULNERABLE ELDERLY RECEIVE WORSE QUALITY CARE FOR GERIATRIC CONDITIONS THAN FOR MEDICAL CONDITIONS. N.S. Wenger¹, D. Solomon², C. Maclean¹, D. Saliba², C. Roth², C. Kamberg², L. Rubenstein³, J. Schnelle⁴, R.T. Young⁵, P.G. Shekelle⁶. ¹UCLA Dept of Medicine, Los Angeles, CA; ²RAND, Santa Monica, CA; ³Sepulveda VA, Sepulveda, CA; ⁴UCLA, Santa Monica, CA; ⁵University of California, Los Angeles, Pacific Palisades, CA; ⁶RAND, Los Angeles, CA

PURPOSE: Measurement of quality of care (QOC) for mainstream medical conditions has received much attention, but no method exists to measure QOC for geriatric conditions. We developed a system to comprehensively measure QOC provided to vulnerable elders and compared QOC for geriatric and medical conditions.

METHODS: Based on structured literature reviews, content experts developed 239 quality indicators (QIs) for the process of medical care provided to "vulnerable elders", community-based persons > 64 years old at increased risk of death or functional decline. These QIs assessed care for 22 conditions. We applied these QIs to a community sample of vulnerable elders by abstracting their outpatient and inpatient charts, comparing QOC provided for medical conditions (depression, diabetes, heart failure, hypertension, coronary disease, osteoarthritis, osteoporosis, pneumonia and stroke) to geriatric conditions (delirium, dementia, falls, geriatric assessment, incontinence, malnutrition and pressure ulcers)

RESULTS: Of 3206 seniors randomly selected from two managed senior health plans, 2,278 (71%) were interviewed and 475 were identified as "vulnerable". Of these, 400 (84%) consented to participate and had abstractable charts. For medical conditions, 52% of 1897 QIs were satisfied (e.g. diabetes 58%, heart failure 72%). This was significantly better ($p < 0.01$) than the 37% of 2036 QIs satisfied for geriatric conditions. Compliance rates for geriatric conditions included falls 26%, dementia 35% and incontinence 29%.

CONCLUSION: QOC for geriatric conditions falls far below that of QOC for medical conditions. Quality assessment and assurance systems should incorporate QIs measuring geriatric syndrome care. Care for geriatric conditions—common among vulnerable elders and associated with functional decline and mortality—must be improved.

MEASURING THE CONTINUITY AND COORDINATION OF CARE AMONG VULNERABLE ELDERLY R.T. Young¹, D. Solomon², D. Saliba², C.P. Roth², C. Maclean³, P.G. Shekelle⁴, N.S. Wenger¹. ¹University of California, Los Angeles, Los Angeles, CA; ²RAND, Santa Monica, CA; ³UCLA Dept of Medicine, Los Angeles, CA; ⁴RAND, Los Angeles, CA

PURPOSE: Continuity and coordination (CC) of care are core elements of general internal medicine. This is particularly important for "vulnerable elders" who may be at high risk of adverse events due to poor CC. Yet there is no method of measuring CC in medical care. We developed and tested a system to measure CC of care among a cohort of vulnerable elders.

METHODS: The Assessing Care of Vulnerable Elders project developed and tested quality indicators (QIs) for community-based persons > 64 years at increased risk of death or functional decline. Based on a systematic literature review, an expert panel rated 13 QIs as valid process measures of care CC. Six of these could be measured by abstracting information from medical records. QIs evaluated medication, test and information continuity, within and between providers, across the inpatient and outpatient settings.

RESULTS: Of 3206 seniors randomly selected from two managed health plans, 2,278 (71%) were interviewed and 475 were identified as "vulnerable". For the 400 (84%) patients who consented to study participation and had abstractable medical records, 67% of 324 CC QIs were satisfied. Of 185 seniors started on new medications during the study, 67% had follow-up of this medication at the next physician visit. Of 72 patients who were hospitalized, 89% received post-discharge follow-up within 6 weeks; 6 of 11 had follow-up of a medication started in the hospital and 10 of 14 had follow-up of a test pending at hospital discharge, but only 41% had a copy of the discharge summary in the outpatient chart.

CONCLUSION: Vulnerable elders at two managed care plans received a high level of CC of care. Measuring CC care identifies aspects that require intervention.

THE "HASSLE FACTOR: HOW PHYSICIANS RESPOND WHEN THIRD PARTY PAYERS DECLINE REIMBURSEMENT FOR IMPORTANT HEALTH CARE SERVICES. R.M. Werner¹, G.C. Alexander¹, A. Fagerlin², P.A. Ubel². ¹University of Pennsylvania, Philadelphia, PA; ²Ann Arbor Veterans Affairs Medical Center, Ann Arbor, MI

PURPOSE: In an effort to control health care costs, many third party payers limited physicians' abilities to order expensive tests for their patients, requiring the use of an appeals process to obtain restricted medical services. Studies suggest that, at times, physicians are willing to misrepresent clinical information to third party payers in order to obtain these services for their patients. Little is known about whether characteristics of the appeals process influence physicians' willingness to sanction deception.

METHODS: We surveyed a random sample of 1,617 physicians in the U.S., and presented each with a hypothetical scenario in which an insurance company refused to pay for a medically important service. We varied the "hassle" of the appeals process by varying the time required to make an appeal and the likelihood of having a successful appeal. In addition, we varied the severity of the patient's health condition from moderate low back pain to severe angina. Physicians were asked if the patient's physician should accept the insurance company's restriction, appeal the restriction or misrepresent the facts to the insurance company in order to obtain coverage for the patient.

RESULTS: Overall, most physicians responded that they would appeal (77%) rather than accept (12%) or misrepresent (11%) a restriction on medically necessary care. Physicians were more likely to misrepresent if the appeals process was longer (OR=1.68, CI 1.09–2.59), less likely to succeed (OR=1.57, CI 1.02–2.42), or involved a more severe condition (OR=2.19, CI 1.39–3.45). Among physicians asked about severe angina, as the appeals process became more cumbersome, they were more likely to choose to misrepresent the facts to the insurer rather than appeal. When asked about a patient with moderate low back pain, physicians' decisions about how to respond to an insurance company restriction were not significantly affected by the hassle associated with the appeals process.

CONCLUSION: Physicians' willingness to sanction deception of insurance companies varies according to length of the appeals process, the likelihood of a successful appeal, and condition severity. When the stakes are high, and the hassle of appeals is great, physicians become increasingly willing to sanction deception. Changes in the difficulty of appeals processes may ease the tension physicians' face between patient advocacy and honesty.

GENERAL INTERNIST SATISFACTION AND RECRUITMENT OF MEDICAL STUDENTS: RESULTS FROM THE PHYSICIAN WORKLIFE STUDY. T.B. Wetternick¹, M. Linzer¹, J. Douglas¹, J.E. McMurray¹. ¹University of Wisconsin-Madison, Madison, WI

PURPOSE: To assess general internists' (GIM) satisfaction and recruitment of medical students and compare these to family physicians (FP).

METHODS: 5704 physicians were surveyed (adjusted response rate 52%) from a national random stratified sample—19% GIM (n=450, 108 women, 75 were "young" (age < 45); 341 men, 109 were young) and 22% FP (n=502). Mean global job and specialty satisfaction as well as likelihood of recommending their specialty to medical students were compared by specialty. All were assessed on a 5 point Likert scale (3 = neither agree nor disagree). For general internists, further analyses were done by age and gender and logistic regression was used to model predictors of medical student recruitment.

RESULTS: GIM had significantly lower job and specialty satisfaction vs FP (job: 3.52 vs 3.77; specialty: 3.18 vs 3.69, both $p < .001$). Global satisfaction did not differ by gender alone for GIM. However, younger women and older men had higher job satisfaction than their same gender counterparts (women 3.75 vs 3.28, $p=.043$, men 3.63 vs 3.28, $p=.015$) with no difference in specialty satisfaction. GIM was less likely than FP to recruit medical students to their specialty (3.26 vs 3.85, $p < .001$). Women GIM recruited with more enthusiasm than men (3.56 vs 3.16, $p=.022$) with younger women the most enthusiastic (young 3.67 vs old 3.35, $p=.038$). Global specialty satisfaction and female gender were positively associated with medical student recruitment by internists.

CONCLUSION: General internists have lower job and specialty satisfaction than family physicians. This dissatisfaction leads to less recruitment of medical students into internal medicine. Increasing numbers of young women in GIM may change this trend but recruitment needs to be seen as an important outcome of physician satisfaction to keep the specialty viable.

DO PATIENTS WITH HIV WHO HAVE MANAGED CARE COVERAGE GET LOWER QUALITY CARE DO PATIENTS WITH HIV WHO HAVE MANAGED CARE COVERAGE GET LOWER QUALITY CARE? J.B. Wilson¹, S. Greenfield², S.H. Kaplan², L.M. Sullivan³. ¹New England Medical Center, Boston, MA; ²Tufts University, Boston, MA; ³Boston University, Boston, MA

PURPOSE: Anecdotes suggest that quality of care is worse for persons with MC coverage compared with FFS coverage. In this study we compared the quality of care received by persons with HIV having these 2 coverage types who were cared for by high volume (> 50 patients) HIV providers in the Boston area.

METHODS: Three types of quality indicators were assessed by mailed survey and chart review: ratings of interpersonal care (IPC, 7 scales, including general communication, HIV-specific

information, and overall satisfaction), ratings of health plan quality (HPQ, 4 scales, including cost ratings and service restrictions), and technical care (TC, 4 indicators, including rates of highly-active antiretroviral therapy (HAART), and rates PPD and flu shots in the last year). To classify patients as MC or FFS, both self-report and care site administrative data were used. For each of the 15 quality indicators, we compared patients with MC and FFS coverage by performing hierarchical multiple linear and logistic regression analysis for continuous and dichotomous outcomes, adjusting for clustering of patients within physicians, age, and health status. Power to detect small differences (5–6 points, .25 SD) for each IPC and HPQ scale was >80%.

RESULTS: Of 29 eligible providers, 22 (78%) participated. Of 774 participating patients, 620 (83%) returned questionnaires. Mean patient age was 41 years, 17% were women, 71% were white, and 49% had MC coverage. Mean scores (SD) for selected IPC and HPQ scales (0–100 point scales) were: general communication, 87 (18); HIV specific communication, 75 (27); plan cost rating, 64 (29); service restrictions 71 (24). TC rates were: HAART, 68%; PPD, 14%; and flu shots, 32%. In multivariable models there were no significant differences between MC and FFS coverage for any of 7 IPC, 4 HPQ, or 4 TC quality indicators. Models controlling for race, education, sexual preference, and duration of physician–patient relationship, which varied by plan type, were no different. Tests for an interaction between physician and FFS/MC differences were non-significant.

CONCLUSION: For patients with HIV in Boston, managed care coverage was not associated with lower quality care. These experienced HIV providers do not appear to practice differently depending on insurance type. Our findings suggest that managed care coverage can be as effective as FFS coverage even for patients with a complex, expensive, chronic illness.

PHYSICIANS WANT EDUCATION ABOUT COMPLEMENTARY AND ALTERNATIVE MEDICINE TO BETTER COMMUNICATE WITH THEIR PATIENTS. L.C. Winslow^{1,2}, H. Shapiro; ¹University of Colorado Health Sciences Center, Denver, CO; ²HealthONE Alliance, Denver, CO

PURPOSE: Studies have revealed that at least one-third of patients in the United States use complementary / alternative medicine (CAM) treatments. Most patients who are using CAM are also seeking conventional medical care. There is, however, little information about how physicians and patients communicate about CAM. We hypothesized that physicians would frequently field questions about CAM from their patients, but, due to a lack of education about these treatments, would feel uncomfortable discussing them. We therefore surveyed a metropolitan area group of physicians to ascertain their patterns of communication with patients regarding CAM and the factors that influence them to discuss CAM with their patients or refer them for CAM treatments.

METHODS: Seven hundred fifty one randomly selected physicians in all specialties in the Denver, Colorado metropolitan area were asked about their exposure to, knowledge of, and belief in 16 CAM modalities and patterns of physician — patient communication regarding CAM. Analyses were conducted using the SAS system (version 6, fourth edition, Cary, NC, 1989) and included the two-tailed t-test, Pearson correlation coefficients, linear regression, and multivariate analysis.

RESULTS: Three hundred and two of 705 deliverable surveys (43%) were returned. Responders averaged 44 years old (+/- 12 years), 63% were male, and 48% were self-designated primary care physicians. Seventy-six percent of physicians reported having at least one patient using a CAM therapy, 59% had received queries from patients about specific CAM treatments, 48% had recommended a specific CAM therapy to at least one patient, and 24% had used a CAM modality for themselves. While patients frequently sought out information about CAM modalities from their physicians, only 31% of physicians routinely ask their patients about their use of CAM modalities, and 17% never ask. Physicians most frequently recommended massage, relaxation, acupuncture, and biofeedback to patients, while the modalities physicians used most frequently for themselves were massage, relaxation, herbs, and yoga. Physician recommendation of CAM use to patients was most strongly associated with physician self-use (odds ratio 6.98, p=0.0001). Only 45% of physicians felt at least somewhat comfortable discussing the risks and benefits of CAM with their patients; the overwhelming majority (84%) felt they needed to learn more about CAM modalities to adequately care for patients.

CONCLUSION: Education about CAM modalities is a significant unmet need among physicians residing in the Denver metropolitan area, and education may help alleviate the discomfort physicians have when fielding questions about CAM from patients. Physicians who already use CAM treatments for themselves are much more likely to recommend CAM for their patients than those physicians who don't personally use CAM.

VA MEDICAL CARE UTILIZATION BY MEMBERS OF MEDICARE HMOs. L.M. Winterbottom¹, S.K. Joos¹, Y. Rozenfeld², F.E. Richards², D.H. Hickam¹; ¹Portland VA Medical Center, Portland, OR; ²OMPRO, Portland, OR

PURPOSE: Medicare HMOs receive capitated payments to provide comprehensive health care to their members. The VA system operates independently of Medicare. We sought to examine concurrent use of VA services and Medicare-funded services among a VA patient population.

METHODS: A questionnaire was mailed to 1766 Portland VAMC patients age 65 or older who had at least one clinic visit from January 1998 through December 1999. Seventy-six percent (1346) responded. The questionnaire included demographic items, the SF-12 to assess physical and psychological function and their satisfaction with, need for and importance of both their VA and non-VA sources of care. Non-VA care was identified using the HCFA enrollment database.

RESULTS: Among the entire population, 303 (17%) belonged to a Medicare HMO for the entire 24 months, and 130 (7%) belonged to an HMO for part of the period. Questionnaire responders did not differ from non-responders by marital status, gender, service connected status or HMO enrollment. Among the HMO enrollees, 66% reported having a VA primary care provider and 46% reported receiving all their medical care from the VA. However, 72% reported having a non-VA primary care provider. When comparing VA users who belonged to a Medicare HMO to those who did not enroll in an HMO, the HMO members reported

significantly higher levels of educational attainment and financial resources. The psychological and physical function scores did not differ amongst the various Medicare enrolled veterans.

CONCLUSION: Users of VA medical care frequently belong to Medicare HMOs, and nearly half of these HMO enrollees report receiving most of their medical care from the VA. VA users who belong to HMOs tend to be better educated and more financially secure. VA care appears to provide a substantial financial benefit to Medicare HMO plans.

ORGANIZATIONAL PREDICTORS OF VARIATIONS IN TOBACCO COUNSELING RATES. E.M. Yano¹, S.E. Sherman², M. Wang², M.L. Lee², L.V. Rubenstein²; ¹Department of Veterans Affairs, Sepulveda, CA; ²VA Greater Los Angeles HSR&D Center of Excellence, Sepulveda, CA

PURPOSE: Tobacco use remains the leading preventable cause of death and disease in the US, despite the availability of a wide range of effective smoking cessation treatments and physician knowledge of the value of even brief counseling. Little is known about the extent to which characteristics of health care organizations foster or hinder such performance. We evaluated variations in tobacco counseling rates among VA medical centers (VAMC's) nationwide and assessed the medical center and primary care program attributes related to high and low performance.

METHODS: We assessed VAMC organizational characteristics (e.g., region, urban location, primary care features) using a previously validated mailout survey fielded to VA primary care directors (100% response). We matched these survey-based measures to two centrally available databases: (1) organizational culture scores derived from the first wave of the National VA Quality Improvement Survey (G. Young, PI) administered to over 100 employees at each VAMC, and (2) tobacco counseling rates for each VAMC from externally-performed chart reviews of randomly sampled users with 1+ primary care and 1+ subspecialty visits (88% match). We used bivariate analyses to examine the relationships between discrete organizational features and counseling rates and multiple linear regression using a forward stepwise algorithm to assess the independent predictors of counseling rates.

RESULTS: Counseling rates varied by region (p<.01), with lowest rates in the East (0.73) and highest in the Western states (0.83). More complex VAMC's had lower tobacco counseling rates (p<.0001), consistent with lower rates among urban (p<.0001), academic (p<.01) VA's, with more internal medicine house officers (p<.01) and larger patient caseloads (p<.001). VA's with higher self-reported levels of primary care implementation, higher proportions of veteran users receiving all or most of their care in primary care, and where primary care physicians had responsibility for their patients as inpatients had higher tobacco counseling rates (all p<.05). Higher rates also occurred in VA's with higher staff ratings of leadership quality (p<.001) and marginally higher group (p=.054) and risk-taking cultures (p=.055), while those VA's with higher staff orientation to quality improvement (p<.05) and more primary care-based process action teams (p<.05) had higher rates.

CONCLUSION: Achieving more guideline-adherent smoking cessation practices in large, academic medical centers may pose significant challenges for primary care managers, but appears more likely among medical centers with more highly organized primary care practice. Improving counseling rates requires systems approaches to address both structural and provider-related constraints.

TIME SPENT WITH CHIROPRACTORS, ACUPUNCTURISTS, AND CONVENTIONAL PHYSICIANS. G.Y. Yeh¹, D.M. Eisenberg¹, R.B. Davis¹, D. Cherkin², R. Phillips¹; Beth Israel Deaconess Medical Center, Boston, MA; Group Health Cooperative, Seattle, WA

PURPOSE: Differences in visit duration between physicians and complementary providers have been suggested as reasons why patients are attracted to complementary therapies. To our knowledge, no prior data have been reported on time spent with complementary providers. Our objective was to describe duration of visit to chiropractors (CP) and acupuncturists (AC), the factors associated with these visit times, and to compare visit duration among physicians (MD), CP, and AC.

METHODS: A random sample of licensed CP (N=130) from Arizona and Massachusetts, and licensed AC (N=133) from Washington and Massachusetts, was surveyed on consecutive office visits in 1998–99 regarding visit time, patient demographics, reason for visit, acuity of problem, payment source, diagnostic and therapeutic techniques performed. (N=2550 CP encounters, mean age 44.9 years, 57% female; N=2561 AC encounters, mean age 46.2 years, 68% female). We used linear regression to analyze factors associated with visit duration and to calculate the minutes attributed to each factor. Models were adjusted for specific therapeutic and diagnostic techniques performed at the visit. MD visit duration was obtained from published 1995–96 National Ambulatory Medical Care Survey results.

RESULTS: Mean visit duration for CP was 21.5 minutes (SE=0.8) and for AC 56.6 minutes (SE=0.7), compared to 18.9 minutes (SE=0.5) for a visit to an internist (p=.006 for CP vs. MD, p<.001 for AC vs. MD). AC spent more time with patients at self-pay visits, new patient visits, and visits for flares of chronic problems, adding 4.7, 4.0, and 2.4 minutes respectively. AC visits for anxiety/depression were 2.6 minutes longer, with a mean duration of 50.2 minutes, compared to 32.6 minutes for MD visits for the same problem (p<.001). For CP visits, significantly more time was spent with new patients, when there was communication with other providers, and with patients receiving concurrent care from an MD, adding 6.7, 3.2 and 2.3 minutes to each visit, respectively. CP visits for wellness and chronic problems were shorter than acute visits, by 3.2 and 4.3 minutes, respectively. Patient age, sex, and reason for visit (i.e. back or neck symptom) did not affect CP visit duration. Visits in Arizona were, on average, 3.0 minutes longer than those in Massachusetts.

CONCLUSION: Acupuncturists spend significantly more time with patients on an average visit compared to physicians. Chiropractors report spending more time than physicians, although this difference is small and may be due to differences in how the data were collected. CP visit duration suggests regional variation. Payment source, communication with other providers, psychosocial reason for visit, and acuity of visit are important correlates of CP or AC visit duration.

THE IMPACT OF VERTICAL INTEGRATION ON PROCESS OF CARE FOR STROKE PATIENTS. D.S. Zingmond¹, G.L. Birbeck², S. Ettner³, D.M. Carlisle³, R.M. Andersen⁴; ¹Division of General Internal Medicine, UCLA, Los Angeles, CA; ²Michigan State University School of Medicine, East Lansing, MI; ³University of California, Los Angeles, California; ⁴UCLA School of Public Health, Los Angeles, CA

PURPOSE: Over the past decade, hospitals have diversified beyond acute care to post-acute care. These trends towards "vertical integration" (VI) have been studied primarily in terms of their financial causes and impact. Effect on clinical care is unknown. This study examines whether hospital VI impacts continuity and timing of clinical process of care for acute stroke patients.

METHODS: A survey was fielded in mid-2000 of the primary stroke providers at all acute care hospitals with acute stroke discharges in California in 1998. Survey findings were linked to California state hospital facility reports and individual patient discharge records for 1998. Multiple regression models were used to examine the association of highest level of integration (internal rehab unit (IR), formal contract of external unit (FER), informal agreement of external unit (IER), and no agreement (NoR)) with service availability, reported interventions, and patient outcome, controlling for other hospital features and patient demographics.

RESULTS: Surveys were completed for 256/374 (68%) of hospitals that admitted stroke patients in 1998 and that were still open in 2000. Non-respondent hospitals were smaller, rural, or independent. Among respondent hospitals, level of integration was 20% IR, 7% FER, 15% IER, and 58% NoR. Greater integration was significantly associated with stated greater use of acute care guidelines, hours of rehab care during acute hospitalization, continuity of providers between acute and post-acute care, management by specialists, and use of common medical information systems. When linked to patient records, level of integration did not predict initiation of rehab during hospitalization or in-hospital mortality, but was associated with increased likelihood of discharge to post-acute rehab.

CONCLUSION: VI appears to have clinical advantages of greater underlying consistency and continuity of care, but does not appear associated with early (acute) rehab intervention. Short-term mortality does not appear to be associated with VI, but other outcome measures such as functional recovery may be more sensitive and appropriate. As trends towards vertical integration continue, a better understanding of VI's clinical impact is imperative.

MEDICAL HUMANITIES-QUALITATIVE RESEARCH

SURVIVORS OF INTIMATE PARTNER VIOLENCE SPEAK OUT: PERSPECTIVES ON TRUST IN THE PATIENT-PROVIDER RELATIONSHIP. I.A. Battaglia¹, J.M. Liebschutz¹; ¹Section of General Internal Medicine, Boston University School of Medicine, Boston, MA

PURPOSE: One barrier to screening and disclosure of Intimate Partner Violence (IPV) is the patient's lack of trust in their health care provider. The objective of this study was to identify characteristics that facilitate trust in the patient-provider relationship among survivors of IPV.

METHODS: Qualitative data were collected using semi-structured, open-ended interviews with a sample of 28 female survivors of IPV. Participants were recruited from community-based IPV organizations in the Boston area. A one hour face-to-face interview explored the woman's beliefs and attitudes about trust in her interactions with health care providers. Interviews were audiotaped and transcribed. Using Grounded Theory Methods, the transcripts were analyzed for significant themes, from which a conceptualization of trust was developed.

RESULTS: Among the participants, ages ranged from 18 to 56, 36% were Black, 32% Hispanic and 18% white. All suffered physical violence, but only 60% ever sought medical care as a result. We identified five dominant concepts that described dimensions of provider behavior on which trust was based: 1) OPEN COMMUNICATION — listening to patients and keeping them well informed of their medical condition; 2) PROFESSIONAL COMPETENCY — appearing knowledgeable, performing thorough medical evaluations and being familiar with the medical and social histories; 3) PERSONAL COMMITMENT — being consistently accessible, both physically and emotionally ("always been there for me"), respecting confidentiality, spending ample time with participants and demonstrating personal concern outside the traditional provider role; 4) EMOTIONAL SAFETY — sharing personal information and establishing a climate of safety for the participants through nonjudgmental and compassionate gestures ("he had tears in his eyes") or comforting nonverbal mannerisms; 5) EMPOWERMENT — promoting self-efficacy and shared decision-making. Gender and ethnic characteristics of the providers had variable effects on trust.

CONCLUSION: These survivors of IPV reported several aspects of provider behavior that facilitate trust in their clinical relationship, including open and explicit communication, professional competence, personal commitment, establishing a safe emotional environment and patient empowerment. Strengthening these provider behaviors may increase trust with patients and thus improve screening and disclosure of IPV.

THE IMPACT OF "TURFING" ON THE QUALITY OF PATIENT CARE: ATTENDING'S PERSPECTIVES. C.V. Caldicott¹, K.A. Dunn¹, R.M. Frankel²; ¹SUNY Upstate Medical University, Syracuse, NY; ²Strong Health, Rochester, NY

PURPOSE: "Turfing" is the transfer or triage of a patient to a physician for reasons the receiving physician deems inappropriate. "Turfing" has been identified as a troubling source of professional and ethical conflict for housestaff and medical students. Prior work has reported that "turfed" patients have a worse hospital experience than appropriately transferred patients. We wondered whether attendings' perceptions of "turfing" supported patients' perspectives.

METHODS: One trained research assistant conducted in-depth, semi-structured, open-ended individual interviews with a convenience sample of 16 attending physicians from internal medicine, emergency medicine and general surgery. The interviews were meant to elicit attendings' views on the quality of "turfed" patients' care as well as other dimensions of this triage

phenomenon. Audiotapes of interviews were subjected to content and thematic analysis. Results presented here include only attendings' comments about the quality of "turfed" patients' care. **RESULTS:** All attendings interviewed in emergency medicine (n=4) and general surgery (n=4) felt that "turfed" patients received inferior care. Of the 8 internal medicine attendings interviewed, 5 felt "turfed" patients received inferior care, 1 was equivocal, and 2 did not comment on this specific issue. Subjects considered quality of care to include length of stay, interpersonal interactions between patients and housestaff, and delays in management plans and bed assignments. Subjects stated: "There is always hostility toward the patient from the housestaff in these situations. If you go watch [the housestaff] and see how they interact with the patient, they are abrupt with them, dismissive toward them." "Patients are put in harm's way." "[Housestaff] probably don't take as inquisitive an approach [to history-taking] which is not in the patient's best interest." "We avoid people who have been in the ED for too long [because no one will accept them on their service]." Three themes emerged as causes for this inferior care: lack of "ownership" of "turfed" patients by physicians, the transient nature of physician relationships with "turfed" patients, and feelings that "turfs" are unwanted.

CONCLUSION: Attendings' interviews substantiate that the care of "turfed" patients suffers. Interviewees recommended remedies for these troubling findings, such as stronger moral leadership from department chairs and improved role modeling by clinical faculty.

DIFFERENCES IN THE WAY AFRICAN AMERICANS AND CAUCASIANS WOULD PRESENT FACTS ABOUT PROSTATE CANCER SCREENING. E.C. Chan¹, S.V. Vernon², F. O'donnell¹, M. Haynes², C. Bachino²; ¹Department of Internal Medicine, University of Texas-Houston Health Science Center, Houston, TX; ²Center for Health Promotion and Prevention Research, School of Public Health, University of Texas-Houston Health Science Center, Houston, TX

PURPOSE: The purpose of this pilot study was to determine whether there are differences in the way Caucasians and African Americans would present facts about prostate cancer screening in a culturally appropriate brochure to men considering screening. Because screening with prostate specific antigen (PSA) is controversial, several professional organizations recommend informing men about the risks and benefits of screening. Compared to Caucasians, African Americans have a higher risk for prostate cancer and are less knowledgeable about screening. In a previous study, experts in prostate cancer and Caucasian and African American couples with screened and unscreened men identified 17 key facts about PSA they believe men ought to know.

METHODS: We convened a focus group of 5 African American couples and a focus group of 5 Caucasian couples who each met twice. At the first meeting couples viewed a videotape about prostate cancer and screening and were asked to discuss how to present the 17 key facts in a culturally-sensitive way in terms of content and format. Mock ups of brochures were developed for discussion at the second meeting. We analyzed transcripts of the focus group meetings using content analysis to identify differences between African American and Caucasian groups in how to present this information.

RESULTS: We found differences in content emphasis and in graphic design choices between the brochures that African Americans and Caucasians would design for members of their racial/ethnic group. Because of the perceived discomfort and embarrassment associated with the digital rectal exam (DRE), African American men felt strongly that it was important to emphasize the advantages and disadvantages of screening with the DRE, as well as the PSA. Caucasian men did not discuss the DRE at all. African Americans believed it was important to emphasize epidemiologic data specific to African American men, such as their higher risk of prostate cancer. They also preferred images and symbols rooted in African American culture (e.g. Kente cloth).

CONCLUSION: African Americans and Caucasians differed in the way they chose to present facts about prostate cancer screening to members of their racial/ethnic group. Cultural differences in format and content need to be considered when designing educational materials such as brochures.

RACIAL PROFILING, CLINICAL EPIDEMIOLOGY, AND CULTURAL COMPETENCE: WHEN IS RISK STRATIFICATION BY RACE OR ETHNICITY JUSTIFIED? M.H. Chin¹, C.A. Humikowski¹; ¹University of Chicago, Chicago, IL

PURPOSE: Population-based, probabilistic approaches to medicine have become more prominent with the rise of the evidence-based medicine movement. Simultaneously, diverse patient populations have spurred development of curricula in cultural competence. Building upon lessons from the racial profiling debate in the legal and political realms, we aimed to develop a conceptual framework for exploring whether race and ethnicity should be used as clinical tools or if dangers of stereotyping are too great.

METHODS: We reviewed key literature pertinent to racial profiling, race and medicine, and curricula in cultural competence. We explored issues of race and medicine through a regular seminar series with medical housestaff, and extensive feedback from colleagues. We developed a conceptual framework that balances clinical epidemiological principles with lessons learned from social and political racial profiling.

RESULTS: We propose that viewing the patient within a wider population-based cultural setting can help guide the initial clinical approach, but individualized care is mandatory. The use of race and ethnicity as clinical tools is on a slippery slope — they may be useful or dangerous. Biological examples are rare in which race or ethnicity are relevant (e.g. — Tay-Sachs, sickle cell). However, ethnicity may have a statistical association with a variety of factors. The potential benefit of using ethnicity as a tool is to aid an individual patient by making medical care better tailored to him or her. The potential detriments are reinforcement and perpetuation of partial myths, encouragement of undesirable behavior by creating negative expectations, and avoidance of addressing the underlying individual factors. Racial profiling has taught us that the costs of using ethnicity as a proxy for behavior are too high. But, ethnicity may appropriately be used as an initial proxy for history, language, culture, and health beliefs, as long as individualization of care is rapid. The predominant approach to cultural competence in medical education today teaches a consideration of individual patients as opposed to a rigid checklist of ethnic traits. For example, providers are advised to inquire about the meaning of illness to the patients, and the social context such as family, literacy, and finances. However, given that physicians have limited time and energy, context including race and ethnicity can be helpful in guiding initial inquiry and raising awareness that an issue (e.g. — strong distrust of the health care system) may be important.

CONCLUSION: In the specific situations outlined, race or ethnicity may be useful clinical tools as long as individualization of care is quick. However, vigilance is required as medicine has not been immune to overt or subconscious racism. Case examples are used to define our conceptual framework in more detail.

THE ROLE OF MUTUAL HELP SUPPORT GROUPS FOR INDIVIDUALS AFFECTED BY MOOD DISORDERS. W. Resnick¹, T. Houston², K. Swartz², L.A. Cooper², DRADA, Baltimore, MD; ¹Johns Hopkins University, Baltimore, MD

PURPOSE: To describe how mutual help support groups are used by individuals affected by mood disorders.

METHODS: Six focus groups, including 39 participants (19 women, 20 men), were conducted. Participants were recruited through Depression and Related Affective Disorders Association (DRADA) support groups. Discussions addressed what if any influence support groups have had on their lives; and the relationship of the groups to medical treatment and/or therapy. Discussions were audiotaped, transcribed verbatim, and reviewed independently by two investigators to group distinct comments into categories with specific themes. Differences were resolved after a joint review of themes. Comments within categories were then checked for relevance and consistency by two second reviewers — a general internist, and a psychiatrist. Adjustments to categories were made until consensus was reached.

RESULTS: Comments fell into 10 categories: 1) Patient Activation/Empowerment in their treatment, 2) Using support groups to complement treatment, 3) Obtaining education and information, 4) Learning to interact positively with family and co-workers, 5) Socializing with support group members, 6) Sharing — to understand each other, 7) Helping others, 8) Increasing self-esteem, 9) Emotional Support, and 10) General comments. All participants were positive about the influence of support groups in their lives. A sense of belonging, talking about problems, and feeling understood by others, were seen as a source of comfort and well-being. Participants felt their family and work relationships, and communication skills were improved by participation in support groups. They also reported an increase in self-esteem and participation in medical decision-making. All agreed that exchange of information about their illness and exchange of ideas with other people about coping, contributed to their health. Emotional support from peers, destigmatization of their illness, and a sense of hope were all benefits from the groups. Most individuals did not use the group as replacement for treatment. **CONCLUSION:** Health professionals should consider recommending mutual-help support groups as an adjunct to treatment. The combination of medical treatment and peer support may enhance patient participation in care, treatment adherence and social skills, and provide patients with positive coping strategies.

BLACK AND BLUE, A CULTURALLY-TAILORED DEPRESSION EDUCATION VIDEOTAPE. A.B. Primm¹, D. Cabot², J. Pettis³, H.T. Vu¹, L.A. Cooper¹; ¹Johns Hopkins Medical Institutions, Baltimore, MD; ²Mental Health Association of Maryland, Baltimore, MD; ³Baltimore Mental Health System, Baltimore, MD

PURPOSE: Studies document underutilization of outpatient specialty mental health services by African Americans (AAs). However, AAs with depression are just as likely as whites to receive care in primary care. Despite their use of primary care, AAs are less likely than whites to be recognized as depressed, offered pharmacotherapy, and to initiate or complete guideline-concordant depression care. AAs express preferences for counseling and negative attitudes toward antidepressant medication, the most common form of depression treatment used by primary care physicians. Few depression education programs and materials target cultural beliefs and values of AAs. The purpose of this study was to determine the acceptability and usefulness of an educational videotape for AAs with depression.

METHODS: After showing the videotape, we held four focus group discussions in two community settings and at an historically black university. Subjects included twenty-four AAs, aged 18–76 years, who screened positive for depression. Thirty-eight percent reported a history of previous treatment. Focus group discussion questions addressed the usefulness of the videotape for helping viewers understand depression and its treatment, the most and least effective parts of the videotape, and the cultural appropriateness of the information presented. Participants took pre- and post-tests on attitudes about depression. Discussions were audiotaped, transcribed, and reviewed independently by two investigators to identify and group comments into specific themes. Two other investigators reviewed the themes and comments for consistency. Changes were incorporated to achieve consensus.

RESULTS: In addition to comments specifically related to positive and negative aspects of the videotape, the following themes were identified: concerns about antidepressant medication; identification with people in the videotape; racial and cultural issues; stigma and stereotypes; spirituality; and validation and support from the focus group. The videotape was well received and rated by most viewers as effective in improving knowledge and alleviating concerns about depression and its treatment. After watching the videotape, attitudes improved in several areas: depression as a medical illness, effectiveness of treatment, negative perceptions of antidepressant medication and reliance upon spirituality alone to heal depression.

CONCLUSION: This culturally tailored videotape about depression is deemed acceptable and effective to AAs with depression participating in focus groups. It also improves knowledge and attitudes about depression. Incorporating culturally-tailored educational materials and messages into community and primary care depression programs may reduce barriers to appropriate depression care for AAs.

THE SIGNIFICANCE OF LONG-TERM DOCTOR-PATIENT RELATIONSHIPS AMONG OLDER PATIENTS. C. Crenner¹, K.A. Greiner²; ¹University of Kansas Medical Center, Kansas City, KS; ²Kansas University Medical Center, Kansas City, KS

PURPOSE: Many older Americans have a personal stake in a relationship to a single physician that extends back decades. Changes in our healthcare system and in chronic care facilities may increasingly disrupt these connections. People who have preserved such long-term relationships have much to tell us about the significance of these relationships and their relevance to care.

METHODS: We invited participation at four sites: a rural nursing home; an ambulatory clinic in an urban medical center; a suburban, assisted-living facility; and a private clinic in a low-income, urban neighborhood. We enrolled subjects who could give an account of past medical care and who had kept the same primary physician for seven years or longer. We sampled to increase variability. We recorded and transcribed 17 semi-structured interviews and conducted a focus group with seven participants including 2 subjects not interviewed individually. We elicited open-ended discussion of each subject's relationship to a long-term doctor, asking about the features that interfered with or helped to sustain the relationship, and the significance that the relationship held. Immersion in the transcribed material and simultaneous reading in social history and anthropology produced a theoretical model based on the distinction between gift-exchange qualities and commodity-exchange qualities in these relationships.

RESULTS: Subjects were 6 men and 13 women, with a mean age of 75 years, and a mean duration of medical relationship of 19 years. Eighteen subjects were white and one was African American. Subjects attributed much of the durability of these relationships to the positive personal and professional characteristics of their doctors. There was variability in characteristics described as important. Few informants recalled their own active role in selecting their current physicians, and some had been in practices that were "handed-down" from doctor to doctor. Distance and transportation were very important considerations for continuing care. Subjects were generally able to describe a rational process for selecting their next physician.

CONCLUSION: Our theoretical model suggested that gift-exchange qualities in these relationships implied mutual obligations between recipients and donors that extended beyond the simple transaction of services. In contrast, commodity-exchange relationships existed solely for the exchange of specific services. Services in gift-exchange relationships had a value partly determined by the value attached to the relationship, while in commodity-exchange the value was independent. Our subjects favored characterizations of their long-term relationships as forms of gift-exchange, but they demonstrated an ability to shift to a commodity-exchange model. Statements consistent with a commodity-exchange model appeared especially in references to relationships of shorter duration, or to the conclusion of relationships to long-term providers.

QUESTIONS OF JUSTICE IN MEDICAL CARE: ETHICAL DILEMMAS FOR THE GENERAL INTERNIST. M. Danis¹, G. Duval², G. Gensler³; ¹National Institutes of Health, Bethesda, MD; ²University of Toronto Joint Centre for Bioethics, Toronto, Ontario; ³EMMES Corporation, Potomac, MD

PURPOSE: The day-to-day practice of medicine entails many ethical quandaries and yet little is known about practicing physicians' encounters with ethical dilemmas or how they resolve them.

METHODS: A randomized, national telephone survey of internists including generalists was conducted to determine the ethical dilemmas encountered, the strategies used to address them, and the use and value of ethics consultation. Coding schemes were used to categorize verbatim descriptions of ethical dilemmas.

RESULTS: Responding generalists (N=95) reflected the demographic characteristics of U.S. general internists (20% female, 23% non-white, 33% foreign born). One quarter of the ethical dilemmas encountered by general internists were related to justice issues concerning uninsured patients, practice in managed care, or allocation of limited resources. During the previous 2 years respondents had requested an average of 2 consultations. Physicians were most hesitant to request consultations because they are time-consuming. While many of the ethical dilemmas encountered were related to justice issues, no justice-related dilemmas were referred for ethics consultation. Physicians handled justice related dilemmas by strategies that included explicitly discussing limits with patients, acquiescing to limits, overstepping limits, searching for alternative sources, or providing free care. Physicians were often dissatisfied with resolution of justice questions and believed institutional or policy changes were required to achieve better resolution.

CONCLUSION: General internists encounter a wide array of ethical dilemmas and address the vast majority of them on their own. Medical ethicists need to be aware that justice related dilemmas are often unsatisfactorily resolved and should explore ways to help in addressing them.

GEOGRAPHIC DISPARITIES IN NATIONAL INSTITUTES OF HEALTH RESEARCH FUNDING: IS THE GAP WIDENING? M. Dansinger¹; ¹New England Medical Center, Boston, MA

PURPOSE: The National Institutes of Health (NIH), which fund approximately half of all biomedical research, distributed over \$130 billion throughout the United States (US) during the past 20 years. Recent data suggest that this funding is becoming more concentrated at the medical schools most active in research, which tend to be in urban centers. I hypothesized that over the past 20 years, NIH research funding has increased in the states and regions with the highest per-capita NIH funding and highest population density, relative to the other states and regions. If present, the hypothesized shift in research funding may hinder efforts to reduce regional disparities in medical care.

METHODS: Data from the NIH and US census bureau were used to calculate the per-capita NIH funding received by each state during 1980–2000. The per-capita NIH funding amount for each state was obtained by dividing the annual funding amount by the population that year. To enable comparisons between years, per-capita funding for each state was standardized by dividing by the mean for all states in the same year. Regional disparities in per-capita NIH funding were evaluated in 2 ways: standardized funding amounts and their changes over 20 years were calculated for each formal US census region (Northeast, Midwest, South, and West), and standardized values and 20-year changes for each state were plotted on US maps. To determine whether per-capita funding shifted from rural to populous states over the past 20 years, the temporal trends in correlation between per-capita funding and population density were examined.

RESULTS: Year 2000 per-capita NIH funding ranged from \$3 in Idaho, to \$247 in Massachusetts. Over the past 20 years, 7 of the 10 states with the highest per-capita NIH funding in 1980 had relative decreases, and 7 of the 10 states with the lowest per-capita NIH

funding had relative increases in per-capita NIH funding. During this time, the Northeastern region of the US received the highest relative per-capita funding (1.6 times the national average), but the proportion (34%) of total research dollars did not significantly change during that time. Analysis according to formal US census region, and by US map inspection, suggested that per-capita funding was relatively stable for large regions, but considerable shifting of per-capita funding occurred within states of every region. States with relatively higher population density tended to receive higher funding per capita ($r=0.59$, $p<0.01$), but this relationship did not substantially change during the past 20 years, suggesting that per-capita NIH funding is not moving toward populous states at the expense of more rural states.

CONCLUSION: During the past 20 years, disparities in relative per-capita NIH funding have shifted substantially among states, but changed little among major regions of the US. There is no evidence to suggest that NIH funding became more concentrated in the states with the highest per-capita funding or population density during that time.

RACE AS A VARIABLE IN THE STUDY OF HEALTH DISPARITIES. I. Denberg¹;

¹University of California, San Francisco, San Francisco, CA

PURPOSE: Repeatedly, race has been shown to be a powerful predictor of health and access to care, but the reasons for racial disparities in health often remain obscure and speculative. Recognizing many concerns about vague and essentialist notions of race in epidemiological and clinical research, the Institute of Medicine, the American Academy of Pediatrics and the National Cancer Institute have recently called for more cautious use of the race variable. The goal of this paper is to provide a much-needed synthesis of key critiques of race, some of which go beyond the concerns expressed by the aforementioned medical bodies. Armed with more complete knowledge about race, investigators should be better able to evaluate, plan, and conduct studies on health differences among human populations.

METHODS: In an iterative fashion, and on the basis of clarity and logic, important critiques of race were identified through online and bibliographic searches and reviews of biomedical and social science publications and through discussions with colleagues. Subsequently, recently published studies in highly-regarded medical journals were selected to illustrate major themes. **RESULTS:** Delineating the causes of health disparities is often limited by misunderstandings of race and inadequate attention to its fundamental confounders, including: **BIOLOGY** (e.g. race is a social construct, not a biological one); **SOCIAL CLASS** (e.g. emphasizing biology diverts attention from considerations of SES, and SES is frequently gauged using an inadequate number of measures); **CULTURE** (e.g. biomedical studies of patients' culturally-based preferences and practices may rely on notions of culture that are overly deterministic, ethnocentric, and unduly constrained by the use of questionnaire data); **ETHNICITY** (e.g. self-reported ethnicity has become a common substitute for race, but although it offers some advantages, researchers should bear the ultimate responsibility for specifying membership in groups, ethnic or otherwise); **RACISM** (e.g. the most disturbing explanation for health disparities, racism is also one of the most difficult to study because it is politically charged and open to competing interpretations).

CONCLUSION: A great deal of important work has been accomplished in terms of documenting the variety and widespread nature of health disparities, generally, and racial disparities in health, more specifically. The next step is extremely challenging: we must discover more precise information regarding causes and mechanisms. This can be accomplished by remaining aware of the shortcomings of overarching categories of race/ethnicity, focusing on more narrowly defined subgroups, and giving more explicit attention to the confounders of race.

TRENDS OF ORGANOPHOSPHATE POISONING IN URBAN ZIMBABWE. X. Dong¹, M.A. Simon²;

¹Yale University, New Haven, Connecticut; ²Yale Medical Center, New Haven, CT

PURPOSE: There have been 3 million reported pesticide poisonings and 200,000 deaths worldwide. In developing countries, poisonings are mostly associated with prescription medications, agrochemicals and household chemicals which have posed major global health problems. Organophosphates are the most commonly formulated, packaged and used pesticides in Zimbabwe. However, there are no studies elucidating the current trends of organophosphate poisoning. The objectives of this study were to examine the mortality, trends and causes of organophosphate poisoning in an urban hospital in Zimbabwe.

METHODS: We conducted a cross sectional descriptive study to examine the occurrence and trends of admissions for organophosphate poisoning in Parirenyatwa, one of the two major urban hospitals in Harare, Zimbabwe. We examined a total 185,828 patients' records and 599 cases on organophosphate poisoning from January 1995 through November 2000. Trends in admissions of the organophosphate poisoning were recorded. Other variables such as sex, age, season and geographic area were examined. Total mortality rates were calculated. The intent of the poisoning was also compared.

RESULTS: These data reveal a steady increase in organophosphate poisoning during these years with the exception of 1999. The numbers of admissions for organophosphate poisoning have increased by more than 320% compared to six years ago. There is no difference in the male and female ratio for admission (48% vs. 52%). The majority of the cases admitted were below age thirty-one (82%) and were from urban settings (86%). Suicide is the predominant reason for poisoning (74%). There was a marked peak in the number of cases of children under age eleven (62%) that were due to accidental organophosphate poisoning. Mortality of organophosphate poisoning is 8.3% over the last six years.

CONCLUSION: These results reveal startling realities. Organophosphate poisoning has been escalating throughout the world, especially in developing countries, such as Africa, South America and Asia, where agriculture demands large supplies of pesticides. Our data, although representing a very small microcosm of the extent of poisoning and in particular organophosphate poisoning, are consistent with multiple prior studies. Organophosphate poisoning is increasing uncontrollably and is being used at high rates as a suicidal agent in the young urban population, and contribute to the accidental ingestion in the pediatric population. Further research should concentrate on behaviors and situations that lead to poisoning.

INTERNAL MEDICINE RESIDENTS ATTITUDES TOWARDS HIGH UTILIZING PATIENTS.

E. Dwamena¹, R. Smith¹, B. Patel¹, J. Lyles¹, K. Lein¹, A. Hodges¹, A. Komareddy¹, M. Kanji¹, F. Hayek¹; ¹Dept. of Medicine, College of Human Medicine, Michigan State University, East Lansing, MI

PURPOSE: Patients with medically unexplained symptoms (MUS) present a major problem for healthcare, yet they have not been adequately described. Physician perspective is especially important in providing the context for these patients' health perceptions, behaviors and outcomes.

METHODS: A focus group of 6 internal medicine resident physicians was conducted to provide preliminary data for subsequent focus groups of primary care physicians of a panel of managed care patients with MUS. The discussion was tape-recorded, transcribed and analyzed via content analysis.

RESULTS: Participants defined high utilization in terms of time and resource utilization and high physician emotional investment and not number of visits. Most participants described a typical high utilizing patient as one with multiple visits for uncontrolled organic disease or chronic, MUS. These patients were considered extremely difficult, demanding and unpleasant with a sense of entitlement. They believed that patients with somatization were often condescending, unappreciative and litigious. Interaction with these patients left the participants feeling frustrated, helpless, inadequate and powerless. Other emotions expressed include anger, guilt and apprehension about the possibility of missing an organic disease. Participants reported positive reactions to high utilizing patients (defined as more than 8 visits/year) with transient, minor acute illness. These patients were described as nice people with simple, straightforward problems, which got better regardless of what they did. Although they believed that the real motivation behind their visit was something other than their physical complaints, (such as stress, or the need for support and validation), they neither saw these patients' visits as problematic nor perceived a real need to address their possible underlying problems. Some participants considered these patients a major source of practice income and not high consumers of resources. Caring for high utilizing patients with organic disease was most gratifying. Most participants would rather not see patients with somatization; and the possibility of having to care for these patients as primary care physicians is a major factor in choosing to pursue subspecialization.

CONCLUSION: This group of internal medicine residents did not consider high utilizing patient with MUS to be problematic if they had multiple transient minor acute illnesses. Conversely they found chronic somatization patients to be extremely problematic and would rather not care for them. Caring for patients with documented organic disease was gratifying regardless of utilization.

METAPHORS OF DEPENDENCY AS BARRIERS TO SUPPLEMENTAL OXYGEN USE — CRUTCHES AND ADDICTION. M.A. Earnest¹; ¹University of Colorado Health Sciences Center, Denver, CO

PURPOSE: Among patients with hypoxemic chronic obstructive pulmonary disease (COPD), supplemental oxygen provides significant physical and cognitive benefits. Despite these benefits to therapy, no trial has demonstrated an improvement in subjective quality of life and patient adherence to therapy is low. The purpose of this study was to describe the experience of using supplemental oxygen therapy to identify barriers to its use and aid in improving adherence to therapy and quality of life among oxygen users.

METHODS: 27 participants with hypoxemic COPD underwent semi-structured interviews to explore their experiences with supplemental oxygen therapy. Interviews were taped, transcribed, and analyzed for thematic content in a manner informed by grounded theory.

RESULTS: The most commonly described barrier to oxygen use among participants was a fear of dependency. Two metaphors were used to describe this fear: crutches and addiction. As described by participants, a crutch was an external device or a behavior used to compensate for some weakness. Achievements and activities accomplished while using a crutch were considered tainted and illegitimate. Subjects feared that using a crutch made them more dependent on the crutch and less able to function without it. They feared that accepting dependency on a crutch was accepting a lesser state of being and a sign of moral weakness. Respondents described addiction as dependency on some substance used to illegitimately induce a sense of well-being or to reduce physical or psychic discomfort. Oxygen was likened to alcohol, psychotropic prescription medications, and illicit drugs. Participants equated using oxygen to relieve breathlessness with using alcohol or other drugs to alleviate psychic or physical discomfort. They feared that their need for oxygen would increase if they allowed themselves to become dependent on it and they considered using a substance to relieve their symptoms a sign of moral weakness.

CONCLUSION: Fear of dependency is a significant barrier to oxygen use among patients who will benefit from supplemental oxygen therapy. In many patients, improving adherence to oxygen therapy may require exploring the metaphors of crutches and addiction with patients to help them see the flaws in these analogies.

THE DYING WILL. J.H. Farber¹, N.J. Farber²; ¹Yale University, New Haven, Connecticut; ²Christiana Care Health System, Wilmington, DE

PURPOSE: The current procedures for procuring tissues and organs for donation have resulted in both organ and tissue shortages and general dissatisfaction with the current system. A new system for allowing individuals to delineate the treatment of their organs and tissues after death could improve donation and assure patient autonomy.

METHODS: We reviewed and analyzed current research and opinions on upholding the wishes of patients regarding the treatment of the body after death. This included a review of literature regarding advance directives, organ and tissue donation, and burial preferences.

RESULTS: It has been shown that for psychological health, individuals need to believe that they can decide what will happen to their bodies after their deaths. Many proposed systems for the procurement of organs and tissues for donation, including both presumed consent and the requirement of familial consent, directly or indirectly deprive the individual of control over the fate of his or her body after death. In addition, the fear caused by misconceptions regarding

organ donation leads otherwise willing people to refuse to donate their own or their relatives' organs.

CONCLUSION: In order to improve the current system of organ and tissue procurement, we propose the concept of the "Dying Will," a legal document in which an individual may delineate his or her wishes regarding organ and tissue donation, funeral procedures, and the like. In order to alleviate the fears of the individual, physicians could be responsible for helping their patients to write their dying wills. This might increase the number of individuals willing to donate their organs or tissues. In addition, with such a system, the permission of family members would be unnecessary for organ donation. This could further increase the number of available donors and prevent family members from overriding the wishes of the deceased. To expedite the transfer of dying will information to medical personnel, computerized databases such as the one recently initiated by Medic Alert could be used.

THE ROLE OF THE FAMILY IN MEDICAL CARE. J.H. Farber¹, N.J. Farber², ¹Yale University, New Haven, CT; ²Christiana Care Health System, Wilmington, DE

PURPOSE: Since the beginning of the practice of medicine, family members have played a role in the treatment of a patient. Although this role has traditionally been limited to providing emotional support, loved ones can also become directly involved in doctor-patient communication or in the medical treatment of a patient. But there may be detriment to patient care due to the involvement of friends or family.

METHODS: We reviewed and analyzed both literary and scholarly opinion on the role of the family in the practice of medicine.

RESULTS: Family members have been described as both helping and hurting the quality of care received by patients. They can provide health care in the home, especially for chronically or terminally ill patients, often facing emotionally or technically trying situations. When a patient is mentally or emotionally compromised, a loved one can speak on the patient's behalf. But this kind of involvement can easily be detrimental to the patient if a friend or relative tries to control the patient or makes decisions based on his or her own values rather than on those of the patient.

CONCLUSION: Friends and relatives can help a patient receive the best care possible; but when these individuals have selfish motivations or feel the need to control their loved one, they can do more harm than good. Physicians should take steps to encourage positive family participation while preventing this involvement from interfering in the patient's care. The physician must be able to tell whether a friend or relative is truly helping the patient or merely acting in his or her own interests. In addition, efforts should be made to contact supportive friends or family members for patients. When no loved one is available, a nurse, social worker, or appointed guardian could play the part of a third party in the doctor-patient relationship. As indicated by the literature, the doctor-patient relationship can greatly benefit from the involvement of a patient advocate.

WHEN THE PATIENT IS NOT A PATIENT. N.J. Farber¹, J.H. Farber², B.M. Aboff¹, J. Weiner³, E.B. Davis⁴, E.G. Boyer⁵, P.A. Ubel⁶, ¹Christiana Care Health System, Wilmington, DE; ²Yale University, New Haven, CT; ³Drexel University, Philadelphia, PA; ⁴George Washington University, Washington, DC; ⁵St. Joseph's University, Philadelphia, PA; ⁶University of Michigan, Ann Arbor, MI

PURPOSE: Physicians may occasionally encounter an ethical dilemma about whether to protect the rights of an individual patient, versus acting in the interests of society. When physicians act on the behest of society, such as in cases of impaired airline pilots, it is usually on the basis of legal imperatives or clearly stated normative guidelines. However, there may be cases where no such guidelines exist, or those in which physicians act to help society despite professional values to the contrary. We examine these situations.

METHODS: We have conducted several studies in which physicians were asked about hypothetical dilemmas involving the potential for harm to the society, but in which there are no clear guidelines; or in which guidelines state that physicians should not act in behest of the society due to harm to the individual. These cases involve the report of a past crime by the patient, the issue of patient-initiated health insurance fraud, and the issue of physician involvement in lethal injection for capital punishment. We also conducted a literature review on this subject.

RESULTS: A number of physicians in each of these studies would be willing to act on behalf of society and against the individual patient despite no clear normative values or those actually contrary to the physicians' decisions. When analyzed for the factors being most likely to influence the respondents, duty to society and the values of the overall population were most associated with the decision to act on behalf of society rather than in the interest of the individual patient.

CONCLUSION: In some settings, a number of physicians are willing to act on behalf of society rather than on behalf of the patient. In such circumstances, physicians rely on their personal and societal values, rather than those promulgated within the medical profession. Some of these decisions may actually be harmful to the patient, and therefore contrary to ethical guidelines established by the medical profession. The impact of societal values on physician decision making needs further exploration.

PERKS AND PITFALLS: THE EXPERIENCE OF PHYSICIAN-PATIENTS WITH CANCER. E.K. Fromme¹, R.S. Hebert¹, J.A. Carrese¹, ¹Johns Hopkins University, Baltimore, MD

PURPOSE: Physicians have been described as the 'worst' patients. Despite this, several prominent medical journals routinely feature writings by physician patients about their illnesses. This qualitative study examined the experience of physician patients treated for cancer.

METHODS: Physicians who had a previous diagnosis of cancer other than non-melanoma skin cancer were recruited by their oncologists or radiation oncologists. 24 physician subjects representing diverse spectra of race, age, specialty, career stage and practice setting were identified. Various disease stages and tumor types were represented, from incidentally identified carcinoma to imminently terminal metastatic carcinoma. Subjects participated in in-depth semi-structured interviews lasting an hour and a half. Interview transcripts were independently coded

by two readers and compared for agreement. Content analysis identified several major categories of themes.

RESULTS: 5 themes representing experiences specific to physician patients are presented. (1) Trusting others to provide care presented a particular challenge to our physician subjects who were well aware of the potential for error and were used to being in control in the medical environment. Physician subjects ranged from total trust in their providers and a desire to 'just be a patient' to getting care only in institutions where they had access to their own lab results and computerized records. (2) Physicians have the ability to doctor themselves, and our physician subjects varied in their willingness to do so. In one instance, a physician found himself ordering his CT scans, referring himself to specialists and administering his own chemotherapy. Others, unable to wait for their physicians, broke their own bad news by checking lab results. (3) Physicians with a serious illness are faced with a choice about disclosing their illness to colleagues. Our physician subjects who chose to do so elicited reactions ranging from strong support to total avoidance. (4) Similarly, physicians are faced with a choice about disclosing their illness to patients. Our physician subjects who chose to tell their patients reported that their patients often responded with interest and support, and that they used self-disclosure to facilitate some aspect of the medical encounter. (5) Our physician subjects perceived clear advantages as patients such as easier access to their records, better understanding of their conditions, and particularly prompt and supportive follow up by their care providers.

CONCLUSION: Our physician subjects identified advantages and challenges in their experiences as patients with cancer. In some cases, their illnesses affected the way that they interacted with colleagues and patients, whereas their physician status affected the way that they interacted with their physicians and the health care system. The insights of these physician patients may be helpful to physicians who are seriously ill and to those who are currently healthy.

THE INNER CURRICULUM: LEARNING AND CHANGE BY PHYSICIAN-PATIENTS WITH CANCER. E.K. Fromme¹, J.A. Carrese¹, R.S. Hebert¹, ¹Johns Hopkins University, Baltimore, MD

PURPOSE: A common notion, exemplified in the 1991 motion picture *The Doctor*, is that physicians who experience serious illness themselves can learn to be more caring and therefore better physicians. This qualitative study examined what physicians with cancer learned from their illnesses and whether they felt the way they practiced medicine changed as a result.

METHODS: Physicians who had a previous diagnosis of cancer other than non-melanoma skin cancer were recruited by their oncologists or radiation oncologists. 24 physician subjects representing diverse spectra of race, age, specialty, career stage and practice setting were identified. Various disease stages and tumor types were represented, from incidentally identified carcinoma to imminently terminal metastatic carcinoma. Subjects participated in in-depth semi-structured interviews lasting an hour and a half. The interview questions were based on a literature review of writings by physicians with cancer and other illnesses in journal citations identified in MEDLINE. Interview transcripts were independently coded by two readers and compared for agreement. Content analysis identified several major learning themes. Relationships between themes were examined and organized conceptually.

RESULTS: Physicians' learning led to perceived changes in 4 main areas: (1) Understanding what patients are going through. Physicians better understood the patient's perspective, for example the anxiety created by waiting for test results or delayed surgery. (2) In some cases, this understanding was accompanied by changes in empathy that manifested as a felt connection to patients or a greater sensitivity to patients' emotions. (3) Changes in communication such as repeating important information, being careful not to minimize illness, and striving to deliver bad news skillfully. (4) Changes in values and priorities such as setting limits with employers and difficult patients, appreciating family, and acknowledging the possibility of death.

CONCLUSION: Experience with a serious illness is a powerful source of learning for many physicians. In some cases, their experiences caused changes in their attitudes, their communication, and other aspects of their medical practice. Educators interested in teaching about patient centered care might utilize the experience and insights of physician patients.

INTEREST IN GENETIC TESTING AMONG RELATIVES OF WOMEN WITH BREAST CANCER: A QUALITATIVE STUDY. M.A. Gilligan¹, M. Schapira¹, A.B. Nattinger¹, ¹Medical College of Wisconsin, Milwaukee, WI

PURPOSE: There is a well-documented interest in genetic susceptibility testing among first-degree relatives of women with breast cancer. Interest in, and uptake of, testing has been shown to remain high despite education regarding risks and limitations of testing suggesting unknown factors influencing the decision to test. The purpose of this study was to explore factors influencing interest in genetic testing for breast cancer.

METHODS: A qualitative study design exploring interest in genetic testing was chosen to allow us to establish the appropriate parameters for future intervention research. We conducted 4 focus groups with 10-12 participants each. Participants were recruited from the general population. Women included in the study were 18-74 years old and had at least one first-degree relative with breast cancer by self-report. Focus groups were videotaped and transcribed. Content analysis was performed including independent extraction of themes by 2 authors and confirmation of themes by a third. A computerized software program was used to systematize the process.

RESULTS: Two critical domains influencing interest in testing were identified: 1) Self-perceived risk, and 2) Perception of testing as loss or gain of control. Accurate self-perceived risk required an understanding of critical genetic concepts such as sporadic vs. hereditary cancer. Women who categorized themselves as low risk for hereditary cancer had less interest in testing while those who categorized themselves as moderate to high risk were more interested in testing. Women who were interested in testing viewed it as gaining control, "knowing what you're facing," and "taking measures to prevent it." Women who were not interested in testing were fearful of losing control over their lives, of genetic determinism. Women were able to understand the probabilistic nature of genetic concepts discussed. Paradoxically, understanding that genetic testing would provide probabilities rather than "black and white answers" made

testing more acceptable to some women by allowing them to maintain “a percentage of control” over their lives. Willingness-to-pay for testing was explored as a proxy for interest in testing. The cost of testing was prohibitive for all but those women who considered themselves at high risk. If cost were not an issue many women would choose to get the test. Concern for genetic discrimination was raised.

CONCLUSION: Interest in genetic susceptibility testing for breast cancer is influenced by self-perceived risk and knowledge and beliefs about genetic testing. Comprehension of critical genetic concepts is necessary to accurately assess risk and understand the risks and limitations of testing. Cost and concern for genetic discrimination will affect uptake of testing.

TRUST AS A FRAMEWORK FOR ETHICAL ANALYSIS. S.D. Goold¹; ¹University of Michigan, Ann Arbor, Michigan

PURPOSE: Examine trust as a sociological and philosophical construct. Argue that trust can serve as a valid and useful conceptual framework to examine ethical issues in healthcare.

METHODS: It is widely acknowledged that trust is a vital component of and basis for relationships between clinicians and patients and that these interpersonal trust relationships have moral content. Trust is especially important in health care, and is present in relationships with organizations (e.g., hospitals, insurers) as well as with clinicians. In this paper I will describe the nature and elements of trust-based relationships, both between individuals (i.e., interpersonal trust) and between individuals and organizations (i.e., institutional trust), and the relationship between interpersonal and institutional trust. I will argue that trust is a valid foundation for healthcare ethics because healthcare is relational and serves vulnerable parties. I will discuss conditions and actions that may increase or decrease actual trust. I will then compare and contrast these with actions and conditions — both for individuals and organizations — that may influence trustworthiness.

VETERANS EXPLANATORY MODELS OF ILLNESS: THE ROLE OF META-NARRATIVES. P. Haidet¹, A. Greisinger², J. Aniol¹, B. Sharf², K. Wristers¹, R. Street²; ¹Houston VAMC/Baylor College of Medicine, Houston, TX; ²Kelsey Research Foundation; ³Texas A&M University, College Station, TX

PURPOSE: Anthropologists have proposed the concept of explanatory models of illness to explain discrepancies between patients' and physicians' views of the illness experience. Because large socioeconomic differences often exist between VA physicians and patients, we performed this qualitative study to explore variations and common tendencies among veterans' explanatory models of illness and the impact that these models have on veterans' perceptions of health and healthcare.

METHODS: We developed an interview guide to elicit patients' perspectives about the cause of, meaning of, severity of, amount of control over, and perceived treatment efficacy for their illness. Using this guide, we conducted 10 in-depth, semi-structured interviews with patients at the Houston VAMC one hour prior to their regularly scheduled primary care visits. We audiotaped and transcribed the interviews. We analyzed these transcripts through 3 iterations of individual reading and team discussion to identify common themes that occurred in our 5 areas of inquiry.

RESULTS: All participants identified themselves as Caucasian, with ages ranging from 52 to 81. Most participants were seeking care for at least one chronic illness. While some veterans' responses to our specific questions could be considered to fit a biomedical paradigm (e.g. Interviewer: “What do you think caused your diabetes?”; Patient: “My pancreas quit working.”), we discovered that broader, non-biomedical stories of experience, or meta-narratives, existed across our categories of questions. These meta-narratives described themes pertaining to patient fatalism and self-efficacy and preferences for physician roles of acknowledgement, action, and communication during the healing process. In cases where veterans perceived their physicians to not attend to these meta-narratives, veterans were dissatisfied and perceived a poorer quality of life.

CONCLUSION: While veterans' answers to specific questions about the experience of illness may follow a biomedical paradigm, broader, non-biomedical narratives emerge when these answers are viewed in aggregate. Physician understanding of these narratives during the medical interview is important for enhancing the patient's ability to cope with chronic illness.

MEDICAL STUDENT REFLECTIONS ON EXERCISES INTENDED TO FOSTER PERSONAL AWARENESS. P. Haidet¹, D.A. Paterniti², J.E. Dains³, J.C. Rogers³; ¹Houston VAMC/Baylor College of Medicine, Houston, TX; ²University of California, Davis School of Medicine, Sacramento, CA; ³Baylor College of Medicine, Houston, TX

PURPOSE: Repeated calls have been made for educators to foster students' personal awareness of attitudes and perspectives that influence their professional practices. As part of an exercise intended to foster such personal awareness, all students at Baylor College of Medicine (BCM) complete several standardized instruments (SI) that measure their views toward the physician-patient relationship, psychosocial care, and stress from uncertainty. Students complete these instruments in their first and third years and, after an explanation of the constructs that are measured by the instruments, receive feedback consisting of their own scores, the average scores of their classes, and published normative data from physicians in various specialties of medicine. We conducted this qualitative study to explore the meanings that students attribute to this feedback.

METHODS: We conducted focus groups with students in each of the classes of 2000, -01, -02, and -03 at BCM (one group for each class). We developed guiding questions that explored the effects of the SI scores on students' perceptions of self-knowledge, abilities, and attitudes (a conceptual framework for personal awareness training proposed by Novack, et al). Focus groups were co-moderated by a sociologist and a physician. All focus groups were audiotaped and transcribed. We developed categories for transcript coding by noting the patterned occurrence of specific behaviors, attitudes, and values in the transcripts.

RESULTS: We conducted 4 focus groups with 22 students; focus groups lasted an average of 90 minutes. Students from different classes framed the information provided by SI scores

differently. Preclinical students expressed a lack of experience with patient care, and therefore completed the questionnaires relative to their best guess of the answers of an 'ideal' physician. This led to anxiety when these 'ideal' physician scores differed unexpectedly from the normative scores of the class or of practicing physicians. Clinical students completed the SIs relative to their own patient-care experiences and devalued scores that differed from norms that they saw as different from their own emerging professional identity.

CONCLUSION: Using a 'trigger' such as SI scores may be a useful means to stimulate student self-reflection. Such feedback, though, should be framed in the context of student experience. Preclinical students struggle with the meaning of feedback given their perceptions of how an 'ideal' physician thinks and acts. Clinical students possess an emerging professional identity and may devalue feedback that is inconsistent with this identity. In order to foster professional attitudes and behaviors using feedback, educators should take into account differences in preclinical and clinical students' perceptions of the meaning and value of feedback.

OLDER HEMODIALYSIS PATIENTS PERSPECTIVES ON QUALITY OF LIFE. S.E. Hardy¹, S.T. Crowley¹, T.R. Fried¹; ¹Yale University, New Haven, CT

PURPOSE: As the dialysis population has aged, quality of life (QOL) has become an increasingly important outcome for both clinicians and researchers in end-stage renal disease. Unfortunately, most QOL scales, whether generic or renal-specific, have been developed with limited input from patients, particularly older patients. The aim of this study was to identify those factors most important to older hemodialysis patients in determining their QOL.

METHODS: Semi-structured interviews lasting 20 to 60 minutes were conducted with 27 hemodialysis patients, aged 65 years or older, who had been on dialysis for at least 90 days. The interviews used open-ended questions, such as “How has kidney disease requiring dialysis affected your quality of life?” Two coders independently reviewed the interview transcripts to identify themes, resolving disagreements by consensus. Themes were compared within and across subjects to identify and organize important concepts. Subjects were enrolled until the point of thematic saturation, when further interviews revealed no new themes.

RESULTS: The 27 subjects interviewed had a mean(±SD) age of 74±7 years and had been on hemodialysis for an average of 3.5±3.1 years. They identified five domains important to QOL: symptoms, burdens of treatment, benefits of treatment, information needs, and relationships with providers. The most common symptoms reported were fatigue, balance problems, weakness, cramps, itching, and nausea. Burdens of treatment included time spent at dialysis, difficulty traveling, problems with vascular access, and giving things up. Benefits of treatment included relief of symptoms, social interaction, and enhanced access to health care providers. Subjects noted the importance of adequate information about how to care for themselves and what symptoms or problems to expect, and of competent and caring providers who treat them with respect and acknowledge their individuality. Symptoms and burdens of treatment affected QOL negatively and benefits of treatment affected QOL positively, while the remaining domains could have either positive or negative effects. The magnitude of these effects was influenced by non-medical factors, such as social support, family relationships, formal instrumental support, finances, and spirituality. Subjects universally noted the necessity of dialysis for sustaining life, and preferred even a low QOL on dialysis to death.

CONCLUSION: When asked about their perspectives, older hemodialysis patients, while acknowledging the necessity of dialysis, identified a variety of domains important to their QOL as dialysis patients. Several of these domains, particularly information needs and the benefits of treatment, are not found in existing generic or renal-specific QOL scales. In developing such scales, it is important to elicit the factors most important to patients and ensure that these are included.

TAKING CARE OF OUR OWN: WHEN A RESIDENT BECOMES A PATIENT. R. Harrison¹, S. Desai¹, D. Webster¹, A.J. Hunter¹, J.L. Bowen¹; ¹Oregon Health Sciences University, Portland, OR

PURPOSE: To date there are no studies discussing the experiences of caring for residents who are hospitalized at their own training institution. While providing care for a hospitalized resident at our institution a great deal of turmoil developed among the housestaff and faculty over the issue of house officers providing care for their peers. Using the critical incident technique we assessed the experiences of the resident patient and the house staff taking care of their peer in the hospital.

METHODS: In order to better understand the origins and nature of these conflicts focus groups were performed that included the resident patient and the house staff who provided the care. We convened 3 focus groups with the house staff and a separate session with the resident patient. All five residents who cared for the hospitalized resident were invited to participate in this study on a voluntary basis. Consent was obtained before audio recording each confidential session. Each session facilitated by the investigators consisted of one to three resident physician volunteers who were asked to describe feelings, both positive and negative surrounding care of their peer. The audiotapes were transcribed, independently analyzed and coded by each investigator. The coded transcripts were then compared and discussed generating a coding scheme. Disagreements were discussed and resolved by consensus. OHSU IRB approved this study.

RESULTS: One hundred percent of eligible participants completed the study. Caring for a colleague had an overwhelmingly negative impact on the residents routine medical decision making, work structure, and team dynamics. Barriers to routine care of the hospitalized resident included intense emotional experiences, ambiguity in the care structure, and concern for confidentiality and privacy. This resulted in paralysis in care decisions leading to reactionary care or even care avoidance. Lack of communication amongst team members, faculty and patient was a commonly cited experience. The housestaff patient perspective was equally negative. Problems with communication and care delivery were emphasized particularly surrounding the issues of objectivity and confidentiality. Themes of vulnerability, frustration and powerlessness dominated the narratives of the resident patient.

CONCLUSION: The critical incident technique offers a unique perspective into the experiences of residents taking care of peers in the hospital. Many of the themes were identical

to both the residents and resident patient including issues of confidentiality, communication and intense emotional conflict. While many residents choose medical care at their own training institutions, this study demonstrates the potential difficulties with this approach and suggests the importance of determining a policy of care for the hospitalized resident.

WOULD YOU BE SURPRISED IF THIS PATIENT DIED: EXPLORING THE TRANSITION BETWEEN AGGRESSIVE AND PALLIATIVE CARE. D.C. Johnson¹, J.S. Kutner¹, J.A. Armstrong¹; ¹University of Colorado Health Sciences Center, Denver, CO

PURPOSE: Multiple studies have shown that physicians inadequately address the suffering of severely ill patients at the end of life. This deficiency, in part, arises from the difficulty in recognizing the dying patient and an overly restricted focus on aggressive care. The purpose of this study was to examine the thought processes of resident physicians taking care of critically ill patients, with a goal to better understand how physicians transition between aggressive and palliative care, and to determine if the explicit consideration of the possibility of death improves attention to the suffering of patients and their families.

METHODS: Internal medicine resident physicians (n=8) caring for severely ill patients were sequentially interviewed over a four week period during Medical Intensive Care Unit Ethics and Discharge Planning rounds. The residents were asked a series of questions based on their response to the core question "Would you be surprised if this patient died?" The interviews were taped, transcribed and qualitatively analyzed for the purpose of identifying common patterns in the decision-making process of physicians when managing severely ill patients.

RESULTS: When asked the core question, there were an equal number of cases in which residents responded that they would (n=4) versus would not (n=4) be surprised if their patient died. Reasons for being surprised that a patient would die included the presence of a reversible disease, the rapid onset of an acute illness, situations where a patient was "doing better" and prior survival under similar circumstances. Among residents who indicated that they would not be surprised, most (n=3) reported that their management would not change knowing that their patient might die. Cited changes, when present, included clarifying goals, improving communication with families, spending more time with patients and ordering fewer labs. Stated or implied barriers to changes in management despite knowing that a patient might die included a lack of time and experience, "not knowing" a patient, an improvement in medical condition ("doing better"), a lack of clear goals ("doing everything"), the presence of uncertainty and an exclusive focus on technical intervention.

CONCLUSION: Asking the question "Would you be surprised if this patient died?" encourages proactive reflection and dialogue about the potential for death in severely ill patients. An active acknowledgment of the possibility of death may lead to changes in management that would potentially diminish the suffering of patients and their families. To effectively impart such changes, it may be necessary to develop and implement techniques to overcome identified barriers.

PERSONAL GROWTH IN MEDICAL FACULTY: A QUALITATIVE STUDY. D. Kern¹, S.M. Wright², J.A. Carrese², M. Lipkin, Jr.⁷, J. Simmons³, D.H. Novack⁴, A. Kalet⁵, R.M. Frankel⁶; ¹Johns Hopkins Bayview Medical Center, Baltimore, MD; ²Johns Hopkins University, Baltimore, MD; ³CenterPoint Human Services of Stokes County, NC, King, NC; ⁴MCP Hahnemann University, Philadelphia, PA; ⁵New York University, New York, NY; ⁶University of Rochester, Rochester, NY

PURPOSE: A physician's effectiveness depends on the application of good communication, cognitive, and technical skills, guided by maturity, wisdom, compassion, and integrity. Development of the latter attributes requires growth in the awareness and management of one's feelings, attitudes, beliefs, and life experiences. Yet there is a dearth of empirical research related to physicians' personal growth. We used qualitative research methods to analyze stories of personal growth obtained from a selected group of medical faculty, in order to increase understanding of personal growth in this group and to develop a conceptual framework that might be useful to medical educators and researchers.

METHODS: Questionnaire survey of facilitators, facilitators-in-training, and members of a personal growth interest group of the American Academy on Physician and Patient, who were chosen because of their interest, knowledge, and experience in the topic area. Respondents were asked to submit one or more stories of growth in their own self-awareness, and how that growth had affected them professionally and personally. Themes were identified in the submitted stories by 5 members of the research team and sorted into categories by 3 members. Slight further revisions were made following discussion by the entire research team and application of the proposed categories to a subset of stories. The findings were verified by two independent reviewers, who reviewed all of the stories.

RESULTS: 32 of 64 subjects returned questionnaires containing 42 stories. There were no significant differences between respondents and non-respondents in age (mean 45.3 vs. 44.6 years), gender (75% vs. 69% male), or specialty (69% vs. 72% internists, 28% vs. 19% behavioral scientists / psychiatrists). Usually personal growth stories began with a powerful experience and/or helping relationship, proceeded to introspection, and ended in a personal growth outcome. Personal growth outcomes included: changes in values, goals, or direction; healthier behaviors; improved connectedness with others; improved sense of self; and increased productivity, energy, or creativity.

CONCLUSION: Powerful experiences, helping relationships, and introspection are processes that preceded personal growth in a subset of medical faculty. The findings, if confirmed in other populations of physicians, may have implications for medical education and practice.

A COMPARISON OF HUMANISTIC QUALITIES BETWEEN MEDICAL STUDENTS, RESIDENT AND ATTENDING PHYSICIANS. M. Lanquill¹, W. Lee¹, S.J. Peterson²; ¹Westchester Medical Center, Valhalla, NY; ²New York Medical College, New York, NY

PURPOSE: The ABIM expects all physicians board certified in internal medicine to possess humanistic qualities of integrity, respect and compassion. To date, no studies have been

done comparing humanistic qualities among physicians at different levels of training. We compared humanistic quality scores of fourth year medical students (SI), internal medicine residents (PGYI and II), and attending physicians on a general medicine ward of a teaching hospital.

METHODS: A validated seventeen-item nursing survey to assess humanistic qualities among physicians was distributed to randomly selected nurses working day shifts on the medicine wards. The survey measured physician relationships with other medical staff, and the physician relationship with the patient and family. Each item was scored on a 5 point Likert scale. Composite scores for physician to staff relationships (PSR) and physician to patient/family relationships (PFR), as well as an overall evaluation score (OE), were compared across levels of physician training. Student's t-test was done to determine statistical significance across training levels.

RESULTS: 33 nurses completed a total of 295 questionnaires. Surveys were performed for each of the 7 sub-interns, 8 PGYI, 4 PGYII, and 11 attending physicians rotating on the medical wards during a month of the ward service. SI's had higher PSR scores (PSR = 24.2) than PGYI (PSR = 21.3, p value = 0.013), PGYII (PSR = 21.9, p = 0.042), and attending physicians (PSR = 23, p < 0.01). SI's also had higher PFR scores (PFR = 33) compared to PGYI (PFR = 29.4, p = 0.025), PGYII (PFR = 28.3, p < 0.001), and attendings (PFR = 31.7, p < 0.01). Finally, SI's had higher OE scores (OE = 4.4) when compared to PGYI (OE = 3.8, p < 0.01), PGYII (OE = 3.8, p < 0.01), and attendings (OE = 4.2, p < 0.001). No statistically significant differences were found comparing PGY I, PGY II, and attending levels among each other.

CONCLUSION: Sub-interns appear to have better perceived qualities of humanism compared to resident and attending physicians. Because resident and attending physicians play an important role in medical education, efforts should be made to better understand the mechanisms of these differences, and to improve the humanistic qualities of both resident and attending physicians.

MODERN CHALLENGES IN ACADEMIC LEADERSHIP IN TODAY'S MEDICAL SCHOOLS. M.L. Lypson¹, D.T. Stern², J.O. Woollicroft²; ¹University of Chicago, Chicago, IL; ²University of Michigan, Ann Arbor, MI

PURPOSE: Academic Medical Center (AMC) leaders must respond to numerous internal and external challenges. Self reports from leaders have documented these challenges; however, little is known of the daily workings of AMC leaders and the tactics used to run an AMC in the 21st century.

METHODS: Ethnographic methods were used to explore the challenges faced by leaders in the top 3 levels of one institution's organizational chart. The study was conducted at a public Midwestern school, with 1600 faculty and \$165 million in NIH funding. The data are: 1)10 hrs. of interviews with leaders and 2)field notes and minutes from over 300 hrs. of participant observations in meetings. Field notes captured process (e.g. deference, control, tolerance etc.) and content (e.g. parking, safety, and budget concerns etc.). The team then identified themes by iterative review of field notes, with validity assessed by comparison with interview responses and subsequent member checking. This report focuses on identified content themes.

RESULTS: Six broad content domains are recurrent: 1)faculty morale, 2)business issues e.g. intellectual property rights and establishing frameworks for joint ventures between AMCs, business and/or government. 3)developing affiliations with others to remain competitive 4)creating plans of action to improve diversity 5)creating tools and metrics to demonstrate "academic productivity" & assure society that graduates are competent. 6)maneuvering AMCs to adjust and survive a unpredictable environment.

CONCLUSION: AMC's are currently facing instability. This study of leaders day to day activities reveals challenges in 3 broad areas: A)human resources, B)institutional fiduciary roles and C)vision setting. Human resources include faculty, student and staff morale, issues of workforce diversity and assessment of "academic productivity". The institution has a fiduciary responsibility in ensuring financial stability, education and assuring graduate competency. Finally vision setting includes the ability to adjust and survive in an unpredictable environment. Improved understanding of leadership challenges will allow better preparation of new generations of leaders, and will better prepare AMCs for future challenges.

PROFESSIONALISM: WHAT DOES IT MEAN TO THIRD YEAR MEDICAL STUDENTS? M.L. Lypson¹, J.M. Hauser¹; ¹University of Chicago, Chicago, IL

PURPOSE: The early 1990's sparked an interest in organized medicine to reclaim and re-evaluate how it promotes professionalism among physicians. The American Board of Internal Medicine (ABIM) launched Project Professionalism as a means to define and evaluate professionalism as a component of clinical competence. During a course (Talking Medicine) developed for third year medical students to teach humanism and professionalism, we asked students to define these concepts and use these definitions to spark small group discussion.

METHODS: At the beginning of each internal medicine clerkship we asked students to define humanism and professionalism anonymously on sheets of paper to be handed to the preceptors. We conducted a content analysis of 3 small groups' definitions (n=14).

RESULTS: In Project Professionalism, the ABIM defines professionalism as: Altruism, Accountability, Excellence, Honor / Integrity, and Respect. The Students, however, saw a broader definition of professionalism. Themes that they identified included: 1) being a physician, student, teacher, philosopher and healer all at once, 2) doing no harm, 3) understanding that "respect for others" must include tolerance of differences; as well as a focus on collegiality among practitioners at various levels of the medical hierarchy 4) having appropriate speech, dress and emotions- which should often be held in "check", 5) honoring the system, but not blindly and 6) stating when a good job has been performed.

CONCLUSION: Third year medical students tend to agree broadly with the ABIM components of Professionalism. However, they focus more on tolerance of difference (e.g. race, socioeconomic status, and varying health beliefs), and the importance of collegiality and collaboration in the new environment of patient care. Their vantage point early in training allows them to look critically at the profession which they are joining and view its shortcomings

and strengths. Future work will focus on how these definitions change as students progress through 3rd and 4th year and into residency.

MINORITIES AND EVIDENCE BASED MEDICINE, DO RANDOMIZED CONTROLLED TRIALS PROVIDE THE ANSWER? S. Mahi¹, S.M. Sandhu¹, H.S. Gurm²; ¹Fairview Hospital, Fairview Park, OH; ²Cleveland Clinic Foundation, Cleveland, OH

PURPOSE: Race has a significant impact on prevalence, presentation and outcome of disease. While the randomized controlled trials (RCTs) have become the gold standard for evidence based medicine, there is a paucity of data on representation of minorities in RCTs. The purpose of this study was to review minority enrollment in RCTs published in a major medical journal.

METHODS: We conducted a systematic search of all articles published in the original articles section of the New England Journal of Medicine from 1993–2000. All randomized control trials done in North America were included for abstraction. We excluded all studies limited to certain ethnic groups, subset analysis of other trials and meta-analyses. Trials were characterized by primary speciality and source of funding. Data were analyzed using Students t test and ANOVA.

RESULTS: 1799 original articles were published in the period between 1993–2000 including 535 Randomized controlled trials. Of the 302 RCTs done in North America, only 106 specified the percentage of minorities enrolled. A total of 180,364 patients were enrolled in these 106 trials of which 37,653 were minorities (20.8%). The maximum number of trials (80) were in cardiovascular medicine. The minority enrollment was specified in only 24, (30%) and in these the minorities constituted only 14% of the enrollees. (4766/34,132). In contrast to cardiovascular medicine, 15 out of 22 trials (68%) in pregnancy related disorders and 12 out of 27 RCTs (44%) related to pediatric population specified minority enrollment. Similarly, the trials in these groups enrolled a larger percentage of minorities: 40.7% (13928/34220) in pregnancy related trials and 51.3% (3253/6343) in the pediatric trials. After trials in pregnant women and children are excluded only 6% (20,472/293,297) of enrolled subjects were minorities.

CONCLUSION: Although minorities now make up almost one third of the American population, they are significantly underrepresented in randomized controlled trials. Most studies fail to specify the ethnic or racial mix of the population and thereby significantly limit the generalizability of the results. At a time when all research is held up to the gold standard of the RCTs, the RCTs fail to provide data on a significant segment of the population. Our data underscores the importance of an ongoing effort to improve minority enrollment and ethnic specific analysis in clinical trials.

FAITH-BASED SUBSTANCE ABUSE REHABILITATION: RELIGIOUS AND SECULAR COMPONENTS. L. McCoy¹, S.M. Frayne¹, J. Hermos¹; ¹Boston University, Boston, MA

PURPOSE: Among the rehabilitation options for persons with substance abuse disorders are the prevalent and increasingly publicized faith-based programs. We sought to describe key structural aspects of substance abuse recovery programs within one major national Christian organization.

METHODS: A convenience sample of 7 administrators, clinical directors, and counselors from 4 New England-based residential recovery programs was identified. Each subject was interviewed for 60 minutes using a semi-structured protocol. Using qualitative methods, key words and phrases were derived from all instances where aspects of the program structure were mentioned. The key words were placed into groups that categorized the major structural elements of the program

RESULTS: The programs in this study provide long-term free services for men and women of all faiths. All programs are self-supporting and receive no governmental funding. Religious (or spiritual) components mentioned were Bible studies, worship and devotional services, vespers and prayer meetings, spiritual growth classes, retreats, and Alcoholics Anonymous or Narcotics Anonymous meetings. Individual and group therapy sessions also included spiritual references and themes. Secular components were medical and dental services, mental health services, Graduate Equivalency Diploma classes and job skills training, work therapy, sponsorship, access to physical fitness facilities, recreational activities, and aftercare program planning.

CONCLUSION: Activities relating to religious study, worship, and spiritual growth are critical components of the faith-based recovery programs studied. However, the religious aspects of these programs appear to complement rather than replace the elements seen in secular programs. Further analyses of how their philosophy of faith is translated into practice are needed.

TAMOXIFEN FOR BREAST CANCER RISK REDUCTION: WOMENS PERSPECTIVES. J. Melnikow¹, D. Patemiti¹, S. Henderson¹, J. Nuovo¹, M. Degregorio¹, M. Kuppermann², R. Nease³; ¹University of California, Davis, Davis, CA; ²University of California, San Francisco, San Francisco, CA; ³Washington University, St Louis, MO

PURPOSE: Tamoxifen was FDA approved for breast cancer risk reduction in 1998 for women who have a 5 year risk > 1.7% (average risk of a 60 year old white woman). Healthy women deciding whether to take it for prevention must weigh its potential benefits and adverse effects. Purpose: To develop an understanding of how ethnically diverse risk-eligible women weigh risks and benefits regarding tamoxifen prophylaxis.

METHODS: We undertook a qualitative intervention study using a focus group-format to provide general information about the risks and benefits of tamoxifen for breast cancer risk reduction and to assess the concerns of ethnically diverse (Caucasian, African-American, and Latina) women about the use of tamoxifen prophylaxis. Group facilitators were matched to the ethnicity of participants. Focus group discussions involved: (1) exploration of knowledge of the risks and benefits of tamoxifen as a preventive therapy; (2) a standardized educational intervention about tamoxifen's risks and benefits based on findings from the Breast Cancer

Prevention Trial; and 3) discussion about participant decision-making and tamoxifen's overall value in prevention. All participants completed a brief questionnaire to assess the impact of the intervention on their understanding of tamoxifen as preventive therapy. Focus groups were audiotaped and later transcribed for analysis. A multi-disciplinary team of investigators identified prominent themes related to health beliefs that emerged from iterative review of focus group transcripts.

RESULTS: 27 women (age range 61–78) participated in the focus groups. Most reported that they felt more informed after the group. In general, fear of breast cancer was not prominent, and participants were not inclined to take tamoxifen as preventive therapy. Decisions about tamoxifen were based on participants' assessment of the balance of competing risks and competing benefits. Specifically, participants expressed limited willingness to take medication with serious side effects for breast cancer risk reduction, uncertainty about its interaction with other medications, and concern about the need to discontinue hormone replacement therapy. Caucasian and Latina groups expressed concerns about reliability of scientific studies. African–American women described faith as important to prevention.

CONCLUSION: Women were wary of taking a drug for a disease they might not develop. Women felt they had options other than Tamoxifen to reduce risk of breast cancer, including early detection, diet, faith, and alternative medicine. Information about tamoxifen decreased interest in use for breast cancer prevention.

SCREENING MAMMOGRAPHY FOR WOMEN UNDER 50: HOW DO WOMEN DECIDE? A QUALITATIVE ASSESSMENT. L. Nekhludov¹, D. Ross-Degnan¹, S.W. Fletcher¹; ¹Harvard Medical School/Harvard Pilgrim Health Care, Boston, MA

PURPOSE: Although shared decision-making has been advocated for screening mammography for women under age 50, little is known about how women decide whether to get screened, what are their preferred sources of information and what are their preferences for involvement in screening decisions. This study explored these issues in patient decision-making.

METHODS: We conducted in-depth, semi-structured telephone interviews with 16 randomly selected English-speaking women ages 38–45 who were enrolled at a large New England HMO, and had no prior history of breast cancer. Interviews were transcribed and coded into major themes using QSR NUD*IST.

RESULTS: Thirteen participants were aged 41–45 and 3 were aged 38–40. Nine were white and seven were African–American. The majority (10/13) of women over age 40 had prior mammograms and all intended to have a screening mammogram before age 50. Factors in women's decisions to undergo screening included knowledge of the “recommendation to begin screening at age 40” obtained either from their physicians or the media and “having a friend or acquaintance with breast cancer.” Most women stated that screening has a number of benefits, including “early detection,” “peace of mind” and “it's better knowing.” Possible risks of mammography, although not considered important, included radiation, pain, and “not being able to catch everything.” None of the women who heard of or experienced false positive mammograms (8/16) were or would be deterred from future screenings. While all women stated that the physician should be the primary source of information about screening, 11/16 had no discussions with their own physicians regarding the mammography procedure, its risks and/or benefits. Many (12/16) learned such information from the media. Preferences for involvement in decision making varied; 4/16 preferred to leave screening decisions mostly to the physician, 9/16 preferred to make them mostly themselves, and the rest were uncertain. We found no major thematic differences among the white and the African–American women.

CONCLUSION: Our results suggest that in this setting the decisions of women under 50 to undergo screening mammography are based on their own perceptions of risks/benefits or on the recommendation of their physicians. However, in these recommendations, discussion of specific information relating to screening rarely occurs, and shared decision-making is limited. These findings can guide future research and lead to the development of policy that facilitates patient–physician communication in this area.

ABSOLUTE VS RELATIVE VALUES: IMPACT ON PHYSICIAN MEDICAL DECISIONS. K.E. Olive¹, J.K. Neumann²; ¹East Tennessee State University, Johnson City, TN; ²James H. Quillen VA Medical Center, Mountain Home, TN

PURPOSE: A previous study indicated that the absolute vs relative ethical construct is a factor influencing physician decisions. The purpose of this study was to confirm, in a different physician population, the previous results — that physicians who believe in ethical values which do not change (absolute values) will respond differently to ethically sensitive clinical scenarios than those who affirm changing ethical values (relative values).

METHODS: A national sample of family physicians and psychiatrists received a mailed survey. The survey included a four-item scale to determine relative vs absolute values, demographic and psychological measures, and three clinical scenarios involving contraception, physician-assisted suicide, and abortion. The primary analysis compared approval of the clinical scenarios for physicians with absolute values to those with relative values. Approval ratings were determined using a seven point Likert-type scale (1=do not approve, 7=do approve). Data were analyzed using analysis of variance.

RESULTS: 317 of 1000 physicians surveyed (32%) responded. Of those responding, 99 met criteria for absolute values and 59 for relative values. Those endorsing absolute values were significantly less approving of all three scenarios than those endorsing relative values: Contraception (5.5 vs 6.4, F=10.17, p=.0017), Physician Assisted Suicide (2.1 vs 4.1, F=36.22, p<.00001), Abortion (3.0 vs 5.0, F=28.45, p<.00001).

CONCLUSION: Physicians with absolute ethical values are less approving of contraception for single women, physician-assisted suicide, and abortion than those endorsing relative ethical values. The absolute vs relative values ethical construct may be useful in examining physician attitudes which impact health care delivery. These survey results replicate a previous study indicating the absolute vs ethical construct is a factor influencing physician decisions. This construct is directly relevant to clinical care and physicians need to be aware of their own biases in discussions with patients, families, and other health care providers.

PHYSICIAN–PATIENT E-MAIL COMMUNICATION: BENEFITS AND BARRIERS. M.R. Patt¹, D.E. Ford¹, D.Z. Sands², T.K. Houston¹; ¹Johns Hopkins University, Baltimore, MD; ²Beth Israel Deaconess Medical Center, Boston, MA

PURPOSE: To explore the perceived benefits of and barriers to physician patient e-mail communication by interviewing physicians frequently using e-mail to communicate with patients.

METHODS: Physicians (n=1,325) responded to an Internet-based questionnaire designed to identify physicians frequently using e-mail with patients. From this sample, we randomly selected 41 primary care physicians, who use e-mail with patients daily, to participate in a 10–15 minute in-depth telephone interview. The interview was designed to explore their experiences through a series of open-ended questions. Transcripts were individually reviewed by each author to develop “themes”, which were then verified by the individual raters through group meetings.

RESULTS: Of the 41 primary care physicians contacted by e-mail, 17 responded, 2 were unable to be reached, and 22 have not responded to 4 separate e-mail requests for an interview, thus far. Prominent themes included: 1) Initiation of communication, 2) Benefits, 3) Barriers, and 4) Establishment of policies. A majority of the physicians reported that patients initially asked them about communicating via e-mail. Physicians appear to be selective in choosing which patients they will communicate with via e-mail, but it is not clear what criteria they use. Benefits of using e-mail included better and more consistent communication with patients who have chronic illnesses and require minimal changes in care, dissemination of educational information (particularly linking to reliable internet sources), ability to respond to non-urgent issues on their own time, and avoidance of phone tag with patients. Barriers to e-mail use are lack of office personnel or nursing staff involvement, potential of increased physician time consumption, difficulty incorporating email into office flow, responding to e-mails in timely manner, tracking e-mail messages, and inappropriate or urgent content in the messages. Confidentiality appeared to be a minor concern among most physicians; however, those who are concerned about confidentiality believe it is a major barrier. Finally, physicians, for the most part, do not have established formal policies/guidelines that they use with patients regarding e-mail communication. Those who do have formal policies in place appear to have fewer concerns about content and overall use.

CONCLUSION: These physicians did perceive benefits to e-mail communication with a select group of patients. Dissemination of formal guidelines, better tracking mechanisms, creation of explicit methodology for selecting patients suitable for e-mail communication, and improved incorporation into office flow are needed before widespread use of this medium can be achieved.

EXPERIENCES OF PEOPLE WHO PURSUE PHYSICIAN-ASSISTED SUICIDE. R. Pearlman¹, C. Hsu², H. Starks², A. Back¹, J. Gordon², A. Bharucha³; ¹VA Puget Sound Health Care System, Seattle, WA; ²University of Washington, Seattle, WA; ³University of Pittsburgh, Pittsburgh, PA

PURPOSE: Rich descriptions of the motivations and experiences of people who pursue physician-assisted suicide (PAS) are lacking.

METHODS: To better understand these motivations and experiences, we recruited (through intermediaries) and interviewed two cohorts multiple times over a three year period: (a) family members of persons who had already hastened their death, and (b) patients who were considering a hastened death and their family members. The interviews were semi-structured, open-ended, and audio-taped. After transcribing them verbatim, we conducted a content analysis to identify themes. Trustworthiness was assured through independent, duplicate coding, review and discussion of discrepancies in coding by the multi-disciplinary research team, and review by professional and lay participants in the hastened death advocacy community.

RESULTS: We interviewed 60 informants about 35 cases. All the patients (cases) were white, 49% were female, 63% had cancer, 13% had AIDS, 71% had received hospice care and/or home care, and 100% had health insurance. None had severe depression. Eighteen cases involved PAS, 7 euthanasia, 8 died of the underlying illness, 1 died from a gunshot wound and 1 was still alive at the end of the study. No single factor on its own led to the pursuit of PAS. Multiple factors interacted with each other and set the stage for individuals' pursuit of PAS. These factors included physical experiences of their illness and dying (e.g., pain, feeling weak and tired), functional limitations (e.g., inability to eat, care for self, use the bathroom), character traits and values (e.g., a strong sense of independence and agency, long-standing beliefs about the value of hastening death to avoid an unacceptable quality of life, assessing and reassessing living and dying with an analytic perspective that weighed the benefits and burdens of available options), fears and distress about the present or future (e.g., discomfort, indignity, cognitive decline), and loss of sense of self (e.g., no longer being the competent adult they knew and valued). Uncontrolled pain and other physical symptoms were rarely mentioned as primary reasons for interest in PAS.

CONCLUSION: These data suggest that for these participants, the pursuit of PAS was a thoughtful attempt to control dying or suffering and not a response to acute depression, inadequate finances, or lack of access to palliative care. These data challenge the assumption that those interested in PAS are motivated by uncontrolled pain or untreated mental illness.

DEFINING MENTAL ILLNESS IN STATE PARITY LEGISLATION. M. Peck¹, D. Gitterman², R. Scheffler³; ¹Stanford University, Stanford, CA; ²University of North Carolina at Chapel Hill, Chapel Hill, North Carolina; ³University of California at Berkeley, Berkeley, CA

PURPOSE: As of December 2000, 31 states passed mental health parity statutes, laws requiring a health plan, insurer, or employer to provide mental health benefits “equal” to physical health benefits. With parity, lawmakers are for the first time legislating their own state definitions of mental illness rather than leaving the definition up to insurance plans or covering all disorders in the American Psychiatric Association's Diagnostic and Statistical Manual (DSM-IV) as legislation has traditionally done. My research examines definitions of “mental illness” used in state parity laws and implications of these definitions including what populations are getting increased access to care.

METHODS: I reviewed the statutory language in 31 state parity bills and conducted a series of interviews with stakeholders participating in the national and state mental health parity process. Interviewees included mental health advocacy groups, provider organizations, legislative bill sponsors, health plans, and employer groups. Interview data was supplemented with extensive literature review.

RESULTS: The definition of mental illness varies significantly across states, but falls into one of three categories: “broad-based mental illness,” “serious mental illness,” and “biologically-based mental illness.” Broad-based mental illness includes all disorders listed in the DSM-IV and/or World Health Organization's ICD-9. “Serious mental illness” and “biologically-based mental illness” include from three to eleven specific DSM-IV mental disorders. Several factors influence a state's definition of mental illness including the cost of parity, the leadership role taken by state advocacy group(s), the strength of insurance interests, the reliance on anti-discrimination arguments, and the political feasibility of parity in the state.

CONCLUSION: With parity, state lawmakers are not relying on clinically accepted definitions or previous federal and state definitions of mental illness. It appears that political and economic factors influence the statutory definition, rather than needs-based strategies or clinical judgement. Insurers argue that legislating parity for all mental disorders results in tremendous healthcare costs and necessitates the development of definitions that target only the most seriously mentally ill. However, using a “pick and choose” approach to the DSM-IV as most states have done has little (if any) clinical basis, and can potentially limit access to care. Increased reliance on clinician judgement and further studies measuring the access implications of current parity laws are needed to identify those who would most benefit from parity.

GENDER-SPECIFIC CULTURAL VIEWS ABOUT ADVANCE CARE PLANNING. H.S. Perkins¹, J.D. Cortez², H.P. Hazuda¹; ¹University of Texas Health Science Center at San Antonio, San Antonio, TX; ²Intercultural Development Research Association, Inc., San Antonio, TX

PURPOSE: Ethnic culture—the basic beliefs an ethnic group uses to interpret experience—affects patients' views on dying. But views within an ethnic group are not all alike. The genders within an ethnic group may share some views but differ on others and, thus, be distinct subcultures for advance care planning purposes.

METHODS: We explored this possibility by asking Mexican Americans (14 men, 12 women), Euroamericans (7 men, 11 women), and African Americans (7 men, 7 women) their views on advance directives (ADs). All subjects were inpatients aged 50 to 79.

RESULTS: While believing ADs improve the chances a patient's wishes would be followed, the genders within all three ethnic groups differed over the power and trustworthiness of the health care system, treatment wishes and the willingness to express them, and beliefs about ADs. Among Mexican Americans most men but few women believed the system controls treatment (79% versus 33%), believed patients should be allowed to die when treatment is futile (57% versus 42%), wanted no life support (71% versus 42%), had not told anyone their wishes (64% versus 42%), and believed ADs are needed for when a patient is brain dead or cannot function (57% versus 33% for both themes). But few men and most women trusted the system to honor written ADs and believed ADs help the staff know a patient's wishes (43% versus 67% for both themes). Among Euroamericans most men but few women distrusted the system (57% versus 18%) and disliked ADs (57% versus 36%). But few men and most women expressed wishes not to suffer during terminal illness (0% versus 55%), believed patients should be allowed to die when treatment is futile (0% versus 55%), and believed the system honors written ADs (43% versus 73%). Among African Americans most men but few women distrusted the system (57% versus 29%) and wanted to wait until very sick to express their wishes (57% versus 43%). And few men but most women said that the patient deserves a say in treatment (43% versus 57%), wanted no life support (29% versus 57%), and believed ADs help staff know the patient's wishes (0% versus 57%) and can prevent life support (14% versus 57%).

CONCLUSION: While sharing some views, men and women of each ethnic group differed on others. Unlike men, women tended to trust the system and to believe ADs will help realize their wishes. Providers should consider gender-specific values within an ethnic culture when doing advance care planning with patients.

AGE-SPECIFIC CULTURAL VIEWS ABOUT ADVANCE CARE PLANNING. H.S. Perkins¹, J.D. Cortez², H.P. Hazuda¹; ¹University of Texas Health Science Center at San Antonio, San Antonio, TX; ²Intercultural Development Research Association, Inc., San Antonio, TX

PURPOSE: Ethnic culture—the beliefs an ethnic group uses to interpret experience—affects patients' views on dying. But views within an ethnic culture are not all alike. Age groups within an ethnic culture may share some views but differ on others and, thus, be distinct subcultures for advance care planning purposes.

METHODS: To explore this possibility we interviewed Mexican Americans, Euroamericans, and African Americans. Younger subjects (ages 50–64) numbered 15, 11, and 10 respectively; older subjects (ages 65–79) numbered 11, 7, and 4.

RESULTS: While all ethnic groups liked some aspects of advance directives (ADs), age groups within each ethnic group differed over trust in the health care system, treatment wishes, and beliefs about ADs. More younger than older Mexican Americans wanted to be allowed to die when treatment is futile (60% versus 36%), had not told their wishes to anyone (60% versus 45%), believed ADs are needed for when a patient cannot voice decisions (60% versus 27%), and trusted the system to honor written ADs (60% versus 45%). Fewer younger than older Mexican Americans expressed wishes unrelated to life support (for example, where or when they wished to die) (33% versus 64%). More younger than older Euroamericans described unacceptable outcomes (55% versus 43%) and believed that ADs help staff know a patient's wishes (91% versus 29%) and are needed for when a patient is brain dead (64% versus 29%). Fewer younger than older Euroamericans disliked some aspects of ADs (36% versus 57%), had not told anyone their wishes (18% versus 71%), and doubted ADs change treatment (27% versus 57%). More younger than older African Americans believed the system controls treatment (80% versus 25%), distrusted the system (50% versus 25%), said the patient deserves a say in treatment (60%

versus 25%), wanted no life support (50% versus 25%), doubted ADs change treatment (50% versus 25%), and said ADs just mean the patient is dying (70% versus 0%). Fewer younger than older African Americans trusted the system to serve patients well (30% versus 50%), had not told their wishes to their physicians (10% versus 50%) but planned to tell someone before becoming too sick (0% versus 75%).

CONCLUSION: While sharing some views, intraethnic age groups differed in their trust of the health system, their willingness to express treatment wishes, and their confidence in ADs. Providers must consider age-specific beliefs within an ethnic culture when conducting advance care planning with patients.

MAY IT PLEASE THE COURTS: THE LEGAL CAMPAIGN AGAINST HOSPITAL DISCRIMINATION. P. Reynolds¹; ¹Johns Hopkins University, Baltimore, MD

PURPOSE: Hospital discrimination was widespread throughout the United States and in many jurisdictions legally sanctioned as late as the mid-1960s. Discrimination most commonly was experienced as the denial of staff privileges to minority physicians and dentists, refusal to admit minority applicants to nursing and residency training positions, and lack of medical and surgical services to minority patients in many urban private and public institutions. A national campaign to eliminate discrimination in hospital clinical services and education programs involved the collaboration among minority medical and activist organizations, and a direct attack against the hospital policies and practices through litigation culminating in two landmark decisions which both emerged from the 4th Circuit Court of the United States.

METHODS: Research was conducted in national archives, in manuscript collections of minority and non-minority physicians, in legal documents with review of relevant court cases, and in primary and secondary law and medical journals.

RESULTS: The first hospital discrimination case brought before the courts was filed in North Carolina in 1956 by three physicians, Dr. Hubert Eaton, Dr. Daniel Roane, and Dr. Samuel Gray, and their lawyer, Conrad Pearson. While the courts initially denied the plaintiffs the relief they sought, the NAACP lawyers used information emerging from Eaton vs James Walker Memorial Hospital when deciding the most essential elements in constructing a "test case". Their attack centered on the federal government's use of public funds to construct "Separate but Equal" hospitals throughout the south. That same year, the City of Chicago passed the Harvey-Campbell Ordinance to prevent racial discrimination in Chicago hospitals.

In 1962 the second hospital discrimination case in North Carolina was filed by George Simkins and the lawyers for the NAACP against the Moses H. Cone Memorial Hospital and Wesley Long Hospital. This case would determine ultimately the legal foundation of "Separate but Equal" in hospital policies and practices. Two years later African-American physicians in Chicago sued 56 hospitals throughout the city for refusal to adhere to the Harvey-Campbell Ordinance charging they had been denied staff appointments to hospitals in the area because of their race.

By the time the Medicare Hospital Certification Program was implemented in the summer of 1966, over 35 suits had been filed by physicians and citizens with the assistance of NAACP lawyers against hospitals and nursing homes located in communities throughout the south. The Medicare hospital racial integration guidelines themselves would be tested in the courts through Cypress versus Newport News Hospital Association.

CONCLUSION: The legal foundation for "Separate but Equal" in hospital policies and practices was eroded through a series of court battles developed and argued by NAACP lawyers culminating in Simkins versus Moses H. Cone Memorial Hospital (1963) and Cypress versus Newport News Hospital Association (1967). The first of these two cases challenged the federal government's use of public funds to expand and maintain segregated hospital care. The second case reaffirmed the federal government's use of public funds to force hospitals to open up patient admission policies, education programs and hospital staff privileges to all citizens and physicians regardless of race, gender, age or nationality. Successful pursuit of a legal strategy against racist hospital policies and practices was a critical cornerstone in a national campaign to eliminate discrimination in health care delivery.

"I THINK MY HEART GOT A LITTLE STONE IN IT NOW: CONSEQUENCES OF VIOLENT VICTIMIZATION AMONG YOUNG AFRICAN AMERICAN MEN. J. Rich¹, C. Grey¹; ¹Boston University School of Medicine, Boston, MA

PURPOSE: Violence remains a leading cause of death and disability for young African American men. Recurrent violent injury is also well documented in the medical literature. The purpose of this study is to understand how the experience of violent injury changes the lives of young black men and predisposes them to recurrent victimization.

METHODS: Participants in the study were 34 African American men, between the ages of 18 and 30, who were admitted to an urban medical center for violent injury. In-depth, semi-structured interviews were conducted within 2 weeks of the initial injury and, in a subset of participants, a follow-up interview was conducted 2 months later. Analysis was performed using grounded theory methods and narrative analysis to identify prominent themes common across interviews.

RESULTS: The narratives of young black men who are victims of violence highlight the physical pain and suffering that they experience. More prominent, however, are accounts of negative interactions with the police, both prior to the injury and in the immediate aftermath, who treat them as perpetrators regardless of the circumstances of their injuries. Also prominent are vivid descriptions of disabling symptoms of traumatic stress, and profound feelings of fear and vulnerability. These extreme feelings of vulnerability, coupled with mistrust of the police and other community institutions, lead these young men to develop their own strategies to remain safe and to deal with their distress. Some of these strategies either hinder their ability to heal from their emotional trauma (such as self-treating with drugs and/or alcohol) or put them at risk for recurrent injury (such as acquiring a weapon). Few young men expressed knowledge of resources to address their psychoemotional issues.

CONCLUSION: The physical and emotional consequences of violent injury, coupled with a social environment that is hostile and unsafe, places young black male victims of violence at risk for recurrent injury. Efforts to reduce recurrent violent injury will be more effective if they

address both the emotional consequences of injury and the often oppressive social environments in which these young men live.

UNDERSTANDING TEAM-BASED QUALITY IMPROVEMENT FOR DEPRESSION IN PRIMARY CARE. L.V. Rubenstein¹, L.E. Parker², L.S. Meredith², N.P. Gordon³; ¹VA Greater Los Angeles, RAND and UCLA, Sepulveda, CA; ²RAND, Santa Monica, CA; ³Kaiser Permanente Division of Research, Oakland, CA

PURPOSE: Team-based quality improvement (QI) methods are an attractive option for designing and implementing improvements in depression care in primary care (PC) practices. QI methods experts disagree, however, about whether teams should be centralized (the CT approach), or local (the LT approach). In CTs a few key experts, with input from local clinicians based in the target practice sites, are responsible for developing the QI intervention design. In LTs the local clinicians develop the QI design through on-site interdisciplinary team meetings. The qualitative analyses presented here compare the success of the two approaches and evaluate the characteristics of teams and organizational environments that affect success.

METHODS: We partnered with two non-profit staff model managed care organizations—VA Greater Los Angeles and Kaiser Permanente of Northern California—to initiate and evaluate five QI teams tasked with improving care for depression. Teams were structured as either CTs or LTs. The three LTs involved three PC practices and the two CTs involved three PC practices. Trained observers transcribed process notes during all team meetings, and carried out structured interviews with team and clinical practice leaders and members at one and two years. Three social scientists with qualitative expertise independently reviewed all study data and rated teams. Six national depression experts independently rated team QI designs (plans). Analysis of team success used qualitative predictor-outcome matrices.

RESULTS: The two CTs ranked higher for depression program planning than two of the three LTs, but one LT received equivalently high ratings. Across all teams, team leadership and primary care and mental health top management support for depression QI had the most effect on plan quality and implementation. CTs, however, had better success than LTs in less supportive environments. For example, PC top management for one of the two PC practices linked to one CT was relatively unsupportive, as was mental health top management for the practice linked to the other CT; the two CTs did better than the two LTs that were based in similarly low support organizational environments. The one LT with favorable team leadership and organizational environment did as well as the best performing CT.

CONCLUSION: CT and LT approaches have equivalent success when team leadership and organizational support are strong, while CTs have better success than LTs in less supportive environments. Our results suggest that depression QI approaches that depend on local interdisciplinary team design and implementation should be reserved for favorable organizational environments. Less favorable environments are likely to be better served by a more centralized, expert directed QI approach.

MISSED OPPORTUNITIES IN PHYSICIAN-PATIENT COMMUNICATION WITH TYPE 2 DIABETES PATIENTS WHO HAVE HEALTH LITERACY PROBLEMS. D. Schillinger¹, J. Piette², K. Leong-Grotz³, C. Wilson¹, C. Castro¹, K. Grumbach¹; ¹UCSF, San Francisco, California; ²Center for Health Care Evaluation, VA Palo Alto Health Care System, Palo Alto, CA; ³Case Western Reserve School of Medicine, Cleveland, OH

PURPOSE: Patients fail to recall and comprehend as many as half of the critical pieces of information conveyed during a clinical encounter. To maximize patients' recall, understanding, and ultimately their treatment adherence, it is recommended that clinicians limit the number of new concepts they present and always elicit patients' comprehension of these points. This educational technique may be particularly important for patients with low functional health literacy (FHL) and chronic conditions such as type 2 diabetes (DM2), since these patients grapple with complex clinical management regimens and often have difficulties processing oral communication. In this study, we measured the extent to which primary care physicians (PCPs) assess these patients' recall and comprehension of their advice.

METHODS: The study was conducted in primary care centers at a public hospital. Outpatient encounters between 30 English-speaking, DM2 patients with low FHL (score <23 on short-form Test of Functional Health Literacy) and 22 PCPs were audiotaped. 2 trained coders measured the duration of each encounter, and identified the following communication events: (1) new concepts raised by physicians (such as an explanation of how blood pressure affects the kidneys, or a change in medication regimen); and (2) whether each new concept was followed up by the physician with an attempt to ensure that patients comprehended the key points, e.g. by a request for patient restatements, or eliciting patient perceptions. The percentage of new concepts for which follow-up assessments occurred was then calculated.

RESULTS: The mean number of new concepts conveyed by PCPs was 2.0 (range 0–5). The mean number of follow-up assessments was 0.27 (range 0–2) or 13%; in over half of these assessments the patient did not recall or comprehend the new concept, allowing the PCP to tailor subsequent information. Visits including one or more follow-up assessments did not differ in duration when compared to visits that lacked such follow-up (18 vs. 21 min, p=0.2).

CONCLUSION: The vast majority of PCPs caring for DM2 patients with low FHL do not assess patient recall or comprehension of new concepts conveyed during the clinical encounter. Overlooking this critical step in the communication process reflects a missed opportunity to enhance patient care. Future studies should evaluate whether more consistent application of this communication technique improves patient recall, adherence, and health outcomes.

EXPLORING BREAKDOWNS AND RESIDENT LEARNING IN CLINIC. C. Smith¹, C. Francovich¹, J. Gieselman¹, M. Morris¹; ¹VA Medical Center, Idaho and University of Washington, Boise, ID

PURPOSE: Breakdown can be described as a disruption in smooth unreflective practice that forces an individual to adopt a more explicit, deliberate and abstract perspective. This study

examined a series of breakdowns in an ambulatory medical clinic. It examined the response of residents to these breakdowns and its effect on patient care and learning.

METHODS: Trained observers collected ethnographic data from waiting areas, workstations and exam rooms. Real-time field notes from sixty-eight separate observations (total 130 hours) were immediately transcribed into the data set, 2919 paragraphs of text. Two analysts coded text units at the paragraph level into recurrent themes using template analysis. Breakdown emerged as the major theme, and was fine coded at the phrase level to create a taxonomy. Using this taxonomy, 156 vignettes were selected to represent the full spectrum of breakdown. Recursive analysis of the vignettes was used to create a structural model of breakdown and learning. Triangulation and member checking were used to validate the model.

RESULTS: Breakdown occurred in 1255 (43%) of the text units, and was clearly important for learning. Breakdowns occurred both from individual actions and from clinic structure and process. Preliminary results indicate that the computer-based patient record is an important source of breakdown; it interferes with communication and structures the interview, more so for residents than for faculty. The structural model of breakdown has three levels: psychomotor, conceptual and cultural. Breakdown begins at the psychomotor level of automatic behaviors (e.g., withdrawal in response to patient anger). If the breakdown persists, it moves to the conceptual level, where models are shared to determine correct action. At conceptual level, breakdowns often involve heuristic biases (e.g., favor curable diagnoses over chronic ones). Unresolved breakdowns become obtrusive and progress to the cultural level, where the entire process of norms, concepts, actions and consequences are reflected upon. At the cultural level, breakdowns reflect failure to engage with cultural norms.

CONCLUSION: Our observations reinforce previous research that indicates a need to provide time and training for communication, cognitive and reflective skills. In addition, new heuristic biases are described, and problems arising from physician interface with computer-based patient records may have implications for clinic structure, process and training needs. Finally, the data suggest a need to align norms, expectations and resources.

MIGHT FUTILITY DETERMINATIONS RELIEVE SURROGATES OF AN UNNECESSARY BURDEN? D.P. Sulmasy¹, L.A. Jansen¹, W.A. Ury¹; ¹St Vincents Hosp & NY Medical College, New York, NY

PURPOSE: Surrogates are often entrusted to make decisions regarding cardiopulmonary resuscitation (CPR) for patients who lack decision making capacity. Not infrequently, surrogates refuse to authorize Do-Not-Resuscitate (DNR) orders, even when clinicians deem CPR futile in a strict biomedical sense. Few states besides NY have legislation to address these cases. In NY, patients or surrogates must generally sign complex DNR consent forms. However, by statute, NY physicians also have the authority to write DNR orders over the objections of surrogates. Before writing such an order, a physician must conduct a "dispute mediation" meeting with the patient's surrogates. The purpose of this paper is to provide a description and moral analysis of a series of such cases.

METHODS: We present a case series of 6 dispute mediation meetings in which surrogates refused to sign for DNR orders when clinicians thought CPR was futile, convened and moderated by the ethics consult service of a single academic medical center. We use casuistic moral analysis to compare what occurred in these cases with the paradigmatic practice of requiring surrogates to provide substituted judgments and to give formal consent for DNR orders.

RESULTS: The average patient age in our series was 60. Two of these patients were from non-white minority groups, 1 from a recent white immigrant group, 2 had malignancies, 2 had HIV, and 2 had cardiovascular disease. In all 6 cases, surrogates expressed their refusals to sign DNR orders in similar terms. Some voiced religious concerns; others "could not give up hope." In 5/6 cases the surrogates refused to sign the DNR order form even after the dispute mediation meeting. Nonetheless, DNR orders were written in all 6 cases with the implicit assent of the surrogates even in the absence of formal consent. After being told that DNR orders would be written even over their objections, surrogates' comments included: "Well, you gotta do what you gotta do," "I no make decision. Doctor decision," "I know he is dying. Just don't make me sign. I feel like I'm signing a death warrant." A psychiatrist involved in 3 of these cases noted denial, anxiety, pathological ambivalence, and guilt in the surrogates. No attempts were made by surrogates to transfer the patients to other physicians or facilities. All of the patients expired within one week. To date, no lawsuits have occurred.

CONCLUSION: These cases suggest that what appear to be "futility" disputes may in fact represent the complex emotional states that surrogates experience when they believe that they are responsible for making these decisions. All of the surrogates seemed relieved when they were told that a DNR order would be written regardless of whether they had chosen to 'sign' the formal document. These cases raise questions for policy and clinical ethics about the extent to which asking surrogates to consent to DNR when CPR is futile imposes unnecessary burdens and no benefits for either patients or their surrogates.

FACTORS INVOLVED IN CLINICAL ETHICAL DECISION-MAKING: RESULTS FROM A STUDY USING Q SORT METHODOLOGY. W. Wong¹, A.R. Eiser¹, R.G. Mrtek¹, P.S. Heckerling¹; ¹University of Illinois at Chicago, Chicago, IL

PURPOSE: A variety of factors influence clinical ethical decision-making by physicians. The process by which physicians weigh these many factors against one another in order to make a treatment decision is poorly understood and documented. This study examines the influential factors involved in clinical ethical decision-making by using Q Sort by-person factor analysis. **METHODS:** A convenience sample of internal medicine attending physicians and housestaff (n = 35) at a university affiliated medical center were presented with a series of four hypothetical case vignettes and asked to select a specific treatment action for each case. Case vignettes involved clinical decision-making near the end-of-life. Based on the selected treatment action, participants arranged twenty-five self-referenced opinion items into a unique quasi-normal

factor array according to a continuum of preferences from most to least salient. Q sort factor analysis was performed by means of standard factor extraction whereby individuals with correlated Q sorts are clustered into homogenous attitude groups. A second order factor analysis was used to cluster factors to reveal their underlying commonalities. Subjective interpretation of the salient factors in relation to the specific clinical case scenario was performed in order to explain the different ethical perspectives utilized in clinical ethical decision-making.

RESULTS: Data analysis revealed four non-correlated salient factors which guided ethical decision making in the four hypothetical cases. The primary salient factors were defined by the perspectives of: 1) beneficence, 2) patient guided decision-making, 3) patient and family focused decision-making, and 4) legally guided decision-making. Normalized factor scores for statements included in each viewpoint had a Z score > 1.00. Non-influential statements noted across all four cases were the economic impact on the physician, expediency in resolving the situation, and the expense of medical treatment (Z < -1.00).

CONCLUSION: Q sort factor analysis is a useful tool with which to study clinical ethical decision-making because it provides an analytical method for discerning salient factors self-identified by physicians in making ethically-challenging treatment decisions. The four factors identified are grounded in current bioethical values.

CASE SENSITIVITY IN CLINICAL ETHICAL DECISION-MAKING. W. Wong¹, A.R. Eiser¹, R.G. Mrtek¹, P.S. Heckerling¹; ¹University of Illinois at Chicago, Chicago, IL

PURPOSE: A variety of factors influence clinical ethical decision-making by physicians. Few studies have been performed which examine the approach by which physicians make clinical ethical decisions. This study evaluated whether physicians approached clinical ethical dilemmas according to a specific "clinical ideology" or not.

METHODS: A convenience sample of internal medicine attending physicians and housestaff (n = 35) at a university affiliated medical center were presented a series of four hypothetical case vignettes involving clinical decision making near the end of life, and were asked to choose between aggressive, conservative, and consultative courses of action. An aggressive course of action was one in which medical and/or surgical treatment was pursued in spite of poor prognostic outcome. A conservative course of action was a treatment decision that withdrew medical therapy and instituted palliative treatment. A consultative course of action was defined by a request for information and guidance from an ethics committee or family members.

RESULTS: Twenty percent (7/35) of respondents demonstrated a consistent pattern of clinical course of action by choosing either an aggressive, conservative, or consultation course of action three or more times. 8.6% (3/35) of respondents chose an aggressive course of action three or more times, whereas 2.9% (1/35) of respondents chose a conservative course of action three or more times. Three respondents (8.6%) chose a consultative course of action three times or more. A majority (80%, 28/35) of respondents did not demonstrate an identifiable pattern of clinical ethical decision-making by choosing an aggressive, conservative, or consultative course of action for more than two out of the four cases presented.

CONCLUSION: When confronted with an ethically challenging clinical case, most physicians adapted their responses to the particulars of the case presented by demonstrating variability in their treatment approaches. Only a minority of physicians demonstrated a consistent pattern of behavior by selecting either an aggressive, conservative, or consultative treatment option. Physician interpretation of the specific ethical situation of each case guides their clinical ethical decision-making.

"THEY TREATED ME LIKE A LEPER" — STIGMATIZATION AND THE EMOTIONAL BURDEN OF HEPATITIS. C. S. Zickmund¹; University of Iowa, Iowa City, IA

PURPOSE: Hepatitis C is a chronic progressive disease that is typically acquired through contaminated blood products or needle sharing. While the widely known association between this illness and intravenous drug use may lead to stigmatization, no study has explored the consequences of such stereotyping. The goal of this investigation was to identify the prevalence and impact of stigmatization on patients with chronic hepatitis C virus (HCV) infection.

METHODS: Patients with HCV infection attending the University of Iowa Health Care liver clinic were randomly chosen to participate. All patients completed a semi-structured interview and the Sickness Impact Profile (SIP). Two blinded coders analyzed the interviews, identifying the frequency of stigmatization, characteristics of relationships with friends, family and colleagues in the work environment, anxiety, depression, and helplessness.

RESULTS: A total of 193 patients (44.8±0.7 years; 33% women) with HCV infection were enrolled. The analysis of qualitative data demonstrated an excellent intercoder reliability (kappa=0.86). Eighty-five patients (44%) experienced stigmatization which they attributed to the disease. Gender, education, professional status and mode of HCV infection did not influence the likelihood of stigmatization. Negative stereotyping was associated with a higher score of the psychological SIP subscore (11.1±1.5 vs. 15.6±1.9; p =0.05). The responses demonstrated that stigmatization affected all facets of social interactions from family life to the work environment, causing a significant emotional burden for the patients. Stereotyping was associated with increased expressions of depression and anxiety (depression: 58% vs. 37%; p <0.01; anxiety: 72% vs. 47%; p <0.01), problems with family (38% vs. 15%; p <0.01), friends (35% vs. 9%; p <0.01) or coworkers (34% vs. 7%; p <0.001). These experiences led to a sense of loss of control (38% vs. 19%; p <0.01) and problems coping with the disease (38% vs. 19%; p <0.01).

CONCLUSION: Stigmatization is a very common experience of patients with HCV infection and erodes important support structures. The resulting distance from family, friends and colleagues increases feelings of anxiety and depression, common emotional disturbances in this group of patients. These data support the need for broad-based educational efforts to reach patients as well as their relatives and friends. Lowering the emotional impact of negative stereotyping may enhance both the quality of life of patients as well as their compliance with the medical therapy and required life style changes.